Oracle® Fusion Cloud EPM

Administering FreeForm

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Oracle Fusion Cloud EPM Administering FreeForm,

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Contents

Documentation Accessibility

Documentation Feedback

1 Creating and Running an EPM Center of Excellence

2 Creating a FreeForm App

Understanding FreeForm 1
FreeForm App Sources 3
Important Considerations for FreeForm Apps 4
Creating a FreeForm App 5
- Creating a FreeForm App Using an Outline File or Snapshot 6
- Creating a FreeForm App Using the Application Creation Wizard 7

Managing Application Ownership 9
Frequently Asked Questions (FAQ) About FreeForm 9

3 Getting Started

About the Home Page 1
About the Standardized Cloud EPM Platform Artifact Pages 5
De-Support of Legacy Features 9

4 Setting Up Access Permissions

About User and Role Management 1
Application Artifacts That Can Be Assigned Permissions 1
Types of Access Permissions 2
Managing Permissions to Artifacts 4
- About Assigning Permissions to Artifacts, Rules, and Folders 4
- Adding, Changing, and Removing Permissions to Artifacts and Folders 5
- Adding, Changing, and Removing Permissions to Rules and Rules Folders 6

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Reporting on Access Permissions 6
Working with Access Permissions Reports 7
Retrieving Changed Information from Access Control 7

5 Managing Applications

Application Overview 2
Refreshing the Application 2
Managing Cubes 2
Viewing and Managing Cubes 3
Adding Cubes 3
Clearing Cubes 4
About Clearing Cubes 4
Creating Clear Cube Jobs 4
Starting and Deleting Clear Cube Jobs 6
Scheduling Clear Cube Jobs 6
Improving Cube Performance 6
Managing Dimensions 8
Filtering the Dimension View by Cube 9
Creating Dimensions 9
Setting the Evaluation Order in the Simplified Interface 10
Setting User Preferences 10
Working with Variables 11
Working with Substitution Variables 11
About Substitution Variables 11
Creating a Substitution Variable 11
Defining the Columns that Display on the Substitution Variables Page 12
Deleting a Substitution Variable 13
Working with User Variables 13
About User Variables 13
Creating a User Variable 14
Defining the Columns that Display on the User Variables Tab 14
Managing User Variables 15
Setting Limits for User Variables on Forms 16
Deleting a User Variable 16
Viewing Activity Reports 16
Importing and Exporting Data and Metadata 17
Importing Metadata 17
Creating the Metadata Import File 18
Loading the Metadata Import File 21
Exporting Metadata 23
Importing and Exporting Data 24

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Importing Data 25
Driver Member Casting Errors 28
Exporting Data 28
Viewing Data Import and Export Status 31
Validating the Essbase Outline 31
Reverting a Hybrid BSO Application to Non-Hybrid 35
Creating and Refreshing Application Databases 36
Creating Application Databases 36
Before Refreshing the Database 36
Refreshing Application Databases 36
Validating Metadata 38
Adding an Aggregate Storage Outline to an Application 39
About Aggregate Storage 39
Aggregate Storage Outline Cube Characteristics 39
Process for Adding an Aggregate Storage Database to an Application 40
Removing an Application 40
Setting the Daily Maintenance Process Start Time 40
Uploading and Downloading Files Using the Inbox/Outbox Explorer 42
Validating Groovy Scripts 42

6 Using the Member Selector

Working with Members 1
Making Selections 2
Member Relationships 5
Using Wildcards in Searches 6
Selecting Attribute Values as Members 7
Selecting Members for Forms 9
Selecting Substitution Variables as Members 10
Selecting User Variables as Members 11
Selecting UDAs as Members 12

7 Connecting Environments in Cloud EPM

About Connecting Cloud EPM Environments 1
Considerations for Migrating Cloud EPM Connections 3
Creating, Editing, and Deleting Connections to Other Cloud EPM Environments 4
Connecting to External Web Services 6
Specifying Advanced Options for External Connections 7
Navigating Across Cloud EPM Environments 8
Customizing Navigation Flows to Access Other Cloud EPM Environments 9
Grouping Cards from Other Cloud EPM Environments into Clusters 9

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Configuring Cards with Tabs from Multiple Cloud EPM Environments 13
Using Direct URLs to Integrate Connected Environments 16
Copying Direct URLs for Artifacts 17
Exporting All URLs to a CSV File 17

8 Designing Custom Navigation Flows

Understanding Navigation Flows 1
What Can Be Customized in the Business Process Interface? 1
Navigation Flow Customization Categories 2
Navigation Flow Permissions 3
Predefined Navigation Flows 3
Understanding Navigation Flow Status after Import 3
Viewing and Working with Navigation Flows 4
Navigation Flow Design Best Practices and Naming Considerations 5
Creating and Duplicating Navigation Flows 6
Editing a Navigation Flow 6
Activating and Deactivating Navigation Flows 7
Previewing the Runtime View of a Navigation Flow for a User 8
Using Validate to Find Missing Artifacts in Navigation Flows 9
Resolving Navigation Flows that Display a Warning Icon 9
Customizing Labels for Cards, Tabs, and Clusters 10
Customizing Icons for Cards and Vertical Tabs 11
Hiding and Unhiding Clusters, Cards, and Tabs 11
Changing the Display Order of Cards on the Home Page 12
Adding Cards 12
Adding Tabs 14
About Using URLs to Embed Third-Party Pages in Cloud EPM Applications 17
Removing Navigation Flows, Cards, and Tabs 17
Grouping Cards into Clusters 18
Reloading a Navigation Flow 20
Switching Navigation Flows at Runtime 21
Sharing the Current Page URL for Cards, Tabs, and Sub Tabs 21

9 Integrating Cloud EPM and Cloud EDM with Oracle Guided Learning

Enabling Context-Based Activation of OGL Guides in Cloud EPM 3

10 Editing Dimensions in the Simplified Dimension Editor

About Editing Dimensions in the Simplified Dimension Editor 2
Dimension Overview 3

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About Dimensions and Members 3
About Sparse and Dense Dimensions 3
About Dimension Hierarchies 3
Data Type 4
About Custom Dimensions 4
Aggregation Options 4
Storage Options 5
About Entities 6
About Accounts 7
Account Types 7
Saved Assumptions 10
Accounts, Entities, Periods, and Cubes 10
Dimension Design 11
Accessing the Simplified Dimension Editor 22
Working in the Edit Member Properties Grid 22
Switching to Another Dimension 23
Customizing the Column Layout 23
Context Menu Options 25
Viewing Ancestors 25
Showing Member Usage in an Application 26
Focusing Your Editing 26
Finding Members 27
Moving Members 27
Sorting Members 28
Moving Members to Another Hierarchy 28
Working with Member Formulas 28
Copying Member Names from Microsoft Excel 29
Editing Member Properties 30
Accessing Edit Member Properties 30
Adding Members 33
Editing Members 34
Deleting Members 34
Adding Shared Members 35
Assigning Access to Dimension Members 36
Adding, Editing, and Removing Access to Members 36
About Effective Access Permissions to Shared Members 37
Editing Dimension Properties 37
Working with Attributes 39
Deleting Attributes 40
Working with UDAs in the Simplified Dimension Editor 41
Working with Attribute Values 42

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11 Administering Tasks with Task Manager

Setting Up Task Manager 1
- Task Manager Terms 2
- Task Manager Overview 4
- Sample Task Flows 7
- Enabling Task Manager in FreeForm 8
- Managing Task Manager Views, Lists, and Filters 9
  - Viewing Task Manager Lists 9
  - Column Definitions for Task Manager 10
- Managing Task Manager System Settings 18
  - Managing Global Integration Tokens 18
  - Managing Task Manager Organizational Units 20
  - Managing Holiday Rules for Tasks 22
  - Changing Configuration Settings 25
  - Using Oracle Cloud Object Storage to Store Attachments 40
- Managing Task Manager Attributes 46
  - Defining Task Manager Attributes 47
  - Defining Calculation Attributes 48
  - Importing List Attributes 51
  - Editing Attributes 52
  - Duplicating Attributes 52
  - Deleting Attributes 53
  - Viewing Attributes 53
  - Searching for Attributes 53
- Managing Artifact Labels in Task Manager 54

Managing Task Types 56
- Task Types 56
- Creating Task Types 57
  - Setting Task Type Properties 57
  - Setting Task Type Parameters 58
  - Specifying Task Type Instructions 59
  - Specifying Task Type Questions 60
  - Assigning Task Type Attributes 61
  - Working With Task Type Rules 62
- Viewing Task Type History 64
- Editing Task Types 65
- Viewing Task Types 65
- Searching for Task Types 66
- Importing Task Types 66
- Exporting Task Types 67
- Deleting Task Types 68

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Managing Task Templates 68
Creating Task Templates 68
Setting Template Properties 69
Specifying Template Instructions 70
Assigning Viewers to Templates 71
Applying Template Attributes 72
Specifying Day Labels 72
Embedding Templates 73
Working With Template Rules 75
Viewing Template History 78
Opening Templates 78
Adding Tasks to Templates 78
Editing Templates 79
Importing Tasks into Templates 79
Task Import File Format 81
Exporting Tasks to Microsoft Excel 83
Reassigning Users in Templates 84
Creating Schedules from Templates 85
Using the Override Parameter Option When Creating Schedules from Templates 86
Manually Validating Templates 87
Viewing Task Manager Templates 88
Searching for Templates 88
Deleting Templates 89
Managing Tasks 89
Creating Tasks 90
Setting Task Properties 90
Setting Task Parameters 92
Specifying Task Instructions 93
Selecting the Workflow 93
Adding Task Questions 96
Setting Task Viewers 98
Setting Task Predecessors 98
Applying Task Attributes 99
Working With Task Rules 100
Viewing Task History 102
Working with the Task Dialog Box 103
Importing and Exporting Tasks 104
Editing Tasks 105
Adding Attachments 106
Sorting Tasks 106
Searching for Tasks 106
Moving Tasks 106

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Cutting, Copying, and Pasting Tasks 107
Reopening Manual and Automated Tasks 107
Submitting Tasks 110
Approving or Rejecting Tasks 110
Claiming or Releasing Tasks 111
Managing Task Reassignments 112
Automated Task Execution 112
Canceling Tasks 113
Deleting Tasks 113
Managing Services 114
Managing Schedules 114
Manually Creating Schedules 115
Setting Schedule Properties 115
Adding Instructions to Schedules 116
Assigning Schedule Viewers 117
Applying Schedule Attributes 118
Adding Day Labels 118
Working With Schedule Rules 119
Setting Required Task Parameters 121
Opening Schedules 121
Editing Schedules 121
Adding Tasks to Schedules 121
Importing Tasks into Schedules 122
Updating Tasks in Schedules 124
Reassigning Users in Schedules 124
Authorizing Process-Automated Tasks 125
Setting Schedule Status 126
Viewing Schedule History 128
Validating Schedules 128
Locking Schedules 128
Viewing Schedules 129
Searching for Schedules 129
Deleting Schedules 130
Managing Task Manager Integrations 130
Managing Integrations with Cloud EPM Services 131
Adding Pre-built Integrations in Cloud EPM 132
End User Integrations for Cloud EPM 133
Automation Integrations for Cloud EPM 137
Copy and Delete Integration Files 172
Managing Integrations with Cloud and On-Premises Applications 174
End User Integrations for Oracle Cloud ERP 174
Event Monitoring Integrations for Oracle Cloud ERP 179

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Creating Custom Integrations 187
Creating End User Integration 187
Creating Custom Process Automation Integrations 187
Creating Custom Event Monitoring Integrations 191
Working with Integrations 195
Creating Integrations 196
Viewing Integrations 198
Editing Integrations 198
Searching for Integrations 199
Validating Integrations 199
Deleting Integrations 199
Managing Connections 199
Adding Connections 200
Editing Connections 200
Deleting Connections 201
Managing Alert Types for Task Manager 201
Creating Alert Types 202
Setting Alert Type Properties 203
Specifying Alert Type Instructions 205
Selecting the Alert Type Workflow 205
Assigning Alert Type Viewers 206
Adding Questions for Alert Types 207
Applying Alert Type Attributes 208
Viewing Alert Type History 209
Viewing Alert Types 209
Column Definitions for Alerts 210
Editing Alert Types 211
Searching for Alert Types 211
Deleting Alert Types 212
Managing Views in Task Manager 212
Viewing Currencies in Task Manager 212
Using Reports 213
Generating Custom Reports for Task Manager 213
Creating a Task Manager Query 214
Creating a Template 218
Setting Up a Report Group 218
Creating a Report 219
Searching and Filtering Queries, Report Groups, or Reports 220
Generating the Report 221
Understanding Reports Security 222
Using Task Manager Report Binders 223
Generating Cloud EPM System Reports for Task Manager 225

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Working with Views and Filters 225
Working with Filters 225
About Filters 225
Viewing the Existing Saved Filters 226
Creating a New Filter 226
Using the Filter Bar to Adjust Your View 227
Clearing Filter Conditions 228
Removing a Filter 228
Editing a Filter 228
Duplicating a Filter 229
Saving a Filter 229
Publishing a Filter 229
Deleting a Filter 230
Displaying Columns for Filters 230
Reordering Columns for Filters 230
Working with Views 231
About Views 231
Working With List Views 232
Working with Pivot Views 233
Working with Chart Views 237
Working With Gantt Chart Views 247
Working With Calendar Views 248
Saving Views 248
Editing Saved Views 249
Deleting Views 249
Publishing Views 249
Setting a Default View 249
Exporting List Views and Pivot Views 250
Managing Operational Dashboards 250
Overview of Operational Dashboards 251
Viewing and Opening Dashboards 251
Operational Dashboards: Design Mode 252
Creating Operational Dashboards 254
Defining Operational Dashboard Properties 254
Defining Access to Operational Dashboards 255
Setting the Object Type Within Operational Dashboards 256
Specifying View Options for Operational Dashboards 256
Editing Operational Dashboards 257
Duplicating Operational Dashboards 257
Deleting Operational Dashboards 257
Exporting Operational Dashboards 258
Importing Operational Dashboards 258

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Running Operational Dashboards 259
Displaying Columns for Viewing Operational Dashboards 259
Reordering Columns for Viewing Operational Dashboards 259
Copying a URL for Operational Dashboards 260
Applying Filters to Data Displayed Within Operational Dashboards 260
Example: Task Manager Operational Dashboard 260
Chart View: Task Completion - Plan vs Actual 261
Chart View: Tasks Completed Late 263
Chart View: Status by Workday 264
Chart View: Completion by Region 265

12 Designing Dashboards

The Power of Dashboards 1
Creating and Managing Dashboards 3
About Dashboard Versions 4
Viewing Dashboards - Standardized Interface 6
Viewing Dashboards - Legacy Interface 9
Converting Dashboard 1.0 Dashboards to 2.0 12
Creating and Managing Dashboards 1.0 13
Concepts in Designing 1.0 Dashboards 13
Creating Dashboard 1.0 Dashboards 15
Creating Dashboards Containing Master Forms and Details 17
About Your Dashboard's Layout - Version 1.0 18
About the Gauge Chart Type 18
About the Tile Chart Type 20
Customizing Dashboard Colors 22
About Global and Local POVs in 1.0 Dashboards 23
Dashboard POVs and Valid Intersections 25
Creating and Managing Dashboards 2.0 25
Concepts in Designing Dashboard 2.0 Dashboards 25
Object Palette 28
Dashboard Toolbar 29
Dashboard Workspace 30
Properties Panel 31
Dashboard Components 34
Considerations for Dashboard 2.0 36
Creating Dashboard 2.0 Dashboards 38
Working with the Member Selector When Designing Dashboards 2.0 39
About Forms 2.0 Grids in Dashboard 2.0 40
Using Tables in Dashboards 41
About Global and Local POVs in Dashboard 2.0 43

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About Quick Analysis 46
About Dashboard Charts 48
About the Geomap Chart Type 49
About the Pyramid Chart Type 50
About the Waterfall Chart Type 51
About the Gauge Chart Type for Dashboard 2.0 54
About the Radar Chart Type for Dashboard 2.0 56
About the Combination Chart Type for Dashboard 2.0 58
About the Tile Chart Type for Dashboard 2.0 60
About the Sankey Chart Type for Dashboard 2.0 62
Dashboard Design Guidelines 64
Dashboard Types 65
Dashboard Layout Guidance 69
Dashboard Styles 71
Color Sequencing in Charts 72
Recommendations for Improving Dashboard Design 73
Summary 78

13 Managing Infolets

About Infolets 1
Anatomy of an Infolet 3
Determining Infolet Content 6
Designing Forms for Infolets 7
Designing Charts for Infolets 7
Using the Infolets Designer 8
Creating Infolets 10
Working with Infolets - Standardized Interface 11
Working with Infolets - Legacy Interface 13
Customizing the Interface to Access Infolets 15

14 Configuring IPM

About IPM 3
Enabling AI in FreeForm 3
About IPM Insights 4
Considerations for IPM Insights 8
About Using Calendars and Events in IPM Insights, Advanced Predictions, Auto Predict, and Predictive Planning 10
Configuring Calendars and Events 13
Configuring IPM Insights 16
Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions 21

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Configuring Additional Insight Settings 24
Applying Scaling and Precision to IPM Insights at the Dimension Level 25
Setting Up Predictions to Run Automatically with Auto Predict 26
About Auto Predict 26
Auto Predict Considerations 28
Creating Auto Predictions for Forecasts 28
Configuring Advanced Auto Predict Options 30
About Auto Predict and Predictive Planning Prediction Accuracy 34
Running Auto Predict Predictions 34
Editing, Deleting, or Duplicating Auto Predict Prediction Definitions 35
Reviewing Detailed Reports for Auto Predict Predictions 35
Setting Up Advanced Predictions in FreeForm 37
About Advanced Predictions 37
Considerations for Advanced Predictions 40
Configuring Advanced Predictions 40
Defining the Model Scope 43
Selecting the Output to Predict 45
Selecting Drivers as Input 46
Preparing the Data 47
Selecting Algorithms for Advanced Predictions in FreeForm 50
Defining Confidence Intervals 52
Including Events in Advanced Predictions 53
Running and Scheduling Advanced Predictions 53
Reviewing Detailed Reports for Advanced Predictions 53
Running and Scheduling Insights 54
Managing Insight Definitions and Generated Insights 55
Managing Tags 55
Bring Your Own ML: About Machine Learning Model Import 56
Considerations for Bring Your Own ML 57
Importing an ML Model 58
Deploying an ML Model to Planners 59

15 Defining Valid Intersections and Cell-Level Security

Defining Valid Intersections 1
Understanding Valid Intersections 1
Valid Intersection Groups 2
Valid Intersection Rules 2
Anchor and Nonanchor Dimensions 3
Valid Intersection Examples 3
Redundancy or Overlap in Valid Intersection Rules 6
Shared Members and Valid Intersection Rules 6

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Substitution Variables and Valid Intersection Rules 6
Evaluation Order 7
Creating Valid Intersections 7
Managing Valid Intersections 8
Viewing Valid Intersections 8
Filtering Valid Intersections 9
Importing and Exporting Intersections 10
Changing the Valid Intersection Group Evaluation Order 12
Disabling and Enabling Valid Intersection Groups 13
Editing Details for a Valid Intersection Group 13
Duplicating Valid Intersection Groups 14
Deleting a Valid Intersection Group 15
Suppressing Invalid Data in Forms 15
Clearing Invalid Data 15
About Invalid Data 16
Working With Invalid Intersection Reports 16
Clearing Data at Invalid Intersections 17
Working with Valid Intersections 17
Working with Valid Intersections in Application Forms 17
Working with Valid Intersections in Calculation Manager Rule Runtime Prompts 18
Defining Cell-Level Security 18
Understanding Cell-Level Security 19
Creating Cell-Level Security Definitions 20
Viewing Cell-Level Security Definitions 21
Filtering Cell-Level Security Definitions 22
Importing and Exporting Cell-Level Security Definitions 22
Testing Cell-Level Security 24
Reordering the Cell-Level Security Definitions List 25

16 Defining Data Maps and Creating File-Based Integrations

17 Managing Jobs

How Jobs Save You Time 1
Viewing the Status of Jobs 1
Scheduling Jobs 2
Job Types 3
Scheduling Jobs to Run Later 5
Scheduling Hourly Jobs 6
Editing and Canceling Jobs 9

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Duplicating Jobs 10
Canceling Rules Jobs and Ruleset Jobs 11
Downloading Export Files From Your Outbox 11

18 Auditing Tasks and Data

Auditing Overview 1
Enabling Audit Tracking 3
Viewing Audit Details 3

19 Managing Data Validation and Format Rules in Forms

About Data Validation Rules 1
Creating and Updating Data Validation Rules in Forms 1
Formatting Cells 5
Viewing Data Validation Rules 6
Order of Evaluation and Execution for Data Validation Rules 6
Conditions Supported by the Rule Builder 6
If Condition Values 7
Then Condition Values 14
Range Condition Values 15
Format Condition Values 15
Data Validation Conditional Operators 16
Data Validation Rule Scenarios 17
About Format Rules in Forms 23
Creating and Updating Format Rules in Forms 25
Cell Formatting Options in Forms 27
Viewing Format Rules 29
Order of Precedence for Format Rules 30
Applying Format Rules to Forms at Runtime 31
Format Rule Scenarios 32

20 Managing Application and System Settings

What Application and System Settings Can I Specify? 1
Defining User Variables 9
Customizing Your Display 10
Announcing Upcoming Events 10
Specifying Artifact Labels 11
Which Artifact Labels Can be Localized? 11
Working With the Artifact Labels Grid 12
Adding Languages and Defining Localized Artifact Labels 12

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Exporting and Importing Artifact Labels for Editing 13

21 Accessing More Administrative Tasks

About the Navigator Menu 1
Administering Data Load Settings 2
Importing Using Data Integration 3
Administering Action Menus 3
Creating and Updating Action Menus 3
Working with Action Menu Items 4
Defining Action Menu Items 5
Enabling Action Menus for Ad Hoc Grids 7
Administering Alias Tables 8
About Aliases 8
About Alias Tables 9
Creating an Alias Table 9
Defining the Columns that Display on the Alias Tables Page 10
Editing, Copying, Clearing, and Deleting Alias Tables 10
Specifying a Default Alias Table and Setting Member and Alias Display Options 11
Administering Dimensions 11
About Dimensions 12
Working with Dimension Hierarchies 12
About Custom Dimensions, Entities, Accounts, Periods, and Cubes 13
Adding or Editing User-Defined Custom Dimensions 14
Working with Members 16
Working with Attributes 23
Working with Attribute Values 26
Customizing Calendars 28
Setting up Dynamic Time Series Members 32
Working with UDAs 34
Working with Member Formulas 35
Administering Forms 40
About Forms 40
Form Components 40
Form Design 42
Understanding Implied Sharing in Forms 45
Creating Forms 46
Defining the Layout 47
About Precision Settings 61
Setting Form Precision and Other Options 61
Moving Data Using Smart Push 62
Creating Asymmetric Rows and Columns 62

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Adding Formula Rows and Columns 63
Defining Form Page and Point of View 63
Designing Specific Types of Forms 64
Working with Forms and Form Components 66
Selecting and Opening Forms and Folders 66
Previewing Forms 67
Printing Form Definitions 68
Searching for Forms 68
Editing Forms 68
Moving, Deleting, and Renaming Forms 69
How Cell Formatting in Smart View Persists in FreeForm 70
Managing Forms and Folders 70
Creating Folders 71
Working with Folders 71
Administering Rules 72
About Rules 72
Adding and Removing Rules in Forms 74
Setting Business Rule Properties 75
Viewing Rules Usage 76
About Runtime Prompts 79
Understanding Runtime Prompts 79
Using Groovy Rules 83
About Groovy Business Rules 83
Java API Reference for Groovy Rules 84
Groovy Business Rule Examples 84
Groovy Business Rule Tutorial Videos 85
Groovy Business Rule Tutorials 85
Groovy Rule Business Scenarios 87
Administering Rules Security 99
Assigning Access to Rules 99
Adding, Editing, and Removing Access to Rules 99
Assigning Access to Groovy Templates 100
Administering Smart Lists 100
Working with Smart Lists 101
Synchronizing Smart Lists in Reporting Applications 102
Setting Smart List Properties 102
Defining Smart List Entries 104
Previewing Smart Lists 105
Displaying #MISSING with Smart Lists 105
Administering Task Lists 105
Working with Task Lists 106
Adding Instructions to Task Lists 106

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Adding Tasks to Task Lists 106
Editing Task Lists 109
Editing Tasks 109
Copying and Moving Tasks 110
Moving and Reordering Task Lists 110
Clearing Task Lists 111
Deleting Tasks and Task Lists 111
Assigning Access to Task Lists 112
Adding Access to Task Lists 112
Changing and Removing Access to Task Lists 112
Clearing Cell Details 113
Copying Data 114
Administering Application Monitoring 116
About Application Monitoring 116
Assumptions 116
How Application Monitoring Works 116
Using Application Monitoring Graphs 117
Launching Application Monitoring 118
Modifying Artifacts for Optimal Performance 119

A Naming Restrictions

Naming Restrictions for Applications and Databases A-1
Naming Restrictions for Dimensions, Members, and Aliases A-2
Dimension and Member Names in Calculation Scripts, Report Scripts, Formulas, Filters, and Substitution Variables A-6
Restrictions for User and Group Names A-6

B Form Formula Functions

About Form Formula Functions B-1
Working with Formulas B-1
Creating Formulas B-1
Editing Formulas B-2
Deleting Formulas B-2
Formula Functions B-2
Arguments B-3
Numeric Arguments B-4
Row, Column, or Cell Reference Arguments B-4
Property Arguments B-5
Embedded Functions as Arguments B-7
Abs B-8

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Average B-8
AverageA B-9
Count B-10
CountA B-11
Difference B-12
Eval B-13
IfThen, If B-13
Notes on Conditions B-15
Complex Conditions B-16
Max B-16
Min B-17
Mod B-17
PercentOfTotal B-18
Pi B-19
Product B-19
Random B-20
Round B-20
Sqrt B-21
Sum B-21
Truncate/Trunc B-22
Variance/Var B-23
VariancePercent/VarPer B-25

C Using Smart View to Manage Applications

About Managing Applications in Smart View C-1
Installing Smart View and the Admin Extension for Application Management C-2
Controlling the Display of Application Management Options in Smart View C-4
Downloading the Application Templates C-4
Downloading the Template in Smart View C-4
Downloading the Application Template Zip File from the Web Interface C-5
Creating an Application C-6
Working with Artifacts in the Application Template C-7
About Working with Artifacts in the Application Template C-8
Application Definition C-8
Dimension Definition C-12
Attribute Dimension Definition C-17
Data Definition C-18
Substitution Variable Definition C-19
Security Definition C-21
Advanced Settings Definition C-22
Updating an Application in Smart View C-26

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Deleting an Application C-27
Planning Admin Extension and Office AutoCorrect C-27

D Using Smart View to Import and Edit Application Metadata

About Using Smart View to Work with Application Metadata D-1
Installing Smart View and the Admin Extension for Editing Dimensions D-2
Using Smart View Grids to Import and Edit Application Metadata D-3
About the Smart View Grid D-3
The Smart View Grid and Ribbon Display D-4
Guidelines for Using the Smart View Grid D-5
Default Metadata Dimension Member Properties D-6
Importing Dimensions in Smart View D-8
Importing Dimensions in Smart View for Office D-9
Importing Dimensions in Smart View (Mac and Browser) D-10
Editing Members in Smart View D-12
Adding Application Members in Smart View D-12
Adding Members in Smart View D-13
Guidelines for Adding Members in Smart View D-14
Moving Members in Smart View D-14
Guidelines for Moving Members in Smart View D-15
Working with Attribute Dimensions D-15
Adding Attribute Dimension Members in Smart View D-15
Adding Attribute Dimension Members in Smart View for Office D-15
Adding Attribute Dimension Members in Smart View (Mac and Browser) D-17
Associating Attribute Dimension Members with Dimension Members D-18
Designating Shared Members in Smart View D-20
Refreshing Databases D-21
Refreshing Databases in Smart View for Office D-21
Refreshing Databases in Smart View (Mac and Browser) D-21
Planning Admin Extension and Office AutoCorrect D-23

E FreeForm Best Practices

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1

Creating and Running an EPM Center of Excellence

A best practice for EPM is to create a CoE (Center of Excellence).

An **EPM CoE** is a unified effort to ensure adoption and best practices. It drives transformation in business processes related to performance management and the use of technology-enabled solutions.

Cloud adoption can empower your organization to improve business agility and promote innovative solutions. An EPM CoE oversees your cloud initiative, and it can help protect and maintain your investment and promote effective use.

The EPM CoE team:

- Ensures cloud adoption, helping your organization get the most out of your Oracle Fusion Cloud Enterprise Performance Management investment
- Serves as a steering committee for best practices
- Leads EPM-related change management initiatives and drives transformation

All customers can benefit from an EPM CoE, including customers who have already implemented EPM.

How Do I Get Started?

Click to get best practices, guidance, and strategies for your own EPM CoE: Introduction to EPM Center of Excellence.

Learn More

- Watch the Cloud Customer Connect webinar: Creating and Running a Center of Excellence (CoE) for Cloud EPM
- Watch the videos: Overview: EPM Center of Excellence and Creating a Center of Excellence.
- See the business benefits and value proposition of an EPM CoE in *Creating and Running an EPM Center of Excellence*.

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2

Creating a FreeForm App

FreeForm apps use an open dimensional cube construct allowing you to create cubes with any dimension combination you need.

Related Topics

- Understanding FreeForm
- FreeForm App Sources
- Important Considerations for FreeForm Apps
- Creating a FreeForm App
- Managing Application Ownership
- Frequently Asked Questions (FAQ) About FreeForm

Understanding FreeForm

FreeForm is a Reporting, Analysis, and Planning application. FreeForm enables you to create an application with the cubes and dimensions of your choice without being constrained by the cube and dimension limitations imposed by standard applications. With FreeForm, you can model and build your own cubes while preserving the ability to leverage business process functionalities.

FreeForm can be accessed using Oracle Smart View for Office or Oracle Fusion Cloud Enterprise Performance Management screens; they support Groovy scripts for custom functions, and business rules for calculations.

You can create applications with multiple cubes or you can create an application with a single cube and then add more cubes later (up to 12 cubes total).

Table 2-1 FreeForm Features at a Glance

What is FreeForm?	What can I create with FreeForm?	What use cases can I solve with multi-cube FreeForm?	What's special about multi-cubeFreeForm?
FreeForm is an EPM application with no dimension prerequisites, to create multiple cubes with full flexibility in dimensions based on business use case needs.	Create up to 12 cubes that can use any of the 29 dimensions in any combination for each cube based on use case needs. (26 custom dimensions plus three account, period, and entity dimension types.)	• Unified Reporting • Essbase SaaS Migration • Excel model migration • Financial and Operational Reporting or Planning across any industry	• Only place in Cloud EPM where you can create a reporting application with only ASO cubes • Only place in EPM where you can import Oracle Essbase files (using Migration or outline load) to create an application

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Table 2-1 (Cont.) FreeForm Features at a Glance

What is FreeForm?	What can I create with FreeForm?	What use cases can I solve with multi-cube FreeForm?	What's special about multi-cubeFreeForm?
Oracle Cloud EPM Platform provides the technology framework guardrails and features to help customers succeed.	Create an application that is all ASO reporting or all Hybrid BSO planning cubes or a combination of ASO and BSO.	• Financial Reporting • Operational Reporting • Revenue Planning • Expense Planning • Demand Planning • Supply Planning • And so on	Gives full control to customers to work with Sales to create a best fit solution that best meets their requirements for reporting or planning

Note

In FreeForm, the Scenario, Version, Period, and Year dimensions can be defined as custom dimensions.

The following features are not available in FreeForm:

- Sandboxes
- Approvals
- Rolling forecasts
- Task Manager

About Multi-Cube FreeForm Apps

- The maximum number of custom dimensions allowed in a FreeForm app is 26.
- Cubes can share dimensions or have standalone dimensionality.
- All dimensions can be custom; or native account, period, and/or entity dimensions can be inherited by the cubes.
- Cross-cube data maps, Smart Push, and Copy Data support are available for FreeForm app types.

Multi-cube FreeForm apps are constrained by these existing business process assumptions:

- Members must be unique across all the cubes within FreeForm.
- Governors are applicable at the application level and not at the cube level.
- The input outlines must not be enabled to allow duplicate member names.

Videos

Your Goal	Learn How
Get an introduction to FreeForm.	Introduction to FreeForm Apps in Cloud EPM

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Your Goal	Learn How
Create FreeForm apps from on-premises Essbase outline (OTL) files and snapshots.	Creating FreeForm applications from on-premises Essbase outline files and snapshots
Create multi-cube FreeForm apps with an Essbase outline	Creating Multi-Cube FreeForm Apps from Essbase Outlines
Create multi-cube FreeForm apps with a BSO or ASO cube	Creating Multi-Cube FreeForm Apps from BSO and ASO Cubes
Learn how to create FreeForm apps using an Excel template. Using the Excel template, you learn how to define application properties, manage cubes, create dimensions and members, attributes, access permissions, and load data into the FreeForm application.	Creating Multi-Cube FreeForm Applications with an Excel Template - Part 1
Learn how to update FreeForm apps using an Excel template and verifying changes in the web interface. After creating your FreeForm application using an Excel template, learn how you can use the same template to make modifications to dimensions, members, substitution variables, and security. Then, in the Planning web interface, assign Application Management options, review the application properties and definition, and the data you imported.	Creating Multi-Cube FreeForm Applications with an Excel Template - Part 2
This overview reviews the best practices in setting up dimensions in FreeForm applications. You will see how to model dimensions and cubes around your business process, use Account, Entity, Period, and Custom dimension types, assign dimensions to BSO and ASO cubes, and review performance considerations using Application Diagnostics.	Best Practices for Setting Up Dimensions in FreeForm

FreeForm App Sources

You can build FreeForm apps by creating dimensions using the application creation wizard. Alternatively, you can use an outline (OTL) file or snapshot from an on-premises release 11.1.2.4.xxx (or later) single cube Oracle Essbase application to create a FreeForm app.

When you use an Essbase cube OTL file or Essbase application snapshot as the source for a FreeForm app, Account, Period (time), and Entity (country) dimensions are created and mapped automatically. Additionally, Version and Scenario dimensions of the Essbase application are created as custom dimensions. As a result, out of the box Workflow functionality is not supported in FreeForm apps.

Service Administrators build a FreeForm app using the application creation wizard or these sources:

- **An Outline file from on-premises release 11.1.2.4.xxx (or later) single cube Essbase application**

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You may use an OTL file from an on-premises release 11.1.2.4.xxx (or later) single cube Essbase application to create the structure of the FreeForm app. New cubes may be added or created for this FreeForm app at a later time.

Because the OTL file name is assigned to the cube created for the FreeForm app, the file name must be eight characters or less. Longer file names will cause the process to fail.

Generally, the OTL file is available in the `EssbaseServer/essbaseserver1/app/<app_Name>` directory within your on-premises Essbase deployment.

Because the OTL file does not contain application data, you must extract Essbase data to a file. Use MaxL, Essbase Studio, or another tool to export data from an Essbase application. The data must be exported in an Essbase data file format.

After creating the application, import the extracted data, selecting Essbase as the source type. You can import from a locally stored data file or from a file that was uploaded to the Oracle Fusion Cloud Enterprise Performance Management inbox. For detailed instructions, see Importing Data.

- **An application snapshot from an on-premises release 11.1.2.4.xxx (or later) single cube Essbase application**

  This option simplifies application creation by migrating an existing Essbase application snapshot (ZIP file) that was created using Migration. This process automates dimensions loading, substitution variables creation, calculation scripts conversion as graphical rules, and data load. New cubes may be added or created for this FreeForm app at a later time.

Important Considerations for FreeForm Apps

> **Note**
>
> Do not attempt to create FreeForm apps by importing Oracle Essbase Migration snapshots into a Oracle Fusion Cloud Enterprise Performance Management environment. Importing snapshots using Migration is supported only for snapshots created from a previous FreeForm app.

- Data maps and other multi-cube features are unavailable if you create a single cube FreeForm app using the application creation wizard.
- Essbase snapshots with Custom Defined Functions (CDF) and CDF references cannot be used to create FreeForm apps.
- Make sure that the OTL file or Essbase application snapshot that you are using does not contain objects with system restricted names; for example, a member named `FY02` in the Year dimension. You must rename or remove such restricted names before generating the OTL file or snapshot that you plan to use to create the application. See Naming Restrictions.
- Ensure that the OTL file or Essbase application snapshot that you are using does not have member and alias names that are longer than 80 characters. If member and alias names are longer than 80 characters, you must remove or shorten them or the import will not be successful. See Naming Restrictions.
- If the OTL file that you are using contains Year Total and Beginning Balance members, the Beginning Balance member must be the first child under the Period dimension, and Year Total must be the second child under Period dimension. These members require special handling and can be moved to appropriate positions by modifying the .csv files in the Migration snapshot before restoring the snapshot.

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- If you plan to use an OTL file or snapshot stored in your Cloud EPM environment as the source, upload the file before starting the application creation process.

  Use the `uploadFile` EPM Automate Utility command or Migration to upload the OTL file or snapshot to a Cloud EPM environment.
- Imported data may not be editable in FreeForm apps. At times, the data is set as Read-only.
- Member outline must be unique in the application.
- If you create a FreeForm app by importing an OTL or snapshot, you can add or create new cubes for this FreeForm app at a later time.
- If you create a FreeForm app by adding new cubes in the application creation wizard, an Essbase cube OTL or Essbase application snapshot cannot be imported into this FreeForm app at a later time to create new cubes.
- Currency symbols don't display in FreeForm applications.

Do not use the following reserved words to name dimensions and members in FreeForm:

Table 2-2 Reserved words that should not be used as member names in FreeForm

Reserved words			
BU Version_1	Users	Attribute Dimensions	HSP_Entity
ConsolidatedData	Groups	Default	HSP_Period
BaseData	Calendars	Task Lists	HSP_Version
SandBoxData	Currencies	Menus	HSP_XCRNCY
Super User	Predefined	CalcMgrRules	HSP_Years
Strategic Planner	FX_Tables	CalcMgrRulesets	HSP_View
Service Administrator	Forms	CalcMgrVariables	HSP_Metric
CalcMgrTemplates	Aliases	HSP_Rates	
Root	Cubes	HSP_Scenario	
Dimensions	Planning Units	HSP_Account	

Creating a FreeForm App

Before you create a FreeForm app, review the important considerations and the reserved words that should not be used as member names in FreeForm apps. See Important Considerations for FreeForm Apps.

The Landing Page

The landing page is your starting point for creating a FreeForm app and for viewing overview videos that help you get started.

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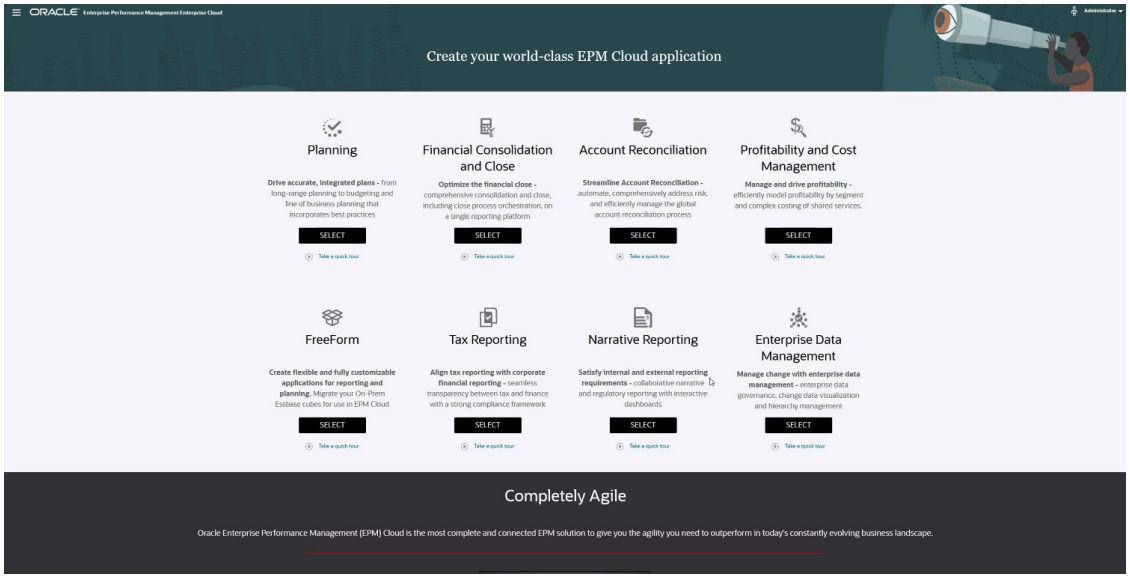
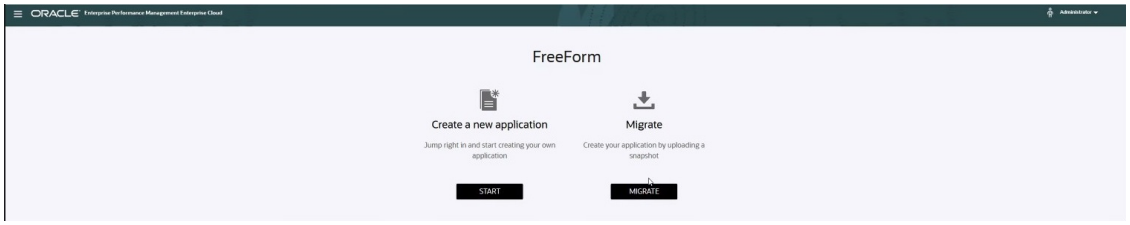


FreeForm App Creation Options

To create a FreeForm app, click **SELECT** under the FreeForm description. You'll see the following options:



- **Create a new application**: Click **START** to manually create a FreeForm app based on your own models using custom dimensions. During this process, you can map the custom dimensions that you want to use as Entity, Period, and Account dimensions, and you can also add other custom dimensions to support your model. See Creating a FreeForm App Using the Application Creation Wizard.
- **Migrate**: Click **MIGRATE** to create a FreeForm application from a snapshot that you previously uploaded to the environment. Only one snapshot is allowed. See Creating a FreeForm App Using an Outline File or Snapshot.

Creating a FreeForm App Using an Outline File or Snapshot

To continue creating a FreeForm app using an outline file or snapshot:

1. Select the location of the source OTL file or the Essbase application snapshot:
   - Select **Local** to access the source OTL file or snapshot from the computer from which you are currently accessing Oracle Fusion Cloud Enterprise Performance Management.

     For **Import File**, click **Choose File** and then select the source OTL file or snapshot.
   - Select **Inbox** to access the source OTL file or snapshot from your Cloud EPM environment.

     From **Select One**, select the source OTL file or snapshot.
2. Click **Next**.

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3. Review application information and then click **Create**.

   When application creation is complete, Cloud EPM Home page is displayed.

4. Make sure that errors were not reported during application creation.
   - Open the Jobs console by clicking **Application**, and then **Jobs**
   - Verify that the **Create Database** and **Process Outline** activities finished without errors. Correct any reported errors.

5. **Optional:** If you created the application using an OTL file as the source, import application data. See Importing Data.

   If you used a snapshot as the application source, the application creation process automatically imports data.

6. Create application users in identity domain and assign application roles as needed. See Managing Users and Roles in *Getting Started Guide for Administrators*.

7. Set up access permissions as needed. See Setting Up Access Permissions.

Creating a FreeForm App Using the Application Creation Wizard

To continue creating a FreeForm app using the application creation wizard:

1. For **No. Of Cubes**, select the number of cubes you'd like to use to create your FreeForm app. You can add up to 12 cubes now, or you can start with one cube and then add more cubes later on the Application Overview page.

   > **Note**
   >
   > If you create a single cube FreeForm app, data maps and other multi-cube features are unavailable until you add cubes to the FreeForm app. Also for a data map to work there needs to be at least one block storage (BSO) cube, and the source for the data map needs to be a BSO cube.

2. In **Cube Name**, enter a name for each cube, maximum eight characters.

3. Select **Is ASO**, if the cube is an aggregate storage cube. If the **Is ASO** checkbox is cleared, then the cube is a Hybrid block storage cube.

   > **Note**
   >
   > Reporting applications that use ASO cubes, generally, store data in aggregate views and can handle a higher number of dimensions than BSO without adversely affecting performance.

4. Click **Next**.

   The **Create Dimensions** page is displayed. From this screen, you can create up to three dimensions and assign one each to Account, Entity, and Period default dimension types. Additional dimensions can be added later in the web interface.

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> **Note**
>
> FreeForm apps do not require standard dimensions. You can create an application with only custom dimensions containing members of your choice. The dimensions you create and its structure are not governed by the constraints imposed on standard application dimensions.

5. In **Create Dimensions**, create and map default application dimensions. Complete a step:
   - To create dimensions and map them to default dimension types, select the **Enabled** check box and then type in dimension names.

     > **Note**
     >
     > Enabling dimensions is not automatic. Just like custom applications, you must manually enable dimensions for FreeForm apps. You can do this now, or you can enable them later.

   - To create a dimension and assign a dimension type, select the check box in a row and then type in a dimension name.
   - To create a shell application, do not change anything in this screen.

     You can add and map dimensions to shell applications after the application creation process is complete.
6. Click **Next**.
7. In **Review**, verify the settings that you selected and then click **Create**.

   The application creation process may take a few minutes.
8. In the Application Creation Status screen, click **OK**.

   The Home page is displayed.
9. **Optional:** Create custom dimensions, if needed.
   a. Click **Application**, then **Overview**, and then select the **Dimensions** tab.
   b. For **Cube**, select a cube that you created for your FreeForm app or select **All**.
   c. Click **Create**.
   d. Complete the dimension details for each dimension you want to add. For descriptions of the dimension properties, see Editing Dimension Properties.

      For FreeForm applications, you can select the dimension type. Dimension types can include:

      - Account
      - Entity
      - Period
      - Custom

      Notes about dimension type:

      - A dimension type is not listed if you've already created it (for Account, Entity, Period) or if you've exceeded the maximum number of dimensions (for Custom).
      - You can't edit the dimension type after you've created the dimension.

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Select **Enabled** to indicate whether the dimension is to be used in this cube.

> **Note**
>
> Enabling dimensions is not automatic. Just like custom applications, you must manually enable dimensions for FreeForm apps.

e. Click **Done**.

f. Click **Action** and then **Refresh Database** to refresh the cube.

10. **Optional:** Import application data. See Importing Data.
11. **Optional:** If you created a shell application with a view to import a snapshot, access Migration, then delete the application, and then complete the import process.
    a. Upload a snapshot from a FreeForm app to your environment. See Uploading Archives to the Service in *Administering Migration* for detailed instructions.
    b. Import the snapshot. See "Importing a Backup to Create a Clone of Another Environment" in Importing Artifacts and Application from a Snapshot in *Administering Migration* for detailed instructions.
12. Create application users in identity domain and assign application roles as needed. See Manage Users and Roles in *Getting Started Guide for Administrators*.
13. Set up access permissions as needed. See Setting Up Access Permissions.

Managing Application Ownership

The Service Administrator who creates the application is auto-assigned the Application Owner function. When a Service Administrator with the Application Owner function is deleted or when the user's application role assignment is downgraded (for example, from Service Administrator to Power User), the Application Owner function is reassigned to the next available Service Administrator user (in alphabetical order) for the application, and then role of the current Application Owner user is deleted or changed to sync up with Access Control.

If there are no available Service Administrators, then the Service Administrator user is not deleted or the role is not changed. Oracle will contact you with next steps to resolve the issue.

Any Service Administrator can use **System Settings** to take over or reassign the Application Owner role to any available Service Administrator.

See What Application and System Settings Can I Specify?.

Frequently Asked Questions (FAQ) About FreeForm

FreeForm Business Process interactive FAQ with documentation links:

- Business and Essbase Users FAQ
  - What is the FreeForm Business Process, and what can you do with it?
  - What are the advantages of using the FreeForm Business Process in Cloud EPM?
  - Are Cloud EPM IPM and GenAI Reporting features available with the FreeForm Business Process?
  - If we are an Essbase customer, why move to the FreeForm Business Process?

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  - Should Excel-focused users move to the FreeForm Business Process?
  - What is the learning curve for SmartView Essbase users moving to FreeForm Apps?
  - What are the web-based benefits for Essbase customers moving to the FreeForm Business Process in Cloud EPM?
  - What is the advantage of using the FreeForm Business Process as your corporate reporting solution?
  - What is the difference between Essbase and FreeForm apps in Cloud EPM?
  - Are all FreeForm apps custom built?
  - Can you create custom dimensions and rules?
  - What admin interfaces are available?
  - Can I use Narrative Reporting with the FreeForm Business Process?
  - Is direct Smart View and Oracle Analytics access supported?
  - Do FreeForm Business Process apps offer extended dimensionality?
  - Do FreeForm Business Process apps offer relational blending?
  - Is there a Task Manager like in Financial Consolidation?
  - Can the FreeForm Business Process be used to back up on-prem HFM Apps?
  - Where to find administrative FAQs?
- Technical FAQ
  - What Essbase version is used in the FreeForm Business Process?
  - How do FreeForm and OCI Essbase 21c work together?
  - Recommended migration approach for Essbase cubes?
  - What hardware limits processing?
  - Does Cube Refresh refresh all cubes at once?
  - Are there size limits?
  - Are hybrid cubes supported?Are slow changing attributes supported?
  - Are multiple FreeForm applications supported in one environment?
  - Is metadata load via Data Integration supported?
  - Do you lose Planning app features?
  - Can Groovy be used?
  - Can Planning Migration files be used like Essbase OTL?
  - Can multiple outline or migration files be imported?
  - How to upload dimensions from legacy Hyperion Planning?
  - Can the database be refreshed from Smart View?
  - How is security configured?
  - How to manage dimensions?
  - Can FreeForm backups migrate to Planning or Close apps?
  - Are business rule scripts portable?
  - Does OTL migration support all Essbase versions?

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- Are there restrictions on Data Integration scenarios?
- Can data be imported after app creation?
- How is EPM Integration Agent used?
- How is tuning handled?
- Is partitioning (transparent, linked, replicated) supported?
- Is MaxL supported?
- Are MDX functions supported?
- Preferred data import/export for large databases?
- Any plans for a CDF to Groovy migration tool?

Business and Essbase Users FAQ

What is the FreeForm Business Process, and what can you do with it?

the FreeForm Business Process lets you import Essbase cubes into Oracle Fusion Cloud Enterprise Performance Management as standalone SaaS apps. They can also be used as a SaaS replacement for Excel-based solutions.

You can create up to twelve cubes with up to twenty-nine dimensions to model, analyze, and report multidimensional data within EPM Cloud.

Learn more: Introduction to the FreeForm Business Process.

What are the advantages of using the FreeForm Business Process in Cloud EPM?

- SaaS benefits with no on-premises infrastructure to manage
- Hybrid architecture combining Essbase cubes and relational metadata
- Flexible multi-cube support (up to twelve cubes)
- Unified security and single place for reporting and planning
- Continuous Oracle updates and integration with EPM features
- No Excel file isolations
- Inbuilt Web based and Excel Reporting

Details: Why Use the FreeForm Business Process

Are Cloud EPM IPM and GenAI Reporting features available with the FreeForm Business Process?

Yes, all IPM features are currently available.

If we are an Essbase customer, why move to the FreeForm Business Process?

FreeForm simplifies administration with SaaS deployment, improves security management, eliminates data latency, and gives access to enhanced cloud features. It simplifies integration with other EPM Business Processes while also providing inbuilt reporting capabilities. This avoids data latency and movement.

See: Essbase Migration Considerations

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Should Excel-focused users move to the FreeForm Business Process?

Yes, FreeForm is an efficient centralized excel replacement with full SmartView compatibility. Smart View integration supports Excel users with shared connections across FreeForm apps, preserving user experience. SmartView also works with Google Sheets and Office 365 Web.

More info: Smart View with FreeForm

What is the learning curve for SmartView Essbase users moving to FreeForm Apps?

Minimal, as Smart View usage remains the same with added capabilities for cloud.

Read: Smart View Usage in FreeForm

What are the web-based benefits for Essbase customers moving to the FreeForm Business Process in Cloud EPM?

Oracle Essbase customers shifting to the FreeForm Business Process in Cloud EPM gain from having centered SaaS advantages—scalability, centralized access, easy cloud deployment, rich analytics via web and Excel, tighter governance, and eliminating IT overhead while preserving Essbase end user capabilities. This enables faster, flexible, and collaborative cloud planning and reporting from a browser or Excel client. Customers do not have to maintain separate licenses and solutions for web-based reports since they can use EPM's native reporting capabilities.

What is the advantage of using the FreeForm Business Process as your corporate reporting solution?

Reporting solutions need a different metadata hierarchy and need a different user access security from other EPM Business Processes. Maintaining your EPM or enterprise reporting in FreeForm empowers organizations to move beyond fragmented siloed reporting systems to a scalable, customizable, flexible, and connected cloud platform that enhances decision-making and operational agility across the enterprise.

What is the difference between Essbase and FreeForm apps in Cloud EPM?

Essbase can be on-premises or OCI IaaS; FreeForm is SaaS in Cloud EPM with hybrid relational Essbase architecture and multi-cube flexibility.

Details: Essbase vs FreeForm Comparison

Are all FreeForm apps custom built?

Yes. It is a fully customizable application. You can customize all dimensions and members fully within limits.

Learn more: FreeForm Customization

Can you create custom dimensions and rules?

Yes, it is fully customizable.

What admin interfaces are available?

Calculation Manager, dimension editor in the FreeForm app, and general Cloud EPM web admin.

Guide: Administration Interface

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Can I use Narrative Reporting with the FreeForm Business Process?

Yes. You can use the native Reports feature within FreeForm or use Narrative Reporting in Cloud EPM across multiple FreeForm apps.

Is direct Smart View and Oracle Analytics access supported?

Yes, fully supported.

See: Smart View and Analytics Integration

Do FreeForm Business Process apps offer extended dimensionality?

Yes, FreeForm apps can have varying dimensionality and dimension hierarchies across cubes,

Do FreeForm Business Process apps offer relational blending?

Yes, you can relationally blend FreeForm apps with extended ADW instance to perform drill through data with relational blending in ADW's tables with extended dimensionality. The ADW instance can also be used as a staging area for data load and extracts fully supported through Cloud EPM Data Integration.

Is there a Task Manager like in Financial Consolidation?

No, only basic task lists are available in FreeForm. We have plans to integrate Task manager with FreeForm apps.

Can the FreeForm Business Process be used to back up on-prem HFM Apps?

Yes. From 26.02 FreeForm Apps will be supported to import HFM apps into Cloud EPM using on-prem FDM.

Where to find administrative FAQs?

See the FreeForm Getting Started Guide for Administrators FAQ.

Technical FAQ

What Essbase version is used in the FreeForm Business Process?

Essbase 21c (version 21.7) is used in all new and updated FreeForm apps.

Details: Essbase Version Info

How do FreeForm and OCI Essbase 21c work together?

They operate on separate platforms. Data movement requires export/import, Smart View connects natively to both.

More info: Hybrid Architecture Explanation

Recommended migration approach for Essbase cubes?

Migrate on-premises Essbase cubes to the FreeForm Business Process in Cloud EPM to unify, simplify, and leverage SaaS benefits.

Guide: Migrating Essbase Cubes

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What hardware limits processing?

Oracle manages hardware in the cloud; customers can request adjustments through support if needed.

Reference: EPM Cloud Performance Policies

Does Cube Refresh refresh all cubes at once?

Yes, refresh is application wide.

Are there size limits?

No fixed limits: scaling managed via Oracle support protocols.

Are slow changing attributes supported?

Not yet, under review based on customer feedback.

Are hybrid cubes supported?

Yes, both hybrid BSO and ASO cubes are supported. You can deploy FreeForm as independent.

Are multiple FreeForm applications supported in one environment?

Each application is separate but can connect and blend data using EPM connection manager and navigation flows.

Is metadata load via Data Integration supported?

Yes, for apps with BSO cubes; ASO-only support is coming soon.

Info: FreeForm Data Management

Do you lose Planning app features?

No significant loss: some Planning-specific features like approvals and task manager are not applicable.

Can Groovy be used?

Yes, fully supported.

Can Planning Migration files be used like Essbase OTL?

FreeForm supports Essbase OTL/Migration imports; direct Planning migration files cannot be imported into FreeForm applications.

Can multiple outline or migration files be imported?

No, only one file per FreeForm app.

How to upload dimensions from legacy Hyperion Planning?

Use Essbase outline files, extracts, web interface, Smart View, or manual creation

Can the database be refreshed from Smart View?

Yes, via right-click > Refresh Databases.

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How is security configured?

Access controls and cell-level security control user permissions. We also support Default Ancestor access settings in App settings page which allows for read write and none access to dimension members in grids and member selector without using cell level security.

How to manage dimensions?

You can do this within theapp through the dimension editor on the web or in SmartView Oracle Cloud Enterprise Data Management.

Can FreeForm backups migrate to Planning or Close apps?

No, FreeForm apps are distinct and not portable.

Are business rule scripts portable?

Yes, Essbase calcscripts can be imported into Calc Manager in the FreeForm App.

Does OTL migration support all Essbase versions?

It supports fom 11.1.2.4.xxx and later.

Are there restrictions on Data Integration scenarios?

Minimum Account, Period, and Scenario dimensions are required.

Can data be imported after app creation?

Yes, using data import, but Essbase OTL/LCM migration imports only during creation.

How is EPM Integration Agent used?

It cnnects on-premises or third-party data sources to Cloud EPM for data and metadata loads.

Read: EPM Integration Agent

How is tuning handled?

Oracle manages tuning and limited property adjustments through Calculation Manager.

Is partitioning (transparent, linked, replicated) supported?

Replicated partitions are supported through Data Maps. Other partitions are not supported due to SaaS infrastructure security restrictions. We will investigate it in the roadmap once we move FreeForm to the Federated Essbase version.

Is MaxL supported?

Use EPM Automate, EPM Agent, or Groovy scripting. MaxL flat file exports using MaxL are in the roadmap.

Are MDX functions supported?

Partial support: ASO calcs, Groovy for custom calc/allocation, admin MDX via future updates.

Preferred data import/export for large databases?

Level 0 Groovy import/export and Essbase file format using data import/export commands. Data Integration is also an option.

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Any plans for a CDF to Groovy migration tool?

No, Groovy is the modern replacement.

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3

Getting Started

Related Topics

- About the Home Page
  After you create the business process, the Home page is the launch point for accessing your business process tasks.
- About the Standardized Cloud EPM Platform Artifact Pages
- De-Support of Legacy Features

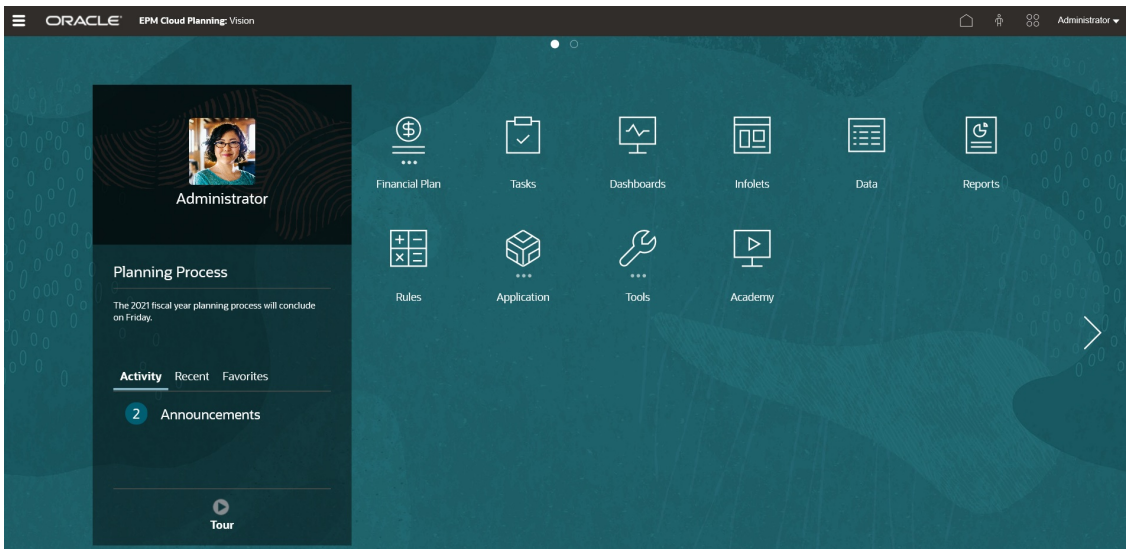
About the Home Page

After you create the business process, the Home page is the launch point for accessing your business process tasks.

The interface provides an intuitive user experience and an overview for quick access to commonly used functions. For example, users can access their tasks, work with data, approve budgets, view reports, and control settings. Service Administrators can manage and customize the business process, create forms, dashboards, and infolets, import and export data and metadata, schedule jobs, define valid intersections, make announcements, and create cross-environment connections.

All Oracle Fusion Cloud Enterprise Performance Management services, business processes, and applications use Redwood Experience. Redwood Experience offers a default Oracle theme and these two additional appearance options: Custom Light and Custom Dark. You can switch between the Oracle, Custom Light, and Custom Dark themes on the **Appearance** page. You can also add a custom branding logo and custom background images to the Home page. To change appearance options for your environment, see Configuring Cloud EPM Appearance in *Getting Started Guide for Administrators*.

Example Home Page



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Global Header

The global header is the area that stretches across the top of the user interface. It contains navigation icons as well as access to accessibility settings and the **Settings and Actions** menu. You can also switch between navigation flows from the global header.

Parts of the global header from left to right:

Global Header Part	Description
(Navigator)	The Navigator icon opens the **Navigator** menu, which serves as a sitemap of the business process and displays links to all of the business process pages to which you have access. **Note:** Some of the links in the **Navigator** menu are available only if you're accessing the business process from the desktop.
ORACLE (Logo Home)	Click the Oracle logo to return to the Home page while working elsewhere in the business process. You can display a custom logo instead of the Oracle logo by selecting a **Logo Image** on the **Appearance** page.
EPM Cloud Planning: Vision	The name of the current business process. You can hide the business process name by selecting **No** for the **Display Business Process Name** option on the **Appearance** page.
(Home)	The Home icon refreshes the Home page or returns you to the Home page while working elsewhere in the business process.
(Accessibility Settings icon)	Click the Accessibility Settings icon to enable accessibility features.
(Navigation Flow)	If you belong to multiple groups or if a navigation flow is assigned to a role, click the navigation flow icon to switch navigation flows at runtime.
Administrator	Click your user name to access the **Settings and Actions** menu.

Work Area

The icons that display in the work area link you to each functional area of the business process to which you have access; for example, **Financials**, **Dashboards**, and **Data**. The three dots above an icon label denotes that the icon opens a grouping of sub-icons, called a cluster.

The **Academy** icon links you to a variety of resources about using the business process.

To replace the default background image in the work area with a custom background image, select **Background Image** on the **Appearance** page. See Customizing Your Display.

Infolet Navigation

If your business process uses infolets to show high-level, essential information, you can explore them by clicking the dots that appear beneath the global header. Arrows are also available on the sides of the Home page to help you navigate easily between the Home page and infolet dashboard pages.

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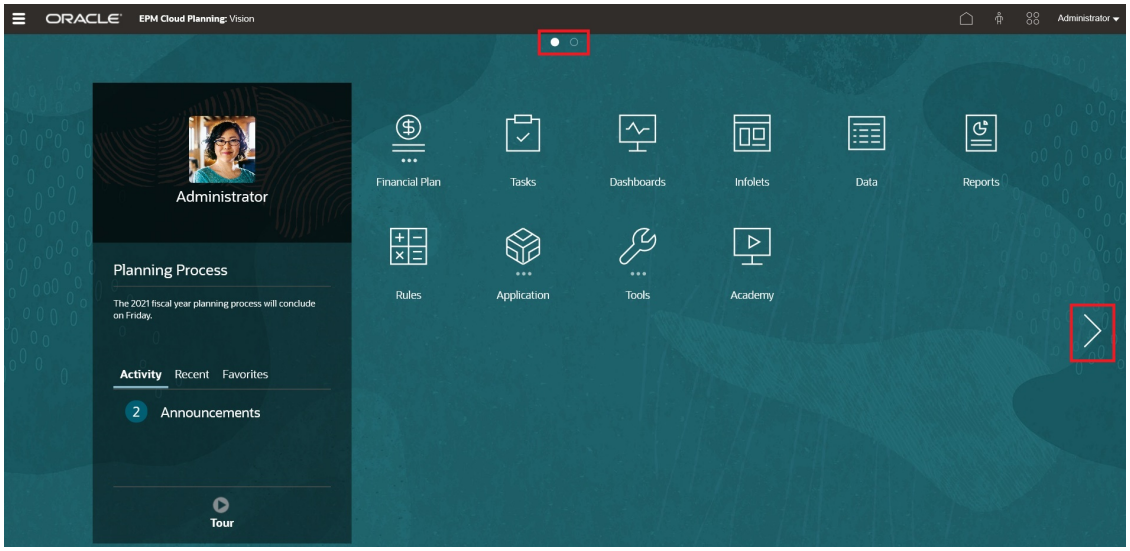

Announcements Panel

The Announcements Panel displays your user name and profile picture (if set), any system announcements entered by the Service Administrator, and helps you track your activity.

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- You can set your profile picture to display at the top of the Announcements panel of the Home page. To set your profile picture, click **Tools**, and then **User Preferences**.

  For more information, see Setting Your Profile Picture in *Working with FreeForm*.
- **Activity**: Summarizes system announcements (the most recent announcement, sorted by effective date, appears at the top) and lists your open tasks. Click **Announcements** to flip the panel and view announcements. Click **Tasks Due Today** to flip the panel and view your tasks.
- **Recent**: Displays a list of links to recently visited forms and dashboards (up to 15 items). Clicking a link will launch the item in a secondary window. Clicking the star next to a link will tag it as a favorite.
- **Favorites**: Displays a list of links to forms or dashboards that were tagged as favorites, and prevents them from being overwritten. Click a link in **Favorites** to launch the item in a

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secondary window. To add items to **Favorites**, click **Recent** to view your recent user activity, and then click the star to the right of the item.

- (Tour): Click to launch a video about key features in the business process.

Videos

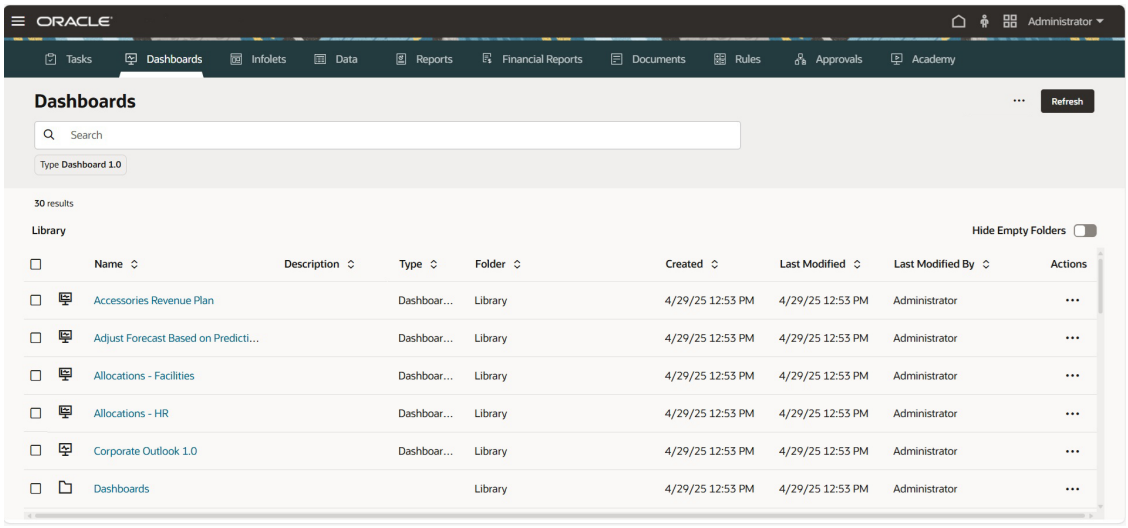
Your Goal	Watch This Video
Explore the default Redwood Experience theme.	Overview: Announcing Cloud EPM's new Redwood Theme
Learn how to customize the interface to streamline workflow.	Overview: Customizing Workflow in Cloud EPM

About the Standardized Cloud EPM Platform Artifact Pages

The following artifact pages (also known as list or listing pages) are standardized to provide a common user experience across Cloud EPM Platform business processes:

- **Dashboards**
- **Infolets**
- **Data**
- **Rules**
- **Documents**

The following image shows the standardized **Dashboards** page.



Viewing Artifact Pages

To view the artifact pages, from the Home page, click the icon for **Rules**, **Dashboards**, **Data**, **Infolets**, or **Documents**.

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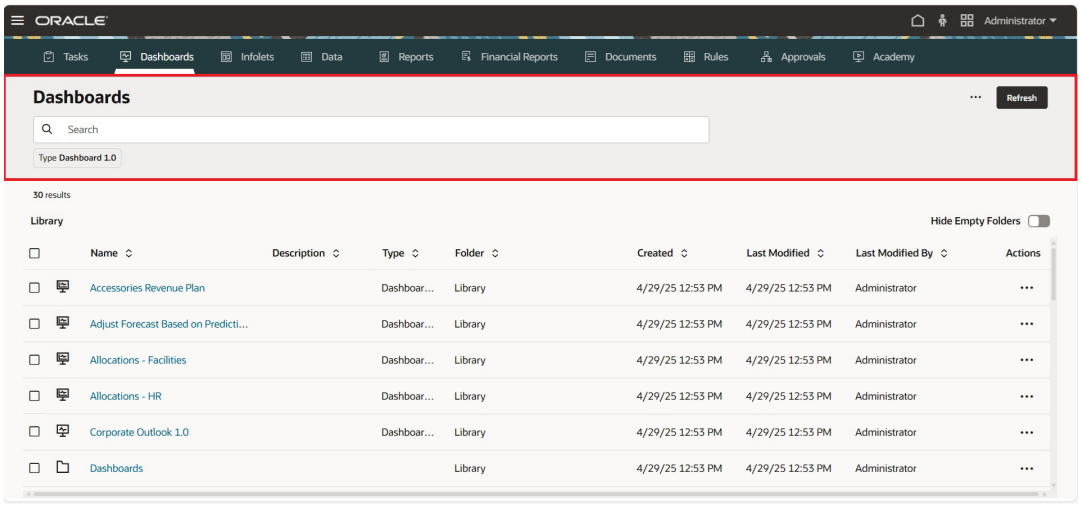
Features of the Standardized Artifact Pages

Standardized artifact pages consist of a header region and a list region.

- Header region features:



  - Global Actions ••• : Actions might include creating an artifact, creating a folder, uploading a file, and converting an artifact, whatever actions are applicable for each artifact. You can also select multiple artifacts in the list region to perform global actions such as **Assign Permission** and **Move to**.

    > **Note**
    >
    > To assign permissions to the Library folder on an artifact page, click ••• (More Actions) without selecting an artifact in the list region, and then select **Assign Permission**.

  - **Refresh**: Updates the artifact list
  - **Search**: Enter search criteria and the search results will display in a drop-down under the **Search** box. Click **Enter** to view the search results in the list region. Search results list all instances of artifacts that include the search criteria (including folder names). Each search result displays the complete folder path. Searching inside a folder narrows the search to that folder and its subfolders.

    > **Note**
    >
    > * The search considers the **Name**, **Description**, and **Folder** columns, wherever applicable, in the search.
    > * If you perform a **Search** and the **Hide Empty Folders** toggle is off, all matching folders and artifacts are displayed. If the toggle is on, only folders containing matching artifacts are displayed. Empty folders are hidden.

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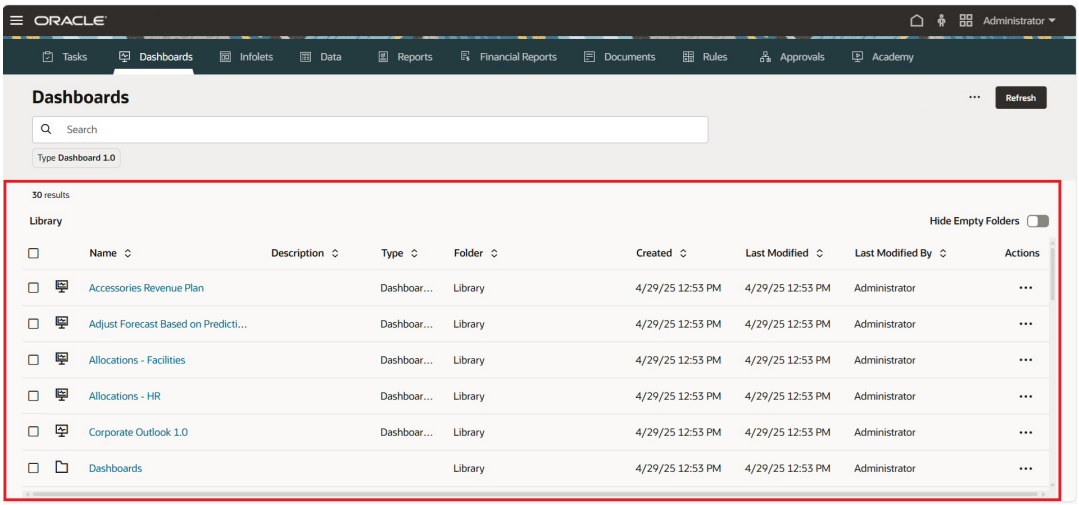
- Filter chips: Filter chips display the search criteria that is entered in the **Search** box. Filter chips can be selected or removed to quickly filter information without changing the current view. We've provided some out-of-the-box filter chips that are combined with a menu so they open a list of selectable options; for example, the **Type** filter chip for Dashboards filters dashboards by type. Click the **Type** filter chip and select **Dashboard 1.0**, **Dashboard 2.0**, or select both.

- List region features:



  - **Hide Empty Folders** (available only for **Dashboards**, **Documents**, **Infolets**, and **Data** artifact pages): Hides folders that don't contain relevant artifacts for the current context. When the toggle is off, all folders are visible in the list region, including empty ones. When the toggle is on, only folders containing relevant artifacts for the specific artifact page (for example, **Data**, **Dashboards**, and so on) are displayed.

    By default this option is off, and will reset to off each time you start a new session. The state of the toggle is retained by the page during the session and is page-specific. Navigating away from the page and returning preserves the toggle state until the tab is closed or the user logs out.

    > **Note**
    >
    > If you create a folder, keep in mind that newly created folders are empty and therefore will not display if the **Hide Empty Folders** toggle is on. New folders become visible after content is added to them, or if the toggle is switched off.

  - Check boxes enable multi-selection to carry out global actions for the selected artifacts.
  - There are new columns for **Type** and **Folder**.
  - Sort items in ascending or descending order using the arrows next to each column heading.
  - Quickly return to root and parent folders by clicking hyperlinks within the **Folder** column or within the breadcrumbs displayed above the artifact list.
  - Each artifact line item has an item-specific **Actions** menu.

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- Assign permissions using the **Assign Permission** option in either the global Actions ••• menu or in the line item-specific **Actions** menu.

  The **Assign Permission** option on the global Actions ••• menu supports assigning permissions to multiple artifacts or folders, or if no artifacts or folders are selected, it assigns permissions to the Library folder.

- Move artifacts using the **Move to** option in either the global Actions ••• menu or in the line item-specific **Actions** menu. The **Move to** option on the global actions menu supports transferring multiple artifacts to a destination folder. While viewing the **Move** interface, you can scroll to find a destination folder or quickly find a destination folder using **Search**.

  > **Note**
  >
  > All folders are displayed and are selectable in the **Move** interface, regardless of the state of the **Hide Empty Folders** toggle.

- For **Dashboards** artifacts, there are new icons to help you differentiate between dashboard versions:
  - : Dashboard 1.0
  - : Dashboard 2.0
- For **Rules** artifacts, the line item-specific **Actions** menus include options to **Edit**, **Launch**, and **Assign Permission**. **Edit** is only available for Service Administrators, and it opens the rule in Calculation Manager within a new dynamic tab.

Reverting to Legacy Artifact Pages

Before standardization, the artifact page options would vary depending on the business process used and the artifacts viewed. By default, all applications now use the standardized artifact pages for **Dashboards**, **Infolets**, **Data**, **Rules**, and **Documents**. However, there are cases when you might prefer to view the legacy artifact pages; for example, if your application uses composite forms and you need to view composite forms on the **Data** page. Use the **Legacy List Pages** application setting to revert the standardized artifact pages to legacy pages.

To revert your application to use legacy artifact pages:

1. From the Home page, click **Application**, and then click **Settings**.
2. Under **Other Options**, find the **Legacy List Pages** setting, and then select **Yes** to revert to legacy artifact pages.
3. Click **Save**.

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> **Note**
>
> Starting with the October 2025 (25.10) update, Oracle plans to de-support the **Legacy List Pages** application setting. For all new and existing environments, the **Legacy List Pages** option will be removed and the following artifact pages will be rendered as standardized:
>
> - **Data**
> - **Dashboards**
> - **Infolets**
> - **Documents**
> - **Rules**

Standardized Artifact Page Considerations

The standardized **Data** page does not support listing composite forms. To view composite forms on the **Data** page, you must revert to the legacy pages using the **Legacy List Pages** application setting. To revert your application to use legacy artifact pages, see Reverting to Legacy Artifact Pages.

De-Support of Legacy Features

Starting with the October 2025 (25.10) update, Oracle is de-supporting the following legacy features:

- Forms 1.0
- Dashboard 1.0
- **Legacy List Pages** application setting
- Infolets

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Legacy Feature	For New Environments (Created on or after October 3, 2025)	For Existing Environments (Created before October 3, 2025)	Details
Forms 1.0	Internally defaulted to Forms 2.0 – No Application Setting option to render Forms in Forms 1.0	Application Setting for **Forms Version** reset to Forms 2.0 – Users can still change it back	New environment users will see all their forms rendered exclusively in Forms 2.0, with no option in Application Settings to switch to Forms 1.0. Existing environment users will also see their Forms rendered in Forms 2.0 by default. However, they will have the option to switch back to Forms 1.0 using Application Settings if needed. See Plan to Discontinue Support for Forms 1.0 and Dashboard 1.0.
Dashboard 1.0	Users will not be able to create Dashboard 1.0 artifacts.	Users will be able to create or modify Dashboard 1.0 artifacts as before.	New environment users will be able to create and modify dashboards using the Dashboard 2.0 framework only. Existing environment users will have the flexibility to create and modify dashboards using either the Dashboard 1.0 or Dashboard 2.0 frameworks. See Plan to Discontinue Support for Forms 1.0 and Dashboard 1.0.
Legacy List Pages application setting	The option to show or hide **Legacy List Pages** in Application Settings is removed. Users will see only Standardized artifact pages (also known as List or Listing Pages)	The option to show or hide **Legacy List Pages** in Application Settings is removed. Users will see only Standardized artifact pages (also known as List or Listing Pages)	Users for all environments will see the following artifact pages rendered as Standardized only: • **Forms** • **Dashboards** • **Infolets** • **Documents** • **Rules** See About the Standardized Cloud EPM Platform Artifact Pages.

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Legacy Feature	For New Environments (Created on or after October 3, 2025)	For Existing Environments (Created before October 3, 2025)	Details
Infolets	Users cannot: • Create new infolets • Include infolets in navigation flows	Users cannot: • Create new infolets • Include infolets in navigation flows There is no impact for existing infolet dots on the Home page.	All environment users are restricted from creating infolets or adding them to navigation flows. See Plan to Discontinue Support for New Infolets and Other Infolet Actions.

Plan to Discontinue Support for Forms 1.0 and Dashboard 1.0

Effective October 2025, all environments will default to the 2.0 setting for Forms and Dashboards with existing environments having the option to switch back to 1.0 with no support for 1.0. All new environments deployed effective October (25.10) will be deployed with Forms 2.0 and Dashboard 2.0 only; the option to switch to Forms 1.0 will no longer be available.

Customers are advised to change application settings from 1.0 to 2.0 and convert their dashboards from 1.0 to 2.0 as soon as possible. Gaps found in 2.0 should be reported using the Service Request process so that Oracle Support can evaluate if the request is justified and, if so, resolve them. Any issues found in 1.0 for either Forms or Dashboards will be resolved in 2.0 if it can be reproduced in 2.0. There will be no bug fixes or enhancements to Forms 1.0 or Dashboard 1.0.

The Task Manager interface will continue to display standalone forms using Forms 1.0 even if the **Forms Version** application setting is set to 2.0.

Plan to Discontinue Support for New Infolets and Other Infolet Actions

Effective October 2025, Oracle will no longer support the following infolet actions:

- Creation of new infolets
- **Copy As**, which duplicates an existing infolet
- Designating an infolet as **Default** and defining infolet dots on the Home page

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4

Setting Up Access Permissions

Assign access permissions to application artifacts such as dimensions, forms, dashboards, and task lists.

Related Topics

- About User and Role Management
- Application Artifacts That Can Be Assigned Permissions
- Types of Access Permissions
- Managing Permissions to Artifacts
- Reporting on Access Permissions
- Retrieving Changed Information from Access Control

About User and Role Management

This business process employs several security layers to ensure security. Infrastructure components, which are implemented and managed by Oracle, create a secure environment for the application.

Business process-level security is ensured by using the following mechanisms that permit only authorized users to access the application:

- Single Sign-on (SSO)
- Role-based access to the business process

Global roles are granted through the Oracle Cloud Identity Console. See *Getting Started Guide for Administrators*.

Access permissions are granted through the application interface and are described in this chapter.

Application Artifacts That Can Be Assigned Permissions

You can assign permissions to:

- Dimensions, including user-defined dimensions

> **Note**
>
> You can assign permissions to members by selecting the dimension property **Apply Security**. If you omit or clear the **Apply Security** setting, all users can access the dimension's members.
>
> See Editing Dimension Properties.

- Navigation flows

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- Launch privileges to rules
- Rule folders
- Forms
- Dashboards
- Infolets
- Reports, Books, and Bursting Definitions
- Form folders
- Dashboard folders
- Infolet folders
- Reports and Documents folders
- Task lists
- Groovy templates

Types of Access Permissions

Access permissions include **Read**, **Write**, **None**, and **Display**.

For descriptions of these options, see Accessing Ancestor Members in Ad Hoc Grids.

You can also set who can launch which rules.

- **Launch**: Allow launch privileges

> **Note**
>
> View user types have no Write permission to dimension members, so can't launch rules having runtime prompts that include members, dimensions, member ranges, or cross-dimension runtime prompt types. They can, however, launch rules having runtime prompts of other types (for example, date type).

- **No Launch**: Disallow launch privileges

> **Note**
>
> If a user inherits Launch permission to a rule by belonging to a group, and is also assigned No Launch permissions by belonging to another group, the more restrictive No Launch assignment takes precedence.

You can specify access permission for individual users and each group. When you assign a user to a group, that user acquires the group's access permissions. If an individual's access permissions conflict with those of a group the user belongs to, user access permissions take precedence.

You can use groups to provide access permissions to your application artifacts such as forms, rules, and dashboards. Oracle Fusion Cloud Enterprise Performance Management recognizes three types of groups:

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- **Predefined**: These groups are automatically created for each application role. All users are assigned to a predefined group based on their application role (for example, Power User).
- **EPM**: These are the groups that you create in **Access Control** in **Tools**.
- **IDCS**: These are the groups that you create in the Oracle Cloud Identity Console. You can view them in **Access Control** and assign them to granular roles and EPM groups.

For more information, see Managing Groups in *Administering Access Control*.

Inheriting Permissions

Inheritance determines the user or group's access permissions. You can specify an attribute that causes the children or descendants of that member to inherit its permissions. Assigned permissions take precedence over inherited permissions. You can include or exclude the member from the permissions setting.

Table 4-1 Options for Inheriting Access Permissions

Inheritance Option	Permission Assignment
Member	Only to the currently selected member
Children	To all children members in the level below the currently selected member
iChildren	To the currently selected member and all children members in the level below it
Descendant	To all descendant members below the currently selected member
iDescendant	To the currently selected member and all descendant members below it

How Permissions are Evaluated

When evaluating permissions, the application gives precedence in this order:

1. Role-level security. Users with the Service Administrator role have permissions to all application elements.
2. For Power User, User, and Viewer user types, permissions that are specifically assigned to users.
3. Permission assignments that are acquired by belonging to a group.

> **Note**
>
> If one member belongs to two groups with different permissions assigned to group members, the least restrictive permission takes precedence. For example, if one group assigns the member Read permission and another group assigns the same member Write permission, Write takes precedence. However if one of the groups assigns no permission (None) to its members, None takes precedence over Read and Write.

4. Parent-level assignments (for example, to parent members or folders).

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Managing Permissions to Artifacts

Related Topics

- About Assigning Permissions to Artifacts, Rules, and Folders
- Adding, Changing, and Removing Permissions to Artifacts and Folders
- Adding, Changing, and Removing Permissions to Rules and Rules Folders

About Assigning Permissions to Artifacts, Rules, and Folders

Service Administrators can assign permissions to artifacts (forms, dashboards, infolets, reports, books, and bursting definitions), rules, and folders.

For information on assigning permissions to dimension members, see Types of Access Permissions and Assigning Access to Dimension Members.

Principles:

- **Artifacts (Forms, Dashboards, Infolets, Reports, Books, and Bursting Definitions):**
  - Users and Power Users can view or enter data only into forms to which they have permissions (and can work only with members to which they have permissions).
  - Service Administrators and Power Users can design artifacts.
  - Power Users can access artifacts they created or to which a Service Administrator assigned them permissions.
  - Service Administrators have Write permissions to all dimension members and to all artifacts.
- **Rules**:
  - Users and Power Users can see and launch only rules to which they are assigned Launch permission.
  - Rulesets inherit launch permissions from the rules included in the ruleset.
- **Folders:**
  - Users who are assigned permissions to a folder can access the items in that folder, unless they are assigned more specific permissions. Likewise, Users have Launch permission to the Calculation Manager rules in folders to which they are assigned permissions, unless they are assigned more specific permissions.
  - When you assign permissions to a folder, all folders under it inherit that permission.
  - If you assign specific permissions (for example, None or Write) to a folder, that permission takes precedence over its parent folder's permissions. For example, if a user has Write permission to Folder1 that contains Folder2 to which the user has None permission, the user can open Folder1, but doesn't see Folder2.
  - If you assign specific permissions (for example, Launch) to a Calculation Manager folder, that permission takes precedence over its parent folder's permissions. For example, if a user has Launch permission to RulesFolder1 that contains RulesFolder2 to which the user has No Launch permission, the user can open RulesFolder1, but doesn't see RulesFolder2.
  - If a user has None permission to a folder called Folder1 that contains a form called Form1 to which the user has Write permission, the user can see Folder1 and Form1.

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- If a user has No Launch permission to a Calculation Manager folder called RulesFolder1 that contains a rule called Rule1 to which the user has Launch permission, the user can see RulesFolder1 and Rule1.

For procedures, see Adding, Changing, and Removing Permissions to Artifacts and Folders and Adding, Changing, and Removing Permissions to Rules and Rules Folders.

Adding, Changing, and Removing Permissions to Artifacts and Folders

The permissions assigned to artifacts (forms, dashboards, infolets, and reports, books, and bursting definitions) and folders refer to the ability to modify the artifact and folder definitions and not the underlying data.

To assign permissions to artifacts and folders:

1. Navigate to the artifact (Data, Dashboards, Infolets, Reports, or Documents) page.
2. Click the Actions icon next to the artifact or folder and click **Assign Permission**, or use the checkboxes to select one or more artifacts or folders, click the global **Actions** menu, and then click **Assign Permission** to assign permissions to the selected artifacts or folders.

> **Note**
>
> - Multi-select only works if you're viewing one of the standardized artifact pages (Dashboards, Infolets, Data, Rules, or Documents). All other artifact pages use the legacy interface, and for legacy artifact pages, you can assign permissions to only one artifact or folder at a time. See About the Standardized Cloud EPM Platform Artifact Pages.
> - To assign permissions to the Library folder on an artifact page, click the global **Actions** button without selecting an artifact in the list region, and then select **Assign Permission**.

3. Perform a task:
   - To add permissions for a user or group, click **Add User/Group**, select from the list of available users and groups, and then use the arrows to move your selections to the **Assigned Permissions** pane.
   - To edit the type of permissions for a user or group, select an option:
     - Click **Read** to allow the selected users and groups to view the artirfact or folder in the listing, but not allow them to create, edit, or delete the artifact or folder information.
     - Click **Write** to allow the selected users and groups to view the artifact or folder in the listing, and to create, edit, or delete the artifact or folder information.
     - Click **None** if you don't want the selected users or groups to view the artifact or folder in the listing.
   - To remove permissions for a user or group, click (Remove) next to the user or group.
4. Click **Save**.

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Adding, Changing, and Removing Permissions to Rules and Rules Folders

The permissions assigned to rules and rules folders control whether users or groups can launch the selected rules.

To assign permissions to rules and rules folders:

1. Click **Rules**.
2. Click the Actions icon next to the rules or rules folder and click **Assign Permission**, or use the checkboxes to select one or more rules or rules folders, click the global **Actions** menu, and then click **Assign Permission** to assign permissions to the selected rules or rules folders.

   > **Note**
   >
   > Multi-select only works if you're viewing the standardized **Rules** page. If you're using the legacy **Rules** page, you can assign permissions to only one rule or folder at a time. See About the Standardized Cloud EPM Platform Artifact Pages.

3. Perform a task:
   - To add permissions for a user or group, click **Add User/Group**, select from the list of available users and groups, and then use the arrows to move your selections to the **Assigned Permissions** pane.
   - To edit the type of permissions for a user or group, select an option:
     - Click **Launch** to allow the selected users and groups to launch the selected rules.
     - Click **No Launch** to prevent the selected users and groups from launching the selected rules.
   - To remove permissions for a user or group, click (Remove) next to the user or group.
4. Click **Save**.

Reporting on Access Permissions

You can view current access permissions and print reports.

To report on current access permissions for users and groups in the application:

1. From the Home page, click Navigator, and then under **Monitor and Explore**, click **System Reports**.
2. Click **Access Control**.
3. On **Select User or Group**, select from the available options.
4. From the left **Available** panel, select and move users or groups on which to report to the **Selected** panel.
5. From the left **Available Objects** panel, select and move objects on which to report to the **Selected Objects** panel.
6. Select reporting options:

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- For **Show Matching Access of Type**, select the access to view: **Read**, **Write**, or **None**.
- For **Group the Results By**, select how to view the report: **Users** or **Objects**.
- From the **Report Type** sections, select **Assigned Access** or **Effective Access**:

Table 4-2 Access Report Types

Report Type	Description	Options
Assigned Access	Summarizes access permissions that Service Administrators assign	Specify whether access permissions are assigned by member selection relation or group membership: • **Show matching access of relation**: **Member**, **Children**, **Children (inclusive)**, **Descendants**, or **Descendants (inclusive)**. • **Show Inherited From Group**: Shows access permissions inherited by users in a group.
Effective Access	Summarizes access permission assignments as the application evaluates them (for example, by member selection relation, such as children, or group membership). This is useful if there are conflicts in access permissions. **Note:** Effective Access reports can't be generated for groups.	Describe the origin of the effective access permissions by selecting **Show effective access origin**. For example, a user named JSomebody may be assigned Write access permissions to Entity1 and belong to a group named Sales that is assigned Read access permissions to Entity1. This setting would show that JSomebody has Write access permissions to Entity1 because individual assigned access permissions supersede access permissions inherited by group membership.

7. Click **Create Report**.

Working with Access Permissions Reports

The report on access permissions displays in Adobe Acrobat. You can use the Adobe Acrobat toolbar to work with the report.

Retrieving Changed Information from Access Control

Changes to group names or group membership from Access Control are not reflected in the application. To apply changes, you must reimport security artifacts.

To synchronize information after modifying group properties:

1. Access the environment as a Service Administrator.
2. Click **Tools**, and then **Migration**.
3. Export Security artifacts:
   a. Click **Core**, and then select **Security**.
   b. Click **Export**.
   c. Enter a file name and click **OK**.
   d. Review the Migration Status Report to ensure that the export is successful.
   e. Click **Close**.
4. Import the snapshot that you created:
   a. On the **Migration** page, click **Snapshots**.

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b. Under **Actions**, click ••• (Actions) next to the snapshot that you created in the preceding step, and then click **Import**.

c. Click **OK**.

d. Review the Migration Status Report to ensure that the import is successful.

e. Click **Close**.

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5

Managing Applications

View and manage applications, cubes, and dimensions using the application console.

Related Topics

- Application Overview
  Start here to learn about managing applications, dimensions, and dimension members.
- Refreshing the Application
  Refresh the application whenever you make changes to the application structure.
- Managing Cubes
- Managing Dimensions
  Start here to learn how to manage dimensions and dimension members.
- Setting User Preferences
- Viewing Activity Reports
  Monitor user activities with activity reports.
- Importing and Exporting Data and Metadata
  Move data and metadata in and out of the application using FreeForm.
- Validating the Essbase Outline
  Pre-validate the Essbase outline to ensure that your application is compatible with a Hybrid Essbase version.
- Reverting a Hybrid BSO Application to Non-Hybrid
- Creating and Refreshing Application Databases
  Things you need to know before you refresh the application database.
- Adding an Aggregate Storage Outline to an Application
  Understand what aggregate storage is and how to use it in an application.
- Removing an Application
  Proceed with caution when removing an application.
- Setting the Daily Maintenance Process Start Time
  Understand how to determine and set the most convenient time to start the hour-long daily maintenance process.
- Uploading and Downloading Files Using the Inbox/Outbox Explorer
  Use the Inbox/Outbox Explorer to upload files to the server or download them to your computer.
- Validating Groovy Scripts
  Validate Groovy business rule scripts to ensure that your application's Groovy scripts can pass the stricter validation rules.

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Application Overview

Start here to learn about managing applications, dimensions, and dimension members.

An application is a related set of dimensions and dimension members used to meet a set of business process needs. Each application has its own accounts, entities, scenarios, and other data elements.

To create an application, see Creating a FreeForm App.

After an application is created, you can view and manage it using the application console. To view the application console, click **Application**, and then click **Overview**.

The application console shows the application properties (including the application type) and application statistics, such as the number of tasks, plans, and rules in your application. It also lists the cubes (for Standard applications) and dimensions (for Reporting applications) used by the application.

- To view the application overview, click **Overview**.
- To manage cubes, click **Cubes**. See Managing Cubes.
- To manage dimensions, click **Dimensions**. See Managing Dimensions.
- To view activity reports, click **Activity**. See Viewing Activity Reports.

Refreshing the Application

Refresh the application whenever you make changes to the application structure.

Changes you've made to the application are not reflected to users performing data entry tasks until you refresh the application. For example, if you modify properties of a dimension member, or add a member, or change access permissions, users will not see the updates you've made until you refresh the application.

To refresh the application:

1. Click **Application**, and then click **Overview**.
2. Click **Refresh**.

> **Note**
>
> The application is automatically refreshed during an application update. Information about the refresh is captured in a job called Content Update. To view the progress and details of the content update process, use the Jobs console, which you can access by clicking **Jobs** in the **Application** cluster in the Home page. See Viewing the Status of Jobs in this guide for information on how to view jobs, and Setting Content Update Start Time in *Getting Started Guide for Administrators* for information about setting the Content Update start time.

Managing Cubes

Related Topics

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- Viewing and Managing Cubes
- Adding Cubes
- Clearing Cubes
- Improving Cube Performance

Viewing and Managing Cubes

To view and manage cubes:

1. Click **Application**, and then click **Overview**.
2. Click **Cubes**.
3. Perform one task:
   - To view the details of a cube, click the name of the cube that you want to view.
   - To add a cube, click **Create**, then complete the cube details, and then click **Done**.

   For cube details, see Adding Cubes.

> **Note**
>
> You can rename a cube only during application creation. After a cube is added, you can't rename or delete it.

Adding Cubes

For FreeForm, you can add up to 12 total cubes with any combination of aggregate storage and block storage. See the chart below for a comparison of cubes by application type. See Understanding FreeForm.

After a cube is added, it behaves like any other application cube. If the cube maps to an aggregate storage database, then aggregate storage limitations apply.

> **Note**
>
> For an aggregate storage cube, you must specify an application name to contain the database since an aggregate storage database must reside in its own application.

Cube Comparison by Application Type

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Cube Comparison by Application Type

** Total combination of 12 Hybrid BSO or ASO*
*** Expanding to 5 per Product Roadmap*

	App type	Open Cubes		Module Cubes		TOTAL
		BSO	ASO	BSO	ASO	
Cloud EPM Enterprise	Modules	3	4	5	2	14
	Custom	6	6	0	0	12
	FreeForm*	12*	12*	0	0	12
	Sales Planning	0	1	3	2	6
	Strategic Workforce	0	1	2	1	4
	Predictive Cash Forecasting (PCF)	0	1	2	1	4
Cloud EPM Standard	Modules	1	1	5	2	9
Legacy	EPBCS	3	4	5	2	14
	PBCS	3	4	0	0	7
NetSuite Planning	Standard	1	1	1	1	4
	Premium	3	1	1**	1	6

Clearing Cubes

Related Topics

- About Clearing Cubes
- Creating Clear Cube Jobs
- Starting and Deleting Clear Cube Jobs
- Scheduling Clear Cube Jobs

About Clearing Cubes

The application enables you to clear specific data within input and reporting cubes. You can start the Clear Cube job right away or schedule it to run later.

> **Note**
>
> The Clear Cube job deletes the data you specify within input and reporting cubes, but it doesn't delete the cube definition in the application's relational tables.

Creating Clear Cube Jobs

To create a Clear Cube job:

1. Click **Application**, and then click **Overview**.
2. Click **Actions**, and then **Clear Cube**.
3. On the **Clear Cube** page, click **Create**, complete the job name, select the cube, then select a clear option:
   - **Clear All**: For both block storage and aggregate storage cubes, delete all data within a cube that is associated with the entities you select:

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  - **Supporting Details**
  - **Comments**
  - **Attachments**
  - **Essbase Data**
- For block storage cubes only:
  - **Upper-level blocks**: Delete data in upper-level blocks
  - **Non-input blocks**: Delete data in non-input blocks
  - **Dynamic blocks**: Delete data in dynamic blocks
  - **Empty blocks**: Delete blocks with #MISSING values
- For aggregate storage cubes only:
  - **All Aggregations**: Delete all aggregate views
  - **Partial Data (Basic Mode)**: Clear the data in the specified region. You can select the point of view by clicking **Member Selector** .
    - **Supporting Details**
    - **Comments**
    - **Attachments**
    - **Essbase Data Logical**: In which the input cells in the specified region are written to a new data slice with negative, compensating values that result in a value of zero for the cells you want to clear. The process for logically clearing data completes in a length of time that is proportional to the size of the data being cleared. Because compensating cells are created, this option increases the size of the database.
    - **Essbase Data Physical**: In which the input cells in the specified region are physically removed from the aggregate storage database. The process for physically clearing data completes in a length of time that is proportional to the size of the input data, not the size of the data being cleared. Therefore, you might typically use this method only when you need to remove large slices of data.
  - **Partial Data (Advanced Mode)**: Clear Essbase data only by using an MDX expression in the text box provided.
    - **Essbase Data Logical**: In which the input cells in the specified region are written to a new data slice with negative, compensating values that result in a value of zero for the cells you want to clear. The process for logically clearing data completes in a length of time that is proportional to the size of the data being cleared. Because compensating cells are created, this option increases the size of the database.
    - **Essbase Data Physical**: In which the input cells in the specified region are physically removed from the aggregate storage database. The process for physically clearing data completes in a length of time that is proportional to the size of the input data, not the size of the data being cleared. Therefore, you might typically use this method only when you need to remove large slices of data.

4. Click **Save and Close**.
5. To view the job on the **Clear Cube** listing page, click **Refresh**.

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To start the Clear Cube job right away or to delete a job, see Starting and Deleting Clear Cube Jobs.

To schedule a Clear Cube job to run immediately, later, or at intervals, go to the Jobs page. See Scheduling Jobs .

> **Tip**
>
> To streamline the job creation process for subsequent **Clear Cube** jobs, you can duplicate an existing job using the **Save As** option, then update it. See Duplicating Jobs.

Starting and Deleting Clear Cube Jobs

To start a Clear Cube job right away or to delete a job:

1. Click **Application**, and then click **Overview**.
2. Click **Actions**, and then **Clear Cube**.
3. On the **Clear Cube** listing page, click ••• (Actions) next to the Clear Cube job that you want to start or delete, and then select one of the following options:
   - Click **Submit** to start the job right away.
   - Click **Delete** to delete the job.

Scheduling Clear Cube Jobs

You can schedule a Clear Cube job to run later or at intervals. To schedule a Clear Cube job, go to the Jobs page. See Scheduling Jobs.

Improving Cube Performance

Jobs You Can Schedule to Improve Cube Performance

The application enables you to schedule jobs that improve cube performance. These jobs include:

- **Restructure Cube**: Performs a full restructure of a block storage cube to eliminate or reduce fragmentation. This will also remove empty blocks. Running this job won't push any changes from the business process to Essbase.
- **Compact Outline**: Compacts the outline file of an aggregate storage cube. Compaction helps keep the outline file at an optimal size. Compacting the outline doesn't clear the data. Running this job won't push any changes from the business process to Essbase.
- **Merge Data Slices**: Merges incremental data slices of an aggregate storage cube. Fewer slices improve a cube's performance. You can merge all incremental data slices into the main database slice or merge all incremental data slices into a single data slice without changing the main database slice. You can optionally remove cells that have a value of zero.
- **Optimize Aggregation**: Generates optimized views based on collected query tracking information in an aggregate storage cube. This job has two actions:
  - **Enable Query Tracking**: Records a meaningful set of queries, and then uses the recorded query data to select the most appropriate set of aggregate views to

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materialize for that database. You must enable query tracking and allow it sufficient time to collect data before executing the aggregation process based on query data.

Once enabled, query tracking continues until query tracking is disabled, the cube is reloaded or refreshed, the outline is compacted, the application is shut down, or additional aggregate views are materialized for the database. Query tracking does not resume automatically after these actions occur.

> **Note**
>
> The following actions clear the stored query tracking information from the aggregate storage cube, but they do not disable query tracking:
>
> * Merging incremental data slices
> * Clearing data
> * Clearing existing aggregations
> * Clearing partial data
> * Loading data into the aggregate storage cube

– **Execute Aggregation Process**: Calculates aggregations for aggregate storage databases that contain data and to which you are granted Calculation permission. To perform an aggregation, you use system recommended views. The selection of views and aggregation processes are combined into one, non-configurable operation performed by the server. You can optionally specify the maximum disk space for the resulting files, base the view selection on user querying patterns, and include rollup hierarchies in the view selection. Select one or more of the following options:

  * **Based on query data?**: Aggregate the views the server selects, based on collected user querying patterns. This option is only available if query tracking is turned on.
  * **Include rollup option?**: Include secondary hierarchies (with default level usage) in the view selection process.
  * **Include growth size option?**: Aggregate the views the server selects, until the maximum growth of the aggregated database exceeds limits you specify. Enter the size (in ratio) beyond which the server should stop the aggregation.

You can start the jobs right away or schedule them to run later.

> **Note**
>
> Users can stay logged in during these actions.

Related Topics

- Scheduling Jobs
- Optimize BSO Cubes in *Operations Guide*
- Optimizing Aggregate Storage Option Cubes in *Operations Guide*

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Managing Dimensions

Start here to learn how to manage dimensions and dimension members.

Dimensions categorize data values.

Dimensions on the **Dimensions** page are listed in order of precedence. The order of dimensions is critical for the structure and performance of an application. The order of your dimensions determines how your data calculations will perform. **Evaluation Order** enables you to specify which data type prevails when a data intersection has conflicting data types.

> **Note**
>
> - Account members that require Text, Date, or Smart List data types, require the Account dimension to have an Evaluation Order set for the cube or cubes in question. See Setting the Evaluation Order in the Simplified Interface.
> - For a Smart List to be viewable in forms, the dimension to which the Smart List type member belongs must be first in the evaluation order. See Working with Smart Lists.

To manage dimensions:

1. Click **Application**, click **Overview**, and then click **Dimensions**.
2. Perform a task:
   - To filter the dimensions by cube, see Filtering the Dimension View by Cube.
   - To view and edit the properties of dimensions and their members, click the name of the dimension. See Editing Dimensions in the Simplified Dimension Editor.
   - To reorder the dimensions, select a cube, and then click the up or down arrow in the **Order** column next to the dimension that you want to move.
   - To set a different evaluation order, select a cube, and then click the **Evaluation Order** field next to the dimension that you want to update. See Setting the Evaluation Order in the Simplified Interface.
   - To rename a dimension, click the name of the dimension that you want to rename, select the **Edit Dimension Properties** tab, and then enter a new name in the **Dimension** field. See Editing Dimension Properties.
   - To create a dimension, click **Create**. See Creating Dimensions.
   - To import metadata, click **Import**. See Importing Metadata.
   - To export metadata, click **Export**. See Exporting Metadata.

Tutorials

Tutorials provide instructions with sequenced videos and documentation to help you learn a topic.

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Your Goal	Learn How
Get an introduction to Dimensions and Members and learn how to manage them in Oracle Fusion Cloud Enterprise Performance Management business processes.	Managing Dimensions in Cloud EPM Business Processes

Filtering the Dimension View by Cube

For a Standard application, you can filter the dimension view by cube. When you select a cube, only dimensions used in that cube are displayed.

To filter the dimension view by cube:

1. Click **Application**, click **Overview**, and then click **Dimensions**.
2. Click the down arrow to the right of **Cube** to filter the list of dimensions by cube.

By default, the application displays all cubes for a standard application. If the dimension list is filtered, the application displays only the dimensions used in the selected cube.

Filtering by cube (rather than viewing all cubes) also enables you to view these details about a dimension:

- Order of precedence
- Dimension density

> **Note**
>
> The **Dense** column is hidden for applications that contain only ASO cubes. If an application contains a BSO cube or a BSO and an ASO cube, the **Dense** column displays for both cubes.

- Evaluation order

For information about editing dimensions, see Editing Dimensions in the Simplified Dimension Editor.

Creating Dimensions

Creating a dimension adds it to the list of dimensions in the dimension view.

To create a dimension:

1. Click **Application**, click **Overview**, and then click **Dimensions**.
2. Click **Create**.
3. On the **Create Dimension** page, enter or select values for the properties that are displayed.

   For FreeForm applications, you can select the dimension type. Dimension types can include:

   - Account
   - Entity
   - Period

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- Custom

Notes about dimension type:

- A dimension type is not listed if you've already created it (for Account, Entity, Period) or if you've exceeded the maximum number of dimensions (for Custom).
- You can't edit the dimension type after you've created the dimension.

For descriptions of the properties, see Editing Member Properties.

For information about editing dimensions, see Editing Dimensions in the Simplified Dimension Editor.

Setting the Evaluation Order in the Simplified Interface

The evaluation order enables you to specify which data type prevails when a data intersection in a form has conflicting data types.

For example, if Account members are set to the Text data type, and Product members are set to the Smart List data type, you can set whether the Account members with the Text data type or the Product members with the Smart List data type prevail at an intersection.

> **Note**
>
> Account members that require Text, Date, or Smart List data types, require the Account dimension to have an Evaluation Order set for the cube or cubes in question.

To set the evaluation order in the Simplified Interface:

1. Click **Application**, click **Overview**, and then click **Dimensions**.
2. Click the down arrow to the right of **Cube** to filter the list of dimensions by cube.
3. Click the **Evaluation Order** field next to the dimension that you want to update.

> **Note**
>
> You need to select only dimensions whose members have specific data types (that is, their data type isn't "Unspecified"). The data type "Unspecified" doesn't conflict with another data type.

Setting User Preferences

Service Administrators specify defaults for the current application. However, users can override these application defaults by setting preferences to control many aspects of the application, such as their profile photo and how numbers display in the application.

To learn more about user preferences options, see Setting Your Preferences in *Working with FreeForm*.

To learn more about application defaults and system settings, see Managing Application and System Settings.

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Working with Variables

Related Topics

- Working with Substitution Variables
- Working with User Variables

Working with Substitution Variables

Related Topics

- About Substitution Variables
- Creating a Substitution Variable
- Defining the Columns that Display on the Substitution Variables Page
- Deleting a Substitution Variable

About Substitution Variables

Substitution variables act as global placeholders for information that changes regularly. For example, you could set the current month member to the substitution variable `CurMnth` so that when the month changes, you need not update the month value manually in the form or the report script. You create and assign values to substitution variables within the application. These substitution variables are then available in the application when you select members for a form.

> **Note**
>
> If you migrated an application that uses Global variables, you can view, but not edit, them in the application.

Creating a Substitution Variable

To create a substitution variable and assign values to it:

1. From the Home page, click **Tools**, and then click **Variables**.
2. Select the **Substitution Variables** tab at the bottom of the page.
3. Click **Create**.
4. In the **Create** dialog box, enter the following information:
   - **Cube**: Cube to which the substitution variable will be added.
   - **Name**: Name of the substitution variable (up to 80 characters).
   - **Value**: Value for the substitution variable (up to 255 characters).

     You can specify a range of values ; for example, `FY16:FY18`.
5. Click **Save**.

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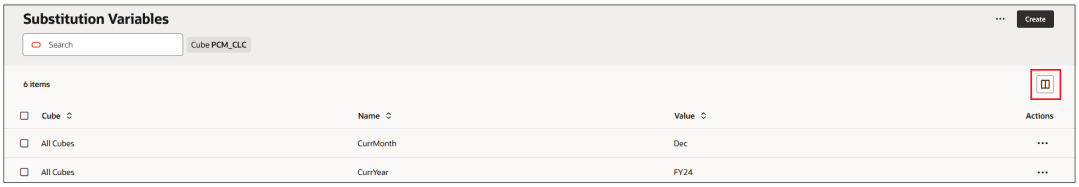
> **Note**
>
> - You can define a substitution variable range for base members and parent members. For base members, you can use a single colon (`:`) or a double colon (`::`) as delimiters; for example, `SunEnT 110::111`. However for parent members, you can only use a single colon (`:`).
> - If the substitution variable will be used in data export job definitions for block storage cubes and the member name is numeric only, the member name must be placed in double quotation marks in the **Value** field; for example, `"1100"` or `"000"`. The double quotation marks aren't necessary for alphanumeric member names; for example, `a1110`. However if this variable will be selected in data export job definitions for aggregate storage cubes, you can't place the member name in double quotation marks in the **Value** field. Therefore if you need to use the same member name in substitution variables for data export jobs defined for both types of cubes, you must define two different substitution variables; one with quotes and one without.
> - After you create a substitution variable, you cannot rename it.

Defining the Columns that Display on the Substitution Variables Page

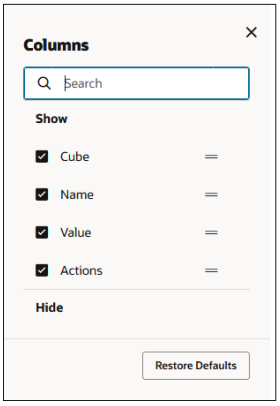
You can define the columns that display on the **Substitution Variables** page, and you can define the order in which the columns display.

To define the column display:

1. From the **Substitution Variables** page, click .



2. Select or de-select the desired columns to show or hide the columns.



3. To change the order in which the columns will display, click and drag to the right of a column to move the column up or down. To restore the columns to their default display, click **Restore Defaults**.

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4. When you have finished defining the columns that will display and the order in which they will display, click to close the **Columns** dialog box.

Deleting a Substitution Variable

To delete a substitution variable:

1. From the Home page, click **Tools**, and then click **Variables**.
2. Select the **Substitution Variables** tab at the bottom of the page.
3. Select a substitution variable, then click ••• in the **Actions** column, and then select **Delete**.
4. Click **Delete** to confirm the deletion.

Working with User Variables

Related Topics

- About User Variables
- Creating a User Variable
- Defining the Columns that Display on the User Variables Tab
- Managing User Variables
- Setting Limits for User Variables on Forms
- Deleting a User Variable

About User Variables

User variables act as filters in forms, enabling users to focus only on certain members, such as a department. Before you can associate a user variable with a form, you must create the user variable. When you create forms with user variables, users must first select values in preferences for the variable before opening forms. Afterward, users can change the variable on the form only if it's a dynamic user variable. Otherwise, they must continue to set the variable in preferences. For example, if you create a user variable called Division, users must select a division before working in the form.

Tip

Attributes are another way to filter forms. But instead of explicitly filtering by an attribute (like Red), you can create a user variable for the attribute dimension, and then use the user variable as the filter. Then you can enable the user variable as a dynamic user variable which allows users to change the value of the filter at runtime. This is a useful technique that allows for dynamic filtering.

Tip

The first time you select a variable for a form, you must do so in preferences. After that, you can update the variable in preferences or in the form.

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Creating a User Variable

To create a user variable:

1. From the Home page, click **Tools**, and then click **Variables**.
2. Select the **User Variables** tab at the bottom of the page.
3. Click **Create**.
4. In the **Create** dialog box, enter the following information:
   - **Dimension**: Dimension for which to create a user variable (including attribute dimensions).
   - **User Variable Name**: Name of the user variable (up to 80 characters).
   - **User Variable Definition**: Click to access the **Select Members** dialog box where you can select dimension members.
   - **Use Context**: Select **Yes** to indicate the value can't be chosen by the user from user preferences. Instead, the value is passed from another form based on its Point of View context and the value will change dynamically based on the context of the form. However if the user variable is used elsewhere, such as in rows and columns, you must create a different user variable so the user can set its own value for it because it won't be set dynamically.
5. Click **Save**.

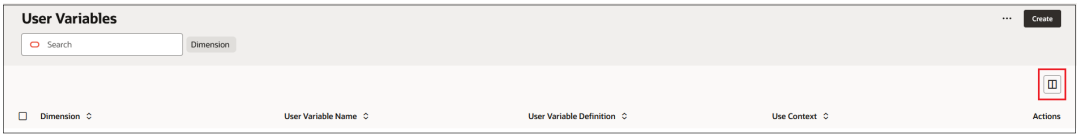
   You can now associate the user variable with a form. You can also set limits for user variables on forms. (See Setting Limits for User Variables on Forms). Afterward, users can select members for the user variable.

Defining the Columns that Display on the User Variables Tab

You can define the columns that display on the **User Variables** page, and you can define the order in which the columns display.

To define the column display:

1. From the **User Variables** page, click .



2. Select or de-select the desired columns to show or hide the columns.

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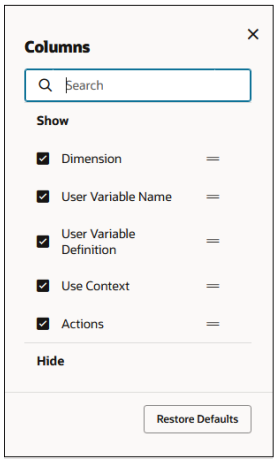

3. To change the order in which the columns will display, click and drag ═ to the right of a column to move the column up or down. To restore the columns to their default display, click **Restore Defaults**.
4. When you have finished defining the columns that will display and the order in which they will display, click × to close the **Columns** dialog box.

Managing User Variables

You can set user variables to limit the number of members displayed on a form, helping users focus on certain members. For example, if you create a user variable called Division for the Entity dimension, users can select a member for their own division. You can create any number of user variables for each dimension (including attribute dimensions), and select user variables for any axis in the form.

The typical sequence of steps:

1. If necessary, create the appropriate parent-level members in the dimension outline.
2. Define user variables for each dimension you want users to be able to filter (including attribute dimensions).
3. When designing the form, associate the user variable with the form.
4. Instruct users to select a member for the user variable associated with the form.

> **Note**
>
> If you are prevented from opening a form (or viewing a form in a dashboard) because it contains missing user variables, you will see a message prompting you to select members for the user variables:
>
> ```
> Before you can open this form, you must select a memaber for each User
> Variable. To select members for missing user variables: Click here.
> ```
>
> When you click the link in the message, the **User Preferences** page opens in a new dynamic tab, and the user variables with missing member values are displayed first. Select a member for each undefined user variable.
>
> See Setting User Variables in *Working with FreeForm*.

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Setting Limits for User Variables on Forms

You can limit which members users can see or select on individual forms by limiting their member selections for a user variable. The user variable limit applies the restriction to all users at the form level, regardless of their access permissions. In addition to setting the limits by members, Service Administrators can select limits by function (for example, "IChildren(Existing Employees)".

To set up user variable limits:

1. During form design, select **Enable Dynamic User Variables** to allow dynamic user variables in the form.
2. Create a user variable.

   See Creating a User Variable.
3. In the **User Variable Definition** field on the **Variables** page under the **User Variables** tab, click (Select Members), and add or remove members to set limits.

   To select members by function, click (Function Selector) and select from the available options.
4. In the form designer, assign the user variable to the page axis.

Deleting a User Variable

> **Note**
>
> You can delete user variables if they are not associated with a form. If you attempt to delete a user variable that is used in a form, you'll see an error message saying it cannot be deleted. The message will also provide the names of the forms where the user variable is used.

To delete a user variable not associated with a form:

1. From the Home page, click **Tools**, and then **Variables**.
2. Select the **User Variables** tab at the bottom of the page.
3. Select a user variable, then click ••• in the **Actions** column, and then select **Delete**.
4. Click **Delete** to confirm the deletion.

Viewing Activity Reports

Monitor user activities with activity reports.

The Activity Report enables Service Administrators to understand application usage. It also helps streamline application design by identifying user requests, calculation scripts, forms, reports, and so on. Two versions of the report; an HTML version and a JSON version, are available.

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Oracle Fusion Cloud Enterprise Performance Management retains Activity Reports for the last 60 days only. Oracle recommends that you download and create backup copies on a local computer to analyze them for historical trends.

The Activity Report is automatically generated in these situations:

- Every day during daily maintenance of the service
- Each time you submit a Provide Feedback submission
- Every time you execute the resetService EPM Automate command to restart an environment.

For details about the information that is provided in activity reports, see Activity Report Contents in *Getting Started Guide for Administrators*.

To view activity reports:

1. Click **Application**, and then click **Overview**.
2. Click **Activity Reports**.
3. To view the report details, click **View** next to the name of the report that you want to view.

Importing and Exporting Data and Metadata

Move data and metadata in and out of the application using FreeForm.

Related Topics

- Importing Metadata
  You can import metadata from a flat file in a comma-delimited, tab-delimited, or other format.
- Exporting Metadata
  You can export metadata to a flat file in a .csv (comma-delimited) or .txt (tab-delimited or other delimiter character) format.
- Importing and Exporting Data

Importing Metadata

You can import metadata from a flat file in a comma-delimited, tab-delimited, or other format.

These artifacts are supported in the imports:

- Dimensions
- Smart Lists
- Exchange rate tables

To import metadata, perform these tasks:

- Create an import file for each artifact that you want to import. See Creating the Metadata Import File.
- Load the import file or files (you can import multiple dimension files at the same time). See Loading the Metadata Import File.

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> **Note**
>
> - You cannot rename members of attribute dimensions during a metadata import. Renaming of members will be ignored.
> - You cannot delete attribute dimensions while importing metadata.

Videos

Your Goal	Watch This Video
Learn how to import metadata.	Importing Metadata in Cloud EPM

Creating the Metadata Import File

Related Topics

- About the Metadata Import File
- Example Entity Dimension Import File
- Other Supported Delimiter Characters

About the Metadata Import File

Before you begin loading, you must create an import file for each artifact you want to import (dimensions, smart lists, and exchange rate tables). The import file must contain a header record, and then below the header record, a list of metadata records that you want to import or update. The file format can be .csv (comma-delimited) or .txt (tab-delimited or other delimiter characters).

1. A header record, the first line in the file:
   - Lists the dimension and any member properties used by subsequent metadata records; the header record and subsequent records don't need to include all properties; properties that are not included are inherited from the corresponding parent's default property value
   - Is case sensitive
   - Can list properties in any order, as long as the subsequent metadata records are in the same order
   - Can use comma or tab delimiters. Other delimiter characters can be used if they are supported and the same delimiter character is used throughout the file. For a list of supported delimiter characters and exceptions, see Other Supported Delimiter Characters.
2. After the header record, a list of metadata records that you want to import or update. Each metadata record contains a delimited list (comma, tab, or other) of property values that matches the order designated in the header record. A metadata record can skip a property that is specified in the header record; in this case, the default property is assumed.

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> **Note**
>
> Only characters that are included in the UTF-8 character set are supported in import or export files. Characters that aren't included in the UTF-8 character set are considered invalid. For dimension name restrictions, see Naming Restrictions for Dimensions, Members, and Aliases.
>
> In order to delete UDAs and Attributes in metadata load files, replace the UDAs and attributes with <none> instead of leaving the property value blank.

For an example import file, see Example Entity Dimension Import File.

Example Entity Dimension Import File

```
Entity,Parent,Data Storage,Plan Type (Plan1),Plan Type (Plan2),Plan Type (Plan3),Data Type,Base Currency,
Alias: Default NoEntity,,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Unspecified Entity
AllEntities,,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Total Entity
1000,AllEntities,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Agencies
1001,1000,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Administration and Finance
1011,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Office of the Secretary of Administration and Finance
1012,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Bureau of State Office Buildings
1101,1012,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Finance and Administration
1102,1012,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,State Office Buildings
1103,1012,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Safety and Security
1104,1012,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,State House ADA/Communications Access
1105,1012,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,State House Events
1013,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Civil Service Commission
1014,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Department of Revenue
1106,1014,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Child Support Enforcement
1107,1014,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Division of Local Services
1108,1014,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Appellate Tax Board
1015,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Developmental Disabilities Council
1016,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Division of Administrative Law Appeals
1017,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Dept. of Revenue
1018,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Executive Office for Administration and Finance
1019,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,George Fingold Library
1020,1001,Never Share,TRUE,TRUE,TRUE,Unspecified,USD,Group Insurance Commission
```

In this example, the import file loads an Entity dimension with the required header record and three data records. This example is comma-delimited. The header record specifies the member to be imported (Entity), the parent member (Parent) into which to import the member, and the Data Storage property to assign to the member.

```
Entity, Parent, Data Storage
e1, Entity,
e2, ,
e1, e2, Shared
```

Using this import file would result in this outline, assuming that no other members exist:

```
Entity
e1
e2
    e1(Shared)
```

The first data record (`e1, Entity`) imports Entity member e1 as a child under the root member Entity. Unspecified values assume the default. For example, if data storage isn't specified, it assumes the default value, Never Share. The next data record (`e2, ,`) imports Entity member e2 under the dimension root member because no parent is specified, and sets data storage to Never Share. The last data record (`e1, e2, Shared`) imports a shared member of e1 under member e2, and sets data storage to Shared.

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Other Supported Delimiter Characters

In addition to commas and tabs, the application supports these delimiter characters in import and export files:

- tilde (~)
- grave accent (`)
- exclamation point (!)
- number sign (#)
- question mark (?)
- dollar sign ($)
- percent sign (%)
- caret (^)
- ampersand (&)
- asterisk (*)
- parentheses ( )
- hyphen-minus (-)
- plus (+)
- colon (:)
- semicolon (;)
- angle brackets (< >)
- backslash (\)
- forward slash (/)
- vertical bar (|)
- apostrophe (')
- braces ({ })
- underscore (_)
- brackets ([ ])
- at sign (@)
- period (.)

Only one character is supported for use as a delimiter. For example, one vertical bar (|) is supported, but two vertical bars (| |) are not supported.

> ⚠ **Caution**
>
> Not all of the characters listed can be used for all import and export scenarios. Note the following exceptions.

Import and Export Metadata Delimiter Exceptions

Don't use these delimiter characters in metadata import and export files.

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Table 5-1 Import and Export Metadata Delimiter Exceptions

Delimiter Character	Reason for Exception
double quotation mark ("")	Creates an empty file
plus (+) minus (-) forward slash (/) percent sign (%)	Causes an error if the metadata import file contains consolidation properties that use these characters
angle brackets (< >)	Causes an error if a property uses the value <none>

> **Note**
>
> Any character that conflicts with a character in a member name will cause an error.

Import and Export Data Delimiter Exceptions

Table 5-2 Import and Export Data Delimiter Exceptions

Delimiter Character	Reason for Exception
apostrophe (')	Causes an error if used in a cube name
parentheses ( )	Causes an error if used in a data import file
double quotation mark ("")	Creates an empty file
hyphen-minus (-)	Causes an error if used in a data import file

Loading the Metadata Import File

To load the metadata import file:

1. Create a metadata import file for each artifact (dimensions, smart lists, and exchange rate tables) that you want to import. See Creating the Metadata Import File.
2. Click **Application**, and then click **Overview**.
3. Click **Dimensions**, and then click **Import**.
4. On the **Import Metadata** page, click **Create**.
5. Select the location of the metadata import file or files:
   - **Local**: Loads the import file or files from a location on your computer. For **Import File**, click **Browse** to select the import file on your computer for the artifact you're importing.
   - **Inbox**: Loads the import file or files from the server. Enter the name of the file in **Import File**. See Uploading and Downloading Files Using the Inbox/Outbox Explorer.
6. For **File Type**, select **Comma delimited**, **Tab delimited**, or **Other**.

   If **Other** is selected, enter the delimiter character that is used in the import file. For a list of supported delimiter characters and exceptions, see Other Supported Delimiter Characters.
7. **Optional:** Select a clear option to delete members not explicitly specified in the load file before performing the import:

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- **Clear Members**: Deletes all members not specified in the load file
- **Clear Shared Members**: Deletes only the shared members not specified in the load file

> **Note**
>
> - Any member not specified is deleted from the application outline after importing the dimension unless it's an ancestor of a member that was specified, or it's a base member of a shared member that was specified.
> - If a clear option isn't selected, the import will only add or update existing members. Oracle recommends first loading the metadata file without selecting a clear option to make sure the file loads successfully. Then, select a clear option and execute the import again
> - The clear options are unavailable in certain dimensions where members must exist and are critical to the application.
> - Members that can't be deleted in the dimension editor for any reason (for example, because they're used in a form or in a validation rule) won't be deleted.
> - Because selecting **Clear Members** automatically deletes all members not specified in the load file (and this includes shared members), selecting **Clear Members** also automatically selects the **Clear Shared Members** option. You cannot deselect the **Clear Shared Members** option if the **Clear Members** option is selected.

8. **Optional:** If the location selected is **Inbox**, click **Save as Job** to save the import operation as a job, which you can schedule to run right away or at a later time. You can't select this option if **Local** is selected. See Managing Jobs.

> **Note**
>
> - Saving an import operation as a job is useful to batch a load sequence; for example, import metadata, then import data, then run rules when the data load is complete.
> - You can run up to five import jobs at one time.
> - If you want to review the data that was rejected during the metadata import job, you can specify an **Error File** on the **Save as Job** dialog. This file will provide information about the metadata records that were not imported for each dimension. If an error zip file is specified, a separate error file is created for each dimension, then the error files are zipped together, and the zip file is stored in the Outbox where you can download the file using Inbox/Outbox Explorer. See Uploading and Downloading Files Using the Inbox/Outbox Explorer.

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> **Tip**
>
> To streamline the job creation process for subsequent **Import Metadata** jobs, you can duplicate an existing job using the **Save As** option, then update it. See Duplicating Jobs.

9. **Optional:** If the location selected is **Local**, click **Validate** to test whether the import file format is correct.
10. Click **Import** to run the import operation.

If the cubes listed in the import file don't match the cube names in the application, the application will ignore unrecognized cube columns.

If the import fails, a **Failed** status will display in the **Last Validate/Import** column. Click **Failed** to view the detailed status. If the import is successful, a **Completed** status will display in the **Last Validate/Import** column. To view details about the successful import, click **Completed**, and then in the **Show** drop-down, select **All**.

When performing metadata loads as a part of cloning environments or importing snapshots using Migration or EPM Automate, Oracle Fusion Cloud Enterprise Performance Management uses the multi-pass solution. This solution automatically initiates another load of the same metadata input file if the previous metadata load resulted in rejected records.

Exporting Metadata

You can export metadata to a flat file in a .csv (comma-delimited) or .txt (tab-delimited or other delimiter character) format.

These artifacts are supported in the exports:

- Dimensions
- Smart Lists
- Exchange rates

The application creates an export file for each artifact (.txt or .csv, depending on the file type), and all export files are consolidated into one zip file. You must extract the .csv or .txt files from the zip file if you want to use the files as import files (for example, when importing into another application).

> **Note**
>
> Only characters that are included in the UTF-8 character set are supported in export or import files. Characters that aren't included in the UTF-8 character set are considered invalid. For dimension name restrictions, see Naming Restrictions for Dimensions, Members, and Aliases.

Videos

Your Goal	Watch This Video
Learn how to export metadata.	Exporting Metadata in Cloud EPM

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To export metadata to a flat file:

1. Click **Application**, and then click **Overview**.
2. Click **Dimensions**, and then click **Export**.
3. On the **Export Metadata** page, click **Create**.
4. Select the target environment of the export file:
   - **Local**: Saves the export file to a location on your local computer.
   - **Outbox**: Saves the export file to the server where you can use it to schedule the export job. See Uploading and Downloading Files Using the Inbox/Outbox Explorer.

     Click **Save as Job** to save the export operation as a job, which you can schedule to run right away or at a later time. You can't select this option if **Local** is selected.

> **Note**
>
> – You can run up to five export jobs at one time.
>
> – When scheduling export jobs, you can specify a unique output file name for each job.
>
> – For more information about scheduling jobs, see Managing Jobs.

> **Tip**
>
> To streamline the job creation process for subsequent **Export Metadata** jobs, you can duplicate an existing job using the **Save As** option, then update it. See Duplicating Jobs.

5. Select the artifact or artifacts to export.
6. Select an option:
   - **Comma delimited**: Creates a comma-delimited .csv file for each artifact.
   - **Tab delimited**: Creates a tab-delimited .txt file for each artifact.
   - **Other**: Creates a .txt file for each artifact. Enter the delimiter character that you want to use in the export file. For a list of supported delimiter characters and exceptions, see Other Supported Delimiter Characters.
7. Click **Export**, and then specify where to save the export file.

Importing and Exporting Data

After the dimensions are imported, you can populate the application data. You can't do a direct data load. You must import data from a file.

Related Topics

- Importing Data
- Driver Member Casting Errors
- Exporting Data
- Viewing Data Import and Export Status

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Importing Data

Before importing data, you must:

- Define or load metadata and refresh the database. See Importing Metadata.
- Prepare the data load file. Source data must be in a business process-specific data file format or in an Oracle Essbase data file format.
- Ensure the data load files do not contain any thousands separators, quotes, and so on, around data values. Up to one single . (decimal point) character is supported.
- Ensure the data import file type is .csv, .txt, or .zip.

You can create a data import file based on a template, or you can generate one based on the source system. To use a template, run the export data process in your business process and use the exported file as a template. See Exporting Data.

> **Note**
>
> When importing data, no validation is performed on intersections to determine which are valid, and data is loaded to all intersections. To review data that has been loaded into invalid intersections, run the Invalid Intersection report before importing data to see and clear invalid intersections. See Working With Invalid Intersection Reports.

To import data:

1. Click **Application**, and then click **Overview**.
2. Click **Actions**, and then click **Import Data**.
3. Click **Create** and make your **Import Data** selections:

Table 5-3 Data Import Options

Data Import Option	Description
Location	Select the location of the data import file: • **Local**: Imports the data import file from a location on your computer. • **Inbox**: Imports the data import file from the server. See Uploading and Downloading Files Using the Inbox/Outbox Explorer. **Note:** Valid data import file types are .csv, .txt, or .zip.

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Table 5-3 (Cont.) Data Import Options

Data Import Option	Description
Source Type	Select the source type for the data import file: • **Default**: Select this source type to load data in a business process-specific data file format. If you select this option, you must choose a **File Type** and a **Date Format**. File type options are **Comma delimited**, **Tab delimited**, and **Other**. Date format options are **MM-DD-YYYY**, **DD-MM-YYYY**, and **YYYY-MM-DD**. **Note:** – If an import error occurs during a business process-specific data load, the import will continue to load valid data (partial load). Any errors will be noted in the import and export status. See Viewing Data Import and Export Status. – For a list of supported delimiter characters and exceptions, see Other Supported Delimiter Characters. – If data is exported to a file in a business process-specific format, you can use Notepad to change the database name in the file; for example, ASOCube, and then import the data file into the aggregate storage outline (provided all dimensions exist in the aggregate storage outline). • **Essbase**: Select this source type to load data in an Essbase data file format. If you select this option, you must choose a cube. For **Delimiter**, specify the delimiter for the Essbase data; options are **Space** or **Tab**. **Note: Stop In Case of Error** is selected by default. If this option is selected and any import errors occur during an Essbase data load, the import will quit and no data will be loaded. If this option is cleared, data will continue to load even if there are errors.
Source File	• For **Local**, click **Browse** to navigate to the source file. • For **Inbox**, enter the name of the file in **Source File**.
Include Metadata (optional)	Clear this option to improve data import performance and to prevent members from changing position in the outline during data import. Regardless of this option setting, the required metadata must already exist in the business process and in the Essbase outline before you can load the application data.
Missing Data Label	Specify how you want missing values to display in forms. Enter an alphabetic value that does not exceed 16 characters. You may begin the value with a # (number sign). If no value is entered, the system defaults to #MISSING. **Note:** If you're using a data import file that's based on a template (in other words, you ran the export data process in your business process and you're using the exported file as a template), this label must match the **Missing Data Label** that was defined during data export. See Exporting Data.

4. Click **Save as Job** to save the import operation as a job, which you can schedule to run later.

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> **Note**
>
> - **Save as Job** is available only when **Inbox** is selected.
> - Saving an import operation as a job is useful to batch a load sequence, for example, import metadata, then import data, then run rules when the data load is complete. See Managing Jobs.
> - You can run up to five import jobs at one time.
> - If you want to review the data that was rejected during the data import job, you can specify an **Error File** on the **Save as Job** dialog. This file will provide information about the data records that were not imported. If an error zip file is specified, the zip file is stored in the Outbox where you can download the file using Inbox/Outbox Explorer. See Uploading and Downloading Files Using the Inbox/Outbox Explorer.

> **Tip**
>
> To streamline the job creation process for subsequent **Import Data** jobs, you can duplicate an existing job using the **Save As** option, then update it. See Duplicating Jobs.

5. **Optional:** Click **Validate** to test whether the import file format is correct.
6. Click **Import** to run the import operation.

> **Note**
>
> There is no import option in this business process to clear data before import. Therefore, the import will not overwrite existing data values with #MISSING values. However, you can run a clear cube job prior to import, or clear data with a script/rule or using the REST API:
>
> - You can run a Clear Cube job that enables you to clear specific business process data within input and reporting cubes. See Clearing Cubes.
> - You can write a business rule that uses the CLEARDATA Essbase command to clear specific data values. See CLEARDATA in *Technical Reference for Oracle Analytics Cloud - Essbase*.
> - You can use the Clear Data Slices REST API to clear business process and Essbase data for a specified region. See Clear Data Slices in *Oracle Fusion Cloud EPM REST APIs*.

> **Caution**
>
> If the evaluation order in the application is changed after exporting data, the import may have unexpected results.

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If a value in a data load input file doesn't match the evaluated driver member type; for example, if the evaluated type of a driver member is "date" and it's corresponding input value isn't a valid date format, a driver member casting error occurs. See Driver Member Casting Errors.

Driver Member Casting Errors

A driver member casting error occurs when a value in a data load input file doesn't match the evaluated driver member type. Here are some examples of driver member casting errors:

- If a driver member is of type "Date" and it's corresponding input value isn't a valid date format
- If a nonexistent Smart List value is specified for a driver member of type "Smart List"
- If a nonnumeric value is supplied for a member of type "Currency," "Data," or "Percent."

A casting error can occur for each driver member specified in the input file and for every occurrence in a data record, so the number of casting errors for a given input record ranges from 0 to the number of drivers. Because successful driver member bindings can occur along with driver member binding errors, the application will not reject the entire input record, but the error will be noted in the import and export status. See Viewing Data Import and Export Status.

Consider the following data record input file where a bad date value is specified:

```
Entity, aDate, aSl, aText, aCurr, aPercent, Point-of-View,Data Load Cube Name
e1, exampleBadDateValue, sl1, text11, 888, 0.99,
"FY15,Current,""BUVersion_1"",Local,u1,Jan",Plan1
```

Error messages displayed:

- After loading the data, the status window will read:

  ```
  Outline data store load process finished (Invalid data values were not
  loaded). 7 data records were read, 7 data records were processed, 7 were
  accepted for loading (verify actual load with Essbase log files), 0 were
  rejected.
  ```

- The error is noted as an error in the import and export status with the following entry details:

  ```
  Category: Data record
  ```

  ```
  Record Index: 2
  ```

  ```
  Message: Unable to load specified data value in Record 2: (aDate:
  exampleBadDateValue)
  ```

- If multiple casting errors occur, each error is recorded in the same error message:

  ```
  Unable to load specified data values in Record 2: (aSl: aaaaaaasl1),
  (aPercent: cc), (aDate: exampleBadDateValue), (aCurr: bb)
  ```

- If the error message count exceeds the maximum limit, it will issue the following message:

  ```
  The warning [Unable to load specified data values] message count limit (100)
  exceeded: no more warnings of this type will be issued.
  ```

Exporting Data

To export data to a file:

1. Click **Application**, and then click **Overview**.

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2. Click **Actions**, and then click **Export Data**.
3. Click **Create** and make your **Export Data** selections:

Table 5-4 Data Export Options

Data Export Option	Description
Location	Select the target environment of the data export file: • **Local**: Saves the data export file to a location on your local computer. • **Outbox**: Saves the data export file to the server. See Uploading and Downloading Files Using the Inbox/Outbox Explorer.
Cube	Select a cube from the drop-down list. **Note:** The driver/column dimension in the Export Data job definition needs to be dense.
File Type	Select from the following types: • **Comma delimited**: Creates a comma-delimited .csv file for each artifact. • **Tab delimited**: Creates a tab-delimited .txt file for each artifact. • **Other**: Creates a .txt file for each artifact. Enter the delimiter character that you want to use in the export file. For a list of supported delimiter characters and exceptions, see Other Supported Delimiter Characters.
Smart Lists	Specify **Export Labels** or **Export Names**.
Dense Dynamic Members (for BSO cubes only)	For a block storage (BSO) cube, select whether to **Include** or **Exclude** dense dynamic calculation members during export. Note the following: • Excluding dynamic calculation members from exported data improves performance during data exports. • Sparse dynamic members are not included in the export even if the **Include** option is selected.
Dynamic Members (for ASO cubes only)	For an aggregate storage (ASO) cube, select whether to **Include** or **Exclude** dynamic calculation members during export. **Note:** Excluding dynamic calculation members from exported data improves performance during data exports.
Decimals	Specify the number of decimal positions (0-16) to format the data after export, or select the default **None** to use the default precision formatting. For example, specifying a decimal value of **3** in the **Decimals** field will result in the exported data displaying three digits to the right of the decimal point, wherever applicable. **Note:** Selecting **None** formats the decimal number as specified in the default precision settings, whereas selecting **0** formats the data to display a whole number. For example, if the data value you are exporting is `123.45678`, selecting **0** will export the data as 123 and selecting **None** will export the data as `123.45678`.

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Table 5-4 (Cont.) Data Export Options

Data Export Option	Description
Disable Implied Share (for BSO cubes only)	For a block storage (BSO) cube, select this option to export the child row in the data file even though both parent and child rows have the same data. **Note:** By default, implied sharing is enabled during data export. This means that when data is exported, the implied sharing relationship is maintained so only the parent row is exported because the child row contains the same data as the parent. But if you want to export the child data row, you can select this option to disable the implied sharing relationship so that the child row can be exported. For more information about implied sharing, see Understanding Implied Sharing in Forms.
Missing Data Label	Specify how you want missing values to display in forms. Enter an alphabetic value that does not exceed 16 characters. You may begin the value with a # (number sign). If no value is entered, the system defaults to #MISSING. **Note:** If you're running the export data process to create a template file for a data import and you define a custom missing data label during data export, the **Missing Data Label** defined during the import data process must match the label defined here. See Importing Data.
Slice Definition	Select the slice of data to be exported. **Note:** If you're exporting data from an aggregate storage cube and you're exporting multiple members simultaneously, make sure that at least one of the members you select is a non-shared member. You can export shared members individually because it's equivalent to selecting a base member. However, selecting level 0 descendants of a parent member whose descendants are all shared members is equivalent to not selecting any members from a dimension and the export will not run. Therefore, you must ensure that one of the members selected is a non-shared member.

4. **Optional:** Click **Save as Job** to save the export operation as a job, which you can schedule to run immediately or at a later time.

> **Note**
>
> - **Save as Job** is available only when **Outbox** is selected.
> - You can run up to five export jobs at one time.
> - When scheduling export jobs, you can specify a unique output file name for each job.
> - For more information about scheduling jobs, see Managing Jobs.

> **Tip**
>
> To streamline the job creation process for subsequent **Export Data** jobs, you can duplicate an existing job using the **Save As** option, then update it. See Duplicating Jobs.

5. Click **Export**, and then specify where to save the data export file.

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> **Note**
>
> To reduce the size of data export files, if a form contains an entire row of missing values (for example, #MISSING), the row will be omitted from the data export file.

Troubleshooting

For help with troubleshooting issues if you get an Oracle Essbase query limit error when exporting a large number of data cells from ASO cubes, see Handling Issues Related to Large Data Export from ASO Cubes in *Operations Guide*.

Viewing Data Import and Export Status

To view the status of a data import and export:

1. Click **Application**, and then click **Jobs**.
2. Click the name of the import or export job under **Recent Activity**.

The status page displays messages for **Errors**, **Warnings**, **Info**, and **All**. If there are errors and you specified an error file name while defining the job, you can download the generated error file using the **Error File** link.

Validating the Essbase Outline

Pre-validate the Essbase outline to ensure that your application is compatible with a Hybrid Essbase version.

As part of an effort to upgrade all environments to an Oracle Essbase version that supports Hybrid Block Storage Option (BSO) cubes, Oracle has added a process to pre-validate Essbase outlines to ensure a smooth transition when your environment is upgraded to a Hybrid Essbase version.

> **Note**
>
> For information about Essbase versions, see About Essbase in Cloud EPM in *Getting Started Guide for Administrators*.

To ensure that your application is compatible with a Hybrid Essbase version, we've implemented a new utility which verifies the member formulas in your environment and provides a report so you can fix any issues.

The utility can be launched manually and it will run automatically during database refresh:

- For information about manually launching the utility, see Manually Launching the Pre-Validation Process.
- For information about automatic launch during database refresh, see Automatic Pre-Validation During Database Refresh.

If errors are found after the utility completes verification of the member formulas, you can view the **Outline Pre-validation Report** and fix the errors it finds. If no errors are found, no additional action is required.

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For a list of sample pre-validation errors and recommended resolutions, see the sections below:

- Addressing Validation Errors
- Sample Validation Errors

Manually Launching the Pre-Validation Process

To manually launch the pre-validation process, select from the following options under the **Essbase Outline Validation** menu option in the **Actions** menu on the application's **Overview** page:

- **Pre validate Outline**: Performs a validation of your application
- **Outline Pre-validation Report**: Displays a list of member formulas that need to be fixed to be compatible with Hybrid Essbase version

> **Note**
>
> - The **Essbase Outline Validation** menu option only displays if your environment is on a non-Hybrid Essbase version. If your environment is on a Hybrid Essbase version, then the **Essbase Outline Validation** menu option is no longer available.
> - Another way to check whether your environment is on a Hybrid or non-Hybrid Essbase version is to view the value of **Essbase Version supports Hybrid Block Storage Option** in the **Activity Report**. If the value is **Yes**, it means that your environment is on a Hybrid Essbase version. If the value is **No**, then your environment is on a non-Hybrid Essbase version.

To validate the Essbase outline in your environment and to fix any errors found in member formulas:

1. From the Home page, click **Application**, and then click **Overview**.
2. Click **Actions**, select **Essbase Outline Validation**, then select from the following options:
   - To perform the validation, select **Pre validate Outline**.
   - To view and download a report of errors found during validation, select **Outline Pre-validation Report**.

After you have fixed the member formulas, you can use the **Pre validate Outline** option again to make sure that all issues are resolved.

> **Note**
>
> After fixing the member formulas, ensure that you refresh the database before viewing the pre-validation report again, otherwise the report won't show that all the errors are resolved.

Automatic Pre-Validation During Database Refresh

If the system detects that your environment uses a non-Hybrid Oracle Essbase version and will need to be upgraded to Hybrid Essbase, the validation utility will automatically launch at the end of the **Refresh Database** process.

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If member formula errors are discovered during the validation process, you'll see a warning telling you to review the report and address any errors. Click **View Outline Pre-validation Report** to view the full report and resolve any errors.

Addressing Validation Errors

Validating the outline may reveal errors like those listed below. Address these issues using the provided recommendations and re-run the validation process until all errors are resolved.

Error Category	Recommendations
ELSE or *ENDIF* without *IF*	The member formula utilizes *if*, *elseif*, *else*, and *endif* constructs, but there are more *endif* tokens than *if* tokens, or the *else* and *if* tokens are mismatched. The number of *if* tokens should correspond with the number of *endif* or *else* tokens, ensuring proper grouping. Correct the formula by addressing the syntax issues.
Invalid Data Type or expected type *NUMBER* found *STRING*	The member formula refers to a non-existing member. Review the formula and remove the non-existing member from it.
Invalid Expression Return Type	The member formula refers to a non-existing member. Review the formula and remove the non-existing member from it.
No Matching Quote Found	The formula contains mismatched quotation marks. Ensure that the number of opening and closing quotation marks are equal and properly grouped. Correct the formula by resolving syntax errors.
Nothing to do	The member formula for the member is commented out using /* and */. Since the member is dynamic, it must have a valid formula. Put a valid member formula for the member or change the member property.
Parenthesis Error	The formula has mismatched parentheses. Ensure that the number of opening and closing parentheses are balanced and correctly grouped. Correct the formula by resolving syntax errors.
Substitution Variable doesn't exist	The member formula references a non-existent Substitution Variable. Create the Substitution Variable with an appropriate value to ensure successful formula validation.
Syntax Error in MDX Query	To fix your syntax error, see MDX Syntax and Grammar Rules.
Missing Semicolon or Unexpected End of Expression	The member formula does not end with a semicolon. The formula may return a value, however the absence of a terminating semicolon is a syntax error. Review and correct the formula.
Unknown Member Name used in function	The member formula references a non-existent member. Review and update the formula to remove any references to members that do not exist.
Environment Variable doesn't exist	The formula refers to an environment variable. You can't create an environment variable in the Oracle Fusion Cloud Enterprise Performance Management applications. Create a substitution variable instead with the same name and value of the environment variable, and use that in the formula instead.

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Error Category	Recommendations
Formula Parsing Error	The member formula references a member name that is not quoted, or the formula does not end with a semicolon. Additionally, the member name imay nclude keywords or function names such as *for*, *year*, and *sum* which are not permitted in the formula. Ensure that the member name is properly quoted and that the formula ends with a semicolon. Also, verify that a member with that name exists in the system.
Unknown Member Used in Query	For an ASO cube, the member formula refers to a non-existent member in the MDX query. Additionally, ensure that the names of dimensions, members, and attribute dimensions do not contain the / (slash) character.
Duplicate Member Name	The member formula refers to a non-existing member. Review the formula and remove the non-existing members from it.
Invalid Assignment	Invalid assignment: The value on the left side of the equal sign must be a member or variable. In the member formula, the left side of the assignment operation using equal to sign (=) should be the name of an existing member. Ensure that the left-hand side of the assignment is a valid member name.
Operator Error	The member formula contains improperly nested *if*, *elseif*, or *endif* constructs, resulting in incorrect and unbalanced nesting. Review and correct the formula to address these syntax issues.
Function Error	The member formula uses functions, but extra parameters have been passed. If the function involves Planning's substitution variables, ensure these variables are available and have values, which should not start with ampersand character (&). Additionally, review the formula to remove any extra & symbols before substitution variables. Ensure that each substitution variable in the formula has a single & before its name for proper resolution.
Missing Expression	Review and fix the formula to remove the expressions related syntax errors.

Sample Validation Errors

In these samples, `[MEMBER_NAME]` indicates the name of the member that failed the validation.

Formula verification error samples: member name, unknown member, missing semi-colon, end of expression, etc.:

- `Formula verification failed for member [MEMBER_NAME]. Error compiling formula for [] (line 3): unknown member name ["[MEMBER_NAME]"] in function [operator @X]`
- `Formula verification failed for member [MEMBER_NAME]. Error parsing member name in formula`
- `Formula verification failed for member [MEMBER_NAME]. Error compiling formula for [] (line 1): semicolon missing`

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- `Formula verification failed for member [MEMBER_NAME]. Error compiling formula for [] (line 19): unexpected end of expression: [...IF(...)]`
- `Formula verification failed for member [MEMBER_NAME]. Error compiling formula for [] (line 1): unexpected end of expression: [@ADD]`

Formula verification errors sample: substitution variables sample:

```
Formula verification failed for member [MEMBER_NAME]. Substitution
variable [[MEMBER_NAME]] doesn't exist
```

Formula verification error sample: parenthesis:
`Formula verification failed for member [MEMBER_NAME]. Error compiling formula for [] (line 1): [)] without [(]`

Formula verification error sample: invalid data type, return type, and assignment:

- `Formula verification failed for member [MEMBER_NAME]. Error compiling formula for [] (line 1): expected type [NUMBER] found [STRING] ([MEMBER_NAME]) in function [@SUB]`
- `Formula verification failed for member [MEMBER_NAME]. Error compiling formula for [] (line 12): invalid expression return type`
- `Formula verification failed for member [MEMBER_NAME]. Error compiling formula for [] (line 4): invalid assignment`

Reverting a Hybrid BSO Application to Non-Hybrid

Upgrading to hybrid-enabled Oracle Essbase allows some subscriptions to enable the use of hybrid BSO cubes in the application. Additionally, newer Enterprise Performance Management Standard Cloud Services, by default, are supported on hybrid-enabled Essbase. Newly created applications for such subscriptions automatically use hybrid BSO cubes. You can disable the use of hybrid BSO cubes.

The availability of the option to disable the hybrid mode depends on whether your subscription supports it. Additionally, only these business processes or application types allow you to disable the hybrid mode:

- Planning Custom applications
- FreeForm

> ⚠ **Caution**
>
> Before disabling the hybrid mode, carefully review the sparse dimension configuration to ensure that the upper level members set to *Dynamic Calc* with hybrid mode enabled are reverted to *Store* or *Never Share*, as appropriate, in non-hybrid mode. Additionally, review rules to include the sparse dimensions that were set to *Dynamic Calc* for parent members for aggregations and other rules, as appropriate.

To disable the use of hybrid BSO cubes:

1. Sign into the application as a Service Administrator.
2. From the Home page, select **Application**, and then **Overview**.

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3. Click **Actions**, and then select **Disable Hybrid Mode**.

For more information about hybrid mode, see About Essbase in Cloud EPM in *Getting Started Guide for Administrators*.

Creating and Refreshing Application Databases

Things you need to know before you refresh the application database.

Related Topics

- Creating Application Databases
- Before Refreshing the Database
- Refreshing Application Databases
  Refresh the database after changing the application structure, and modifying artifacts such as dimensions and members.
- Validating Metadata

Creating Application Databases

The application database, which stores data for each cube in the application, is created when the application is created.

To create a FreeForm application, see Creating a FreeForm App.

If you need to clear the data, you can do so using Calculation Manager. To refresh the data, see Refreshing Application Databases.

Before Refreshing the Database

You must refresh the database after changing the application structure, and modifying artifacts such as dimensions and members. Oracle recommends that you notify all users, asking them to save their work and close the application, before you refresh the database.

Before refreshing the database:

- Back up your application and export data from all databases.
- Decide if you want to enable all users or just the current Service Administrator to use the application in administration mode during the refresh
- Decide if you want to log off all users
- Terminate any active application requests

Refreshing Application Databases

Refresh the database after changing the application structure, and modifying artifacts such as dimensions and members.

The application database stores data for each cube in the application. You can start the Refresh Database process right away or schedule it to run later as a job.

To refresh the database:

1. See Before Refreshing the Database.
2. Click **Application**, and then click **Overview**.

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3. Click **Actions**, and then **Refresh Database**.
4. On the **Refresh Database** page, click **Create**, and then select from the following options:

   Before Refresh Database

   - **Enable use of the application for**: Lets **All users** or **Administrators** (or the current logged in Service Administrator) access the application in administration mode during the refresh
   - **Log off all users**: Logs off all application users before starting the refresh
   - **Stop all active requests**: Terminates any active requests in the Essbase application before starting the refresh

   After Refresh Database

   - **Enable Use of the Application for**: Lets **All users** or **Administrators** use the application after the refresh
   - **Validate Metadata**: By default, this option is not selected. When this option is selected, the system scans for member identity (ID) mismatches between the business process and Oracle Essbase and then automatically resolves them after the refresh process. Selecting **Validate Metadata** might add additional time to the refresh process. For more information, see Validating Metadata.

     > **Note**
     >
     > For the first cube refresh after upgrading from non-hybrid to Hybrid Essbase, enable the **Validate Metadata** option (either immediately or later as a scheduled job). For examples of potential errors during validation and how to address them, see Validating the Essbase Outline in *Getting Started Guide for Administrators*.

     The **Validate Metadata** option is available for cubes with these storage options running on the Oracle Essbase version that supports Hybrid Block Storage Option (BSO) cubes: Hybrid BSO, ASO, and BSO. This option is not available for cubes running on the legacy Essbase version that does not support Hybrid BSO cubes.

   - **Execute Application Diagnostics**: By default, this option is not selected. When this option is selected, Application Diagnostics automatically runs during the Refresh Database process.
     The Execute Application Diagnostics setting is saved with a database refresh job and retained when the job is edited or rerun. If enabled, the job progress includes a Generating Application Diagnostics Report step. After the job completes, the Jobs console displays the refresh and diagnostics results, along with the report name if generated successfully.

     > **Note**
     >
     > Application Diagnostics is an optional validation step. It does not change the outcome of the database refresh. If the refresh fails, the overall job is reported as failed even if diagnostics completes successfully.

5. To refresh the database now, click **Refresh Database**. To run the refresh later as a job, click **Save as Job**. All of the options you've selected on the **Refresh Database** page, including **Validate Metadata**, will persist to the saved job. See Managing Jobs.

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> **Tip**
>
> To streamline the job creation process for subsequent **Refresh Database** jobs, you can duplicate an existing job using the **Save As** option, then update it. See Duplicating Jobs.

Troubleshooting

For help with troubleshooting database refresh issues, see Troubleshooting Database Refresh Issues in *Operations Guide*.

If your system detects Essbase outline pre-validation errors during database refresh, see Validating the Essbase Outline.

Validating Metadata

About Validate Metadata

Use the **Validate Metadata** setting (formerly called **Autocorrect Errors**) as part of the database refresh process to scan for and automatically resolve member identity (ID) mismatches between the business process and Oracle Essbase.

> **Caution**
>
> Selecting **Validate Metadata** might add additional time to the refresh process.

In certain cases, the identity of a member changes in a way that causes it to be not fully recognized in the multi-dimensional database used by the application. When a member's identity is not fully correctly established within all parts of the system, it could sometimes cause hierarchy issues like incorrect member ordering, or it could miss children underneath that member, or miss some siblings, or miss certain properties, and so on. Selecting **Validate Metadata** detects such members and automatically resolves their identity, allowing these members as well as their hierarchies along with properties and siblings, to be fully and correctly recognizable and usable within all the parts of the system.

Viewing the Report Detailing Member Movement and Deletion

After making member ID corrections, members might be deleted or moved. A report file detailing any member movement (to a different parent) or deletion is generated and placed in the Outbox in a csv (comma-delimited) file format. Review the report to reaggregate some of the stored parent members where the child hierarchy may have changed.

For example:

Action	Member	Parent	Dimension	Cube	New Parent
Delete	Member1	Parent1	Scenario	Plan1	
Move	Member2	Parent2	Account	Plan2	Parent3

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If a correction is made, the system generates a CSV file to the Outbox folder, and the filename is automatically generated with the application name followed by `_autocorrect.csv`; for example, `vision_autocorrect.csv`. No file is created if no issue was addressed. If a previous `*_autocorrect.csv` file exists and a correction is made during a subsequent run, this file is overwritten.

After running the **Refresh Database** job with the **Validate Metadata** setting, you can download the export file from the Outbox.

To download files:

1. Click **Application**, and then click **Jobs**.
2. Under **Recent Activity**, click the job.
3. At the top of **Job Details**, click the export file option to select a download location.

Adding an Aggregate Storage Outline to an Application

Understand what aggregate storage is and how to use it in an application.

Related Topics

- About Aggregate Storage
- Aggregate Storage Outline Cube Characteristics
- Process for Adding an Aggregate Storage Database to an Application

About Aggregate Storage

Aggregate storage is the database storage model that supports large-scale, sparsely distributed data that is categorized into many, potentially large dimensions. Selected data values are aggregated and stored, typically with improvements in aggregation time. Aggregate storage is an alternative to block storage (dense-sparse configuration).

Aggregate Storage Outline Cube Characteristics

- The application doesn't generate XREFs on aggregate storage databases. XREFs can only be generated on block storage databases.
- Dynamic time series members are not applicable for the Period dimension in an aggregate storage application.
- Creating and refreshing security filters are not applicable for aggregate storage databases.

Troubleshooting

Performance of aggregate storage reporting cubes is governed by many factors, including the number of slices in the cube, the outline of the cube, and the type of dimensions in the cube. See Optimizing Aggregate Storage Option Cubes in *Operations Guide*.

Aggregate Storage Outline Cube Characteristics

- The application doesn't generate XREFs on aggregate storage databases. XREFs can only be generated on block storage databases.
- Dynamic time series members are not applicable for the Period dimension in an aggregate storage application.
- Creating and refreshing security filters are not applicable for aggregate storage databases.

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Process for Adding an Aggregate Storage Database to an Application

To add an aggregate storage database to an application:

1. Create an aggregate storage cube. Perform one task:
   - Create an aggregate storage cube during application creation.

     To create a FreeForm application, see Creating a FreeForm App.
   - Add a new cube using the cube editor.

     See Managing Cubes.
2. Add dimensions to the aggregate storage cube.

   See Editing Dimensions in the Simplified Dimension Editor.
3. Add dimension members. See Adding Members.
4. Refresh the outline for the application. See Refreshing Application Databases.
5. Create a form using the dimensions associated with the aggregate storage database. See Administering Forms.

Removing an Application

Proceed with caution when removing an application.

Because removing an application will delete all of its contents and any scheduled jobs, make a backup of your application.

To remove an application:

1. Click the **Application**, then click **Overview**.
2. Click **Actions**, and then **Remove Application**.

Setting the Daily Maintenance Process Start Time

Understand how to determine and set the most convenient time to start the hour-long daily maintenance process.

A business process instance requires one hour every day to perform routine maintenance. Service Administrators can select (and change) the most convenient time to start the hour-long daily maintenance process.

For information about daily maintenance operations and scheduling the maintenance start time for an environment, see Managing Daily Maintenance in *Getting Started Guide for Administrators*.

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> **Note**
>
> There is a system setting called **Export Cloud EPM Smart List textual data during daily maintenance for incremental data import** that, depending on if this setting is enabled, could impact the duration of the export. For more information and for recommendations about using this setting, see the following topics:
>
> - Exporting Smart List Textual Data During Daily Maintenance for Incremental Data Import in *Operations Guide*
> - What Application and System Settings Can I Specify?

To prevent automatic backup from failing due to certain jobs running during daily maintenance, Oracle Fusion Cloud Enterprise Performance Management disallows certain jobs from starting while the daily maintenance process is running.

The following jobs are prevented from starting during daily maintenance:

- Import Data
- Import Metadata
- Export Data
- Export Metadata
- Refresh Database
- Clear Cube
- Restructure Cube
- Compact Outline
- Merge Data Slices
- Optimize Aggregation

These jobs are prevented from starting whether they are jobs that are scheduled or jobs that are started on an ad hoc (unscheduled) basis.

For unscheduled jobs, this means that if you attempt to start a job during the daily maintenance process using the **Run Now** option on the **Schedule Jobs** page or if you submit a job from the listing page of your job definitions, the job will be prevented from starting. If the system prevents an ad hoc job from starting, an error message will display.

For example, if you view the jobs listed on the **Clear Cube** page while daily maintenance is running and then you click **Actions** and then **Submit**, an error message will display and the job will not start. Similarly, if you create a **Clear Cube** job on the **Schedule Job** page during daily maintenance and select **Run Now**, the system will prevent the job from starting and you'll see an error message.

If the system prevents a job from starting, the reason will be stated in the Job Details. If you've enabled email notifications for the Job Console, you'll receive an email notification when a job does not start. If you have a job that is scheduled to start during the daily maintenance process, it is recommended that you reschedule your job to start outside of the daily maintenance window. See Scheduling Jobs.

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Uploading and Downloading Files Using the Inbox/Outbox Explorer

Use the Inbox/Outbox Explorer to upload files to the server or download them to your computer.

To schedule an import or export job, see Scheduling Jobs.

To upload or download files:

1. Click **Application**, and then click **Overview**.
2. Click **Actions**, and then **Inbox/Outbox Explorer**.
3. **Optional:** Filter by name, refresh the listing, and upload files from the server.
4. To upload a file from the server, click **Upload**. In the **Upload File** dialog box, click **Browse**, select the file to upload, and then click **Upload File**.
5. To download a file from the server, select the desired file, then click ••• (Actions), and then click **Download File**.
6. Click **Close** to close the **Inbox/Outbox Explorer** and return to the **Overview** tab.

> **Note**
>
> There is an upload file size limit of 2GB. If your upload file exceeds the size limit, either select a smaller file or use EPM Automate to upload the larger file. For information about EPM Automate, see *Working with EPM Automate*.

Validating Groovy Scripts

Validate Groovy business rule scripts to ensure that your application's Groovy scripts can pass the stricter validation rules.

As you migrate to an upgraded version of Groovy, you might encounter certain issues that need to be addressed. A program called the **Groovy Script Validator** aids in finding the rules that need to be adjusted and provides a report so you can fix any issues.

> **Note**
>
> The **Groovy Script Validator** is only available only when your application contains user-created Groovy business rules.

To validate Groovy scripts:

1. Sign into your application as a Service Administrator.
2. From the Home page, select **Application**, and then **Overview**.
3. Click **Actions**, and then select **Groovy Script Validator**.

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The **Groovy Script Validator** option is only available when your application contains user-created Groovy business rules.

After the program completes verification of the Groovy scripts, you can view the Groovy validation report and fix any errors it finds. If no errors are found, no additional action is required.

To view the Groovy validation report:

1. From the Home page, select **Application**, and then **Jobs**.
2. Under **Recent Activity**, click the **Groovy validation report Job**.
3. Next to **Groovy Script Validator Report**, click (Download File).
4. Optionally, you can download the `GroovyValidationReport.html` file from the **Inbox/Outbox Explorer** by selecting **Application**, then **Overview**, click **Actions**, and then select **Inbox/Outbox Explorer**.

> **Note**
>
> For a list of validation errors and recommended resolutions, see Resolving Groovy Business Rule Validation Issues in *Designing with Calculation Manager*.
>
> To learn more about Groovy business rules, see Creating a Groovy Business Rule in *Designing with Calculation Manager*.

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6

Using the Member Selector

Use the member selector to select members and use wildcards in searches. You can also select substitution variables, user variables, and user-defined attributes (UDAs).

Related Topics

- Working with Members
- Making Selections
- Member Relationships
- Using Wildcards in Searches
- Selecting Attribute Values as Members
- Selecting Members for Forms
- Selecting Substitution Variables as Members
- Selecting User Variables as Members
- Selecting UDAs as Members

Working with Members

Use the **Select Members** dialog box to select the members to use with the following features:

- Business rule runtime prompts
- User variables
- Dynamic variables
- Data export
- Point of view and page axis where valid intersection rules are applied
- Valid intersections

> **Note**
>
> Forms 2.0, Dashboards 2.0, and ad hoc grids use a different version of the member selector. See Using the Member Selector in *Working with FreeForm*.

If variables and attributes are defined, you can select variables and attributes. You can display and select members by member name or alias. The display options that you define for the **Member Selection** dialog box override those defined as an application default by a Service Administrator, and those specified as an application preference.

If drop-down member selectors are defined for row dimensions on a form, you can select members directly from the list, enter data, and add rows to a form.

See Entering Data for Suppressed or Excluded Members in *Working with FreeForm*.

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Making Selections

Only members, substitution variables, and attributes to which you have access are displayed. The **Selections** pane only displays if you invoke the member selector for multiple member selection.

If valid intersection rules are used, only valid members are displayed in application forms and in runtime prompts for Calculation Manager rules. In ad hoc forms, both on the Web and in Oracle Smart View for Office, the application doesn't filter by valid intersections rules in point of view and page axis. Only cells in grids honor valid intersection rules in ad hoc forms.

To make selections:

1. Click (Member Selector).
2. **Optional:** Perform these tasks:
   - To enter search criteria (member name or alias only), press **Enter** (from the desktop) or click **Search** (on mobile).

     > **Note**
     >
     > The search isn't case-sensitive. You can search for a word, multiple words, or wildcard characters. See Using Wildcards in Searches.

   - To change display options such as viewing variables and attributes, showing alias names, showing member counts, sorting alphabetically, refreshing the member list, or clearing selections, click (Settings) next to **Search**, and then select from the list of display options.
   - To filter the members that are displayed in the member list, under (Settings) next to **Search**, select **Add Filter**, and then select from the list of filter options.

     > **Note**
     >
     > Android users can click (Filter) to view and select filter options.

3. Make selections by clicking a member in the member list.

   To understand how related members are selected, see Member Relationships.

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> **Note**
>
> - Selected members display a check mark and are moved to the **Selections** pane, if applicable.
> - To expand a parent member to see its child members, click the expansion icon to the right of the parent member name. Clicking the expansion icon will not select the parent member.
> - To clear selections you have made, click (Settings) and then select **Clear Selection**.
>
> At times, you may find that a point of view and page axis member you want to select is suppressed. This occurs because a dimension selection in the point of view and page axis has invalidated other dimensions due to valid intersections that were applied. To resolve this issue, use the **Clear Selection** option to clear the point of view and page axis members you previously selected. Then you can use the point of view and page axis again to select members that were previously suppressed.
> - To show all the members that are suppressed due to valid intersection rules, click (Settings) and then select **Show Invalid Members**. Invalid members are displayed but are unavailable for selection.

4. **Optional:** Perform these tasks:
   - To further refine which related members are selected in the **Selections** pane, click (Function Selector) to the right of the member to display the member relationships menu:

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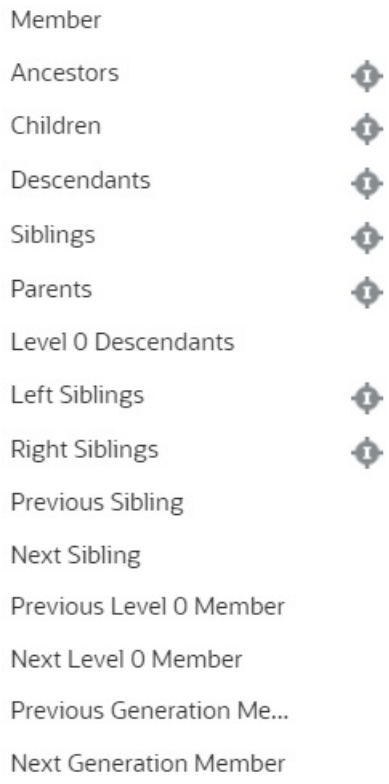

Click the relationship name to select the related members, excluding the selected member. Click **Include** to the right of the relationship name to select the related members, including the selected member.

For descriptions of the relationships, see Member Relationships.

- If substitution variables or attributes are defined, below the member selection area, click (Down Arrow) next to **Members**, and then select **Substitution Variables** or **Attributes** to select members for substitution variables or attributes. Members are displayed as children. Only members to which the user has read access are displayed in forms.
- To move or remove members in the **Selections** pane, click (Settings) next to **Selections**.
- To highlight the location of a selected member within the dimension hierarchy, double-click the member name in the **Selections** pane.

> **Note**
>
> To highlight the location of a selected member on a mobile device, tap the member name in the **Selections** pane, and then tap **Locate** under (Settings) next to **Selections**.

5. When you're done making selections, click **OK**.

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Member Relationships

This table describes which members and related members are included during member selection.

Table 6-1 Member Relationships

Relationship	Members Included
Member	The selected member
Ancestors	All members above the selected member, excluding the selected member
Ancestors (inc)	The selected member and its ancestors
Children	All members in the level immediately below the selected member
Children (inc)	The selected member and its children
Descendants	All descendants of the selected member, excluding the selected member
Descendants (inc)	The selected member and its descendants
Siblings	All members from the same level in the hierarchy as the selected member, excluding the selected member
Siblings (inc)	The selected member and its siblings
Parents	The member in the level above the selected member
Parents (inc)	The selected member and its parent
Level 0 Descendants	All descendants of the selected member that have no children
Left Siblings	The members that appear before the selected member with the same parent
Left Siblings (inc)	The selected member and its left siblings
Right Siblings	The members that appear after the selected member with the same parent
Right Siblings (inc)	The selected member and its right siblings
Previous Sibling	The member that appears immediately before the selected member with the same parent
Next Sibling	The member that appears immediately after the selected member with the same parent
Previous Level 0 Member	The previous level zero member that appears before the selected member Examples: PrevLvl0Mbr(Jan) returns BegBalance; PrevLvl0Mbr(Jul) returns Jun; PrevLvl0Mbr(BegBalance) returns an empty result
Next Level 0 Member	The next level zero member that appears after the selected member Examples: NextLvl0Mbr(Mar) returns Apr; NextLvl0Mbr(Dec) returns an empty result
Previous Generation Member	The member that appears immediately before the selected member within the same generation
Next Generation Member	The member that appears immediately after the selected member within the same generation

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Table 6-1 (Cont.) Member Relationships

Relationship	Members Included
Relative	Returns a member relative to the specified member at the same generation with the specified offset. Examples: Relative("Jan", 4) returns May, which is January plus 4 months in a standard monthly Period dimension; Relative("Jan", -2) returns Nov, which is January minus 2 months in a standard monthly Period dimension.
Relative Range	Returns all of the members from the starting member to the offset member when going forward or from the offset member to the starting member when going backwards. Examples: RelativeRange("Jan", 4) returns Jan, Feb, Mar, Apr, May; RelativeRange("Jan", -2) returns Nov, Dec, Jan.

> **Note**
>
> Member relationships work for any dimension, not just the Period dimension. We've used the Period dimension in our examples because it's easier to explain as time is linear.

Using Wildcards in Searches

You can use these wildcard characters to search for members.

Table 6-2 Wildcard Characters

Wildcard	Description
?	Match any single character
*	Match zero or multiple characters. For example, enter "sale*" to find "Sales" and "Sale" because * includes zero or more characters after the word "sale." The default search uses the * wildcard. For example, entering "cash" searches for "*cash*" and returns "Restricted Cash", "Cash Equivalents", "Cash", and "Noncash Expenses" because the word "cash" appears within each matched item.
#	Match any single number (0-9)
[list]	Match any single character within a specified list of characters. You can list specific characters to use as wildcard. For example, enter [plan] to use all the letters within the brackets as a single wildcard character. You can use the "-" character to specify a range, such as [A-Z] or [!0-9]. To use the "-" character as part of the list, enter it at the beginning of the list. For example, [-@&] uses the characters within the brackets as wildcard characters.
[!list]	Match any single character not found within a specified list of characters. The "-" character can also be used to indicate a range, such as [!A-Z] or [!0-9].

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Selecting Attribute Values as Members

If attribute members are defined, you can select attribute values on the **Select Members** dialog box. For attribute members, selecting a non-level 0 attribute selects all level 0 descendants and applies the operator to each. For attributes of type numeric, date, and Boolean (where false = 0 and true = 1), evaluation is based on the minimum and maximum values. For text attributes, evaluation is based on the position from top to bottom in the hierarchy. The top position has the lowest value, and the bottom position has the highest value.

Example: Numeric attribute

In this example, the selected operator is applied to each level 0 descendant, based on the numeric value. For example, selecting NotEqual and Small in the Member Selection dialog box includes all values not equal to 1 and not equal to 2, so the selection includes 3, 4, 5, and 6. Selecting Greater and Small includes all values greater than 1 or greater than 2, so the selection includes 2, 3, 4, 5, and 6.

```
Size

Small

1

2

Medium

3

4

Large

5

6
```

Table 6-3 Example: Numeric Attribute Evaluation

Selected Operator	Selected Attribute Value	Result	Explanation
Equal	Large	5, 6	The Equal operator is applied to all level 0 descendants of Large, which includes 5 and 6.
Less	Medium	1, 2, 3	The Less operator is applied to all level 0 descendants of Medium. This includes values < 3 OR < 4, which results in 1, 2, and 3.
Greater	Medium	4, 5, 6	The Greater operator is applied to all level 0 descendants of Medium. This includes values > 3 OR > 4, which results in 4, 5, and 6.
GreaterOrEqual	Medium	3, 4, 5, 6	The GreaterOrEqual operator is applied to all level 0 descendants of Medium. This includes values >=3 OR >= 4, which results in 3, 4, 5, and 6.
LessOrEqual	Medium	1, 2, 3, 4	The LessOrEqual operator is applied to all level 0 descendants of Medium. This includes values <=3 OR <=4, which results in 1, 2, 3, and 4.

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Table 6-3 (Cont.) Example: Numeric Attribute Evaluation

Selected Operator	Selected Attribute Value	Result	Explanation
NotEqual	Medium	1, 2, 5, 6	The NotEqual operator is applied to all level 0 descendants of Medium. This includes values not equal to 3 AND not equal to 4, which results in 1, 2, 5, and 6.

Example: Text attribute

For text attributes, the selected operator is applied to each level 0 descendant based on its position in the hierarchy, from top (lowest value) to bottom (highest value).

In this example, Envelope is at the top position and has the lowest value. Packet has the next higher value, followed by Box, Carton, Barrel and Crate. Crate is at the bottom position and has the highest value.

For this text attribute, selecting Less and Small includes values that are less than Envelope or less than Packet. Because Envelope is less than Packet, the resulting selection includes only Envelope. Likewise, selecting Greater and Large includes values that are greater than Barrel or greater than Crate, so the resulting selection includes only Crate.

```
Containers

Small

Envelope

Packet

Medium

Box

Carton

Large

Barrel

Crate
```

Table 6-4 Example: Text Attribute Evaluation

Selected Operator	Selected Attribute Value	Result	Explanation
Equal	Medium	Box, Carton	The Equal operator is applied to all level 0 descendants of Medium, which includes Box and Carton.
NotEqual	Medium	Envelope, Packet, Barrel, Crate	The NotEqual operator is applied to all level 0 descendants of Medium. This includes values not equal to Box AND not equal to Carton, which results in Envelope, Packet, Barrel, and Crate.

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Table 6-4 (Cont.) Example: Text Attribute Evaluation

Selected Operator	Selected Attribute Value	Result	Explanation
Less	Medium	Box, Packet, Envelope	The Less operator is applied to all level 0 descendants of Medium. This includes everything at a lower position than Carton OR a lower position than Box, which results in Box, Packet, and Envelope.
LessOrEqual	Medium	Envelope, Packet, Box, Carton	The LessOrEqual operator is applied to all level 0 descendants of Medium. This includes everything at the same position as Carton OR at a lower position than Carton, which results in Envelope, Packet, Box, and Carton.

Selecting Members for Forms

When selecting members for forms:

- To filter members from certain users, restrict their access permissions to members, and then refresh the plan.
- In the **Layout** tab of the **Form Management** or **Edit** dialog box, you can open the **Member Selection** dialog box by clicking the member selection icon, or by right-clicking a row or column and selecting **Select Members**.
- The order of members in the **Selected Members** list determines the order on forms. To change the order, select a member and click the Up or Down Arrow above the selected members list.

> **Note**
>
> – If you select members individually and select their parent first, the parent displays in the form at the top of its member hierarchy. (Note that depending on the number of hierarchy levels, calculating totals for the parent of individually selected members could take several passes, slowing calculations).
>
> – The parent of members selected by relationship; for example, when you select the **Descendants (inc)** function in the **Function Selector**, displays the parent at the bottom of the hierarchy. For a list of member relationships in the **Function Selector**, see Member Relationships.
>
> To use the Parent First (PF) functions to place the parent members above the children in forms, see Defining Parent First Functions on Forms.

- To select different sets of members across the same dimension, see Creating Asymmetric Rows and Columns.
- For forms with multiple dimensions in a row or column, you can set member selection options for a dimension by selecting that dimension from the **Dimensions** drop-down list that is displayed in the Member Selection dialog box for multiple dimensions in a row or column.
- If you click the member selection icon, an option is displayed for **Place Selection in Separate Rows** or **Place Selection in Separate Columns**. This adds the selection to the

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rows or columns after the last existing row or column on the form. For example, for a form that contains members Acct1, Acct2, and Acct3 in Column A, if you select these members with **Place Selection in Separate Columns** selected, Acct1 is selected for column A, Acct2 for column B, and Acct3 for column C. If you select the members without this option, all of the members are selected for column A.

This feature is available for single members only, not for members selected with functions, such as Children (inc). For example, if you select Q/IChildren for Column A and select **Place Selection in Separate Columns**, the form layout isn't changed.

- All settings except Count are retained after the Member Selection dialog box is closed, and members in the Member Selection dialog box are displayed based on user-defined settings. Members displayed on the Layout tab don't inherit the display settings defined in Member Selection dialog box. Instead, they are displayed using the member name.
- To define different sets of members for a dimension, see Creating Asymmetric Rows and Columns.
- To set display, functionality, and printing options, see Setting Form Precision and Other Options.

Selecting Substitution Variables as Members

Substitution variables act as global placeholders for information that changes regularly. When you select substitution variables as members on the form, their values are based on dynamically generated information. For example, you could set the current month member to the substitution variable `CurMnth` so that when the month changes, you need not update the month value manually in the form or the report script.

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Note

- When you open or calculate values on forms, the application replaces substitution variables with values assigned to them.
- You create and assign values to substitution variables within the application. These substitution variables are then available in the application when you select members for a form. For instructions on creating and assigning values to substitution variables using the application, see Working with Substitution Variables.
- Substitution variables must be appropriate for the context in forms. For example, you could select a substitution variable named `CurrQtr` with a value of Qtr2 as a member of the Time Period dimension. It's not valid to select a substitution variable named `CurrYr` for the Years dimension if its value is Feb. You can set substitution variables at the application or database level.

  The same substitution variable can exist on multiple levels; the application uses the first one it finds as it searches in this order:

  1. Database
  2. Application
- You can select from substitution variables if they are enabled for runtime prompts in business rules, and their values match a member set in the runtime prompt for a business rule.
- The application checks the validity of substitution variables when they are used (for example, when the form is opened). It doesn't check when you design forms, so you should test substitution variables by saving and opening forms.
- If you migrated an application that uses Global variables, you can view, but not edit, them in this business process.

To specify substitution variables in forms:

1. Create the form (see Creating Forms).
2. In **Member Selection**, click the **Variables** tab, expand **Substitution Variables**, and then select substitution variables the same way you select members, to move substitution variables to and from **Selected Members**.

   When selected, a substitution variable is preceded by an ampersand (&). For example:

   `&CurrentScenario`
3. Click **OK**.

Selecting User Variables as Members

User variables act as filters in forms, enabling users to focus only on certain members, such as a department. Before you can associate a user variable with a form, you must create the user variable. See Managing User Variables.

When you create forms with user variables, users must select values for the variable before opening forms. For example, if you create a user variable called Division, users must select a division before working in the form. The first time you select a variable for a form, you do it in preferences. Afterward, you can update the variable in preferences or in the form.

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To select user variables for forms:

1. Create the form (see Creating Forms).
2. On **Member Selection**, click the **Variables** tab, expand **User Variables**, and then select user variables the same way you select members, using the arrows to move user variables to and from **Selected Members**.

   User variables are displayed for the current dimension. For example, user variables for the Entity dimension might display as follows:

   ```
   Division = [User Variable]
   ```

   When selected, a user variable is preceded by an ampersand. For example:

   ```
   Idescendants(&Division)
   ```

3. Click **OK**.

Selecting UDAs as Members

You can select members for forms based on a common attribute, which you have defined as a user-defined attribute (UDA). Before you can associate the UDA with a form, you must create the UDA. See Working with UDAs.

When you create forms with UDAs, any members that are assigned to the UDA are dynamically added to the form. For example, if you create a UDA called New Products and assign this UDA to the new products in the Product dimension hierarchy, the form will automatically display the new products at runtime.

To select UDAs for forms:

1. Create the form (see Creating Forms).
2. On **Member Selection**, click the **Variables** tab, expand the **UDAs**, and then select the UDAs the same way you select members, using the arrows to move UDAs to and from the **Selected Members**.

   UDAs are displayed for the current dimension only. When selected, a UDA is preceded by `UDA`. For example:

   ```
   UDA(New Products)
   ```

3. Click **OK**.

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7

Connecting Environments in Cloud EPM

Learn how to connect environments so users who have access across Oracle Fusion Cloud Enterprise Performance Management can use a single access point with one login. Understand use cases, scenarios, and considerations, and learn how to connect to external web services and use direct URLs to integrate environments.

Related Topics

- About Connecting Cloud EPM Environments
- Considerations for Migrating Cloud EPM Connections
- Creating, Editing, and Deleting Connections to Other Cloud EPM Environments
- Connecting to External Web Services
- Specifying Advanced Options for External Connections
- Navigating Across Cloud EPM Environments
- Customizing Navigation Flows to Access Other Cloud EPM Environments
- Using Direct URLs to Integrate Connected Environments

About Connecting Cloud EPM Environments

Overview

Service Administrators can connect multiple Oracle Fusion Cloud Enterprise Performance Management environments of the following types:

- Account Reconciliation
- Enterprise Profitability and Cost Management
- Financial Consolidation and Close
- FreeForm
- Narrative Reporting
- Planning
- Planning Modules
- Profitability and Cost Management
- Sales Planning
- Strategic Workforce Planning
- Tax Reporting

Once Service Administrators set up the connections, users who have access across Cloud EPM environments can navigate across them from a single access point with one login. Also artifacts such as forms and dashboards, from across environments can be co-mingled within a cluster or within tabs on a card in navigation flows. Artifacts in the target environment are accessible based on the user's role.

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> **Note**
>
> You can also connect directly Oracle Analytics Cloud Enterprise Edition or Professional Edition 5.6 to Cloud EPM Platform, provided you have both services. When you've configured the connection, you can visualize data from Cloud EPM business processes in Oracle Analytics Cloud. You no longer have to model EPM data in a metadata repository (RPD) file to create visualizations and dashboards in Oracle Analytics Cloud. For more information, see the Oracle Analytics Cloud documentation.
>
> Additionally, you can connect Oracle® NetSuite Account Reconciliation and Oracle NetSuite Planning and Budgeting.

Which environments can I connect?

The source environment is the environment from which you're creating the connection. The target environment is the environment to which you're connecting from the source environment.

You can connect these source environments (these environments can also be target environments):

- Enterprise Profitability and Cost Management
- FreeForm
- Financial Consolidation and Close
- Oracle NetSuite Planning and Budgeting
- Planning
- Planning Modules
- Sales Planning
- Strategic Workforce Planning
- Tax Reporting

Source environments can also connect to these target environments (these environments can't be source environments):

- Account Reconciliation
- Narrative Reporting
- Oracle NetSuite Account Reconciliation
- Profitability and Cost Management

What are the ways I can connect to other Cloud EPM environments?

- Toggle between the source environment and the target environment on the **Navigator** menu. See Navigating Across Cloud EPM Environments.
- Customize navigation flows in the source environment to access clusters, cards, and artifacts in other target environments from the Home page. See Customizing Navigation Flows to Access Other Cloud EPM Environments.
- Use direct URLs to seamlessly integrate connected environments. See Using Direct URLs to Integrate Connected Environments.

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Considerations

- Only Service Administrators create cross-environment connections.

  Users click a navigation link to open the linked environment. Access within the linked environment is determined by the application role and access permissions, if any, assigned to the user.

- For cross-environment navigation to be seamless, all environment instances to which cross-environment navigation flows are setup must belong to the same identity domain.

  > **Note**
  >
  > If the target and source environment instances are not on the same identity domain, then you'll not be able to establish a connection between them.

- Service Administrators cannot configure cross-environment connections using corporate SSO (identity provider) credentials.

  If your environments are configured for SSO, ensure that identity domain credentials are maintained for the Service Administrators who configure cross-environment connections. For information about how to maintain the identity domain credentials for these Service Administrators, see Managing User Credentials in SSO-Enabled Cloud EPM Environments in *Getting Started Guide for Administrators*.

- Migrating cross-environment connections between test and production environments can cause issues in certain use case scenarios. For more information, see Considerations for Migrating Cloud EPM Connections.

- Vanity URLs are not supported in cross-environment connections.

Troubleshooting

For help with information on fixing common errors while connecting environments, see Handling Issues with Navigation Flows in *Operations Guide*.

Videos

Your Goal	Watch This Video
Watch this video to learn how to customize a Cloud EPM workflow.	Overview: Configure Navigation Flows to Integrate Cloud EPM Business Processes

Considerations for Migrating Cloud EPM Connections

Oracle recognizes that it's common practice for Service Administrators to try out new features, such as connecting environments, on test environments and then migrate to production environments. However, in doing so, it could cause some issues after migration. Here are some use case scenarios that you need to be aware of.

In the following scenarios, assume you have environments for Financial Consolidation and Close and Planning.

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Use Case Scenario 1: Test to Production

When migrating connections from test environments to production environments, ensure that connections that were defined in the test environment are changed to point to the corresponding production environments.

For example, a Service Administrator has defined a connection between the test environments of Planning and Financial Consolidation and Close. The Service Administrator then uses this connection to build a navigation flow in Planning that refers to a card in the Financial Consolidation and Close. The snapshot that the Service Administrator creates for migrating the Planning test environment will include connections and navigation flows, including the connection to Financial Consolidation and Close test environment.

On migrating the snapshot into the Planning production environment, Planning will have an undesirable connection to the Financial Consolidation and Close test environment. You must manually change undesirable connections to point to the corresponding production environment either before or after migrating the environment.

Use Case Scenario 2: Production to Production or Test to Test

This scenario doesn't have any caveats.

Use Case Scenario 3: Production to Test

In this scenario, the Service Administrator might be trying to migrate a snapshot from a production environment into a test environment to resolve an issue. Because the connections created in the test environment still points to a production environment, it is important for the Service Administrator to modify connections so that they point to a test environment. Connections in test environments that point to a production environment may inadvertently tamper with the production environment.

Creating, Editing, and Deleting Connections to Other Cloud EPM Environments

Before you can create connections to other Oracle Fusion Cloud Enterprise Performance Management environments, you must ensure you have access to the source and target environments you're connecting. You must also provide URLs for the other environments you're connecting and login details for each environment such as user ID (Service Administrator) and password.

> **Note**
>
> Oracle requires the secure HTTPS protocol (`https://`) for new connections. Existing connections may continue to use the HTTP (`http://`) protocol. Also, attempts to import a non-secure HTTP connection using Migration will be blocked and the connection artifact will not be imported.

To create, edit, duplicate, and delete connections:

1. Login to the source environment.
2. From the Home page, click **Tools**, and then click **Connections**.
3. Choose an action:

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- To add a connection:
  a. On the **Manage Connections** page, click **Create**.
  b. On the **Select Provider to Create Connection** page, select the target environment you want to add.
  c. Enter the target environment connection details:
     – Click **Change Provider** to select a different target environment.
     – In **Connection Name** enter name for this navigation link; for example, `Consolidation Application`.
     – Enter an optional description for the link.
     – In **URL**, enter the URL of the target environment instance; for example, `https://your-target-host-url.com`. This is the URL that you normally use to sign in to the target environment instance.

       > **Note**
       >
       > Oracle requires the secure HTTPS protocol (`https://`) for new connections. Verification is performed to ensure that the provided URL is allowed.

     – Use **Service Administrator** and **Password** to specify the credentials of a Service Administrator.

       > **Note**
       >
       > * These credentials are used only to define and validate the navigation link. When a user logs in, their own role and access will be applied to access the target environment.
       > * Do not prefix user names with the domain name for connections to other Cloud EPM environments. However, the domain name is still needed for connections to other external Web services. See Connecting to External Web Services.

     – The **Domain** field is automatically populated based on the URL you enter. If there is no domain in the URL, then the **Domain** field is left blank.
  d. Click **Validate**.
  e. If the validation is successful, click **Save and Close**.
- To edit connections:
  a. On the **Manage Connections** page, click the name of a connection.
  b. Edit connection details.

     > **Note**
     >
     > If you edit the URL to connect to a new service type, you could cause navigation flows to break. If you want to connect to a different service, Oracle recommends creating a new connection instead.

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c. Click **Validate**.

d. If the validation is successful, click **Save and Close**.

- To duplicate a connection:

  a. On the **Manage Connections** page, in the **Action** column next to the connection you want to duplicate, click ••• (Action).

  b. Click **Duplicate**.

  c. Enter a name for the duplicate connection, then click **OK**.

- To delete a connection:

  a. On the **Manage Connections** page, in the **Action** column next to the connection you want to delete, click ••• (Action).

  b. Click **Delete**.

When target environments are connected to a Cloud EPM source environment, they are listed in the **My Connections** pane on the **Navigator** menu of the source environment. The **My Connections** pane on the **Navigator** menu is where you can navigate across environments. See Navigating Across Cloud EPM Environments.

Troubleshooting

For troubleshooting assistance, see Handling Issues with Navigation Flows in *Operations Guide*.

Connecting to External Web Services

Service Administrators can also connect to external Web services for the purpose of reading data from and writing to an external Web service.

This connection can be referenced or used in a Groovy script to create a communications link between the Groovy script and the external HTTP/HTTPS resource. For more details and examples of how this connection can be used in a Groovy script, see the EPM Groovy Rules documentation for the Connection and the HttpRequest objects in the EPM Groovy object model.

> **Note**
>
> The **Other Web Service Provider** connection type is only available for use with those business processes that allow Groovy Rules to be created. See the Oracle Enterprise Performance Management Cloud, Groovy Rules Java API Reference.

Before you can create connections to external Web services, you must ensure you have access to the Web service you're connecting. You must also have a secure URL for the Web service and any login details, if required.

To create a connection to an external Web service:

1. Login to the source environment.
2. From the Home page, click **Tools**, and then click **Connections**.
3. On the **Manage Connections** page, click **Create**.

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4. Click **Other Web Service Provider**.
5. Enter a **Connection Name** and a **Description** for the connection.
6. Enter the secure **URL** for the target connection.

> **Note**
>
> - Oracle requires the secure HTTPS protocol (`https://`) for new connections.
> - Only port number 443 is allowed for secure URLs. Either provide a URL with port number 443 or provide no port number at all.
> - Attempts to import a non-secure HTTP connection using Migration will also be blocked and the connection artifact will not be imported.

7. Enter optional advanced options for the URL.

> **Note**
>
> The optional advanced options enable you to specify query or header parameters when defining an external connection. See Specifying Advanced Options for External Connections.

8. Enter **User** and **Password** login credentials for the connection, if required. In some cases, such as connecting to Oracle Fusion Cloud Enterprise Performance Management services, the domain name may need to be prefixed to the user name; for example, `<Identity Domain>.<User Name>`.

   To understand basic authentication for Cloud EPM REST APIs and for instructions on finding your identity domain, see Basic Authentication - for Classic and OCI in *Oracle Fusion Cloud EPM REST APIs*.
9. Click **Save and Close**.

Specifying Advanced Options for External Connections

The optional advanced options enable you to specify query or header parameters when defining an external connection.

> **Note**
>
> The ability to define query parameters for an external connection is only available for use with those business processes that allow Groovy Rules to be created. See the Oracle Enterprise Performance Management Cloud, Groovy Rules Java API Reference.

To specify advanced options for external connections:

1. Create an external connection or open an existing external connection.

   See Connecting to External Web Services.
2. Enter connection details, and then click **Show Advanced Options**.

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3. Specify query details as follows:

   - (Add): Add query
   - (Delete): Delete query
   - **Type**: Select **Header** or **Parameter**.

     Header sets a default header that will be sent on every request made for this connection. **Parameter** sets a default query parameter that will be sent on every request made for this connection.
   - **Secure**: If selected, the value entered in the **Value** field will be encrypted. Clearing the **Secure** check box for a query will remove the value.

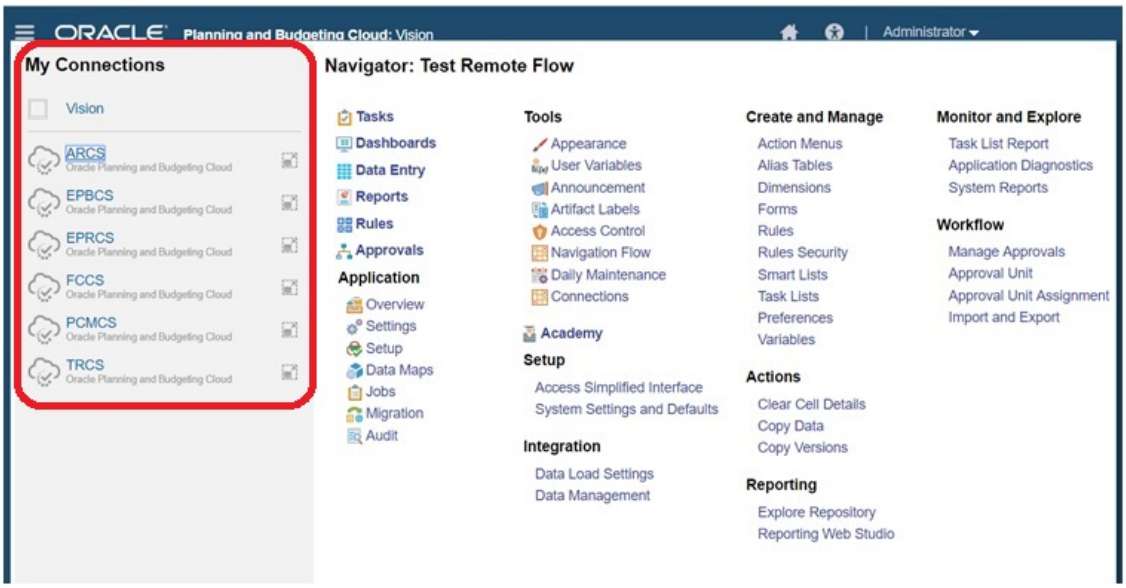
     An example header that one would secure is the Bearer Token for external Web services that supports Bearer Authentication, or the API Key query parameter for external Web services that supports API keys for authentication.

     > **Note**
     >
     > Queries that are marked **Secure** for **Other Web Service Provider** connection types cannot be exported or imported.

   - **Name**: Enter a name for the header or query parameter.
   - **Value**: Enter the value for the header or query parameter.

Navigating Across Cloud EPM Environments

Once a Service Administrator creates connections to other Oracle Fusion Cloud Enterprise Performance Management environments, the connections are listed in the **My Connections** pane on the **Navigator** menu.



You can toggle between the environments from this location. You must have access to the other environments in order to open them. Artifacts are accessible based on the user's role.

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To open another Oracle Fusion Cloud Enterprise Performance Management environment:

1. From the Home page, click **Navigator** .
2. If environments are connected and you have access to those environments, you'll see a list of connected environments in the **My Connections** pane. Click an environment to open it.

> **Note**
>
> Click the icon to the right of the environment name to open the environment in a new window.

Customizing Navigation Flows to Access Other Cloud EPM Environments

You can customize the business process interface to access other Oracle Fusion Cloud Enterprise Performance Management environments from the Home page of a source environment. For example, you can add artifacts to the Home page, such as forms or reports, from other Cloud EPM environments. You can group these artifacts (called cards) into clusters by customizing navigation flows. Clusters and cards from target Cloud EPM environments can be directly included in the navigation flows of source Cloud EPM environments. You can also use the Navigation Flow Designer to customize cards to have tabular pages where each tab is an artifact from a different environment.

These two use cases describe in detail how to customize navigation flows to access other Cloud EPM environments:

- Grouping Cards from Other Cloud EPM Environments into Clusters
- Configuring Cards with Tabs from Multiple Cloud EPM Environments

To learn more about designing navigation flows, see Designing Custom Navigation Flows.

Grouping Cards from Other Cloud EPM Environments into Clusters

You can group cards from various Oracle Fusion Cloud Enterprise Performance Management environments into a cluster that is accessible from the Home page of a source environment. For example, you can create a cluster within Financial Consolidation and Close consisting of cards with pre-built external reports from Narrative Reporting.

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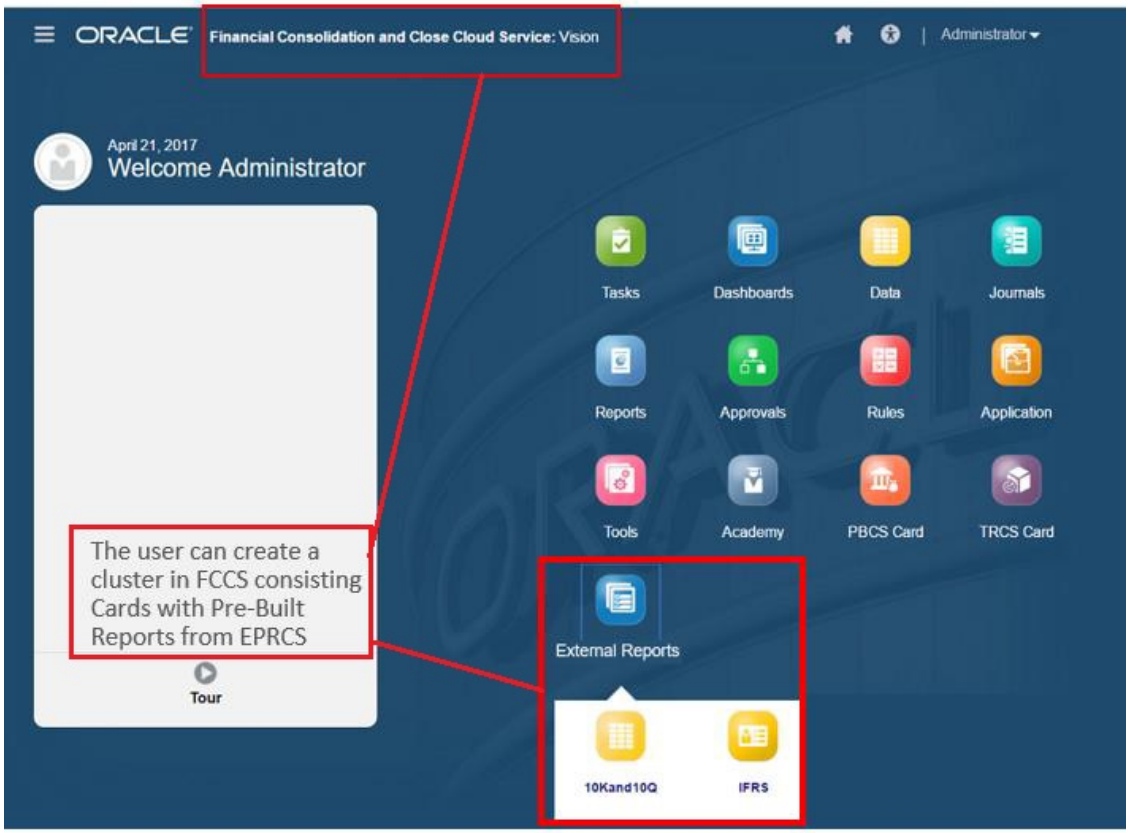
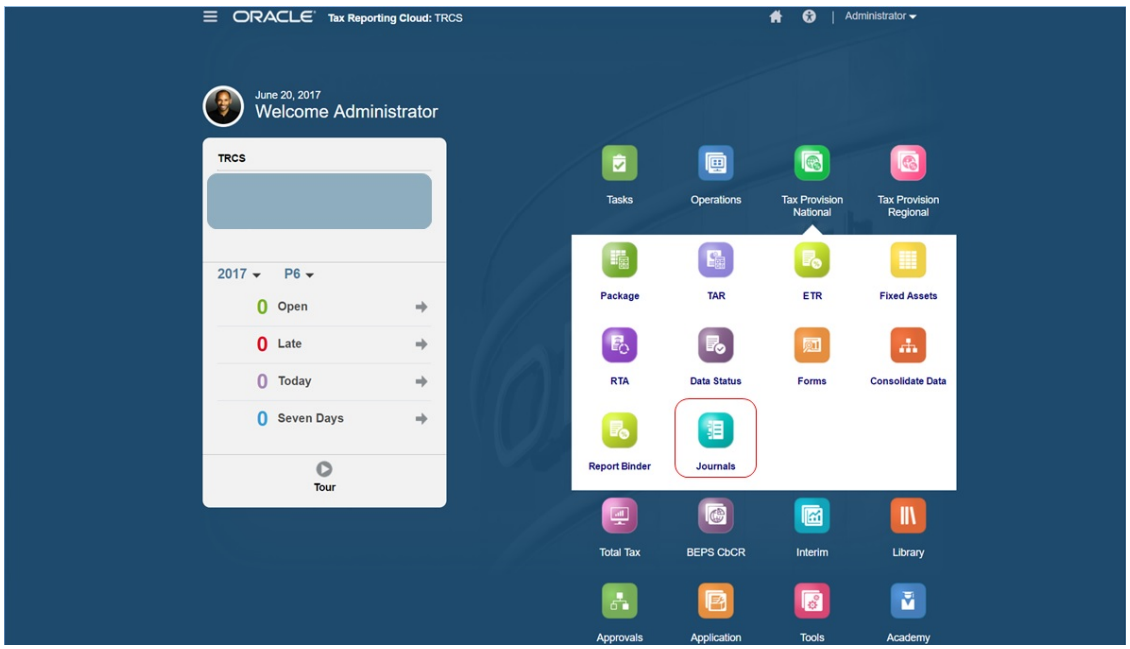


Cards from multiple environments can also be included within the same cluster on a source environment. For example, a Tax Reporting user can launch a Journals icon from Financial Consolidation and Close without leaving Tax Reporting.



You create clusters and add cards to clusters by customizing navigation flows.

For general information about navigation flows, see Designing Custom Navigation Flows.

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To create a cluster made up of cards from other Cloud EPM environments:

1. Launch the Navigation Flow page and create a navigation flow or edit an existing navigation flow:

   > **Note**
   >
   > To create a navigation flow, you must first select an existing navigation flow and make a copy of it. Then edit the duplicate navigation flow details and save them.

   a. Click **Tools**, and then click **Navigation Flows**.

   b. To create a navigation flow, select the navigation flow you want to duplicate, then in the upper right-hand corner of the page, click (Actions), and then select **Create Copy**. Enter a name for the navigation flow, and then click **OK**.

      > **Note**
      >
      > New flows are marked **Inactive** until they are activated by the Service Administrator. To activate or deactivate a navigation flow, in the **Active** column, click **Active** or **Inactive**. Only one navigation flow at a time can be active.

   c. To edit an existing navigation flow, click the name of the navigation flow you want to edit.

      > **Note**
      >
      > Editing is only possible if the navigation flow is inactive. If the navigation flow you want to edit is active, ensure you mark it **Inactive** before editing.

2. Create a cluster or add an existing cluster:

   a. If it isn't already open, from the **Navigation Flow** page, click the name of the navigation flow in which you want to add a cluster.

   b. To create a new cluster, right-click a card or cluster or click (Actions), click **Add Cluster**, enter or select the cluster details, and then choose an icon for the cluster.

   c. If there is an existing cluster you want to add from another environment, right-click a card or cluster (or click **Add Existing Card/Cluster**), select the target environment under **My Connections**, and then choose the cluster you want to add to your navigation flow.

      Note the following:

      - Clusters can't be directly selected from Narrative Reporting and Profitability and Cost Management using the **Add Existing Card/Cluster** option.
      - Clusters that are added from another navigation flow or from another environment will display the localized labels that were defined in the source navigation flow. To update the cluster labels in your navigation flow, on the Home page, click **Tools**, and then click **Artifact Labels**.

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See Specifying Artifact Labels.

3. Select the cards to include in the cluster using one of these options:
   - Navigate to the card you want to add to the cluster. If the card is within another environment, first select the environment under **My Connections**, and then navigate to the card in that environment. Assign the card to a cluster using one of these options:
     - To the right of the card that you want to move, in the **Order** column, click (Move). Select the cluster, and then click **OK**.
     - Click the name of the card to view the card details, then for **Cluster** select a cluster for the card, and then click **OK**.
   - Navigate to the cluster in which you want to add the new card. If the cluster is within another environment, first select the environment under **My Connections**, and then navigate to the cluster in that environment. Right-click the cluster or click (Actions), click **Add Card In Cluster**, and then select an option:
     - Select **Add Existing Card**, to select an existing card or to add existing cards from another cluster to the selected cluster.
     - Select **Add Card**, and then enter card details to add a new card to the selected cluster.

   > **Note**
   >
   > You cannot add a card to a cluster if the card or the cluster is already referenced from another navigation flow.

   The cards will appear in the listing as children of the cluster. Use the up and down arrows next to the cards to reorder the cards within the cluster, if needed.
4. Click **Save and Close**.

You must activate the navigation flow and reload it to view your design time changes. To reload a navigation flow, click the down arrow next to your user name. Then on the **Setting and Actions** menu, click **Reload Navigation Flow**.

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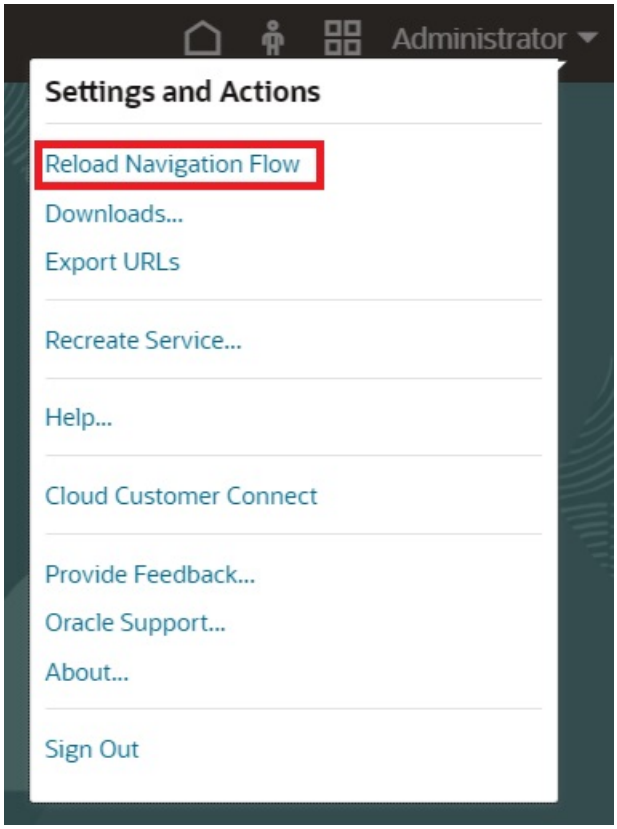

Troubleshooting

If you can't see your referenced artifacts after activating and reloading the navigation flow, see Handling Issues with Navigation Flows in *Operations Guide*.

Configuring Cards with Tabs from Multiple Cloud EPM Environments

You can also customize cards in navigation flows to have tabular pages where each tab is an artifact from a different environment. For example, a Planning user can click a Revenue icon which launches a card with horizontal tabs showing reports from Narrative Reporting.

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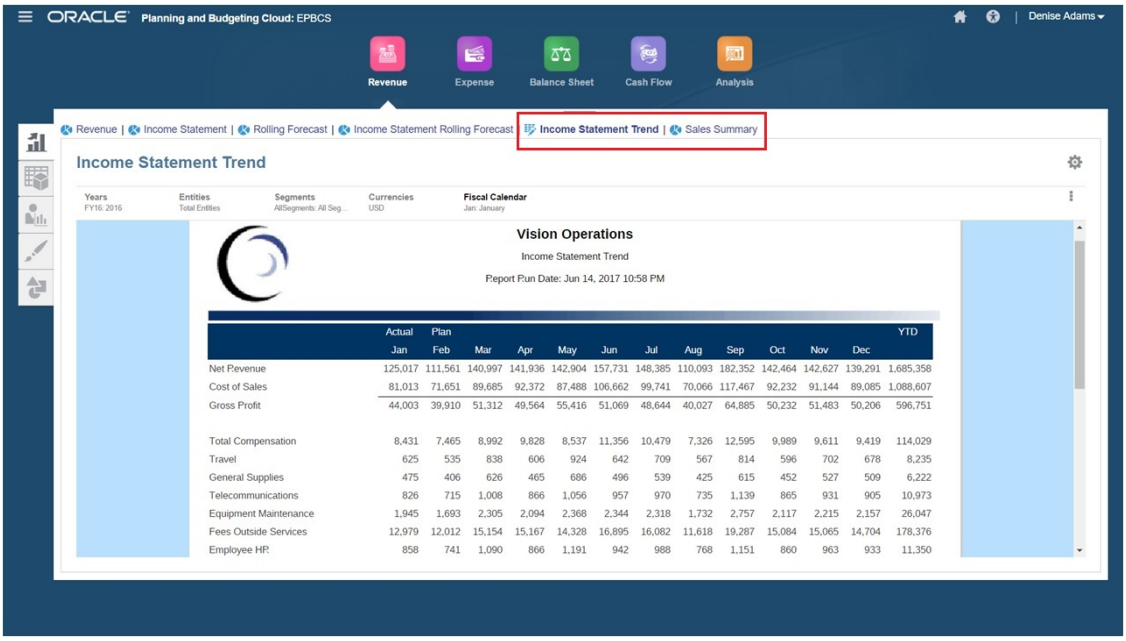

You create tabular cards by customizing navigation flows.

For general information about navigation flows, see Designing Custom Navigation Flows.

To configure a card made up of tabs and sub tabs from other Oracle Fusion Cloud Enterprise Performance Management environments:

1. Launch the **Navigation Flow** page and create a navigation flow or edit an existing navigation flow:

   > **Note**
   >
   > To create a navigation flow, you must first select an existing navigation flow and make a copy of it. Then edit the duplicate navigation flow details and save them.

   a. Click **Tools**, and then click **Navigation Flows**.

   b. To create a navigation flow, select the navigation flow you want to duplicate, then in the upper right-hand corner of the page, click (Actions), and then select **Create Copy**. Enter a name for the navigation flow, and then click **OK**.

      > **Note**
      >
      > New flows are marked **Inactive** until they are activated by the Service Administrator. To activate or deactivate a navigation flow, in the **Active** column, click **Active** or **Inactive**. Only one navigation flow at a time can be active.

   c. To edit an existing navigation flow, click the name of the navigation flow you want to edit.

2. Add a tabular card with artifacts from various target environments:

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a. If there is an existing card you want to add from another environment, from the **Navigation Flow** page, right-click a card or cluster or click (Actions), click **Add Existing Card/Cluster**, select the target environment under **My Connections**, and then choose the card you want to add to your navigation flow.

   Note the following:

   - Cards can't be directly selected from Narrative Reporting and Profitability and Cost Management using the **Add Existing Card/Cluster** option.
   - Cards that are added from another navigation flow or from another environment will display the localized labels that were defined in the source navigation flow. To update the card labels in your navigation flow, on the Home page, click **Tools**, and then click **Artifact Labels**.

     See Specifying Artifact Labels.

b. To add a new tabular card to the navigation flow, from the **Navigation Flow** page, right-click a card or cluster or click (Actions), click **Add Card**, and then select details for the card:

   - **Name**: Enter a label for the card.
   - **Visible**: Select whether the card is visible to users on the Home page.
   - **Cluster**: If clusters exist, select a cluster for the card or select **None**.
   - **Icon**: Select the icon that will be displayed for the card you're creating. Choose from the available icons provided in the icon library.
   - **Content**: Select from the following options:
     - **Page Type**: Select a multiple page (tabular) format.
     - **Orientation**: Select **Vertical** or **Horizontal**.

3. Add tabs and sub tabs to the tabular card:

   a. To add an existing tab, right-click a tab, click **Add Existing Tab** (or click the **Add Existing Tab** button), and then select a tab from the Object Library.

   b. To add a new tab, right-click a tab, click **Add New Tab** (or click the **Add New Tab** button), and then edit tab details.

   c. Right-click a tab, click **Add New Sub Tab** or **Add Existing Sub Tab** (or click the **Add New Sub Tab** or **Add Existing Sub Tab** buttons), and then choose a sub tab from the Object Library or edit sub tab details.

   d. For **Artifact**, click (Choose Artifact) to select an artifact in the Artifact Library; for example, if the artifact is a form, then select the specific form from the artifact listing. Available artifacts include forms, dashboards, and reports. To select an artifact from another environment, select the target environment under **My Connections**, and then choose the artifact you want to add to your sub tab.

   e. Repeat adding tabs and sub tabs until the card is complete.

4. Click **Save and Close**.

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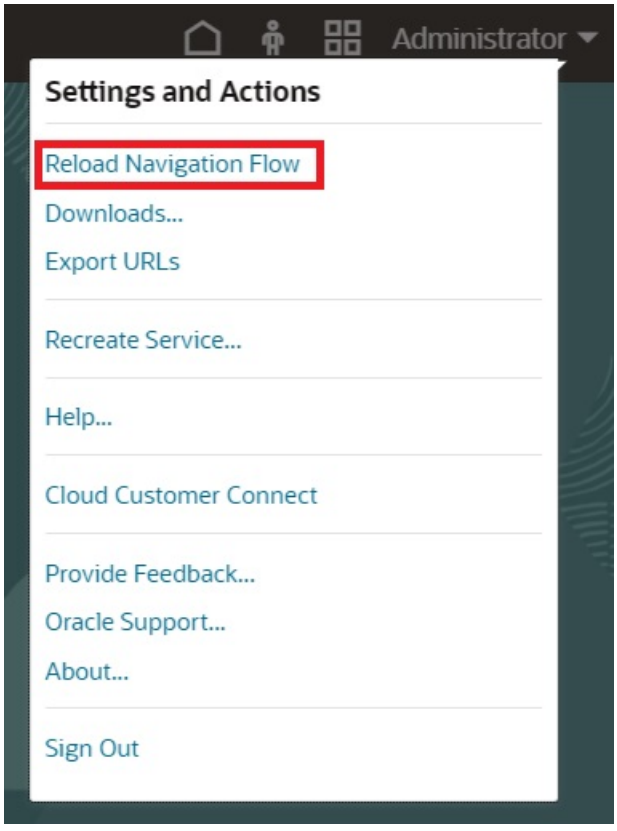
> **Note**
>
> - For cards with multiple tabs or sub tabs, the last tab accessed by a user will be retained the next time the user accesses the card in the same session. If the user logs out and then logs back in, the default tab will be displayed.
> - Tabs or sub tabs that are added from another navigation flow or from another environment will display the localized labels that were defined in the source navigation flow. To update the tab labels in your navigation flow, on the Home page, click **Tools**, and then click **Artifact Labels**.
>
> See Specifying Artifact Labels.

You must reload the navigation flow to view your design time changes. To reload a navigation flow, click the down arrow next to your user name. Then on the **Setting and Actions** menu, click **Reload Navigation Flow**.



If you can't see your referenced artifacts after reloading the navigation flow, see Handling Issues with Navigation Flows in *Operations Guide*.

Using Direct URLs to Integrate Connected Environments

Other source systems like 'Oracle® ERP Cloud' can embed URLs to directly link to artifacts contained in cards, tabs, and sub-tabs within connected Oracle Fusion Cloud Enterprise Performance Management environments.

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Other Cloud environments like 'ERP' use direct URL links to open connected Cloud EPM content like forms, dashboards, infolets, and reports. To make the integration between Cloud EPM and other systems seamless, you can copy the unique URLs for the artifacts within a connected Cloud EPM business process. You can copy the unique URLs in one of two ways:

- Copy the individual URL for an artifact in the business process. See Copying Direct URLs for Artifacts.
- Export all of the URLs in the business process to a CSV file, then find and copy the unique URLs. See Exporting All URLs to a CSV File.

Videos

Your Goal	Watch This Video
Learn how to use direct links to embed Oracle Fusion Cloud Enterprise Performance Management content in other systems such as Cloud ERP and NetSuite.	Overview: Embed Content Using Direct Links

Copying Direct URLs for Artifacts

Use the **Copy URL** option on the artifact listing page to copy the unique URL for an artifact (dashboards, forms, infolets, and reports) in your Oracle Fusion Cloud Enterprise Performance Management business process.

> **Note**
>
> **Copy URL** is available to all users so that direct URLs to artifacts can be shared. However, only users with access to the targeted artifact can perform actions on it.

To copy the unique URL for an artifact:

1. From the Home page, open the listing page for the artifact.

   For example, click **Dashboards**, **Data**, **Infolets**, or **Reports**.
2. From the listing page, click • • • (Actions) next to the artifact, and then select **Copy URL**.
3. The **Copy URL** dialog displays the unique URL for the artifact. Copy the URL.

Copied URLs can, at times, become invalid. Possible reasons for invalid URLs include:

- The artifact is deleted.
- User access for the artifact is revoked.
- The artifact is renamed, which breaks the URL.
- The artifact is moved to a new folder, resulting in a new URL, which breaks the URL.

Exporting All URLs to a CSV File

Use the **Export URLs** option to create a CSV file that provides the unique URLs for each card, tab, or sub-tab within a connected Oracle Fusion Cloud Enterprise Performance Management business process. URLs are grouped by navigation flow and cluster, so the URLs are easier to find within the CSV file. You can open the CSV file with a text editor or Microsoft Excel and

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embed the relevant URL within the source system pages to serve as a launch point into Cloud EPM.

To export Cloud EPM URLs to a CSV file:

1. Log into a Cloud EPM environment.
2. From the Home page, click the down arrow next to the user name (upper right corner of the screen).
3. On the **Settings and Actions** menu, click **Export URLs**, and then click **Save**.

The system saves a CSV file to the default download folder on your local machine, and the filename is automatically generated with the current date and time of the server; for example, `19_Feb_2021 13_15_38 Navigation Flow URLs.csv`. Find the file in your download folder, and open it with a text editor or with Microsoft Excel.

Viewing the Exported URLs File

The CSV file lists all of the URLs in the business process. Each card, tab (vertical tab), and sub-tab (horizontal tab) has a unique URL. When viewed in a text editor like Notepad or in Microsoft Excel, it identifies the unique URL for each card, tab, and sub-tab, so the URLs for each artifact can be more easily found. URLs are grouped by navigation flow and by cluster.

> **Note**
>
> Only cards, tabs, and sub-tabs have URLs. Navigation flows and clusters don't have URLs.

Table 7-1 Direct URLs Export File Headers

Header	Description
Navigation Flow Name	The name of the navigation flow; for example, Default or Financial Flow.
Status	Status of the navigation flow; for example, Active or Inactive.
Type	Type of entry; for example, cluster, card, tab, or sub tab
Name	The cluster, card, tab or sub-tab name where the artifact is contained. This entry will be empty for clusters or cards which do not contain an artifact directly.
Artifact Type	The type of artifact; for example, Form, Dashboards, Financial Reports, and URL type artifacts.
Artifact Name	The name of the artifact or, in the case of an URL type artifact, the direct URL of the target page. **Caution:** If a direct URL is displayed, do not confuse this URL with the unique URL you'll use to integrate the connected environments.
URL	This is the unique URL you will use to integrate connected environments.
Visible	Indicates whether the artifact in the navigation flow is visible on the Home page to users or groups; for example, Y or N.
Role/Group	The role or group who can view the navigation flow. If a navigation flow is Global, then it can be seen by all users.
Description	The description of the navigation flow, if provided.

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The URLs export file provides the information separated by a vertical bar or pipe (|) delimiter character. See the following example direct URLs export file as viewed in Notepad:

```
Navigation Flow Name|Status|Type|Name|Artifact Type|Artifact Name|URL|Visible|Role/Group|Description
Dashboard 2.0 Test|Inactive|cluster|Strategic Modeling||||Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Model View|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_30=EPM_CA_158|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Consolidation View|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_30=EPM_CA_159|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Templates|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_30=EPM_CA_160|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Tasks|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_2|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Dashboards|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_1|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Infolets|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_204|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Data|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_3|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Reports|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_6|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|tab|Reports|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_6=EPM_TA_606|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|tab|Financial Reports|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_6=EPM_TA_617|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|tab|Documents|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_6=EPM_TA_633|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Rules|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_4|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Approvals|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CA_5|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|cluster|Application||||Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Overview|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_120|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Settings|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_121|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Valid Intersections|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_122|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|sub tab|Setup|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_122=EPM_TA_427=EPM_TA_9|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|sub tab|Reports|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_122=EPM_TA_427=EPM_TA_10|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Data Exchange|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_208|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|sub tab|Data Integration|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_208=EPM_TA_599=EPM_TA_598|Y|Global|Default Navigation
Dashboard 2.0 Test|Inactive|sub tab|Data Maps|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_208=EPM_TA_599=EPM_TA_11|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Jobs|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_214|Y|Global|Default Navigation Flow
Dashboard 2.0 Test|Inactive|card|Cell Level Security|||http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?EC_HL=Y&EC_NF=Dashboard+2.0+Test&EC_PAGE_ID=EPM_CL_22=EPM_CA_214|Y|Global|Default Navigation Flow
```

To view the URLs export file in Microsoft Excel:

1. Open Excel, and then click the **Data** menu.
2. Click **New Query**, then **From File**, and then click **From CSV**.
3. Find and select the CSV file you exported, and then click **Import**. A new window displays the data in the CSV file.
4. To make the first row of the CSV file the header row, click **Edit**, click **Use First Row as Headers**, and then click **Close and Load**.

The resulting Excel file will look like the following example:

	A	B	C	D	E	F	G	H	I	J
1	Navigation Flow Name	Status	Type	Name	Artifact Type	Artifact Name	URL	Visible	Role/Group	Description
2	Dashboard 2.0 Test	Inactive	cluster	Strategic Modeling				Y	Global	Default Navigation Flow
3	Dashboard 2.0 Test	Inactive	card	Model View			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
4	Dashboard 2.0 Test	Inactive	card	Consolidation View			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
5	Dashboard 2.0 Test	Inactive	card	Templates			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
6	Dashboard 2.0 Test	Inactive	card	Tasks			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
7	Dashboard 2.0 Test	Inactive	card	Dashboards			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
8	Dashboard 2.0 Test	Inactive	card	Infolets			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
9	Dashboard 2.0 Test	Inactive	card	Data			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
10	Dashboard 2.0 Test	Inactive	card	Reports			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
11	Dashboard 2.0 Test	Inactive	tab	Reports			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
12	Dashboard 2.0 Test	Inactive	tab	Financial Reports			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
13	Dashboard 2.0 Test	Inactive	tab	Documents			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
14	Dashboard 2.0 Test	Inactive	card	Rules			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
15	Dashboard 2.0 Test	Inactive	card	Approvals			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
16	Dashboard 2.0 Test	Inactive	cluster	Application				Y	Global	Default Navigation Flow
17	Dashboard 2.0 Test	Inactive	card	Overview			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
18	Dashboard 2.0 Test	Inactive	card	Settings			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
19	Dashboard 2.0 Test	Inactive	card	Valid Intersections			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
20	Dashboard 2.0 Test	Inactive	sub tab	Setup			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
21	Dashboard 2.0 Test	Inactive	sub tab	Reports			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
22	Dashboard 2.0 Test	Inactive	card	Data Exchange			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
23	Dashboard 2.0 Test	Inactive	sub tab	Data Integration			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
24	Dashboard 2.0 Test	Inactive	sub tab	Data Maps			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
25	Dashboard 2.0 Test	Inactive	card	Jobs			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow
26	Dashboard 2.0 Test	Inactive	card	Cell Level Security			http://slcar261.usdv1.oraclecloud.com:10151/HyperionPlanning?E	Y	Global	Default Navigation Flow

Find and copy the unique URL (found in the URL column) for the card, tab, or sub-tab that you wish to have integrated into the other connected environment. Only the URL target will open and users with access to the targeted artifact can perform the same actions as if they're working within the target business process.

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8

Designing Custom Navigation Flows

Customize the business process interface using navigation flows. Navigation flows enable designers to control how roles or groups interact with the business process.

Related Topics

- Understanding Navigation Flows
- Viewing and Working with Navigation Flows

Understanding Navigation Flows

Navigation flows give business process designers control over how various roles, or groups, interact with the business process. The Navigation Flow Designer enables you to customize the business process interface. For example, you can change the names of the cards and clusters that display on the Home page and the order in which they are displayed. You can hide cards, create new cards, and group cards into clusters. You can also customize the vertical and horizontal tabs that display on a card.

Videos

Your Goal	Watch This Video
View the highlights of customizing workflows for connected environments.	Overview: Configure Navigation Flows to Integrate Cloud EPM Business Processes

Related Topics

- What Can Be Customized in the Business Process Interface?
- Navigation Flow Customization Categories
- Navigation Flow Permissions
- Predefined Navigation Flows
- Understanding Navigation Flow Status after Import
- Viewing and Working with Navigation Flows

What Can Be Customized in the Business Process Interface?

Navigation flows are made up of cards. Each card contains content with information that is displayed as one or more tabbed pages. The content included on cards can be URLs or artifacts such as forms, dashboards, and reports. Cards can be grouped into clusters.

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> **Note**
>
> If Dashboard 2.0 and Forms 2.0 are enabled in your business process:
>
> - Navigation flows support the features included in Dashboard 2.0 and Forms 2.0, including the ability to edit a form from within the runtime dashboard, and editing it with or without data. Dashboard 2.0 component actions such as **Open Form** and **Edit Form** also work directly within the navigation flow.
> - If a dashboard contains multiple components, when you edit a dashboard from within a navigation flow, the focus will be set to the last component in the dashboard.
>
> See Working with 2.0 Dashboards in *Working with FreeForm*.

Navigation flows can be customized in the following ways:

- Labels for cards or tabs
- Icons that are used for cards or tabs
- Hide and unhide cards and tabs
- Display order of cards and tabs
- Add new cards
- Add existing cards
- Add new horizontal or vertical tabs
- Remove navigation flows, cards, and tabs
- Group cards into clusters
- Add existing clusters

See Viewing and Working with Navigation Flows.

Navigation Flow Customization Categories

Navigation flows are categorized as follows for customization:

1. Global: Navigation flows are seen by all users
2. Role: Navigation flows are seen only by users in a specific role; for example, Service Administrator or Power User
3. Group: Navigation flows are seen only by users belonging to a specific group; for example, Sales

Navigation flows can be defined at any of these levels. In cases where navigation flows exist at multiple levels, updates are applied in the order of highest (global) to lowest (groups).

For example, if you create a navigation flow that displays an icon on the Home page named "My Tasks", and then another Service Administrator duplicates the navigation flow, makes the following changes to the card, and then associates the navigation flow with a group:

- At the global level, they rename "My Tasks" to "Company Tasks"
- At the group level, for a group named Sales, they rename "My Tasks" to "Sales Tasks"

Users who belong to the group called Sales will see the label "Sales Tasks" in the navigation flow instead of "My Tasks," and all other users will see the label "Company Tasks".

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Navigation Flow Permissions

The business process offers three levels of permissions for navigation flows:

- Role-based: Permissions are granted to users or groups assigned to a specific role; for example, a User will see different cards displayed on the Home page than a Service Administrator
- Artifact-based: Permissions are granted to users or groups who can see certain artifacts; for example, a User will see only the forms to which they have been assigned permission
- Global: Permissions are granted to all users

Predefined Navigation Flows

The business process comes with one predefined navigation flow, called Default. The Default navigation flow is read only; therefore, you can't make any modifications to it.

These are the operations you can and cannot perform on the Default navigation flow:

- Name: You can't modify the name.
- Delete: You can't delete the navigation flow.
- Edit: You can view the navigation flow details, but you can't change anything.
- Activate or Deactivate: You can activate or deactivate the navigation flow.
- Duplicate: You can make a copy of the navigation flow.

Understanding Navigation Flow Status after Import

When you import a snapshot containing an active navigation flow into an environment containing an active navigation flow of the same category (role, group, or global), then the new navigation flow being imported is deactivated.

For example, the following navigation flows are currently active on your system:

- NF Sales Q2 (Sales group)
- NF Administrator (Service Administrator role)
- NF Custom Default (global)

Let's say you import a navigation flow called NF Sales Q3, which uses the same Sales group as NF Sales Q2. The NF Sales Q3 navigation flow will be imported as inactive and the existing NF Sales Q2 will remain active for the Sales group. If you want the new NF Sales Q3 to be active, you must manually activate it (which will inactivate the old NF Sales Q2). We deactivate the imported Q3 navigation flow because the import might occur ahead of the Q3 start date, and we want to ensure the Q3 navigation flow isn't activated before the Q3 start date.

If a new application is created from a Migration snapshot that includes an active custom global navigation flow (for example, NF Custom), the NF Custom global navigation flow will become inactive after Migration import. The same is true if the application is cloned. The Default global navigation flow will become the active global navigation flow and you'll need to ensure you activate the global NF Custom navigation flow after the new application is created. If all users are using the NF Custom global navigation flow, they might experience a service interruption until the NF Custom navigation flow is activated.

When designing navigation flows:

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- Create navigation flows for specific groups or classes of users. Then associate the navigation flow to corresponding user groups.
- Avoid creating one global navigation flow for all users.

Viewing and Working with Navigation Flows

Users assigned the Service Administrator role can view all navigation flows, including the predefined navigation flow, on the Navigation Flows page. Users assigned to other roles and groups will only see the navigation flows to which they have access.

The Navigation Flow page lists each navigation flow by name, indicates the role or the group that has access to the navigation flow (if assigned), and provides a description of the navigation flow (if provided). The listing also indicates whether the navigation flow is active or not.

Viewing Navigation Flows

To view the Navigation Flow page:

1. On the Home page, click **Tools**.
2. Click **Navigation Flows**.

Working with Navigation Flows

To work with a navigation flow, perform an action:

- For design best practices and considerations, see Navigation Flow Design Best Practices and Naming Considerations.
- To create and duplicate navigation flows, see Creating and Duplicating Navigation Flows.
- To edit a navigation flow, see Editing a Navigation Flow.
- To activate or deactivate a navigation flow, see Activating and Deactivating Navigation Flows.
- To preview the runtime view of a navigation flow for a user, see Previewing the Runtime View of a Navigation Flow for a User.
- To validate navigation flows and to learn how to find and resolve navigation flow elements with missing artifacts, see Using Validate to Find Missing Artifacts in Navigation Flows.
- To resolve a navigation flow in the listing that is displaying a warning icon , see Resolving Navigation Flows that Display a Warning Icon.
- To rename cards and tabs, see Customizing Labels for Cards, Tabs, and Clusters.
- To customize the graphics used for cards and tabs, see Customizing Icons for Cards and Vertical Tabs.
- To hide and unhide cards and tabs, see Hiding and Unhiding Clusters, Cards, and Tabs.
- To change the display order of cards on the Home page, see Changing the Display Order of Cards on the Home Page.
- To add cards, see Adding Cards.
- To add tabs, see Adding Tabs.
- To embed third-party pages in navigation flows, see About Using URLs to Embed Third-Party Pages in Cloud EPM Applications.

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- To remove navigation flows, cards, and tabs, see Removing Navigation Flows, Cards, and Tabs.
- To group cards into clusters, see Grouping Cards into Clusters.
- To reload a navigation flow, see Reloading a Navigation Flow.
- To switch navigation flows at runtime, see Switching Navigation Flows at Runtime.
- To copy the current page URL for sharing, see Sharing the Current Page URL for Cards, Tabs, and Sub Tabs.

Troubleshooting

For help with resolving navigation flow issues, see Handling Issues with Navigation Flows in *Operations Guide*.

Navigation Flow Design Best Practices and Naming Considerations

Design Best Practices

To provide an optimal user experience and to avoid excessive scrolling on the Home page and within cards and tabs, you must adhere to the following guidelines when designing navigation flows:

- Keep the top-level items (cards and clusters) to no more than 20 visible items.
- Add no more than 16 visible cards to a cluster.
- Add no more than 10 visible vertical tabs within a card.
- Add no more than 20 visible sub tabs (horizontal tabs) within a vertical tab.
- Label names on sub tabs (horizontal tabs) display only the first 30 characters at runtime. Hovering your cursor over the tab reveals the entire label.

> **Note**
>
> If you attempt to exceed the visibility limits, you will see a warning message telling you that you've exceeded the limit.

Naming Considerations

There are naming restrictions for navigation flows, cards, clusters, tabs, and infolets (if your business process uses infolets) in navigation flows. You cannot use these special characters:

- ampersand (&)
- less than sign (<)
- greater than sign (>)
- quotation mark (")
- backslash (\ )
- plus sign (+)

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Creating and Duplicating Navigation Flows

To create a navigation flow, you must first select an existing navigation flow and make a copy of it. Then edit the duplicate navigation flow details and save them.

To create and duplicate a navigation flow:

1. Open the **Navigation Flow** page. See Viewing and Working with Navigation Flows.
2. In the upper right-hand corner of the page, click (Actions), and then select **Create Copy**.
3. Enter a name for the navigation flow, and then click **OK**.

> **Note**
>
> Ensure that you adhere to the navigation flow naming restrictions outlined in Navigation Flow Design Best Practices and Naming Considerations.

4. Edit details for the navigation flow. See Editing a Navigation Flow.

> **Note**
>
> New flows are marked **Inactive** until they are activated by the Service Administrator. To activate a navigation flow, see Activating and Deactivating Navigation Flows.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Editing a Navigation Flow

To edit a navigation flow:

1. Open the **Navigation Flow** page. See Viewing and Working with Navigation Flows.
2. Click the name of the navigation flow you want to edit.

> **Note**
>
> The predefined navigation flow isn't editable. However, you can create a copy of a predefined navigation flow and edit the copy. See Predefined Navigation Flows.

You'll see a page listing the cards and clusters in the navigation flow. On this page you can edit the role or goup assignation, designate which clusters and cards are visible on the Home page, change the order in which the navigation flow clusters and cards are displayed, add cards to clusters or remove them, and remove clusters and cards from the navigation flow.

- **Assign To**: Click (Select) to assign the navigation flow to a group of users or to a role.

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- **Visible**: Edit the visibility of the navigation flow clusters and cards on the Home page by selecting or deselecting them in the **Visible** column.

> **Note**
>
> Ensure that you adhere to the navigation flow guidelines for visibility outlined in Navigation Flow Design Best Practices and Naming Considerations.

- **Order**: Clusters and cards are listed in the order in which they are displayed on the Home page, if visible. Selecting an up or down arrow option repositions the clusters and cards in the listing and changes the display order of the clusters and cards on the Home page. Selecting the right arrow moves a card into a cluster.
- **Remove**: Removes the cluster or card from the navigation flow.

3. Click a cluster or card to edit details. For descriptions of card details, see the following topics:
   - Adding Cards
   - Adding Tabs

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Activating and Deactivating Navigation Flows

You can create multiple navigation flows for each category (global, role, or group), but only one navigation flow can be active in each category. Whenever you make a navigation flow active, the other navigation flows in the same category will become inactive.

> **Note**
>
> Each business process requires one active global navigation flow. To make a different global navigation flow active, select another global navigation flow and activate it.

For information about categories, see Navigation Flow Customization Categories.

These are the operations users can and cannot perform on an active navigation flow:

- Name: Users can't modify the name.
- Delete: Users can't delete the navigation flow.
- Edit:
  - Users can view the navigation flow definition, but they can't change anything.
  - If the business process is in administration mode, then users can save any modifications.
- Activate or Deactivate: Users can activate or deactivate a navigation flow.
- Duplicate: Users can make a copy of a navigation flow.

To activate or deactivate a navigation flow:

1. Open the **Navigation Flow** page. See Viewing and Working with Navigation Flows.

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2. In the **Active** column, click **Active** or **Inactive**. An active flow will be marked inactive. Conversely, an inactive flow will be marked active.

Previewing the Runtime View of a Navigation Flow for a User

Navigation flow designers can use the **User Access View** option to preview the runtime view for any user assigned to a group associated with a navigation flow.

This preview option is only available when the navigation flow is associated with a group. The **User Access View** option is not enabled for the Default navigation flow or for any navigation flow that is not associated with a group.

To preview the runtime view of a navigation flow for a user:

1. Open the **Navigation Flow** page. See Viewing and Working with Navigation Flows.
2. Select the row for the navigation flow you want to preview.

> **Note**
>
> The navigation flow must be associated with a group and it cannot be the Default navigation flow.

3. In the upper right-hand corner of the page, click (Actions), and then select **User Access View**.

   The navigation flow opens and displays the name of the navigation flow followed by **(User Access View)**.
4. Select from the following options:
   - **Users**: Select a user to preview the runtime view for that user. By default, the **Users** box displays the first name in the user list for the group associated with the navigation flow. To select a different user, click (Users), select another user, and then click **OK**.
   - **Display**: Select **All nodes** or **Visible nodes**. **All nodes** is the default option and it displays all the nodes for the navigation flow in the **Navigation** list. If some of the nodes are not visible to the user (possible reasons being they don't have access to an artifact within the node or an artifact within a node was removed or renamed), an error icon will indicate which nodes include artifacts that are not visible for that user. **Visible nodes** displays only the nodes that are visible for that user in the **Navigation** list.
5. If **All nodes** is selected, the **Navigation** list displays error icons next to the nodes that are fully or partially inaccessible to the user:
   - (No Access): This entire node is not accessible to the user. Possible reasons for limited access: the user doesn’t have access to the page or to the referenced node, the navigation flow might’ve been deleted or renamed, the connection might be unavailable or deleted.
   - (Partial Access): One or more nodes are not accessible to the user. Possible reasons for limited access: the user doesn’t have access to the page or to the referenced node, the navigation flow might’ve been deleted or renamed, the connection might be unavailable or deleted.

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If a node displays a (Partial Access), expand the node to view the artifacts within each node that are inaccessible to the user.

6. Click **Close** to exit **User Access View**.

Using Validate to Find Missing Artifacts in Navigation Flows

While viewing the Navigation Flow listing, you might see navigation flow nodes or artifacts displaying an error icon. This error occurs because artifacts that were associated with the navigation flow were either renamed or removed and they are now considered missing. You'll need to edit the navigation flow to associate it with either a renamed artifact or with a different artifact before you can activate the navigation flow. You might not realize that artifacts are considered missing, so it's recommended that you validate your navigation flows before setting them to Active status.

> **Note**
>
> To resolve a navigation flow displaying a warning icon, see Resolving Navigation Flows that Display a Warning Icon.

To validate navigation flows to find missing artifacts and reassociate them:

1. Open the **Navigation Flow** page. See Viewing and Working with Navigation Flows.
2. If they aren't already deactivated, set the navigation flows you want to validate to **Inactive** status. See Activating and Deactivating Navigation Flows.
3. Highlight the row or rows of the navigation flows you want to validate.
4. Click (Actions), and then select **Validate**.

   If artifacts are missing, you'll see an error message indicating which navigation flows are referencing artifacts that cannot be found.
5. Click the name of each navigation flow with the error, and then expand the nodes which display the error icon until you reach the Manage page which displays the missing artifact.
6. For **Artifact**, click (Choose Artifact) to select the renamed artifact or a different artifact in the Artifact Library.
7. Click **Save and Close**.
8. Repeat the validation on your navigation flows and drill down to make corrections, as needed, until you see a message indicating the navigation flows are valid.
9. Activate the navigation flows. See Activating and Deactivating Navigation Flows.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Resolving Navigation Flows that Display a Warning Icon

While viewing the Navigation Flow listing, you might see a navigation flow displaying a warning icon. This occurs because the group that was associated with the navigation flow was

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deleted. You'll need to edit the navigation flow to associate it with a group or role before you can activate the navigation flow.

> **Note**
>
> To resolve navigation flows displaying an error icon , see Using Validate to Find Missing Artifacts in Navigation Flows.

To resolve the navigation flow:

1. Open the **Navigation Flow** page. See Viewing and Working with Navigation Flows.
2. Click the name of the navigation flow displaying the warning icon .
3. For **Assign To**, click (Select) to assign the navigation flow to a group of users or to a role, then click **Save and Close**.
4. Activate the navigation flow. See Activating and Deactivating Navigation Flows.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Customizing Labels for Cards, Tabs, and Clusters

You can customize the labels for cards (the icons that display on the Home page), tabs, and clusters. Labels are limited to 25 characters or less. For vertical tabs, there is no character limitation since the label for vertical tabs displays as hover text.

> **Note**
>
> Do not update the labels for cards, tabs, or clusters against the Default navigation flow. Only customize labels for custom navigation flows.

To customize labels for cards, tabs, and clusters:

1. Open the **Navigation Flow** page and click the name of the navigation flow you want to edit. See Viewing and Working with Navigation Flows.
2. If customizing the label for a card or cluster:
   a. Click the name of the card or cluster you want to edit.
   b. Enter a new name and save it.

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> **Note**
>
> - You can edit the label here. However, if the label is defined in the **Artifact Labels** page on the **Tools** cluster, that definition will take precedence and will display during runtime. To change a label permanently, redefine it in the **Artifact Labels** page.
>
> See Specifying Artifact Labels.
> - Ensure that you adhere to the naming restrictions outlined in Navigation Flow Design Best Practices and Naming Considerations.

3. If customizing the label for a tab:
   a. Click the name of the card you want to edit.
   b. In the tab listing on the **Manage Tab** page, click the name of the tab you want to edit.
   c. Enter a new name for the tab and save it.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Customizing Icons for Cards and Vertical Tabs

You can change the icons used for cards and vertical tabs. You must pick from the available icons provided in the icon library.

To customize the icons for cards and vertical tabs:

1. Open the **Navigation Flow** page and click the name of the navigation flow you want to edit. See Viewing and Working with Navigation Flows.
2. If customizing the icon for a card:
   a. Click the name of the card you want to edit.
   b. Click the icon for the card, select a new icon from the library, and then save it.
3. If customizing the icon for a tab:
   a. Click the name of the icon you want to edit.
   b. In the tab listing on the **Manage Tab** page, click the name of the tab you want to edit.
   c. Click the icon for the tab, select a new icon from the library, and then save it.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Hiding and Unhiding Clusters, Cards, and Tabs

You cannot hide the following navigation elements:

- The **Application** cluster and the **Settings** icon in the **Application** cluster.
- The **Tools** cluster and these icons in the **Tools** cluster:
  - **Access Control**
  - **Navigation Flows**
  - **Daily Maintenance**
  - **Migration**

Ensure that you adhere to the navigation flow guidelines for visibility outlined in Navigation Flow Design Best Practices and Naming Considerations.

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To hide and unhide clusters, cards, and tabs:

1. Click the **Navigation Flow** icon and click the name of the navigation flow you want to edit. See Viewing and Working with Navigation Flows.
2. Edit the Home page visibility of the navigation flow clusters and cards by selecting or deselecting them in the **Visible** column.
3. If hiding or unhiding a tab:
   a. Click the name of the card you want to edit.
   b. In the tab listing on the **Manage Tab** page, select or deselect the check box in the **Visible** column.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Changing the Display Order of Cards on the Home Page

You can change the display order of cards in the Navigation Flow Designer. Cards display on the Home page in the order they appear within the listing

To change the display order of the cards on the Home page:

1. Open the **Navigation Flow** page. See Viewing and Working with Navigation Flows.
2. In the listing, use the up and down arrows in the **Order** column to move cards up or down in the navigation flow order.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Adding Cards

The icons you see on the Home page represent cards. Cards are specific to each functional area of the business process. Each card navigates users to the corresponding area, where other information is displayed as one or more tabbed pages. You can create single page or multiple page (tabular) cards.

To ensure an optimal user experience, review the navigation flow design best practices. See Navigation Flow Design Best Practices and Naming Considerations.

You can also group cards into clusters. See Grouping Cards into Clusters.

To add cards to a navigation flow:

1. Open the **Navigation Flow** page and click the name of the navigation flow you want to edit. See Viewing and Working with Navigation Flows.
2. To add an existing card to the navigation flow, right-click a card or cluster in the list or click (Actions), click **Add Existing Card/Cluster**, and then select a card. If there is an existing card you want to add from another environment, right-click a card or cluster in the list or click (Actions), click **Add Existing Card/Cluster**, select the target environment under **My Connections**, and then choose the card you want to add to your navigation flow.

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> **Note**
>
> - Cards can't be directly selected from Narrative Reporting and Profitability and Cost Management using the **Add Existing Card/Cluster** option.
> - Cards that are added from another navigation flow or from another environment will display the localized labels that were defined in the source navigation flow. To update the card labels in your navigation flow, on the Home page, click **Tools**, and then click **Artifact Labels**.
>
>   See Specifying Artifact Labels.
> - A reference card is a card that is already referenced from another navigation flow. References to already referenced cards are not supported in navigation flows and will not be available for selection in the Object Library when adding an existing card; for example:
>   - A card referring to a remote artifact or remote tab will not be available from the Object Library when adding an existing card.
>   - A card referring to a tab from another navigation flow will not be available from the Object Library when adding an existing card.

A card is added to the list as a sibling under the card or cluster that is currently selected. To add a card to a cluster, see Grouping Cards into Clusters.

> **Note**
>
> Cards that are added without first selecting a card or cluster are added to the end of the list.

3. To add a new card to the navigation flow, right-click a node in the list or click + (Actions), click **Add Card**, and then select details for the new card:

Table 8-1 New Card Details

Label	Description
Name	Enter a label for the card. Ensure that you adhere to the naming restrictions outlined in Navigation Flow Design Best Practices and Naming Considerations.
Visible	Select whether the card is visible to users on the Home page.
Cluster	If clusters exist, select a cluster for the card or select **None**.
Icon	Select the graphic that will be displayed for the card you're creating. Choose from the available graphics provided in the graphics library.
Page Type	Select **Single Page** or **Tabular Page** format.

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Table 8-1 (Cont.) New Card Details

Label	Description
Content Source	If you selected the **Single Page** format, select **Artifact** or **URL**: • For **Artifact**, click (Choose Artifact) to select an artifact in the Artifact Library; for example, if the artifact is a form, then select the specific form from the artifact listing. Available artifacts include forms (including hidden forms), dashboards, and reports. To select an artifact from another environment, select the target environment under **My Connections**, and then choose the artifact you want to add. • For **URL**, enter a complete URL; for example, a URL to embed an Oracle Analytics Cloud dashboard in a card, then click **Preview** to validate the URL in a popup window. Insert only external site URLs starting with the `https://` security protocol. Don't use internal or relative URLs or URLs for unconsenting third-party sites. See About Using URLs to Embed Third-Party Pages in Cloud EPM Applications.
Orientation	If you selected the **Tabular Page** format, select **Vertical** or **Horizontal**, and then add new or existing tabs and sub tabs. See Adding Tabs.

A card is added to the list as a sibling under the card or cluster that is currently selected. To add a card to a cluster, see Grouping Cards into Clusters.

> **Note**
>
> Cards that are added without first selecting a card or cluster are added to the end of the list.

4. Click **Save and Close**.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Adding Tabs

Tabs can be horizontal or vertical. For example, the **Valid Intersections** card (under the **Application** cluster) is a tabular page with two horizontal tabs: **Setup** and **Reports**.

> **Note**
>
> The horizontal tabs are at the bottom of the page.

You can also create tabular pages with vertical tabs. Vertical tabs display a graphic and text appears when the cursor is hovered over the tab. Horizontal tabs display labels with text only or text with icons.

To ensure an optimal user experience, review the navigation flow design best practices. See Navigation Flow Design Best Practices and Naming Considerations.

To add tabs to a tabular page:

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1. Open the **Navigation Flow** page and click the name of the navigation flow you want to edit. See Viewing and Working with Navigation Flows.
2. Edit an existing card by clicking the name of the card you want to edit, or add a new card by right-clicking a card or clicking (Actions), and then clicking **Add Card**.
3. On the **Manage Card** page, select the following options:
   - For **Page Type**, select **Tabular Page**.
   - For **Orientation**, select **Vertical** or **Horizontal**.

   A tab listing displays at the bottom of the **Manage Card** page.
4. To edit an existing tab, click a tab name from the tab listing, and edit tab details.
5. To add a new or existing tab:
   a. To add an existing tab, right-click a tab in the listing at the bottom of the **Manage Card** page, click **Add Existing Tab** (or click the **Add Existing Tab** button), select a tab from the Object Library, and then click **OK**.

      > **Note**
      >
      > A reference tab is a tab that is already referenced from another navigation flow. References to already referenced tabs are not supported in navigation flows and will not be available for selection in the Object Library when adding an existing tab; for example:
      >
      > - A tab referring to a remote artifact or remote sub-tab will not be available from the Object Library when adding an existing tab.
      > - A tab referring to a sub tab from another navigation flow will not be available from the Object Library when adding an existing tab.

   b. To add a new tab, right-click a tab, click **Add New Tab** (or click the **Add New Tab** button), and then edit tab details.
   c. Select the content for the new tab:
      - For **Artifact**, click (Choose Artifact) to select an artifact in the Artifact Library; for example, if the artifact is a form, then select the specific form from the artifact listing. Available artifacts include forms (including hidden forms), dashboards, and reports. To select an artifact from another environment, select the target environment under **My Connections**, and then choose the artifact you want to add.
      - For **URL**, enter a complete URL; for example, a URL to embed an Oracle Analytics Cloud dashboard in a tab, then click **Preview** to validate the URL in a popup window.

        Insert only external site URLs starting with the `https://` security protocol. Don't use internal or relative URLs or URLs for unconsenting third-party sites. See About Using URLs to Embed Third-Party Pages in Cloud EPM Applications.

      A tab is added to the list as a sibling under the tab that is currently selected.

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> **Note**
>
> Tabs that are added without first selecting a tab are added to the end of the list.

6. To add new or existing sub tabs to a tab:
   a. Click the name of a tab in the in the tab listing.
   b. For **Page Type**, select **Tabular Page**.
   c. Right-click a tab, and click **Add New Sub Tab** or **Add Existing Sub Tab** (or click the **Add New Sub Tab** or **Add Existing Sub Tab** button), and then edit sub tab details.
   d. Select the content for the new sub tab:

      - For **Artifact**, click (Choose Artifact) to select an artifact in the Artifact Library; for example, if the artifact is a form, then select the specific form from the artifact listing. Available artifacts include forms (including hidden forms), dashboards, and reports. To select an artifact from another environment, select the target environment under **My Connections**, and then choose the artifact you want to add.
      - For **URL**, enter a complete URL; for example, a URL to embed an Oracle Analytics Cloud dashboard in a sub tab. Click **Preview** to validate the URL in a popup window.

        Insert only external site URLs starting with the `https://` security protocol. Don't use internal or relative URLs or URLs for unconsenting third-party sites. See About Using URLs to Embed Third-Party Pages in Cloud EPM Applications.

   A sub tab is added to the list as a sibling under the tab that is currently selected.

> **Note**
>
> Sub tabs that are added without first selecting a tab are added to the end of the list.

7. Click **Save and Close**.

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

> **Note**
>
> - For cards with multiple tabs or sub tabs, the last tab accessed by a user will be retained the next time the user accesses the card in the same session. If the user logs out and then logs back in, the default tab will be displayed.
> - Tabs or sub tabs that are added from another navigation flow or from another environment will display the localized labels that were defined in the source navigation flow. To update the tab labels in your navigation flow, on the Home page, click **Tools**, and then click **Artifact Labels**.
>
>   See Specifying Artifact Labels.

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About Using URLs to Embed Third-Party Pages in Cloud EPM Applications

Oracle Fusion Cloud Enterprise Performance Management uses IFrame to embed third-party URLs. IFrame requires that the page that is being embedded must give approval to the page that is embedding it. For example, if we want to embed a page from sharepoint.com into a Cloud EPM application, then sharepoint.com must allow oraclecloud.com to embed the page from sharepoint.com.

This can be achieved by adding oraclecloud.com in the Content Security Policy of the web application whose page you need to embed.

When embedding a third-party page you also must consider if the page is available to the public or if it requires a login. For example, pages from wikipedia.org do not need any authentication.

If you are embedding a page that requires authentication, you need to see if SSO can be enabled for the page. If not, then your page may not load inside IFrame. As a workaround, log in to that web application in another browser tab, and then if you access the same page from the Cloud EPM application, it will open.

This URL support feature allows you to embed the following types of pages:

- Other Oracle products (would need to enable SSO)
- Web applications owned by customer (would need to allow the Cloud EPM application by updating the Content Security Policy and enabling SSO, or likewise)
- Pages from the public domain (for example, wikipedia.org)

> **Note**
>
> - Do not use this feature to embed a URL for an entire other Cloud EPM instance; for example, do not embed a URL for Narrative Reporting into Planning Modules.
> - Though not supported by Oracle, you can embed links to the following resources within navigation flow cards, tabs, and sub tabs:
>   - Google Sheets
>   - Files stored in Google Drive (for example, PDF and Excel)
>   - Files stored in Microsoft Office 365
>
> Use the instructions provided by the third-party sites to generate a URL which can then be used in your Cloud EPM application.

Removing Navigation Flows, Cards, and Tabs

You can't remove the following navigation elements:

- The **Application** cluster and the **Settings** icon in the **Application** cluster.
- The **Tools** cluster and these icons in the **Tools** cluster:
  - **Access Control**
  - **Navigation Flows**
  - **Daily Maintenance**

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- **Migration**

To remove navigation flows, cards, and tabs:

1. Open the **Navigation Flow** page. See Viewing and Working with Navigation Flows.
2. If removing a navigation flow:
   a. Select the navigation flow you want to remove.
   b. In the upper right-hand corner of the page, click (Actions), and then select **Delete**.

> **Note**
>
> You can't delete the predefined navigation flow, called Default.

3. If removing a card:
   a. Click the name of the navigation flow you want to edit.
   b. In the **Remove** column for the card you want to remove, click (Remove).
4. If removing a tab:
   a. Click the name of the navigation flow you want to edit.
   b. Click the name of the card you want to edit.
   c. In the tab listing at the bottom of the **Manage Tab** page, in the **Remove** column for the tab you want to remove, click (Remove).

To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Grouping Cards into Clusters

A cluster is a grouping of cards. You must first create a cluster and then you can assign cards to it. You can also add existing clusters to navigation flows.

To ensure an optimal user experience, review the navigation flow design best practices. See Navigation Flow Design Best Practices and Naming Considerations.

To assign cards to clusters:

1. Create a new cluster or add an existing cluster:
   a. Open the **Navigation Flow** page and click the name of the navigation flow in which you want to add a cluster. See Viewing and Working with Navigation Flows.
   b. To create a new cluster, right-click a card or cluster in the list or click (Actions), click **Add Cluster**, enter or select the cluster details, and then choose a graphic for the cluster.

> **Note**
>
> Ensure that you adhere to the visibility and naming restrictions outlined in Navigation Flow Design Best Practices and Naming Considerations.

A cluster is added to the list as a sibling under the cluster that is currently selected.

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> **Note**
>
> Clusters that are added without first selecting a card or cluster are added to the end of the list.

c. To add an existing cluster, right-click a card or cluster in the list or click (Actions), click **Add Existing Card/Cluster**. If there is an existing cluster you want to add from another environment, right-click a card or cluster in the list or click (Actions), click **Add Existing Card/Cluster**, select the target environment under **My Connections**, and then choose the cluster you want to add to your navigation flow.

> **Note**
>
> - Clusters can't be directly selected from Narrative Reporting and Profitability and Cost Management using the **Add Existing Card/Cluster** option.
> - Clusters that are added from another navigation flow or from another environment will display the localized labels that were defined in the source navigation flow. To update the cluster labels in your navigation flow, on the Home page, click **Tools**, and then click **Artifact Labels**.
>
> See Specifying Artifact Labels.
> - A reference cluster is a cluster that is already referenced from another navigation flow. References to already referenced clusters are not supported in navigation flows and will not be available for selection in the Object Library when adding an existing cluster.

A cluster is added to the list as a sibling under the card or cluster that is currently selected.

> **Note**
>
> Clusters that are added without first selecting a card or cluster are added to the end of the list.

d. Click **Save and Close**.

2. Select the cards to include in the cluster using one of these options:

a. Navigate to the card you want to add. If the card is within another environment, first select the environment under **My Connections**, and then navigate to the card in that environment. Assign the card to a cluster using one of these options:

- To the right of the card in the **Order** column, click (Move to Cluster), select the cluster, and then click **OK**.
- Click the name of the card to view the card details, then for **Cluster** select a cluster for the card, and then click **OK**.

b. Navigate to the cluster in which you want to add a card, then right-click the cluster or click (Actions), click **Add Card In Cluster**, and then select an option:

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- Select **Add Existing Card**, to select an existing card or to add existing cards from another cluster to the selected cluster.
- Select **Add Card**, and then enter card details to add a new card to the selected cluster.

> **Note**
>
> You cannot add a card to a cluster if the card or the cluster is already referenced from another navigation flow.

c. Click **Save and Close**.

The cards will appear in the listing as children of the cluster. Use the up and down arrows next to the cards to reorder the cards within the cluster, if needed.

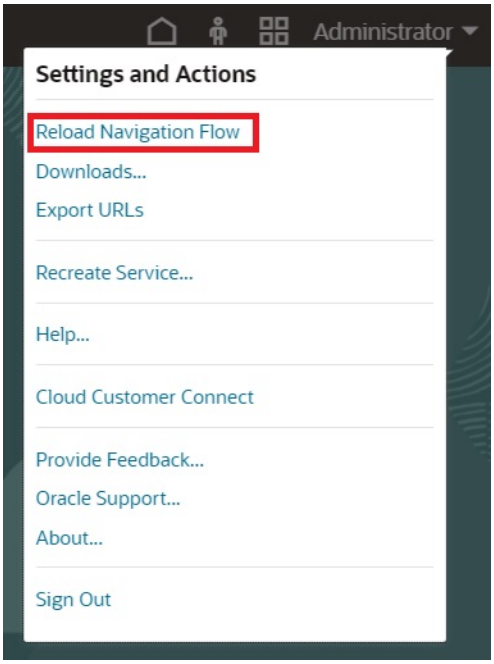
To reload a navigation flow to view design time changes, see Reloading a Navigation Flow.

Reloading a Navigation Flow

To display design changes while you're working with a navigation flow, you can reload the navigation flow.

There are two ways to reload a navigation flow after making design changes. From the Home page:

1. Click the down arrow next to the user name (upper right corner of the screen). Then on the **Settings and Actions** menu, click **Reload Navigation Flow**.



2. Click (Navigation Flows). Then on the **Navigation Flows** menu, click (Reload Navigation Flow).

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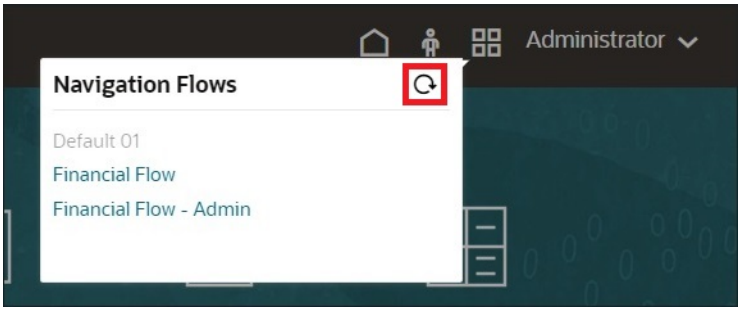


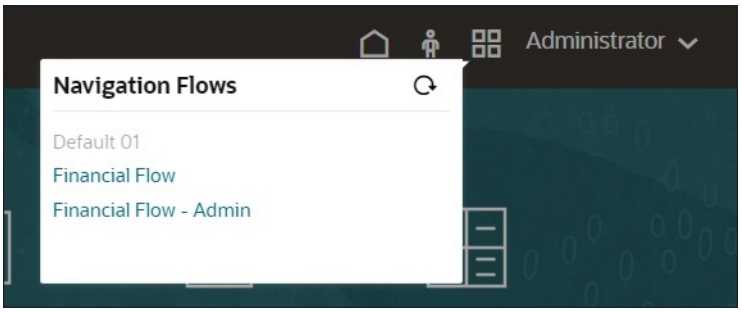
Switching Navigation Flows at Runtime

If you belong to multiple groups or if a navigation flow is assigned to a role, you might have access to more than one navigation flow.

> **Note**
>
> Users assigned the Service Administrator role can access all navigation flows.

To switch navigation flows at runtime:

1. From the Home page, click (Navigation Flows).
2. Select the navigation flow you want to view.



> **Note**
>
> Click (Reload Navigation Flow) to refresh the list of navigation flows.

Sharing the Current Page URL for Cards, Tabs, and Sub Tabs

Copy the URL for the currently displayed card, tab, or sub tab and share it with other users who have access.

Use the **Share Current Page URL** option on the **Settings and Actions** menu to copy the URL for the currently displayed card, tab, or sub tab in your Oracle Fusion Cloud Enterprise Performance Management business process.

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URLs can be copied and then shared with other users in an email, Slack, or any other messaging application. Also, copied URLs can be used in a slide show presentation such as PowerPoint. Note that users can only access cards, tabs, or sub tabs if they are authorized to access them.

When the copied URL is launched the page will be rendered with the full header, meaning they will be able to navigate to all of the pages to which they have access within the navigation flow. If the user launching the URL is not logged in, they will be redirected to a login page, then after login the URL page will open.

To copy a URL for sharing:

1. Navigate to the page (card, tab, or sub tab) in your application that you want to copy.

   For example, from the Home page, click Dashboards to view the Dashboards page.
2. Click the down arrow next to your user name (upper right corner of the screen).
3. On the **Settings and Actions** menu, select **Share Current Page URL**.

   Note: The **Share Current Page URL** menu option only displays if you're currently viewing a card, tab, or sub tab. If you're currently viewing the Home page, the **Share Current Page URL** menu option won't display on the **Settings and Actions** menu.
4. Select the displayed URL and copy it by pressing Ctrl+C.

   After pasting it into a browser, the URL will enable direct access to the current page in the application.

Considerations

Note the following considerations when using the **Share Current Page URL** option:

- The page specified in the URL will not open if the navigation flow was deleted, modified, or made inactive. Instead, you'll be redirected to the home page of the currently active navigation flow.
- If you're using a classic environment, you cannot share URLs. URLs are not supported for classic pages.
- URLs cannot be embedded into dashboards as the content source.
- URLs cannot be embedded into navigation flows as the content source.
- URLs cannot be shared for pages with dynamic tabs. Dynamic tabs are local and limited to the current user session, and therefore cannot be shared.
- URLs cannot be shared for infolets. URLs are not supported for infolets.
- URLs cannot be embedded into Task Lists.

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9

Integrating Cloud EPM and Cloud EDM with Oracle Guided Learning

Oracle Guided Learning (OGL) offers a robust framework for developing personalized, guided, and contextual training and user onboarding experiences. Beyond the EPM documentation, this versatile toolset includes the options to create custom process guides, display groups, message guides, smart tips, and more. It offers a comprehensive learning interface tailored to meet specific needs within Oracle Fusion Cloud Enterprise Performance Management and Oracle Enterprise Data Management Cloud environments, enhancing user proficiency and operational efficiency. For a detailed overview, see Introduction & Overview of Oracle Guided Learning in *Oracle Guided Learning User Guide*.

The users have access to OGL Console, which allows them to define content that streamlines adoption by enabling them to design and activate in-application guidance tailored to specific workflows and requirements. For instance, you can create OGL guides for your Navigation Flows.

Configuring Settings in the Environment

Before proceeding, ensure you have an active OGL account and at least one Application ID (App ID) set up. An App ID is a logical grouping of guides (contents). Each App ID can group guides customized for EPM workflows for a particular application. For setup details, see Getting Started in *Oracle Guided Learning User Guide*.

> **Note**
>
> When integrating an OGL App ID to use OGL Help within a Navigation Flow, configure it solely at the source environment. OGL setup is unnecessary on any of the connected environments.

Follow these steps to integrate an OGL application into your environment:

1. Navigate to **Applications**, then **Settings**
2. Enter the following Oracle Guided Learning Settings:
   - **Application ID**
   - **Server URL** - Depending on the location of your environment, enter the URL:
     - NA: `https://guidedlearning.oracle.com`
     - EMEA: `https://guidedlearning-emea.oracle.com`
     - APAC: `https://guidedlearning-apac.oracle.com`
3. Click **Save**. You'll need to log off and log back in for the OGL widget to display on your screen.

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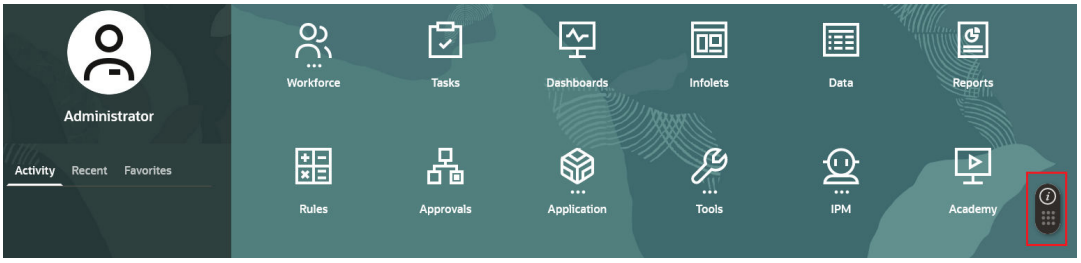


Configuring settings in the OGL Environment

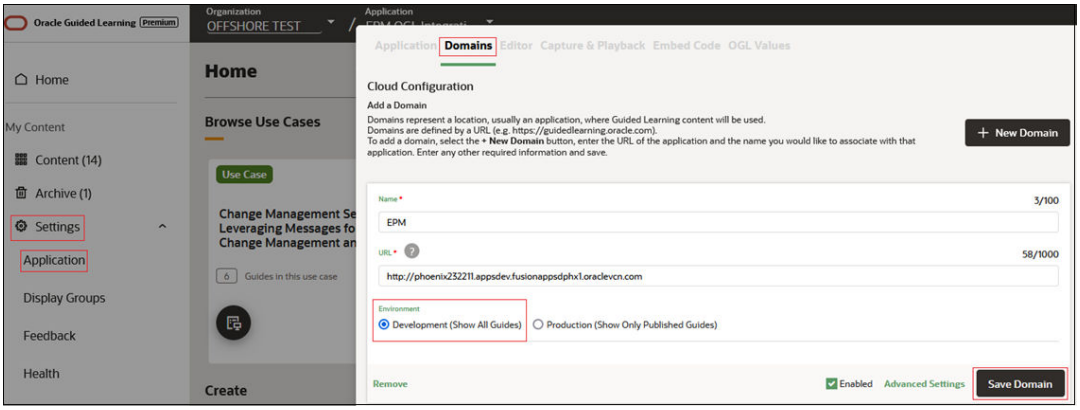
Once the OGL Application is integrated into the Cloud EPM and Cloud EDM environments, the OGL widget will be accessible to the users. Only published guides are visible since the domains by default is configured to Production, which hides the guides that are in development mode.

To allow users to view both published and in-development guides from the environment, change the application's settings as follows:

1. Access the OGL Console application settings. See OGL Console Left Panel in the *Oracle Guided Learning User Guide*.
2. Navigate to the **Domains** tab.
3. Select the **Environment** to **Development** to display all guides.
4. Check the domain status to **Enabled.**

> **Note**
>
> Please make sure to complete the domain entry configuration correctly so that OGL content displays correctly on all applications. Remember OGL content won't appear for domains not listed in OGL Cloud Configuration.

5. Click **Save Domain**.



Key Considerations

- Cloud EPM and Cloud EDM environments are required to perform daily maintenance. Oracle generates a maintenance snapshot, known as an Artifact Snapshot, which captures existing artifacts and data. It's important to note that OGL configuration in Cloud EPM and Cloud EDM is not integrated, and therefor is not included in the Artifact Snapshot. For

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more information, see Exporting OGL Guide Content in *Oracle Guided Learning User Guide*.

- If the Cloud EPM and Cloud EDM environments have Network Perimeter setup, the IP address of the OGL server or the outbound IP address of the region that hosts the OGL server must be added to the Network Perimeter. See Outbound IP Addresses of Cloud EPM Data Centers and Regions in *Operations Guide*.

Enabling Context-Based Activation of OGL Guides in Cloud EPM

Oracle Guided Learning (OGL) seamlessly integrates with applications to capture their workflows, enabling precise, in-application guidance tailored to user needs. The **Advanced Setting** feature in the **OGL Editor** further enhances this capability by allowing Service Administrators to:

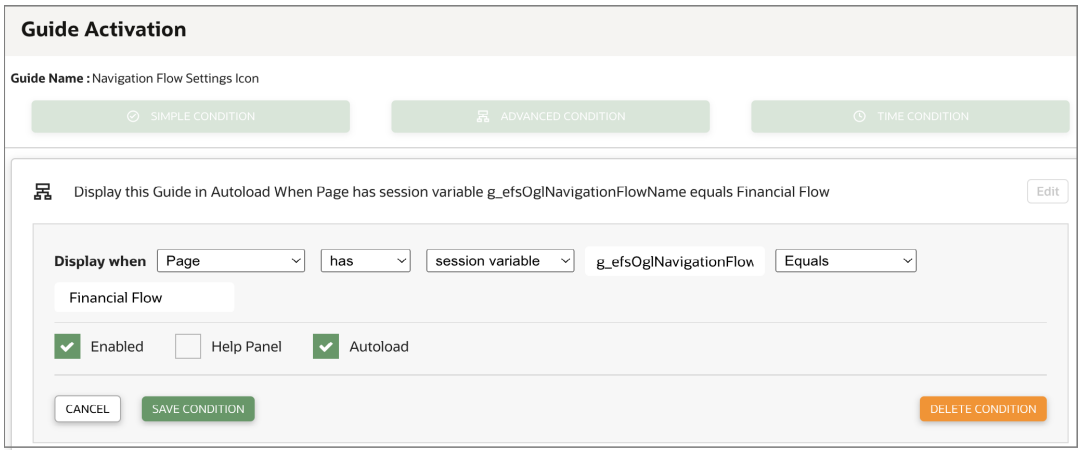
- Configure conditional activation based on Navigation Flow name, cluster ID, card ID, tab, or sub-tab ID.
- Incorporate dynamic content and define custom triggers for guide activation.
- Control visibility and placement of guidance on-screen.

This flexibility ensures that OGL guidance can be fine tuned to meet diverse business requirements within the application environment. Upon integration, Oracle Fusion Cloud Enterprise Performance Management offers the following attributes for activating guides conditionally on pages or components, allowing advanced settings for context-based activation of OGL Guides in the Cloud EPM environments:

Attributes for Context-Based Activation in Cloud EPM

For more information on updating attributes, see Working with the Editors in *Oracle Guided Learning User Guide*.

- **Activating Guides by Navigation Flow Name**: Use the attribute `g_efsOglNavigationFlowName` to activate a guide for specific Navigation Flow. For instance, the following condition activates a guide for all pages within the "Financial Flow" Navigation Flow.



- **Activating Guide Names by Page ID:** The attribute `g_efsOglFqId` enables activation at various levels within the navigation hierarchy:
  - `g_efsOglFqId/<SUB_TAB_ID>` - Activates the guide on the specified sub-tab.

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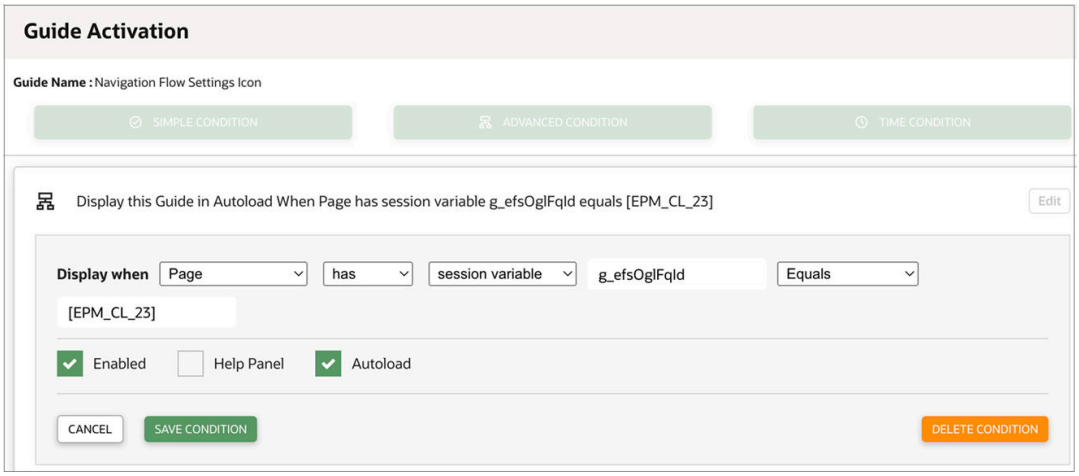
- `g_efsOglFqId/<TAB_ID>/<SUB_TAB_ID>>` - Activates the guides on a sub-tab within a specific tab.
- `g_efsOglFqId/<CARD_ID>/<TAB_ID>/<SUB_TAB_ID>` - Activates the guide on a sub-tab within a tab within a specified card.
- `g_efsOglFqId/<TAB_ID>` - Activates the guides for all the pages within the specified tab.
- `g_efsOglFqId/<CARD_ID>` - Activates the guides for all the pages within the specified card.
- `g_efsOglFqId/<CLUSTER_ID>` - Activates the guides for all the pages within the specified cluster.

For instance, the following activation condition for OGL enables a guide for all pages defined in the cluster ID EPM_CL_23.



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10

Editing Dimensions in the Simplified Dimension Editor

Work with application dimensions and members in the simplified grid interface.

Related Topics

- About Editing Dimensions in the Simplified Dimension Editor
  The Simplified dimension editor displays dimensions and members in a grid format.
- Dimension Overview
- About Dimensions and Members
- About Sparse and Dense Dimensions
- About Dimension Hierarchies
- Data Type
- About Custom Dimensions
- Aggregation Options
- Storage Options
- About Entities
- About Accounts
- Account Types
- Saved Assumptions
- Accounts, Entities, Periods, and Cubes
- Dimension Design
  Designing dimensions effectively is important to ensure accurate reporting, analysis, and performance management.
- Accessing the Simplified Dimension Editor
- Working in the Edit Member Properties Grid
  Work with elements in the **Edit Member Properties** grid interface.
- Switching to Another Dimension
- Customizing the Column Layout
- Context Menu Options
- Viewing Ancestors
- Showing Member Usage in an Application
- Focusing Your Editing
- Finding Members
- Moving Members
- Sorting Members

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- Moving Members to Another Hierarchy
- Working with Member Formulas
- Copying Member Names from Microsoft Excel
- Editing Member Properties
- Accessing Edit Member Properties
- Adding Members
- Editing Members
- Deleting Members
- Adding Shared Members
- Assigning Access to Dimension Members
- Adding, Editing, and Removing Access to Members
- About Effective Access Permissions to Shared Members
- Editing Dimension Properties
- Working with Attributes
- Deleting Attributes
- Working with UDAs in the Simplified Dimension Editor
- Working with Attribute Values

About Editing Dimensions in the Simplified Dimension Editor

The Simplified dimension editor displays dimensions and members in a grid format.

> **Note**
>
> This section documents how to edit dimensions using the Simplified dimension editor which was released in the 17.05 (May 2017) update.
>
> To use the Classic Dimension Editor, which is accessed by using the **Dimensions** link in the Navigator menu, see Administering Dimensions.

For information about using the Simplified Interface to filter the dimension view by cube, create dimensions, or set the evaluation order, see Managing Dimensions.

With the grid format, dimensions and members are editable on a single page. You can edit member properties directly on the grid and you can perform ad hoc operations such as zooming in, zooming out, keep selected, remove selected, and freeze. In addition:

- The grid provides the ability to do a filtered search within the grid and to reorder columns using drag-and-drop. The expansion states of the member hierarchy in the grid are saved for the next session.
- The column selector interface enables you to select the columns that will display within the grid. The column selector groups properties according to their functionality and saves the column selections that you add to the grid for the next session. There is also an option to resize the columns so that all columns are visible on the grid without scrolling.

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Dimension Overview

Related Topics

- About Dimensions and Members
- About Sparse and Dense Dimensions
- About Dimension Hierarchies
- Data Type
- About Custom Dimensions
- About Entities
- About Accounts
- Accounts, Entities, Periods, and Cubes

About Dimensions and Members

Dimensions are data categories used to organize business data for retrieval and preservation of values.

Dimensions contain hierarchies of related *members* grouped within them. For example, a Years dimension often includes members for each time period, such as quarters and months.

FreeForm apps can have up to 29 total dimensions across all cubes: 26 custom dimensions plus three Account, Period, and Entity dimension types.

About Sparse and Dense Dimensions

Sparse dimensions lack data values for the majority of member combinations. Dense dimensions have data values for the majority of member combinations. At least one dense dimension is required. Custom attributes can't be assigned to dense dimensions. The application designates the Account and Period dimensions as dense, and the remaining dimensions as sparse. To optimize performance for sparse dimensions, the application searches for and calculates only occupied data values in each dimension combination, reducing calculation time and disk usage. You can modify these settings.

About Dimension Hierarchies

Dimension hierarchies define structural and mathematical relationships, and consolidations between members in the application. Relationships are represented graphically in a collapsible hierarchy diagram. The levels below the cube name are dimensions, and the levels below each dimension are members.

The Period dimension can contain the member YearTotal, which contains members Q1, Q2, Q3, and Q4. Members Q1, Q2, Q3, and Q4 contain their own members for the corresponding months in the year. To consolidate data values in the Period dimension, roll up monthly data values to get quarterly data values, and quarterly data values to get yearly data values.

Members of the same level that belong to the same dimension or member are called siblings. For example, Q1, Q2, Q3, and Q4 are siblings because they are at the same level in the hierarchy, and are members of the same member, YearTotal.

The members of a dimension are called children of the dimension. Members that belong to a member are called children of that member. The member YearTotal is a child of Period, the

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members of Q1, Q2, Q3, and Q4 are children of YearTotal, and Jan, Feb, and Mar are children of Q1. Q1 is the parent of Jan, Feb, and Mar, YearTotal is the parent of Q1, Q2, Q3, and Q4, and Period is the parent of YearTotal.

Data Type

Data type determines how values are stored for dimension members.

Available data types for dimension members' values:

- **Unspecified**
- **Percentage**: Stores a numeric value and displays as a percent.
- **Smart List**
- **Date**: Displays as a date.
- **Text**: Displays as text.
- **Transient Text**: Displays as text without storing.

The **Transient Text** data type displays text in forms which doesn't need to be persisted. For example, real-time inventory data can be pulled from an external system (REST API), and then it can be displayed in the form using a post-load Groovy rule for that form.

> **Note**
>
> **Transient Text** is not stored in Essbase or in the relational database. If an existing member is changed to use the **Transient Text** data type, then the existing data will not be displayed in the form and the data will not be deleted from Essbase.

About Custom Dimensions

In FreeForm apps, you can have up to 26 custom dimensions. The Scenario, Version, Period, and Year dimensions can be defined as custom dimensions. See Understanding FreeForm.

> **Caution**
>
> You can't delete custom dimensions after you create them.

- Aggregation Options
- Storage Options

Aggregation Options

You can define calculations within dimension hierarchies using aggregation options. Aggregation options determine how child member values aggregate to parent members:

- + Addition
- - Subtraction
- * Multiplication

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- / Division
- % Percent
- ~ Ignore
- Never (don't aggregate, regardless of hierarchy)

Storage Options

When working with dimensions, it's important to understand the different data storage options and how to use them in the business process.

- Storage Options
- About Dynamic Calc
- Dynamic Calc Versus Dynamic Calc and Store
- About Store Data Storage
- About Shared Data Storage
- About Never Share Data Storage
- About Label Only Data Storage

Storage Options

Table 10-1 Storage Options

Option	Impact
Dynamic Calc and Store	Calculates data values of members, and stores values.
Store	Stores data values of members.
Dynamic Calc	Calculates data values of members, and disregards the values.
Never Share	Prohibits members in the same dimension from sharing data values.
Shared	Allows members in the same dimension to share data values.
Label Only	Displays the data of the first child member regardless of the aggregation setting of the children.

About Dynamic Calc

With dynamically calculated members, the application calculates data values of members, and disregards these values. As a best practice, Oracle recommends a limit of 100 children under a **Dynamic Calc** parent. Changing a member's storage to **Dynamic Calc** may result in loss of data, depending on how the data was originally derived. You may need to update outlines, calculations, or both to get the dynamically calculated value.

Dynamic Calc Versus Dynamic Calc and Store

In most cases, you can optimize calculations and lower disk usage by using **Dynamic Calc** instead of **Dynamic Calc and Store** when calculating members of sparse dimensions. Use **Dynamic Calc and Store** for members of sparse dimensions with complex formulas, or that users retrieve frequently.

For members of dense dimensions, use **Dynamic Calc**. **Dynamic Calc and Store** provides only a small decrease in retrieval time and regular calculation time, and doesn't significantly lower disk usage. For data values accessed concurrently by many users, use **Dynamic Calc**. Retrieval time may be significantly lower than for **Dynamic Calc and Store**.

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> **Note**
>
> - Don't use **Dynamic Calc** for base-level members for which users enter data.
> - Don't use **Dynamic Calc** for a parent member if you enter data for that member in a target version. Parent members set to **Dynamic Calc** are read-only in target versions.
> - Data values are not saved for **Dynamic Calc** members.

About Store Data Storage

Don't set parent members to **Store** if their children are set to **Dynamic Calc**. With this combination, new totals for parents are not calculated when users save and refresh forms.

About Shared Data Storage

Use **Shared** to allow alternate rollup structures in the application.

About Never Share Data Storage

The default data storage type is **Never Share** when you add user-defined custom dimensions. You can use **Never Share** for parent members with only one child member that aggregates to the parent, to apply access to the child member.

About Label Only Data Storage

Label Only members are virtual members; they are typically used for navigation and have no associated data.

> **Note**
>
> - You can't assign level 0 members as **Label Only**.
> - **Label Only** members can display values.
> - Making dimension members **Label Only** minimizes database space by decreasing block size.
> - You can't assign attributes to **Label Only** members.
> - Data storage for children of **Label Only** parents is set to **Never Share** by default.
>
> > **Caution**
> >
> > Don't design forms in which **Label Only** parents follow their first child member, as you can't save data in the first child member. Instead, create forms with **Label Only** parents selected before their children, or don't select **Label Only** parents for forms.

About Entities

Entities typically match your organization's structure, such as geographical regions, departments, or divisions.

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About Accounts

Account dimension members specify the information needed from users. Create an account structure that lets budget preparers input data for budget items. You can define calculations in the account structure.

Related Topics

- Account Types
- Saved Assumptions
- Data Type

Account Types

Account type defines accounts' time balance (how values flow over time) and determines accounts' sign behavior for variance reporting with member formulas.

Examples of Using Account Types

Table 10-2 Using Account Types

Account Type	Purpose
Expense	Cost of doing business
Revenue	Source of income
Asset	Company resource
Liability and Equity	Residual interest or obligation to creditors
Saved assumption	Centralized planning assumptions ensuring consistency across the application

Summary of Account Types

Table 10-3 Summary of Account Types

Account Type	Time Balance	Variance Reporting
Revenue	Flow	Non-Expense
Expense	Flow	Expense
Asset	Balance	Non-Expense
Liability	Balance	Non-Expense
Equity	Balance	Non-Expense
Saved Assumption	User-defined	User-defined

Variance reporting and time balance settings are system-defined; only Saved Assumption is user-defined.

Time Balance Property

Time balance specifies how the application calculates the value of summary time periods.

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Table 10-4 Time Balance Properties

Time Balance Property	Description	Example
Flow	Aggregate of all values for a summary time period as a period total.	Jan: 10 Feb: 15 Mar: 20 Q1: 45
First	Beginning value in a summary time period as the period total.	Jan: 10 Feb: 15 Mar: 20 Q1: 10
Balance	Ending value in a summary time period as the period total.	Jan: 10 Feb: 15 Mar: 20 Q1: 20
Average	Average for all the child values in a summary time period as the period total.	Jan: 10 Feb: 15 Mar: 20 Q1: 15
Fill	The value set at the parent is filled into all its descendents. If a child value changes, the default aggregation logic applies up to its parent. Consolidation operators and member formulas overwrite Fill values when the members are recalculated.	Jan: 10; Feb: 10; Mar: 10; Q1: 30
Weighted Average - Actual_Actual	Weighted daily average, based on the actual number of days in a year; accounts for leap year, in which February has 29 days. In the example, the average for Q1 is calculated: (1) Multiply each month’s value in Q1 by the number of days in the month, (2) Sum these values, (3) Divide the total by the number of days in Q1. Assuming it's a leap year, the result is calculated: (10 * 31 + 15 * 29 + 20 * 31) / 91 = 15 Note that this time balance property is only supported for dimensions bound to a block storage cube. Aggregate storage cubes do not support the Weighted Average - Actual_Actual time balance property.	Jan: 10 Feb: 15 Mar: 20 Q1: 15
Weighted Average - Actual_365	Weighted daily average, based on 365 days in a year, assuming that February has 28 days; doesn't account for leap years. In the example, the average for Q1 is calculated: (1) Multiply each month’s value in Q1 by the number of days in the month, (2) Sum these values, (3) Divide the total by the number of days in Q1. Assuming it's not a leap year, the result is calculated: (10 * 31 + 15 * 28 + 20 * 31) / 90 = 15 Note that this time balance property is only supported for dimensions bound to a block storage cube. Aggregate storage cubes do not support the Weighted Average - Actual_365 time balance property.	Jan: 10 Feb: 15 Mar: 20 Q1: 15
Custom	Spreading is disabled and the application designer is expected to add customized spreading. For instance, you can feed a value into the period total (Q1) using Groovy rules and it will calculate the summary time periods using the Flow method.	NA
Disable	Spreading is disabled and the summary time period is read-only. Disable prevents data from being entered on non-level zero periods. The Flow method is used to aggregate into the period total (Q1), but it won’t spread down to the summary time periods.	NA

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> **Note**
>
> - You can use the Weighted Average - Actual_Actual and Weighted Average - Actual_365 time balance properties only with a standard monthly calendar that rolls up to four quarters.
>
> For information on how the application calculates and spreads data with the different Time Balance settings, see How Spreading Data Works in *Working with FreeForm*.
> - For Custom and Disable time balance properties, the application designer needs to be aware of the storage characteristics of the member that they write to, be it aggregate storage or block storage. For example, you can only save to level zero members in aggregate storage and if you try to save to a dynamic calc member, it'll be ignored and overwritten when recalculated. Also note that spreading only happens in the grid prior to save, be it automatic or using a Groovy rule. After the grid is saved, normal Essbase behavior will take over with regard to saving and reading data (that is, normal outline math will apply, member formulas, time balance, and so on).

Account Types and Variance Reporting

An account's variance reporting property determines whether it's treated as an expense when used in member formulas:

- Expense: The actual value is subtracted from the budgeted value to determine the variance
- Non-Expense: The budgeted value is subtracted from the actual value to determine the variance

Setting Account Calculations for Zeros and Missing Values

With time balance properties First, Balance, and Average, specify how database calculations treat zeros and missing values with the Skip options.

Table 10-5 Effect of Skip Options When Time Balance is Set to First

Skip Option	Description	Example
None	Zeros and #MISSING values are considered when calculating parent values (the default). In the example, the value of the first child (Jan) is 0, and zeros are considered when calculating the parent value, so Q1 = 0.	Jan: 0 Feb: 20 Mar: 25 Q1: 0
Missing	Excludes #MISSING values when calculating parent values. In the example, the value of the first child (Jan) is #MISSING, and #MISSING values are not considered when the calculating parent values, so Q1 = second child (Feb), or 20.	Jan: #MISSING Feb: 20 Mar: 25 Q1: 20
Zeros	Excludes zero values when calculating parent values. In the example, the value of the first child (Jan) is 0, and zero values are not considered when calculating parent values, so Q1 = the second child (Feb), or 20.	Jan: 0 Feb: 20 Mar: 25 Q1: 20

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Table 10-5 (Cont.) Effect of Skip Options When Time Balance is Set to First

Skip Option	Description	Example
Missing and Zeros	Excludes #MISSING and zero values when calculating parent values. In the example, the value of the first child (Jan) is zero, and the value of the second child (Feb) is missing. Because missing and zero values are not considered when calculating parent values, Q1 = the third child (Mar), or 25.	Jan: 0 Feb: #MISSING Mar: 25 Q1: 25

Saved Assumptions

Use saved assumptions to centralize planning assumptions, identifying key business drivers and ensuring application consistency. You select time balance and variance reporting properties.

- Variance reporting determines the variance between budgeted and actual data, as an expense or non-expense.
- Time balance determines the ending value for summary time periods.

Examples of how time balance and variance reporting properties are used with saved assumption account members:

- Create a saved assumption of an expense type for variance reporting, assuming that the actual amount spent on headcount is less than the amount budgeted. To determine the variance, the application subtracts the actual amount from the budgeted amount.
- Determine the value for office floor space by using the time period's last value.
- Make an assumption about the number of product units sold at the end of the time period. Determine the final value for the summary time period by aggregating the number of units sold across time periods.

Accounts, Entities, Periods, and Cubes

By assigning cubes for Account, Entity, and Period members, you set to which cubes the members' children have access. For example, Total Sales Account may be valid for Revenue and P&L, but Fixed Assets Account may be valid for only Balance Sheet. Not assigning a cube to a member prevents that member's children from accessing that cube.

When moving members, if the new parent is valid for different cubes, members remain valid only for cubes they have in common with the new parent. If the new parent of an account member has another source cube, the member's source cube is set to the first new valid cube of that member.

Entities and Cubes

Typically, entity members prepare different plans. When defining entity members, specify cubes for which they are valid. Because forms are associated with cubes, you can control which entity members can enter data for each cube.

Accounts and Cubes

If accounts are valid for multiple cubes, specify the source cube to determine which cube's database stores the account value for them.

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Dimension Design

Designing dimensions effectively is important to ensure accurate reporting, analysis, and performance management.

Follow these best practices during dimension design.

- Design Dimensions to Create the Application Structure
- Follow this Process to Identify Dimensions
- Consider Common Use Cases for Dimensions
- Review Example Design Strategies
- Know the Top Ten Best Practices for Dimensions
- Plan the Entity Dimension
- Plan the Account Dimension
- Plan the Version Dimension
- Plan the Currency Dimension
- Plan Exchange Rates
- Plan the Period Dimension
- Plan Years and Substitution Variables
- Design Custom Dimensions
- Additional Best Practices

Design Dimensions to Create the Application Structure

Add accounts, entities, and other dimensions to support your business process.

Dimensions categorize data values. Planning includes these dimensions: Account, Entity, Scenario, Version, Period, and Years. If you plan in multiple currencies, your application also has a Currency dimension.

You can use the Custom dimension to define your own values, such as Product, Customer, or Market. You can have up to 32 total dimensions. However, the best practice recommendation is to include fewer than 12. You can add dimensions using a load file, or you can build them in Oracle Smart View for Office.

Videos

Your Goal	Watch This Video
Learn how to export and import data in the application.	Exporting and Importing Data in Oracle Planning and Budgeting Cloud
Learn how to load dimensions using a file.	Importing Metadata in Oracle Planning and Budgeting Cloud

Follow this Process to Identify Dimensions

Use this process to identify which dimension to include in the application.

1. **Identify unique planning processes based on requirements**.

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Example: Marketing Planning, Sales Planning, Overhead Planning, Capital Planning, Cash flow Planning, Workforce Planning

2. **Identify dimensions for each planning process**.
   Example: Product, Market, Channel, Product Segment, Customer Segment
3. **Define how dimensions are related to each other**.
   Example: Product has a many-to-many relationship with Market. Product Segment and Product have a one-to-many relationship. Labor Resources and Material Resources have no relationship.
4. **Separate dimensions into buckets for planning and reporting**.
   Example: Product, Market, and Channel are Planning dimensions, while Product Segment and Customer Segment are reporting dimensions.
5. **Map planning processes to Planning Modules**.
   Example: Configure Marketing Planning using Projects or Financials, and configure Overhead Planning and Cash flow Planning using Financials and custom cubes.

Consider Common Use Cases for Dimensions

Review these common use cases for dimensions, and understand the guidelines for how to address them.

- **Standard (mandatory) dimensions**
  Most of the high level and common Financial Planning use cases can be addressed with standard dimensions, which can also be renamed.
- **Custom or optional dimensions**
  Extend dimensionality with custom or optional dimensions that can be added (enabled) or renamed per your requirements.
- **Multiple hierarchies in a dimension**
  Combine two or more unrelated dimensions into a single dimension to avoid inter-dimensional irrelevance.
- **Alternate Hierarchies for Planning or Reporting**
  Use alternate hierarchies when the same members can be grouped under different parents for top-down allocation or reporting purpose.
- **Attribute dimensions for reporting**
  Attribute dimensions are useful to meet reporting requirements if they are related to one dimension and the relationship between the two dimension does not change over time.
- **Smart Lists and ASO dimensions for reporting**
  Using Smart Lists and ASO dimensions for reporting is useful when the relationship between the reporting dimension and other dimensions changes with time.
- **Smart Lists and multi-cube BSO dimensions for planning**
  This strategy is useful when the planning dimension is not a primary dimension for a planning process and there is a need to break it into sub-processes.

This sample worksheet shows an example of how to plan dimensions, including indentifying dimensions and listing their use cases.

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	A	B	C	D	E	F
1	**Planning Process**	**Dimensions**	**Description**	**Is Planning vs Reporting?**	**Relation with other dimensions**	**Dimensionality Options**
17	Marketing Planning	Activity	Standard Activities for Campaigns	Planning	Plan activities by Campaign and Marketing expenses at activity level. Many to many relation with Product, Campaign, Customer & IO dimensions	Custom
18	Marketing Planning	Product	Hierarchy/List of Products (not SKU)	Planning	Plan marketing expenses by Product. Many to many relation with Campaign, Activity, Customer & IO dimensions	Custom
19	Marketing Planning	Customer	List of Major Customers	Planning	Plan marketing expenses by Customer. Many to many relation with Campaign, Activity, Product & IO dimensions	Custom
20	Marketing Planning	IO	Member on the fly generated for marketing expenses	Planning	Generate IO for marketing expenses. Many to many relation with Campaign, Activity, Customer & Product dimensions	Custom
21	Revenue Planning	Product	Hierarchy/List of Products (not SKU)	Planning	Plan Revenue by Product. Many to Many relation with Channel & Customer. Many to one relation with Product Segment dimension. No relation with customer segment	Custom
22	Revenue Planning	Channel	List of Sales Channels	Planning	Plan Revenue by Channel. Many to Many relation with Product & Customer. No relation with Product Segment & Customer Segment	Custom

Review Example Design Strategies

Review these examples to understand additional design strategies for dimensions.

Using attribute dimensions for reporting

Using attribute dimensions for reporting can be useful to meet reporting requirements. The attribute is related to one dimension, and the relationship between the two dimension doesn't change with time.

Example:

- Define an attribute called Program on the Project dimension.
- You can then create a hierarchy of members in the Program dimension. Top level members in the Program dimension will be automatically aggregated.
- Associate each Project member to a leaf level member in the Program dimension in an Add Project Groovy rule.
- This allows you to filter projects by program.
- Reporting forms can show program level Expenses and Revenue.

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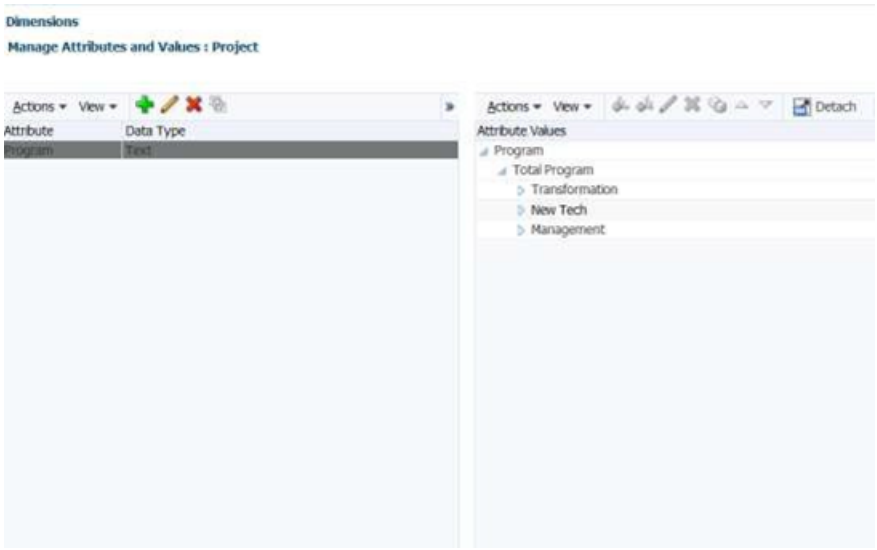


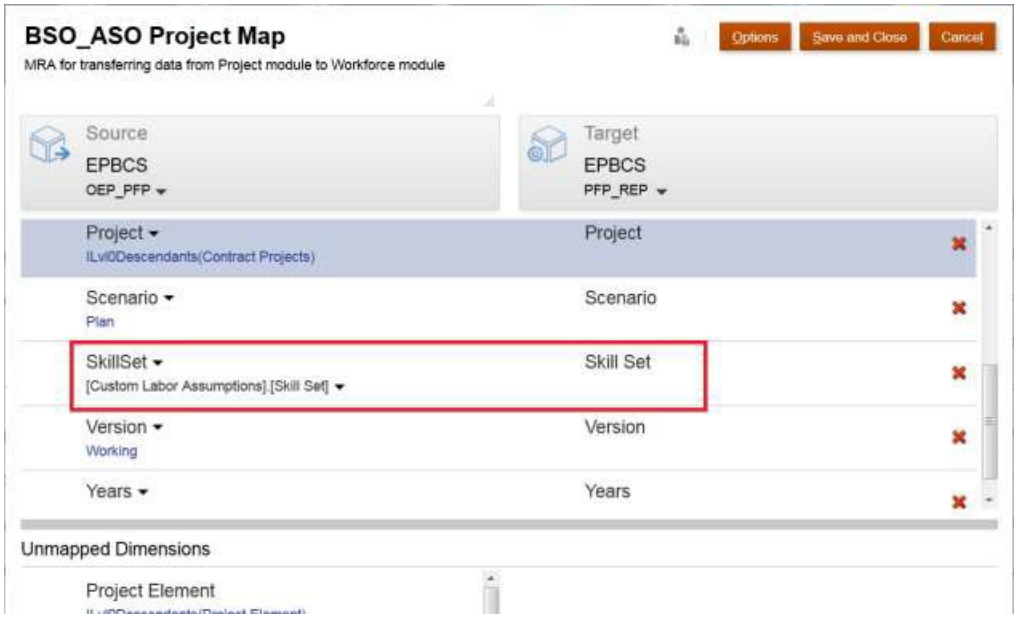
Note that using too many attribute dimensions on a form or report may degrade performance. This should be included in the performance and load testing aspects of an implementation.

Using Smart Lists and ASO dimensions for reporting

This strategy is useful when the relationship between the reporting dimension and other dimensions changes with time.

Example:

- Add a Skillset dimension in ASO and map the Skillset Smart List to the Skillset dimension.
- Create a data map for moving data from BSO to the ASO cube.
- The data map can be run as a batch process, or you can implement Smart Push in forms or Groovy rules.
- You can use reporting forms on the ASO cube to showing the labor requirement by Skillset across projects.



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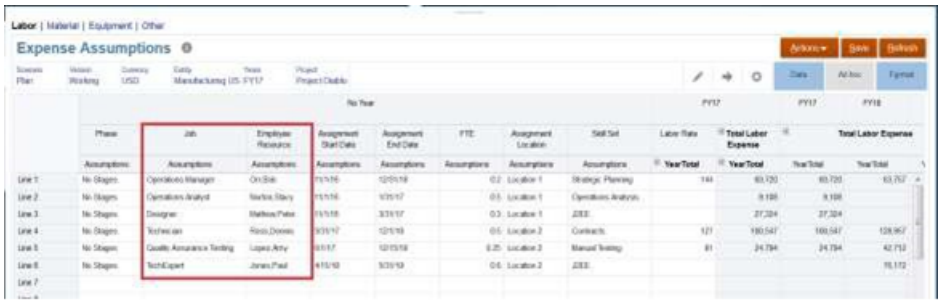
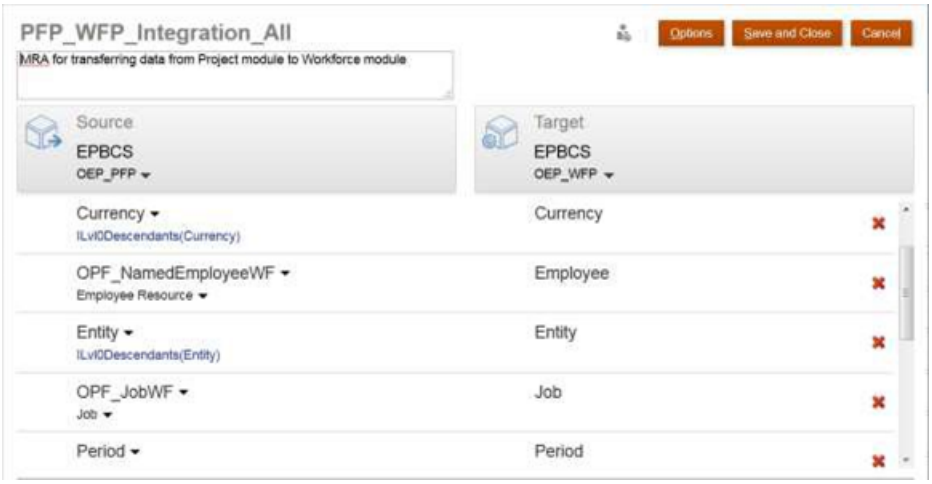
Using Smart Lists and multi-cube BSO design

This strategy is useful when the planning dimension is not a primary dimension for a planning process and there is a need to break it into sub-processes.

Example:

- Employee and Job are dimensions in Workforce and Smart Lists in Project Planning.
- Project Planning uses line items to plan labor expenses at the Job and Employee level.





Multiple hierarchies in a dimension

You can use multiple hierarchies in a dimension. Combine two or more unrelated dimensions into a single dimension to avoid inter-dimensional irrelevance.

Example:

- Jobs, Equipment, and Material are unrelated in Projects, so they are combined into a single Resource Class dimension.
- Profit Centers and Cost Centers are unrelated in Financials, so they can be combined into a single Entity dimension.

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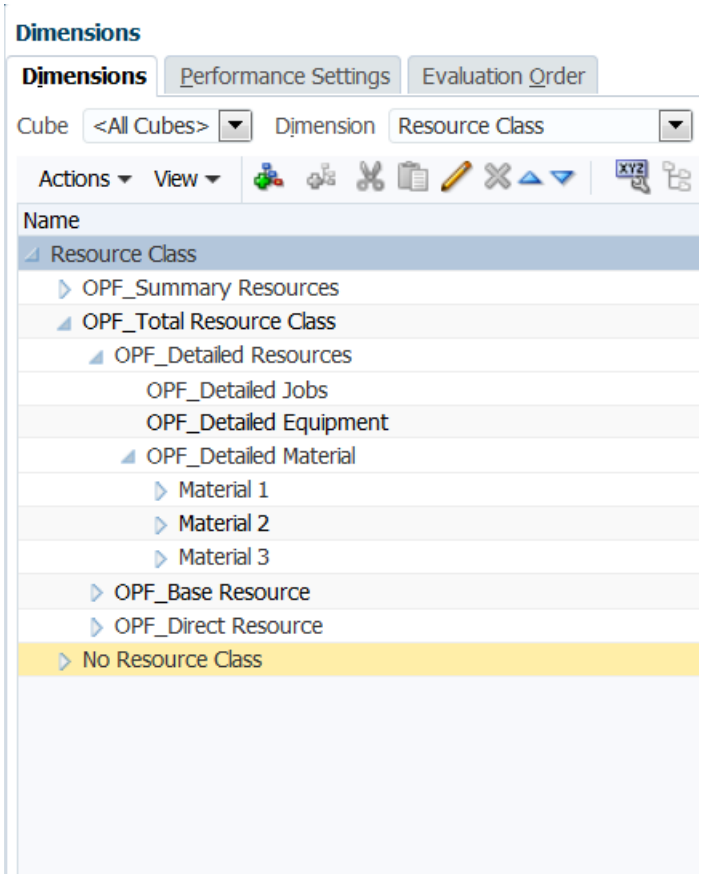

Alternate hierarchies for planning and reporting

Use alternate hierarchies when the same members can be grouped under different parents for top-down allocation or reporting purposes.

Example: Create alternate rollups in the Product dimension for planning and reporting by the Brand and Product categories.

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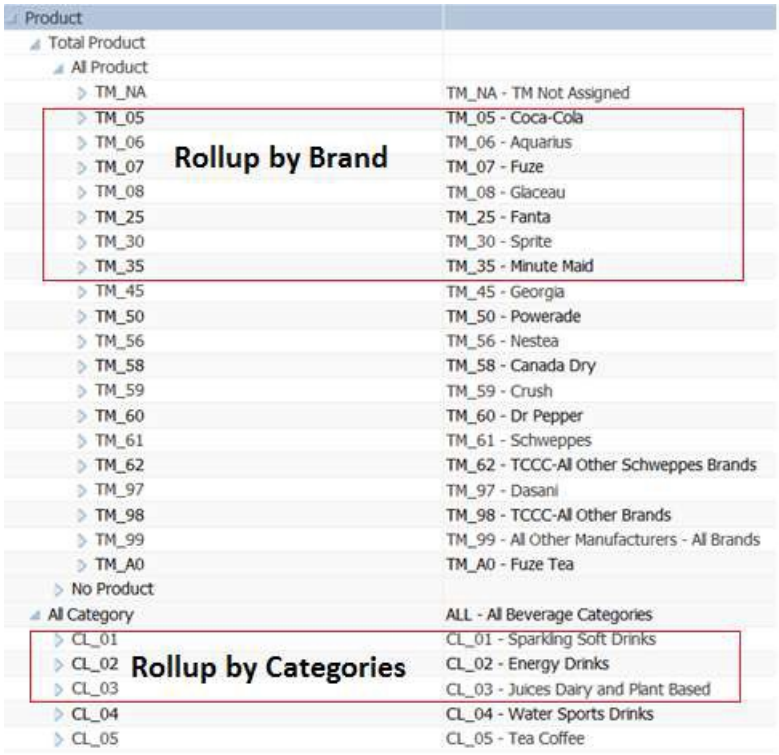

Know the Top Ten Best Practices for Dimensions

Follow these important best practices when designing dimensions.

1. Dimension order should follow a modified hour glass format.
   With this format, the most dense dimension is the first dimension, followed by less dense dimensions. This should be followed by sparse dimensions, with aggregating sparse dimensions before non-aggregating sparse dimensions. Within sparse dimensions, you should have the most dense sparse dimensions before the least dense sparse dimensions.

   With hybrid BSO, the order is the same, except that you should have non-dynamic sparse dimensions before dynamic sparse dimensions.

   Learn more about dense and sparse dimensions in this tutorial: Managing Dimensions in Cloud EPM.

2. Large block size has a major impact on performance.
   Block size is determined by the number of dense dimensions and the stored members in those dense dimensions. Optimal block size is between 8 KB to 500 KB. Reduce the number of dense dimensions to a maximum of 3. Level 1 and above should be label only or dynamic calc for dense dimensions.

3. Text, Smart List, Date, and Stored Percentage type accounts should be set to Never for the consolidation property.
   Aggregating these values, unless the accounts are explicitly excluded in an aggregation script, will create useless data and make the cube unnecessarily large, which likely will degrade performance.

4. All generation two members should be set to Ignore.
   You cannot include these root members in your forms because you cannot define security for those members. So there is no point in aggregating the generation two members to the root. This will also increase the number of blocks in your application.

5. Long or flat dimensions will lead to an issue with the performance of aggregation.

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If there are more than 200 members under a parent member, add intermediate parents.

6. Enabling dimension members for multiple cubes will create dynamic X-Refs and lead to performance issues.
   Use the HSP_NOLINK UDA to avoid creating dynamic X-Refs. Use data maps or Smart Push for moving data between cubes.
7. For simple calculations, leverage outline math instead of writing member formulas.
   An example of a simple calculation is Account C = Account A – Account B.
8. When possible, avoid single child parent members.
   Single children parent members lead to implied shares or duplicate blocks and data on disk if the parent member is made Never Share.
9. Aggregate large dimensions in ASO instead of BSO whenever possible.
   This improves performance for example during cube refresh and maintenance time.
10. Store historical data beyond two years in ASO instead of BSO.
    If you have 5 or 10 past years of historical data, not all of the data is needed for calculations. If needed, you can have a couple of years of historical data for calculations in your BSO cube, and you can move other historical data to the ASO cube. For optimum performance, it is a best practice to keep the BSO cube light and ensure that it is focused on the calculations for data entry.

Plan the Entity Dimension

The Entity dimension represents your organizational structure, such as Cost Centers, Departments, Business Units, Divisions, and so on.

You can group Cost Centers by creating rollup members, called parents, to reflect how your organization is viewed. For example, rollups can be by business unit, division, or other functional structure. As an example, you could create Cost Centers that roll up to Business Units that roll up to Divisions.

You can also create multiple reporting structures. For example, an alternate structure could be created to support regional reporting. If you plan in multiple currencies, set the base currency of each entity.

The Entity dimension is one of the primary dimensions used for the budgeting process. Together with the Scenario and Version dimensions, the Entity dimension is used to define an approval unit, a discrete component that can be promoted or demoted for approval or review by a user’s peers. Members of all dimensions outside the approval unit are promoted and demoted along with the approval unit itself. For example, all twelve months are promoted together when an approval unit is promoted. Individual months can't be promoted independently.

A secondary dimension is also supported, as described in Approvals Design Considerations. In addition, you can include phased approvals, as described in Managing Approval Phases.

After each dimension is loaded or updated, it is a best practice to refresh the application.

Plan the Account Dimension

The Account dimension is the place for your chart of accounts. It should include the members to which you plan or forecast. It doesn't necessarily include every account in your chart.

For example, your Account dimension could include accounts for Income Statement, Balance Sheet, and Cash Flow. Or, it could include accounts for KPIs and Ratios. In some cases, your accounts may have sub accounts, but this is not typical.

The Account dimension includes financial intelligence. The following account types are supported:

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- **Expense**: Cost of doing business
- **Revenue**: Source of income
- **Asset**: Company resources
- **Liability and Equity**: Residual interest or obligation to creditors
- **Saved assumption**: Centralized planning assumptions ensuring consistency across the application

The account type settings are used to report Quarterly and Year Total values and for variance analysis.

Planning uses a hierarchical structure to create Account grouping subtotals and totals. Each account group is assigned a consolidation operator that determines how it rolls up to its parent.

Example:

Net Income = Total Revenues - Total Expenses

In this example, the consolidation operator for Total Revenues is Addition, and the consolidation operator for Total Expenses is Minus. Intelligence determines, in part, the sign in which data is loaded, entered, or calculated.

The Account dimension can be populated either by loading data or using Smart View. To load data from a file, the file format must meet specific requirements.

After each dimension is loaded or updated, it's a best practice to refresh the application.

Best practices:

- Upper level members should be set to Dynamic Calc or Label Only.
- For member formulas used to calculate Ratios and other types of KPIs or percentages, set them to Dynamic Calc, Two Pass. The Two Pass setting properly calculates Percentages at upper levels.

Plan the Version Dimension

You can use versions to preserve different iterations of the planning process. Versions are also useful for controlling data access to Read or Write.

These two types of versions are available:

- **Standard Target**: Input data can be entered to upper levels.
- **Standard Bottom Up**: Input data can be entered to level 0 only.

Approvals and workflow functionality can be enabled only for Bottom Up versions.

As a best practice, these versions are recommended:

- **Working**: Where users perform their tasks, including reviewing Actual Results and developing Plan and Forecast.
- **1st Pass**: If you want to maintain multiple iterations of your Plan, you can preserve a pass of it in this version. You can create other members if you require more than one saved iteration. You can leverage the Copy Data functionality to move data to this version. Copy data copies data and textual input.
- **What If**: Provides a placeholder where users can change assumptions and analyze the outcome.

After each dimension is loaded or updated in the build process, it's a best practice to refresh the application.

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Plan the Currency Dimension

If you enabled multiple currencies for your application, you can add the currencies you use to plan and report.

You can then define exchange rates by scenario and year to be used in conversions. A calculation script is created that enables you to perform currency conversion. To enter exchange rates, follow the process in Specifying Exchange Rates in *Administering Planning*.

Best practices:

- Limit the number of reporting currencies. Typically, customers have only one.
- Enter exchange rates for each valid scenario and year combination.
- From this point on, currency conversion can be calculated by running the Calculate Currencies business rule that is associated by default with each form.

Run the currency conversion calc script prior to:

- Reviewing any updated local data in reporting currencies
- Running certain calculations that may be dependent on reporting currency data

Plan Exchange Rates

Each application has a default currency that you specify when creating the application. When you set up exchange rate tables, you enter exchange rates from all source currencies to the default. Triangulation is used to convert to all other Reporting currencies.

Exchange rates are set by Scenario by year for Average and Ending Rates.

Plan the Period Dimension

Use the Period dimension to establish the calendar’s range within a given year, for example, by month.

Best practices:

- Use substitution variables for this dimension to support reporting and calculations. Potential substitution variables are: CurrMo, CurrQtr, and PriorMo. These variables must be updated on a monthly basis.
- To use time period calculations such as Year to Date (Y-T-D) or Quarter to Date, select the dynamic time series icon in the Period dimension. You can then select which time period calculations you need to support your process.
- Summary time periods such as quarter totals and a year total should be set to dynamic calculate to reduce calculation time.
- After each dimension is loaded or updated, refresh the application.

Plan Years and Substitution Variables

Years are incorporated into the application in many places, including forms, calculations, reports, and Smart View. Because you'll use the application for many years into the future, the best practice to referencing this dimension is by using a substitution variable.

Substitution variables act as global placeholders for information that changes regularly. The variable and value correspond to the year, and the value can be changed at any time.

The value of the substitution variable is displayed on forms and reports as a placeholder. This reduces maintenance for the application.

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As a best practice, create substitution variables for each year that is included in your process. For example:

- CurrY, Current Year
- NextYr, Budget (Plan) Year
- PriorYr, Prior Year

Plan the HSP_Metric Dimension

The HSP_Metric dimension enables you to view Total Values associated with approval units within forms. The HSP_Metric dimension is automatically created and seeded with dimension members when you create an application. You can create additional members to view information about approval units such as Name, Date, Status, or Owner, and create hierarchies to group commonly used Approvals metrics under a parent member.

By default, when you create an application, the HSP_Metric dimension is created with the Metric Dimension Type. It includes these members, which cannot be edited:

- HSP_Approval Unit Submission Group
- HSP_Approval Unit Hierarchy
- HSP_Cell Level Security Access

To use HSP_Metric dimension members in forms, when you create a form, on the **Properties** tab, select the **Use Metrics** option.

Design Custom Dimensions

You can use a custom dimension to further categorize your data. For example, custom dimensions might include Product or Markets.

Keep in mind that access permissions can't be granted at the dimension level, also called generation one. For example, access permissions can't be assigned directly to the Product member for all descendants. If you enable security for your custom dimension, it is recommended that you design generation two for all custom dimensions to which security will be applied with security access assignments in mind.

After each dimension is loaded or updated, it is a best practice to refresh the application.

Additional Best Practices

Complete these tasks after you add or update dimensions.

- **Refresh the application.**
  You must refresh the application whenever you change the application structure.

  Changes made to the application are not reflected to users performing data entry and approvals tasks until the application is refreshed.

  For example, when you modify properties of an Entity member or add a Scenario, these changes are reflected to users after you refresh the application.
- **Load historical data.**
  After you load all of your structures, such as accounts and entities, you can load historical data. This can include data from prior year actual results and current year plan and budget.

  Loading historical data provides users a way to analyze results, review trends, and make meaningful comparisons.

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This also helps verify the structures that you have built into your application. For example, you can verify that data ties to previously created reports. If the data doesn't reconcile, you must verify if this is caused by a data issue or if there is an issue with the structures.

Create an aggregation rule to see consolidated data in your application.

- **Plan valid intersections.**
  Valid intersections let Service Administrators define rules called valid intersection rules that filter dimensional intersections for users when they enter data or select runtime prompts. For example, you can specify that certain programs are valid only for specific departments. Leverage valid intersections to control data entry only for valid intersections.

  During form design, keep these points in mind for valid intersections.

  – If dimensions with valid intersections are on the Page, the user will only be presented with valid combinations in the member selector.

  – If dimensions with valid intersections are on the column or row, the form designer can completely suppress invalid intersections. When the suppression option is not selected, invalid intersections are set to read only.

Accessing the Simplified Dimension Editor

To access the Simplified dimension editor:

1. From the Home page, click **Application**, and then click **Overview**.
2. Click **Dimensions**.
3. The **Cube** drop-down list defaults to **All**. Click the down arrow to the right of the **Cube** drop-down list to filter the list of dimensions by cube.
4. Click the name of the dimension that you want to view.
5. Select from the following tabs:

   - : Click **Edit Member Properties** to view and edit dimension members. See Editing Member Properties.

   - : Click **Edit Dimension Properties** to view and edit dimension details. See Editing Dimension Properties.

Working in the Edit Member Properties Grid

Work with elements in the **Edit Member Properties** grid interface.

Related Topics

- Switching to Another Dimension
- Customizing the Column Layout
- Context Menu Options
- Viewing Ancestors
- Showing Member Usage in an Application
- Focusing Your Editing
- Finding Members

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- Moving Members
- Sorting Members
- Moving Members to Another Hierarchy
- Working with Member Formulas
- Copying Member Names from Microsoft Excel

Switching to Another Dimension

To switch to another dimension while viewing the **Edit Member Properties** grid:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.

2. Click the down arrow next to the dimension name at the top of the page.

> **Note**
>
> The dimension drop-down displays all of the dimensions, regardless of the cube you are currently viewing.

Customizing the Column Layout

Each column in the grid represents a member property (**Member Name**, **Parent Member**, **Default Data Storage**, and so on). The columns that initially display on the grid can be different based on which dimension type you're editing.

You can customize the layout of columns in the **Edit Member Properties** grid by selecting columns, clearing columns, and resizing or reordering columns. You can also display the complete set of properties (all columns) by selecting **Select All** in the **Select Columns** dialog box.

The properties listed in the column selector are not a flat list of properties. Instead, properties are grouped in the column selector according to their functionality. For example, Account-specific properties such as **Account Type**, **Variance Reporting**, and **Time Balance**, are grouped under **Account** in the column selector. When you select **Account** in the column selector, all properties in the group are selected for inclusion in the grid. Then you can clear the checkboxes for any properties that you don't want added to the grid.

To customize the column layout in the **Edit Member Properties** grid:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.

2. On the grid, right-click any column heading.

> **Tip**
>
> Alternatively, you can display this dialog box by clicking (Select Columns) or by clicking **Actions**, and then **Select Columns**.

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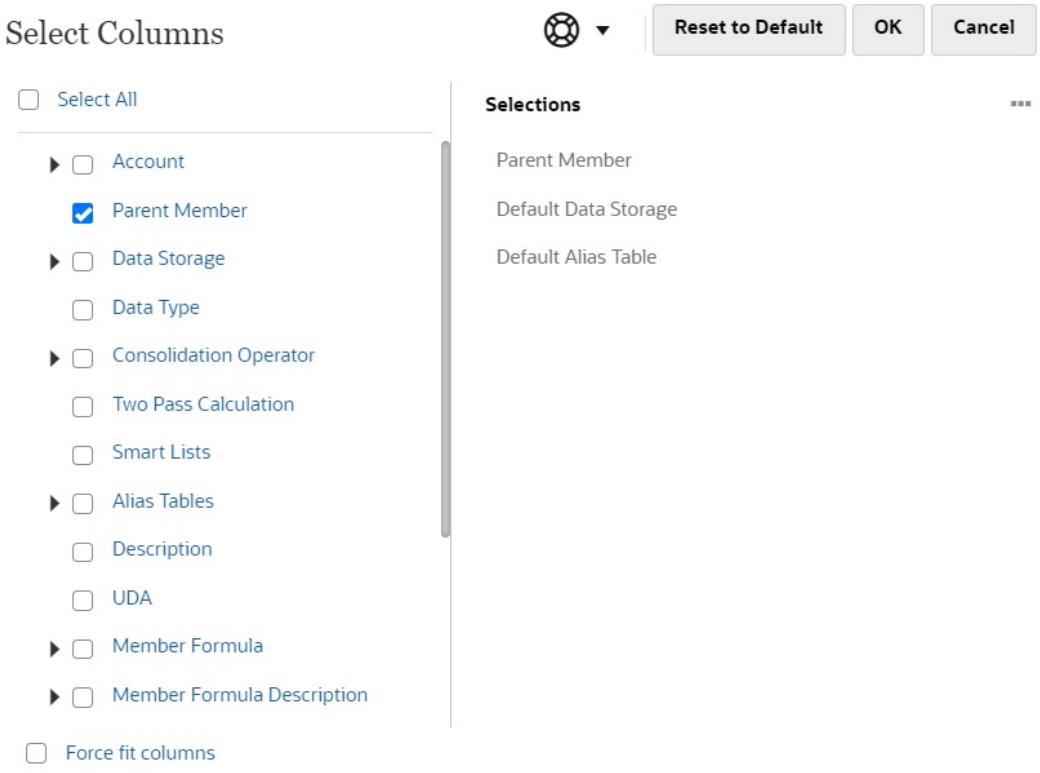
The **Select Columns** dialog box displays.



3. Select or clear the check boxes for the columns (properties) you want to add to or remove from the grid. Select a top-level group such as **Account** or **Data Storage** to select all of the properties grouped under those categories.
   - To display all property columns on the grid, click **Select All**.
   - In the **Selections** pane of the **Select Columns** dialog box, columns appear in the order they are selected, and this is the order in which they will be displayed on the grid. To reorder the columns on the grid, hover over a column in the **Selections** pane, then select the up or down arrows to move the column up or down in the order.

     > **Note**
     >
     > You can also reorder columns by dragging and dropping them directly on the grid.

   - To remove a column from the **Selections** pane, hover over a column and click (Delete).
   - To remove all columns from the **Selections** pane, click (Selections Actions), and then select **Remove All**.
   - To view the columns that initially displayed on the grid, click **Reset to Default**.

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> **Note**
>
> After making your selections, the columns you select will be saved for the next session.

Context Menu Options

In the **Edit Member Properties** grid, right-click on a cell in the grid to view the following context menu options:

- **Move** options:
  - **Move Up**: Moves the member up one position
  - **Move Down**: Moves the member down one position
- **Zoom In** options:
  - **Zoom in Next Level**: Displays all members one level below the selected member
  - **Zoom in All Levels**: Displays all descendant members below the selected member
  - **Zoom in Bottom Level**: Displays all descendant members of the selected member that have no children
- **Zoom Out**: Displays the member one level above the selected member
- **Keep Selected**: Displays only the selected row or column on the grid
- **Remove Selected**: Removes the selected row or column from the grid
- **Delete**: Deletes the selected member, its descendants, and associated shared members

> **Note**
>
> The context menus might differ depending on the dimension you are viewing. For some members, certain commands like **Move** or **Delete** are disabled.

Viewing Ancestors

Ancestors are all the members above the selected member in the dimension hierarchy.

To view the ancestors for the selected member in the Simplified dimension editor grid:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. Select a member on the **Edit Member Properties** grid.
3. Click **Actions**, and then select **Show Ancestors**.

> **Tip**
>
> Alternatively, you can view a member's ancestors by clicking (Show Ancestors).

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Showing Member Usage in an Application

Before performing such operations as deleting members, it's important to understand where in the application the members are used (in which forms, exchange rates, and so on) by using **Show Usage**.

To view where members are used in an application:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. Select a member on the **Edit Member Properties** grid.
3. Click **Actions**, and then **Show Usage**.

> **Tip**
>
> Alternatively, you can click (Show Usage) to view where a member is used in the application.

Focusing Your Editing

Use zoom in, zoom out, keep selected, remove selected, and freeze to focus your editing while working in the **Edit Member Properties** grid.

To perform these operations while viewing the **Edit Member Properties** grid:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. To focus your editing on specific members in the grid, select a member, and then click one of the following zoom operations:
   - Zoom in options:
     - : **Zoom in Next level** displays all members one level below the selected member.
     - : **Zoom in All levels** displays all descendant members below the selected member.
     - : **Zoom in Bottom level** displays all descendant members of the selected member that have no children.
   - : **Zoom Out** displays the member one level above the selected member.
3. To focus your editing on specific rows or columns in the grid, select a row or column, and then choose from the following operations:
   - : **Keep Selected** displays only the selected row or column on the grid.
   - : **Remove Selected** removes the selected row or column from the grid.

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- : **Freeze** (for columns only) keeps the selected column and all columns to the left of the selected column stationary so the column or columns can't be scrolled. Only columns to the right of the frozen column can be scrolled. For example, you can freeze the first column that includes the member name so that you can scroll and edit that member's properties and still see the member name. To unfreeze columns, click **Freeze** once again.

> **Tip**
>
> Alternatively, you can select these operations on the **Actions** menu. You can also view some of these options by right-clicking on a cell in the grid.

Finding Members

To find dimension members in the **Edit Member Properties** grid:

1. View **Edit Member Properties** for a dimension.

   See Accessing the Simplified Dimension Editor.
2. For **Search**, select **Name**, **Alias**, or **Both**, then enter the search text (member name, alias, or partial string) for which to search.
3. Select one of the following search options:
   - : **Search Up**
   - : **Search Down**
   - : **Filtered Search** searches for all members that match the search term and returns a list of members. Click **Clear Filter** to display the unfiltered grid again.

Moving Members

To move members:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. Select the member to move.
3. To move the member up one position, click (Move Up). To move the member down one position, click (Move Down).

> **Tip**
>
> Alternatively, you can click **Actions**, then select **Move Up** or **Move Down** to move a member up or down one position in the hierarchy. You can also right-click on a cell in the grid to view the **Move** options.

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Sorting Members

You can sort members in ascending or descending order, by children or descendants. Sorting members affects the outline. For example, you can alphabetize a list of child members by sorting in descending order to help users quickly locate members in hierarchies in the **Edit Member Properties** grid.

To sort members using the **Edit Member Properties** grid:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. On the grid, select the member whose children or descendants you want to sort.
3. For **Sort**, select **Children** or **Descendants**.

> **Note**
>
> - If you do not see the **Sort** options at the top of the page, click » (More Options).
> - Sorting by children affects only members in the level immediately below the selected member. Sorting by descendants affects all descendants of the selected member.

4. Click (Sort Ascending) or (Sort Descending).

Moving Members to Another Hierarchy

To move members to another hierarchy in the **Edit Member Properties** grid:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. In the **Parent Member** column of the grid, type a new parent name for the member you want to move.
3. Click **Save**.

Working with Member Formulas

You can define or edit member formulas directly in the **Edit Member Properties** grid, in the formula bar, or in the **Member Formula** dialog where you can validate member formulas.

You can define member formulas to combine operators, calculation functions, dimension and member names, and numeric constants to perform calculations on members. Member formulas can also include:

- Operator type, function, value, member name, UDA, and so on allowed in formulas.
- Predefined formula expressions, including Smart List values, that expand into a formula or value upon database refresh.

To define or edit member formulas in the Simplified dimension editor:

1. View **Edit Member Properties**.

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See Accessing the Simplified Dimension Editor.

2. In the **Default Formula** column of the grid, select the member for which you want to define or edit a formula. Define or edit the formula for the member using one of the following options:
   - Click within the formula bar above the dimension editor grid, and then enter or edit the formula.
   - Click (Member Formula) next to the formula bar, and then enter or edit the formula.

> **Note**
>
> - Double-clicking the formula cell used to display a multi-line popup with **OK** or **Cancel**. We've removed the popup, and instead it's a direct edit within the formula cell.
> - The formula bar displays when a formula cell is highlighted, and is hidden when you navigate away from the formula cell.
> - We recommend you edit formulas using the formula bar or using the script editor that opens when you click (Member Formula).

> **Tip**
>
> - If you don't see the **Default Formula** column, you'll need to add it to the grid. Click to open the **Select Columns** dialog, expand **Member Formula**, select **Default Formula**, and then click **OK**. After selecting it, the **Default Formula** column will be included in the grid for all subsequent sessions.
> - To include member names in formulas, keep the focus on the formula cell in the grid. Press Ctrl while clicking the member name you want to include in the formula. The member name will display in the formula bar.

3. **Optional**: To check the validity of a member formula, click (Member Formula) next to the formula bar, and then click **Validate**.
4. Click **Save**.

Copying Member Names from Microsoft Excel

To copy and paste member names from Microsoft Excel:

1. In Excel, highlight the member names in one cell or in a range of cells, and press Ctrl+C to copy the data onto the clipboard.
2. Highlight and select the target cell or cells in the Simplified dimension editor, and then press Ctrl+V.
3. When the Clipboard helper is displayed, press Ctrl+V again. The data is pasted to the Clipboard helper.

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4. Click **Paste** to paste the data into the Simplified dimension editor.

Editing Member Properties

Related Topics

- Accessing Edit Member Properties
- Adding Members
- Editing Members
- Deleting Members
- Adding Shared Members
- Assigning Access to Dimension Members

Accessing Edit Member Properties

To access the **Edit Member Properties** tab in the Simplified dimension editor:

1. From the Home page, click **Application**, and then click **Overview**.
2. Click **Dimensions**, and then click the name of the dimension for which you want to view member properties.

> **Note**
>
> The **Cube** drop-down list defaults to **All**. Click the down arrow to the right of the **Cube** drop-down list to filter the list of dimensions by cube.

3. Click the (Edit Member Properties) tab.
4. To edit member properties in the **Edit Member Properties** grid, click within a grid cell to edit text or to view a drop-down menu from which you can choose member properties. You can also drag and drop member property values over to rows and columns to fill in properties that are the same. See the Member Properties table below for property options and descriptions.

   See Working in the Edit Member Properties Grid.

> **Note**
>
> Each column in the **Edit Member Properties** grid represents a member property. The list of properties (columns) that initially displays on the grid can be different based on which dimension type you're editing. You can customize which columns are displayed on the grid by right-clicking a column heading to display the **Select Columns** dialog box and then selecting columns or clearing them. You can also display the complete set of properties (all columns) by selecting **Select All** in the **Select Columns** dialog box.
>
> To customize the columns in the **Edit Member Properties** grid, see Customizing the Column Layout.

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Member names must conform to guidelines listed in Naming Restrictions.

Shared members must be consistent with Adding Shared Members.

Table 10-6 Member Properties

Property	Value
Member Name	A name that is unique across all dimension members
Parent Member	If the member has a parent in the hierarchy, the parent name.
Description	**Optional**: A description of the member (you can use the same description for multiple members) **Note:** Editing the description of a seeded member is not allowed. Descriptions for all other members added by users can be edited.
Alias Table	**Optional**: The alias table which stores the alias name. Enter an alternate name for the member in **Alias**. See Administering Alias Tables.
For Account members only: **Account Type**	Select **Expense**, **Revenue**, **Asset**, **Liability**, **Equity**, or **Saved Assumption**. For descriptions, see Account Types.
For Account members only: **Variance Reporting**	If the account type is **Saved Assumption**, select **Expense** or **Non-Expense**. Designate the saved assumption as a revenue, asset, liability, or equity account.
For Account members only: **Time Balance**	Select **Flow**, **First**, **Balance**, **Average**, **Fill**, **Weighted Average - Actual_Actual**, or **Weighted Average - Actual_365**. For descriptions, see Time Balance Property.
For Account members only: **Skip**	If the account type is **Asset**, **Equity**, or **Liability**, select **None**, **Missing**, **Zeros**, or **Missing and Zeros**. For descriptions, see Setting Account Calculations for Zeros and Missing Values.
For Account members only: **Exchange Rate Type**	Select **Average**, **Ending**, or **Historical**. For descriptions, see Data Type.
Data Type	Select **Percentage**, **Date**, or **Text**. For descriptions, see Data Type.
For Account members only: **Distribution**	Sets the weekly distribution. Available for leaf Account members if the option was selected when creating the application and the base time period is 12 months.
Hierarchy Type	Available for dimensions bound to an aggregate storage cube. Aggregate storage dimensions are automatically enabled to support multiple hierarchies. The first hierarchy in a multiple hierarchy dimension must be stored. For members with a stored hierarchy type, the only valid cube aggregation options are Addition or Ignore. In a stored hierarchy, the first member must be set to Addition. For members with a dynamic hierarchy type, all cube aggregation options are valid. Stored hierarchy members that are not children of Label Only members must have Addition set as the consolidation operator. Children of Label Only members can be set to Ignore.
Data Storage	The data storage property. Never Share is the default for new custom dimension members (except root members).
Two Pass Calculation	Recalculate values of members based on values of parent members or other members. Available for Account and Entity members with **Dynamic Calc** or **Dynamic Calc and Store** properties.

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Table 10-6 (Cont.) Member Properties

Property	Value
Cube Consol op	Select an aggregation option for each cube: • Not used for Cube (member will be invalid for cube) • Addition • Subtraction • Multiplication • Division • Percent • Ignore • Never (member doesn't aggregate, regardless of hierarchy) You can select a source cube only if multiple cubes are valid for the member. Only cubes and aggregation options for which the member's parent is valid are available. If the parent isn't valid for a cube or aggregation option, neither is the child member. Deselecting a cube for an account or entity parent member deselects it for all descendents of that parent. For members with a stored hierarchy type, the only valid aggregation options are Addition or Ignore. You can set usage by cube for members of custom dimensions and the Period dimension, similar to the Account and Entity dimensions. **CAUTION!** Deselecting a cube for dimension members after data is entered into an application may result in loss of data when an application is refreshed. For account members, data is lost if the deselected cube is the source cube.
For Account members only: **Source Cube**	Select the source cube for the member. A shared member is a pointer to the base member and isn't stored; this is disabled for shared members. The source cube of a shared Account member matches the source cube of the base member, even though Source Plan field is unavailable because it doesn't apply to shared members.
Smart Lists	**Optional**: Select a Smart List to associate with the member.
Solve Order	Specifies the order in which formulas are evaluated. Enter a whole number between 0 and 100000. The formulas for members that have a specified solve order are calculated in order from the lowest solve order to the highest. The default is 0. **Solve Order** is available for aggregate storage cubes and also for block storage cubes that are enabled for Hybrid. Solve Order for block storage cubes is only editable using the Simplified dimension editor.
Enable for Dynamic Children	Enables users to create children for this member by entering a member name in the runtime prompt for a business rule that has been configured with a dynamic parent member.
Number of Possible Dynamic Children	This option is available if **Enable for Dynamic Children** is selected. Enter the maximum number of dynamically-added members that users can create. The default is 10.

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Table 10-6 (Cont.) Member Properties

Property	Value
Access Granted to Member Creator	This option is available if **Enable for Dynamic Children** is selected. Determines the access that member creators have to dynamic members that they create with a runtime prompt: • **Inherit**: The member creator will inherit the closest parent's access to the newly-created member. • **None**: The member creator will not be assigned any access to the newly-created member. (A Service Administrator can later assign the member creator access to the members.) • **Read**: The member creator will be assigned Read access to the newly-created member. • **Write**: The member creator will be assigned Write access to the newly-created member. If a Service Administrator changes these settings, they affect only future dynamic members; they don't retroactively affect dynamic members.

Adding Members

Members must conform to guidelines listed in Naming Restrictions.

To add shared members, see Adding Shared Members.

To add members in the **Edit Member Properties** grid:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.

2. Add members:

   - To add a child member, select the parent level member, click **Actions**, and then click **Add Child**.

     > **Note**
     >
     > Child members inherit the dimension properties of the parent member.

   - To add a sibling member, select a member, click **Actions**, and then click **Add Sibling**.
   - To add years to the calendar, select the Year dimension, click **Actions**, and then click **Add Years**. Enter the number of years to add, click **Apply**, and then confirm whether to add the years to the end of the calendar. Click **Yes** to add years after the End year. Click **No** to add years before the Start year.
   - To add an All Years parent member that includes all members of the Year dimension, select the Year dimension, click **Actions**, and then click **Add "All Years"**.

     The All Years parent member enables users to view the accumulated data across multiple years, for example, a project's total cost up to its end date. The All Years member doesn't include the No Year member, if one is defined for the application.

3. To set or change member properties, click a cell in the **Edit Member Properties** grid and make updates. See Editing Members.

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4. To undo all changes that were made since the last save, click (Refresh).
5. To save your changes, click **Save**.
6. To apply your changes across the application after a save, click (Refresh Database).
7. After creating a dimension member, you typically complete these tasks:
   - Assign access. Click **Actions**, and then click **Assign Permission**.
   - Assign custom attributes. See Editing Dimension Properties.

Editing Members

You can edit members directly in the **Edit Member Properties** grid.

To edit members:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. Take an action:
   - To add members, see Adding Members
   - To navigate to the **Edit Member Properties** grid and to focus your editing on certain members, rows, or columns, see Working in the Edit Member Properties Grid.
   - To modify member properties, click within a cell on the **Edit Member Properties** grid and select an option from the drop-down list. For example, to edit the Account Type for an Account dimension member, click within a cell in the Account Type column. Click the down arrow that appears within the cell, and then select an Account Type option. For descriptions of member properties, see Editing Member Properties.

     > **Note**
     >
     > To view all property columns in the grid, right-click any heading on the grid, select the **Select All** option in the **Select Columns** dialog box, then click **OK**.

   - To delete members, see Deleting Members.
3. To undo all changes that were made since the last save, click (Refresh).
4. To save your changes, click **Save**.
5. To apply your changes across the application after a save, click (Refresh Database).

Deleting Members

Each data value is identified by a set of dimension member values and a cube. Deleting dimension members or deselecting the cube results in data loss when refreshing an application.

Before deleting members, understand where in the application they are used (in which forms, exchange rates, and so on) by using **Show Usage**. See Showing Member Usage in an Application.

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You must delete the entity member throughout the application before deleting it from dimension. For example, if the entity member is used in a form, you must delete it from the form before deleting it from dimensions.

To delete members:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. In the **Edit Member Properties** grid, select the member to delete.
3. Click **Actions**, and then click **Delete Member**.

> **Note**
>
> Deleting a base member also deletes its shared members.

4. In the confirmation box, click **Yes**.
5. Update and validate business rules and reports.

Adding Shared Members

Sharing members allow alternate rollup structures within an application. A base member must exist before you can create a shared member. You can create multiple shared members for the base member. A base member must display before its shared members in position from top to bottom.

Shared members are available for Entity, Account, and user-defined custom dimensions. Shared member values can be ignored to avoid double-counting values when you roll up the outline.

Shared members share some property definitions with base members, such as member name, alias name, and cubes for which members are valid. Shared members must have unique parent members and different rollup aggregation settings. Custom attributes, custom attribute values, and member formulas are not allowed for shared members. Renaming base members renames all shared members.

> **Note**
>
> Shared members must have the same names as their base member. Also, base and shared member names are case-sensitive.

Shared members can't be moved to another parent member. You must delete shared members and recreate them under different parent members. The base member need not be level zero. You can enter data in shared members, and values are stored with base members.

To add shared members to the **Edit Member Properties** grid:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.
2. Add shared members:
   a. Add a child member with the same name as the base member.

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b. For **Parent Member** of the new member, type a new parent name.

c. For **Default Data Storage** of the new member, select **Shared**.

3. Click **Save**.

Data storage properties for all other cubes will automatically be set to **Shared**.

Assigning Access to Dimension Members

Before you can assign access to members of user-defined custom dimensions, you must select the **Apply Security** check box on the dimension's **Edit Dimension Properties** page. See Editing Dimension Properties.

To assign access to dimension members:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.

2. Select the member, click **Actions**, and then click **Assign Permission**.
3. Click **Permissions**.
4. See the following topics to add, edit, or remove permissions:
   - Adding, Editing, and Removing Access to Members
   - About Effective Access Permissions to Shared Members

Adding, Editing, and Removing Access to Members

You can specify which users and groups can access the selected member.

To assign, edit, and remove access permissions to members:

1. View **Edit Member Properties**.

   See Accessing the Simplified Dimension Editor.

2. Select the member, click **Actions**, and then click **Assign Permission**.
3. Click **Permissions**.
4. To add access, click **Add User/Group**, then click (Search) to select the users and groups to access the selected member.

   Click **Users** to display all users; click **Groups** to display all groups.

5. To modify the type of access:

   a. Select a **Permission** for the displayed users or groups:
      - **Read**
      - **Write**
      - **None**
      - **Display**

      For descriptions of the preceeding options, see Accessing Ancestor Members in Ad Hoc Grids.

   b. **Optional**: Select a relationship.

      For example, select **Children** to assign access to the children of the selected member.

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6. To remove access, click (Remove Access).
7. Click **OK**.
8. Click **Close**.

About Effective Access Permissions to Shared Members

You can't assign access directly to a shared member. A shared member inherits access permissions from its base member, parent, or ancestor.

The application checks access permissions at each level, first by user, then by group, based on the member's access permissions inheritance relationship. If multiple access permissions exist, the least restrictive access permission is applied (for example, Write access takes precedence over Read access).

Sample Parent and Child Entity Members

Table 10-7 Example of Inherited Access to Shared Members

Case	Access Permission	Effective Access for Base and Shared Member CA	Explanation
Case 1	CA (base) = None iDescendants (West) = Read	Read	CA inherits Read access from its West parent because Read is less restrictive than None.
Case 2	iDescendants (United States) = None iDescendants (West) = Read iDescendants (Sales Region 1) = Write	Write	CA inherits Write access from its Sales Region 1 parent because Write is less restrictive than Read or None.
Case 3	iDescendants (United States) = Write iDescendants (West) = None iDescendants (Sales Region 1) = Read	Write	CA inherits Write access from its United States parent because Write is less restrictive than Read or None.

Editing Dimension Properties

To access the properties of a dimension in the Simplified dimension editor:

1. From the Home page, click **Application**, and then click **Overview**.
2. Click **Dimensions**.
3. The **Cube** drop-down list defaults to **All**. Click the down arrow to the right of the **Cube** drop-down list to filter the list of dimensions by cube.

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4. Click the name of the dimension that you want to view.
5. Select the **Edit Dimension Properties** tab .

Dimension properties must conform to guidelines listed in Naming Restrictions.

Table 10-8 Dimension Properties

Property	Value
Dimension	Enter a name that is unique across all dimensions.
Description	**Optional**: Enter a description.
Alias Table and **Alias**	**Optional**: Select an alias table. Enter an alternate name for the dimension. See Administering Alias Tables.
Cube	Select the cubes for which the dimension is enabled. Clearing this option disables all members of the dimension for the deselected cube.
Two Pass Calculation	Recalculate values of members based on values of parent members or other members. Available for Account and Entity members with Dynamic Calc or Dynamic Calc and Store properties.
Apply Security	Allow security to be set on the dimension members; must be selected before assigning access rights to dimension members. Otherwise, dimensions have no security and users can access members without restriction. See Assigning Access to Dimension Members.
Data Storage	Select a data storage option. The default is **Never Share**. • **Store**: Stores data values of members. • **Dynamic Calc and Store**: Calculates data values of members, and stores values. • **Dynamic Calc**: Calculates data values of members, and disregards the values. • **Never Share**: Prohibits members in the same dimension from sharing data values. • **Label Only**: Has no data associated with the member. • **Shared**: Allows members in the same dimension to share data values. See Storage Options.
Display Option	Set application default display options for the **Member Selection** dialog box. Select **Member Name** or **Alias** to display members or aliases. **Member Name:Alias** displays members on the left and aliases on the right. **Alias:Member Name** displays aliases on the left and members on the right.
Hierarchy Type	Available for dimensions bound to an aggregate storage cube. Aggregate storage dimensions are automatically enabled to support multiple hierarchies. The first hierarchy in a multiple hierarchy dimension must be **Stored**. For members with a **Stored** hierarchy type, the only valid cube aggregation options are **Addition** or **Ignore**. In a stored hierarchy, the first member must be set to **Addition**. For members with a **Dynamic** hierarchy type, all cube aggregation options are valid. **Stored** hierarchy members that are not children of **Label Only** members must have **Addition** set as the consolidation operator. Children of **Label Only** members can be set to **Ignore**.

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Table 10-8 (Cont.) Dimension Properties

Property	Value
Custom Attributes	Click to **Create** or **Synchronize** custom attributes for a dimension.

Working with Attributes

Attributes describe characteristics of data, such as the size and color of products. You can use attributes to group and analyze members of dimensions based on their characteristics. For example, you can analyze product profitability based on size or packaging, and you can make more effective conclusions by incorporating market attributes, such as the population size of each market region, into your analysis.

You can assign attributes to sparse dimensions only. You can't assign attributes to label-only members. Attribute dimensions don't have aggregation properties because parents are dynamically calculated.

The Account dimension is usually defined as dense, so you can't assign attributes to it unless it's changed to sparse for all cubes. If you change a dimension from sparse to dense, all attributes and attribute values for that dimension are automatically deleted.

Attributes can have data types of text, date, Boolean, and numeric, as described in Understanding Attribute Data Types.

Attribute names must conform to guidelines listed in Naming Restrictions.

To create or edit attributes in the Simplified dimension editor:

1. From the Home page, click **Application**, click **Overview**, and then click **Dimensions**.
2. The **Cube** drop-down list defaults to **All**. Click the down arrow to the right of the **Cube** drop-down list to filter the list of dimensions by cube, and to see which cubes are marked as **Dense**.
3. Select a sparse dimension for the attribute.

> **Note**
>
> Only sparse dimensions can contain attributes.
>
> See About Sparse and Dense Dimensions.

4. Select the **Edit Dimension Properties** tab .
5. Select options:
   - To create an attribute, next to **Custom Attributes**, click **Create**. Type an attribute name, and select a data type: **Text**, **Date**, **Boolean**, or **Numeric**.

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> **Note**
>
> – You can't modify the type after the attribute is created.
>
> – Before working with date attributes, you must select the **Attribute Dimension Date Format** in **Application Settings** and save it.

See Understanding Attribute Data Types.

- To modify attributes, under **Custom Attributes**, click ••• (Actions) next to the attribute you want to modify, select **Edit**, and then update the attribute name.

6. Click **Close**.

   When you click **Close**, the hierarchy is validated and an error displays if issues are detected. For example, date attribute values must be entered in the correct format, and numeric and date attribute dimensions must have at least one attribute value defined.

7. Update and validate business rules and reports.

> **Tip**
>
> Instead of explicitly filtering by an attribute (like Red, for instance), you can create a user variable for the attribute dimension, and then use the user variable as the filter. Then you can enable the user variable as a dynamic user variable which would allow users to change the value of the filter at runtime. This is a useful technique that allows for dynamic filtering.
>
> See Managing User Variables.

Deleting Attributes

When you delete an attribute, all attribute values associated with the attribute are also deleted. Attribute values are removed from members to which they had been assigned, and the attribute is removed from dimensions to which it was assigned.

To delete attributes:

1. From the Home page, click **Application**, click **Overview**, and then click **Dimensions**.
2. Click the name of the dimension with the attribute you want to delete, and then click the **Edit Dimension Properties** tab.
3. Under **Custom Attributes**, click ••• (Actions) next to the attribute you want to edit, and then select **Delete**.
4. Click **Yes**.
5. Update and validate business rules and reports.

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Working with UDAs in the Simplified Dimension Editor

You can use user-defined attributes (UDAs), descriptive words or phrases, within calc scripts, member formulas, reports, and forms. UDAs return lists of members associated with the UDA. For example:

- You can use the `HSP_UDF` UDA to prevent a formula from being overwritten when the application is refreshed. You must log on to each database associated with the business process and create the member formula with a UDA. The syntax for this UDA is: `(UDAs: HSP_UDF)`.
- If you use the `@XREF` function to look up a data value in another database to calculate a value from the current database, you can add the `HSP_NOLINK` UDA to members to prevent the `@XREF` function from being created on all cubes that are not the source cube selected for that member.
- For a Product dimension with several product members, you can create a UDA called New Products and assign this UDA to the new products in the Product dimension hierarchy. Then you can base certain calculations on the designation New Products.
- When designing forms, you can use a UDA to select members for forms based on a common attribute. When you create forms with UDAs, any members that are assigned to the UDA are dynamically added to the form. For example, if you create a UDA called New Products and assign this UDA to the new products in the Product dimension hierarchy, the form will automatically display the new products at runtime. When selected in the form designer, a UDA is preceded by `UDA`; for example, `UDA(New Products)`.
- You can use the `HSP_NOMOF_<PlantypeName>` UDA to make sure that dynamic members (or "on the fly members") are not created in the specified plan type, `<PlantypeName>`, even if the parent member is valid for this plan type. This prevents dynamic children from affecting aggregation on select cubes.

UDAs are specific to dimensions. For example, creating a UDA for an Account member makes it available for non-shared Account members. Deleting it removes it for all Account members. To make UDAs available for multiple dimensions, create the same UDA for multiple dimensions. For example, create a UDA named New for Account and Entity dimensions to make the UDA named New available for Account and Entity members.

> **Note**
>
> This procedure describes how to work with UDAs in the Simplified dimension editor. To work with UDAs in the Classic Dimension Editor, see Working with UDAs.

To work with UDAs in the **Edit Member Properties** grid:

1. From the Home page, click **Application**, click **Overview**, and then click **Dimensions**.
2. Click on the name of the dimension for whose members to associate the UDA.
3. Select the **Edit Member Properties** tab.
4. On the dimension grid, right-click any column heading, select **UDA**, and then click **OK** to make the UDA column visible in the grid.
5. Locate the member to add the UDA, and then scroll to the UDA column.
6. Click in the UDA cell to display the **UDA** management dialog.

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7. Select a task:
   - To create a UDA, click
     (Create), enter a name, and then click **Save**.

     > **Note**
     >
     > Use no more than 60 characters when naming UDAs.

   - To modify a UDA, select the UDA, click
     (Edit), change the name, and then click **Save**.
   - To delete a UDA, select the UDA, click
     (Delete), and then confirm deletion.

     > **Note**
     >
     > Deleting the UDA removes it for the whole dimension. If you delete UDAs, you must update all member formulas, calc scripts, and reports that reference them.

   - To clear UDA selections, click
     (Clear).
8. To add or remove UDAs for the member, use the arrows to move UDAs to and from the **Selected UDA** panel.
9. Click **OK**.

Working with Attribute Values

Attribute values provide users with another way to select dimension members when using forms. Data values for attributes are dynamically calculated but not stored.

You can define attribute values for sparse dimensions, which are typically the Entity and user-defined custom dimensions. After you define an attribute value for a dimension, you can assign it to members of that dimension.

Creating Attribute Values

To create attribute values:

1. On the Home page, click **Application**, and then click **Overview**.
2. Click the **Dimensions** tab.
3. For **Cube**, select **All**.
4. Select an attribute dimension.

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> **Note**
>
> Attribute dimensions are listed at the end of the dimensions listing and at the bottom of the dimension drop-down.

5. In **Edit Member Properties**, add members:
   - To add a child member, select the parent level member, click **Actions**, and then click **Add Child**.

     > **Note**
     >
     > Child members inherit the dimension properties of the parent member.

   - To add a sibling member, select a member, click **Actions**, and then click **Add Sibling**.
6. To set or change member properties, click a cell in the **Edit Member Properties** grid and make updates. See Editing Member Properties.
7. To save your changes, click **Save**.

Assigning Attribute Values to Base Dimensions

When an attribute dimension is associated with a base dimension, it displays in the Simplified Dimension Editor in the same way as a regular dimension. To view attribute dimensions in the listing, select **All** in the **Cube** drop-down list on the main dimension page. Then you can select the attribute dimension to add values to the grid.

To assign attribute values to base dimensions:

1. Open the **Edit Member Properties** page for the base dimension and find the column with the name of the attribute dimension.
2. Click within the grid for a specific member to display a drop-down list with attribute values, and then select an attribute value from the list.
3. Click **Save** on the **Edit Member Properties** page to assign the value to the base member.

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11

Administering Tasks with Task Manager

Task Manager provides centralized monitoring of all tasks and provides a visible, automated, repeatable system of record for running a business process.

Related Topics

- Setting Up Task Manager
- Managing Task Types
- Managing Task Templates
- Managing Tasks
- Managing Schedules
- Managing Task Manager Integrations
- Managing Alert Types for Task Manager
- Managing Views in Task Manager
- Viewing Currencies in Task Manager
- Generating Custom Reports for Task Manager
- Working with Views and Filters
- Managing Operational Dashboards

Setting Up Task Manager

Related Topics

- Task Manager Terms
- Task Manager Overview
- Sample Task Flows
- Enabling Task Manager in FreeForm
- Managing Task Manager Views, Lists, and Filters
- Managing Task Manager System Settings
- Managing Task Manager Attributes
- Managing Artifact Labels in Task Manager

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Task Manager Terms

Table 11-1 Task Manager Terms

Term	Description
Tasks	Tasks are activities performed during a business process, for example, Enter Form Data, or Review New Accounts. Service Administrators and Power Users define the tasks for the process, and include details for each task, such as start and end dates, assignees, and approvers. Users open assigned tasks, read instructions, answer questions, and after completing tasks, submit them.
Templates	Templates define a repeatable set of tasks required for a business process. Administrators can save a set of defined tasks as a template to use for future periods, such as monthly or quarterly, rather than recreating the tasks each time. For example, an administrator can save a set of tasks required for a monthly process and use the same template for that process every month.
Schedules	A schedule is an ordered set of tasks for a business process. Schedules can be created from a template of predefined tasks, or manually created. When administrators create a schedule, they map the relative dates that were defined for tasks such as Day 1 and 2 of the business process to calendar dates, such as March 1 and 2. When they change the status of the schedule to Open, the tasks run according to their scheduled start date and times. For example, an administrator creates a monthly process schedule for March, with the Review New Accounts task assigned to March 1. When the administrator changes the schedule from Pending to Open, on March 1, the Review New Accounts task starts and an email notification is sent to the task assignee.
Task Types	Task Types enable the standardization of task details across all tasks of the same type. Details defined on the Task Type are inherited by all tasks created from the Task Type. For example, when an administrator defines Task Type settings such as instructions or rules for the Approve Data Task Type, those settings are shared by all tasks created using the Approve Data Task Type. • Basic Task: a default Task Type, which does not contain any task details • Parent Task: a Task Type that enables you to create parent tasks for task hierarchies. For example, the Load Data parent task can have child tasks of Load EMEA Sales Data and Load NA Sales Data. • Integration Task: a Task Type that defines an action and includes the application component, such as a form, or parameters needed to link to an external application. Predefined Integrations and Task Types are automatically loaded for your service. After you set up a connection to a different service, you can include Integration Task Types from that service in your tasks. For example, an administrator using Task Manager in one service can enable and synchronize connections to a different service, such as Account Reconciliation, which loads the predefined Account Reconciliation Integrations and Task Types.

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Table 11-1 (Cont.) Task Manager Terms

Term	Description
Integrations	Integrations are used to automate tasks across Oracle Fusion Cloud Enterprise Performance Management business processes and external applications. You can define tasks that include Integrations with other applications. For Cloud EPM Services, the service is automatically configured with local predefined Integrations for the business process. If you have subscriptions to other Cloud EPM services, you can create connections between services and enable integrations using Task Manager. Pre-built integrations allow you to perform Task Manager tasks that access other Cloud EPM functionality. You can also create custom Integrations, for example, custom Event Monitoring tasks. Integration tasks run based on one of these Execution Types defined in the task: • End User - a task that requires the user to interact with a user interface to complete the task. When a user launches an end user task, the task automatically opens the user interface component needed to perform the task. For example, the Enter Form Data task requires a user to enter data. When the Enter Form Data task is launched, it opens the form UI. • Process Automation - a task that runs in the background without any user interaction required. These integrations are automatically executed in external applications when their start date and time are reached, and any predecessor tasks are completed. For example, a Load Data task can be defined to run automatically overnight based on its start date and time. • Event Monitoring - a task used to monitor a particular event that happened on an external system. When the action or status occurs in the external application, the task is changed to Closed. For example, if you are working with Task Manager in one application, you can monitor the status of a Copy Data task in an external application. When the Copy Data task completes and you see that the task is Closed, you can start the next task in your application, for example, Approve Data.
Alerts	Alerts are notifications from users on issues that they encounter during the process, such as hardware or software issues. When users encounter an issue and create an alert for it, they select from a list of predefined Alert Types to identify the issue. The Alert is then sent to the assignee defined in the Alert Type to be resolved. For example, a user has a task due and can't access their assigned tasks. The user selects an Alert Type, such as Access Issue, which sends the alert to the predefined assignee, for example, the System Administrator. The System Administrator resolves the access issue, then closes the alert.
Alert Types	Alert Types are categories of possible issues that might occur during the business process. Administrators can create Alert Types such as Missing Data or Access Issue. Each Alert Type defines a procedure that is used to capture critical information about the issue and assigns key personnel to resolve it. For example, an administrator can create an Alert Type called Access Issue, which contains questions to users for more detailed information. The administrator defines a workflow for the Alert Type with an assignee to resolve the issue such as a System Administrator, and a backup assignee if desired.

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Table 11-1 (Cont.) Task Manager Terms

Term	Description
System Settings	System Settings enable Service Administrators to specify defaults for Task Manager settings. Administrators can define Global Integration Tokens to manage parameters for URLs, for example, in task instructions. They can determine Holiday Rules for which holiday dates apply to schedules, and create geographical or business unit Organizational Units to associate with task settings such as holidays and time zones. They can also configure settings such as comments and task deletions, email notifications, task governors for maximum values of list and file sizes, reassignment request approvals, permissions to reopen tasks, and specify how tasks display in Smart View, the Worklist, and the Welcome panel.
Worklist	A Worklist is the main end-user screen where users can easily work on their assigned tasks.
Schedule Tasks	Schedule Tasks page displays a list of schedules and the tasks they contain. This enables users to see all the tasks within a schedule. Filtering and Saved View options enable users to quickly review their tasks.
Overview Dashboard	The Overview Dashboard provides a summarized view of all tasks within a POV. You can filter tasks by Year, Period, and Schedule. You can view tasks by Schedule, Priority, Task Type, or Organizational Unit, and drill into task details.
Compliance Dashboard	The Compliance Dashboard provides metrics on how well the company achieved their compliance objectives, distinguishing tasks that were completed on time from those that were late, and providing a count of those that were approved or rejected. You can filter tasks by Year, Period, and Schedule, view user performance metrics based on user and organizational unit, and drill into task details.
Operational Dashboard	The Operational Dashboard provides an overall summary of data for a Task Manager process using a variety of displays. Operational dashboards can combine charts, lists, or pivot views. For example, you can create an Operational dashboard called Task Performance that contains a Bar chart for On Time Performance, a Pie chart for Task Status, and a Pivot view for Status by User.
Task Manager Report	The Task Manager Report is a report provided by default that reports on task execution. For example, it displays tasks that are At Risk, Early, Late, or Rejected by an approver. It also displays Task Compliance, Performance by Organization Unit, Users with Late Tasks, and an Audit Trail of task changes. Administrators can define their own custom reports as needed.

Task Manager Overview

Task Manager helps you define, execute, monitor, and report the interdependent activities of a business process. It provides centralized monitoring of all tasks and provides a visible, automated, repeatable system of record for running a business process. A task represents the unit of action performed in the application. Administrators or Power Users can define integrated task flows with interdependent relationships and hierarchies within their business processes. They can track progress, identify bottlenecks, and correct any errors or delays that may arise throughout the business process.

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With an EPM Enterprise subscription, the service is preloaded with integrated end user and process automation tasks for use within the business process. Administrators can add additional integrated tasks with other business processes by making connections with other Oracle Fusion Cloud Enterprise Performance Management services. The service can be further extended to include integrations with Non-Cloud EPM applications. EPM Standard subscription do not include any local integrated tasks and can not be extended to include any other integrated tasks.

> **Note**
>
> During application creation, you were presented with two task flow type options: **Task List** and **EPM Task Manager**. **EPM Task Manager** is the default option for new applications. For information about administering Task Lists, see Administering Task Lists.

With Task Manager, you can:

- Define the tasks and schedule to ensure an efficient task flow
- Automate the business process, track its status, and provide notifications and alerts
- Monitor a business process and quickly act on possible errors to avoid delays.
- Provide notifications by email for delinquencies, due dates, and status changes.
- Analyze the effectiveness of the business process

How Task Manager Works

Task Manager provides a framework where the following key components for a business process can be defined:

- Organizational units are used to model regions, entities, business units, divisions, and other organizational or geographical divisions that are associated with the task. Model these organization units as hierarchies to create the actual structure and facilitate reporting.
- Holiday rules is a collection of holiday dates and is used by scheduling the task.
- Task types are set up by the administrator to ensure consistency across tasks and to leverage predefined product integrations. If the predefined task types doesn't meet the requirement, then they can create a task type. The task types can be either automated or manual with a defined assignee.
- Task templates are created to use a repeatable set of tasks across multiple periods. These schedules represent the chronological sequence of tasks created by selecting a template and assigning calendar dates.
- Tasks can be assigned to a user or a group.

The business process starts when the schedule status is changed from Pending to Open. During the business process, user receives email notifications of assigned tasks, and can click links in the email for direct access to the assigned task. Administrators have to enable the email notifications to allow this to happen. Users can also create alerts for issues that may occur, and these are forwarded to assignees and approvers for the resolution.

Roles in Task Manager

Task Manager includes three key roles:

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- **Service Administrator** or **Power User**: Sets up templates and tasks for the business process and monitors the status of dashboard activities.
- **End User**: Reviews and completes assigned tasks. They can view the list of assigned tasks in the application or from the assigned task email.
- **Approver**: Reviews tasks and determines if the tasks can be approved or not.

For sample task flows, see Sample Task Flows.

Service Administrator or Power User tasks:

1. Set up users. See Managing Users and Roles in *Getting Started Guide for Administrators*.
2. Completes the required setup procedures:
   - Set up organizational units. See Managing Task Manager Organizational Units.
   - Set up global integration tokens. See Managing Global Integration Tokens.
   - Set up holiday rules. See Managing Holiday Rules for Tasks.
   - Set up attachment size and other system settings. See Setting Task Manager Governors.
   - Enable email notifications. See Setting Up Email for Notifications in *Working with FreeForm*.
3. Review the tasks required for a business process and set up Task Types to ensure consistency across tasks and leverage predefined product integrations. See Managing Task Types.
4. Define and save a set of tasks as a template to use for future periods because many business processes are repetitive. See Managing Task Templates.
   For example, an administrator can set up a monthly or quarterly business process once and then use it repeatedly for all months or quarters. Tasks are defined with task predecessors, Assignees, and Approvers.
5. Generate a schedule (a chronological set of tasks) by selecting a template and assigning calendar dates to initiate a business process. The generic tasks in the template are applied to calendar dates. See Managing Schedules.
6. Change the schedule status from Pending to Open to begin a business process.
7. Modify and monitor the schedule as needed.

End Users and Approver Tasks:

After the administrator changes the schedule status (Open) to begin a business process, users can read task instructions, answer questions, submit, reassign, approve, and reject tasks, and access tasks from email notifications or by logging on to the application.

During the business process, users can:

1. Receive email notifications of assigned tasks and click links in the email for direct access to assigned tasks.
2. Alternatively, log on to review and access assigned tasks in different types of views.
3. Complete tasks and the tasks are sent to approvers and can be viewed by other users.
4. Create alerts for issues, such as hardware or software issues, that they encounter. Alerts are forwarded to Assignees and Approvers for resolution.

Watch this overview video to learn more.

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Overview Video

UI Navigation for Task Manager

- Service Administrator or Power users access Task Manager from **Applications** > **Task Manager**.
- Users access all Task Manager features from **Home** > **Tasks**. This includes all task and alert listing screens, dashboards, and reports.

Sample Task Flows

Scenario 1: Power User

The Power User sets up a template and tasks for an upcoming business process.

- The Power User logs on and opens the Manage Templates page.
- The Power User selects the template for the business process.
- The Power User selects the calendar dates on which to line up the template tasks, and creates a schedule.
- The Power User then adds a task to the template.
- The Power User opens the schedule, which begins the process.

Scenario 2: Power User

The Power User monitors the status of activities through the Dashboard.

- The Power User logs on and checks the status of activities through Day 3.
- The Power User drills down to see the details of incomplete tasks.
- The Power User reviews open tasks for comments or attachments provided by assigned users.

Scenario 3: Approver

A user assigned as an Approver reviews a current task to determine whether it can be approved.

- The assigned Approver receives an email notification of an outstanding task.
- From the email, the Approver selects the link for the task.
- The Task Actions page is launched outlining the process.
- The Approver reviews the document that the user submitted when completing the task to ensure completeness.
- The Approver enters additional comments and approves the submission.
- If another level of approval is required, the task is forwarded to the next approver. If the task was the last requiring approval, then the task completes, and the system runs the next task if it is ready.
- The Approver can reject a task instead of approving it, and the task is reassigned to the Assignee.

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Scenario 4: User

A user logs on to the application and reviews and completes an assigned task.

- The user logs on and reviews their tasks.
- The user clicks the link for an assigned task.
- The task page is launched with instructions for the task, and a reference document.
- The user reviews the instructions and the reference document, processes the updates, enters a comment about the task, and submits it for approval.
- The system automatically updates the task status and sends a notification to the assigned approver.

Scenario 5: User

A user responds to an email notification of a task to load data, clicks a link in the email, and then completes the task.

- The user receives an email notification of an outstanding task.
- From the email, the user selects the link for the page where the required process is documented.
- The user reviews the instructions for the task and opens the task.
- The user loads data into the system.
- The user enters a comment about the task and submits it for approval.

Enabling Task Manager in FreeForm

You can enable Task Manager for a FreeForm application that currently uses Task List as its task flow type. This is a one-way process that replaces Task List functionality with Task Manager. Existing Task Lists and Tasks aren't migrated and must be deleted before you enable Task Manager. Reverse enablement from Task Manager back to Task List isn't supported.

Prerequisites

Before enabling Task Manager:

- Make sure that the application has separate **Year** and **Period** dimensions. If you can't map any existing dimensions to Year and Period, create new dimensions for them.
- Delete all existing Task Lists and Tasks.
- (Recommended) Create an LCM export as a backup before deleting Task Lists.

Enabling Task Manager

To enable Task Manager:

1. Go to **Applications** > **Overview** and select **Actions** > **Enable Task Manager**.
2. Select the **Year** and **Period** dimensions when prompted.
3. Click **Confirm** to start the enablement process. The selected dimensions become permanent and can’t be changed later.

During enablement, the application is placed in Admin Mode, preventing other users from accessing it. The process may take several minutes. When it completes, you are automatically logged out. Log in again to begin using Task Manager.

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After enabling Task Manager, administrators can manage tasks from **Applications** > **Task Manager**, and users can access Task Manager through **Tasks** on the Home page.

Managing Task Manager Views, Lists, and Filters

Related Topics

- Viewing Task Manager Lists
- Column Definitions for Task Manager

Viewing Task Manager Lists

The Task Manager list provides a central place to manage all your tasks. It displays a list of tasks and their status for a selected Year, Period, and Schedule.

If you are an Administrator, you can see all the available tasks in the system, and the user who is working on them.

If you are a user other than an Administrator, you can see all active tasks for your role that need your attention.

To view tasks:

1. On the Home page, click **Tasks**.
2. Click **Tasks** on the left navigation.
3. Select a **Year**, **Period**, and **Schedule**.
4. **Optional:** To search for a task, in the **Search** field, enter any text or numeric for which to search the task list.
5. **Optional:** To filter the list of tasks, click **Add a Filter** in the filter bar and specify filter criteria.

> **Note**
>
> To find late tasks, you can use the `Ever Been Late` attribute to easily identify any task that missed one or more of the scheduled due dates within its workflow. Even if the task is not currently late, but if the Assignee or any Approver has ever been late with their workflow, the `Ever Been Late` attribute indicates the task was missed for the scheduled due date.

Sample Task List

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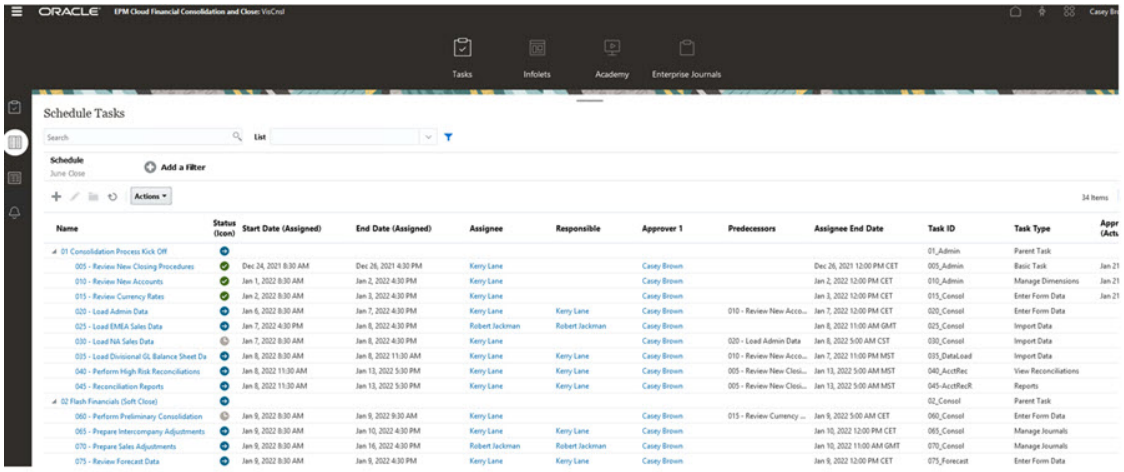

Column Definitions for Task Manager

This section provides the column definitions for the following Task Manager objects:

- Task Manager – Schedule Tasks
- Task Manager – Templates
- Task Manager - Template Tasks
- Task Manager – Schedules
- Task Manager – Task Types
- Task Manager – Integrations
- Task Manager – Alert Types
- Task Manager – Filters
- Task Manager – Views
- Task Manager – Attributes
- Task Manager – Currencies

Task Manager – Schedule Tasks

This section provides the definitions for the Task Manager Schedule Tasks columns.

Table 11-2 Column Definitions - Task Manager – Schedule Tasks

Column Name	Description
Approver 1-10	User, Group, or Team responsible for approving the task
Approver 1-10 (Actual)	User who approved the task
Approver 1-10 (Backup)	User who can approve the task if the Approver (Primary) is unavailable
Approver 1-10 (Primary)	User, Group, or Team that was specified to approve the task
Approver 1-10 End Date (Actual)	Actual end date when the role was completed
Approver 1-10 End Date (Actual, Local)	Actual end date when the role was completed, in the local time zone of the org unit (if assigned an org unit)
Approver 1-10 End Date (Assigned)	Assigned end date for the approver

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Table 11-2 (Cont.) Column Definitions - Task Manager – Schedule Tasks

Column Name	Description
Approver 1-10 End Date (Assigned, Local)	Assigned end date for the approver of the task, in the local time zone of the org unit (if assigned an org unit)
Approver 1-10 Rejections (Count)	Total count of scheduled tasks rejected by the approver
Approver 1-10 Scheduled End Date	Effective end date for the approver of the task
Approver 1-10 Scheduled End Date (Local)	Scheduled end date for the approver of the task, in the local time zone of the org unit (if assigned an org unit)
Approver 1-10 Scheduled Start Date	Effective start date for the approver of the task
Approver 1-10 Scheduled Start Date (Local)	Scheduled start date for the approver of the task, in the local time zone of the org unit (if assigned an org unit)
Approver 1-10 Start Date	Assigned start date of the task
Approver 1-10 Start Date (Local)	Assigned start date of the task, in the local time zone of the org unit (if assigned an org unit)
Approvers (Count)	Total count of assigned approvers
Assignee	User, Group, or Team currently responsible for working on the task
Assignee (Actual)	User who submitted the task
Assignee (Backup)	User who can work on the task if the Assignee (Primary) is unavailable
Assignee (Primary)	User, Group, or Team who was specified to work on the task
Assignee End Date	Effective end date for the assigned user
Assignee End Date (Actual)	Actual end date when the assignee role was completed
Assignee End Date (Actual, Local)	Actual end date when the assignee role was completed, in the local time zone of the org unit (if assigned an org unit)
Assignee End Date (Assigned)	Effective end date for the assignee role to be completed
Assignee End Date (Assigned, Local)	Actual end date when the role was completed, in the local time zone of the org unit (if assigned an org unit)
Assignee End Date (Local)	assigned End Date of the task, in the local time zone of the org unit (if assigned an org unit)
Comments (Count)	Total count of comments for the scheduled task
Created By	Name of the user who created the scheduled task
Created On	Date and time when the scheduled task was created
Description	Description of the scheduled task
End Date	Effective end date of the scheduled task
Ever Been Late (Assignee)	Indicates whether the date when the assignee submitted the task was past the due date
Ever Been Late (Approver 1-10)	Indicates whether approver role has ever been late
Execution Type	Execution type such as End User, System Automated, or Event Monitoring
Instructions	Set of instructions for completing the scheduled task
Integration	Integration associated with the scheduled task
Last Updated By	Name of the user who last updated the scheduled task
Last Updated On	Date and time when the scheduled task was last updated

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Table 11-2 (Cont.) Column Definitions - Task Manager – Schedule Tasks

Column Name	Description
Late (Approver 1-10)	Indicates whether the approver completed the task past the due date
Late (Assignee)	Indicates whether the assignee completed the task past the due date
Open Task	Indicates whether the task is opened
Organizational Unit	Assigned organizational unit
Owner	Name of the scheduled task owner, which must be a Service Administrator or Power User
Parent	Task ID of the parent task
Predecessors	Predecessor for the current task. It can be from another template.
Priority	Priority of the task
References (Count)	Count of references added for the scheduled task (URL and content server references)
Rules (Count)	Total count of rules directly on the scheduled task
Run As	Run the task as the user. Only applicable for system-automated tasks.
Run As Authorized	User who can authorize this task. If the Run As user is the user who also created the task, then the user can authorize the task, without being prompted to re-enter their password.
Start Date	Effective start date of the scheduled task
Start Task Early	Indicates whether the assignee is allowed to open a pending task prior to the scheduled start time
Status (Icon)	Icon indicating the status of the scheduled task
Successors	Successor for the current task
Task ID	Unique identifier for the task
Task Type	Type of task

Example

The following example shows the task workflow between Assignee or Approver, Actual Assignee or Approver, and Primary Assignee or Approver.

Sl No	Action	Assignee / Approver	Assignee / Approver (Actual)	Assignee / Approver (Primary)	Comments
1	Initial state	PREP_TEAM		PREP_TEAM	No one has claimed or submitted the task
2	Barry Mills claims the task	Barry Mills		PREP_TEAM	User information based on claim action
3	Barry Mills submits the task	Barry Mills	Barry Mills	PREP_TEAM	Actual set on workflow action

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Sl No	Action	Assignee / Approver	Assignee / Approver (Actual)	Assignee / Approver (Primary)	Comments
4	Admin re-assigns task to Susan Chang	Susan Chang	Barry Mills	Susan Chang	Both Assignee and Assignee (Primary) are changed, but Assignee (Actual) remains with the user who submitted the task
5	Task is rejected	Susan Chang		Susan Chang	Assignee (Actual) is cleared. Workflow goes back to the user who was re-assigned in the previous step
6	Susan Chang submits the task	Susan Chang	Susan Chang	Susan Chang	Actual set on new user's workflow action.

Task Manager – Templates

This section provides the definitions for the Task Manager templates columns.

Table 11-3 Column Definitions - Task Manager Templates

Column Name	Description
Created By	Name of the user who created the template
Created On	Date and time when the template was created
Description	Description of the template
Embedded In	A list of templates that the current template is embedded in
Embedded Only	Template is only to be used as an embedded template and isn't allowed to be deployed. This prevents supporting templates from being deployed by mistake.
Embedded Templates	A list of templates that are embedded in the current template
Last Schedule	Last schedule created from each template
Last Updated By	Name of the user who last updated the template
Last Updated On	Date and time when the template was last updated
Name	Name of the template
Number of Days After Day 0	Number of days a template tracks after day zero. Days after Day Zero are represented as Day 3, Day 4, and so on.
Number of Days Before Day 0	Number of days a template tracks before day zero. Days before Day Zero are represented as Day-5, Day-4, and so on.
Organizational Unit	Assigned organizational unit

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Table 11-3 (Cont.) Column Definitions - Task Manager Templates

Column Name	Description
Owner	Name of the template owner, which must be a Service Administrator or Power User. The template owner is also the default owner of all tasks in the template unless overridden at the task level.
Schedules	Total number of schedules created from each template

Task Manager - Template Tasks

This section provides the definitions for the Task Manager template task columns.

Table 11-4 Column Definitions - Task Manager Template Tasks

Column Name	Description
Active	Status of the task template
Approver 1-10	User, Group, or Team responsible for approving the task
Approver 1-10 (Backup)	User who can approve the task if the Approver (Primary) is unavailable
Approver 1-10 (Primary)	User, Group, or Team that was specified to approve the task
Approver 1-10 End Date (Assigned)	Assigned end date for the approver
Approver 1-10 End Date (Assigned, Local)	Assigned End Date of the task, in the local time zone of the org unit (if assigned an org unit)
Approver 1-10 Scheduled End Date	Scheduled End Date of the task
Approver 1-10 Scheduled End Date (Local)	Scheduled End Date of the task, in the local time zone of the org unit (if assigned an org unit)
Approver 1-10 Scheduled Start Date	Scheduled Start Date of the task
Approver 1-10 Scheduled Start Date (Local)	Scheduled Start Date of the task, in the local time zone of the org unit (if assigned an org unit)
Approver 1-10 Start Date	Assigned Start Date of the task
Approver 1-10 Start Date (Local)	Assigned Start Date of the task, in the local time zone of the org unit (if assigned an org unit)
Approvers (Count)	Total count of assigned approvers
Assignee	User, Group, or Team currently responsible for working on the task
Assignee (Backup)	User who can work on the task if the Assignee (Primary) is unavailable
Assignee (Primary)	User, Group, or Team who was specified to work on the task
Assignee End Date	End date for the assigned user
Assignee End Date (Local)	Assigned End Date of the task, in the local time zone of the org unit (if assigned an org unit)
Created By	Name of the user who created the template task
Created On	Date and time when the template task was created
Description	Description of the template task
Execution Type	The execution type such as End User, System Automated, or Event Monitoring
Instructions	Set of instructions for completing the task

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Table 11-4 (Cont.) Column Definitions - Task Manager Template Tasks

Column Name	Description
Integration	The execution type, End User, System Automated or Event Monitoring
Last Updated By	Name of the user who last updated the template task
Last Updated On	Date and time when the template task was last updated
Organizational Unit	Assigned organizational unit
Owner	Name of the template task owner, which must be a Service Administrator or Power User. The template task owner is also the default owner of all tasks in the template unless overridden at the task level.
Parent	Task ID of the parent task
Predecessors	Predecessor for the current task. It can be from another template.
Priority	Priority of the task
References (Count)	Count of references added for the template task (URL and content server references).
Rules (Count)	Total count of rules directly on the enterprise journal
Run As	Run the task as the user. Only applicable for system-automated tasks.
Run As Authorized	User who can authorize this task. If the Run As user is the user who also created the task, then the user can authorize the task, without being prompted to re-enter their password.
Start Task Early	Whether the assignee is allowed to open a pending task prior to the scheduled start time
Successors	Successor for the current task
Task ID	Unique identifier for the task
Task Type	Type of task
Template	Template associated with the task

Task Manager – Schedules

This section provides the definitions for the Task Manager Schedules columns.

Table 11-5 Column Definitions - Task Manager – Schedules

Column Name	Description
Created By	Name of the user who created the schedule
Created On	Date and time when the schedule was created
Date Zero Date	Date to assign as day zero
Description	Description of the schedule
End Date	Effective end date for the schedule
Last Updated By	Name of the user who last updated the schedule
Last Updated On	Date and time when the schedule was last updated
Name	Name of the schedule
Organizational Unit	Organizational unit associated with the schedule
Owner	Name of the schedule owner, which must be a Service Administrator or Power User
Period	Period for the schedule

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Table 11-5 (Cont.) Column Definitions - Task Manager – Schedules

Column Name	Description
Start Date	Effective start date for the schedule
Status	Status of the schedule
Tasks	Tasks associated with the schedule
Year	Year when the schedule was created

Task Manager – Task Types

This section provides the definitions for the Task Manager Task Types columns.

Table 11-6 Column Definitions - Task Manager – Task Types

Column Name	Description
Created By	Name of the user who created the task type
Created On	Date and time when the task type was created
Description	Description of the task type
Execution Type	Type of execution associated with the task type
Integration	Integration associated with the task type
Last Updated By	Name of the user who last updated the task type
Last Updated On	Date and time when the task type was last updated
Name	Name of the task type
Task Type ID	Unique identifier of the task type

Task Manager – Integrations

This section provides the definitions for the Task Manager Integrations columns.

Table 11-7 Column Definitions - Task Manager – Integrations

Column Name	Description
Connection	Application to which the task belongs
Created By	Name of the user who created the integration
Created On	Date and time when the integration was created
Description	Description of the integration
Execution Type	Execution type for the integration
Integration Cloud Deployment	Associated integration cloud deployment for the integration
Last Updated By	Name of the user who last updated the integration
Last Updated On	Date and time when the integration was last updated
Name	Name of the integration

Task Manager – Alert Types

This section provides the definitions for the Task Manager Alert Types columns.

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Table 11-8 Column Definitions - Task Manager – Alert Types

Column Name	Description
Alert Type ID	Unique identifier for the alert type
Associated With	Object associated with the alert type such as a task
Created By	Name of the user who created the alert type
Created On	Date and time when the alert type was created
Description	Description of the alert type
Enabled	Indicates whether the alert type is enabled
Last Updated By	Name of the user who last updated the alert type
Last Updated On	Date and time when the alert type was last updated
Name	Name of the alert type

Task Manager – Filters

This section provides the definitions for the Task Manager Filters columns.

Table 11-9 Column Definitions - Task Manager – Filters

Column Name	Description
Created By	Name of the user who created the filter
Created On	Date and time when the filter was created
Description	Description of the filter
Last Updated By	Name of the user who last updated the filter
Last Updated On	Date and time when the filter was last updated
Name	Name of the filter
Public	Whether the filter is public
Type	Type of the filter

Task Manager – Views

This section provides the definitions for the Task Manager Views columns.

Table 11-10 Column Definitions - Task Manager – Views

Column Name	Description
Created By	Name of the user who created the view
Created On	Date and time when the view was created
Default	Indicates whether the view is set as the default view
Description	Description of the view
Last Updated By	Name of the user who last updated the view
Last Updated On	Date and time when the view was last updated
Name	Name of the view
Public	Indicates whether the view is public
Type	Type of the view

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Task Manager – Attributes

This section provides the definitions for the Task Manager Attributes columns.

Table 11-11 Column Definitions - Task Manager – Attributes

Column Name	Description
Attachments	Indicates whether the attribute has attachments
Calculation	Indicates if the attribute is a calculation attribute
Created By	Name of the user who created the attribute
Created On	Date and time when the attribute was created
Last Updated By	Name of the user who last updated the attribute
Last Updated On	Date and time when the attribute was last updated
Lines	Number of lines of text that are visible, without scrolling, on the Actions dialog boxes
Locked	Indicates whether the attribute is locked
Name	Name of the attribute
Type	Type of the attribute

Task Manager – Currencies

This section provides the definitions for the Task Manager Currencies columns.

Table 11-12 Column Definitions - Task Manager – Currencies

Column Name	Description
Currency Code	Country's currency code that identifies the currency
Currency Symbol	Currency symbol for the currency code
Decimal Places	Indicates how to display numbers with decimal values
Description	Name of the country associated with the currency code
Enabled	Indicates whether the currency code is enabled

Managing Task Manager System Settings

Related Topics

- Managing Global Integration Tokens
- Managing Task Manager Organizational Units
- Managing Holiday Rules for Tasks
- Changing Configuration Settings
- Using Oracle Cloud Object Storage to Store Attachments

Managing Global Integration Tokens

The Global Integration Tokens feature enables creation of parameterized URLs. The URLs can be used for a variety of purposes.

When creating the URL, the parameters are inserted into the URL. When the URL is clicked, the parameters are replaced with the appropriate values.

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For example, the following is an excerpt of the parameters:

```
…$YearName$%22%22Period%20Name%22&val6=%22$PeriodName$
%22&col17=%22Logical%20Schedules%22.%22Year%20Name%22&val7=%22$Y earName$
%22&col8=%22Logical%20Schedules%22.%22Deployment%20Name%22&val8=%22$
ScheduleName $%22
```

where

```
$YearName$ = 2012
$PeriodName$ = Jan12
$ScheduleName$ = DemoSchedule
```

The URL becomes:

```
…
%222012%22%22Period%20Name%22&val6=%22Jan2012%22&col7=%22Logical%20Schedules%2
2.%22Year%20Name%22&val7=%222012%22&col8=%22Logical%20Schedules%22.%22Schedule
%20Name%22&val8=%22DemoSchedule%22
```

Parameters can be configured from static parameters defined in your application, attributes of type Text and List, and the following native attributes assigned to Tasks, Templates, and Schedules:

- Period Name
- Schedule Name
- Task Name
- Task ID
- Year Name

You can access URLs from the following locations:

- **Templates**: After an administrator or power user adds a reference URL to a template in the Instructions section, the URL is clickable on the Instruction tab.
- **Schedule**: After an administrator or power user adds a reference URL to a schedule in the Instructions section, the URL is clickable in Instructions.
- **Task Details**: After an administrator or power user or task owner adds a reference URL to a task in the Instructions section, the URL is clickable in Instructions.
- **Task Types**: After an administrator adds a reference URL to a Task Type in the Instructions section, the URL is clickable on the **Instruction** tab.
- **Task Actions**: Viewers of **Task Actions** can click the reference URLs.

Creating a Global Integration Token

Use these guidelines for creating Global Integration Tokens:

- The token name must be unique.
- Tokens can't be modified.
- Tokens shouldn't be deleted. If you try to delete a token, a warning will display: "Deleting a Global Integration Token will invalidate the URLs that are referencing it. Are you sure you want to continue?"

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To create a Global Integration token:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left, and then select **Global Integration Tokens**.
3. Click **New**.
4. Enter:
   - **Name**: A unique token name
   - **Type**: Static or Task Attribute
   - **Token Value**:
     - If **Task Attribute** is selected as the Parameter Type, select the value that is passed when the URL is rendered.
     - If **Static Attribute** is selected as the Parameter Type, enter the value that is passed when the URL is rendered.
5. Click **Save**.

Deleting a Token

To delete a token:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left, and then select **Global Integration Tokens**.
3. Select a token, and then click **Delete**.

   A warning message is displayed: "Deleting a Global Integration Token will invalidate the URLs that are referencing it. Are you sure you want to continue?"
4. Click **Yes** to confirm deletion.

Managing Task Manager Organizational Units

Organizational Units enable administrators to define regions, entities, business units, divisions, departments, or any other entity that might be affiliated with a task. For example, you could define a geographical Organizational Unit called "North America", or a business unit Organizational Unit called "Sales". Organizational Units are hierarchical to facilitate reporting. The "Sales" Organizational Unit could contain "East Sales Region" and "West Sales Region" as part of its hierarchy.

When you create a schedule, you select the Organizational Unit to be used for the schedule, and the selected Organizational Unit is applied to all tasks in the schedule.

For example, you may have defined an Organizational Unit called "East Sales Region". When you select "East Sales Region" as the Organizational Unit for the schedule, the selected Organizational Unit is applied to all tasks in the schedule. The settings that you defined for that Organizational Unit, such as Timezone, Holiday Rules, and Work Days, apply to all tasks in the schedule for the "East Sales Region".

Related Topics

- Adding Organizational Units
- Importing Organizational Units
- Selecting an Organizational Unit

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Adding Organizational Units

To add an organization:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Organizations** tab on the left and then select **Organizational Units**.
3. Click **Add Sibling**.
4. On the **Properties** tab, enter:
   - **Name**: You can enter a maximum of 255 characters.

     Administrators can change the name, and it need not be unique.
   - **Organizational Unit ID**: A unique ID used to identify the transaction for Migration purposes.

     The Organizational Unit ID can't be changed after an organizational unit is set. You can enter a maximum of 255 characters.
   - Optional: **Parent Organization**: Enables Administrators to change the hierarchy.
   - **Description**: You can enter a maximum of 255 characters.
   - **Timezone**: Determines which time zone applies to the organizational unit. Selection of a time zone is optional. If a time zone is not selected, the task time zone will revert to the user's time zone.
   - **Holiday Rule**: Determines which list of holidays applies to the organizational unit. Selection of a Holiday Rule is optional.
   - **Work Days**: Determines which days of the week are work days.
5. The **Access** tab enables Administrators to assign viewer and commentator access in a centralized location, rather than having to assign it to each task.

   To select a user:

   a. Click **Add**.
   b. In **Select Viewers**, enter the First Name and Last Name, or click **Search Users**, then select **Users** or **Groups**.
   c. In **Search Results**, select the users and click **Add** or **Add All** to move them to the Selected list.

Importing Organizational Units

To import organizational units:

1. On the Home page, click **Application** and then click **Task Manager**.
2. Click the **Organizations** tab on the left and select **Organizational Units**.
3. Click **Import** .
   - Click **Choose File** to navigate to the CSV import file.

     Sample Organizational import format:

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> **Note**
>
> The following section shows the import format. This example requires a pre-existing parent organizational unit named "Americas", a holiday rule named "US", and a Viewer named "fm_user5".

```
"OrganizationalUnitID","Name","ParentOrganization","Description","TimeZone
","H olidayRule","Calendar","Workdays","Viewer1","Commentator1"
"US2","US2","Americas","Import Organization US2 Example","ET
(UTC-05:00)","US","",2-3-5,"fm_user5",""
```

- For **Import Type**, select one:
  - **Replace**: Replaces the Organizational Unit detail with the Organizational Unit that is in the file that you are importing. It does not affect other units that are not specified in the import file.
  - **Replace All**: Imports a new set of Organizational Units that replaces the existing Units. This option is useful when you want to replace a unit in one system with an updated definition from another system. Organizational Units that are not specified in the import file are deleted.
- For **File Delimiter**, select **Comma**, **Tab**, or **Other**.

4. Click **Import**.

Selecting an Organizational Unit

Administrators define organizational units in Organizations to help organize and schedule tasks.

To select an organization:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Organizations** tab on the left, and then select **Organizational Units**.
3. Select an organization. An arrow indicates a child organization exists. Expand the parent to select a child organization.

Managing Holiday Rules for Tasks

Holiday Rules are collections of holiday dates that Administrators use when scheduling tasks.

After you create a Holiday Rule, you can apply the rule to an Organizational Unit and then apply the Organizational Unit to the schedule template. For example, you might have defined a Holiday Rule for all the yearly holiday dates for the "East Sales Region" Organizational Unit. When you select the "East Sales Region" as the Organizational Unit for a schedule, the predefined holiday dates apply to the tasks in the schedule.

See Applying a Holiday Rule to an Organizational Unit and Applying a Holiday Rule to a Schedule Template.

Creating Holiday Rules

To create holiday rules:

1. On the Home page, click **Application**, and then click **Task Manager**.

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2. Click the **Organizations** tab on the left.
3. Click **Holiday Rules**.
4. Click the **New** icon.
5. Enter:
   - **Holiday Rule ID**: You can enter a maximum of 255 characters. The **Holiday Rule ID** is mandatory and must be unique.
   - **Name**: You can enter a maximum of 50 characters. The **Name** does not need to be unique.
   - **Year**: The Year attribute behaves as a filter option. You don't need to select a value for this attribute. If you do, the table is filtered to display the dates associated with the selected year.
6. To name a holiday, in the **Holiday** section, click the **Import** icon to import the list of holidays, or click the **New** icon and enter the date and name of the holiday.

Applying a Holiday Rule to an Organizational Unit

After you create a holiday rule, you can apply it to an organizational unit, and then apply it to a schedule.

To apply a holiday rule:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Organizations** tab on the left.
3. Click **Organizational Units**.
4. Create or edit an organization.
5. On **Properties**, from the **Holiday Rule** drop-down, select a holiday rule.
6. Click **Save**.

Applying a Holiday Rule to a Schedule Template

After you create a holiday rule and apply it to an organizational unit, you can apply the organizational unit to a schedule template.

To apply a holiday rule:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Select a template, click the **Actions** icon to the right of the template, and then select **Create Schedule**.
4. For **Schedule Parameters**, specify the **Schedule**, **Year**, **Period**, and **Day Zero Date**.
5. Under **Date Mapping**, click **Advanced Configuration**.
6. For **Organizational Unit**, select **Use Task Value** or **Set Value To**. For **Set Value To**, click the **Search** icon to select an organizational unit.
7. Select the organizational unit that you created or edited that contains the holiday rule and click **OK**.

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8. Click **OK**.

   The Schedule Parameters Date Mapping now shows the holiday.

Importing Holiday Dates

You can import dates into a holiday rule.

To import holiday dates:

1. On the Home page, click **Application** and then click **Task Manager**.
2. Click the **Organizations** tab on the left.
3. Click **Holiday Rules**.
4. Create or select a holiday rule.
5. Before the **Holiday** section, click the **Import** icon.
6. Click **Choose File** to navigate to the CSV import file.
7. For **Import Type**, select one:
   - **Replace**: Replaces the holiday dates with the holiday dates in the file that you are importing. Does not affect other units that are not specified in the import file.
   - **Replace All**: Imports a new set of holiday dates that replaces the existing holiday dates. Use this option to replace unit in one system with an updated definition from another system. Holiday dates that are not specified in the import file are deleted.
8. For **Date Format**, select a format from the drop-down list of allowed date formats. Date formats are not translated. By default, the date format is set to the locale date format of the exported file location.
9. For **File Delimiter**, select **Comma**, **Tab**, or **Other**.
10. Click **Import**.

Editing Holiday Rules

To edit holiday rules:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Organizations** tab on the left and then select **Holiday Rules**.
3. Select a rule.
4. Edit the following fields:

   > **Note**
   >
   > The **Holiday Rule ID** is not editable.

   - **Name**: The name can be a maximum of 50 characters. It does not need to be unique.
   - **Year**: The Year attribute behaves as a filter option. You do not need to select a value for this attribute. If you do, the table is filtered to display the dates associated with the selected year.
5. Click **Save**.

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Duplicating Holiday Rules

To duplicate holiday rules:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Organizations** tab on the left.
3. Click **Holiday Rules**.
4. Select a rule and click the **Duplicate** icon.
5. Edit the rule as needed.

Deleting Holiday Rules

You cannot delete a holiday rule associated with an organizational unit that is not logically deleted. A list of organizational units to which the holiday rule was assigned is displayed.

To delete holiday rules:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Organizations** tab, and then select **Holiday Rules**.
3. Select a rule, and then click the **Delete** icon.

Changing Configuration Settings

Related Topics

- Allowing Editing and Deleting Comments
- Allowing Task Deletions
- Enabling Application Dimension Alias
- Enabling Artifact Labels
- Enabling Email Notifications
- Setting Task Manager Governors
- Allowing Reassignment Request Approvals
- Reopening Tasks
- Settings for Task Manager Reports
- Setting Export to Excel Format
- Specifying Task Display in Smart View
- Displaying Upcoming Tasks in Worklist and Welcome Panel

Allowing Editing and Deleting Comments

A Service Administrator can decide to allow editing and deleting comments in tasks by using the **Allow Comment Editing and Deleting** configuration settings option. By default, this option is Off.

To allow users to edit and delete comments:

1. On the Home page, click **Application**, and then click **Task Manager**.

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2. Click the **System Settings** tab on the left.
3. Click **Comments**.
4. For **Allow Comment Editing and Deleting**, select **Turn On** to enable editing and deleting comments.

Allowing Task Deletions

Service Administrators can specify whether to allow tasks to be deleted. By default, this option is Off.

> **Note**
>
> The setting affects only running and closed tasks.

To allow users to delete tasks:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left.
3. Select **Allow Task Deletion**.
4. Select **Turn On** to enable deletions.

Enabling Application Dimension Alias

You can enable Application Dimension Alias support for Task Manager . Within the application, administrators can assign alternate names, or aliases, to system dimensions and user-defined dimension members. The application allows up to 30 aliases per dimension member, including the default alias. See Working with Alias Tables in *Administering FreeForm*.

When enabled, the Year and Period dimensions will display to the user their alternative alias table labels based on the user's preferred alias table selection and member name display preference. The alias labels will be displayed within the listing page columns, filter bars, and dashboard displays. Users can set their alias table selection and member name display preference within the **User Preferences** > **General** > **Alias Settings**. See Setting the Display of Member Names or Aliases in *Working with FreeForm*.

> **Note**
>
> The Year dimension will be renamed from Year to Years and its members will have the fiscal year format (FY00).

If the Dimension Alias support is disabled, then Years will revert to Year and display as numeric values (example, 2024, 2025, and so on) and Periods will be shown with their default Member Name. They will be displayed this way regardless of the user's preferred Alias table selection.

To enable or disable dimension alias:

1. Go to **Applications** > **Task Manager** > **System Settings**.
   The Manage System Settings page is displayed.
2. Click **Translation** and then select the **Enable Dimension Aliases** option. To disable dimension alias, clear this option.

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> **Note**
>
> Dimension Alias support is configured for within the Task Manager System Settings.

After enabling or disabling this option, the database is refreshed on the application. The Refresh Dimensional Metadata job is initiated to copy the metadata.

The following jobs may be queued if they are triggered:

- LCM Import
- LCM Export
- Import Tasks
- Import Reports

As the Year value can be modified from FY00 format to numerical format and vice versa, the preceding jobs are blocked until the Refresh Dimensional Metadata job is completed.

After the database is refreshed, you must sign-off and sign-in to view the alias support. All dimensions will be realigned with the application dimension definitions. All references to the year dimension and members will automatically be updated. Custom Report definitions may need to be revised depending on how the Year dimension was referenced.

Enabling Artifact Labels

Enabling Artifact Labels for Task Manager allows Administrators to enter translated labels in different languages for artifact properties such as a Task's name or description. These labels will be displayed to users based on their language preference. If an Artifact Label hasn't been entered for the user's preferred language, then the original label entered in the artifact property, or the default artifact label, is shown.

To enable artifact labels:

1. Go to **Applications** > **Task Manager** > **System Settings**.
   The Manage System Settings page is displayed.
2. Click **Translation** and then select the **Enable Artifact Labels** option.
   The setting is saved and will be effective after you re-login.

   When you re-login, to access Artifact Labels, go to **Application** > **Task Manager** and click the **Artifact Labels** icon on the left. The Artifact Labels page is displayed.

Enabling Email Notifications

Service Administrators can set up Email Notifications for components such as Alerts, Task Manager, and Consolidation Journals. When you enable email notifications, batch notifications are generated. When the notification is no longer required, you can deactivate email notifications and reminders.

> **Note**
>
> By default, email notifications aren't enabled.

Administrators can also assign the number of days before a due date to send reminder notifications.

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Due Date Reminder notifications are emailed to Assignees and Approvers in these conditions:

- Responsibility for a component action changes: Sent based on information in the action
- A due date is missed: Sent based on information in the action
- A due date is approaching: Sent based on the number of days in advance that you specify

To enable email notifications:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left.

   The Manage System Settings page is displayed.
3. Click **Email Notifications**.
4. Specify the following information:
   - **Email Notifications**: Select **Turn On** to enable the notification. Click **Turn Off** to disable the notification.
   - If you can edit the **From Address** field, enter an email address to indicate from where the user is receiving the notification. You can also include a product acronym. If you can't edit the **From Address** field, the default email address is displayed as `no.reply@epm.oraclecloud.com`.
     Email notifications also include the service name in the footer of the email. For example: "Note - This is an automated email notification generated from EPM Task Manager on efim.trcs".
   - **Number of days before due date to send reminder**
   - **User Specified**: Select this checkbox to allow users to customize their email notification settings. Users can personalize their email notification settings within their user preferences (**Tools** > **User Preferences** > **Notifications**).
     If you don't select this checkbox, then the user inherits the settings made by the administrator.
   - **Email Start Time** and **Email End Time**: Specify the time period in a day when scheduled emails will be sent. This setting will take effect in each user's individual time zone.
5. For each component, you can specify the **Notification Type**. The types of notifications are:
   - **Late Notifications** (for Task Manager)
   - **Status Change Notifications** (for Alerts, Task Manager and Consolidation Journals)
   - **Due Date Reminder Notifications** (for Task Manager)
6. Select an option in the **Notify Me** column for each notification or notification type to indicate the frequency of the notifications:
   - **Immediately**: Email notifications are sent immediately.

     > **Note**
     >
     > If you set **Notify Me** to **Immediately**, the email notifications are sent immediately even when it is outside of the user's email start/end time.
   - **Once a day**
   - **Every 2 hours**

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- **Every 3 hours**
- **Every 4 hours**
- **Every 6 hours**
- **Mixed**
- **Never**: Select this option to deactivate a notification. This option is only applicable for **Status Change Notification** and **Due Date Reminder Notification**. For **Late Notification**, only the following notification types get this option:
  - You are the task owner and task assignee is late
  - You are the task owner and task approver is late

For each type of notification, expand the required component, and choose when to be notified. For example, for **Status Change Notification**, expand **Task Manager**. For **You are the task approver and task approver is late**, select **Every 2 hours** as a choice of notification.

If an Administrator disables the **Email Notifications** in Services, then notifications are suspended.

For more information, see Managing Notification Types and Settings.

Enable email notifications. See Setting Up Email for Notifications in *Working with FreeForm*.

Managing Notification Types and Settings

Service Administrators can define the default settings for the notifications for all users for the following components:

- Alerts
- Task Manager
- Consolidation Journals

Alert Notifications

Status Change Notification Options

Notification Type	Notify Me Options
You are the alert approver and alert status changes to open with approver	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the alert assignee and alert status changes to open with assignee	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

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Notification Type	Notify Me Options
You are the alert owner and alert status changes to open with assignee	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the alert owner and alert status changes to open with approver	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the alert owner and alert status changes to completed	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the alert owner and alert status changes to open with owner	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are an alert viewer and alert status changes to open with owner	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are an alert viewer and alert status changes to open with assignee	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are an alert viewer and alert status changes to open with approver	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

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Notification Type	Notify Me Options
You are an alert viewer and alert status changes to completed	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

> **Note**
>
> When selecting an option other than **Immediately**, any Alerts that are associated with Pending or Locked Schedules will not be emailed.

Task Manager Notifications

Late Notification Options

Notification Type	Notify Me Options
You are the task approver and task approver is late	• Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task assignee and task assignee is late	• Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task owner and task assignee is late	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task owner and task approver is late	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

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Status Change Notification Options

Notification Type	Notify Me Options
You are the schedule owner and schedule is created	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the schedule owner and schedule status changes to open	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the schedule owner and schedule status changes to closed	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the schedule owner and schedule status changes to locked	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the schedule viewer and schedule is created	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the schedule viewer and schedule status changes to open	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the schedule viewer and schedule status changes to closed	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

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Notification Type	Notify Me Options
You are the schedule viewer and schedule status changes to locked	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task assignee and task status changes to open with assignee	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task approver and task status changes to open with approver	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the backup assignee, and the primary assignee is unavailable	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the backup approver, and the primary approver is unavailable	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task authorizer and schedule status changes to open	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task owner and task status changes to open with assignee	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

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Notification Type	Notify Me Options
You are the task owner and task status changes to open with approver	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task owner and task status changes to closed	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task owner and task status changes to force closed	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task owner and task status changes to aborted	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task viewer and task status changes to open with assignee	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task viewer and task status changes to open with approver	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task viewer and task status changes to closed	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

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Notification Type	Notify Me Options
You are the task viewer and task status changes to force closed	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task viewer and task status changes to aborted	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the journal submitter, and the journal needs to be posted	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the journal approver, and the journal needs approval	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

Due Date Reminder Notification Options

Notification Type	Notify Me Options
You are the task approver and task is "At Risk"	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task assignee and task is "At Risk"	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task owner and task is "At Risk"	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

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Notification Type	Notify Me Options
You are the task assignee and assignee due date is within 1 days	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the task approver and approver due date is within 1 days	• Never • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

Consolidation Journals

Status Change Notification Options

Notification Type	Notify Me Options
You are the journal approver, and the journal needs approval	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day
You are the journal submitter, and the journal needs to be posted	• Never • Immediately • Every 2 hours • Every 3 hours • Every 4 hours • Every 6 hours • Once a day

Email notifications for Journal actions are based on entity access, not on the Consolidation Journal roles. Users that do not have the Consolidation Journal roles and that should not be receiving emails should disable Consolidation Journal notifications from User Preferences.

To disable Consolidation Journal notifications:

1. From the Home page, select **Tools**, then **User Preferences**.
2. Select **Notifications**, then **Status Change Notification,** then select **Consolidation Journals**.
3. Change the setting for **You are the journal submitter and the journal needs to be posted** to **Never**.
4. Change the setting for **You are the journal approver and the journal needs approval** to **Never**.
5. Save the changes.

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Setting Task Manager Governors

In Task Manager, you can set governors on various system settings and the system provides default settings which you can decrease or increase up to the maximum value. For example, you can specify the maximum number of rows to display in lists.

- **Maximum Number of Items displayed in a List**: Determines the maximum number of rows displayed in a List view; for example, Schedule Tasks.
- **Maximum Approver Levels**: Determine the number of levels that a task might be reviewed.
- **Select maximum file upload size (MB)**: Determines the maximum attachment file size that users can upload. This setting helps administrators adapt to work within limitations of the environment in which the application is installed.

To set Task Manager governors:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left.
3. On the **Manage System Settings** page, click **Governors**.
4. Specify a maximum value or leave the default setting:
   - **Maximum Number of Items displayed in a List**: The default is 10000.
   - **Maximum Approver Levels**: Select a value from 1 to 10.
   - **Select maximum file upload size (MB)**: Select a value from 5 MB, with incremental values of 5 MB, up to 100 MB.
5. In the **Reports** section, change the setting only if instructed by Oracle Support.
6. In the **Dashboards** section, set **Execute Queries in Parallel** to **Turn On** to enable parallel execution for operational dashboard queries. The default is **Turn Off**.

> **Note**
>
> This setting affects all operational dashboard objects such as Task Manager, Supplemental Data, Enterprise Journals, and Alerts.

7. Click **Save**.

Allowing Reassignment Request Approvals

As an Administrator, you can specify which workflow users can directly perform and approve reassignment requests rather than submitting requests to an Administrator or Power User for approval.

Users can then approve reassignment requests from their Worklist.

Service Administrators should also see the reassignment requests if they are required to approve the requests in case the Task Owner is unavailable.

To allow workflow users to perform reassignments:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left.

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3. Select **Reassignment Requests**.
4. For **Allow reassignment request approval by**, select one or more options:
   - **Administrator**
   - **Power User**
   - **Users**

Reopening Tasks

In Task Manager, there might be times when users need to reopen a task. Administrators can specify whether to allow open or closed tasks to be reopened.

To allow reopening of tasks:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left.
3. Select **Reopen**.
4. Set the conditions to allow reopening of a task that is Open with an Assignee or Approver. For **Task is Open with Assignee or Approver**:
   - **Not Allowed**: This is the default option and assignees or approvers cannot change the workflow of an open task after it has been submitted or approved.
   - **Allowed for all approvers**: Allows approvers who have approved a task to return the workflow to themselves.
   - **Allowed for all assignees and approvers**: Allows an assignee who has submitted a task or an approver who has approved a task to return the workflow to themselves.
5. Set the conditions to allow reopening closed tasks. For **Task is Closed**:
   - **Not Allowed**: This is the default option and users will not be able to reopen a closed task.
   - **Allowed for final approver**: Allows only the final approver to reopen and return the workflow to themselves.
   - **Allowed for all approvers**: Allows an approver to reopen and return the workflow to themselves.
   - **Allowed for all assignees and approvers**: Allows assignees and approvers of a task to reopen and return the workflow to themselves.

Settings for Task Manager Reports

When generating reports in Microsoft Excel format, a row that spans multiple lines may result in empty rows displayed after the row that contains data. This may cause a problem if you've scripts that perform calculations on a row value. Task Manager provides a setting to control the display of these extra rows.

To resolve this issue, you must modify the template file of the report by performing the following:

- Remove any empty lines from the table row
- Change the row height of the row to 0

Alternatively, Task Manager provides a setting to control the display of these extra rows. This setting applies to all the reports.

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1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left.
3. Click **Reports**.
4. Unselect **Enable Row Split** to ensure that multiple rows aren't displayed for each row of data in Microsoft Excel.

Setting Export to Excel Format

To export data to Microsoft Excel, you can set the format to .xls or .xlsx.

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab on the left.
3. Click **Reports**.
4. In **Export Excel Format**, select one of the following options:
   - **Excel 97 - 2003 (.xls)**: Exports data in tables to .xls format.
   - **Excel (.xslx)**: Exports data in tables to .xlsx format.

Specifying Task Display in Smart View

You can configure the display of Task Manager tasks when they are launched from Oracle Smart View for Office by selecting columns to display; for example, Task Attributes.

To select columns to display in Smart View:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **System Settings** tab.
3. Select **Smart View**.
4. Under **Column Selection**, from the **Available** list, select the columns to display, then click the **Add** arrow keys to move them to the **Selected** column, or **Remove** to remove them.

   You can select a maximum of 30 columns, and reorder the columns as needed.
5. Click **Save** to save your selection.

Displaying Upcoming Tasks in Worklist and Welcome Panel

As an administrator, you can decide whether to show only Current Tasks, or Current and Upcoming Tasks in the Workflow of the Worklist and Welcome Panel.

If you enable Current and Upcoming Tasks, users can see items currently available to work on, and also items in the upcoming workflow. For example, users can see an upcoming task that is due for them to approve, but not immediately available, as the preparer has not yet finished the task.

You also have the option to display the tasks with description (default) or by the Task ID/ Organizational Unit.

By default, the system displays only Current Tasks.

To set the display for tasks:

1. On the Home page, click **Application**, and then click **Task Manager**.

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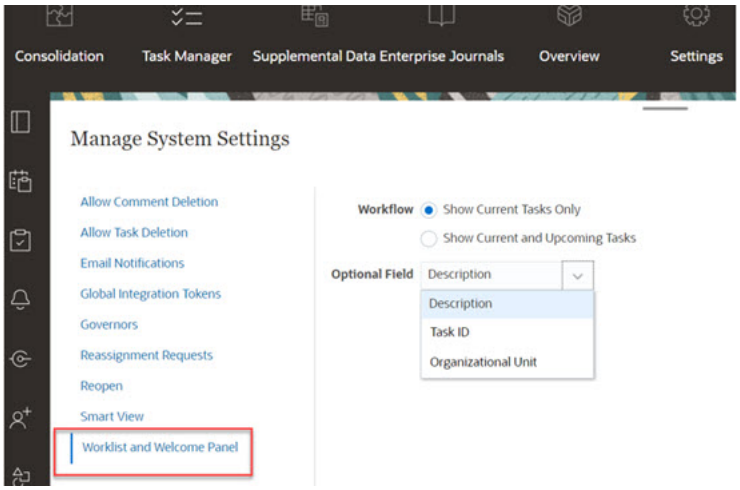
2. Click the **System Settings** tab on the left.
3. Select **Worklist and Welcome Panel**.
4. Select an option for **Workflow**:
   - **Show Current Tasks Only**
   - **Show Current and Upcoming Tasks**
5. Select an option for **Optional Field**:
   - **Description** (default)
   - **Task ID**
   - **Organizational Unit**



Using Oracle Cloud Object Storage to Store Attachments

Related Topics

- About Using Oracle Cloud Object Storage to Store Attachments
- Understanding What is Stored on OCI Object Storage
  FreeForm automatically handles moving files to and from OCI Object Storage so users can upload and view attachments seamlessly.
- Steps to Use OCI Object Storage with FreeForm

About Using Oracle Cloud Object Storage to Store Attachments

Customers who have a large FreeForm database mainly due to the number of uploaded attachments should use **Oracle Cloud Infrastructure (OCI) Object Storage** to store the attachments. Using a separate storage option will allow a large reduction in the size of the LCM snapshot so the backup, download and restore steps will be more streamlined (for example, when restoring a snapshot from test to production, or when downloading a snapshot for offline storage). Another advantage is that using OCI Object Storage allows you to increase the maximum attachment size from 20 MB to 100 MB. It is a highly recommended best practice to use OCI Object Storage.

OCI Object Storage requires its own subscription and configuration so this feature is optional. If you choose to set it up, attachments are stored in OCI Object Storage going forward and existing attachments will be moved to OCI Object Storage once configured.

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When you create an application using a snapshot or by cloning, and the source application uses OCI Object Storage to store attachments, the password required to access the OCI Object Storage bucket is not exported in the new application for security reasons. The recommended best practice is to separately clone your source OCI Object Storage bucket, and configure the new bucket on the new application. See Setting Up OCI Object Storage in FreeForm.

Note

- Once you start using OCI Object Storage, you cannot go back to using the FreeForm database for these attachments.
- During cloning if the source is set to use object storage in between, then after migration, sometimes the overall statistics about the attachments on the target (OCI) doesn't match the source because of the following:
  - **Orphan records**: When orphan references at source (that are considered for statistics at source) are migrated, they get lost at the target system because they aren't associated with any objects. This causes a mismatch in the statistics at target when compared to source. For example, orphan references are created when an alert has comment references and the comments are deleted but the references aren't deleted at source.
  - **Duplicate attachments**: For duplicate attachments at source, only the distinct reference for statistics at source are considered. However, after migration, each duplicate record has its own unique reference at target because they are migrated as part of different artifacts. This causes mismatch in the statistics.

Note

This is applicable only for Task Manager.

Understanding What is Stored on OCI Object Storage

FreeForm automatically handles moving files to and from OCI Object Storage so users can upload and view attachments seamlessly.

Some examples of what FreeForm stores in OCI Object Storage are attachments for Task Manager.

A unique ID is generated for each attachment that is created in Task Manager. This ID is used as the file name when the attachment is stored in OCI Object Storage. The name actual name of the attachment file is stored as a property of the object storage file. For example, assume that you add an attachment with the file name `fx_translations.xlsx` to Task Manager. A unique ID, say `fpbh-2765`, is generated for this attachment. This attachment is stored in OCI Object Storage with the name `fpbh-2765` and the actual name of the attachment file, `fx_translations.xlsx`, is stored as a property of `fpbh-2765` in OCI Object Storage. To determine the actual name of an attachment file, use **View Details** for the file in OCI Object Storage.

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> **Note**
>
> FreeForm does not delete any files from OCI Object Storage. Instead, OCI Object Storage attachments are purged based on the Retention Policy set in OCI Object Storage. See Using Retention Rules to Preserve Data.

Steps to Use OCI Object Storage with FreeForm

The following are the high-level steps for using OCI Object Storage with FreeForm.

1. Subscribe to OCI Object Storage. See Oracle Cloud Infrastructure - Cloud Storage.

   To review the user assistance for OCI Object Storage, see Overview of Object Storage.

2. Create the environment required to store attachments in OCI Object Storage. This includes setting up the bucket (logical storage location in OCI Object Storage) in which attachments will be stored.

   See Setting Up OCI Object Storage.

3. Configure FreeForm to store attachments in OCI Object Storage.

   See Setting Up OCI Object Storage in FreeForm.

4. Configure Backup and Disaster Recovery for object storage. For more information, see *How can I set up backup and disaster recovery for my own Object Storage bucket?* section in Backup and Disaster Recovery in *Getting Started Guide for Administrators*.

> **Caution**
>
> If an Object Storage bucket is accidentally deleted, it cannot be recovered. Therefore, it is vital that you configure backup and disaster recovery for buckets that store your data.

Setting Up OCI Object Storage

A separate subscription to **OCI Object Storage** is required to use this feature. Note that a Bucket is a logical container in **OCI Object Storage** for storing objects. In the context of Task Manager , your attachments are considered objects.

The high level steps in **OCI Object Storage** are listed here:

1. Create a Bucket in **OCI Object Storage** to store your attachments. For instructions, see Creating a Bucket.

> **Note**
>
> You can create a bucket in an existing compartment or create a new compartment for the attachments.

Here's an example of a set up Bucket.

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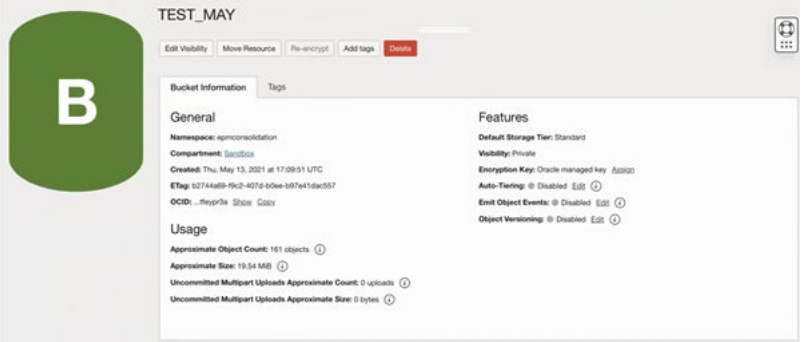

To enable you to test access to the bucket and to switch seamlessly from a test environment to a production environment, it is recommended that you create two folders within the bucket – one folder for the test environment and another folder for the production environment. For example, if you create a bucket named `task_manager_rec_data` to store your attachments, create folders `test` and `prod` within this bucket. During the implementation phase, use the `test` folder by specifying the bucket URL as `<bucket_url>/test`. When moving to the production environment, switch the bucket configuration to `<bucket_url>/prod`. The folder `test` can subsequently be deleted without any impact in the production environment.

2. Ensure that Auto-Tiering is disabled for the bucket. See Managing Auto-Tiering for an Object Storage Bucket.
3. You need to keep the **Lifecycle Policy Rules** as is in **OCI Object Storage**. Do not change this.
4. Optional: **Retention Rules** in **OCI Object Storage** follow your company's audit requirements (for example five to seven years).
5. In **Oracle Cloud Infrastructure** (OCI), you need to create a user for FreeForm and grant that user at least READ and WRITE access but do not grant DELETE access. The user can be an **Identity and Access Management** (IAM) user or a Federated user.
   We recommend a separate user be created for accessing Object Storage for FreeForm. This user has to be granted privileges to access to the attachment storage bucket and to manage objects in the bucket.
6. You need to create a group to assign policies.

   Access to Object Storage is managed by **Identity and Access Management (IAM)** policies. Common object storage policies can be found in Let users write objects to Object Storage buckets

   To create **IAM** policies, see Getting Started with Policies

   Here is an example of the policy that is required.
7. An auth token has to be created for the user. For details, see Working with Auth Tokens

   > **Note**
   >
   > The auth token will not be displayed after it has been created so make a note of the token since it will be used later in the configuration process.
8. Once you have created the Bucket and created a user, you need to set up **OCI Object Storage** in FreeForm so that the connection is made using the **Bucket URL**, **Username** , and **Password**. See Setting Up OCI Object Storage in FreeForm.

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Setting Up OCI Object Storage in FreeForm

To set up **OCI Object Storage** in FreeForm for Task Manager attachments:

1. Go to **Navigator** and click **Connections** under **Tools**.
   The Manage Connections page is displayed.
2. Click **Create** to create a connection.
   The Select Provider to Create Connection dialog box is displayed.
3. Click **Other Web Service Provider**.
   The Enter Connection Details dialog box is displayed.
4. Specify the following details:
   - **Connection Name**: Name of the Object Storage connection.
   - **Description**: Brief description about the connection.
   - **URL**: Object Storage bucket URL. It's the URL of the Oracle Object Storage Cloud bucket.
   - **User** and **Password**: User name and password you created for FreeForm in **Oracle Cloud Infrastructure**.

   > **Note**
   >
   > If the user was created in the Federated Identity Provider, then the user name has to be prefixed with the Federated Identity Provider name. For example: `oracleidentitycloudservice/username`. For password, you must use the auth token as the password, not the login password.
5. Click **Save and Close** to save the Object Storage connection details.
6. Go to **Home** > **Applications** > **Task Manager**.
7. Click **System Settings** tab.
   The Manage System Settings page is displayed.
8. Click **Object Storage** and from the **Connections** drop-down list, select the Object Storage connection.
   The **Bucket URL** is populated automatically.

   The URL format is:`https://swiftobjectstorage.region_identifier.oraclecloud.com/v1/namespace/bucket_name` where:
   - `region_identifier` is the hosting Oracle Cloud Infrastructure (OCI)
   - `region.namespace` is the top-level container for all buckets and objects. Each Oracle Cloud Infrastructure tenant is assigned a unique system-generated and immutable Object Storage namespace name at account creation time. Your tenancy's namespace name, for example, `axaxnpcrorw5`, is effective across all regions
   - `bucket_name` is the name of a logical container where you store your data and files. Buckets are organized and maintained under compartments.

   An example URL is: `https://swiftobjectstorage.us-ashburn-1.oraclecloud.com/v1/epmcloud/fccsAttachments`

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If you created folders within your bucket, the URL format is `https://swiftobjectstorage.us-ashburn-1.oraclecloud.com/v1/epmcloud/bucket_name/folder_name`, where `folder_name` is the name of the folder containing the data.

9. Click **Validate and Save**. The system validates to make sure the connection is working properly and creates and reads a test file.
   The Object Storage confirmation dialog box is displayed.
10. Click **Continue**. A batch job starts immediately that will move attachments currently in the database to the specified Object Storage location. Henceforth, all attachments will be saved in the Object Storage location instead of the database. You can't revert to the database once the attachments are stored in Object Storage location.
11. Statistics display depending on whether you are an existing customer or a new customer first setting up **OCI Object Storage** and FreeForm:
    - Number of migrated attachments: How many attachments are migrated.
    - Size of migrated attachments (MB): Size of attachments that are migrated.
    - Number of pending attachments: How many attachments are still pending for migration.
    - Size of pending attachments (MB): Size of the attachments pending for migration.
    - Number of all the attachments: How many total attachments are in the system (includes migrated attachments plus pending migration attachments plus newly added attachments).
    - Size of all the attachments (MB): Size of all attachments in the system

    > **Note**
    >
    > After migration from test environment to production environment, if you do not update the manual step of connections details, then uploading and downloading attachments within the application results in an error. Download will happen with 0 bytes and upload will result in validation error.

    - If you are an existing customer, you will see statistics relating to migration of the attachments from the FreeForm database to **OCI Object Storage**. You will see statistics increment as attachments are moved.
      FreeForm takes the attachments from the database and moves files to **OCI Object Storage** in the Bucket you have set up. This is performed based on the **Bucket URL**, along with the user name and password you created for FreeForm in **Oracle Cloud Infrastructure**. The attachments are moved from the database into **OCI Object Storage** and then removed from the database.

      After the first time migration, uploads and downloads of attachments are stored and retrieved from **OCI Object Storage**.
    - If you are new to FreeForm, many of the statistics shown will display as zero since you are not moving existing attachments to **OCI Object Storage**.

    Here's a sample screenshot of the statistics.

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Clicking **Reset** will reset to the last connection that worked.

Troubleshooting Configuration Problems

When setting up **OCI Object Storage**, if you see the following error, there may be problem with the bucket URL or credentials:

```
Error: Unable to connect to the Object Storage service using the specified URL
and credentials.
```

Use any REST client such as cURL or Postman to test the bucket URL and credentials.

The format of the cURL command is as follows: `curl <bucket URL> -u "<username>:<password>"`

Following is an example of a cURL command to test your URL and credentials:

```
curl https://swiftobjectstorage.us-ashburn-1.oraclecloud.com/v1/epmcloud/
taskAttachments -u "username:password"
```

Managing Task Manager Attributes

Attributes are user-defined fields defined centrally by administrators and used in many places.

You can specify different value types for attributes: Date, Date/Time, Integer, List, Multi-Line Text, Number, True/False, Text, User, and Yes/No. For example, you can define a custom attribute named Auto Submit with a value type of Yes or No. When you assign the Auto Submit attribute to an item, you can set it to Yes for tasks required by external users.

If you select the List type, you can define a pick list of values. For example, you can define a List attribute named Sales Region, with North, South, East, and West as values.

You can create, edit, and delete attributes.

> **Note**
>
> The maximum number of Task Manager attributes that you can create or import is 10,000.

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Defining Task Manager Attributes

To define a Task Manager attribute:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Attributes** tab on the left.
3. Click the **New** icon, then on the **Properties** tab specify the following properties:
   - **Name**: You can enter a maximum of 80 characters.

     > **Note**
     >
     > You can create an object using the same name that you used earlier and deleted. Any objects that were previously linked to the deleted object will be associated with the new object.

   - **Description**: You can enter a maximum of 255 characters.
   - **Type**:
     - **Date**
     - **Date/Time**
     - **Integer**: Allows a value between -2147483648 and 2147483647. For **Total**, specify the totaling method for the attribute:
       - **None** (default): No total is calculated.
       - **Average**: The average of the rows with data. Rows without data are not counted in the denominator
       - **Count**: The count of the rows with data.
       - **Sum**: Additive total.
     - **List**: Click **List Values** to enter a list of valid responses to the question. You can enter a maximum of 255 characters.
     - **Multi-Line Text**: Click **Format** and specify the **Number of Lines**, from 3 to 50 lines. Multi-Line Text determines how many lines of text are visible, without scrolling, on the Actions dialog boxes. The maximum length should be less than 4,000 characters.

       Select **Include Attachments** if you want to include an **Attachments** section on the Actions dialog box.
     - **Number**: Enter a value xxxxxxxxxxxxxxxxx.xxxxxxxxx (17 digits and 9 decimal places). Click **Format** and select number formatting options:
       - For **Decimal Places**, enter a value for the number of decimal places to display.
       - For **Display As Percentage**, select if you want a percentage to display.
       - Select the **Thousands Separator** option if you want numbers to display a thousands separator (for example, 1,000.00).
       - For **Currency Symbol**, select a currency symbol; for example, $ (United States of America Dollar).

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    - For **Negative Number**, select how to display negative numbers; for example, (123).
    - For **Scale**, select a scale value for numbers; for example, 1000.
  - **Text**: Enter a maximum of 4000 characters.
  - **True/False**
  - **User**
  - **Yes/No**
- **Calculation**: Select to add logical and mathematical functions for attributes in the **Calculation** tab.

  See Defining Calculation Attributes.

Defining Calculation Attributes

Calculation attributes enable you to add logical and mathematical functions to derive values for attributes. They can be embedded within other calculated attributes to produce complex functions. Calculated attributes are read-only. They can only be evaluated against Schedules.

Schedules: You can add any Calculation and the calculation is evaluated, regardless of whether or not it was explicitly assigned to the task.

Templates: You cannot add a custom attribute that has the Calculation option selected as a column or a filterable attribute.

When administrators add attributes to the Attributes sections in the Actions dialogs, they are viewable by workflow users. Administrators can restrict access to certain roles by setting the access to Do Not Display. For example, for calculated attribute XYZ, an administrator could set the access to `Viewer: Do Not Display` so that the XYZ attribute is not displayed for users with only the Viewer role.

Any user role can add calculated attributes as columns in views and portlets. They can also be added as filterable attributes in the Filter Panel.

To define a calculation attribute:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Attributes** tab on the left.
3. Click **New** icon, and then on the **Properties** tab enter a **Name** for the attribute, and then select an attribute **Type**.
4. Select the **Calculation** option.
5. On the **Calculation** tab, select a **Calculation Type**.

You can select from the following calculation types. The available types are determined by the attribute **Type** (**Text**, **Integer**, and so on).

- **Assign Value to List**: Assign custom values to a List type attribute
- **Assign List To Value**: Assign custom list values to the values of a different attribute. Only available for List type attributes.
- **Conditional**: A conditional calculation (If – Then – Else)
- **Scripted**: A free-form scripted calculation. Only available for attributes of type Integer, Multi-line Text, Number, or Text.

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The following table lists the Calculation types available for each Attribute type.

Attribute Type	Assign Value to List	Conditional	Scripted	Assign List to Value
Date				
Date/Time				
Integer	X	X	X	
List		X		X
Multi-Line Text	X	X	X	
Number	X	X	X	
Text	X	X	X	
True/False		X		
User				
Yes/No		X		

Calculation Definition

The **Scripted** Calculation type enables you to enter a free-form calculation equation. You can use **Add Attribute** and **Add Function**:

- **Add Attribute**: Select an attribute and insert the attribute into the Calculation Definition box at the location of the cursor. If the cursor is in the middle of a word or attribute, the word/attribute will be replaced in the definition. The script format adds brackets {} around the name of the new attribute.

  The only available attributes are Schedule attributes.

- **Add Function**: Select a function and add the function to the Calculation Definition. The Function is added with placeholders for each parameter.

For example: Insert the DATE_DIFF function in the calculation definition:

```
DATE_DIFF(<Date1>, <Date2>, <Type>)
```

Then replace the placeholders with attributes:

```
DATE_DIFF( {Start Date}, {End Date}, 'DAYS')
```

Numeric and Date Functions

- **Absolute Value**: Returns the absolute value of a specified number. If a number is less than zero, the positive value of the number is returned. If the specified number is equal to or greater than zero, the specified number is returned.

  ```
  ABS(<Number>)
  ```

- **Add Month**: Returns a date offset a specified number of months from the starting date. The date will always fall in the specified month offset. If the starting date has a day value beyond what is in the offset month, the last day of the offset month will be used. For example, EDate (31-Jan-2017, 1) returns (28-Feb-2017). For Months, enter the number of months before or after the start date. A positive value for months yields a future date. A negative value yields a past date.

  ```
  ADD_MONTH(<Start Date>, <Months>, <Length>)
  ```

  Example: `ADD_MONTH(DATE(2017, 2, 15) 3)`

- **Date**: Returns a date value based on specified integer values for the year, month and day.

  ```
  DATE(<Year>, <Month>, <Day>)
  ```


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- **Date Difference**: Returns the difference in years, months, days, hours, minutes, or seconds between two dates. For DATE 1 and DATE 2, the values TODAY and NOW can be used, which denote the current date (with no time component) and date-time, respectively.

  `DATE_DIFF(<Date1>, <Date2>, <Type>)`

  Example: `DATE_DIFF( {Start Date}, 'TODAY', 'DAYS')`

- **Day**: Returns the day value of a date as an integer number

  `DAY(<DATE>)`

- **Extract Text**: Returns the substring within the value, from the specified positions.

  `SUBSTRING(<Value>, <Location>, <Length>)`

  Example: `SUBSTRING( {Name}, 4, 10)`

- **If Then Else**: Allows you to insert a conditional calculation into the scripted calculation. IF_THEN_ELSE calculations can also be nested to support ELSE IF type calculations.

  `IF_THEN_ELSE(<Condition>, <Value1>, <Value2>)`

  Example: `IF_THEN_ELSE( ( {Priority} = 'Low'), 1,IF_THEN_ELSE( ( {Priority} = 'Medium'), 2, IF_THEN_ELSE( ( {Priority} = 'High'), 3, 0)))`

  Example: `IF_THEN_ELSE( ( {Priority} = 'Low'), 'Good'`
  `IF_THEN_ELSE( ( {Priority} = 'Medium'), 'Better'. 'Best') )`

- **Instring**: Returns the index of the substring within the value.

  `INSTRING(<Value1>, <Value to Search>)`

  Example: `INSTRING({Name}, 'a')`

- **Length**: Takes a text value as a parameter and returns an integer which is the number of characters in the text. If the value is empty/null, the calculation will return 0.

  `Length ({<attribute>})`

  Example: `LENGTH("Value")` would return 5, and `LENGTH({Name})` would return the number of characters in the name of the object.

  Use the calculation with SUBSTRING to extract the last four characters of a text value.

  `SUBSTRING( {MyString}, LENGTH ({MyString}) - 4`

- **Lowercase**: Returns the value in lower case

  `LOWERCASE(<Value>)`

  Example: `LOWERCASE ({Task Code})`

- **Maximum**: Returns the maximum value from a list of attributes. There can be any number of parameters.

  `MAX <Value1>, <Value2>, <ValueN>)`

  Example: `MAX( {Scripted Substring Loc a}, {Scripted Substring Loc s}, {Scripted Substring Loc t} )`

- **Minimum**: Returns the minimum value from a list of attributes. There can be any number of parameters.

  `MIN (<Value1>, <Value2>, <ValueN>)`

  Example: `MIN( {Scripted Substring Loc a}, {Scripted Substring Loc s}, {Scripted Substring Loc t} )`

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- **Month**: Returns the month value of a date as an integer number (1-12)

  `MONTH (<DATE>)`

- **Power**: Raises one number to the exponential power of another.

  `POWER(x,y) where x=BASE NUMBER,and y=EXPONENT` and x and y can be attributes or calculations, as long as they are numeric.

  Example: `POWER(3,4)=81`

  > **Note**
  >
  > Fractional values will reduce the number to its root. For example, `POWER(27, 1/3) = 3` the cube root.
  >
  > Negative values will perform an inverse of the exponential calculation. For example, `POWER(2, -2) = 1 / (2^2) = 1 / 4 = .25`.

- **Round**: Returns the value rounded to the decimal places specified

  `ROUND <Attribute>, <Decimal Places>)`

  Example: `ROUND( ({Scripted Substring Loc t} / 7), '4')`

- **Text Location**: Returns the index of the substring within the value, starting at 1 as the first position.

  `INSTRING(<Value>, <Value To Search>)`

  Example: `INSTRING( UPPERCASE( {Name} ), 'TAX' )`

- **Uppercase**: Returns the value in upper case.

  `UPPERCASE(<Value>)`

  Example: `UPPERCASE( {Name} )`

- **Year**: Returns the year value of a date as an integer number.

  `YEAR (<DATE>)`

Calculation Validation

The following validation checks are performed when you save the Calculation:

- The syntax of the Calculation is correct.
- The Attributes and Functions specified in the Calculation exist.
- Parameters to Functions are correct.
- No Circular Loops exist.

Importing List Attributes

To import Task Manager attributes of the List type:

1. Create an import file of List attributes in a TXT file format, with each value on a separate line.

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For example:

```
Blue
Yellow
Red
Green
```

The import option is always "Replace All".

2. On the Home page, click **Application**, and then click **Task Manager**.
3. Click the **Attributes** tab on the left.
4. Select an attribute of type **List**, and click **Edit**.
5. Click the **Import** icon.
6. For **File**, click **Choose File**, find a TXT import file, and then select a **File Delimiter**.
7. Click **Import**. **Import List Values** displays the values: Total List Values, Completed, Errored, List Values Created, and List Values Updated.

   If **Completed Successfully**, click **OK**.

   If **Completed with Errors**, the errors are listed. To export the error list, click **Export to Excel** .

Editing Attributes

You can edit the name of a custom Task Manager attribute. If the attribute type is a List, you can also add, rename, or delete list values.

To edit Task Manager attributes:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Attributes** tab on the left.
3. Select an attribute and click **Edit**.
4. Edit the attribute name.

> **Note**
>
> If the attribute is a List type, you can add, rename, or delete list values. After a custom attribute is saved, you can't change its type.

5. Click **OK**.

   All related templates, schedules, task types, or tasks are updated.

Duplicating Attributes

You can duplicate attributes.

To duplicate attributes:

1. On the Home page, click **Application**, and then click **Task Manager**.

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2. Click the **Attributes** tab on the left.
3. Select an attribute to duplicate, and click **Duplicate**.
4. Click **Close**.

Deleting Attributes

You can delete attributes that you no longer need. When you delete an attribute, the system removes all references to the attribute.

To delete attributes:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Attributes** tab on the left.
3. Select the attributes to delete, and click **Delete**.
4. At the confirmation prompt, click **Yes**.

Viewing Attributes

In Attributes, you can specify which columns to display for the list of attributes, or show all. You can also reorder columns, sort columns by ascending or descending order, and change the column widths.

To display columns:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Attributes** tab on the left.
3. Do one or more of the following tasks:
   - To display all columns, select **View**, then **Columns**, and then select **Show All**.
   - To display specific columns, select **View**, then **Columns**, and then select or deselect the column names.
   - To reorder columns, select **View**, and then **Reorder Columns**, select columns and use the Up or Down arrows or drag them to change the order.
   - To sort columns, hover over a column header until the Sort icons are displayed, and then click **Sort Ascending** or **Sort Descending**.
   - To change column widths, hover over the column header dividers until the arrows display, and drag the columns to the desired width.

Searching for Attributes

You can use the Search function in the Attributes list to find attributes for Task Manager tasks. You can enter full or partial names on which to search. Using the filter bar, you can control the attributes that you see in the list. By default, all attributes are displayed.

To search for Task Manager attributes:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Attributes** tab on the left.
3. To search for an attribute, enter search criteria in the **Search** text box.

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4. **Optional:** From the filter bar, click on a category to display additional search operators such as **Equals**, **Does Not Equal**, **Contains**, **Does Not Contain**, **Starts With**, and **Ends With**.

   You can filter attributes using these categories: **Name**, **Type**, **Calculation**, **Created By**, **Created On**, **Last Updated By**, or **Last Updated On**.

> **Note**
>
> - Click **Add a Filter** to view all categories.
> - To hide the filter bar, click the **Filter** icon.
> - To clear all filters, click **Clear All Filters** in the
>
> icon.

Managing Artifact Labels in Task Manager

Administrators can enter translated labels in different languages for artifact properties like a task's Name, Description, and Task Type Name using the Artifact Labels page. These labels will be displayed to users based on their language preference. If an Artifact Label hasn't been entered for the user's preferred language, then the original label entered in the artifact property, or default artifact label, is shown.

The Artifact Labels list is automatically updated when a new artifact is added to the application.

Artifacts that have the exact label for a property appear only once within the listing. For example, a task named "Review Plan Operating Expenses" appears in multiple monthly schedules. The Artifact Labels list shows only one entry for a Task with the Name - "Review Plan Operating Expenses" and shares all entered language labels.

Editing an artifact property, such as renaming a task, creates a new entry within the artifact label list, which requires new language labels to be entered.

An artifact label is removed from the Artifact Label list when all instances of the artifact referring that label have been removed from the application. For example, all tasks named "Review Plan Operating Expenses" need to be deleted from all the monthly schedules before it's removed from the Artifact Label list.

> **Note**
>
> You must have the **Task Manager - Artifacts – Manage** role to access the Artifact Labels. Additionally, the **Enable Artifact Labels** option must be enabled. For more information, see Enabling Artifact Labels .

View Artifact Labels

To view artifact labels:

1. Go to **Application** > **Task Manager** and click the **Artifact Labels** icon on the left. The Manage Artifact Labels page displays a list of artifact labels with the following information:
   - **Artifact Type**: Type of artifact such as Schedule, Task, and Template.

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- **Property**: The artifact property such as Name or Description.
- **Default**: Default label entered on the artifact.
- **English**: Text to be displayed in English language.
- **Created By**: Indicates the user that created the default artifact label.
- **Last Updated By**: Indicates the last user to update the artifact label.

Optionally, you can click the **Customize Columns** icon and add or remove the required columns. Columns for each supported language is available. When you click **Customize Columns**, the Columns window is displayed. To show a column, in the Hide section, mouse grab the icon for the column name that you want to show and then drag and drop to the required position in the Show section. To show a column, in the Hide section, mouse grab the icon for the column name that you want to hide and then drag and drop to the required position in the Hide section. You can also use the following shorcut keys:

- Focus on a row, then use F2 followed by Space to select or deselect it.
- With focus on a row, use **CTRL+X**, followed by the Up or Down arrow keys to move to another row, then use **CTRL+V** to move the initial row above the current row.

2. Use the **Search** field and the filter clips to search for a required artifact label.

Edit a Artifact Label

To edit an artifact label:

1. On the Artifact Labels page, select the artifact label that you want to edit and click **Edit**. The Edit Languages panel is displayed with a field for each supported language.
2. Specify the translated text for the required language and click **Save**.

Edit Artifact Labels in Bulk

To edit the artifact labels in bulk, you can export the artifact list, update the file, and then import the file. Any translations in the import file updates the existing translations in the UI.

To export the list of artifact labels:

1. On the Artifacts page, click **Export**.
   The displayed artifact labels are exported to a CSV file and downloaded to your system.

   Optionally, you can limit the export to the artifact labels you want to edit by using the search bar and filter chips.
2. In the exported file, add new artifact labels within the language columns, and save the file. The exported file includes four columns that indicate the Artifact Type Id (ArtifactTypeID), Object Type (ObjectType), Property (Property), and the Default Label (DefaultLabel). The values within the rows in these columns can't be modified. Also, you can't remove these four columns.

   Additionally, there are columns for each supported language (LangXx). You can remove the language column that you won't update with artifact labels.

   The following table provides the language column codes.

Language Column Codes		
LangAr = Arabic	LangHe = Hebrew	LangPtBr = Portuguese (Brazil)
LangCs = Czech	LangHu = Hungarian	LangRo= Romanian

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Language Column Codes		
LangDa= Danish	LangIt = Italian	LangRu= Russian
LangDe= German	LangJa = Japanese	LangSv = Swedish
LangEn = English	LangKo= Korean	LangTh= Thai
LangEs = Spanish	LangNl = Dutch	LangTr = Turkish
LangFi = Finnish	LangNo = Norwegian	LangZhCn = Chinese (Simplified)
LangFr = French	LangPl = Polish	LangZhTw = Chinese (Traditional)
LangFrCa = French (Canada)	LangPt = Portuguese	

3. To import the translated artifact labels, click **Import** and select the updated CSV file that contains the translated artifact labels.
Initially a message at the bottom of the page is displayed indicating import has started.

   After some time, import completed successfully is displayed. You can click **View Details** to view the following import details:

   - File name
   - Type of import
   - Importer's name
   - Start time of the import operation
   - End time of the import operation
   - Status of the import:
     - If Success, the number of objects created, deleted, and updated are displayed
     - If Failed, the error and the details are displayed

4. Click **Refresh** to refresh the list of artifact labels.

Managing Task Types

Related Topics

- Task Types
- Creating Task Types
- Viewing Task Type History
- Editing Task Types
- Viewing Task Types
- Searching for Task Types
- Importing Task Types
- Exporting Task Types
- Deleting Task Types

Task Types

Task Types are saved definitions of commonly performed tasks. They identify and categorize tasks commonly performed during a business process, for example, Data Entry or General Ledger Extract. Task Types enable you to set default information, such as settings that must be

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input by the user, and questions or instructions that are shared by all tasks of that type. For example, if you have a set of required questions to be answered by users responsible for data loads, you can create a Data Load Task Type with the required questions. When you create data load tasks, you can select the Data Load Task Type and the questions are automatically included in the tasks.

Two predefined Task Types are installed by default:

- Basic Task: Basic Task Type that does not contain instructions, questions, or attributes.
- Parent Task: Enables you to create parent tasks to define task hierarchies.

To create and manage Task Types, you must be assigned the Service Administrator or Power User security role. Power Users can create their own Task Types, but can only view those of others.

> **Note**
>
> EPM Enterprise subscriptions are configured with additional Task Types for all local integrations for the business process. New Integration Task Types will be automatically added to the service when new predefined integrations are enabled.

See Managing Integrations with Cloud EPM Services and Managing Integrations with Cloud and On-Premises Applications.

Creating Task Types

Task Types enable you to set default information for a task, such as settings that need to be input by the user, or questions or instructions that are shared by all tasks of that type.

To create a Task Type:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Click **New**.
4. See also:
   - Setting Task Type Properties
   - Setting Task Type Parameters
   - Specifying Task Type Instructions
   - Specifying Task Type Questions
   - Working With Task Type Rules
   - Viewing Task Type History

Setting Task Type Properties

The Properties tab enables you to set the name, ID, and description, and associate an Integration Type.

For End User Types, you can allow an Assignee to open a pending task prior to the scheduled time, if all task predecessor conditions have been met.

To set Task Type properties:

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1. Create a new Task Type.

   The **Properties** tab is displayed by default.
2. Enter a **Task Type Name**. You can enter a maximum of 80 characters.
3. Enter a **Task Type ID** that can be used to identify the Task Type. You can enter a maximum of 80 characters.
4. Enter a **Description** for the Task Type. You can enter a maximum of 255 characters.
5. To specify an Integration, click the magnifying glass to search for and select an Integration, then click **OK**. Selecting an Integration enables the Task Type to inherit the parameters from that Integration.
6. **Optional:** For an End-User Type, select **Allow Early Start** to allow the Assignee to open the task before the scheduled start time.
7. See also:
   - Setting Task Type Parameters
   - Specifying Task Type Instructions
   - Specifying Task Type Questions
   - Working With Task Type Rules
   - Viewing Task Type History

Setting Task Type Parameters

The Parameters tab enables you to set the parameters for the Task Type.

> **Note**
>
> The Parameters tab is available only if you have selected an Integration that has parameters that must be defined. Parameter values can be set at the Task Type level or at the task level.

Some tasks contain parameters that need to be changed each time that they are applied to a schedule. For example, date-driven parameters for Data Entry tasks may need to be set to the current month each time they are added to a schedule. When you set Task Type parameters, you can specify which parameters can be overridden during the scheduling process.

To set Task Type parameters:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab, and then click **New**.
3. Enter **Task Type Name** and **Task Type ID**.
4. Select the **Parameters** tab.
5. Enter parameter values as required.
6. **Optional:** To enable the parameter to be overridden at scheduling time, select **Override at Scheduling**.
7. See also:
   - Setting Task Type Properties

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- Specifying Task Type Instructions
- Specifying Task Type Questions
- Working With Task Type Rules
- Viewing Task Type History

Specifying Task Type Instructions

You can create a set of instructions for completing tasks. You can also add references to documents, files, or URLs to websites. For example, you can attach a policy document as a reference, or a link to a product instruction guide.

To specify instructions:

1. On the Home page, click **Application**, and then **Task Manager**.
2. Click the **Task Types** tab, and then click **New**.
3. Enter the **Task Type Name** and **Task Type ID**.
4. Select the **Instructions** tab.
5. In **Instructions**, enter the instruction text.

To add a reference:

1. In the **References** section, click **Add**.
2. From the **Type** list, select one of these types:
   - **Local File**: Click **Choose File** to select and attach the file, enter a **Name**, and then click **OK**.
   - **URL**: Enter the **URL** and a URL **Name**; for example, https://www.oracle.com, Oracle, and then click **OK**.

> **Note**
>
> You can add one or more attachments by using drag and drop functionality available from the **Add Attachments** dialog box. You can rename the attachment in the **Name** field, if desired. If you drag and drop multiple attachments, you can upload them at one time.
>
> You must access the **Add Attachments** dialog box to properly drag and drop attachments.

> **Tip**
>
> To delete a reference, select the reference, and click **Delete**.

3. See also:
   - Setting Task Type Properties
   - Setting Task Type Parameters
   - Specifying Task Type Questions
   - Working With Task Type Rules

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- Viewing Task Type History

Specifying Task Type Questions

You can specify Task Type questions that apply to all tasks with that Task Type.

> **Note**
>
> The Questions tab is not available for a Task Type that uses an automated Integration.

To specify questions:

1. Create a new Task Type.
2. Select the **Questions** tab.
3. Click **New**.
4. From the **New Question** dialog box, for **Question**, enter the text for the question.
5. From the **Type** list, select a question type:
   - **Date**
   - **Date/Time**
   - **Integer**
   - **List**: Enter a list of valid responses to the question.
   - **Multi-Line Text**: The maximum length should be less than 4,000 characters.

     For **Number of Lines**, enter a value between 3 and 50 lines. Multi-Line text determines how many lines of text are visible, without scrolling, on the Actions dialog boxes.

     Select **Attachments** if you want the custom attribute to include an attachments section.
   - **Number**: Select formatting options to override the defaults set in the **Preferences** section of **System Settings**:
     - For **Decimal Places**, enter the number of decimal places to display.
     - Select **Display as Percentage** to display a percent sign.
     - Select **Thousands Separator** to display a thousands separator (for example, 1,000.00). The system displays the thousands separator symbol for the user locale.
     - For **Currency Symbol**, select the currency; for example, $ (United States of America Dollar).
     - For **Negative Number**, select how to display negative numbers; for example, (123).
     - To scale a number, in **Scale**, select from 1000 to 1000000000000.
   - **True/False**
   - **Text**: The maximum length is 255 characters.
   - **User**
   - **Yes/No**

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6. Assign a **Role**. The purpose of assigning a role is to determine which role can answer the question:
   - **Assignee**
   - **Approver**
   - **Owner**
   - **Viewer**

   When re-ordering questions, you can only re-order within a role.
7. If the question is required, select **Required**.

   The **Required** checkbox is disabled for Questions assigned to Owner or Viewer roles.
8. Click **OK** to save the question.
9. **Optional:** To change the order of questions, select a question, then click **Move to Top, Move Up** **Move Down**, or **Move to Bottom.**
10. **Optional:** To edit a question, select the question and click **Edit**. To remove a question, select the question and click **Delete**.
11. Click **Save and Close** to save the Task Type.
12. See also:
    - Setting Task Type Properties
    - Setting Task Type Parameters
    - Specifying Task Type Instructions
    - Working With Task Type Rules
    - Viewing Task Type History

Assigning Task Type Attributes

To locate Task Types in the system, you can apply attributes to the Task Type. When you select an attribute, you can set a value for the attribute based on the attribute type. You can later filter by the attribute value.

For example, you may have a List attribute named Sales Region with the values of North, South, East, and West. The current Task Type applies only to the West Sales Region, so you can add the Sales Region attribute and set it to "West".

To apply an attribute:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left, and then click **New**.
3. Enter the **Task Type Name** and the **Task Type ID**.
4. Select the **Attributes** tab.

   Enables administrators to assign attributes and provide values for the attributes.

   To add an attribute, click **Add**. Enter:
   - **Attribute**: Select an attribute from the list of defined attributes.
   - **Type**: This field is not editable; it's populated by the Attribute.

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- **Value**: Select a value associated with the type of attribute; for example, a numeric value for Formatted Number attribute, a List for List attribute, multiple lines of displayed text without scrolling for Multi-Line Text, a name of a person, User, or Yes or No for the Yes/No attribute.
- **Access**: All roles have view access unless otherwise specified in the Access box.

  To add an access, for each of the roles:

  a. Click ➕ **Add**.
  b. Select a role: **Owner**, **Assignee**, **Approver**, with separate roles for each Approver level currently in use in the application, or **Viewer**.
  c. Select one of the Role access types.
     - **Do Not Display**: Does not see this attribute on the Task Actions dialog box, or in any of the dashboards, list views, or reports.
     - **Allow Edits**: Has the ability to add, change, and remove values for the attribute, but subject to the editability rules.
     - **Required**: Requires a value for the attribute. The **Required** option is available for Assignees and Approvers. Until a value is provided, Assignees are prevented from submitting, and Approvers are prevented from approving.
  d. Click **OK**.

5. Click **OK**.
6. See also:
   - Setting Task Type Properties
   - Setting Task Type Parameters
   - Specifying Task Type Instructions
   - Specifying Task Type Questions
   - Working With Task Type Rules
   - Viewing Task Type History

Working With Task Type Rules

Task Type Rules apply to all Tasks of the same Task Type and thus serve as a convenient mechanism to apply rules to groups of Tasks. Rules configured on the Task Type dialog propagate down to the Edit Task Dialog, appearing on the Rules tab in that dialog, in read-only form for template tasks, and copied to schedule tasks.

Available task type rules:

- **Auto Approve Task** : Automatically completes specified approvals only if specified conditions have been met.
  Example of conditions that could apply to this rule: If attributes have specified values (including calculated attributes)

  When conditions are satisfied, the specified approver levels are marked as complete, thus progressing workflow to the next approval level, or to Closed if no additional approval levels exist.

  Auto Approve Task runs when the Task status changes to Open with Approver.
- **Auto Submit Task**: Causes a task to be submitted automatically if specified conditions are met.

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When conditions are satisfied, the assignee role is marked as complete, thus progressing workflow to the first approval level, or to Closed if no approval levels exist.

Auto-Submit Task rule runs when the Task status changes from Pending to Open with Assignee.

- **Prevent Task Approval**: This rule prevents approval of a task based on attribute values, or other characteristics. This rule runs when the Approver clicks **Approve**.
- **Prevent Task Submission**: This rule prevents submission of a task based on attribute values, or other characteristics. This rule runs when the Assignee clicks **Submit**.
- **Prevent Task Rejection**: This rule prevents rejection of a task based on attribute values, or other characteristics. This rule runs when the Assignee clicks **Submit**.
- **Send Email on Update**: This rule runs based on user actions on the task. Hence, you must set the conditions for the rule appropriately. The most common condition is to have a condition with the Status attribute or the Status Detailed attribute:
  - Status attribute: The only valid value that you can select is Closed. Pending, Open, and Error are all statuses that result from automated actions on the task.
  - Status Detailed attribute: The valid statuses are with the approver.

To work with task type rules:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Click a task type and then click **Edit**.
4. Select the **Rules** tab. You can view the following information:
   - **Order**: The order of precedence
   - **Rule**: Name of the Rule
   - **Conditions**: The choice of what conditions must exist before running the rule
5. To edit a rule, on the Rules tab, click **Edit** and update:
   - **Rule**: Select a rule.
   - **Description**: **Optional.** Explain why you configured the rule and how it should be used.
   - Message (for some rules as noted below):
     - **Message to Approver**: Define an optional message to approver in a **Prevent Task Approval** rule.
     - **Message to Assignee**: Define an optional message to assignee in a **Prevent Task Submission** rule.
   - **Approver Level**: For certain rules, select the rule for **All Levels** or select the Approver levels.

> **Note**
>
> The **Approver Level** must be set on the **Auto Approve Task** rule, and the **Prevent Task Approval** rule.

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- Select **Create Filter** and populate the conditions section or select **Use Saved Filter**, and then select a filter. The filter selected and configured for the rule determines the conditions that trigger the rule to apply.
- For **Conditions**, select one:
  - **Use Saved Filter**: The **Condition** section displays a read-only version of the conditions associated with the saved filter.
  - **Create Filter**: The **Condition** section is enabled.

    Conjunction, **Source**, **Attribute**, **Operator**, and **Value** behave as they do for the existing advanced filter feature.
- For **Filter Task**, specify on which task the conditions should be selected:
  - **Current Task**
  - **Any Predecessor**
  - **Specific Task**: Provide the **Task ID**.

> **Note**
>
> When you make changes to a rule definition, the additional information is added to the audit log. You can view the changes both in the **History** tab as well as the **Audit Report**.

6. See also:
   - Setting Task Type Properties
   - Setting Task Type Parameters
   - Specifying Task Type Instructions
   - Specifying Task Type Questions
   - Viewing Task Type History

Viewing Task Type History

The system maintains a history of Task Type actions, which you can view from the View Task Types dialog box. The History tab displays the components that were updated, the modification type, the old and new values, the user who made the modification, and the change date. The information on the History tab is read-only and cannot be changed.

To view Task Type history:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Select a Task Type and click **Edit**.
4. Select the **History** tab.
5. When you finish, click **Save and Close** or **Cancel**.

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Editing Task Types

From the Task Types dialog box, you can edit Task Types. For example, you can add or delete attributes, edit parameters, add or delete questions, edit instructions, or change the order of questions.

Editing Task Types may have an effect on tasks that are created from the Task Type. For tasks in a template, the effect is immediate. Changes to the Task Type (instructions, questions, and so on), are automatically updated in tasks of that type. There is no effect if you edit a Task Type in a schedule.

To edit a Task Type:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Select the Task Type that you want to edit.
4. Use one of these methods:
   - Select the **Actions** icon to the right of the Task Type, and then select **Edit**.
   - Click the **Edit** icon.
   - Right-click and select **Edit**.
5. Select the tab for the information you want to edit and edit the Task Type.
6. Click **Save and Close**.

Viewing Task Types

In Task Types, you can specify columns to display for the list of Task Types, or show all. You can reorder columns, sort columns by ascending or descending order, or change the column widths.

To display columns:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Do one or more of the following tasks:
   - To display all columns, select **View**, then **Columns**, and then select **Show All**.
   - To display specific columns, select **View**, then **Columns**, and then select or deselect the column names.
   - To reorder columns, select **View**, and then **Reorder Columns**. Select columns and use the Up or Down arrows or drag them to change the order.
   - To sort columns, hover over a column header until the Sort icons are displayed, and then click **Sort Ascending** or **Sort Descending**.
   - To change column widths, hover over the column header dividers until the arrows display, and drag the columns to the desired width.

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Searching for Task Types

You can use the Search function in the Task Types list to find Task Types. You can enter full or partial names on which to search. Using the filter bar, you can control the task types that you see in the list. By default, all task types are displayed.

To search for task types:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. To search for a task type, enter search criteria in the **Search** text box.
4. **Optional:** From the filter bar, click on a category to display additional search operators such as **Equals**, **Does Not Equal**, **Contains**, **Does Not Contain**, **Starts With**, and **Ends With**.

   You can filter task types using these categories: **Name**, **Task Type ID**, **Integration**, **Execution Type**, **Created By**, **Created On**, **Description**, **Last Updated By**, **Last Updated On**, and **User Created**.

> **Note**
>
> - Click **Add a Filter** to view all categories.
> - To hide the filter bar, click the **Hide Filter Bar** icon.
> - To clear all filters, click the **Filters** icon, and then click **Clear All Filters**.

Importing Task Types

You can import Task Types or partial Task Type information from text files. The process is similar to importing tasks into a template.

> **Note**
>
> Task Type rules cannot be imported. Use Migration to import Task Type rules.

To import Task Types, you must have the Service Administrator or Power User security role.

To import Task Types:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Click **Actions**, and then **Import**.
4. Click **Choose File** and navigate to the file.
5. Select an **Import Type**:

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- **Replace**: Completely replaces the definition of a Task Type with the definition in the import file. This option replaces all Task Type detail with the information that is in the file that you are importing. It does not affect Task Types that are not specified in the import file.

  The system displays a warning that task types matching a task type ID in the import file will be modified. If you do not want to overwrite the task type, click **Cancel**.

  > **Note**
  >
  > You can select the **Keep Attachments** checkbox to retain attachments to task types that are being replaced.

- **Update**: Updates partial information for Task Types. This option is not a full replacement of the Task Type details. Only details for the Task Type properties specified in the file are updated.

6. Select a **Date Format**.

   Select a format from the drop-down list of allowed date formats. Date formats are not translated. By default, the date format is set to the locale date format of the exported file location.

7. Select a **File Delimiter** for the import file. Choose **Comma**, **Tab**, or choose **Other** to specify any single character for the **Other Delimiter**.

8. Click **Import**.

Exporting Task Types

You can export Task Types to a comma-separated values (CSV) file, which can be read by Excel, then modified and reimported.

> **Note**
>
> Task Type rules can't be exported. Use Migration to export Task Type rules.

To export Task Types:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Select one or more task types, click **Actions**, and then select **Export**.
4. In the **Export Task Types** dialog, select from the following export options:
   - For **Rows**, select **All Task Types** or **Selected Task Types**.
   - For **Format**, select **Formatted data** or **Unformatted data for future import**.
5. Click **Export**. The file is downloaded in Excel or CSV format depending on the selection: **Formatted data** or **Unformatted data for future import**.
6. Click **Close** to close the Export Task Types dialog.

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Deleting Task Types

You can delete Task Types that you no longer need. To delete a Task Type, you must have Service Administrator or Power User security rights.

> **Note**
>
> You cannot delete a Task Type if tasks belong to it.

To delete a Task Type:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Select the Task Type that you want to delete.
4. Use one of these methods:
   - Select **Actions**, and then **Delete**.
   - Click the **Delete** icon.
   - Right-click and select **Delete.**
5. At the confirmation prompt, click **Yes.**

Managing Task Templates

A task template defines a repeatable set of tasks required for a business process. It is used as a basis for creating schedules. You can create templates for different types of processes.

Template tasks are not assigned to specific days on a calendar, but are defined using generic days, for example, day-3, day-2, day-1, day 0, based on the process activities. You apply the task flow defined in the template to calendar dates when you create schedules.

If you are a Service Administrator or Power User, you can view, create, edit, and delete templates.

> **Note**
>
> The maximum number of task templates that you can create or import is 10,000.

Creating Task Templates

Administrators create templates for repeatable tasks in a business process. You can create templates for different types of business processes, such as monthly or quarterly. For example, you can save a set of tasks required for a monthly close and use the same template every month.

When you create a template, you set template properties such as a name, description and time span. You can specify instructions, attributes, rules, and day labels, and assign owners and viewers.

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Watch this video to learn about creating task templates.

Creating Task Templates.

To create a template:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Click **New**.
4. Enter the necessary information in the template sections:
   - Setting Template Properties
   - Specifying Template Instructions
   - Assigning Viewers to Templates
   - Applying Template Attributes
   - Specifying Day Labels
   - Embedding Templates
   - Working With Template Rules

Setting Template Properties

The Properties tab enables you to specify the template name, description, owner, and time span.

You must assign a template owner to every template, which must be a Service Administrator or Power User. The current ID is the default owner. The template owner is also the default owner of all tasks in the template unless overridden at the task level.

Tasks in a template are organized by numeric template days. Each template has a Day Zero. You can specify the number of days a template tracks before and after the Zero day. Days before Day Zero are represented as Day-5, Day-4, and so on. Days after Day Zero are represented as Day 3, Day 4, and so on. A template has at least one day (Day Zero), even if you do not specify days before or after it.

To set template properties:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. From the **New** or **Edit Template** dialog, on the **Properties** tab, enter:
   - **Name**: You can enter a maximum of 80 characters.
   - **Description**: You can enter a maximum of 255 characters.
   - **Organizational Unit**
4. Enter the time span for the template.
   - For **Number of Days Before Day 0,** select a number.
   - For **Number of Days After Day 0,** select a number.
5. **Optional:** Select **Embedded Only** if the template is only to be used as an embedded template.

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If this option is enabled, the template is not allowed to be deployed. This prevents supporting templates from being deployed by mistake.

6. For **Owner**, use the default owner or click **Select Owner** . Administrators, or Power Users configured with the Administrator or Power User roles are the only roles that will display when you click **Search** to select a user to assign as an Owner. Any member of the Shared Services group can perform the role, but the same person cannot perform more than one role.
7. Enter the necessary information in the template sections:
   - Specifying Template Instructions
   - Assigning Viewers to Templates
   - Applying Template Attributes
   - Specifying Day Labels
   - Embedding Templates
   - Working With Template Rules

Specifying Template Instructions

You can specify instructions in the template for completing the business process. Users can view the instructions from tasks in the template. The instructions are also transferred to each schedule.

You may want to supply additional detail to help users understand the purpose of the template and how to complete the tasks. You can attach additional references.

You can easily see if a task includes instructions within the Schedule Task and Template Task lists. You can add the Instructions column to your list view and see the textual instructions for the task.

To specify template instructions:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. From the New or Edit Template dialog, select the **Instructions** tab.
4. In **Instructions**, enter instruction text for each task in the template to include.
5. Click **OK**.

To add a reference:

1. In the **References** section, click **Add**.
2. From the **Reference Type** list, select one of these types:
   - **Local File**: Click **Choose File** to select and attach the file, enter a name, and click **OK**.
   - **URL**: Enter the URL, then enter a URL name; for example: https://www.oracle.com., Oracle, and then click **OK**.
3. Enter the necessary information in the template sections:
   - Setting Template Properties
   - Assigning Viewers to Templates
   - Applying Template Attributes

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- Specifying Day Labels
- Embedding Templates
- Working With Template Rules

Assigning Viewers to Templates

The Viewers tab enables you to assign viewer rights to users who may view tasks in the schedule and schedule tasks that are generated from the template. You can assign multiple users to a template as viewers. Viewers can be a team or group assignment. A Power User viewer can view the template details and template tasks as read-only. A Viewer has no template access, and only read-only access to schedules produced from the template.

> **Note**
>
> Only users who have the task Viewer role can respond to questions.

To assign viewer rights:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. From the New or Edit Template dialog, select the **Viewers** tab.
4. Click **Add**.
5. To search by users or groups, click **Search Users**, then select **Users** or **Groups**.
6. Enter a full or partial user name, then click **Search**.
7. To specifically identify a user, click **Advanced**, then enter a **User ID** or **Email** address.
8. From the **Search Results** list, select users, then move them to the **Selected** pane.
9. For additional details about the user, such as groups or roles, click **Details**.
10. Click **Add** or **Add All** to move users to the **Selected** list.

> **Tip**
>
> To remove users, select them, and then click **Remove** or **Remove All**.

11. Enter the necessary information in the template sections:
    - Setting Template Properties
    - Specifying Template Instructions
    - Applying Template Attributes
    - Specifying Day Labels
    - Embedding Templates
    - Working With Template Rules

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Applying Template Attributes

To locate templates in the system, you can apply attributes to the template. When you select an attribute, you can set a value for the attribute based on the attribute type. You can later filter by the attribute value.

For example, you may have a List attribute named Sales Region with the values of North, South, East, and West. The current template applies only to the West Sales Region, so you can add the Sales Region attribute and set it to "West".

To apply an attribute:

1. On the Home page, click **Application,** and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. From the **New Template** or **Edit Template** dialog, select the **Attributes** tab.
4. Click **Add**.
5. From the **Attribute** list, select an attribute. The **Type** field is not editable; it's populated by the Attribute.
6. For **Value**, depending on the attribute, select a value for the attribute from a drop-down list, or enter a value.
7. Click **OK**.
8. Enter the necessary information in the template sections:
   - Setting Template Properties
   - Specifying Template Instructions
   - Assigning Viewers to Templates
   - Specifying Day Labels
   - Embedding Templates
   - Working With Template Rules

Specifying Day Labels

You can customize the names of days in the template. Editing the name of the day does not change its sequential order.

Day Labels created in a template are copied to any schedule created from the template.

To specify day labels:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. From the **New Template** or **Edit Template** dialog, select the **Day Labels** tab.
4. Select a day to rename.
5. Rename the day from its default value to a custom name. You can enter a maximum of 20 characters.
6. Click **OK**.
7. Enter the necessary information in the template sections:

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- Setting Template Properties
- Specifying Template Instructions
- Assigning Viewers to Templates
- Applying Template Attributes
- Embedding Templates
- Working With Template Rules

Embedding Templates

You can reuse tasks from one task template in another by embedding one template into another template. When you schedule tasks for a template, tasks for embedded templates are also scheduled. For example, you may have two templates that are essentially the same, with some additional tasks for one of the templates. You can specify that one template embeds the second template so if you change one template, you don't need to update the other one. When you generate a schedule, it will include the tasks from the other template.

For example, you can have a core set of monthly tasks and you can extend that for your quarterly close by creating a new template called quarterly tasks and embed the monthly tasks and add a few additional tasks. Additionally, they can create an Annual template and embed the quarterly template and add additional new tasks as needed.

When you select a template to embed within your template, you can have the option to make the embedded template's tasks editable within your template. When you embed a template as editable, you can edit the embedded template tasks allowing overrides on the task properties within the containing template. You can create an embed only template with the core set of common tasks, which multiple groups can leverage for their business process. Each group can then create a master template and embed the core template in their master template. Additional tasks specific to their group can be added in each of the master templates. They can make changes to the embedded tasks in the master template. The overridden fields are tracked in each master template.

Any changes to the core tasks that are not already overridden will appear immediately in the master template. Instead of deletion, the user can inactivate a core task in a particular master template so that it will not be deployed to a schedule. If any tasks are added to the core set, those will automatically flow into the different group's templates within the organization.

The following image shows how the fields are edited and overridden from the core template to the master templates:

- **Core Template**: Updates to the first field in the Load Data Task and the last field in the Consolidate Task will be carried through to the master Templates unless otherwise overridden.
- **Western Template**: First field of the Consolidate task is overridden.
- **Central Template**: Override to the first field of the Load Data Task will be used even though it has been updated in the Core Template.
- **Eastern Template**: All fields of the Consolidate Task have been overridden and these values will be used.

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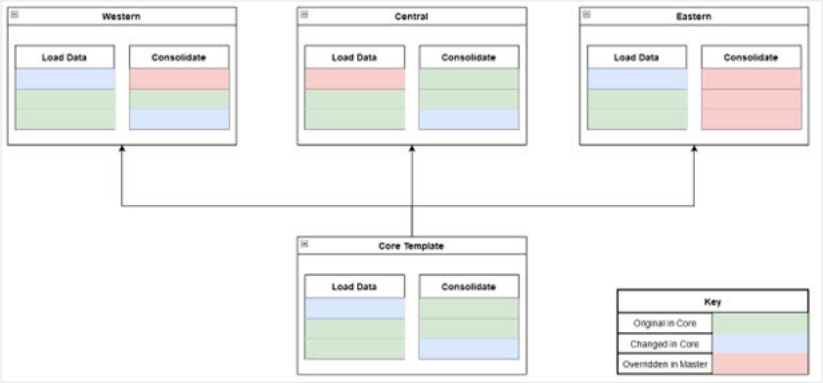

> **Note**
>
> You can embed multiple levels of templates. For example, if Template A is embedded in Template B, then Template B can be embedded in another template. However, you can only edit the template up one level.

Embedded templates are task templates that are not designed to be deployed directly, but rather to be used by other templates that are directly deployed. Use the following columns to identify embedded templates:

- **Embedded**: A list of templates that are embedded in the current template. Blank if none.
- **Embedded In**: A list of templates that the current template is embedded in. Blank if none.

> **Note**
>
> These columns are not displayed by default, but are available from the **View** menu.

To embed templates:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. To identify previously embedded templates, click **View**, then **Columns**, and then select **Embedded In** or **Embedded Templates**.
4. In the **New Template** or **Edit Template** dialog, select the **Embedded Templates** tab.
5. From the **Templates** list, select a template to embed.
6. From **Embedded Mode** list, select one of the following:
   - View Only
   - Editable

> **Tip**
>
> To remove a template or templates, select the template from the **Embedded Template Name** list and click **Remove**.

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7. Click **OK**.
8. Enter the necessary information in the template sections:
   - Setting Template Properties
   - Specifying Template Instructions
   - Assigning Viewers to Templates
   - Applying Template Attributes
   - Specifying Day Labels
   - Working With Template Rules

   When editing a task within an embedded template:
   - On the **Properties** tab, you can change the attribute value for all the properties except the **Task ID** and **Task Type**. When you change a value, the **Overridden** column displays **Yes** for the attribute.
   - On the **Instructions** tab, if the **References** section displays references from the core set of tasks, then you can't modify the references. However, you can add new references.
   - On the **Workflow** tab, if the embedded task has two approvers, you can't delete them because two approvers are mandatory. If you add a new approver, then you can delete the one of the existing approvers.
   - On the **Questions** tab, any questions from the core set are displayed as locked and you can't modify them. But you can add new questions. The same applies to the **Viewers** and **Rules** tabs.
   - On the **Predecessors** tab, all predecessors are defined in the core template, and you can't add new predecessors.
   - On the **Attributes** tab, you can add new and edit existing attributes. You can't reorder the attributes inherited from the core tasks, but the newly added attributes could be re-ordered.
   - You can't change the children for an embedded parent task.

   To clear overrides, select the required template task and click **Actions** > **Clear overrides**.

Working With Template Rules

Template rules apply to all tasks in the template, and rules are also applied to groups of tasks. Rules that you configure in the **Template** dialog box apply to the **Schedule** dialog box and the **Edit Task** dialog box, and they are displayed on the **Rules** tabs, in read-only form.

Template rules:

- **Auto Approve Task**: Automatically completes specified approvals if specified conditions have been met.
  Examples of conditions that could apply to this rule:
  - Attributes have specified values (including calculated attributes)
  - Predecessor task contains an attribute with a specified value

  When conditions are satisfied, the rule causes the specified approver levels to be marked as complete, and workflow progresses to the next approval level or is closed if no additional approval levels exist.

  This rule runs when the Task status changes to Open with Approver.

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- **Auto Submit Task**: Automatically submits a task if specified conditions are met. When conditions have been satisfied, the rule causes the assignee role to be marked as complete, thus progressing workflow to the first approval level, or to Closed if no approval levels exist.

  > **Note**
  >
  > – Auto Submit Task runs when the Task status changes from Pending to Open with Assignee. Auto Submit Task rule honors predecessor relationship. It will only run when Finish-To-Finish predecessors have completed without error and Finish Error-To-Finish predecessors have completed. When a Finish-To-Finish predecessor is Closed by either user or rules, it should check for its Running successors and trigger Auto Submit Task rules if necessary.
  >
  > – Auto Submit Task rule is not triggered when the task has missing parameters.

- **Prevent Task Approval**: This rule prevents approval of a task based on attribute values, or other characteristics. This rule runs when the Approver clicks **Approve** or when the approver clicks **Actions** > **Update**.
- **Prevent Task Submission**: This rule prevents submission of a task based on attribute values, or other characteristics. This rule runs when the Assignee clicks **Submit** or when the assignee clicks **Actions** > **Update**.
- **Prevent Task Rejection**: This rule prevents rejection of a task based on attribute values, or other characteristics. This rule runs when the Assignee clicks **Submit** or when the assignee clicks **Actions** > **Update**.
- **Send Email on Update**: This rule runs based on user actions on the task. Hence, you must set the conditions for the rule appropriately. The most common condition is to have a condition with the Status attribute or the Status Detailed attribute:
  - Status attribute: The only valid value that you can select is Closed. Pending, Open, and Error are all statuses that result from automated actions on the task.
  - Status Detailed attribute: The valid statuses are with the approver.

  The rule runs when a user using the task action dialog:
  - Claims, releases, submits, approves, rejects, reopens, force-starts, or force-closes a task
  - Updates an attribute value
  - Answers a question
  - Adds a comment
  - Deletes a comment
  - Adds a reference attachment

To work with template rules:

1. On the Home page, click **Application**, then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Open a template.
4. Click the **Rules** tab. You can view the following information about the rules that are associated with a template:
   - **Order**: Order of precedence

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- **Rule**: Name of the rule
- **Conditions**: Conditions that must exist before running the rule

5. To create or edit a rule, on the **Rules** tab, click the **New** icon or select an existing rule and click the **Edit** icon.
6. On the **Properties** tab:
   - **Rule**: Select a rule.

     > **Note**
     >
     > When you make changes to a rule definition, the additional information is added to the audit log. You can view the changes both in the **History** tab as well as the **Audit Report**.

   - **Description**: **Optional.** Explain why you configured the rule and how it should be used. You can enter a maximum of 255 characters.
   - **Run When**: Indicates the events associated with the rule.
   - **Approver Level**: Select **All Levels** or select the Approver levels.
7. On the **Filter** tab:
   - Select **Create Filter** and populate the conditions section or select **Use Saved Filter**, and then select a filter. The filter selected and configured for the rule determines the conditions that trigger the rule to apply.
   - **Filter Task**: Specify for which task the condition should be selected:
     - **Current Task**
     - **Any Predecessor**
     - **Specific Task**: Provide the **Task ID**
   - **Conditions**:
     - **Use Saved Filter**: The **Conditions** section displays a read-only version of the conditions associated with the saved filter.
     - **Create Filter**: The **Conditions** section is enabled and you can click **Create Condition** or **Create Condition Group** to specify conditions.

       Conjunction, **Source**, **Attribute**, **Operator**, and **Value** behave as they do for the existing advanced filter feature.
8. To delete, duplicate, or reorder a rule, click the appropriate button and follow the instructions.
9. Enter the necessary information in the template sections:
   - Setting Template Properties
   - Specifying Template Instructions
   - Assigning Viewers to Templates
   - Applying Template Attributes
   - Specifying Day Labels
   - Embedding Templates

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Viewing Template History

The system maintains a history of template actions, which you can view from the Template dialog box. The History tab displays the components that were created or updated, the modification type, the old and new values, the user who made the modification, and the change date. The information on the History tab is read-only.

To view template history:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Select a template for which to view history and click **Edit**.
4. Select the **History** tab.
5. Click **OK**.

Opening Templates

You can open and work with templates from the Templates dialog box.

To open a template:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. From **Templates**, select a template.
4. Use one of these methods:
   - Click the **Actions** icon to the right of the template, and then select **Edit**.
   - Click the template.
   - Right-click and then select **Edit**.

Adding Tasks to Templates

You can add tasks to Task Manager templates. Each task has a start date and time. End-User tasks also have an end date and time when added to a template. The start day in a template corresponds to template days, rather than to calendar days.

> **Note**
>
> The maximum number of tasks that you can add to a Task Manager template is 500,000.

To add a task to a Task Manager template:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Select and open a template.

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If you have a large list of templates, enter a full or partial name in the **Search** box to search for templates from a list in the selection dialog.

4. Add tasks as required.
5. Click **OK**.

Editing Templates

You can edit Task Manager templates to change the properties, such as the name of a template and the number of days before and after Day Zero. If you reduce the number of days before or after the Day Zero, tasks associated with those days are removed. Editing a template has no effect on schedules previously generated from that template.

You can edit, copy, and delete tasks in a template.

You must be working in the source template to edit tasks from embedded templates.

To edit a Task Manager template:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Select a template and click **Edit**.
4. Edit the template properties.
5. Click **OK**.

Importing Tasks into Templates

You can import tasks, or partial task data, from text files such as a Microsoft Excel spreadsheet, into a template. For example, you can enter task definitions as rows in Excel, save the file as a comma-separated values (CSV) file, and then import it into a template. You can also use the Import feature to quickly add many repetitive tasks by editing the fields in a text file and importing it, rather than creating individual tasks.

> **Note**
>
> Before you import tasks into a template, you must specify task IDs that are unique in the template to ensure that no two tasks have the same ID.

To import tasks, you must have security rights to edit the template.

These options are available for importing tasks into a template:

- **Replace**: Replaces the definition of a task with the definition in the import file. This option replaces the task detail with the detail that is in the file that you are importing. It does not affect other tasks in the template that are not specified in the import file.
- **Update**: Updates partial information for tasks. For example, in the import file, you might have made changes to task instructions, reassigned Owners, Assignees, and Approvers, or removed some attributes and added new attributes. You might also have made the same change to a large number of tasks, for example, adding a new attribute to 100 of 400 tasks. The update option is not a full replacement of the task details. Only details for the task properties specified in the file are updated. For example, if the import file has only a

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column for a task instructions, the task name, Assignee, attributes, and other properties are not affected.

- **Delete**: Deletes tasks and associated information based on a list of task IDs provided in a file. The required information to perform a delete is a file with a column of task IDs.

To import tasks into a template:

1. On the Home page, click **Application**, and then click **Task Manager**,
2. Click the **Templates** tab on the left.
3. Select a template into which to import tasks.
4. Click **Import Tasks**.
5. Enter the name of the file to import, or click **Choose File** to find the file.
6. Select an import option:
   - **Replace**: Replaces the definition of a task with the definition in the import files. It does not affect other tasks that are not specified in the import file.

     > **Note**
     >
     > You can select the **Keep Attachments** checkbox to retain attachments to tasks that are being replaced.

   - **Update**: Updates partial information for tasks. For example, in the import file, you might have made changes to task instructions, reassigned Owners, Assignees, and Approvers, or removed some attributes and added new attributes. You might also have made the same change to a large number of tasks, for example, adding a new attribute to 100 of 400 tasks. The update option is not a full replacement of the task details. Only details for the task properties specified in the file are updated. For example, if the import file has only a column for a task instructions, the task name, Assignee, attributes, and other properties are not affected.

     > **Note**
     >
     > The system displays a warning that tasks in the template that match a task ID in the import file will be modified. If you do not want to overwrite the task, click **Cancel** to cancel the import process.

   - **Delete**: Deletes tasks and associated information based on a list of task IDs provided in a file. The required information to perform a delete is a file with a column of task IDs.
7. Select a **Date Format**.

   Select a format from the drop-down list of allowed date formats. Date formats are not translated. By default, the date format is set to the locale date format of the exported file location.
8. Select a **File Delimiter** for the import file. Select **Comma** or **Tab**. Select **Other** to specify any other single character as the **Other Delimiter**.
9. Click **Import**.

- If the import is successful, the Import Success dialog box is displayed, indicating the template name, the name of the file containing the tasks, and the total number of tasks imported. Click **OK**.

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- If errors are found, the import process is not run, and the Import Errors dialog box displays errors. View the errors, then click **OK** to return to the Templates page.

Table 11-13 Troubleshooting Import Errors

Error	Resolution
Duplicate Task ID	Check for duplicate Task IDs. Task IDs must be unique within the template or schedule. See Creating Tasks.
Task Name is too long	The Task Name can contain a maximum of 80 characters. See Setting Task Properties.
Organizational Value - "Invalid Value"	The Organizational Value cannot be blank, contain bullet points or multiple lines.

Task Import File Format

The task import file enables you to specify the task information that you want to import into a template. The first row contains only the column headings that identify the data in that column, for example, TaskName. Separate rows are used to define each task.

Each task in the file must be given a unique ID in the TaskID column. You use the numeric IDs to uniquely identify predecessors and parent tasks. For example, when you enter a parent task in the Parent column, you enter the numeric ID of the task rather than the task name.

When you use the Replace or Replace All options, you can add, remove, and shift the columns but you cannot remove required columns. When you use the Update option, there are no required columns except TaskID, or TaskTypeID when you are importing Task Types. The data in each column must match the column ID in the first row.

Many column IDs may have an indeterminate amount of values. For example, a task may have many viewers. You can add parameters by modifying the ending number, for example, Viewer4, Viewer5, and so on.

The numbers that you append to the column IDs for Approvers, Questions, or References only represent the relative order with the other approvers, questions, or references. When you update these, the system matches by the name of the approver, question, or reference without regard to the number in the file format column. To replace question text, approvers, or reference names, you must use Replace mode. When you select Update mode, the system only adds a new question, approver, or reference if the file names do not match.

All Text Lookup rows in the file must match values stored in the application. For example, if you are importing Task Types, you must have specified values for them when you set up the application. For groups of column headings, if you have a certain number of one of the group, you must have the same number for other members. For example, if you have QText5, there must be a corresponding QType5.

> **Note**
>
> When you need to create multiple lines of text in the import file (for example, in Instructions), use this syntax to indicate a new line: '\n'. Do not use hard returns to separate lines of text.

Column ID	Description	Type
TaskID	ID of task. Maximum of 80 characters.	Text

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Column ID	Description	Type
TaskName	Name of task	Text
TaskType	Task Type	Text Lookup
Description	Full description of task. Maximum of 1000 characters.	Text
Priority	Task priority (High, Medium, Low)	Text Lookup
StartDate	Start date, in template days (for example, -5, 0, 1, 12) For schedules, the start date must be a calendar date in the locale format.	Integer
StartTime	Start time in HH:MM (24-hour), for example: 13:00	Integer
EndDate	User tasks only, in template days (for example, -5, 0, 1, 12) For schedules, the end date must be a calendar date in the locale format.	Integer
EndTime	User tasks only, in HH:MM (24-hour), for example, 13:00	Integer
Duration	Task duration in minutes (automated tasks only)	Number
Owner	Task owner	Text Lookup
Assignee	Task Assignee (user tasks only)	Text Lookup
Active	Whether the task is active. Only applies to template import.	Boolean
StartEarly	Whether the Assignee is allowed to open a pending task prior to the scheduled start time	Boolean
RunAs	RunAs user name. Only applicable for system-automated tasks.	Text Lookup
Instruction	Instruction text. Maximum of 2 GB.	Text
Reference#	Instruction reference (URL and Content Server references)	URL
RefText#	Reference name, for URL references	Text
RefType#	Reference type (URL, Document)	Text Lookup
RefURL#	Reference URL, for URL references	URL
RefDocId#	Reference document ID, for document references	Integer
Approver#	Approver Level	User Name
Viewer#	Task viewer	User Name
QText#	Required for question. Text of question. Maximum of 255 characters.	Text
QType#	Required for question. Data type of question (Text, Member, True/False, and so on)	Text Lookup
QReq#	Required for question. Question required? (Yes or No)	Boolean

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Column ID	Description	Type
QList#	List of values for question, separated by | symbol, for example, Red|Green|Blue|Yellow.	Text Lookup
QCurrSymbol#	Currency format for question X	Text
QDecPlaces#	Number of significant digits after the decimal places for question X	Number
QNegNumFmt#	Negative number format for question X	Text: - or ( )
QScale#	Scale format for question X	Number, in magnitude: 1000 and so on
QThouSep#	Thousand separator indicator for question X	Boolean
Attribute#	Attribute name, for example, Color	Text Lookup
AttrVal#	Attribute value, for example, Red. Maximum of 255 characters.	Text
AttrDisplay	Display to user (in task action dialog box) for attribute	Boolean
Param#	Parameter name	Text Lookup
ParmVal#	Parameter value	Text
Parent	Task ID of the task parent	Task ID
Predecessor#	Predecessor task ID (from column A)	Task ID
PredType#	Required. Predecessor type (for example, Finish to Start).	Text Lookup
PredTemplate#	Name of the predecessor template or schedule, if the predecessor belongs to a different template or schedule	Text

When you import Task Types, these columns are used:

- TaskTypeID
- IntegrationType
- ParamOverride# - whether the parameter can be overridden

A sample task import file is provided. It contains these dynamic parameters, which you can modify:

- Two questions
- Two attributes
- Two instructional references
- Three predecessor tasks
- Three approver levels

Exporting Tasks to Microsoft Excel

You can export tasks from a template to a flat file in CSV format, then use Microsoft Excel to modify the file. For example, you can export the tasks from a template in a test installation and import them to a template in a production installation. When you export tasks from a template, they are saved to a CSV file.

To export tasks:

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1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Click the **Tasks** tab.
4. Select one or more tasks and from the **Actions** menu, click **Export**. The file is downloaded in Excel or CSV format depending on the selection: Formatted data (visible columns only) or Unformatted data for future import.
5. Click **Close** to close the Export Tasks dialog box.

Reassigning Users in Templates

You may periodically need to reassign users to different tasks. For example, you may create a template and assign a user to certain tasks; however, later that employee leaves the company and another employee assumes those tasks. You can use the Reassign feature in the Templates dialog box to automatically change assignments instead of manually searching for, opening, and editing individual tasks. The Reassign feature enables you to quickly find the tasks associated with one user and reassign them to another.

You can reassign users for multiple templates at a time. This can be helpful if your application has a large number of templates.

You can select the roles for which to reassign users. For example, if you select the Owner role, the system reassigns the user only in tasks in which the user was assigned as the owner.

When tasks are reassigned, email notifications are sent immediately to reassigned users.

To reassign users:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Select one for more templates for which you want to reassign users.
4. Select **Actions**, and then **Reassign User**.
5. For **Find User**, enter the search criteria for the user that you want to replace:
   a. Click **Find User**, then from the **Select User** dialog box, enter the first or last name for the user, and then click **Search**.
   b. Select the user from the results, then click **OK**.
6. Click **Replace With** and enter search criteria for the user to whom you want to reassign tasks.
7. Select roles for the user that must be reassigned:
   - **Owner**
   - **Assignee**
   - **Approver**
   - **Viewer**
8. Click **Reassign**.

   When the process ends, the system displays a "Reassign Users - Success" message that indicates that the user reassignment is complete, and displays the template name and total number of user reassignments made.

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Creating Schedules from Templates

You can create schedules from templates. Creating a schedule from a template populates the schedule with the tasks in the template. If the source template has embedded templates, the tasks in the embedded templates are included. Template instructions, viewers, and attributes are also added to the schedule.

To create a schedule from a template, you must have the Service Administrator or Power User role.

When tasks are added from a template to a schedule, they are assigned calendar dates based on their relationship to the Day Zero template date. Calendar dates can account for non-working days of the schedule if required. For example, if you exclude weekend days in your schedule, if Day 0 is a Friday, the tasks for Day 1 are applied to the following Monday instead of Saturday. The Date Map displays the calendar date assigned to each template day and you can use it to adjust date assignments.

> **Tip**
>
> You can create one global template and use that template to create the appropriate schedules for each regional entity, taking into consideration the holiday calendars, time zones, and work day configurations of the region.

A task in one template can have predecessor tasks in other templates. When you create a schedule from a template, you can select predecessor tasks in schedules as predecessors for tasks with predecessors in other templates. You use the Predecessor Task Links tab to select predecessor tasks in schedules.

By default, a new schedule has a status of Pending, which enables you to make changes as needed.

You can also use a REST API to create Schedules from Templates. This enables you to create an automated routine to generate all of your periodic task schedules. For more information, see Deploy Task Manager Templates in *REST APIs* .

To create a schedule from a template:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Select the template that you want to schedule.
4. If the template was successfully validated, create the schedule. Click the **Actions** icon on the right side, and then select **Create Schedule**, or click the **Create Schedule** icon.
5. On the **Schedule Parameters** tab, enter the following information:
   - **Schedule**: The name of the schedule.
   - **Year**: Select the schedule year.
   - **Period**: Select a period for the schedule.
   - **Day Zero Date**: Select the calendar date to assign to the template Day Zero.
   - **Date Mapping**:
     - **Organizational Unit** - Select an option:

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    - **Use Task Value**: Select a value from the Organizational Unit.
    - **Set Value To**: The selected organizational unit is applied to all tasks in the Schedule (even if the task in the template contained a different Organizational Unit value). Since all tasks have the same Organizational Unit, the Organizational Unit in the Date Mapping panel on the Create Schedule from Template dialog box is Read Only.
  - **Calendar days**
  - **Working days** - By default, Monday - Friday are working days, and Saturday and Sunday are non-working days.
  - **Advanced Configuration** - Select this option to override the **Organizational Unit** and **Work Days** configuration for all tasks in the schedule.
  - **Date Map**: To modify a date assignment, enter a new date in the **Date** column.

    Date mapping may vary by Organization Unit. If there is no Organization Unit assigned to the template level, the "Unassigned" day mapping will be applied to the schedule. If the template has an Organization Unit assigned, then the day mappings for that Organization Unit are assigned to the schedule.

    For multiple days mapped to the same date, the Day Labels will be concatenated with a comma. Non-working days have a blank Day Label.

6. **Optional:** Select the **Predecessor Task Links** tab, select the schedule that contains the predecessor task, then in the **Assigned Task** list, select the predecessor task.
7. **Optional:** Select the **Override Parameters** tab, review the parameters, and in the **New Value** column, for each parameter that you want to modify, enter a new value.

   > **Note**
   >
   > The **Override Parameters** tab displays only tasks that use a Task Type in which at least one parameter has the **Override** option enabled.

8. Click **Create Schedule**.

   A schedule is created and populated with the tasks from the template, and a status of Pending.

   You can view a list of schedules on the Schedule Tasks page.

   When deploying a template, only rules of the main template is copied to the deployed schedule. Rules of the embedded template isn't copied to the schedule or schedule tasks. Autosubmit rules are automatically run when schedule tasks are changing from Pending to Open with Assignee status, or when schedule tasks are re-opened. To understand the rules that will be run after deploying a template to schedule, you can click **Edit** to view the rules of a specific schedule task in **Tasks** > **Schedule Tasks** list.

   When deploying template tasks, by default, the timezone of the current user is used for scheduling unless the template task is associated with an Organizational Unit. When the Organizational Unit has a specified timezone, that timezone is used for scheduling the task.

Using the Override Parameter Option When Creating Schedules from Templates

It’s common for an integrated task to have parameters that you must update periodically. Many integrated tasks have a **Period** parameter that varies between task executions in different

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schedules. For example, the **Run Data Rule** integrated task's **Start Period** and **End Period** parameters may need to be modified within different schedules created from the template.

The **Override Parameter** option allows a quick way to update these parameters, which you must update periodically.

Initially, designate the integration parameters that you must update at scheduling time by selecting the **Override at Scheduling** property in the task type. For new task types, see Setting Task Type Parameters. For existing task types, see Editing Task Types.

For example, edit the **Run Data Rule** task type. On the **Parameters** tab, select the **Override at Scheduling** option for **Start Period** and **End Period**, and then save.

Secondly, when adding the integrated tasks to your template, select the task type and populate the required parameters. For the parameters designated for override, you can provide the default values or leave them blank. See Adding Tasks to Templates in *Administering Financial Consolidation and Close*.

For example, add one or more **Run Data Rule** tasks to your template by selecting the **Run Data Rule** task type. On the **Parameters** tab, enter **Jan** for **Start Period** and **Feb** for **End Period**.

Lastly, when creating a schedule from template, select the **Override at Scheduling** tab. Within the table, all parameters that are designated for an override at scheduling will be listed with their current values. You can enter a new parameter value, which will be set for all tasks using the task type.

For example, when creating a schedule from the template, the **Override at Scheduling** tab will show the Run Data Rule **Start Period** and **End Period** parameters with their current values. You can override the parameter by updating the values within the table.

Table 11-14 Override Parameters at Scheduling

Task Type Name	Parameter Name	Current Value	New Value
Run Data Rule	Start Period	Jan	Feb
Run Data Rule	End Period	Feb	Mar

When the schedule is created, the Run Data Rule tasks will be set with **Start Period** as **Feb** and **End Period** as **Mar**.

Manually Validating Templates

When you generate a schedule from a template, the system automatically validates the template first to check for uniqueness violations, tasks that directly or indirectly depend on each other, or a date-precedence mismatch. For example, you cannot have circular predecessors.

You can manually validate templates. Validation is particularly useful to check for problems after you import tasks into a template from a file.

The validation results show the template names, status, and error messages. You can sort the list of validation results by Template Name or Status.

To manually validate templates:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.

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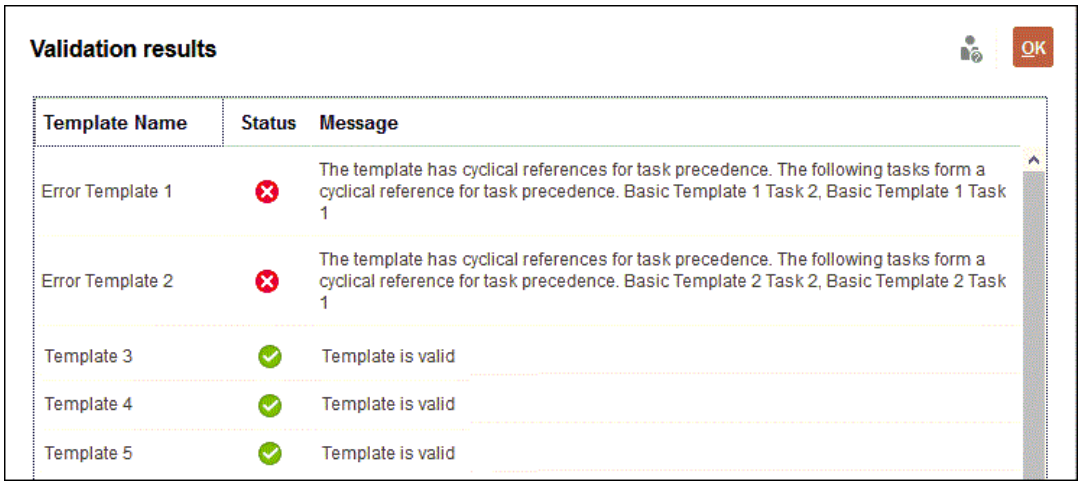
3. From **Templates**, select one or more templates to validate.
4. Click **Validate**.

   If no errors exist, the validation results show a "Template is valid" message. If errors exist, it shows the error details.



Viewing Task Manager Templates

In the Templates dialog box, you can specify which columns to display for the list of templates, or show all. You can also reorder columns, sort columns by ascending or descending order, or change the column widths.

To display columns:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Do one or more of the following tasks:
   - To display all columns, select **View**, then **Columns**, and then select **Show All**.
   - To display specific columns, select **View**, then **Columns**, and then select or deselect the column names.
   - To reorder columns, select **View**, and then **Reorder Columns**. Select columns and then use the Up or Down arrows or drag them to change the order.
   - To sort columns, hover over a column header until the Sort icons are displayed, and then click **Sort Ascending** or **Sort Descending**.
   - To change column widths, hover over the column header dividers until the arrows display, and drag the columns to the desired width.

Searching for Templates

You can use the Search function in a Templates list to quickly find Task Manager templates. You can enter full or partial names on which to search. Using the filter bar, you can control the templates that you see in the list. By default, all templates are displayed.

To search for templates:

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1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. To search for a template, enter search criteria in the **Search** text box.
4. **Optional:** From the filter bar, click on a category to display additional search operators such as **Equals**, **Does Not Equal**, **Contains**, **Does Not Contain**, **Starts With**, and **Ends With**.

   You can filter templates using these categories: **Name**, **Owner**, **Organizational Unit**, **Days After Day 0**, **Days Before Day 0**, **Embedded In**, **Embedded Only**, **Embedded Templates**, **Description**, **Created By**, **Created On**, **Last Updated By**, or **Last Updated On**.

> **Note**
>
> - Click **Add a Filter** to view all categories.
> - To hide the filter bar, click the **Hide Filter Bar** icon.
> - To clear all filters, click the **Filters** icon, and then click **Clear All Filters**.

Deleting Templates

You can delete Task Manager templates that you no longer need. To delete a template, you must have security rights to the template.

Deleting a template removes it from the list of available templates and removes tasks associated with it. It does not directly affect a running schedule, however some reporting and dashboard metrics may use the template to link various schedule runs together, which are not possible if the template is deleted.

To delete a Task Manager template:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left.
3. Select the template, and click **Delete**.
4. At the confirmation prompt, click **Yes**.

Managing Tasks

Tasks are the core units of action in a business process, such as data entry or data consolidation.

Each task has different parameters depending on the Task Type. If you have Service Administrator or Power User security rights, you can create, edit, or delete tasks.

See Creating Tasks.

See also: Managing Services

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Creating Tasks

You can add tasks to templates or schedules. If you create a task in a template, you assign the start and end date as days before or after Day Zero. If you create a task in a schedule, you select calendar dates for the start and end dates.

You can group tasks under parent tasks to provide a simpler view of the business process. After viewing upper-level parent tasks, you can then drill into the underlying tasks. Child tasks of parent tasks may have different owners than the parent task.

You can create a task using any of these methods:

- Drag and drop the Task Type onto a template or schedule in a view.
- Right-click on a task and select **New**.

Then enter the task information:

- Setting Task Properties
- Setting Task Parameters
- Specifying Task Instructions
- Selecting the Workflow
- Adding Task Questions
- Setting Task Viewers
- Setting Task Predecessors
- Applying Task Attributes
- Working With Task Rules
- Viewing Task History

To learn more about creating tasks, watch the video Creating Tasks.

For additional information on creating tasks, see the tutorial Viewing, Updating, and Adding Tasks in Task Manager.

Setting Task Properties

The Properties dialog enables you to set the task name, task ID, description, task type, priority, start and end date or duration.

You also use Properties to specify task Owners and Assignees. If an Owner or Assignee is not available to work on the task (due to sick time, vacation, has left the company, or is simply too busy), you can reassign users associated with a task, template, or schedule. Template, schedule, and task Owners, Assignees, and Approvers may also reassign their tasks.

For end-user tasks, you can allow an Assignee to open a pending task prior to the scheduled time, if all task predecessor conditions have been met.

Automated tasks have no Assignees, however you can specify Run As users for users under which tasks of that type are run.

Event Monitoring tasks have no Assignees and Run As users.

To set task properties:

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1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left and then click the **Tasks** tab.

> **Note**
>
> The **Tasks** tab is displayed at the bottom of the page.

3. Open a task.
4. Click **Properties** and enter this information:
   - **Task Name**: Enter a maximum of 80 characters or less.
   - **Task ID**: Identifies the task (required). Task IDs must be unique within the template or schedule. You can enter a maximum of 80 characters or less.
   - **Description**: Enter a maximum of 255 characters or less.

> **Note**
>
> You can use carriage returns and URLs in Descriptions, Questions, and Instructions.

   - **Organizational Unit**: Represents a hierarchical entity-type structure that you can use to model your organization. Define a separate organizational unit for each entity for which separate reporting is required, or for entities that require different configurations for any of the following: time zones, holidays, work days, or viewer or commentator assignments. Organizational Units are defined in system settings.

     When deploying template tasks, by default, the timezone of the current user is used for scheduling unless the template task is associated with an Organizational Unit. When the Organizational Unit has a specified timezone, that timezone is used for scheduling the task.
   - **Task Type**: Click the magnifying glass and select a Task Type (required).
   - For **Priority**, select from the following options:
     - **High**
     - **Medium**
     - **Low**
   - **Owner**: Use the default owner or click **Select Owner** to select a user or group.
   - Select **Active** to include this task in schedules generated from the template.

> **Note**
>
> This option is not available for tasks created in schedules.

5. Click **Save and Close**.
6. See also:
   - Setting Task Parameters

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- Specifying Task Instructions
- Selecting the Workflow
- Adding Task Questions
- Setting Task Viewers
- Setting Task Predecessors
- Applying Task Attributes
- Working With Task Rules
- Viewing Task History

Setting Task Parameters

The Parameters dialog only contains data for tasks that are associated with an Integration Type and have parameters. Integration Types provide links to external applications. From Parameters, you can set specific information about the task and how it is run. The parameters are passed to the external application. For example, if the task contains a link to a data grid, you might use the Parameters dialog to select a point of view for the grid.

To set task parameters:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left and then click the **Tasks** tab.

> **Note**
>
> The **Tasks** tab is displayed at the bottom of the page.

3. From the **New Task** or **Edit Task** dialog box, click **Parameters**.
4. If the parameter values are not entered by the task type, you may enter the required parameter values.

> **Note**
>
> If the parameters are not defined when the task is ready to start, an email is sent to the task owner. The task waits to start until the values are provided.

5. See also:
   - Setting Task Properties
   - Specifying Task Instructions
   - Selecting the Workflow
   - Adding Task Questions
   - Setting Task Viewers
   - Setting Task Predecessors
   - Applying Task Attributes
   - Working With Task Rules
   - Viewing Task History

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Specifying Task Instructions

Task owners can create a set of instructions for completing the task. All other users who have access to the task are only able to read the instructions. If there are task instructions that are inherited from the Task Type, template, or schedule, that instruction text is displayed above the instruction text box and is read-only.

For template tasks, the Template and Task Types instructions are viewable but not editable.

To specify instructions:

1. On the Home page, click **Application** and then click **Task Manager**.
2. Click the **Templates** tab on the left, and then click the **Tasks** tab.

> **Note**
>
> The **Tasks** tab is displayed at the bottom of the page.

3. From the **New Task** or **Edit Task** dialog box, click **Instructions**.
4. Enter the **Instructions** which can have unlimited characters.

To add a reference:

1. In the **References** section, click **Add**.
2. From the **Reference Type** list, select one of these types:
   - **Local File**: Click **Choose File** to select and attach the file, enter a **Name**, and then click **OK**.
   - **URL**: Enter the **URL** and URL **Name**; for example, https://www.oracle.com, Oracle.
3. See also:
   - Setting Task Properties
   - Setting Task Parameters
   - Selecting the Workflow
   - Adding Task Questions
   - Setting Task Viewers
   - Setting Task Predecessors
   - Applying Task Attributes
   - Working With Task Rules
   - Viewing Task History

Selecting the Workflow

To select the Task Manager workflow:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left and then click the **Tasks** tab.

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> **Note**
>
> The **Tasks** tab is displayed at the bottom of the page.

3. From the New Task or Edit Task dialog box, click **Workflow**.

   The Workflow section contains the Assignee and Approver assignments.

> **Note**
>
> You cannot assign the same user to multiple roles within the workflow.

4. Enter the following information for the Assignee:

   - **Assignee**: To assign an Assignee to a named user or Shared Services Group, click **Select Assignee**.
     The **Actual** attribute is available for each workflow stage, showing the actual user for the completed workflow. For example, **Assignee (Actual)**. In the workflow, the **Actual** column shows who performed work on the task, regardless of the user status.

     > **Note**
     >
     > The additional user information only appears if the task is reassigned after it was completed by the initial user.

     For an End-User task, for **Assignee,** click **Select Assignee** to find a user.

     If you do not select an Assignee, the owner becomes the default Assignee. Parent and automated tasks have no Assignees.

     > **Note**
     >
     > For a Process-Automated task, click **Run As**, search for and select a user under which tasks of this type are run, and click **OK**.
     > If the selected **Run As** user is the user who also created the task, then the user can authorize the task, without being prompted to re-enter their password.

   - **Backup**: If you assigned a user for the primary Assignee, you can assign a backup user authorized as an Assignee:

     a. Click **Select Backup Assignee** .
     b. Enter the **First Name** and **Last Name** or click **Search** to select a backup user.
     c. Click **OK**.

   - **Require Action By**: When a group or team is assigned as the assignee, the **Backup** option is replaced with the **Require Action By** option with the following values:
     – **Any Assignee** (default)
     – **All Assignees**

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If **All Assignees** is selected, then everyone in the group or team must submit the task for it to be sent to the approvers.

- **Starts**: Select a start date, then select the time of day, in increments of 15 minutes, for the task to start.
- **Ends**: Select an end date, then select the time of day, in increments of 15 minutes, for the task to end.
- **Optional:** For **Minimum Duration**, enter the minimum duration of a task in the form of **Day(s)**, **Hour(s)**, and **Minute(s)**. An At Risk criteria is based on the condition if the Start date was missed **and** (the End_date minus the Current_date is less than the minimum task duration **or** the End_Date minus the Start_Date is less than the minimum duration). The At Risk tasks display in the Needs Attention Graph in the Status Chart of the Dashboard.
- **Optional:** For an End-User task, select **Allow Early Start** to allow the Assignee to open the task before the scheduled start time.

5. Optional: Enter this information for the Approver:
   a. Click the + to add an approval level
   b. Assign an **Approver** by selecting a User, Group, or Team.
   c. **Backup**: If you assigned a user for the Approver, you can assign a backup user authorized as an Approver.
      i. Click **Select Backup Approver** .
      ii. Enter the **First Name** and **Last Name** or click **Search** to select a backup user.
      iii. Click **OK**.
   d. Specify the **End Date**.
   e. **Require Action By**: When a group or team is assigned as the approver, the **Backup** option is replaced with the **Require Action By** option with the following values:
      - **Any Approver** (default)
      - **All Approvers**

      If **All Approvers** is selected, , then all approvers are required to approve the task prior to the task moving to the next level of approval or being marked complete. If any one of the approvers rejects the task, the task is assigned back to the **Assignee**.
   f. Repeat to add additional approval levels.
   g. Use the arrows icons to reorder the approval levels.
6. See:
   - Setting Task Properties
   - Setting Task Parameters
   - Adding Task Questions
   - Setting Task Viewers
   - Setting Task Predecessors
   - Applying Task Attributes
   - Working With Task Rules
   - Viewing Task History

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Adding Task Questions

> **Note**
>
> The Questions tab is not displayed for automated tasks or parent tasks.

When you create a task, you may want the Assignee to answer questions about their actions before they indicate a task is complete. For example, you may ask if a certain process was followed when completing that task.

You can specify various types of questions, such as Text, Number, or True/False, and indicate whether they are required. If a question is required, the user must respond or they cannot submit the task for approval. You can also order the questions by using the Move Up and Move Down buttons.

For schedule tasks, users can respond to the questions from the Task Actions page. If you are creating or updating a task in a template, an additional column displays, showing whether the row is locked or unlocked to indicate whether the user can remove or update the questions. Rows that are locked cannot be edited or deleted because they are inherited from the Task Type.

In templates, the Task Type questions are displayed, but are not editable.

> **Note**
>
> Only users who have the task Viewer role can respond to questions.

To add a question:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left, and then click the **Tasks** tab.

   > **Note**
   >
   > The **Tasks** tab is displayed at the bottom of the page.

3. From the **New Task** or **Edit Task** dialog box, click **Questions**.
4. Click **Add**.
5. Enter the **Question** with a maximum of 2000 characters or less.
6. For **Type**, select from the following options:
   - **Date**
   - **Date/Time**
   - **Integer**: Enter a value between -2147483648 and 2147483647.
   - **List**: Enter a list of valid responses to the question (up to 255 characters).

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- **Multi-Line Text**: Enter the **Number of Lines**, from 3 to 50 lines. Enter a maximum of 4000 characters. Multi-Line Text determines how many lines of text are visible without scrolling, on the Actions dialog boxes. Select **Attachments** if you want to include an attachments section.
- **Number**: Enter a value xxxxxxxxxxxxxxxxx.xxxxxxxxx (17 digits and 9 decimal places).

  Select number formatting options:

  - For **Decimal Places**, enter a value for the number of decimal places to display.
  - For **Display As Percentage**, check the box if you want a percentage to display.
  - Select the **Thousands Separator** option if you want numbers to display a thousands separator (for example, 1,000.00).
  - For **Currency Symbol**, select a currency symbol; for example, $ (United States of America Dollar).
  - For **Negative Number**, select how to display negative numbers; for example, (123).
  - For **Scale**, select a scale value for numbers; for example, 1000 or 1000000000000.
- **True/False**
- **Text**: You can enter a maximum of 4000 characters.
- **User**
- **Yes/No**

7. Assign a **Role** (**Assignee**, **Approver**, **Owner**, **Viewer**) to the question. Access is based on responsibilities.
8. If the question is required, select **Required**.

   The **Required** checkbox is disabled for Questions assigned to Owner or Viewer roles.
9. Click **OK**.
10. **Optional:** To change the order of questions, select a question, then click **Move to Top, Move Up** **Move Down**, or **Move to Bottom.**.
11. **Optional:** To edit a question, select the question and click **Edit**. To remove a question, select the question and click **Delete**.
12. See also:
    - Setting Task Properties
    - Setting Task Parameters
    - Specifying Task Instructions
    - Adding Task Questions
    - Setting Task Viewers
    - Setting Task Predecessors
    - Applying Task Attributes
    - Working With Task Rules
    - Viewing Task History

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Setting Task Viewers

The Viewers tab enables you to add or remove Viewers for the current task. You can specify an unlimited number of viewers.

To set task access:

1. From the **New Task** dialog box, select the **Viewers** tab.
2. Enter the first or last name of the user, then click **Search**.

   To specifically identify a user, click **Advanced**, then enter a **User ID** or **Email** address.

   > **Tip**
   >
   > For additional details about the user, such as groups and roles, click **Details**.

3. To remove a user from the Viewers list, select the user, then click **Remove**.
4. See also:
   - Setting Task Properties
   - Setting Task Parameters
   - Specifying Task Instructions
   - Selecting the Workflow
   - Adding Task Questions
   - Setting Task Predecessors
   - Applying Task Attributes
   - Working With Task Rules
   - Viewing Task History

Setting Task Predecessors

The Predecessors tab enables you to set predecessors for the current task. For template tasks, you can choose a predecessor from another template, and for schedule tasks, you can choose a predecessor from another schedule. You must assign a condition to the predecessor/successor relationship.

A task does not need predecessor tasks as long as a start date and time is specified. If both are specified, the task begins when the predecessor tasks are started or complete (for Finish to Start) and the start date and time is reached.

You can set these conditions under which the predecessor relationship is established:

Condition	Description
Finish to Start	Default. The task starts as soon as the predecessor task completes or finishes with warning.
Finish Error to Start	The task starts as soon as the predecessor task completes, even if it is in error.

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Condition	Description
Finish Error to Finish	The task completes as soon as the predecessor task completes, even if it is in error.
Finish to Finish	The task cannot be marked as Complete until the predecessor task is marked as complete. This is primarily used for user tasks. For example, a user can start a document summary, but cannot mark it complete until the consolidated data is final.

To set task predecessors:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left and then click the **Tasks** tab.

> **Note**
>
> The **Tasks** tab is displayed at the bottom of the page.

3. From the **New Task** or **Edit Task** dialog box, select the **Predecessors** tab.
4. Click **Add**.
5. To search for a Predecessor task, click **Expand Search**, enter the **Task Name**, then click the **Search** button. To refine the search, click **Advanced**, and enter advanced search criteria.
6. Select a predecessor task(s) and click **OK**.
7. See also:
   - Setting Task Properties
   - Setting Task Parameters
   - Specifying Task Instructions
   - Selecting the Workflow
   - Adding Task Questions
   - Setting Task Viewers
   - Applying Task Attributes
   - Working With Task Rules
   - Viewing Task History

Applying Task Attributes

To locate tasks in the system, you can apply attributes to the task. When you select an attribute, you can set a value for the attribute based on the attribute type. You can later filter by the attribute value.

For example, you may have a List attribute named Sales Region with the values of North, South, East, and West. The current task applies only to the West Sales Region, so you can add the Sales Region attribute and set it to "West".

To apply task attributes:

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1. Open a task and select the **Attributes** tab.

   Click **Add**, then enter the following:

   - **Attribute**: Select an attribute from the list of defined attributes.
   - **Type**: This field is not editable; it is populated by the Attribute.
   - **Value**: Select a value associated with the type of attribute; for example, a numeric value for Formatted Number attribute, a List for List attribute, multiple lines of displayed text without scrolling for Multi-Line Text, a name of a person, User, or Yes or No for the Yes/No attribute.
   - **Access**: All roles have view access unless otherwise specified below.

     To add an access, for each of the Text Box and Attachments tabs:

     a. Click **Add**.

     b. Select a role.

     c. Select one of the Role access types:

        – **Do Not Display**: Does not see this attribute in any of the dashboards, list views, or reports.

        – **Allow Edits**: Has the ability to add, change, and remove values for the attribute, but subject to the editability rules.

        – **Required**: Requires a value for the attribute.

2. Click **OK**.
3. See also:
   - Setting Task Properties
   - Setting Task Parameters
   - Specifying Task Instructions
   - Selecting the Workflow
   - Adding Task Questions
   - Setting Task Viewers
   - Setting Task Predecessors
   - Working With Task Rules
   - Viewing Task History

Working With Task Rules

Task rules affect task behavior. These rules enable users to assign rules directly to tasks when assigning rules isn't appropriate at the Template, Schedule, or Task Type level. Rules apply to the tasks for which the rules were configured.

Available task rules:

- **Auto Approve Task**: Automatically completes specified approvals only if specified conditions have been met.
  Examples of conditions that could apply to this rule include:
  – Attributes have specified values (including calculated attributes)
  – Predecessor task contains an attribute with a specified value

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When conditions are satisfied, the specified approver levels are marked as complete, thus progressing workflow to the next approval level, or to Closed if no additional approval levels exist.

This rule runs when the Task status changes to Open with Approver.

- **Auto Submit Task** - Automatically submits a task if specified conditions are met. When conditions are satisfied, the assignee role is marked as complete, thus progressing workflow to the first approval level, or to Closed if no approval levels exist.

  This rule runs when the Task status changes from Pending to Open with Assignee.

- **Prevent Task Approval**: Prevents approval of a task based on attribute values, or other characteristics. This rule runs when the Approver clicks **Approve** or when the approver clicks **Actions** > **Update**.
- **Prevent Task Submission**: Prevents submission of a task based on attribute values, or other characteristics. This rule runs when the Assignee clicks **Submit** or when the assignee clicks **Actions** > **Update**.
- **Prevent Task Rejection**: This rule prevents rejection of a task based on attribute values, or other characteristics. This rule runs when the Assignee clicks **Submit** or when the assignee clicks **Actions** > **Update**.
- **Send Email on Update**: This rule runs based on user actions on the task. Hence, you must set the conditions for the rule appropriately. The most common condition is to have a condition with the Status attribute or the Status Detailed attribute:
  - Status attribute: The only valid value that you can select is Closed. Pending, Open, and Error are all statuses that result from automated actions on the task.
  - Status Detailed attribute: The valid statuses are with the approver.

  The rule runs when a user using the task action dialog:
  - Claims, releases, submits, approves, rejects, reopens, force-starts, or force-closes a task
  - Updates an attribute value
  - Answers a question
  - Adds a comment
  - Deletes a comment
  - Adds a reference attachment

To work with task rules:

1. On the Home page, click **Tasks**, and then **Tasks**.
2. Open a task.
3. Click the **Rules** tab. You can view the following information:
   - **Order**: The order of precedence.
   - **Rule**: Name of the rule.
   - **Conditions**: The choice of what conditions must exist before running the rule.
4. To create or edit a rule, on the **Rules** tab, click the **New** icon or select an existing rule and click the **Edit** icon.
5. On the **Properties** tab:
   - **Rule**: Select a rule.

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> **Note**
>
> When you make changes to a rule definition, the additional information is added to the audit log. You can view the changes both in the **History** tab as well as the **Audit Report**.

- **Description**: **Optional.** Explain why you configured the rule and how it should be used. You can enter a maximum of 255 characters.
- **Run When**: Indicates the events associated with the rule.
- **Approver Level**: Select **All Levels** or select the Approver levels.

6. On the **Filter** tab:
   - Select **Create Filter** and populate the conditions section or select **Use Saved Filter**, and then select a filter. The filter selected and configured for the rule determines the conditions that trigger the rule to apply.
   - **Filter Task**: Specify for which task the condition should be selected:
     - **Current Task**
     - **Any Predecessor**
     - **Specific Task**: Provide the **Task ID**
   - **Conditions**:
     - **Use Saved Filter**: The **Conditions** section displays a read-only version of the conditions associated with the saved filter.
     - **Create Filter**: The **Conditions** section is enabled and you can click **Create Condition** or **Create Condition Group** to specify conditions.

       Conjunction, **Source**, **Attribute**, **Operator**, and **Value** behave as they do for the existing advanced filter feature.
7. See also:
   - Setting Task Properties
   - Setting Task Parameters
   - Specifying Task Instructions
   - Selecting the Workflow
   - Adding Task Questions
   - Setting Task Viewers
   - Setting Task Predecessors
   - Applying Task Attributes
   - Viewing Task History

Viewing Task History

For each task, the system retains a history of the changes made to it; for example, a shift in dates or change in ownership. Each change record includes the field, modification type such as added, created, or changed, the old and new values, the user who made the change, and the date on which the change was made. The information on this tab is read-only.

To view task history:

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1. On the Home page, click **Tasks**, and then **Tasks**.
2. Select a task.
3. Select the **History** tab on the right.
4. When you finish, click **Save and Close**.

Working with the Task Dialog Box

You can view a list of tasks from the Schedule Tasks page. You can filter the available schedule list by date or status.

To view the Tasks from the Schedule Tasks page:

1. On the Home page, click **Tasks**, and then **Tasks**.
2. For **Schedule**, select a schedule, or select **All** if you do not want to filter the list.
3. **Optional:** To filter the schedule list by year or period, select a **Year** and **Period**.
4. **Optional:** To filter the schedule list by status, from **Schedule Status**, select a status.
5. Select a task.
6. Click the **Properties** tab on the right to view the following fields:
   - **Name**
   - **Task ID**
   - **Status**
   - **Schedule**
   - **Priority**
   - **Task Type**
   - **Task Type**
   - **Description**
   - **Owner**
   - **Start Date**
   - **End Date**
   - **Duration**
   - **Actual Start Date**
   - **Actual End Date**
   - **Actual Duration**
7. Click **Instructions** to display any instructions.
8. Click **Alerts** to display current alerts.
9. Click **Workflow** to view assignees for the tesk.
10. Click **Attributes** to view any attributes for the task.
11. Click **Questions** to view any questions that have been posted for the task.
12. Click **Comments** to view any comments that have been posted for the task.
13. Click **Related Tasks** to view any related tasks. You can view Predecessors or Successor tasks.

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14. Click **Parameters** to view dashboard, cluster, application, and so on.
15. Click **History** to capture an audit trail of changes to the task. You can view the activity for the task, including **All Activity**, **Last 7 Days**, and **Today**.
16. Click **Actions**, then select one of the following actions:
    - **Submit Task**
    - **Abort Task**: Cancels a task. The purpose of canceling a task (instead of performing a forced close) is to prevent successors from continuing and to keep the process moving.
    - **Force Close Task**
    - **Refresh**
17. Click **Close**.

Importing and Exporting Tasks

Importing Tasks

To import tasks:

1. On the Home page, click **Tasks**, and then **Tasks**.
2. Select a task, click **Actions**, and then **Import Tasks**.
3. Click **Browse** to find the import file.
4. Select an **Import Type**:
    - **Replace**: Replaces the definition of a task with the definition in the import file. This option replaces the task detail with the detail that is in the file that you are importing. It does not affect other tasks in the schedule that are not specified in the import file.

      The system displays a warning that tasks in a schedule that match a task ID in the import file will be modified. If you do not want to overwrite the task, click **Cancel**.

      You can select the **Keep Attachments** checkbox to retain attachments to tasks that are being replaced.
    - **Update**: Updates partial information for tasks. For example, in the import file, you might have made changes to task instructions, reassigned Owners, Assignees, and Approvers, or removed some attributes and added new attributes. You might also have made the same change to a large number of tasks, for example, adding a new attribute to 100 of 400 tasks. The update option is not a full replacement of the task details. Only details for the task properties specified in the file are updated. For example, if the import file has only a column for a task instructions, the task name, assignee, attributes, and other properties are not affected.
    - **Delete**: Deletes tasks and associated information based on a list of task IDs provided in a file. The required information to perform a delete is a file with a column of task IDs.
5. Select a **Date Format**.

   Select a format from the drop-down list of allowed date formats. Date formats are not translated. By default, the date format is set to the locale date format of the exported file location.
6. Select a **File Delimiter** for the import file: **Comma** or **Tab**. Choose **Other** to specify any single character as the **Other Delimiter**.
7. Click **Import**.

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Exporting Tasks

To export tasks:

1. On the Home page, click **Tasks**, and then **Tasks**.
2. Select a task, click **Actions**, and then select **Export Tasks**.
3. Select Export options:
   - Select **All Tasks** or **Selected Tasks**.
   - For **Format**, select **Formatted data (visible columns only)** or **Unformatted data for future import**.

> **Note**
>
> If you are an Administrator or Power User, you can select the format. If you have User security rights, by default the Format option is **Formatted data (visible columns only)**, and it cannot be changed.

4. Click **Export**. The file is downloaded in Excel or CSV format depending on the selection: Formatted data (visible columns only) or Unformatted data for future import.
5. Click **Close**.

Editing Tasks

You can edit tasks depending on their status and your security rights. For example, you can edit a task description, attributes, or end date.

For an Open task in a Task List, you cannot edit the start date, because the task has started. You can only change the duration or the end date. You also cannot edit instructions, questions, Assignee, or approver, and you cannot add, delete, or edit predecessors. You can reassign the Assignee or approver from the Task Actions workflow.

To edit a task:

1. On the Home page, click **Tasks**, and highlight the task that you want to edit. The **Edit** icon is enabled.
2. Click the **Edit** icon.
3. To edit task attributes:

   Click the **Attributes** tab.

   To add an attribute, click the **Add** icon to display the **Add Attribute Assignment** dialog box, and then make the following selections:

   - **Attribute**: Select an attribute from the list of defined attributes.
   - **Type**: This noneditable field is populated by the Attribute.
   - **Value**: Select a value associated with the type of attribute; for example, a numeric value for Formatted Number attribute, a List for List attribute, multiple lines of displayed text without scrolling for Multi-Line Text, a name of a person, User, or Yes or No for the Yes/No attribute.
   - **Access**: Select the access to the attribute.

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4. Click **OK**.

Adding Attachments

If you have an attachment you want to include with your task, such as a Word file, Excel spreadsheet or other document, you can use one of the following methods to attach the document to the task. You can attach multiple items at the same time. When you click OK, the system uploads all the documents that you attached.

To add an attachment to a task:

1. On the Home page, click **Tasks**, and then **Tasks**.
2. Select a task.
3. Select the **Comments** tab.
4. Click the **Attachment** icon to open the **Add Attachment** dialog box.
5. Select one of the following options to attach the document:
   - Browse to the location of the document.
   - Drag and drop the document directly from your local directory.
6. **Optional:** Rename the document.
7. Click **OK**.

Sorting Tasks

From the Task List, you can sort tasks by ascending or descending order. You can sort by Schedule Name, Status, Owner, Start or End Date, or Duration.

To sort tasks:

1. On the Home page, click **Tasks**, and then click **Tasks**.
2. Hover over a column header until the Sort icons display, and then click **Sort Ascending** or **Sort Descending**.

Searching for Tasks

You can use the Task Find feature to find specific tasks. For example, you might enter "load" to find load tasks. You can search using full or partial words. You can scroll through the results using the Previous and Next buttons.

To search for a task:

1. On the Home page, click **Tasks**, and then click **Tasks**.
2. Enter a task name in the **Search** field.
3. Click **Previous** or **Next** to find the previous or next task that matches the search criteria.

Moving Tasks

You can use views to move tasks. For example, you can move tasks in the Gantt view.

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> **Note**
>
> Move is not available for parent tasks.

To move a task:

1. On the Home page, click **Tasks**, and then click **Tasks**.
2. Select a task.
3. Right-click on the task and drag it to a new location, or cut, copy, and paste the task.
4. Navigate to the target destination for the task and click **OK**.

Cutting, Copying, and Pasting Tasks

You can cut and copy tasks in templates and schedules. For example, you can copy a task from one schedule or template and paste it into another.

When you paste a task, the Task Details dialog box opens and enables you to modify the task that you are pasting.

If you paste a task onto a parent task, the pasted task becomes a child of the parent task. If you paste a task onto a child task, the pasted task becomes a sibling of the child task.

> **Note**
>
> Cut, Copy, and Paste features are not available for parent tasks. The Cut function is not available for scheduled (Open or Closed) tasks.

To cut, copy, or paste a task:

1. On the Home page, click **Tasks** and select a task.
2. Right-click on the task and select an action, or from the main toolbar, click **Actions**, and then select:
   - **Cut**: Cuts the task and stores it on the clipboard.
   - **Copy**: Copies the task and stores it on the clipboard.
3. Right-click on the task, then select **Paste**, or from the main menu toolbar, click **Actions**, and then select **Paste**.

   The **Task Details** dialog box opens and enables you to make changes to the task that you are pasting.
4. Click **OK** to complete the paste operation.

   The task is inserted and the views and filter views are refreshed to display the pasted task.

Reopening Manual and Automated Tasks

You can reopen tasks and choose to reestablish the predecessor and successor relationships.This is available on both manual end user tasks as well as automated tasks. Reopening an automated task will execute the task with the same parameters. If modifications

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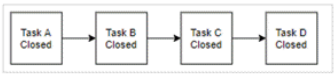
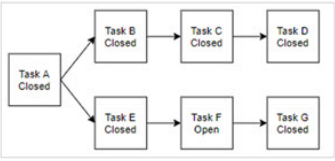
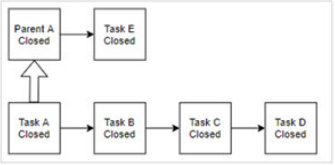
to the automated task parameters are required, edit the closed automated task and select **Reopen Task**.

Reopening a Manual Task

To reopen a task:

1. On the Home page, click **Tasks**, and then **Tasks**.
2. Select a closed task and click **Open** or click on the task name.
3. On the **Action** dialog, select **Reopen Task** from the **Actions** menu.
   You can choose which successors to automatically reopen and restore the predecessor links when a task is reopened. From the list of these successor tasks, you can choose all of them, none (default), or some subset for reopening.

Table 11-15 Successor Type - Use Case Example

Successor Type	Description	Use Case Example
Direct Successors	The simplest case is a single line of successors and is likely to be the most common case. Each Closed or Errored (less likely) task that is a successor of the task to be reopened is eligible to be reopened	 If Task A is reopened, 'Task B', 'Task C' and 'Task D' are available to be reopened.
Mixed Status Successors	Once an Open or Pending task is encountered, it and any subsequent tasks are not eligible to be reopened.	If Task A is reopened, only 'Task B' is available to be reopened.
Multiple Successors	A task can have more than one successor. Each branch should be followed to determine eligible tasks. This should extend all the way down the tree.	 If Task A is reopened, 'Task B', 'Task C', 'Task D' and 'Task E' are available to be reopened.
Parent Tasks	Parent tasks will be automatically be reopened if they were closed prior to their child task re-opening. In this case, any successor tasks of the parent are eligible to be reopened.	 If Task A is reopened, 'Task B', 'Task C', 'Task D' and 'Task E' are available to be reopened. 'Parent A' is automatically reopened.

- If there are no eligible successor tasks, the **Reopen Tasks** dialog displays a confirmation message prompting you to reopen the specified task.

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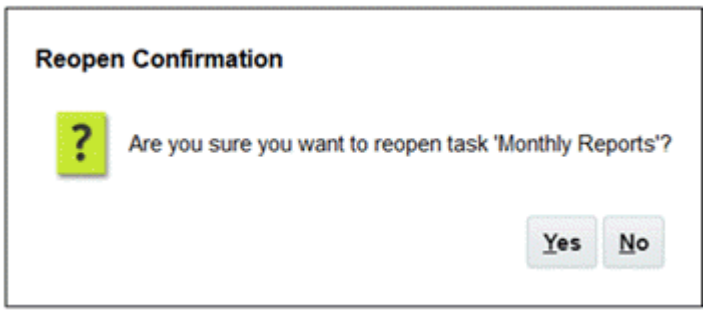
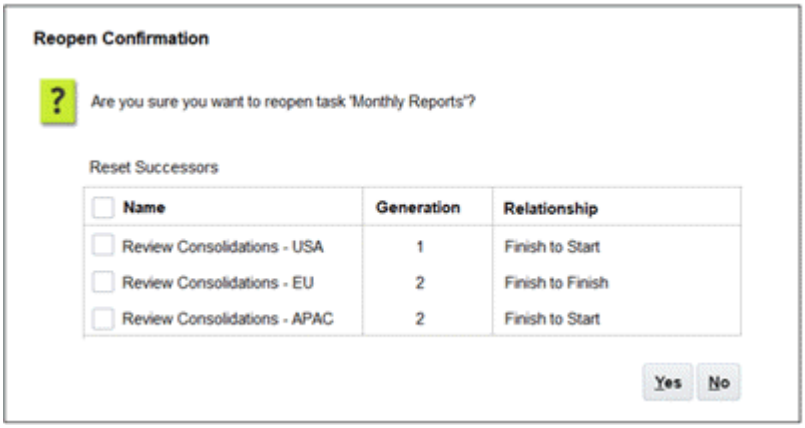


  - Click **Yes**, to close the dialog and reopen the task.
  - Click **No**, to close the dialog without reopening the task.
- If there are eligible successors to the task to be reopened, the **Reopen Tasks** dialog displays a confirmation message listing the successors with checkboxes, and you can select them to reopen.

> **Note**
>
> Automated tasks that didn't complete successfully due to an error, warning, or were force closed won't be listed in the successors list.



  - **Name** checkbox: Selecting the **Name** checkbox toggles between selecting all or selecting none of the successors. By default, this is unchecked and all of the tasks are unchecked.
  - **Successor Name**: Selecting a checkbox next to a successor task will cause it to reset when the task reopens.
  - **Generation**: Read-only field indicating the distance from the task being reopened. Parent tasks will not count as part of this calculation.
  - **Relationship**: Read-only field displaying the relationship to the task being reopened.
  - Click **Yes** to close the dialog and reopen the task.
  - Click **No**, to close the dialog without re-opening the task

4. Add or delete predecessors and then click **Save and Close**.
   Designate relationships:
   - Designate predecessor tasks. Choose one:

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  - If there are predecessors, then you can choose this option: Re-establish predecessor relationships - Tasks will open in original order.
  - Ignore predecessor relationships - All tasks will re-open immediately.
- Designate successor tasks:
  a. Select **Re-Open Successor Tasks**. The successor list is displayed.
  b. Select the successor tasks.

5. Click **OK** to close. The task details dialog box closes and the task reopens based on your selections. Click **OK**, otherwise, click **Cancel**.

Reopening an Automated Task

To reopen an automated task and change the automation parameters:

1. On the Home page, click **Tasks**, and then **Tasks**.
2. Select a closed automated task and click the **Edit** icon.
3. On the **Task Details** dialog, click **Reopen**. The Task is now editable.
4. Click on the parameters tab and update parameters as needed.
5. Click **Save and Close**.
   The service will display a confirmation message to reopen the automated task.
   - Click **Yes**, to close the dialog and reopen the task.
   - Click **No**, to close the dialog without reopening the task.

   If there are eligible successors to the task to be reopened, the **Reopen Tasks** dialog displays a confirmation message listing the successors with checkboxes, and you can select them to reopen. See the previous section Reopening a Manual Task.

Submitting Tasks

To submit tasks:

1. On the Home page, click **Tasks**, and then click **Tasks**.
2. Select the tasks that you need to submit.

   To multiselect, press **Ctrl** for random selection or press **Shift** while you click the first and last row in a range.
3. Select **Submit Task**. A warning is displayed that this will complete the task. Click **Yes** to proceed.
4. Review any errors, and click **OK**.

Approving or Rejecting Tasks

To approve or reject tasks:

1. On the Home page, click **Tasks**, and then click **Tasks**.
2. Select the tasks that you need to approve or reject.

   To select multiple tasks, press **Ctrl**, or press **Shift** while you click the first and last row in a range.

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3. Highlight the selected task, and under **Actions**, select **Set Status**, and then select **Approve** or **Reject**.
4. Review any errors, and click **OK**.

Claiming or Releasing Tasks

You can claim and release a single task or multiple tasks at a time in Task ManagerTask Manager. Users who are associated with tasks can claim or release tasks.

The ability to claim a task from another user is useful when a user has claimed a task, but can't complete a task, for example, because of an absence.

When a task is claimed, the Assignee changes to the person who claimed the task.

When a task is released, the Assignee is no longer associated with the task.

> **Note**
>
> If the bulk update feature is enabled, you can claim or release multiple tasks at a time.

Claiming or Releasing a Task Using Task Actions Menu

To claim or release a task from the task **Actions** menu:

1. On the Home page, click **Tasks**, and then click **Schedule Tasks**.
2. Click the task that you want to claim or release, or select the task and click the **Open** icon.
3. From the **Actions** menu, click **Claim** or **Release**.
   A pop-up message asking for confirmation is displayed.
4. Click **Yes**.
   A confirmation message is displayed indicating the task was claimed or released successfully.

   If any errors occur, review and fix the errors, and then perform the same steps to claim or release the tasks.
5. Click **Close** to close the dialog box.

Claiming or Releasing Tasks from the Task List

To claim or release tasks from the Task List:

1. On the Home page, click **Tasks**, and then click **Schedule Tasks**.
2. Select the task that you want to claim or release.
   To select multiple tasks, press **Ctrl**, or press **Shift** while you click the first and last row in a range. The selected tasks are highlighted.
3. From **Actions**, click **Update**, and then click **Claim** or **Release**.
   A pop-up message asking for confirmation is displayed.
4. Click **Yes**.
   The Task Action Results dialog box displays information about the status, number of tasks selected, tasks considered, tasks processed successfully, and tasks that were unsuccessful. These are the possible statuses for the process:

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- Processing
- Completed Successfully
- Completed with Errors

A visual indicator shows the completion percentage. For tasks completed with errors, the dialog box displays the errors. You can click **Export to Excel** and export the displayed errors to an Excel file. Review and fix the errors, and then perform the same steps to claim or release the tasks.

5. Click **Close** to close the dialog box.

Managing Task Reassignments

You may periodically need to reassign users to different tasks. For example, you may assign a task to an employee, but later that employee leaves the company and another employee assumes those tasks. You can use the Reassign feature to quickly find the tasks associated with one user and reassign them to another.

From the Task Actions dialog box, workflow users (Assignees and Approvers) can request reassignment of their workflow role for a single task. These requests require approval. Administrators and schedule/template Owners can reassign the task using the Edit Task dialog box without requiring approval.

When tasks are reassigned, email notifications are sent immediately to reassigned users.

To request a reassignment:

1. On the Home page, click **Tasks**, and then click **Tasks**.
2. Click on a task to open the Task Properties.
3. Click the **Actions** menu and select **Request Reassignment**.
4. Enter or click **Search** to find the task to be reassigned.
5. For **Create Reassignment Request**, click the **Select User** icon and select **To User**.
6. Under **Reassign**, select the tasks to be reassiged:
   - **Selected Tasks**
   - **Selected and Future Tasks**
7. Enter a **Justification** for the reassignment.
8. Click **OK** and then **Close**.

Automated Task Execution

You can view the status of the automated tasks on the Schedule Tasks page.

To view the status of the tasks:

1. On the Home page, click **Tasks**, and then click **Tasks**. For more information, see Working with the Task Dialog Box
2. Click the **Refresh** icon to update the task details within the task list.

For Process Automation tasks, Task Manager executes the business process when the start time is met. When the business process completes, the task is marked complete.

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During the execution of this process, Task Manager submits the job to the target application which returns a Job ID as most of the tasks execute in the background. Task Manager polls the target application to check the status of the job at regular intervals. Initially, the polling interval is less than a second and gradually it increases to a maximum polling interval of 15 mins. The polling interval is proportional to the time taken by the task to complete as described in the following table. The polling interval reduces the overhead related to frequent querying for job completion status in both Task Manager and the underlying application which ensures optimal system performance.

Process Automation Execution Time	Additional Time to Update Task Completion Status
1 minute	Maximum 1 second
10 minutes	Maximum 15 seconds
1 hour	Maximum 2 minutes
5 hours	Maximum 9 minutes
10 hours	Maximum 15 minutes

> **Note**
>
> If the task status has not been updated after the job is completed, don't force close the task immediately. It's recommended to wait for up to 15 minutes for long running tasks or as per the preceding timings.

For event monitoring tasks, Task Manager executes the event monitor when the start time is met. The task status is updated once Task Manager receives the event notification from the other application.

Canceling Tasks

The purpose of canceling a task, instead of performing a Forced Close, is to prevent successor tasks from continuing and to keep the process moving:

- When a Finish-to-Start or Finish-Error-to-Start predecessor is canceled, its successor tasks remain as Pending. To keep the process moving, you must edit the successor tasks to remove the canceled predecessor and to start the successors.
- When a Finish-to-Finish or Finish-Error-to-Finish predecessor is canceled, to keep the process moving, the task owner/schedule owner/administrator must force a close of the successor task.

Deleting Tasks

You can delete tasks that you no longer need. To delete a task, you must have Service Administrator or Power User security rights.

In schedules, you can delete only tasks that have a status of Pending. You cannot delete tasks that have a status of Open or Closed. If you delete a parent task, you can choose to delete just the parent, or the parent and its children.

To delete a task:

1. On the Home page, click **Tasks**, and then click **Tasks**.
2. Click **Actions**, and then select **Delete**.

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3. At the confirmation prompt, click **Yes**.

Managing Services

The Services (**Homepage** > **Application** > **Services**) option allows you to view the status of certain service related tasks. Green indicates that it's running. Actions to take from Services are **Run Now** or **Restart**.

> **Note**
>
> You can use **Run Now** to take an immediate action such as run open tasks. You should only use **Restart** if there was a service shutdown and this option is not green. Restart resets the service without running the service itself.

- **System Maintenance**: Refers to actions that Task Manager performs routinely within Tasks, such as the user synch process and cleanup of the database and is not the same as actions taken at the Cloud level called Daily Maintenance. Daily Maintenance is accessed by clicking Tools, then Daily Maintenance, includes actions such as operational maintenance and backup snapshots performed in the Cloud on test or production environments.
- **Task Email Notifications**: Used to notify users that work has been assigned to them. A Service Administrator can use the option on Services to **Restart** or **Run Now**.

  > **Note**
  >
  > – There is a separate option that a Service Administrator can use to turn off email notifications under **Application** → **Configuration** → **Settings** → **System Maintenance Process**.
  >
  > – Email notifications contain service and tenant information in the footer to help identify where an email is coming from.

- **[Schedule Name] Monitor**: Available for each schedule that is open and with tasks in the future. This option checks if any tasks need to be open in a schedule.
- **Task Status Update**: Background service which executes and monitors the process automated tasks.

Managing Schedules

A schedule defines a chronologically ordered set of tasks that must be executed for a specific business process, and is the application of a template into the calendar. For example, you can apply the Quarterly template as Q1FY19 for the first Quarter, then apply the template again as Q2FY19 for the second quarter.

The maximum number of schedules that you can create, import, or deploy is 100,000.

Schedules have a status of Pending, Open, Closed, or Locked. You can change the status of a schedule from Pending to Open, or from Open to Closed or Locked.

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Note

- After a schedule is set to Open, it cannot be reset to Pending.
- After a schedule is set to Locked, its status cannot be changed.
- Schedule Owners and Viewers are notified by an email when a Schedule is deleted.
- An immediate reassignment request notification is sent out to administrators and schedule owners when an assignee or approver requests a reassignment.

Manually Creating Schedules

To create a schedule, you must be an Administrator or Power User. A Power User can run Create Schedule on a template or select New from Manage Schedules. You can manually define a schedule from the Manage Schedules page, or create a schedule from a template. When you create a schedule from a template, all values are inherited from the template definition.

Watch this video to learn more about creating schedules.

Creating Schedules

To manually create a schedule:

1. On the Home page, click **Application**.
2. Click **Task Manager**.
3. Click the **Schedules** tab on the left.
4. Click **New**.
5. Enter schedule information on the schedule tabs:
   - Setting Schedule Properties
   - Adding Instructions to Schedules
   - Assigning Schedule Viewers
   - Applying Schedule Attributes
   - Adding Day Labels
   - Working With Schedule Rules
   - Viewing Schedule History

Setting Schedule Properties

The Properties tab enables you to set the name, description, start and end dates, and owners for schedules. The schedule owner must be an Administrator or Power User. The default owner is the current user. The start and end dates specify the initial date range for the schedule, however, after the schedule is created, you can add tasks with dates earlier or later and the properties are updated to reflect the earliest and latest dates.

To set schedule properties:

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1. Create a new schedule and select the **Properties** tab:
   - **Name**: You can enter a maximum of 80 characters.
   - **Description**: You can enter a maximum of 255 characters.
   - **Organizational Unit**
   - **Start Date**: Starting date for the schedule
   - **End Date**: Ending date for the schedule
   - **Year**
   - **Period**
   - **Day Zero Date**: The date to assign as day zero
   - **Owner**: Use the default owner or click **Select Owner**.
   - **Status**

     > **Note**
     >
     > The **Status** field does not display until the new schedule is saved. The schedule status is initially set to Pending and is display only.

   - **Source Template**
2. To save and close, click **OK** or click another tab; all entries are saved.
3. Enter schedule information on the schedule tabs:
   - Adding Instructions to Schedules
   - Assigning Schedule Viewers
   - Applying Schedule Attributes
   - Adding Day Labels
   - Working With Schedule Rules
   - Viewing Schedule History

Adding Instructions to Schedules

You can specify instructions and supporting documents for a schedule, which are then inherited by all tasks in the schedule.

To add instructions to a schedule:

1. Create a new schedule and select the **Instructions** tab.
2. In **Instructions**, enter instructions for the schedule.
3. To add a reference:
   a. In the **References** section, click **Add**.
   b. From the **Reference Type** list, select one of these types:
      - **Local File**:

        Click **Choose File** to select and attach the file, enter a **Name**, and click **OK**.
      - **URL**:

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Enter the **URL**, then enter a URL **Name**; for example, https://www.oracle.com, Oracle, and then click **OK**.

4. To save and close, click **OK** or click the **Viewers** tab; all entries are saved.
5. Enter schedule information on the schedule tabs:
   - Setting Schedule Properties
   - Assigning Schedule Viewers
   - Applying Schedule Attributes
   - Adding Day Labels
   - Working With Schedule Rules
   - Viewing Schedule History

Assigning Schedule Viewers

The Viewers tab enables you to assign viewer rights for schedules. A schedule can have multiple viewers, however they must have the proper security roles. Viewers receive read-only access to all tasks in the schedule.

> **Note**
>
> Only users who have the task Viewer role can respond to questions.

To assign viewer rights:

1. Create a new schedule and select the **Viewers** tab.
2. Click **Add**.
3. To search by users or groups, click the **Search Users** icon, then select **Users** or **Groups**.
4. Enter a user name, or part of the name, then click **Search**.
5. To identify a user, click **Advanced**, then enter a **User ID** or **Email**.
6. From the **Search Results** list, select users.
7. For additional details about the user, such as groups, roles, and assignments, click **Details**.
8. Click **Add** or **Add All** to move users to the Selected list.

> **Tip**
>
> To remove users, select users, then click **Remove** or **Remove All.**

9. To save and close, click **OK** or click the **Attributes** tab; all entries are saved.
10. Enter schedule information on the schedule tabs:
    - Setting Schedule Properties
    - Adding Instructions to Schedules

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- Applying Schedule Attributes
- Adding Day Labels
- Working With Schedule Rules
- Viewing Schedule History

Applying Schedule Attributes

To locate schedules in the system, you can apply attributes to the schedule. When you select an attribute, you can set a value for the attribute based on the attribute type. You can later filter by the attribute value.

For example, you may have a List attribute named Sales Region with the values of North, South, East, and West. The current schedule applies only to the West Sales Region, so you can add the Sales Region attribute and set it to "West".

To apply an attribute:

1. Create a new schedule and select the **Attributes** tab.
2. Click ➕ **Add**.
3. From the **Attribute** list, select an attribute.
4. For **Value**, depending on the attribute, select a value for the attribute from a drop-down list, or enter a value.
5. To save and close, click **OK** or click the **Day Labels** tab; all entries are saved.
6. Enter schedule information on the Schedule tabs:
   - Setting Schedule Properties
   - Adding Instructions to Schedules
   - Assigning Schedule Viewers
   - Adding Day Labels
   - Working With Schedule Rules
   - Viewing Schedule History

Adding Day Labels

You use business day labels in the business activity for a calendar day. Labels can mark a milestone day or specify the purpose of the day.

To add a day label to a schedule:

1. Create a new schedule and select the **Day Labels** tab. You can enter a maximum of 20 characters.
2. Add a day label to a specific date.
3. To save and close, click **OK** or click the **Rules** tab; all entries are saved.
4. Enter schedule information on the schedule tabs:
   - Setting Schedule Properties
   - Adding Instructions to Schedules
   - Assigning Schedule Viewers
   - Applying Schedule Attributes

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- Working With Schedule Rules
- Viewing Schedule History

Working With Schedule Rules

Schedule rules apply to all tasks in the schedule and thus apply rules to groups of tasks. Rules configured in the **Schedule** dialog box are copied to new schedule tasks.

Available schedule rules:

- **Auto Approve Task**: Automatically completes specified approvals only if specified conditions have been met.
  Example of conditions that could apply to this rule include: Attributes have specified values (including calculated attributes).

  When conditions are satisfied, the specified approver levels are marked complete, and workflow progresses to the next approval level, or closes if no additional approval levels exist.
- **Auto Submit Task**: Automatically submits a task if specified conditions are met.
  When conditions are satisfied, the Assignee role is marked complete, and workflow progresses to the first Approval level, or closes if no Approval levels exist.
- **Prevent Task Approval**: Prevents approval of a task based on attribute values, or other characteristics. This rule runs when the Approver clicks **Approve** or when the approver clicks **Actions** > **Update**.
- **Prevent Task Submission**: Prevents submission of a task based on attribute values, or other characteristics. This rule runs when the Assignee clicks **Submit** or when the assignee clicks **Actions** > **Update**.
- **Prevent Task Rejection**: Prevents rejection of a task based on attribute values, or other characteristics. This rule runs when the Assignee clicks **Submit**. This rule runs when the Assignee clicks **Submit** or when the assignee clicks **Actions** > **Update**.
- **Send Email on Update**: Runs based on user actions on the task. Hence, you must set the conditions for the rule appropriately. The most common condition is to have a condition with the Status attribute or the Status Detailed attribute:
  - Status attribute: The only valid value that you can select is Closed. Pending, Open, and Error are all statuses that result from automated actions on the task.
  - Status Detailed attribute: The valid statuses are with the approver.

  The rule runs when a user using the task action dialog:
  - Claims, releases, submits, approves, rejects, reopens, force-starts, or force-closes a task
  - Updates an attribute value
  - Answers a question
  - Adds a comment
  - Deletes a comment
  - Adds a reference attachment

To view schedule rules:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.

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3. Double-click a schedule.
4. Click the **Rules** tab. You can view the following information about the rules that are associated with a template:
   - **Order**: Order of precedence
   - **Rule**: Name of the rule
   - **Conditions**: Conditions that must exist before running the rule
5. To create or edit a rule, on the **Rules** tab, click the **New** icon or select an existing rule and click the **Edit** icon.
6. On the **Properties** tab:
   - **Rule**: Select a rule.

     > **Note**
     >
     > When you make changes to a rule definition, the additional information is added to the audit log. You can view the changes both in the **History** tab as well as the **Audit Report**.

   - **Description**: **Optional.** Explain why you configured the rule and how it should be used. You can enter a maximum of 255 characters.
   - **Run When**: Indicates the events associated with the rule.
   - **Approver Level**: Select **All Levels** or select the Approver levels.
7. On the **Filter** tab:
   - Select **Create Filter** and populate the conditions section or select **Use Saved Filter**, and then select a filter. The filter selected and configured for the rule determines the conditions that trigger the rule to apply.
   - **Filter Task**: Specify for which task the condition should be selected:
     - **Current Task**
     - **Any Predecessor**
     - **Specific Task**: Provide the **Task ID**
   - **Conditions**:
     - **Use Saved Filter**: The **Conditions** section displays a read-only version of the conditions associated with the saved filter.
     - **Create Filter**: The **Conditions** section is enabled and you can click **Create Condition** or **Create Condition Group** to specify conditions.

       Conjunction, **Source**, **Attribute**, **Operator**, and **Value** behave as they do for the existing advanced filter feature.
8. Enter schedule information on the schedule tabs:
   - Setting Schedule Properties
   - Adding Instructions to Schedules
   - Assigning Schedule Viewers
   - Applying Schedule Attributes
   - Adding Day Labels

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- Viewing Schedule History

Setting Required Task Parameters

The task is in a pending state until the required parameters are filled in. If they are not filled in before the specified start date, then an email notification is sent to the task owner. In addition, the task shows up in views under **Needs Attention**. After the required parameters are filled in, the task starts.

Opening Schedules

You open schedules to add, edit, or work on tasks.

When you create a schedule, it has a status of Pending by default. When you select the View menu option for a schedule, you can make final adjustments to it, and add, edit, or delete tasks.

To run a schedule, you must change the schedule Status from Pending to Open. See Setting Schedule Status.

To open a schedule for editing:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Use one of these methods:
   - Select **Actions** to the right of the schedule in the listing, and then select **View**.
   - Highlight a schedule, then click the **View** icon.
   - Right-click the highlighted schedule, and then select **View**.

Editing Schedules

You can edit schedules to change the properties, such as the name or the schedule or the start and end dates. You cannot change the start date to a date later than the first task in the schedule, or the end date to a date earlier than the last task in the schedule. You can make changes to an Open or Pending schedule to modify pending tasks. You cannot add, change, or delete tasks in a Closed or Locked schedule. To edit a schedule, you must be the schedule owner or an Administrator.

To edit a schedule:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Select a schedule and click **Edit**.
4. Edit the schedule.
5. Click **OK**.

Adding Tasks to Schedules

You can add tasks to a schedule if it is Pending or Open. You cannot add tasks to a Closed or Locked schedule.

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> **Note**
>
> The maximum number of tasks that you can add to a Task Manager schedule is 500,000.

To add a task to a schedule:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Open a schedule.
4. Add tasks.

Importing Tasks into Schedules

You can import tasks, or partial task data, from text files into a schedule with a status of Pending or Open. For example, if you have a Microsoft Excel spreadsheet with task definitions, you can save the file as a CSV file, then import it into a schedule. You can also use the Import feature to quickly add many repetitive tasks by editing the fields in a CSV file and importing it, rather than creating individual new tasks.

You cannot import tasks into schedules that have a Closed or Locked status.

> **Note**
>
> Before you import a file to a schedule using the Update option for a closed task, remove the Owner, Assignee, Approver, and Start Date and Time fields from the import file, or an error occurs.

If you import information on an End-User task that has started running, it is reset to Open with Assignee status, and the answers to questions are cleared.

To import tasks into a schedule:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Click **Import Tasks**.
4. Enter the name of the file to import, or click **Choose File** to find the file.
5. Select an import option:
   - **Replace**: Replaces the definition of a task with the definition in the import file. This option replaces the task detail with the detail that is in the file that you are importing. It does not affect other tasks in the schedule that are not specified in the import file.

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> **Note**
>
> You can select the **Keep Attachments** checkbox to retain attachments to tasks that are being replaced.

- **Update**: Updates partial information for tasks. For example, in the import file, you might have made changes to task instructions, reassigned Owners, Assignees, and Approvers, or removed some attributes and added new attributes. You might also have made the same change to a large number of tasks, for example, adding a new attribute to 100 of 400 tasks. The update option is not a full replacement of the task details. Only details for the task properties specified in the file are updated. For example, if the import file has only a column for a task instructions, the task name, Assignee, attributes, and other properties are not affected.

> **Note**
>
> The system displays a warning that tasks in the schedule that match a task ID in the import file will be modified. If you do not want to overwrite the task, click **Cancel**.

- **Delete**: Deletes tasks and associated information based on a list of task IDs provided in a file. The required information to perform a delete is a file with a column of task IDs.

6. **Optional**: Select **Keep Attachments** to retain any attachments associated with a task that is being replaced.
7. Select a **Date Format**.

   Select a format from the drop-down list of allowed date formats. Date formats are not translated. By default, the date format is set to the locale date format of the exported file location.
8. For **File Delimiter** for the import file, select **Comma** or **Tab**. Select **Other** to specify any single character as the **Other Delimiter**.
9. Click **Import**.
   - If the import is successful, the Import Success dialog box is displayed, indicating the schedule name, the name of the file containing the tasks, and the total number of tasks imported. Click **OK**.
   - If errors are found, the import process is not run, and the Import Errors dialog box displays the errors. View the errors, then click **OK** to return to the **Schedules** page.

Table 11-16 Troubleshooting Import Errors

Error	Resolution
Duplicate Task ID	Check for duplicate Task IDs. Task IDs must be unique within the template or schedule. See Creating Tasks.
Task Name is too long	The Task Name can contain a maximum of 80 characters. See Setting Task Properties.
Organizational value - "Invalid Value"	The Organizational value cannot be blank, contain bullet points or multiple lines.

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Updating Tasks in Schedules

You may need to manually update information on a task that is running, and in this case, you can reopen it. When you reopen a task, it is reset to Open with Assignee status, and you can edit the information. For example, you can change the instructions and references, attributes, and questions. If you make changes, previous answers to questions are cleared.

Reopening a series of tasks does not reestablish the predecessor relationships. All end-user tasks are reset to Open with Assignee. No tasks are reverted to Pending status.

You can reopen tasks under these conditions:

Table 11-17 Conditions for Reopening Tasks

Task Status	End-User
With Assignee/Running	You can edit or import data into the Instruction, Attribute, or Question sections. When you save the task, it is reset to the Assignee, and the answers to questions are cleared.
With Approver	You can edit or import data in to the Instruction, Attribute, or Question sections. When you save the task, it is reset to the Assignee, and the answers to questions are cleared.
Closed/Error	From Task Details, the task owner can click Reopen to reopen the task and make changes. When you save the task, it is reset to the Assignee.

To update tasks in a schedule:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Open a schedule.
4. Select a task with a **Closed** or **Error** status, right-click, and then select **View**.

   The View Task dialog box is displayed.
5. Click **Reopen**.
6. Edit the task.
7. If the system displays a warning that the Assignee must complete the task again, or that the service will be executed again, click **Yes** to continue or **No** to cancel.
8. Perform an action:
   - For a Closed task, click **Close**.
   - For an Error task, click **Save and Close**.

Reassigning Users in Schedules

You may periodically need to reassign users to different schedules. For example, you may create a schedule and assign a user to certain rights; however, later that employee leaves the company and another employee assumes those schedules. You can use the Reassign feature in the Schedules dialog box to automatically change assignments instead of manually searching for, opening, and editing individual schedules. The Reassign feature enables you to quickly find the schedules associated with one user and reassign them to another.

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You can reassign users for multiple schedules at a time. This can be helpful if your application has a large number of schedules.

To reassign users:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Select one or more schedules for which you want to reassign users.
4. Click **Actions**, and then **Reassign User**.
5. For **Find User**, click **Find User** and enter search criteria for the user that you want to replace:
   a. From the **Select User** dialog box, enter the first or last name for the user, and click **Search**.
   b. Select the user from the results, then click **OK**.
6. For **Replace With**, click the **Replace With** icon and enter search criteria for the user to whom you want to reassign tasks:
   a. From the **Select User** dialog box, enter the first or last name for the user, and click **Search**.
   b. Select the user from the results, then click **OK**.
7. Select the **Ending Between** dates.
8. Select roles for the user that must be reassigned:
   - **Owner**
   - **Assignee**
   - **Approver**
   - **Viewer**
9. Click **Reassign**.

   When the process ends, the system displays a Reassign Users - Success message that indicates that the user reassignment is complete, and displays the schedule name and total number of user reassignments made.

Authorizing Process-Automated Tasks

When you create Process-Automated tasks, for security purposes, you specify a user account under which the task is run. To preserve security, you may require authorization to perform the task if any of these conditions occurs:

- If you add a Process-Automated task with an alternate runtime user known as the Run As user to a schedule, from a template or by manually adding the task
- When you set a schedule to Open status, the system automatically issues a request for authorization if it has not been completed.

If the parameters of a task are modified by a user other than the Assignee (or task owner if the owner is also the Assignee), the authorization is reset to Unauthorized and must be obtained by entering a password. For Process-Automated tasks, if a user other than the specified or default Run As user modifies the parameters, the task is reset to Unauthorized.

Authorization ensures that the user performing the Process Automated task has security privileges for the application and data for which the task runs. An administrator who knows the

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credentials of the runtime user can perform the authorization or issue a request to the user to obtain authorization.

When the Process Automated task is scheduled to run, if authorization is not provided, the task is not run, and its status changes to Needs Attention. If an owner or Assignee edits the task, the task details indicate that authorization is required. In this case, only the Run As user can authorize the task.

A user who receives a request for authorization can access the authorization on the Worklist from a link in the email or by logging on to the application.

> **Note**
>
> From a Schedule or from a Worklist, before the task Start Date, the Administrator, Schedule Owner, or Task Owner can authorize the task. After the task Start Date, only the Run As user can authorize the task.

To authorize a task:

1. On the Home page, click **Application**, and then click **Task Manager**
2. Select the **Schedules** tab on the left.
3. Select a schedule.
4. Select **Actions**, and then select **Authorize Tasks**.

   The system displays a **Users** selection list and a table of unauthorized tasks for the selected schedule. The user selection list is populated with users that have pending Process-Automated tasks assigned to them that require authorization.
5. From the user selection list, select a user.

   The system displays a list of unauthorized tasks for that user. Your user name displays first on the list in bold by default. If you have no unauthorized tasks, the list is blank.
6. Select a task that needs authorization.
7. To view task details, click a task name and review the task parameters.

> **Tip**
>
> To contact the task owner by email, click the Owner name next to the task and view the user details.

Setting Schedule Status

You manage the schedule lifecycle by setting the schedule status. You can set the status of a schedule to Open, Closed, or Locked, depending on its current status. To set schedule status, you must be the schedule owner or Service Administrator.

You can set the status for multiple schedules at a time. This can be helpful if your application has a large number of schedules.

These are the available statuses:

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- **Pending**: The schedule is not yet active. When you create a schedule, it has a status of Pending by default so that you can make final adjustments to it, and add, edit, or delete tasks. When a schedule has a Pending status, you cannot close or lock the schedule.
- **Open**: To run a schedule, change the status from Pending to Open. When the schedule is opened, tasks begin to execute according to their definition. Status for tasks that have met their start date, time and conditions are set to Open, and task notifications are sent to their Assignees.
- **Closed**: When work on the schedule has reached a stage when the schedule is no longer active, and follow-up work is all that is required, you set the status to Closed. You cannot add new tasks to a Closed schedule. However, users can continue to work on tasks that are not complete. You can reopen a Closed schedule if needed, which changes its status to Open.
- **Locked**: When all tasks are completed, you set the status to Locked. After a schedule is locked, it cannot be modified. You cannot edit a Locked schedule, but you can set the status back to Open, if required.

Automated tasks execute based on their defined workflow within the schedule and don't require manual intervention. If you want the automated task execution to be dependent on a manual user action, there are two scheduling options:

- Add a basic or an end user task as a **Start to Finish** predecessor of the automated task. The automated task will execute after the preceding task is completed, and the start time is met. See Setting Task Predecessors in *Administering Financial Consolidation and Close*.
- Assign a **Run As** user within the workflow on the automated task. See Selecting the Workflow in *Administering Financial Consolidation and Close*. When a **Run As** user is specified for the automated task, then the task's execution will be dependent on the designated **Run As** user authorization. The automated task will only execute after the start time, predecessors, and **Run As** authorization have been met. See Authorizing Process-Automated Tasks in *Administering Financial Consolidation and Close*.

> **Note**
>
> If a schedule task start time is reached and authorization has not been provided for a system-automated task, the task remains in the Pending status and requires authorization.

To set the schedule status:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Select one or more schedules for which to set status.
4. Click **Actions**, and then **Set Status,** or select the **Set Status** dropdown.
5. Click one of these status options, depending on the current status:
   - **Open**
   - **Closed**
   - **Locked**
6. A system warning is displayed about the change, for example, that setting the status to Open will begin execution of all tasks for the current date. To confirm the status change, click **Yes**.

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Viewing Schedule History

The system maintains a history of schedule actions, which you can view from the Edit Schedules dialog box. The History tab displays the components that were updated, the modification type, the old and new values, the user who made the modification, and the change date. The information is read-only.

To view schedule history:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Select a schedule.
4. Select the **History** tab and review the schedule history.
   - **Modification Type**: Indicates the type of change: Created, Changed, Added, Removed
   - **Modified On**: Date of the modification
   - **Modified By**: Name of the user who modified the schedule
   - **Old Value**
   - **New Value**
5. Click **OK**.

Validating Schedules

You can validate schedules with a status of Pending or Open. Validating a schedule checks for problems with start and end dates, predecessor relationships, parent-child relationships, and missing task parameters for product integrations. You cannot change a schedule status from Pending to Open until all validation errors are resolved. To validate schedules, you must be the schedule owner or Service Administrator.

The validation results show the schedule names, status, and error messages. You can sort the results by Schedule Name or Status.

To validate schedules:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Select one or more schedules to validate.
4. Click **Actions**, and then select **Validate**.

   If no errors exist, the validation results show a "Schedule is valid " message. If errors exist, it shows the error details.

Locking Schedules

You can lock a schedule to prevent users from making further changes.

To lock a schedule:

1. On the Home page, click **Application**, and then click **Task Manager**.

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2. Click the **Schedules** tab on the left.
3. From the list of schedules, select a schedule.
4. Select the **Actions** icon to the right of the schedule in the listing, and then select **Set Status**, or from the **Set Status** drop-down list, select **Locked**.
5. **Optional:** If you need to reopen the Locked schedule, select **Actions**, and then **Set Status**, or from the **Set Status** drop-down list, select **Open**.

Viewing Schedules

In Schedules, you can specify which columns to display for the list of schedules, or show all. You can also reorder columns, sort columns by ascending or descending order, or change the column widths.

To display columns:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. Do one or more of the following tasks:
   - To display all columns, select **View**, then **Columns**, and then select **Show All**.
   - To display specific columns, select **View**, then **Columns**, and then select or deselect the column names.
   - To reorder columns, select **View**, then **Reorder Columns**, select columns, and then use the Up or Down arrows or drag them to change the order.
   - To sort columns, hover over a column header until the Sort icons are displayed, and then click **Sort Ascending** or **Sort Descending**.
   - To change column widths, hover over the column header dividers until the arrows display, and drag the columns to the desired width.

Searching for Schedules

You can use the Search text box in a Schedules list to quickly find schedules. You can enter full or partial names on which to search. Using the filter bar, you can control the schedules that you see in the list. By default, all schedules are displayed.

To search for a schedule:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. To search for a schedule, enter search criteria in the **Search** text box.
4. **Optional:** From the filter bar, click on a category to display additional search operators such as **Equals**, **Does Not Equal**, **Contains**, **Does Not Contain**, **Starts With**, and **Ends With**.

   You can filter schedules using these categories: **Name**, **Year**, **Period**, **Status**, **Start Date**, and **End Date**. Click **Add a Filter** to add additional filters: **Created By**, **Created On**, **Day Zero Date**, **Description**, **Last Updated By**, **Last Updated on**, **Organizational Unit**, **Owner**, and **Tasks**.

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Note

- Click **Add a Filter** to view all categories.
- To hide the filter bar, click the **Hide Filter Bar** icon.
- To clear all filters, click the **Filters** icon, and then click **Clear All Filters**.

Deleting Schedules

You can delete a schedule that you no longer need. To delete a schedule, you must be the schedule owner or Service Administrator. Deleting a schedule removes it and all references to it from the system.

To delete a schedule:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Schedules** tab on the left.
3. From **Schedules**, select the schedule that you want to delete.
4. Click **Delete**.

   The system displays a warning that if you delete a schedule, it will also permanently delete all tasks within the schedule, and the only way to recover is from a backup.
5. To delete the schedule, click **Yes**.

Managing Task Manager Integrations

You can enable Task Manager tasks to include integrations within the Oracle Fusion Cloud Enterprise Performance Management and other external applications.

Task Manager allows you to incorporate integrated tasks within your business processes. These simplify the process by automating manual tasks or including embedded application pages or links.

To manage Task Manager integrations, you must have the Service Administrator role.

Task Manager supports the following integration types:

End User

An End User integrated task requires the user to interact with an application web page. These application web pages are displayed within the task or included as links within the task. An Integration requires an execution URL for user tasks and an optional set of parameters. The execution URL launches the external program, and the parameters pass information required for the task to the external program. For example, the Approve Journals Integration contains parameters such as the Point of View dimension values for the journal.

Users must perform and validate the task. For example, the task may be a generic task such as submitting data, or it may require product integration to facilitate or validate its completion.

Process Automation

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A Process Automation task initiates an action within the connected application. These integrations are automatically executed in external applications when their Start date and time are reached, and their predecessor tasks are completed, for example, an overnight feed from a General Ledger. These tasks are often executed after working hours. They require limited user interaction and do not have Assignees.

Event Monitoring

An Event Monitoring task is a task which is passive. It does not initiate any action but monitors another application awaiting for an action or status to occur. Once the action or status occurs, the task is marked complete. These are based on events that occur in external applications, for example Approve Journals.

Task Manager provides these pre-built integrations:

- Cloud Integrations: End User and Process Automation integrations for Cloud EPM Connections. See Managing Integrations with Cloud EPM Services.
- Cloud Integrations: Oracle Cloud ERP. See Managing Integrations with Cloud and On-Premises Applications.
- On-Premises End User and Event Monitoring Integrations. See Managing Integrations with Cloud and On-Premises Applications.
- On-Premises Oracle E-Business Suite General Ledger, Accounts Payable, and Accounts Receivable Integrations. See Managing Integrations with Cloud and On-Premises Applications.

If you require any other Integrations for Cloud or On Premises, you can create Custom Integrations. See Creating Custom Integrations.

Managing Integrations with Cloud EPM Services

If you are using Task Manager and have subscriptions to other Oracle Fusion Cloud Enterprise Performance Management services, you can create connections between services and enable integrations using Task Manager functionality.

Pre-built integrations enable you to perform Task Manager tasks that access other Cloud EPM functionality.

Pre-built integrations are provided within Task Manager for these Cloud EPM services:

- Account Reconciliation
- Enterprise Data Management
- Enterprise Profitability and Cost Management
- Financial Consolidation and Close
- FreeForm
- Planning and Planning Modules
- Profitability and Cost Management
- Tax Reporting

To learn more about how to set up integrations, see Adding Pre-built Integrations in Cloud EPM.

To learn more about which end user integrations are available, see End User Integrations for Cloud EPM.

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To learn more about which pre-built automated integrations are available, see Automation Integrations for Cloud EPM.

Watch the video Managing Task Manager Integrations

Adding Pre-built Integrations in Cloud EPM

Make sure you have the prerequisites and follow these steps to set up an integration between Task Manager and other Oracle Fusion Cloud Enterprise Performance Management services.

For an overview of the Task Manager integrations, see Managing Task Manager Integrations.

Prerequisite

To create integrations between Cloud EPM services, you need to obtain a subscription to the Cloud EPM service you want to integrate with.

To Create the Integration Type and Task Type

For Cloud EPM services where Task Manager has been enabled (for example, Financial Consolidation and Close), the service is automatically configured with the Integrations and Task Types for that business process.

Task Manager can be extended to include integrations with your other business processes by defining a connection to the business process and then deploying the Integrations and Task Types.

To add pre-built integrations within the Cloud EPM:

1. Create the connection between the service that contains Task Manager and the other service:
   - On the Home page, click **Tools**, and then click **Connections**.
   - Click **Create** to create a new connection:
     - For Enterprise Data Management, select **Other Web Service Provider**.
     - For all other Cloud EPM services, select **Oracle Enterprise Performance Management Cloud** provider.
   - Name the EPM Connect connection. For example, if integrating with FreeForm, a possible **Connection Name** could be **Freeform**.
   - Specify the connection URL.
   - Specify the user credentials.
   - Additionally for **Other Web Service Provider**, you must specify advanced options. Click **Show Advanced Options**.
     - For **Type**, select **Parameter**.
     - For **Name**, enter SERVICE_TYPE (this is a fixed value).
     - For **Value**, enter EDMCS (this is a fixed value).
2. On the Home page, click **Application**, and then click **Task Manager**.
3. Click the **Integrations** tab on the left.
4. From the **Integrations** page, click **Manage Connections**.

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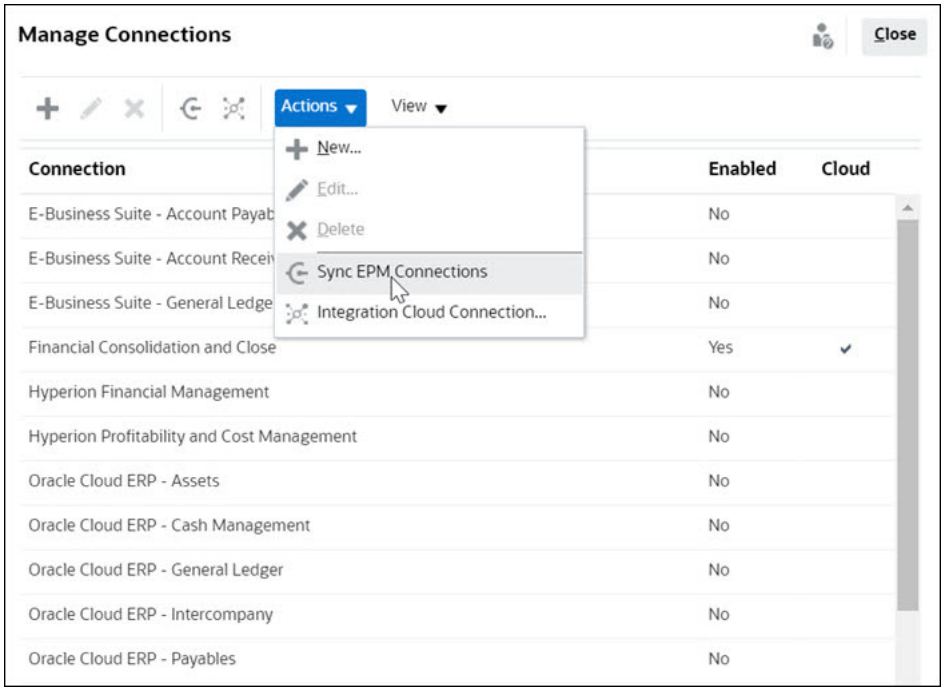
5. On **Manage Connections**, from **Actions**, select **Sync EPM Connections**.



A message displays the progress of the synchronization, and details the integrations added to the service. An Integration Type and Task Type is added for each Integration.

See also:

- End User Integrations for Cloud EPM
- Automation Integrations for Cloud EPM

End User Integrations for Cloud EPM

End user integrations allow you to access functionality in other remote Oracle Fusion Cloud Enterprise Performance Management environments while using Task Manager.

For an overview of the Task Manager integrations, see Managing Task Manager Integrations.

This section lists the available end user integrations for each Cloud EPM service:

- End User Integrations for Account Reconciliation
- End User Integrations for Enterprise Profitability and Cost Management
- End User Integrations for Financial Consolidation and Close
- End User Integrations for FreeForm
- End User Integrations for Planning and Planning Modules
- End User Integrations for Profitability and Cost Management
- End User Integrations for Tax Reporting

For a description of the remote Cloud EPM functionality, see the documentation for that Cloud EPM service.

End User Integrations for Account Reconciliation

- Console
- Data Exchange

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- Reconciliation List (Period, Saved List)
- Reports
- Transaction List (Period, Saved List)

End User Integrations for Enterprise Profitability and Cost Management

- Calculate Model
- Data Exchange
- Enter Form Data
- Export Data
- Export Metadata
- Import Data
- Invalid Intersections Report
- Manage Dimensions
- Manage Forms
- Manage Valid Intersections
- Profit Curves
- Refresh Application
- Reports
- Update Rules
- Validation Model
- View Calculation Analysis
- View Dashboard
- View Operational Dashboard
- View Rule Balancing
- View Trace Allocations

End User Integrations for Financial Consolidation and Close

- Approvals
- Configure Application
- Data Exchange
- Enter Form Data (Form)
  Required parameter is Form (EPM Artifact Type) displayed in drop down list in Task Details dialog.
- Export Data
- Export Metadata
- Export Journal
- Generate Financial Report
  Required parameter is the choice of report from drop down.
- Import Data
- Import Metadata

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- Import Journal
- Invalid Intersections Report
- Generate Intercompany Matching Report
- Generate Journal Report
- Manage Approvals
- Manage Dimensions
- Manage Forms
- Manage Journals
- Manage Ownership
- Manage Periods
- Manage Valid Intersections
- Refresh Application
- Reports
- View Dashboard (Dashboard)
  Required parameter is Dashboard (EPM Artifact Type) displayed in the drop down list in the Task Details dialog.
- View Financial Reports
- View Operational Dashboard

End User Integrations for FreeForm

- Configure Application
- Data Exchange
- Enter Form Data (Form)
  Required parameter is Form (EPM Artifact Type) displayed in drop down list in Task Details dialog.
- Export Data
- Export Metadata
- Generate Financial Report
  Required parameter is the choice of report from drop down.
- Import Data
- Import Metadata
- Invalid Intersections Report
- Manage Dimensions
- Manage Forms
- Manage Valid Intersection
- Refresh Application
- Reports
- View Dashboard (Dashboard)
  Required parameter is Dashboard (EPM Artifact Type) displayed in the drop down list in the Task Details dialog

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- View Financial Reports
- View Operational Dashboard

End User Integrations for Planning and Planning Modules

- Approvals
- Data Exchange
- Enter Form Data (Form)
  Required parameter is Form (EPM Artifact Type) displayed in drop down list in Task Details dialog.
- Generate Financial Report
  Required parameter is the choice of report from drop down.
- Invalid Intersections Report
- Manage Approvals
- Manage Dimensions
- Manage Rules
- Manage Valid Intersections
- Reports
- View Dashboard (Dashboard).
  Required parameter is Dashboard (EPM Artifact Type) displayed in the drop down list in the Task Details dialog
- View Financial Reports
- View Operational Dashboard

End User Integrations for Profitability and Cost Management

- Dashboards
- Data Exchange
- Generate Profitability Report
- Profit Curves (Profit Curve)
  Required parameter is Profit Curve (EPM Artifact Type) displayed in drop down list in Task Details dialog.
- View Dashboard (Dashboard).
  Required parameter is Dashboard (EPM Artifact Type) displayed in the drop down list in the Task Details dialog.

  View Report (Report)

End User Integrations for Tax Reporting

- Approvals
- Configure Application
- Data Exchange
- Enter Form Data (Form)
  Required parameter Form (EPM Artifact Type) displayed in drop down list in Task Details dialog.
- Export Data

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- Export Metadata
- Generate Financial Report
  Required parameter is the choice of report from drop down.
- Import Data
- Import Metadata
- Invalid Intersections Report
- Manage Approvals
- Manage Dimensions
- Manage Forms
- Manage Periods
- Manage Valid Intersections
- Refresh Application
- Reports
- View Dashboard (Dashboard).
  Required parameter is Dashboard (EPM Artifact Type) displayed in the drop down list in the Task Details dialog
- View Financial Reports
- View Operational Dashboard

Automation Integrations for Cloud EPM

These pre-built integrations are available

- Automation Integrations for Cloud EPM Platform
- Automation Integrations for Account Reconciliation
- Automation Integrations for Enterprise Data Management
- Automation Integrations for Enterprise Profitability and Cost Management
- Automation Integrations for Financial Consolidation and Close and Tax Reporting
- Automation Integrations for FreeForm
- Automation Integrations for Planning and Planning Modules
- Automation Integrations for Profitability and Cost Management

To use the pre-built Oracle Fusion Cloud Enterprise Performance Management Integrations, you must specify parameters for the integration. Many parameters for automated integrations are selectable from drop-down lists, which eliminates the need to manually enter values. For example, to run a rule or ruleset, you can select from a list of business rules, such as ForceConsolidate or ForceTranslate.

For an overview of the Task Manager integrations, see Managing Task Manager Integrations.

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Automation Integrations for Cloud EPM Platform

Integration Name / Module	Module	Description	Parameters/ Description
Copy File from FreeForm	All Cloud EPM Services except Enterprise Data Management	Copies a file from current service where Task Manager is configured to another Cloud EPM Service. For example, if you have configured Task Manager in FreeForm and set up Account Reconciliation connection, **Copy File from** FreeForm copies the file from FreeForm to Account Reconciliation.	File Name: Name of the file that you want to copy. Save File As: Name of the file that you want to save. This can be different than the original file name. External Directory Name (Optional): Name of the directory.
Copy File to FreeForm	All Cloud EPM Services except Enterprise Data Management	Copies a file to current service where Task Manager is configured from another Cloud EPM service.	File Name: Name of the file that you want to copy. Save File As: Name of the filethat you want to save. This can be different than the original file name. External Directory Name (Optional): Name of the directory.
Delete File From FreeForm	All Cloud EPM Services except Enterprise Data Management	Deletes a file from a Cloud EPM service.	File Name: Name of the file that you want to delete.

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Integration Name / Module	Module	Description	Parameters/ Description
Lock Unlock Data Integration	All Cloud EPM Services except Enterprise Data Management	Locks or unlocks an integration for a location, category, and period in Data Exchange. This is a process-automated integration.	Operation: Choose from lock or unlock. Lock Type: Choose whether the Lock/ Unlock operation is for an application or a location. Period: Specify the period of the POV from the integration or data load rule defined in Data Exchange, for example, "Jan-21". Category: Specify the predefined Scenario value based on the POV Category from the integration (data rule) definition. The categories available are those that are created in the Data Integration set-up, such as "Actual." Application (Optional): If selected Lock Type is application, specify the name of application, for example, "Vision". Location (Optional): If selected Lock Type is location, specify the name of the location. If the location is locked, data cannot be loaded to it. Unlock By Location (Optional): This parameter can be specified when selected operation is lock and selected location is application. If checked when locking the target application, then the system locks all rules present in the location under target application and not the application-level lock. For more information, see Lock and Unlock POV in *REST APIs*.
Run Data Integration	All Cloud EPM Services except Enterprise Profitability and Cost Management	Execute an integration or data load rule based on how periods are processed and source filters. This allows the integration of data loads defined in Data Exchange easily into the monthly processing schedule.	Job Type: Integration is the job type. Integration Name: The name of the integration defined in Data Integration. Period Name: Name of the period. Import Mode: Determines how the data is imported into Data Integration. Export Mode: Determines how the data is exported into Data Integration. File Name: Applicable only for native file-based data loads and ignored if specified for other loads. Source Filters: A parameter used to update the source filters defined for the data load rule or integration. Target Options: A parameter used to update the target options defined for the data load rule or integration. Execution Mode: Applicable only for Quick Mode integrations. For more details about these parameters, see Running Integrations in *REST APIs*.

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Integration Name / Module	Module	Description	Parameters/ Description
Run Pipeline	All Cloud EPM Services except Enterprise Data Management	Executes a pipeline based on job parameters and variables that you select.	Job Type: Pipeline is the job type. Job Name: Pipeline code defined for the Pipeline in Data Integration. Start Period: The first period for which data is to be loaded. This period name must be defined in Data Integration period mapping. End Period: The last period for which data is to be loaded. This period name must be defined in Data Integration period mapping. Import Mode: Determines how the data is imported into Data Integration. Export Mode: Determines how the data is exported into Data Integration. Attach Logs: Indicated whether logs are included as an attachment to an email. Send Email: Determines when an email is sent when a Pipeline is run. Send To: Determines the recipient email ID for the email notification. Variables: Used to specify custom variables for the pipeline job in either JSON or key-value format. For more details about these parameters, see Running a Pipeline in *REST APIs*.

See also Copy and Delete Integration Files

Automation Integrations for Account Reconciliation

Integration Name / Module	Module	Description	Parameters/ Description
Change Period Status	Reconciliation Compliance	Changes the status of a period (Open, Closed, Pending, Locked).	Period: The name of the period Status: Pending, Open, Closed, Locked
Create Period End Reconciliations	Reconciliation Compliance	Copies all selected profiles to a period and returns success or failure status.	Period: The name of the period Filter: The name of the filter that matches the reconciliation
Import Balances	Reconciliation Compliance	Imports balance data using Data Management from a previously created Data Load definition.	Period: The name of the period dl_Definition: The name of a previously saved data load using the format DL_name such as DL_test

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Integration Name / Module	Module	Description	Parameters/ Description
Import Pre-Mapped Balances	Reconciliation Compliance	Imports pre-mapped balances.	Period: The name of the period BalanceType: SUB|SRC for sub system or source system CurrencyBucket: Currency bucket, such as Functional File: The name of the file relative to the inbox, for example, balances.csv. The file has to be uploaded to Account Reconciliation using EPM Automate or REST API.
Import Pre-Mapped Transactions	Reconciliation Compliance	Imports pre-mapped transactions for a particular period.	TransactionType: Allowed Transaction Types are BEX (Explained Balance), SRC (Adjustment to Source System), and SUB (Adjustment to Subsystem) File: The name of the file relative to the inbox, for example, transactions.csv. The file has to be uploaded to Account Reconciliation using EPM Automate or REST API. DateFormat: Date Format, such as MM/dd/yyyy, dd/MM/yyyy, dd-MMM-yy, MMM d,yyyy, or All.
Import Pre-Mapped Transactions	Transaction Matching	Imports a file of pre-mapped transactions into Transaction Matching.	DataSource: Text ID of the data source where the transaction will be imported to File: The name of the file relative to the inbox, for example, transactions.csv. The file has to be uploaded to Account Reconciliation using EPM Automate or REST API. ReconciliationType: Text ID of the reconciliation type where the transaction file will be imported to, such as Bank to GL. DateFormat: Date Format, such as MM/dd/yyyy, dd/MM/yyyy, MM-dd-yyyy, d-M-yyyy, dd-MMM-yy, MMM d, yyyy

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Integration Name / Module	Module	Description	Parameters/ Description
Import Profiles	Reconciliation Compliance	Imports profiles for a particular period.	ImportType: The import type. Supported values are Replace and ReplaceAll Period: The period for which to import ProfileType: The profile type. Supported values are Profiles and Children File: The name of the file relative to the inbox, for example, profiles.csv. The file has to be uploaded to Account Reconciliation using EPM Automate or REST API. DateFormat: Date Format, such as MM/dd/yyyy, dd/MM/yyyy, d-M-yyyy, dd-MMM-yy, MMM d, yyyy, or All
Import Rates	Reconciliation Compliance	Imports rates for a particular period and rate type.	Period: The name of the period RateType: The rate type, such as Accounting Import Rates (Reconciliation Compliance) File: The name of the file relative to the inbox, for example, rates.csv. The file has to be uploaded to Account Reconciliation using EPM Automate or REST API. ImportType: Supported import types are Replace and ReplaceAll
Monitor Reconciliations	Reconciliation Compliance	Monitors list of reconciliations in ARCS.	Period: The name of the period Filter: Filter string used to query list of reconciliations
Run Auto Match	Transaction Matching	Runs the auto match process in Transaction Matching.	ReconTypeId: The Text ID of the Reconciliation type to be auto matched
View Reconciliations	Reconciliation Compliance	View reconciliations for a specified period.	Period: The name of the period Saved List: The name of a Public saved list

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Integration Name / Module	Module	Description	Parameters/ Description
View Transactions	Transaction Matching	View transactions for a specified period.	Period: The name of the period Saved List: The name of a Public saved list

Automation Integrations for Enterprise Data Management

Integration Name	Description	Parameters/ Description
Export Dimension	Exports a dimension from Enterprise Data Management to a configured connection. This is a process-automated integration. See Adding Pre-built Integrations in Cloud EPM.	Application: The name of the Enterprise Data Management application from which to export the dimension. Dimension: The name of the dimension to export. Connection: **Optional.** The name of the connection to which to export the dimension. File Name: The file and path from which to export the dimension.
Export Dimension Mapping	Exports a Dimension Mapping from Enterprise Data Management to a configured connection. This is a process-automated integration.	Application: The name of the Enterprise Data Management application from which to export the Dimension Mapping. Dimension: The name of the Dimension Mapping to export. Connection: **Optional**. The name of the connection to which to export the Dimension Mapping. Mapping Location: The location to which to export the Dimension Mapping. File Name: The file and path from which to export the Dimension Mapping.
Import Dimension	Imports a Dimension from a configured connection to an Enterprise Data Management application. This is a process-automated integration. See Adding Pre-built Integrations in Cloud EPM.	Application: The name of the Enterprise Data Management application to which to import the dimension. Dimension: The name of the dimension to import. Connection: The name of the connection from which to import the dimension. File Name: The file and path from which to import the dimension. Import Option: **Optional**. Determines how the data is imported into Enterprise Data Management.

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Integration Name	Description	Parameters/ Description
Extract Dimension	Extracts a dimension from Enterprise Data Management to a configured connection. This is a process-automated integration.	Application: The name of the Enterprise Data Management application from which to extract the dimension. Dimension: The name of the dimension to extract. Extract: The name of the extract. Connection: The name of the connection to which to extract the dimension. File Name: The file and path from which to extract the dimension.

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Automation Integrations for Enterprise Profitability and Cost Management

Integration Name	Description	Parameters/ Description
Calculate Model	Calculates a model for one or more points of view.	Job Type: Calculate Model Job Name: Name of the job POV Delimiter: Delimiter used in POV values. The default delimiter is _ (under score). The delimiter must be enclosed in double quotation marks. Only these delimiters are supported: • _ (under score) • # (hash) • & (ampersand) • ~ (tilde) • % (percentage) • ; (semicolon) • : (colon) • - (dash) POV Name: Name of the POV to calculate. You can pass one or more POVs separated by a comma (,). Model Name: Name of the model to calculate Execution Type: Identifies the rule execution type Monitoring Task: Monitors another application awaiting an action or status to occur Rule Name: Name of the single rule to run First Ruleset Sequence Number: Sequence number of the first rule in the rule set to run Last Ruleset Sequence Number: Sequence number of the last rule in the rule set to run Clear Existing Calculations: Whether to clear existing calculations Execute Calculations: Whether to execute calculations Optimize for Reporting: Whether to optimze the calculation process for reporting Generate Debug Scripts: Whether to generate debug scripts

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Integration Name	Description	Parameters/ Description
		Comment: Comments to describe the job
Clear Cube	Clears specific data within the PCM_CLC and PCM_REP cubes.	Job Type: Clear Cube Job Name: Name of the job
Clear Data by Point of View	Clears data from a point of view without removing the point of view.	Job Type: Clear POV Job Name: Name of the job POV Delimiter: Delimiter used in POV values. The delimiter must be enclosed in double quotation marks. Other than a comma, only these delimiters are supported: • _ (under score) • # (hash) • & (ampersand) • ~ (tilde) • % (percentage) • ; (semicolon) • : (colon) • - (dash) POV Name: Name of the POV to clear Cube Name: Name of the cube on which clear operation is to be executed Clear Input Data: Whether to clear input data Clear Allocated Data: Whether to clear allocated data Clear Adjustment Data: Whether to clear adjustent data

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Integration Name	Description	Parameters/ Description
Copy Data by Point of View	Copies data from one point of view to another.	Job Type: Copy POV Job Name: Name of the job POV Delimiter: Delimiter used in POV values. The delimiter must be enclosed in double quotation marks. Other than a comma, only these delimiters are supported: • _ (under score) • # (hash) • & (ampersand) • ~ (tilde) • % (percentage) • ; (semicolon) • : (colon) • - (dash) Source POV: Name of the source POV Destination POV: Name of the destination POV Copy Type: Specifies the data to copy from the source Source Cube Name: Name of the source cube Destination Cube Name: Name of the destination cube
Cube Refresh	Refreshes the OLAP cube.	Job Type: Cube Refresh Job Name: Name of the job
Export Data	Exports application data into a file using the export data settings, including file name, specified in a job of type export data. The file containing the exported data is stored in the repository.	Job Type: Export Data Job Name: Name of the job Export File Name: File name to which data is to be exported
Export Data Mapping	Exports a Data Mapping defined in Data Management to a specified location. This is a process-automated integration. For more information, see Adding Pre-built Integrations in Cloud EPM. Member mappings define relationships between source members and target dimension members within a single dimension.	Job Type: Export Data Mapping Dimension: The dimension name for a specific dimension to import, such as ACCOUNT, or ALL to import all dimensions File Name: The file and path from which to export mappings. The file format can be .CSV, .TXT, .XLS, or .XLSX. Include the outbox in the file path, for example, outbox/BESSAPPJan-06.csv. Location Name: Name of the location to which to export

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Integration Name	Description	Parameters/ Description
Export Metadata	Exports metadata from a file in the repository into the application using the export metadata settings specified in a job of type export metadata.	Job Type: Export Metadata Job Name: Name of a batch defined in export metadata Export Zip File Name: Name of the zip file for the exported metadata
Import Data	Imports data from a file in the repository into the application using the import data settings specified in a job of type import data.	Job Type: Import Data Job Name: Name of the job Import File Name: File name from which data is to be imported
Import Data Mapping	Imports a Data Mapping defined in Data Management to a specified location. This is a process-automated integration. Member mappings define relationships between source members and target dimension members within a single dimension. You can import member mappings from a selected Excel, .CSV or .TXT file.	Job Type: Import Data Mapping Dimension: Dimension name for a specific dimension to import, such as ACCOUNT, or ALL to import all dimensions File Name: The file and path from which to import mappings. The file format can be .CSV, .TXT, .XLS, or .XLSX. The file must be uploaded prior to importing, either to the inbox or to a sub-directory of the inbox. Include the inbox in the file path, for example,inbox/BESSAPPJan-06.csv. Import Mode: MERGE to add new rules or replace existing rules, or REPLACE to clear prior mapping rules before import Validation Mode: Whether to use validation mode: true or false. An entry of true validates the target members against the target application; false loads the mapping file without any validations. Note that the validation process is resource intensive and takes longer than the validation mode of false; the option selected by most customers is false. Location: Data Management location where the mapping rules should be loaded. Mapping rules are specific to a location in Data Management.

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Integration Name	Description	Parameters/ Description
Import Metadata	Imports metadata from a file in the repository into the application using the import metadata settings specified in a job of type import metadata.	Job Type: Import Metadata Job Name: Name of a batch defined in import metadata Import Zip File Name: Name of the zip file for the imported metadata
Lock Unlock Data Integration	Locks or unlocks an integration for a location, category and period in Data Exchange. This is a process-automated integration.	Job Type: Lock Unlock Data Integration Operation: Lock or unlock Lock Type: Whether the Lock/ Unlock operation is for an application or a location Period: Period of the POV from the integration or data load rule defined in Data Exchange, for example, "Jan-21" Category: Predefined Scenario value based on the POV Category from the integration (data rule) definition. The categories available are those that are created in the Data Integration set-up, such as "Actual." Application (Optional): If the selected Lock Type is application, the name of the application; for example, "Vision". Location (Optional): If the selected Lock Type is location, the name of the location. If the location is locked, data cannot be loaded to it. Unlock By Location (Optional): This parameter can be specified when the selected operation is lock and selected location is application. If checked when locking the target application, then the system locks all rules present in the location under target application and not the application-level lock. For more information, see Lock and Unlock POV in *REST APIs*.

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Integration Name	Description	Parameters/ Description
Run Batch	Executes a batch of jobs that have been defined in Data Management	Job Type: Run Batch Batch Name: Name of the batch to be executed, such as Dimension Map For POV (Dimension, Cat, Per) Path Report Format Type: The file format of the report - PDF, XLSX, or HTML Parameters: Can vary in count and values based on the report Location: The location of the report, such as Comma_Vision Run As: Specify this parameter in the Workflow tab
Run Business Rule	Launches a business rule.	Job Type: Run Business Rule Rule Name: Name of the business rule. Parameters: Run time prompts in JSON syntax. The parameter name should be the same as th name defined in rule definition. For example, {"MyScenario1":"Current", "MyVersion1":"BU Version_1", "ToEntity":"CA", "Rule_Level_Var":"AZ", "planType":"Plan1"} The following format is also supported; for example: "Scenario=Actual" "Entity=Total Geography" "Year=FY21" "Period=Apr"
Run Business Rule Set	Launches a business rule set. Rule sets with no runtime prompts or runtime prompts with default values will be supported.	Job Type: Run Business Rule Set Job Name: Name of the job Ruleset Name: Name of the business rule set Parameters: Run time prompts in JSON syntax. The Parameter name should be the same as the name defined in rule definition. For example, {"MyScenario1":"Current", "MyVersion1":"BU Version_1", "ToEntity":"CA", "Rule_Level_Var":"AZ", "planType":"Plan1"} The following format is also supported; for example: "Scenario=Actual" "Entity=Total Geography" "Year=FY21" "Period=Apr"

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Integration Name	Description	Parameters/ Description
Run Data Integration	Execute an integration or data load rule based on how periods are processed and source filters. This allows the integration of data loads defined in Data Exchange easily into the monthly processing schedule.	Job Type: Run Data Integration Integration Name: Name of the integration defined in Data Integration Period Name: Name of the period Import Mode: Determines how the data is imported into Data Integration Export Mode: Determines how the data is exported into Data Integration File Name: Applicable only for native file-based data loads and ignored if specified for other loads Source Filters: Parameter used to update the source filters defined for the data load rule or integration Target Options: Parameter used to update the target options defined for the data load rule or integration Execution Mode: Applicable only for Quick Mode integrations For more details about these parameters, see Running Integrations in *REST APIs*.

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Integration Name	Description	Parameters/ Description
Run Data Rule	Executes a Data Management data load rule based on the start period and end period, and import or export options that you specify.	Job Type: Run Data Rule Data Rule Name: Name of a data load rule defined in Data Management Start Period: The first period for which data is to be loaded. This period name must be defined in Data Management period mapping. End Period: The last period for which data is to be loaded. This period name must be defined in Data Management period mapping. Import Mode: Determines how the data is imported into Data Management: • APPEND to add to the existing rule • POV data in Data Management • REPLACE to delete the POV data and replace it with the data from the file • RECALCULATE to skip importing the data, but re-process the data with updated Mappings and Logic Accounts. • NONE to skip data import into Data Management staging table Export Mode: Determines how the data is exported into Data Management: • STORE_DATA to merge the data in the Data Managementstaging table with the existing Financial Consolidation and Close or Tax Reporting data • ADD_DATA to add the data in the Data Management staging table to Financial Consolidation and Close or Tax Reporting • SUBTRACT_DATA to subtract the data in the Data Management staging table from existing Financial Consolidation and Close or Tax Reporting data • REPLACE_DATA to clear the POV data and replace it with data in the Data

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Integration Name	Description	Parameters/ Description
		Management staging table. The data is cleared for Scenario, Version, Year, Period, and Entity • NONE to skip data export from Data Management to Financial Consolidation and Close or Tax Reporting File Name: If you do not specify a file name, the API imports the data contained in the file name specified in the load data rule. The data file must already reside in the INBOX prior to data rule execution. Run As: Specify this parameter in the Workflow tab
Run Pipeline	Executes a pipeline based on job parameters and variables that you select.	Job Type: Pipeline Job Name: Pipeline code defined for the Pipeline in Data Integration Start Period: The first period for which data is to be loaded. This period name must be defined in Data Integration Period mapping. End Period: The last period for which data is to be loaded. This period name must be defined in Data Integration period mapping. Import Mode: Determines how the data is imported into Data Integration Export Mode: Determines how the data is exported into Data Integration Attach Logs: Whether logs are included as an attachment to an email Send Email: Determines when an email is sent when a Pipeline is run Send To: Determines the recipient email ID for the email notification For more details about these parameters, see Running a Pipeline in *REST APIs*.

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Automation Integrations for Financial Consolidation and Close and Tax Reporting

Integration Name	Description	Parameters/ Description
Clear Cube **Note:** This integration is applicable only for Tax Reporting.	Clears specific data within input and reporting cubes.	Name: Name of the clear cube job.
Copy Ownership Data to Next Year	Automates task to copy the ownership data from the last period of a year to the first period of the next year. For more information, see copyOwnershipDataToNextYear in *Working with EPM Automate*.	Scenario: The name of the scenario, such as Actual, selectable Years: Selectable
Cube Refresh	Refreshes the OLAP cube.	Name: Name of the refresh cube job.
Clear Data	Executes a Clear Data job using the profile name. For more information about using Clear Data in Financial Consolidation and Close, see Clear Data in *REST APIs*. For more information about using Clear Data in Tax Reporting, see Clear Data in *REST APIs*.	Profile Name: Clear data profile name.
Copy Data	Executes a Copy Data job using the profile name.	Profile Name: Copy data profile name.
Export Data	Exports application data into a file using the export data settings, including file name, specified in a job of type export data. The file containing the exported data is stored in the repository.	Name: Name of the export data job. Export File Name: **Optional**. File name to which data is to be exported.
Export Data Mapping	Exports a Data Mapping defined in Data Management to a specified location. This is a process-automated integration. For more information, see Adding Pre-built Integrations in Cloud EPM. Member mappings define relationships between source members and target dimension members within a single dimension.	Dimension: The dimension name for a specific dimension to import, such as ACCOUNT, or ALL to import all dimensions. File Name: The file and path from which to export mappings. The file format can be .CSV, .TXT, .XLS, or .XLSX. Include the outbox in the file path. For example, outbox/BESSAPPJan-06.csv. Location Name: The name of the location to which to export.

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Integration Name	Description	Parameters/ Description
Export Ownership Data	Automates task to export ownership data from a entity to a comma-delimited CSV file. For more information, see exportOwnershipData in *Working with EPM Automate*.	Entity: The name of the entity. Scenario: The name of the scenario, such as Actual. Selectable. Years: Selectable Period: The name of the period, such as January. Selectable. File Name: The name of the file to export.
Import Data	Imports data from a file in the repository into the application using the import data settings specified in a job of type import data.	Name: Name of the import data job. Import File Name: **Optional**. File name from which data is to be imported.
Import Data Mapping	Imports a Data Mapping defined in Data Management to a specified location. This is a process-automated integration. Member mappings define relationships between source members and target dimension members within a single dimension. You can import member mappings from a selected Excel, .CSV or .TXT file.	Job Type: The job type, MAPPINGIMPORT. Job Name: The dimension name for a specific dimension to import, such as ACCOUNT, or ALL to import all dimensions. File Name: The file and path from which to import mappings. The file format can be .CSV, .TXT, .XLS, or .XLSX. The file must be uploaded prior to importing, either to the inbox or to a sub-directory of the inbox. Include the inbox in the file path, for example,inbox/BESSAPPJan-06.csv. Import Mode: MERGE to add new rules or replace existing rules, or REPLACE to clear prior mapping rules before import. Validation Mode: Whether to use validation mode: true or false. An entry of true validates the target members against the target application; false loads the mapping file without any validations. Note that the validation process is resource intensive and takes longer than the validation mode of false; the option selected by most customers is false. Location Name: The Data Management location where the mapping rules should be loaded. Mapping rules are specific to a location in Data Management.

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Integration Name	Description	Parameters/ Description
Import Metadata	Imports metadata from a file in the repository into the application using the import metadata settings specified in a job of type import metadata.	Name: The name of a batch defined in import metadata.
Import Ownership Data	Automates task to import ownership data from a CSV file available in the environment into a period. For more information, see importOwnershipData in *Working with EPM Automate*.	Scenario: The name of the scenario, such as Actual. Selectable. Years: Selectable Period: The name of the period, such as January. Selectable. File Name: The name of the file to import.
Journal Period	Opens or closes a journal period automatically. The system will close the period only if there are no Approved and Unposted journals. If there are Approved and Unposted journals, the system will not close the period, and returns an error. If there are Unposted journals in Working and Submitted status, the system will close the period, with a warning.	Scenario: The name of the scenario, such as Actual Year: The year, such as FY20 Period: The name of the period, such as January Action: Open or Close
Monitor Enterprise Journals **Note:** This integration is applicable only for Financial Consolidation and Close.	Monitors the completion status of Journals within a Year/ Period or filtered list.	Year: Optional. The year, such as 2022. Selectable. Period: Optional. The name of the period, such as January. Selectable. Filter Name: Optional.The name of the Filter you created to monitor the status of the Enterprise Journals. **Note:** Though all the parameters are optional, you must specify at least a Filter Name or Year and Period.
Recompute Ownership Data	Automates task for recomputing of ownership data. For more information, see recomputeOwnershipData in *Working with EPM Automate*.	Scenario: The name of the scenario, such as Actual Years: The year, such as FY20 Period: The name of the period, such as January

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Integration Name	Description	Parameters/ Description
Run Batch Rule	Executes a batch of jobs that have been defined in Data Management.	Name: The name of the report to be executed, such as Dimension Map For POV (Dimension, Cat, Per) Path Report Format Type: The file format of the report - PDF, XLSX, or HTML Parameters: Can vary in count and values based on the report Location: The location of the report, such as Comma_Vision Run As: You must specify this parameter in the Workflow tab.
Run Business Rule	Launches a business rule.	Name: The name of a business rule exactly as it is defined. Parameters: Run time prompts in JSON syntax. Parameter name should exactly be same as defined in rule definition. For example, { "MyScenario1":"Current", "MyVersion1":"BU Version_1", "ToEntity":"CA", "Rule_Level_Var":"AZ", "planType":"Plan1"} Following format is also supported, example: "Scenario=Actual" "Entity=Total Geography" "Year=FY21" "Period=Apr"
Run Business Rule Set	Launches a business rule set. Rule sets with no runtime prompts or runtime prompts with default values will be supported.	Name:The name of a business rule set exactly as it is defined. Parameters: Run time prompts in JSON syntax. Parameter name should exactly be same as defined in rule definition. For example, { "MyScenario1":"Current", "MyVersion1":"BU Version_1", "ToEntity":"CA", "Rule_Level_Var":"AZ", "planType":"Plan1"} Following format is also supported, example: "Scenario=Actual" "Entity=Total Geography" "Year=FY21" "Period=Apr"

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Integration Name	Description	Parameters/ Description
Run Consolidation	This task is a utility task to run consolidation. Task will prompt user to enter parameters for running the tasks such as Scenario, Year, Period, and Entity.	Scenario Year Period Entity: Multiple entities can be added with comma separator.

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Integration Name	Description	Parameters/ Description
Run Data Rule	Executes a Data Management data load rule based on the start period and end period, and import or export options that you specify.	Job Name: The name of a data load rule defined in Data Management. Start Period: The first period for which data is to be loaded. This period name must be defined in Data Management period mapping. End Period: The last period for which data is to be loaded. This period name must be defined in Data Management period mapping. Import Mode: Determines how the data is imported into Data Management. APPEND to add to the existing rule POV data in Data Management REPLACE to delete the POV data and replace it with the data from the file RECALCULATE to skip importing the data, but re-process the data with updated Mappings and Logic Accounts. NONE to skip data import into Data Management staging table Export Mode: Determines how the data is exported into Data Management. STORE_DATA to merge the data in the Data Management staging table with the existing Financial Consolidation and Close or Tax Reporting data ADD_DATA to add the data in the Data Management staging table to Financial Consolidation and Close or Tax Reporting SUBTRACT_DATA to subtract the data in the Data Management staging table from existing Financial Consolidation and Close or Tax Reporting data REPLACE_DATA to clear the POV data and replace it with data in the Data Management staging table. The data is cleared for Scenario, Version, Year, Period, and Entity NONE to skip data export fromData Management to

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Integration Name	Description	Parameters/ Description
		Financial Consolidation and Close or Tax Reporting File Name: **Optional**. If you do not specify a file name, this API imports the data contained in the file name specified in the load data rule. The data file must already reside in the INBOX prior to data rule execution. Run As: You must specify this parameter in the Workflow tab.
Run Force Consolidation	This task is a utility task to run force consolidation. The task will prompt the user to enter parameters for running the tasks such as Scenario, Year, Period, and Entity.	Scenario Year Period Entity: Multiple entities can be added using a comma separator.
Run Force Translation	This task is a utility task to run force translation. The task will prompt user to enter parameters for running the tasks such as Scenario, Year, Period, and Entity.	Scenario Year Period Entity: Multiple entities can be added with comma separator.
Run Translation	This task is a utility task to run translation. The task will prompt user to enter parameters for running the tasks such as Scenario, Year, Period, and Entity.	Scenario Year Period Entity: Multiple entities can be added with comma separator.

Automation Integrations for FreeForm

Integration Name	Description	Parameters/ Description
Clear Cube	Clears specific data within input and reporting cubes.	Name: Name of the clear cube job.
Cube Refresh	Refreshes the OLAP cube.	Name: Name of the refresh cube job.
Export Data	Exports application data into a file using the export data settings, including file name, specified in a job of type export data. The file containing the exported data is stored in the repository.	Name: Name of the export data job. Export File Name: **Optional**. File name to which data is to be exported.

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Integration Name	Description	Parameters/ Description
Export Data Mapping	Exports a Data Mapping defined in Data Management to a specified location. This is a process-automated integration. For more information, see Adding Pre-built Integrations in Cloud EPM. Member mappings define relationships between source members and target dimension members within a single dimension.	Job Type: Export Data Mapping Dimension: The dimension name for a specific dimension to import, such as ACCOUNT, or ALL to import all dimensions File Name: The file and path from which to export mappings. The file format can be .CSV, .TXT, .XLS, or .XLSX. Include the outbox in the file path, for example, outbox/BESSAPPJan-06.csv. Location Name: Name of the location to which to export.
Export Metadata	Exports metadata from a file in the repository into the application using the export metadata settings specified in a job of type export metadata.	Dimension: The dimension name for a specific dimension to import, such as ACCOUNT, or ALL to import all dimensions. File Name: The file and path from which to export mappings. The file format can be .CSV, .TXT, .XLS, or .XLSX. Include the outbox in the file path. For example, outbox/BESSAPPJan-06.csv. Location Name: The name of the location to which to export.
Import Data	Imports data from a file in the repository into the application using the import data settings specified in a job of type import data.	Name: Name of the import data job. Import File Name: **Optional.** File name from which data is to be imported.

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Integration Name	Description	Parameters/ Description
Import Data Mapping	Imports a Data Mapping defined in Data Management to a specified location. This is a process-automated integration. Member mappings define relationships between source members and target dimension members within a single dimension. You can import member mappings from a selected Excel, .CSV or .TXT file.	Job Type: Import Data Mapping Dimension: Dimension name for a specific dimension to import, such as ACCOUNT, or ALL to import all dimensions File Name: The file and path from which to import mappings. The file format can be .CSV, .TXT, .XLS, or .XLSX. The file must be uploaded prior to importing, either to the inbox or to a sub-directory of the inbox. Include the inbox in the file path, for example,inbox/BESSAPPJan-06.csv. Import Mode: MERGE to add new rules or replace existing rules, or REPLACE to clear prior mapping rules before import Validation Mode: Whether to use validation mode: true or false. An entry of true validates the target members against the target application; false loads the mapping file without any validations. Note that the validation process is resource intensive and takes longer than the validation mode of false; the option selected by most customers is false. Location: Data Management location where the mapping rules should be loaded. Mapping rules are specific to a location in Data Management.
Import Metadata	Imports metadata from a file in the repository into the application using the import metadata settings specified in a job of type import metadata.	Name: The name of a batch defined in import metadata.
Run Batch	Executes a batch of jobs that have been defined in Data Management.	Name: The name of the report to be executed, such as Dimension Map For POV (Dimension, Cat, Per) Path Report Format Type: The file format of the report, PDF, XLSX, or HTML Parameters: Can vary in count and values based on the report Location: The location of the report, such as Comma_Vision

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Integration Name	Description	Parameters/ Description
Run Business Rule	Launches a business rule.	Name: The name of a business rule exactly as it is defined. Parameters: Run time prompts in JSON syntax. Parameter name should exactly be same as defined in rule definition. For example, { "MyScenario1":"Current", "MyVersion1":"BU Version_1", "ToEntity":"CA", "Rule_Level_Var":"AZ", "planType":"Plan1"} Following format is also supported, example: "Scenario=Actual" "Entity=Total Geography" "Year=FY21" "Period=Apr"
Run Business Ruleset	Launches a business ruleset. Rulesets with no runtime prompts or runtime prompts with default values will be supported.	Name: The name of a business ruleset exactly as it is defined. Parameters: Run time prompts in JSON syntax. Parameter name should exactly be same as defined in rule definition. For example, { "MyScenario1":"Current", "MyVersion1":"BU Version_1", "ToEntity":"CA", "Rule_Level_Var":"AZ", "planType":"Plan1"} Following format is also supported, example: "Scenario=Actual" "Entity=Total Geography" "Year=FY21" "Period=Apr"

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Integration Name	Description	Parameters/ Description
Run Data Rule	Executes a Data Management data load rule based on the start period and end period, and import or export options that you specify.	Job Name: The name of a data load rule defined in Data Management. Start Period: The first period for which data is to be loaded. This period name must be defined in Data Management period mapping. End Period: The last period for which data is to be loaded. This period name must be defined in Data Management period mapping. Import Mode: Determines how the data is imported into Data Management. APPEND to add to the existing rule POV data in Data Management REPLACE to delete the POV data and replace it with the data from the file RECALCULATE to skip importing the data, but re-process the data with updated Mappings and Logic Accounts. NONE to skip data import into Data Management staging table exportMode: Determines how the data is exported into Data Management. STORE_DATA to merge the data in the Data Management staging table with the existing Oracle Hyperion Planning data ADD_DATA to add the data in the Data Management staging table to Planning SUBTRACT_DATA to subtract the data in the Data Management staging table from existing FreeForm data REPLACE_DATA to clear the POV data and replace it with data in the Data Management staging table. The data is cleared for Scenario, Version, Year, Period, and Entity NONE to skip data export from Data Management to FreeForm File Name: **Optional**. If you do not specify a file name, this API imports the data contained in the file name specified in the

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Integration Name	Description	Parameters/ Description
		load data rule. The data file must already reside in the INBOX prior to data rule execution.

Automation Integrations for Planning and Planning Modules

Integration Name	Description	Parameters/ Description
Clear Cube	Clears specific data within input and reporting cubes.	Name: Name of the clear cube job.
Cube Refresh	Refreshes the OLAP cube.	Name: Name of the refresh cube job.
Export Data	Exports application data into a file using the export data settings, including file name, specified in a job of type export data. The file containing the exported data is stored in the repository.	Name: Name of the export data job. Export File Name: **Optional**. File name to which data is to be exported.
Import Data	Imports data from a file in the repository into the application using the import data settings specified in a job of type import data.	Name: Name of the import data job. Import File Name: **Optional.** File name from which data is to be imported.
Import Metadata	Imports metadata from a file in the repository into the application using the import metadata settings specified in a job of type import metadata.	Name: The name of a batch defined in import metadata.
Run Batch	Executes a batch of jobs that have been defined in Data Management.	Name: The name of the report to be executed, such as Dimension Map For POV (Dimension, Cat, Per) Path Report Format Type: The file format of the report, PDF, XLSX, or HTML Parameters: Can vary in count and values based on the report Location: The location of the report, such as Comma_Vision

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Integration Name	Description	Parameters/ Description
Run Business Rule	Launches a business rule.	Name: The name of a business rule exactly as it is defined. Parameters: Run time prompts in JSON syntax. Parameter name should exactly be same as defined in rule definition. For example, { "MyScenario1":"Current", "MyVersion1":"BU Version_1", "ToEntity":"CA", "Rule_Level_Var":"AZ", "planType":"Plan1"} Following format is also supported, example: "Scenario=Actual" "Entity=Total Geography" "Year=FY21" "Period=Apr"
Run Business Ruleset	Launches a business ruleset. Rulesets with no runtime prompts or runtime prompts with default values will be supported.	Name: The name of a business ruleset exactly as it is defined. Parameters: Run time prompts in JSON syntax. Parameter name should exactly be same as defined in rule definition. For example, { "MyScenario1":"Current", "MyVersion1":"BU Version_1", "ToEntity":"CA", "Rule_Level_Var":"AZ", "planType":"Plan1"} Following format is also supported, example: "Scenario=Actual" "Entity=Total Geography" "Year=FY21" "Period=Apr"

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Integration Name	Description	Parameters/ Description
Run Data Rule	Executes a Data Management data load rule based on the start period and end period, and import or export options that you specify.	Job Name: The name of a data load rule defined in Data Management. Start Period: The first period for which data is to be loaded. This period name must be defined in Data Management period mapping. End Period: The last period for which data is to be loaded. This period name must be defined in Data Management period mapping. Import Mode: Determines how the data is imported into Data Management. APPEND to add to the existing rule POV data in Data Management REPLACE to delete the POV data and replace it with the data from the file RECALCULATE to skip importing the data, but re-process the data with updated Mappings and Logic Accounts. NONE to skip data import into Data Management staging table exportMode: Determines how the data is exported into Data Management. STORE_DATA to merge the data in the Data Management staging table with the existing Oracle Hyperion Planning data ADD_DATA to add the data in the Data Management staging table to Planning SUBTRACT_DATA to subtract the data in the Data Management staging table from existing Planning data REPLACE_DATA to clear the POV data and replace it with data in the Data Management staging table. The data is cleared for Scenario, Version, Year, Period, and Entity NONE to skip data export from Data Management to Planning File Name: **Optional**. If you do not specify a file name, this API imports the data contained in the file name specified in the

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Integration Name	Description	Parameters/ Description
		load data rule. The data file must already reside in the INBOX prior to data rule execution.

Automation Integrations for Profitability and Cost Management

Integration Name	Description	Parameters/ Description
Apply Data Grants	Applies data grants for a given Profitability and Cost Management application. This API submits a job to create and apply the data grants in Essbase. This API removes all existing data grants in Oracle Essbase and recreates them with the latest information from the application. It can also be used to repair data grants if there are any issues.	None
Deploy ML Cube	Deploy or redeploy the calculation cube for a selected Profitability and Cost Management application.	isKeepData: Specify whether to preserve existing data isReplacecube: Specify whether to replace existing comment: Any user comments

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Integration Name	Description	Parameters/ Description
Run ML Calc	Run or clear calculations for a selected application. Use with Management Ledger.	povGroupMember: The POV group member for which to run calculations, such as 2015_January_Actual isClearCalculated: Whether to clear the calculation data, true or false subsetStart: Rule Set Starting Sequence Number subsetEnd: Rule Set Ending Sequence Number Rule: Rule Name for a SINGLE_RULE ruleSetName: Rule Set Name for a SINGLE_RULE option exeType: The execution type specifies which rules to run; possible values are ALL_RULES, RULESET_SUBSET, SINGLE_RULE. Other parameters are required based on the exeType value. exeType: ALL_RULES overrides all other options such as subsetStart, subsetEnd, ruleSetName, ruleName, and so on. exeType: RULESET_SUBSET considers only subsetStart and subsetEnd. exeType: SINGLE_RULE considers only ruleSetName and ruleName. Comment: Use comment text. Delimiter: String delimiter for POV group members, such as an underscore (_).
Clear ML POV	Clear model artifacts and data from a POV combination for any application.	POV GroupMember: The POV group member for which to run calculations, such as 2015_January_Actual isManageRule: Whether to clear the program rule details isInputData: Whether to clear input data IsAllocatedValues: Whether to clear allocated values stringDelimiter: String delimiter for POV group members

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Integration Name	Description	Parameters/ Description
Copy ML POV	Copy model artifacts and data from a Source POV combination to a Destination POV combination for any application. Use with Management Ledger applications.	POVs: Included in the path srcPOVMemberGroup: Source POV member group, such as 2014_January_Actual destPOVMemberGroup: Destination POV member group, such as 2014_March_Actual isManageRule: Whether to copy the program rule details isInputData: Whether to copy input data modelViewName: To copy a slice of data from source POV to destination POV Create Dest POV: Whether to create the destination POV if it does not already exist String Delimiter: String delimiter for POV group members

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Integration Name	Description	Parameters/ Description
Run Data Rule	Executes a Data Management data load rule based on the start period and end period, and import or export options that you specify.	Job Name: The name of a data load rule defined in Data Management. Start Period: The first period for which data is to be loaded. This period name must be defined in Data Management period mapping. End Period: The last period for which data is to be loaded. This period name must be defined in Data Management period mapping. Import Mode: Determines how the data is imported into Data Management. APPEND to add to the existing rule POV data in Data Management. REPLACE to delete the POV data and replace it with the data from the file RECALCULATE to skip importing the data, but re-process the data with updated Mappings and Logic Accounts. NONE to skip data import into Data Management staging table Export Mode: Determines how the data is exported into Data Management. STORE_DATA to merge the data in the Data Management staging table with the existing Profitability and Cost Management data. ADD_DATA to add the data in the Data Management staging table to Profitability and Cost Management. SUBTRACT_DATA to subtract the data in the Data Management staging table from existing Profitability and Cost Management data REPLACE_DATA to clear the POV data and replace it with data in the Data Management staging table. The data is cleared for Scenario, Version, Year, Period, and Entity NONE to skip data export from Data Management to

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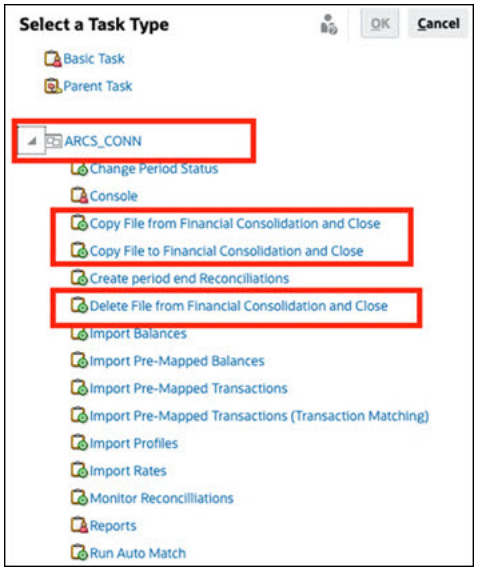
Integration Name	Description	Parameters/ Description
		Profitability and Cost Management File Name: **Optional**. If you do not specify a file name, this API imports the data contained in the file name specified in the load data rule. The data file must already reside in the INBOX prior to data rule execution.
Run Batch Rule	Executes a batch of jobs that have been defined in Data Management.	Job Name: The name of a batch defined in Data Management.
Update Dimension	Uploads a new dimension flat file for an application created using a flat file. This is a process-automated integration. For more information, see Update Dimensions As a Job in *REST APIs*.	File Name: Data file name Seperator Character: **Optional** Parameter

Copy and Delete Integration Files

The following integrations are common for Oracle Fusion Cloud Enterprise Performance Management:

- **Copy File from** <Cloud EPM Service>
- **Copy File to** <Cloud EPM Service>
- **Delete File from** <Cloud EPM Service>

Example



To perform these integrations:

1. On the Home page, click **Application**, and then click **Task Manager**.

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2. Click the **Integrations** tab on the left.
3. From the list of pre-built integrations, select **Copy File from** <Cloud EPM Service>. This copies a file from the Current Task Manager service to a Remote Cloud EPM Service. For example, if you have configured Task Manager in FreeForm and set up an Account Reconciliation connection, **Copy File from FreeForm** copies the file from FreeForm to Account Reconciliation

> **Note**
>
> This applies to all remote Cloud EPM services except Enterprise Data Management.

- Enter the following Parameters:

  - **File Name**: Browse to and select the file that you want to copy.
  - **Save File As**: Enter a file name (which can be different than the original file name).
  - **External Directory Name** (Optional): Select the name of a directory.
- Click **Save and Close**.

4. Select **Copy File to** <Cloud EPM Service>. This copies a file to the current service where Task Manager is configured from another Cloud EPM service.

> **Note**
>
> This applies to all remote Cloud EPM services except Enterprise Data Management.

- Enter the following Parameters:
  - **File Name**: Browse to and select the file that you want to copy.
  - **Save File As**: Enter a file name (which can be different than the original file name).
  - **External Directory Name** (Optional): Select the name of a directory.
- Click **Save and Close**.

5. Select **Delete File from** <Cloud EPM Service>. This deletes a file from a Oracle Fusion Cloud Enterprise Performance Management service.

> **Note**
>
> This applies to all remote Cloud EPM services except Enterprise Data Management.

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- For **File Name**, browse to and select the file that you want to delete.

- Click **Save and Close**.

See also, Automation Integrations for Cloud EPM.

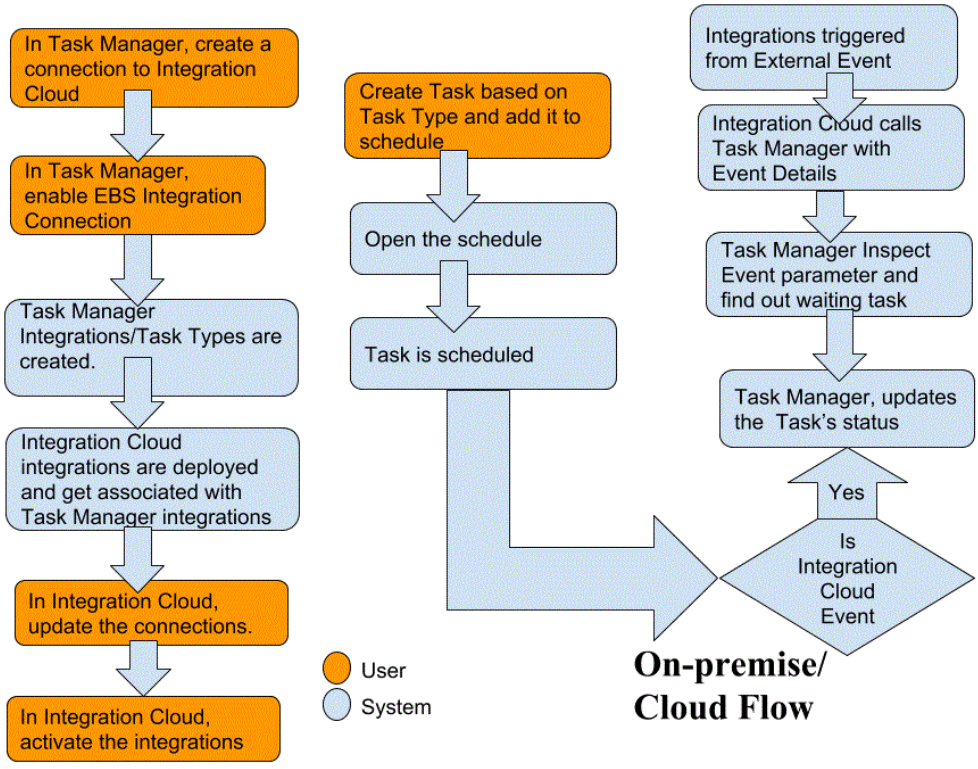
Managing Integrations with Cloud and On-Premises Applications

This appendix describes how to set up these pre-built Task Manager Integrations:

- Oracle Cloud ERP Integrations
- On-Premises Integrations
- On-Premises Oracle E-Business Suite Event Monitoring Integrations

Oracle Fusion Cloud Enterprise Performance Management uses Oracle Integration cloud as an integration platform for Task Manager to perform automated tasks with non-Cloud EPM services.

The following diagram shows the system and user flow that applies to FreeForm:



End User Integrations for Oracle Cloud ERP

Task Manager end user integrations allow you to access functionality in remote Cloud environments. This section lists the available Task Manager end user integrations available for Oracle Cloud ERP.

For a description of the remote Cloud functionality, see the documentation for that Cloud service.

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End User Integrations for Oracle Cloud ERP - Assets

- Asset Balances Reconciliation with General Ledger
- Asset Cost Summary
- Asset Reserve Summary
- Calculate Depreciation
- Calculate Deferred Depreciation
- Capitalize CIP Assets
- Create Assets Accounting
- Journal Entry Reserve Ledger Report
- Manage Asset Assignments
- Manage Asset Financial Transactions
- Manage Mass Retirements
- Manage Mass Transfers
- Periodic Mass Copy for Tax Books
- Period Close Exception Report
- Prepare Source Lines and Submit Post Mass Additions
- Reinstate Assets
- Retire Assets

End User Integrations for Oracle Cloud ERP - Cash Management

- Bank Statement Reconciliation
- Create Accounting
- Cash to General Ledger Reconciliation Report
- Subledger Period Close Exceptions Report

End User Integrations for Oracle Cloud ERP - General Ledger

- AutoPost Journals
- AutoReverse Journals
- Close General Ledger Periods
- Create Allocation Rules
- Create Balance Sheet Closing Journals
- Create Income Statement Closing Journals
- Extract Payables to General Ledger Reconciliation Data
- Extract Receivables to General Ledger Reconciliation Data
- Financial Reporting Center Work Area
- General Accounting Dashboard
- Generate Allocation Rules
- General Ledger Trial Balance

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- General Ledger Average Trial Balance
- General Ledger Journals Report
- General Ledger General Journals Report
- General Ledger Journals Details Report
- General Ledger Journals Day Book Report
- General Ledger Journals Batch Summary Report
- General Ledger Account Details Report
- General Ledger Account Analysis for Contra Account Report
- General Ledger Average Balance Audit Account Analysis Report
- General Ledger Account Analysis Report
- General Ledger Trial Balance Report
- Import Journals
- Manage General Ledger Accounting Periods
- Manage Journals
- Open General Ledger Periods
- Period Close Dashboard
- Reconcile Payables to General Ledger
- Reconcile Receivables to General Ledger
- Revalue Balances
- Translation
- Transfer Balances Cross Ledgers
- Transfer Balances to Secondary Ledger

End User Integrations for Oracle Cloud ERP - Intercompany

- Intercompany Account Details Report
- Intercompany Reconciliation
- Intercompany Transaction Summary Report
- Intercompany Transactions Work Area
- Manage Intercompany Period Status
- Transfer Intercompany Transactions to General Ledger
- Transfer Intercompany Transactions to Receivables
- Transfer Intercompany Transactions to Payables

End User Integrations for Oracle Cloud ERP - Payables

- Apply Missing Conversion Rates
- Complete or Cancel Incomplete Payment Process Requests
- Create Payables Accounting
- Create Mass Additions
- Extract Payables to General Ledger Reconciliation Data

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- Import Payables Invoices
- Import Payables Payment Requests
- Manage Payables Period
- Open Items Revaluation Report
- Payables Unaccounted Transactions Report
- Payables Invoices Landing Page
- Payables Unaccounted Transactions Report with Sweep Option
- Payables Trial Balance Report
- Period Close Exceptions Report
- Review Payables to General Ledger Reconciliation Report
- Transactions Pending Approvals
- Transfer Cost to Cost Management
- Update Matured Bills Payable Status
- Update Matured Bills Payable Status
- Validate Unvalidated Transactions

End User Integrations for Oracle Cloud ERP - Receivables

- Approve or Reject Customer Credits
- Billing Work Area
- Clear Receipts Automatically
- Create Automatic Billing Adjustments
- Create Automatic Receipt Batch
- Create Receipts Remittance Batch
- Create Late Charge Batch
- Create Automatic Receipt Write-Offs
- Create Receivables Accounting
- Extract Reconciliation Data From Receivables to General Ledger
- Import AutoInvoice: Master
- Manage Receivables Accounting Periods
- Manage Revenue Adjustments
- Receivables Balances Work Area
- Process Receipts Through Lockbox
- Recognize Revenue
- Receivables Aging by General Ledger Account Report
- Submit Subledger Period Close Exceptions Report

End User Integrations for Oracle Cloud ERP - Tax

- Tax Reconciliation Report
- Tax Reconciliation by Taxable Account Report

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Application Token Values for Cloud ERP Integrations

Token Name: SERVER

Token Description: Property used for URL-based integration, for example, as in this fictitious URL: `https://customer_chosen_domain_name_fa.DC.oraclecloud.com`

> **Note**
>
> Do not specify forward slash (/) at the end of the URL.

Pre-requisites to Display ERP Cloud Task in Task Manager

To display Cloud ERP task in Task Manager, Cloud ERP application must be enabled to display the site within iFrame. If you see a blank page with an error "refused to connect <Cloud ERP URL>" when opening the Cloud ERP, make sure you complete these pre-requisite.

1. Identity Cloud Service (IDCS) - Allow Authentication Via an iFrame
   a. In the IDCS Admin Console, in the navigation tray, click **Settings** > **Session Settings**.
   b. Enable the **All Cross-Origin Resource Sharing (CORS)** slider.
   c. In **Allowed CORS Domain Names**, enter the EPM root URL
   d. Click **Save**. See IDCS - How to Allow Authentication Via an an IFrame.

   > **Note**
   >
   > If you are not using IDCS and using some other application for single sign on such as Okta, you must enable Cross-Origin Resource Sharing (CORS) in that application for EPM.

2. Request the Oracle Cloud ERP Applications administrator to enable Cross-Origin Resource Sharing (CORS) by adding the Oracle Fusion Cloud Enterprise Performance Management root URL to the Oracle Cloud Application for the profile options:
   - ORA_CORS_ORIGINS
   - ORACLE.ADF.VIEW.ALLOWED_ORIGINS

   To update the profile value:
   a. Login to Cloud ERP applications as an administrator.
   b. From the **User Admin** menu, go to **Setup And Maintenance**.
   c. Search for **Manage Administrator Profile Values**.
   d. Enter ORACLE.ADF.VIEW.ALLOWED_ORIGINS as the profile code and click **Search**.
   e. In the **Profile Values** section, specify the value. To access third party URL, include 'self' along with third party URL (use white space as separator). For example: 'self' <white space> <url>

      See
      - Valid Value for ORACLE.ADF.VIEW.ALLOWED_ORIGINS to Avoid Fetching Data Issue
      - How to use the ORACLE.ADF.VIEW.ALLOWED_ORIGINS profile option

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- Administering Visual Builder Studio

f. Repeat the same steps for the ORA_CORS_ORIGINS profile option.

Event Monitoring Integrations for Oracle Cloud ERP

This section lists the out of box event monitoring integrations for Oracle Cloud ERP.

> **Note**
>
> For custom process automation or event monitoring integration to Oracle Cloud ERP, see Creating Custom Task Manager Integrations.

Supported Events for Oracle Cloud ERP General Ledger

The following events are supported for General Ledger:

Source	Event	Description
General Ledger	Accounting Period Closed	Signals when a general ledger accounting period is closed.
General Ledger	Accounting Period Opened	Signals when a general ledger accounting period is opened.
General Ledger	Accounting Period Reopened	Signals when a general ledger accounting period is reopened.
General Ledger	Journal Batch Approved	Signals when a journal batch is approved.
General Ledger	Journal Batch Posting Completed	Signals when a journal batch is posted.

Integrations Added to Existing Out of Box Connection

The table lists integrations that are added to monitor these business events from Task Manager, which includes:

- Integration Types in Tax Reporting
- Task Types in Tax Reporting
- Integration Flows in Integration Cloud

Connection Name	Integration Name	Integration Code	Event Name	Description	Parameter(s)
Oracle Cloud ERP - General Ledger	Period Close Event	R13GLPeriodClose	Accounting Period Closed	Oracle Cloud ERP General Ledger Period Close Event Monitoring	**LedgerName**: Name of the Ledger. For example, US Primary Ledger. **Period**: Name of the Period. For example, 01-19.

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Connection Name	Integration Name	Integration Code	Event Name	Description	Parameter(s)
Oracle Cloud ERP - General Ledger	Period Open Event	R13GLPeriodOpen	Account Period Opened	Oracle Cloud ERP General Ledger Period Open Event Monitoring	**LedgerName**: Name of the Ledger. For example, US Primary Ledger. **Period**: Name of the Period. For example, 01-19.
Oracle Cloud ERP - General Ledger	Period Reopen Event	R13GLPeriodReopen	Accounting Period Reopened	Oracle Cloud ERP General Ledger Period Reopen Event Monitoring	**LedgerName**: Name of the Ledger. For example, US Primary Ledger. **Period**: Name of the Period. For example, 01-19.
Oracle Cloud ERP - General Ledger	Journal Approve Event	R13GLJournalBatchApprove	Journal Batch Approved	Oracle Cloud ERP General Ledger Journal Batch Approve Event Monitoring	**BatchName**: Journal Batch Name **Period**: Name of the Period. For example, 01-19.
Oracle Cloud ERP - General Ledger	Journal Post Event	R13GLJournalBatchPost	Journal Batch Posted	Oracle Cloud ERP General Ledger Journal Batch Post Event Monitoring	**BatchName**: Journal Batch Name **Period**: Name of the Period. For example, 01-19.

Setting Up the Integration in Task Manager

In Task Manager, you can create an Event Monitoring integration for Oracle Cloud ERP - General Ledger. The Event Monitoring integration is triggered when an external event occurs in another Cloud service or on-premises application.

Make sure the business events are enabled in Oracle Cloud ERP. You can verify the events using REST API. See these topics in the "ERP Business Events REST Endpoints" section of the *REST API for Oracle Financials Cloud* guide available at docs.oracle.com.

To verify the events, see "Get all business event records".

If any event is not enabled, see "Update the enabled indicator for a business event".

Task Manager uses Oracle Integration Cloud for all the integrations to external applications. You can setup a connection to Oracle Integration Cloud in Task Manager using Basic or OAuth 2.0 authentication.

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> **Note**
>
> - For Oracle Integration Cloud Generation 2, both Basic Auth and OAuth 2.0 is supported.
> - For Oracle Integration Cloud Generation 3, only OAuth 2.0 is supported.
> - Basic Auth user must have the Service Administrator role for Oracle Integration Cloud.
> - For OAuth 2.0, only client credentials are supported. OAuth 2.0 client application should have the Service Administrator role for Oracle Integration Cloud and allowed scope should be all.

Before setting up the OAuth 2.0 connection, make sure you've the client credentials such as access token URL, client ID, client secret, and scope. For setting up the OAuth client application, see **OAuth Authentication in Oracle Integration** in *Oracle Integration Cloud Generation 3* documentation.

To setup the Integration Cloud connection in Task Manager:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Click **Manage Connections**, and then from **Actions**, select **Integration Cloud Connection**.
4. Select one of the following authentication mechanisms:
   - **Basic Auth**: Specify the Integration Cloud URL, service administrator user ID, and password.
   - **OAuth 2.0**: Provide the Integration Cloud URL, access token URL, client ID, client secret, and scope.

   > **Note**
   >
   > - For Gen 2, enter the URL: `https://<SERVICE_NAME>-<TENANT_NAME>.integration.ocp.oraclecloud.com`
   > - For Gen 3, enter the URL: `https://<SERVICE_INSTANCE>.integration.<REGION>.ocp.oraclecloud.com`. To find your service instance, log into Integration Cloud and open the About window.

5. Click **Validate**. After validation is successful, click **Save**. This saves the server and credentials of the connection.

Enabling the Oracle Cloud ERP - General Ledger Connection

To enable the Oracle Cloud ERP - General Ledger connection:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left, then click **Manage Connections**.
3. Select **Oracle Cloud ERP - General Ledger** from the list, click **Actions**, and then select **Edit**.

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4. In the **Edit Connection** dialog, select **Enabled**, and then click **OK**.

> **Note**
>
> The first time you enable the connection to Integration Cloud, all Integration Cloud deployments will fail during activation. The Administrator must login to Integration Cloud, test and save the deployed connections, and then activate the deployed integrations in Integration Cloud.

Setting Up the Integration Flow in Integration Cloud

To set up the integration flow in Integration Cloud:

1. Log in to Integration Cloud Services.
2. Navigate to **Connections**.

   EPM Fusion Connection and **EPM Connection** are automatically created.
3. Edit the connections by providing the environment and credential information:
   - **EPM Fusion Connection** is the connection to the Oracle Cloud ERP. You must update the credentials, Service Catalog Service WSDL URL, and Event Catalog URL. See Prerequisites for Creating a Connection for more details.
   - **EPM Connection** is the connection to the Oracle Fusion Cloud Enterprise Performance Management Service in Task Manager. See Oracle Enterprise Performance Management Cloud Adapter with Oracle Integration for more details.

   After configuring, testing, and saving the connections, you see a green check mark next to the connections.

4. Return to Task Manager and select **Integrations**.
5. Select the integration from the list, then click **Deploy to Integration Cloud** from the toolbar.
6. In the **Deploy to Integration Cloud** dialog, click **Generate**.

   The Cloud Integration deployment is successful. Verify the activation status in the Integration Cloud by logging into the Oracle Integration Cloud service, then Integrations.

Adding the Event Monitoring Task to a Template or a Schedule

After the set up is complete, you can add the event monitoring tasks to a schedule or a template.

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Templates** tab on the left and create a new template; for example, **ERP Event Template**.
3. Create the event monitoring task. See Creating Tasks.

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In the Task Type, make sure to select **Oracle Cloud ERP - General Ledger**, then select the event monitoring task; for example, **Period Open Event**.

4. In Task Manager, click **Template**, select **ERP Event Template**, and create a schedule. See Creating Schedules from Templates.

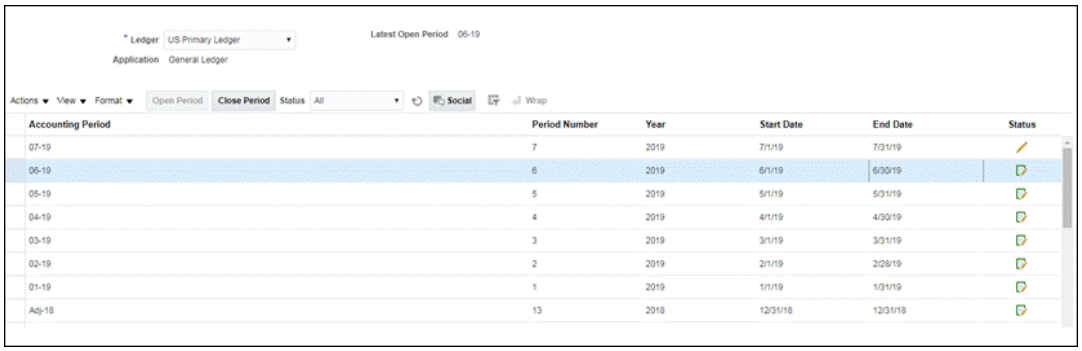
5. In Task Manager, click **Schedules**, select the schedule, and change the status to **Open**.

   You can monitor the schedule task in the **Schedule Tasks** screen.

Triggering the Period Close Event in Oracle Cloud ERP

To raise the Period Close event in Oracle Cloud ERP:

1. Login to Oracle Cloud ERP.
2. Select the **General Accounting** tab, then **Period Close**.
3. Click **General Ledger** and select the period, then click **Close Period**.



4. Wait for a few minutes for the task to be completed. Then navigate to the **Schedule Tasks** screen in **Tasks** to verify the task is closed.

Triggering the Period Open/Reopen Event in Oracle Cloud ERP

To raise the Period Open event in Oracle Cloud ERP:

1. Login to Oracle Cloud ERP.
2. Select the **General Accounting** tab, then **Period Close**.
3. Click **General Ledger** and select the period, then click **Open Period**.
4. Wait for a few minutes for the task to be completed. Then navigate to the **Schedule Tasks** screen in **Tasks** to verify the task is closed.

> **Note**
>
> If you open a period that has never been opened, the Period Open Event is triggered. If you open a period that has been closed before, the Period Reopen Event is triggered.

Triggering the Journal Batch Approved Event in Oracle Cloud ERP

To raise the Journal Batch Approved event in Oracle Cloud ERP:

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1. Login to Oracle Cloud ERP.
2. Select the **General Accounting** tab, then **Journals**.
3. Click **Requiring My Approval** and select the journal batch, then click **Approve**.
4. Wait for a few minutes for the task to be completed. Then navigate to the **Schedule Tasks** screen in **Tasks** to verify the task is closed.

> **Note**
>
> You can trigger the Journal Batch Approved event only if it is the last approval in the multi level approval process.

Triggering the Journal Post Event in Oracle Cloud ERP

To raise the Journal Batch Approved event in Oracle Cloud ERP:

1. Login to Oracle Cloud ERP.
2. Select the **General Accounting** tab, then **Journals**.
3. Click **Tasks** and select **Create Journal**.
4. In **Create Journal** screen, specify the required information, then click **Save**.

> **Note**
>
> The journal batch name and accounting period should match the parameters of the task in the Task Manager schedule.

5. Wait for a few minutes for the task to be completed. Then navigate to the **Schedule Tasks** screen in **Tasks** to verify the task is closed.

Troubleshooting

This section provides the troubleshooting information for the event monitoring integrations for Oracle Cloud ERP.

Issue	Root Cause	Resolution
Communication error importing to Oracle Integration Cloud	The configured OAuth application doesn't have Service Administrator role for Oracle Integration Cloud.	Create a separate OAuth application and grant Service Administrator role for the OAuth application. Add OIC scope all. See Setting Up the Integration in Task Manager. This is required so that you can deploy the out of box OIC flow from Task Manager to OIC.

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Issue	Root Cause	Resolution

Posting the journal did not trigger the flow in Oracle Integration Cloud.	Business event not enabled in ERP .	Perform the following:
		• Retrieve a list of available business events
		• Get specific business event
		• Update the enable indicator for business event
		• Check the subscription to business event
		You can call the `PublicEvent` subscription's REST resource using any REST client to check the subscription status.
		`Method : GET`
		`URL : https://fa_host.oraclecloud.com/soa-infra/PublicEvent/subscriptions`
		The response contains the endpoint URL for the related subscription.
		For troubleshooting issues related to business events, see Business Event Troubleshooting Issues

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Issue	Root Cause	Resolution
Posted Journal does not close the task in Financial Consolidation and Close	Journal batch name is updated with additional information when the Journal is posted	Perform the following: 1. Create a new connection in task manager. For example, Custom_XXX_ERP. 2. Create a custom integration and task type in task manager. **Note** Create the integration under new connection. Use the same parameters as out of box integration. Prefix the integration name, code name, and event name. For example, Custom_XXX_JournalPostEvent. 3. In Oracle Integration Cloud: a. Create a new connection for Oracle Cloud ERP and Oracle Fusion Cloud EPM. b. Duplicate the out of box OIC flow. c. Replace the connections to point to new connections created for Cloud ERP and Cloud EPM. d. Customize the OIC flow to remove information from the batch name. e. Send the modified batch name to the Update Task for Event Monitoring API. f. Test the end-to-end flow from Oracle Cloud ERP to FreeForm. For more information, see Creating Custom Integrations.

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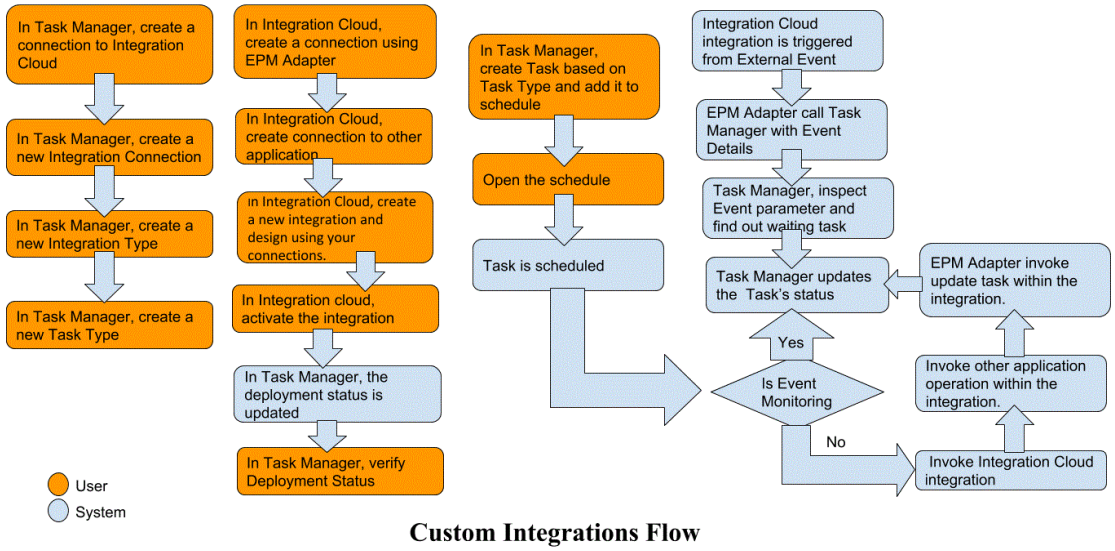
Creating Custom Integrations

In Task Manager, you can create and manage custom Process Automation or Event Monitoring integrations with external applications. An EPM Adapter enables you to create connections and integration flows in Integration Cloud Service using Oracle Oracle Fusion Cloud Enterprise Performance Management and other cloud and on-premises applications.

The following diagram shows the user and system flow for creating custom integrations:



Custom Integrations Flow

See these topics:

- Creating Custom Process Automation Integrations
- Creating Custom Event Monitoring Integrations

Creating End User Integration

When you create Integrations, you select End-User as the Execution type. Users can then create task types based on the defined Integrations, or validate the Integrations.

To create an Integration:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Click **New**.

See also, Setting Integration Properties and Setting Integration Parameters.

Creating Custom Process Automation Integrations

In Task Manager, you can create custom process automation integrations with external applications. In a Process Automation integration, a task is automatically executed in an external application when its start date and time are reached, and when any predecessor tasks are completed, for example, an overnight feed from a General Ledger.

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When you create a Process Automation task, if you have set up email notifications, the task Owner automatically receives an email notification when the task starts or is completed. The Assignee that is specified when you set up a workflow receive an action notification when there is a change in the task status and an action needs to be performed, such as an approval, and the Owner receives a notification when the approval is completed.

Make sure you have the prerequisites and follow these steps to set up an integration between Task Manager and an external application.

For an overview of the custom integrations flow, see Creating Custom Integrations.

Prerequisites

To integrate Task Manager with an external application, you need:

- A subscription to Oracle Integration Cloud Service, to integrate with on-premises and non-Oracle Fusion Cloud Enterprise Performance Management services.

> **Note**
>
> You need one Integration Cloud instance per Oracle Cloud EPM instance.

- The external application set up.

1. Subscribe to **Oracle Integration Cloud/Oracle Autonomous Integration Cloud**. See Integration Cloud Service Documentation for detailed information.
2. Install Integration Cloud Agent in your application environment to communicate with your Oracle Cloud EPM instance. For details on setting up Integration Cloud Agent, see Managing Agent Groups.

> **Note**
>
> If the on-premises application services are deployed in an environment set up in a demilitarized zone (DMZ) configuration so that these services are publicly accessible through the Internet, you do not need to install Integration Cloud Agent.

Creating a Connection in Task Manager

1. On the Home page, click **Application** and then click **Task Manager**.
2. Click the **Integrations** tab on the left, and then click **Manage Connections**.
3. Click **New**.
4. For **Connection** enter a name for the connection.
5. Select **Enabled** to enable the connection.
6. Select **Cloud** if the external application is a Cloud service.
7. Click **OK** to save the connection.

Setting Up the Integration in Task Manager

1. On the Home page, click **Applications**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left, and then click **New**.
3. On the **Properties** tab, specify the required information:

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a. For **Name**, enter a name for the Integration.

b. For **Code**, enter an Integration Code for the integration task.

   The code is used to execute the Integration and to map updates to an Integration from a file import.

c. **Optional:** In **Description**, enter a description for the integration task.

d. For **Connection**, select an application to which the task belongs.

   You can maintain the list of applications from the Manage Connections icon in the Manage Connections dialog box.

e. For **Execution Type**, select **Process Automation**.

4. On the **Parameters** tab, click **New** and then specify the required information:

   a. For **Name**, enter a name for the parameter.

   b. For **Parameter Code**, enter a parameter code.

   c. **Optional:** Enter a parameter **Tooltip**.

   d. From the **Parameter Type** list, select a type, and enter additional information for the parameter:

      - **Checkbox**: Boolean value
      - **Date**: Date value
      - **Integer**: Numeric value in whole numbers
      - **Number**: Numeric value in whole numbers or fractions
      - **Options Group**: Check box for a predefined set of values
      - **Static List**: Predefined set of text values
      - **Task Information**: Task information, for example, Assignee, duration, start and end dates
      - **Text**: Free-form text value.
      - **EPM Artifact**: Name of the artifact, such as the form or report.

   e. If the parameter requires a value, select **Required**.

   f. Click **OK** to save the parameter.

   g. Click **Save and Close** to save the Integration.

Creating a Task Type for the Integration

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Click **New**.
4. On the **Properties** tab, specify a **Task Type Name** and **Task Type ID**.
5. For **Integration**, click **Search**, select the Integration, and then click **OK**.
6. On the **Parameters** tab, set the Task Type parameters.

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Setting Up the Integration Cloud Connection in Task Manager

Task Manager uses Oracle Integration Cloud for all the integrations to external applications. You can setup a connection to Oracle Integration Cloud in Task Manager using Basic or OAuth 2.0 authentication.

> **Note**
>
> - For Oracle Integration Cloud Generation 2, both Basic Auth and OAuth 2.0 is supported.
> - For Oracle Integration Cloud Generation 3, only OAuth 2.0 is supported.
> - Basic Auth user must have the Service Administrator role for Oracle Integration Cloud.
> - For OAuth 2.0, only client credentials are supported. OAuth 2.0 client application must have Service Administrator role for Integration Cloud and allowed scope should be all.

Before setting up the OAuth 2.0 connection, make sure you've the client credentials such as access token URL, client ID, client secret, and scope. For setting up the OAuth client application, see **OAuth Authentication in Oracle Integration** in *Oracle Integration Cloud Generation 3* documentation.

To provide Service Administrator role for Integration Cloud:

1. Log in to the Identity Cloud Service Administrator console as an Administrator.
2. From Oracle Cloud Service, select the Oracle Integration Cloud Service.
3. On the **Application** tab, from the row menu for Service Administrator, click **Assign Application** and select the OAuth application you setup in the previous step.

To setup the Integration Cloud connection in Task Manager:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Click **Manage Connections**, and then from **Actions**, select **Integration Cloud Connection**.
4. Select one of the following authentication mechanisms:
   - **Basic Auth**: Specify the Integration Cloud URL, service administrator user ID, and password.
   - **OAuth 2.0**: Provide the Integration Cloud URL, access token URL, client ID, client secret, and scope.

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> **Note**
>
> - For Gen 2, enter the URL: `https://<SERVICE_NAME>-<TENANT_NAME>.integration.ocp.oraclecloud.com`
> - For Gen 3, enter the URL: `https://<SERVICE_INSTANCE>.integration.<REGION>.ocp.oraclecloud.com`. To find your service instance, log into Integration Cloud and open the About window.

5. Click **Validate**. After validation is successful, click **Save**. This saves the server and credentials of the connection.

Creating a Connection and Integration in Integration Cloud

1. Log in to Integration Cloud.
2. Create a connection using Oracle Enterprise Performance Management Cloud adapter to your EPM server. Refer to this link for more details: Create a Connection to Oracle Enterprise Performance Management Cloud.
3. Create one or more connections to the other application as needed.
4. Create Process Automation Integration using the connections created and activate it. See the example in Using the Oracle Enterprise Performance Management Cloud Adapter with Oracle Integration.

Completing the Integration Setup in Task Manager

Use these steps to complete the Integration setup in Task Manager.

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left and verify that the new Integration is displayed on the Integrations list.
3. In Task Manager, select **Schedules**, and create a new Schedule in Pending state. See Manually Creating Schedules.
4. Create the process automation task and add it to the schedule. See Creating Tasks.
5. From **Schedules**, select and open the schedule.

Creating Custom Event Monitoring Integrations

In Task Manager, you can create an Event Monitoring integration for an external application. The Event Monitoring integration is triggered when an external event occurs in another Cloud service or on-premises application. An example of an Event Monitoring integration is a trigger from Oracle E-Business Suite - General Ledger when a period, for example, January 2018, is closed.

Make sure you have the prerequisites and follow these steps to set up an integration between Task Manager and an external application.

For an overview of the custom integrations flow, see Creating Custom Integrations.

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Prerequisites

> **Note**
>
> Event monitoring can also be triggered through any integration tool capable of invoking REST APIs. In that case you do not need the these prerequisites.

To integrate Task Manager with an external application, you need:

- A subscription to Oracle Integration Cloud Service, to integrate with on-premises and non-Oracle Fusion Cloud Enterprise Performance Management services.

  > **Note**
  >
  > You need one Integration Cloud instance per Oracle Cloud EPM instance.

- The external application set up.

1. Subscribe to Oracle Integration Cloud / Oracle Autonomous Integration Cloud. See Integration Cloud Service Documentation for detailed information.
2. Install Integration Cloud Agent in your application environment to communicate with your Oracle Cloud EPM instance. For details on setting up Integration Cloud Agent, see Managing Agent Groups.

   > **Note**
   >
   > If the on-premises application services are deployed in an environment set up in a demilitarized zone (DMZ) configuration so that these services are publicly accessible through the Internet, you do not need to install Integration Cloud Agent.

Setting Up the Integration in Task Manager

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left, and then click **New**.
3. On the **Properties** tab, specify the required information:
4. a. For **Name**, enter a name for the Integration.
   b. For **Code**, enter an Integration Code for the integration task.

      The code is used to execute the Integration and to map updates to an Integration from a file import.

   c. **Optional:** In **Description**, enter a description for the integration task.
   d. For **Connection**, select an application to which the task belongs.

      You can maintain the list of applications from the Manage Connections icon in the Manage Connections dialog box.

   e. For **Execution Type**, select **Event Monitoring**.
   f. For **Event Name**, enter a name for the event.

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5. On the **Parameters** tab, click **New** and specify the required information:
   a. For **Name**, enter a name for the parameter.
   b. For **Parameter Code**, enter a parameter code.
   c. **Optional:** Enter a parameter tooltip.
   d. From the **Parameter Type** list, select a type, and enter additional information for the parameter:
      - **Checkbox**: Boolean value
      - **Date**: Date value
      - **Integer**: Numeric value in whole numbers
      - **Number**: Numeric value in whole numbers or fractions
      - **Options Group**: Check box for a predefined set of values
      - **Static List**: Predefined set of text values
      - **Task Information**: Task information, for example, Assignee, duration, start and end dates
      - **Text**: Free-form text value
      - **EPM Artifact**: Name of the artifact, such as the form or report
   e. If the parameter requires a value, select **Required**.
   f. Click **OK** to save the parameter.
   g. Click **Save and Close** to save the Integration.

Creating a Task Type for the Integration

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Task Types** tab on the left.
3. Click **New**.
4. On the **Properties** tab, specify a **Task Type Name** and **Task Type ID**.
5. For **Integration**, click **Search**, select the Integration, and then click **OK**.
6. On the **Parameters** tab, set the Task Type parameters.

Setting Up the Integration Cloud Connection in Task Manager

Task Manager uses Oracle Integration Cloud for all the integrations to external applications. You can setup a connection to Oracle Integration Cloud in Task Manager using Basic or OAuth 2.0 authentication.

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> **Note**
>
> - For Oracle Integration Cloud Generation 2, both Basic Auth and OAuth 2.0 is supported.
> - For Oracle Integration Cloud Generation 3, only OAuth 2.0 is supported.
> - Basic Auth user must have the Service Administrator role for Oracle Integration Cloud.
> - For OAuth 2.0, only client credentials are supported. OAuth 2.0 client application should have the Service Administrator role for Integration Cloud and allowed scope should be all.

Before setting up the OAuth 2.0 connection, make sure you've the client credentials such as access token URL, client ID, client secret, and scope. For setting up the OAuth client application, see **OAuth Authentication in Oracle Integration** in *Oracle Integration Cloud Generation 3* documentation.

To setup the Integration Cloud connection in Task Manager:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Click **Manage Connections**, and then from **Actions**, select **Integration Cloud Connection**.
4. Select one of the following authentication mechanisms:
   - **Basic Auth**: Specify the Integration Cloud URL, service administrator user ID, and password.
   - **OAuth 2.0**: Provide the Integration Cloud URL, access token URL, client ID, client secret, and scope.

   > **Note**
   >
   > - For Gen 2, enter the URL: `https://<SERVICE_NAME>-<TENANT_NAME>.integration.ocp.oraclecloud.com`
   > - For Gen 3, enter the URL: `https://<SERVICE_INSTANCE>.integration.<REGION>.ocp.oraclecloud.com`. To find your service instance, log into Integration Cloud and open the About window.

5. Click **Validate**. After validation is successful, click **Save**. This saves the server and credentials of the connection.

Setting Up the Integration in Integration Cloud

> **Note**
>
> If you are not planning to use Integration Cloud, skip this step.

To set up the integration in Integration Cloud:

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1. Log in to Integration Cloud.
2. Create a connection using Oracle Enterprise Performance Management Cloud adapter to your EPM service. Refer to this link for more details: Create a Connection to Oracle Enterprise Performance Management Cloud.
3. Create one or more connections to the other application as needed.
4. Create Event Monitoring Integration using the connections created and activate. Refer to this example: Using the Oracle Enterprise Performance Management Cloud Adapter with Oracle Integration.

Setting Up the Integration Without Integration Cloud

> **Note**
>
> If you are planning to use Integration Cloud, skip this step.

Configure the integration tool which you are planning to use.

1. Review the REST API, Update Task Status for Event Monitoring in *REST APIs*
2. Configure the integration tool to call the REST API to close the Event Monitoring task passing the parameters for the event.

For more information, see Update Task Status for Event Monitoring in *REST APIs*.

Completing the Integration Setup in Task Manager

Use these steps to complete the Integration setup in Task Manager.

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left, and verify that the new Integration is displayed.
3. In Task Manager, click **Schedules**, and create a new Schedule in Pending state. See Manually Creating Schedules.
4. Create the event monitoring task and add it to the schedule. See Creating Tasks.
5. From **Schedules**, select and open the schedule.
6. Trigger the action that generates the event in the external application (for example, EBS). If you are not using Integration Cloud but other integration tool, invoke the REST API from the integration tool.

> **Note**
>
> Pass the parameters as expected by the integration type, event, parameters, otherwise the task will not be considered if they do not match.

7. Wait for the task to be completed.
8. **Optional:** To monitor the status of the Integration in Integration Cloud Services, log in to Integration Cloud and navigate to **Monitoring**.

Working with Integrations

Related Topics

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- Creating Integrations
- Viewing Integrations
- Editing Integrations
- Searching for Integrations
- Validating Integrations
- Deleting Integrations

Creating Integrations

When you create Integrations, you select End-User as the Execution type. Users can then create task types based on the defined Integrations, or validate the Integrations.

To create an Integration:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Click **New**.

Add information to the following sections:

- Setting Integration Properties
- Setting Integration Parameters

Setting Integration Properties

You can set properties for the Integration such as the associated application and End User tasks.

For an End User task, you can select the single sign-on (SSO) Parameter option to enable users to access an external web application without being prompted for authentication. You can use an SSO parameter for an external application if that application is integrated with the Oracle EPM System SSO framework.

To set Integration properties:

1. Select the Integration to edit.
2. In **Properties**, for **Name**, enter a name for the Integration. You can enter a maximum of 80 characters.
3. Enter a **Code**, for example, `HFM_CONS` for the Hyperion Financial Management Consolidate integration task. You can enter a maximum of 90 characters.

   The code is used to execute the Integration and to map updates to an Integration.
4. **Optional:** In **Description**, enter a description for the integration task. You can enter a maximum of 255 characters.
5. In **Connection**, select an application to which the task belongs.

> **Note**
>
> You can maintain the list of applications from the Manage Connections icon in the Manage Connections dialog box.

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6. For **Execution Type**, select **End User** and select options:
   - **End User Endpoint**: To enter parameters for an **End User** task, the **End User Endpoint** should contain these parameters in these formats:

     $*Parameter Type Code*$, for example `$COLORS$`. The system replaces the parameter tokens in the **End User Endpoint** with the information you specified for the task parameters.
   - **Optional: Show in-line**: Select whether to display the URL in line within the Task Actions dialog.
   - **Optional: SSO Parameter**: Specify the name of the SSO parameter for your application to include when executing the End User task URL to the external application. If you do not specify an SSO parameter, the system uses the End User URL.
7. **Optional:** Click **Parameters**.

Setting Integration Parameters

Task Manager integration parameters enable the application to pass information to the end point to control how the end point should perform its action. For example, when you run a consolidation, your program can pass which application to run the consolidation against, and the dimension selections for the consolidation. You set the parameter values defined in the Integrations in the task or Task Type using the integration.

Each parameter definition requires a name, description, unique code, type, and whether a value is required. The parameter code is the token that replaces the parameter in the execution URL for user tasks, or the parameter name that is passed to the execution web service for system-automated integrations. Required values must have a value for all tasks in a schedule before the schedule can be set to Open.

The Parameter Type controls the parameter value and how the user enters the value in the task field.

To set Integration parameters:

1. In the **New Integration** dialog box, click **Parameters**.
2. Click **New**.
3. Enter a **Name** for the parameter.
4. Enter a **Parameter Code**.
5. Enter a parameter **Tooltip**.
6. From the **Parameter Type** list, select a type, and enter additional information for the parameter:
   - **Checkbox**: Boolean value
   - **Date**: Date value
   - **Integer**: Numeric value in whole numbers
   - **Number**: Numeric value in whole numbers or fractions
   - **Options Group**: Checkbox for a predefined set of values
   - **Static List**: Predefined set of text values
   - **Task Information**: Task information. For example, Assignee, duration, start and end dates

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- **Text**: Free-form text value.
- **EPM Artifact**: Name of the artifact, such as the form or report

7. If the parameter requires a value, select **Required**.
8. Click **OK** to save the parameter.
9. Click **Save and Close** to save the Integration.

Viewing Integrations

You can view the properties and parameters of imported Integrations. You can specify which columns to display, or show all. You can also reorder columns, sort columns by ascending or descending order, or change the column widths.

To view the properties and parameters of imported integrations:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
   The Integrations page is displayed.
3. Click the integration name for which you want to view the properties and parameters.
   The View Integration dialog box displays the **Properties** tab by default. Click the **Parameters** tab to view the associated parameters for the integration. For information about the fields, see Setting Integration Properties and Setting Integration Parameters.

To display columns, on the Integrations page, click **View**, then **Columns**, and then select an option:

- To display all columns, select **Show All**.
- To display specific columns, select or deselect the column names.

To reorder columns, on the Integrations page, click **View**, and then **Reorder Columns**. Select columns and use the **Up** and **Down** arrows to change the order.

To sort columns, on the Integrations page, hover over a column header until the **Sort** icons display, then click **Sort Ascending** or **Sort Descending**.

To change column widths, on the Integrations page, hover over the column header dividers until the arrows display. Drag the columns to the desired width.

Editing Integrations

You can't edit the pre-built Integrations provided by default by the system. You can only edit custom-built integrations that you created.

For an End User type, you can edit the endpoint, change the point of view, or change the list of values.

To edit an Integration:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Select an Integration, and click **Edit**.
4. Edit the Integration.
5. Click **Save and Close**.

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Searching for Integrations

You can use the Search box on the Integrations page to find Integrations. You can enter full or partial names on which to search.

To search for Integrations:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. In the **Search** box, enter full or partial search criteria for the Integration.
4. Press **Enter**.

   To reset the list to display all Integrations, clear the **Search** box and then press **Enter**.

Validating Integrations

You can test and validate Task Manager integration definitions through a test schedule and adding a task to the schedule.

To validate an Integration:

1. In Task Manager, click **Schedules**, then click **New** to create a Schedule in Pending state. See Manually Creating Schedules.
2. Add a task based on the integration to be validated to the schedule. See Creating Tasks.
3. From **Schedules**, select and open the schedule.
4. From **Tasks**, click **Schedule**. Select the schedule and view the Task.
5. For Event Monitoring Task, trigger the action that generates the event in the external application (for example, EBS). If you are not using Integration Cloud but other integration tool, invoke the REST API from the integration tool.

Deleting Integrations

You can delete Integrations that you no longer need. However, you can't delete pre-built Integrations provided by default and you can't delete an Integration while it is associated with a Task Type.

To delete an Integration:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Select an Integration to delete.
4. Click **Delete**.
5. At the confirmation prompt, click **Yes**.

Managing Connections

Integrations are assigned connections for the external products to which they link. From the Manage Connections module, you can maintain a list of connections associated with the Integrations. You can search on and sort the list by connection.

- Adding Connections

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- Editing Connections
- Deleting Connections

Adding Connections

You can add connections to associate with an Integration Type. You can also specify the security policy to use for an Integration Type.

> **Note**
>
> Multi-factor authentication (MFA) is not supported for Connections.

In addition, you can specify application-level tokens and values for all Integration Types in an application. When you specify an application-level token in an end-user URL or web service WSDL, the system replaces the token with the value defined for that token in the application. For example, you can specify tokens with values for server and port, and the system automatically applies those values to the Integration Types in the application.

To add a connection:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Click **Manage Connections**.
4. Click **New**.
5. Complete the **New Integration Properties** and **Parameters**, then click **Save and Close**.
6. **Optional:** To add an application-level token, click **Manage Connections**. Click **New** or select a Connection and click **Edit**. In the **Application Tokens** table, click **Add**. Enter a **Token Name** and optionally a **Token Value**, and then click **OK**.

   > **Tip**
   >
   > To remove a token, click **Delete**.

Editing Connections

You can edit the name of a connection, the security information, and application-level tokens.

> **Note**
>
> You cannot add or modify application tokens for seeded applications. You can only edit the token values.

You also use the Edit Connections dialog to enable pre-built Integrations. After you enable the Integration and fill in the parameters, you can then create tasks using Task Types for the enabled Integrations.

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To edit a connection:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Click **Manage Connections**.
4. Select a connection, and click **Edit**.
5. To enable a pre-built Integration, select the connection, select **Enabled**, and fill in the parameters.

   You can enable or disable a connection at any time.
6. Edit the settings or tokens as needed, and click **OK**.

Deleting Connections

You can delete connections that you no longer need for Integrations. You cannot delete a connection while it is associated with an Integration Type. You must modify the properties for each Integration Type that references the connection before you can delete the connection.

> **Note**
>
> You cannot delete seeded Integration connections.

To delete a connection:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Integrations** tab on the left.
3. Click **Manage Connections**.
4. Select a connection, and click **Delete**.

Managing Alert Types for Task Manager

Alert Types

While performing a task, users might encounter issues such as a hardware failure, access problem, system failure, and so on.

Administrators can create Alert Types for categories of possible issues. Each Alert Type defines a procedure that is used to capture critical information about the issue, and assigns key personnel to resolve it.

For example, an administrator can create an Alert Type called Access Issue, which contains questions to users for more detailed information. The administrator defines a workflow for the Alert Type with an assignee to resolve the issue, and a backup assignee if desired.

Using Alert Types to Create Alerts

When users encounter an issue and create an alert for it, they select from a list of predefined Alert Types to identify the issue. The alert is then sent to the assignee defined in the Alert Type.

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For example, a user has a task due and can't access their assigned tasks. The user selects an Alert Type, such as Access Issue, which sends the alert to the predefined assignee to resolve their issue.

To manage Alert Types, see these topics:

- Creating Alert Types
- Editing Alert Types
- Viewing Alert Types
- Searching for Alert Types
- Deleting Alert Types

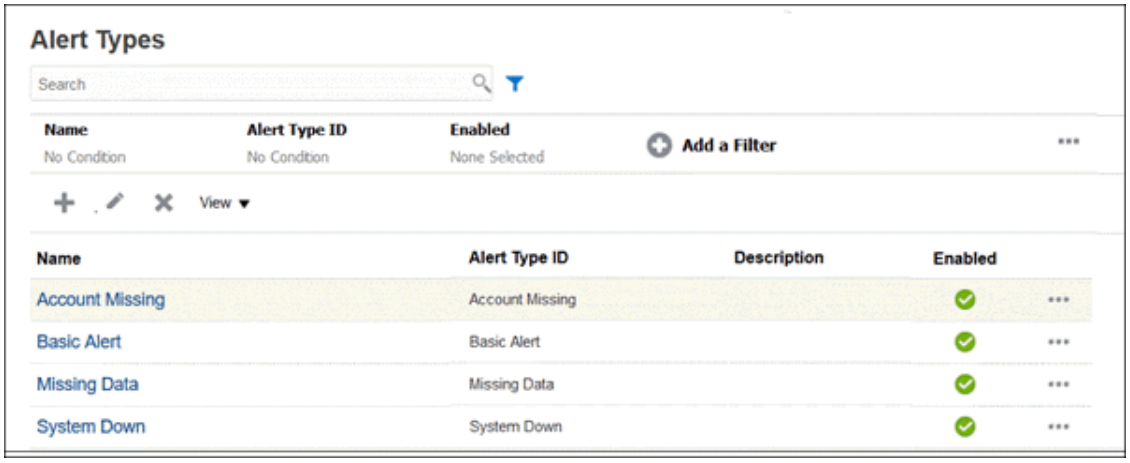
Creating Alert Types

When users encounter issues that prevent them from performing a task such as a hardware problem or system failure, they can create alerts for the issue.

Administrators can create Alert Types for categories of possible issues, such as System Down, and so on. When you create an Alert Type, you define the properties such as name, description, Alert Type ID, any instructions that you want to include for users, and additional questions to users for more details. You set up the workflow with an assignee, optional backup assignee, and any additional viewers for the Alert Type.

Using information gathered from Alert Types that users select, you can analyze the types of issues that users encounter during the business cycle and make changes to prevent them in future cycles.

You can view the list of the Alert Types that you have created, and edit and delete them from the Alert Types page.



To create an alert type:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. Click **New**.
4. Enter the necessary information on the Alert Type tabs:
   - Setting Alert Type Properties
   - Specifying Alert Type Instructions

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- Selecting the Alert Type Workflow
- Assigning Alert Type Viewers
- Adding Questions for Alert Types
- Applying Alert Type Attributes
- Viewing Alert Type History

Setting Alert Type Properties

The **Properties** tab enables you to specify the alert type name and description, and associate it with Task Manager object, such as a task or schedule. An individual alert can be associated with multiple objects.

You can place restrictions on the relationship between the alert and its associated object. Not all restrictions can be applied to all objects.

Table 11-18 Alert Restrictions

Restriction	Description	Example
None	No restrictions on the status of the object and the status of the alert	A user raises a 'slow performance' alert while working on a reporting task. While this affects how long the task takes, it does not prevent the task from completing normally. Even if the task completes, the user still wants the alert open until the performance issue is resolved.
Prevent Workflow	Workflow on the object cannot proceed forward (no submits, approvals, and so on) until the alert is closed. This does not prevent claims or rejections (workflow moving backward). It also does not prevent an Administrator or Owner from forcing the workflow forward.	A user raises an alert that the reporting system is down. This will prevent any work on reporting tasks until the alert is resolved. Preventing workflow includes preventing status changes from Pending to Open and Open to Closed.
Prevent Close	The object cannot be moved into a closed state until the alert is closed. However, intermediate workflow may proceed. This does not prevent an Administator or Owner from closing or 'force closing'.	A user raises an alert that some comparison data is missing for a reporting task. While this does not prevent the report from being created and going through initial approvals, the report should not be fully signed off until it can be compared to the missing data.

An object may have multiple alerts with different restrictions. If this is the case, the following rules will apply in order of precedent:

1. If any open alert associated with the object has a Prevent Workflow restriction, Prevent Workflow will stop the object's (for example, a Task) workflow until the alert is closed.
2. If any open alert associated with the object has a Prevent Close restriction, then the object cannot be closed until the alert(s) is closed.

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In addition, an alert may be associated with multiple objects. If it has more than one Prevent Close restriction to different objects, the alert will only be closed when the last object is closed.

To set alert type properties:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. Click **New** to open **New Alert Type**, which will default to the **Properties** tab.
4. For **Name**, enter an alert type name. You can enter a maximum of 80 characters.
5. Enter an **Alert Type ID**. You can enter a maximum of 80 characters.

   An Alert Type ID is required, and must be unique.
6. **Optional:** For **Description**, enter an alert type description. You can enter a maximum of 255 characters.
7. **Optional:** From the **Associated With** drop-down list, select an object, such as a task, with which to associate the alert.
8. **Optional:** For **Restrictions**, enter any restrictions for the alert. For example, if you select **Prevent Close** for an alert on a task, the user can't complete the close task until the alert is complete.

   If you select **All Types** for **Associated With**, no restrictions are available.
9. Select from the following values for both **Period Selection** and **Year Selection**:
   - **Required** – The user is required to associate the Alert with a Year or Period.
   - **Visible** – The user may associate the Alert with a Year or Period.
   - **Hidden** – The property is hidden and the Alert will not be associated with a Year or Period.

   These options indicate how Periods and Years are associated with an Alert.
10. **Optional:** In **Automatic Priority Escalation**, specify when priority should be automatically updated for each level:
    - **Priority**: Every priority level except the lowest should be displayed.
    - **Enabled**: Indicate whether the automatic escalation is turned on or off. This is off by default and off for all upgraded alerts.
    - **Days**: Specify the number of days before the automatic escalation occurs.
11. Select **Allow Remove Associations** option to remove an existing association from an Alert. If you unselect this option, you will not be able to remove existing associated artifacts from the Alert.
12. Click **Enabled** to enable the alert type.

    Only alert types that are Enabled are displayed in the list of available Alert Types and available for selection when creating new alerts.
13. Click an Alert Type tab and continue entering information.

    When you are done entering the Alert Type information, click **Save and Close**.

> **Note**
>
> You cannot create an alert on an existing alert.

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- Specifying Alert Type Instructions
- Selecting the Alert Type Workflow
- Assigning Alert Type Viewers
- Adding Questions for Alert Types
- Applying Alert Type Attributes

Specifying Alert Type Instructions

You can specify instructions in an alert type to help users understand what they need to do for the alert. You can add additional references from File and URL attachments.

To specify instructions for an alert type:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. From the New or Edit dialog, select the **Instructions** tab.
4. In **Instructions**, enter instruction text for the alert type.

To add a reference:

1. In the **References** section, click **Attach a file** or **Attach a link**.
   - **Attach a file**

     Click **Choose File** to select and attach the file. For **Name**, enter a name or use the filename. Click **OK.**
   - **Attach a link**

     Enter the **URL**, then enter a **Name** for the URL; for example: https://www.oracle.com, Oracle. Click **OK**.
2. Click an Alert Type tab and continue entering information.

   When you are done entering the alert type information, click **Save and Close**.

   - Selecting the Alert Type Workflow
   - Assigning Alert Type Viewers
   - Adding Questions for Alert Types
   - Applying Alert Type Attributes

Selecting the Alert Type Workflow

The Workflow section contains the Assign ee and Approver assignments. You can also select Backup users.

To select the Alert Type workflow:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. From the New or Edit dialog, click the **Workflow** tab.
4. For **Assignee**, click **Select a User** and select an assignee.

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The assignee is the user or group assigned to work on the alert when one is created of this type. If you do not specify an assignee, the user who creates the alert will need to fill one in.

5. **Optional: For Backup User**, select a backup user for the assignee:
   a. On the Home page, select **Tools**, then **Access Control**, and then **Task Manager Users**.
   b. Edit the assignee ID and change the **Status** field to **Unavailable**.
   c. Then from the Alert Types **Workflow** tab, you can select a backup user.

   The Backup User is the backup individual assigned to work on the alert if the main assignee is out of office. You must have previously assigned an assignee and a backup assignee.
6. **Optional:** For **Notification Assignee**, specify a user to be notified of the alert. This is usually an external user. All notifications sent to the Alert Assignee will be sent to this user.
7. Click **Add** to add an Approver, and enter this information for the Approver:
   - **User Name**
   - **Backup User**: You can specify a default backup Approver user for the alert if the main Approver is out of the office. This is not required.

   Level displays the approvers in the order they were entered. You can reorder the Approver list by selecting an approver and clicking the arrow buttons to move them up or down in the list.

   You can add one or more levels of default Approver users or groups for the alert when one is created of this type. These are not required. If you do not specify Approvers, the user who creates the alert has the option to add them.
8. Click an Alert Type tab and continue entering information.

   When you are done entering the Alert Type information, click **Save and Close**.

   - Assigning Alert Type Viewers
   - Adding Questions for Alert Types
   - Applying Alert Type Attributes

Assigning Alert Type Viewers

The Viewers tab enables you to assign Viewer rights for alert types. Viewers have read-only access.

To assign Viewer rights:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. Click **New** or **Edit** and click on the **Viewers** tab.
4. Click **Add** and select the name of the user or group that will have View access to the alert.

   The system opens the Member Selector. It is editable for external Viewers.
5. Select a **Notification Priority** to indicate at what alert priority the users will be alerted via email.

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Notifications will be sent for the priority level or higher. So if set to High, notifications will be sent only when the alert is set to High. If set to Low, then they will be notified for all priority types (Low, Medium, High). If this is blank (default), no notification will be sent.

6. Click an Alert Type tab and continue entering information.

   When you are done entering the Alert Type information, click **Save and Close**.

   - Adding Questions for Alert Types
   - Applying Alert Type Attributes

Adding Questions for Alert Types

When you create an Alert Type, you may want a user to answer questions about their actions before they indicate the Alert Type is complete. For example, you may ask if a certain process was followed when completing that task. Questions can be set to the Administrator, Assignee, Approver, Owner, or Viewer roles. This allows key details to be gathered from the user experiencing the issue.

You can specify various types of questions, such as Text, Number, or True/False, and indicate whether they are required. If a question is required, the user must respond or they cannot submit the task for approval. You can also order the questions by using the Move Up and Move Down buttons.

To add a question:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. From the **New** or **Edit** dialog, select the **Questions** tab.
4. Click **New**.
5. For **Question**, enter text for the question, with a maximum of 4000 characters.
6. From the **Type** list, select a question type:
   - **Date**
   - **Date/Time**
   - **Integer**
   - **List**

     Enter a list of valid responses to the question.
   - **Multi-Line Text**

     The maximum length should be less than 4,000 characters.

     Select Multi-Line Text, and then enter the **Number of Lines**, from 3 to 50 lines. Multi-Line Text determines how many lines of text are visible without scrolling, on the Actions dialog boxes.
   - **Number**

     If you select **Number**, select number formatting options:

     – For **Decimal Places**, enter a value for the number of decimal places to display.

     – Select the **Thousands Separator** option if you want numbers to display a thousands separator (for example, 1,000.00)

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  - From the **Currency Symbol** list, select a currency symbol; for example, $ (United States of America Dollar).
  - From the **Negative Number** list, select how to display negative numbers; for example, (123).
  - From the **Scale** list, select a scale value for numbers; for example, 1000.
- **Text**
- **True/False**
- **User**
- **Yes/No**

7. Assign the **Role** (Owner, Assignee, Approver, Viewer) of the user who should answer the question.
8. If the question is required, select **Required**.

   Required indicates that the question must be answered by the user before they can proceed. Required questions can be set for Assignees, Approvers, and Owners. This is disabled for the System Administrator and Viewer role.
9. Click **OK**.
10. **Optional:** To change the order of questions, select a question, then click **Move To Top**, **Move Up**, **Move Down**, or **Move To Bottom**.
11. **Optional:** To edit a question, select the question and click **Edit**. To remove a question, select the question and click **Delete**.
12. Click an Alert Type tab and continue entering information.

    When you are done entering the Alert Type information, click **Save and Close**.

- Applying Alert Type Attributes
- Viewing Alert Type History

Applying Alert Type Attributes

When you select an attribute, you can set a value for the attribute based on the attribute type. You can later filter by the attribute value.

For example, you may have a List attribute named Sales Region with the values of North, South, East, and West. The current alert type applies only to the West Sales Region, so you can add the Sales Region attribute and set it to "West".

To apply an attribute:

1. On the Home page, click **Application,** and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. Click **New** or **Edit**, and select the **Attributes** tab.
4. Click **Add**.
5. From the **Attribute** list, select an attribute.
6. From **Value**, depending on the attribute, select a value for the attribute from a drop-down list, or enter a value.
7. **Optional:** To change access for the attribute, select a **Role** and **Access**.

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8. Click **Save and Close**.
9. To edit other alert type information, see these topics:
    - Setting Alert Type Properties
    - Specifying Alert Type Instructions
    - Selecting the Alert Type Workflow
    - Assigning Alert Type Viewers
    - Adding Questions for Alert Types

Viewing Alert Type History

The system maintains a history of alert type actions. The History tab displays the components that were created or updated, the modification type, the old and new values, the user who made the modification, and the change date. The information on the History tab is read-only.

To view alert type history:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. Edit an alert type and select the **History** tab.
4. View the history, then click **Save and Close**.

Viewing Alert Types

On the Alert Types page, you can specify which columns to display in the list or display all columns. You can reorder columns, sort columns by ascending or descending order, or change the column widths.

To display columns on the Alert Types page:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. Do one or more of the following tasks:
    - To display all columns, select **View**, then **Columns**, and then select **Show All**.
    - To display specific columns, select **View**, then **Columns**, and select or deselect the column names.
    - To reorder columns, select **View**, and then **Reorder Columns**, select columns and use the Up or Down arrows or drag them to change the order.
    - To sort columns, hover over a column header until the Sort icons are displayed, and then click **Sort Ascending** or **Sort Descending**.
    - To change column widths, hover over the column header dividers until the arrows display, and drag the columns to the desired width.

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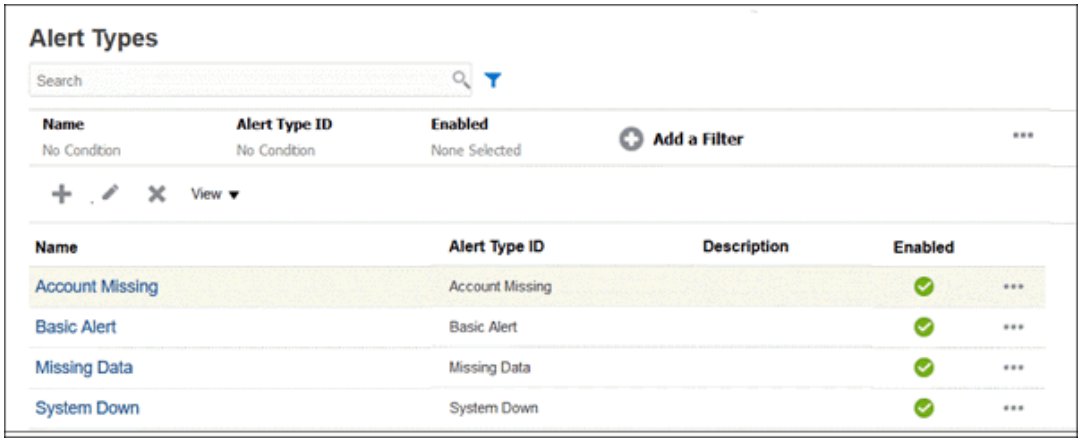

Column Definitions for Alerts

This section provides the definitions for the Alerts columns.

Table 11-19 Column Definitions - Alerts

Column Name	Description
Alerts (Count)	Total count of alerts
Alerts Open (Count)	Total count of open alerts
Alert Type	Type of the alert. Based on the type, you can analyze the types of issues that users encounter during the business cycle and make changes to prevent them in future cycles
Approver 1-10	Approver level
Approver 1-10 (Backup)	Backup approver for the alert
Approver 1-10 (Primary)	Primary approver for the alert
Approver 1-10 End Date (Actual)	Actual end date for each approver level
Approver 1-10 Rejections (Count)	Count of alerts rejected by approvers
Assignee	Assigned user for the alert
Assignee (Backup)	Backup user assigned for the alert
Assignee (Primary)	Primary user assigned for the alert
Assignee End Date (Actual)	Actual end date for the assigned user
Associated Object Type	Type of object associated with the alert
Associated Objects (Count)	Total count of the associated objects for the alert
Closed Date	Date when the alert was closed
Comments (Count)	Total count of comments for the alert
Created By	Name of the user who created the alert
Created On	Date and time when the alert was created
Description	Description of the alert
Duration	Defined alert duration
Duration (Actual)	Actual alert duration
End Date	Effective end date for the alert
ID	Unique identifier for the alert
Last Updated By	Name of the user who last updated the alert
Last Updated On	Date and time when the alert was last updated

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Table 11-19 (Cont.) Column Definitions - Alerts

Column Name	Description
Name	Name of the alert
Owner	Name of the alert owner, which must be a Service Administrator or Power User.
Owner End Date (Actual)	Actual end date for the alert owner
Period	Period for the alert
Priority	Priority assigned for the alert
Start Date	Effective start date for the alert
Status	Status of the alert
Status (Detailed)	Detailed status of the alert
Status (Icon)	Status of the alert indicated as icon
Year	Year when the alert was created

Editing Alert Types

You can edit the names and descriptions of alert types, and change whether or not they are Enabled. When you enable an alert type, it is displayed in the list of available Alert Types.

To edit an alert type:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. Select the alert type and click **Edit**.
4. Edit the required information. For details about the fields, see the following:
   - Setting Alert Type Properties
   - Specifying Alert Type Instructions
   - Selecting the Alert Type Workflow
   - Assigning Alert Type Viewers
   - Adding Questions for Alert Types
   - Applying Alert Type Attributes
   - Viewing Alert Type History
5. Click **Save and Close**.

Searching for Alert Types

You can use the Search function in the Alert Types list to quickly find alert types. You can enter full or partial names on which to search. Using the filter bar, you can control the alert types that are displayed in the list. By default, all alert types are displayed.

To search for alert types:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. To search for an alert type, enter search criteria in the **Search** text box.

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4. **Optional:** From the filter bar, click on a category to display additional search operators such as **Equals**, **Does Not Equal**, **Contains**, **Does Not Contain**, **Starts With**, and **Ends With**.

   You can filter alert types using these categories: **Name**, **Alert Type ID**, **Enabled**, **Description**, **Created By**, **Created On**, **Last Updated By**, or **Last Updated On**.

> **Note**
>
> - Click **Add a Filter** to view all categories.
> - To hide the filter bar, click the **Hide Filter Bar** icon.
> - To clear all filters, click the **Filters** icon, and then click **Clear All Filters**.

Deleting Alert Types

You can delete alert types that you no longer need. When an alert type is deleted, alerts that were created with that alert type are not deleted, but lose their alert type assignment.

To delete an alert type:

1. On the Home page, click **Application**, and then click **Task Manager**.
2. Click the **Alert Types** tab on the left.
3. Select the alert type.
4. Click **Delete**, then at the confirmation message, click **OK**.

Managing Views in Task Manager

You can edit, delete, and publish views within Task Manager. Also, you can set a view as the default. For more information, see:

- Editing Saved Views
- Deleting Views
- Publishing Views
- Setting a Default View
- Working with Views

Viewing Currencies in Task Manager

To view the configured currency codes within Task Manager, from the Home page, click **Application**, **Task Manager**, and then **Currencies**. A list of configured currency codes are displayed.

For each currency, you can view the **Currency Code**, **Currency Symbol**, **Decimal Places**, **Description**, and whether the currency is **Enabled**. For information about these columns, see Task Manager – Currencies.

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Using Reports

You can generate custom reports and Cloud EPM system reports for Task Manager.

For more information, see:

- Generating Custom Reports for Task Manager
- Generating Cloud EPM System Reports for Task Manager

Generating Custom Reports for Task Manager

You can create reports for Task Manager. The reports can be generated in PDF, CSV, HTML, or XLSX format.

> **Note**
>
> Task Manager provides sample reports to assist you in getting started. You can use them as templates and can duplicate them as needed, then make modifications or extend access to users. To duplicate a report, select the report, then select **Duplicate** from the ellipsis button ▪▪▪. You can then edit the newly created report.

You can also create custom reports. Building custom reports involves four steps and can only be performed by a Service Administrator:

1. Creating a Task Manager Query.
2. Creating a Template.
3. Setting Up a Report Group
4. Creating a Report

To build custom reports, you create a query whose definition can then be exported as an XML file. Import the XML file into Word and use it to create a template document, which must be in RTF format. Then you can set up a report group to group individual reports. Finally, generate the report, which uses both the query and the template you created and can be set with a default output of CSV, PDF, HTML, or XLSX.

After you have built a custom report, the Service Administrator or other users who have been granted access can generate the report. See Generating the Report.

> **Note**
>
> If you are managing your migration from Oracle Classic environment to Oracle Cloud Infrastructure (OCI) and have any custom reports, you may have to update the custom report's query definitions in the Target Service. This is because the internal IDs for objects (for example, Tasks, Journals, Attributes, etc.) may change during the migration. However, in case of Oracle managed migration, there will not be any change in the internal object IDs, so changes to custom report definitions is not required.

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Creating a Task Manager Query

Creating a query is the first step in creating a custom report.

To create queries:

1. From the Home Page, click **Application**, and then click **Task Manager**.
2. Select the **Task Manager Reports** tab on the left.
3. On the **Queries** tab, click **New**.
4. On the **New Query** screen, enter a **Name** and an optional **Description**.
5. From **Type**, select an option:
   - **Parameter Query**: A Parameter Query type is used to present a list of options that you can specify for this parameter's value. Parameter Query allows you to present a list of options used when filling in a parameter's value for a Report Query, where the list of options is not a simple Attribute already defined, but is instead a complex query that you need to define.

     This parameter query example provides a list of all Task Manager periods:

     ```
     SELECT PERIOD_ID, PERIOD_NAME FROM FCC_PERIODSWHERE APP_ID=1
     ```

   - **Report Query**: Select the records to be included in the report. You can apply a security filter, so users see only the data that they are authorized to see based on their roles and the reports to which they are assigned. To apply a Security Filter to a report query, add the following syntax to the end of the query WHERE CLAUSE statement.

     ```
     $FCC_SECURITY_CLAUSE$
     ```

     When using`$FCC_SECURITY_CLAUSE$` in the query, the `FCC_TASKS` must be aliased to `TaskEO`

     Because many predefined queries included with Task Manager have the Security Filter applied, you can use them as examples when building your own.

     For example, the following Report Query specifies scheduled tasks to display Task Code, Name, Assignee and Schedule for all High Priority tasks in the selected schedule.

     ```
     SELECT TaskEO.TASK_CODEAS "$TASK_CODE$",

     TaskEO.TASK_NAMEAS "$NAME$" ,

     ((SELECT CASE WHEN FIRST_NAME IS NULL AND LAST_NAME IS NULL THEN
     USER_LOGIN ELSE FIRST_NAME||' '||LAST_NAME END FROM FCM_USERS WHERE
     USER_ID = (coalesce(AssigneeEO.ACTIVE_USER_ID, AssigneeEO.USER_ID))))AS
     "$ASSIGNEE$" ,

     DeploymentEO.DEPLOYMENT_NAMEAS "$SCHEDULE$"

     FROM FCC_TASKS TaskEO

     LEFT OUTER JOIN FCC_ACCESS AssigneeEO ON (TaskEO.TASK_ID =
     AssigneeEO.SOURCE_ID AND AssigneeEO.ACCESS_TYPE = "AS")

     LEFT OUTER JOIN FCC_DEPLOYMENTS DeploymentEO ON (TaskEO.SOURCE_ID =
     DeploymentEO.DEPLOYMENT_ID)

     WHERE (TaskEO.SOURCE_TYPE = "DEPLOYMENT")

     AND ((((TaskEO.PRIORITY=3 )
     ```


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```
AND (TaskEO.SOURCE_ID=~SCHEDULE~ ))))
```

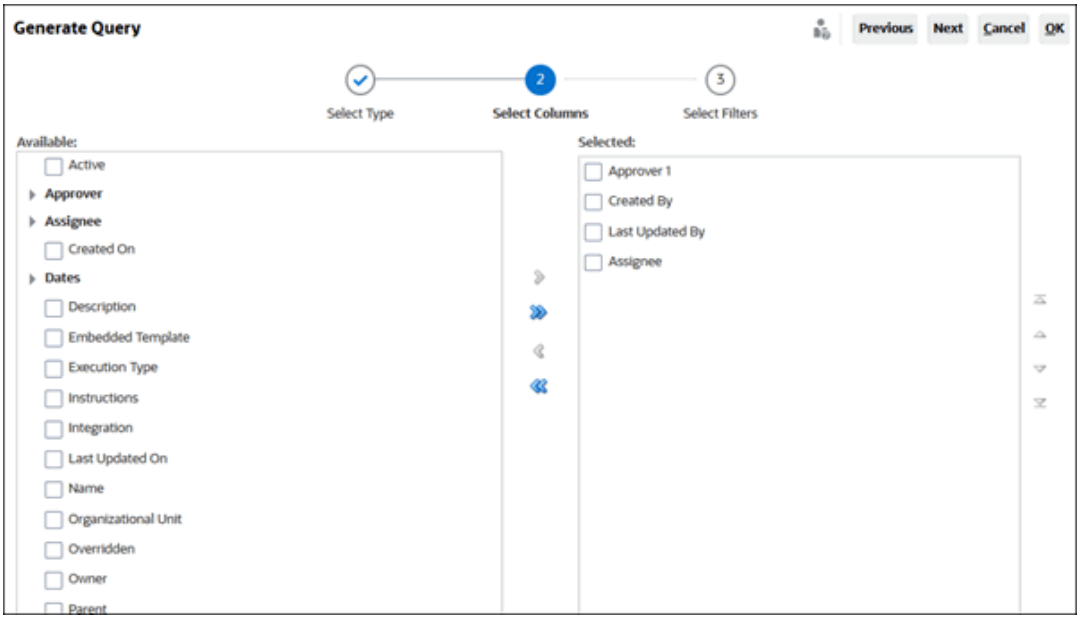
6. Click **Generate Query** to build the query from the New Query dialog. The dialog assists you in creating a query against the database by allowing you to select any existing attribute in the product to be queried and/or filtered against. The system then generates the SQL to match the specified attributes and filters, at which time you can modify and enhance it.

   a. On the **Select Type** screen, select options:

      - From **Query**, select **Template Tasks** or **Schedule Tasks**.
      - **Optional**: Select the **Apply Security** checkbox to automatically apply the user security filter to the generated query. This applies the Security token to the query, to be filled in with the correct SQL when the report is generated.

   b. Click **Next**.

7. From **Select Columns**, select the columns to display in the query, and then click **Next**.



8. From **Select Filters**, click **Create Condition** or **Create Condition Group** and select the conditions to create the filters to create the query.

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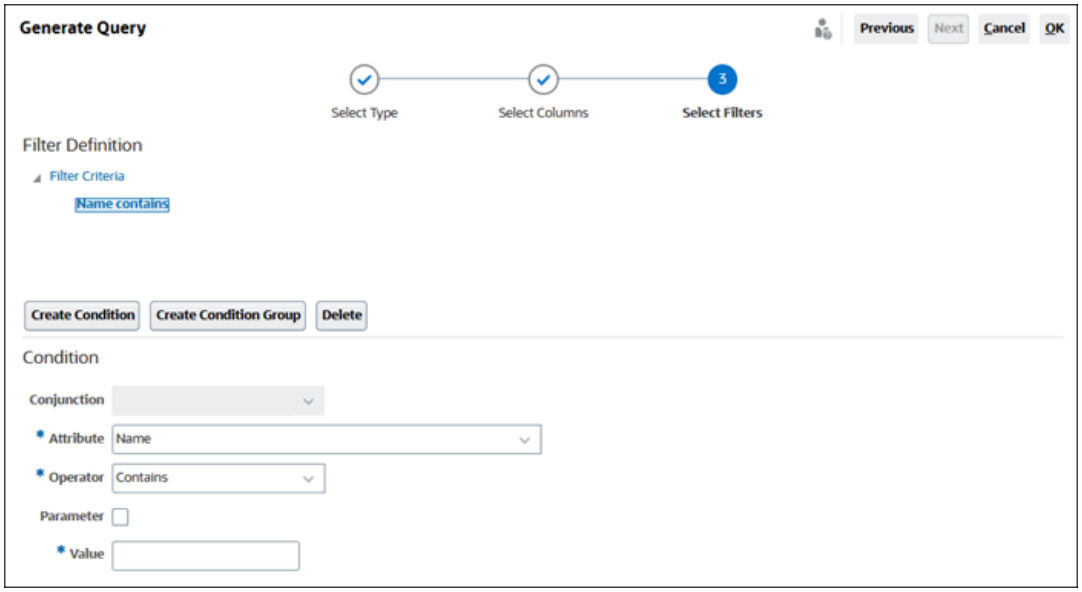
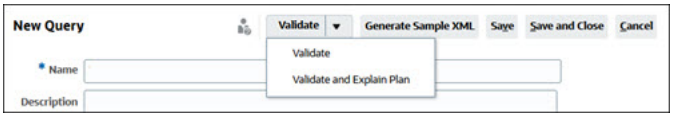


9. Click **OK**.
10. **Optional**: If you want to use the report in the future, click **Generate Sample XML**.
11. Click **Save**.
12. To test the query for errors:
    - Click **Validate** from the New Query dialog. You can also select **Validate** from the drop-down list.

      The query displays in the Queries tab.

      > **Note**
      >
      > You can easily delete a query, or duplicate a query using the Action menu.

    - Click **Validate and Explain Plan** from the drop-down to validate the query and also generate the execution plan for this query. A .txt file that uses the same name as the query is generated. You can view or download the generated plan.
      Open this file to view the generated plan which contains the sequence of steps that is performed to run this query. If the performance of a query is suboptimal, you can use this plan to understand the cause of the problem and determine the best way to improve the query performance.

      > **Note**
      >
      > By default generation of execution plans is available only for OCI (Gen 2) environments. To enable it for Classic environments, contact Oracle using a service request.

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Using Date Parameters

When defining queries for task manager reports, you can define runtime parameters and then pass these parameters to the report query. If your report query references a database column that's defined as a `date` data type, you may need to use the `to_date` function to make sure that the report parameter is correctly interpreted when the system runs the report query.

In the following sample, the system captures a date from the user, and the `to_date` function is used in the where clause to convert the date to the correct format for the query. The where clause entry for the date field is coded as `(TDATASEG.PERIODKEY)=to_date('~POVPERIOD~','YYYY-MM-DD')` to make sure that the date selected from the parameters is interpreted by the system in the correct format.

The full query is:

```
SELECT ENTITY, ENTITYX, ACCOUNT, PERIODKEY, tPOVPartition.PartName AS
Partitionkey,  tPOVCategory.CatKey,  tPOVCategory.CatName,  ACCOUNTX, UD1,
UD1X, UD2, UD2X, UD3, UD3X, UD4, UD4X, UD5, UD5X, UD6, UD6X, UD7, UD7X, UD8,
UD8X, UD9, UD9X, UD10, UD10X, UD11, UD11X, UD12, UD12X, UD13, UD13X, UD14,
UD14X, UD15, UD15X, UD16, UD17X, UD18, UD18X, UD19, UD19X,
UD20,UD20X,DESC1,DESC2,DATAVIEW,CURKEY,AMOUNT,AMOUNTX  from TDATASEG INNER
JOIN tPOVCategory  ON tDataSeg.CatKey = tPOVCategory.CatKey  INNER JOIN
tPOVPartition  ON tDataSeg.PartitionKey = tPOVPartition.PartitionKey
WHERE (((TDATASEG.PERIODKEY)=to_date('~POVPERIOD~','YYYY-MM-DD')) AND
((tPOVCategory.CatKey)=~POVCAT~) AND (~POVRULE~ is NULL or RULE_ID =
~POVRULE~) AND ((tPOVPartition.PartitionKey)=~POVLOC~ ))
```

The Period query is defined as:

```
select periodkey, perioddesc from TPOVPERIOD
```

Modifying A Query

A system administrator can edit a Task Manager query.

1. From the Home Page, click **Application**, and then click **Task Manager**.
2. Select the **Task Manager Reports** tab on the left.
3. Select the **Queries** tab if it's not already selected, and then **Edit** from ▪▪▪ next to the query you are editing.

> **Note**
>
> If it is locked, ask the System Administrator to unlock.

4. Select Task Manager query.
5. In **Edit Query**, edit the information as needed.
6. After you are done with your changes, click **Save and Close**.

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Creating a Template

Creating report templates is the second step in generating custom reports. Report templates are created in Microsoft Word with Oracle BI Publisher Desktop installed.

Make sure that you've already generated the sample XML when you created the query. The sample XML file is generated in the Sample XML folder.

To create a report template:

1. Open Microsoft Word with a new document.
2. Select the **BI Publisher** tab, then select the **Sample XML** folder and click **Load Data**.
3. Locate the `SampleQuery.xml` that was generated when you created the query and click **Open**.
   When the message displays, "*Data Loaded Successfully*", click **OK**.
4. Select **Insert**, and then **Table Wizard**.
5. Select **Table** and click **Next**.
6. Select the default data set and click **Next**.
7. Select the desired fields to show in the report and click **Next**.
8. Select **Group By**, select the fields to group by, and then click **Next**.
9. Select **Sort By**, select the fields to sort by, and then click **Finish**.
10. Save the template as an `.rft` file; for example: `SampleQuery.rtf`.

For more information about creating RTF templates, see Creating RTF Templates.

Setting Up a Report Group

Creating report groups is the third step in generating custom reports. A report group enables you to group individual reports together so that you can organize reports in folder structures.

> **Note**
>
> Nested Report Groups are not supported.

After a report group has been created, you can modify it if necessary. You can duplicate a report group but its name must be unique. You can also delete a report group, however, deleting a report group deletes all reports associated with that group.

To create report groups:

1. From the Home Page, click **Application**, and then click **Task Manager**.
2. Select the **Task Manager Reports** tab on the left.
3. From the **Report Groups** tab, click **New**.
4. In the **New Report Group** window, enter:
   - **Name**: Enter a group name for the group of reports.
   - **Description**
   - **Display to User**: Select if you want this report group displayed to the user.

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The **Display to User** option enables report writers to have a group of reports hidden while they are working on them.

5. On the **Reports** tab, reorder or edit reports that have been added to the report group using the Move icons

.
6. Click **Save and Close**.

Creating a Report

Creating report definitions is the fourth step in generating reports. Users can create reports from the queries and assign them to groups.

To create report definitions:

1. From the Home Page, click **Application**, and then click **Task Manager**.
2. Select the **Task Manager Reports** tab on the left.
3. Select **Reports**, and then click **New**.
4. In **New Report**, enter:
   - **Name**
   - **Description**
   - **Query**: Select a query.
   - **Template**: Click **Browse** and then browse to a report template. You can upload any supported Oracle Business Intelligence Publisher template format. See Creating a Template.
   - **Report Group**: Select the Group Name for the report from the drop-down menu.
   - **Display to User**: Select if you want the report displayed to a user.
   - **Output Format**: Select a report output format that BI Publisher supports from one of the following options:
     - **xlsx**: Not supported for graphs.
     - **html**: Not supported for graphs and charts.
     - **pdf**
     - **csv**: Does not use a template and is better at faster data retrieval.
     - **csv (Formatted)**: Is best suited for a simple data table with formatted data and does not support images, graphics, or styling in the template.

> **Note**
>
> The **csv (Formatted)** output format takes additional time to generate the report to honor template formatting when compared to the **csv** format. Therefore, you can select **csv** to generate the data quickly or **csv (Formatted)** to generate formatted template based data.

5. To complete the report definition, you must set the parameters and access:
   a. For Parameters, select the **Parameters** tab.

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Parameters from the query are identified and added to the list. Update the following:

- **Display Name**
- **Parameter Type**: The available options are:
  - **Task Manager Attribute**
  - **Text**
  - **Date**
  - **Date/Time**
  - **Number**
  - **Query**
  - **Integer**
  - **True/False**
  - **Yes/No**
  - **User**
- **Display To User**: Clear the checkbox if you do not want the parameter to be displayed and want the value to be used. You can use a single query to create multiple reports. In such scenarios, few parameters are hidden and values defined in the **Parameters** tab are used.
- **Attribute/Query**: For Attributes, the drop-down lists the attributes for Task Manager. For Query, the drop-down lists all queries of the **Parameter Type**. This can be used to make parameters such as dynamic LOVs. See Creating a Task Manager Query
- **Parameter Value**: The default value is displayed.

b. For Access, select the **Access** tab.

c. From the drop-down lists, select the **Application Module**, then select a **Role** for the Application Module.
For example, you can select Task Manager for the Application Module, then select to assign access to the Power User role.

You can use **Add** or **Remove** to change the list of users who can access the report.

> **Note**
>
> The report must be granted access to at least one application module for the report to display in the corresponding **Reports** tab.

6. Click **Save and Close**.

Searching and Filtering Queries, Report Groups, or Reports

Using the search field and filter options, you can control the records that you see in the list of queries, report groups, or reports.

To filter queries, report groups, or reports:

1. From the Home Page, click **Application**, and then click **Task Manager**.
2. Select the **Task Manager Reports** tab on the left.

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3. Select the **Queries**, **Report Groups**, or **Reports** tab.
4. Enter search criteria in the **Search** text box.
5. **Optional:** From the filter bar, click on a category to display additional search operators such as **Equals**, **Does Not Equal**, **Contains**, **Does Not Contain**, **Starts With**, and **Ends With**.

> **Note**
>
> - Click **Add a Filter** to view all categories.
> - To hide the filter bar, click the **Hide Filter Bar** icon.
> - To clear all filters, click the **Filters** icon, and then click **Clear All Filters**.

Generating the Report

Generating reports is the final step in the process. After a Service Administrator has built the custom report, any user or group granted access can generate the report.

A Service Administrator can also use REST API commands to generate a report. For detailed information, see Generate Report for Financial Consolidation and Close and Tax Reporting in *REST APIs*. For large reports, the `runAsync` parameter for report generation in REST API allows reports to run asynchronously (value of true).

To generate reports:

1. From the Home Page, click **Tasks**, and then select the **Reports** tab on the left.
2. Click the **Task Manager Reports** tab.
3. Select the report you want to generate. You can use the quick **Search** option to find a report under a selected group.
4. For **Output Format**, select from the following:
   - **xlsx**: Not supported for graphs.
   - **html**: Not supported for graphs and charts.
   - **pdf**
   - **csv**: Does not use a template and is better at faster data retrieval.

     > **Note**
     >
     > For reports that contain more than 10,000 records, it is recommended to use the **csv** format.

   - **csv (Formatted)**: Is best suited for a simple data table with formatted data and does not support images, graphics, or styling in the template.

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> **Note**
>
> The **csv (Formatted)** output option takes additional time to generate the report to honor template formatting when compared to the **csv** format. Therefore, you can select **CSV** to generate the data quickly or **csv (Formatted)** to generate formatted template-based data.

5. Click **Generate**.
6. Enter the **Name**, and then select the **Schedule** and **Period** from the drop down menus.
7. Click **Generate**.
   When done, the system displays a "Completed Successfully" status message.

> **Note**
>
> - Click **Back** to preserve the parameter values that you have selected and regenerate the report.
> - Click **Reset** to clear the parameter values that you have selected and regenerate the report.

8. Select **Open with** or **Save File** to save the ZIP file.

The report generation process uses a backend job framework that runs report jobs in the background. If the report has an error, for example, if the size is greater than the recommended size, you see a message stating the error in the Generate Report dialog before the report is generated. For reports with very large number of records, it is recommended to use filtering to break the report down to fewer records.

Understanding Reports Security

This topic describes the security for standard reports and custom reports in Task Manager.

> **Note**
>
> A Service Administrator can also use REST API commands to generate user details report. For detailed information, see Generate User Details Report for Financial Consolidation and Close and Tax Reporting in *REST APIs*.

Standard (Predefined) Reports Security

A security clause is in all standard report queries. This means that the reports have a security filter enforced on these reports by default. For example, Power User security (established under **Access Control**), to allow access to certain tasks is enforced on reports. A Power User with access to only certain tasks can only see reports for those tasks.

Administrators can edit reports to give users access to a report using the **Access** tab in the **Edit Report** dialog. This ensures that if you allow users to see reports, they can only see data appropriate for them as determined by the Administrator.

The security clause is also included as part of the following predefined **Task Manager** reports:

- At Risk Tasks

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- Task Compliance Analysis
- Early Tasks
- Late Tasks
- Task Performance Analysis by Organizational Unit
- Rejected Tasks
- Task Audit Trail
- Users with Late Tasks

Assigning User Access to Standard Reports

An Administrator can decide to give access to standard reports to various users. This is done by using the **Access** tab on the **Edit Report** dialog.

To grant users access to a report:

1. From the Home Page, click **Application**, and then click **Task Manager**.
2. Select the **Task Manager Reports** tab on the left.
3. From **Reports**, select the report you want, and from **Actions**, select **Edit**.
4. On the **Edit Report** dialog, under **Access**, you can use **Add** or **Remove** to change the list of users who can access that report.
5. Click **Save** or **Save and Close**.

Custom Reports Security

For custom reports, you can determine when you create your query whether you want to have a security clause inserted into the query and then assign access to the report to users. This means that the creator of the report determines who should be given access.

Using Task Manager Report Binders

Task Manager report binders enable you to create a history of all activities that occurred as part of a schedule, for example, the Corporate Q1 Close. You use filters to specify the schedule, tasks to include, and which task information to include.

Information for all tasks that meet the binder filter criteria are output in HTML to a ZIP file. They contain HTML and other files. When you extract the ZIP file, a directory structure is formed, in which you can access the HTML file to view the report binder. Because the report is self-contained in one file, it can easily be copied, printed, or emailed for internal or external reviews.

Generating Report Binders

The Generate Report Binder dialog box enables you to specify the parameters needed to create a report. When you open a schedule in a view and run Generate Report Binder, the report binder is applied to the current view and incorporates only the tasks within the current filtered view.

The report binder is generated and returned through the browser as a document. When you run Generate Report Binder, you have the option to open it immediately or to save it to disk as a ZIP file.

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You can generate report binders for multiple schedules at a time. This can be helpful if your application has a large number of schedules.

To generate a report binder:

1. From the list of Schedules, select one or more schedules for which to generate a report binder.
2. From the **Actions** dropdown, select **Generate Report Binder**.
3. For **Report Binder Name**, enter a name.
4. For **Description**, enter a binder description.
5. From **Optional Components**, select the task components that you want to display in the report:
   - **Alerts**
   - **Comments**
   - **Attachments**

   > **Note**
   >
   > Including attachments in the report greatly increases the size of the report and may impact the performance.

6. Click **Generate**.
7. From File Download, select **Save**.

   The Save As dialog box is displayed, and the ZIP file name is displayed at the bottom.
8. Select a directory for the ZIP file, click **Save**, then **Close**.

Viewing Report Binders

When you generate a report binder, it is output in one zipped file. The ZIP file name is the name that you specified for the Report Binder. The report pages are merged into an HTML report, with page breaks for sections as required, so that the report can be printed with a print command. If you choose to include attachments, a separate attachment appendix, containing links to attachments with corresponding tasks and alerts, is created, which you can print separately. All attachments are downloaded to separate folders.

If you saved the report binder as a ZIP file, you can extract everything from the ZIP, which creates a directory structure with the same name as the report binder. You can see the report binder by opening the HTML page in the directory. The first report page contains information on the report binder, the schedule, and displays a list of tasks and alerts available in the report binder. You can navigate to the Tasks section to see task details such as status, Assignee, start and end dates, and duration. If a task has not started, the projected (scheduled) date is displayed. Each task is a link to a Task Detail page. By default, it includes sections for Attributes, Instructions, Questions, Workflow, Predecessors, and History. If you selected the options to include Alerts and Comments when you generated the report binder, those sections are also displayed.

The first page also contains a list of alerts into which you can drill further. From the Alert Detail page, you can navigate to the associated task.

To view report binders:

1. Navigate to the directory in which you downloaded the ZIP file, and double-click the file.

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2. Extract the ZIP files to the desired directory.
3. Navigate to that directory and locate the subdirectory that matches the ZIP file name.
4. From the subdirectory, double-click *report_binder_name*.html to view the report binder.

Generating Cloud EPM System Reports for Task Manager

Reports provides a user-friendly, robust report development framework along with an enriched report viewer experience. Using Reports' Tables, you can incorporate Task Manager data within the Cloud EPM Reports platform – Reports, Books, and Bursting.

For more information, see:

- Designing Reports in *Designing with Reports*
- Task Manager Attributes List in *Cloud EPM System Reporting Schema Reference Guide*

Working with Views and Filters

This chapter describes how to create, manage, and use views and filters.

Related Topics

- Working with Filters
- Working with Views

Working with Filters

Use filters to limit the records displayed in your views and reports.

Related Topics

- About Filters
- Viewing the Existing Saved Filters
- Creating a New Filter
- Using the Filter Bar to Adjust Your View
- Clearing Filter Conditions
- Removing a Filter
- Editing a Filter
- Duplicating a Filter
- Saving a Filter
- Publishing a Filter
- Deleting a Filter
- Displaying Columns for Filters
- Reordering Columns for Filters

About Filters

Filters control the records that you see in list views and reports.

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You can provide values for the attributes that you want to filter on and the operator to use for filtering. Typical operators include the following: equal to, not equal to, starts with, ends with, contains, greater than, less than, and so on. The available operators depends on the attribute's data type. For example, operators for filtering text values are different than the operators for filtering numeric values. Filters are combined together using and logic, which means only those records that meet all filter criteria are displayed.

You can also create more complex filters using and as well as or logic and grouping logic to determine the order in which filters are applied.

You can save a filter for future use. However, for setting up your views, saved lists can be used. See Working With List Views.

Service Administrators and power users can publish filters so that they are accessible to other users and these are then marked as Public. Service Administrators and power users can use saved filters to test rules.

Where Are Filters Used in Task Manager

There are many locations within Task Manager where filters can be used.

The filters functionality remains consistent throughout these different areas:

- Schedule Tasks
- Templates
- Schedules
- Task Types
- Alert Types
- Views
- Attributes

Viewing the Existing Saved Filters

Filter definitions that were saved are available on the **Filters** tab within **Configuration**.

To view saved filters:

1. From **Home**, click **Application**, and then **Task Manager**.
2. Click **Filters** to see existing filters.
   Public filters are marked with a green check mark.

Creating a New Filter

You create a new filter to control the display of rows as per your requirement.

To create a new filter:

1. From **Home**, click **Application**, then **Task Manager**, and then **Filters**. You can also go to any of the following pages to create a filter:
   - **Tasks > Schedule Tasks**. Click **...** on the right side and then **Manage Filters**.
   - **Tasks > Supplemental Data Tasks**. Click **...** on the right side and then **Manage Filters**.
   - **Tasks > Alerts**. Click **...** on the right side and then **Manage Filters**.

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2. Click New (+) to add a new filter. The New Filter dialog appears.
3. In **Name**, enter a unique name for the filter.
4. In **Description**, enter a description (optional) for the filter.
5. In **Type**, select the type of filter being created.
6. In the Filter Definition section, click **Create Condition** to create a condition that will be used to filter data. You can specify multiple conditions and condition groups.
   For each condition, specify the following:
   - (Condition Groups only) Conjunction: Select **And** or **Or**. These state how this condition or group relate to prior sibling conditions or groups. This field is enabled only if the selected node is not the first child of its parent node.
   - Attribute: An attribute is a field or value that a condition will be comparing to a value for inclusion in the result set of a filter. However, for the case of a filter, an attribute denotes more than the list of user-defined attributes.
   - Operator: States the type of evaluation to perform against the attribute. For example, Equals, Between, Does not Equal, Greater than, Is blank, Is not blank, Less than, Not between, Starts With, Ends With.
   - Value: Specifies what values to compare the attribute against. The type of attribute determines what input field is available.

Using the Filter Bar to Adjust Your View

The filter bar works across many different areas to filter data displayed in a list. You can add multiple filters to a list.

Here's an example of the filter bar on the Filter configuration. Note that the default value **All** is used. This means that all objects are displayed unless you choose to make a selection of filter attributes.

To add a filter to your list:

1. Click **+ Add a Filter** to display a list of attributes that you can use to filter data.
   The list of attributes is different, depending on the list page on which you are adding the filter.
2. Select the filter attribute you'd like to add.
   The selected attribute appears on the filter bar.

Using a Date Range

Some filter attributes have a date range. For example, if you want to add another filter attribute to the filter bar, such as **Created On**, you can use the date range feature. Narrow your view by using a date value and then use the **Operator** field to set the conditions for dates values that meet this criteria.

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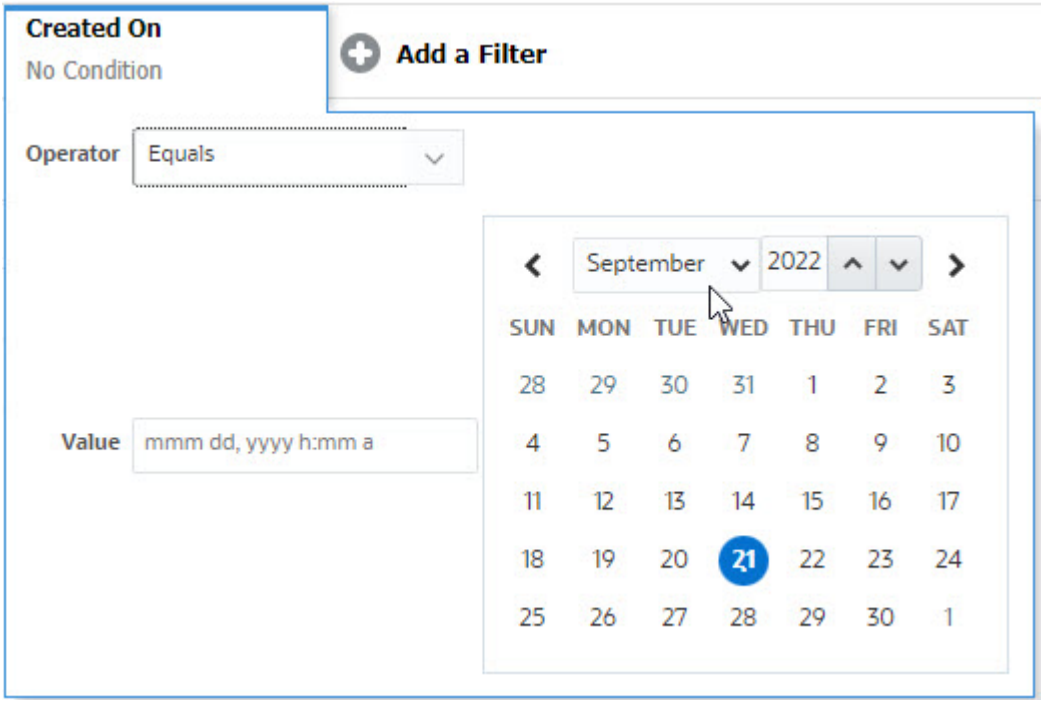

Clearing Filter Conditions

When you clear a filter, the conditions applied by this filter on the displayed data are removed.

After adding the filter to the Filter bar, you can do one the following to clear the filter conditions:

- Hover on each filter, click the ••• icon, and click **Clear** to clear and reset the condition for the specific filter.
- Click the ••• icon on the right side of the Filter bar and select **Clear All Filters**. The conditions for all the filters in the Filter bar are cleared and reset.

Removing a Filter

You can remove a filter from the Filter bar.

Removing a filter also clears the conditions applied by the filter on the listed data.

To remove a filter from the filter bar:

1. Click the icon ••• adjacent to the filter.
2. Click **Remove** to remove the filter from the Filter bar.

On a few pages, you can't remove the default filters.

Editing a Filter

Edit a filter to modify its definition.

To edit a filter:

1. Open the Filters page.

   From the Home page, click **Application**, then **Task Manager**, and then **Filters**.

   You can also edit filters when you click **…** and then **Manage Filters** on a page.

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2. Select the filter you want to edit and click the icon.
   The Edit Filter dialog is displayed.
3. Update the required information.
4. Click **OK**.
   The filter details are updated.

Duplicating a Filter

You duplicate a filter to create a copy of an existing filter.

To duplicate a filter:

1. On the Filters page, select the filter you want to duplicate and click the icon.
   A copy of the filter is displayed on the page.
2. Click the copied filter.
   The Edit Filter window is displayed.
3. Update the required information.
4. Click **OK**.
   The filter details for the copied filter are updated.

Saving a Filter

Saving a filter definition enables you to easily reuse these filter settings when required (instead of manually setting up filters when required).

By default, filters are saved as private filters. To enable other users to use the same saved filters, you can publish the filters.

To save a filter:

1. Use the **Add a Filter** to create a filter for the required attributes.
2. From the Actions menu, select **Save Filter**.
   The Save Filter dialog appears.
3. In **Name**, enter a unique name for the filter.
4. In **Description**, enter an optional description for the filter.
   Notice that the Filter Definition section displays the selected filter conditions.
5. Click **OK**.

Publishing a Filter

Service Administrators can publish a filter to make it available to other users. Publishing a filter marks it as **Public**.

To publish a filter:

1. On the Filters page, select the filter you want to publish, and click the icon.
   A message asking for confirmation is displayed.
2. Click **Yes**.
   A copy of the filter is created and the **Public** column displays icon indicating the filter is published.

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> **Note**
>
> Only Power Users and Administrators can publish filters.

Deleting a Filter

Delete a filter to remove its definition from the system.

To delete a filter:

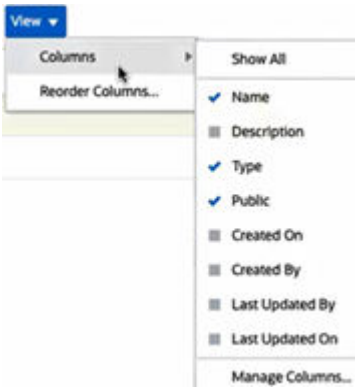
1. On the Filters page, select the filter you want to delete and click the icon.
   A message is displayed asking for confirmation.
2. Click **Yes**.
   The filter is deleted.

Displaying Columns for Filters

To display the columns for the filters:

1. On the Filters page, click **View** and then **Columns**.
2. Select the columns you want to display or click **Show All** to display all the columns.

You can click **Manage Columns** and hide or show the columns that are displayed in **View > Columns**.



Reordering Columns for Filters

To reorder the columns:

1. On the Filters page, click **View** and then click **Reorder Columns**.
2. Select the column name that you want to reorder and use the following icons to move it to the required position:
   - : Move selected items to top of the list
   - : Move selected items up one in the list
   - : Move selected items down one in the list
   - : Move selected items to bottom of the list

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3. Click **OK**.
   The columns are displayed based on the specified order.

Working with Views

Views provide multiple ways to visualize and analyze your data.

Related Topics

- About Views
- Working With List Views
- Working with Pivot Views
- Working with Chart Views
- Working With Gantt Chart Views
- Working With Calendar Views
- Saving Views
- Editing Saved Views
- Publishing Views
- Setting a Default View

About Views

Oracle Fusion Cloud Enterprise Performance Management products provide you with great flexibility in the ways in which you can visually view your data.

Types of Views

- **List view**: Provides a table display of information in columns and rows for the objects which meet the filter conditions on the page.
- **Gantt chart**: Displays a timeline for a schedule that graphically represents the duration of tasks and the dependencies between them. You can use the Gantt chart to see the status of predecessor tasks to other tasks.
- **Calendar**: Displays schedule tasks in traditional calendar format by Month, Week, or Day. After you select the Calendar View, use the calendar icons to select Month, Week, or Day. In the Calendar View, you can also see the Day Label in the Day header.
- **Pivot view**: Allows you to create summaries of numeric attribute data by attribute value.
- **Chart view**: Displays data in the form of charts.

Use the View selector ( ) to move between views.

Choosing the Compact Display of Views

When you're viewing items in a List, Gantt, or Pivot view, you've the ability to view the data in a compact format. The **Compact** option works as a toggle switch and displays more rows of information. Click **Compact** from the **View Selector** to display the records in a compact format.

If you save a view, your choice of Compact will also be saved.

Objects for Which Views are Available

Views are available for the following objects in Task Manager:

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- Schedule Tasks: Contains the list of scheduled tasks for Task Manager
- Alerts: Contains the list of alerts and provides drill-down to the alert details

About Filtering Data in Views

You can use filters to restrict the data displayed in the views. Filters can be created at run time (after the data is displayed), or when designing the chart or pivot view.

You can filter data in the following ways:

- Filter bar in a view:
  Using filters from the Filter bar enables users to dynamically decide the criteria that must be applied to the view data. The specified filter condition is applied at run time, after the view is generated and the display is updated to reflect the applied condition. See Using the Filter Bar to Adjust Your View.

  For example, add a filter to the Filter bar in the Tasks list with the Assignee column selected. You can use this filter to dynamically filter the chart data by selecting one or more assignees.
- **Filter** in the **Layout** tab of the Edit Pivot or Edit Chart dialogs:
  This filter is added by the view designer. The specified conditions are applied on the data before the data is included in the pivot view or chart view. You can create a filter for each data set listed in the **Legend (Series)** section. See Setting the Chart Layout.

  For example, if you create a filter condition for a data set with Status set to Open, only open tasks are included when charting that data set.

Working With List Views

Use list views to display records in a simple tabular form.

Related Topics

- About List Views
- Customizing the List View
- Task Manager Example: Customizing the List View for Schedule Tasks

About List Views

List views present records for on screen viewing and provide drill-down capabilities to the record details. They display objects in a vertical list, in the form of rows and columns.

To see more rows displayed in the list view, use **Compact** from the **View Selector** at the right hand side.

List views provide the following reporting features:

- Columns can be added to or removed from the view and re-ordered.
- Filters can be applied to limit the records included in the list.
- List views can be printed or exported to Excel for ad-hoc reporting.

Customizing the List View

The list view displays with the default columns selected. You can easily change the default columns to customize your view.

To customize a list view:

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1. Display the list view for the required object.
2. In the **Actions** drop-down, select **Select Columns**.

   The Select Columns dialog appears. The columns that are currently displayed are listed in the **Selected** section.
3. In the **Available** section, select the attributes that must be included in the list view and use the arrow keys to move them to the **Selected** section.
4. Click **OK**.

Task Manager Example: Customizing the List View for Schedule Tasks

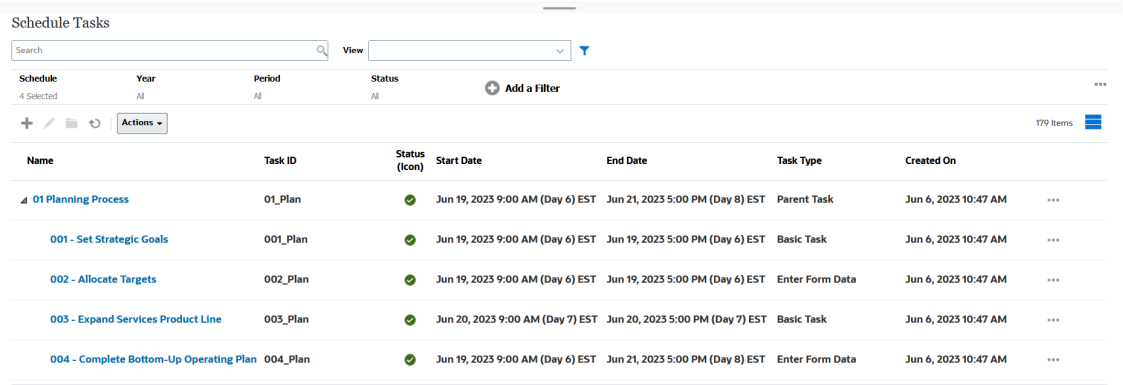
Assume that you want to see more information such as when was the task created.

Use the following steps:

1. From **Home**, click **Tasks** and then click **Schedule Tasks**.
2. From the **Actions** drop-down, click **Select Columns**.

   The Select Columns dialog appears. Use this to add additional columns to view information about the tasks.
3. In **Available**, scroll down the list, select **Created On**, and use the arrow keys to move it to **Selected** on the right side.
4. Click **OK**.

   The Schedule Tasks page displays the **Created On** column for the tasks.



Working with Pivot Views

Use pivot views to summarize your data.

Related Topics

- About Pivot Views
- Customizing a Pivot View
- Example: Viewing Organizational Units Based on Task Status

About Pivot Views

A pivot view enables you to create summaries of the attribute data.

There is a default layout for each object that supports pivot views. However, you can customize the pivot view based on your requirement.

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Click each summary value displayed in the pivot view to drill down and view the records that comprise the total.

Customizing a Pivot View

You can customize a pivot view to summarize the data based on your business requirement. Data can be grouped by a maximum of two attributes.

To customize a pivot view:

1. Display the pivot view for the required object.
2. Click the **Edit** icon at the top-right.

   The Edit Pivot dialog appears. It contains three tabs: **Properties**, **Layout**, and **Columns**.
3. In the **Properties** tab, enter the basic view properties. See Setting Pivot View Properties.
4. In the **Layout** tab, specify the details of how data must be grouped. See Setting Pivot View Layout.
5. In the **Columns** tab, enter the data labels and the sorting order for the data. See Specifying Pivot View Columns.
6. Click **OK**.

> **Note**
>
> If you drill down into a record, update one or more details, save it, and return to the pivot view, these changes are not visible. You must refresh the pivot view to see the latest updates.

Setting Pivot View Properties

Use the **Properties** tab of the Edit Pivot dialog to set the basic properties of a pivot view.

To set the pivot view properties:

1. In **Background**, select one of the following:
   - **No Fill**: No background color is used
   - **Solid**: Select the color to be used as the background color. You can optionally select a color in **Gradient**.
2. In **Header Color** and **Header Size**, select a color and font size for the pivot view header.
3. In **Data Color** and **Data Size**, select a color and font size for the data displayed in the pivot view.
4. Select **Missing Values as Zero** to display a zero when data values are missing.
   This includes missing data values and values that are missing because of an applied filter.
5. In **Period**, select one of the following:
   - **Selection**: Use the selection from the Period filter as-is.
   - **Latest in Selection**: Use the latest Period chronologically in the filter.
   - **Prior from Selection**: Create a new filter of the prior X periods based on the latest period in the filter.

     The Period selection allows you to fine tune the list of Periods for the particular pivot view. This can be done so that pivot views sharing the same global period filter can

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display different amounts of data, or to make it easier to show a distribution of period data based on a single local period selection.

Setting Pivot View Layout

The **Layout** tab specifies how data in the pivot view must be summarized. Filters can also be applied on the data that is considered for summarizing.

Specify the following layout details:

1. In the **X Axis** section, specify details about the primary grouping to use in the view.
   - In **Type**, select one of the following: None, Attribute, Date, User, Yes/No, or True/False.

     If you set the **Type** to Attribute, in **Attribute 1**, select the attribute that must be used to group data. For example, Organizational Unit.

     Optionally, you can create another subgroup, within the **Attribute 1** group, by selecting another attribute in **Attribute 2**.
   - In **Sort**, select the attribute and the order on which data must be sorted.
   - Select **Swap Rows and Columns** to swap the display of rows and columns in the grid.
2. In the **Legend (Series)** section, click the **Add** icon to create one row for each attribute for which you want to summarize data. This summary is created within the specified **Type**. There must be at least one row in this section.

   For each row, specify the following details:
   - In **Value**, select the attribute whose data must be summarized. For example, Approvers (Count).
   - In **Aggregation**, select the aggregation operation to be performed on the summarized data. It will default to the aggregation method specified on the attribute that you selected. The supported aggregation methods are Average, Count, Maximum, Minimum, and Sum.

     > **Note**
     >
     > If you select a non-numeric attribute in **Value**, the only **Aggregation** available is **Count**.
   - In **Categories**, optionally select the attribute by which the data is summarized, within the **Value** group.
   - In **Group By**, select an attribute from the list. This is applicable only when **Type** is set to a value other than Attribute.
   - Click **Filter** to apply a filter on the data that is included in the pivot view. This filter is first applied on the data and then the data is summarized using the specified criteria.

     By default, the first clause in a filter condition is displayed as the label for this filter.
3. Click **OK**.

Considerations When Using Aggregation

For Task Manager example, see Example: Viewing Organizational Units Based on Task Status

Specifying Pivot View Columns

Use the **Columns** tab to customize the labels and order of data displayed in the pivot view.

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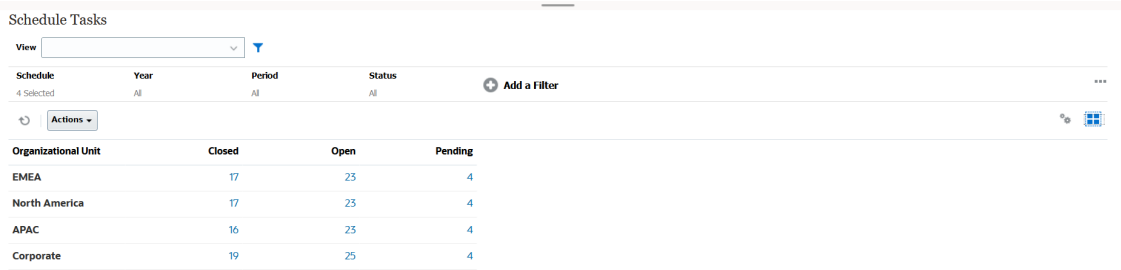
Each row in this tab will be displayed as a column in the pivot view. The number of rows is determined by what is specified in **X Axis** and **Categories** in the **Layout** tab.

To customize the display of the pivot view:

- In **Label**, enter the label that must be displayed for this column.
- Select **Sort** to sort the pivot data by this column. Click the icon next to this option to sort in ascending or descending order.
- Use the arrows on the right to change the order in which the columns are displayed. This list of columns are displayed from left to right in the pivot view.

Example: Viewing Organizational Units Based on Task Status

This example describes how to create a pivot view that groups organizational units based on the status of the tasks. The following image displays the Pivot view.



Use the following steps to create the above Pivot view:

1. In the Home page, click **Tasks** to display the Schedule Tasks page.
2. Click the icon and select **Pivot View**.
3. Click the **Edit** icon to configure the Pivot view as per your requirement.
4. On the **Properties** tab:
   - **Background**: No Fill
   - **Header Size**: 13
   - **Data Size**: 14
   - **Period**: Selection
5. On the **Layout** tab:
   - In **X-Axis**:
     - In **Type**, select **Attribute**.
     - In **Attribute1**, select **Organizational Unit**.
     - In **Sort**, select **Organizational Unit** from the first drop-down list and then select **Descending** from the second drop-down list.
   - In **Legend (Series)**:
     - In **Value**, select **Task**.
     - In **Categories**, select **Status**.
6. On the **Columns** tab, select the displayed labels.
7. Click **OK**.

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Working with Chart Views

Chart views enable you to visualize your data.

Related Topics

- Overview of Chart Views
- Creating Chart Views
- Example: Creating a Chart to Display Task Status by Month
- Example: Creating a Chart to Display Plan versus Actual End Date
- Example: Creating a Chart to Display Task Execution by Organization Unit

Overview of Chart Views

Related Topics

- About Chart Views
- About Chart View Properties
- Types of Chart Views

About Chart Views

A chart provides a visual representation of data.

Charts are generated dynamically based on your business process data. Click an area within a chart, for example, an area within a pie chart, to drill-down and view details about the information summarized by that area. The chart and pivot view layouts are linked, allowing you to quickly switch between the chart or pivot view.

When you configure settings for a chart view, the system remembers these settings. If you close the chart view, perform other actions in your business process, and then return to the chart view, the chart is displayed using the previous settings.

Multiple types of chart views are available to provide a graphical representation of your tasks. See Types of Chart Views.

About Chart View Properties

Chart properties include basic details such as the type of chart, chart orientation, and background. The properties that you can set for a chart depend on the type of chart.

Chart Property	Applicable Chart Types	Description
Type	All	Type of chart See Types of Chart Views.

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Chart Property	Applicable Chart Types	Description
Background	Area, Bar, Column, Combination, Doughnut, Line, Pie, Tile	Background color used in chart This color applies to the entire chart area, from border to border. You can choose not to use a background color, in which case a white background is used for the chart. If you specify a background color, an optional gradient also can be specified.
Label Position	Area, Bar, Column, Combination, Line	Location of the data labels in relation to the elements The options available depend on the type of chart. For example, bar charts have the following label options: No Label, Center, Inside Edge, or Outside Edge.
Bar Label Position	Combination	Location of the data labels in the bar chart
Grid Lines	Area, Bar, Column, Combination, Line	Grid lines used to divide the chart area
Orientation	Area, Bar, Column	Orientation of the chart - Vertical or Horizontal
Missing Values as Zero	Area, Combination, Line, Tile	Displays a zero (instead of a blank) when the data for a particular element is missing When you select this option, all missing values are displayed as zero. This includes missing data values and values that are missing because of an applied filter.

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Chart Property	Applicable Chart Types	Description
Dual-Y Axis	Area, Bar, Column, Combination, Line	Appears on the opposite side of the chart from the primary Y-axis The scale of the secondary Y-axis reflects the values for the associated data series. While the secondary Y-axis can be used with any line and bar chart type, it is more commonly used with the combination chart type, which helps to distinguish the data series that are plotted on the secondary Y-axis. For example, select **Bar** for the primary Y-axis and **Line** for the secondary Y-axis. When the data values in a chart vary widely from data series to data series, or when you have mixed types of data (for example, currency and percentages), you can plot one or more data series on a secondary (vertical) Y axis. You can also plot the secondary Y-axis as a split dual chart, where the secondary Y-axis appears with its data series below the original chart. In this case you can use any bar or line chart type.
Line Weight	Combination, Line	Width of the line in the chart The default is 5 pixels.
Bar Label Position	Combination	Position of the label for the data set displayed as a bar chart
Display as Percentage	Doughnut, Pie	Values for each data set displayed in terms of percentages (instead of as a numerical value)
3D chart	Pie, Tile	3-dimensional chart format

Types of Chart Views

You can create the following types of chart views:

- Area

  Displays a shaded area to represent each data set
- Bar

  Displays a graphical summary of multiple data values for comparison purposes

  Bar charts can be plotted vertically or horizontally
- Column

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Displays stacked bars that represent different data sets on top of each other

The height of the resulting bar shows the combined result of the data sets.

- Combination

  Provides a visualization that lets you merge into one chart any combination of bar, line, and area charts.

  Use this type when you have two different sets of data that are plotted on the y-axis. You can add a secondary Y-axis on the opposite side from the primary Y-axis. Adding a secondary Y-axis lets you display the scale for one measure that doesn't scale appropriately with the other measures on the chart; for example, showing a percentage measure and a currency measure on the same chart.

- Doughnut

  Displays a circular graph which is divided into segments to compare data sets to one another

  The blank center displays the sum of all data sets.

- Line

  Enables users to visualize a trend in data over intervals of time

- Pie

  Displays a circular graph which is divided into slices to compare data sets with one another

- Tile

  Enables users to select specific values from a data set, to display in separate tiles

  It emphasis a small number of data points or an individual data point.

The default chart view is a Bar chart. You can customize your chart display and change the default chart view.

Creating Chart Views

Create charts to portray your application's data graphically.

To create a chart view:

1. Display the chart view for the required object.
2. From the **View Selector** at the top-right of the page, select **Chart View**.

   If this is the first time you're accessing the chart view, the default chart view layout is displayed.
3. Click **Edit** at the top-right to display the Edit Chart dialog. It contains three tabs: **Properties**, **Layout**, and **Legend**.
4. In the **Properties** tab, specify the required chart properties, as described in About Chart View Properties.
5. Enter the information required to create the chart, as described in the following topics:
   - Setting the Chart Layout
   - Setting the Chart Axis
   - Setting the Chart Legend
6. Click **OK** to save the chart definition and then display the chart.

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Click any area within the chart view to drill-down and display all the records that are aggregated by the selected area. The records are displayed in the Details dialog. You can click any link within this dialog to access the details of that individual record.

> **Note**
>
> If you drill down into an individual record, update one or more details, save it, and return to the chart view, these changes are not visible. You must refresh the chart view to see the latest updates.

Setting the Chart Layout

Use the **Layout** tab to specify the data sets that must be plotted on the x-axis and y-axis of the chart.

To set the chart layout:

1. Open the **Layout** tab of the Edit Chart dialog.
2. In the **X Axis** section, specify the data that must be plotted on the x-axis of the chart.
   - In **Type**, select the type of data. Options include None, Attribute, Date, User, Yes/No, and True/False.
   - If you set the **Type** to **Attribute**, you must specify at least one attribute that is plotted on the x-axis. You can specify a maximum of two attributes.

     In **Attribute 1**, select the first attribute to plot on the x-axis. Optionally, in **Attribute 2**, select the second attribute to plotted on the X axis. The various combinations of Attribute 1 and Attribute 2 values are plotted.
   - In **Sort**, select **Ascending** or **Descending** to sort your X axis data.
3. In the **Legend (Series)** section, specify one or more data sets that must be plotted along the Y axis of the chart view.

   For example, if you define three data sets, then three values are plotted on the y-axis.

   Within each data set, you can perform aggregation operations or group data. Optionally, you can apply a filter to restrict the data that is selected when computing the aggregation on the data set.

   For each data set in this section, specify the following:
   - In **Value**, select the value that must be plotted. For example, Amount or Status.
   - In **Aggregation**, select the aggregation that must be performed on the selected value. The supported aggregation methods are Average, Count, Maximum, Minimum, and Sum. See "Considerations When Using Aggregation" at the end of this topic.
   - (Optional) In **Categories**, select the attribute that must be used to categorize data within the data set. Use Categories to break your graph into more smaller details.
   - (Optional) In **Group By**, select the attribute by which Date type data is grouped.

     If **Group By** is set to a date, a new column **Date Aggregation** is displayed.
   - (Optional) In **Filter**, click the Filter icon to add a filter condition. This condition further refines the data shown for the data set within the chart view. It is applied before the data is included in the chart view. For example, if your **Value** is set to Tasks, you can create a filter to only plot tasks with a status of Open.

     By default, the first clause in a filter condition is displayed as the label for this filter.

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4. Click **OK** to save the chart view settings.

Examples

For Task Manager, see:

- Example: Creating a Chart to Display Task Status by Month
- Example: Creating a Chart to Display Plan versus Actual End Date
- Example: Creating a Chart to Display Task Execution by Organization Unit

Setting the Chart Axis

Chart Axis properties affect the visual display of the chart's axis labels. You can create a title and label for the X axis, and the primary and secondary Y axis of the chart. You can also edit the text and number format of the Y axis, and define a custom range for both Y axes.

> **Note**
>
> This tab doesn't display for chart views with **Type** set to Doughnut, Pie, or Tile.

To define the chart axis:

1. Open the **Axis** tab of the Edit Chart dialog.
2. Select **X Axis** and specify the following (only for chart views with **Type** set to Date in the **Layout** tab):
   - **Interval:** Select **Auto**, **Daily**, **Monthly**, or **Yearly**.
   - **Maximum:** Select **Auto** or **Fixed**. For **Fixed**, select a date.
   - **Minimum:** Select **Auto** or **Fixed**. For **Fixed**, select a date.
3. Click **Y Axis** and specify **Auto** or **Fixed** for the following: **Minimum**, **Maximum**, **Major Unit**, and **Minor Unit.**
4. If **Dual Y Axis** is selected for the chart in the **Properties** tab, click **Dual-Y Axis** and specify the **Minimum**, **Maximum**, **Major Unit**, and **Minor Unit**.
5. Click **OK** to save the chart view settings.

Setting the Chart Legend

Legend properties affect the visual display of the chart's legend.

You can change the position of the legend, as well as display a title for the legend, change the background and border colors, and edit the font and format of chart labels.

For a Combination chart, the **Legend** tab contains one new column named **Type**. You can select the type for each configured legend (series). For example, **Type** = **Bar** for one legend and **Type** = **Line** for the other.

To define the chart legend:

1. Open the **Legend** tab of the Edit Chart dialog.
2. In **Legend Position**, select the position of the data labels in relation to the elements in a chart.

   The available options are **No Legend**, **Left**, **Right**, **Top**, or **Bottom**.

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3. In **Palette**, select the color palette used when displaying the chart view.
4. In the **Series** table, one row is displayed for each legend specified in the **Layout** tab.

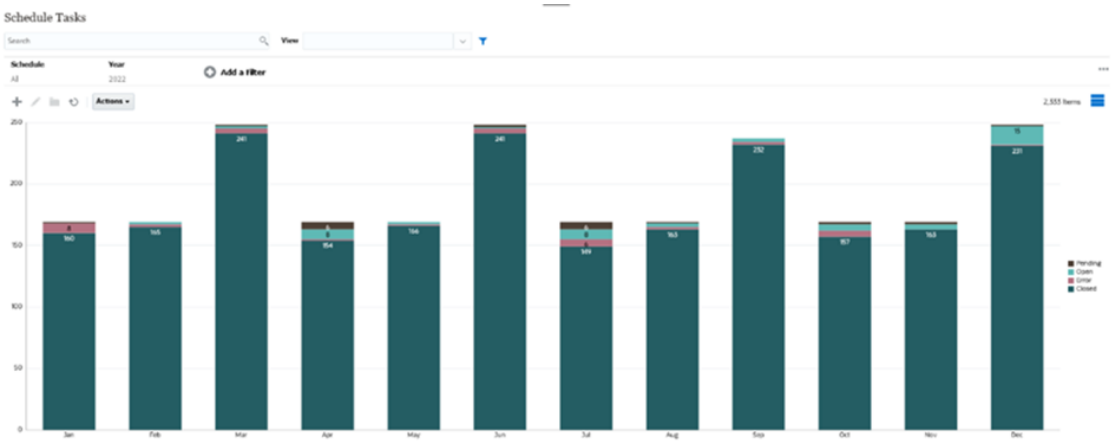
   For each data set:

   - In the **Label**, enter the label that must be displayed for this data series in the chart.
   - Select **Dual-Y** for the data series that must be displayed on the secondary Y axis.
5. Click **OK** to save the chart view settings.

Example: Creating a Chart to Display Task Status by Month

This example creates a chart which summarizes the completion status of all tasks by period. This can help identify if there are certain tasks which are not completed period over period.

The generated chart is displayed below. Click any area within the chart to drill down and view of the individual records that comprise that aggregated area.



To create this chart:

1. From the Home page, click **Tasks**. Click **Schedule Tasks** to display the Task List.
2. From the **View Selector** at the top-right of the page, select **Chart View**.

   The Chart view appears, with a default chart displayed.
3. Click **Edit** at the top-right to display the Edit Chart dialog.
4. In the Properties tab, specify the following:
   - In Type, select **Column**.
   - Deselect **Grid Lines**.
   - In Period, select **Selection**.
5. In the **Layout** tab, specify the following:

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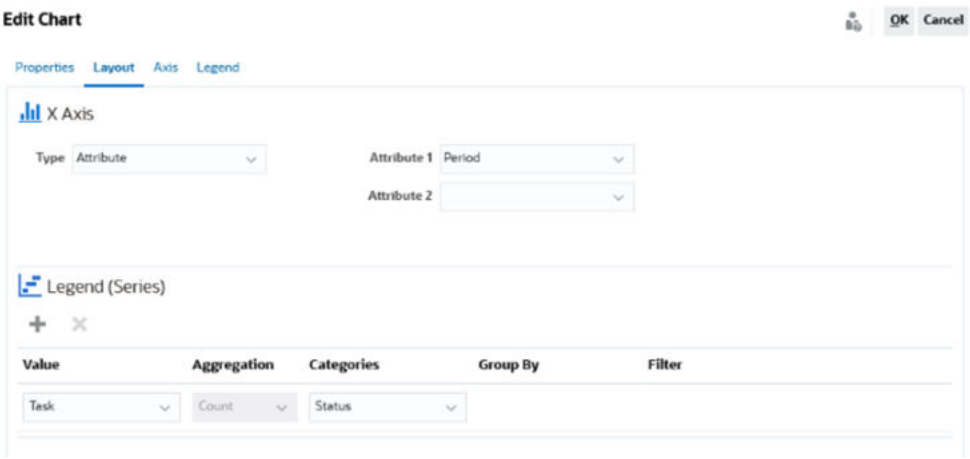


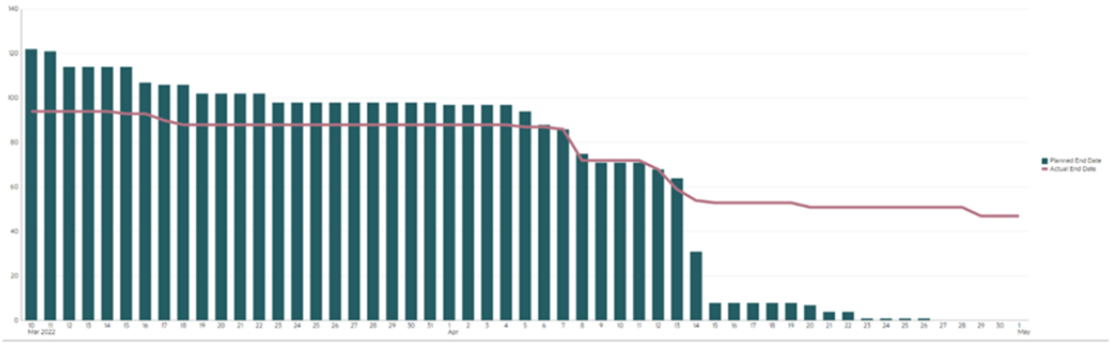
- In the **X-Axis** section:
  - In **Type**, select **Attribute**.
  - In **Attribute1**, select **Period**.
- In the **Legend (Series)** section:
  - In **Value**, select **Task**.
  - In **Categories**, select **Status**.

6. In the **Axis** tab, leave the default settings as is.
7. In the **Legend** tab, specify the following:
   - In **Legend Position**, select **Right**.
   - In **Palette**, select **Redwood**.
8. Click **OK**.

Example: Creating a Chart to Display Plan versus Actual End Date

This example visualizes Tasks planned vs actual completion over time - allowing you to see if your schedules are running ahead or behind schedule. The chart shows all of the tasks end dates that fall on or after a certain date. This will show a large number of tasks within the beginning of the schedule and gradually go down until the final task in the schedule.

The generated chart is displayed below. Click any area within the chart to drill down and view of the individual records that comprise that aggregated area.



To create this chart:

1. From the Home page, click **Tasks**, and then click **Schedule Tasks** to display the Task List.

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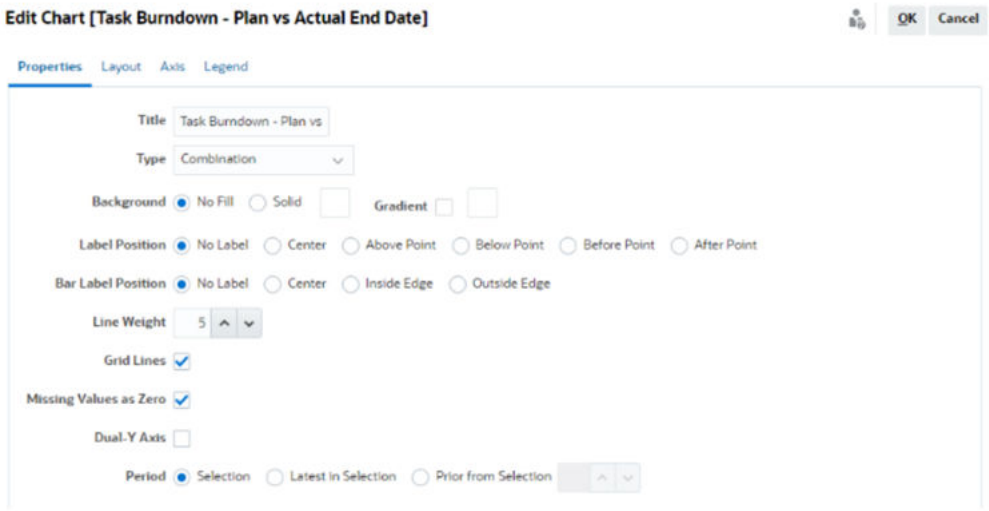
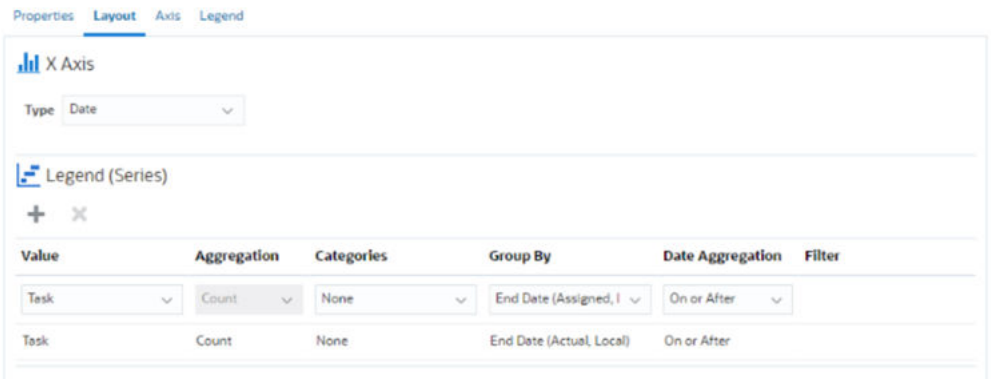
2. From the **View Selector** at the top-right of the page, select **Chart View**.
   The Chart view appears with a default chart displayed.
3. Click the **Edit** icon at the top-right to display the Edit Chart dialog.
4. In the **Properties** tab, specify the following:



   - In **Type**, select **Combination**.
   - In **Background**, select **No Fill.**
   - In **Label Position**, select **No Label**.
   - In **Bar Label Position**, select **No Label**.
   - Select **Line Weight** as **5**.
   - Select **Grid Lines**.
   - Deselect **Dual Y Axis**.
   - In **Period**, select **Selection**.
5. In the **Layout** tab, specify the following:



   - In the **X-Axis** section, for **Type**, select **Date**.
   - In the **Legend (Series)** section, enter 2 values:
     For the first entry:
     - In **Value**, select **Task**.
     - In **Categories**, select **None**.

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- In **Group By**, select **End Date (Assigned)**
- In **Date Aggregation**, select **On or After**.

For the second entry:

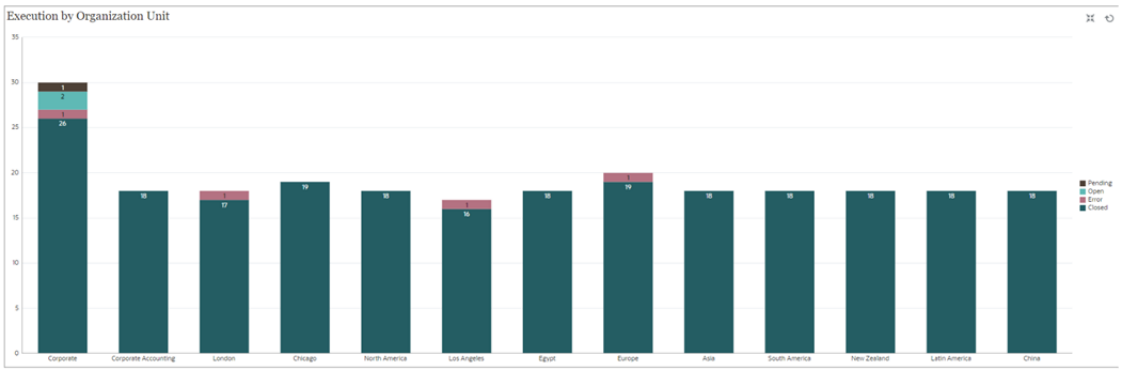
- In **Value**, select **Task**.
- In **Categories**, select **None**.
- In **Group By**, select **End Date (Actual, Local)**
- In **Date Aggregation**, select **On or After**.

6. In the **Axis** tab, leave the default settings as is.
7. In the **Legend** tab, specify the following:
   - In **Legend Position**, select **Right**.
   - In **Palette**, select **Redwood**.
8. Click **OK**.

Example: Creating a Chart to Display Task Execution by Organization Unit

This example creates a chart which shows task completion by organization unit allows you to monitor how each organization is doing completing their tasks.

The generated chart is displayed as shown. Click any area within the chart to drill down and view of the individual records that comprise that aggregated area.



To create this chart:

1. From the Home page, click **Tasks.** Click **Schedule Tasks** to display the task list.
2. From the **View Selector** at the top-right of the page, select **Chart View**.

   The Chart view appears, with a default chart displayed.
3. Click **Edit** at the top-right to display the Edit Chart dialog.
4. In the **Properties** tab, specify the following:
   - In **Type**, select **Column**.
   - Deselect **Grid Lines**.
   - In **Period**, select **Selection**.
5. In the **Layout** tab, specify the following:

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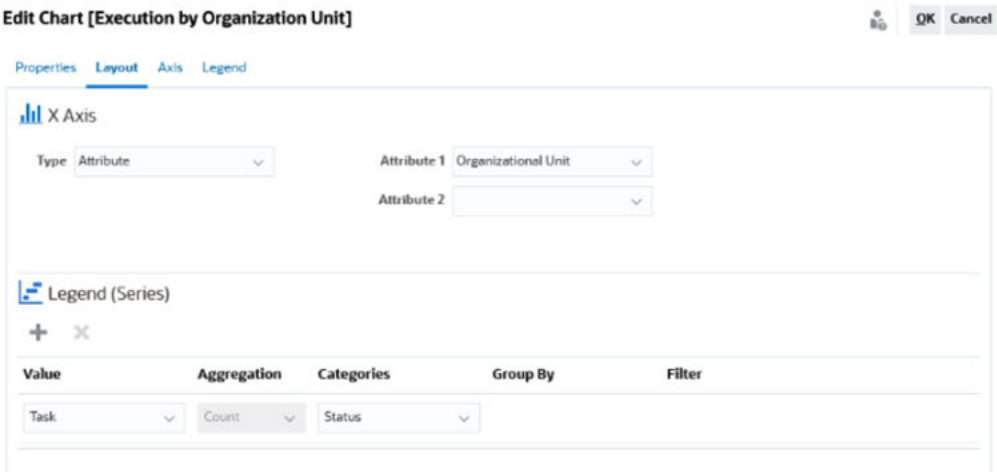

- In the **X-Axis** section:
  - In **Type**, select **Attribute**.
  - In **Attribute1**, select **Organizational Unit**.
- In the **Legend (Series)** section:
  - In **Value**, select **Task**.
  - In **Categories**, select **Status**.

6. In the **Axis** tab, leave the default settings as is.
7. In the **Legend** tab, specify the following:
   - In **Legend Position**, select **Right**.
   - In **Palette**, select **Redwood**.
8. Click **OK**.

Working With Gantt Chart Views

Gantt charts displays a timeline for a schedule that graphically represents the duration of tasks and the dependencies between them. You can view the predecessor relationships between tasks. Bars represent duration, and arrows represent dependencies. You can also see the parent-child task relationships in the hierarchy.

You might choose to use the Gantt Chart view when you create a template to see the predecessor relationships between the tasks, or to verify the precedence order after adding tasks to a schedule.

Gantt Chart view is available for **Scheduled Tasks**.

To view the data in a Gantt Chart:

1. On the Home page, click **Tasks**, and then click **Schedule Tasks**.
2. Click the **View Selector** icon and choose **Gantt Chart**.
   The schedule tasks data is displayed in the Gantt Chart view as shown.

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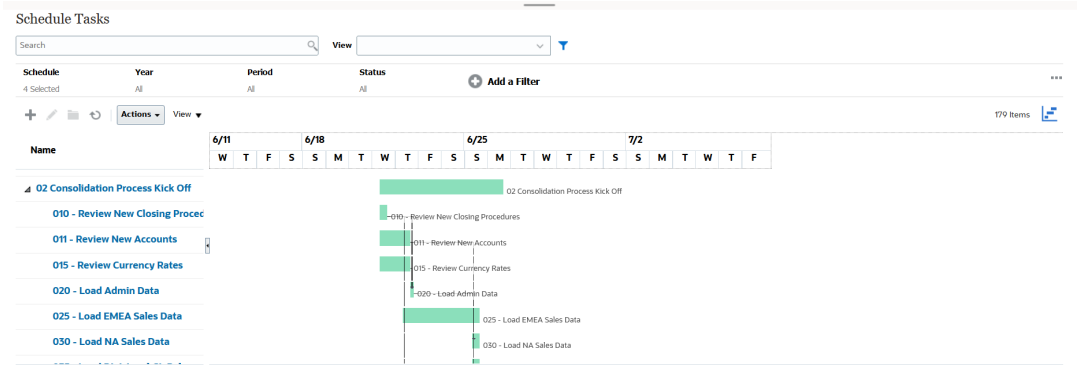


Working With Calendar Views

Calendar views display all the scheduled tasks in a traditional calendar format based on month, week, or day.

The Month view displays the days in standard calendar format.

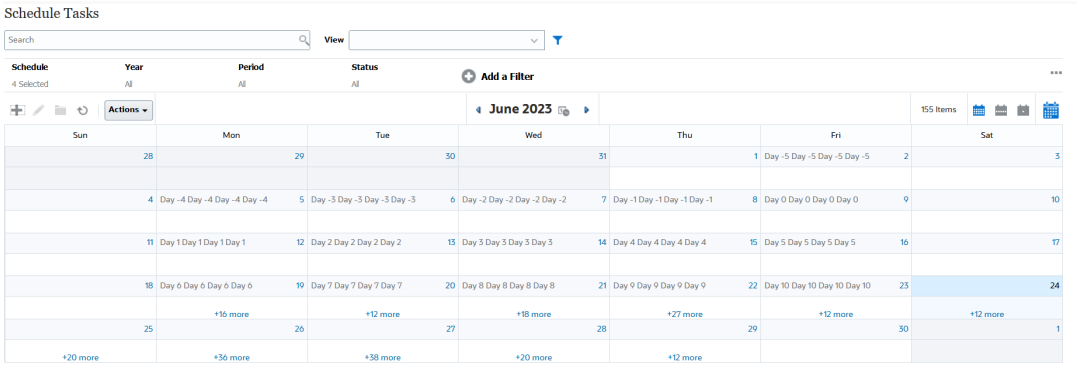
The Week view displays the tasks of the specified week. The top section displays tasks that span multiple days. The second section displays tasks that are contained within one day. The tasks are displayed vertically, spanning the hours of the day.

The Day view is focused on one 24-hour day.

To view the scheduled tasks in Calendar View:

1. On the **Home** page, click **Tasks**, and then click **Schedule Tasks**.
2. Click the **View Selector** icon and choose **Calendar View**.
   The schedule tasks data is displayed in the Calendar view. Use the calendar icons to select Month, Week, or Day. In the Calendar View, you can also see the Day Label in the Day header.

   The following image shows the scheduled tasks in a Calendar View for a month.



Saving Views

Saving a view enables you to save your view definition and settings. When you need to display data in the format set by this view, you can use the saved view, instead of again setting the required view properties.

The scope of a saved view is limited to the object on which it is based. For example, if you create a saved view with Tasks, you can't use it with journals. A saved view on a particular supplemental data set can't be used with a different data set. So, you can only use it with certain forms or within a collection.

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To save a view:

1. Adjust the view to see the data you would like by adjusting filters, columns or sorting on your list. See Using the Filter Bar to Adjust Your View.
2. From **Actions**, click **Save View**.
3. In the Save View dialog, enter a Name (can enter a maximum of 80 characters) and an optional description (can enter a maximum of 255 characters).
4. Click **OK**.

You can see all the saved lists by clicking the **View** drop-down selector.

Editing Saved Views

Edit a saved list view, pivot view, or chart view to modify its name and definition.

To edit a view in Task Manager :

1. From the Home page, click **Application**, **Task Manager**, and then **Views**.
   A list of available views is displayed.
2. Select a particular view and click the **Edit** icon. You can also click the **...** icon for a view and select **Edit**.
3. Update the required information and click **OK**.

Deleting Views

An Administrator or Power User can delete a view.

To delete a view in Task Manager:

1. From the Home page, click **Application**, **Task Manager**, and then **Views**.

   A list of available views is displayed.
2. Select a view that you want to delete and click the **Delete** icon. You can also click the **...** icon for a view and select **Delete**.

   The deleted view no longer appears in the list of views.

Publishing Views

An Administrator or Power User can publish a view to make a private view available to other users. The saved view is duplicated and there is now a private and public version.

To publish a view in Task Manager:

1. From the Home page, click **Application**, **Task Manager**, and then **Views**.

   A list of available views is displayed.
2. Select a particular view and click the **Publish** icon. You can also click the **...** icon for a view and select **Publish**.

   The saved view is duplicated and there is now a private and public version. The public version will show a check mark in the **Public** column.

Setting a Default View

A Service Administrator or Power User can set a published, saved, view as the default view.

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The first time a user opens a view, the default view is displayed. The system then remembers the view settings that were used. So, when you open a view subsequently, it is displayed with the settings that were used most recently.

To set a default view in Task Manager:

1. From the Home page, click **Application**, then **Task Manager**.
2. Click the **Views** tab to display the Views page.
3. Select the view that you want to set as default and then click the **Set Default** icon. Or, click the **Actions** menu for the view that you want to set as default and select **Set Default**.

> **Note**
>
> You can use the **Set Default** action for a published list view. The Administrator or Power User sets the view as the default so that the saved list view will not only be available to all users of that view type, but it will also be selected by default when a user opens that view type.

Exporting List Views and Pivot Views

You can export the records in list views or pivot views to Microsoft Excel.

To export a list view or pivot view:

1. Navigate to the list view or pivot view that must be exported.
2. Click **Actions**, and then **Export to Excel**.

The exported data is stored in an Excel file and can be downloaded. The format of the Excel file is .xls or .xlsx, depending on the **Export to Excel** setting.

Managing Operational Dashboards

Related Topics

- Overview of Operational Dashboards
- Viewing and Opening Dashboards
- Operational Dashboards: Design Mode
- Creating Operational Dashboards
- Setting the Object Type Within Operational Dashboards
- Specifying View Options for Operational Dashboards
- Editing Operational Dashboards
- Duplicating Operational Dashboards
- Deleting Operational Dashboards
- Exporting Operational Dashboards
- Importing Operational Dashboards
- Running Operational Dashboards
- Displaying Columns for Viewing Operational Dashboards

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- Reordering Columns for Viewing Operational Dashboards
- Copying a URL for Operational Dashboards
- Applying Filters to Data Displayed Within Operational Dashboards
- Example: Task Manager Operational Dashboard

Overview of Operational Dashboards

Dashboards provide enhanced reporting capabilities. They enable users to chart, evaluate, and (in some cases) even change key business data.

About Operational Dashboards

Operational dashboards provide greater visibility into your business process.

Each dashboard can have a maximum of four views. This includes list views, pivot views, or chart views. You can open a maximum of 10 dashboards at a time. Service Administrators, or any user who is granted either the Operational Dashboards - Manage role or Dashboard - Manage role can create and edit dashboards. The dashboard creator can also control access to the dashboard by granting permissions to one or more users, groups, or teams. Users who have been granted access to a dashboard can run the dashboard and view its data. The access only allows a user to view the dashboard. It doesn't extend their access to the underlying data. Therefore, a user can only view records based on their own access within the service.

Features of Operational Dashboards

Dashboards provide the following features:

- Visualize data interactively with a wide variety of chart types
- Obtain both high-level and detailed insights into the performance of your business processes
- Create filters at the dashboard level and at the individual view level
  Filters created at the dashboard level are applied to all the views within the dashboard. After this, filters defined at the view level (if any) are applied.
- Control access to viewing dashboards

Viewing and Opening Dashboards

Viewing Existing Dashboards

To view the dashboards, from **Home**, click **Tasks**, then **Dashboards**.

The Operational Dashboards page appears with the list of existing dashboards.

Opening a Dashboard

To open a dashboard:

1. Access the Operational Dashboards page. This page displays the list of existing Operational dashboards
2. (Optional) Use the **Search** box to search for a particular dashboard.
3. Click the name of the dashboard to open.

   The dashboard opens within a new dynamic tab next to the Operational Dashboard tab.

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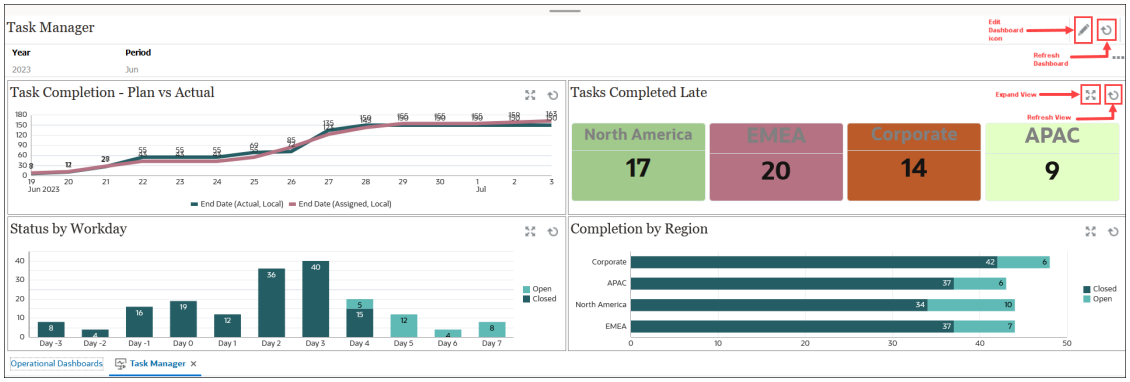
You can perform the following actions on the dashboard:

- Click any area or total, in any view, to drill down into the individual records that comprise that area or total.
- Edit the dashboard settings using the icon.
- Refresh all the views in the dashboard using the icon (at the dashboard level).
- Use filters that were created at the dashboard level. See Applying Filters to Data Displayed Within Operational Dashboards.

On each view in the dashboard, you can:

- Refresh all the views in the dashboard using the icon (at the dashboard level).
- Expand or collapse views in dashboard views using the icon.
- Use filters that were defined at the view level. See Applying Filters to Data Displayed Within Operational Dashboards.

The following image shows the options available when you open a dashboard.



Operational Dashboards: Design Mode

The design mode of a dashboard enables you to configure and manage your dashboard and its components (chart views, list views, and pivot views).

This image shows the design mode of an operational dashboard. This dashboard contains one chart view and one pivot view.

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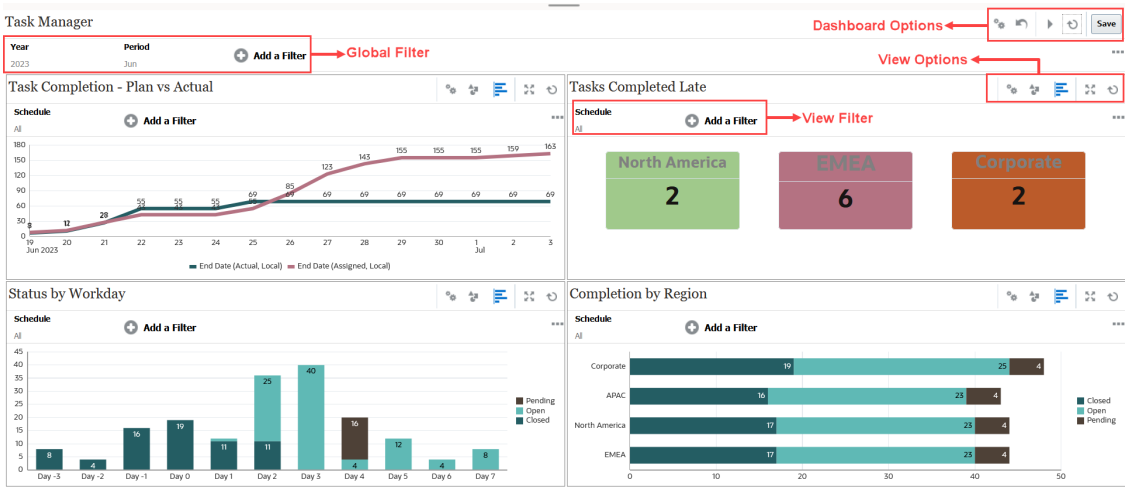

At the top-right are these options to manage the dashboard:

- **Settings**
  Edit the properties of the dashboard.
- **Revert**
  Revert changes made to the dashboard properties.
- **Run**
  Run the dashboard.

  Dynamically fetches and displays data in the dashboard. Only users who have been granted permissions can view the dashboard's data.

  In Run mode, users will see only the **Refresh** and **Expand** options.
- **Refresh**
  Refreshes the data displayed in the dashboard, including all the contained views.
- Save **Save**
  Saves the changes made to the dashboard properties.

Each view (list, chart, or pivot) within the dashboard contains these options:

- **Settings**
  Edit the properties of the chart, list, or pivot view.
- **Object Type**
  Options are: Tasks and Alerts
- **View Selector**
  Options are: Chart View, List View, or Pivot View.
- **Expand**
  Displays the view in the entire dashboard area.

  This option is shown only when a dashboard contains more than one view.

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- **Refresh**
  Refreshes the data displayed in the view.

Creating Operational Dashboards

Service Administrators and users who have been granted the Operational Dashboards - Manage granular role can create dashboards.

To create a dashboard:

1. From **Home**, click **Tasks**, then **Dashboards**.

   The Operational Dashboards page appears, with the list of existing dashboards.

2. Click the **New** icon.

   The new dashboard is opened in a new tab. It contains the following:

   - A default name, in the format New Dashboard #.
   - One empty view (list view, pivot view, or chart view)
   - Two sets of icons (at the top-right) that correspond to the options for the dashboard and the view within the dashboard. See Operational Dashboards: Design Mode.
   - A filter for filtering data at the dashboard level. Click **Add a Filter** to add filters.

3. In the set of icons that correspond to the dashboard, click the **Settings** icon.

   The Edit Dashboard dialog appears.

4. In the **Properties** tab, specify the required dashboard properties.

   See Defining Operational Dashboard Properties.

5. Click the **Access** tab to give access to this dashboard for users, groups, or teams. Users who are granted access can run this dashboard and view its data.

   See Defining Access to Operational Dashboards.

6. Click **OK**.

   A separate area, with its corresponding icons, is shown for each area specified by the dashboard layout. For example, if you set **Layout** to **Grid**, the page shows four separate areas.

7. For each area within the dashboard, set the object type based on which you want to visualize the data. See Setting the Object Type Within Operational Dashboards.

8. For each area within the dashboard, click the select the **View Selector** icon and specify the type of view. See Specifying View Options for Operational Dashboards.

9. For each view, click **Settings** and specify the settings for that view.

   - See Creating Chart Views.
   - See Working With List Views.
   - See Working with Pivot Views.

10. Click **Save** to save the dashboard settings.

Defining Operational Dashboard Properties

Dashboard properties include name, description, borders, background, and layout.

To define a dashboard's properties:

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1. Open the **Properties** tab of the Edit Dashboard dialog.
2. In **Name**, enter a unique name for the dashboard.
3. In **Description**, enter an optional description for the dashboard.
4. Select **Borders** to display borders around each view in the dashboard.
5. Select **Global Filter Bar** to include a global filter bar in the dashboard. This filter bar is applicable to all the views within the dashboard.
6. Select **Show Filter Bars** to display the view-level filters when you run the dashboard.
7. In **Background**, select one of the following:
   - No Fill
   - Solid: Select the color that must be used as the background color. Optionally, select **Gradient**, and then select a color.
8. In **Layout**, specify the number of regions (one for each view) into which the dashboard is divided. The maximum number of regions is four.

   Select one of the following:
   - Single
     Displays a single view
   - Horizontal
     Displays two views, one below the other

     Use **Horizontal Split** to specify how the screen area is split between the two views.
   - Vertical
     Displays two views, one beside the other

     Use **Vertical Split** to specify how the screen area is split between the two views. For example, if you set **Vertical Split** to 30, the chart displayed at the top of the dashboard uses 30% of the dashboard space. The second chart uses the remaining 70% of the dashboard.
   - Grid
     Displays four views in the dashboard
9. Click **OK** to save the dashboard settings.

Defining Access to Operational Dashboards

Service Administrators and users who have been granted the Operational Dashboards - Manage role can grant access to this dashboard for specific users, groups, or teams. Such users can run the dashboard and view its data.

To add access for running a dashboard:

1. Open the **Access** tab of the Edit Dashboard dialog.

   By default, there is one entry for the user who created the dashboard.
2. Click the **Add** icon.

   The **Select User** dialog is displayed.
3. Click the **Select User** icon, and select one of the following: **Users**, **Groups**, or **Teams**.

   The default selection is **Users**.
4. Enter the name or description and click **Search**.

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- You can click **Search** to display the entire list of available users, groups, or teams. For example, if you don't know the name of a group, select Group using the **Select User** icon and then click **Search**. The list of available groups is displayed.
- When selecting users, click **Advanced** to display the **User ID** and **Email** fields. You can also use these attributes while searching for users.

5. Double-click the username, group name, or team name to provide access.

   The selected name appears on the **Access** tab. You can click this name to view the details.

6. Click **OK** to save the dashboard settings.

> **Note**
>
> Service Administrators and users who have been granted the Task Manager - Manage Dashboards granular role can view and edit all dashboards.

To remove access for running a dashboard:

1. Open the **Access** tab of the Edit Dashboard dialog for the dashboard whose access is being set.

   The users, groups, and teams who have been granted permissions to run the dashboard are listed under **Name**.

2. Click the name of the user, group, or team for whom permissions to run the dashboard must be revoked.

   To select multiple items, press and hold the **Ctrl** key and then select the required items.

3. Click **Remove** and then **OK**.

Setting the Object Type Within Operational Dashboards

After setting the properties and access for the dashboard, you can select the data type to report on or visualize for each view.

All the data, metadata, filters, and so on in the view will be based on this object type. Each view can have only one object type, but it may vary between views.

In FreeForm, these object types are available:

To specify the object type:

1. Open an operational dashboard from the Operational Dashboard page.
2. Click the **Object** icon and select the required object.
3. Click **Save** to save the settings.

If you change the object type, all the settings in the region will be reset.

Specifying View Options for Operational Dashboards

Within a dashboard, after specifying the object type, you can select the view such as List View, Pivot View, or Chart View to display the data within the dashboard.

To specify the view option:

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1. On the Operational Dashboards page, open the dashboard for which you want to specify the view options.
2. Click the **View Selector** icon and choose any of the following:
   - List View
   - Pivot View
   - Chart View
3. Click the **Settings** icon corresponding to the **View Selector** icon and specify the view options.
   - For List Views, see Working With List Views .
   - For Pivot Views, see Working with Pivot Views.
   - For Chart Views, see Creating Chart Views.

   The data is displayed within the dashboard based on the specified view settings.
4. Click **Save**.

Editing Operational Dashboards

Service Administrators can edit a dashboard and modify its properties and access permissions.

To edit a dashboard:

1. From **Home**, click **Tasks**, then **Dashboards**.
2. Click the dashboard that you want to edit and then click the **Edit** icon or you can click the **Actions** icon and then click **Edit**.
   The Dashboard Designer appears.

   Additionally, if you're viewing the dashboard in run mode, you can click the **Edit** icon to switch to design mode.
3. Click **Settings** to display the Edit Dashboard dialog.
4. Modify the required properties using the **Properties** and **Access** tabs.
   See Defining Operational Dashboard Properties and Defining Access to Operational Dashboards.
5. Click **OK**.

Duplicating Operational Dashboards

To duplicate a dashboard:

1. From **Home**, click **Tasks**, then **Dashboards**.
2. Select the dashboard that you want to duplicate and then click the **Duplicate** icon. You can also click the **Actions** icon and then click **Duplicate**.
   The Copy of <Dashboard name> tab opens in the Dashboard Designer.
3. Specify the properties, access, and view options.
4. Click **Save** to save the dashboard.

Deleting Operational Dashboards

To delete a dashboard:

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1. From **Home**, click **Tasks**, then **Dashboards**.
2. Click the dashboard that you want to delete and then click the **Delete** icon. You can also click the **Actions** icon and then click **Delete**.

   A message is displayed asking for confirmation.
3. Click **Yes**.

   The dashboard is deleted and no longer appears on the Operational Dashboard page.

Exporting Operational Dashboards

You can export dashboard definitions to an XML file. Using the export functionality, you can copy a dashboard definition from one business process to another.

> **Note**
>
> You must have the Manage Dashboard privilege to export the operational dashboards.

To export a dashboard:

1. From **Home**, click **Tasks**, then **Dashboards**.

   The Operational Dashboards page is displayed.
2. Select one or more dashboards that you want to export and then click the **Export** icon. The Export Dashboards dialog is displayed.
3. Select one of the following:
   - **All Rows**: Export all existing operational dashboards
   - **Selected Rows**: Export the dashboards that you had selected in the previous step.
4. Click **Export**.

   The specified dashboards are exported as a JSON file, with the name *Dashboard.json*.

Importing Operational Dashboards

Service Administrators can import dashboards that were previously exported using the **Export** option

> **Note**
>
> You must have the Manage Dashboard privilege to import the operational dashboards.

To import a dashboard:

1. From **Home**, click **Tasks**, then **Dashboards**.

   The Operational Dashboards page is displayed.
2. Click **Import**.
   The Import Operational Dashboards dialog is displayed.
3. In **File**, click **Browse** and specify the JSON file that contains the dashboard that you want to import..

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4. Click **Import**.
   The **Status** field displays the status of the import.
5. If the import completed successfully, click **OK**.

The imported dashboard is displayed in the Operational Dashboards page.

If the import file contains dashboards that have the same name as those in the Dashboards page, then these dashboards are updated.

Running Operational Dashboards

Users can run dashboards for which they have been granted access.

When a dashboard is run, the latest data is displayed in the various charts and pivot views in the dashboard.

To run a dashboard:

1. From **Home**, click **Tasks**, then **Dashboards**.

   The Operational Dashboards page is displayed.
2. Do one of the following:
   - Click the dashboard you want to run and then click the **Run** icon.
   - Click the **Actions** icon for the dashboard you want to run and select **Run**.
   - Open a dashboard and click the **Run** icon.

     The dashboard is displayed in runtime mode in a new tab with the latest data.
3. If required, filter the displayed data. See Applying Filters to Data Displayed Within Operational Dashboards.

Displaying Columns for Viewing Operational Dashboards

To display the columns to view the dashboards:

1. On the Operational Dashboard page, click **View** and then **Columns**.
2. Click **Show All** to display all the columns or select specific columns.

To show or hide displayed columns:

1. Click **View** > **Columns** > **Manage Columns.**
2. Move the required columns between **Hidden Columns** and **Visible Columns** using the move icons to show or hide the displayed columns.

Reordering Columns for Viewing Operational Dashboards

You can reorder columns to display them in a different order.

To reorder columns:

1. On the Operational Dashboard page, click **View** and then click **Reorder Columns**.
2. Select the column name that you want to reorder and use the following icons to move it to the required position:
   - : Move selected items to top of the list

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- : Move selected items up one in the list
- : Move selected items down one in the list
- : Move selected items to bottom of the list

3. Click **OK**.
   The columns are displayed based on the specified order.

Copying a URL for Operational Dashboards

You can copy a URL for an operation dashboard to enable access to the dashboard.

To copy a URL for an operational dashboard:

1. From **Home**, click **Tasks**, then **Dashboards**.
   The Operational Dashboards page is displayed.
2. Click the **Actions** icon for the dashboard to copy the URL, and then click **Copy URL**.
3. Copy the displayed URL. The URL will enable direct access to the dashboard.

Applying Filters to Data Displayed Within Operational Dashboards

You can use filters to restrict the data displayed in a dashboard and its views.

Use the **Show/Hide** toggle icon next to the **Search** box to display or hide the filter bar.

In the Filter bar, you can do the following to filter the displayed dashboards:

- Add additional filters
- Specify the condition for each filter
- Clear or Remove a filter
- Clear All Filters

To restrict data displayed in a dashboard:

1. Run the dashboard and display its data.
2. If a global filter was configured for this dashboard, use this to filter data displayed in all the views in the dashboard.
3. For any view, use the filter defined on a view to filter data displayed in that view.

The global filter is first applied to all the views in the dashboard. Then, for each view, the filter defined for that view is applied.

For more information about each of these features, see Working with Filters.

Example: Task Manager Operational Dashboard

This example shows how to create a dashboard.

The dashboard uses these views for Task Manager:

- Chart View: Task Completion - Plan vs Actual
- Chart View: Tasks Completed Late

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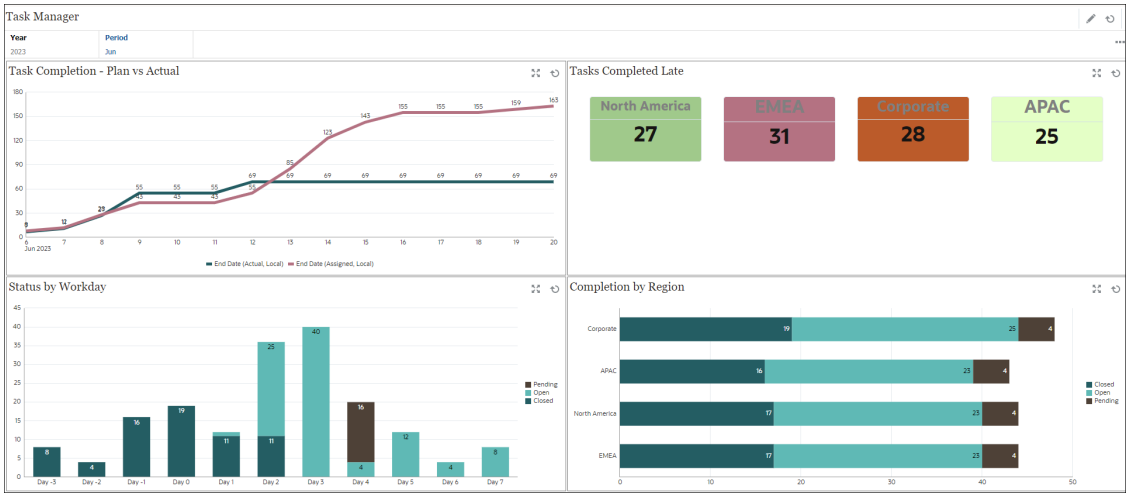
- Chart View: Status by Workday
- Chart View: Completion by Region

The completed dashboard is shown in the screenshot. You can click any area in any view to drill down into the individual records that comprise that area.



In this example, to create the dashboard:

1. From **Home**, select **Tasks**, and then click **Operational Dashboard**.

   The existing operational dashboards are displayed.

2. Click the **New** icon to create a new dashboard.

   A new dashboard is created with a default name and is opened in a new tab.

3. Click the **Settings** icon for the dashboard and update the properties on the **Properties** tab:
   - **Name**: Task Manager
   - **Borders**: Select this checkbox
   - **Global Filter Bar:** Select this checkbox
   - **Background**: No fill
   - **Layout**:
     - **Grid**: Select this check box.
     - **Horizontal Split:** 50
     - **Vertical Split**: 50
4. To create the chart view on the top left, see Chart View: Task Completion - Plan vs Actual.
5. To create the chart view on the top right, see Chart View: Tasks Completed Late.
6. To create the chart view on the bottom left, see Chart View: Status by Workday.
7. To create the chart view on the bottom right, see Chart View: Completion by Region.
8. Click **Save** to save the dashboard.

Chart View: Task Completion - Plan vs Actual

This example shows task completion over time, allowing you to see if your tasks are running ahead or behind schedule.

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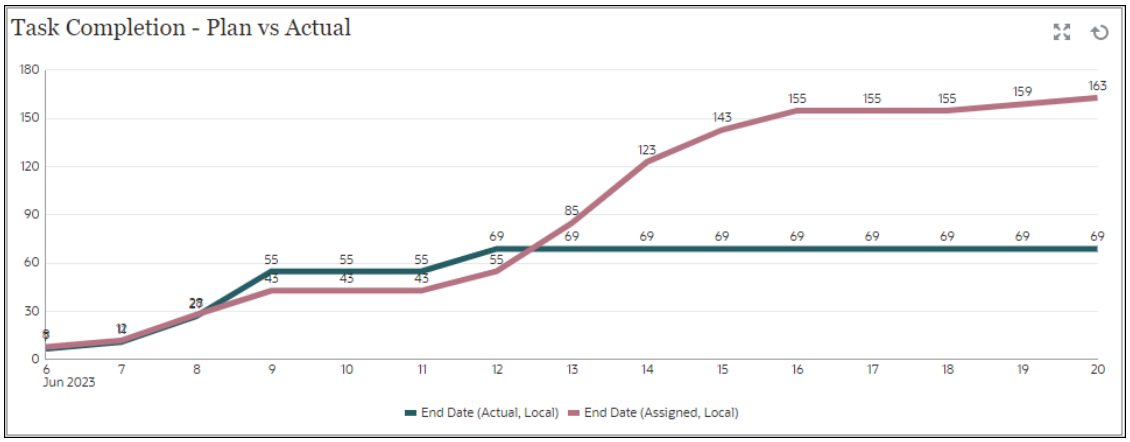
The chart includes all of the task end dates that fall on or before a certain date. It also shows the status of completing those tasks.



Within the dashboard, top left view:

1. Click the **Object Type** icon and select **Tasks**.
2. Click the **View Selector** icon and select **Chart View**.
3. Click the **Settings** icon and specify the following:
   - On the **Properties** tab:
     - **Title**: Task Completion - Plan vs Actual
     - **Type**: Line
     - **Background**: No fill
     - **Label Position**: Above Point
     - **Line Weight**: 5
     - **Grid Lines**: Select this check box
     - **Period**: Selection
   - On the **Layout** tab:
     - **X-Axis**:
       - **Type**: Date
       - **Sort**: X-Axis from the first drop-down menu and Descending from the second drop-down menu
     - **Legend (Series)**:
       - First row, add:
         - **Value**: Task
         - **Aggregation**: Count
         - **Categories**: None
         - **Group By**: End Date (Actual , Local)
         - **Date Aggregation**: On or Before
       - Click **New** and add another row. Specify the following:
         - **Value**: Task
         - **Aggregation**: Count

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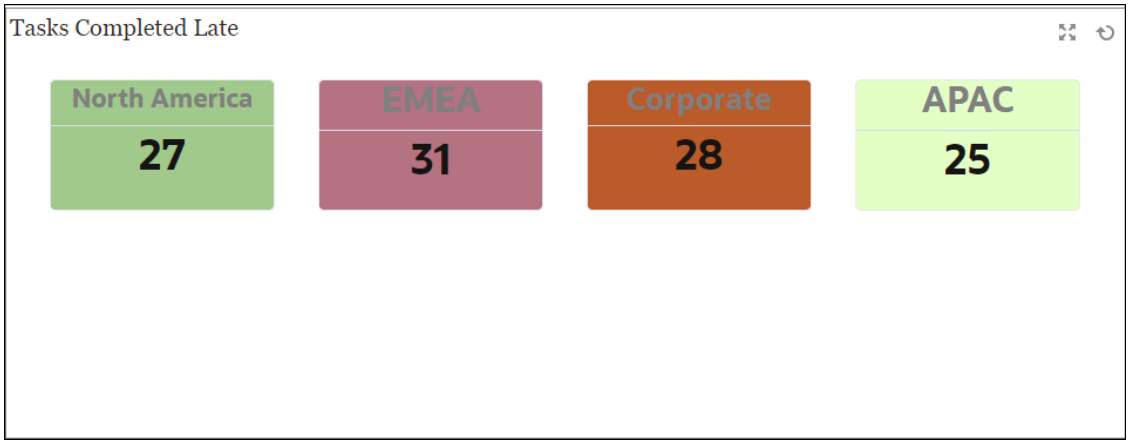
        - **Categories**: None
        - **Group By**: End Date (Assigned, Local)
        - **Date Aggregation**: On or Before
    - On the **Axis** tab:
        - Select **X Axis**.
        - **Interval** : Daily
        - **Minimum**: Auto
        - **Maximum**: Auto
    - On the **Legend** tab:
        - **Legend Position**: Bottom
        - **Palette**: Redwood
        - Select the default labels for each series: **End Date (Actual , Local)** and **End Date (Assigned, Local)**
        - Select the default colors for each series.
4. Click **OK**.

Chart View: Tasks Completed Late

This example shows the tasks completed behind schedule in various organizational units.



Within the dashboard, top left view:

1. Click the **Object Type** icon and select **Tasks**.
2. Click the **View Selector** icon and select **Chart View**.
3. Click the **Settings** icon and specify the following:
    - On the **Properties** tab:
        - **Title**: Tasks Completed Late
        - **Type**: Bar
        - **Background**: No fill
        - **Label Position**: Inside Edge
        - **Orientation**: Vertical

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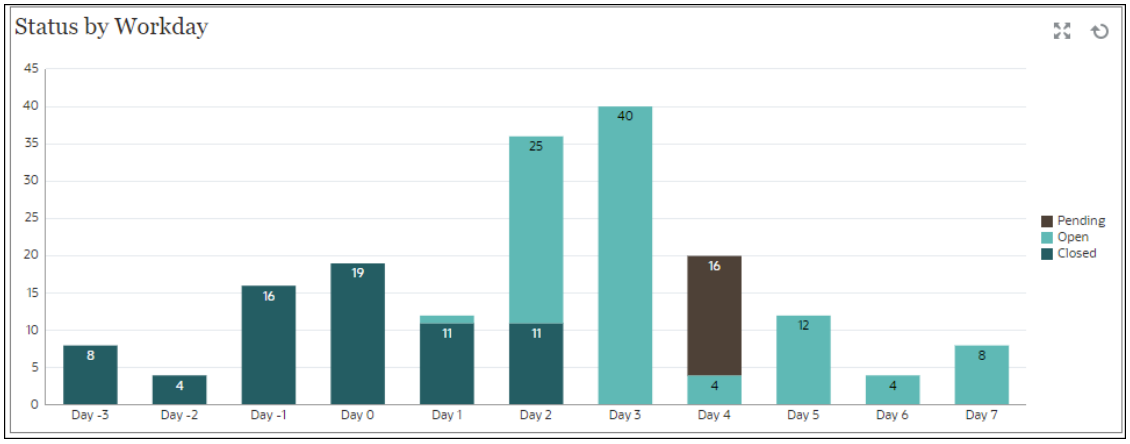
  - **Period**: Selection
- On the **Layout** tab:
  - **X-Axis**:
    - **Type**: Date
    - **Sort**: X-Axis from the first drop-down menu and **Ascending** from the second drop-down menu
  - **Legend (Series)**:
    - **Value**: Task
    - **Aggregation**: Count
    - **Categories**: Organizational Unit
    - **Filter**: Ever Been Late (Assignee) does not equal No
- On the **Axis** tab, select the default options for X axis.
- On the **Legend** tab, select the default options.

4. Click **OK**.

Chart View: Status by Workday

This example chart shows task completion based on a custom Workday attribute.

In this example, the custom attribute was added previously and defined with values for days -15 through 35. All tasks were then assigned a work day based on their planned completion.



Within the dashboard, bottom left view:

1. Click the **Object Type** icon and select **Tasks**.
2. Click the **View Selector** icon and select **Chart View**.
3. Click the **Settings** icon and specify the following:
   - On the **Properties** tab:
     - **Title**: Status by Workday
     - **Type**: Column
     - **Background**: No fill
     - **Label Position**: Inside Edge

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- **Orientation**: Vertical
- **Grid Lines**: Select this check box
- **Period**: Selection

- On the **Layout** tab:
  - **X-Axis**:
    - **Type**: Attribute
    - **Attribute 1**: Workday (This custom attribute must have been added previously for it to display in the list.)
    - **Sort**: Workday from the first drop-down menu and Ascending from the second drop-down menu.
  - **Legend (Series)**: In first row, add:
    - **Value**: Task
    - **Aggregation**: Count
    - **Categories**: Status
  - **Y Axis**: Select this checkbox.
  - **Minimum**: Auto
  - **Maximum**: Auto
  - **Major Unit**: Auto
  - **Minor Unit**: Auto
- On the **Legend** tab:
  - **Legend Position**: Right
  - **Palette**: Redwood
  - Select the default labels for each series: **Closed**, **Error**, **Open**, and **Pending**.
  - Select the default colors for each series.

4. Click **OK**.

Chart View: Completion by Region

This example chart shows task completion based on region. In this example, the custom Organizational Unit attribute was added previously and defined with values for tasks completed based on the region.

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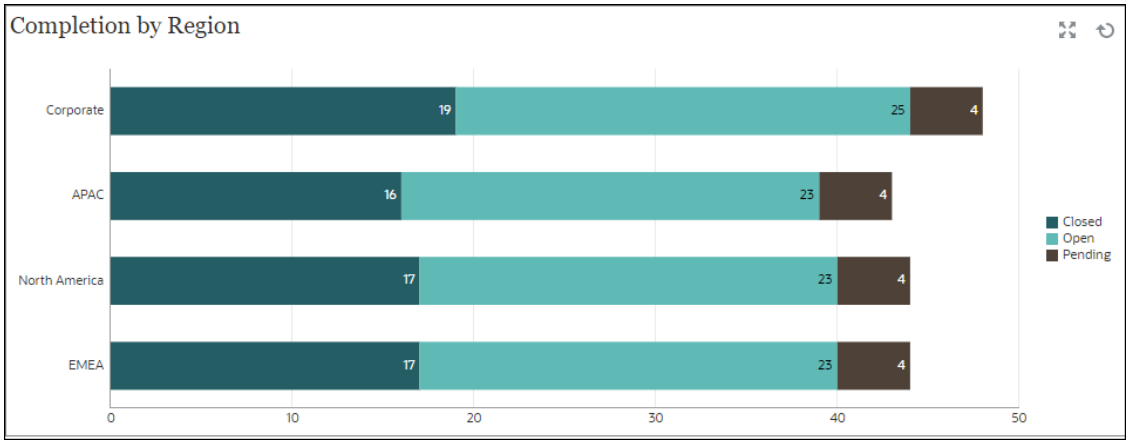

Within the dashboard, bottom left view:

1. Click the **Object Type** icon and select **Tasks**.
2. Click the **View Selector** icon and select **Chart View**.
3. Click the **Settings** icon and specify the following:
   - On the **Properties** tab:
     - **Title**: Completion by Region
     - **Type**: Column
     - **Background**: No fill
     - **Label Position**: Inside Edge
     - **Orientation**: Horizontal
     - **Grid Lines**: Select this check box
     - **Period**: Selection
   - On the **Layout** tab:
     - **X-Axis**:
       - **Type**: Attribute
       - **Attribute 1**: Organizational Unit

         (This custom attribute must have been added previously for it to display in the list.)
       - **Sort**: Organizational Unit from the first drop-down menu and Descending from the second drop-down menu.
     - **Legend (Series)**:
       - **Value**: Task
       - **Aggregation**: Count
       - **Categories**: Status
   - On the **Axis** tab:
     - **Y Axis**: Select this checkbox.
     - **Minimum**: Auto
     - **Maximum**: Auto

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- **Major Unit**: Auto
- **Minor Unit**: Auto

- On the **Legend** tab:
  - **Legend Position**: Right
  - **Palette**: Redwood
  - Select the default labels for each series: **Closed**, **Error**, **Open**, and **Pending**.
  - Select the default colors for each series.

4. Click **OK**.

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12

Designing Dashboards

Design dashboards that show summary data to users. Dashboards enable users to chart, evaluate, highlight, comment on, and even change key business data, as well as add and edit rules.

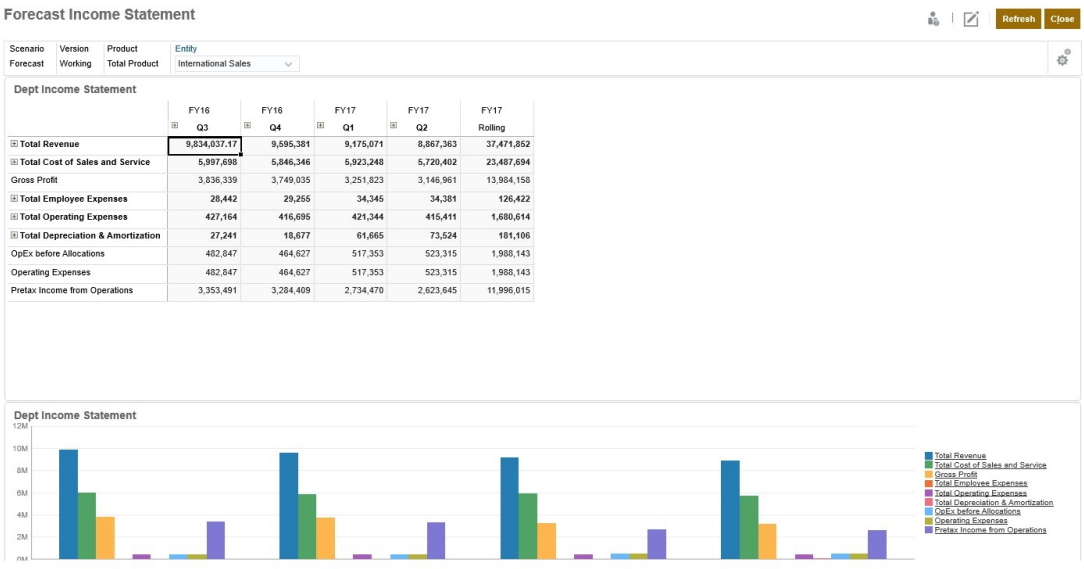
Related Topics

- The Power of Dashboards
- Creating and Managing Dashboards
- Creating and Managing Dashboards 1.0
- Creating and Managing Dashboards 2.0

The Power of Dashboards

Dashboards typically provide an overview to users at the beginning of their planning and forecast process by showing summary data. The versatility of dashboards enables users to chart, evaluate, highlight, comment on, and even change key business data. They can change a driver in a form that's in a dashboard and immediately see its impact in other forms and charts:

Sample Dashboard 1.0



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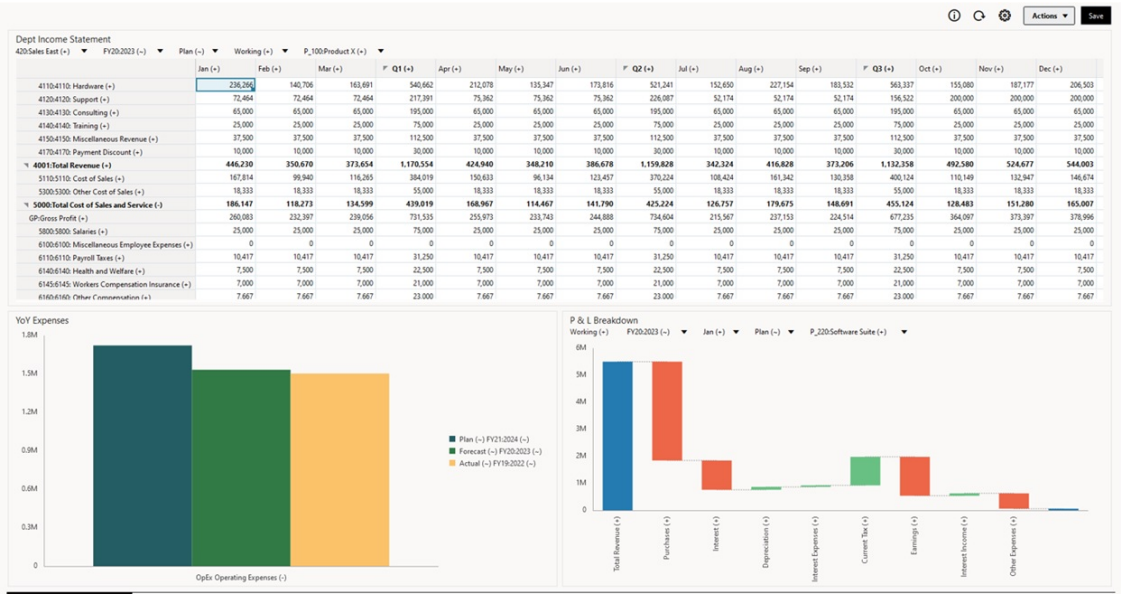
Sample Dashboard 2.0



You create dashboards by simply dragging and dropping a variety of objects from the library panel on the left to the dashboard workspace.

With dashboards, you can:

- Include multiple forms that dynamically update, including their associated charts, as users change data in the form.
- Include tiles that display a specific value from the cube. To provide the value for each tile, you can specify a form or a cell intersection as a data source.
- Switch easily between using the design environment and the runtime environment, so you can see exactly what the dashboard user will see.
- Depict data interactively with a wide variety of chart types such as area, bar, bubble, column, combination bar and line, doughnut, funnel, gauge, scatter, radar, and so on.
- Control the dashboard's layout.
- Depending on the form design, enable users to drill down into underlying detail and select which members to work with.
- Include user variables in the global POV bar and the local POV.
- In certain charts, customize the colors and line width, and show or hide gridlines.
- Add links to dynamically display external Web pages.
- Include explanations of data called *Commentary* in the dashboard.
- Tag a form as master and then filter members from the master form to detail forms or charts within the same dashboard.
- Change legend and label positions in charts.
- In Dashboard 2.0:
  - Add up to 12 components within a dashboard
  - Add up to 10 tiles in a tile chart type
  - Add hierarchical labels in charts

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- Use a logarithmic scale in relevant chart types
- Add a secondary Y-axis in relevant chart types
- Render the dashboard in Default, Light, and Dark background colors using the Style property
- Use grid display with write-enabled grids (used with Forms 2.0)
- Enjoy other usability improvements such as more screen space, options to open and edit a form directly, option to edit a dashboard without data, automatically apply POV changes, and hide dimension names
- Drill down on more than one row or column dimension. For example, in a dashboard chart with two or more row dimensions, you can drill down to the next level of members for the primary row dimension, and then drill into the secondary row dimension. This enables you to view data for a particular cell, including data for the selected child members of the primary dimension.
- Improved Rules on Save with dashboards: with/without runtime prompts before/after Load/Save
- Migrate easily from Dashboard 1.0

> **Note**
>
> Dashboard 1.0 dashboards are deployed with Forms 1.0, even when the **Forms Version** setting is **Forms 2.0**.

When users use a dashboard (referred to as *runtime*), they can set many aspects of the components, such as the type of chart displayed, the dashboard's title, and so on. A toolbar is available for each component. Dashboard users can change and save data, run rules, and so on. However, changes made in runtime to the chart type options aren't saved for the next session. Clicking **Save** in runtime saves the data, but not the dashboard definition.

Service Administrators create, redesign, delete, and assign permissions to dashboards and dashboard folders.

Related Topics

- About Dashboard Versions
- Concepts in Designing 1.0 Dashboards
- Concepts in Designing Dashboard 2.0 Dashboards
- Creating and Managing Dashboards

Creating and Managing Dashboards

Related Topics

- About Dashboard Versions
- Viewing Dashboards - Standardized Interface
- Viewing Dashboards - Legacy Interface
- Converting Dashboard 1.0 Dashboards to 2.0
  Learn how to convert Dashboard 1.0 dashboards to Dashboard 2.0 dashboards.

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About Dashboard Versions

Dashboards are currently available in two versions:

- **Dashboard 1.0**: Uses Oracle Application Development Framework (Oracle ADF) technology
- **Dashboard 2.0**: Uses Oracle JavaScript Extension Toolkit (Oracle JET) technology

Both dashboard versions currently coexist in the same business process at the same time. Dashboard 1.0 uses Oracle ADF technology and the behavior of the Oracle ADF-based dashboards remains unchanged. You can still create and update these dashboards in your business process as you did before. Dashboard 2.0 uses the existing Dashboard functionality and enhances it with Oracle JET technology for faster rendering, sophistication, ease of use, improved flexibility, better visualization, and additional designs.

> **Note**
>
> Starting with the October 2025 (25.10) update, Oracle plans to de-support Dashboard 1.0. Users on new environments created on or after October 3, 2025 won't be able to create Dashboard 1.0 artifacts. You'll only be able to create and modify dashboards using the Dashboard 2.0 framework. Users on environments created before October 3, 2025 will still have the flexibility to create and modify dashboards using either the Dashboard 1.0 or 2.0 frameworks.

To help you differentiate between the dashboard versions on the **Dashboards** page, we've provided icons in front of each dashboard.

Table 12-1 Dashboard Version Icons

Icon	Dashboard Version
	Dashboard 1.0 (Standardized list page interface)
	Dashboard 1.0 (Legacy list page interface)
	Dashboard 2.0 (Standardized list pages interface)
	Dashboard 2.0 (Legacy list pages interface)

- See Creating and Managing Dashboards 1.0.
- See Creating and Managing Dashboards 2.0.

Dashboard 2.0 Designer Features

Dashboard 2.0 Designer supports:

- More chart types than Dashboard 1.0; for example, Waterfall, Pyramid, and Geomap and sub-chart types for Radar, Combination, and Gauge charts are all Dashboard 2.0 features

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- Creating tables in dashboards with visual inline views of data, called Spark Charts
- Quick Analysis, which lets you add components to dashboards without having a prebuilt form

  You create a Quick Analysis ad hoc query by typing member names in the Search bar or using the Member Selector, and then you control the layout of the query in the Layout tab of the Properties panel.
- Working directly with data in form grids if the underlying form is a Forms 2.0 form
- Creating an ad hoc grid directly from a chart built on forms in Dashboard 2.0 dashboards

  You create an ad hoc grid by right-clicking a chart element; for example, right-clicking on a bar within a Bar chart, and then clicking **Analyze** to open ad hoc in a dynamic tab.
- A URL type of dashboard component
- A Commentary type of dasboard component with a text editor
- A custom formatted dashboard title
- Custom formatted dashboard instructions
- Background colors for URL, Commentary, and Grid component types
- Grid chart type option to render Forms 2.0 inside Dashboards 2.0
- A new **Associations** tab in the Properties panel for associating global rules
- Multiple charts for Gauge, Pie, and Doughnut chart types
- New menu options for directly opening and editing forms from within dashboard components
- An Information icon for forms and dashboards. Drop any form into a dashboard and see its information.
- The ability to resize the Properties panel
- The ability to set the background color for each dashboard component
- An option to remove the component title, using the saved space for visualizations
- An increased number of dashboard components (up to 12)
- An increased number of tiles (up to 10)
- The ability to edit without data
- A customized cell visualization limit of up to 10,000 cells with any row x column combination, such as:
  - 100 rows x 100 columns=10,000 cells
  - 1000 rows x 10 columns=10,000 cells
  - 250 rows x 40 columns= 10,000 cells
- The ability to select a dashboard **Style**. Options are **Default**, **Light**, and **Dark**.
- Options to show and hide borders between dashboard components and POV bars
- The new Oracle JET member selector
- A surface Designer in Dynamic Tabs

To review considerations for Dashboard 2.0, see Considerations for Dashboard 2.0.

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Viewing Dashboards - Standardized Interface

About the Dashboards Interface

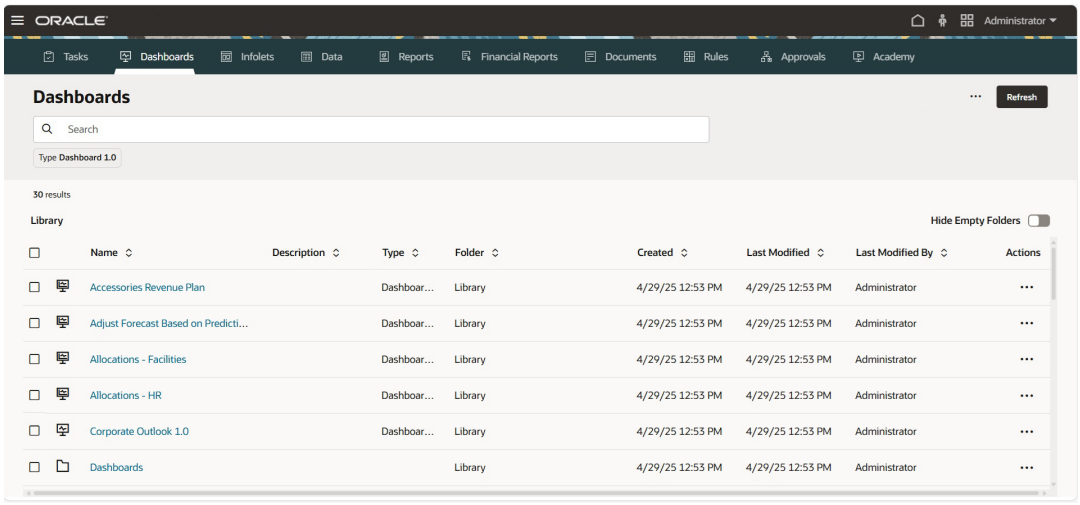
This topic provides details about working with the **Dashboards** page in the default standardized interface. To learn more about the standardized interface, see About the Standardized Cloud EPM Platform Artifact Pages.

However, there are cases when the legacy artifact pages might still be in use. If you're viewing the legacy **Dashboards** page, see Viewing Dashboards - Legacy Interface.

Viewing the Dashboards Page

To view the **Dashboards** page:

1. On the Home page, click **Dashboards**.



2. To open a dashboard, click the dashboard name.

This opens the dashboard in the *runtime* environment.

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> **Note**
>
> - If you've selected a Dashboard 2.0 dashboard and you see an `Error during Dashboard 2.0 component initialization` message, the dashboard has Forms 2.0 forms associated with it. You must ensure that Forms 2.0 is enabled before you can view the dashboard.
>
> To enable Forms 2.0, click **Application**, then **Settings**, and then under **Other Options**, find **Forms Versions**, select **Forms 2.0**, and then click **Save**.
> - A dashboard form might be prevented from opening if it includes undefined user variables. If you're prevented from viewing a dashboard form, you'll see a message prompting you to select members for the user variables:
>
>  ```
>  Before you can open this form, you must select a member for each
>  User Variable. To select members for missing user variables: Click
>  here
>  ```
>
> When you click the link in the message, the **User Preferences** page opens in a new dynamic tab and the user variables with missing member values are displayed first. Select a member for each undefined user variable.
>
> See Setting User Variables in *Working with FreeForm*.

Features of the Standardized Dashboards Page

On the standardized **Dashboards** page:

- Folders are supported. The **Dashboards** page uses the same folder hierarchy as infolets and data entry forms and all artifacts and folders reside under a root folder called **Library**. You cannot delete or rename the **Library** folder.

  > **Note**
  >
  > Only administrators can add files (for example, dashboards, infolets, forms, reports, and so on) to the **Library** root folder.

- To help you differentiate between the dashboard versions on the **Dashboards** page, we've provided an icon in front of each dashboard:
  - : Dashboard 1.0
  - : Dashboard 2.0
- The **Dashboards** page can be filtered by type. In the Standardized interface, click the **Type** filter chip in the **Search** box and select **Dashboard 1.0**, **Dashboard 2.0**, or select both.
- **Hide Empty Folders**: Hides folders that don't contain relevant artifacts for the current context. When the toggle is off, all folders are visible in the list region, including empty ones. When the toggle is on, only folders containing relevant artifacts for the **Dashboards** artifact page are displayed.

  By default this option is off, and will reset to off each time you start a new session. The state of the toggle is retained by the page during the session and is page-specific.

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Navigating away from the **Dashboards** page and returning preserves the toggle state until the tab is closed or the user logs out.

- To search for a specific dashboard, enter search criteria and the search results will display in a drop-down under the **Search** box. Click **Enter** to view the search results in the list region. Search results list all instances of dashboards that include the search criteria (including folder names). Each search result displays the complete folder path. Searching inside a folder narrows the search to that folder and its subfolders.

> **Note**
>
> – The search considers the **Name**, **Description**, and **Folder** columns, wherever applicable, in the search.
>
> – If you perform a **Search** and the **Hide Empty Folders** toggle is off, all matching folders and artifacts are displayed. If the toggle is on, only folders containing matching artifacts are displayed. Empty folders are hidden.

Standardized Interface Dashboards Actions

The Standardized interface **Dashboards** page features a global Actions menu ••• and a line item-specific **Actions** menu.

The global Actions menu ••• (within the header region of the **Dashboards** page) includes the following options:

- **Create Dashboard 1.0**: Creates a Dashboard 1.0 dashboard and places it in the folder that is currently open. If no folder is open, new dashboards are placed in the Library folder.
- **Create Dashboard 2.0**: Creates a Dashboard 2.0 dashboard and places it in the folder that is currently open. If no folder is open, new dashboards are placed in the Library folder.
- **Create Folder**: Creates a new folder in the folder that is currently open. If no folders are open, new folders are created in the Library folder.

> **Note**
>
> Keep in mind that newly created folders are empty and therefore will not display if the **Hide Empty Folders** toggle is on. New folders become visible after content is added to them, or if the toggle is switched off.

- **Convert Dashboards to 2.0**: Finds any Dashboard 1.0 dashboards in the folder that is currently open, including any subfolders, and converts them all to Dashboard 2.0 dashboards. Selecting a folder in the listing and then selecting the global action for **Convert Dashboards to 2.0**, finds any Dashboard 1.0 dashboards within the folder, including any subfolders, and converts them to Dashboard 2.0 dashboards. Selecting one or more Dashboard 1.0 dashboards in the listing and then selecting the global action for **Convert Dashboards to 2.0**, converts the selected dashboards to Dashboard 2.0 dashboards. This action cannot be undone.
- **Assign Permission**: Assigns access permissions to the folder that is currently open. If no folder is open, permissions are assigned to the Library folder.
- **Move to**: (This option is only visible if one or more artifacts in the listing are selected.) Moves the selected artifacts to another location in the folder hierarchy.

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Each dashboard line item also has an item-specific **Actions** menu which includes the following options:

- **Open**: Opens the dashboard in the *runtime* environment so you can view it. The dashboard opens in a dynamic tab.
- **Edit**: Opens the dashboard in the Dashboard Designer with the underlying form data. The dashboard opens in a dynamic tab.
- **Delete**: Deletes a folder or a dashboard. The folder must be empty to delete it.
- **Rename**: Renames a folder or a dashboard. The folder cannot be renamed if it contains other folders.
- **Copy As**: Copies a dashboard with a new name. The dashboard is copied to the listing directly under the original dashboard.
- **Move to**: Moves the folder to another location in the folder hierarchy or moves a dashboard to another folder.

> **Note**
>
> All folders are displayed and are selectable in the **Move** interface, regardless of the state of the **Hide Empty Folders** toggle.

- **Copy URL**: Enables you to copy and share the direct URL for a dashboard.

  See Copying Direct URLs for Artifacts.
- **Convert Dashboards to 2.0** (Dashboard 1.0 option only): For folders, finds any Dashboard 1.0 dashboards within the folder hierarchy, and converts them to Dashboard 2.0 dashboards. For dashboards, converts the dashboard to a Dashboard 2.0 dashboard. This action cannot be undone.
- **Assign Permission**: Assigns access permissions to the folder or dashboard.

Viewing Dashboards - Legacy Interface

About the Dashboards Interface

This topic provides details about working with the **Dashboards** page in the legacy interface. If you're viewing the default standardized **Dashboards** page, see Viewing Dashboards - Standardized Interface.

> **Note**
>
> Starting with the October 2025 (25.10) update, Oracle plans to de-support legacy artifact pages. For all new and existing environments, the **Legacy List Pages** application setting will be removed and the **Dashboards** page will only render as standardized.

To learn more about the standardized interface, see About the Standardized Cloud EPM Platform Artifact Pages.

Viewing the Dashboards Page

To view the **Dashboards** page:

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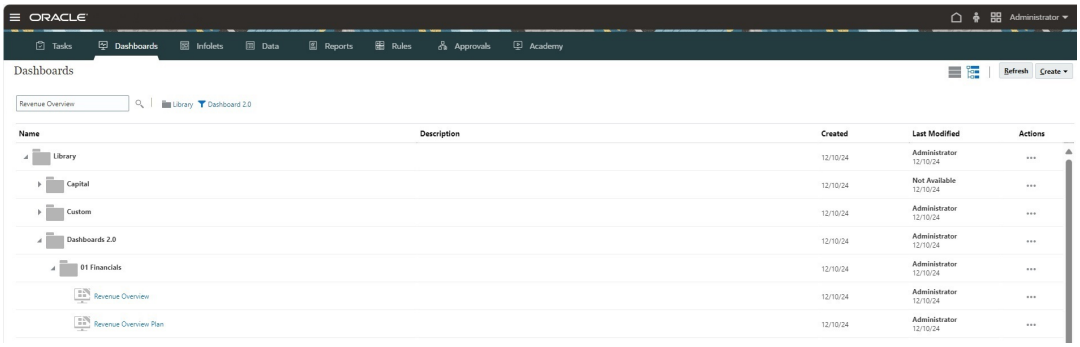
1. On the Home page, click **Dashboards**.



2. To open a dashboard, click the dashboard name.

   This opens the dashboard in the *runtime* environment.

> **Note**
>
> - If you've selected a Dashboard 2.0 dashboard and you see an `Error during Dashboard 2.0 component initialization` message, the dashboard has Forms 2.0 forms associated with it. You must ensure that Forms 2.0 is enabled before you can view the dashboard.
>
>   To enable Forms 2.0, click **Application**, then **Settings**, and then under **Other Options**, find **Forms Versions**, select **Forms 2.0**, and then click **Save**.
> - A dashboard form might be prevented from opening if it includes undefined user variables. If you're prevented from viewing a dashboard form, you'll see a message prompting you to select members for the user variables:
>
>   ```
>   Before you can open this form, you must select a member for each
>   User Variable. To select members for missing user variables: Click
>   here
>   ```
>
>   When you click the link in the message, the **User Preferences** page opens in a new dynamic tab and the user variables with missing member values are displayed first. Select a member for each undefined user variable.
>
>   See Setting User Variables in *Working with FreeForm*.

Features of the Legacy Dashboards Page

On the legacy **Dashboards** page:

- Folders are supported. The **Dashboards** page uses the same folder hierarchy as infolets and data entry forms and all artifacts and folders reside under a root folder called **Library**. You cannot delete or rename the **Library** folder.

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> **Note**
>
> Only administrators can add files (for example, dashboards, infolets, forms, reports, and so on) to the **Library** root folder.

- To help you differentiate between the dashboard versions on the **Dashboards** page, we've provided an icon in front of each dashboard:
  -
    : Dashboard 1.0
  -
    : Dashboard 2.0
- You can toggle between viewing dashboards by a Flat View or a Tree View:

- The **Dashboards** page can be filtered by type. Click (Filter), and then select from the following filter options:
  - **All Types** (default)
  - **Dashboard 1.0**
  - **Dashboard 2.0**
- To search for a specific dashboard, click

  (Search) to search by keywords or (Library) to narrow the search to specific library folders.

Legacy Interface Dashboards Actions

On the right side of the **Dashboards** page there is an **Actions** column. Clicking ••• (Actions) next to the dashboard or folder, displays the following actions:

- Folder Actions:
  - **Assign Permission**: Assigns access permissions to the folder. Folder permissions apply to all items within that folder, even to items that may not be visible in the list.
  - **Create Dashboard**: Creates a Dashboard 1.0 dashboard.
  - **Create Dashboard 2.0**: Creates a Dashboard 2.0 dashboard.
  - **Create Folder**: Creates a new folder.
  - **Delete**: Deletes a folder. The folder must be empty to delete it.
  - **Rename**: Renames a folder. The folder cannot be renamed if it contains other folders.
  - **Move to**: Moves the folder to another location in the folder hierarchy.
  - **Convert All Dashboards to 2.0**: Finds any Dashboard 1.0 dashboards within the folder hierarchy, and converts them to Dashboard 2.0 dashboards.
- Individual Dashboard Actions:

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- **Edit**: Opens the dashboard in the Dashboard Designer with the underlying form data. The dashboard opens in a dynamic tab.
- **Edit without Data** (Dashboard 2.0 option only): Opens the dashboard in the Dashboard Designer without the underlying form data so you can more easily perform such actions as realigning the dashboard components and tiles. The dashboard opens in a dynamic tab.
- **Rename**: Renames a dashboard.
- **Copy As**: Copies a dashboard with a new name. The dashboard is copied to the listing directly under the original dashboard.
- **Delete**: Deletes a dashboard.
- **Move to**: Moves a dashboard to another folder.
- **Assign Permission**: Assigns access permissions to the dashboard.
- **Convert to Dashboards 2.0** (Dashboard 1.0 option only): Converts the dashboard to a Dashboard 2.0 dashboard.
- **Copy URL**: Enables you to copy and share the direct URL for a dashboard.

  See Copying Direct URLs for Artifacts.

Converting Dashboard 1.0 Dashboards to 2.0

Learn how to convert Dashboard 1.0 dashboards to Dashboard 2.0 dashboards.

If your application currently uses Dashboard 1.0, you can convert them to Dashboard 2.0.

You convert dashboards on the **Dashboards** listing page at the individual dashboard level or at the folder level. For example, if you wish to convert all dashboards in the application from 1.0 to 2.0, you can select the top level Library folder and convert them all.

> **Note**
>
> - After converting to Dashboard 2.0, you cannot convert your dashboards back to Dashboard 1.0.
> - The conversion process does not change the dashboard names. Navigation flows will still work as they did before the conversion.
> - After converting to Dashboard 2.0, dynamic POVs are displayed first instead of fixed POVs.

To convert Dashboard 1.0 to Dashboard 2.0:

1. On the Home page, click **Dashboards**.
2. Navigate to the folder or to the individual Dashboard 1.0 dashboard that you'd like to convert, then select from the following options:
   - To convert an individual dashboard, click ••• (Actions) to the right of the dashboard, and then select **Convert to Dashboards 2.0**.
   - To convert all dashboards within a folder, click ••• (Actions) to the right of the folder, and then select **Convert All Dashboards to 2.0**.

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Note

- Very old dashboards might include a dashboard definition with a component width value that is no longer supported. This could result in an unexpected dashboard rendering after converting an old dashboard to Dashboard 2.0.

  To resolve this issue, we recommend any of the following workarounds:

  – Open the original dashboard in Dashboard 1.0, edit the dashboard, and then save it. The dashboard will be automatically saved with the corrected component width values, and then it can be converted to Dashboard 2.0.

  – Create a new dashboard in Dashboard 1.0 with the same set of forms and charts and the same settings, then convert the dashboard to Dashboard 2.0.

  – Create a new dashboard in Dashboard 2.0 with the same artifacts and settings.

- Dashboards containing master forms and details that were converted to Dashboard 2.0 prior to the 23.11 update might display two **Apply Context** menu options in a grid's context menu. This issue was fixed in 23.11.

Creating and Managing Dashboards 1.0

- Concepts in Designing 1.0 Dashboards
- Creating Dashboard 1.0 Dashboards
- Creating Dashboards Containing Master Forms and Details
- About Your Dashboard's Layout - Version 1.0
- About the Gauge Chart Type
- About the Tile Chart Type
- Customizing Dashboard Colors
- About Global and Local POVs in 1.0 Dashboards
- Dashboard POVs and Valid Intersections

Concepts in Designing 1.0 Dashboards

Helpful information as you design 1.0 dashboards:

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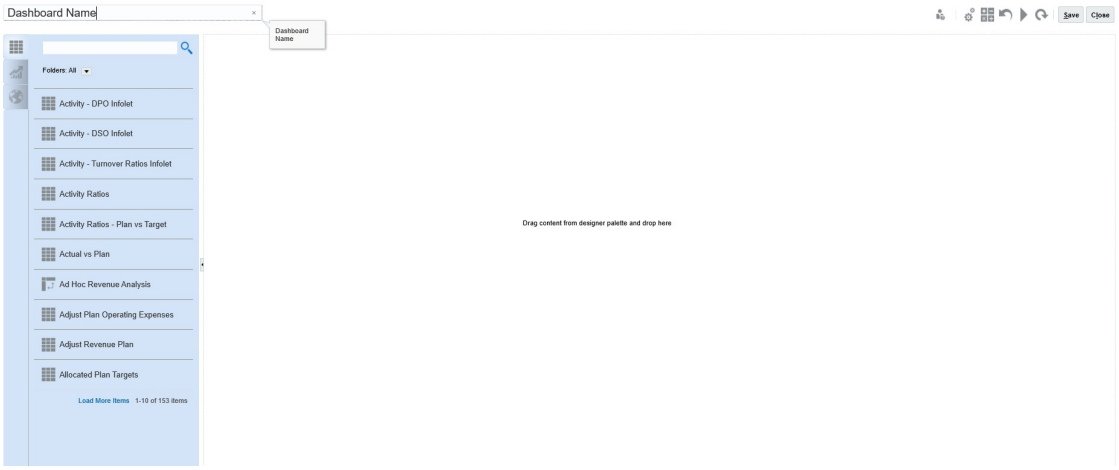


- On the left is the design palette. Simply drag and drop objects from the palette to the canvas.

  > **Tip**
  >
  > Drag an object to a border line. The drag icon changes to a plus sign when you can drop the object in an allowed space. See About Your Dashboard's Layout - Version 1.0.

- On the top right are settings for the entire dashboard:

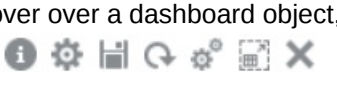
- Click **Settings**

  to set these general aspects of the dashboard:

Table 12-2 Dashboard General Settings

Option	Description
Use name as title	By default, this option is selected. Clear this option to give the dashboard a title with custom formatting.
Borders	When you create a new dashboard, borders are hidden by default. To show borders in newly-created dashboards, select **Show**.
Layout	Select **Fixed** (default) or **Flexible**.
POV Bars	Select **Show** (default) or **Hide**.
Global POV Bar	Select **Enable** (default) or **Disable**.

For information about POVs, see About Global and Local POVs in 1.0 Dashboards.

- When you hover over a dashboard object, a toolbar for that object displays in the upper right corner: 

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> **Note**
>
> Dashboard 1.0 hover icons provide options such as **Instructions** (available only if instructions are configured for a form), **Actions**, **Save**, **Refresh**, **Settings**, and **Maximize**, depending on the type of object.

- As you create a dashboard, click **Runtime** so that you can immediately see how the dashboard looks and works to a dashboard user. To return to designer mode to continue designing the dashboard, click (Dashboard Designer).
- By default, missing or suppressed data is plotted as zeros in graphs. You can clear the **Plot Missing Values as Zero** setting to ignore missing or suppressed data in certain chart types so it's no longer plotted as zeros.
- The listing page for dashboards supports folders. Folders enable you to assign permissions to all dashboards within a folder rather than assigning permissions to each individual dashboard. The dashboard listing page uses the same folder hierarchy as infolets and data entry forms and all artifacts and folders reside under a root folder called **Library**.

> **Note**
>
> Only administrators can add files (for example, dashboards, infolets, forms, reports, and so on) to the **Library** root folder.

- On the listing page for dashboards, you can toggle between viewing dashboards by a Flat View or a Tree View:

  Then you can search for dashboards using **Search**. The flat view displays only the dashboards that meet the search criteria, not the folders that contain them. The tree (or hierarchical) view displays dashboards in the context of the folders that contain them.

  To search on another keyword, clear the search criteria by clicking in the **Search** box.

Creating Dashboard 1.0 Dashboards

Need help deciding which dashboard version to choose? See About Dashboard Versions.

> **Note**
>
> To create a Dashboard 2.0 dashboard, see Creating Dashboard 2.0 Dashboards.

1. From the Home page, click **Dashboards**, and then click **Create**.

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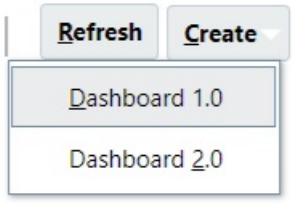
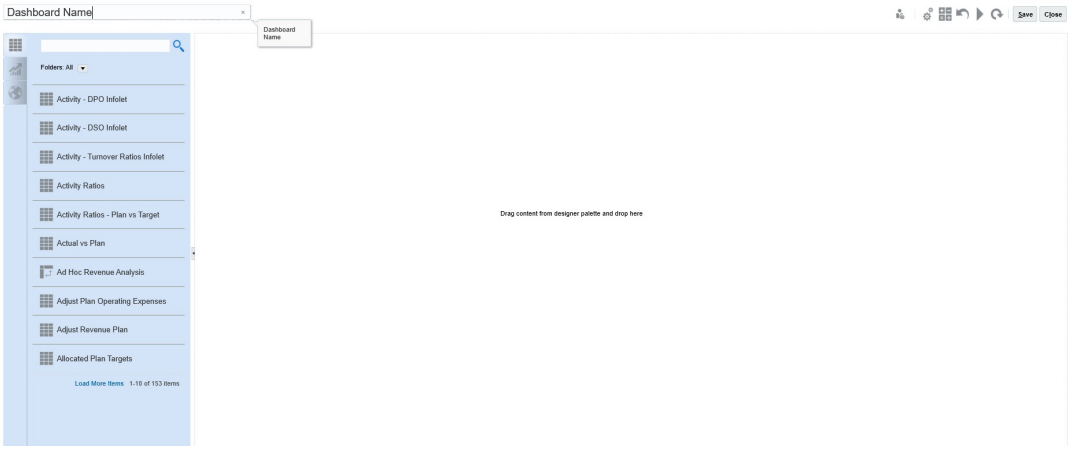


2. From the options provided, select **Dashboard 1.0**.



3. Enter a name by clicking the default dashboard name and entering a new name in the input box.

   You can give the dashboard a title with custom formatting, click **Settings**, clear **Use name as title**, and then enter the title and set formatting the dialog box.

4. From the design palette on the left, drag and drop objects onto the dashboard canvas.

   Select from these objects:

Table 12-3 Dashboard Objects

Object	Description
Forms	Select forms to include in the dashboard by navigating to the forms folders or by searching for them by name. To view instructions for forms after they've been added to the dashboard, hover over the form and click **Instructions** . The access permissions set for forms are honored in dashboards.

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Table 12-3 (Cont.) Dashboard Objects

Object	Description
Chart Types	Select the chart types to include in the dashboard. When first added, a selected chart has sample data. You then associate it with a form as its data source. When you link a chart to a form, users can immediately see the impact of changing data in the form on the associated charts. By default, missing or suppressed data is plotted as zeros. For selected chart types (Area, Bubble, Combination, Line, Radar, and Scatter), you can disable this setting by clearing the **Plot Missing Values as Zero** option in the chart's settings. When this option is cleared, the missing or suppressed data is ignored and is no longer plotted for those chart types. The Combination chart type alternates displaying row data with vertical bars and lines in the chart. For example, the data in row 1 of a form is displayed as a bar and the data in row 2 as a line, with alternating chart types for even and odd-numbered rows. Although the Combination chart type can display up to 20 rows of data, it's particularly useful when you want to compare two categories of data. For example, you want to compare Germany and France's average exchange rates over several years, so the form has Germany rates in row 1 of the form, and France's rates are in row 2. For information on the Gauge chart type, see About the Gauge Chart Type. The Tile chart type, sometimes called *performance tiles*, lets you select specific values from the cube to display. See About the Tile Chart Type.
Commentary	Select **External Artifacts**, and then **Commentary**. Enter text that explains the data or charts. Note that the text limit is 2000 characters, including any formatting tags that are added for rich text.
URL	Dynamic web page summary. Select **External Artifacts**, and then **URL**. Insert only external site URLs starting with the `https://` security protocol. Don't use internal or relative URLs or URLs for unconsenting third party sites such as google.com.

5. Customize the dashboard using the dashboards settings and the objects' hover toolbar, and then click **Save**.

   See About Your Dashboard's Layout - Version 1.0.

You can easily duplicate and modify a dashboard using **Copy As** on the Dashboard list page. Select the Actions icon next to the dashboard you want to copy.

Creating Dashboards Containing Master Forms and Details

You can design dashboards that have one master form and multiple detail (or target) objects (forms or charts). When you do so, the selection of members in the master form automatically filters to the members in the target objects, and the target objects show only the details that are relevant to the members highlighted in the master form.

Considerations when designing dashboards with master forms and target objects:

- Master forms must always be a form (you cannot tag a chart as a master form).
- The target objects can be forms or charts.

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- The dashboard containing a master form must always contain more than one object.
- Only one form on a dashboard can be tagged as master. If you want to tag a different form on a dashboard as master, you must first untag the existing master form.

To create a dashboard containing a master form and target objects:

1. Create a dashboard with at least one form and one or more target objects.

   See Creating and Managing Dashboards.
2. Ensure the dashboard is in designer mode by clicking **Actions**, and then clicking **Edit**.
3. Select the form on the dashboard that you want to designate as master, click the Settings icon for that form, and then select **Tag as Master**.

To filter the data in the target forms or charts that is relevant to the data in a master form, right-click the relevant data in the master form and select **Apply Context**.

About Your Dashboard's Layout - Version 1.0

About setting up a dashboard layout in version 1.0:

- The first object you drag occupies the whole canvas.
- Then drag objects to the left, the right, the top, or the bottom of the existing object.
- The dashboard canvas provides two types of drop zones: One to place two objects next to each other, each occupying half the space, and the other to place three objects, each occupying a third of the space.
- You can design a dashboard with objects to display vertically, horizontally, and each can have its own size.
- To resize an object that uses **Flexible** layout, drag the object's border. You can also click an object's handle to expand it, and then click the handle again to restore the object to its original size.
- To resize an object that uses **Fixed** layout, set its width or height percentage in **Settings**.
- The form layout can be asymmetric.
- In runtime mode, if a user doesn't have access to a form or if the form is missing, then an adjacent object takes its space. In designer mode, all empty objects are displayed so that the designer can choose to remove them.

About the Gauge Chart Type

Gauge chart types are handy for showing whether data values fall within an acceptable range or not. You set the maximum value, the range maximums, and the gauge displays ranges as red, yellow, and green to help you quickly assess a current value. So, gauge chart types help you identify problems in important data points or measures. For example, you could use a gauge to display the current sales, where the thresholds are set to represent the sales targets.

If the form has multiple values, you can display multiple gauges, up to a maximum of 36 (the values in the first 6 rows and the first 6 columns in the form). The remaining values in the form are ignored. If you want the gauge chart to display only one value, then associate it with a form that has only one cell value.

You can select either a dial gauge or a status meter gauge. You can display a status meter gauge using either horizontal or vertical bars.

Dashboard designers can set:

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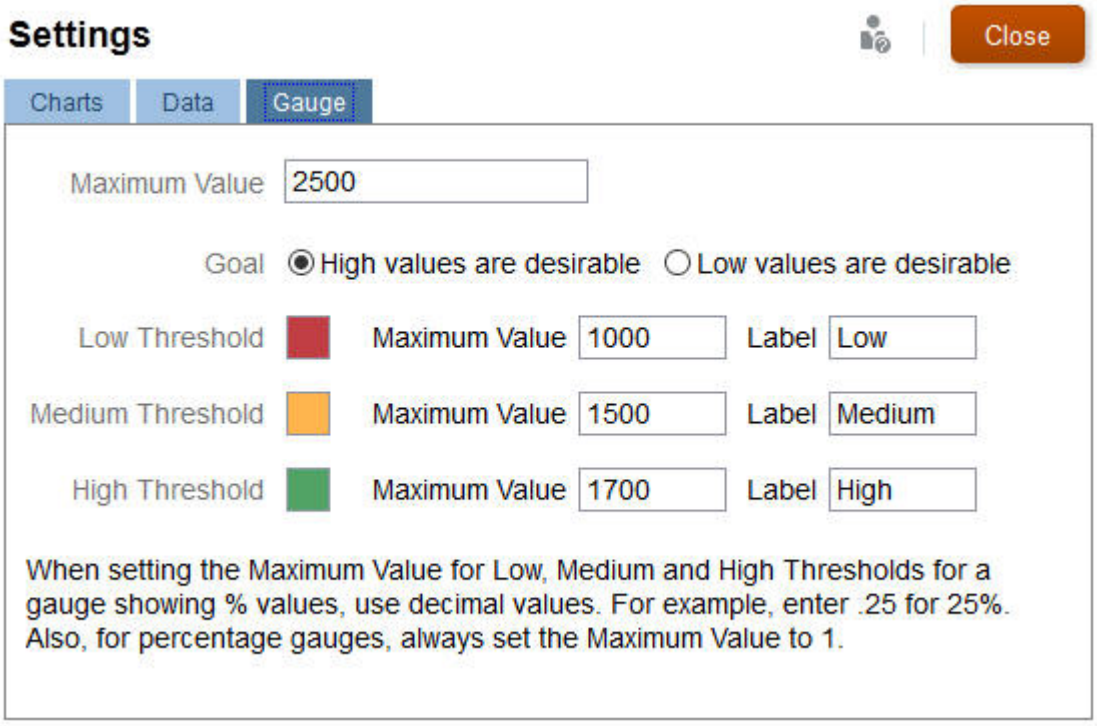
- **Maximum Value**: The highest value on the gauge. The dashboard designer sets the **Maximum Value** as a default, and then users can temporarily change it at runtime. If the dashboard designer doesn't specify a maximum value, the application automatically sets the maximum value as greater than the value on the gauge.
- **Thresholds**:
  - Low, Medium, and High thresholds: To visually indicate whether a measure lies in the acceptable range or not, these thresholds enable you to display the gauge in red, yellow and green based on the specified values.
  - Thresholds where low values are desirable.
  - Appropriate labels for the thresholds that are displayed when hovering over the thresholds in the gauge.



For example, here's a form's data:

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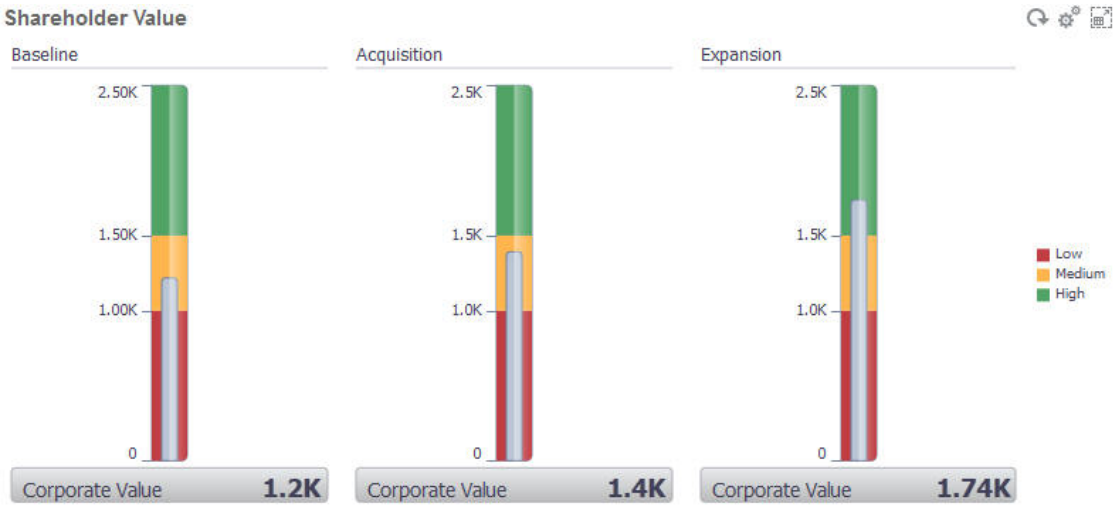
Here's the resulting dial gauge:

Here's the resulting status meter gauge with vertical bars:



> **Note**
>
> If a cell in the form is missing a value, no gauge is displayed for that cell. Also, you must specify at least 2 consecutive thresholds. The application needs the middle threshold value to calculate the chart.

About the Tile Chart Type

A tile is a chart type that lets you select specific values from the cube to display. In addition to using a form as a data source, you can directly enter a cell intersection that provides the value for a tile. You can have up to 6 tiles across, and 4 rows down in a dashboard, and give them a title. Until you associate a tile with data, it displays sample data.

With a form as the data source for a tile:

- You can have up to six tiles per object.
- The values in the first column (up to the sixth row) are used to create the tiles.

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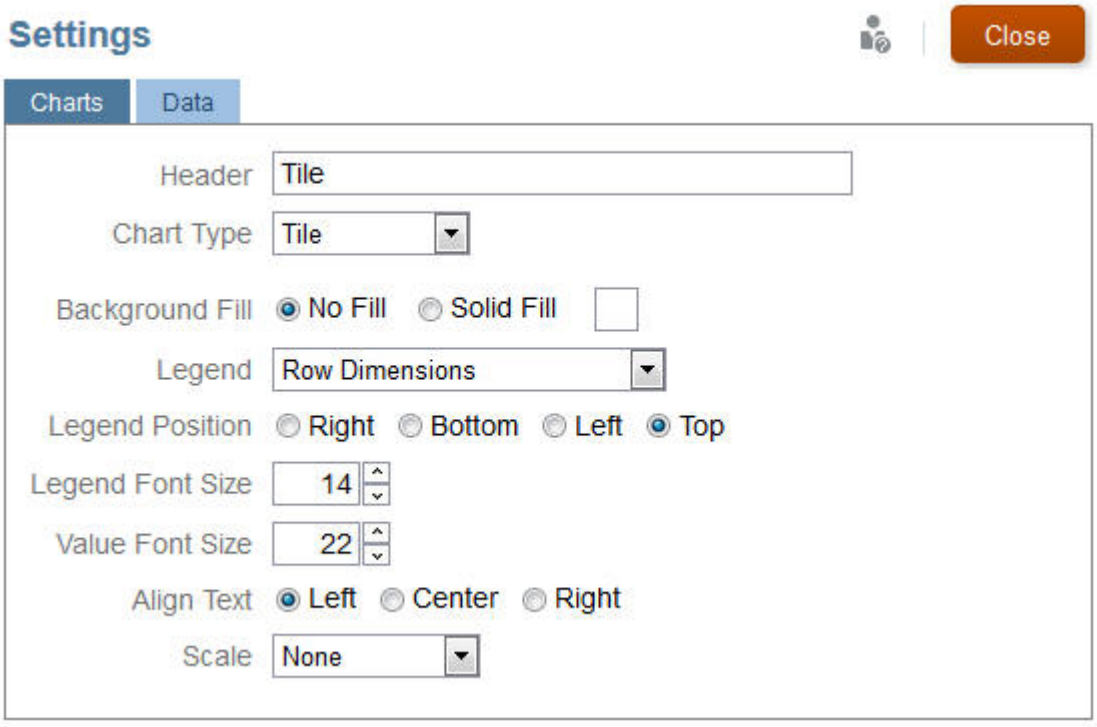
> **Note**
>
> Sometimes in a form, the first column may be collapsed (hidden) when viewed as a grid. But the collapsed column is still considered when the tile gets its values from the form.

- The tile's title is the row's title, and it gets its values from the first column, then by row.
- You can set the tile's title, the tile's height percentage, legend, and can select which axes from the form to include. For example, if you select a form with three rows, the tile displays three values.

With a cell intersection as the data source for a tile, you can have only one tile per object.

> **Tip**
>
> To select a tile chart type, expand the list of chart types by clicking the link at the bottom of the list.

Here are options you can set for a tile chart type. Note that you can set whether the displayed value is horizontally aligned in the tile to the left, the center, or the right.



Scaling Large Numbers

Especially useful for large numbers, you can scale how a value is displayed. For example, if the tile value is 1,689,000 and you select **K** as the scaling option, the tile displays the value as 1689K. Your scaling options:

- **None**: No scaling is applied.

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- **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T.
- **K**: The value is displayed as thousands units. For example, 1689000 displays as 1689K.
- **M**: The value is displayed as millions units. For example, 123,456,789 displays as 123M.
- **B**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B.
- **T**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.

Customizing Dashboard Colors

Maybe your company uses a standard set of colors in charts to denote different types of data. For example, dark blue might represent actual data versus light blue for budget data. When you customize dashboard colors, you select colors in the order of rows on the form. Series 1 represents the first row of data, and so on. You can assign each row in the form a color that represents its data in the chart.

You can customize dashboard colors in Bar, Line, Area, Bubble, Column, Combination, Doughnut, Pie, Radar, and Scatter chart types.

1. With the chart on the dashboard's design palette, click **Settings** .
2. Click **Colors**.
3. Clear the **Default** check box, and then click the down arrow for the Series you want to change.

   Check the chart in the background to see the data type that each series represents.

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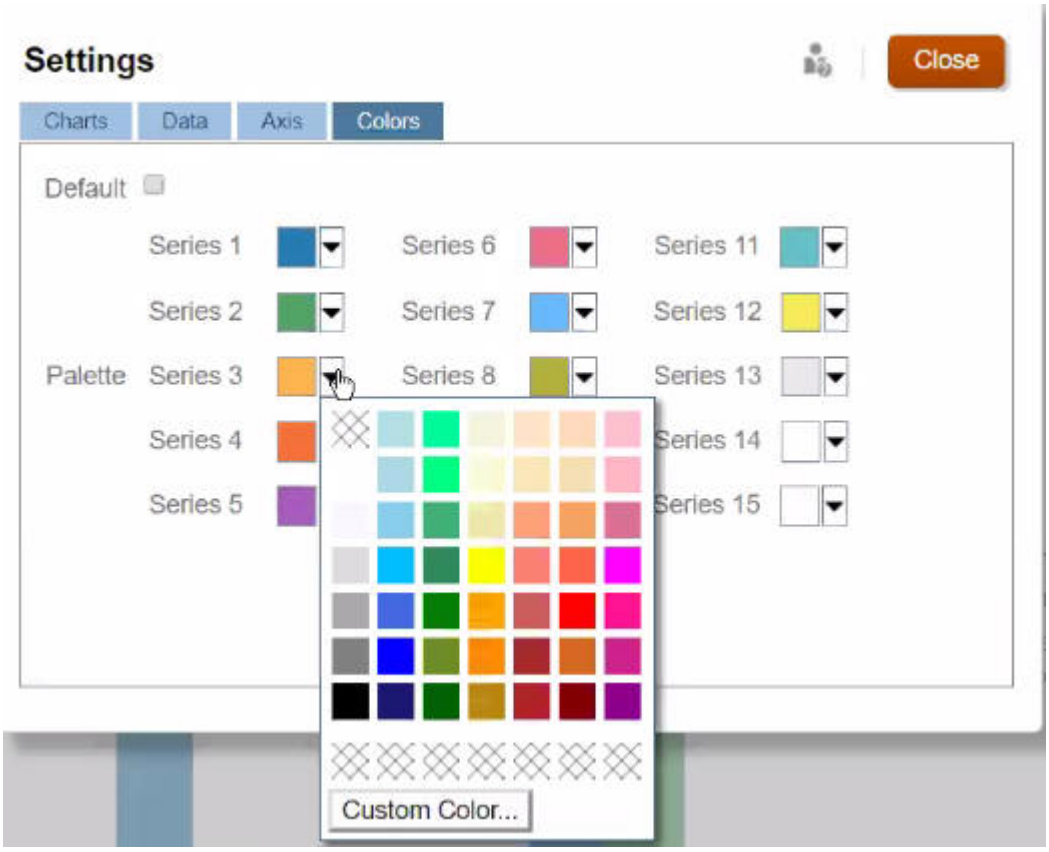

4. Click the colors you want for the selected Series, and then click **Close**.

> **Note**
>
> To select more shades of colors than the ones initially displayed, click **Custom Color...**.

Your selections apply only to the current chart. Follow these steps to change the colors of other charts in the dashboard.

About Global and Local POVs in 1.0 Dashboards

A local POV on a form reflects the dimension members the form designer selected for that form. Dashboards also support *global POV bars*, so that the local POVs that are common are combined in the global POV bar to avoid repeating them in each object.

Dashboard 1.0 POVs Example

Here's a 1.0 dashboard showing a global POV bar (showing Entity, Product, and Year) and a local POV (the Plan drop-down list showing Q2):

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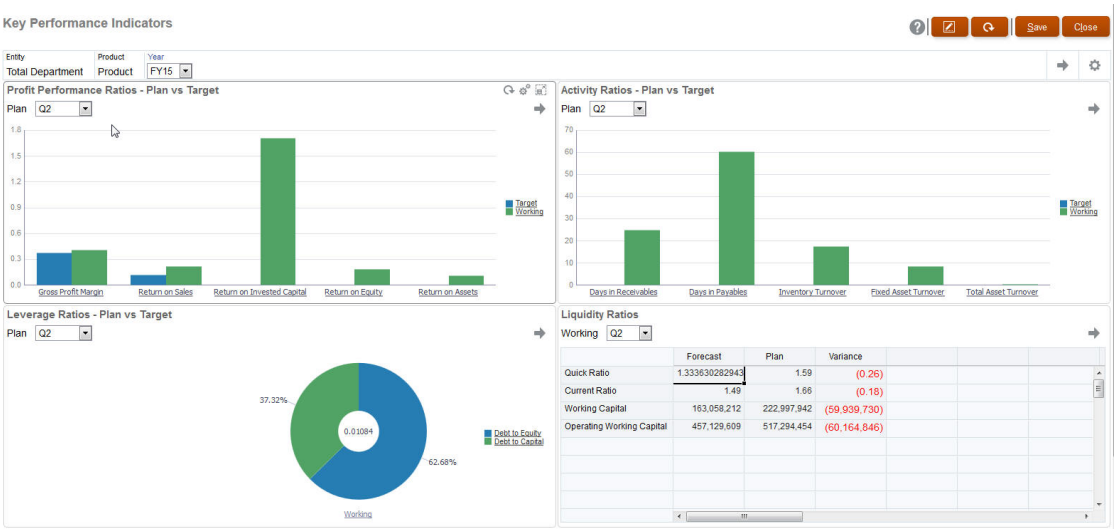

With a global POV bar, if you change a page in the global POV bar and then click **GO**, the page changes for all objects that are based on forms. The global POV bar displays at the top of the dashboard above all the objects, while the local POV bar displays within the object. User variables are supported in both global and local POVs in dashboards and you can change dynamic user variables directly from the POV bar.

In dashboard **Settings**, you can set whether to show or hide POVs and whether to enable or disable global POV bars. (If you select **Hide** for **POV Bars**, and **Enable** for **Global POV Bar**, the **Hide** option overrides the **Enable** option.) The global POV bar is enabled by default; if you disable it, the complete POV is displayed for each local POV as applicable.

About global POV bars:

- The POV Bar is made up of local POV dimensions, Page dimensions and user variables.
- Depending on the local POV and page of each form on the dashboard, the global POV bar is automatically calculated.
- They are reflected in the other objects using forms in that dashboard. That is, they apply to forms in a dashboard, to charts that are linked to forms, and to tiles that use forms as a data source. So if the dashboard doesn't include a form as a data source, then neither the local nor global POV bar is available.

Here's an example of how the global POV bar is calculated, based on the local POV dimensions for two forms:

The global POV bar is disabled:

- Form A local POV: Year, Entity, Product
- Form B local POV: Year, Entity, Project

The global POV bar is enabled:

- Global POV bar: Year, Entity
- Form A local POV: Product
- Form B local POV: Project

Because not all dimensions and page selections may be common to all forms on a dashboard, the complete POV for a dashboard object may get split between the local and global POV bar. The global POV bar and the local POV together contain the complete intersection information for each form on a dashboard.

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If there is only one object on the dashboard that uses a form as a data source, then the entire POV/page of the form can be moved to the global POV bar.

If there is more than one object on a dashboard that use forms as a data source, then this is how the application determines which dimensions go in the global POV bar or stay in the local POV:

- If the dimension is in the POV or page of all the forms, and the member selection is the same in all the forms, the dimension goes in the global POV bar.
- If the dimension is in the POV on one form and in the page of another form, then the dimension stays in the local POV.
- If the dimension is in the POV, then the same members must be selected in all the forms for the dimension.
- If the dimension is a page dimension, then the selected page members must be the same and display in the same order in all the forms.

The POVs in 1.0 dashboards honor valid intersections by hiding invalid Page members. See Dashboard POVs and Valid Intersections.

Dashboard POVs and Valid Intersections

The POVs in dashboards honor the valid intersections by hiding invalid Page members. Just like in forms, the Page drop-down list is filtered for all selected members in the POV and Page dimensions. Because dashboards support both global and local POVs, the context for filtering the Page drop-down list depends on which POV the members are located. If the Page drop-down list is on a global POV, the filtering context is only the global POV dimensions. If the Page drop-down list is on the local POV, the filtering context is all the global dimensions plus the dimensions on a chart's local POV.

See also About Global and Local POVs in 1.0 Dashboards.

Creating and Managing Dashboards 2.0

Related Links:

- Concepts in Designing Dashboard 2.0 Dashboards
- Considerations for Dashboard 2.0
- Creating Dashboard 2.0 Dashboards
- Working with the Member Selector When Designing Dashboards 2.0
- About Forms 2.0 Grids in Dashboard 2.0
- Using Tables in Dashboards
- About Global and Local POVs in Dashboard 2.0
- About Quick Analysis
- About Dashboard Charts
- Dashboard Design Guidelines

Concepts in Designing Dashboard 2.0 Dashboards

This topic contains helpful information as you design Dashboard 2.0 dashboards:

- About the Dashboard 2.0 Designer

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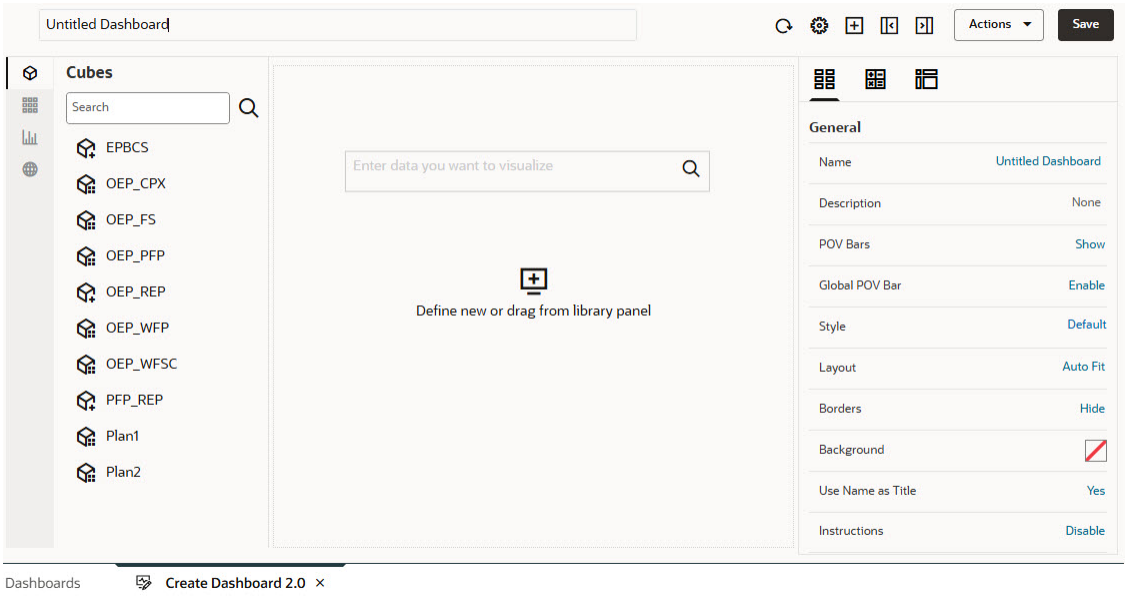
- Opening the Dashboard 2.0 Designer
- Data Sources for Charts
- More Helpful Information About Dashboard 2.0
- Videos

About the Dashboard 2.0 Designer



The Dashboard Designer consists of the following dashboard components:

- Object Palette
- Dashboard Toolbar
- Dashboard Workspace
- Properties Panel
- Dashboard Components

Opening the Dashboard 2.0 Designer

As a reminder, Dashboard 2.0 dashboards on the **Dashboards** page are identified by one of the following icons:

- (Dashboard 2.0 icon in the standardized interface)
- (Dashboard 2.0 icon in the legacy interface)

To open the Dashboard 2.0 Designer:

1. From the Home page, click **Dashboards**.
2. Select one of the following options:
   - To create a dashboard, click **Create**, and then select **Dashboard 2.0**.
   - To edit a dashboard, in the **Actions** column to the right of the Dashboard 2.0 dashboard you'd like to edit, click ••• (Actions), and then click **Edit**.

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If you don't need to work with data while editing a Dashboard 2.0 dashboard; for example, you'd like to realign the dashboard components and tiles, click ••• (Actions), and then click **Edit without Data**.

3. If the Dashboard 2.0 dashboard is open in the runtime environment, you can switch from the runtime environment to the Dashboard Designer environment.

   Click **Actions**, and then select **Edit** or **Edit without Data**.

Data Sources for Charts

Dashboard 2.0 supports ad hoc grids, forms, and cubes for choosing the data. You pick the data sources from the object palette.

> **Note**
>
> You can include a maximum of 12 components in a Dashboard 2.0 dashboard.

- The properties panel provides full data source information for a dashboard component on the **Chart** tab . Select the dashboard component, then click (Chart). The data source information is displayed under the **Visualization** heading.
- When using a cube as the data source for quick analysis, drag and drop a cube from the object palette to the dashboard workspace, then click in the Search box and either enter the member name or launch the member selector to choose members. The **Layout** tab will then display in the properties panel where you can pivot the dimensions and choose which dimensions appear on the POV, row, or column axis. See About Quick Analysis.

  > **Note**
  >
  > There is an upper limit of 50 members across all dimensions that can be selected to create a chart using quick analysis. This limit together with the number of dimensions in the cube will determine the maximum size of the grid that is supported for a chart based on quick analysis.

- When the data source for a dashboard component is a form or an ad hoc grid, the data is fetched using the form definition.
  - You can change the source form in the properties panel by clicking the **Form** property in the (Chart) tab of the properties panel.
  - The total amount of form data that's allowed to be visualized in a chart is 10,000 cells. By default, this is 100 rows and 100 columns. However, you can change the number of rows and columns to any combination that totals 10,000 cells or less; for example, if you reduce the number of rows to 5, then you can increase the number of columns to 2,000 (5 x 2000=10,000).
  - If a form's precision is set, displayed values after conversion to Dashboard 2.0 are based on the form's precision setting. If the precision value on a form is not set, then the values displayed after conversion to Dashboard 2.0 might change.

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> **Note**
>
> When the data source for a dashboard component is a form, and the form has rows that contain blank header cells or empty formula rows that are not configured, then the corresponding chart legends will not display.

- For Doughnut, Pie, and Gauge chart types, you can add multiple charts to one component. To add multiple charts, click **Single** or **Multiple** in the chart's **Display** property. If **Single** is selected, the chart displays only the first column of the underlying form's data. If **Multiple** is selected, a chart is displayed for each column of the underlying form's data.

> **Note**
>
> For dashboards with Doughnut, Pie, and Gauge chart types that are converted from Dashboard 1.0 to Dashboard 2.0, the default option for the **Display** property is **Multiple**. For newly created Dashboard 2.0 dashboards, the default option for **Display** is **Single**.

More Helpful Information About Dashboard 2.0

- As you create a dashboard, click **Actions** and then select **Save and Run** to immediately see how the dashboard looks and works to a dashboard user. To return to designer mode to continue designing the dashboard, click **Actions** and then select **Edit** or **Edit without Data**.
- By default, missing or suppressed data is plotted as zeros. For selected chart types (Area, Bubble, Combination, Line, Radar, and Scatter), you can disable this setting by selecting the **Plot Missing Values as Zero** option in the chart's settings. When this option is disabled, the missing or suppressed data is ignored and is no longer plotted as zeros for those chart types.
- To review Dashboard 2.0 considerations, see Considerations for Dashboard 2.0.

Videos

Your Goal	Watch This Video
Learn how to create a dashboard with multiple components in the Dashboard Designer for Dashboard 2.0.	Creating Dashboards in Dashboards 2.0

Object Palette

On the left side of the Dashboard Designer is the object palette where you select the content for dashboard components. Simply drag and drop objects from the object palette to the workspace or search for content using the Search box.

The object palette can be resized. To resize the object palette, hover over the right side of the palette and drag it. To hide the palette, hover over the palette until you see the handle icon , then click it to hide the palette. Click (Handle) again to reopen the palette.

Content in the object palette is grouped into these categories:

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Table 12-4 Object Palette Content Categories

Icon	Description
	Cubes: Perform quick analysis by adding components of a cube to a dashboard without having a prebuilt form. See About Quick Analysis.
	Library: Select ad hoc and standard forms to include in the dashboard by navigating to the forms folders or by searching for them by name. • Click **List** to view forms in a **List View** or **Tree View**. • Click (Search) to type search keywords. The access permissions set for forms are honored in dashboards.
	Visualizations: Select a chart to include in the dashboard. To view data in a chart, you must select components of a cube or associate the chart with a form as its data source. When you link a chart to a form, users can immediately see the impact of changing data in the form on the associated charts.
	Other: • **Commentary**: Adds the comment you enter below the graph or chart and to the bottom of the **Chart** tab in the properties panel. **Commentary** includes a WYSIWYG text editor. • **URL**: Adds the URL you enter below the graph or chart and to the bottom of the **Chart** tab in the properties panel. **URL** includes a Reference URL link and the URL must be IFrame supported. Insert only external site URLs starting with the `https://` security protocol. Don't use internal or relative URLs or URLs for unconsenting third party sites such as google.com. Don't imbed into a dashboard direct URLs to other EPM artifacts such as forms and dashboards on either the same Oracle Fusion Cloud Enterprise Performance Management instance or to a different Cloud EPM instance. **Note:** If a URL added to a dashboard doesn't work in the runtime environment, right-click the URL and open the link in a new tab.

Dashboard Toolbar

In the upper right corner of the Dashboard Designer is the dashboard toolbar where you can perform these dashboard tasks:

Table 12-5 Dashboard Toolbar

Icon	Description
	The **Information** icon displays the folder in which the dashboard is located.
	The **Refresh Data** icon refreshes all data associated with the entire dashboard.

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Table 12-5 (Cont.) Dashboard Toolbar

Icon	Description
	The **POV Settings** icon performs these POV actions: • **POV : Hide Dimension Labels**: If your dashboard contains a global POV bar, select to hide the dimension names on the global POV bar. Clearing this option once again displays dimension names on the global POV bar. • **POV : Auto-Apply**: If your dashboard contains a global or local POV, enable this action to automatically apply changes to the POV. If you don't want to automatically apply changes, clear this option. • **POV : Clear All Page Selections**:
	The **Add** icon adds a new empty dashboard component to the design canvas (this is an alternative to drag and drop).
	The **Properties** icon hides and unhides the Properties Panel.
Actions menu	The **Actions** menu enables you to perform these actions: • **Revert to Last Saved**: Discards edits made since the last time the dashboard was saved • **Save and Run**: Saves any edits made and opens the dashboard in the runtime environment • **Create Form**: If you have the correct permissions, allows you to create a form associated with the dashboard component in the forms editor. The forms editor opens in a dynamic tab next to the current tab. • **Open Jobs**: Opens the Jobs console in a dynamic tab next to the current tab. • **Show Details**: Displays the underlying form details, including the form name, the folder path of the form, and the cube
Save	Clicking **Save** saves the dashboard definition.

Dashboard Workspace

The large area in the middle of the page between the object palette and Properties panel is the dashboard workspace. In addition to drag and drop, you can use the Search box to specify or search for the data you want to display for each component (a *component* is an empty object in the dashboard workspace into which you can add a dashboard object, such as a chart or a table). You can add up to 12 components to the dashboard workspace.

If the forms version is Forms 2.0, any forms dropped into the workspace are rendered initially as editable grids. Ad hoc grids dropped into the workspace are rendered initially as tables.

- The first object that you drag and drop occupies the entire workspace, then drag additional objects from the object palette to the left, the right, the top, or the bottom of the existing component.
- To move a component to another part of the dashboard, click the component, and then hover your cursor over the upper edge of the component until you see the drag icon. Drag and drop the component to the left, right, top, or bottom of another component and the other components on the dashboard will move and realign.
- To maximize a component so it fills the workspace, hover over the component, click (Actions), and then click **Maximize**. The component will fill the workspace, and the other

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components on the workspace will be hidden. Clicking ••• (Actions), and then **Restore** resizes the component to its original size and makes the other components in the workspace visible again.

> **Note**
>
> The **Maximize** option only enlarges the dashboard component. If the dashboard component is a grid and you want to perform grid actions on the component such as **Adjust**, **Grid Spread**, **Comment**, and so on, select **Open Form** (in the **Actions** menu) to open the underlying form associated with the component. The form will open in a dynamic tab next to the current tab.

- To change the height and the width of components in the dashboard, perform one of the following actions:
  - Drag the borders of the components within the dashboard workspace
  - Select a dashboard component, then in the Properties panel on the right, click (Charts) to view and change the component's height and width.
- To maximize the overall size of the dashboard workspace, you can resize or hide the object palette and properties panel on either side of the workspace.
- For a dashboard with a URL component, clicking on the body of the URL component does not switch focus to the URL component. To switch focus to the URL component, either click Alt + W or hover over the URL title bar until you see the cursor change to the **Move** icon , then click to change the focus.

> **Note**
>
> In the runtime environment, if a user doesn't have access to an object like a form, or if the form is missing, then an adjacent component will take its space. In the designer environment, all empty components are displayed so that the designer can choose to remove them.

Properties Panel

On the right side of the Dashboard Designer is the Properties panel where you can set general properties for the entire dashboard as well as component-specific properties.

The Properties panel can be resized. To resize the panel, hover over the left side of the panel and drag it. To hide the panel, click (Properties) at the top of the page. Click (Properties) again to reopen the panel.

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Table 12-6 Properties Panel

Icon	Description
	The **General** tab includes the following properties for the entire dashboard: • **Name**: Edit the title of the entire dashboard. • **Description** (optional): Describe the dashboard. • **POV Bars**: Show or hide POV bars. • **Global POV Bar**: Enable or disable the global POV bar. • **Style**: Select the dashboard color scheme. Options are **Default**, **Light**, and **Dark**. **Note:** If you select **Dark**, dashboard error messages and legend text will be more difficult to view. • **Layout**: Options are **Scrollable**, which enables vertical scrolling while viewing a dashboard, and **Auto Fit**, which is the default. Note that scrolling can only be selected when creating a new dashboard, and the limit for components on dashboards with **Scrollable** selected is 16, and with **Auto Fit** is 12. • **Borders**: Show or hide component borders. • **Background**: Select a new dashboard background from a menu of colors. **Note:** If your dashboard contains a grid, only the background color will change. The grid color will remain the same. • **Use Name as Title**: Selecting **No** lets you customize the dashboard's title. You can further customize the title by changing the font family, font size, bolding, italicizing, underlining, font color, and font background color. • **Instructions**: Enable or disable instructions for the entire dashboard. Choose **Enable** to enter instructional text, format it, and then display it to runtime dashboard viewers. Note that the text limit for instructions is 2000 characters. The instructions can be viewed by clicking (Instructions) in the runtime dashboard's global toolbar. Choose **Disable** to hide instructions. **Note:** To add instructions for individual dashboard components, open the dashboard component and edit it. For information about POVs, see About Global and Local POVs in Dashboard 2.0.

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Table 12-6 (Cont.) Properties Panel

Icon	Description
	The **Associations** tab lists the rules associated with the dashboard and the underlying form data. **Note:** You can associate rules which are required to be run during a global **Save**. These rules can be associated or removed at any time. You can add rules and edit them: • If no rules are associated, click (Add/Edit business rules) to view a list of rules from which to choose. Rules already associated with forms in the dashboard are listed first and are named as follows: Business Rules for <name of form> • When you add rules, you'll need to specify when the rule should run: **Before Load**, **After Load**, **Before Save**, **After Save**, **Use Members on Form**, **Hide Prompt**. • Under **Actions**, you can delete rules or change the order in which they will run: **Move to Top**, **Move Up**, **Move Down**, **Move to Bottom**, and **Delete**. • If the rules have runtime prompts, they will be displayed when the rule is run. See Setting Business Rule Properties.
	The **Chart** tab includes properties for the selected components in the dashboard such as chart title, component height, width, and background color. You can also change the chart type, the data source type, and which underlying form is used for each component. Using different background colors for each component enables you to uniquely differentiate each dashboard component. Removing the chart title frees space that is then adjusted for the local POV bar and for the chart itself. For rows and columns, the total amount of form data that's allowed to be visualized in a chart is 10,000 cells. By default, this is 100 rows and 100 columns. However, you can change the number of rows and columns to any combination where the multiplication totals 10,000 cells or less; for example, if you reduce the number of rows to 5, then you can increase the number of columns to 2,000 (5 x 2000=10,000). **Note:** The 10,000 cell visualization limit does not apply to grids. If the form associated with a chart exceeds the 10,000 cell visualization limit, it might result in discrepancies between the data you see in the form and the way the data is visualized in the dashboard chart. It also might result in some other runtime issues such as drilling through on charts and legends. If you've added **Commentary** or **URL** information types to this component, you can edit the content of those information types at the bottom of this tab.

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Table 12-6 (Cont.) Properties Panel

Icon	Description
(this icon will change depending on the component that's selected in the dashboard)	This tab lets you customize visualizations for the selected chart or table. Set display properties for the selected components such as how to scale values, show or hide row dividers or grid lines, choose different font sizes and colors for headers and data, set orientation vertically or horizontally, define legend positions, add spark charts to tables, add a secondary Y-axis, and so on. See the chart type topics for detailed property descriptions. **Note:** For data visualizations in Dashboard 2.0, the custom color palette now supports only 12 colors instead of 15. If you created a dashboard in an earlier update that used 15 colors, you will continue to see 15 colors at runtime. However when you try to change the custom colors, the dashboard designer will only display 12 colors from which you can select and save.
	The **Layout** tab includes the **POV**, **Row**, and **Column** properties for the selected cube for quick analysis. See About Quick Analysis.

Dashboard Components

You can add up to 12 components to the dashboard workspace.

When you hover over a component in the workspace, a toolbar for that component displays in the upper right corner:

Table 12-7 Dashboard Components Toolbar

Icon	Description
	Instructions: Displays instructions for the form associated with the dashboard component (only available if the form designer added instructions to the form)
	Save: Saves any changed data and runs any rules associated with the dashboard component
	Refresh: Refreshes the data associated with the dashboard component
	Chart Type: Changes the chart type associated with the dashboard component

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Table 12-7 (Cont.) Dashboard Components Toolbar

Icon	Description
Actions menu	**Actions** menu: • **Save and Run**: Saves any edits made and opens the dashboard in the runtime environment • **Maximize**: Enlarges the dashboard component. Once maximized, click **Restore** to minimize the dashboard component to its original size. **Note:** The **Maximize** option only enlarges the dashboard component. If the dashboard component is a grid and you want to perform grid actions on the component such as **Adjust**, **Grid Spread**, **Comment**, and so on, select **Open Form** (in the **Actions** menu) to open the underlying form associated with the component. The form will open in a dynamic tab next to the current tab. • **Rules & Smart Push**: Lists any rules associated with the form. Click the rules to launch them. Analyze data coming from different cubes with Smart Push. • **Analyze**: Perform ad hoc analysis on a form. This menu option only displays if the chart type is a grid and if the data source is a form. See Get Started Performing Ad Hoc in *Working with FreeForm*. • **Open in Smart View**: Work with a form in Oracle Smart View for Office. This menu option only displays if the chart type is a grid and if the data source is a form. See Opening Forms and Ad Hoc Grids from Web Applications in *Smart View for Office User's Guide 23.100*. • **Spreadsheet Export**: Export data from a form to Microsoft Excel so you can explore "what-if" scenarios in Excel before copying and pasting values back to your business process. This menu option only displays if the chart type is a grid and if the data source is a form. See Exporting Data to Microsoft Excel in *Working with FreeForm*. • **Open Form**: Opens the runtime form associated with the dashboard component in a dynamic tab next to the current tab • **Edit Form**: If you have the correct permissions, opens the form associated with the dashboard component in the forms editor. The forms editor opens in a dynamic tab next to the current tab. • **Create Form**: If you have the correct permissions, allows you to create a form associated with the dashboard component in the forms editor. The forms editor opens in a dynamic tab next to the current tab. • **Show Details**: Displays the underlying form details, including the form name, the folder path of the form, and the cube • **Open Jobs**: Opens the Jobs console in a dynamic tab next to the current tab.

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Considerations for Dashboard 2.0

> **Note**
>
> Starting with the October 2025 (25.10) update, Oracle plans to de-support Dashboard 1.0. All Oracle Fusion Cloud Enterprise Performance Management environments will default to Dashboard 2.0 with existing environments created before October 3, 2025 having the option to switch back to Dashboard 1.0. All new environments created on or after October 3, 2025 will be deployed with Dashboard 2.0 only.

Note the following considerations when using Dashboard 2.0:

- When viewing dashboards in Dashboard 2.0, data for each form is visualized based on the data type of the form's first row, first cell. For example, if the first row, first cell in a form uses a percentage data type, then the data displayed for that form in Dashboard 2.0 is visualized as a percentage. If another form in the same dashboard uses a numeric data type in the first row, first cell, then the data will display as a decimal.
- For charts, each column must have a unique name in order for the data to be rendered properly.
- In some Dashboard 2.0 charts (such as Line and Area charts), data markers are defaulting to always display. In a future update, dashboard designers will have the ability to control the visibility of data markers.
- 3D visualization is not supported in Dashboard 2.0. This affects the following Dashboard 2.0 chart types:
  - Area
  - Bar
  - Bubble
  - Column
  - Combination
  - Doughnut
  - Gauge
  - Grid
  - Line
  - Radar
  - Scatter

  > **Note**
  >
  > Dashboard 1.0 still supports 3D visualizations for charts, as applicable.

- In a future update, Dashboard 2.0 will support Smart Push.
- Funnel charts are useful for viewing data for stages of a process, such as the stages of a sales process. The area of a funnel slice is proportional to its value for the corresponding

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stage. Dashboard 1.0 renders the funnel chart very differently than the way in which Dashboard 2.0 renders the funnel chart. In Dashboard 1.0, the funnel chart supports multiple columns, and the first two rows of the form are used to compare Actual with Target. The chart then plots the differences between the two in the funnel corresponding to the respective column. In Dashboard 2.0 the funnel chart supports only the first column and the area of the funnel corresponding to each cell on the row is colored and is sized proportionally to its numeric value.

- When accessing ad hoc grids from Dashboard 2.0 chart types:
  - **Analyze** only works for chart elements built on a form.
  - **Analyze** is not supported for Geomap chart types or for Quick Analysis charts.
  - For charts built on a form with a formula row or column, after right-clicking the chart element which corresponds to the formula row/column (like a bar in a Bar chart), clicking **Analyze** will not perform any action.
- All future dashboard enhancements will be in Dashboard 2.0.

Forms Versions and Dashboard 2.0

Runtime forms are also available in two versions:

- **Forms 1.0**: Uses Slick Grid with Oracle ADF Fuse components
- **Forms 2.0**: Uses Slick Grid with Oracle JET components

Certain features in Dashboard 2.0 only work with Forms 2.0; for example, Dashboard 2.0 only lets you work directly with data in form grids if the underlying form is a Forms 2.0 form. The **Forms Version** can be updated in your application settings.

> **Note**
>
> Starting with the October 2025 (25.10) update, Oracle plans to de-support Forms 1.0. Users on new environments created on or after October 3, 2025 will see all their forms rendered exclusively in Forms 2.0, with no option in application settings to switch to Forms 1.0. Existing environment users will also see their forms rendered in Forms 2.0 by default. However, users on environments created before October 3, 2025 will have the option to switch back to Forms 1.0 using the **Forms Version** application setting, if needed.

You'll need to be aware of some differences in the way the dashboard versions interact with the forms versions.

Forms Version 1.0

When the **Forms Version** application setting is set to **Forms 1.0**:

- You can continue to use Dashboard 1.0 with Forms 1.0.
- You cannot use Dashboard 2.0 with Forms 1.0.

  Forms 2.0 is supported with Dashboard 2.0 only. Use Forms 2.0 with Dashboard 2.0 for all interactions.

  If you've selected a Dashboard 2.0 dashboard and you see an `Error during Dashboard 2.0 component initialization` message, the dashboard has Forms 2.0 forms associated with it and you'll need to ensure that Forms 2.0 is enabled before you can view the dashboard.

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- When Dashboard 2.0 artifacts are created, the **Grid** visualization option will not be available.
- To use forms as grids in Dashboard 2.0, update the **Forms Version** application setting to Forms 2.0.
- Existing Dashboard 2.0 dashboards (Dashboard 2.0 dashboards created before November 2023) will continue to display forms as tables and charts, not grids.

Forms Version 2.0

When the **Forms Version** application setting is set to **Forms 2.0**:

- You can continue to use Dashboard 1.0 with Forms 2.0.
- Dashboard 2.0 will display the **Grid** visualization option, which uses Forms 2.0 with read, write, and calculate abilities.
- Forms 2.0 within Dashboard 1.0 will appear as Forms 1.0.

> **Note**
>
> Saved forms definitions can be used as a data source for dashboard charts regardless of the **Forms Version** application setting: **Forms 1.0** or **Forms 2.0**.

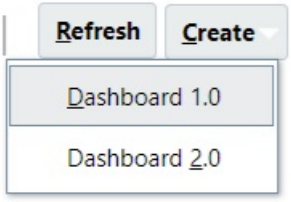
Creating Dashboard 2.0 Dashboards

Need help deciding which dashboard version to choose? See About Dashboard Versions.

> **Note**
>
> To create a Dashboard 1.0 dashboard, see Creating Dashboard 1.0 Dashboards.

To create a Dashboard 2.0 dashboard:

1. From the Home page, click **Dashboards**, and then click **Create**.



2. From the options provided, select **Dashboard 2.0**.

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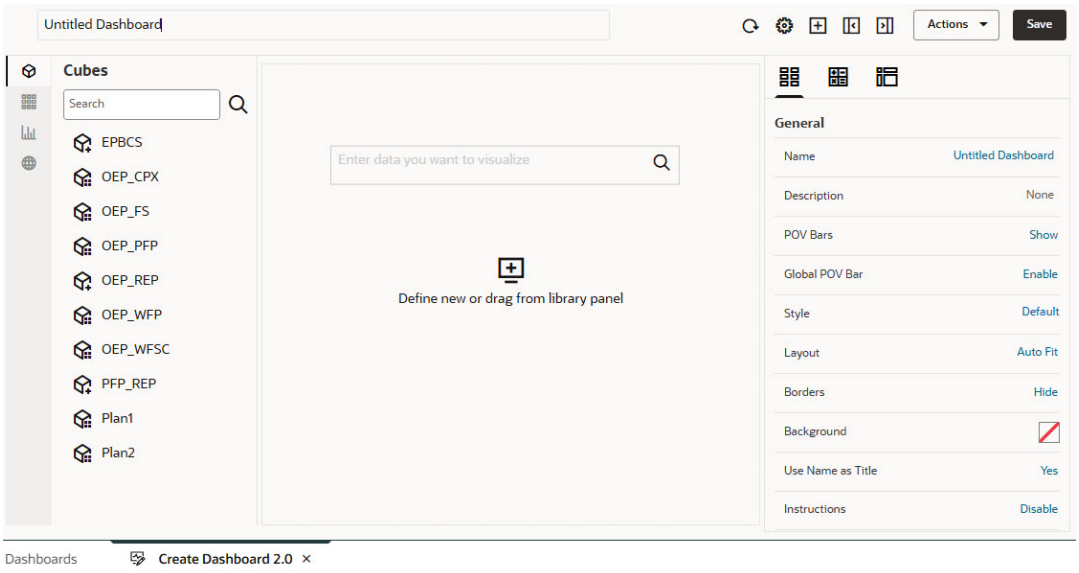

3. Enter a name for the dashboard by clicking the default dashboard name and entering a new name in the input box.
4. Drag and drop content from the object palette on the left onto the dashboard workspace. Customize the dashboard using the dashboard settings and properties.

   For a detailed description of the Dashboard Designer, see Concepts in Designing Dashboard 2.0 Dashboards.

   While working on the dashboard, you can save it as a new dashboard, including any unsaved changes. To do so, click **Actions**, then **Save As,** and then save the dashboard with the new name.
5. Click **Save**.

You can easily duplicate and modify a dashboard using **Copy As** on the **Dashboards** listing page. Select **Actions** ••• next to the dashboard you want to copy.

Working with the Member Selector When Designing Dashboards 2.0

When designing Dashboard 2.0 dashboards, you can easily search for and select new dimension members using the member selector.

When you're creating a Quick Analysis in your dashboard, after you add a cube, you can select members for each dimension using the member selector.

To open the member selector for Quick Analysis:

Click the member selector icon in the Quick Search bar:

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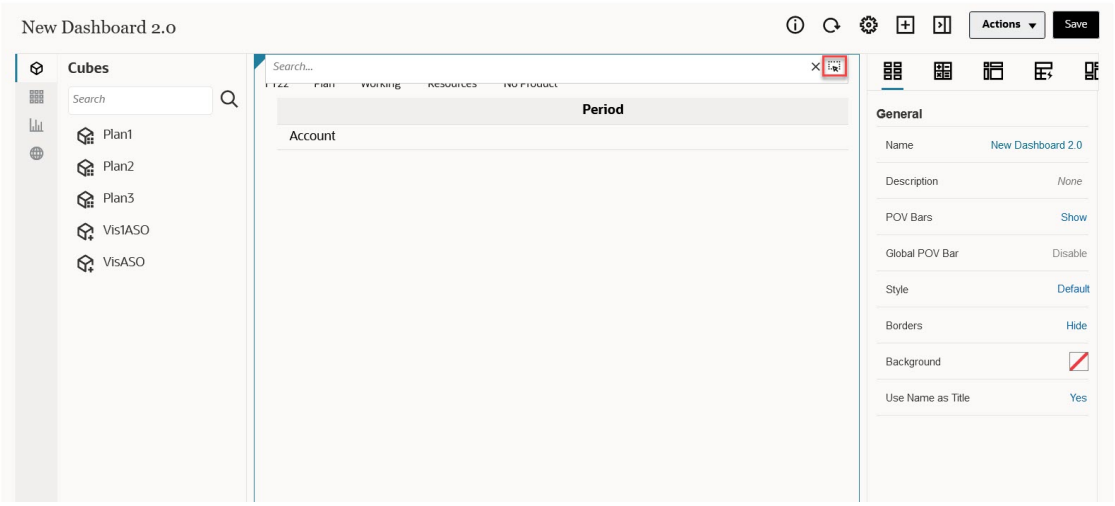
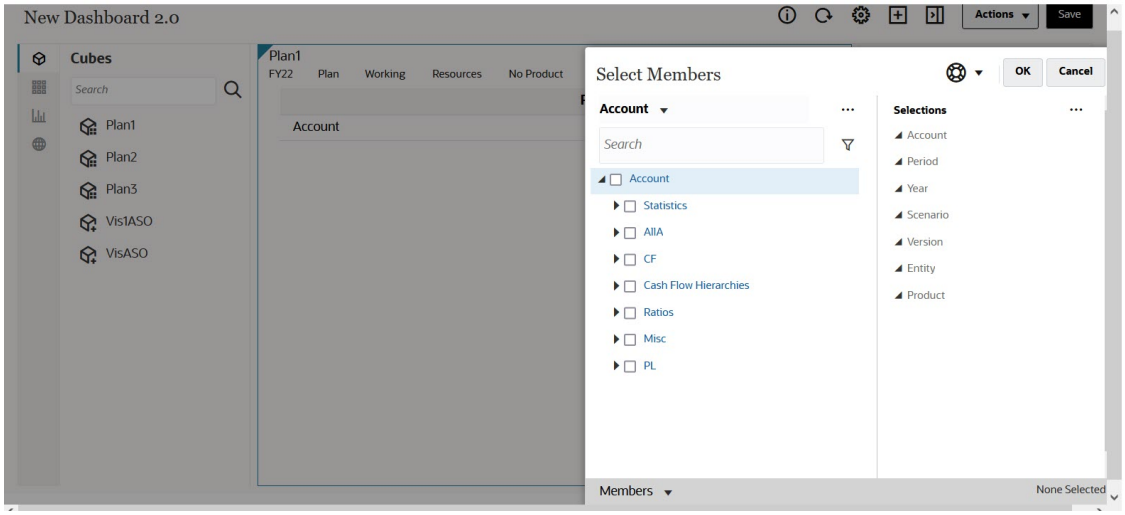


Select members for each dimension using the member selector.



For more information about working with the member selector, see Using the Member Selector.

About Forms 2.0 Grids in Dashboard 2.0

Newly added components in Dashboard 2.0 default to displaying the form grid, and the grids are writable if the associated form is a Forms 2.0 form. Grids in dashboards enable you to work directly with data in forms if the underlying form is a Forms 2.0 form. You can write-back data updates and invoke rules in dashboards with grid components. You can also analyze form data, work with a form in Oracle Smart View for Office, and export data from a form to a spreadsheet.

Right-clicking inside a grid enables you to view a context menu where you can perform additional grid actions such as grid spread, and adding comments and attachments.

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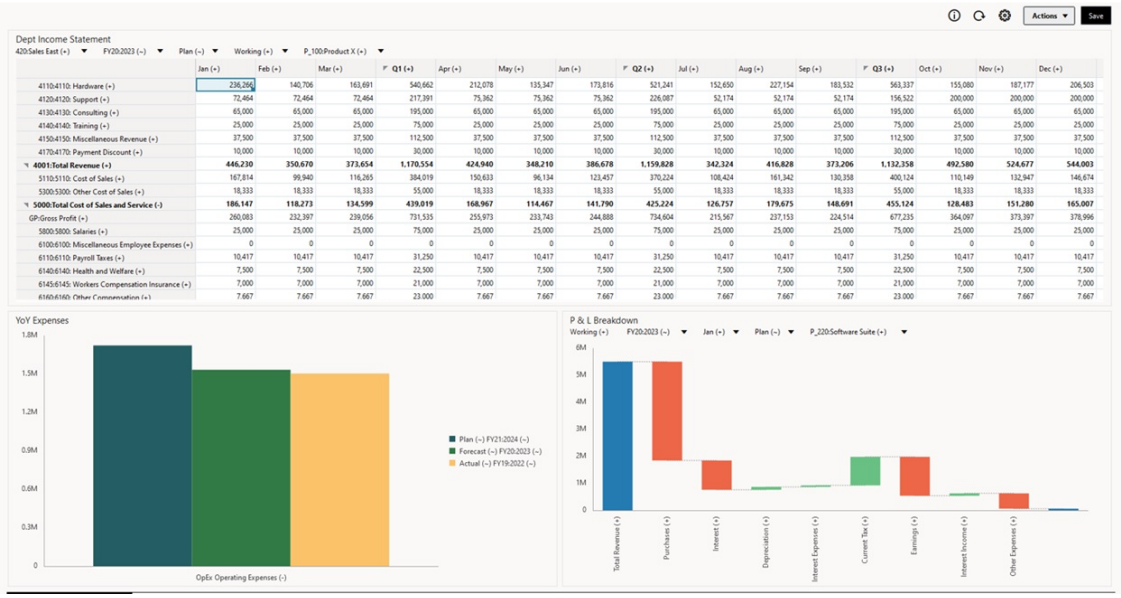
Example Dashboard with a Grid and Charts



Using Tables in Dashboards

Tables in dashboards provide a list view of the data showing multiple values from a form or grid. You can embed mini-charts into each row of the table, called Spark Charts, which plot column data. Tables are also searchable.

Example Dashboard Showing Plain Tables

Balance Sheet - Plan

	Q1	Q2	Q3	Q4
Current Assets	535 M	557 M	588 M	625 M
Fixed Assets	54 M	50 M	45 M	41 M
Other Assets Total	223 M	223 M	223 M	223 M
Total Assets	812 M	830 M	857 M	890 M
Current Liabilities	335 M	336 M	337 M	345 M
Long Term Liabilities	44 K	44 K	44 K	44 K
Total Owners Equity	477 M	494 M	519 M	545 M
Total Liabilities and Owners Equity	812 M	830 M	857 M	890 M

Balance Sheet - Forecast

	FY16 | Q3	FY16 | Q4	FY17 | Q1	FY17 | Q2
Current Assets	479.40 M	515.61 M	488.15 M	500.23 M
Fixed Assets	61.17 M	59.33 M	53.98 M	50.52 M
Other Assets Total	223.09 M	223.09 M	223.09 M	223.09 M
Total Assets	763.66 M	798.02 M	765.21 M	773.85 M
Current Liabilities	331.48 M	338.24 M	333.92 M	335.93 M
Long Term Liabilities	43.77 K	43.77 K	43.77 K	43.77 K
Total Owners Equity	432.07 M	459.66 M	431.25 M	437.87 M
Total Liabilities and Owners Equity	763.59 M	797.94 M	765.21 M	773.85 M

Example Dashboard Showing a Table With a Spark Chart

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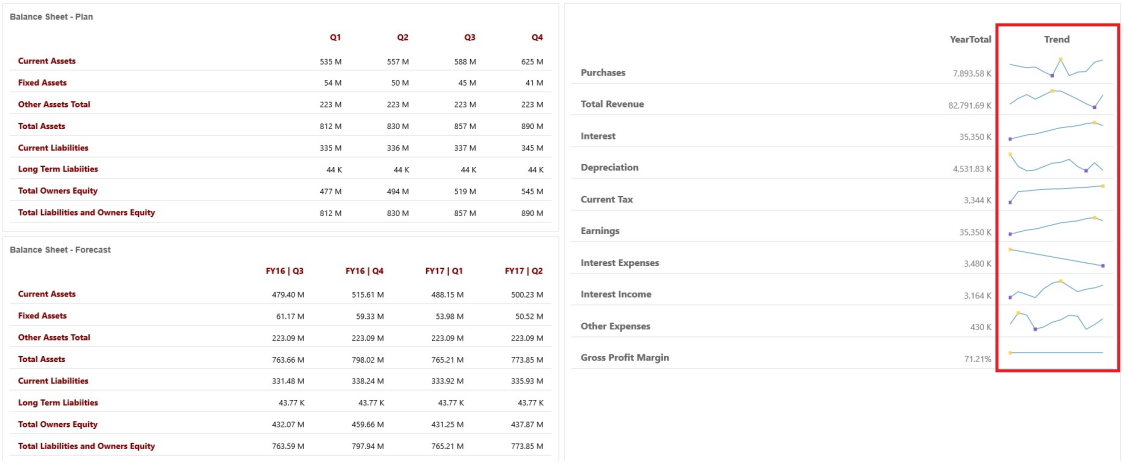

Dashboard designers can set these table properties:

Table 12-8 Table Properties

Setting	Description
Value Scale	Especially useful for large numbers, you can scale how a value is displayed. For example, if the value is 1,689,000 and you select **K - Thousand** as the scaling option, the table displays the value as 1,689K. Your scaling options: • **None**: No scaling is applied. • **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T. • **K - Thousand**: The value is displayed as thousands units. For example, 1,689,000 displays as 1689K. • **M - Million**: The value is displayed as millions units. For example, 123,456,789 displays as 123M. • **B - Billion**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B. • **T - Trillion**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.
Row Divider	Choose **Show** or **Hide**.
Font	Specify size and color options for headers and data.
Spark Chart	• **Chart Type**: Select **Bar**, **Line**, **Area**, and **Line with Area** with associated options. Or you can select **None**. • **Header**: Column header for the Spark Chart column • **Starting Index**: Column index where to start charting data • **Ending Index**: Column index where to stop charting data

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Videos

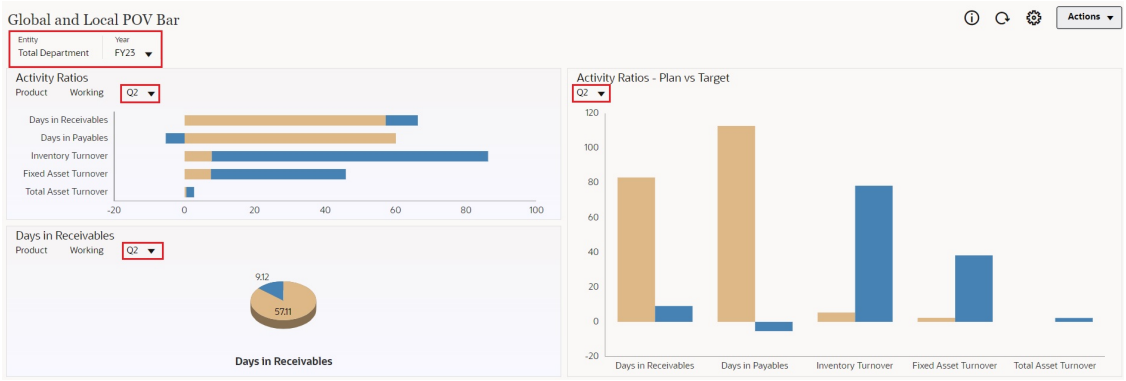
Your Goal	Watch This Video
See how to create a dashboard with multiple components in the Dashboard Designer for Dashboard 2.0. You create a dashboard with a table, a table with a spark chart, and a tile chart.	Creating Table, Spark, and Tile Charts in Dashboards 2.0

About Global and Local POVs in Dashboard 2.0

A local POV on a form reflects the dimension members the form designer selected for that form. Dashboards also support *global POV bars*, so that the local POVs that are common are combined in the global POV bar to avoid repeating them in each component.

Dashboard 2.0 POVs Example

This Dashboard 2.0 dashboard shows a global POV bar (showing Entity and Year) and local POVs (the drop-down lists for Q2):



With a global POV bar, if you change a page in the global POV bar and **POV Bar : Auto-Apply** is enabled, the page changes for all components in the dashboard that are based on forms. The global POV bar displays at the top of the dashboard above all the components, while the local POV bar displays within the component. User variables are supported in both global and local POVs in dashboards and you can change dynamic user variables directly from the POV bar.

In the dashboard toolbar **POV Settings** menu (in both the runtime and edit modes), you can set whether to hide dimension labels in the global POV (**POV Bar : Hide Dimension Labels**) and whether to enable or disable automatically applying changes (**POV Bar : Auto-Apply**).

About global POV bars:

- The POV Bar is made up of local POV dimensions, Page dimensions, and user variables.
- Depending on the local POV and page of each form on the dashboard, the global POV bar is automatically calculated.
- Global POVs reflect the content of the components using forms in that dashboard. That is, they apply to forms in a dashboard, to charts that are linked to forms, and to tiles that use

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forms as a data source. So if the dashboard doesn't include a form as a data source, then neither the local nor global POV bar is available.

Here's an example of how the global POV bar is calculated, based on the local POV dimensions for two forms:

The global POV bar is disabled:

- Form A local POV: Year, Entity, Product
- Form B local POV: Year, Entity, Project

The global POV bar is enabled:

- Global POV bar: Year, Entity
- Form A local POV: Product
- Form B local POV: Project

Because not all dimensions and page selections may be common to all forms on a dashboard, the complete POV for a dashboard component may get split between the local POV and the global POV bar. The global POV bar and the local POV together contain the complete intersection information for each form on a dashboard.

If there is only one component on the dashboard that uses a form as a data source, then the entire POV/page of the form is moved to the global POV bar.

If there is more than one component on a dashboard that uses forms as a data source, then this is how the application determines which dimensions go in the global POV bar or stay in the local POV:

- If the dimension is in the POV or page of all the forms, and the member selection is the same in all the forms, the dimension goes in the global POV bar.
- If the dimension is in the POV on one form and in the page of another form, then the dimension stays in the local POV.
- If the dimension is in the POV, then the same members must be selected in all the forms for the dimension.
- If the dimension is a page dimension, then the selected page members must be the same and display in the same order in all the forms.

Searching for POV Members in Dashboard 2.0 Dashboards

When working in Dashboard 2.0 dashboards, you can easily find new POV members. To search, click a POV link. The search interface displays a list of the most recently used members from which you can choose.

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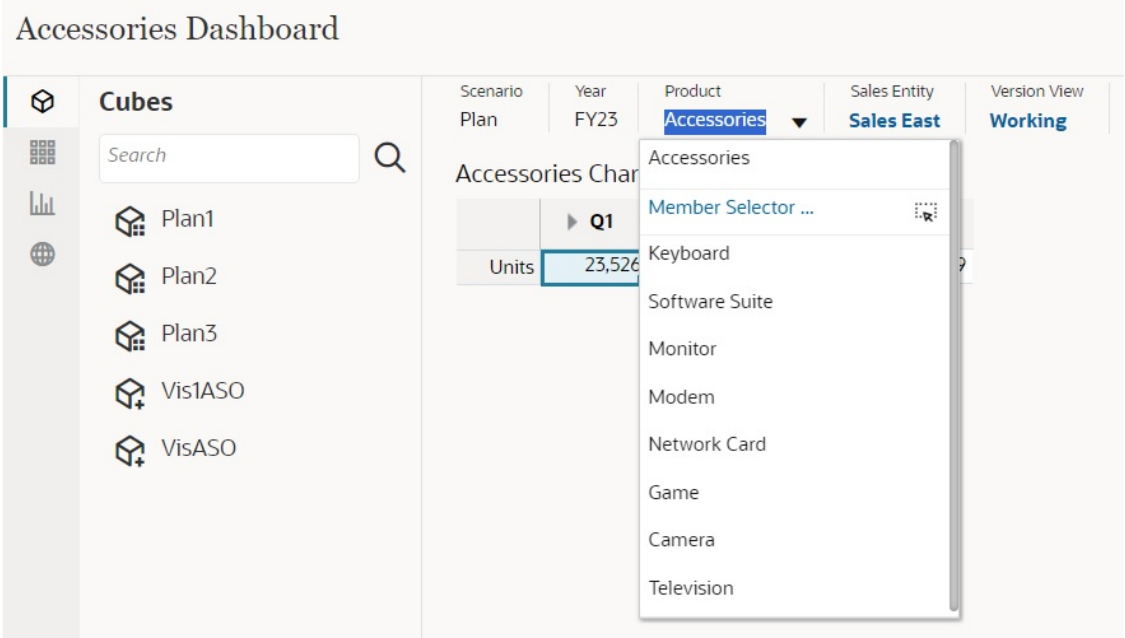
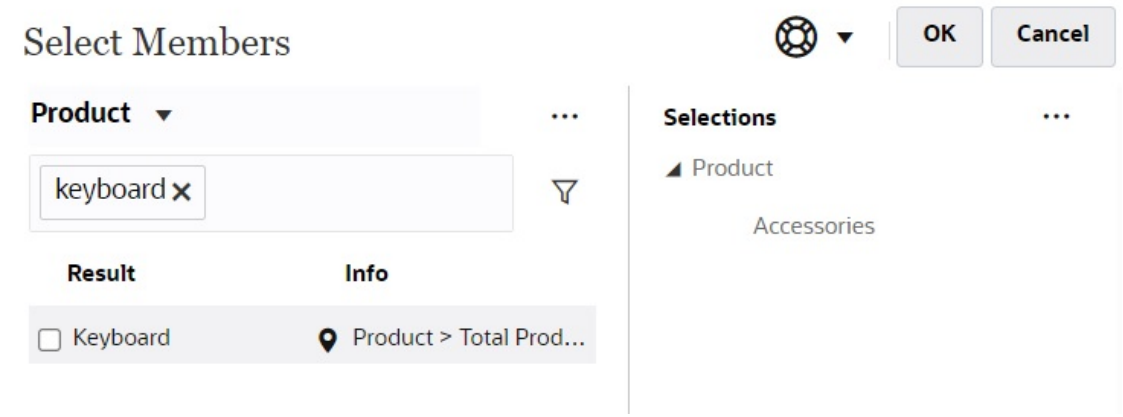
Figure 12-1 Dashboard 2.0 POV Most Recently Used Members List



You also can search using the member selector. Click a POV link, then click the member selector icon to launch the member selector. Members in the member selector are displayed as a hierarchy or you can type a member name in the **Search** box. If a member is located using the **Search** box, you'll see a result displayed under the Search box.

Figure 12-2 Dashboard 2.0 Member Selector Search Results



You can view where the member is located within the hierarchy by hovering over the result's **Info** column and clicking (Locate Member).

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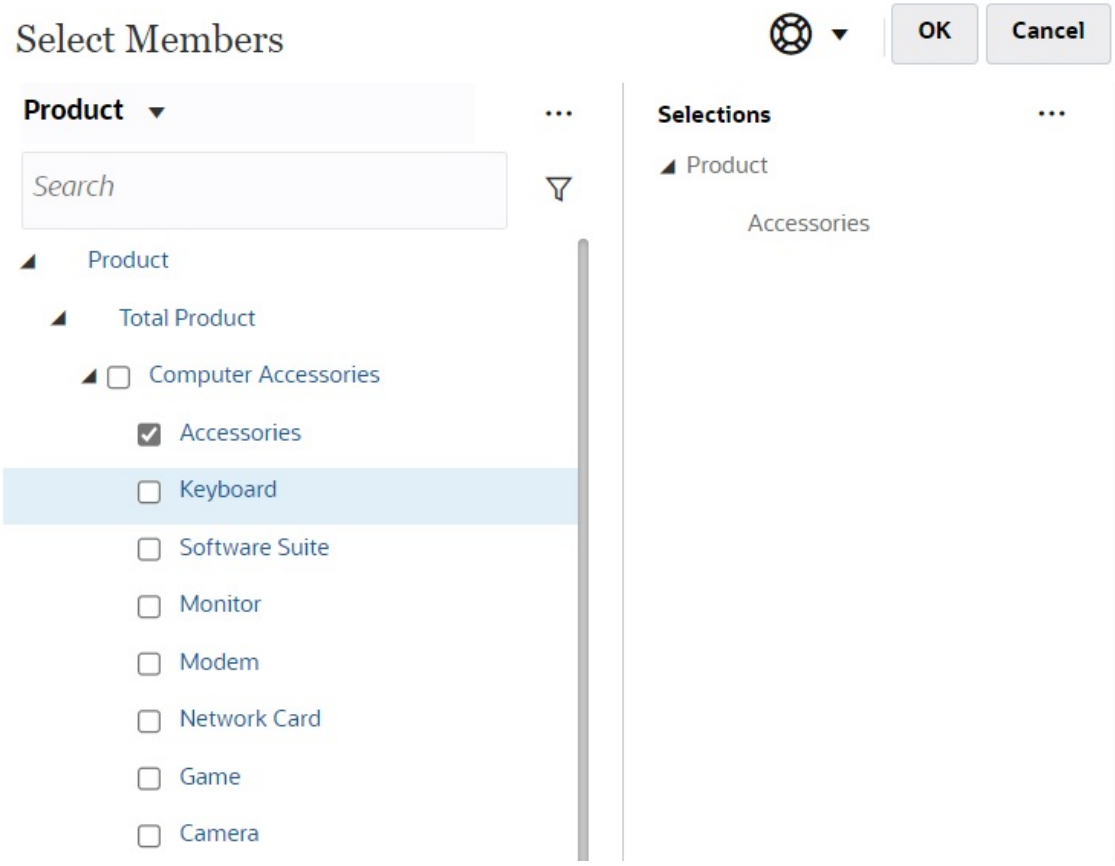
Figure 12-3 Dashboard 2.0 Member Selector Search Results in Hierarchy



Dashboard 2.0 POV Bar Considerations

- For POV bar Page drop-down lists, the number of members shown in the page drop-down is controlled in **User Preferences** under **Page Options**. (From the Home page, click **Tools**, then **User Preferences**, and then click **Display** to view the **Page Options**.) If the number of items in the page drop-down exceeds this limit, then users cannot view them. Note that this will be fixed in a future release with the ability to use the member selector to browse through the rest of the members, similar to Dashboards 1.0.
- If your dashboard includes multiple forms, but some forms have different properties for the same dimension; for example, the Version dimension is hidden on some forms but it's visible on other forms in the same dashboard, the global POV might not display any Version dimension members. We recommend you ensure that all properties for all dimensions are the same for all forms included within a dashboard.

About Quick Analysis

Quick Analysis lets you add components to dashboards without having a prebuilt form. You create an ad hoc query by typing member names in the Search bar or using the Member Selector, and then you control the layout of the query in the Layout tab of the Properties panel.

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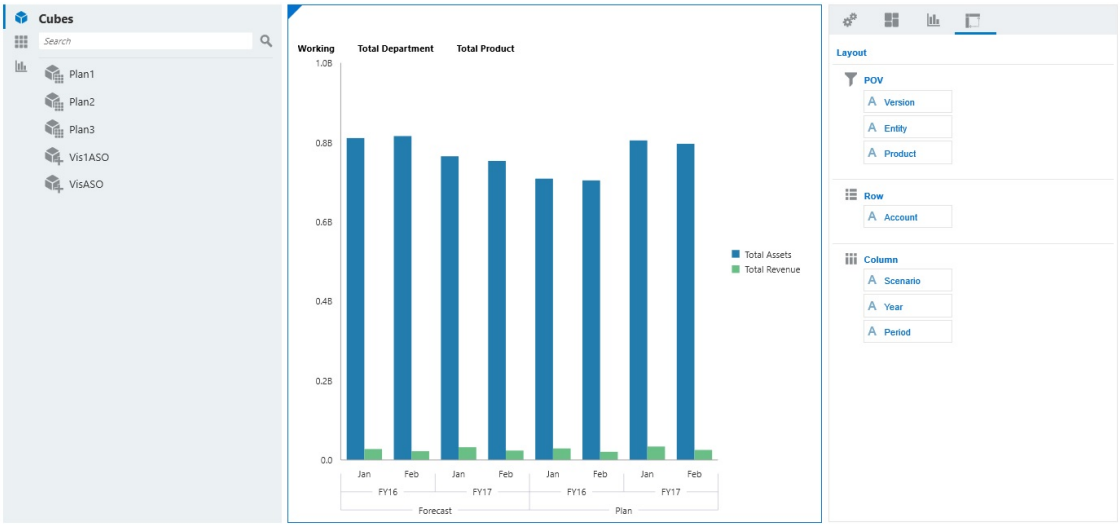
Example Dashboard 2.0 Quick Analysis Chart in Design Mode



To create a quick analysis in your dashboard:

1. Select a cube for the query. You can do this in one of two ways:
   - Click in the workspace Search box to display a list of cubes, then select a cube.
   - Drag and drop a cube from the object palette to the dashboard workspace.

> **Note**
>
> With either method, the system starts with a default query; a single intersection in the cube driven from your Most Recently Used, or MRU. The default query varies based on what you've recently been looking at in the cube.
>
> If the MRU intersection in the cube is invalid, you'll get an error. If you see an error, open another form and change the page to a valid intersection so that the MRU gets updated with a valid intersection.

2. Refine the query by specifying members for the query:
   - In the Search box, type member names to display an auto-complete list from the cube.
   - Launch the member selector to select members, and then click **OK**.

> **Note**
>
> When using the Search box to find members for quick analysis and more than 25,000 members are present in a cube, the Search box only searches the first 3000 members in the dimension for the cube. If the cube contains less than 25,000 members, then the Search box will search through all members. To search a cube with more than 25,000 members, launch the member selector to make your quick analysis selections.

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3. Select a chart type for the query.

   Click (Chart Type) to select a chart type; for example, Bar .

4. In the Properties panel, click the Layout tab .

5. Drag and drop the dimensions in the Layout tab of the Properties panel to other sections of the Layout tab to change the POV, Row, and Column layout of the dashboard component.

Videos

Your Goal	Watch This Video
Discover how to add quick analysis components to dashboards without having a prebuilt form. You learn how to create an ad hoc query in the Dashboard Designer and format the results by changing the dimension layout and applying visualizations.	Creating Quick Analyses in Dashboards 2.0

About Dashboard Charts

Dashboard charts are the visual elements that make up a dashboard. They take complex data and translate it into easily understandable visuals like:

- **Bar Charts:** Used to compare data across different categories or to show changes over time.
- **Line Charts:** Used to display trends and patterns over time.
- **Pie Charts:** Used to show the proportion of different parts to a whole.
- **Other types:** There are many other specialized charts like geomap, tile, gauge, radar, and combination charts, each with their own specific uses.

See the following topics for chart descriptions:

Chart Type	Icon	Want to know more?
Geomap		See About the Geomap Chart Type.
Pyramid		See About the Pyramid Chart Type.
Waterfall		See About the Waterfall Chart Type.
Gauge		See About the Gauge Chart Type for Dashboard 2.0.
Radar		See About the Radar Chart Type for Dashboard 2.0.
Combination		See About the Combination Chart Type for Dashboard 2.0.
Tile	50	See About the Tile Chart Type for Dashboard 2.0.
Sankey	Sankey	See About the Sankey Chart Type for Dashboard 2.0.

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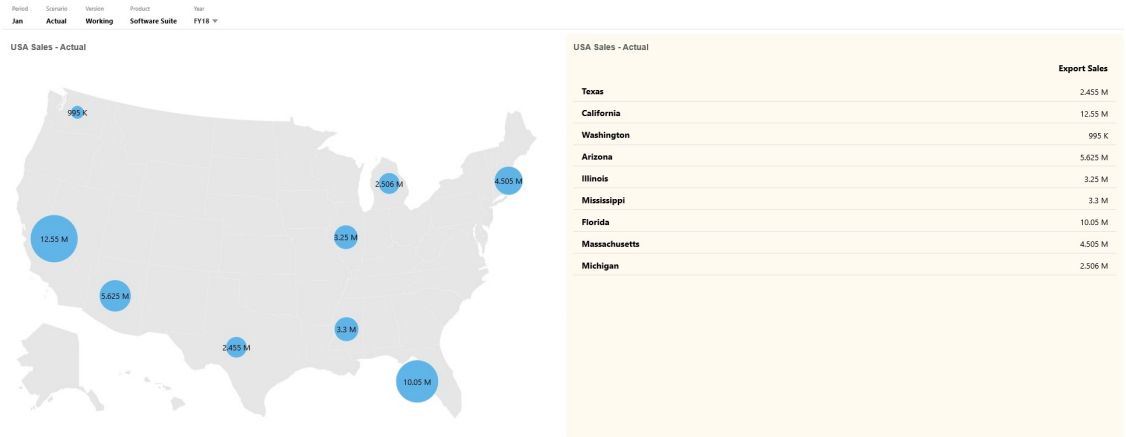
About the Geomap Chart Type

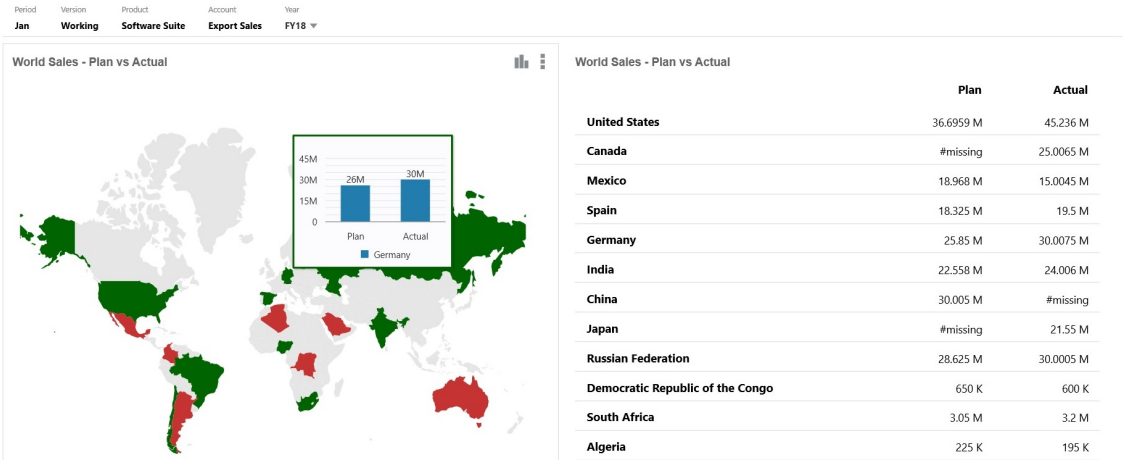
Geomaps are geographical maps with overlays and charts that display some measure about each of the geographical areas that are displayed. Hovering over the bubbles or the color highlight in the Geomap displays a data label.

> **Note**
>
> Color highlight geomaps are designed to show any variances (red or green, depending upon your **Desired Variance Trend** setting) based on the values in the underlying data. If a geomap region has missing values in the 1st or 2nd column or if the values in those columns are equal and there is no variance, the region will be highlighted blue. Hovering over a highlighted region displays a bar chart with all the values for the row except for any missing values.

Example Geomap - USA Sales Using Data Bubbles



Example Geomap - World Sales Using Color Highlight



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Dashboard designers can set these geomap chart properties:

Table 12-9 Geomap Chart Properties

Setting	Description
Map Layer	Select **World** or narrow the scope of the map to a continent such as **Asia**, **North America**, or **Africa**.
Type	The **Data Bubbles** geomap displays data as sized bubbles over the map layer. The **Color Highlight** geomap compares two data values, displaying geographical areas that have an increase in green, and those with a decrease in red. Select **Data Bubbles** or **Color Highlight** and then specify the following additional options: • For **Data Bubbles**, select the desired settings for **Label** (**Show** or **Hide**), **Label Color**, and **Bubble Color**. • For **Color Highlight**, specify the **Desired Variance Trend** (**Increase** or **Decrease**).
Geography Names	We've provided you with a metadata load file that contains the geographical elements the geomap uses like countries, states, and cities. Click to download a CSV file which you can then upload into a dimension as new members or (with some manual changes) as aliases to existing members in your application. Note that geomaps only work if the geography names in the forms match exactly the geography names that Dashboard 2.0 uses for its map layers. So if your entity/geography dimension says, for example, USA, that won't be recognized as a valid name in the map layer, which uses United States.

Videos

Your Goal	Watch This Video
Learn how to download geography members for geomap charts, then import them into your dimension. You learn how to set up both data bubble and color highlight geomap charts.	Creating Geomaps in Dashboards 2.0

About the Pyramid Chart Type

Pyramid chart types use a triangle diagram to represent hierarchies visually. The triangle is divided into sections, illustrating vertically top-down and broad-narrow structures and processes.

Example 3D Pyramid Chart

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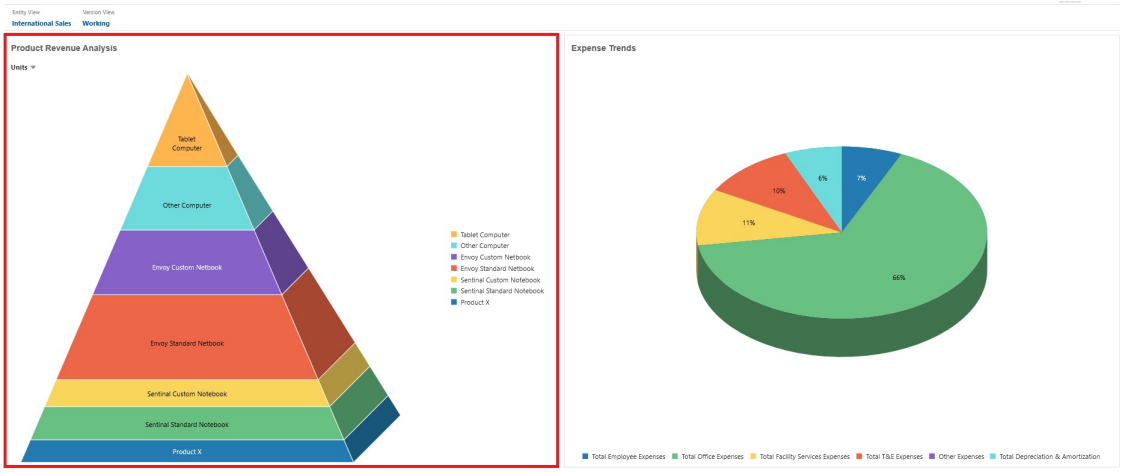

Dashboard designers can set these pyramid chart properties:

Table 12-10 Pyramid Chart Properties

Setting	Description
Legend Position	Choose **Left**, **Right**, **Top**, **Bottom**, or **None**.
Value Scale	Especially useful for large numbers, you can scale how a value is displayed. For example, if the value is 1,689,000 and you select **K - Thousand** as the scaling option, the chart displays the value as 1,689K. Your scaling options: • **None**: No scaling is applied. • **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T. • **K - Thousand**: The value is displayed as thousands units. For example, 1,689,000 displays as 1689K. • **M - Million**: The value is displayed as millions units. For example, 123,456,789 displays as 123M. • **B - Billion**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B. • **T - Trillion**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.
3D Graph	Choose **Yes** to visualize the data as a 3D chart.
Color	Select **Default** or click a section of the pyramid chart and choose a custom color for that section.

About the Waterfall Chart Type

Waterfall charts display a running total of positive and negative values, which is helpful in showing how you arrived at a net value from an initial value.

Waterfall charts are used to portray how an initial value is affected by a series of intermediate positive or negative values. Usually the initial and the final values (end points) are represented by whole columns, while the intermediate values are shown as floating columns that begin based on the value of the previous column. The columns can be color-coded to distinguish between positive and negative values.

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Example Waterfall Chart - Inventory Analysis

This example visualizes changes in a single account over time.

Example Waterfall Chart - P & L Statement

This example has expense accounts that are shown as a negative change to Total Revenue, and income accounts that are shown as a positive change, for a final Net Income value.

Dashboard designers can set these waterfall chart properties:

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Table 12-11 Waterfall Chart Properties

Setting	Description
Value Scale	Especially useful for large numbers, you can scale how a value is displayed. For example, if the value is 1,689,000 and you select **K - Thousand** as the scaling option, the chart displays the value as 1,689K. Your scaling options: • **None**: No scaling is applied. • **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T. • **K - Thousand**: The value is displayed as thousands units. For example, 1,689,000 displays as 1689K. • **M - Million**: The value is displayed as millions units. For example, 123,456,789 displays as 123M. • **B - Billion**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B. • **T - Trillion**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.
Grid Lines	Select **Hide** or **Show** to display the x-axis lines of the grid.
Connecting Lines	Select **Hide** or **Show** to display the lines connecting the end of each column to the beginning of the next column. This helps visualize the flow of the data in the chart.
Legend Position	Choose **Left**, **Right**, **Top**, **Bottom**, or **None**.
Plot Deltas	Select **Yes** to automatically calculate and plot the common start and end points from one bar to the next based on the values provided. This is useful for showing the trend for balance type of accounts. Otherwise, select **No** to manually set increase and decrease points. When **No** is selected, you can manually set these increase/decrease options on a bar-by-bar basis: • Bar Direction: Select a bar to manually set the bar direction to increase/decrease. • All bars: Display a list of all bars in the chart to manually set bar directions to increase/decrease. The manual options you can set for each bar are **Auto**, **Reverse**, and **Total**. **Auto** will consider the positive/negative values from the data source and plot accordingly. **Reverse** will change the sign, and **Total** will make the chart restart at the corresponding bar. When you change the POV, the Bar direction that you selected for the chart column is retained. which eliminates the need to manually reset the chart direction.
Increase **Decrease** **Total**	Select a color for each property.
End Bar	Select whether to **Hide** or **Show** the end bar on the chart. In the preceding example, the end bar is shown. If the end bar is shown, you can specify a label.
Axes	Select a **Custom Range** for the **X-Axis** and the **Y-Axis**, or select **Off**.

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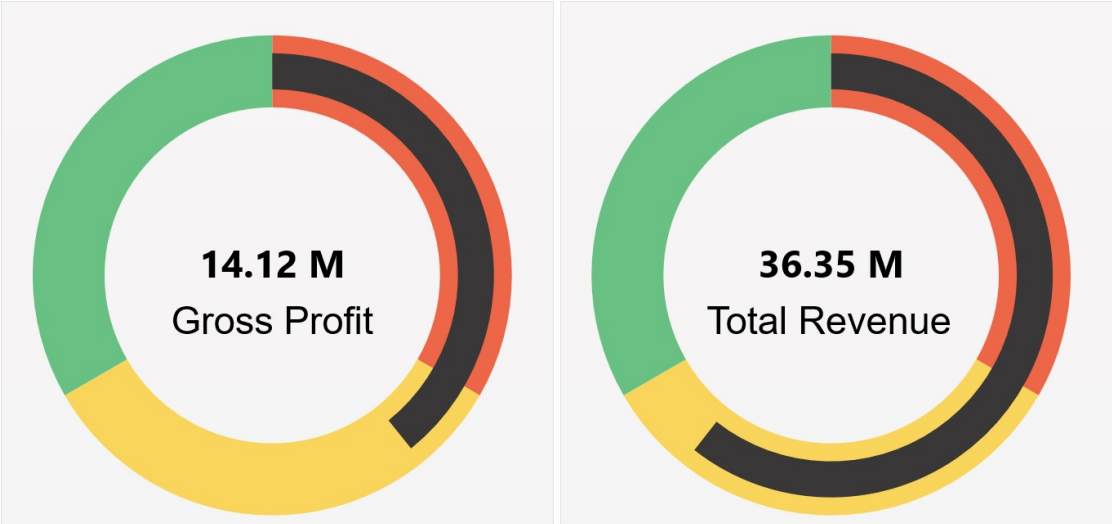
Videos

Your Goal	Watch This Video
Learn how to create waterfall charts to track the changes in a single account over time and to track how multiple accounts interact to produce a final total.	Creating Waterfall Charts in Dashboards 2.0

About the Gauge Chart Type for Dashboard 2.0

Gauge chart types show you whether data values fall within an acceptable range or not. You set the minimum and maximum values, the range maximums, and the gauge uses red, yellow, and green traffic lighting to help you quickly assess a current value and identify problems in important data points or measures. For Dashboard 2.0 dashboards, you can select circular, horizontal, and vertical gauge types.

Dashboard 2.0 Example - Circular Gauge Type



Dashboard 2.0 Example - Horizontal and Vertical Gauge Types

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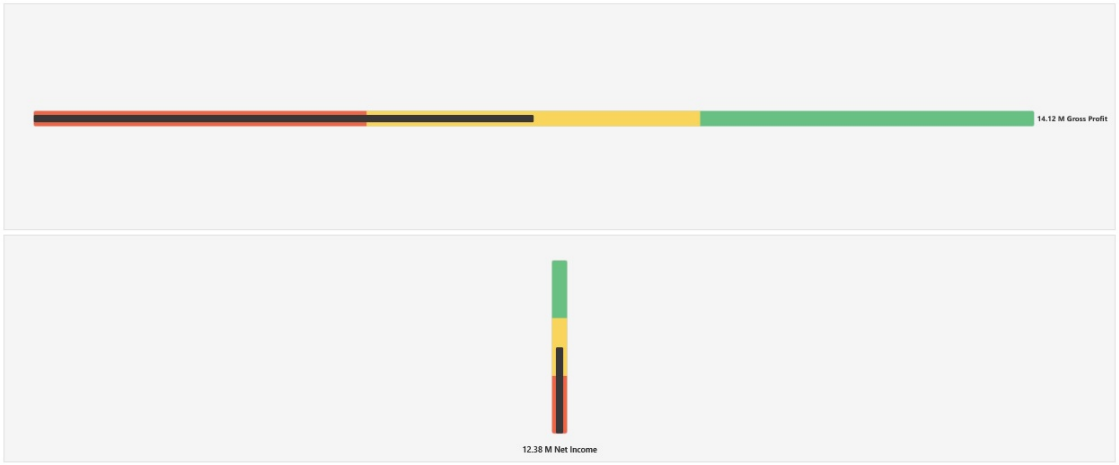
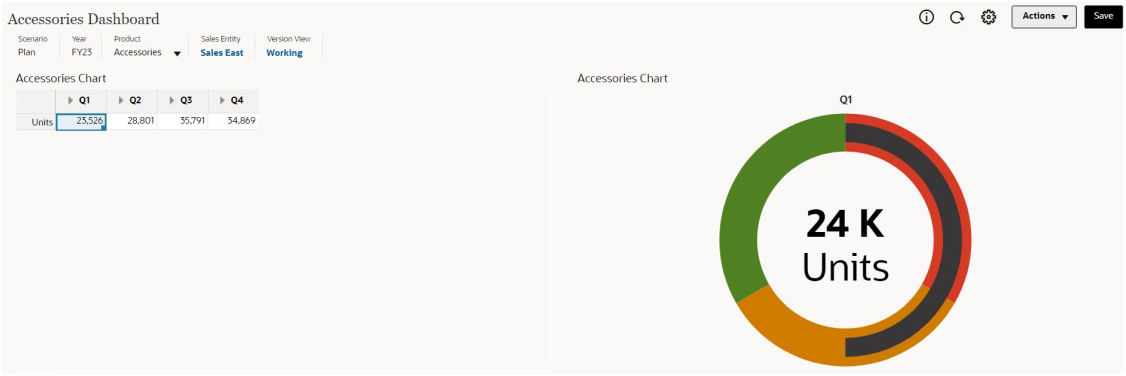


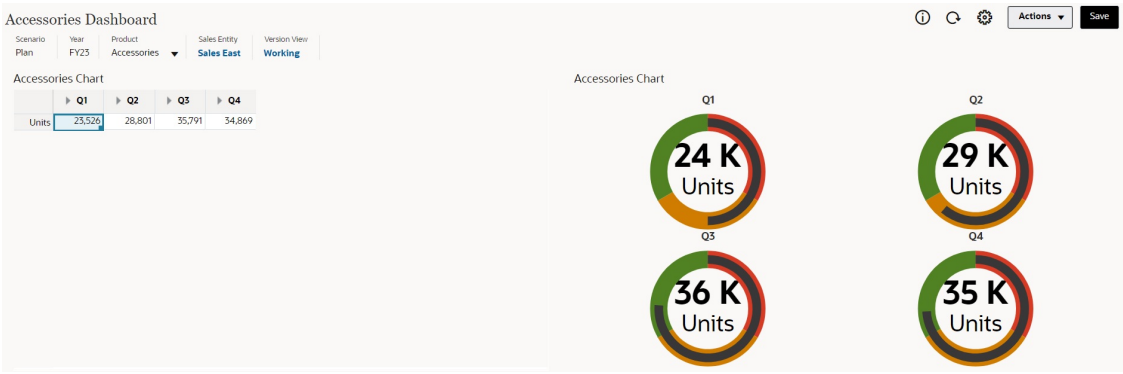
Dashboard 2.0 Example - Single Gauge Display

For **Display**, if **Single** is selected, then the gauge chart displays only the first column of the underlying form's data.



Dashboard 2.0 Example - Multiple Gauge Display

For **Display**, if **Multiple** is selected, then a gauge chart is displayed for each column of the underlying form's data.



Dashboard designers can set these gauge chart type properties:

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Table 12-12 Gauge Chart Type Properties

Setting	Description
Scale	Especially useful for large numbers, you can scale how a value is displayed. For example, if the value is 1,689,000 and you select **K - Thousand** as the scaling option, the chart displays the value as 1,689K. Your scaling options: • **None**: No scaling is applied. • **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T. • **K - Thousand**: The value is displayed as thousands units. For example, 1,689,000 displays as 1689K. • **M - Million**: The value is displayed as millions units. For example, 123,456,789 displays as 123M. • **B - Billion**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B. • **T - Trillion**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.
Display	You can display **Single** or **Multiple** gauge charts in a component. If **Single** is selected, then the gauge chart displays only the first column of the underlying form's data. If **Multiple** is selected, then a gauge chart is displayed for each column of the underlying form's data.
Gauge Type	Select from the following options: • **Circular** • **Horizontal** • **Vertical**
Minimum Value **Maximum Value**	The lowest and highest values on the gauge. The dashboard designer sets the values as a default, and then users can temporarily change them at runtime. If the dashboard designer doesn't specify these values, the application automatically sets the minimum value as zero and the maximum value as greater than the value on the gauge.
Desired Goal	Specify whether **High** values or **Low** values are desirable.
Low Threshold **Medium Threshold** **High Threshold**	To visually indicate whether a measure lies in the acceptable range or not, these thresholds enable you to display the gauge in red, yellow and green based on the specified values. Specify a text **Label** and a number **Value** for each threshold.

About the Radar Chart Type for Dashboard 2.0

A radar chart is a two-dimensional chart type designed to plot one or more series of values over multiple quantitative variables. Radar charts are best for determining which variable in the data is doing better than the rest. Hence, they are mostly used for performance analysis.

For Dashboard 2.0, you can select bar, line, area, scatter, and bubble radar types.

Dashboard 2.0 Example - Radar Bar Chart Type

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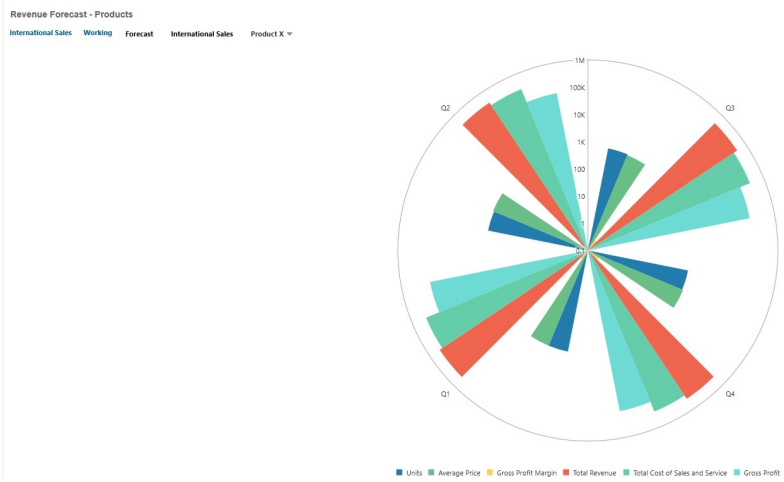
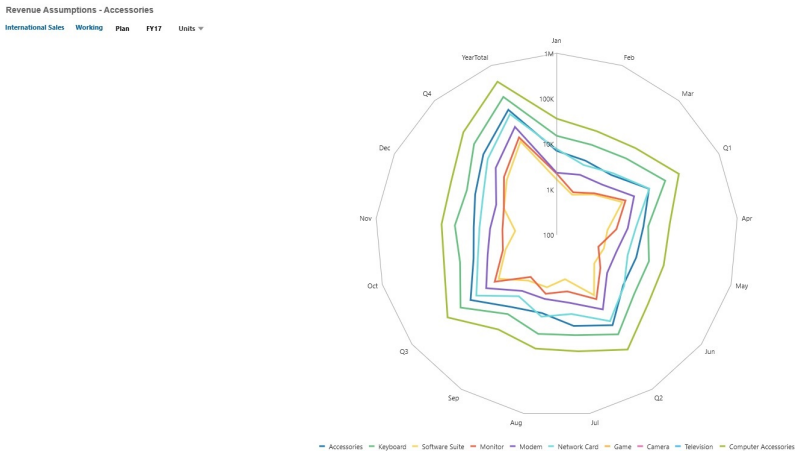


Dashboard 2.0 Example - Radar Line Chart Type



Dashboard designers can set these radar chart type properties:

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Table 12-13 Radar Chart Type Properties

Setting	Description
Value Scale	Especially useful for large numbers, you can scale how a value is displayed. For example, if the value is 1,689,000 and you select **K - Thousand** as the scaling option, the chart displays the value as 1,689K. Your scaling options: • **None**: No scaling is applied. • **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T. • **K - Thousand**: The value is displayed as thousands units. For example, 1,689,000 displays as 1689K. • **M - Million**: The value is displayed as millions units. For example, 123,456,789 displays as 123M. • **B - Billion**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B. • **T - Trillion**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.
Type	Select from the following options: • **Bar** • **Line** • **Area** • **Scatter** • **Bubble**
Legend Position	Choose **Left**, **Right**, **Top**, **Bottom**, or **None**.
Grid Lines	Select **Hide** or **Show**.
Logarithmic Scale	Especially useful for displaying numerical data that covers a wide range of values (typically, the largest numbers in the data are hundreds or even thousands of times larger than the smallest numbers), a logarithmic scale displays these values in a way that is more compact and readable. Select whether to use a logarithmic scale in your radar chart.
Color	Select **Default** or click data in the chart and choose a custom color for that data.

About the Combination Chart Type for Dashboard 2.0

The combination chart is a visualization that lets you merge into one chart any combination of bar, line, and area charts.

You can also add a secondary Y-axis on the opposite side from the primary Y-axis. Adding a secondary Y-axis lets you display the scale for one measure that doesn't scale appropriately with the other measures on the chart; for example, showing a percentage measure and a currency measure on the same chart.

Dashboard 2.0 Example - Area Bar Line Combination Chart

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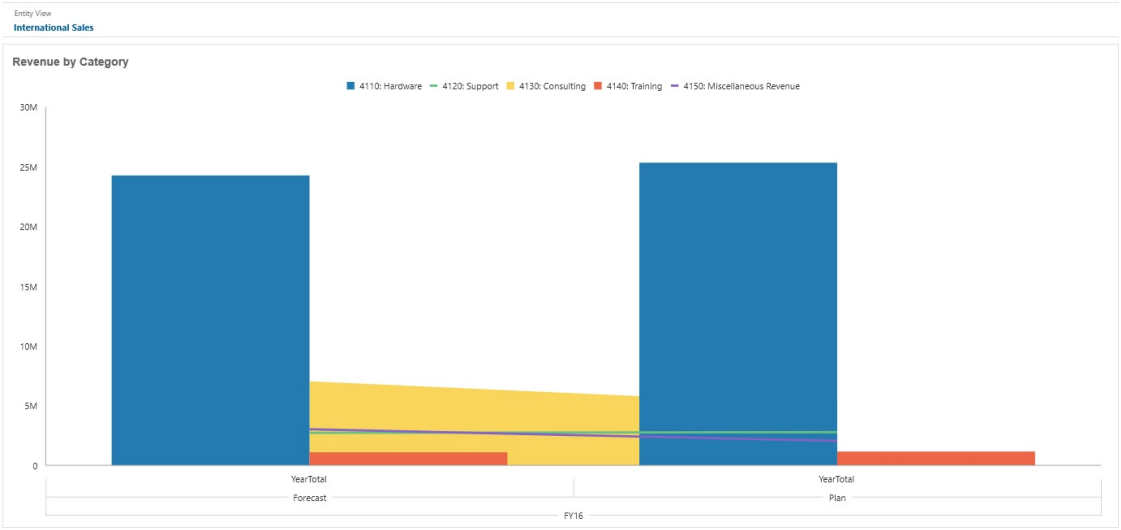

Dashboard designers can set these combination chart type properties:

Table 12-14 Combination Chart Type Properties

Setting	Description
Value Scale	Especially useful for large numbers, you can scale how a value is displayed. For example, if the value is 1,689,000 and you select **K - Thousand** as the scaling option, the chart displays the value as 1,689K. Your scaling options: • **None**: No scaling is applied. • **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T. • **K - Thousand**: The value is displayed as thousands units. For example, 1,689,000 displays as 1689K. • **M - Million**: The value is displayed as millions units. For example, 123,456,789 displays as 123M. • **B - Billion**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B. • **T - Trillion**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.
Line Weight	Click the counter to set the line width. Note that the default width of a line is 5 pixels. You can select from 1 to 12 pixels.
Legend Position	Choose **Left**, **Right**, **Top**, **Bottom**, or **None**.
Label Position	Set the position of the data labels on the chart. Choose **Outside Bar Edge**, **Center**, **Inside Bar Edge**, or **None**.
Grid Lines	Select **Hide** or **Show**.
Logarithmic Scale	Especially useful for displaying numerical data that covers a wide range of values (typically, the largest numbers in the data are hundreds or even thousands of times larger than the smallest numbers), a logarithmic scale displays these values in a way that is more compact and readable. Select whether to use a logarithmic scale in your combination chart.

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Table 12-14 (Cont.) Combination Chart Type Properties

Setting	Description
Secondary Y	Select which data to plot on a secondary Y-axis which will appear on the opposite side of the chart from the primary Y-axis. When the data values in a chart vary widely from data series to data series, or when you have mixed types of data (for example, currency and percentages), you can plot one or more data series on a secondary vertical (Y) axis. While the secondary Y-axis can be used with any line and bar chart type, it is more commonly used with the combination chart type, which helps to distinguish the data series that are plotted on the secondary Y-axis. For example, use a bar for the primary Y- axis, and line for the secondary Y-axis.
Series	Select the bar/line/area combination for the chart.
Color	Select **Default** or click data in the chart and choose a custom color for that data.
Axes	Select a **Custom Range** for the **X-Axis**, **Y-Axis**, and **Secondary Y-Axis**, or select **Off**.

About the Tile Chart Type for Dashboard 2.0

A tile is a chart type that lets you select specific values from the cube to display. In addition to using a form as a data source, you can directly enter a cell intersection that provides the value for a tile. By default, six tiles are initially displayed in a component, but you can display up to 10 tiles in a component. Until you associate a tile with data, it displays sample data.

With a form as the data source for a tile:

- You can have up to 10 tiles per component (six by default).
- The values in the first column of the form (up to the 10th row) are used to create the tiles.
- By default, the row title in the form is the tile's title, and it gets its values from the first column, then by row.
- You can set the tile's component title, the tile's height percentage, legend labels, and can select which axes from the form to include. For example, if you select a form with three rows, the tile displays three values.

> **Note**
>
> Sometimes in a form, the first column may be collapsed (hidden) when viewed as a grid. But the collapsed column is still considered when the tile gets its values from the form. We recommend that you update the form definition and use a form as a data source that contains only the cells to be shown on the tile.

With a cell intersection as the data source for a tile, you can have only one tile per object.

The table below lists the options you can set for the tile chart type. Note that you can set whether the displayed value is horizontally aligned in the tile to the left, the center, or the right.

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Table 12-15 Tile Chart Type Properties

Setting	Description
Value Scale	Especially useful for large numbers, you can scale how a value is displayed. For example, if the value is 1,689,000 and you select **K - Thousand** as the scaling option, the chart displays the value as 1,689K. Scaling options: • **None**: No scaling is applied. • **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T. • **K - Thousand**: The value is displayed as thousands units. For example, 1,689,000 displays as 1689K. • **M - Million**: The value is displayed as millions units. For example, 123,456,789 displays as 123M. • **B - Billion**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B. • **T - Trillion**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.
Align Text	Text alignment options: • **Left** • **Center** • **Right**
Count	The number of tiles displayed in the component (up to 10)
Display	Legend display options: • **Row and Column Dimensions** • **Row Dimensions** • **Column Dimensions** • **None**
Position	Legend positioning options: • **Left** • **Right** • **Top** • **Bottom**
Legend Font Size	Decrease or increase the legend font size.
Value Font Size	Decrease or increase the value font size.

> **Note**
>
> Dashboard 2.0 does not support cell details. Instead we recommend you use Quick Analysis as a data source, which is a more advanced version of cell details. If you convert a dashboard to Dashboard 2.0 that has a component using cell details, delete the component using cell details as a data source, and recreate it using Quick Analysis as a data source.

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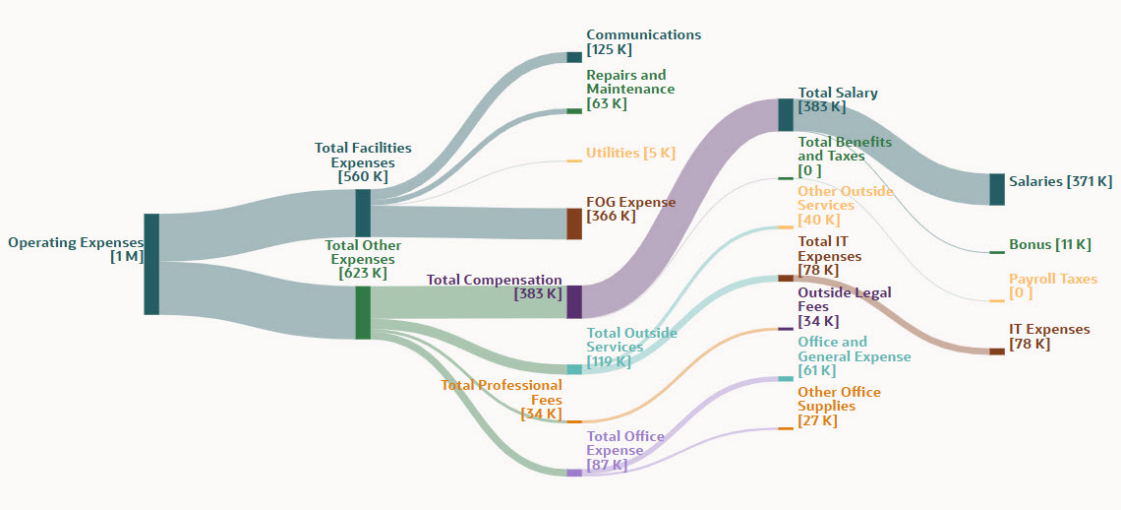
About the Sankey Chart Type for Dashboard 2.0

A sankey chart is a flow diagram that shows how values flow across dimensions. It helps visualize business data distribution and makes it easy to analyze value contribution, hierarchy relationships, and cross-dimensional movement.

Categories are represented as nodes, and the nodes are connected by pathways. The width of each flow is proportional to its value and this helps identify the largest flows, distributions, and points of movement. Using different colors for pathways can help easily distinguish between categories, flow types, or measure values.

Example: Operating Expenses Breakup

This chart represents the flow of a planned income statement. It shows how operating expenses are distributed across two major categories - Facilities and Other, each represented as a node in the chart. These two categories are further distributed across various expense types, and are represented using two additional levels in the chart.



Example: Cost of Sales Distribution

This chart provides a visual representation of how total product costs are distributed across cost categories (Labor, Material, Overhead, and so on) and then across market classifications. Each flow shows the relationship between the total cost, the relevant cost category, and its market classification. The width of the flow indicates the relative magnitude of that category.

Because the market classification values are repeated for each cost category, this chart uses consolidation at the node level. Amounts associated with the same category or classification are aggregated into a single node and represented as a combined flow. This reduces duplication and provides a clearer, more compact view of the overall cost structure and allocation relationships.

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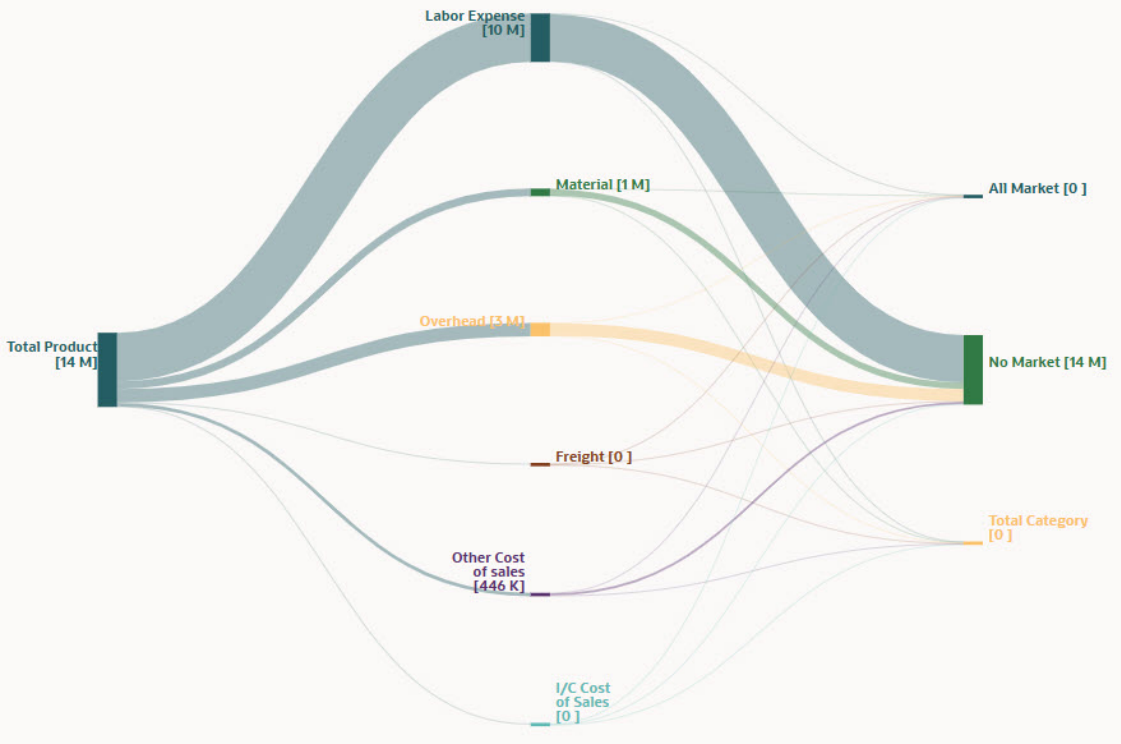

Dashboard designers can set these sankey chart type properties:

Table 12-16 Sankey Chart Properties

Setting	Description
Scale	Especially useful for large numbers, you can scale how a value is displayed. For example, if the value is 1,689,000 and you select **K - Thousand** as the scaling option, the chart displays the value as 1,689K. Your scaling options: • **None**: No scaling is applied. • **Auto**: The value is displayed based on its range. For example, 1,500 displays as 1.5K, 1,689,000 displays as 1.69M, 42,314,531,211 displays as 42.31B, and 1,234,567,891,234 displays as 1.23T. • **K - Thousand**: The value is displayed as thousands units. For example, 1,689,000 displays as 1689K. • **M - Million**: The value is displayed as millions units. For example, 123,456,789 displays as 123M. • **B - Billion**: The value is displayed as billions units. For example, 12,345,678,912 displays as 12B. • **T - Trillion**: The value is displayed as trillions units. For example, 1,234,567,891,234,567 displays as 1,234T.
Legend Position	Choose **None**, **Left**, **Right**, **Top**, or **Bottom**.

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Table 12-16 (Cont.) Sankey Chart Properties

Setting	Description
Flip Display	Flips the chart display by 180 degrees. Possible values are: • **No**: Dimensions and nodes flow from left to right. This is the default setting. • **Yes**: Dimensions and nodes flow from right to left.
Link Color	Links can use source-node colors or simple neutral link color. Options are **Show** or **Hide**.
Consolidate Nodes	Specify if amounts must be aggregated at the node level. Possible values are: • **Yes**: Aggregates amounts at the node level, thereby combining repeated records into a clearer and more readable display. This is useful for multi-dimensional charts. This is the default setting. • **No**: Displays the nodes in a distributed manner, without consolidation.
Link Opacity	Specify the opacity level of the links by selecting a value between 1 and 9.
Link Width	Allows links to show contribution weight or render as simple, fixed-width links. Options are **Value** or **Uniform**.
Color	Select **Default** or click data in the chart and choose a custom color for that data.

Dashboard Design Guidelines

Provides guidance to dashboard designers on how to build visually appealing and functionally effective Oracle Fusion Cloud Enterprise Performance Management dashboards.

Dashboards must visually communicate clearly and immediately a vast array of information in a small amount of space and in a manner that meets the expectations of various audiences and user groups. This requires good design.

The guidelines provided here will assist you in learning how to design effective Cloud EPM dashboards for all types of users.

Related Topics

- Dashboard Types
- Dashboard Layout Guidance
- Dashboard Styles
- Color Sequencing in Charts
- Recommendations for Improving Dashboard Design
- Summary

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Dashboard Types

Dashboards are a powerful interactive interface used by various stakeholders with varying use cases. Dashboard designers must understand who is using the dashboard and for what purpose in order to create a dashboard that is appealing and functional to the intended user.

Every dashboard user has unique expectations, and these expectations must be kept in mind while designing dashboards. For Oracle Fusion Cloud Enterprise Performance Management, we've identified two general dashboard types:

- Read-only dashboards: Read-only dashboards are visual tools that aggregate and display critical metrics and key performance indicators (KPIs) in a single screen, enabling users to monitor and examine business performance at a glance. See Read-Only Dashboards.
- Write-enabled dashboards: Write-enabled dashboards are unique to Cloud EPM and they support the ability to update data directly in the dashboard and immediately visualize it. See Write-Enabled Dashboards.

Read-Only Dashboards

Read-only dashboards are visual tools that aggregate and display critical metrics and key performance indicators (KPIs) in a single screen, enabling users to monitor and examine business performance at a glance.

When designing read-only dashboards, keep the following factors in mind so that the dashboards are appealing and functional to the intended users:

- Know your audience

  Different dashboards have different audiences and different uses. Managers and Executives use read-only dashboards strategically to monitor essential business KPIs and metrics so they can effectively make decisions. Data Analysts and Interactive Users use read-only dashboards analytically to visualize important high-level summary data with the ability to drill through to detailed data.

  > **Note**
  >
  > Another type of dashboard user are Data Input and Impact Analyzers. They are data entry users who use dashboards as a data input interface, but also have the ability to immediately view the newly input data impact on a chart in real-time. These types of dashboards are unique to Oracle Fusion Cloud Enterprise Performance Management and are called write-enabled dashboards. For information about designing write-enabled dashboards, see Write-Enabled Dashboards.

- Keep it simple

  The fundamental challenge of dashboard design is to squeeze as much information as possible into a small amount of space. Nobody wants to leave out critical information, but there's a tradeoff between complexity and usefulness. Be clear in your objective for a dashboard, and narrow the information to that which directly applies to meeting that objective. If there is deeper information under the initial data presented, provide an option to drill down and seek ever more focused and useful information.

- Prioritize elements

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Prioritize elements top left to bottom right. Place high-level, easy-to-read, actionable components near the top left, and place components with supporting information lower. If your audience has limited time or attention, place important elements where they will be noticed.

- Aesthetics matter

  The aesthetics of a dashboard (or lack therof) can impact its adoption. Use only one typeface, and don't overuse colors. Each color in a data visualization should serve a purpose. Use different dashboard colors only when you're communicating different things. Be consistent. If your dashboard includes multiple charts, ensure to match colors between charts when they refer to the same group or entity. If colors change their meaning between charts, this can make it harder for users to understand the chart.

Read-Only Dashboard Examples by User

Provided in this topic are examples of read-only dashboards by user. Also provided are recommendations for the information to display in dashboards, the chart types to use, and number of components.

- Read-Only Dashboards for Managers and Executives
- Read-Only Dashboards for Data Analysts and Interactive Users

Read-Only Dashboards for Managers and Executives

Managers and Executives are typically decision makers. They expect to see tables and charts that give them clear information. Make sure to include expected details in one dashboard consisting of multiple components.

Table 12-17 Read-Only Dashboard Recommendations for Managers and Executives

Dashboard Content	Recommendation
Information to display	• Key Performance Indicators (KPIs) • High-level trends • Variances • YoY and/or QoQ comparisons
Chart types	Tile, Gauge, Table, Pie/Doughnut, Line/Bar, Geomap, Radar, Waterfall
Number of dashboard components	3 to 6

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Read-Only Dashboards for Data Analysts and Interactive Users

Data Analysts and Interactive Users typically perform data analysis and provide insights to senior management for important decisions.

Table 12-18 Read-Only Dashboard Recommendations for Data Analysts and Interactive Users

Dashboard Content	Recommendation
Information to display	• Drill-enabled summary level chart • Page-level and chart-level POV switchers to review various data combinations • Variances • YoY and/or QoQ comparisons • Charts with a Master-Detail relationship
Chart types	Tile, Gauge, Table with Spark Chart, Pie/Doughnut, Line/Bar, Geomap, Radar, Waterfall, Bubble, Scatter, Funnel, Pyramid, Grids (with summary data)
Number of dashboard components	3 to 6

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Write-Enabled Dashboards

Write-enabled dashboards are unique to Oracle Fusion Cloud Enterprise Performance Management because they enable users to visualize, read, write, and calculate in one dashboard.

Write-enabled dashboards facilitate bi-directional data interactions, both to display data as well as update it. Dashboards typically enable users to visualize data in an easily understandable manner by representing data in different chart types with scales for comparison. Write-enabled dashboards also enable users to make updates to data values and have the visualization rendered instantaneously.

Interactions in Cloud EPM dashboards are not just limited to changing POVs, drill-through, different chart type visualizations, and so on. Write-enabled dashboards enable users to update data directly in the dashboards. And with the ability to associate Business Rules in dashboards, complex calculations can be done on the fly and values are updated.

Write-Enabled Dashboard Examples

With write-enabled dashboards, Oracle Fusion Cloud Enterprise Performance Management supports the ability to update data and immediately visualize it.

Provided in the table below are recommendations for the components to display in write-enabled dashboards, the chart types to use, and the number of components.

Dashboard Content	Recommendation
Information to display	• Enterable grids • Charts with a Master-Detail relationship • Page-level and chart-level POV switchers to review various data combinations • Reference data stacked side-by-side
Chart types	Grids, Tables, Line/Bar/Area, Waterfall, Pie, Gauge, Doughnut
Number of dashboard components	2 to 4

In the examples below, note that the dashboards have both visual charts to represent data as well as grids (wherein data updates are made). When the grid values are updated and saved, the changes are reflected in the chart(s) in real-time.

Example Write-Enabled Dashboard: KPI Tiles with Writeable Grids

In the dashboard example below, the top component of the dashboard uses tiles. The middle left and bottom components are writeable grids, and the middle right component is a bar chart.

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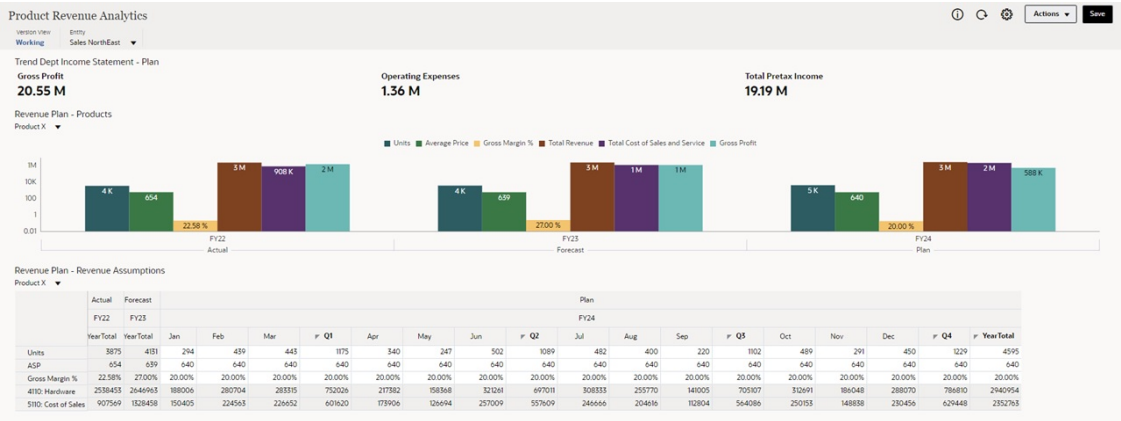

Example Write-Enabled Dashboard: Sales Forecast

In the Sales Forecast dashboard example below, the Actuals vs Forecast chart at the top is a combination chart, and the Sales Forecast with Actuals grid at the bottom is a writeable grid.

Dashboard Layout Guidance

Keep the following in mind while choosing a layout for a dashboard:

- Aim for layouts that adhere to an evenly spaced 3-column or 2-column arrangement of charts. We do not recommend dashboards exceeding three columns.
- Avoid using multiple scroll bars on a dashboard page. Ideally, place any scrolling components in a panel that touches the bottom of the screen. See the yellow panels in the examples below for the recommended placement of scrolling components.

1-Panel Dashboard Layout

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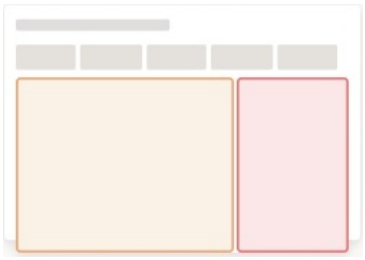
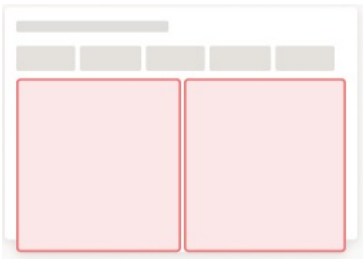
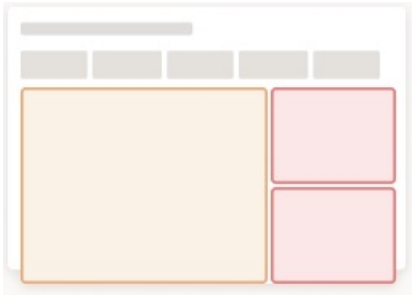
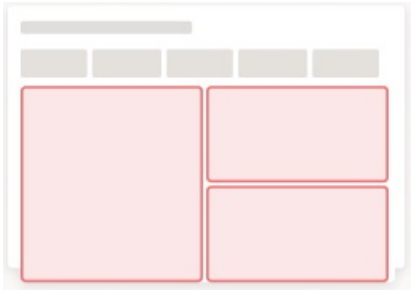
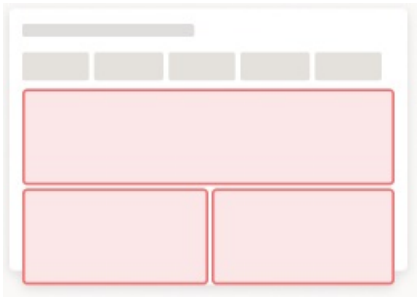
2-Panel Dashboard Layouts

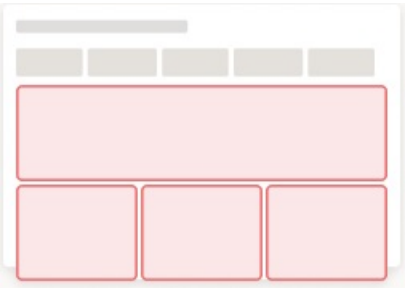
Example 1	Example 2	Example 3
		

3-Panel Dashboard Layouts

Example 1	Example 2
	

Example 3	Example 4
	

4-Panel Dashboard Layout



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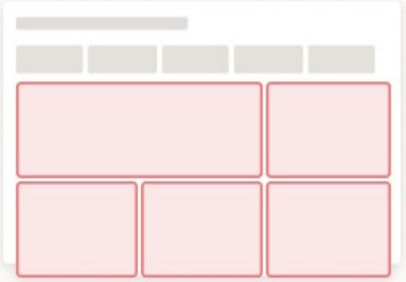
5-Panel Dashboard Layout



Dashboard Styles

For Oracle Fusion Cloud Enterprise Performance Management, you design dashboards with a color theme called a style. Each dashboard style provides you with a different set of colors and backgrounds from which you can choose to make your dashboard visually pleasing, easy to read, and impactful.

The dashboard styles are:

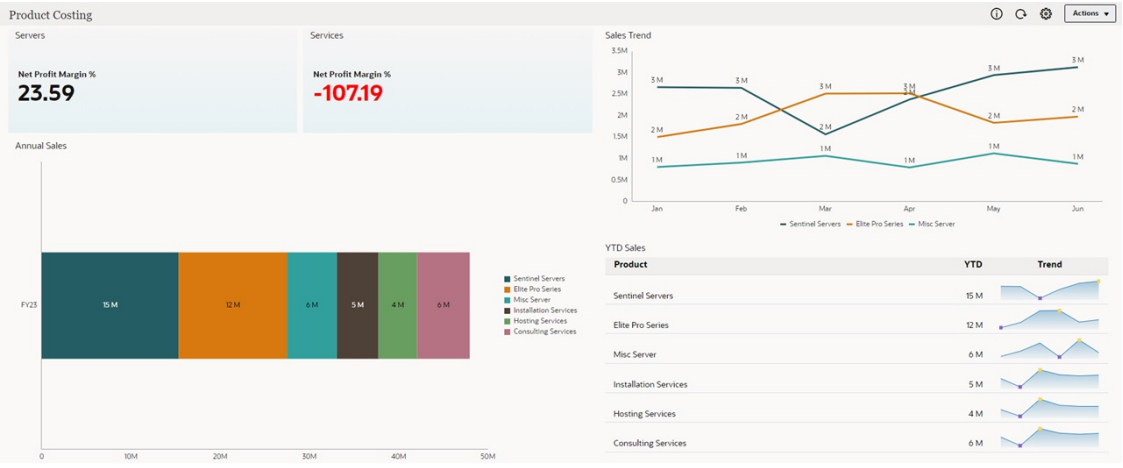
- Light (Default)
- Dark
- White

The categorical color palettes for each dashboard style have been intentionally designed to maximize adjacent color differences.

The following examples show dashboards customized with the Light, Dark, and White styles.

Light Dashboard Style

The Light dashboard style provides a light background and uses dark fonts, dark images, and dark icons. If a white background is needed, choose the alternative White dashboard style.



Dark Dashboard Style

The Dark dashboard style provides a dark background and uses light fonts, light images, and light icons.

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White Dashboard Style

The White dashboard style provides a pure white background and uses dark fonts, dark images, and dark icons. If a white background is needed, choose this style as an alternative to Light.

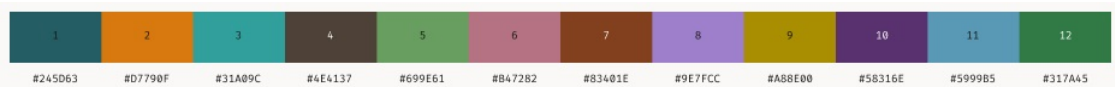
Color Sequencing in Charts

The categorical color pallette is intentionally designed to maximize adjacent color differences. We recommend applying pallette colors to data in the order specified below.

Across charts, color sequences can be repeated or you can pick up the sequence where the last chart ended.

Color Sequence for Charts with Light Backgrounds

For charts with light backgrounds, we recommend using the following color sequence:



When colors are needed to convey negative, neutral, and positive, we recommend using the following set of colors. These colors can be reserved from the sequence on that page.

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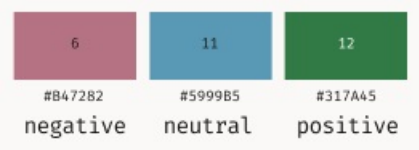
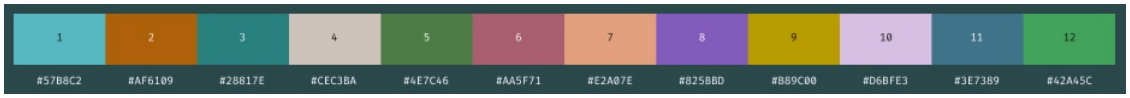


Color Sequence for Charts with Dark Backgrounds

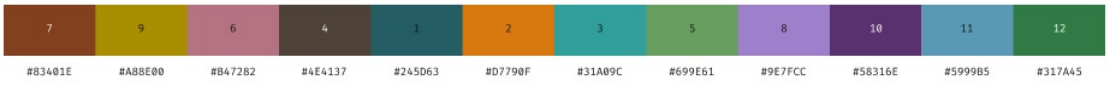
For charts with dark backgrounds, we recommend using the following color sequence:



When colors are needed to convey negative, neutral, and positive, we recommend using the following set of colors. These colors can be reserved from the sequence on that page.

Color Sequence for Charts with White Backgrounds

For charts with white backgrounds, we recommend using the following color sequence:



When colors are needed to convey negative, neutral, and positive, we recommend using the following set of colors. These colors can be reserved from the sequence on that page.

Recommendations for Improving Dashboard Design

In the topics to follow, we've identified some best practices for improving your dashboard design and we've provided example dashboards.

- Chart Sizing
- Readability
- Spacing and Alignment
- Consistency

Chart Sizing

Here are some best practices to follow when sizing charts in dashboards:

- The dashboard layout should be no more than three panels wide.
- Avoid using pie, ring, gauge, and doughnut charts in widths other than one third.

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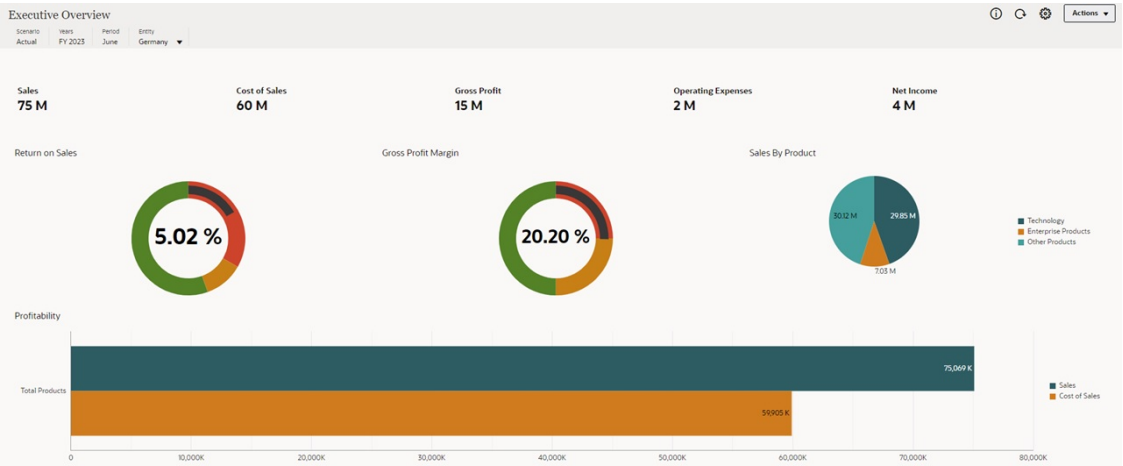
- If using multiple pie, ring, gauge, and doughnut charts in a dashboard, ensure that they're all similarly sized.

Figure 12-4 Executive Overview



Readability

Here are some best practices to follow to make dashboards more readable:

- Left-align text rather than centering text or setting it to align right.
- Use grid lines in all charts.
- Use red for negative numbers.
- Use the same chart types for similar information; for example, Inflow and Outflow should both be visualized as the same type of chart.

Figure 12-5 Example Dashboard: Periodic Summary

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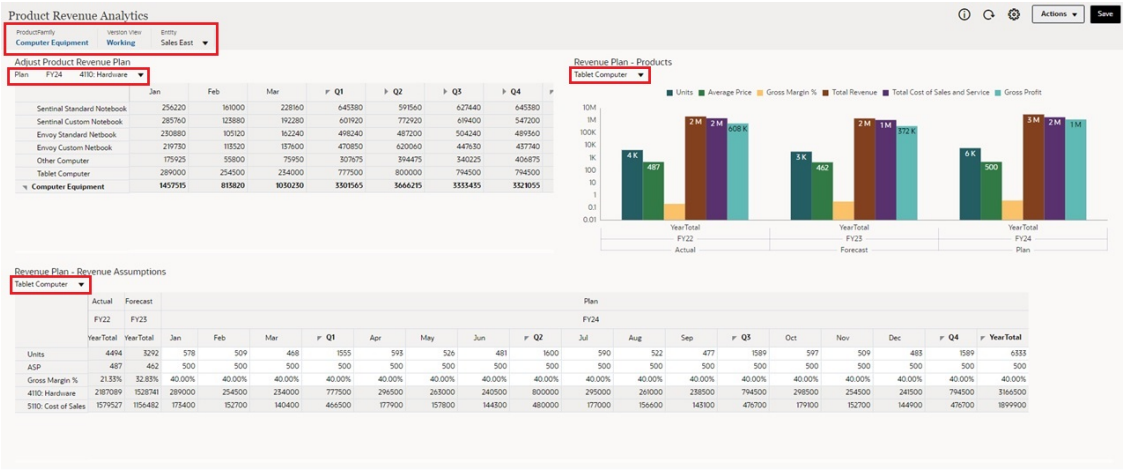
Improving Dashboard Readability with POVs

POVs are useful tools for improving dashboard readability. Follow these best practices for using POVs in dashboards:

- Enable **POV : Auto-Apply** in the **POV Settings** menu. Every time POV members are selected, the associated data and charts are automatically refreshed in the dashboard. This improves efficiency and avoids unnecessary clicks.
- Enable the **Global POV Bar** in the Dashboard **General Settings**. If multiple dashboard components use the same Page and POV dimensions, the dimensions are automatically moved to the global POV instead of repeating them within each component. Using global POVs eliminates clutter and saves space.
- Consider enabling **POV : Hide Dimension Labels** in the **POV Settings** menu. If the dimension member names are self-explanatory, then dimension labels are unnecessary. For example, product names and currencies might not need a "Product" or "Currency" label. Hiding the dimension labels further reduces clutter and improves dashboard readability.

Figure 12-6 Example Dashboard POVs: Product Revenue Analytics



Using Master Details Dashboards

You can design dashboards that have one master form and multiple detail (or target) forms or charts. This is also referred to as a Master Details dashboard. The selection of members in the master form automatically filters to the members in the target forms or charts, and the target forms or charts show only the details that are relevant to the members highlighted in the master form.

Here are some best practices when designing Master Details dashboards to improve efficiency and provide faster visualizations:

- The **Apply Context** right-click menu option filters the data in the target forms or charts that is relevant to the data in a master form. Master Details dashboards help you visualize faster without having to change local POVs. Select a data row in the master form and then right-click to select **Apply Context** to refresh the data for the selected member automatically in all the detail forms or charts.

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- The **Apply Context** option supports context-sensitive user and substitution variables. User variables must have the **Use Context** variable option enabled, and can be used on Rows, Columns and Pages. Substitution variables can only be used on Pages.
- Master Details dashboards can be used in various combinations that include grids with write-back capability. This helps you to review and update details for different data sets and effectively enables multitasking.
- Master Details dashboards are not a different type of dashboard, but they can be designed differently from other dashboards. This is accomplished by placing dimension members in rows for the master form, and then repeating the same dimensions at the page level for the other detail forms or charts.

Example Master Details Dashboards

In the examples below, note that the master form in the top left displays a list of products, while the detailed grid and chart display the products as page dimensions. When you select a product in one of the rows and then right-click **Apply Context**, every target grid and chart that includes the selected member in the local POV is automatically and instantly updated, and the data is refreshed.

Figure 12-7 Example Master Details Dashboard Before Applying Context

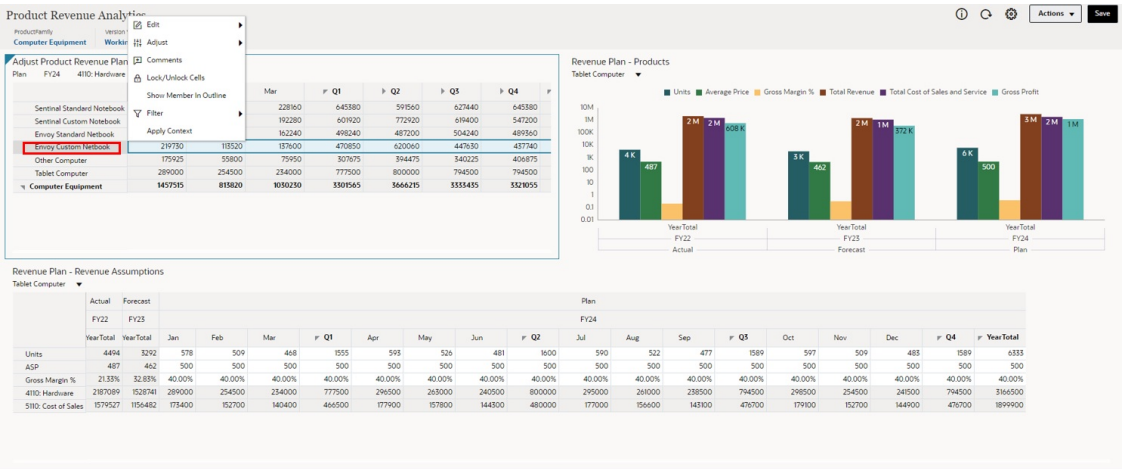
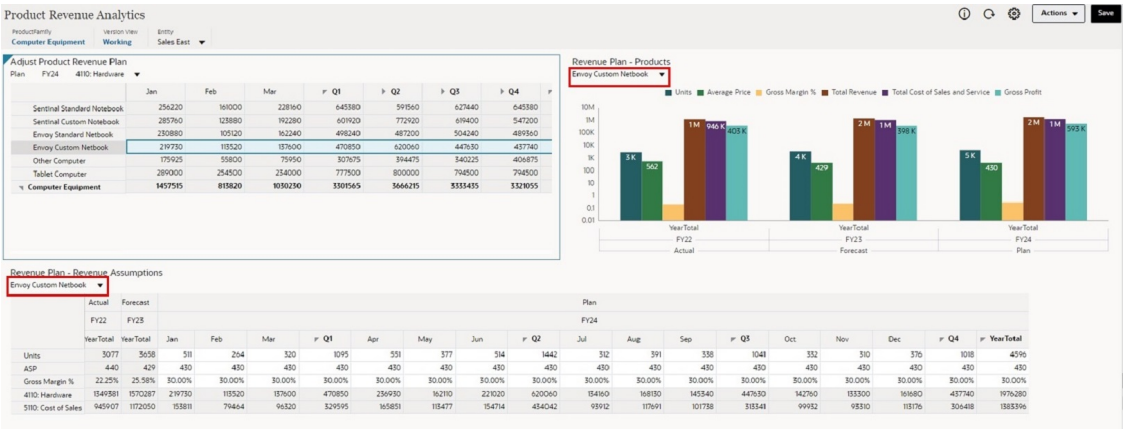


Figure 12-8 Example Master Details Dashboard After Applying Context



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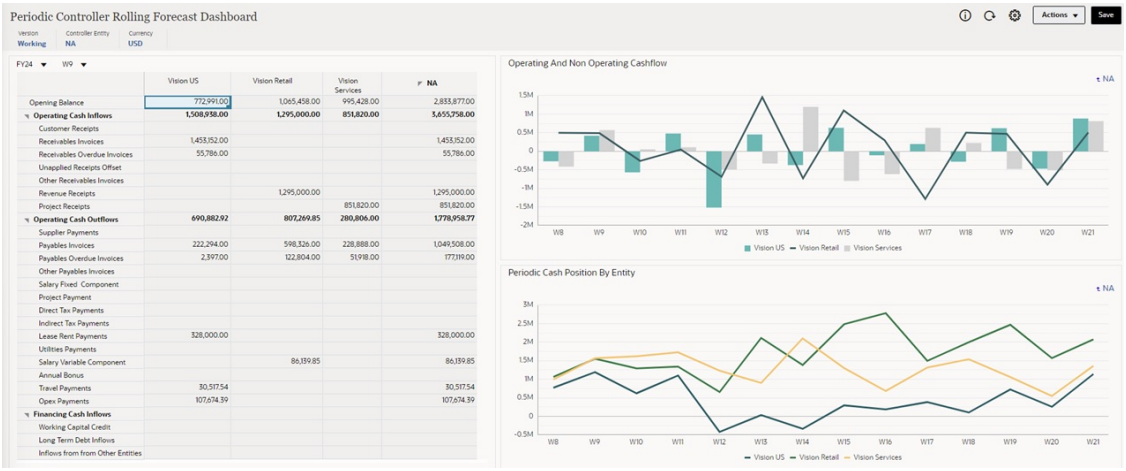
Spacing and Alignment

Here are some best practices to follow for spacing and alignment of charts and grids in dashboards:

- Ensure that grids fill the entire designated area.
- Grid background colors should denote active and read-only cells, if applicable.
- Place legends in a location that enables charts to align. For example in the dashboard below, both of the legends are placed below the charts so the x-axis line in the bottom chart aligns with the x-axis line in the chart above it. If one of the legends was placed to the right of the chart, the x-axis lines would not align.
- To reduce clutter, consider combining charts with similar information (such as Operating Cashflow and Non Operating Cashflow) into one chart.
- Another way to reduce clutter is to hide local and global POVs.

Figure 12-9 Example Dashboard: Periodic Controller Rolling Forecast



Consistency

Here are some best practices to follow to improve consistency in dashboards:

- Use labels in the same order if they're used in multiple places. For example, if using **Inflow** and **Outflow**, use them in the same order everywhere.
- Ensure that all charts have titles.
- Use a single, consistent typeface (for example, Oracle Sans) and no more than three sizes of type; one for titles, one for labels, and another for other content.

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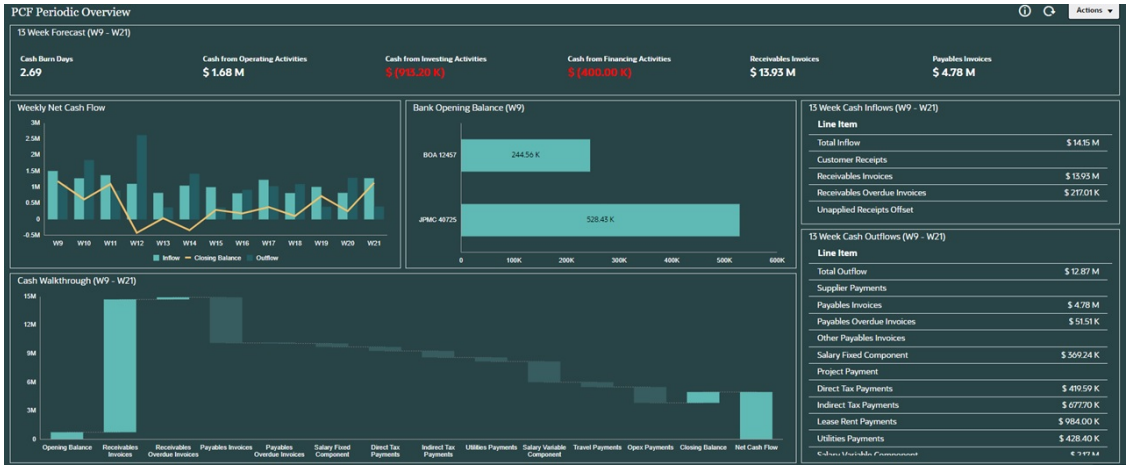
Figure 12-10 Example Dashboard: PCF Periodic Overview (Dark)



Summary

To design dashboards that your users will want to work with, focus on the fundamental goal of communicating. Ensure that your users can easily and quickly view the dashboards and understand them.

For additional information about using color to focus attention on what your data means, we recommend the following resource:

Communicating with color in data visualizations

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13

Managing Infolets

Infolets enable users to view and interact with high-level, essential information from different sources. This helps users quickly assess where to direct their attention.

Related Topics

- About Infolets
- Anatomy of an Infolet
- Determining Infolet Content
- Using the Infolets Designer
- Creating Infolets
- Working with Infolets - Standardized Interface
- Working with Infolets - Legacy Interface
- Customizing the Interface to Access Infolets

About Infolets

Infolets enable users to view and interact with high-level, essential information generated from different sources so that they can quickly assess where to direct their attention. Service Administrators create, redesign, delete, and assign permissions to infolets.

> **Note**
>
> Starting with the October 2025 (25.10) update, Oracle will no longer support the following infolet actions:
>
> - Creation of new infolets
> - **Copy As**, which duplicates an existing infolet
> - Designating an infolet as Default and defining infolet dots on the Home page
>
> Users for all new and existing environments will be restricted from creating infolets or adding them to navigation flows.

What is an Infolet?

An infolet is a self-contained, interactive box-shaped container used to display information using text and charts. Infolets are interactive and use progressive disclosure to display high-level, aggregated, essential information for quick consumption at a glance, and then can be acted upon as needed. Infolets can be flipped and resized to display up to three charts or sets of values.

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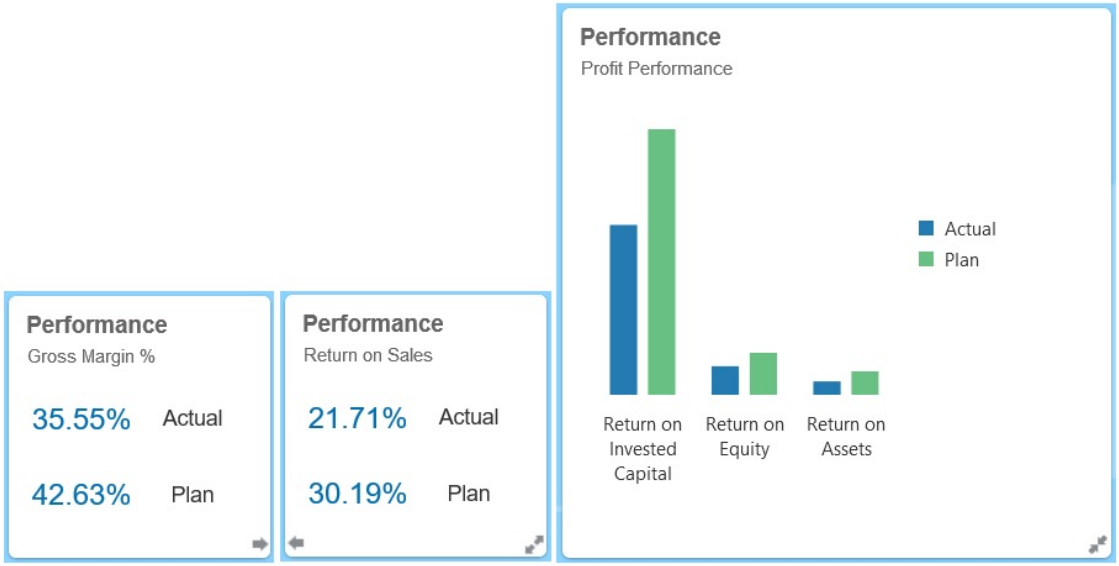

For more information about infolets, see Anatomy of an Infolet.

How Can I Use Infolets?

Use infolets to:

- Promote essential, easily consumable information
- Provide answers to your most critical questions:
  - What is new or what has changed?
  - What is the most important information that supports my work?
- Group key information by user role in a way that helps users quickly assess and prioritize their work
- Progressively display essential details and actions

  Display these additional details in the various infolet views accessed by flipping or expanding an infolet. However, a single infolet view is acceptable.
- Provide a visually rich means of displaying essential or summary information

Do not use infolets to feature highly complex information, such as reporting functions, or to present detailed visuals.

See Determining Infolet Content.

What is an Infolet Page?

An infolets page is a page that contains one or more infolets. It houses a container that manages the space occupied by the infolets and rearranges them depending on the size of the browser and the size of the infolets. Each infolet you create belongs to an infolet page. The **Infolets** card on the Home page provides a list of infolet pages.

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> **Note**
>
> Not all features pictured in the preceding image are supported in this update. Oracle plans to support these features in a later update.

See Working with Infolets - Standardized Interface.

Anatomy of an Infolet

Infolet Views

An infolet supports up to three views:

1. Front view (required)

2. Back view (optional)

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3. Expanded view (optional)

The front view is required, and:

- Provides a quick look or glimpse at high-level information that has a direct effect on your work; for example, the front view can display status, counts, totals, or the most recent updates
- Promotes a glancing action that helps you identify important information that you may want to explore a bit more
- Uses all infolet sizes except 3x2 (see information about infolet sizes below)
- Returns the expanded view to its original size in the front view or back view
- Includes an Actions menu icon available only on hover and either a (optional) flip to back icon or an expand icon in the lower-right corner

> **Note**
>
> If only one view is presented, it must be a front view.

The back view is optional, and:

- Presents analytical information (for example, a graph)

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- Promotes a scanning action that helps you explore or become more familiar with the information represented on the front view
- Is sized the same as the front view
- Includes an Actions menu icon available only on hover, a flip to front icon in the lower-left corner, and an (optional) expand icon in the lower-right corner

The expanded view is optional, and:

- Presents even more detailed information about the single data point or interdependent data set presented in the front and back views; for example, the expanded view can display more details about an object or a list of recent items than what's displayed on either the front or back view
- Provides enough information to help you decide if you're ready to take action and move to a focused context on a work area page
- Transitions smoothly from other views. You see a smooth, seamless expansion, as one infolet pushes others into new positions.
- Must be sized bigger than the front or back views
- Includes an Actions menu icon available only on hover, and a collapse icon in the lower-right corner

Infolet views honor the access permissions assigned to the underlying forms and dimensions. Therefore, the same infolet may display varying views from user to user if they have different access permissions.

Infolet Sizes

Infolets can be sized as follows:

> **Note**
>
> 1x1 refers to a box that spans one column and row width (170 pixels).

- 1x1
- 2x1
- 3x1
- 2x2
- 3x2 (expanded view only)

The size of the front and the back views are always the same. Changing the size of the front view will automatically reset the size of the back view. Because the size of the expanded view must always be greater than the size of the front/back views, if the size of the front/back view of an infolet is enlarged, the expanded view automatically resets to a larger size than the front/back view.

> **Note**
>
> Front and back views cannot use the 3x2 size. This size is applicable for the expanded view only.

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The size, title, and subtitle of a view is set by the designer in the properties panel. See Using the Infolets Designer.

Navigating Between Infolet Views

An infolet can be created with one of the following view combinations:

1. Front view only
2. Front and back views
3. Front and expanded views
4. Front, back, and expanded views

Switching from one view to another is controlled by clicking a flip icon, an expand icon, or a collapse icon available in the bottom-right or bottom-left corner of the infolet. Hovering your cursor over the bottom corners reveals a flip, expand, or collapse icon.

Determining Infolet Content

Consider the following general tips when determining infolet content:

- Look to existing dashboards and work area pages.

  Dashboards and work area pages are excellent starting points because they display collections of information summaries.

- Apply the 10/90/90 principle.

  Seek out the most essential, easily consumable information that addresses frequently asked questions garnered from the top 10 percent of use cases. These use cases can come from across your enterprise—from business intelligence, social, transactional, external, and so on.

  Then focus this information to reveal what 90 percent of your users would benefit from viewing 90 percent of the time. You can apply this 10/90/90 percent principle to existing dashboard content, to existing work area page content, or generally, to percolate eligible information for an infolet.

- Restate your top use cases in the form of frequently asked business questions.

  Present the corresponding infolet content in such a way as to answer these business questions; for example, how many orders are in jeopardy, listed by status?

- Look for one point or a tightly related, interdependent set of points, instead of multiple points of information.

  The process of determining content for an infolet is similar to the process that is used to yield dashboard content—but to a deeper level of analysis. Look for information within a data point or data set that is suitable to be displayed in no more than three views of information hierarchy and that answers a critical business question.

- Start with the single most important point.

  An infolet displays aggregated information about a single data aspect or point of information in relation to an event that the user needs to know about or a task that the user needs to address.

  If a dashboard contains multiple aspects about one or more objects (for example, numeric totals and currency totals), start with the single most important point and add that as a simple overview (for example, as a total using a stylized numeric value) to the front view of an infolet. Then determine the content for the back view, if needed. Finally, determine the content for the expanded view, if needed.

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An infolet should have no more than three views. If there is only a single data point or only one tightly related, interdependent data set to display on an infolet, use only the front view.

Related Links

Designing Forms for Infolets

Designing Charts for Infolets

Designing Forms for Infolets

Use only forms with small data sets in infolets. Permissions that are set for forms are honored in infolets.

You can create forms that are specifically used in infolets:

- Forms used in infolets should have fewer cells than a traditional data entry form; for example, forms used in infolets should only have up to 12 cells.
- Forms used in infolets should have no more than 12 rows and columns. If a form has more than 12 rows and columns, the infolet will only display the first 12 rows and columns.
- Infolets currently don't support Page dimensions or POVs, therefore forms used in infolets should not contain Page dimensions.
- If a form that is used in infolets contains a grid with members that expand, the infolet will display all the members in the form, including the expanded members.

Designing Charts for Infolets

Use the title and subtitle in charts to show static context.

There are six types of charts that you can use in infolets:

- Bar: Displays a graphical summary of multiple data values for comparison purposes. Bar charts can be plotted vertically or horizontally. Up to eight bars are recommended for bar chart infolets.
- Column: Displays stacked bars that represent different data sets on top of each other. The height of the resulting bar shows the combined result of the data sets.
- Doughnut: A circular graph which is divided into segments to compare data sets to one another. The blank center displays the sum of all data sets. Up to six segment values are recommended for doughnut chart infolets.
- Line: Use to visualize a trend in data over intervals of time.
- Pie: A circular graph which is divided into slices to compare data sets to one another. Up to six slices are recommended for pie chart infolets.
- Tile: Lets you select specific values from a data set to display. No more than three values are recommended for tile chart infolets.

> **Note**
>
> Tile charts can only use the 1x1 size. You cannot resize an infolet using the tile chart until the chart type is changed. If you drag and drop a tile chart to an infolet that is greater than 1x1, you will be prompted to change either the size of the infolet or the chart type.

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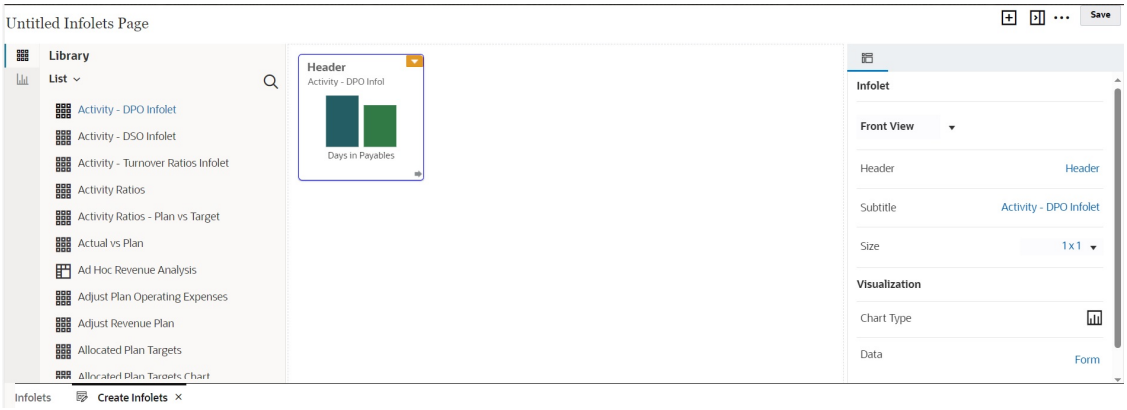
Using the Infolets Designer

Service Administrators use the Infolets Designer to create infolets and infolet pages. The Infolets Designer enables easy toggling between runtime and designer views.

To access the Infolets Designer, click the **Infolets** icon on the Home page, then create an infolet or edit an existing infolet.

Clicking the infolet name in the list region, launches the runtime version of the infolet page. You can toggle from runtime view to designer view by clicking (Infolet Designer).

Infolets Designer



Infolets Toolbar

On the upper right is the Infolets Toolbar.

: Adds a new infolet to the Infolets Designer

: Hides and unhides the Properties panel

: Click to perform these actions:

- **Reset**: Resets the Infolets Designer to a previously saved state
- **Refresh**: Refreshes the data from Essbase and updates the infolet definition from the database
- **Runtime**: Hides all Infolets Designer elements and displays the infolet as it would appear to users during runtime

: From runtime mode, displays the Infolets Designer

Infolets Designer Palette

On the left is the Designer Palette. The Designer Palette has two tabs: Forms and Chart Types. Highlight objects in the Designer Palette and then drag and drop them onto the drop zone.

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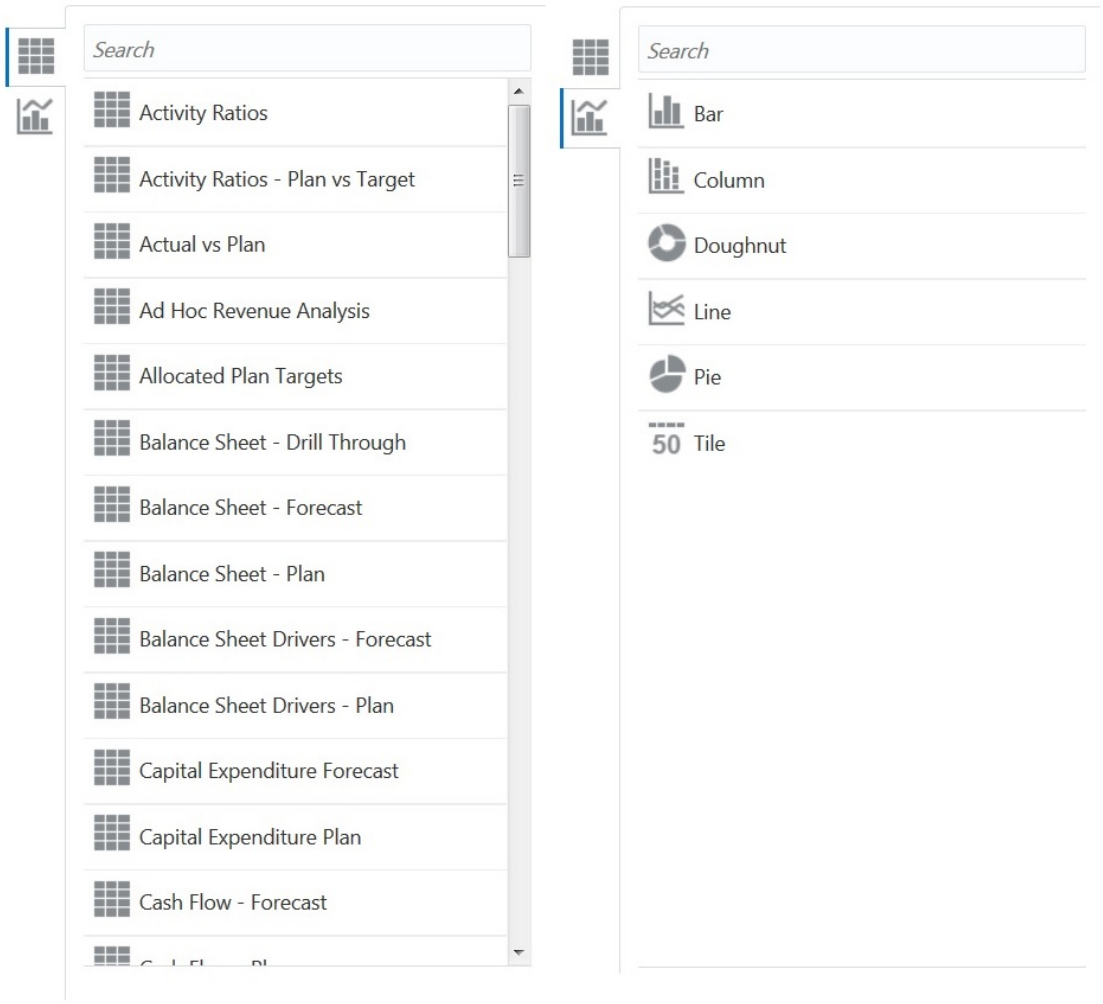

Designer Palette objects:

- Forms: Select forms to include in the infolet by scrolling through the forms or by searching for them by name. For details about forms, see Designing Forms for Infolets.

> **Note**
>
> The access permissions set for forms are honored in infolets.

- Chart Types: Select the chart types to include in the infolet. Infolets display sample data in the charts until you associate the chart with a form as its data source. When you link a chart to a form, users can see the impact of changing data in the form on the associated charts. To associate a chart with a form, highlight a chart and drag and drop it onto the drop zone, in the Properties panel, click **Sample**, and then click **Forms** to select the data source. For details about charts, see Designing Charts for Infolets.

Infolet Menu

The infolet menu contains the infolet delete and clear actions. To view the infolet menu, hover over the upper right corner of the infolet, and then click the down arrow to display the menu options:

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- **Delete**: Removes the infolet from the page
- **Clear**: Clears infolet details

Properties Panel

The properties panel on the right side of the Infolets Designer enables you view and work with these infolet properties:

> **Note**
>
> The header you specify is the same for all views of an infolet, but you can specify a different subtitle for each view; for example, the front, back, and expanded views of an infolet can each have different subtitles, but they must have the same header.

- **Header**
- **Subtitle**
- **Size**: Displays the infolet in the selected size
- **Chart Type**: Displays the infolet data as the selected chart type
- **Data**: Displays the associated data source (Sample or Form)
- **Form**: Displays the selected infolet form

Actions such as delete and clear are on the infolet menu.

By default, the front view of an infolet is displayed in the properties panel. You can see the other views by selecting them from the drop-down. If you flip or expand an infolet to view the back or expanded views, the properties for those views display in the properties panel. In addition, the properties for the corresponding chart type are also displayed in the properties panel.

Creating Infolets

To create infolets:

1. From the Home page, click **Infolets**.
2. Click **Actions**, then **Create Infolets** (or if you're using the legacy interface, click **Create**).
3. Click **Untitled Infolets Page**, and enter a title for the new infolets page you are creating.
4. From the designer palette on the left, choose either the **Library** tab or the **Visualizations** tab, highlight an object, and then drag and drop it onto the infolets drop zone.

   Alternatively, you can click

   (Add Infolet) at the top of the page to add a new infolet to the workspace.
5. If the **Properties** panel isn't already displaying, click

   (Properties) at the top of the page to reveal it. Customize the highlighted infolet using selections made in the **Properties** panel.
6. Click **Save**.

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You can easily duplicate and modify an infolet using the **Copy As** action on the Infolet list page. From the list page, click the Actions icon next to the infolet you want to copy, and then click **Copy As**.

Working with Infolets - Standardized Interface

About the Infolets Interface

This topic provides details about working with the **Infolets** page in the default standardized interface. To learn more about the standardized interface, see About the Standardized Cloud EPM Platform Artifact Pages.

However, there are cases when the legacy artifact pages might still be in use. If you're viewing the legacy **Infolets** page, see Working with Infolets - Legacy Interface.

After you've created an infolets page, it is displayed in the list region on the **Infolets** page.

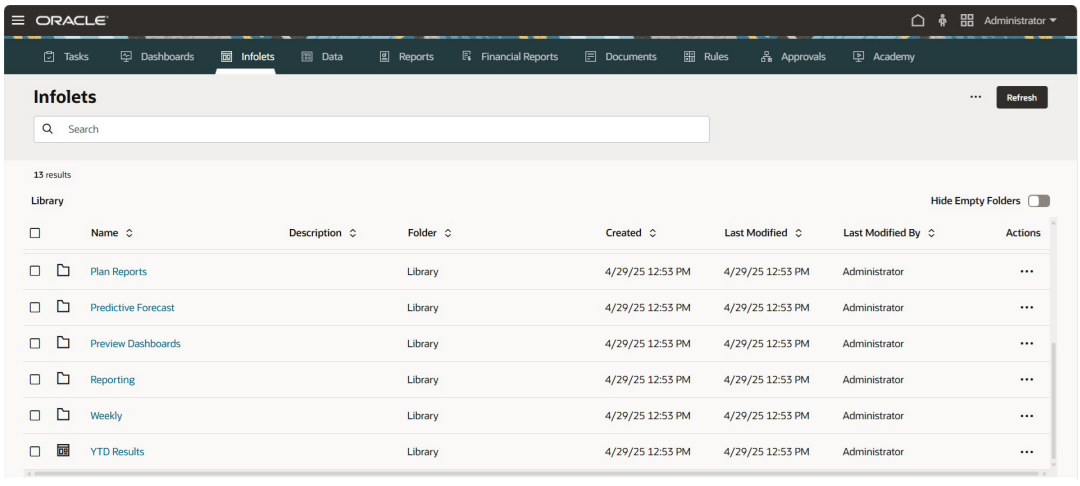
The **Infolets** page supports folders. Folders enable you to assign permissions to all infolets within a folder rather than assigning permissions to each individual infolet. The **Infolets** page uses the same folder hierarchy as dashboards and data entry forms and all artifacts and folders reside under a root folder called **Library**.

> **Note**
>
> Only administrators can add files (for example, dashboards, infolets, forms, reports, and so on) to the **Library** root folder.

Viewing the Infolets Page

To view the infolets page:

1. On the Home page, click **Infolets**.



2. To open an infolet, click the infolet name.

   This opens the infolet in the *runtime* environment.

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Features of the Standardized Infolets Page

On the standardized **Infolets** page:

- To search for a specific infolet, enter search criteria and the search results will display in a drop-down under the **Search** box. Click **Enter** to view the search results in the list region. Search results list all instances of infolets that include the search criteria (including folder names). Each search result displays the complete folder path. Searching inside a folder narrows the search to that folder and its subfolders.

  > **Note**
  >
  > – The search considers the **Name**, **Description**, and **Folder** columns, wherever applicable, in the search.
  >
  > – If you perform a **Search** and the **Hide Empty Folders** toggle is off, all matching folders and artifacts are displayed. If the toggle is on, only folders containing matching artifacts are displayed. Empty folders are hidden.

- **Hide Empty Folders**: Hides folders that don't contain relevant artifacts for the current context. When the toggle is off, all folders are visible in the list region, including empty ones. When the toggle is on, only folders containing relevant artifacts for the **Infolets** artifact page are displayed.

  By default this option is off, and will reset to off each time you start a new session. The state of the toggle is retained by the page during the session and is page-specific. Navigating away from the **Infolets** page and returning preserves the toggle state until the tab is closed or the user logs out.

- To refresh the infolets list, click **Refresh**.

- The global Actions menu ••• (within the header region of the **Infolets** page) includes the following options:

  – **Create Infolets**: Creates a new infolet. See Creating Infolets.

  – **Create Folder**: Creates a new folder.

  > **Note**
  >
  > Keep in mind that newly created folders are empty and therefore will not display if the **Hide Empty Folders** toggle is on. New folders become visible after content is added to them, or if the toggle is switched off.

  – **Assign Permission**: Assigns access permissions to the selected infolets or folders. Folder permissions apply to all items within that folder, even to items that may not be visible in the list. You can assign permissions to multiple infolets or folders at one time. If no infolets or folders are selected, the global **Assign Permission** action assigns permissions to the Library folder.

  – **Move to**: If one or more infolets or folders is selected in the list region, this option moves the folder to another location in the folder hierarchy or moves an infolet to another folder.

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> **Note**
>
> All folders are displayed and are selectable in the **Move** interface, regardless of the state of the **Hide Empty Folders** toggle.

- Each infolet line item also has an item-specific **Actions** menu which includes the following options:
  - **Open**: Opens the infolet in the *runtime* environment so you can view it. The infolet opens in a dynamic tab.
  - **Edit**: Opens the infolet in the Infolets Designer with the underlying form data.The infolet opens in a dynamic tab.
  - **Delete**: Deletes a folder or an infolet. The folder must be empty to delete it.
  - **Rename**: Renames a folder or an infolet. The folder cannot be renamed if it contains other folders.
  - **Copy As**: Copies an infolet with a new name. The infolet is copied to the listing directly under the original infolet.
  - **Move to**: Moves the folder to another location in the folder hierarchy or moves an infolet to another folder.

    > **Note**
    >
    > All folders are displayed and are selectable in the **Move** interface, regardless of the state of the **Hide Empty Folders** toggle.

  - **Copy URL**: Enables you to copy and share the direct URL for an infolet.

    See Copying Direct URLs for Artifacts.
  - **Default** or **Unmark**: **Default** marks an infolet page as default and makes it accessible directly from the Home page by clicking the second infolet dot that is displayed beneath the global header on the Home page. Only one infolet page can be marked as default, and a "(Default)" prefix displays before the name of that infolet in the listing. **Unmark** removes the default designation from the infolet page.
  - **Assign Permission**: Assigns access permissions to the folder or infolet.

Working with Infolets - Legacy Interface

> **Note**
>
> Starting with the October 2025 (25.10) update, Oracle plans to de-support legacy artifact pages. For all new and existing environments, the **Legacy List Pages** application setting will be removed and the **Infolets** page will only render as standardized.

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About the Infolets Interface

This topic provides details about working with the **Infolets** page in the legacy interface. If you're viewing the default standardized **Infolets** page, see Working with Infolets - Standardized Interface.

To learn more about the standardized interface, see About the Standardized Cloud EPM Platform Artifact Pages.

After you've created an infolets page, it is displayed in the list region on the **Infolets** page.

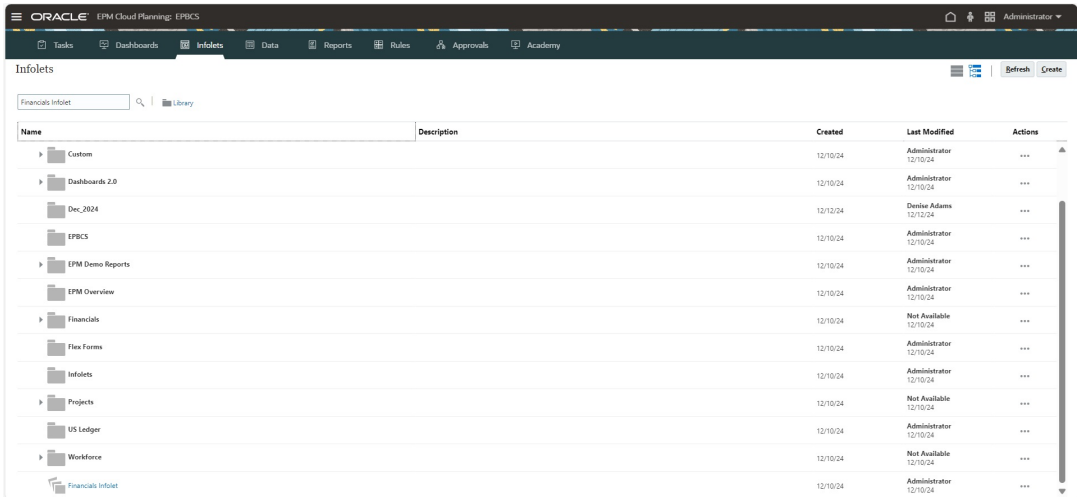
The **Infolets** page supports folders. Folders enable you to assign permissions to all infolets within a folder rather than assigning permissions to each individual infolet. The **Infolets** page uses the same folder hierarchy as dashboards and data entry forms and all artifacts and folders reside under a root folder called **Library**.

> **Note**
>
> Only administrators can add files (for example, dashboards, infolets, forms, reports, and so on) to the **Library** root folder.

Viewing the Infolets Page

To view the infolets page:

1. On the Home page, click **Infolets**.



2. To open an infolet, click the infolet name.

   This opens the infolet in the *runtime* environment.

Features of the Legacy Infolets Page

On the legacy **Infolets** page:

- You can toggle between viewing infolets by a flat view or a tree view:

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- You can search for infolets using **Search** . The flat view displays only the artifacts that meet the search criteria, not the folders that contain them. The tree (or hierarchical) view displays artifacts in the context of the folders that contain them.
- To refresh the infolets list, click **Refresh**.
- To create infolets, click **Create**. See Creating Infolets.
- To perform the following actions on infolet pages, click the **Actions** icon next to the infolet page, and then select:
  - **Create Folder**: Creates a folder in the listing
  - **Edit**: Opens the infolet page in the Infolets Designer
  - **Rename**: Renames the infolet page
  - **Copy As**: Duplicates an infolet page
  - **Delete**: Deletes the infolet page from the listing
  - **Move To**: Moves an infolet page to another folder
  - **Default** or **Unmark**: **Default** marks an infolet page as default and makes it accessible directly from the Home page by clicking the second infolet dot that is displayed beneath the global header on the Home page. Only one infolet page can be marked as default, and a "(Default)" prefix displays before the name of that infolet in the listing. **Unmark** removes the default designation from the infolet page.
  - **Assign Permission**: Enables you to assign Read, Write, and None access permissions to infolet pages and folders for individual users or groups
  - **Copy URL**: Enables you to copy and share the direct URL for an infolet.

    See Copying Direct URLs for Artifacts.

Customizing the Interface to Access Infolets

You can customize the business process interface to add links to infolet pages from the Home page using the Navigation Flow Designer. When you customize your interface to access infolet pages, dots will appear on the Home page beneath the global header and arrows will display to help you navigate easily between the Home Page and infolet dashboard pages. Each dot that appears on the Home page represents an infolet page and hovering over each dot displays the name of the infolet page. Clicking an infolet dot launches the infolet page associated with that dot. You can define up to seven infolet dots on the Home page. If you've created connections to other Oracle Fusion Cloud Enterprise Performance Management environments, you can also add links to infolet pages in other Cloud EPM environments.

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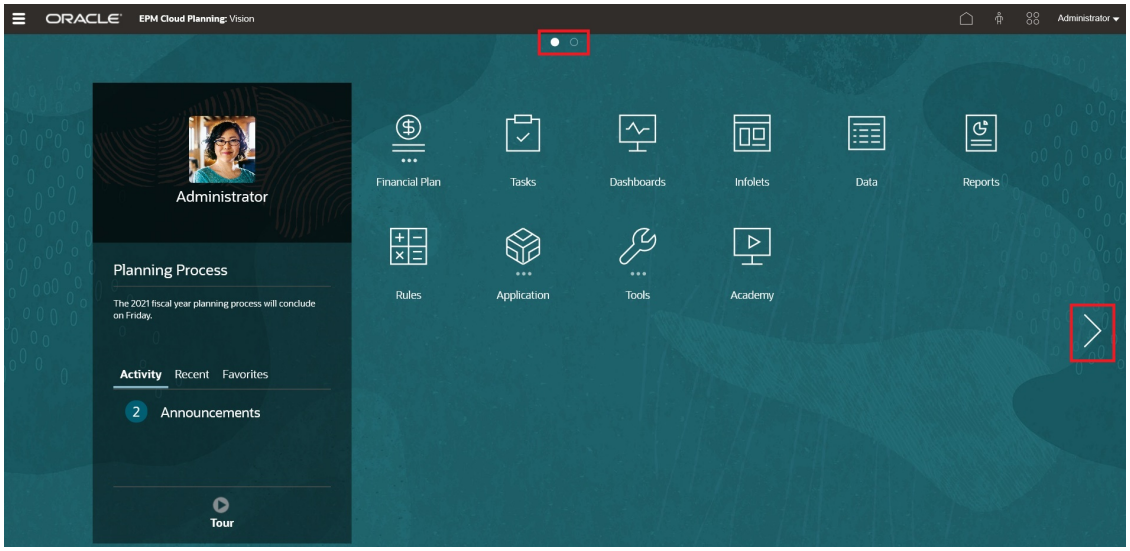

Users will only see dots displayed on the Home page for infolet pages to which they have access. The types of infolet dots displayed are as follows:

- Home dot: This dot always appears first and it links to the Home page. There can only be one home dot. If you are not viewing the Home page, clicking the home dot will bring you back to the Home page.
- User dot: Links to an infolet page marked by an end user as the default infolet page. There can only be one user dot and it always appears after the Home dot on the user's Home page. User dots cannot be added using the Navigation Flow Designer. For more information about marking an infolet as the default, see Working with Infolets - Standardized Interface.
- Customizable dot: Links to infolet pages created by Service Administrators. Customizable dots can be integrated into navigation flows and their visibility and the order in which they appear is determined by the navigation flow designer. There can be up to seven customizable dots and they always appear after the home and user dots.

To add infolet dots to your business process interface using the Navigation Flow Designer:

1. From the Home page, click **Tools**, and then click **Navigation Flows**.
2. Select an inactive navigation flow in the listing, and then click the **Infolets** tab on the Navigation Flow Designer.
3. Click .
4. In **Manage Infolet**, name the infolet dot, set visibility, and then click (Choose Artifact) to select an infolet in the Artifact Library.

   > **Note**
   >
   > You can select an infolet from another Cloud EPM environment if you've created connections to other environments. First select the environment under **My Connections**, and then navigate to the infolet in that environment.

5. Click **Save and Close**.

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> **Note**
>
> Infolets can also be associated with a tab or a card in a navigation flow. While adding or updating a tab or a card, select an infolet in the Artifact Library.

To view design time changes to the navigation flow, activate the navigation flow, and then from the Home page, click the down arrow next to the user name (upper right corner of the screen) and click **Reload Navigation Flow**.

To learn more about designing navigation flows, see Designing Custom Navigation Flows.

To learn more about connecting Cloud EPM environments and for information about the **Copy URL** feature, see Connecting Environments in Cloud EPM.

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14

Configuring IPM

Use Intelligent Performance Management (IPM) to inform your planning decisions.

Related Topics

- About IPM
  Oracle Fusion Cloud Enterprise Performance Management provides Intelligent Performance Management—IPM—integrated directly in your applications to inform your planning decisions with predictive capabilities and accelerate insight discovery with financial pattern recognition.
- Enabling AI in FreeForm
  Enable AI in FreeForm to empower users with data science optimized for multidimensional planning and forecasting use cases and to utilize Reports to generate narrative summaries.
- About IPM Insights
  IPM Insights empowers finance and operational users with data science optimized for multidimensional planning, and forecasting use cases.
- About Using Calendars and Events in IPM Insights, Advanced Predictions, Auto Predict, and Predictive Planning
  When defining IPM Insights, Advanced Predictions, Auto Predict predictions, and when running Predictive Planning from forms on the web, you must specify a calendar to associate with the insight definition or prediction. Optionally, you can also include events in the insight definition or prediction.
- Configuring Calendars and Events
  You can define calendars and events to associate with insights or predictions to improve the accuracy of the insights or predictions.
- Configuring IPM Insights
  Configure IPM Insights in FreeForm to define the insights that planners see on the Insights dashboard.
- Setting Up Predictions to Run Automatically with Auto Predict
  Use Auto Predict in FreeForm to predict future performance based on historical data.
- About Auto Predict
  With Auto Predict, administrators can define a prediction to predict future performance based on historical data and schedule a job to run that prediction definition, automating the prediction process.
- Auto Predict Considerations
  The following are some of the considerations for Auto Predict.
- Creating Auto Predictions for Forecasts
  Create an Auto Predict prediction definition to automate predictions.
- Configuring Advanced Auto Predict Options
  Use Auto Predict advanced options for additional options for data screening, data attributes, prediction methods, and missing value treatment.
- Data Screening Options
  Data Screening options enable you to select several ways of handling missing values, identifying and adjusting outliers, and including events in predictions.

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- Data Attributes Options
  Select from these options for data attributes and seasonality.
- Methods Options
  Select the method to use for predictions.
- Missing Value Treatment for Last Periods
  You have the option when defining the prediction to determine how Auto Predict should handle a missing value for the last historical period.
- Handling Decimal Values and Negative Values in Auto Predict and Predictive Planning
  Auto Predict and Predictive Planning handle decimal values and negative values so that they are appropriately applied to members in predictions.
- About Auto Predict and Predictive Planning Prediction Accuracy
  The prediction accuracy calculation for Auto Predict and Predictive Planning predictions is a simple and generic calculation for all algorithms.
- Running Auto Predict Predictions
  You can run an Auto Predict prediction definition from the Auto Predict page or the IPM page, or you can schedule a job to run the prediction automatically.
- Editing, Deleting, or Duplicating Auto Predict Prediction Definitions
  You can edit, delete, or duplicate existing Auto Prediction prediction definitions.
- Reviewing Detailed Reports for Auto Predict Predictions
  You can review Auto Predict prediction result details in a downloadable report in an Excel worksheet format, including details such as prediction status, method used, and prediction accuracy.
- Running and Scheduling Insights
  After configuring insights, you can run them any time. You can also set up a job to run insights at a regular interval, for example, when you have a new set of actuals.
- Managing Insight Definitions and Generated Insights
  You can edit, delete, or duplicate existing insight definitions. Administrators can delete insights that have been generated.
- Managing Tags
  You can collaborate with your colleagues more effectively and leverage the power of automated analysis with IPM Insights using tags.
- Bring Your Own ML: About Machine Learning Model Import
  With Bring Your Own ML, EPM administrators can import a fully trained Machine Learning (ML) model and deploy it to a FreeForm application. Planners can then leverage robust, ML-based forecasting that uses advanced predictive modeling techniques to generate more accurate forecasts.
- Considerations for Bring Your Own ML
  The following are some of the considerations for Bring Your Own ML.
- Importing an ML Model
  Import a fully trained ML model into a FreeForm application to prepare it for use by business users.
- Deploying an ML Model to Planners
  Deploy an imported ML model to make it available to planners on forms and dashboards. Planners can use machine learning to generate a prediction.

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About IPM

Oracle Fusion Cloud Enterprise Performance Management provides Intelligent Performance Management—IPM—integrated directly in your applications to inform your planning decisions with predictive capabilities and accelerate insight discovery with financial pattern recognition.

Learn more about the Cloud EPM IPM offerings:

- **Advanced Predictions** enable you to use machine learning algorithms to generate predictions, to improve your forecasting accuracy. See Setting Up Advanced Predictions in FreeForm.
  Supported for FreeForm and Planning.
- **Auto Predict** enables you to jumpstart your planning process and improves the accuracy of your forecasts with powerful and automated predictive capabilities. See About Auto Predict.
  Supported for FreeForm and Planning.
- **IPM Insights** reduces time spent in data analysis with financial pattern recognition that offers insights to planners, which they can evaluate and take action on. See About IPM Insights.
  Supported for Enterprise Profitability and Cost Management, Financial Consolidation and Close, FreeForm, Planning, and Tax Reporting.
- **Machine Learning Model Import** lets business users harness the power of data science by performing what-if analysis on a trained ML model that has been imported into a Planning application. See Bring Your Own ML: About Machine Learning Model Import.
  Supported for FreeForm and Planning.

Videos

Your Goal	Watch This
This overview shows FreeForm Applications now support Intelligent Performance Management (IPM). FreeForm also supports IPM calendars and events with IPM Insights, Auto Predict, and Predictive Planning. Using these powerful features in FreeForm enables you to spend less time in analysis, saving time in the overall planning process.	Introduction to IPM in FreeForm

Enabling AI in FreeForm

Enable AI in FreeForm to empower users with data science optimized for multidimensional planning and forecasting use cases and to utilize Reports to generate narrative summaries.

To enable AI in FreeForm:

1. From the Home Page, click **Application**, and then click **Settings**.
2. In the **Enable AI** section, select **Generative AI** to enable these abilities:
   - Summarize IPM Insights with Generative AI for a single insight or group of insights.
   - When viewing insights in forms and dashboards, generate a GenAI-enhanced summary of the insights.

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- Use GenAI to determine how to handle negative values and decimal values in Auto Predict and Predictive Planning.
- Enable Gen AI narrative summaries for Reports.
- Enable the **EPM Assistants** option in the **Downloads** menu.

3. In the **Enable AI** section, select **Advanced Predictions** to enable AI data analysis for advanced, multivariate predictions.
4. Click **Save**.

> **Note**
>
> The **Enable AI** option is available only for EPM Enterprise subscriptions and additionally, it is available only for Insights and prediction features when Hybrid mode is enabled. For more information, see Considerations for IPM Insights.

Generative AI features are available in many OCI realms. To view the realms where it is available, refer to Availability of Generative AI in the *Operations Guide*. If AI isn't available in your home region, then the **Enable AI** option is not displayed.

About IPM Insights

IPM Insights empowers finance and operational users with data science optimized for multidimensional planning, and forecasting use cases.

IPM Insights analyzes past data and predicted data, helping you find patterns and insights into data that you might not have found on your own. Insights can be trends, anomalies, forecast bias, or variations. With IPM Insights, the insight discovery phase of the planning process is automated with data science and financial pattern recognition, enhancing your forecast effectiveness. Using IPM Insights, you can analyze and explore data across any account. IPM Insights automates processing large amounts of multidimensional data, so that as new actuals come into the system, you quickly detect patterns in your data or hidden correlations, streamlining reporting, improving your forecasting, and strengthening your decision making. You spend less time in analysis, saving time in the overall planning process.

IPM Insights offers these benefits:

- Reduces time spent in data analysis
- Minimizes hidden biases
- Improves forecast accuracy and reliability, enhancing forecast effectiveness with informed decision-making
- Encourages collaboration with your colleagues
- Helps you detect trends early to enhance strategic decision-making

IPM Insights provides these types of insights, using historical and predicted data:

- **Forecast variance and bias insights** reveal hidden bias in forecasts submitted by planners by analyzing historical data. This type of insight measures the variance or bias between two historical scenarios such as Forecasts and Actuals, and calculates the percentage of risk associated with the current forecast.
  Forecast variance indicates the level accuracy of the forecasts submitted by planners. Forecast bias indicates the tendency of the direction of forecast error. For example, a

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tendency to forecast in excess of the actuals is called the over-forecasting bias; a tendency to forecast below the actuals is called the under-forecasting bias.

IPM Insights triggers an insight when either the variance or risk crosses the specified threshold.

- **Prediction insights** help you uncover significant deviations in forecasts compared to predicted values. This type of insight looks for variances between two future scenarios such as forecasts and computer-generated predictions, and can calculate a measure of risk, for example the risk of meeting a target forecast.
  By comparing a planner's forecast with a machine-generated prediction, prediction insights measure the variance as well as the presence of any hidden bias (a planner's tendency towards over-forecasting or under-forecasting).

  IPM Insights triggers an insight when either the variance or bias crosses the specified threshold.

- **Anomaly insights** detect unusual patterns in historical data that deviate from expected results. This type of insight detects outlier values that vary widely from other values, for example, a sudden volatility in actuals compared to the base level, or missing actuals data. Anomalies can result from key business actions or they could contain valuable information about a hidden trend.

  Anomalies in historical data can also result from events that historically led to spikes or declines in the data. You can choose to take these spikes or declines into consideration during insight generation, so that historical spikes or declines are also reflected in future predictions.

  IPM Insights triggers an insight when an outlier measure crosses the specified threshold.

- **Period Movement Variance insights** detect significant variance between the current period and a benchmark period. This type of insight allows you to monitor an account member, measure, KPI, or metric of interest to you. For example, you might want to generate an insight when % change in Sales over the previous month is greater than or equal to 10%.
  Insights are triggered when the absolute variance exceeds the administrator-defined threshold. Thresholds can be set for:

  – Absolute Variance (%)

  – Absolute Variance (Numbers)

  Several benchmarks can be used for comparison, depending on how the insight definition was configured:

  – Year-on-year

  – Period-on-period for the previous period

  – Period-on-period for a custom period

  – Historical average

  Period movement variance insights compare single time-series data with the previous period to analyze variance. Future period analysis is not yet available.

  By automatically detecting significant variances against historical, prior-period, and year-over-year benchmarks, period movement variance insights allow organizations to quickly surface emerging opportunities and risks, reduce manual monitoring efforts, and focus attention on the areas that require immediate action. Configurable thresholds ensure that alerts are tailored to business priorities, improving the relevance and accuracy of insights delivered to users.

To get started with IPM Insights, administrators configure insights and then run them, and can schedule them to run on a regular basis.

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Planners can then review and analyze insights in a dashboard, and take appropriate action to improve the forecast.

Figure 14-1 Forecast Variance and Bias Insight

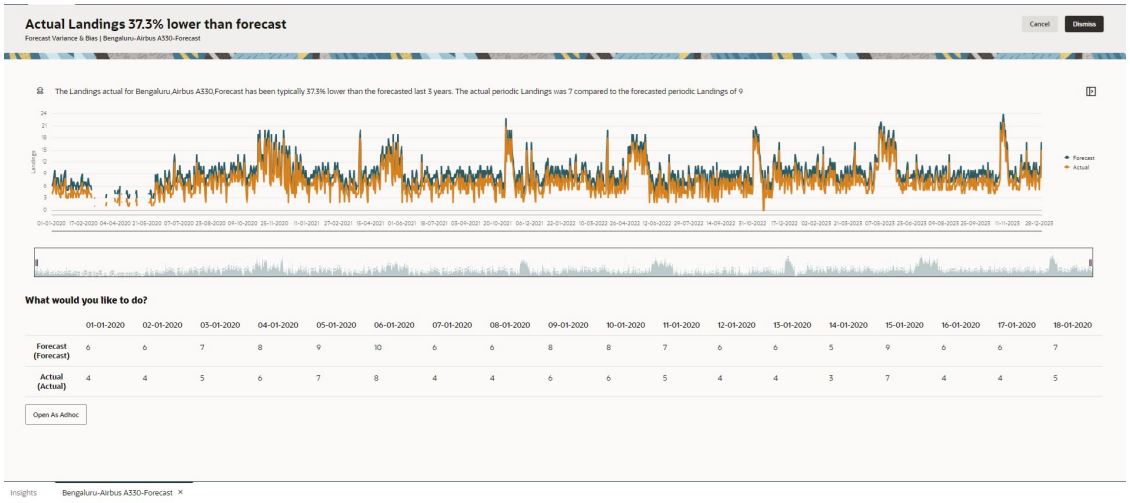

Figure 14-2 Anomaly Insight

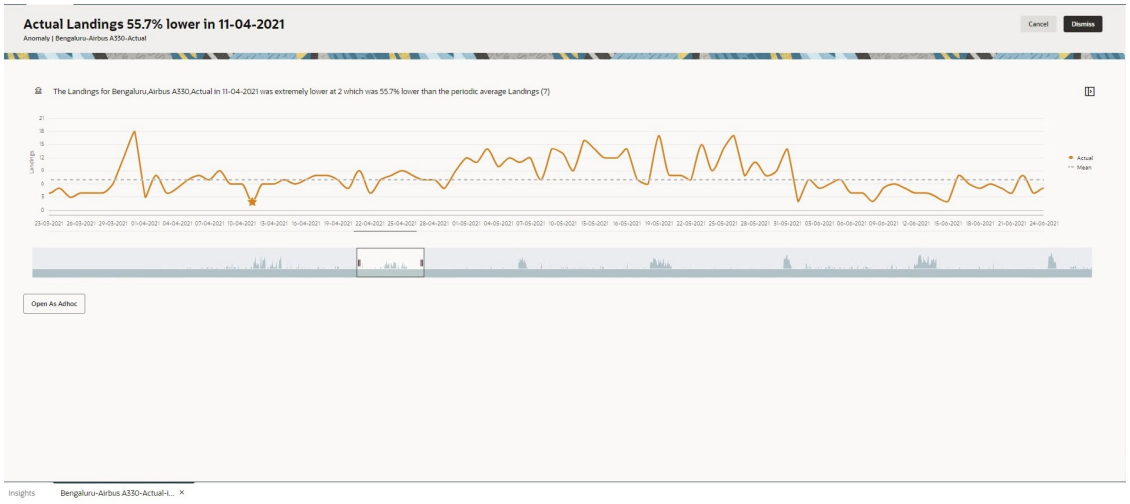

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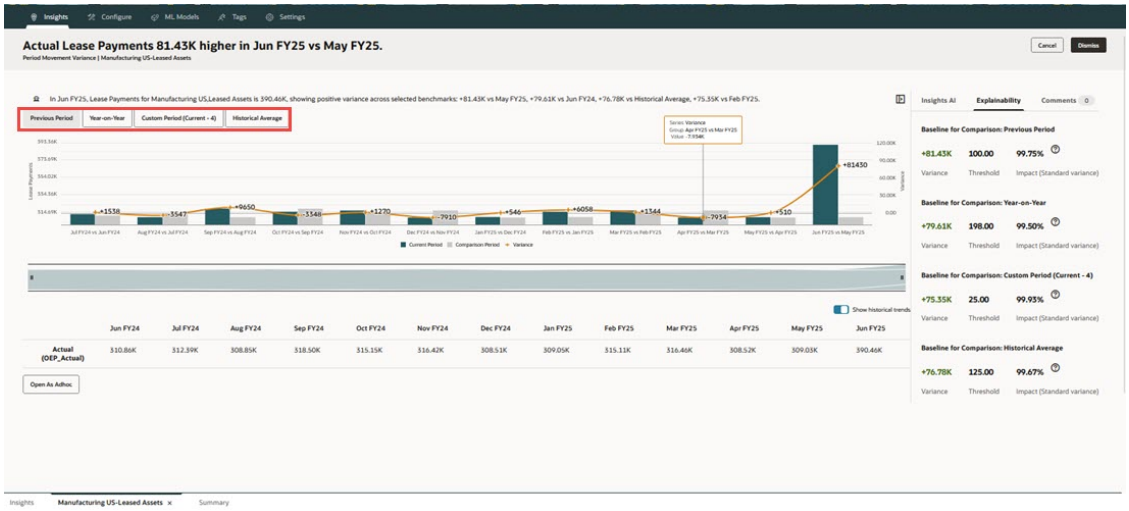
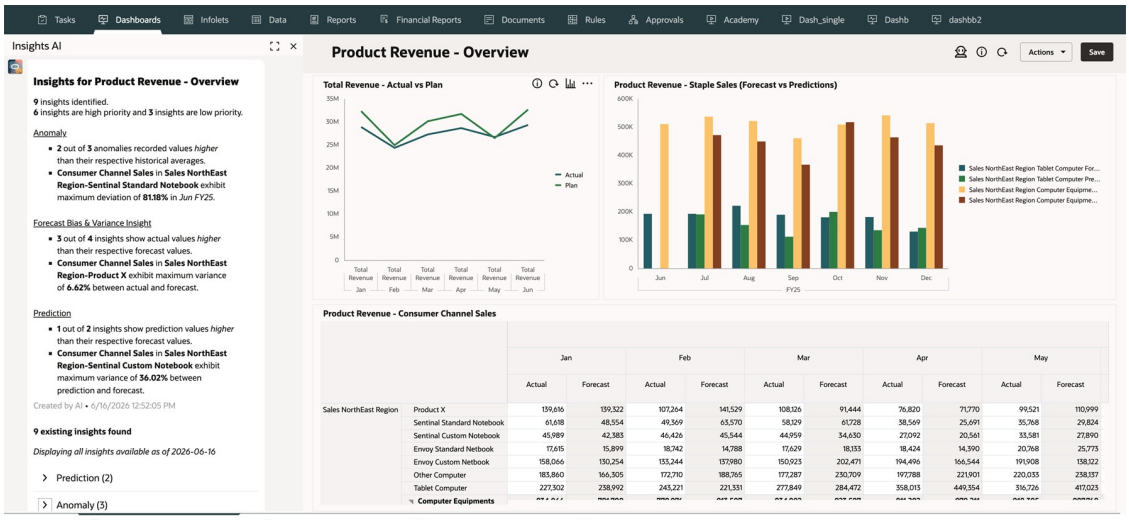
Figure 14-3 Period Movement Variance Insight



You can also view insights directly from a form or dashboard.



Tip

You can also work with IPM Insights in Oracle Smart View for Office. For more information, see Using IPM Insights in Smart View.

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Videos

Your Goal	Watch This
This overview introduces you to Insights. Insights empowers finance and operational users with data science optimized for multidimensional planning and forecasting use cases. Insights analyzes past data and predicted data, helping you find patterns and insights into data that you might not have found on your own. You can evaluate those insights and take action on them.	Overview: Insights in Cloud EPM

Considerations for IPM Insights

The following are some of the considerations for IPM Insights and Auto Predict.

Considerations:

- IPM Insights requires an EPM Enterprise Subscription.
- IPM Insights works with these business processes:
  - Planning. IPM Insights works with these Planning application types:
    - Custom
    - Modules
    - FreeForm
    - Sales Planning
    - Strategic Workforce Planning
    - Cash Forecasting
  - FreeForm.
  - Oracle Sales Planning Cloud and Oracle Strategic Workforce Planning Cloud.
  - Enterprise Profitability and Cost Management.
  - Tax Reporting.
  - Financial Consolidation and Close.
- IPM Insights works only with applications with Hybrid mode enabled.
- IPM Insights honors both member level security and cell level security, so planners see insights only for slices of data to which they have access. However, note that insights are not generated for a user when cell level security is defined on Year/Period dimensions.
- If you add dimensions to the application after the job is created, you must reconfigure the IPM job to include the additional dimensions.
- Note the following considerations for the Insights Generative AI summary feature:
  - Insight summaries use GenAI. Output from GenAI may not always be factual, accurate, or appropriate. You are responsible for reviewing and using GenAI output at your discretion.
  - The Insights Generative AI summary feature requires an EPM Enterprise Cloud service subscription.

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- Narratives are generated in English only.
- Summaries are available for all insight types.
- Administrators need to re-run existing insights for users to be able to see the AI summary.
- The GenAI feature is not yet available in all OCI regions. To view the regions where it is available, refer to Availability of Generative AI in the *Operations Guide*.

IPM Insights Support Matrix

Review the table for a summary of the supported SKUs, business processes, and application types for IPM Insights.

Table 14-1 IPM Insights Support

SKU	Business Process	Application Type	Supported?	Comments
EPM Enterprise Subscription	Planning	Modules	Supported	Hybrid Essbase must be enabled.
	Planning	Custom	Supported	Hybrid Essbase must be enabled.
	Planning	Freeform	Supported	Hybrid Essbase must be enabled.
	Planning	Sales Planning	Supported	Hybrid Essbase must be enabled.
	Planning	Strategic Workforce Planning	Supported	Hybrid Essbase must be enabled.
	Planning	Predictive Cash Forecasting	Supported	Hybrid Essbase must be enabled.
	FreeForm		Supported	Hybrid Essbase must be enabled.
	Enterprise Profitability and Cost Management		Supported	
	Tax Reporting		Supported	Hybrid Essbase must be enabled.
	Financial Consolidation and Close		Supported	Hybrid Essbase must be enabled.
	Other business processes		Not supported	
Legacy SKU	Planning	Enterprise Planning and Budgeting Cloud	Not supported	
	Planning	Planning and Budgeting Cloud	Not supported	
	Other business processes		Not supported	

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Table 14-1 (Cont.) IPM Insights Support

SKU	Business Process	Application Type	Supported?	Comments
EPM Standard Subscription	Planning	Standard Module	Not supported	
	Other business processes		Not supported	
Sales Planning	Planning	Sales Planning	Supported	Hybrid Essbase must be enabled
Strategic Workforce Planning	Planning	SWP	Supported	Hybrid Essbase must be enabled

Best practices:

- **Historical Data**: Prediction results are more accurate the more historical data you have. There should be at least twice the amount of historical data as the number of prediction periods. For example to predict 12 months in the future, you should have at least 24 months of historical data. At the time of prediction, if there is not enough historical data available, a warning is displayed.
- **Time Granularity of Predictions**: Because the lowest Period dimension member level in an Insights job determines the time granularity of the prediction, create the Insights job using the lowest level of Period members possible so that the greatest amount of historical data can be used.

About Using Calendars and Events in IPM Insights, Advanced Predictions, Auto Predict, and Predictive Planning

When defining IPM Insights, Advanced Predictions, Auto Predict predictions, and when running Predictive Planning from forms on the web, you must specify a calendar to associate with the insight definition or prediction. Optionally, you can also include events in the insight definition or prediction.

If you are using Advanced Predictions, you must define a calendar to use with the prediction definition.

Events are typically one-off or recurring events that historically led to spikes or declines in the data. You can choose to take these spikes or declines into consideration during insight analysis, so that historical spikes or declines are also reflected in future analyses. For example, a North American calendar might include an event for Christmas, or an APAC calendar might include an event for Diwali, when sales would typically spike. By including the historical data spikes in the analysis, you see the spikes in the insight data, so you can plan ahead for volume or to make use of the opportunity.

Without events, spikes or falls in data are normalized and distributed over the analysis period, potentially leading to less accurate analyses. Using events helps you plan ahead for the events and take advantage of opportunities by allowing you to see anticipated spikes and falls in the insight data for specific events. Using events with IPM Insights, Advanced Predictions, Auto Predict, and Predictive Planning simplifies analysis and significantly improves the insight accuracy for complex and volume based industry verticals such as air travel, public transportation, retail, E-commerce, brick and mortar stores, FMCG, and automobiles.

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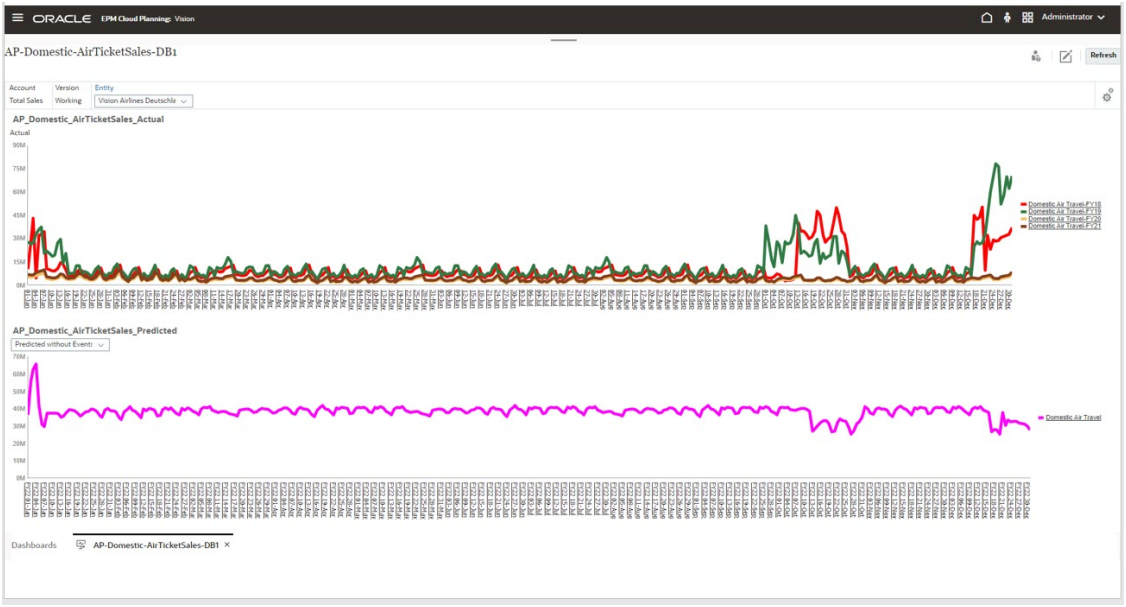
Alternatively, you can choose to skip the historical data spikes or falls, so that the data for the event is completely excluded from the analysis.

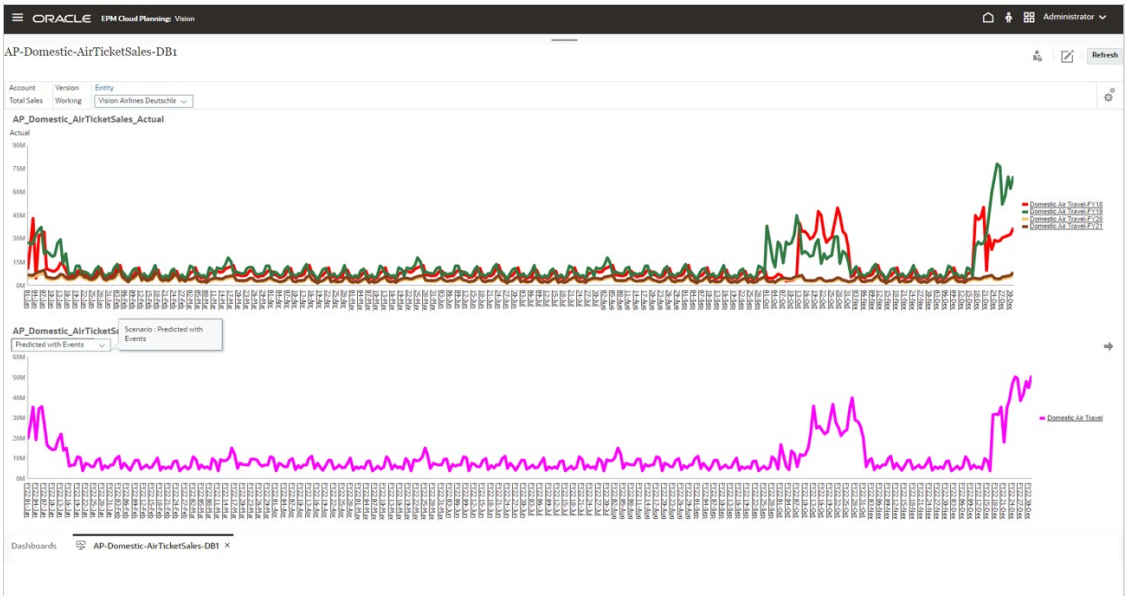
To associate calendars and events with insight definitions, Advanced Predictions, and Auto Predict prediction definitions, configure calendars and events first. See Configuring Calendars and Events.

These dashboards illustrate the effect of including events in the prediction.

In this example, the upper chart shows the historical data, which includes several spikes. The lower chart shows the prediction results without events. Notice that the predicted data is spread and normalized so that there are no spikes.



In this example, the upper chart shows the same historical data, which includes several spikes. The lower chart shows the prediction results with events. Notice that the spikes are evident in the predicted data. In this example, using events significantly improves the prediction accuracy.



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To use calendars and events with IPM Insights, Advanced Predictions, and Auto Predict:

1. Define calendars.
2. Define events.
3. When configuring an IPM Insights job, Advanced Predictions, or an Auto Predict prediction definition, associate a calendar with the insight, Advanced Prediction, or Auto Predict job. Then, select whether to include events in the analysis.

For details on these procedures, see Configuring Calendars and Events and Configuring IPM Insights.

You can use calendars and events with Predictive Planning on forms:

- When designing a form, use **Prediction Options** on the Form Designer Layout tab to specify a calendar to associate with the prediction for the form, and to select whether to include events in the prediction by default. This selection is saved with the form, so it is the default option for all users of the form. See Setting Prediction Options.
- When running a prediction from a form, planners can use the **Settings** option to select a different calendar to use for the prediction and whether to include events in the prediction. They can use the **Chart Settings** to specify whether to show events on the chart, and can also choose to exclude Skip events. Changes to these settings are for the current user session. These settings revert to defaults for each new login session.
  See Predicting Future Values Based on Past Performance in *Working with FreeForm*.
- In FreeForm applications, forms must have a calendar associated with them, and the calendar must map to the granularity defined in the form. If you don't associate the calendar with the form in the Form Designer, a user who opens the form and runs a prediction is prompted to select a calendar for use with the prediction on the form. This selection is saved with the form.

Notes about calendars and events:

- Defining events is optional. Calendars are required for Insights, Advanced Predictions, and Auto Predict jobs, and must be associated with the form when running Predictive Planning from a form.
- For Predictive Planning, events are available only on the web, not in Oracle Smart View for Office. Because events aren’t supported in Smart View, if you run a prediction on a web form that includes events, and then you open the same form in Smart View and run a prediction, you might see different prediction results on the web and in Smart View.
- You can use calendars with each insight type, with Advanced Predictions, and with Auto Predict. Events are an option for the Anomaly insight job type, Advanced Predictions, and Auto Predict.

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Videos

Your Goal	Watch This Video
This video introduces you to how to include events into your prediction process. Considering events when running predictions improves the accuracy of the predictions and helps you plan well in advance for the impact on sales or volume demand caused by events. By incorporating seasonal, cultural and other types of events, you can take advantage of opportunities that allow you to see anticipated spikes and falls in the predicted data for those specific events. Historical spikes or declines due to events are also reflected in future predictions.	Introduction to Predicting with Events in Cloud EPM Planning

Configuring Calendars and Events

You can define calendars and events to associate with insights or predictions to improve the accuracy of the insights or predictions.

To use calendars and events with IPM Insights, Advanced Predictions, and Auto Predict:

1. Define calendars. From the Home page, click **IPM**, then click **Configure**, and then click the **Calendar** tab. See Defining Calendars.
2. Define events. From the Home page, click **IPM**, then click **Configure** and then click the **Events** tab. See Defining Events.
3. When configuring an IPM Insights job, Advanced Predictions, or an Auto Predict prediction definition, associate a calendar with the insight, Advanced Prediction, or Auto Predict job. Events are an option for the Anomaly insight job type, Advanced Predictions, and Auto Predict. See Configuring IPM Insights, Creating Auto Predictions for Forecasts, and Configuring Advanced Predictions.

You can also define calendars and events from **Auto Predict**. See Defining Calendars and Events from Auto Predict.

Defining Calendars

A calendar defines the scope and range of the time series for selected cubes. For example, you can create a daily calendar, a weekly calendar, and an alternate calendar. You might want to create a calendar for a specific entity, because not all events are applicable to all entities. For example, Diwali applies to APAC and Thanksgiving applies to North America, so you can create an APAC calendar with its own events and a North America calendar with its own events.

You can use calendars on their own to quickly set up the time frame for insight and prediction definitions, or use calendars with events.

> **Note**
>
> If you are defining calendars for use with Advanced Predictions, note that predictions do not work if security is enabled on the Year/Period or dimensions defined in the calendar.

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To define calendars:

1. On the **Calendar** tab, click **Add Calendar**.
2. Enter a name and description for the calendar.
3. From the **Cube** column, select the cubes the calendar applies to.
4. From the **Dimensions** column, select one or more dimensions (up to three) that store date-related information, for example, Year, Quarter, and Month. If you are selecting multiple dimensions, select the dimensions in order of granularity, with the highest granularity first. For example, select the dimension that stores years before the dimension that stores months.
5. From the **Time** column, specify the members from each dimension that define time for the start of the calendar. You can select specific members or use substitution variables.

   Time includes the entire range of historical and future periods required for the predictions.
6. From the **Current** column, specify the members from each dimension that define the start of the prediction period. The previous period is the end of historical data.
7. Enter the **Number of Historical Periods** and **Number of Future Periods**.
   The number of historical periods and the number of future periods you enter must result in member selections that are within the range defined for calendar. For example, if the calendar range is defined as FY20 to FY25, Lev0Descendants(YearTotal), and the current period is defined as Jan->FY24, then the maximum number of historical periods you can enter is 48 periods (4 years). Any number higher than 48 results in an error. Similarly, the maximum number of future periods you can enter is 24 periods (2 years). Any number higher than 24 results in an error, because Jan FY26 or higher is not part of the calendar range.
8. For Advanced Predictions, click **Date Mapping** to define the frequency and date format for the predictions. Date Mapping is required for data science engine preprocessing and training the ML model.
   Select the **Frequency** for the data (**Daily**, **Weekly**, **Monthly**, **Quarterly**, or **Custom**) and then select the options for the selected frequency, for example the date, the month and year, and so on. When you set the first date for the prediction period, the system derives the rest of the dates. For **Custom**, click **Import** to import custom values from a `.csv` file.

   > **Note**
   >
   > To see how the dates are generated based on your selections, click **Export Mappings**, and then open the `.csv` file that is generated.

   When the date mapping is done, click **Save** .
9. Click **Save** .

Editing Calendars

To edit a calendar, from the **Calendar** list, double-click the row with the calendar you want to edit to make the options editable. Make any required changes, and then click **Save** . To discard your changes, click **Cancel** .

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For example, you can change the **Number of Historical Periods** and **Number of Future Periods**.

Alternatively, click the Actions menu next to a calendar, and then select **Edit** or **Delete**.

For Date Mapping for Advanced Predictions, you can also change the frequency. Click **Date Mapping** and change the frequency, for example from weekly to monthly. Click **Export Mappings** to export a list of time periods from the application mapped to an actual date, in `.csv` format. If needed, make any required changes and then click **Import Mappings** to import the updated data to the application.

To delete a calendar, from the **Calendar** list, click **Actions** ••• next to the calendar you want to delete, and then click **Delete Calendar**.

Defining Events

Events are associated with calendars.

To define events:

1. On the **Events** tab, click **Add Event**.
2. Enter a name and description for the event.
3. From the **Type** column, select the type of event:
   - **One-off**: An event you don't expect to be repeated, but you want to include it in your data.
   - **Repeat**: An event that is regularly repeated, where increases or decreases will also be repeated. For example, holidays such as Diwali or Thanksgiving are repeating events.
   - **Skip**: Similar to a one-off event, it's an event you don't expect to be repeated, however, you don't want to include it in your data. Data for Skip events is removed from the series.

   > **Note**
   >
   > If you define a Skip event that is shorter than your seasonality, it can affect the accuracy of the historical data. For example, you might define an event for Covid that is three months long, but your season is twelve months, so you have only nine months of data. You lose seasonality because the data is skewed. In these cases, it might be more effective to skip the entire season.
4. From the **Calendar** column, select the calendar that the event applies to.
5. From the **Starts on** column, specify the start date for the event.
6. From the **Duration** column, enter the event duration (the number of periods the event is valid). For example, in a daily calendar, if duration is 4, the event occurred for 4 days continuously from the start of the event. In a weekly calendar, if duration is 1, the event occurred for a week.
7. For regularly repeating events, from the **Interval** column, enter the interval for the repeat. For regularly repeating events, the interval should be a number that indicates after how many days, weeks, or months the event repeats. For example, in a weekly calendar, Christmas repeats after 52 weeks.
   You don't need to specify an interval for one-off or skip events.

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8. For custom repeating events, from the **Occurrences** column, click + and enter specific details about when the event occurs. For example, US Thanksgiving occurs annually, but on a different date each year.
9. Click **Save** ✓.

Events are automatically set in **Active** status (indicated by the check mark in the **Active** column). You can clear the check box to make an event inactive. Only active events are taken into consideration during predictions.

To edit an event, from the **Events** list, double-click the row with the event you want to edit to make the options editable. Make any required changes, and then click **Save** ✓. To discard your changes, click **Cancel** ✕.

To delete an event, from the **Events** list, click **Actions** ••• next to the event you want to delete, and then click **Delete Event**.

Importing and Exporting Events

You can export events to a .csv that you download, and then add or modify events. You can then import the events from the csv file. This process is a fast way to add many events, and can be valuable for auditing and troubleshooting event definitions.

To export events, on the **Events** tab of IPM Configure, from the **Actions** menu, select **Export Events**.

To import events, on the **Events** tab of IPM Configure, from the **Actions** menu, select **Import Events**, select the file and then click **Import**. You can then review the **Import Events** job. From the Home page, click **Application**, then click **Jobs** and then click the **Import Events** job that was run.

If you import an event with the same name as an existing event, the existing event is overwritten.

Defining Calendars and Events from Auto Predict

1. From the Home page, click **Application**, then click **Overview**, and then from the **Actions** menu, select **Auto Predict**.
2. From the **Actions** menu, select **Calendar** to define calendars or select **Events** to define events. See Defining Calendars and Defining Events.

Configuring IPM Insights

Configure IPM Insights in FreeForm to define the insights that planners see on the Insights dashboard.

You configure IPM Insights by selecting the type of analysis to perform, specifying the slice definitions for analysis, configuring the analysis, and then defining the settings for the insights displayed on the Insights dashboard. You can define insights for as many data intersections as you need. Planners see insights only for slices of data to which they have access.

You can also define a prediction using Auto Predict to predict future performance based on historical data and schedule a job to run that prediction definition, automating the prediction process. You can use these prediction results as input for generating insights.

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To associate calendars and events with insight definitions, Advanced Predictions, and Auto Predict prediction definitions, configure calendars and events first. See Configuring Calendars and Events.

To configure IPM Insights:

1. From the Home page, click **IPM** and then click **Configure**.
2. Click **Create**.
3. In IPM Configurator, on the **Types** page, enter a name and description.
4. Select either **Auto Predict** to predict future performance based on historical data using time series forecasting techniques, or **Advanced Predictions** to use machine learning algorithms to generate predictions. Or select **None** if you don't want to generate any predictions. For more information on setting up Auto Predict and Advanced Predictions, see Creating Auto Predictions for Forecasts and Configuring Advanced Predictions.
5. Select the types of insights to generate, and then click **Next**. You can select as many as you need.
   - **Forecast Bias Insights**. Reveals hidden bias in forecasts submitted by planners by analyzing historical data. Measures the variance or bias between two historical scenarios such as Forecasts and Actuals.
   - **Prediction Insights**. Looks for variance between future scenarios such as forecasts made by planners and computer-generated predictions.
     In **Do you have prediction data available?**, if you have prediction data available and know where prediction data is, select **Yes**. If not, select **No** and **Auto Predict** is automatically selected. Auto Predict uses the historical data for the intersection you defined to generate prediction data that is used for the insight.

     You can also use a prediction based on Machine Learning instead of auto prediction.
   - **Anomaly Insights**. Detects outlier values that vary widely from other values.
   - **Period Movement Variance Insights**. Detects significant variance between the current period and a benchmark period. This type of insight allows you to monitor an account member, measure, KPI, or metric of interest to you.

   You can also choose to just run Auto Predict by selecting **Auto Predict**.
6. On the **Select Calendar** page, select a calendar to use, and then click **Next**. When you select a calendar, the **Historical Data** section and the **Future Data** section are populated with details from the selected calendar.
   Calendars are required for Insights, Advanced Predictions, and Auto Predict jobs.

   For **Future Data**, you can define insights that analyze at the parent level or at the leaf level for Period members. For example, for **Future Periods**, if you have monthly data, you can perform the analysis at the quarterly level. Note that leaf level and middle level parent members are allowed. Top level parent members, anything directly below Period, are not supported. For Auto Predict, analysis must always be at the leaf level.
7. On the **Define Slice** page, define the slice of data to analyze for historical data (Actual and Forecast) and future data and then click **Next**.
   You can use sibling-based functions when defining the slice, which allows you to select members relative to a given member within a hierarchy. The following functions are supported:
   - LSIBLINGS—Returns siblings to the left of the member.
   - ILSIBLINGS—Returns the member and its siblings to the left.
   - RSIBLINGS—Returns siblings to the right of the member.

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- IRSIBLINGS—Returns the member and its siblings to the right.

For considerations on using dynamic calc parent members in the slice definition, see Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions.

> **Note**
>
> After completing the IPM configuration for prediction insights, you must also set the **POV** for the **Source** cube in the **Predictive Planning** settings in **Application Settings** to be the same **Scenario/Version** as the slice you defined for **Historical Data**.

Note:

- In the **Historical Data** section, select the cube and then define the slice of data for the historical data—both actuals and the forecasts provided by the planners. The slice definitions show all the dimensions for the cube, except for the Year and Period dimensions. All members start out with their root member selected.
- In the **Future Data** section, select the cube and then define the slice of data for the future data—both the forecast provided by planners, and the base prediction (most likely scenario).

  If the data is available, click ⊕ to add a definition for the slice of data for the best case and worst case scenarios.

  If you selected Auto Predict, prediction results are stored in this location.

8. On the **Configure** page, configure the insights by defining the error tolerance and thresholds for insights, and then click **Next**. IPM Configurator selects default metrics for analysis.

   - **Auto Predict**: Select whether to include events in the prediction.
     If you select **Show Advanced Options**, select additional options to define an auto predict job. For more information, see Configuring Advanced Auto Predict Options.
   - **Forecast Variance & Bias Insights**: Select an error tolerance percentage, which defines the percentage variance between historical forecasts submitted by planners and actuals that lies within acceptable range. If the percentage variation crosses the error tolerance limit, it is considered for bias calculation.
     For example, if you enter .05, any difference of 5% or above between historical forecasts and historical actuals is considered a significant variance (and considered either under-forecasting or over-forecasting) and triggers an insight.

     You can select or modify the metric and threshold for deviation on the **Advanced Options** page. See Configuring Additional Insight Settings.

     An insight is generated when either deviation or bias crosses the specified threshold.
   - **Prediction Insights**: Select a threshold percentage, which defines the acceptable percentage variance between future forecasts submitted by planners and computer-generated predictions.
     An insight is generated when either deviation or risk crosses the specified threshold.

     For example, if you enter 25%, any difference of 25% or above between future forecasts and future predictions would be considered a significant variance (and considered either under-forecasting or over-forecasting) and would trigger an insight.
   - **Anomaly Insights**: Select whether to include events in the insights calculation.

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Select a threshold, which defines the acceptable threshold for Z-score value (the standard deviation from the mean of the values). Anything that is too far from zero (the threshold is usually a Z-score of 3 or -3) should be considered an outlier.

An insight is generated when the specified outlier detection metric crosses the specified threshold.

For example, if you enter 3, anything that is three standard deviations or higher from the mean would be considered an outlier, and would trigger an insight.

- **Period Movement Variance Insights**. Select the variance metric to use, **By Percentage** or **By Absolute Value**, and then define the threshold for variance for the default benchmark, **Previous Period**. Insights are triggered when the absolute variance exceeds the threshold. For example, you might want to generate an insight when % change in Sales over the previous month is greater than or equal to 10%.

> **Note**
>
> If you later edit the variance metric, rerunning the Insights job dismisses the previous insights that were generated and generates new insights, even though the details are different for the previous and new insights. Because the POV is the same, the insights are always regenerated in this scenario.

If you want to configure additional options, such as selecting the metrics to use or selecting additional benchmarks to use for period movement variance insights, click **Show Advanced Options**. See Configuring Additional Insight Settings.

9. On the **Settings** page, configure the settings that define how to display the insights to planners, and then click **Save**.
   - **Display Dimensions**: Select the dimensions that planners will use to navigate and analyze the insights. The dimensions you select will display on the Insights dashboard.
   - **Impact Magnitude Thresholds**: When insights are displayed on the Insights dashboard, this setting categorizes insights into High, Medium, and Low groups based on the percentage or absolute value impact calculated for each insight. This helps planners focus attention on the insights that have more extreme variations.
     - Anything less than the low value is in the Low category.
     - Anything higher than the high value is in the High category.
     - Anything between the low and high value is in the Medium category.

     For example, if you specify 30% as the Low threshold and 60% as the High threshold, any insight with a percentage impact less than 30% is in the Low category; any insight with a percentage impact greater than 60% is in the High category. Any insight with a percentage impact between 30% and 60% is in the Medium category.

     When entering an absolute value for impact magnitude thresholds, the currency defined in the slice definition applies to the absolute value. If you have multiple currencies defined, you can't apply an absolute value magnitude threshold; the option is not available.
10. Now you're ready to run the insights you configured. See Running and Scheduling Insights.

Insight definitions are saved as global artifacts (in the Auto Predict folder under Global Artifacts) and are backed up with the maintenance snapshot.

For prediction insights that are generated, the historical data is displayed based on the **POV** selection for the **Source** cube in the **Predictive Planning** settings in **Application Settings**.

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About Using Events with Insights, Advanced Predictions, Auto Predict, and Predictive Planning

Note the following about using events:

- Using events can impact the anomaly insights that are generated. For example, if you create a one-off or repeating event for a specific date where sales spike, and you generate an analysis, anomalies in data for that date are not reported as anomaly insights. Instead, the analysis includes the sales spike for that date. If you generate an analysis without events, an anomaly insight is reported for that date.
- In Auto Predict jobs, note the following behavior for skip and one-off events:
  - For skip events, once the skip event goes into the prediction range, the skip event is not mentioned or listed in the downloaded job report.
  - For one-off events, once the one-off event reaches the last period of the historical data (the period which bumps up against the prediction range), the one-off event is not mentioned or listed in the downloaded job report.
- In prediction on forms using Predictive Planning, note the following behavior for skip and one-off events:
  - For skip events, once the skip event goes into the prediction range, the skip event is no longer applied.
  - For one-off events, once the one-off event reaches the last period of the historical data (the period which bumps up against the prediction range), the one-off event is no longer applied.

Videos

Your Goal	Watch This Video
Learn how to configure insights to automate data analysis. Administrators configure insights to define the insights that planners see on the Insights page. You select the type of analysis to perform, specify the slice definitions for analysis, configure the analysis, and then define the settings for the insights displayed on the Insights page. You can run the insight or schedule it to run on a regular basis.	Oracle Fusion Cloud EPM - Configuring Insights
Learn how Period Movement Variance insights help identify significant changes between current and benchmark periods. Learn how thresholds and benchmark comparisons make variance results easier to understand and review.	Generating Period Movement Variance Insights in Cloud EPM

Tutorials

Tutorials provide instructions with sequenced videos and documentation to help you learn a topic.

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Your Goal	Learn How
Learn how to configure IPM Insights by selecting the type of analysis to perform, specifying the slice definitions for analysis, configuring the analysis, and then defining the settings for the insights displayed on the Insights dashboard. You can define insights for as many data intersections as you need. Planners see insights only for slices of data to which they have access.	Configuring Insights

Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions

Using Dynamic Calc members allows you to view insights and predict values at the parent level or any level in the hierarchy.

You can generate insights and predictions for **Dynamic Calc** parent and leaf members. Predictions include Predictive Planning, Auto Predict, and Advanced Predictions.

- Rather than having to sort through too many leaf level-insights, you can focus your analysis on parent-level or any level insights.
- Generating insights and predictions at the parent level improves accuracy, especially when data is missing or not accurate in lower levels of the hierarchy.
- Prediction accuracy is improved, because rather than predicted data getting aggregated at the parent level, the parent level is also predicted.

  For example, predicting for overall Operating Expenses might be better than predicting each of the expense categories under Operating Expenses such as Travel, Entertainment, and Staff Welfare, because the child member data could be sparse.

Generating Insights for Dynamic Calc Members

To generate insights for **Dynamic Calc** members for Forecast Variance & Bias and Anomaly insights, simply include the **Dynamic Calc** members in the insight slice definition.

Generating Prediction insights for **Dynamic Calc** members requires you to specify a slice definition that includes members in Scenario, Version, Plan Element or relevant dimensions associated with the UDA **HSP_DYN_PREDICT**.

To generate Prediction insights for **Dynamic Calc** members:

1. Associate the UDA **HSP_DYN_PREDICT** with the Scenario/Version/Plan Element or similar dimension used to store predicted values. (This step is not needed for Forecast Variance & Bias insights and Anomaly insights.)
2. If you associated the UDA **HSP_DYN_PREDICT** with dimensions other **Scenario** or **Version**, update the **Predictive Planning** application settings:

   From the Home page, click **Application** and then click **Settings**. In the **Predictive Planning** section, update the **POV** to include the members from the other dimensions that are associated with the UDA **HSP_DYN_PREDICT**.

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> **Note**
>
> The **Source Cube** option defines the cube where historical actuals are stored. Members selected for **Source Cube** should not be associated with the UDA **HSP_DYN_PREDICT**.

3. Include the **Dynamic Calc** members in the Insight slice definition. Make sure to include only values with the UDA **HSP_DYN_PREDICT** in the slice definition.

Generating Auto Predict and Advanced Predictions for Dynamic Calc Members

To define Advanced Predictions for **Dynamic Calc** members, include the **Dynamic Calc** members in the Advanced Prediction definition. **Dynamic Calc** members can be defined in the source slice (historical data slice) and target slice (future data slice) and in the target measure to predict.

To define Auto Predict predictions for **Dynamic Calc** members, include the **Dynamic Calc** members in the Auto Predict slice definition, as either the Historical Data slice or the Future Data slice. Additionally, you can select **Dynamic Calc** members as the POV in the Source.

Additionally, associate the UDA **HSP_DYN_PREDICT** with non-aggregating sparse dimensions such as Scenario, Version, Plan Element and so on.

To specify whether to perform a prediction when the form loads: In the Form Designer, in the **Prediction Options** settings, select **Prediction for Dynamic Calc**. A validation confirms that the **HSP_DYN_PREDICT** UDA is associated with the members on the form.

For Advanced Predictions, selecting **Explain Prediction** (either by right-clicking a cell or from the **Actions** button on the form) on the **Dynamic Calc** member associated cell intersections renders predicted data in the chart.

Generating and Viewing Predictions at the Parent Level and Loading the Form with Predicted Values

Because predicted values can't be stored for **Dynamic Calc** members, you can specify to generate or view predictions or view the already-run prediction) at the parent level at run time and load the form with predicted values, by using a form option and by associating members in Scenario, Version, Plan Element, or relevant dimensions associated with the UDA **HSP_DYN_PREDICT**. You can review predicted results in a web form, a dashboard, or in Oracle Smart View for Office. Planners can use Predictive Planning on the form to see explainability for the parent level predicted values.

To generate or view predictions at the parent level at run time and load the form with predicted values:

1. Associate the UDA **HSP_DYN_PREDICT** with the Scenario/Version/Plan Element or similar dimension that is used to stored predicted values. Any members associated with this UDA will display predicted values rather than aggregated values.
2. If you associated the UDA **HSP_DYN_PREDICT** with dimensions other **Scenario** or **Version**, update the **Predictive Planning** application settings:

   From the Home page, click **Application** and then click **Settings**. In the **Predictive Planning** section, update the **POV** to include the members from the other dimensions that are associated with the UDA **HSP_DYN_PREDICT**.

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> **Note**
>
> The **Source Cube** option defines the cube where historical actuals are stored. Members selected for **Source Cube** should not be associated with the UDA **HSP_DYN_PREDICT**.

3. Use the Form Designer to specify whether to perform a prediction when the form loads. In the Form Designer, in the **Prediction Options** settings, select **Prediction for Dynamic Calc**. A validation confirms that the **HSP_DYN_PREDICT** UDA is associated with the members on the form.

> **Note**
>
> - This feature is supported only for **Dynamic Calc** members; it is not supported for **Dynamic Calc and Store** members.
> - **Dynamic Calc** parent level values can't be stored or exported.
> - You can use a Groovy script to allocate the **Dynamic Calc** parent-level predicted values to level zero members.
> - Associate the UDA **HSP_DYN_PREDICT** only with Scenario/Version/Plan Element or similar sparse dimension members.
> - For Predictive Planning, if predicting values for **Dynamic Calc** parent members at runtime would make loading the form take too long, the prediction is not run and an error message displays.
> - Generating predictions for **Dynamic Calc** members is not supported in ad hoc grids.
> - For Predictive Planning, when you generate predictions at run time and load a form with predicted values, the predictions for **Dynamic Calc** members always use the default prediction options for the form. Any modifications you make to the prediction options (in the prediction results area, click **Settings**) are not reflected in the predicted values for **Dynamic Calc** members when you load the form. This setting does not apply for Advanced Predictions. For Advanced Predictions, the prediction is already run and the job is not run again. However, if the **Dynamic Calc** members don’t have data from the Advanced Prediction job, a univariate prediction runs to predict values.

Videos

Your Goal	Watch This Video
Learn how you can generate dynamic parent-level or any level predictions and insights to provide more accurate data analysis when subcategory data may be missing or inconsistent. Generate predictions and insights at a higher, aggregated level – whether it's a parent, a department, or any level in your hierarchy, to improve planning and forecasting accuracy.	Generating Dynamic Predictions and Insights in Cloud EPM

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Your Goal	Watch This Video
In this tutorial, you learn the unique considerations for generating Dynamic parent-level Predictions and Insights in FreeForm's flexible multi-cube environment. Generating Dynamic Predictions and Insights helps you analyze data more accurately when subcategory data is missing or inconsistent.	Generating Dynamic Predictions and Insights in FreeForm

Tutorials

Your Goal	Learn How
This tutorial shows you how to generate dynamic predictions and insights in Oracle Fusion Cloud Enterprise Performance Management.	Generating Dynamic Predictions and Insights in Cloud EPM

Configuring Additional Insight Settings

Use IPM Configurator advanced options to define the metrics and trigger criteria for insights.

On the IPM Configurator **Configure** page, click **Show Advanced Options**. For information about Auto Predict Advanced Options, see Configuring Advanced Auto Predict Options.

- For **Forecast Variance & Bias Insights**:
  - **Deviation Metric**
    - **Metric**: Select the metric to use for analysis for the variations between the historical actuals and the historical forecasts submitted by planners.
    - **Threshold**: Select the deviation percentage at which to trigger an insight. For example, if you specify 15 as the threshold, any time the deviation between historical actuals and the historical forecasts submitted by planners is 15% or greater, an insight is generated.
  - **Bias Metric**
    - **Error Tolerance Limit**: Specify a percentage for the error tolerance limit. For example, if you enter 5%, any difference of 5% or greater between historical forecasts and historical actuals is considered a significant variance.
    - **Bias Metric**: Specify the metric to use to measure forecast bias.
    - **Bias Threshold**: Select the deviation percentage at which to trigger an insight.
- For **Prediction Insights**:
  - **Deviation Metric**
    - **Metric**: Select the metric to use for analysis for the variations between the future forecasts submitted by planners and computer-generated predictions.
    - **Threshold**: Select the deviation percentage between the future forecasts submitted by planners and computer-generated predictions at which to trigger an insight.
  - **Risk %** and **Threshold**: Select the risk level you are willing to tolerate of not meeting your forecast before triggering an insight. This setting evaluates the probability of meeting or missing the target for your current forecast numbers.

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- For **Anomaly Insights**:
  - **Method**: Select the anomaly detection metric to use for analysis.
  - **Threshold**: Select the threshold at which to trigger an insight.
- For **Period Movement Variance Insights**
  - **Variance Metric**: Select the variance metric to use, **Absolute Variance (%)** or **Absolute Variance (Numbers)**.
  - **Baseline for Comparison**: Select the benchmarks you want to evaluate, and for each benchmark you select, specify the threshold. Several benchmarks can be used for comparison. Prior periods are defined based on the calendar selected for the insight definition.
    - **Compare Year-on-Year**
    - **Compare Period-on-Period (Previous Period)**
    - **Compare Period-on-Period (Custom Period)**. For this benchmark, you also select the number of periods to evaluate relative to the current period.
    - **Compare with Historical Average**

    For more information about how variance is calculated for performance benchmarks, see Understanding Period Movement Variance Insight Comparison Benchmarks.

For details on the metrics used in IPM Insights, see IPM Insights Metrics in *Working with FreeForm*.

Applying Scaling and Precision to IPM Insights at the Dimension Level

Administrators can specify scaling settings for IPM Insights at the dimension level by associating a UDA with the **Account** dimension.

Scaling and precision settings ensure consistent formatting for all data values displayed for an Insight.

While users can apply scaling and precision for IPM Insights in **User Preferences** or in **IPM Settings**, if you use a UDA, it takes precedence over **User Preferences** and **IPM Settings** for scaling.

Before you apply a UDA, you must first enable the option to apply scaling:

1. From the Home page, click **IPM** and then click **Settings**. (This option is available for all users.)
2. To turn on scaling, select **Apply Scaling (K, M, B, T) to values displayed in IPM Insights**. Scaling is applied to currency values and non-currency values.
   You can also specify the precision to apply after scaling for currency values and non-currency values and percentage values.

   The default values are the precision settings that are set in **User Preferences**. The **Example Preview** panel shows how values will appear with the specified settings.
3. Click **Save**.

To apply a UDA:

1. Open the Dimension Editor.
2. Select the **Account** dimension and member to edit.
3. Click the **UDA** tab.

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4. Apply one of the following UDAs and then click **Save**.
    - HSP_DISPLAY_SCALE_K
    - HSP_DISPLAY_SCALE_M
    - HSP_DISPLAY_SCALE_B
    - HSP_DISPLAY_SCALE_T

The settings are applied throughout the Insights list, and in the Insights Analyzer View, for example, in the insight description, the graph, the grid, the Explainability tab, and the Insights AI tab.

Note the following:

- By default, the **Apply Scaling (K, M, B, T) to values displayed in IPM Insights** option is enabled, and insights are displayed for each user using the precision set in their **User Preferences**.
- You can apply a UDA only to the **Account** dimension.
- Scaling and precision settings are not applied in ad hoc grids.
- To apply the precision settings to narrative summaries for existing insights, you must re-run the insight.

Setting Up Predictions to Run Automatically with Auto Predict

Use Auto Predict in FreeForm to predict future performance based on historical data.

Related Topics

- About Auto Predict
  With Auto Predict, administrators can define a prediction to predict future performance based on historical data and schedule a job to run that prediction definition, automating the prediction process.
- Creating Auto Predictions for Forecasts
  Create an Auto Predict prediction definition to automate predictions.
- Running Auto Predict Predictions
  You can run an Auto Predict prediction definition from the Auto Predict page or the IPM page, or you can schedule a job to run the prediction automatically.
- Editing, Deleting, or Duplicating Auto Predict Prediction Definitions
  You can edit, delete, or duplicate existing Auto Prediction prediction definitions.
- Reviewing Detailed Reports for Auto Predict Predictions
  You can review Auto Predict prediction result details in a downloadable report in an Excel worksheet format, including details such as prediction status, method used, and prediction accuracy.

About Auto Predict

With Auto Predict, administrators can define a prediction to predict future performance based on historical data and schedule a job to run that prediction definition, automating the prediction process.

Auto Predict uses the same forecasting and statistical methods as Predictive Planning, but unlike Predictive Planning, you aren't limited to running a prediction on a form. You can predict values for thousands of cells at once, when needed, by scheduling a job, and automatically write prediction results to a scenario/version—in the same cube or in a different cube from the

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historical data. You can also include Best and Worst Case prediction results. When you run the prediction, historical data for each member in the Auto Predict definition is retrieved and then analyzed using time series forecasting techniques to predict the future performance for these members.

Auto Predict is useful:

- When you have large amounts of data to predict.
- To jumpstart the forecasting process by pre-filling a prediction scenario with predictions based on historical data. Then, planners can compare predictions to forecasts.
- For variance analysis. Seed a prediction scenario with prediction results, and compare predictions to forecast. To incorporate variance thresholds, you can define a groovy rule that requires planners to add explanatory commentary at a defined threshold, for example if the forecast is 5% greater than the prediction.
- To pre-populate your Forecast or Plan scenarios before starting a forecast or planning cycle. Planners can use these prediction results as a starting point for their forecasts, and make adjustments as needed.
- To keep predictions up-to-date as actuals come in by scheduling Auto Predict jobs. When new actuals come in, the job can run automatically to populate the cube; planners can see updated predictions and compare to forecast and plan.

Use Auto Predict results in forms, dashboards, and in ad-hoc analysis. You can also use Auto Predict prediction results within Smart View.

You define a prediction to read historical data from a cube (typically an ASO cube) and write prediction results to the same cube or to a different cube, for example to a BSO cube where planning and forecasting is done.

You can also use calendars and events with Auto Predict. For more information, see About Using Calendars and Events in IPM Insights, Advanced Predictions, Auto Predict, and Predictive Planning.

To work with Auto Predict:

1. Define the Auto Predict prediction. See Creating Auto Predictions for Forecasts.
2. Modify application settings if needed to set the **Prediction Interval**. These settings are shared with Predictive Planning. By default, the prediction interval for worst and best cases is 2.5% and 97.5%. See Managing Application and System Settings.
3. Run the Auto Predict prediction: See Running Auto Predict Predictions.
    - Run the Auto Prediction from the Auto Predict or IPM list.
    - Schedule a job to run the Auto Predict prediction automatically.

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Videos

Your Goal	Watch This Video
Learn how to set up an auto predict definition, and schedule a job to run that definition. See how to prefill a prediction scenario to jump start the forecasting process, and how to directly compare the system's predictions to your ongoing forecast in real time. You compare predictions to Forecast and incorporate variance thresholds that require you to enter explanatory commentary. Also see how to seed plans or forecasts prior to starting the forecast cycle.	Forecasting with Auto Predict in Cloud EPM Planning

Auto Predict Considerations

The following are some of the considerations for Auto Predict.

- Auto Predict works with EPM Standard Subscription and EPM Enterprise Subscription applications for Custom and Module application types when Hybrid Essbase is enabled.
- Auto Predict works with EPM Enterprise Subscription applications for Sales Planning, Strategic Workforce Planning, and Cash Forecasting application types when Hybrid Essbase is enabled.
- In addition, you can use Auto Predict with Oracle Sales Planning Cloud and Oracle Strategic Workforce Planning Cloud.
- Auto Predict is available in FreeForm applications.
- For legacy applications, Auto Predict works with legacy Enterprise Planning and Budgeting Cloud Service and Planning and Budgeting Cloud Service Plus One (PBCS +1), when the Essbase version is upgraded to the version that supports Hybrid Essbase and Hybrid Essbase is enabled.
- If you add dimensions to the application after the job is created, you must reconfigure the IPM job to include the additional dimensions.
- Prediction results are more accurate the more historical data you have. There should be at least twice the amount of historical data as the number of prediction periods.

  When series have less than six historical data points, Auto Predict does not use forecasting, instead it uses a straight line fit for the prediction series. In this case there are results only for the base case. Best case and worst case are not calculated, because the values would be the same as the base case.

  Auto Predict can give straight line results when the historical data has no pattern.
- Calendars are always required for running Auto Predict jobs in FreeForm applications.
- For details about the statistical forecasting methods used in Auto Predict predictions, see Predictive Planning Forecasting and Statistical Descriptions in *Working with FreeForm*.
- Planners can do predictions directly on a form. For more information, see Improving Forecasting Accuracy with Predictive Planning in *Working with FreeForm*.

Creating Auto Predictions for Forecasts

Create an Auto Predict prediction definition to automate predictions.

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To associate calendars and events with Auto Predict prediction definitions, configure calendars and events first. See Configuring Calendars and Events.

To create an Auto Predict prediction definition:

1. From the Home page, click **Application**, then click **Overview**, and then from the **Actions** menu, select **Auto Predict**.
   Alternatively, if you are using IPM Insights, from the Home page, click **IPM** and then click **Configure**.
2. Click **Create**.
3. On the **Types** page, enter a name and a description for the Auto Predict definition. Additionally, if you are using IPM Insights, select **Auto Predict** in the **Generate Predictions** area.

   Then click **Next**.
4. On the **Select Calendar** page, select a calendar to use, and then click **Next**. When you select a calendar, the **Historical Data** section and the **Future Data** section are populated with details from the selected calendar.
   Calendars are required for Auto Predict jobs.
5. On the **Define Slice** page, define the slice of data to analyze for historical and future data, and then click **Save**.
   - In the **Historical Data** section, select the cube and then define the slice of data for the historical data—both actuals and the forecasts provided by the planners. The slice definitions show all the dimensions for the cube, except for the Year and Period dimensions. All members start out with their root member selected.
   - In the **Future Data** section, select the cube and then define the slice of data for future data. Auto Predict results are stored in this location. You can click ⊕ to add a definition for the slice of data for the best case and worst case scenarios.

     The prediction interval used to define the **Best Case** and **Worst Case** results is defined in **Settings** for the application and is shared with Predictive Planning. See What Application and System Settings Can I Specify?.
   - To define Auto Predict predictions for **Dynamic Calc** members, include the **Dynamic Calc** members in the Auto Predict slice definition, as either the Historical Data slice or the Future Data slice. While predictions for **Dynamic Calc** members are supported without having to setup an Auto Predict job, this feature allows you to use **Dynamic Calc** members while setting up an Auto Predict job, which can be used to generate prediction insights and also allows it to be configured with **Store** members. You also use a form option and associate members in Scenario, Version, Plan Element or relevant dimensions with the UDA **HSP_DYN_PREDICT**. For more information about predictions for **Dynamic Calc** members, see Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions.

     Additionally, you can select Dynamic Calc members as the POV in the Source. This option supports cases where you want to predict using a parent member such as Total Plan in the Plan Element dimension, where the Base forecast and adjustments are both in Actuals.
   - You can use sibling-based functions when defining the slice, which allows you to select members relative to a given member within a hierarchy. The following functions are supported:
     - LSIBLINGS—Returns siblings to the left of the member.
     - ILSIBLINGS—Returns the member and its siblings to the left.

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- RSIBLINGS—Returns siblings to the right of the member.
- IRSIBLINGS—Returns the member and its siblings to the right.

> **Tip**
>
> When setting the **Start** for a prediction, when the start is **Specific Period**, for the **Select Period** Member Selector, you can select by substitution variable for the Current Period. In this way, job definitions are updated each time the value for the substitution variable is updated—change the value for the substitution variable and you don't have to edit every job to update the current period. To select for Substitution Variables, in the lower left corner of the Member Selector, select **Substitution Variable** from the list.

6. On the **Configure** page, select whether to include events in the prediction. If you want to define advanced options, click **Show Advanced Options**. For details on advanced auto predict options, see Configuring Advanced Auto Predict Options.
7. Click **Save**.

After defining the Auto Predict prediction, you can run the prediction from the Auto Predict page or from the IPM page, or you can schedule a job to run the prediction automatically. See Running Auto Predict Predictions.

Auto Predict definitions are saved as global artifacts and are backed up with the maintenance snapshot.

Configuring Advanced Auto Predict Options

Use Auto Predict advanced options for additional options for data screening, data attributes, prediction methods, and missing value treatment.

If you haven't yet created an Auto Predict prediction definition, see Creating Auto Predictions for Forecasts.

You can use default prediction options, or on the **Configure** page, click **Show Advanced Options**.

Showing advanced options gives you additional options for:

- Data Screening: See Data Screening Options
- Data Attributes: See Data Attributes Options
- Prediction Methods: See Methods Options
- Missing value treatment for last periods: See Missing Value Treatment for Last Periods
- Prediction Constraints for handling negative predicted values: See Handling Decimal Values and Negative Values in Auto Predict and Predictive Planning.

Data Screening Options

Data Screening options enable you to select several ways of handling missing values, identifying and adjusting outliers, and including events in predictions.

Historical data can have missing values and outliers, which are data points that differ significantly from the rest of the data, or can include events, which are typically one-off or recurring events that historically led to spikes or declines in the data. Because adjusted outliers are treated as missing values, both of these situations are discussed and handled together.

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Select from these options for **Data Screening**.

- **Adjust Outliers**—When this option is selected, when an outlier is detected in the series, outlier values are replaced with the prediction trend line value to avoid the impact of outliers.
- **Fill in missing values**—When this option is selected, if there are missing values in the time series, the missing values are populated with the prediction trend line value to continue with the prediction.
- **Minimum missing threshold**—When this option is selected, missing values in the time series are filled until the threshold is met. If the number of missing values is above the threshold provided, the prediction is not done. The maximum value can't be greater than 50%.

Data Attributes Options

Select from these options for data attributes and seasonality.

- **Override prediction range**: When this option is selected, the prediction overrides the minimum number of historical data points typically required to run a prediction. Instead, it predicts the number of periods based on the number of historical data points that are available. For example, it predicts y periods when 2y historical data points are available. If this option is not selected, if the prediction range is set to x periods, a minimum of 2x historical data points are required to run a prediction. If the minimum number of historical data points are not available, the prediction does not proceed.
- **Auto Detect**: When this option is selected, the prediction automatically detects seasonality based on the data.
- **Yearly Seasonal Override (12 periods)**: When this option is selected, seasonality is defined by the number of level 0 periods from the selected hierarchy. In cases where the number of historical data points is less than 2x of yearly seasonality, the prediction reverts to using **Auto Detect** for seasonality.
- **Custom**: When this option is selected, you can override the seasonality with a custom value by providing it in **Periods per cycle**. You can select this option when seasonality is not yearly.
- **Periods per cycle**: When **Custom** is selected, select the exact number of periods to be considered. If not enough data points are available, the prediction reverts to using **Auto Detect** for seasonality. The maximum allowed value for **Periods per cycle** is half of the total number of historical data range periods.

Methods Options

Select the method to use for predictions.

- **Apply seasonal methods**: Seasonal forecasting methods extend the nonseasonal forecasting methods by adding an additional component to capture the seasonal behavior of the data.
  For information on seasonal methods, see Classic Seasonal Forecasting Methods in *Working with FreeForm*. The prediction selects the best seasonal forecasting method to use.
- **Apply non-seasonal methods**: Nonseasonal methods attempt to forecast by removing extreme changes in past data where repeating cycles of data values are not present.
  For information on non-seasonal methods, see Classic Nonseasonal Forecasting Methods in *Working with FreeForm*. The prediction selects the best nonseasonal forecasting method to use.

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- **Use ARIMA**:
  For more information about Autoregressive Integrated Moving Average (ARIMA), see ARIMA Time-series Forecasting Methods in *Working with FreeForm*.
- **Use Extended ARIMA Search**: Performs extended model search on ARIMA models by comparing with historical data. Results are accurate, but analysis will take noticeably more time.

Missing Value Treatment for Last Periods

You have the option when defining the prediction to determine how Auto Predict should handle a missing value for the last historical period.

You can select from the following options:

- **Skip Series**—Skips the Time Series with the missing last historical period. The default option.
- **Mean**—Fills the missing last historical period with the Mean value generated using data from earlier historical periods.
- **Zero**—Fills the missing last historical period with Zero.
- **Last observed value**—Fills the missing last historical period with the last available value in the time series.
- **Cubic Spline Interpolation**—Fills the missing last historical period using cubic spline interpolation technique.

Handling Decimal Values and Negative Values in Auto Predict and Predictive Planning

Auto Predict and Predictive Planning handle decimal values and negative values so that they are appropriately applied to members in predictions.

Handling Decimal Values

Auto Predict and Predictive Planning automatically handle decimal values in predictions based on the account and whether it's valid for the account to have decimal predicted values. When needed, the prediction engine uses GenAI to decide how to process values. This ensures that decimal values are handled properly for all predictions without any intervention required from users.

Not all accounts are valid to have decimal predicted values, for example, accounts such as Headcount, New Hires, or Number of Stores. Auto Predict and Predictive Planning automatically check the account to determine whether predictions can have decimal values, and then apply adjustments during the post prediction process if needed.

GenAI must be enabled for this feature to work. If GenAI is not enabled, then no post prediction processing is done, and decimal values can show up for accounts or measures such as head count, even though historical data has no decimal values. See Enabling AI in FreeForm.

Once a prediction is generated, if decimal values are found, Auto Predict and Predictive Planning use this process for determining how to handle the decimal values:

1. Check historical data and see how it is handled. If historical data has decimal values, then decimal values are valid for predictions, and no further are steps taken.

   If historical data does not have decimal values, move to the next step.
2. Check the data type. If the data type is currency or percentage, it is valid to have decimals for the prediction, and no further steps are taken.

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If the data type is not currency or percentage, move to the next step.

3. Use GenAI and LLM to validate the account details such as account type, alias name, and account name and determine whether it's valid to have decimal prediction values.
4. Apply adjustments during post prediction processing.

   For an account that can't have decimal predictions, values are rounded off to the nearest integer.

Auto Predict and Predictive Planning first process negative predictions if negative values are found, and then process predictions with decimal values.

Handling Negative Prediction Values

Auto Predict and Predictive Planning handle negative values in predictions based on the account and whether it's valid for the account to have negative predicted values. Optionally, the prediction engine can use GenAI when needed to decide how to process values. This ensures that negative values are handled properly for predictions.

Not all accounts are valid to have negative predicted values, for example, accounts such as headcount or units sold. Other accounts, such as marketing expense, could have negative predicted values.

Auto Predict and Predictive Planning check the account to determine whether predictions can have negative values, and then apply adjustments during the post prediction process if needed.

Optionally, you can use GenAI if needed to help determine whether predictions can have negative values. Using this option requires you to enable GenAI. See Enabling AI in FreeForm.

To select options for handling negative predictions:

1. For Auto Predict:
   a. From the Home page, click **IPM**, and then click **Configure**.
   b. Select an Auto Predict job or create a new one.
   c. On the **Configure** page of the IPM Configuration wizard, click **Show Advanced Options**, and in the **Prediction Constraints** area, select **Adjust Invalid Negative Predictions**. With this option selected, the prediction engine checks historical data to determine how to handle negative prediction values.
   d. Optionally, select **Detect non-negative measures using GenAI** to use GenAI if needed to determine how to handle negative prediction values.
   GenAI must be enabled for **Detect non-negative measures using GenAI** to work. If GenAI is not enabled, only historical values are checked for negative prediction values.
2. For Predictive Planning:
   a. After you run a Predictive Planning prediction on a form, in the results area, click (**Settings**) and then click **Prediction Options**.
   b. In the **Prediction Constraints** area, select **Adjust Invalid Negative Predictions**. By default this option is off.
   c. Optionally, select **Detect non-negative measures using GenAI** to use GenAI if needed to determine how to handle negative prediction values.

Once a prediction is generated, if negative predicted values are found, Auto Predict and Predictive Planning use this process for determining how to handle the negative values:

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1. If **Adjust Invalid Negative Predictions** is selected: Check historical data and see how it is handled. If historical data has negative values, then negative values are valid for predictions, and no further are steps taken.

   If historical data does not have negative values, and **Detect non-negative measures using GenAI** is selected, move to the next step.

   If historical data does not have negative values, and **Detect non-negative measures using GenAI** is **not** selected, then negative predictions are adjusted to zero.
2. If **Detect non-negative measures using GenAI** is selected, use GenAI and LLM to validate the account details such as account name, alias name, account type, hierarchy type, parent name and determine whether it's valid to have negative predicted values. (If GenAI is not enabled, only historical values are checked for negative prediction values.)
3. Apply adjustments during post prediction processing.

   For an account that can't have negative predictions, the negative values are adjusted to zero.

Auto Predict and Predictive Planning first process negative predictions if negative values are found, and then process predictions with decimal values.

The downloadable Auto Predict report shows details about adjusted negative predictions. See the columns: **Adjusted Negative Predictions Count** and **Updated Negative Predictions Period**.

About Auto Predict and Predictive Planning Prediction Accuracy

The prediction accuracy calculation for Auto Predict and Predictive Planning predictions is a simple and generic calculation for all algorithms.

The prediction accuracy calculation is **1-sMAPE** (Symmetric Mean Absolute Percentage Error) for all univariate predictions in Auto Predict and Predictive Planning.

For Auto Predict, you can see the Accuracy value for a prediction in the downloadable report.

For Predictive Planning, you see the Accuracy value for a prediction in the **Prediction Details** area (click (**Details**) in the **Prediction** box in the Prediction results area.)

Running Auto Predict Predictions

You can run an Auto Predict prediction definition from the Auto Predict page or the IPM page, or you can schedule a job to run the prediction automatically.

Running an Auto Predict Prediction

To run an Auto Prediction prediction definition:

1. From the Home page, click **Application**, then click **Overview**, and then from the **Actions** menu, select **Auto Predict**.
   Alternatively, if you are using IPM Insights, from the Home page, click **IPM** and then click **Configure**.
2. From the list, click **Actions** next to an Auto Predict definition, and then select **Run**.
   For Auto Predict predictions with a large number of cells to calculate, in **Run Parameters**, select a dimension and then click **Run Now**.

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This setting is used to speed up an Auto Predict job by running predictions in parallel in separate prediction threads. For these parallel jobs to be efficient, choose a dimension that will result in evenly spread data for each prediction thread.

3. To review a summary of the prediction, review details in the Jobs Console: Click **Application**, then **Jobs**, and then click the prediction definition name on the **Jobs** page.

If a long-running prediction job is interrupted by an automated maintenance window, when the maintenance is complete, the prediction job picks up where it left off.

Scheduling an Auto Predict Job

For information about scheduling Auto Predict jobs, see Scheduling Jobs. Auto Predict has its own job type, called **Auto Predict**.

For Auto Predict predictions with a large number of cells to calculate, in **Run Parameters**, select a dimension and then click **OK**. This setting is used to speed up an Auto Predict job by running predictions in parallel in separate prediction threads. For these parallel jobs to be efficient, choose a dimension that will result in evenly spread data for each prediction thread. For recurring jobs, the selected dimension is used for jobs until you change it.

Before running a scheduled prediction job, Auto Predict analyzes certain factors to determine if prediction results would change. If the prediction results would not change, the Auto Predict job is not run. For example, if the historical range was set for the last three years and there has been no change in the start period/year for prediction, the historical data for analysis would be the same and the prediction results wouldn't change. In this case, to optimize performance and prevent unnecessary jobs from running, the Auto Predict job is skipped. A message is displayed indicating that the job will not run. If you want to run the prediction anyway, you can run the job from the Auto Predict page.

Editing, Deleting, or Duplicating Auto Predict Prediction Definitions

You can edit, delete, or duplicate existing Auto Prediction prediction definitions.

To edit, delete, or duplicate an Auto Predict prediction definition:

1. From the Home page, click **Application**, then click **Overview**, and then from the **Actions** menu, select **Auto Predict**.
   Alternatively, if you are using IPM Insights, click **IPM** and then click **Configure**.
2. From the list, click **Actions** next to an Auto Predict definition, and then select **Edit**, **Duplicate**, or **Delete**.

Reviewing Detailed Reports for Auto Predict Predictions

You can review Auto Predict prediction result details in a downloadable report in an Excel worksheet format, including details such as prediction status, method used, and prediction accuracy.

The Excel workbook contains several worksheets:

- **Summary**: A summary of the prediction including details about the prediction and POV used, and any events that were considered during the prediction.
- **Report**: A detailed report of the prediction displayed by series using the alias for member names. Includes details such as status, accuracy, prediction method, seasonality, adjusted outliers, error measure, and other prediction statistics.
- **SV_Series_Data_*CubeName*** (Smart View Series Data) data access sheet: Data access for the cube used for the prediction using Smart View. Also allows you to perform ad hoc

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analysis on prediction data. For multiple cube jobs, includes a worksheet for the source cube and a worksheet for the prediction cube.

The detailed report is automatically generated when an Auto Predict job runs and is available in compressed format in the Outbox when the job is complete. The report is overwritten the next time the Auto Predict prediction job runs.

Reviewing the Report

To review the report from the Auto Predict page:

1. From the Home page, click **Application**, then click **Overview**, and then from the **Actions** menu, select **Auto Predict**.

   Alternatively, if you're using IPM Insights, from the Home page, click **IPM** and then click **Configure**
2. From the **Actions** menu next to an Auto Predict definition, select **Download Report**.

To download the report from the Outbox:

1. From the Home page, click **Application**, and then click **Overview**.
2. From the **Actions** menu, select **Inbox/Outbox Explorer**.
3. Look for AP_*JobName*.zip and download the file.

Analyzing Prediction Data

On the **SV_Series_Data_*CubeName*** worksheet, you can perform ad hoc analysis using Smart View and drill into prediction data:

1. Select the **SV_Series_Data_*CubeName*** worksheet. The worksheet is populated with basic information including the historical date range, the prediction date range, and a member to start analysis.
2. Connect to the server where the Auto Predict prediction was run.
3. Select the cube used for the prediction (the cube name is appended to the worksheet name) and then select **Set Active Connection for this Worksheet**.
   If the prediction definition uses two cubes, a source cube and a prediction cube, repeat this step on both **SV_Series_Data_*CubeName*** worksheets, selecting the appropriate cube to associate with each worksheet.
4. Click **Refresh**.

Tips for analyzing prediction data:

- For members you want to analyze, copy and paste the members from the **Report** worksheet (from the Series Dimension columns) to the **SV_Series_Data_*CubeName*** worksheet.
- On the **Report** worksheet, you can sort to analyze data in different ways. For example, sort on the **Accuracy** column to review cases with low accuracy. Then, copy and paste the members you want to analyze further to the **SV_Series_Data_*CubeName*** worksheet, for example, the ten members with the lowest accuracy.
- Use Excel charting features to visualize prediction results in different ways.
- If you want to use Predictive Planning on the worksheet data, you must first pivot all of the dimensions from the columns, other than the month and year, to the POV using the **Planning Ad Hoc** menu.

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Setting Up Advanced Predictions in FreeForm

Related Topics

- About Advanced Predictions
  Use the power of Advanced Predictions in FreeForm to improve your forecasts using multiple input drivers and sophisticated forecasting and machine learning algorithms.
- Considerations for Advanced Predictions
  The following are some of the considerations for Advanced Predictions.
- Configuring Advanced Predictions
  You can use machine learning algorithms to generate advanced multivariate predictions.
- Running and Scheduling Advanced Predictions
  After configuring Advanced Predictions, you can run them any time.
- Reviewing Detailed Reports for Advanced Predictions
  You can review Advanced Prediction prediction result details in a downloadable report in an Excel worksheet format, including details such as prediction status, method used, and prediction accuracy.

About Advanced Predictions

Use the power of Advanced Predictions in FreeForm to improve your forecasts using multiple input drivers and sophisticated forecasting and machine learning algorithms.

Advanced Predictions are an extension of Auto Predict, expanding the power of the prediction engine by using machine learning backed by OCI Data Science Cloud. While Auto Predict and Predictive Planning offer univariate forecasting based on a single measure, Advanced Predictions provide multivariate predictions, looking at multiple input drivers to make the prediction. You train the ML model, which improves the quality of the predictions.

Univariate predictions used in Auto Predict and Predictive Planning create predictions based on *one* variable, for example, analyzing historical data for sales volume to predict future sales volume.

Multivariate predictions used in Advanced Predictions use advanced statistical and machine learning algorithms to generate multivariate predictions to improve your forecasting accuracy. Predictions are based on *multiple* variables. For example, to predict future sales volume, you can analyze several drivers or variables that correlate with your target variable. For example, you can analyze Industry Volumes, Average Sales Price, Advertising and Promotions, GDP Rate, and so on. The algorithms take the historical data for that variable and the drivers you included, as well as their what-if projections, to create a more accurate prediction.

With Advanced Predictions, you can predict using any of these algorithms or use Oracle AutoMLx, which runs all of these algorithms and selects the best option with the best results for you:

- **Light GBM**
- **XGBoost**
- **Prophet**
- **SARIMAX**

Administrators use the familiar IPM Configuration wizard to easily configure Advanced Predictions to provide insights into solutions.

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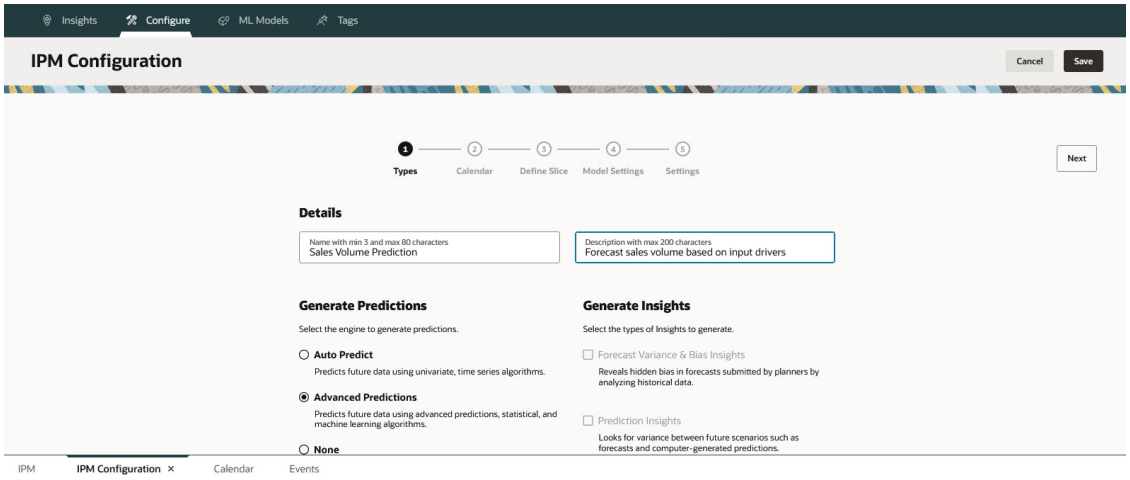

Depending on the implementation, planners can use a dashboard to see an overview of the prediction, which can use multiple external and input drivers, such as industry volumes, advertising and promotions, sales price, and GDP rates.

Depending on the implementation, planners can also review:

- **Prediction**—Review prediction details, historical data, and input driver details.
- **Input Drivers**—Review and edit the input driver details.
- **Forecast Accuracy**—Review and ensure the accuracy of the prediction.
- **Predict by Algorithms**—Explore prediction results using different algorithms.

Explain Prediction opens prediction details that contain a line chart for historical trends and prediction results considering the best case, worse case, and most likely scenarios. You can also review details about the prediction including accuracy, the error measure, and the algorithm that was used.

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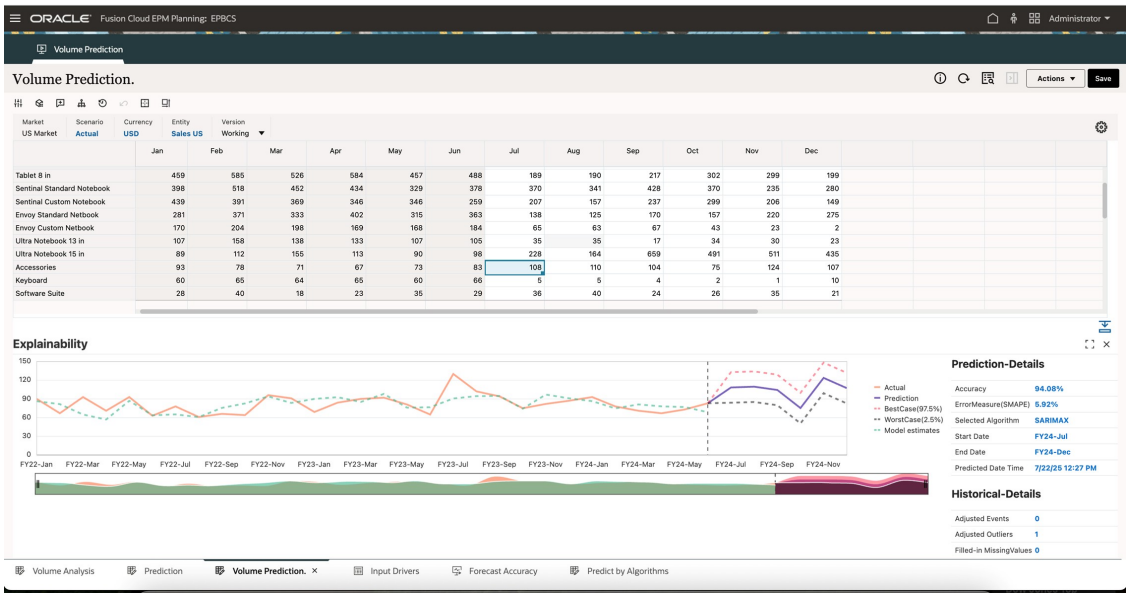

Your current investment in Cloud EPM is delivering AI to you:

- The AI features are already built into the business processes and is powered by OCI AI and leading large language models.
- Your results are based on the data you provided and the relevant context surrounding it, making sure that your predictions are accurate and aligned with your specific data sources.
- It honors Cloud EPM's multi-layer security structure.
- Its extensible framework allows for new capabilities and functionalities.

Advanced Predictions:

- Enables more powerful predictions by leveraging more sophisticated algorithms
- Improves accuracy as it correlates with multiple provided data points
- Easy to configure and use with the IPM Configuration wizard that can automatically select the best algorithm to use
- Designed for business finance, embedded in Cloud EPM so it can be used in context, without needing to learn data science

> **Tip**
>
> You also work with Advanced Predictions in Oracle Smart View for Office. For more information, see Using Advanced Predictions in Smart View.

Videos

Your Goal	Watch This
Learn how Advanced Predictions can perform AI-assisted data analysis that helps you streamline reporting and strengthen decision making. Advanced Predictions takes advantage of machine learning models to generate multivariate predictions to improve your forecasting accuracy.	Introducing Advanced Predictions in Cloud EPM

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Your Goal	Watch This
Learn how Advanced Predictions can perform AI-assisted data analysis in FreeForm to help you streamline reporting, analysis and strengthen decision-making. Advanced Predictions takes advantage of Machine Learning models to generate multivariate predictions to improve your forecasting accuracy.	Introduction to Advanced Predictions for FreeForm

Considerations for Advanced Predictions

The following are some of the considerations for Advanced Predictions.

- Advanced Predictions requires an EPM Enterprise Cloud Service subscription.
- Advanced Predictions is available in FreeForm and in all Planning application types: Modules, Custom, FreeForm, Sales Planning, Strategic Workforce Planning, Predictive Cash Forecasting.
- You must opt in to use Advanced Predictions.
- There is no additional cost required to deploy OCI Data Science. OCI Data Science is embedded in Oracle Fusion Cloud Enterprise Performance Management.
- Advanced Predictions works with both BSO and ASO cubes.
- If you add dimensions to the application after the job is created, you must reconfigure the IPM job to include the additional dimensions.

Configuring Advanced Predictions

You can use machine learning algorithms to generate advanced multivariate predictions.

If you have not already created a calendar, create a calendar to use with your advanced predictions. You can also configure events to associate with Advanced Prediction prediction definitions. Configure calendars and events first, before you define an advanced prediction. See Configuring Calendars and Events.

Advanced Predictions require you to use a predefined calendar.

1. Enable Advanced Predictions. See Enabling AI in FreeForm.
2. Prepare, review, and modify your input driver data as needed.
3. From the Home page, click **IPM** and then click **Configure**.
4. Click **Create**.
5. In IPM Configurator, on the **Types** page, enter a name and description.
6. Click **Advanced Predictions** and then click **Next**.
7. On the **Calendar** page, select a calendar to use, and then click **Next**. The **Historical Data** section and the **Future Data** section are populated with details from the selected calendar. The cube selection also comes from the calendar definition.
8. On the **Historical Data** page, define the slice for the prediction, the output to predict, and the drivers to use for input for historical data, and then click **Next**. This information is used to train the model that drives the predictions. These selections define what you want to predict and what the input drivers are.

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- **Define Model Scope**. See Defining the Model Scope.
- **Select Output to Predict**. See Selecting the Output to Predict.
- **Select Drivers as Input**. See Selecting Drivers as Input.

> **Note**
>
> To define Advanced Predictions for **Dynamic Calc** members, include the **Dynamic Calc** members in the Advanced Prediction definition. **Dynamic Calc** members can be defined in the source slice (historical data slice) and target slice (future data slice) and in the target measure to predict.
>
> Additionally, associate the **HSP_DYN_PREDICT** with non-aggregating sparse dimensions such as Scenario, Version, Plan Element and so on. For more information about predictions for **Dynamic Calc** members, see Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions.
>
> Selecting **Explain Prediction** (either by right-clicking a cell or from the **Actions** button on the form) on the **Dynamic Calc** member associated cell intersections renders predicted data in the chart.

9. On the **Future Data** page, in the **Define Model Scope** section, define where to store the prediction results and the data slice for drivers in future periods, and then click **Next**. See Defining the Model Scope.
   You also use a form option and associate members in Scenario, Version, Plan Element or relevant dimensions with the UDA **HSP_DYN_PREDICT**. For more information about predictions for Dynamic Calc members, see Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions.
10. On the **Prepare Data** page, define how to assess the data before performing the prediction, and then click **Next**. See Preparing the Data.
11. On the **Model Settings** page, select the algorithm, forecast error method, and confidence intervals to use for the prediction, and specify whether to include events in the prediction, and then click **Save**.
    Advanced Predictions automatically detect seasonality and perform data screening.
    - **Select Algorithms**—Select the algorithms to use for the advanced prediction. See Selecting Algorithms for Advanced Predictions in FreeForm.
    - **Confidence Intervals**—Specify the confidence intervals to use for the advanced prediction. See Defining Confidence Intervals.
    - **Include Events**—Select whether to include events that are associated with the calendar in the prediction. If you select this option, the prediction takes events into account in the data used for training the ML model. See Including Events in Advanced Predictions.
12. Click **Save**.
    You see a message when the Advanced Prediction is saved.
13. Now you're ready to run the Advanced Predictions you configured. See Running and Scheduling Advanced Predictions.
14. When the job completes successfully, you can download a report with details about the prediction. See Reviewing Detailed Reports for Advanced Predictions.
15. Extend the implementation to include forms, dashboards, navigation flows, and so on, for planners to review and analyze advanced prediction results and to review explainability for

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a prediction. You can also prepare an Oracle Smart View for Office sheet so planners can review prediction results.
Explain Prediction, which allows planners to see details about predicted values, is available by default.

Oracle provides a navigation flow for a volume prediction use case. You can use it as an example to customize your implementation for your business use case.

> **Tip**
>
> You can run the advanced prediction using different algorithms, and save each set of results in a different Version. Planners can then select different Versions on a form to see the prediction results from each algorithm.

Advanced Prediction definitions are saved as global artifacts (in the Auto Predict folder under Global Artifacts) and are backed up with the maintenance snapshot.

Additionally, you can export the explainability artifacts of an Advanced Prediction using **Migration**. From the Home page, click **Tools** and then click **Migration**. Artifacts are saved with the **Essbase Data** in the **Plan Type** with the prediction.

Videos

Your Goal	Watch This
In this tutorial, you learn which tasks are required before configuring Advanced Predictions. Advanced Predictions takes advantage of machine learning models to generate multivariate predictions to improve your forecasting accuracy.	Setting Up Cloud EPM for Advanced Predictions
In this tutorial, you learn how to configure Advanced Predictions and perform AI-assisted data analysis. Advanced Predictions takes advantage of machine learning models to generate multivariate predictions to improve your forecasting accuracy.	Configuring Advanced Predictions in Cloud EPM
In this tutorial, you learn how to set up and configure Advanced Predictions in FreeForm. First, you learn which pre-requisite tasks are required for setting up Advanced Predictions in FreeForm. Then, you learn how to configure Advanced Predictions to perform AI-assisted data analysis in FreeForm.	Setting Up and Configuring Advanced Predictions in FreeForm

Tutorials

Tutorials provide instructions with sequenced videos and documentation to help you learn a topic.

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Your Goal	Learn How
This tutorial supports customers and partners in configuring Advanced Predictions with ML capability to generate more accurate predictions by considering key business drivers. This tutorial shows you how to configure Advanced Predictions and provides implementation recommendations based on business considerations and IPM's solution capabilities.	Configuring and Reviewing Predictions with Advanced Predictions

Defining the Model Scope

Define the slice definition, or model scope, for both historical data and future data for Advanced Predictions.

Define the model scope for the Advanced Prediction:

- On the **Historical Data** page, you select members from each dimension to define the location of the historical data that should be used to train the ML model for the advanced predictions. See Defining the Model Scope for Historical Data.
- On the **Future Data** page, you define where you want to store the output from predictions and the data slice for drivers in future periods. See Defining the Model Scope for Future Data.

Tip

You can use sibling-based functions when defining the model scope, which allows you to select members relative to a given member within a hierarchy. The following functions are supported:

- LSIBLINGS—Returns siblings to the left of the member.
- ILSIBLINGS—Returns the member and its siblings to the left.
- RSIBLINGS—Returns siblings to the right of the member.
- IRSIBLINGS—Returns the member and its siblings to the right.

Note

To define Advanced Predictions for **Dynamic Calc** members, include the **Dynamic Calc** members in the Advanced Prediction definition. **Dynamic Calc** members can be defined in the source slice (historical data slice) and target slice (future data slice) and in the target measure to predict.

Additionally, associate the **HSP_DYN_PREDICT** with non-aggregating sparse dimensions such as Scenario, Version, Plan Element and so on. For more information about predictions for **Dynamic Calc** members, see Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions.

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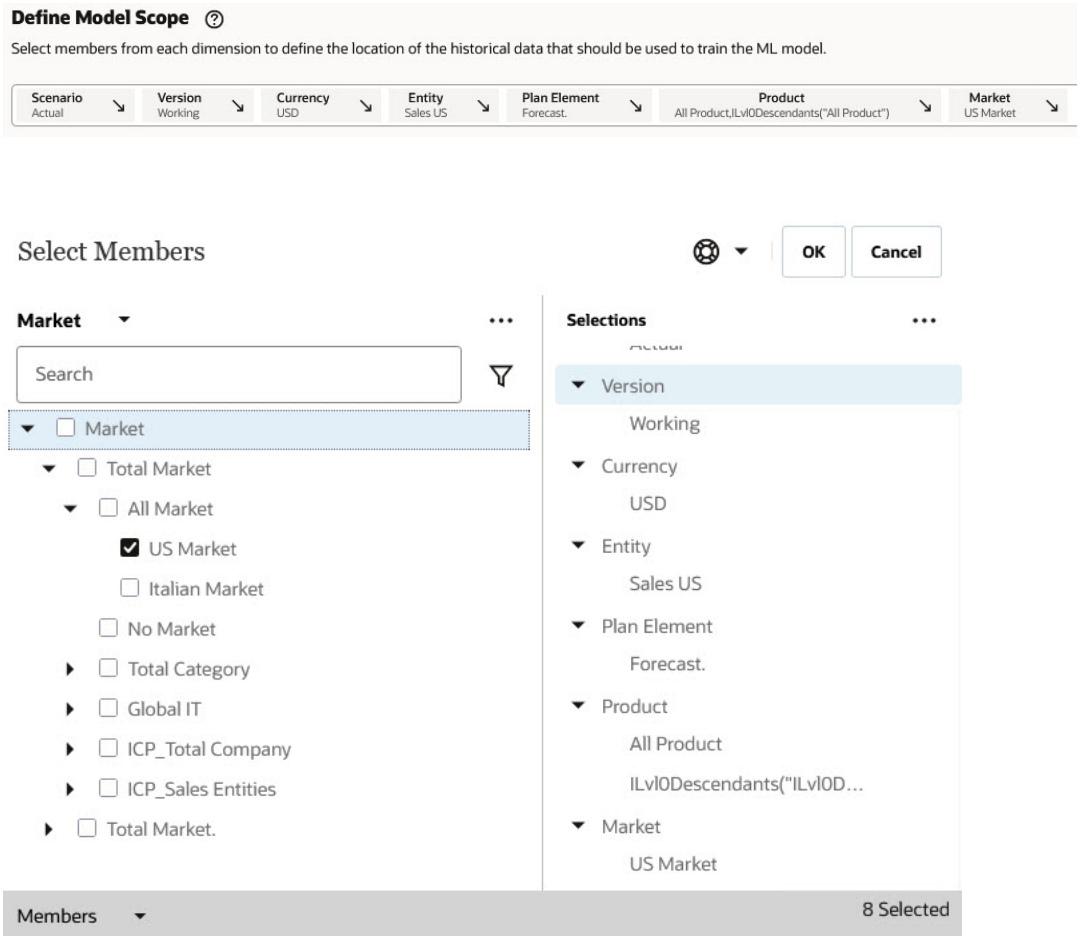
Defining the Model Scope for Historical Data

To define the model scope for **Historical Data**:

1. In the **Define Model Scope** section, first determine which dimension has the measure you want to predict. Click the down arrow ↘ next to this dimension name.
   The dimension is now listed in **Select Output to Predict** and **Select Drivers as Input** .
2. In the **Define Model Scope** section, for the remaining dimensions, select the members that define the location of the historical data that should be used to train the advanced prediction model.
   To select the members, click a dimension, and then use the member selector to select the members. You can also use functions to select members.



Defining the Model Scope for Future Data

To define the model scope for **Future Data**:

1. In the **Define Model Scope** section, select members from each dimension to define where to store the prediction data.

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> **Note**
>
> Member selections are pre-filled with the member selections you made for historical data, so you need to change only the specific members where future data exists and where the predictions are to be stored. For example, you might only need to change **Scenario** from **Actual** to **Forecast**.

2. If needed, make any required changes to **Select Output to Predict** or **Select Drivers as Input**.
3. Click **Next**.

Selecting the Output to Predict

Define the measure to predict for Advanced Predictions.

To define the output to predict:

1. On the **Historical Data** page, if you haven't already done so, from the dimension list in the **Define Model Scope** section, click the down arrow ↘ next to the dimension name for the dimension that has the measure you want to predict. The dimension is now listed in **Select Output to Predict** and **Select Drivers as Input**.
2. In the **Select Output to Predict** section, enter a name for the measure, click the dimension name, and then use the Member Selector to select the member or members you want to predict.
   You can select multiple members for **Output to Predict**. For example, you can select:
   - Member functions, such as @LEV0DESCENDANTS("Operating Expenses")
   - Explicit member lists, such as LIST(Account1, Account2, Account3)
   - Individual members

   Multiple member selection allows you to predict multiple target accounts or members using the same driver setup. For example, you can predict all level-0 operating expense accounts using @LEV0DESCENDANTS("Operating Expenses"), instead of creating separate prediction definitions for each account. This reduces duplicate setup and makes Advanced Predictions easier to configure and maintain.

   This measure is also automatically used as an input driver to drive the prediction. For example, if you want to predict volume, historical volume is automatically taken into account in the prediction.

   On the **Future Data** page, you can make changes if needed for **Select Output to Predict**.

> **Note**
>
> To define Advanced Predictions for **Dynamic Calc** members, include the **Dynamic Calc** members in the Advanced Prediction definition. **Dynamic Calc** members can be defined in the source slice (historical data slice) and target slice (future data slice) and in the target measure to predict.
>
> Additionally, associate the **HSP_DYN_PREDICT** with non-aggregating sparse dimensions such as Scenario, Version, Plan Element and so on. For more information about predictions for **Dynamic Calc** members, see Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions.

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Selecting Drivers as Input

Select the drivers to use for Advanced Predictions to predict the selected measure.

The input drivers are factors that are used to train the ML model to generate the prediction.

The measure you selected as the output to predict is automatically used as an input driver to drive the prediction. For example, if you want to predict volume, historical volume is automatically taken into account in the prediction.

Define as many input drivers as you need:

1. On the **Historical Data** page, in the **Select Drivers as Input** section, enter a name for the driver.
2. Select the **Input Type** for the measure:
   - **Smart List**—When the input driver is a categorical, string-type value, for example, an attribute for the Product dimension.
   - **Cell Value**—When the input driver is a numeric variable, or continuous variable.
3. Click the dimension name, and then use the Member Selector to select the member you want to drive the prediction.
   You can select a driver from a specific member intersection, even when the **Output to Predict** is selected for multiple members. For example, the **Output to Predict** can be selected as @LEV0DESCENDANTS("Total Product"), however, one of the input drivers might not have data by product. A macroeconomic driver such as GDP, inflation rate, or interest rate might be stored only at "No Product" because it applies across all products rather than to each individual product.

   In this case, you can select the driver at a specific intersection, such as "No Product", while still predicting results for multiple product-level targets.

   This allows you to use drivers that exist at a different POV or intersection from the prediction target. This is especially useful for broad business or macroeconomic drivers, such as GDP, inflation, or interest rates, where the driver data is not available at the same detailed level as the target being predicted.
4. To add another driver, click **Add Driver**, and then enter the driver details.

> **Note**
>
> The prediction definition must have at least two input drivers. If you edit an existing job that has only one driver, you must select an additional driver in order to proceed and save the updated job definition.

In this example, you see some of the possible input drivers that can drive a prediction for Volume, such as Industry Volume, Average Selling Price, and Discount Rate.

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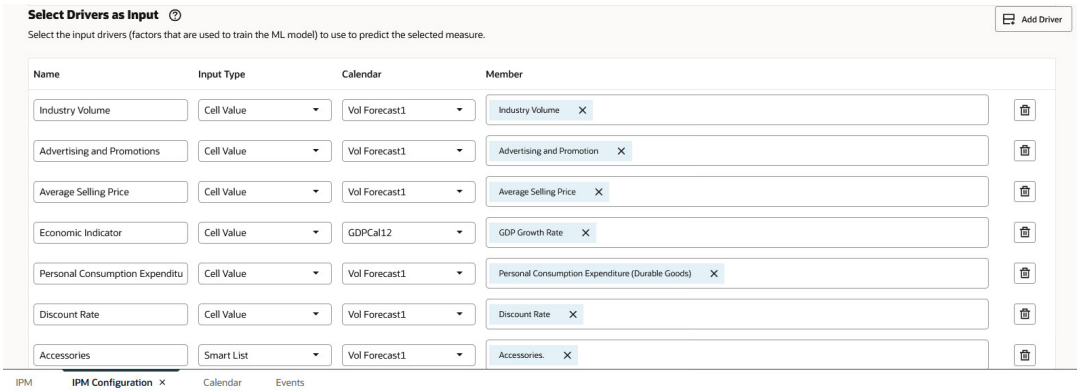

5. Click **Next**.

> **Note**
>
> No changes are needed on the **Future Data** page for **Select Drivers as Input**.

Preparing the Data

Define how to assess and manage the data before performing the prediction, for example, how to handle outliers or missing values. These options for preparing the data can improve the quality of the data used for the prediction.

Define how to manage the data for each driver. Note that a **Target** value of **Yes** indicates the target measure that is being predicted.

1. For future input driver data, select **Predict missing input driver values** if you want to predict values where input driver values are missing.
   Missing values are predicted using statistical forecasting (univariate predictions), and are used for ML model training.
2. Optionally, select **Enable automated feature engineering and selection**. For more information, see Feature Engineering and Selection.
3. For each driver, define how to handle missing values: click the Edit icon in the **Actions** column, and then, from the **Missing Values** list, select an option for handling missing values for the driver.
   Data can contain missing values for a several reasons, such as measurement failures, formatting problems, human errors, or lack of information. You define how to fill these missing values, which adds standardized values to missing entries in the dataset.
   - **None**—Take no action and send the data as is.
   - **Zero**—Replace missing values for any column with zero.
   - **Replace with Mean**—Replace with the mean across the historical series.
   - **Replace with Median**—Replace with the median point of the historical series.
   - **Replace with Mode**—Replace with the most common value in the historical data.
   - **Replace with Next Observed Value**—Replace missing values with the value that was observed in the next period.
   - **Replace with Last Observed Value**—Replace missing values with the value that was observed in the previous period.

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4. For each driver, from the **Outliers** list, select an option to use to handle outlier values, which are any values that fall out of range `mean +/- 3*Standard Deviation` for the driver:
   - **None**—Take no action and send the data as is.
   - **Replace with Zero**—Replace with zero.
   - **Replace with Mean**—Replace with the mean.
   - **Replace with Z_score**—Replace with the z_score.

     For any numerical column, any value falling out of `mean +/- 3*Standard Deviation` (std dev) is treated as an outlier. A value less than `mean - 3*std dev` is replaced with `mean -3*std dev`. A value greater than `mean + 3*std dev` is replaced with `mean + 3*std dev`.
5. Click **Next**.

Feature Engineering and Selection

Advanced Predictions can perform Feature Engineering and Feature Selection on data before running predictions, which improves prediction accuracy.

When defining an Advanced Prediction, administrators can select a new option on the **Data Preparation** page, **Enable automated feature engineering and selection**. This option is selected by default.

Feature Engineering is the process of preparing data for machine learning by transforming existing features or creating new ones to improve model performance. Automated Feature Engineering accelerates this process, enabling better models with improved accuracy.

Without Feature Engineering, all of the drivers are treated just as they are. With Feature Engineering, additional information is derived from the features, leading to more accurate predictions. The following types of transformations are applied:

- Time-based features. For example, does a particular day of the week have more of an impact?
- Lag effects. For example, what is the lag effect of a business driver on a target, such as the impact of Marketing Spend for June on Sales Volume for August?
- Aggregate transformations. For example, what is the effect of the rolling average value on a business driver, rather than a single point of data?

When **Enable automated feature engineering and selection** is selected, multiple engineered features are created, and the top 15 (or fewer) features are selected for model generation and predictions. The **Feature Importance** chart shows the importance of the selected top 15 features for the output variable.

When **Enable automated feature engineering and selection** is not selected, no engineered features are created. However, feature importance is still assessed, and any features found to be unimportant are excluded from the model. All other features are used as-is for model generation.

The **Feature Importance** chart shows the importance of the given features for the output variable. The chart shows the top nine features; any remaining features are aggregated by their percentage values and grouped under the heading **Others**.

You can see the impact of Feature Engineering and Feature Selection in the **Explain Prediction** screen: In a form with Advanced Prediction predicted values, right-click a cell with predicted data and then select **Explain Prediction**. Review details about the prediction on the **Feature Importance** tab and on the **Prediction Analysis** tab.

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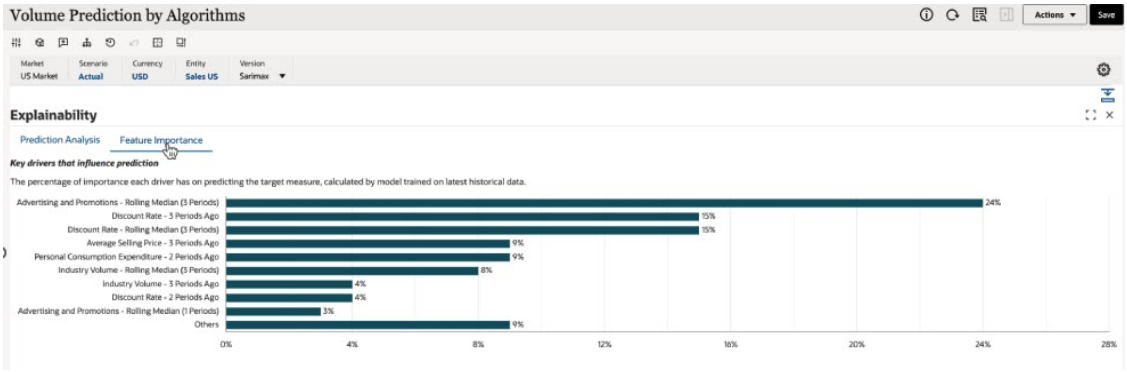
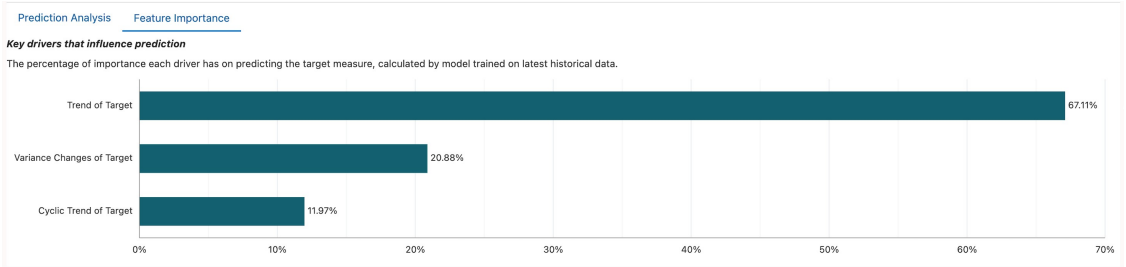


If you selected **AutoMLx** as the algorithm, note that **AutoMLx** automatically selects either a univariate or a multivariate model based on accuracy.

If a univariate model is selected, the **Feature Importance** graph displays only output/target-related features, such as **Trend of Target**, **Target N Time Periods Ago**, **Variance Changes of Target**, and **Cyclic Trend of Target** (as applicable).

The **Feature Importance** graph for a univariate model would look something like this:



Feature Engineering finds the hidden relationship between input features and the output variable. Well-engineered features allow models to capture more relevant information, leading to improved model performance and better predictions.

Feature Selection identifies the most relevant business drivers that impact forecast accuracy, and filters out "noisy" or low-impact variables to avoid over-fitting. It also improves performance by reducing complexity and processing time. It supports explainability by ranking features based on predictive power.

Note the following:

- **Enable automated feature engineering and selection** is selected by default.
- Selecting **Enable automated feature engineering and selection** increases the time required to run the prediction.

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Videos

Your Goal	Watch This
Learn how to incorporate multiple engineering features into your advanced predictions. With Feature Engineering, additional information is derived from the features leading to more accurate predictions. Transformations that are applied include time-based features, lag effects, and aggregate transformations. You can select the option to enable automated feature engineering and selection on the **Data Preparation** page when defining an advanced prediction.	Analyzing Advanced Predictions with Feature Engineering

Selecting Algorithms for Advanced Predictions in FreeForm

Select the algorithms to use for the advanced prediction in FreeForm.

To define the algorithm to use for the advanced prediction:

1. In the **Select Algorithms** section, select the algorithm to use:
   - **Oracle AutoMLx**—A proprietary suite of ten algorithms (including both univariate and multivariate); automatically selects the best model for the given error measure. Runs all of these algorithms and selects the best option with the best results for you.
     - Runs various statistical models and machine learning algorithms on your data.
     - Tunes and validates the models.
     - Finds the best model for your data.
     - Fits your data to the best model.
   - **Light GBM**—Light Gradient Boosting, an ensemble- and tree-based, speed efficient algorithm suited for larger data sets. Best suited for data sets where time has less weight as compared to other features.
   - **XGBoost**—Extreme Gradient Boosting, an ensemble- and tree-based algorithm, best suited for data sets where time has less weight as compared to other features.
   - **SARIMAX**—Arima with exogeneous algorithms.
   - **Prophet**—Time series algorithm best suited for data with strong seasonal effects and several seasons of historical data.
   - **Linear Regression**—Simple linear relationships.
     - This method is simple and easy to understand. For example, "I want to forecast monthly sales based primarily on marketing spend and pricing. The relationship is straightforward; higher marketing drives higher sales, higher prices reduce sales."
     - This method is best when there are one to five drivers with clear linear relationships.
   - **Lasso Regression**—Linear with many drivers.
     - This method is useful when you have many drivers. It automatically identifies and selects the most impactful drivers, and removes weak drivers. For example, "I have eight potential cost drivers for our operating expenses (labor, materials,

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utilities, maintenance, and so on), but I'm not sure which ones actually drives the variance. I want the system to automatically identify the key drivers."

  - This method is best when you have many drivers and aren't sure which drivers matter.

- **Ridge Regression**—Linear with correlated drivers.
  - This method retains all drivers while weighting their relative importance, and handles correlation well. For example, "I want to forecast fee revenue using assets under management, market index performance, and trading volume. These metrics are correlated with each other but all are important drivers of our revenue."
  - This method when is best when drivers are correlated but all matter.

> **Note**
>
> In Cloud EPM OCI realms OC2, OC3, and OC4, only SARIMAX and Prophet algorithms are supported.

2. Select the **Forecast Error Metric** to use for the selected algorithm to define how the algorithm should select the best model. It optimizes the model training based on the selected error metric to determine the best option to use for the prediction. The ML engine learns the patterns from the data, and looks for the best option to minimize errors to the extent possible. The ML engine evaluates each iteration against the error metric you select, and selects the iteration when the error metric is the lowest.
   - sMAPE—Symmetric Mean Absolute Percentage Error
   - MAPE—Mean Absolute Percentage Error
   - RMSE—Root Mean Squared Error

   Using your choice of error measure, Advanced Predictions:
   - Chooses the model with the least error as the best model.
   - For the best model:
     - Generates fitted series corresponding to the input series.
     - Generates forecast for the horizon.

About Regression Algorithms

For regression models, the error measure reported is for the holdout period by doing an 90:10 split. A portion of historical data is set aside and not used during model training and is instead used to validate model performance:

1. Historical data is split into **training data** (used to build the model) and **holdout data** (used for validation).
2. The model is trained on the training dataset.
3. Predictions are generated for the holdout period.
4. Predictions are compared with actual values to measure accuracy. The accuracy and fitted values are calculated on the holdout period.

For example, if you have five years of monthly historical data (60 data points):

1. These 60 data points are split 90:10. 90% of the data points (54) are used for training the model and 10% of the data points (6) are used as test data points.
2. The test model is created by using 54 data points. The prediction is for 6 data points.

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3. The predictions and test data are compared and the following error measures are calculated: SMAPE, MAPE and RMSE.
4. Accuracy is calculated as 100 – SMAPE.

Regression algorithms are useful in some scenarios where time series forecasting is not applicable. For example, use regression algorithms in these cases:

- Specific drivers matter more than time patterns.
- You have clear causal relationships (marketing drives sales, price affects demand).
- You want to understand "what happens if we change X."
- You need to explain driver impacts to stakeholders.
- Time patterns are weak or irregular.

Videos

Your Goal	Watch This
Learn how the new regression algorithms in Oracle EPM Advanced Predictions—Linear Regression, Lasso Regression, and Ridge Regression—help you forecast outcomes based on business drivers. You'll also learn a simple decision guide to choose between these new regression algorithms.	Overview of New Regression Algorithms in Advanced Predictions

Defining Confidence Intervals

In addition to calculating a base prediction, the most likely scenario, Advanced Predictions can also calculate additional scenarios—the best case and worst case scenarios, also known as a confidence interval.

Based on confidence intervals, Advanced Predictions generate multiple scenarios of ML Predictions and stores the results and in the scenario you select.

- The confidence intervals in a prediction provide an upper and lower bound for predicted output values.
- For example, using the confidence intervals of 10% (P10) and 90% (P90) provides a range of values known as an 80% confidence interval. The observed value is expected to be lower than the P10 value 10% of the time, and the P90 value is expected to be higher than the observed value 90% of the time.

By generating forecasts at P10 and P90, you can expect the true value to fall between those bounds 80% of the time.

To define confidence intervals:

1. Select whether to generate confidence intervals.
2. Select the prediction interval to use for the prediction. The prediction interval, or confidence interval, defines the best case and worst case values for the prediction. For example, if you select **5% and 95%**, then the best case scenario is the 95th percentile of the prediction, the maximum value for the prediction, and the worst case scenario is the 5th percentile of the prediction, the lowest value for the prediction. Select the prediction interval to use for the prediction.

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3. For **Model estimates (fitted values) for historical data**, **Best Case**, and **Worst Case**, select members from each dimension to define where to store prediction data for fitted values, best case, and worst case scenarios.

Including Events in Advanced Predictions

You can include events in Advanced Predictions for advanced tuning and improving accuracy in cases where events in the past should be considered for future predictions.

To include events in Advanced Predictions, click **Include Events**.

For more information about using events in predictions, see About Using Events with Insights, Advanced Predictions, Auto Predict, and Predictive Planning.

Running and Scheduling Advanced Predictions

After configuring Advanced Predictions, you can run them any time.

You can also set up a job to schedule and run Advanced Predictions at a regular interval.

> **Note**
>
> Users with the **IPM Manage** role must perform the steps in this topic before running an Advanced Prediction: Allowing Non-Administrators to Execute Jobs in the Pipeline in *Administering Data Integration*.

To run an Advanced Prediction:

1. From the Home page click **IPM** and then click **Configure**.
2. From the **Actions** menu next to the Advanced Prediction you want to run, click **Run**.
3. To review the Advanced Prediction jobs, from the Home page, click **Application**, then click **Jobs**, and then select the Advanced Prediction. You can see the job status and the associated subjobs.

> **Note**
>
> You can see the pipeline for an Advanced Prediction job to review the progress of the job. From the Home page, click **IPM** and then click **Configure**. From the **Actions** menu next to the Advanced Prediction, select **Pipeline Status**. Or, from the Home page, click **Application**, then click **Data Exchange**, and then on the **Data Integration** tab, select **Advanced Predictions**. The pipeline is for review purposes only.

Reviewing Detailed Reports for Advanced Predictions

You can review Advanced Prediction prediction result details in a downloadable report in an Excel worksheet format, including details such as prediction status, method used, and prediction accuracy.

To review the report:

1. From the Home page, click **IPM**, then click **Configure**.

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2. From the **Actions** menu next to an Advanced Prediction definition, select **Download Report**.

The Excel workbook contains several worksheets:

- **Overview**—A summary of the prediction including details about the prediction and POV used, and any events that were considered during the prediction.
- **Feature Summary**—A summary list of the target and input drivers for the prediction, along with details about how missing values and outliers were treated.
- **ML Model Summary**—For each of the time series in the prediction, summarizes the input drivers, the pre-processing that was applied, and metrics from the model evaluation job.
- **Imputation Values**—Future value predictions for future data.

You can also download the report from the Outbox:

1. From the Home page, click **Application**, and then click **Overview**.
2. From the **Actions** menu, select **Inbox/Outbox Explorer**.
3. Look for `ML_JobName.zip` and download the file.

Running and Scheduling Insights

After configuring insights, you can run them any time. You can also set up a job to run insights at a regular interval, for example, when you have a new set of actuals.

When you run an insight, IPM Insights analyzes the data at each available intersection and looks for patterns and anomalies.

To run insights:

1. From the Home page click **IPM** and then click **Configure**.
2. From the **Actions** menu next to the insight you want to run, click **Run**.
3. To review the insight jobs, from the Home page, click **Application**, then click **Jobs**, and then select the insight. You can see the job status and the associated subjobs for each type of insight or auto prediction.

> **Note**
>
> You must have already run the Insights job in order for users to see insights in forms and dashboards.

For information about scheduling IPM Insights jobs, see Scheduling Jobs.

IPM Insights has its own job type, called **Insights**.

The **Insights** job indicates details about insight generation:

- How many insights were generated for each insight type.
- How many new insights were generated since the last job.
- How many insights were excluded because they were dismissed. (Any insights that have been dismissed are not regenerated.)

Insights are available on the Insights dashboard for two years, after which they are purged the next time you run an IPM Insights job.

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Tutorials

Tutorials provide instructions with sequenced videos and documentation to help you learn a topic.

Your Goal	Learn How
Learn how to run and schedule Insights in Oracle Fusion Cloud Enterprise Performance Management	Running and Scheduling Insights

Managing Insight Definitions and Generated Insights

You can edit, delete, or duplicate existing insight definitions. Administrators can delete insights that have been generated.

To edit, delete, or duplicate an insight definition:

1. From the Home page, click **IPM** and then click **Configure**.
2. From the **Actions** menu next to an insight, click **Edit**, **Duplicate**, or **Delete**.

Administrators can delete insights that have been generated. To delete an insight, on the Insights dashboard, from the **Actions** menu for the insight you want to delete, click **Delete Insight**.

Managing Tags

You can collaborate with your colleagues more effectively and leverage the power of automated analysis with IPM Insights using tags.

You can add tags in the **Comments** box for an insight:

- Use # to tag insights with a custom label. For example, you can tag a set of insights for team review by adding #TeamReview.
- Use @ to tag a user or group.

> **Note**
>
> A tag is automatically added to **Comments** when you dismiss an insight, including the dismissal reason, for example **#lowpriority** or **#duplicateInsight**, along with the associated remark.

Then, filter your Insights dashboard to see tagged insights. For more details about adding tags, and filtering and searching on tags, see Collaborating with Insights Using Tags in *Working with FreeForm*.

As an administrator, you can manage tags by adding more tags or deleting tags that are no longer needed.

To manage tags:

1. From the Home page, click **IPM** and then click **Tags**.
2. To add a new tag, click **Add Tag** and then enter a tag name.

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3. To delete a tag, click the **Actions** menu next to a tag and then click **Delete**.

Videos

Your Goal	Watch This
Learn about tagging insights to help your team collaborate and practice management by exception by bringing attention to insights that require attention.	Collaborating on Insights with Tags in EPM Planning

Bring Your Own ML: About Machine Learning Model Import

With Bring Your Own ML, EPM administrators can import a fully trained Machine Learning (ML) model and deploy it to a FreeForm application. Planners can then leverage robust, ML-based forecasting that uses advanced predictive modeling techniques to generate more accurate forecasts.

Data scientists gather and prepare historical data related to a business problem, train the algorithm, and generate a PMML file (Predictive Model Markup Language, a standard language used to represent predictive models) using a third party tool. These predictive analytic models and machine learning models use statistical techniques or ML algorithms to learn patterns hidden in large volumes of historical data. Predictive analytic models use the knowledge acquired during training to predict the existence of known patterns in new data.

EPM administrators can then import and configure the fully trained ML model, which generates two Groovy rules. Adminstrators attach the rule to a form or dashboard, or schedule a job to generate prediction results on a regular basis. This puts the benefits of machine learning and the power of data science into the hands of business users, enhancing the planning and budgeting process and leading to better business decisions.

For example, you can predict product volume for an entity, using key drivers such as average sales price, planned spend on promotions and advertising, historical volumes, and estimated industry volumes.

You can import ML Models and use them to predict numeric values in other finance use cases, for example:

- Trade promotion impacts on sales uplifts
- Marketing mix modeling to drive better ROMI
- Internal and external driver impacts on revenue forecasts
- Predictive cash forecasting for better cash position

Overview of Steps

Prerequisite: Data scientists build and train the ML model in a data science tool (any third party tool or Oracle Data Science Cloud) and save it as a PMML file.

Next, EPM administrators put the model to work to get business value from the trained model:

1. Administrators import the ML model in PMML format to a FreeForm application and define how the input variables and target variable maps to dimension members or cell values in the FreeForm application. This step generates automatic Groovy rules that connect the ML model to the FreeForm application. Two Groovy rules are generated for each ML model definition: one rule to associate with a form or dashboard, which allows users to make

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predictions on demand, and another to generate large scale predictions in a scheduled job for bulk processing. See Importing an ML Model.

2. Administrators deploy the ML model in a FreeForm application by associating the Groovy rule to relevant action menus, forms, or dashboards. See Deploying an ML Model to Planners. Administrators can also create a job to run the Groovy rule in a batch process.
3. Planners leverage ML-powered business rules in forms to generate predicted values, which are saved on the form. Planners can perform what-if analysis using the generated predictions, or modify predicted values on the form. Planners add value with their expertise and judgement, and then finalize the forecast.
4. This is an iterative process. As planners make predictions based on the ML model, administrators can measure the performance of the model, and can work with data scientists to update or replace the ML model when needed. Then, administrators re-import and deploy the retrained ML model.

   When you re-import the retrained ML model, the Groovy rules are regenerated.

Videos

Your Goal	Watch This
This overview introduces you to Bring Your Own ML (Machine Learning), where EPM Administrators can import a fully trained ML model and deploy it to a FreeForm application. Planners can then leverage robust, ML-based forecasting that uses advanced predictive modeling techniques to generate more accurate forecasts.	Overview: Bring Your Own Machine Learning (ML)
Learn how to configure ML model import for Bring Your Own Machine Learning. You import a fully trained ML model into FreeForm. You follow the steps in a wizard to map, analyze, and test the model. After saving the model, two Groovy rules are created. To complete the integration process, see the related video for deploying an ML model to FreeForm.	Configuring Machine Learning (ML) Model Import
Learn how to deploy an ML Model to FreeForm. After configuring the ML Model Import, you integrate the PMML file into your application by creating an Action Menu with the Groovy rule generated from the configured ML model. Then you associate the Action Menu with a form. When planners run the rule from the form, the rule returns the set of predicted values.	Deploying a Machine Learning (ML) Model to Planning

Considerations for Bring Your Own ML

The following are some of the considerations for Bring Your Own ML.

Considerations:

- Bring Your Own ML requires an EPM Enterprise Subscription; it is not available in EPM Standard Subscription. It is also available with Enterprise Planning and Budgeting Cloud and FreeForm.

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In addition, you can use Bring Your Own ML with Oracle Sales Planning Cloud and Oracle Strategic Workforce Planning Cloud.

- Bring Your Own ML works with Custom, FreeForm, Modules, Sales Planning, Strategic Workforce Planning, and Cash Forecasting application types.
- Bring Your Own ML requires Groovy rules.

Importing an ML Model

Import a fully trained ML model into a FreeForm application to prepare it for use by business users.

Prerequisite: Before you can import the ML model, the data science team must build, train, and save the ML model as a PMML file.

To import an ML model to a FreeForm application:

1. From the Home page, click **IPM** and then click **ML Models**.
2. Click **Import**, and then drag and drop the PMML file, or browse to it and select it.
   On the **Import Model** page, you see information about the PMML file, such as the target column (the variable to be predicted using the ML model) and the training date.
3. Enter a model name and description, and then click **Next**.
4. On the **Generate Rule** page, enter information that will generate a Groovy rule to associate with forms or dashboards:
   In **Model Mapping**, select the cube where the ML model will be used and define the scope of data in which to use the ML model by selecting a member or set of members from each dimension.
5. Map **Input** and **Output** to the appropriate dimensions and members in the cube and then click **Next**.
   The **Input** and **Output** sections contain the list of input features (features/columns that are used to make predictions) and target feature (column that is expected to be predicted). FreeForm analyzes the PMML file to generate the list of inputs and outputs.

   Input features are independent variables, similar to drivers, that act as input to your system. When you make predictions, the model uses input features to predict your output. In this step, you map the input from the ML model to the output in the FreeForm cube. **Input** describes how to extract the data from the ML model. **Output** defines the target measure you want to predict and where to paste the predicted values in the FreeForm application.

   For example, product, price, and industry volume, the input features, might be used to predict volume, the output.

   - In the **Input** area, for each input feature, select an **Input Type** and if you select **Cell Value** or **Member**, select the members or dimensions in the Planning application to map to. Input types:
     - **Prompt**: If you don't have a member or dimension in FreeForm that maps to this input value from the ML model, when the predicion is made, prompt the user to enter an estimate for the value.
     - **Cell Value**: Map an input feature to one or more dimension members in the FreeForm cube. For example, the input feature called Price maps to an account member called Price in the FreeForm application.
     - **Member**: Map an input feature to a dimension in the FreeForm cube. For example, the Input feature called Product maps to the Product dimension in the FreeForm application.

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- In the **Output** area, select an **Input Type** and if you select **Cell Value** or **Member**, select the members or dimensions in the FreeForm application to map to to store prediction results.

6. In **Analyze Model**, review the ML model and then click **Next**.
   This step represents MLX (Machine Learning Explainability), and extracts additional information about the ML model. For example, review Regression Coefficients to see how the relative impact of key input features is used to predict the output. The height of the bar represents the incremental effect of one unit increase in an input feature on the target variable.
7. In **Test Model**, test the ML model by generating a prediction for a set of sample values. For each **Input**, enter a sample input value and then click **Predict**.
8. Review the predicted **Output** value, and then click **Save and Close**.
9. Click **Yes** to confirm the creation of Groovy Rules.

Two Groovy rules are generated for each ML model definition:

- `ML_`*`MLModelName`*`_Form`: Use this rule to associate with a form or dashboard, which allows users to make predictions on demand.
- `ML_`*`MLModelName`*: Use this rule to generate large scale predictions in a scheduled job for bulk processing.

You can review the generated rules in Calculation Manager. The Groovy rules define the name and location of the PMML file, along with input and output based on the mapping you defined. For more information on using these generated Groovy rules, see Deploying an ML Model to Planners.

Tutorials

Tutorials provide instructions with sequenced videos and documentation to help you learn a topic.

Your Goal	Learn How
Learn how to import a fully trained ML model and deploy it to a FreeForm application. Planners can then leverage robust, ML-based forecasting that uses advanced predictive modeling techniques to generate more accurate forecasts.	Importing ML Models

Deploying an ML Model to Planners

Deploy an imported ML model to make it available to planners on forms and dashboards. Planners can use machine learning to generate a prediction.

After importing a PMML model, EPM administrators integrate the PMML file into the FreeForm application by associating the generated Groovy rule with FreeForm forms.

You can first review the Groovy rules in Calculation Manager to validate the mappings and make changes if needed. When the Groovy rule is finalized, you can associate it with relevant forms or dashboards.

Make the Groovy rule available to planners by creating an Action Menu that you associate with a form.

To associate an ML Groovy rule with a form:

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1. Create an Action Menu item:
   a. From the **Navigator**, under **Create and Manage**, click **Action Menus**.
   b. Click , enter the menu's name, and then click **OK**.
   c. Select the new menu item, click , and then click **Add Child**.
   d. Enter a **Menu Item** name, and give it a label, for example, **Predict Volume**, of type **Business Rule**, and select the appropriate cube and Groovy rule with the `Form` suffix, for example: `ML_`*`MLModelName`*`_Form`. Then click **Save**.

   For more information, see Creating and Updating Action Menus.
2. Associate the action menu with a form:
   a. From the **Navigator**, under **Create and Manage**, click **Forms**.
   b. Navigate to the form, click , and under **Other Options**, click the Action Menu item you created. Associate the menu with the form by moving it from **Available Menus** to **Selected Menus**. Then click **Save**.

   For more information, see Administering Forms.

When a planner selects the Action Menu item from the form, it triggers the Groovy rule. The rule picks the data for all input drivers, sends it to the PMML file for processing, returns the set of prediction values, and pastes them to the output, as defined in the **Import Model** wizard. The rule runs in the context of the form, predicting values only for cells on the form. Security is honored so that planners see predictions only for intersections to which they have access. Planners can run through various what-if scenarios to adjust their forecasts and plans, or adjust the predicted values.

Note that planners must have access to the model, form, rule, and members in order to run the rule.

Running an ML Groovy Rule as a Batch Job

You can run the ML Groovy rule as a job. The batch rule runs for all cells for the full scope of the model, not just for cells on a particular form. You can set it up to run as a recurring job, for example, to load predictions so they are available to planners on a regular basis.

To run or schedule a batch job, in **Jobs**, select a **Job Type** of **Rules**, and then select the Groovy batch version of the business rule for the PMML model—`ML_`*`MLModelName`*.

For information about scheduling jobs, see Scheduling Jobs.

Tutorials

Tutorials provide instructions with sequenced videos and documentation to help you learn a topic.

Your Goal	Learn How
Learn how to take the Groovy rule created from an imported ML Model, and add it to a form. Then you make predictions with the ML model in FreeForm.	Making Predictions with ML Models

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15

Defining Valid Intersections and Cell-Level Security

Define rules that restrict who can enter or view data in your application.

You can restrict who can enter data in your application by creating rules that mark certain member intersections as valid (or invalid) for data entry. These rules are called valid intersections. You can also restrict who can view data in your application by creating rules that remove read or write access to cells that a user would normally have access to due to their regular security. These rules are called cell-level security.

- To define valid intersections, see Defining Valid Intersections.
- To define cell-level security, see Defining Cell-Level Security

Defining Valid Intersections

Define valid intersection rules which filter certain cell intersections to users when they enter data or select runtime prompts.

Related Topics

- Understanding Valid Intersections
- Creating Valid Intersections
- Managing Valid Intersections
- Suppressing Invalid Data in Forms
- Clearing Invalid Data
- Working with Valid Intersections

Understanding Valid Intersections

Valid intersections are cell intersections that are filtered based on rules you define, called valid intersection rules, which filter certain cell intersections to users when they enter data or select runtime prompts. For example, you can specify that certain programs are valid only for some periods or departments. After valid intersections are defined, cells containing invalid data are read-only. This restriction speeds the planning process and optimizes the information available to users.

Conversely, you might have a use case where data entry is allowed in most cell combinations and you only need to prevent data entry to a select few cell combinations. In this case you can ease the definition process by defining invalid intersections. You define invalid intersections in the same way that you define valid intersections except you can specify the invalid intersection definition type while creating the intersection group.

To better understand how valid intersections affect behavior in forms and in runtime prompts, see Working with Valid Intersections.

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Videos

Your Goal	Watch This Video
Learn how to manage valid intersections.	Managing Valid Intersections in Cloud EPM

Related Topics

- Valid Intersection Groups
- Valid Intersection Rules
- Anchor and Nonanchor Dimensions
- Valid Intersection Examples
- Redundancy or Overlap in Valid Intersection Rules
- Shared Members and Valid Intersection Rules
- Substitution Variables and Valid Intersection Rules
- Evaluation Order

Valid Intersection Groups

Valid intersection groups define:

- Dimensions to be included
- One of those dimensions as the anchor dimension
- Whether nonanchor dimensions are required or not
- Whether the anchor dimension members not specified or referenced will be valid or invalid

Valid Intersection Rules

Valid intersection rules:

- Must use the same dimensions that were defined within their valid intersection group
- Valid intersection rules within the same valid intersection group that produce an apparent conflict or overlap, are marked valid if either valid intersection rule condition is met
- Valid intersection rules in different valid intersection groups that produce an apparent redundancy or overlap, are marked valid if they satisfy the requirements of all valid intersection groups

  Thus, if any valid intersection group marks an intersection invalid, regardless of other valid intersection groups making it valid, the system will mark the intersection invalid. Invalid groups override valid group results.

> **Note**
>
> If you want to remove valid intersections regardless of what other valid intersection groups allow, then this rule must be in a different valid intersection group.

See Valid Intersection Examples.

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Anchor and Nonanchor Dimensions

Anchor and nonanchor dimensions:

- Anchor dimensions are always required dimensions in the cube that is used in the valid intersection evaluation.
- Nonanchor dimensions are either required or not:
  - If a nonanchor dimension is required, any cube that doesn't use that dimension will ignore any valid intersection group where that dimension is tagged as required as it evaluates the valid intersections.
  - If a nonanchor dimension isn't required, any cube that doesn't use that dimension will still evaluate any valid intersection group that includes that dimension as not required and evaluate the intersections of any other dimensions in the valid intersection group in use in the cube.
- Unselected anchor dimension members are valid by default, but you can mark them invalid by clearing the **Unselected Members are Valid** option. This option marks all intersections with anchor dimension members not selected in this rule as invalid.

See Valid Intersection Examples.

Valid Intersection Examples

This section provides valid intersection group and valid intersection rule examples to illustrate a few simple, complex, and edge-case scenarios.

Example: Anchor and Nonanchor Dimensions

The choice of the anchor dimension is critical. Consider the following example, which produces a different result based on the anchor dimension definition:

- Valid intersection group 1 defines Entity as the anchor dimension and Product as a nonanchor dimension.
- Valid intersection group 2 reverses this definition with Product as the anchor dimension and Entity as the nonanchor dimension.

Table 15-1 Example - Anchor Dimension is Entity

Valid Intersection Group	Anchor Dimension - Entity	Nonanchor Dimension - Product
1	DESC(500 - Manufacturing) - Unselected members are valid	DESC(P_TP1 - Computer Equipment)

Group 1 means entities that are descendants of Manufacturing are valid only with descendant products of Computer Equipment. No other products are valid with descendants of Manufacturing. All other entities besides descendants of Manufacturing are valid with all products, including descendants of Computer Equipment.

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Table 15-2 Example - Anchor Dimension is Product

Valid Intersection Group	Anchor Dimension - Product	Nonanchor Dimension - Entity
2	DESC(P_TP1 - Computer Equipment) - Unselected members are valid	DESC(500 - Manufacturing)

Group 2 means products that are descendants of Computer Equipment are only valid with descendant entities of Manufacturing. No other entities are valid with descendants of Computer Equipment. All other products besides descendants of Computer Equipment are valid with all entities, including descendants of Manufacturing.

> ⚠ **Caution**
>
> The choice of anchor dimension is significant. You'll get dramatically different results if you choose the wrong anchor dimension.

Example: Required Dimension

In the following example, if a nonanchor dimension isn't required, then the application evaluates all remaining dimension intersections in the valid intersection group for a cube that doesn't contain the nonrequired dimension. This behavior could result in the evaluation of a valid intersection group with only one effective dimension.

Table 15-3 Example - Required vs. Non-Required Nonanchor Dimensions

Valid Intersection Group	Anchor Dimension - Entity	Nonanchor Dimension - Product
1	DESC(500 - Manufacturing) - Unselected members are valid	DESC(P_TP1 - Computer Equipment) - Not required

In Group 1, the product dimension isn't required, and unselected entities are valid. Therefore, if the cube of the form or business rule, at runtime, doesn't include the product dimension, the application evaluates the entity dimension selections to mark all entities as valid for a cube that doesn't contain the product dimension.

Table 15-4 Example - Required vs. Non-Required Nonanchor Dimensions

Valid Intersection Group	Anchor Dimension - Entity	Nonanchor Dimension - Product
2	DESC(500 - Manufacturing) - Unselected members are invalid	DESC(P_TP1 - Computer Equipment) - Not required

In Group 2, the product dimension isn't required, and unselected entities are invalid. Therefore, if a cube doesn't include the product dimension, then the application evaluates the entity dimension selections to mark all entities except descendants of Manufacturing as invalid. Thereafter, any cube that doesn't use the product dimension will only allow data entry in the descendants of Manufacturing entities.

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> **⚠ Caution**
>
> Carefully consider whether a nonanchor dimension is required or not, especially if the result leaves a valid intersection group with only one effective dimension. Additionally, selecting the **Unselected Members are Valid** option for anchor dimension members also plays a significant role in the system behavior for valid intersections. See Valid Intersection Examples.

Example: Unselected Members are Valid

In the following example, two intersection groups are valid. In one group, the anchor dimension unselected members are invalid (this option is cleared). In the other group, the anchor dimension unselected members are valid (this option is selected).

Table 15-5 Example - Unselected Members are Valid

Valid Intersection Group	Anchor Dimension - Account	Nonanchor Dimension - Entity
1	IDESC(BS - Balance Sheet) - Unselected members are invalid	000 - No Department
2	IDESC(GP - Gross Profit) - Unselected members are valid	IDESC(403 - Sales)

Because Group 1 defines all unselected members as invalid, the application marks noninclusive descendants of Balance Sheet invalid. Gross Profit isn't an inclusive descendant of Balance Sheet. So even though Group 2 explicitly states inclusive descendants of Gross Profit are valid with inclusive descendants Sales entities, the invalid definition from Group 1 overrides any further valid intersections of the same anchor dimension member set.

Example: Redundant or Overlapping Valid Intersection Rules Within the Same Valid Intersection Group

When valid intersection rules are within the same valid intersection group and produce any redundancy or overlap, the system marks an intersection valid if either of the valid intersection rule conditions are met.

Table 15-6 Example - Redundant or Overlapping Valid Intersection Rules Within the Same Valid Intersection Group

Valid Intersection Rule	Anchor Dimension - Account	Nonanchor Dimension - Entity
1	IDESC(GP - Gross Profit) - Unselected members are valid	IDESC(403 - Sales)
2	IDESC(NI - Net Income) - Unselected members are valid	IDESC(TD - Total Department)

Because Gross Profit is a descendant of Net Income and Sales is a descendant of Total Department, inclusive descendants of Gross Profit are valid with any inclusive Descendant of Total Department. Rule 1 is a subset of Rule 2, so Rule 1 is effectively a "No operation" rule and is unnecessary. There is no restriction on inclusive descendants of Gross Profit accounts only being valid for inclusive descendants of Sales Entities.

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Example: Redundant or Overlapping Valid Intersection Rules in Different Valid Intersection Groups

When valid intersection rules are in different valid intersection groups and produce any redundancy or overlap, the system marks an intersection valid only if it satisfies the requirements of all valid intersection groups.

In the following example, there are redundant or overlapping rules in different groups:

Table 15-7 Example - Redundant or Overlapping Valid Intersection Rules in Different Valid Intersection Groups

Valid Intersection Group	Anchor Dimension - Account	Nonanchor Dimension - Entity
1	IDESC(GP - Gross Profit) - Unselected members are valid	IDESC(403 - Sales) - Required
2	IDESC(NI - Net Income) - Unselected members are valid	IDESC(TD - Total Department) - Not required

Because Group 1 is further restrictive for inclusive descendants of Gross Profit accounts being valid with inclusive descendants of Sales entities, the application enforces this group for these intersections. Other, non-Gross Profit accounts can still use all inclusive descendants of Total Department entities, but inclusive descendants of Gross Profit accounts must use inclusive descendants of Sales entities.

Redundancy or Overlap in Valid Intersection Rules

Valid intersection rules within the same valid intersection group, which produce any apparent conflict or overlap, are marked valid if either valid intersection rule condition is met.

If different valid intersection groups share the same attributes, including the anchor dimension, required and not required nonanchor dimensions, and **Unselected Members are Valid** attribute, they will be treated as rules of the same valid intersection group.

Shared Members and Valid Intersection Rules

Shared members are supported in valid intersection rules. If a base member is selected for a valid intersection rule, any shared members are also included in the rule. Conversely, if a shared member is selected for a valid intersection rule, the base member is also included in the rule.

Substitution Variables and Valid Intersection Rules

You can use substitution variables in valid intersection rules. User variables are not supported. Substitution variables can be set on the Essbase server, application, or database level. The same substitution variable can exist on multiple levels; the application uses the first one it finds as it searches in this order:

1. Database (cube)
2. Application
3. Server

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Evaluation Order

Evaluation order for valid intersection groups orders invalid results sets as quickly as possible, increasing the speed and efficiency of the overall valid intersection evaluation.

For example, the application evaluates the first valid intersection group in the list, then the second group, and so on. If the application finds an invalid intersection in the second group in the list, it will stop evaluating the rest of the list because, once an intersection is defined as invalid, it will override other valid intersection rule results.

To change the order in which the groups are evaluated, see Changing the Valid Intersection Group Evaluation Order.

Creating Valid Intersections

You can define rules to filter certain cell intersections to users when they enter data, select members, or select runtime prompts.

To create a valid intersection:

1. Click **Application**, and then click **Valid Intersections**.
2. If it isn't already selected, click the **Setup** tab.

> **Note**
>
> The **Setup** tab is at the bottom of the page.

3. Create the valid intersection group:
   a. Click **Create**.
   b. Enter a name and description for the new intersection.
   c. The **Enabled** checkbox is selected by default. To disable the valid intersection group, clear the **Enabled** checkbox. You can also enable or disable a valid intersection group directly on the **Valid Intersections** page. See Disabling and Enabling Valid Intersection Groups.
   d. For **Definition Type**, select one of the following:
      - **Valid Intersection**
      - **Invalid Intersection**
   e. To select the anchor dimension for the intersection group, click next to **Select Anchor Dimension**.
   f. **Optional:** By default, the anchor dimension members that are not specified in the valid intersection rule are marked valid. To clear this option, click (Edit) next to the anchor dimension, and then click **Unselected members are valid**.

      For an explanation of this option and for an example of how it's used, see Anchor and Nonanchor Dimensions.
   g. To select additional dimensions (called nonanchor dimensions), click **Add Dimension**.

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h. **Optional:** By default, nonanchor dimensions are not required. To make a nonanchor dimension required, click (Edit) next to the nonanchor dimension, and click **Required**.

4. Define the valid intersection rule:
   a. Click **Add Rule**.
   b. To select the range of members to include, exclude, or remove in the valid intersection, click (Edit) next to the dimensions in the new rule:
      - Click **Edit** to open the **Select Members** page and select the members, substitution variables, and attributes to include in the valid intersection rule. See Using the Member Selector.
      - Click **Exclude** or **Exclude All** to define the dimension members you want to exclude from the rule:
        – **Exclude**: Selecting this option excludes members by ID. Only the specified members (base or shared) will be excluded.
        – **Exclude All**: Selecting this option excludes members by name. If a base member is specified, then the base and all of its shared members will be excluded. If a shared member is specified, then this member, its base member, and all other shared members of this member will be excluded.
      - Click **Clear** to clear the selection.

      To delete a rule, click (Remove).
5. Click **Save and Close**.

The new valid intersection group is added to the end of the valid intersections list. To reorder the rules in the list, see Changing the Valid Intersection Group Evaluation Order.

Managing Valid Intersections

Related Topics

- Viewing Valid Intersections
- Filtering Valid Intersections
- Importing and Exporting Intersections
- Changing the Valid Intersection Group Evaluation Order
- Disabling and Enabling Valid Intersection Groups
- Editing Details for a Valid Intersection Group
- Duplicating Valid Intersection Groups
- Deleting a Valid Intersection Group

Viewing Valid Intersections

To view valid intersections:

1. Click **Application**, and then click **Valid Intersections**.
2. If it isn't already selected, click the **Setup** tab.

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> **Note**
>
> The **Setup** tab is at the bottom of the page.

3. To search the entire listing for certain intersections, enter search criteria in the **Search** box and then click to display only the intersections that meet the search criteria.
4. To filter the listing so it displays only the intersections that meet the filter criteria, specify the following filter options:
   - **Filter** : Click to display the **Filter** page where you can select from a robust list of filter options. For a complete list of filter options and their descriptions, see Filtering Valid Intersections.
   - **Clear All** : Click to clear all filters.
   - **Enabled**: Filters the list so you view only the intersections that are enabled (**Yes**), not enabled (**No**), or select **All**.
   - **Definition Type**: Filters the list so you view only the intersections that are the **Valid Intersection** or **Invalid Intersection** types, or select **All**.
5. You can also perform these tasks on the **Valid Intersections** page:
   - **Actions** menu: To import and export valid intersections, see Importing and Exporting Intersections .
   - **Create**: See Creating Valid Intersections.
   - **Refresh**: Click to refresh the intersections listing.
6. For each listing on the intersections page you can perform the following tasks:
   - **Enabled** column: Indicates whether an intersection is enabled. Click the check mark next to the intersection to disable or enable it. A green check mark indicates that the definition is enabled. See Disabling and Enabling Valid Intersection Groups.
   - **Action** column: Click ••• (Action) to edit, duplicate, or delete an intersection, or to move it up or down in the listing order. See the following topics:
     - Changing the Valid Intersection Group Evaluation Order
     - Editing Details for a Valid Intersection Group
     - Duplicating Valid Intersection Groups
     - Deleting a Valid Intersection Group

Filtering Valid Intersections

You can filter the list of intersections by certain criteria such as whether the intersection is enabled or not, whether the definition is valid or invalid, when it was modified, and by whom. When you filter, only the intersections that meet the filter criteria are displayed on the **Valid Intersections** page.

To filter intersections:

1. Click **Application**, and then click **Valid Intersections**.
2. Click (Filter) to specify filter criteria:

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- **Enabled**: View only the intersections that are enabled (**Yes**), not enabled (**No**), or select **All**.
- **Definition Type**: View only the intersections that are the **Valid Intersection** or **Invalid Intersection** types, or select **All**.
- **Modified**: View only the intersections that were modified before or after a certain date and time, or the intersections modified within a range of dates or times. Select **After**, **Before**, or **Between** and then click (Select Date and Time) to specify the date and time criteria.
- **Modified By**: View only the intersections modified by selected users.

3. Click **More** to further refine the filter criteria:
   - **Dimensions**
   - **Anchor Dimensions**
   - **Unselected members are valid**: Choose **Yes**, **No**, or **All**.
   - **Additional Dimensions Required**: Choose **Yes**, **No**, or **All**.
4. Click **Apply**.

Importing and Exporting Intersections

About Importing and Exporting Valid Intersections

You can export the filtered list of valid intersections from the listing page to a location on your local computer, or you can export them to the server. If no filter is defined on the listing page, then all intersections are exported.

Use the Import action to import intersections into the application from a location on your local computer or import them from the server. When you perform an import, the system tests the import file for anything that might break a definition; for example, if a cube is missing, or if an anchor dimension doesn't exist, or if a subrule is not found, errors are logged in the error file. Import jobs will only complete successfully if the import file has no errors.

> **Note**
>
> Subrules provide information about the content of the rules such as the member selection or exclusion for anchor and non anchor dimensions, and the restriction type applied for each subrule.

Depending on the export or import location you choose, the intersections are exported or imported in either an Excel file format (XLSX) or a zip file format.

> **Note**
>
> Locked valid intersection rules are not exported when you export valid intersections. Locked valid intersection rules (and rules that start with restricted prefixes such as FCCS_, OCX_, OEP_, OFS_, OPF_, OWP_, TRCS_) are not imported when you import valid intersections.

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Valid Intersections Import File

The Excel import file must have two sheets with the following names for the first and second sheets:

1. `Rules`
2. `Sub Rules`

The `Rules` sheet has the following column headings:

- **Name**
- **Position**
- **Description**
- **Enabled**
- **Definition Type**
- **Anchor Dim Name**
- **Anchor Dimension Apply to Selected Members**
- **Dim1**
- **Dim1 Required**
- **Dim2**
- **Dim2 Required**
- **Dim***X*
- **Dim***X* **Required**

The `Sub Rules` sheet has the following column headings:

- **Name** - This column must contain the name of the Rule from the first sheet
- **Anchor Members**
- **Anchor Exclusion**
- **Anchor Exclusion All**
- **Dim1 Members**
- **Dim1 Exclusion**
- **Dim1 Exclusion All**
- **Dim2 Members**
- **Dim2 Exclusion**
- **Dim2 Exclusion All**
- **Dim***X* **Members**
- **Dim***X* **Exclusion**
- **Dim***X* **Exclusion All**

Exporting and Importing Valid Intersections

To export and import intersections:

1. Click **Application**, and then click **Valid Intersections**.

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2. Apply filters to the listing, as needed. See Filtering Valid Intersections.
3. To export, click **Actions**, then **Export**, and then select the target location for the export file:
   - **Local**: Saves the export file to a location on your local computer. If you choose this option, click **Export**, and then specify where to save the export file.
   - **Outbox**: Runs a job that saves the export file in a zip format to the server which you can download now or use to schedule an export job later. If you choose this option, click **Save and Run Job**.

     To download the export file from the Outbox:

     a. Click **Application**, and then click **Jobs**.

     b. Under **Recent Activity** click the export job.

     c. On the **Job Details** page, click (Download) to select a download location for the export file. You can also download the export file from the Inbox/Outbox Explorer for your business process.
4. If editing the export file in Excel, note that the Excel file has two sheets: `Rules` and `Sub Rules`.
5. To import, click **Actions**, then **Import**, and then select the location of the import source file:
   - **Local**: Loads the import file from a location on your computer. For **Source File**, click **Browse** to select the import file on your computer, and then click **Import**.
   - **Inbox**: Runs a job that loads the import file from the server. The import file must be in a zip file format. Enter the name of the file in **Source File**, click **Save and Run Job**, and then specify the **Name** and **Error File** on the **Save as Job** dialog. The error file provides information about the intersections that were not imported. You can download the error file from the Inbox/Outbox Explorer for your business process.

Changing the Valid Intersection Group Evaluation Order

Evaluation order for intersection groups orders invalid results sets as quickly as possible, increasing the speed and efficiency of the overall intersection evaluation.

To learn more about evaluation order, see Evaluation Order.

To change the position of a valid intersection group in a list:

1. Click **Application**, and then click **Valid Intersections**.
2. If it isn't already selected, click the **Setup** tab.

> **Note**
>
> The **Setup** tab is at the bottom of the page.

3. To the right of the intersection group in the listing, click ••• (Actions).
4. Select **Move Up** or **Move Down**.

> **Tip**
>
> You can also drag intersection groups to move them up and down in the list.

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Disabling and Enabling Valid Intersection Groups

Intersection groups, by default, are enabled at the time of creation. If you don't want an intersection group to be evaluated or used, you can disable it on the **Valid Intersections** page. When an intersection group is disabled, the intersection rule for that group no longer applies when viewing application forms, business rules, or runtime prompts. You can also reenable a disabled intersection group.

To disable and enable an intersection group:

1. Click **Application**, and then click **Valid Intersections**.
2. If it isn't already selected, click the **Setup** tab.

> **Note**
>
> The **Setup** tab is at the bottom of the page.

3. In the Enabled column of the intersection list, click the check mark next to the intersection group that you're disabling or enabling.

> **Note**
>
> The check mark is green if the group is enabled.

4. Ensure that any remaining groups that are enabled are still listed in the correct evaluation order in the intersections list. If they are not, then move them up or down in the order.

   See Changing the Valid Intersection Group Evaluation Order.

Editing Details for a Valid Intersection Group

To edit intersection group details, you work with dimension members in the member selector. You can also define exclusions in intersection rules.

To edit intersection group details:

1. Click **Application**, and then click **Valid Intersections**.
2. If it isn't already selected, click the **Setup** tab.

> **Note**
>
> The **Setup** tab is at the bottom of the page.

3. Click the name of the intersection group you want to edit.

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> **Note**
>
> A pagination bar displays at the bottom of the rules page. If you have a large number of rules, you can use the pagination bar to easily navigate between the rules pages and to jump to a specific page. The page size options (number of rules per page), which you can select on the pagination bar, are **25**, **50**, **100**, and **All**. (Note that the **All** option isn't available if you have more than 300 rules.)

- To edit dimension details, next to the dimension, click (Edit) to select the members to include, exclude, or remove in the intersection rule:
  - Click **Edit** to open the **Select Members** page and select members, substitution variables, and attributes to include in the intersection rule. You can also type in the members or functions.

    See Using the Member Selector.
  - Click **Exclude** or **Exclude All** to define the dimension members you want to exclude from the rule:
    - **Exclude**: Selecting this option excludes members by ID. Only the specified members (base or shared) will be excluded.
    - **Exclude All**: Selecting this option excludes members by name. If a base member is specified, then the base and all of its shared members will be excluded. If a shared member is specified, then this member, its base member, and all other shared members of this member will be excluded.
  - Click **Clear** to clear the selection.
- To delete a dimension from an intersection group, next to the dimension, click (Edit), and then click (Remove).
- To remove a rule from an intersection group, click (Remove).
- To add a dimension or a rule to an intersection group, click **Add Rule** or **Add Dimension**.

4. Click **Save and Close**.

Duplicating Valid Intersection Groups

To speed intersection group creation, you can duplicate an existing intersection group and then edit it.

To duplicate an intersection group:

1. Click **Application**, and then click **Valid Intersections**.
2. If it isn't already selected, click the **Setup** tab.
3. Click (Actions) to the right of the intersection group you want to duplicate, and then select **Duplicate**.

   The duplicate group is added to the end of the intersections list with the word "Copy" appended to the name.
4. Open the intersection group and edit it.

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5. Reorder the intersection groups, if needed. See Changing the Valid Intersection Group Evaluation Order.

Deleting a Valid Intersection Group

After a group is deleted, the intersection groups are reordered. If there are three intersection groups, and the second one in the order is deleted, the third intersection group becomes number two.

To delete an intersection group:

1. Click **Application**, and then click **Valid Intersections**.
2. If it isn't already selected, click the **Setup** tab.

> **Note**
>
> The **Setup** tab is at the bottom of the page.

3. Click ••• (Actions) to the right of the intersection group that you want to remove, and then select **Delete**.
4. Reorder the remaining intersections, if needed. See Changing the Valid Intersection Group Evaluation Order.

To delete an intersection rule from an intersection group, see Editing Details for a Valid Intersection Group.

Suppressing Invalid Data in Forms

Suppressing invalid data hides rows or columns in application forms that contain invalid data. If this option isn't selected, then the application displays rows or columns that contain cells with data that are invalid. Cells with invalid data are read-only.

To suppress invalid data in forms:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Forms**.
2. Select the form, click (Edit), and then click **Layout**.
3. Under **Grid Properties**, select **Suppress invalid data - Rows** and/or **Suppress invalid data - Columns**.
4. Click **Save**.

Clearing Invalid Data

Related Topics

- About Invalid Data
- Working With Invalid Intersection Reports
- Clearing Data at Invalid Intersections

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About Invalid Data

If data already exists at intersections, then adding or modifying valid intersections invalidates the existing data. Creating a valid intersection rule or modifying an existing valid intersection rule doesn't clear data in the invalid intersections. You must generate a valid intersection rule report, which will show where data exists at invalid intersections, and then determine whether to clear the data.

> **Note**
>
> Data may remain at an invalid intersection for historical purposes or for use in forward looking scenarios. Therefore, it's not a requirement to clear data at invalid intersections.

Working With Invalid Intersection Reports

You can view reports that show data at invalid intersections on the **Reports** tab of the **Valid Intersections** page. **Invalid Intersection Reports** lists existing reports, their status, and the last time they were run.

To work with invalid intersection reports:

1. Click **Application**, then **Valid Intersections**, and then click the **Reports** tab.

   > **Note**
   >
   > The **Reports** tab is at the bottom of the page.

2. Perform a task:
   - To refresh the listing, click **Refresh**.
   - To create a report, click **Create**, name the report, select the cube, choose the scope of the report, and then select when to run the report. You can run the report now or run the report later by saving it as a job. To save a report without running it, click **Save and Close**. To remove a report after it's run, click **Remove Reports**.

     > **Note**
     >
     > Invalid intersection reports are not supported for aggregate storage cubes. Aggregate storage cubes are not listed in the **Cube** drop-down list.

   - To edit a report, click (Actions), and then **Edit**.
   - To copy a report, click (Actions), and then **Duplicate**.
   - To run a report, click (Actions), and then **Run**.
   - To delete a report, click (Actions), and then **Delete**.

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Clearing Data at Invalid Intersections

Users with appropriate permissions can clear the invalid data if the data isn't needed. To clear invalid data, run the report, and then click **Clear Invalid Intersections**.

Working with Valid Intersections

Related Topics

- Working with Valid Intersections in Application Forms
- Working with Valid Intersections in Calculation Manager Rule Runtime Prompts

Working with Valid Intersections in Application Forms

Using valid intersections prevents data entry for invalid intersections as defined in the applicable valid intersection group. The affected cells in the form display as read-only following standard, read-only color coding. If you hover the cursor over an invalid intersection, a tool tip displays indicating the cell is read-only because it's defined as an invalid intersection.

The valid intersection group applies first to the form point of view and page axis. If the point of view intersections are all invalid, then a warning message is displayed, and the form doesn't render a data grid until a valid intersection is selected.

If the point of view has valid intersections, then the rows and columns are filtered to restrict data entry at invalid intersections. If the **Suppress Invalid Data** option for the form is enabled, then the form suppresses invalid rows, columns, or both, as appropriate.

Any rows or columns, which consist of a mix of valid and invalid intersections, display those intersections as valid or invalid, as appropriate. Invalid intersections are displayed with standard, read-only shading and preclude data entry.

Application users can reset the point of view to the default, unfiltered list without closing and reopening the form by clearing the selections. Application users can also clear a selection, thus opening up more selections for other dimensions. They will not be able to render a form with a dimension cleared, because valid members must be selected for each dimension.

In the member selector, invalid members are suppressed due to valid intersection rules. Application users can display invalid members in the member selector using the **Show Invalid Members** option. Invalid members are displayed but are unavailable for selection.

> **Note**
>
> Valid intersection groups don't grant access to dimension members. Valid intersection groups further restrict the valid intersections of dimension members already granted to an application user.

Table 15-8 Form Behavior if Valid Intersections are Applied

Action	Behavior
Open a form	The form renders with member selections as defined in the form definition, adhering to the user's access rights for dimensions, and applies valid intersection groups with the most recently used as current selections.

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Table 15-8 (Cont.) Form Behavior if Valid Intersections are Applied

Action	Behavior
Select members from a point of view dimension	The application: • Enables users to select a member on the point of view • In the member selector for a point of view dimension, enables users to select from a filtered list of remaining valid intersections, which is based on the members that were selected for the other point of view dimensions • Ignores the order in which point of view dimension members are selected because selecting a member from any dimension included in a valid intersection group dynamically filters the remaining dimension members lists for those dimensions included in the valid intersection group, as appropriate, when that dimension is selected • Provides the option to hide invalid members from dimension lists or display them as unselectable in the point of view • Provides the ability to reset the point of view to the fully unfiltered list without closing and reopening the form by clearing the selections Note that ad hoc forms, both in Web and Oracle Smart View for Office, will not filter page or point of view members according to valid intersection groups.
Select **Go** to render a form based on point of view selections. You can also click the right arrow in the form point of view.	The form renders as defined based on the valid point of view intersections.
Enter and save data	The form data is entered and saved.

Working with Valid Intersections in Calculation Manager Rule Runtime Prompts

Valid intersection groups apply to runtime prompts when launched from within the context of the application. Runtime prompts will prevent users from selecting invalid intersections as defined in the valid intersection groups.

Filtering according to valid intersection groups isn't supported in Oracle Smart View for Office forms. The rule, however, will not launch if you choose an invalid intersection in the runtime prompts both in the Web and in Smart View.

Table 15-9 Runtime Prompt Behavior if Valid Intersections are Applied

Action	Behavior
Launch a Calculation Manager rule runtime prompt	The application: • Prevents the user from selecting invalid intersections within the runtime prompt based on the valid intersection group • Prevents the Calculation Manager rule from executing if there are invalid intersections in the runtime prompts
Enter valid intersections	The valid intersection is allowed to be entered.

Defining Cell-Level Security

Define security rules which restrict users and groups from viewing data in certain cell intersections in forms.

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Related Topics

- Understanding Cell-Level Security
- Creating Cell-Level Security Definitions
- Viewing Cell-Level Security Definitions
- Filtering Cell-Level Security Definitions
- Importing and Exporting Cell-Level Security Definitions
- Testing Cell-Level Security
- Reordering the Cell-Level Security Definitions List

Understanding Cell-Level Security

About Cell-Level Security

Service Administrators applying cell-level security can deny access to cells that a user would normally have access to due to their regular security. Cell-level security is therefore defined as an exception to the existing member security. For example, a Department Manager requires access to all accounts in their own department, but only a certain account in all other departments. With the usual metadata security the Manager would have access to all accounts across all departments, but using cell-level security enables the Service Administrator to control the intersection of all accounts with the Manager's department and only the specific account in all other departments.

Cell-level security uses rules, similar to valid intersection rules, to deny read or write access to users viewing certain cell intersections anywhere a cell is shown (for example, forms, Smart View, reports, dashboards, infolets, and so on). When cell-level security rules are applied, users with read access can see the data value in a cell but the cell is not editable. If users are denied read access to a cell, the value displayed in the cell is #noaccess.

If you are a Service Administrator, you can define and assign cell-level security rules to any user or group. Cell-level security doesn't affect you.

Anchor and Nonanchor Dimensions

Cell-level security definitions use anchor and nonanchor dimensions:

- Anchor dimensions are always required dimensions in the cube that is used in the cell-level security definition.
- Nonanchor dimensions are either required or not:
  - If a nonanchor dimension is required, any cube that doesn't use that dimension will ignore any cell-level security definitions where that dimension is tagged as required.
  - If a nonanchor dimension isn't required, any cube that doesn't use that dimension will still evaluate any cell-level security definition that includes that dimension as not required and evaluate the definitions of any other dimensions in the definition in use in the cube.
  - By default, nonanchor dimensions aren't required. To make a nonanchor dimension required, click (Edit) next to the nonanchor dimension, and click **Required**.
- By default, the anchor dimension members that are not specified in the rule are included in the security definition, but you can clear this option by clicking (Edit) next to the anchor dimension, and then clicking **Apply to Selected Members Only**.

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Creating Cell-Level Security Definitions

To create a cell-level security definition:

1. Click **Application**, and then click **Cell-Level Security**.
2. Create the definition:
   a. Click **Create**.
   b. Enter a name and description for the definition.
   c. The **Enabled** checkbox is selected by default. To disable the definition, clear the **Enabled** checkbox. You can also enable or disable a definition directly on the **Cell-Level Security Definitions** page.
   d. To define cube-specific security, click **Cubes** and select from the list of cubes or select **All**.
   e. An anchor dimension is required. To select the anchor dimension, click **Select Anchor Dimension**. For information about anchor and nonanchor dimensions, see Understanding Cell-Level Security
   f. **Optional:** By default, the anchor dimension members that are specified in the rule are included in the security definition. If you click (Edit) next to the anchor dimension, and then click **Apply to Selected Members Only**, cell-level security is disabled for the selected members and cell-level security is applied to the rest of the members, excluding the selected members.
   g. To select additional dimensions (called nonanchor dimensions), click **Add Dimension**.
   h. **Optional:** By default, nonanchor dimensions are not required. To make a nonanchor dimension required, click (Edit) next to the nonanchor dimension, and click **Required**.
3. Define the cell-level security rule:
   a. Click **Add Rule**.
   b. In the **Users, Groups** column, click (Search) to find the users and groups to include in the cell-level security rule.
   c. For **Restriction**, choose **Deny Read** (default) or **Deny Write**. **Deny Read** is the default option because it is the most restrictive. If users are denied read access to a cell, the value displayed in the cell is #noaccess. Users with **Deny Write** access can see the data value in a cell but the cell is not editable.
   d. Click (Edit) next to the dimensions in the new rule:
      - Click **Edit** to open the **Select Members** page and select the members, substitution variables, and attributes to include in the cell-level security rule.
      - Click **Exclude** or **Exclude All** to define the dimension members you want to exclude from the rule:
        – **Exclude**: Selecting this option excludes members by ID. Only the specified members (base or shared) will be excluded.
        – **Exclude All**: Selecting this option excludes members by name. If a base member is specified, then the base and all of its shared members will be excluded. If a shared member is specified, then this member, its base member, and all other shared members of this member will be excluded.

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- Click **Clear** to clear the selection.

To delete a rule, click (Remove).

4. Click **Save**.

The new cell-level security definition is added to the end of the list. Definitions are evaluated in the order they appear in the list. To reorder the definitions list, see Reordering the Cell-Level Security Definitions List.

After a definition is created, you can test it in a form to see how the form will look from a user's perspective. See Testing Cell-Level Security.

Viewing Cell-Level Security Definitions

To view cell-level security definitions:

1. Click **Application**, and then click **Cell-Level Security**.
2. To search the entire listing for certain definitions, enter search criteria in the **Search** box and then click (Search) to display only the definitions that meet the search criteria.
3. To filter the listing so it displays only the definitions that meet the filter criteria, specify the following filter options:
   - **Filter** : Click to display the **Filter** page where you can select from a robust list of filter options. For a complete list of filter options and their descriptions, see Filtering Cell-Level Security Definitions.
   - **Clear All** : Click to clear all filters for **Effective Assignment**, **Enabled**, and **Restriction**.
   - **Effective Assignment**: Filters the list so you view only the definitions that effect certain users or groups.
   - **Enabled**: Filters the list so you view only the definitions that are enabled (**Yes**), not enabled (**No**), or select **All**.
   - **Restriction**: Filters the list so you view only the definitions that are assigned the **Deny Read** or **Deny Write** restriction, or select **All**.
4. You can also perform these tasks on the **Cell-Level Security Definitions** page:
   - **Actions** menu: To import and export definitions, see Importing and Exporting Cell-Level Security Definitions.
   - **Test**: See Testing Cell-Level Security.
   - **Create**: See Creating Cell-Level Security Definitions.
   - **Refresh**: Click to refresh the definitions listing.
5. For each listing on the definitions page you can perform the following tasks:
   - **Enabled** column: Indicates whether a definition is enabled. Click the check mark next to the definition to disable or enable it. A green check mark indicates that the definition is enabled.
   - **Action** column: Click ••• (Action) to edit, duplicate, or delete a definition, or to move it up or down in the listing order.

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Filtering Cell-Level Security Definitions

You can filter the list of cell-level security definitions by certain criteria such as by cube, by restriction, or by date. When you filter, only the definitions that meet the filter criteria are displayed on the **Cell-Level Security Definitions** page.

To filter cell-level security definitions:

1. Click **Application**, and then click **Cell-Level Security**.
2. Click (Filter) to specify filter criteria:
   - **Cubes**: View only the definitions in the selected cube or cubes, or select **All**.
   - **Enabled**: View only the definitions that are enabled (**Yes**), not enabled (**No**), or select **All**.
   - **Restriction**: View only the definitions that are assigned the **Deny Read** or **Deny Write** restriction, or select **All**.
   - **Modified**: View only the definitions that were modified before or after a certain date and time, or the definitions modified within a range of dates or times. Select **After**, **Before**, or **Between** and then click (Select Date and Time) to specify the date and time criteria.
   - **Modified By**: View only the definitions modified by selected users.
   - **Effective Assignment**: View only the definitions that effect the selected users or groups.
3. Click **More** to further refine the filter criteria:
   - **Dimensions**
   - **Anchor Dimensions**
   - **Anchor Dimension: Apply to Selected Members Only**: Choose **Yes**, **No**, or **All**.
   - **Additional Dimensions Required**: Choose **Yes**, **No**, or **All**.
4. Click **Apply**.

Importing and Exporting Cell-Level Security Definitions

About Importing and Exporting Cell-Level Security Definitions

You can export the filtered list of cell-level security definitions from the listing page to a location on your local computer, or you can export them to the server. If no filter is defined on the listing page, then all cell-level security definitions are exported.

Use the Import action to import cell-level security definitions into the application from a location on your local computer or import them from the server. When you perform an import, the system tests the import file for anything that might break a definition; for example, if a cube is missing, or if an anchor dimension doesn't exist, or if a subrule is not found, errors are logged in the error file. Import jobs will only complete successfully if the import file has no errors.

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> **Note**
>
> Subrules provide information about the content of the rules such as the user assignment for each member of the rule (subrule), the member selection or exclusion for anchor and non anchor dimensions, and the restriction type applied for each subrule.

Depending on the export or import location you choose, the application definitions are exported or imported in either an Excel file format (XLSX) or a zip file format.

Cell-Level Security Import File

The Excel import file must have two sheets with the following names for the first and second sheets:

1. `Rules`
2. `Sub Rules`

The `Rules` sheet has the following column headings:

- **Name**
- **Position**
- **Description**
- **Enabled**
- **Valid Cubes** - This column can contain either `All` or a list of comma-separate names of cubes, such as `Plan1, Plan2`
- **Anchor Dim Name**
- **Anchor Dimension Apply to Unselected Members**
- **Dim1**
- **Dim1 Required**
- **Dim2**
- **Dim2 Required**
- **Dim***X*
- **Dim***X* **Required**

The `Sub Rules` sheet must have the following column headings:

- **Name** - This column must contain the name of the Rule from the first sheet
- **Users**
- **User Groups**
- **Restriction** This column can contain `Deny Read` or `Deny Write`
- **Anchor Members**
- **Anchor Exclusion**
- **Anchor Exclusion All**
- **Dim1 Members**
- **Dim1 Exclusion**

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- **Dim1 Exclusion All**
- **Dim2 Members**
- **Dim2 Exclusion**
- **Dim*X* Members**
- **Dim*X* Exclusion**
- **Dim*X* Exclusion All**

Exporting and Importing Cell-Level Security Definitions

To export and import cell-level security definitions:

1. Click **Application**, and then click **Cell-Level Security**.
2. Apply filters to the listing, as needed. See Filtering Cell-Level Security Definitions.
3. To export, click **Actions**, then **Export**, and then select the target location for the export file:
   - **Local**: Saves the export file to a location on your local computer. If you choose this option, click **Export**, and then specify where to save the XLSX export file.
   - **Outbox**: Runs a job that saves the export file in a zip format to the server which you can download now or use to schedule an export job later. If you choose this option, click **Save and Run Job**.

     To download the export file from the Outbox:

     a. Click **Application**, and then click **Jobs**.

     b. Under **Recent Activity** click the **Export CLS** job.

     c. On the **Job Details** page, click (Download) to select a download location for the cell-level security export file. You can also download the export file from the Inbox/Outbox Explorer for your business process.
4. If editing the export file in Excel, note that the Excel file has two sheets: `Rules` and `Sub Rules`.
5. To import, click **Actions**, then **Import**, and then select the location of the import source file:
   - **Local**: Loads the import file from a location on your computer. For **Source File**, click **Browse** to select the import file on your computer, and then click **Import**.
   - **Inbox**: Runs a job that loads the import file from the server. The import file must be in a zip file format. Enter the name of the file in **Source File**, click **Save and Run Job**, and then specify the **Name** and **Error File** on the **Save as Job** dialog. The error file provides information about the definitions that were not imported. You can download the error file from the Inbox/Outbox Explorer for your business process.

Testing Cell-Level Security

After defining cell-level security and enabling it, you can test it to see what an effected user would see when they view a form. Testing ensures that users are seeing only the cell values they are allowed to see and nobody is seeing more than they should. To test a definition, you must enable it.

To test cell-level security:

1. Click **Application**, and then click **Cell-Level Security**.
2. Enable the definition you want to test.

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3. Click **Test**, and then select the form that you want to test.
4. In the **Select or Enter a User Name** text box, specify a user name or click (Search) to select a user, and then click OK.

> **Note**
>
> You can test only one user at a time.

The form displays the user access to each cell as Read, Write, or None.

Reordering the Cell-Level Security Definitions List

Cell-level security definitions are evaluated in the order they appear on the definitions list; for example, the application evaluates the first cell-level security definition in the list, then the second definition, and so on.

To change the position of a cell-level security definition in the list:

1. Click **Application**, and then click **Cell-Level Security**.
2. To the right of the cell-level security definition, click ••• (Action).
3. Select **Move Up** or **Move Down**.

> **Tip**
>
> You can also drag cell-level security definitions to move them up and down in the list.

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16

Defining Data Maps and Creating File-Based Integrations

Define data maps to move data, comments, attachments, and supporting detail from source cubes and Smart Lists to target reporting cubes to consolidate data.

Data Integration is the mechanism by which integration processes are performed in Oracle Fusion Cloud Enterprise Performance Management. You can define file-based and direct integration sources, create mapping rules to translate source data in the required target format, and execute and manage the periodic data loading process. To learn how to create a file-based integration see Creating File-Based Integrations in *Administering Data Integration*.

To learn how to define data maps, see Defining Data Maps in *Administering Data Integration*.

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17

Managing Jobs

Schedule jobs to perform common administrative tasks and lighten your workload. You can start jobs right away or schedule jobs to run later at intervals.

Related Topics

- How Jobs Save You Time
- Viewing the Status of Jobs
  View pending jobs and recent activity on the Jobs console.
- Scheduling Jobs
- Editing and Canceling Jobs
- Duplicating Jobs
- Canceling Rules Jobs and Ruleset Jobs
- Downloading Export Files From Your Outbox

How Jobs Save You Time

Jobs are actions such as exporting data or refreshing the database, which you can start right away or schedule to run later at intervals. To lighten your workload, define jobs to perform common administrative tasks such as:

- Importing and exporting metadata and data
- Refreshing the database
- Mapping cubes

Viewing the Status of Jobs

View pending jobs and recent activity on the Jobs console.

The Jobs console lists jobs that are in a pending state under **Pending Jobs**. Jobs that are processing, have run and are completed, or have errors are listed under **Recent Activity**.

Jobs are retained in the Jobs console for 90 days.

To view the status of jobs:

1. Click **Application**, and then click **Jobs**.
2. Perform a task:
   - To filter jobs and activity by criteria such as date or job type, click (Filter).
   - To search for a job by name, enter text in the **Search**, and then click (Search).
   - To view a job's details, click the job.

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> **Note**
>
> - The application is automatically refreshed during an application upgrade. If any refresh errors occur during an application upgrade, you can view the errors on the Jobs page. These errors are captured in the job called **Content Update**.
> - The **Recent Activity** list on the **Jobs** page defaults to display only the last day's data. Similarly, the job filter's **Start Time** and **End Time** dates default to yesterday and today to have only a one day difference. Click (Filter) to change the default display.
> - Jobs older than 90 days are purged.

Scheduling Jobs

To schedule jobs:

1. Click **Application**, then **Jobs**, and then click **Schedule Jobs**.
2. Specify the following:
   - The type of job you're creating. For a list of jobs and descriptions, see Job Types.
   - When to run the job. You can schedule a job to run now or to run later. If scheduling the job to run later, see Scheduling Jobs to Run Later.
3. Click **Next**.
4. Select a job from the job list. Depending on the job type, there may be additional options and considerations. See Job Types.

   > **Note**
   >
   > - You can run up to five import or export jobs at one time.
   > - For export jobs, you can specify a unique output file name for each job. From the job listing page, click (Edit) next to the selected job and specify the output file name with a zip extension. The unique file name you specify will override the default export file name when the job runs.
   > - To prevent automatic backup from failing due to certain jobs running during daily maintenance, Oracle Fusion Cloud Enterprise Performance Management disallows certain jobs from starting while the daily maintenance process is running. If the system prevents a job from starting, the reason will be stated in the Job Details. If you've enabled email notifications for the Job Console, you'll receive an email notification when a job does not start. If you have a job that is scheduled to start during the daily maintenance process, it is recommended that you reschedule your job to start outside of the daily maintenance window. See Setting the Daily Maintenance Process Start Time.

5. Click **Next**.
6. Review your choices. If satisfactory, click **Finish**.

To edit or cancel a job after scheduling it, see Editing and Canceling Jobs.

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Job Types

Table 17-1 Job Types

Job Type	Description
Rules	Runs a rule that you select. Note the following: • You can filter the rules list by cube and by rule type. • You must click the check mark next to the rule you want to run before you can proceed. • For rules jobs with runtime prompts, clicking the check mark next to a rule will display the runtime prompt parameters. Set the runtime prompt values with which to run the rule in the job scheduler, and then click **OK**. • Hidden runtime prompts will pick up the default values that were set at design time in Calculation Manager. See About Rules.
Import Data*	Performs a data import that was saved as a job. See Importing Data.
Import Metadata*	Performs a metadata import that was saved as a job. See Importing Metadata.
Import Cell-Level Security Definition	Imports a cell-level security definition that was saved as a job. See Importing and Exporting Cell-Level Security Definitions.
Import Valid Intersections	Imports valid intersections that were saved as a job. See Importing and Exporting Intersections.
Export Data*	Performs a data export that was saved as a job. You can specify a unique output file name for each **Export Data** job. From the job list, click (Edit) next to the selected job and specify the **Output File Name** using a zip extension. You can verify the export file name on the **Review** page. The unique file name will override the default export file name when the job runs. See Exporting Data. **Note:** The driver/column dimension in the Export Data job definition needs to be dense.
Export Metadata*	Performs a metadata export that was saved as a job. You can specify a unique output file name for each **Export Metadata** job. From the job list, click (Edit) next to the selected job and specify the **Output File Name** using a zip extension. You can verify the export file name on the **Review** page. The unique file name will override the default export file name when the job runs. See Exporting Metadata.
Export Cell-Level Security Definitions	Exports a cell-level security definition that was saved as a job. See Importing and Exporting Cell-Level Security Definitions.
Export Valid Intersections	Exports valid intersections that were saved as a job. See Importing and Exporting Intersections.
Refresh Database*	Refreshes the application database. See Refreshing Application Databases.

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Table 17-1 (Cont.) Job Types

Job Type	Description
Data Map	Performs a data mapping operation. See Defining Data Maps in *Administering Data Integration*.
Invalid Intersection Reports	Runs a report that shows where data exists at invalid intersections. See Working With Invalid Intersection Reports.
Clear Cube*	Performs a cube clearing operation. See Creating Clear Cube Jobs.
Restructure Cube*	Performs a full restructure of a block storage cube to eliminate or reduce fragmentation. This will also remove empty blocks. Running this job won't push any changes from the business process to Essbase. See Improving Cube Performance.
Compact Outline*	Compacts the outline file of an aggregate storage cube. Compaction helps keep the outline file at an optimal size. Compacting the outline doesn't clear the data. Running this job won't push any changes from the business process to Essbase. See Improving Cube Performance.
Merge Data Slices*	Merges incremental data slices of an aggregate storage cube. Fewer slices improve a cube's performance. You can merge all incremental data slices into the main database slice or merge all incremental data slices into a single data slice without changing the main database slice. You can optionally remove cells that have a value of zero. See Improving Cube Performance.
Optimize Aggregation*	Generates optimized views based on collected query tracking information in an aggregate storage cube. For additional option descriptions, see Improving Cube Performance.
Administration Mode	Changes the login level for a business process. If you select **Administrators**, all non-administrative users will be logged off from the application after job execution. To restore access to an application for all users, select **All users**. See What Application and System Settings Can I Specify?
Execute Bursting Definition	Runs a report bursting job If you're using the next-generation Reports reporting solution, the bursting feature enables you to run a single report or book for more than one member of a single dimension for one data source, and publish a PDF output for each member. See "Working with Bursting" in *Oracle Fusion Cloud EPM Designing with Reports*.
Integration	Runs a data rule in Data Management or integration in Data Integration) based on the parameters defined in the data rule or integration definition. For information about running a data rule or integration using the Job Scheduler, see: Scheduling a Data Rule, Integration, or Batch Definition to Run in the Job Scheduler in *Oracle Fusion Cloud EPM Administering Data Integration*

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Table 17-1 (Cont.) Job Types

Job Type	Description
Integration Pipeline*	Runs a Pipeline definition. This job supports running a Pipeline based on the parameters and variables that were defined for the Pipeline in the Data Integration user interface; for example, **Start Period** and **Import Mode**. For descriptions of the job parameters and variables, see the following topics: • Using an Integration Job Type in *Oracle Fusion Cloud EPM Administering Data Integration* • Running a Pipeline in *Oracle Fusion Cloud EPM REST APIs* For information about creating a Pipeline definition, see Using the Pipeline in *Oracle Fusion Cloud EPM Administering Data Integration*.
Integration Batch	Runs a batch definition based on parameters defined in Data Management. For information about running a batch definition using the Job Scheduler, see: Scheduling a Data Rule, Integration, or Batch Definition to Run in the Job Scheduler in *Oracle Fusion Cloud EPM Administering Data Integration*
Integration Maintenance Script	Runs a Maintenance Script based on parameters and variables in Data Management For information about running Maintenance Scripts, see: Scheduling Maintenance Scripts in the Job Scheduler in *Oracle Fusion Cloud EPM Administering Data Integration*.

*This job is prevented from starting during daily maintenance, whether it's a scheduled job or a job that is started on an ad hoc (unscheduled) basis. Oracle recommends that you start this job outside of the daily maintenance window. See Setting the Daily Maintenance Process Start Time.

Scheduling Jobs to Run Later

Table 17-2 Scheduling Jobs Options

Option	Description
Schedule starting from	Select the starting date and time, including the time zone.
Name	Specify a name for the scheduled job; for example, **MyDailyCubeRefresh**. The name you specify is displayed along with the job name (which you'll choose on the next screen); for example, **MyDailyCubeRefresh : Refresh Database**.

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Table 17-2 (Cont.) Scheduling Jobs Options

Option	Description
Recurrence Pattern	Specify the frequency with which to run the job: • **Hourly**: Hourly jobs run according to a timetable which is based on the values you set in the **Schedule starting from** and **Hour** fields. The schedule for **Hourly** jobs restarts each day during the midnight hour and the recurring job starts at the first hour, second hour, third hour (and so on up to 12 hours) after the midnight hour, depending on the value you select in the **Hour** field. So for example, if you specify a value of **5** in the **Hour** field, the possible times the job could start are during the 12:00 AM hour, the 5:00 AM hour, the 10:00 AM hour, the 3:00 PM hour, and the 8:00 PM hour each day. If the scheduled start time for a job is 12:48 PM and you set the **Hourly** recurrence for **5 Hours**, the job will start at 3:48 PM, which is the first scheduled time available after 12:48 PM in which to start a 5-hour recurring job. Then the job will run again at 8:48 PM, 12:48 AM, 5:48 AM, and 10:48 AM. For the default recurrence of **1 Hour**, the job will start running at the start time you specify and continue to run each hour until the ending date and time; for example, if the scheduled start time for a job is 12:48, the job will run at 12:48, 1:48, 2:48, 3:48 and so on. For a schedule of times based on the values selected in the **Hour** field, see Scheduling Hourly Jobs. • **By Minute**: Set the **Frequency** for 15 or 30 minutes. The job will start running in 15 or 30 minutes and continue to run at the selected frequency until the ending date and time; for example, if you set the job to run every 15 minutes and the starting time is 3:15, the job will start to run at 3:30, then 3:45, and so on. Jobs cannot be scheduled to run in increments smaller than 15 minutes. • **Run Once**: The job will run once at the starting date and time. • **Yearly**: The job will run at the starting date and time and continue to run each year thereafter until the end date. • **Monthly**: The job will run at the starting date and time and continue to run each month thereafter until the end date. • **Weekly**: The job will run at the starting date and time and continue to run each week until the end date. • **Daily**: The job will run at the starting date and time and continue to run each day until the end date.
End Date	If the job is recurring, select an ending date and time.

Scheduling Hourly Jobs

Hourly jobs run according to a timetable which is based on the values you set in the **Schedule starting from** and **Hour** fields. The schedule for **Hourly** jobs restarts each day during the midnight hour and the recurring job starts at the first hour, second hour, third hour (and so on up to 12 hours) after the midnight hour, depending on the value you select in the **Hour** field.

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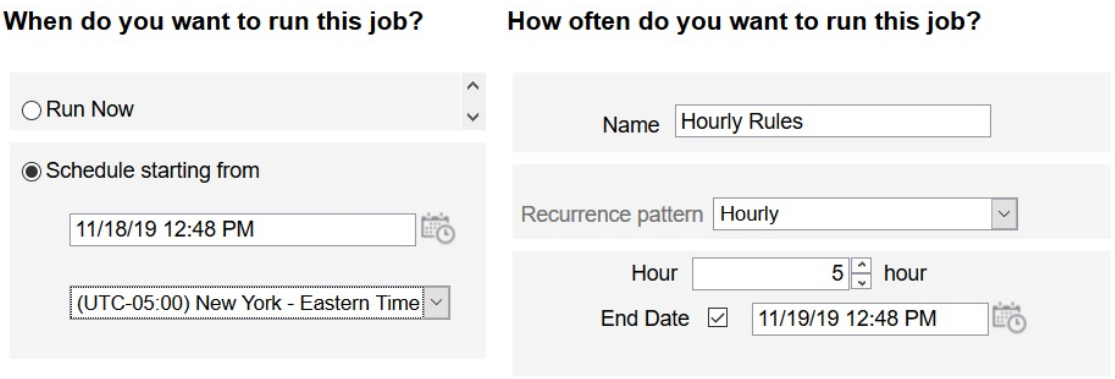

For example, if you specify a value of **5** in the **Hour** field, the hours during which the job will run each day are 12 AM, 5 AM, 10 AM, 3 PM, and 8 PM. The start time indicates the hour and minute when the system should start honoring the schedule, and the end time indicates when the system should stop. So if the start time is 12:48 PM on 11/18, then that means 12:48 AM, 5:48 AM, and 10:48 AM have already occurred on the current day (11/18) and the next available timeslot is 3:48 PM, which is when the scheduled job will run for the first time. After that it will run at 8:48 PM on 11/18. However on 11/19 it starts over again and will run at 12:48 AM, 5:48 AM, 10:48 AM, and so on until the system reaches the specified end time (in this case, 12:48 PM on 11/19).

> **Note**
>
> When a job spans two days, the job runs during the midnight hour the next day and resets the recurrence pattern.

To schedule hourly jobs, see Scheduling Jobs.

Table 17-3 Timetable for Hourly Jobs

Hourly Recurrence	Timetable	Examples
1 (default)	AM: 12:00, 1:00, 2:00, 3:00, 4:00, 5:00, 6:00, 7:00, 8:00, 9:00, 10:00, 11:00 PM: 12:00, 1:00, 2:00, 3:00, 4:00, 5:00, 6:00, 7:00, 8:00, 9:00, 10:00, 11:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 1 hour, the job will run for the first time at 12:48 PM, and then again at 1:48 PM, 2:48 PM, 3:48 PM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 1 hour, the job will run for the first time at 7:48 AM, and then again at 8:48 AM, 9:48 AM, 10:48 AM, and so on.
2	AM: 12:00, 2:00, 4:00, 6:00, 8:00, 10:00 PM: 12:00, 2:00, 4:00, 6:00, 8:00, 10:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 2 hours, the job will run for the first time at 12:48 PM, and then again at 2:48 PM, 4:48 PM, 6:48 PM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 2 hours, the job will run for the first time at 8:48 AM, and then again at 10:48 AM, 12:48 PM, 2:48 PM, and so on.

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Table 17-3 (Cont.) Timetable for Hourly Jobs

Hourly Recurrence	Timetable	Examples
3	AM: 12:00, 3:00, 6:00, 9:00 PM: 12:00, 3:00, 6:00, 9:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 3 hours, the job will run for the first time at 12:48 PM, and then again at 3:48 PM, 6:48 PM, 9:48 PM, 12:48 AM, 3:48 AM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 3 hours, the job will run for the first time at 9:48 AM, and then again at 12:48 PM, 3:48 PM, 6:48 PM, 9:48 PM, 12:48 AM, 3:48 AM, and so on.
4	AM: 12:00, 4:00, 8:00 PM: 12:00, 4:00, 8:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 4 hours, the job will run for the first time at 12:48 PM, and then again at 4:48 PM, 8:48 PM, 12:48 AM, 4:48 AM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 4 hours, the job will run for the first time at 8:48 AM, and then again at 12:48 PM, 4:48 PM, 8:48 PM, 12:48 AM, and so on.
5	AM: 12:00, 5:00, 10:00 PM: 3:00, 8:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 5 hours, the job will run for the first time at 3:48 PM, and then again at 8:48 PM, 12:48 AM, 5:48 AM, 10:48 AM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 5 hours, the job will run for the first time at 10:48 AM, and then again at 3:48 PM, 8:48 PM, 12:48 AM, 5:48 AM, and so on.
6	AM: 12:00, 6:00 PM: 12:00, 6:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 6 hours, the job will run for the first time at 12:48 PM, and then again at 6:48 PM, 12:48 AM, 6:48 AM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 6 hours, the job will run for the first time at 12:48 PM, and then again at 6:48 PM, 12:48 AM, 6:48 AM, and so on.
7	AM: 12:00, 7:00 PM: 2:00, 9:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 7 hours, the job will run for the first time at 2:48 PM, and then again at 9:48 PM, 12:48 AM, 7:48 AM, and so on. If the scheduled start time is 8:48 AM and you set the hourly recurrence for 7 hours, the job will run for the first time at 2:48 PM, and then again at 9:48 PM, 12:48 AM, 7:48 AM, and so on.

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Table 17-3 (Cont.) Timetable for Hourly Jobs

Hourly Recurrence	Timetable	Examples
8	AM: 12:00, 8:00 PM: 4:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 8 hours, the job will run for the first time at 4:48 PM, and then again at 12:48 AM, 8:48 AM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 8 hours, the job will run for the first time at 8:48 AM, and then again at 4:48 PM, 12:48 AM, and so on.
9	AM: 12:00, 9:00 PM: 6:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 9 hours, the job will run for the first time at 6:48 PM, and then again at 12:48 AM, 9:48 AM, 6:48 PM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 9 hours, the job will run for the first time at 9:48 AM, and then again at 6:48 PM, 12:48 AM, and so on.
10	AM: 12:00, 10:00 PM: 8:00	For example, if the scheduled start time is 12:48 PM and you set the hourly recurrence for 10 hours, the job will run for the first time at 8:48 PM, and then again at 12:48 AM, 10:48 AM, 8:48 PM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 10 hours, the job will run for the first time at 10:48 AM, and then again at 8:48 PM, 12:48 AM, and so on.
11	AM: 12:00, 11:00 PM: 10:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 11 hours, the job will run for the first time at 10:48 PM, and then again at 12:48 AM, 11:48 AM, 10:48 PM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 11 hours, the job will run for the first time at 11:48 AM, and then again at 10:48 PM, 12:48 AM, and so on.
12	AM: 12:00 PM: 12:00	If the scheduled start time is 12:48 PM and you set the hourly recurrence for 12 hours, the job will run for the first time at 12:48 PM, and then again at 12:48 AM, and so on. If the scheduled start time is 7:48 AM and you set the hourly recurrence for 12 hours, the job will run for the first time at 12:48 PM, and then again at 12:48 AM, and so on.

Editing and Canceling Jobs

You can edit the schedule for pending jobs, and delete pending and complete jobs. You can't modify or delete jobs that are processing.

To edit or delete jobs:

1. Click **Application**, and then click **Jobs**.

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2. To change when a job runs, click ••• (Actions), then **Edit** and modify the schedule.

   If you do not change the date and time of the edited schedule to a future time, the job will honor the currently set date and time (and will not consider the current time or system time). If this is in the past, the job will run once immediately and continue to run at the existing frequency.

   > **Note**
   >
   > When changing the frequency of a job, the job will not run at the new frequency until the next time it runs at the previous frequency; for example, if the job is set up to recur daily and you change the frequency to hourly, the job won't start running at the new hourly frequency until after the scheduled daily job runs the next day. If you want the job to begin running at the new frequency sooner, Oracle recommends that you delete the scheduled job and create a new one.

3. To delete jobs, select them, click ••• (Actions), and then **Delete**.

Duplicating Jobs

Use the **Save As** option to create a duplicate of an existing job, and then update it to avoid starting over making selections each time you create a new job.

Save As is supported for the following job types:

- Export Data
- Import Data
- Export Metadata
- Import Metadata
- Refresh Database
- Clear Cube
- Import Exchange Rates

> **Note**
>
> For descriptions of the job types, see Job Types.

To duplicate a job:

1. Click **Application**, and then click **Overview**.
2. Click **Actions**, and then select one of the following actions:
   - **Export Data**
   - **Import Data**
   - **Export Metadata**
   - **Import Metadata**
   - **Refresh Database**

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- **Clear Cube**
- **Import Exchange Rates**

3. On the listing page, click ••• (Actions) in the **Actions** column next to the job that you want to duplicate, and then select **Save As**.
4. Enter a name for the new job, and then click **OK**.

After the duplicate job is created, you can then open the job and update it. The error file for the duplicate job will automatically use the new job name as a prefix.

Canceling Rules Jobs and Ruleset Jobs

You can cancel ruleset or rule jobs that are processing and display in **Recent Activity**. To cancel these jobs, click **Application**, then click **Jobs**.

To cancel a rules job, click (Rule) beside the job, then (Actions), and then **Cancel**.

To cancel a ruleset, click (Ruleset), then (Actions) on **Job Details**, and then **Cancel**.

Downloading Export Files From Your Outbox

After running a metadata or data export job, you can download the export file from the Outbox.

To download files:

1. Click **Application**, and then click **Jobs**.
2. Under **Recent Activity**, click the job.
3. At the top of **Job Details**, click the export file option to select a download location.

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18

Auditing Tasks and Data

Understand how to audit tasks performed by users and how to view audit details.

Related Topics

- Auditing Overview
- Enabling Audit Tracking
- Viewing Audit Details

Auditing Overview

Use the Audit feature to view tasks performed by users. You can filter audited tasks by audit type (for example, Data or Clear Cell Details), date range (for example, Yesterday or Last 60 Days), and user name.

You must be a Service Administrator to enable audit tracking, and to view and export audit information.

These are the types of user activities the system can log in the task audit:

Table 18-1 Tasks That Can Be Audited

Audit Types	Tracked Changes
Dimension Administration	• Dimension hierarchy: Adding a member or dimension; moving, deleting, and changing properties; renaming a member and dimension • Performance settings: Resetting a dimension's dense or sparse setting, changing the order of dimensions
Alias Table Administration	Changes to alias tables: Creating, copying, renaming, deleting, and clearing
Data	• Cell values • Supporting detail • Account annotations • Cell-level documents **Note:** The audit log only captures modifications if the data change occurs within a data form. If a modification occurs as part of, for example, a business rule calculation, a data map, or a direct data load to Oracle Essbase, it will not be captured in the audit log.
Launch Business Rules	Updates from calculation scripts and business rules (including runtime prompts)
Form Definition	Forms: Creating, moving, deleting forms (The audit record doesn't record how the design changed.)
Form Folder Administration	Folders: Created, moved, and deleted
Security	Access permissions to dimension members, forms, form folders, business rules, and task lists

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Table 18-1 (Cont.) Tasks That Can Be Audited

Audit Types	Tracked Changes
Users Administration	Users added, changed, and deleted **Note:** Information on Users Administration is not included in audit reports. You can get this information using EPM Automate and Oracle Fusion Cloud Enterprise Performance Management REST APIs. See the following topics: • roleAssignmentAuditReport in *Working with EPM Automate* • Role Assignment Report in *Oracle Fusion Cloud EPM REST APIs* • Role Assignment Audit Report for OCI in *Oracle Fusion Cloud EPM REST APIs*
Groups Administration	Groups added, changed, and deleted; users added and removed **Note:** Information on Groups Administration is not included in audit reports. You can get this information using EPM Automate and Cloud EPM REST APIs. See the following topics: • groupAssignmentAuditReport in *Working with EPM Automate* • Group Assignment Audit Report in *Oracle Fusion Cloud EPM REST APIs*
Tasklist	Task lists: Created, updated, saved, moved, and deleted
Copy Data	Users' selections for Static Dimensions, Source Dimension, and Destination Dimension, including supporting detail, cell text, cell attachments, and data without any details
Clear Cell Details	Users' selections for clearing supporting details, comments, and attachments
Variables	Variables (substitution variables and user variables): Added, changed, and deleted
Application Administration	• Changes to the login level for an application, including: – Running the **Administration Mode** job – Changing the system setting for **Enable Use of the Application for** (both when it's changed manually and when it's changed automatically during a database refresh) – Running the EPM Automate command `setApplicationAdminMode` – Using REST API to execute the **Administration Mode** job • Changes to the hybrid mode (**Enable Hybrid Mode** or **Disable Hybrid Mode**) for Planning Custom applications and FreeForm.
Report Artifacts	For Report, Shared Reports, Snapshot, Book, and Bursting Definition types: • Create: Name, Parent Name • Modify: Name, Parent, Content • Delete: Name
Documents	Documents added, modified, and deleted

The Audit page displays the following information:

- Audit (task)
- Source
- Action
- User
- Name

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- Date
- Details
- Property
- Old Value
- New Value

All columns can be sorted in ascending or descending order.

Enabling Audit Tracking

Audit tracking isn't enabled by default. Service Administrators must enable auditing so data changes can be tracked.

To enable auditing:

1. On the Home page, click **Tools**, and then click **Audit**.
2. From the **Audit** page, click **Enable Auditing**.
3. From the **Enable Auditing** page, select the audit types, and then click **Save and Close**.

   For a list of audit types and descriptions, see Auditing Overview.

Viewing Audit Details

You can view and export up to 180 days of audit details in the business process interface.

To view audit details:

1. On the Home page, click **Tools**, and then click **Audit**.

   If auditing is enabled, by default the **Audit** page displays the audit records for **Data** tasks for the **Last 7 Days**. To enable auditing, see Enabling Audit Tracking.

2. To filter, click (Filter), and select filter criteria:
   - **Audit Types**: Select one or more, or **All**.

     For a list of audit types and descriptions, see Auditing Overview.
   - **Date Range**: Select **Today**, **Yesterday**, **Last 7 Days**, **Last 30 Days**, **Last 60 Days**, or **Last 180 Days**.

     > **Note**
     >
     > – You can't select a date range prior to 180 days from today. The date range you select must occur between today and 180 days prior to today (including today).
     >
     > – The business process retains up to 365 days of audit details from the current system date. To export more than 180 days of audit details, you can use the EPM Automate `exportAppAudit` command or the REST API Export Audit job. See *Working with EPM Automate* or *REST APIs*.

   - **User Name**: Enter a user name or click (Search) to search for the user.

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3. When you're done selecting filter criteria, click **Apply**.

   The grid displays the first 200 records from the audit table that match the filter criteria. You can view a legend at the bottom of the table if the filter criteria has more than 200 records in it.

4. To export the audit information to a Microsoft Excel spreadsheet, click **Export**, and follow the download instructions.

   When you select the **Export** option, the system exports all of the records matching the filter criteria to an `xlsx` file (format for Microsoft Excel versions 2007 and later).

> **Note**
>
> If data auditing is enabled, users can see what data has changed by selecting **Actions**, and then **Change History**.

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19

Managing Data Validation and Format Rules in Forms

Learn how to build data validation and format rules that are checked when conditions are met in forms.

Related Links

- About Data Validation Rules
- About Format Rules in Forms

About Data Validation Rules

To implement business policies and practices, you can build data validation rules that are checked when conditions are met in forms. Validation messages can be generated if entered data violates validation rules.

Sample scenarios that can be addressed using data validation rules are described in Data Validation Rule Scenarios.

Defining data validation rules involves these main tasks:

- Identifying the data cells or location that you want to display with validation messages or in different colors when conditions are met.
- Identifying the cells that need to participate during rule evaluation, and defining the rule accordingly.
- Creating the data validation rule at the location identified, as described in the topics linked below.

Related Links

- Creating and Updating Data Validation Rules in Forms
- Formatting Cells
- Viewing Data Validation Rules
- Order of Evaluation and Execution for Data Validation Rules
- Conditions Supported by the Rule Builder
- Data Validation Conditional Operators
- Data Validation Rule Scenarios

Creating and Updating Data Validation Rules in Forms

Data validation rules are defined by Administrators on the **Layout** tab of the Form Designer. You can add validation rules at the grid, row heading, column heading, or cell level. Areas on the **Layout** tab with a green background, indicate locations where rules have been defined.

To create and update validation rules:

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1. On the Home page, click **Navigator**, and then under **Create and Manage**, click **Forms**.
2. Create or edit a form, and then on the **Form and Ad Hoc Grid Management** page, click **Layout**.
3. In **Layout**, right-click the row, column, or cell, for which you want to add or update the validation rule. Depending on the location and whether rules were already defined at that location, you'll see some or all of the following context menu options:
   - **Add/Edit Format/Validation Rules**: Create or modify rules for individual data cells or all data cells in a grid, row, or column in the **Format/Validation Rule Builder**.
   - **Copy | Paste Format/Validation Rules**: Copy and paste the rules at the selected locations.
   - **Add/Edit Format/Validation Rules for Metadata**: Create or modify format rules for the row metadata cells or column metadata cells in the **Format/Validation Rule Builder**.
   - **Copy | Paste Format/Validation Rules for Metadata**: Copy and paste the rules for row metadata cells or column metadata cells at the selected locations.
4. Select **Add/Edit Format/Validation Rules** to create or update rules.

> **Note**
>
> To add rules for the entire grid, expand the **Format/Validation Rules** pane on the right. For **Rules for location**, select **Grid**, and then select (Add Validation Rule).

5. In the **Format/Validation Rule Builder**, create a new rule or update an existing rule.
   - If necessary, move the rule by selecting an option from **Location**. To create a rule similar to an existing rule, click (Duplicate) and then update the rule. To view rules in the current location, click **View Rule**. See Viewing Data Validation Rules.
   - **Enable format/validation rule**: Select to make the format/validation rule available in forms. You can temporarily disable a format/validation rule by clearing this checkbox.
   - **Format Only**: Select to mark a rule as format only. This option is automatically enabled when a format-only rule is created. See Creating and Updating Format Rules in Forms.
6. Update the rule.
   a. For **Condition**, select an option to begin the condition statement: **If**, **Else If**, **Else**, **Then**, **Range**, **Check Range**, or **Format**.

> **Note**
>
> For the **Format** condition, the **Source Type** field will automatically display the **Process Cell** option. There are no other **Source Type** options for the **Format** condition. See About Format Rules in Forms.

For validation rules, the first part of a validation rule must include an **If** condition. Validation rules must also contain some form of **Then** condition. See Conditions Supported by the Rule Builder.

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b. For **Source Type**, select an option for evaluation by the rule.

   The **Source Type** list displays the appropriate options for the Condition. For example, **If** conditions can include the following **Source Type** options: **Current Cell Value**, **Cell Value**, **Column Value**, **Row Value**, **Member Name**, **Member**, **Cross-Dim Member**, **Account Type**, **Version Type**, **Var Reporting Type**, **UDA**, **Attribute**, **Generation**, **Has Children**, or **Dimension Name**. For detailed information about each type of condition, see Conditions Supported by the Rule Builder.

c. If applicable for the selected **Source Type**, enter a value in **Source Value** by selecting an option or entering a free form value.

d. Select an operator for the evaluation: **=**, **!=**, **<**, **<=**, **>**, **>=**, **Equals**, **Not Equals**, **Contains**, **Starts With**, or **Ends With, In, or Not In**.

   For examples, see Data Validation Conditional Operators.

e. Select an option for the appropriate **Target Type** for the rule.

f. Update conditions by clicking an icon in the **Actions** column to the right of the condition builder:

   - **Add Condition** : Add a condition next to the current row.
   - **Delete Condition** : Delete a condition at the current row.

g. Select conditions or condition blocks to update.

   To update condition blocks, click an icon in the **Condition** area, at the top of the condition builder:

   - **Add Condition Block**: Add a condition block within the validation rule starting with If. You can expand or collapse the condition. See Conditions Supported by the Rule Builder.
   - **Delete Condition Block**: Delete the selected condition block.
   - **Cut Conditions**: Delete selected conditions, and copy them to paste in a new location.
   - **Copy Conditions**: Copy selected conditions.
   - **Paste Conditions**: Paste selected conditions to a new location.
   - **Group Conditions**: Group the selection within a condition, and add a grouping parenthesis. In addition to groupings that you set, the If statement in the grouping block is grouped when a condition is grouped, and the **Custom Grouping** option is enabled.
   - **Ungroup Conditions**: Ungroup the selected conditions. The grouping parentheses are removed from the selected condition. One grouping is removed from the condition each time Ungroup is selected.

     You can also set your own grouping for conditions by selecting **Custom Grouping**, then setting up the grouping in the rule definition area.

     Selected conditions are displayed as shaded. To clear selected conditions, click once more to the left of the **Condition** column.

7. Click (Process Cell) in the rightmost column to add processing instructions.

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> **Note**
>
> If the icon isn't displayed, ensure that the rule is valid and that it permits cell processing instructions. For example, cell processing instructions are included for Else, Range, Then, and Format conditions. The icon isn't displayed until all required columns are selected for a rule.

See Formatting Cells.

8. As a reminder, when you're ready to enable the rule to make it available in the form, select **Enable format/validation rule**.

> **Tip**
>
> While you're building a rule, you can save the rule without enabling it. After any errors are resolved and the rule is ready to be validated and used, you can enable and save the rule to make it available in the form. You can temporarily disable a rule by clearing **Enable format/validation rule**.

9. When you finish updating the rule, validate the rule:
   a. Click **Validate**.

      The validation status displays at the top of the dialog box. You must correct errors before you save changes. If you close the dialog box without validating rules and fixing errors noted during validation, updates are not saved.
   b. After fixing any errors noted during validation, ensure that **Enable format/validation rule** is selected above the rule definition area to enable the rule for the application.
   c. After the rule is validated, click **OK**.
10. **Optional**: In the **Form and Ad Hoc Grid Management** page, view and update rules:
    - In the **Format/Validation Rules** pane on the right side of the **Form and Ad Hoc Grid Management** page, add, edit, or delete rules at any level, including the grid level, by clicking (Add Validation Rule), (Edit Validation Rule), or (Delete Selected Validation Rule).
    - If multiple rules are defined at the same location, you can change the order in which rules are processed when rules have the same precedence. To move a rule up, down, or to the top or bottom of the list, select the rule and click the arrows. See Order of Evaluation and Execution for Data Validation Rules.
    - Select form validation rules options:

Table 19-1 Form Validation Rules Options

Option	Description
Validate only for pages with existing blocks	When enabled, the system figures out which page combinations have potential blocks and runs the validations only for those page combinations. There are a few exceptions to this. If a page combination has any Dynamic Calc, Dynamic Calc and Store, Label only, or Store with one child member, then that page is always loaded.

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Table 19-1 (Cont.) Form Validation Rules Options

Option	Description
Validate only for cells and pages the user has access to	When enabled, validations are run as the currently logged-in user and not as the Service Administrator, which means the user's security will be applied to the form members.

11. In the **Form and Ad Hoc Grid Management** page, preview and validate the form, resolve any validation errors, and then save changes.
    - For forms that have data validation rules enabled, rules are validated when the form is loaded or saved. Data validation rules are saved when the form is saved.

      See Creating Forms.
    - When users open the form, they can view and resolve validation messages using the **Format/Validation Messages** pane.

      See Resolving Data Validation Errors in *Working with FreeForm*.

Formatting Cells

After a rule is set up, use the Process Cell dialog box to set how cells display in forms.

To format cells:

1. In the **Format/Data Validation Rule Builder** dialog box, click (Process Cell) in the right-most column.

   > **Note**
   >
   > If the Process Cell icon isn't displayed, ensure that the rule is valid and that it permits cell processing instructions. For example, cell processing instructions are included for Else, Range, Then, and Format conditions. The icon isn't displayed until all required columns are selected for a rule. See Creating and Updating Data Validation Rules in Forms.

2. In the Process Cell dialog box, set how the cell should appear in forms if the conditions defined by this rule are fulfilled.

   When the rule is enabled, it doesn't validate unless you specify at least one of the formatting options or a validation message.
   - To add formatting options, click options in **Format Cell** for **Font and Alignment** and **Border and Shading**. As format options are selected, the **Current Format** displays how it would appear when displayed on the form.
   - To display a validation message for the cell, enter the message in the **Message** field. Users see this text when they select **Show Data Validation Messages** in the context menu that appears when you hover over the cell in the form. It also appears as a link in the **Data Validation Messages** pane if data cells are flagged by the validation rules and the **Show in Format/Validation Messages pane** check box is selected. For information on viewing and resolving data validation errors, see *Working with FreeForm*.
3. Click **OK**.

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Updates for the rule are displayed in the **Process** column for the rule. If you specified formatting, that formatting is displayed. You can preview a validation message by hovering the cursor over the Process column.

Viewing Data Validation Rules

After data validation rules are defined on the **Layout** tab, you can view rules that apply to the selected grid, row, column, or cell, or you can opt to view all rules on the form. Areas on the **Layout** tab with a green background, indicate locations where rules have been defined.

To view data validation rules:

1. Open the form, and then click **Layout**.
2. Expand the **Format/Data Validation Rules** pane on the right, and then click the drop-down list next to **Rules for location** to view rules for a specific location (grid, row, column, cell) or select **All Rules** to view all rules in the form.
3. Double-click a rule name to view rule details.

Order of Evaluation and Execution for Data Validation Rules

For data validation rules in forms, the precedence for rule evaluation depends on condition priority, location of the rule, and position of the rule in the rule list (if multiple rules exist in the same location). First, cell-level rules are processed. Next, rules at the column level are processed, and then row-level rules are processed. Finally, rules at the grid level are processed. The rules are evaluated based on their position in the rule list within each level.

Location and position determine the order in which the rule will be processed. However, the priority of the processing instructions determines which rule is applied to the data cell. So, if a cell-level rule contains processing instructions with priority 4 and a grid-level rule contains processing instructions with priority 5, the grid-level rule is applied to the data cell. If all rules have processing instructions with the same priority, the first processed rule wins. The priority is based on whether the cell processing instructions specify a validation message, a color, the **Do Not Promote** promotional path option, or a combination of these settings.

Table 19-2 Priority for Rules in Forms

Default Condition Priority	Validation Message	Color	Do Not Promote
1 (lowest)	X		
1		X	
1	X	X	
2			X
3	X		X
4		X	X
5 (highest)	X	X	X

Conditions Supported by the Rule Builder

These conditions are supported by the format/data validation rule builder: If, Else, Else If, Then, Range, Check Range, and Format.

For details and examples of the values supported by these conditions, see these sections:

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- **If, Else, Else If**: If Condition Values
- **Then**: Then Condition Values
- **Range, Check Range**: Range Condition Values
- **Format**: Format Condition Values

If Condition Values

For information on other conditions, see Conditions Supported by the Rule Builder.

Current Cell Value

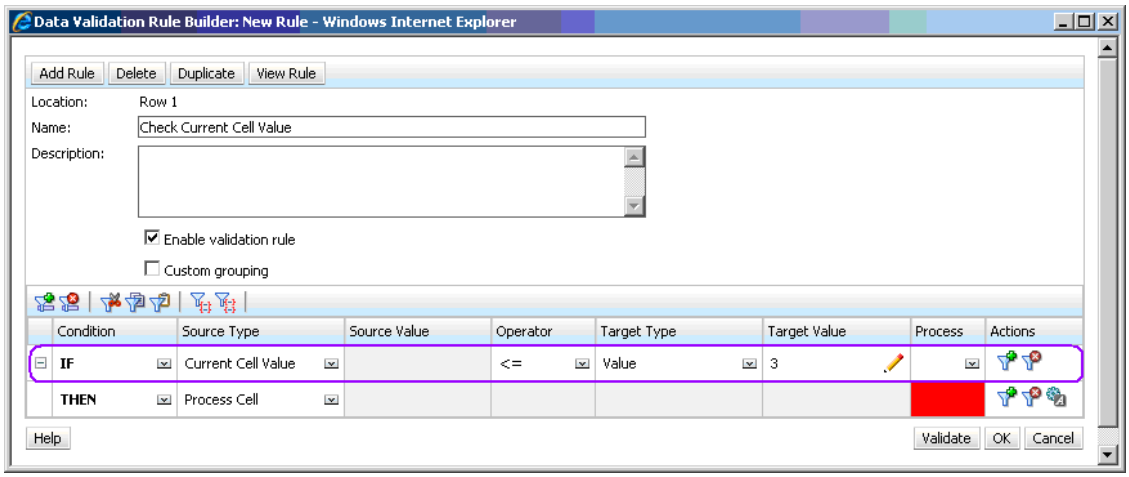
Action:

The action is performed when the value in the current data cell on which the rule is invoked satisfies this condition.

Operators:

The operators available for this function: =, !=, <, <=, >, >=, Equals, Not Equals, Contains, Starts With, or Ends With These operators act on the selected target value, which can be a free form value, cell value, column value, row value, or cross-dimension member.

Condition Definition:



Condition Evaluation:

With the condition shown in the previous figure, the cells in Row 1 with member Row_Member1 will turn red when the condition is evaluated.

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		A	
		Column_Member 1	Column_Member 2
1	Row_Member 1	1.0	2.0
	Row_Member 2	5.0	6.0
	Row_Member 3	9.0	10.0
2	Row_Member 4	13.0	14.0
	Row_Member 5	17.0	18.0

Cell Value

Action:

The action is performed when the value for the specified cell satisfies the condition.

Operators:

The operators available for this function: =, !=, <, <=, >, >=, Equals, Not Equals, Contains, Starts With, Ends With. These operators act on the target value selected, which can be a free form value, cell value, column value, row value, or cross-dimension member.

Condition Definition:

Condition Evaluation:

A design-time cell can expand to one or more data cells at data entry time, as shown in the following figure. The value for the cell is the sum of values in all the expanded data cells. For example, the value for cell A1 is the sum of the values in the cells outlined in purple (1+2+5+6+9+10=33), and the value for cell A2 is the sum of the values in the cells outlined in blue (13+14+17+18=62).

		A	
		Column_Member 1	Column_Member 2
1	Row_Member 1	1.0	2.0
	Row_Member 2	5.0	6.0
	Row_Member 3	9.0	10.0
2	Row_Member 4	13.0	14.0
	Row_Member 5	17.0	18.0

Column Value

Action:

The action is performed when the value for the specified column satisfies the condition.

Operators:

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The operators available for this function: =, !=, <, <=, >, >=, Equals, Not Equals, Contains, Starts With, or Ends With. These operators act on the target value selected, which can be a free form value, cell value, column value, row value or cross dim member.

Condition Definition:

Condition Evaluation:

A design-time column can expand to data cells at data entry, as shown in the following figure. The value for a column is the sum of the values in all the expanded data cells in that column at the current row location. The current row changes as the current cell for which the rule is being evaluated changes within the grid.

For example, the value for column A is the sum of the values in the cells outlined in purple (1+2=3), when the rule is evaluated for any cell in row 1 with member Row_Member1. The value for column A is the sum of values in cells outlined in blue (9+10=19) when the rule is evaluated for any cell in row 1 with member Row_Member3. Similarly, the value for column A is the sum of values in cells outlined in green (17+18=35) when the rule is evaluated for any cell in row 2 with member Row_Member5, and so on.

		A	
		Column_Member 1	Column_Member 2
1	Row_Member 1	1.0	2.0
	Row_Member 2	5.0	6.0
	Row_Member 3	9.0	10.0
2	Row_Member 4	13.0	14.0
	Row_Member 5	17.0	18.0

Row Value

Action:

The action is performed when the value for the specified row satisfies the condition.

Operators:

The operators available for this function: =, !=, <, <=, >, >=, Equals, Not Equals, Contains, Starts With, Ends With. These operators act on the target value selected, which can be a free form value, cell value, column value, row value, or cross-dimension member.

Condition Definition:

Condition Evaluation:

A design-time row can expand to one or more data cells at data entry time, as shown in the following figure. The value for a row is the sum of the values in all of the expanded data cells in

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that row at the current column location. The current column changes as the current cell for which the rule is being evaluated changes within the grid.

For example, the value for row 1 is the sum of the values in the cells outlined in purple (1+5+9=15), when the rule is evaluated for any cell in column A with member Column_Member1. Similarly, the value for row 2 is the sum of values in cells outlined in blue (14+18=32) when the rule is evaluated for any cell in column A with member Column_Member2, and so on.

		A	
		Column_Member 1	Column_Member 2
1	Row_Member 1	1.0	2.0
	Row_Member 2	5.0	6.0
	Row_Member 3	9.0	10.0
2	Row_Member 4	13.0	14.0
	Row_Member 5	17.0	18.0

Cross-Dim Member

Action:

The action is performed when the value in the data cell referenced by the cross-dimension member satisfies the condition. The current data cell members are used to fully qualify the cell for dimensions whose members are not specified in the cross-dimension. This member name is a free form entry.

> **Note**
>
> If a form validation rule uses the cross-dim operator, the cross-dimension cell must be represented on the form. The column or row containing the cross-dimension cell might be hidden on the form if the user does not want to see it there.

Operators:

The operators available for this function: =, !=, <, <=, >, >=, Equals, Not Equals, Contains, Starts With, or Ends With. These operators act on the target value selected. The target value can be a free form value, cell value, column value, row value or cross-dimensional member. It can include one member only from each dimension, and must include only members for dimensions on rows or columns.

Condition Definition:

Condition Evaluation:

When the previous rule is applied at grid level, the rule is invoked on each cell in the form, and the value in that cell is compared with the value in the cell outlined in purple. Thus, the cell at Row_Member 5->Column_Member 2 will turn red.

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		A	
		Column_Member 1	Column_Member 2
1	Row_Member 1	1.0	2.0
	Row_Member 2	5.0	6.0
	Row_Member 3	9.0	10.0
2	Row_Member 4	13.0	14.0
	Row_Member 5	17.0	18.0

Member Name

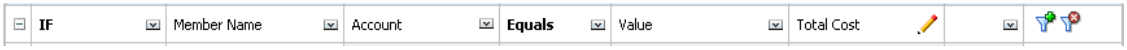
Action:

The action is performed if the current data cell on which the rule is invoked has the specified dimension member in its intersection. The member name for the selected dimension should be in the cube for which the form is created.

Operators:

The operators can be Equals, Not Equals, Contains, Starts With, or Ends With. The target value, which is the member name, is selected or entered free form.

Condition Definition:



Member

Action:

The action is performed if the current data cell on which the rule is invoked has the specified dimension member (or any one of the members that result from evaluating the specified function) in its intersection.

Operator:

The available operators are In and Not In. The target value, which is the member, is selected or entered free form.

Including Attributes

Rules can include attribute values. If Source Type is **Attribute**, the available operator is Is, and you can type an attribute value directly in the Target Value field. If Source Type is **Member**, and you select the In or Not In operator in the Target Value field, you can select an attribute by clicking (Rule Selection), and then clicking **Variables** in the Member Selection dialog box. You can use the member selection function selector to select functions for the attribute, such as NotEqual and GreaterOrEqual.

When using attribute values in data validation rules, keep in mind the way attributes are evaluated. If a rule references one or more attributes from one or more dimensions, they are evaluated as an OR for attribute values from the same attribute dimension, and as an AND for attributes from different attribute dimensions. For example, if the rule includes attributes IN

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Red, Blue, True, Big, then all members are selected that are either (Red OR Blue) AND True AND Big. For additional information, see Selecting Attribute Values as Members.

Condition Definition for Member Source Type

Condition Definition for Attribute Source Type

Account Type

Action:

The action is performed if the current data cell on which the rule is invoked has an account with the specified account type in its intersection. Refers to all the currently supported Account Types: Expense, Revenue, Asset, Liability, Equity, and Saved Assumption.

Operator:

The available operator is Is.

Condition Definition:

Version Type

Action:

The action is performed if the current cell on which the rule is invoked has a version with the specified version type in its intersection. It refers to the version types currently supported, standard bottom-up and standard top down.

Operator:

The available operator is Is.

Condition Definition:

Variance Reporting Type

Action:

The action is performed if the current cell on which the rule is invoked has an account with the specified variance reporting type in its intersection. Refers to the available variance reporting types, Expense and Non-Expense.

Operator:

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The available operator is Is.

Condition Definition:

UDA

Action:

The action is performed if the current cell on which the rule is invoked has this UDA associated with the specified dimension's member in its intersection. The UDA reference is selected based on the selected dimension. The condition is based on the UDA for this dimension being equal to the selected value. You must select the UDA value from the drop-down list.

Operator:

The available operator is Is.

Condition Definition:

Attribute

Action:

The action is performed if the current cell on which the rule is invoked has this attribute associated with the specified dimension's member in its intersection. The attribute reference is selected based on the selected dimension. The condition is based on the attribute for this dimension being the selected target value.

Operator:

The available operator is Is.

Condition Definition:

Generation

Action:

The action is performed if the current data cell or header (metadata) cell on which the rule is invoked has a member from the dimension specified in Source Value (Account) belonging to the generation specified in the Target Value (1) in the dimension hierarchy.

Operator:

The operators available for this function: =, !=, <, <=, >, >=

Condition Definition:

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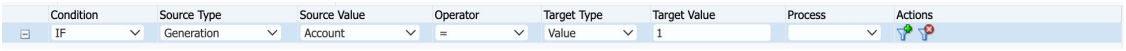


Has Children

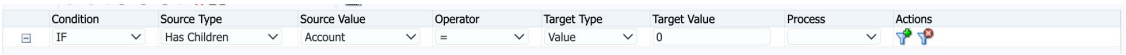
Action:

The action is performed if the current data cell or header (metadata) cell on which the rule is invoked has a level 0 member if the Target Value is 0 (false) or is a non-level 0 members if the Target Value is 1 (true) from the dimension specified in the Source Value (Account).

Operator:

The operators available for this function: =, !=

Condition Definition:



Dimension Name

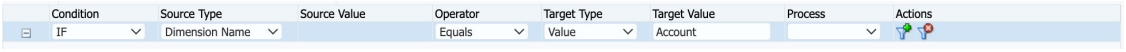
Action:

The action is performed if the current header (metadata) cell on which the rule is invoked belongs to the dimension specified in the Target Value.

Operator:

The operators can be Equals, Not Equals, Contains, Starts With, or Ends With. The target value, which is the dimension name, is selected or entered free form.

Condition Definition:



Then Condition Values

For information on other conditions, see Conditions Supported by the Rule Builder.

Then

Action:

The Then conditions supported by the data validation rule builder support Process Cell conditions only. To enter Process Cell Conditions, see Formatting Cells.

Condition Definition:

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Range Condition Values

The **Check Range** and **Range** conditions are used together. These conditions can be used in the data validation rule builder in a Then clause or standalone.

For information on other conditions, see Conditions Supported by the Rule Builder.

Check Range

Action:

Defines the value that needs to be in a specific range.

Value:

This value can be the Current Cell Value or the value in a particular Row, Column, or Cell.

Range

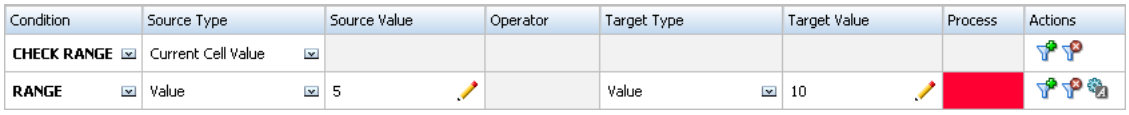
Action:

Defines a valid range for the value defined in the Check Range condition. This range includes all values that are >= the minimum value and < the maximum value. If the value specified in the Check Range condition is within this range, then the processing instructions defined by this condition are applied to the data cell on which the rule is being invoked. You can define multiple ranges of values and provide different processing instructions for each range.

Value:

The minimum and maximum values for the range can be defined using Cell Value, Current Cell Value, Row Value, Column Value, Cross-dimension Value, or by entering a free-form value. For example, the following rule ensures that the current cell value is >= 5 and < 10. If this condition is met, the cell is turned red.

Condition Definition:



Format Condition Values

For information on other conditions, see Conditions Supported by the Rule Builder.

Format

Action:

The Format conditions supported by the format/data validation rule builder support Process Cell conditions only. To enter Process Cell Conditions, see Formatting Cells.

Condition Definition:

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Data Validation Conditional Operators

Conditional operators in the data validation rule builder can include these types of comparisons:

- Numeric comparisons, using these operators: =, !=, <, <=, >, >=.
- String value comparisons, using these operators: Equals, Not Equals, Contains, Starts With, Ends With, In, and Not In.

Rules can compare cells with different data types; for example, text and Smart List. The data type of the cell is honored if the referenced value always comes from one cell. This is the case when using Current Cell Value and Cross Dim Member to refer to a cell value. In cases where the value being compared comes from multiple cells (such as row value, column value, and cell value), the data type is defaulted to double.

When comparing values for these data types:

- For double, a string representation is used for the double value, such as "123.45." If the double is a whole number with no fractional part, such as 123.00, the integer value is used, for example, "123."
- For text, rules use only the text value for comparison.
- All other data types (percentage and date) are treated as double.

Table 19-3 Examples of Results for Starts With, Ends With, and Contains

Operator	Compare Value	Compare To Value
Starts With	2.0	2
	1234.0	12.0
	101.0	10
	2.0	2.0
	2.5	"2."
	"YearTotal"	"Year"
Ends With	2.0	2.0
	2.0	2
	2.5	5
	2.5	".5"
	"YearTotal"	"al"
	"YearTotal"	"Total"
Contains	2.0	2.0
	2.0	2
	2.5	5
	2.5	".5"
	2.5	2.5
	23.567	3.5
	23.567	67
	23.567	"23."
	23.567	".56"
	"YearTotal"	"al"

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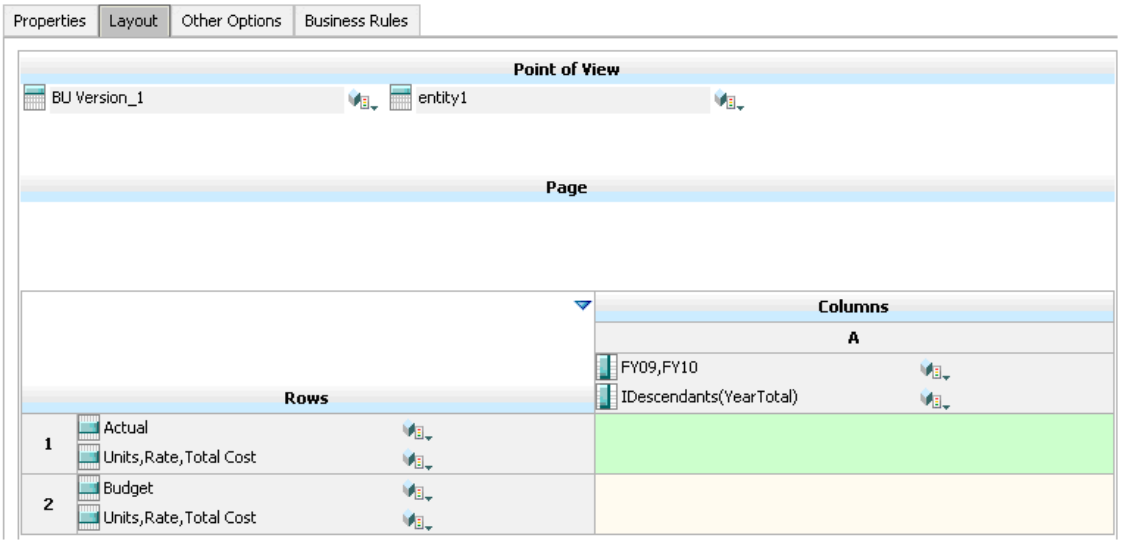
Data Validation Rule Scenarios

These scenarios provide examples of how data validation can help implement business policies.

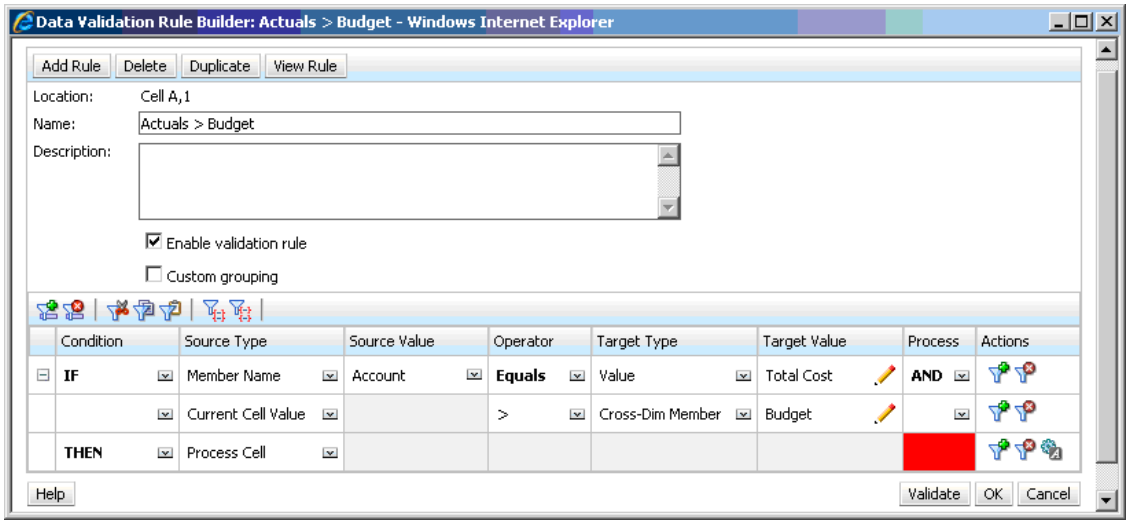
Scenario 1

John is hired by a company called Acme, Inc. as a consultant to design forms and implement data validation rules that enforce some of the company policies. He is asked to implement a validation rule that flags Actual amounts in red if the Total Cost in actuals exceeds the budgeted amount. This test must be repeated for each year and time period in the application. John designs the form and adds a data validation rule at cell level using a cross-dimension member, as shown in the following figures.

Form Layout at Design Time:



Data Validation Rule at Design Time:



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Form at Data Entry Time with Data Validations Applied:

Version: BU Version_1 | Entity: entity1

		FY09								FY10			
		Jan	Feb	Mar	⊟Q1	⊞Q2	⊞Q3	⊞Q4	⊟YearTotal	Jan	Feb	Mar	⊟Q1
Actual	Units	3	4	6	13	12	24	21	70	5	14	7	26
	Rate	5	5	5	15	15	15	9	54	4	4	4	12
	⊟Total Cost	15	20	30	195	180	360	189	3780	20	56	28	312
Budget	Units	3	4	6	13	12	24	21	70	5	13	7	25
	Rate	4	6	3	13	15	15	9	52	5	4	4	13
	⊟Total Cost	12	24	18	169	180	360	189	3640	25	52	28	325

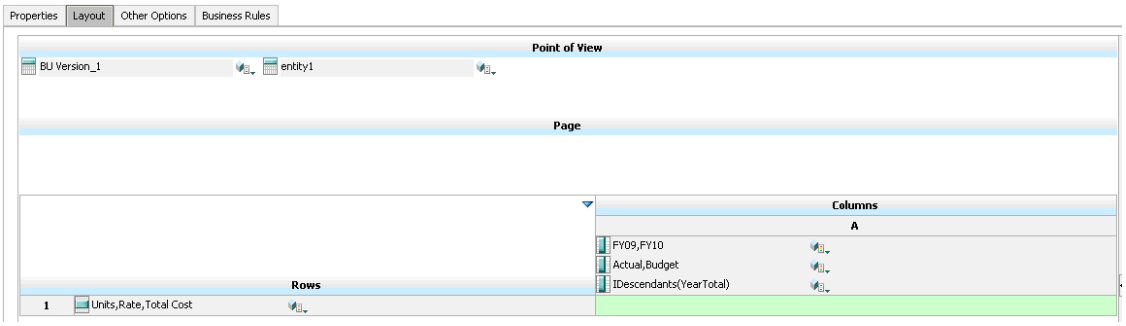
Tips:

- John can split Total Cost into its own segment and apply the data validation rule at that segment for a slight performance gain. However, doing so would increase maintenance as new accounts and scenarios were added to the form.
- If the requirements changed such that only the YearTotal Period in Actual had to be flagged in red, John would have two options. The best option is to add an IF entry to check if the Period member is YearTotal. Another option is to split the YearTotal member into a separate column for better performance. However, doing so would break the spreading logic, the column header for Year would repeat, and the form would be harder to maintain as new years were added.

Scenario 2

After reviewing the form designed by John in Scenario 1, Acme decides that they want Budget on the column instead of the row. To implement this requirement, John can move members within the axes to change the form layout. However, he doesn't need to update the data validation rules. John updates the form as shown in the following figure.

Form Layout at Design Time:



Form at Data Entry Time with Data Validations Applied:

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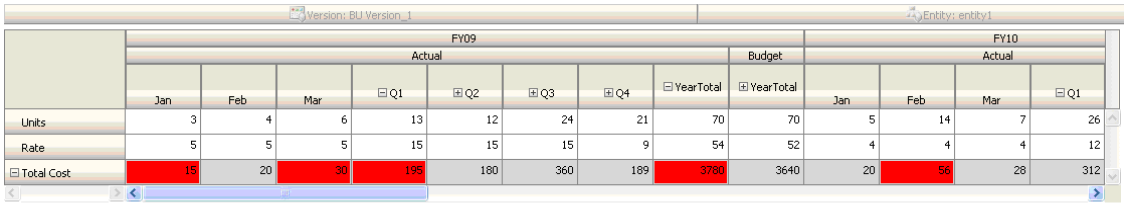


	FY09									FY10			
	Actual								Budget	Actual			
	Jan	Feb	Mar	Q1	Q2	Q3	Q4	YearTotal	YearTotal	Jan	Feb	Mar	Q1
Units	3	4	6	13	12	24	21	70	70	5	14	7	26
Rate	5	5	5	15	15	15	9	54	52	4	4	4	12
Total Cost	15	20	30	195	180	360	189	3780	3640	20	56	28	312

Scenario 3

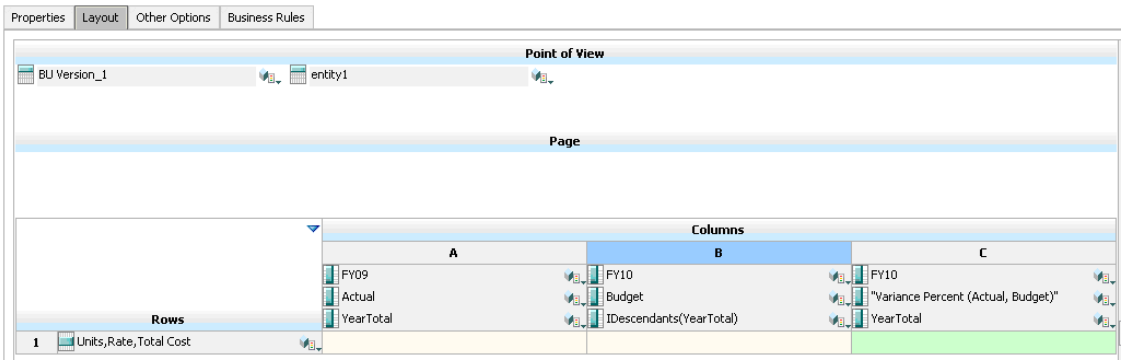
Following the successful rollout of these forms, John is asked to implement the next policy, which is to ensure that this year's Budget amounts are not significantly higher than previous year's Actual amounts. If the difference is greater than 5%, then flag the difference in red.

John decides to use a member with a member formula to calculate the variance between this year's Budget and the previous year's Actual amount. He adds this member formula:

```
@varper(@Prior("Actual", 1, @Relative("Year", 0)), budget)/100;
```

John designs the form and adds a data validation rule at cell level, as shown in the following figure. He uses Member Name to apply the validation only to Total Cost.

Form Layout at Design Time:



Data Validation Rule at Design Time:

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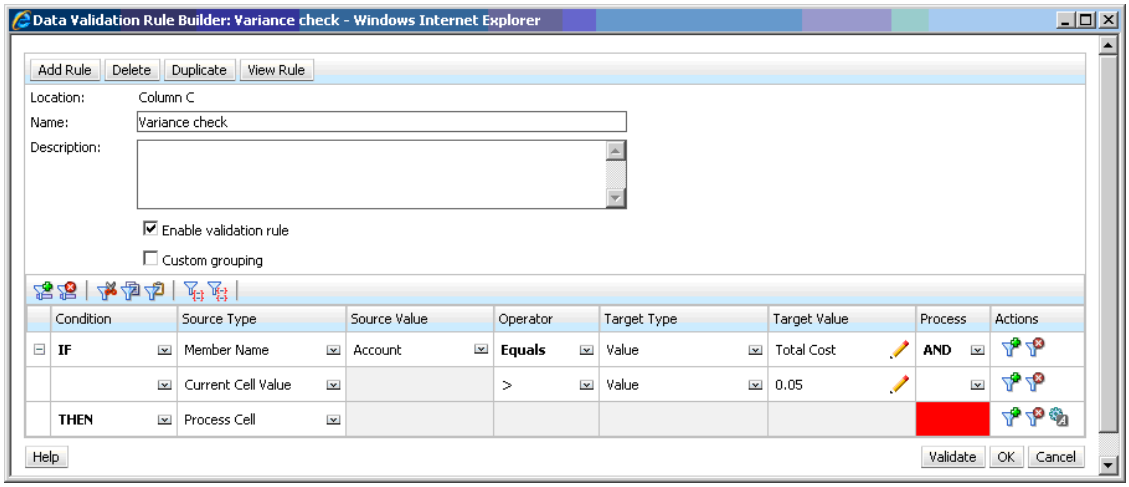
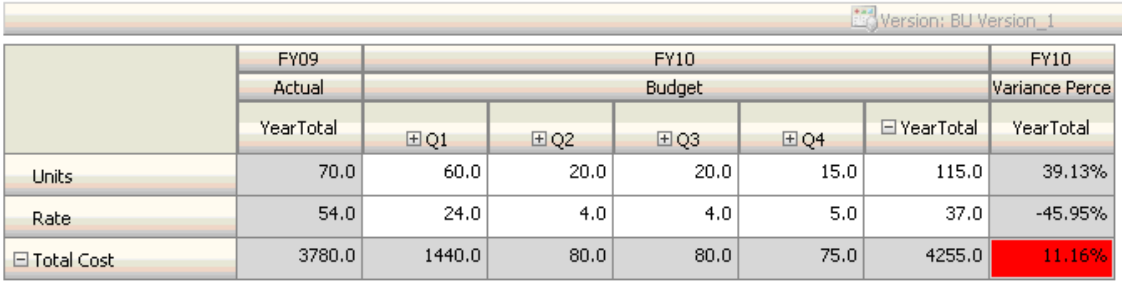


Form at Data Entry Time with Data Validations Applied:



	FY09	FY10				FY10
	Actual	Budget				Variance Perce
	YearTotal	Q1	Q2	Q3	Q4	YearTotal	YearTotal
Units	70.0	60.0	20.0	20.0	15.0	115.0	39.13%
Rate	54.0	24.0	4.0	4.0	5.0	37.0	-45.95%
Total Cost	3780.0	1440.0	80.0	80.0	75.0	4255.0	11.16%

Tips:

- If John isn't allowed to change the outline, or if he experiences performance issues related to member formulas, he can use a formula column.

  See Designing Forms with Formula Rows and Columns.

- John defines the rule at the Variance Percent column for these reasons.
  - It improves performance. The rule is evaluated only on the cells in the Variance Percent column. If the rule had been assigned to YearTotal, it would have to be evaluated for all Time Periods for the current year budget.
  - It helps users respond to the data validation message. John can add a message to the Variance Percent column stating that the variance is higher instead of adding it to YearTotal. This way, users don't have to look for Variance Percent to determine the difference.
- John could have flagged both YearTotal and Variance Percent in red if this had been part of the requirement.

Scenario 4

In addition to flagging the cell in red, the rule is also required to prevent anyone from promoting the approval unit if this year's Budget is significantly higher (> 5%) than the previous year's Actual amounts. To implement this requirement, all John needs to do is edit the data validation rule's processing instructions and select **Do Not Promote**, as shown in the following figure.

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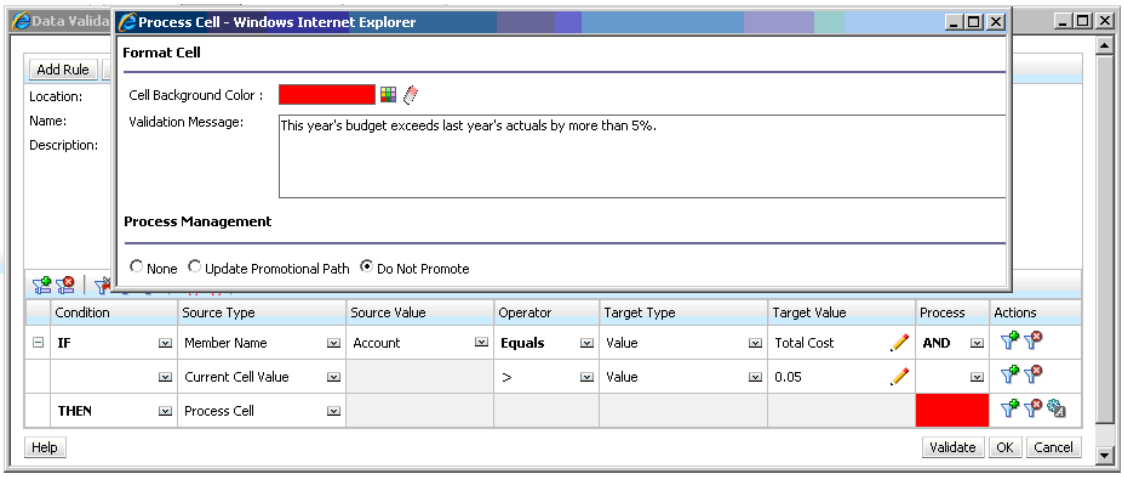
Data Validation Rule at Design Time:



Scenario 5

Finally, John is asked to design a data validation rule to validate that the total compensation for employees in a particular department is within the allowed range. The rule evaluates Existing Employees in the Operations department. It validates that, if Total Compensation is > than Min allowed, and is <= ¾ of the compensation range for the employee's grade, no action is needed.

If Total Compensation is greater than ¾ of the compensation range, a validation message is provided, and the approval units must be approved by a human resource manager. If the value is less than Min and greater than Max, an error is generated, and users can't promote their approval units.

John opens the Employee Expenses Summary form in the Form Management dialog box. The form has employees and departments on the page, accounts (such as Total Compensation) on the row, and time period on the column. To make validations easier to build, John adds a calculated row to calculate ¾ of the compensation range, and adds Min Compensation and Max Compensation members to the form, as shown in the following figures. Min Compensation and Max Compensation for the employee's grade are calculated using member formulas.

Form Layout at Design Time:

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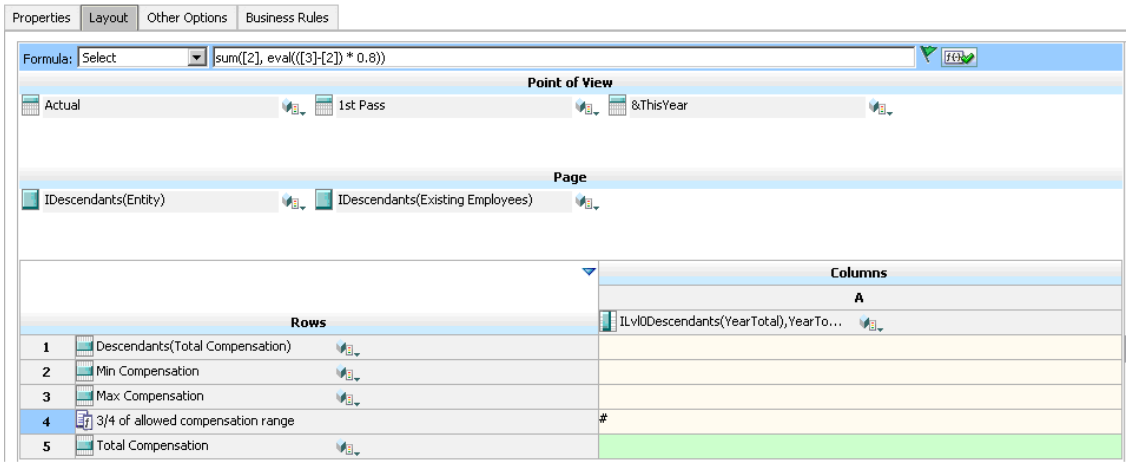
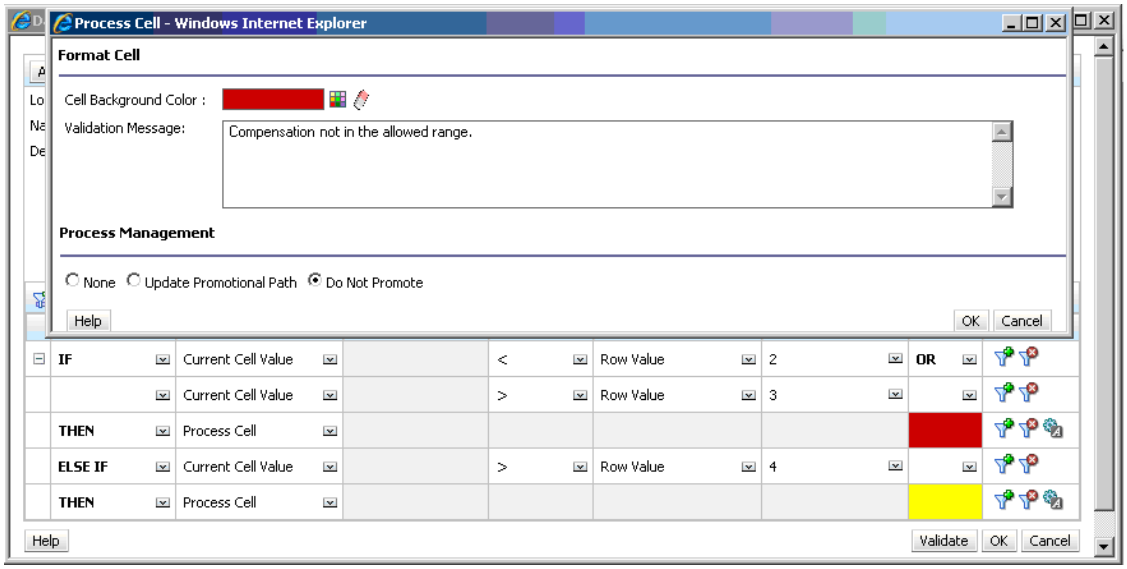


Data Validation Rule to Stop Promotion of Approval Units:



Data Validation Rule to Add the Human Resources Manager as Reviewer:

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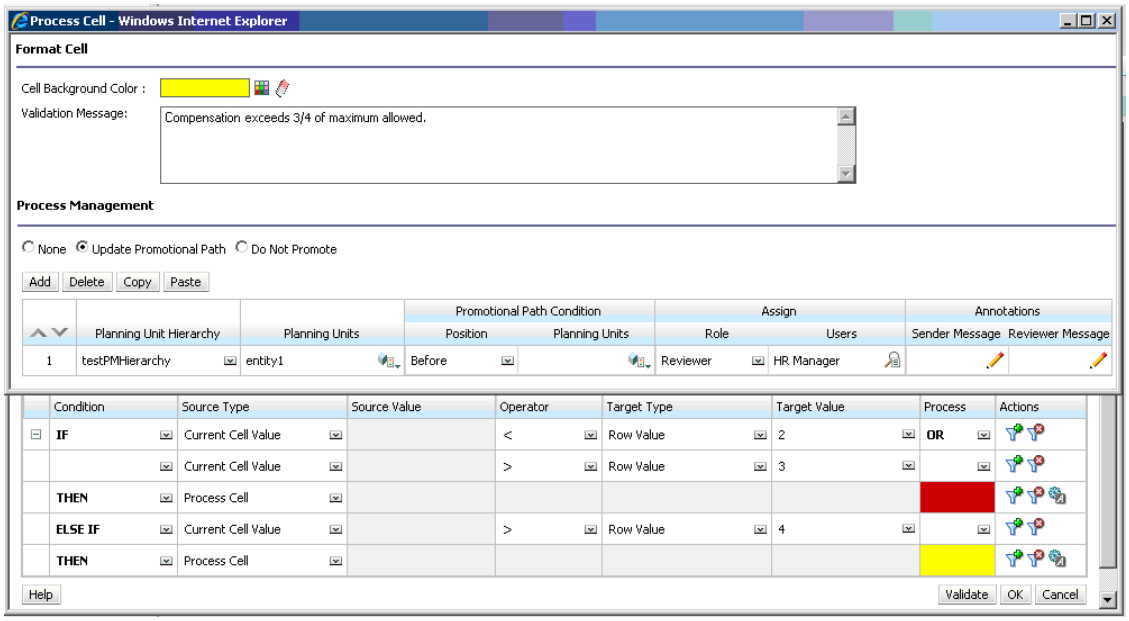
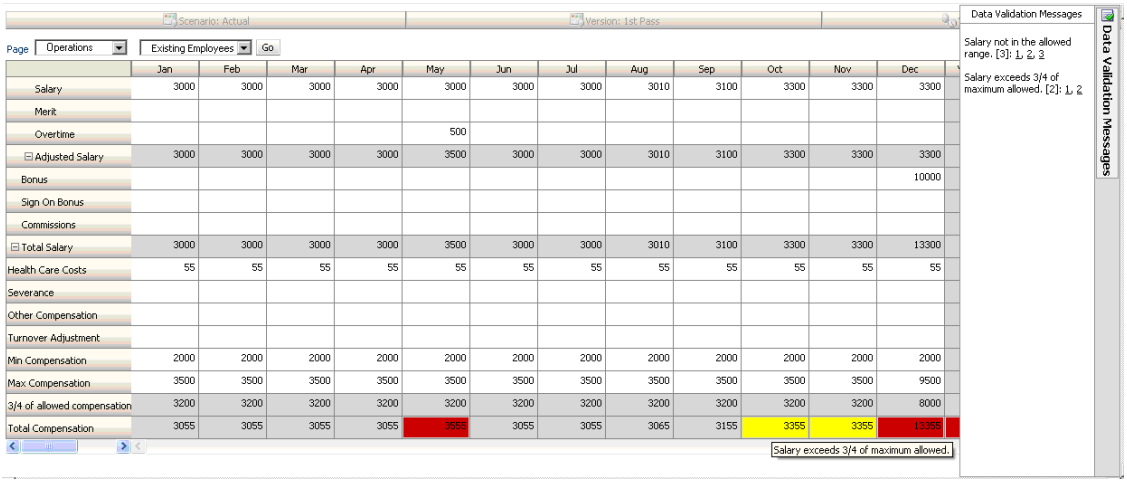


Form at Data Entry Time with Data Validations Applied and Validation Messages Shown:



About Format Rules in Forms

You can define format rules for a form at design time, which enables users to view formatting at runtime when custom styles are applied to the form.

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Sample scenarios that can be addressed using format rules are described in Format Rule Scenarios.

Defining format rules involves these main tasks:

- Identifying the cells or location that you want to display with formatting
- Creating the format rule at the location identified
- Applying styles at runtime

Related Links

- Creating and Updating Format Rules in Forms
- Cell Formatting Options in Forms
- Viewing Format Rules
- Order of Precedence for Format Rules
- Applying Format Rules to Forms at Runtime
- Format Rule Scenarios

Tutorials

Tutorials provide instructions with sequenced videos and documentation to help you learn a topic.

Your Goal	Learn How
Get an introduction to Design Time Formatting (DTF). You'll navigate to the menus for important user interface elements in Format Rules. You'll modify the Format Rules to show Form Data Grids with distinctive display elements.	Data Forms Design Time Formatting – Introduction
Get an introduction to the Design Time Formatting (DTF) options in web forms. You'll use the style selector and review the differences in runtime on applying the available style options. You'll see that Groovy Rule takes precedence over all other rules.	Data Forms in Web - Format Display in Runtime

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Your Goal	Learn How
Learn about creating a Format Rule and using the Design Time Formatting (DTF) options in web forms. You'll format various sections for an Income Statement using multiple Format Rules. You'll also duplicate and modify Format Rules.	Data Forms Design Time Formatting - Advanced
Get an introduction to the Design Time Formatting (DTF) options in Oracle Smart View for Office. You'll review the differences in runtime on applying the available style options. You'll see the order of precedence for the different types of Rules and Cell Styles in Smart View.	Data Forms in Smart View - Format Display Options

Creating and Updating Format Rules in Forms

Format rules are defined by Administrators on the **Layout** tab of the Form Designer. You can add format rules at the grid, row, row metadata cell, column, column metadata cell, or cell level.

Areas on the **Layout** tab with a green background, indicate locations where rules have been defined.

To create and update format rules:

1. On the Home page, click **Navigator** , and then under **Create and Manage**, click **Forms**.
2. Create or edit a form, and then on the **Form and Ad Hoc Grid Management** page, click **Layout**.
3. In **Layout**, right-click the row, column, or cell level for which you want to add or update the format rule. Depending on the location and whether rules were already defined at that location, you'll see some or all of the following context menu options:
   - **Add/Edit Format/Validation Rules**: Create or modify rules for individual data cells or all data cells in a grid, row, or column in the **Format/Validation Rule Builder**.
   - **Copy | Paste Format/Validation Rules**: Copy and paste the rules at the selected locations.
   - **Add/Edit Format/Validation Rules for Metadata**: Create or modify format rules for the row metadata cells or column metadata cells in the **Format/Validation Rule Builder**.
   - **Copy | Paste Format/Validation Rules for Metadata**: Copy and paste the rules for row metadata cells or column metadata cells at the selected locations.
4. Select **Add/Edit Format/Validation Rules** or **Add/Edit Format/Validation Rules for Metadata** to create or update rules.

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> **Note**
>
> To add rules for the entire grid, expand the **Format/Validation Rules** pane on the right. For **Rules for location**, select **Grid**, and then select (Add Validation Rule).

5. In the **Format/Validation Rule Builder**, create a new rule or update an existing rule.
6. To add a rule, enter a name and description for the rule.
   - If necessary, move the rule by selecting an option from **Location**. To create a rule similar to an existing rule, click (Duplicate) and then update the rule. To view rules in the current location, click **View Rule**. See Viewing Format Rules.
   - **Enable format/validation rule**: Select to make the format/validation rule available in forms. You can temporarily disable a format/validation rule by clearing this checkbox.
   - **Format Only**: Select to mark a rule as format only. This option is automatically enabled when a format-only rule is created.
7. Enter conditional logic for the rule:
   a. For **Condition**, select **Format**.

      You can use conditional logic similar to data validation rules when defining or updating format rules. In that case the THEN condition will have the Process Cell Source Type similar to FORMAT condition.

      > **Note**
      >
      > For descriptions of all conditions, see Conditions Supported by the Rule Builder.

   b. The **Source Type** field will automatically display the **Process Cell** option. There are no other **Source Type** options for the **Format** condition.

   c. In the **Actions** column on the right, click (Process Cell) to add formatting instructions. See Cell Formatting Options in Forms.
   d. After you finish adding format instructions, click **OK**.
8. As a reminder, when you're ready to enable the rule to make it available in the form, select **Enable format/validation rule**.

> **Tip**
>
> While you're building a rule, you can save the rule without enabling it. When the rule is ready to be used, you can enable and save the rule to make it available in the form. You can temporarily disable a rule by clearing **Enable format/validation rule**.

9. **Optional**: In the **Form and Ad Hoc Grid Management** page, you can view and update rules:

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- In the **Format/Validation Rules** pane on the right side of the **Form and Ad Hoc Grid Management** page, add, edit, or delete rules at any level, including grid level and metadata cells, by clicking (Add Validation Rule), (Edit Validation Rule), or (Delete Selected Validation Rule).
- If multiple rules are defined at the same location, you can change the order in which rules are processed when rules have the same precedence. To move a rule up, down, or to the top or bottom of the list, select the rule and click the arrows. You cannot reorder rules when you are viewing **All Rules**. For information about precedence of rules, see Order of Precedence for Format Rules.

> **Note**
>
> The Validation Rules options at the bottom of the pane are for data validation rules only. For descriptions of those options, see Creating and Updating Data Validation Rules in Forms.

10. In the **Form and Ad Hoc Grid Management** page, click **Finish** to save your updates.

    For forms that have formatting rules enabled, formatting is displayed when the form is loaded.

> **Note**
>
> Currently the preview only displays Forms 1.0 runtime. Hence the formatting rules will not be rendered.

See Including Data Validation and Format Rules in Forms.

Cell Formatting Options in Forms

Use the **Format/Validation Rule Builder** dialog box to enable format/validation rules, mark rules as format only, and to define formatting for cells in forms.

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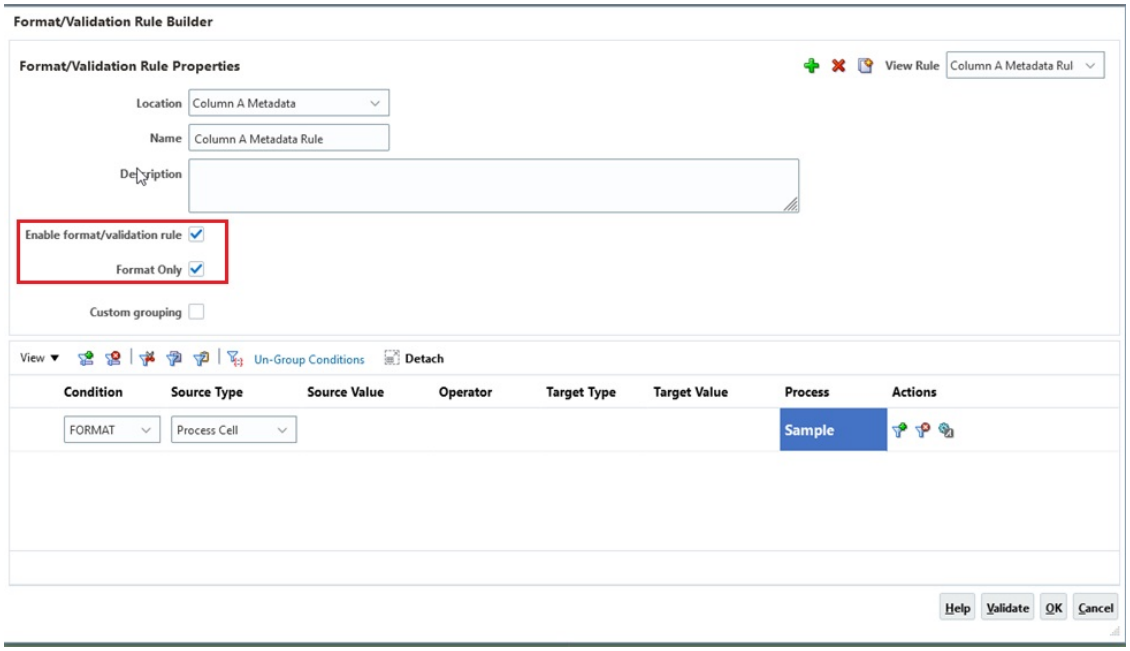

- **Enable format/validation rule**: Select to make the format/validation rule available in forms. You can temporarily disable a format/validation rule by clearing this checkbox.
- **Format Only**: Select to mark a rule as format only. This option is automatically enabled when a format-only rule is created.

In the **Actions** column for a rule, click (Process Cell) to add formatting instructions.

> **Note**
>
> If the Process Cell icon isn't displayed, ensure that the rule is valid and that it permits cell processing instructions. For example, cell processing instructions are included for Else, Range, Then, and Format conditions. The icon isn't displayed until all required columns are selected for a rule.
>
> See Creating and Updating Data Validation Rules in Forms.

There are options under **Format Cell** for **Font and Alignment** and **Border and Shading**.

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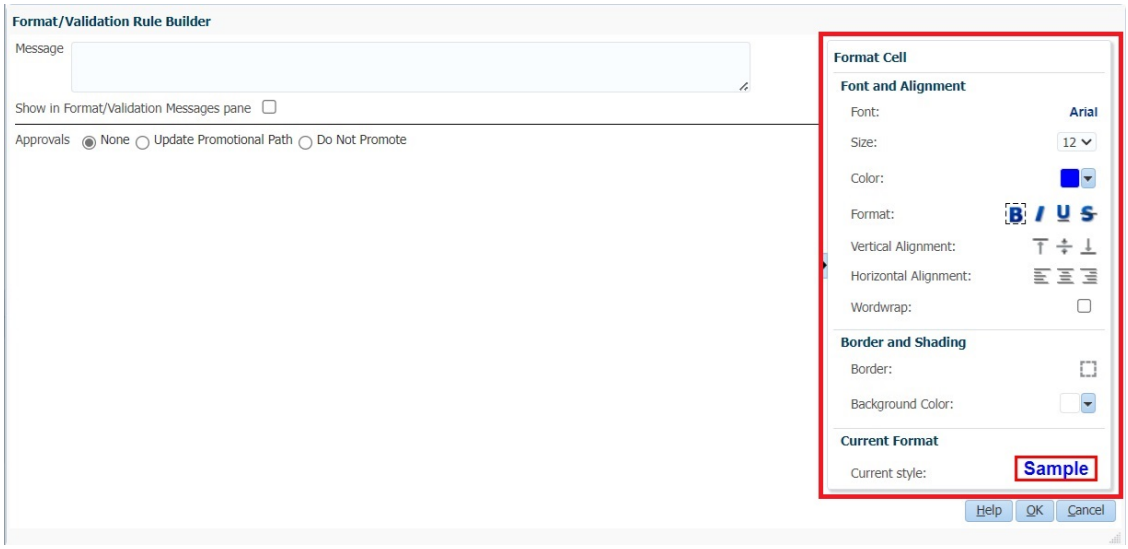
Figure 19-1 Format Cell Pane in the Format/Validation Rule Builder Dialog Box



To format cells:

1. In the **Format Cell** pane, select options to set how the cell should appear in forms. Select options for **Font and Alignment** and **Border and Shading**.

   Click options on the right for **Font and Alignment**, including **Font**, **Size**, **Color**, **Format**, **Vertical Alignment**, **Horizontal Alignment**, and **Wordwrap**.

   Click options on the right for **Border and Shading**, including **Border** (**Left**, **Top**, **Right**, **Bottom**, and **All**) and **Background Color**.

   As options are selected, the **Current Format** displays how it would appear when displayed on the form.

2. Optionally, you can add tooltip text to the **Message** field. At runtime when a user hovers their mouse over the location on the form that is formatted, they'll see the tooltip.

   > **Note**
   >
   > The **Show in Format/Validation Messages page** checkbox is disabled for Format only rules and, consequently, messages will not appear in the **Data Validation Messages** pane in forms.

3. Click **OK**.

   Cell formatting for each rule is displayed in the **Process** column for that rule.

Viewing Format Rules

After format rules are defined on the **Layout** tab, you can view rules that apply to the selected grid, row, row metadata cell, column, column metadata cell, or cell level, or you can opt to view all rules on the form. Areas on the **Layout** tab with a green background, indicate locations where rules have been defined.

grid, row, row metadata cell, column, column metadata cell, or cell level

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To view format rules:

1. Open the form, and then click **Layout**.
2. Expand the **Format/Data Validation Rules** pane on the right, and then click the drop-down list next to **Rules for location** to view rules for a specific location (grid, row, row metadata cell, column, column metadata cell, or cell level) or select **All Rules** to view all rules in the form.

> **Note**
>
> Click on a specific location on the grid area to refresh that location's rules (**Rules for location**) in the **Format/Data Validation Rules** pane.

3. Double-click a rule name to view rule details.

Order of Precedence for Format Rules

It's possible to have multiple rules specified in the same location on a form which could impact how formatting is displayed. For example, a form could include a Groovy rule, a data validation rule, Web form styles, and a format rule all at the same location. Therefore, there's an order of precedence for displaying rules in forms.

The order of precedence is as follows:

1. (P1) Groovy rule
2. (P2) Data Validation rule in the Format/Data Validation Rule Builder

> **Note**
>
> For a cell that inherits data validation rules that are defined in multiple places (grid, row, column, or cell level), the rule that is closest to the location of the rule definition wins priority. So if there is a rule applied at the cell level, that rule wins. Next is column level, then row level, and then grid level.

3. (P3) Oracle Smart View for Office cell styles
4. (P4) Format rule in the Format/Data Validation Rule Builder

> **Note**
>
> Same as the note above. For a cell that inherits rules that are defined in multiple places (grid, row, column, or cell level), the rule that is closest to the location of the rule definition wins priority. So if there is a rule applied at the cell level, that rule wins. Next is column level, then row level, and then grid level.

5. (P5) Smart View Custom formatting (not applicable on Web forms)

Considerations When Merging Rules

When merging formats from different rules (format and data validation rules), the rule with merged formatting is uplifted to the priority of a data validation rule (P2). In other words, if there is a P4 format rule, a P2 data validation rule, and a P3 style defined for the same cell, the

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formatting for the P4 format rule will display instead of the P3 style because the merging of the formatting and validation rules uplifted the format rule to a P2 priority level.

Tooltips and valid flags will be that of the higher priority data validation rule. Only formats are merged.

If a format is not set for a property, it will take on the properties of an incoming format regardless of the type of rule it is (format only or data validation). For example, if a left border is not set, it will be set with the incoming left border value. Each border is treated as a separate property. Also, left border thickness/style count as the same property.

Applying Format Rules to Forms at Runtime

After you define format rules for forms at design time, users can apply the format rules to forms at runtime using styles. Styles control how format rules are displayed for grids, row headings, column headings, and cells.

> **Note**
>
> You can only view runtime forms with format-only rules in Forms 2.0. To work in Forms 2.0, the **Forms Version** setting in application settings must be set to **Forms 2.0**. Also, format rules are supported in Oracle Smart View for Office 24.200. Versions earlier than 24.200 do not support this feature. If you define format-only rules for a form and then view the runtime form in Forms 1.0 or in a previous version of Smart View, the format rules will be ignored.

To apply styles to forms:

1. Open a form in Forms 2.0 (or in Smart View 24.200).
2. Click the style icon on the quick toolbar that is displayed above the form, and select one of the following style options:
   - **Cell Style**: Applies Web form styles (such as read-only cells, locked cells, supporting detail), Smart View cell styles defined in Smart View 24.200, data validation rules, and Groovy rules to the form. No custom (format-only) styles are applied.
   - **Custom Style**: Applies format-only rules that were defined in the rule builder at design-time, and any Excel custom formatting that was added by administrators and users in Smart View. No cell styles are applied.
   - **Custom and Cell Style**: Applies all cell styles and custom styles to the form. This is the default style option.

> **Note**
>
> Format rules have an order of precedence in cases where multiple rules are defined in the same location on a form. Order of precedence could impact how formatting is displayed on the runtime form. For more information, see Order of Precedence for Format Rules.

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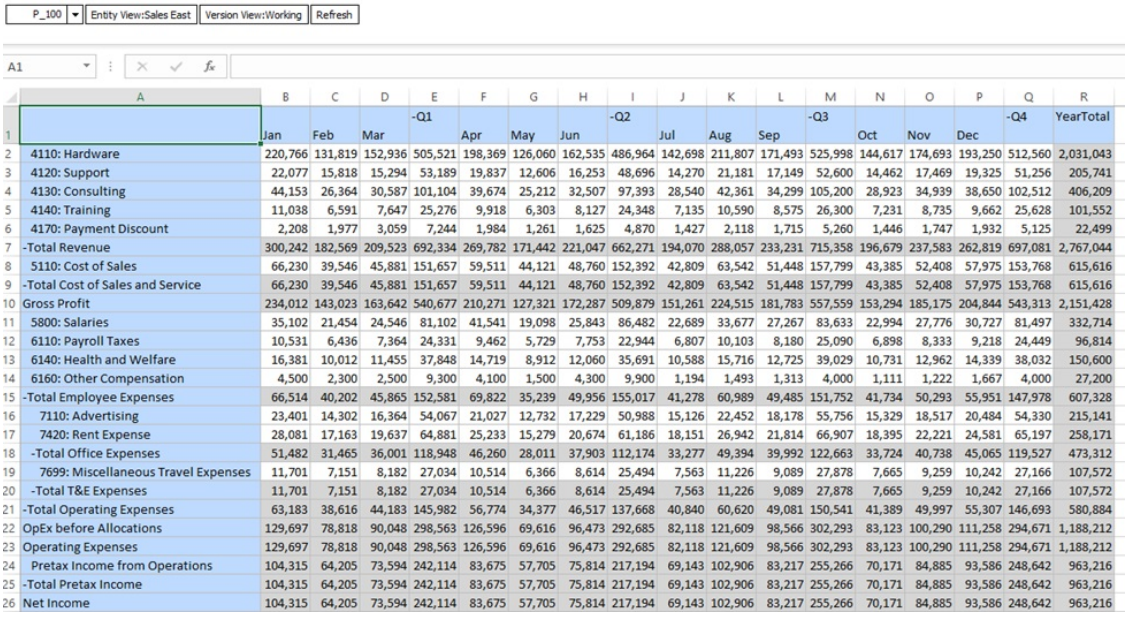
Format Rule Scenarios

These scenarios provide examples of how format/validation rules and Groovy rules that are defined at design time affect the display in runtime forms when cell styles and custom styles are applied.

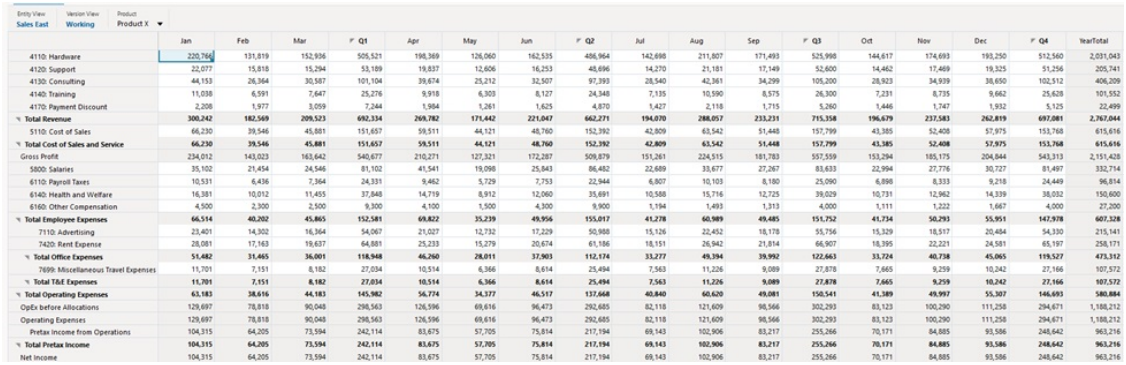
Scenario 1: Apply Cell Styles

Smart View runtime data grid with Smart View cell styles applied:



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1		Jan	Feb	Mar	-Q1	Apr	May	Jun	-Q2	Jul	Aug	Sep	-Q3	Oct	Nov	Dec	-Q4	YearTotal
2	4110: Hardware	220,766	131,819	152,936	505,521	198,369	126,060	162,535	486,964	142,698	211,807	171,493	525,998	144,617	174,693	193,250	512,560	2,031,043
3	4120: Support	22,077	15,818	15,294	53,189	19,837	12,606	16,253	48,696	14,270	21,181	17,149	52,600	14,462	17,469	19,325	51,256	205,741
4	4130: Consulting	44,153	26,364	30,587	101,104	39,674	25,212	32,507	97,393	28,540	42,361	34,299	105,200	28,923	34,939	38,650	102,512	406,209
5	4140: Training	11,038	6,591	7,647	25,276	9,918	6,303	8,127	24,348	7,135	10,590	8,575	26,300	7,231	8,735	9,662	25,628	101,552
6	4170: Payment Discount	2,208	1,977	3,059	7,244	1,984	1,261	1,625	4,870	1,427	2,118	1,715	5,260	1,446	1,747	1,932	5,125	22,499
7	-Total Revenue	300,242	182,569	209,523	692,334	269,782	171,442	221,047	662,271	194,070	288,057	233,231	715,358	196,679	237,583	262,819	697,081	2,767,044
8	5110: Cost of Sales	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
9	-Total Cost of Sales and Service	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
10	Gross Profit	234,012	143,023	163,642	540,677	210,271	127,321	172,287	509,879	151,261	224,515	181,783	557,559	153,294	185,175	204,844	543,313	2,151,428
11	5800: Salaries	35,102	21,454	24,546	81,102	41,541	19,098	25,843	86,482	22,689	33,677	27,267	83,633	22,994	27,776	30,727	81,497	332,714
12	6110: Payroll Taxes	10,531	6,436	7,364	24,331	9,462	5,729	7,753	22,944	6,807	10,103	8,180	25,090	6,898	8,333	9,218	24,449	96,814
13	6140: Health and Welfare	16,381	10,012	11,455	37,848	14,719	8,912	12,060	35,691	10,588	15,716	12,725	39,029	10,731	12,962	14,339	38,032	150,600
14	6160: Other Compensation	4,500	2,300	2,500	9,300	4,100	1,500	4,300	9,900	1,194	1,493	1,313	4,000	1,111	1,222	1,667	4,000	27,200
15	-Total Employee Expenses	66,514	40,202	45,865	152,581	69,822	35,239	49,956	155,017	41,278	60,989	49,485	151,752	41,734	50,293	55,951	147,978	607,328
16	7110: Advertising	23,401	14,302	16,364	54,067	21,027	12,732	17,229	50,988	15,126	22,452	18,178	55,756	15,329	18,517	20,484	54,330	215,141
17	7420: Rent Expense	28,081	17,163	19,637	64,881	25,233	15,279	20,674	61,186	18,151	26,942	21,814	66,907	18,395	22,221	24,581	65,197	258,171
18	-Total Office Expenses	51,482	31,465	36,001	118,948	46,260	28,011	37,903	112,174	33,277	49,394	39,992	122,663	33,724	40,738	45,065	119,527	473,312
19	7699: Miscellaneous Travel Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
20	-Total T&E Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
21	-Total Operating Expenses	63,183	38,616	44,183	145,982	56,774	34,377	46,517	137,668	40,840	60,620	49,081	150,541	41,389	49,997	55,307	146,693	580,884
22	OpEx before Allocations	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
23	Operating Expenses	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
24	Pretax Income from Operations	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
25	-Total Pretax Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
26	Net Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216

Web runtime data grid with Web cell styles applied:



Scenario 2: Apply Custom Style

In the Format/Validation Rule Builder, a format-only rule is created on each of the row and column header cells and on Column A and Column B as shown in the screen shots below.

Format/Validation Rule Builder Properties:

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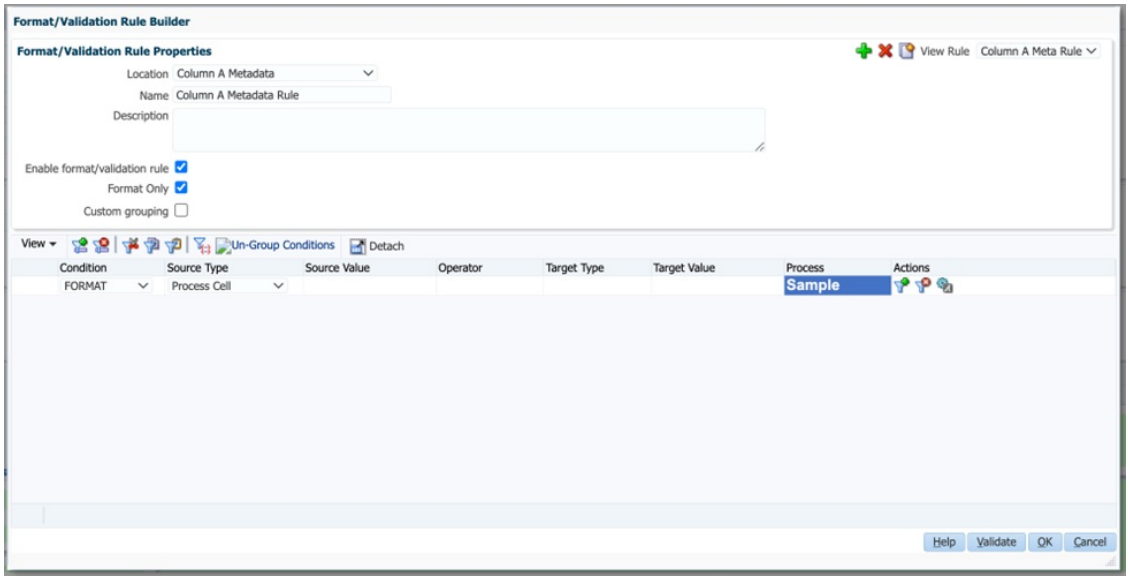
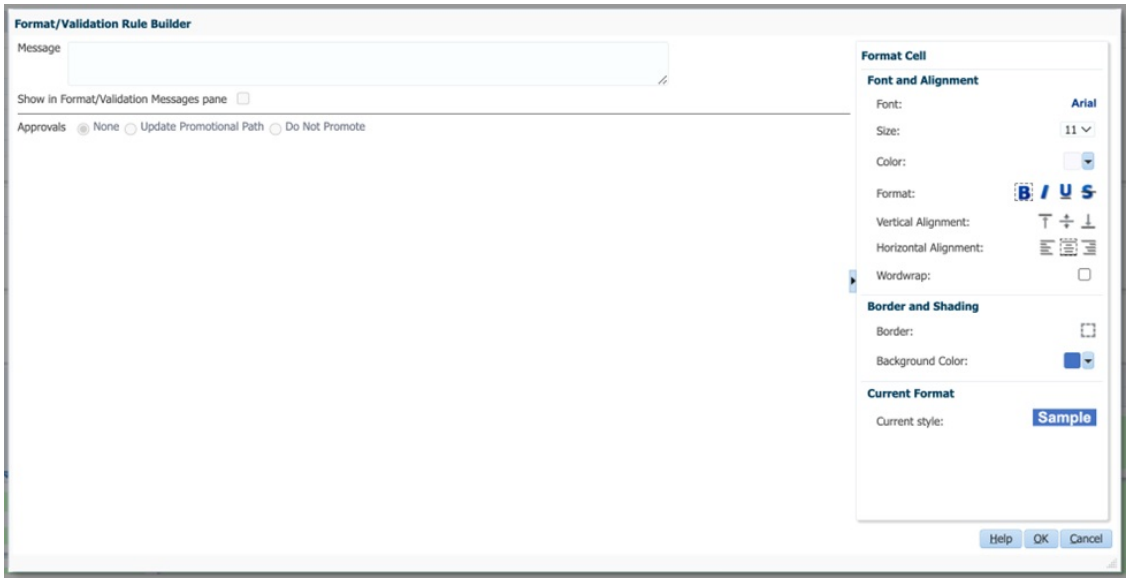


Format/Validation Rule Builder Format Cell Options:



Smart View runtime data grid with custom styles applied so only design-time format rules display:

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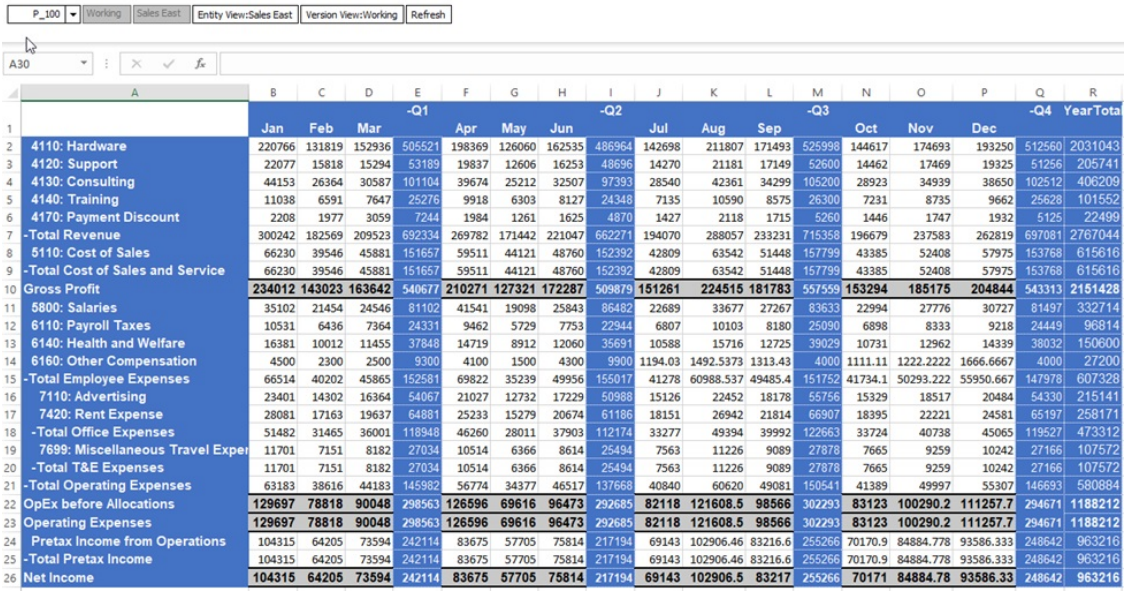
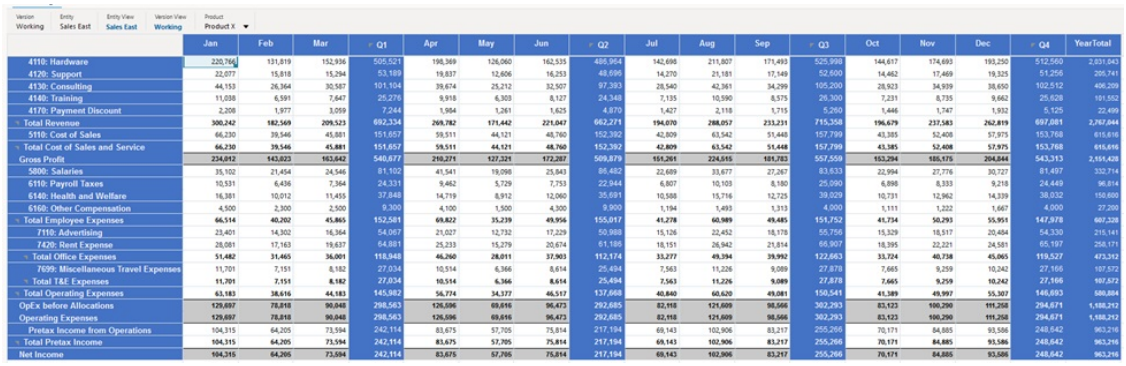


	Jan	Feb	Mar	-Q1	Apr	May	Jun	-Q2	Jul	Aug	Sep	-Q3	Oct	Nov	Dec	-Q4	YearTotal
4110: Hardware	220766	131819	152936	505521	198369	126060	162535	486964	142698	211807	171493	525998	144617	174693	193250	512560	2031043
4120: Support	22077	15818	15294	53189	19837	12606	16253	48696	14270	21181	17149	52600	14462	17469	19325	51256	205741
4130: Consulting	44153	26364	30587	101104	39674	25212	32507	97393	28540	42361	34299	105200	28923	34939	38650	102512	406209
4140: Training	11038	6591	7647	25276	9918	6303	8127	24348	7135	10590	8575	26300	7231	8735	9662	25628	101552
4170: Payment Discount	2208	1977	3059	7244	1984	1261	1625	4870	1427	2118	1715	5260	1446	1747	1932	5125	22499
-Total Revenue	300242	182569	209523	692334	269782	171442	221047	662271	194070	288057	233231	715358	196679	237583	262819	697081	2767044
5110: Cost of Sales	66230	39546	45881	151657	59511	44121	48760	152392	42809	63542	51448	157799	43385	52408	57975	153768	615616
-Total Cost of Sales and Service	66230	39546	45881	151657	59511	44121	48760	152392	42809	63542	51448	157799	43385	52408	57975	153768	615616
Gross Profit	234012	143023	163642	540677	210271	127321	172287	509879	151261	224515	181783	557559	153294	185175	204844	543313	2151428
5800: Salaries	35102	21454	24546	81102	41541	19098	25843	86482	22689	33677	27267	83633	22994	27776	30727	81497	332714
6110: Payroll Taxes	10531	6436	7364	24331	9462	5729	7753	22944	6807	10103	8180	25090	6898	8333	9218	24449	96814
6140: Health and Welfare	16381	10012	11455	37848	14719	8912	12060	35691	10588	15716	12725	39029	10731	12962	14339	38032	150600
6160: Other Compensation	4500	2300	2500	9300	4100	1500	4300	9900	1194.03	1492.5373	1313.43	4000	1111.11	1222.2222	1666.6667	4000	27200
-Total Employee Expenses	66514	40202	45865	152581	69822	35239	49956	155017	41278	60988.537	49485.4	151752	41734.1	50293.222	55950.667	147978	607328
7110: Advertising	23401	14302	16364	54067	21027	12732	17229	50988	15126	22452	18178	55756	15329	18517	20484	54330	215141
7420: Rent Expense	28081	17163	19637	64881	25233	15279	20674	61186	18151	26942	21814	66907	18395	22221	24581	65197	258171
-Total Office Expenses	51482	31465	36001	118948	46260	28011	37903	112174	33277	49394	39992	122663	33724	40738	45065	119527	473312
7699: Miscellaneous Travel Expe	11701	7151	8182	27034	10514	6366	8614	25494	7563	11226	9089	27878	7665	9259	10242	27166	107572
-Total T&E Expenses	11701	7151	8182	27034	10514	6366	8614	25494	7563	11226	9089	27878	7665	9259	10242	27166	107572
-Total Operating Expenses	63183	38616	44183	145982	56774	34377	46517	137668	40840	60620	49081	150541	41389	49997	55307	146693	580884
OpEx before Allocations	129697	78818	90048	298563	126596	69616	96473	292685	82118	121608.5	98566	302293	83123	100290.2	111257.7	294671	1188212
Operating Expenses	129697	78818	90048	298563	126596	69616	96473	292685	82118	121608.5	98566	302293	83123	100290.2	111257.7	294671	1188212
Pretax Income from Operations	104315	64205	73594	242114	83675	57705	75814	217194	69143	102906.46	83216.6	255266	70170.9	84884.778	93586.333	248642	963216
-Total Pretax Income	104315	64205	73594	242114	83675	57705	75814	217194	69143	102906.46	83216.6	255266	70170.9	84884.778	93586.333	248642	963216
Net Income	104315	64205	73594	242114	83675	57705	75814	217194	69143	102906.5	83217	255266	70171	84884.78	93586.33	248642	963216

Web runtime data grid with custom styles applied so only design-time format rules display:



Scenario 3: Apply Custom and Cell Styles

In the screen shots below, the Oracle Smart View for Office cell styles take precedence over design-time format rules.

Smart View runtime data grid with custom and cell styles applied:

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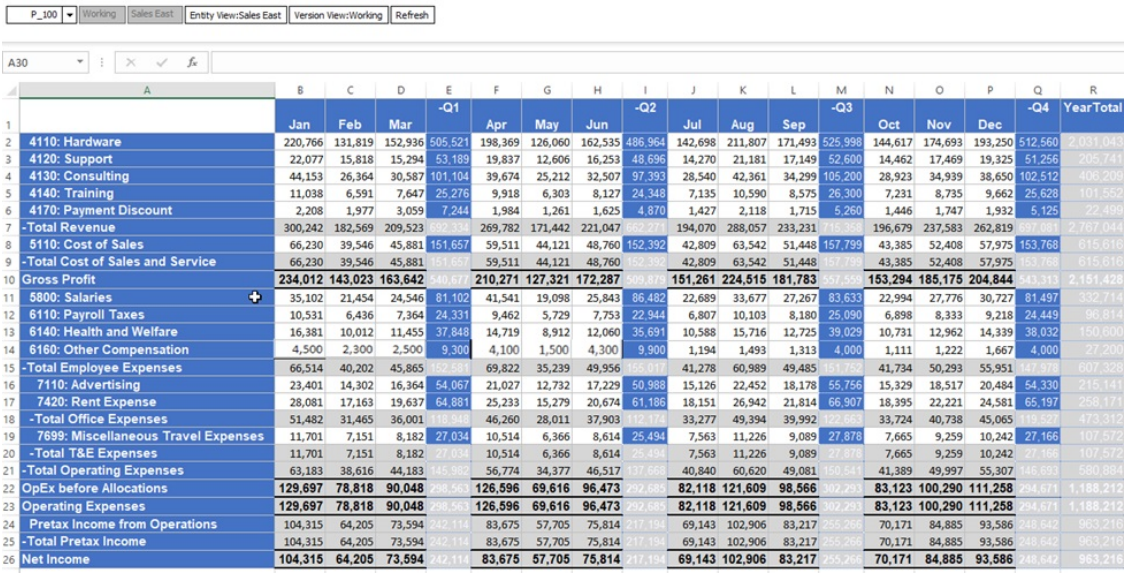
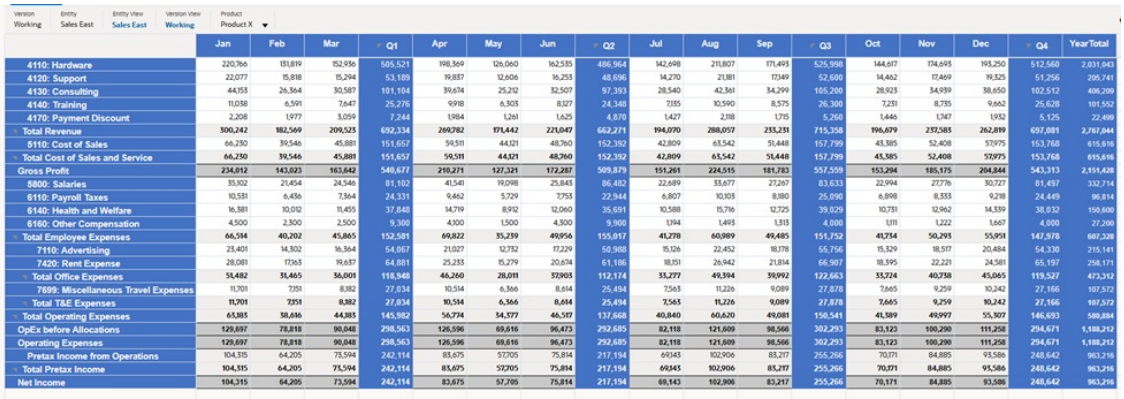


	Jan	Feb	Mar	-Q1	Apr	May	Jun	-Q2	Jul	Aug	Sep	-Q3	Oct	Nov	Dec	-Q4	YearTotal
4110: Hardware	220,766	131,819	152,936	505,521	198,369	126,060	162,535	486,964	142,698	211,807	171,493	525,998	144,617	174,693	193,250	512,560	2,031,043
4120: Support	22,077	15,818	15,294	53,189	19,837	12,606	16,253	48,696	14,270	21,181	17,149	52,600	14,462	17,469	19,325	51,256	205,741
4130: Consulting	44,153	26,364	30,587	101,104	39,674	25,212	32,507	97,393	28,540	42,361	34,299	105,200	28,923	34,939	38,650	102,512	406,209
4140: Training	11,038	6,591	7,647	25,276	9,918	6,303	8,127	24,348	7,135	10,590	8,575	26,300	7,231	8,735	9,662	25,628	101,552
4170: Payment Discount	2,208	1,977	3,059	7,244	1,984	1,261	1,625	4,870	1,427	2,118	1,715	5,260	1,446	1,747	1,932	5,125	22,499
-Total Revenue	300,242	182,569	209,523	692,334	269,782	171,442	221,047	662,271	194,070	288,057	233,231	715,358	196,679	237,583	262,819	697,081	2,767,044
5110: Cost of Sales	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
-Total Cost of Sales and Service	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
Gross Profit	234,012	143,023	163,642	540,677	210,271	127,321	172,287	509,879	151,261	224,515	181,783	557,559	153,294	185,175	204,844	543,313	2,151,428
5800: Salaries	35,102	21,454	24,546	81,102	41,541	19,098	25,843	86,482	22,689	33,677	27,267	83,633	22,994	27,776	30,727	81,497	332,714
6110: Payroll Taxes	10,531	6,436	7,364	24,331	9,462	5,729	7,753	22,944	6,807	10,103	8,180	25,090	6,898	8,333	9,218	24,449	96,814
6140: Health and Welfare	16,381	10,012	11,455	37,848	14,719	8,912	12,060	35,691	10,588	15,716	12,725	39,029	10,731	12,962	14,339	38,032	150,600
6160: Other Compensation	4,500	2,300	2,500	9,300	4,100	1,500	4,300	9,900	1,194	1,493	1,313	4,000	1,111	1,222	1,667	4,000	27,200
-Total Employee Expenses	66,514	40,202	45,865	152,581	69,822	35,239	49,956	155,017	41,278	60,989	49,485	151,752	41,714	50,293	55,951	147,978	607,328
7110: Advertising	23,401	14,302	16,364	54,067	21,027	12,732	17,229	50,988	15,126	22,452	18,178	55,756	15,329	18,517	20,484	54,330	215,141
7420: Rent Expense	28,081	17,163	19,637	64,881	25,233	15,279	20,674	61,186	18,151	26,942	21,814	66,907	18,395	22,221	24,581	65,197	258,171
-Total Office Expenses	51,482	31,465	36,001	118,948	46,260	28,011	37,903	112,374	33,277	49,394	39,992	122,663	33,724	40,738	45,065	119,527	473,312
7699: Miscellaneous Travel Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
-Total T&E Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
-Total Operating Expenses	63,183	38,616	44,183	145,982	56,774	34,377	46,517	137,668	40,840	60,620	49,081	150,541	41,389	49,997	55,307	146,693	580,884
OpEx before Allocations	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
Operating Expenses	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
Pretax Income from Operations	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
-Total Pretax Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
Net Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216

Web runtime data grid with custom and cell styles applied:



	Jan	Feb	Mar	Q1	Apr	May	Jun	Q2	Jul	Aug	Sep	Q3	Oct	Nov	Dec	Q4	YearTotal
4110: Hardware	220,766	131,819	152,936	505,521	198,369	126,060	162,535	486,964	142,698	211,807	171,493	525,998	144,617	174,693	193,250	512,560	2,031,043
4120: Support	22,077	15,818	15,294	53,189	19,837	12,606	16,253	48,696	14,270	21,181	17,149	52,600	14,462	17,469	19,325	51,256	205,741
4130: Consulting	44,153	26,364	30,587	101,104	39,674	25,212	32,507	97,393	28,540	42,361	34,299	105,200	28,923	34,939	38,650	102,512	406,209
4140: Training	11,038	6,591	7,647	25,276	9,918	6,303	8,127	24,348	7,135	10,590	8,575	26,300	7,231	8,735	9,662	25,628	101,552
4170: Payment Discount	2,208	1,977	3,059	7,244	1,984	1,261	1,625	4,870	1,427	2,118	1,715	5,260	1,446	1,747	1,932	5,125	22,499
Total Revenue	300,242	182,569	209,525	692,334	269,782	171,442	221,047	662,271	194,070	288,057	233,231	715,358	196,679	237,585	262,819	697,081	2,767,044
5110: Cost of Sales	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
Total Cost of Sales and Service	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
Gross Profit	234,012	143,023	163,642	540,677	210,271	127,321	172,287	509,879	151,261	224,515	181,783	557,559	153,294	185,175	204,844	543,313	2,151,428
5800: Salaries	35,102	21,454	24,546	81,102	41,541	19,098	25,843	86,482	22,689	33,677	27,267	83,633	22,994	27,776	30,727	81,497	332,714
6110: Payroll Taxes	10,531	6,436	7,364	24,331	9,462	5,729	7,753	22,944	6,807	10,103	8,180	25,090	6,898	8,333	9,218	24,449	96,814
6140: Health and Welfare	16,381	10,012	11,455	37,848	14,719	8,912	12,060	35,691	10,588	15,716	12,725	39,029	10,731	12,962	14,339	38,032	150,600
6160: Other Compensation	4,500	2,300	2,500	9,300	4,100	1,500	4,300	9,900	1,194	1,493	1,313	4,000	1,111	1,222	1,667	4,000	27,200
Total Employee Expenses	66,514	40,202	45,865	152,581	69,822	35,239	49,956	155,017	41,278	60,989	49,485	151,752	41,714	50,295	55,951	147,978	607,328
7110: Advertising	23,401	14,302	16,364	54,067	21,027	12,732	17,229	50,988	15,126	22,452	18,178	55,756	15,329	18,517	20,484	54,330	215,141
7420: Rent Expense	28,081	17,163	19,637	64,881	25,233	15,279	20,674	61,186	18,151	26,942	21,814	66,907	18,395	22,221	24,581	65,197	258,171
Total Office Expenses	51,482	31,465	36,001	118,948	46,260	28,011	37,903	112,174	33,277	49,594	39,992	122,663	33,724	40,738	45,065	119,527	473,312
7699: Miscellaneous Travel Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
Total T&E Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
Total Operating Expenses	63,183	38,616	44,183	145,982	56,774	34,377	46,517	137,668	40,840	60,620	49,081	150,541	41,389	49,997	55,307	146,693	580,884
OpEx before Allocations	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
Operating Expenses	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
Pretax Income from Operations	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69343	102906	83,277	255,266	70,171	84,885	93,586	248,642	963,216
Total Pretax Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69343	102,906	83,277	255,266	70,171	84,885	93,586	248,642	963,216
Net Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216

Scenario 4: Apply Cell Style with One Data Validation Rule

In the Format/Validation Rule Builder screen shots below, a data validation rule is added that sets the background color and tooltip for level 0 Other Compensation cells if the value exceeds $4000.

Format/Validation Rule Builder Properties:

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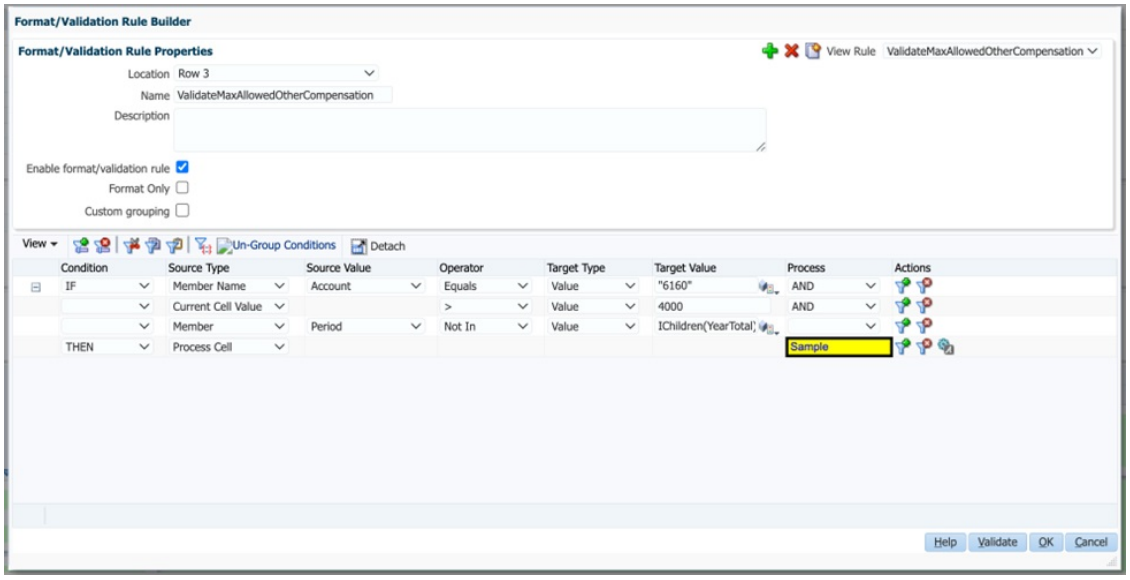
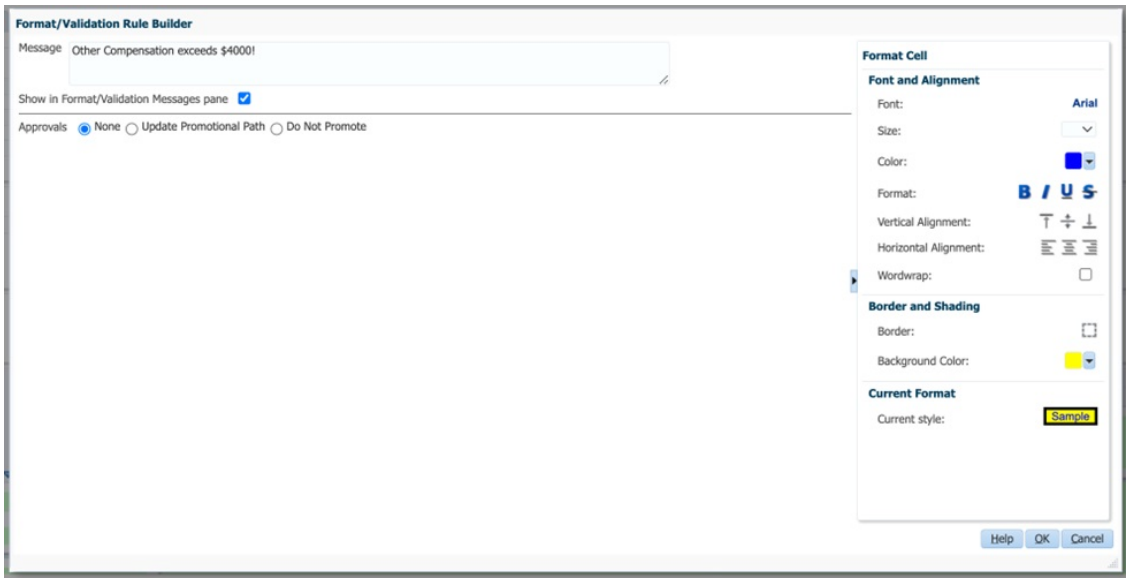


Format/Validation Rule Builder Format Cell Options:



Smart View runtime data grid with cell styles applied and one data validation rule:

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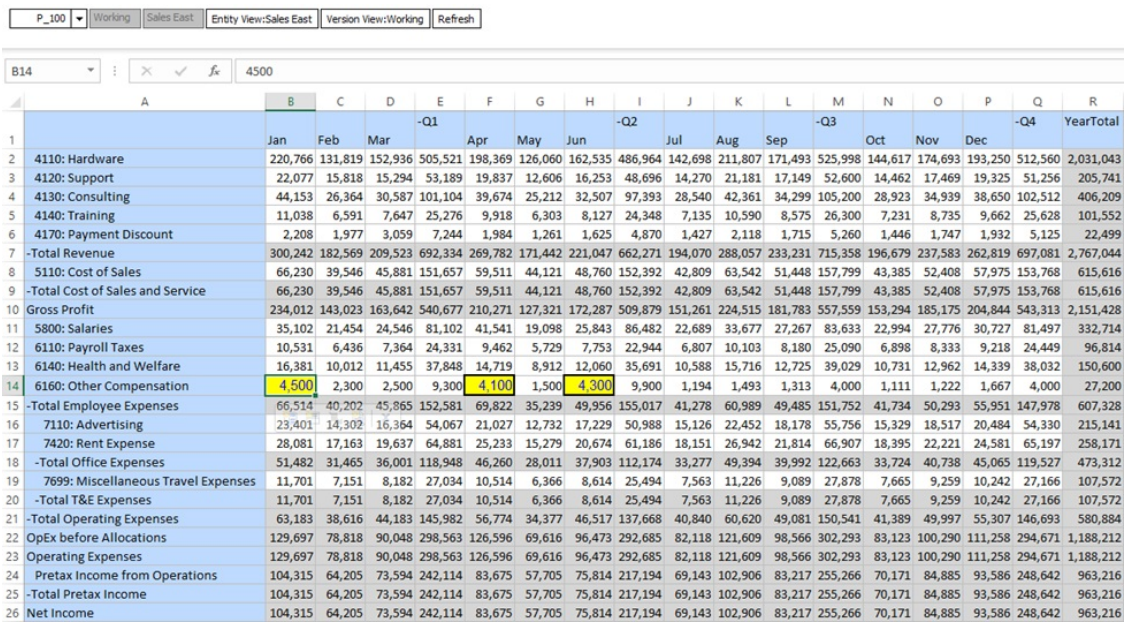
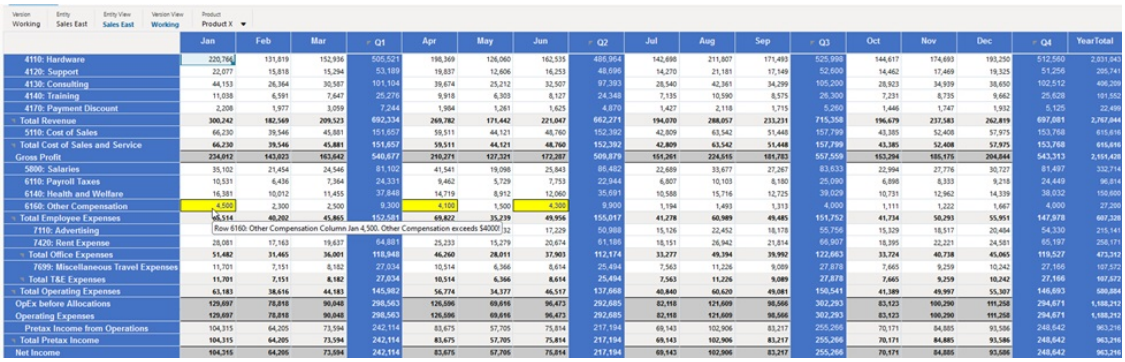


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1		Jan	Feb	Mar	-Q1	Apr	May	Jun	-Q2	Jul	Aug	Sep	-Q3	Oct	Nov	Dec	-Q4	YearTotal
2	4110: Hardware	220,766	131,819	152,936	505,521	198,369	126,060	162,535	486,964	142,698	211,807	171,493	525,998	144,617	174,693	193,250	512,560	2,031,043
3	4120: Support	22,077	15,818	15,294	53,189	19,837	12,606	16,253	48,696	14,270	21,181	17,149	52,600	14,462	17,469	19,325	51,256	205,741
4	4130: Consulting	44,153	26,364	30,587	101,104	39,674	25,212	32,507	97,393	28,540	42,361	34,299	105,200	28,923	34,939	38,650	102,512	406,209
5	4140: Training	11,038	6,591	7,647	25,276	9,918	6,303	8,127	24,348	7,135	10,590	8,575	26,300	7,231	8,735	9,662	25,628	101,552
6	4170: Payment Discount	2,208	1,977	3,059	7,244	1,984	1,261	1,625	4,870	1,427	2,118	1,715	5,260	1,446	1,747	1,932	5,125	22,499
7	-Total Revenue	300,242	182,569	209,523	692,334	269,782	171,442	221,047	662,271	194,070	288,057	233,231	715,358	196,679	237,583	262,819	697,081	2,767,044
8	5110: Cost of Sales	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
9	-Total Cost of Sales and Service	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
10	Gross Profit	234,012	143,023	163,642	540,677	210,271	127,321	172,287	509,879	151,261	224,515	181,783	557,559	153,294	185,175	204,844	543,313	2,151,428
11	5800: Salaries	35,102	21,454	24,546	81,102	41,541	19,098	25,843	86,482	22,689	33,677	27,267	83,633	22,994	27,776	30,727	81,497	332,714
12	6110: Payroll Taxes	10,531	6,436	7,364	24,331	9,462	5,729	7,753	22,944	6,807	10,103	8,180	25,090	6,898	8,333	9,218	24,449	96,814
13	6140: Health and Welfare	16,381	10,012	11,455	37,848	14,719	8,912	12,060	35,691	10,588	15,716	12,725	39,029	10,731	12,962	14,339	38,032	150,600
14	6160: Other Compensation	4,500	2,300	2,500	9,300	4,100	1,500	4,300	9,900	1,194	1,493	1,313	4,000	1,111	1,222	1,667	4,000	27,200
15	-Total Employee Expenses	66,514	40,202	45,865	152,581	69,822	35,239	49,956	155,017	41,278	60,989	49,485	151,752	41,734	50,293	55,951	147,978	607,328
16	7110: Advertising	23,401	14,302	16,364	54,067	21,027	12,732	17,229	50,988	15,126	22,452	18,178	55,756	15,329	18,517	20,484	54,330	215,141
17	7420: Rent Expense	28,081	17,163	19,637	64,881	25,233	15,279	20,674	61,186	18,151	26,942	21,814	66,907	18,395	22,221	24,581	65,197	258,171
18	-Total Office Expenses	51,482	31,465	36,001	118,948	46,260	28,011	37,903	112,174	33,277	49,394	39,992	122,663	33,724	40,738	45,065	119,527	473,312
19	7699: Miscellaneous Travel Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
20	-Total T&E Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
21	-Total Operating Expenses	63,183	38,616	44,183	145,982	56,774	34,377	46,517	137,668	40,840	60,620	49,081	150,541	41,389	49,997	55,307	146,693	580,884
22	OpEx before Allocations	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
23	Operating Expenses	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
24	Pretax Income from Operations	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
25	-Total Pretax Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
26	Net Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216

In the screen shot above, the data validation rule that was defined in the rule builder is applied to cells [B,14], [F,14], and [H,14]. There's a black border on all four sides, yellow background color, and blue font.

Web runtime data grid with cell styles applied and one data validation rule:



Note the following about the screen shot above:

- Formatting: Black border on all four sides, yellow background color, and blue font.
- Tooltip Message: Other Compensation exceeds $4000!

Scenario 5: Apply Cell Style with One Data Validation Rule and One Groovy Rule

The Groovy rule definition sets the background color and a tooltip for the Other Compensation cells if the cell value exceeds 10% of Salaries.

Groovy Rule definition:

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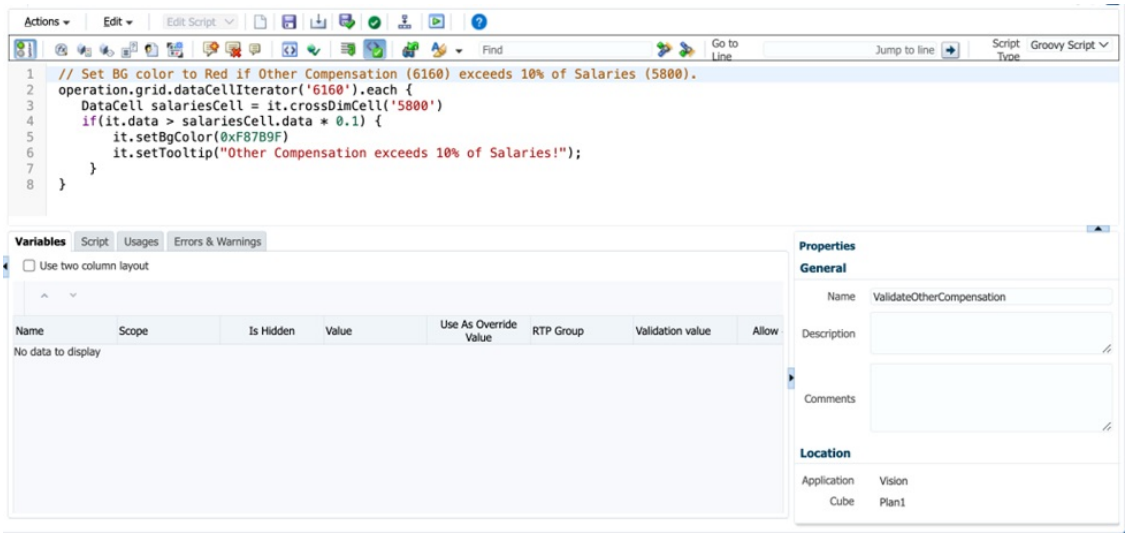
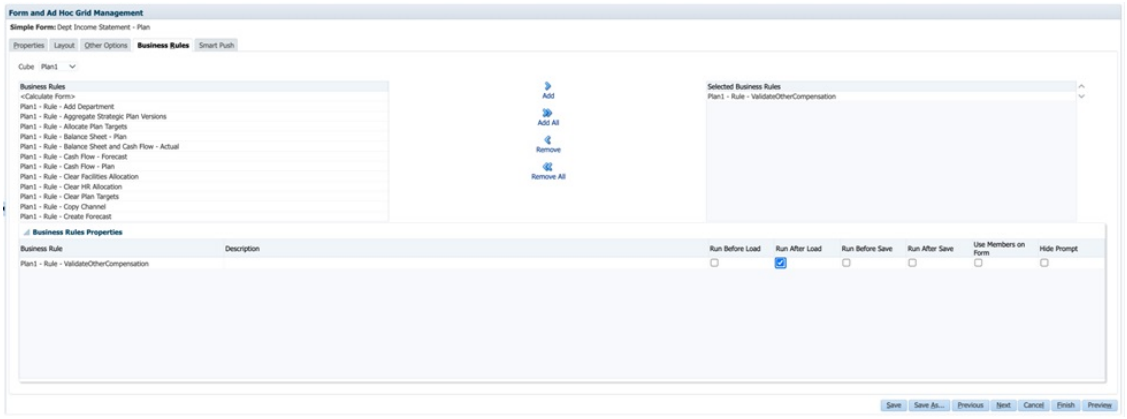


Adding the Groovy rule to the form on the Business Rules tab of the Form Designer:



Smart View runtime data grid with Groovy rules and data validation rules applied:

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P_100 | Working | Sales East | Entity View:Sales East | Version View:Working | Refresh

B2 220766

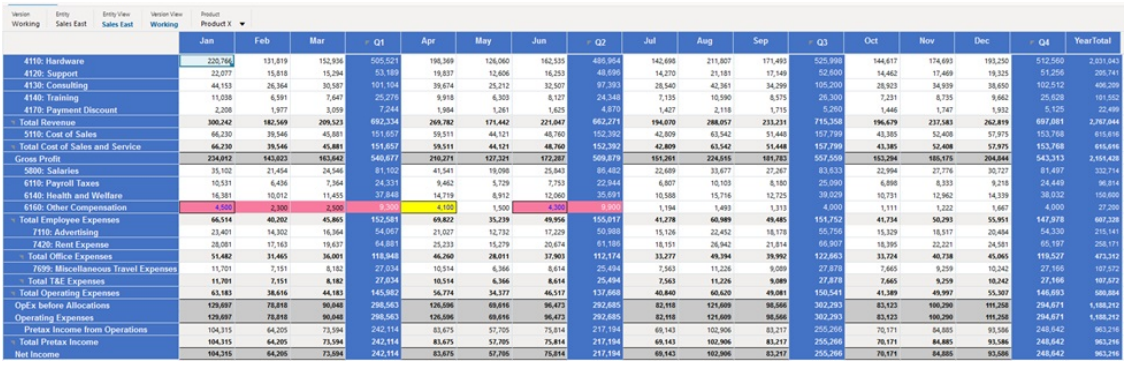
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1		Jan	Feb	Mar	-Q1	Apr	May	Jun	-Q2	Jul	Aug	Sep	-Q3	Oct	Nov	Dec	-Q4	YearTotal
2	4110: Hardware	220,766	131,819	152,936	505,521	198,369	126,060	162,535	486,964	142,698	211,807	171,493	525,998	144,617	174,693	193,250	512,560	2,031,043
3	4120: Support	22,077	15,818	15,294	53,189	19,837	12,606	16,253	48,696	14,270	21,181	17,149	52,600	14,462	17,469	19,325	51,256	205,741
4	4130: Consulting	44,153	26,364	30,587	101,104	39,674	25,212	32,507	97,393	28,540	42,361	34,299	105,200	28,923	34,939	38,650	102,512	406,209
5	4140: Training	11,038	6,591	7,647	25,276	9,918	6,303	8,127	24,348	7,135	10,590	8,575	26,300	7,231	8,735	9,662	25,628	101,552
6	4170: Payment Discount	2,208	1,977	3,059	7,244	1,984	1,261	1,625	4,870	1,427	2,118	1,715	5,260	1,446	1,747	1,932	5,125	22,499
7	-Total Revenue	300,242	182,569	209,523	692,334	269,782	171,442	221,047	662,271	194,070	288,057	233,231	715,358	196,679	237,583	262,819	697,081	2,767,044
8	5110: Cost of Sales	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
9	-Total Cost of Sales and Service	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
10	Gross Profit	234,012	143,023	163,642	540,677	210,271	127,321	172,287	509,879	151,261	224,515	181,783	557,559	153,294	185,175	204,844	543,313	2,151,428
11	5800: Salaries	35,102	21,454	24,546	81,102	41,541	19,098	25,843	86,482	22,689	33,677	27,267	83,633	22,994	27,776	30,727	81,497	332,714
12	6110: Payroll Taxes	10,531	6,436	7,364	24,331	9,462	5,729	7,753	22,944	6,807	10,103	8,180	25,090	6,898	8,333	9,218	24,449	96,814
13	6140: Health and Welfare	16,381	10,012	11,455	37,848	14,719	8,912	12,060	35,691	10,588	15,716	12,725	39,029	10,731	12,962	14,339	38,032	150,600
14	6160: Other Compensation	4,500	2,300	2,500	9,300	4,100	1,500	4,300	9,900	1,194	1,493	1,313	4,000	1,111	1,222	1,667	4,000	27,200
15	-Total Employee Expenses	66,514	40,202	45,865	152,581	69,822	35,239	49,956	155,017	41,278	60,989	49,485	151,752	41,734	50,293	55,951	147,978	607,328
16	7110: Advertising	23,401	14,302	16,364	54,067	21,027	12,732	17,229	50,988	15,126	22,452	18,178	55,756	15,329	18,517	20,484	54,330	215,141
17	7420: Rent Expense	28,081	17,163	19,637	64,881	25,233	15,279	20,674	61,186	18,151	26,942	21,814	66,907	18,395	22,221	24,581	65,197	258,171
18	-Total Office Expenses	51,482	31,465	36,001	118,948	46,260	28,011	37,903	112,174	33,277	49,394	39,992	122,663	33,724	40,738	45,065	119,527	473,312
19	7699: Miscellaneous Travel Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
20	-Total T&E Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
21	-Total Operating Expenses	63,183	38,616	44,183	145,982	56,774	34,377	46,517	137,668	40,840	60,620	49,081	150,541	41,389	49,997	55,307	146,693	580,884
22	OpEx before Allocations	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
23	Operating Expenses	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
24	Pretax Income from Operations	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
25	-Total Pretax Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
26	Net Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216

The data validation rule and the Groovy rule are applied to cells [B,14] and [H,14].

Note the following in the above Smart View runtime data grid:

- The background color is applied from the Groovy rule as it has higher precedence than the data validation rule. The formatting (border and blue font) is merged from the data validation rule since the Groovy rule does not apply any formatting.
- In cells [C,14], [D,14], [E,14], and [I,14], only the Groovy rule definition is displayed.
- Only cell [F,14] applies the data validation rule

Web runtime data grid with Groovy rules and data validation rules applied:



Note the following in the above Web runtime data grid:

- The background color is applied from the Groovy rule as it has higher precedence than the data validation rule. The formatting (border and blue font) is merged from the data validation rule since the Groovy rule does not apply any formatting.
- In some cells, only the Groovy rule definition is displayed.
- Only one cell applies the data validation rule

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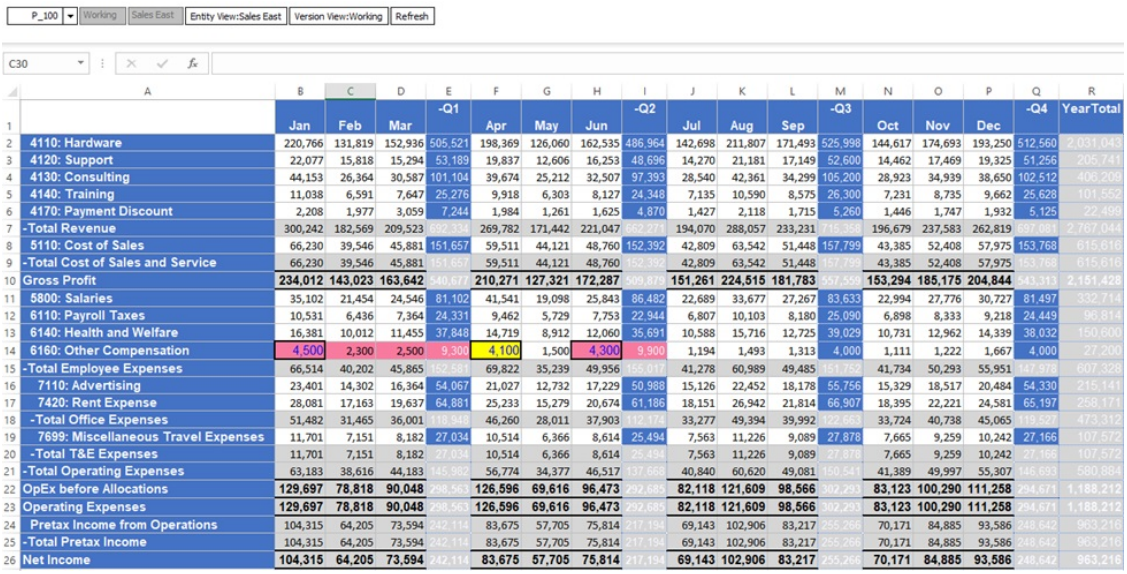
Scenario 6: Apply Custom and Cell Styles with One Data Validation Rule and One Groovy Rule

In the screen shots below, the order of precedence is applied with P1 being the highest precedence:

- (P1) Groovy rule
- (P2) Data Validation rule in the Format/Data Validation Rule Builder
- (P3) Smart View cell styles
- (P4) Format rule in the Format/Data Validation Rule Builder
- (P5) Smart View Custom formatting

Smart View runtime data grid with custom and cell styles applied and one data validation rule and one groovy rule:



	Jan	Feb	Mar	-Q1	Apr	May	Jun	-Q2	Jul	Aug	Sep	-Q3	Oct	Nov	Dec	-Q4	YearTotal
4110: Hardware	220,766	131,819	152,936	505,521	198,369	126,060	162,535	486,964	142,698	211,807	171,493	525,998	144,617	174,693	193,250	512,560	2,031,043
4120: Support	22,077	15,818	15,294	53,189	19,837	12,606	16,253	48,696	14,270	21,181	17,149	52,600	14,462	17,469	19,325	51,256	205,741
4130: Consulting	44,153	26,364	30,587	101,104	39,674	25,212	32,507	97,393	28,540	42,361	34,299	105,200	28,923	34,939	38,650	102,512	406,209
4140: Training	11,038	6,591	7,647	25,276	9,918	6,303	8,127	24,348	7,135	10,590	8,575	26,300	7,231	8,735	9,662	25,628	101,552
4170: Payment Discount	2,208	1,977	3,059	7,244	1,984	1,261	1,625	4,870	1,427	2,118	1,715	5,260	1,446	1,747	1,932	5,125	22,499
-Total Revenue	300,242	182,569	209,523	692,334	269,782	171,442	221,047	662,471	194,070	288,057	233,231	715,358	196,679	237,583	262,819	697,081	2,767,044
5110: Cost of Sales	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
-Total Cost of Sales and Service	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
Gross Profit	**234,012**	**143,023**	**163,642**	540,677	**210,271**	**127,321**	**172,287**	509,879	**151,261**	**224,515**	**181,783**	557,559	**153,294**	**185,175**	**204,844**	543,313	2,151,428
5800: Salaries	35,102	21,454	24,546	81,102	41,541	19,098	25,843	86,482	22,689	33,677	27,267	83,633	22,994	27,776	30,727	81,497	332,714
6110: Payroll Taxes	10,531	6,436	7,364	24,331	9,462	5,729	7,753	22,944	6,807	10,103	8,180	25,090	6,898	8,333	9,218	24,449	96,814
6140: Health and Welfare	16,381	10,012	11,455	37,848	14,719	8,912	12,060	35,691	10,588	15,716	12,725	39,029	10,731	12,962	14,339	38,032	150,600
6160: Other Compensation	4,500	2,300	2,500	9,300	4,100	1,500	4,300	9,900	1,194	1,493	1,313	4,000	1,111	1,222	1,667	4,000	27,200
-Total Employee Expenses	66,514	40,202	45,865	152,581	69,822	35,239	49,956	155,017	41,278	60,989	49,485	151,752	41,734	50,293	55,951	147,978	607,328
7110: Advertising	23,401	14,302	16,364	54,067	21,027	12,732	17,229	50,988	15,126	22,452	18,178	55,756	15,329	18,517	20,484	54,330	215,141
7420: Rent Expense	28,081	17,163	19,637	64,881	25,233	15,279	20,674	61,186	18,151	26,942	21,814	66,907	18,395	22,221	24,581	65,197	258,171
-Total Office Expenses	51,482	31,465	36,001	118,948	46,260	28,011	37,903	112,174	33,277	49,394	39,992	122,663	33,724	40,738	45,065	119,527	473,312
7699: Miscellaneous Travel Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
-Total T&E Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
-Total Operating Expenses	63,183	38,616	44,183	145,982	56,774	34,377	46,517	137,668	40,840	60,620	49,081	150,541	41,389	49,997	55,307	146,693	580,884
OpEx before Allocations	**129,697**	**78,818**	**90,048**	298,563	**126,596**	**69,616**	**96,473**	292,685	**82,118**	**121,609**	**98,566**	302,293	**83,123**	**100,290**	**111,258**	294,671	1,188,212
Operating Expenses	**129,697**	**78,818**	**90,048**	298,563	**126,596**	**69,616**	**96,473**	292,685	**82,118**	**121,609**	**98,566**	302,293	**83,123**	**100,290**	**111,258**	294,671	1,188,212
Pretax Income from Operations	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
-Total Pretax Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
Net Income	**104,315**	**64,205**	**73,594**	242,114	**83,675**	**57,705**	**75,814**	217,194	**69,143**	**102,906**	**83,217**	255,266	**70,171**	**84,885**	**93,586**	248,642	963,216

Smart View runtime data grid with read-only disabled:

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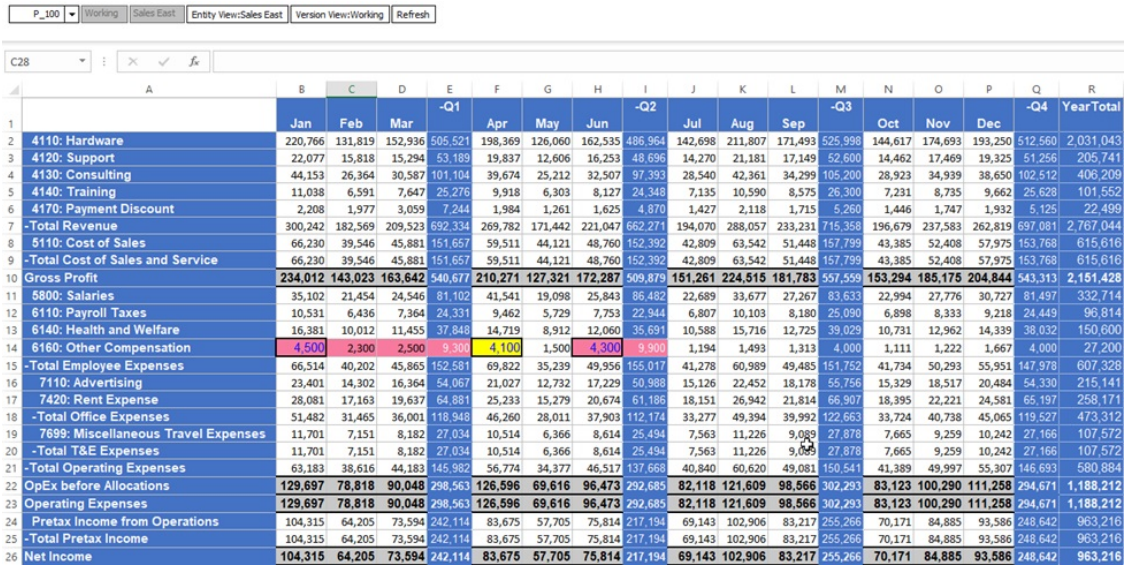

P_100 | Working | Sales East | Entity View:Sales East | Version View:Working | Refresh

C28

	A	Jan	Feb	Mar	-Q1	Apr	May	Jun	-Q2	Jul	Aug	Sep	-Q3	Oct	Nov	Dec	-Q4	YearTotal
2	4110: Hardware	220,766	131,819	152,936	505,521	198,369	126,060	162,535	486,964	142,698	211,807	171,493	525,998	144,617	174,693	193,250	512,560	2,031,043
3	4120: Support	22,077	15,818	15,294	53,189	19,837	12,606	16,253	48,696	14,270	21,181	17,149	52,600	14,462	17,469	19,325	51,256	205,741
4	4130: Consulting	44,153	26,364	30,587	101,104	39,674	25,212	32,507	97,393	28,540	42,361	34,299	105,200	28,923	34,939	38,650	102,512	406,209
5	4140: Training	11,038	6,591	7,647	25,276	9,918	6,303	8,127	24,348	7,135	10,590	8,575	26,300	7,231	8,735	9,662	25,628	101,552
6	4170: Payment Discount	2,208	1,977	3,059	7,244	1,984	1,261	1,625	4,870	1,427	2,118	1,715	5,260	1,446	1,747	1,932	5,125	22,499
7	-Total Revenue	300,242	182,569	209,523	692,334	269,782	171,442	221,047	662,271	194,070	288,057	233,231	715,358	196,679	237,583	262,819	697,081	2,767,044
8	5110: Cost of Sales	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
9	-Total Cost of Sales and Service	66,230	39,546	45,881	151,657	59,511	44,121	48,760	152,392	42,809	63,542	51,448	157,799	43,385	52,408	57,975	153,768	615,616
10	Gross Profit	234,012	143,023	163,642	540,677	210,271	127,321	172,287	509,879	151,261	224,515	181,783	557,559	153,294	185,175	204,844	543,313	2,151,428
11	5800: Salaries	35,102	21,454	24,546	81,102	41,541	19,098	25,843	86,482	22,689	33,677	27,267	83,633	22,994	27,776	30,727	81,497	332,714
12	6110: Payroll Taxes	10,531	6,436	7,364	24,331	9,462	5,729	7,753	22,944	6,807	10,103	8,180	25,090	6,898	8,333	9,218	24,449	96,814
13	6140: Health and Welfare	16,381	10,012	11,455	37,848	14,719	8,912	12,060	35,691	10,588	15,716	12,725	39,029	10,731	12,962	14,339	38,032	150,600
14	6160: Other Compensation	4,500	2,300	2,500	9,300	4,100	1,500	4,300	9,900	1,194	1,493	1,313	4,000	1,111	1,222	1,667	4,000	27,200
15	-Total Employee Expenses	66,514	40,202	45,865	152,581	69,822	35,239	49,956	155,017	41,278	60,989	49,485	151,752	41,734	50,293	55,951	147,978	607,328
16	7110: Advertising	23,401	14,302	16,364	54,067	21,027	12,732	17,229	50,988	15,126	22,452	18,178	55,756	15,329	18,517	20,484	54,330	215,141
17	7420: Rent Expense	28,081	17,163	19,637	64,881	25,233	15,279	20,674	61,186	18,151	26,942	21,814	66,907	18,395	22,221	24,581	65,197	258,171
18	-Total Office Expenses	51,482	31,465	36,001	118,948	46,260	28,011	37,903	112,174	33,277	49,394	39,992	122,663	33,724	40,738	45,065	119,527	473,312
19	7699: Miscellaneous Travel Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
20	-Total T&E Expenses	11,701	7,151	8,182	27,034	10,514	6,366	8,614	25,494	7,563	11,226	9,089	27,878	7,665	9,259	10,242	27,166	107,572
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22	OpEx before Allocations	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
23	Operating Expenses	129,697	78,818	90,048	298,563	126,596	69,616	96,473	292,685	82,118	121,609	98,566	302,293	83,123	100,290	111,258	294,671	1,188,212
24	Pretax Income from Operations	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
25	-Total Pretax Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216
26	Net Income	104,315	64,205	73,594	242,114	83,675	57,705	75,814	217,194	69,143	102,906	83,217	255,266	70,171	84,885	93,586	248,642	963,216

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20

Managing Application and System Settings

Set application defaults and system settings such as number formatting, notifications, number of items displayed on a page, aliases, date formatting, and assigning application ownership.

Related Topics

- What Application and System Settings Can I Specify?
- Defining User Variables
- Customizing Your Display
  Change the theme of your display or add your company logo or a background image to the Home page.
- Announcing Upcoming Events
- Specifying Artifact Labels

What Application and System Settings Can I Specify?

You can control many aspects of the application and the system, such as:

- How to display thousands, decimals, and negative numbers in forms
- Define the actions about which you want to be notified
- Display the full names of users rather than user IDs
- Set reporting options
- Assign application ownership to another Service Administrator

To change application and system settings:

1. Click **Application**, and then click **Settings**.
2. Specify defaults for the current application. For descriptions of the settings, see the Application Settings table below.
3. Specify system settings. For descriptions of the settings, see the System Settings table below.
4. Click **Save**.

Table 20-1 Application Settings

Application Setting	Description
Alias Setting	For option descriptions, see Specifying a Default Alias Table and Setting Member and Alias Display Options.
Notifications	Enable notifications for task lists, and job console.

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Table 20-1 (Cont.) Application Settings

Application Setting	Description
Page	Set defaults for indenting members on a page and setting the number of items on the page drop-down. Note that the **Number of Items on the Page Drop-down** option lets you shorten the member list so that the **Search** box can be more easily seen. If the list is shortened to 10 members, for example, then you won't need to scroll to see the **Search** box.

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Table 20-1 (Cont.) Application Settings

Application Setting	Description
Other Options	Set these other configuration options: • **Date Format** • **Attribute Dimension Date Format** • **UI Display**: Choose **Standard Interface** or **Simplified Interface**. • **Partial Grid Fetch Size (Rows,Columns)**: Enter number of rows and columns in the format `nn,nn` • **Suppress Application Management Options in Smart View** • **Enable Data Load for Ad Hoc Read-Only Role**: Default is **No**. Select **Yes** to enable data load for users with Adhoc read-only role. • **Set Number of Seconds Before Rules Run in Background**: Enter a value between 0 and 600. **Note:** Rules that are executed from within the form's context are executed in synchronous mode and will not run in background. This applies to rules that are automatically run after save, run before load, or rules that are run from the form's context menu or **Actions** menu. • **Client Log Level**: All error, warnings, and informational messages from the connected data source are displayed when they occur, but you can choose which of these message levels to record in a browser console log file. This setting is only available for web interfaces using Oracle JET technology. Select a message level to display and record: – **None**: Suppress all messages. – **Information**: All messages, including warnings and errors—recommended to diagnose problems. May adversely impact performance. – **Warning**: Warnings and error level messages. May adversely impact performance. – **Error**: Error messages only—recommended for general use. Has minimal impact on performance. – **General**: Information-level messages plus all server responses and requests. Adversely impacts performance. • **Enable User Formulas in Ad Hoc**: If you select **Yes**, in an ad hoc grid, when you hover over a cell with a member formula attached, you see the member formula associated with the cell and you can perform calc on the fly. • **Filter Out Excluded Members in Segment Drop-down**: Default is **Yes**. Choose **No** to display the excluded members in the row drop-down POV in forms. • **Forms Version**: Specify a forms version. For existing applications, the default is **Forms 1.0**. Newly created applications will default to **Forms 2.0**. This setting is application-specific and can vary across applications. • **Legacy List Pages**: Reverts the standardized artifact pages (also known as list or listing pages) to legacy artifact pages. The default is **No**, which displays the standardized artifact pages. Choose **Yes** to revert to the legacy artifact pages. See About the Standardized Cloud EPM Platform Artifact Pages.

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Table 20-1 (Cont.) Application Settings

Application Setting	Description
Predictive Planning	Predictive Planning and IPM Insights options for specifying: • The **Prediction Interval** to use for worst case and best case values. Also applies to Auto Predict. • The **Prediction Cube** to select the cube the forms are associated with, where you want to perform predictions. • The **Source Cube** to select the cube where historical actuals are stored. This can be the same cube as the Prediction cube, or a different cube. • The **POV** for the source cube. You can select members from all dimensions when defining the POV for the historical data source. If you don't specify a member for a dimension, during predictions, Predictive Planning uses the members in the form's POV. The **POV** option is also used to define members associated with predictions and prediction insights for **Dynamic Calc** members. See Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions. • For prediction insights that are generated, the historical data is displayed based on the **POV** selection for the **Source** cube. • The **Measure Dimension**. For FreeForm applications, to select the dimension that stores measures, for use with IPM Insights. You can select the **Measure Dimension** for each cube in the application. Note the following: • If you are using different cubes for source data and predictions, they must be in the same application. • For each prediction cube, you can specify a source cube (which can be the same as the prediction cube or a different cube) and the POV for the source cube. • You can select the POV for each cube in the application.
Digital Assistant Settings	If you have implemented Oracle Digital Assistant for Enterprise Performance Management, specify configuration settings for the assistant. **Note**: These settings are part of a larger configuration process to enable you to work with Oracle Digital Assistant for Enterprise Performance Management. You must complete the configuration steps before using the assistant. See Configuring Applications in *Getting Started with the Digital Assistant*. • **Channel ID**: Enter the channel ID generated when you created the Oracle Web channel for Oracle Digital Assistant for Enterprise Performance Management. • **Service Name**: Enter the Digital Assistant URL, which is the Oracle Digital Assistant Service URL that you see when you log on to the service. Enter the URL without either http:// or https:// preceding it.

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Table 20-1 (Cont.) Application Settings

Application Setting	Description
Smart View Add-on	Enable these Smart View-related options, as required for your environment: • **Google Sheets**: Select to enable support for Smart View in Google Sheets. When selected, users can connect to their applications from Google Sheets and use Smart View features to view and analyze data in Google Sheets. • **Keep Invalid Members/Comments**: Select to allow users to retain invalid members and comments on ad hoc grids in Smart View after a refresh. For example, users may enter member names incorrectly in ad hoc grids or add comments within or outside the grid. Selecting this option for the application allows users to refresh the grid and retain the invalid members and the comments, along with any data they may have already entered. Smart View users can set a cell style for comments, which will call out the invalid members and comments. When this option is not selected, then the invalid member reverts to the closest previous member in the grid, with the exception of the first row or first column in the grid, which is retained as a comment. Blank rows or columns inserted into the grid remain blank.
Oracle Guided Learning Settings	If you have implemented Oracle Guided Learning, specify configuration settings. **Note:** These settings are part of a larger configuration process to enable you to work with Oracle Guided Learning. You must complete the configuration steps before using Oracle Guided Learning. See Integrating Cloud EPM and Cloud EDM with Oracle Guided Learning. • **OGL Application ID**: Enter the application ID. • **OGL Server URL**: Enter the URL, which depends on your location: – NA: `https://guidedlearning.oracle.com` – EMEA: `https://guidedlearning-emea.oracle.com` – APAC: `https://guidedlearning-apac.oracle.com`
Enable AI	Enable the use of AI in applications: Select **Generative AI** to enable these abilities: • Automatically generate narrative summaries for a single insight or group of insights. • Use GenAI to determine how to handle negative values and decimal values in Auto Predict and Predictive Planning. • Enable Gen AI narrative summaries for Reports. Select **Advanced Predictions** to enable AI data analysis for advanced, multivariate predictions.

Table 20-2 System Settings

System Setting	Description
Display Users' Full Names	When selected, the system displays the user's full name (for example, Max Hennings). When cleared, the system displays the user's ID (for example, VHennings).

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Table 20-2 (Cont.) System Settings

System Setting	Description
Include Shared Members in Cube Refresh	When selected, shared members will inherit the highest security access based on a combination of the access assigned to the base member and parent of shared member. When cleared, shared members will inherit the security access assigned to the base member.
Upper Level Write Back (FreeForm only)	In FreeForm applications only: When selected, you can enter and adjust data directly at parent (upper-level) members in FreeForm application grids without custom scripting or workarounds. This accelerates enterprise planning by enabling fast, intuitive top-down data entry directly at summary levels, improving efficiency, flexibility, and alignment with organizational targets.
Default Ancestor Access	Select an option to specify the default user access to ancestor members in ad hoc grids and the Member Selector: • **None**: By default, users are not able to see the ancestor members of members that they have access to unless they are explicitly granted access to them. When this option is selected, the system works the same way it did prior to the 24.03 update. • **Read**: By default, users are granted Read Only access to the ancestor members of members that they have access to unless they are explicitly granted **Write** or **Display** access, which would override this default. • **Write**: By default, users are granted Write access to the ancestor members of members that they have access to unless they are explicitly granted **Read** or **Display** access, which would override this default. • **Display**: By default, users are granted Display Only access to the ancestor members of members that they have access to unless they are explicitly granted **Read** or **Write** access, which would override this default. **Note:** Members with Display Only access will be displayed, but instead of data values in the cells associated with those members, users will see #NoAccess. See Accessing Ancestor Members in Ad Hoc Grids in *Working with Oracle Smart View for Office* for more information on working with the **Default Ancestor Access** setting.
Email Character Set	Select **UTF-8** or **Regional Setting**.
Business Rules Notification	If set to **Yes**, notifies users or groups when rules (which are enabled for notification in Calculation Manager) are completed or encounter errors. In **Notify These Users**, select the users or groups to notify.
Allow Drill Down on Shared Members in Ad Hoc	• **Yes** enables drilling on shared members in an ad hoc grid. When set to **Yes**, the **Drill to All Levels in Base** check box is displayed. Note that block suppression is not supported when this option is set to **Yes**. To disable block suppression: – In Oracle Smart View for Office, in **Options**, **Data Options**, disable **Suppress Missing Blocks**. – In the web, in **Preferences**, **Ad Hoc Options**, disable **Missing blocks on rows**. • **No** disables drilling on shared members in an ad hoc grid.

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Table 20-2 (Cont.) System Settings

System Setting	Description
Drill to All Levels in Base	Allows ad hoc grid users to drill down (or zoom in) from shared members to all levels within the base hierarchy. This check box option applies to Standard-mode applications only. The **Drill to All Levels in Base** check box appears when **Allow Drill Down on Shared Members in Ad Hoc** is set to **Yes**. A shared member is often a parent member in its base hierarchy. With the **Drill to All Levels in Base** check box enabled, users can drill down from the shared member to the base hierarchy using the zoom options, **All Levels** or **Bottom Level**. When the **Drill to All Levels in Base** check box is disabled, the user may zoom in to the various levels of the base hierarchy one level at a time using the **Next Level** zoom option. Multi-cell zoom-in on shared members is not supported.
Enable Use of the Application for	Determines whether users can access the application in administration mode, such as during backups. When you select **Administrators**, if any non-administrative users are logged on to the application, they are forced off the system and will not be able to log on. To restore access to an application for all users, select **All users**.
Assign Application Owner	Assigns ownership of the application to another Service Administrator. For more information about application ownership, see Managing Application Ownership.
Enable the Display of Substitution Variables	Set how substitution variables display in the Member Selection dialog box when users respond to runtime prompts in business rules. **Display All** displays all substitution variables. **Display None** displays no substitution variables. **Enable Filtering** displays only substitution variables that are valid for the runtime prompt.
Suppression Mode	Choose a suppression behavior for ad hoc grids in the web and in Smart View for cases where rows and columns contain missing data or zeroes. • **Suppress Missing values only** (default): Suppresses rows, or columns, or both that contain No Data/Missing. • **Suppress Missing also Suppresses Zeros**: Suppresses rows, or columns, or both that contain both No Data/Missing and Zero.
Smart View Ad Hoc Behavior	All new and recreated applications are automatically set to **Standard**. For existing and migrated applications, choose to enable enhanced ad hoc features and behaviors in Smart View: • **Native** (default): Does not enable enhanced ad hoc features. Supported for all Smart View releases. **Note:** Starting in 25.01, the **Native** mode option may be used, but is no longer supported. Oracle recommends converting your application to **Standard** mode. • **Standard**: Enables enhanced ad hoc features. Supported for Smart View release 11.1.2.5.900 and later. For a complete description of the enhanced ad hoc features available in Standard mode, see Smart View Behavior Options in Cloud EPM in *Working with Oracle Smart View for Office*.

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Table 20-2 (Cont.) System Settings

System Setting	Description
Use All Alias Tables on Refresh	Specify whether alias names entered into ad hoc grids are evaluated using the currently selected alias table or against all alias tables • **Yes**: The input is evaluated against all alias tables and all member names. • **No**: The input is evaluated against the currently selected alias table and all member names. This is the default setting. If the system cannot identify the input as a valid member name or alias, the input will be displayed as a comment.
Export Cloud EPM Smart List textual data during daily maintenance for incremental data import	Choose whether to perform a complete export during the daily maintenance process or to create an application backup: • **Yes**: Performs a complete export, such that data, including the business process Smart List data, can be incrementally imported to an application (this option may lengthen the maintenance process duration) • **No** (default): Creates an application backup during the maintenance process, such that data can be used as part of a full restoration Note that this setting applies only to Oracle Essbase that does not support hybrid. For more information, see Exporting Smart List Textual Data During Daily Maintenance for Incremental Data Import in *Oracle Fusion Cloud EPM Operations Guide*.
Link Accounts by Default	For block storage (input) cubes, select whether to XREF linked account members by default: • **Yes** (default): XREFs will be created on account members, and the application will work the same way it has in earlier releases. • **No**: XREFs will not be created for account members, which may improve the application's performance. With **No** selected, after **Cube Refresh** is run, all existing XREFs on account members will be deleted, and non-source cubes will no longer show data from the source cube. **Note:** `HSP_LINK` and `HSP_NOLINK` UDAs on specific account members override the XREF setting for those account members. For example, if this option is set to **No** and you use the `@XREF` function to look up a data value in another cube to calculate a value from the current cube, you can add the`HSP_LINK` UDA to such members to create the `@XREF` function only for these specific members. If this option is set to **Yes**, HSP_NOLINK works the same way it worked in earlier releases and prevents XREFs from being created on specific members.
Attribute Dimension Reorder Threshold	Enter a threshold value between 0 and 500 (500 is the default value). For attribute dimensions, when the number of members under a given parent exceeds the specified threshold value, then the members will be reordered at the end of a metadata load instead of during the load. Depending on the specific shape of the attribute dimension, adjusting this number can sometimes have an impact on performance. In general, this setting can be ignored unless attribute dimension load times degrade beyond acceptable levels.

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Table 20-2 (Cont.) System Settings

System Setting	Description
Apply Valid Intersection and Approvals Filtering on Hidden RTPs	Select whether to apply valid intersection and approvals filtering on hidden run-time prompts. • **Yes** (default): Applies valid intersection and approvals filtering on hidden run-time prompts. • **No**: Does *not* apply valid intersection and approvals filtering on hidden run-time prompts. This is an application level setting and applies to all hidden prompts in all rules and rulesets.
Set Reporting Options	If you're using the next-generation Reports reporting solution, Oracle recommends you use this setting to upload any TrueType fonts that your company uses to produce reports. To upload TrueType fonts, click **Report Settings**, click **Manage Fonts**, click (Create), and then select the font files to upload. See Working with the Reports Reporting Solution in *Working with FreeForm*. To access the Gen AI ML Template files, click Report Settings, click Machine Learning Files, and then select one the following Machine Learning (ML) template files: • Describe • Compare • Compare Columns • Compare Column Dimension • Causality • Causality Two Level For more information, see Working with the Machine Learning Template Files in *Designing with Reports* .

Defining User Variables

You can define user variables to help users focus on particular members, such as their department's expenses. For example, you can create a form with entities on the rows, and a user variable called Department. You can limit the number of rows displayed on the form by selecting a member for the Department user variable, such as Sales. Later, you can select another value for Department, such as Marketing.

To add or update user variables:

1. Click **Tools**, and then click **Variables**.
2. Click the **User Variables** tab.
3. To add a user variable, click (Add), to add the new variable to the end of the variables list.
4. To define a member value for a user variable, click (Member Selector) next to the variable, and then select members.
5. Select **Use Context** to use the user variable in a form's POV. With this setting, the value of the user variable changes dynamically based on the context of the form in which the variable is used.

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> **Note**
>
> For Forms 2.0, the server automatically sets default values for dynamic and context-sensitive user variables. You won't need to select values for dynamic and context-sensitive user variables before working in a form. For Forms 1.0, you must select values for all user variables, including dynamic and context-sensitive user variables, before working in the form.

Customizing Your Display

Change the theme of your display or add your company logo or a background image to the Home page.

On the **Appearance** page, you can change the general look and feel of your Oracle Fusion Cloud Enterprise Performance Management environment.

All Cloud EPM services, business processes, and applications use Redwood Experience. The Redwood Experience offers a default Oracle theme and these two additional appearance options: Custom Light and Custom Dark.

In addition to changing the Redwood Experience appearance options, you can customize your Home page by adding a custom branding logo, a custom background image, applying a custom color to the band that displays below the global header, and hiding the business process name. For general information about using the Home page, see About the Home Page.

> **Note**
>
> You can set your profile picture to display at the top of the Announcements panel of the Home page in **User Preferences**. Click **Tools**, and then **User Preferences**.
>
> For more information, see Setting Your Profile Picture in *Working with FreeForm*.

To customize your display, see Configuring Cloud EPM Appearance in *Getting Started Guide for Administrators*.

Announcing Upcoming Events

Create and send announcements to alert users about upcoming events, such as system maintenance or the running of jobs. Announcements are displayed in the Announcements area on the application's Home page.

For more information about the Announcements area, see About the Home Page

To create an announcement:

1. Click **Tools**, and then click **Announcement**.
2. Click **Create**, and enter information such as:
   - A subject that summarizes the purpose of the announcement
   - The start date; when to send the announcement. The end date is optional.

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- The content. You may need to select an editing mode (rich text or source code) first.

Specifying Artifact Labels

The **Artifact Labels** page on the **Tools** cluster enables Service Administrators to customize artifact labels (artifact names, descriptions, and so on) based on the user's browser locale.

A few examples:

- If you create a form with a cryptic name that you don't want displayed to the user, you can define a meaningful name for the form that is displayed in the language of the user.
- If you want to create a useful instruction for an artifact that only displays in the language of the user, for example:

  "This formula calculates the number of regular employees away on a Leave of Absence."

Related Links

- Which Artifact Labels Can be Localized?
- Working With the Artifact Labels Grid
- Adding Languages and Defining Localized Artifact Labels
- Exporting and Importing Artifact Labels for Editing

Which Artifact Labels Can be Localized?

The application supports changing the language for the following artifacts:

- Card
- Cluster
- Dashboard
- Data Map
- Data Validation Rule
- Dimension
- Folder
- Form
- Member
- Menu
- Menu Item
- Navigation Flow
- Cube
- Approval Unit Hierarchy
- Report
- Rule
- Ruleset
- Smart List
- Smart List Entry

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- Tab
- Task
- Task List
- Template
- User Variable
- Valid Intersection

Working With the Artifact Labels Grid

The **Artifact Labels** page displays an Excel-style spreadsheet grid that is filtered by artifact and property type.

The row axis of the grid displays the artifacts and their properties.

The column axis of the grid displays the following columns:

- **Artifact**: The type of artifact (for example, Task List or Rule)
- **Property**: The artifact's property type (for example, Name, Description, and so on)
- **Default**: Displays the artifact labels that were defined when the artifact was created.

When a language is added, a new column displays to the right of the **Default** column.

To view and filter the **Artifact Labels** grid:

1. Click **Tools**, and then click **Artifact Labels**.
2. To filter:
   a. Click (Filter), and then select the artifacts you want to work with. For some artifacts, you can further filter by property type.
   b. Click **Apply** to close the **Filter** window and display the artifact grid filtered by artifact type and property type.

Adding Languages and Defining Localized Artifact Labels

Service Administrators can add a language for a given artifact to the **Artifact Labels** grid from a list of supported languages. You can select only one language at a time. When you add a language, a new column for that language is added to the grid to the right of the **Default** column. The cells in the language-specific column are editable.

> **Tip**
>
> Use this method to add labels directly in the artifact labels grid. This method is ideal if you only need to add or update a few labels at a time. For bulk changes or edits on artifact labels; for example, terminology changes that affect multiple labels, use the export feature to edit in Excel, then import. See Exporting and Importing Artifact Labels for Editing.

To add a language:

1. Click **Tools**, and then click **Artifact Labels**.

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2. Click (Filter), and then select the artifacts you want to work with. For some artifacts, you can further filter by property type.
3. Click **Apply**.
4. Click **Add Language**.
5. Select from the list of supported languages.
6. In the language-specific column, enter artifact labels into the editable cells for each artifact property (Name, Description, and so on).

> **Note**
>
> Using Ctrl+C (Copy) and Ctrl+V (Paste) are not supported in the artifact labels grid.

7. Click **Save**.

> **Note**
>
> When you define a localized artifact label for the Default navigation flow (for example, editing the name of an icon on the Home page), your update will automatically propagate to all navigation flows. However, if you define a localized artifact label for another navigation flow that isn't the Default flow, then that update will override the label coming from the Default flow.

Exporting and Importing Artifact Labels for Editing

You can export all the artifact labels in a given language to edit them. Only artifacts that already have labels will be exported. The labels are exported in an Excel file format (XLSX). After you edit the labels, you can import them back into the application.

> **Tip**
>
> Use this method for bulk changes or edits on artifact labels by language; for example, terminology changes that affect multiple labels. For updates to individual artifact labels, you can edit them directly in the artifact grid. See Adding Languages and Defining Localized Artifact Labels.

To export all artifact labels by language for editing and then import them:

1. Click **Tools**, and then click **Artifact Labels**.
2. Export the XLSX file containing all of the artifact labels:
   a. Click **Actions**, and then **Export**.
   b. Select the target location for the export file:
      - **Local**: Saves the export file to a location on your local computer.
      - **Outbox**: Saves the export file to the server. See Uploading and Downloading Files Using the Inbox/Outbox Explorer.

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c. Choose a language.

d. Click **Export**.

3. Edit the labels in the XLSX file.

4. Import the XLSX file:

a. Click **Actions**, and then **Import**.

b. Select the location of the import file:

- **Local**: Loads the import file from a location on your computer. For **Source File**, click **Browse** to select the import file on your computer for the artifact you're importing.
- **Inbox**: Loads the import file from the server. Enter the name of the file in **Source File**. See Uploading and Downloading Files Using the Inbox/Outbox Explorer.

c. Click **Import**.

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21

Accessing More Administrative Tasks

Use the information in this chapter to perform additional administrative tasks.

Related Links

- About the Navigator Menu
- Administering Data Load Settings
- Importing Using Data Integration
- Administering Action Menus
- Administering Alias Tables
- Administering Dimensions
- Administering Forms
- Administering Rules
- Administering Rules Security
- Administering Smart Lists
- Administering Task Lists
- Setting User Preferences
- Working with Variables
- Clearing Cell Details
- Copying Data
- Administering Application Monitoring

About the Navigator Menu

You can access additional administrative tasks from the Navigator Menu.

Clicking **Navigator** on the Home page displays a list of links that connect you to more business process functionality.

> **Note**
>
> Some of the links are available only if you're accessing the business process from the desktop.

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Administering Data Load Settings

Specify parameters to enable data to be loaded directly into an application database. Optionally, you can use advanced settings if you want to load details to child members of parent dimension members based on unique identifiers for the driver dimension.

For example, a company might load the Employee dimension member with account data for Start Date, Position, Salary Basis, and Pay Type. Because the human resource data includes placeholders for new and existing employees, the company could set up the following advanced settings:

- Data load dimension parents: New Employees, Existing Employees
- New Employees unique identifiers: Start Date, Position
- Existing Employees unique identifiers: Salary Basis, Pay Type

During data load, child members of New Employees and Existing Employees are evaluated for data updates. The unique identifiers Start Date, Position, Salary Basis, and Pay Type determine if existing data load dimension values are updated, or if new values are added: If the unique identifier's data values are the same, data is updated. If the data value is different, the next available child member is used.

To specify parameters for loading data:

1. From the Home page, click **Navigator** , and then under **Integration**, click **Data Load Settings**.
2. For **Data Load Dimension**, select the dimension (such as Employee) for which data is loaded for the application.
3. For **Driver Dimension**, click (Select Members) to select the dimension into which data is loaded.

   For example, if you're loading data to Employee, the driver dimension might be Account.
4. Select the members of the driver dimension.

   For example, if the driver dimension is Account, driver dimension members might include Start Date, Grade, Position, Salary Basis, and Pay Type.
5. **Optional:** To use advanced settings, complete these steps.

   a. Add a row by clicking (Add New Row).

   b. To the right of the new field, click (Select Data Load Dimension Parent) and select a parent member.

      For information about selecting members, see Using the Member Selector.

   c. To the right of the parent member, under **Driver Dimension Unique Identifiers**, select members as unique identifiers. (Members selected for this field must be included in the list of selected Driver Dimension members at the top of the page.)

      Each parent member must include at least one unique identifier member. These members determine if existing data load dimension values are updated, or if new values are added.

   d. If necessary, continue adding rows by repeating the previous steps.

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e. To duplicate or delete a row, click within a row, and click (Duplicate Row) or (Remove Row).

Importing Using Data Integration

Data Integration is the mechanism by which integration processes are performed in Oracle Fusion Cloud Enterprise Performance Management. Designed for busy administrators and users, you can define file-based and direct integration sources, create mapping rules to translate source data into the required target format, and execute and manage the periodic data loading process. Common integration tasks are done using an easy-to-navigate interface that supports and conforms to how you work.

For a general understanding of the integration process, see Defining a Data Integration in *Administering Data Integration*.

Tutorials

Tutorials provide instructions with sequenced videos and documentation to help you learn a topic.

Your Goal	Learn How
Learn how to register applications, define period mappings, define category mappings, and then create and run a file-based data integration.	Loading Data Using Data Integration

Administering Action Menus

Related Topics

- Creating and Updating Action Menus
- Working with Action Menu Items
- Defining Action Menu Items
- Enabling Action Menus for Ad Hoc Grids

Creating and Updating Action Menus

Service Administrators can create right-click menus (also called action menus or context menus) and associate them with forms, enabling users to click rows or columns in forms and select menu items to:

- Launch another application, URL, or business rule, with or without runtime prompts
- Move to another form
- Add Dashboard

You can also create global action menus that can be enabled for ad hoc grids. See Enabling Action Menus for Ad Hoc Grids.

The context of the right-click is relayed to the next action: the POV and the Page, the member the user clicked on, the members to the left (for rows), or above (for columns).

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When designing forms, use **Other Options** to select menus available for Form menu item types. As you update an application, update the appropriate menus. For example, if you delete a business rule referenced by a menu, remove it from the menu.

To create, edit, or delete action menus:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Action Menus**.
2. Perform an action:
   - To create an action menu, click **Actions**, then **Create Menu**, enter the **Menu Name**, and then click **OK**.

     Select **Enable for Ad Hoc** to create an ad hoc menu which is a global action menu that can be used in ad hoc grids. See Enabling Action Menus for Ad Hoc Grids.
   - To add menu items to an action menu or to edit details of an action menu, select a menu, click **Actions**, and then click **Edit Menu**. See Working with Action Menu Items.
   - To delete action menus, select a menu or menus, click **Actions**, click **Delete Menu**, and then click **OK**.

> **Note**
>
> You cannot delete an action menu that is associated with a form. If you try to delete the menu, you'll be prompted to remove the menu from the form. Then you can delete the menu.

Working with Action Menu Items

The **Edit Menu** page displays menu items on the current action menu, including names, labels, required dimensions, icon, and type, such as URL, Form, Business Rule, Dashboard, and Menu Header. You can also edit ad hoc menus or convert existing action menus into ad hoc menus.

To work with action menu items:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Action Menus**.
2. Select a menu, click **Actions**, and then click **Edit Menu**.
3. **First time only:** To add the first item to the menu, click **Add Child** and **Save**.
4. Select a menu item and:
   - To add menu items below the selected item, click **Actions**, then **Add Child** (available for Menu Header menu types).
   - To add menu items at the same level as the selected item, click **Actions**, then click **Add Sibling**.
   - To edit menu items and define the menu item properties, click **Actions**, then click **Edit Menu Item**. For descriptions of the menu item properties, see Defining Action Menu Items.

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- To delete menu items, click **Actions**, then click **Delete Menu Item**.
- To change the order of menu items within the same level, click **Move Up** or **Move Down**. You can move multiple items.

5. Click **Save**.

   Click **Save As** to save the current selections under a new menu name.

Defining Action Menu Items

To define action menu items:

1. From the Home page, click **Navigator** , and then under **Create and Manage**, click **Action Menus**.
2. Select a menu, click **Actions**, and then click **Edit Menu** .
3. Select the menu item, click **Actions**, and then click **Edit Menu Item** or **Add Sibling**.
4. Define the menu item:

Table 21-1 Edit Menu Item Options

Item	Description
Menu Item	Enter a unique name containing only alphanumeric and underscore characters, with no special characters or spaces
Label	Enter text to be displayed when the menu is selected. Spaces and special characters are allowed. Menu labels display in the user interface. Labels can be text or can reference a resource variable by name. For example, to set a menu's label to File, set it to `File` directly or to the name of a resource, such as LABEL_FILE, which can be localized.
Icon	This option is not currently supported.
Type	Select the menu item type to determine available properties. **Note:** No properties are available for Menu Header. • **URL**: Create a menu that opens the specified URL. • **Form**: Create a menu that launches the selected form. The member selection context for the member, page, and POV is retained when users right-click in the source form. If the target form contains these dimension members on the page, its page is set to match the context. When launched from the web, the form opens in the web; when launched from Oracle Smart View for Office, the form opens in Smart View. Action menus referencing forms enabled as flex forms will open these forms as flex forms in Smart View and as simple forms in the web. • **Business Rule**: Create a menu that launches the selected business rule. • **Dashboard**: Create a menu that launches the selected dashboard. The cell context is retained when users right-click in the source form, and the dashboard is opened in a new dynamic tab. • **Menu Header**: Create a menu under which you can create children menu items. To display a separator bar on the menu at this item, enter one hyphen as the Label. In this case, the Required Dimension list isn't available.

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Table 21-1 (Cont.) Edit Menu Item Options

Item	Description
Required Parameters	Select a dimension and member, or select an option for where the menu item displays: Point of View, Page, Row, Column, Members Only, Cell Only. For example, if you select Account, users can right-click Account members on a form to open the menu. If you select Row, the menu is available when users right-click a row. Selecting None makes the menu available whenever the user right-clicks in the form. **Note:** In Smart View, for action menus attached to forms (simple forms or flex forms), the Page and POV options are not supported.

5. Define menu item properties, which differ for menu item types:

Table 21-2 Options for Menu Item Types

Type	Options

URL	a. In **URL**, enter the complete URL to which to direct the user. For example: `https://`*server name*`/HFM/Logon/HsvLogon.asp`. URL type action menus automatically launch a new tab.
	b. Select **Use Form Context** to replace the bracketed dimension name in the URL (for example, `<Entity>` or `<Account>`) with the member name of the corresponding dimension from the form's page or POV.
	For example, to return Entity, Scenario, Version, and Account dimension's members in the URL, enter the URL and enable **Use Form Context**:
	`https://yourcompanyurl/EntDim=<Entity>&test['VERSION']=<Version>&Acc=<Account>`
	The URL will open in a new tab as:
	`https://yourcompanyurl/EntDim=<410>&test['VERSION']=<working>&Acc=<1110>`
	If **Use Form Context** is disabled, then the URL will be launched in a new tab as is without the context replacements.

Form	a. In **Form Folder**, select the folder containing the destination form. b. In **Form**, select the form.

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Table 21-2 (Cont.) Options for Menu Item Types

Type	Options
Business Rule	a. In **Cube**, select the cube for which the business rule is available. b. In **Business Rules**, select the business rule to launch. c. In **View Type**, select how to display runtime prompt pages: • **Classic View**: Use the default application view • **Streamline View**: Display each runtime prompt on a different line d. **Optional**: In **Window Title**, enter a title to display instead of Runtime Prompts. e. **Optional**: In **OK Button Label**, enter the text to display for the OK button. f. **Optional**: In **Cancel Button Label**, enter the text to display for the Cancel button. g. **Optional**: In **Launch Confirmation Message**, enter text to display when the business rule is invoked, but before it's launched. This option enables Service Administrators to provide meaningful messages to users about the consequences of launching business rules.
Dashboard	In **Dashboard**, select the dashboard.

6. Click **Save**.

Enabling Action Menus for Ad Hoc Grids

You can create a global action menu, called an ad hoc menu, that is enabled for all ad hoc grids. An ad hoc menu works within the context of one of more cubes. You must associate the ad hoc menu with one or more cubes, and then the menu can be viewed in all ad hoc grids that use those cubes. For example, you can define an ad hoc menu that launches a business rule. When you right-click a data cell in any ad hoc grid that uses the selected cube, the business rule menu option displays and you can run the rule.

You can also designate an existing action menu as a global ad hoc menu.

> **Note**
>
> Global ad hoc menus are also enabled for Oracle Smart View for Office ad hoc grids. See Right-Click Context Menu in *Working with Oracle Smart View for Office*.

To create an ad hoc menu:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Action Menus**.
2. Perform an action:
   - To create a new ad hoc menu:
     a. Click **Actions**, then **Create Menu**, and then enter the **Menu Name**.

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b. Select **Enable for Ad Hoc**, and then either keep the default **All** cubes, or select one or more cubes for the ad hoc menu. Ad hoc menus will only work within the context of the defined cubes.

c. Click **OK**.

- To make an existing menu an ad hoc menu or to save an existing menu as an ad hoc menu:

  a. Select a menu, click **Actions**, and then click **Edit Menu** .

  b. Select **Enabled for Ad Hoc**, and then either keep the default (whereby all cubes are enabled for ad hoc menus), or click **Select Cube** to select one or more cubes for the ad hoc menu. Ad hoc menus will only work within the context of the defined cubes.

  c. Click **Save** to update the current selection. Click **Save As** to save the current selections under a new menu name.

3. The new menu is added to the **Action Menus** list.

   To add menu items such as URL, Form, Business Rule, Dashboard, and Menu Header to ad hoc menus, see Working with Action Menu Items.

Administering Alias Tables

Related Topics

- About Aliases
- About Alias Tables
- Creating an Alias Table
- Defining the Columns that Display on the Alias Tables Page
- Editing, Copying, Clearing, and Deleting Alias Tables
- Specifying a Default Alias Table and Setting Member and Alias Display Options

About Aliases

You can assign alternate names, or aliases, to Account, Currency, Entity, Scenario, Period, Version, Years, and user-defined dimension members. The application allows up to 30 aliases per dimension member, including the default alias. Aliases can have the same name within an alias table and across alias tables.

Aliases can also have:

- The same name as a member
- The same alias on members that are parent and child
- The same name for members from different dimensions or from the same dimension

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> **Note**
>
> - You can't have the same alias for two members that are siblings because there would be no way to uniquely identify the member. This rule is enforced by the application for base members, but not for shared members.
>
> This rule isn't enforced for shared members because you can't directly set aliases for shared members; shared member aliases are inherited from the alias of their base member. It's possible to create an alternative hierarchy where you can have two shared members that have the same alias and are siblings. However, this situation is discouraged if you want to reference these members by their aliases from an adhoc grid because, when you type the alias into the grid and submit it to the application, the application can't uniquely resolve this member and it will return an error. If you're using the alias for display purposes only then there will be no issue, but this design is discouraged because, visually, there is no way to differentiate between these two members.
> - Although aliases can have the same name as a member, be careful not to set the alias of member1 to be the same name as member2. This can lead to unintended results and will cause confusion on the form grids.
> - Member names must be unique so that they can be used in rules and form designs.

About Alias Tables

You can create and update alias tables, and set a default alias table for the application.

Follow naming conventions in Naming Restrictions.

Multiple alias tables support, for example, these language combinations:

- English, French, German, Spanish, and Italian
- Japanese and English
- Korean and English
- Turkish and English

> **Note**
>
> Alias table support isn't restricted to these language combinations.

You can set alias tables to display members in an application. Users can set alias tables in preferences.

Creating an Alias Table

To create an alias table:

1. From the Home page, click **Manage Artifacts**, and then click **Alias Tables**.

   You can also access the **Alias Tables** page from the Navigator by clicking the **Alias Tables** links under **Manage Artifacts**.

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2. On the **Alias Tables** page, click **Create**.
3. In the **Add - Alias Table** dialog box, enter a name for the alias table.

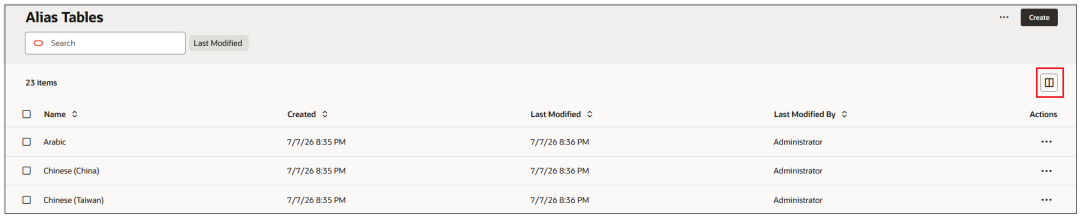
   Follow the naming conventions in Naming Restrictions.
4. Click **Save**.

Defining the Columns that Display on the Alias Tables Page

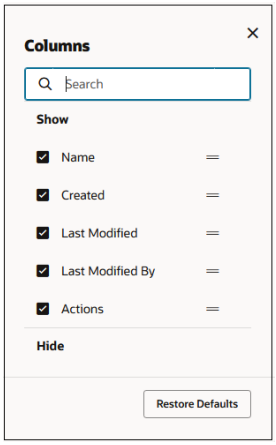
You can define the columns that display on the **Alias Tables** page, and you can define the order in which the columns display.

To define the column display:

1. From the **Alias Tables** page, click ◫.



2. Select or de-select the desired columns to show or hide the columns.



3. To change the order in which the columns will display, click and drag ═ to the right of a column to move the column up or down. To restore the columns to their default display, click **Restore Defaults**.
4. When you have finished defining the columns that will display and the order in which they will display, click ✕ to close the **Columns** dialog box.

Editing, Copying, Clearing, and Deleting Alias Tables

To edit, copy, clear, or delete an alias table:

1. Select an alias table and click ••• in the **Actions** column.
2. Select an action:
   - **Edit**

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- **Copy**
- **Clear**
- **Delete**

> **Note**
>
> - When copying an alias table, the destination alias table must exist. Copying does not create a table.
> - Clearing an alias table removes the contents of the table, but does not remove the table.
> - You cannot delete the default alias table. (See Specifying a Default Alias Table and Setting Member and Alias Display Options).

Specifying a Default Alias Table and Setting Member and Alias Display Options

If you create alias tables with aliases for Account, Currency, Entity, Scenario, Period, Version, Years, and user-defined dimensions and members, you can select a default alias table for the application. Users can set preferences for which set of aliases (stored in an alias table) to use for displaying member and dimension names.

To select the application's default alias table:

1. From the Home page, click **Application**, and then click **Settings**.
2. For **Alias Table**, select an alias table.
3. For **Display Member Label as**, select the option that enables the kind of member data to be displayed in the member selector throughout the application:
   - **Default**: The data determined by the form, grid, or dimension settings
   - **Member Name** : Just member names
   - **Alias**: Just member aliases, if defined
   - **Member Name:Alias**: Names followed by aliases, if defined
   - **Alias:Member Name**: Alias, if defined, followed by the names
4. Click **Save** or **Reset**.

Administering Dimensions

This section documents how to edit dimensions using the Classic Dimension Editor which is accessed by using the **Dimensions** link in the Navigator menu. In the 17.05 (May 2017) update, we released the Simplified Dimension Editor.

For detailed information about using the Simplified Dimension Editor, see Editing Dimensions in the Simplified Dimension Editor.

Related Links

About Dimensions

Working with Dimension Hierarchies

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About Custom Dimensions, Entities, Accounts, Periods, and Cubes

Adding or Editing User-Defined Custom Dimensions

Working with Members

Working with Attributes

Working with Attribute Values

Customizing Calendars

Setting up Dynamic Time Series Members

Working with UDAs

Working with Member Formulas

About Dimensions

Dimensions categorize data values.

FreeForm enables you to create an application with the cubes and dimensions of your choice without being constrained by the cube and dimension limitations imposed by standard applications. You can add up to 12 total cubes with any combination of aggregate storage and block storage. See Understanding FreeForm.

Members are components of dimensions.

For complete information about dimensions, see Dimension Overview.

Working with Dimension Hierarchies

Related Topics

- Filtering the Dimension View by Cube
- Sorting Members
- Viewing a Member's Ancestors
- Determining Where Members Are Used in an Application

Filtering the Dimension View by Cube

You can filter the dimension view by cube. When you select a cube, only dimensions used in that cube are displayed on the **Dimensions** page.

To filter the dimension view by cube:

1. From the Home page, click **Navigator** , and then under **Create and Manage**, click **Dimensions**.
2. For **Cube**, select the cube.

The application displays only the dimensions used in the selected cube.

Sorting Members

You can sort members in ascending or descending order, by children or descendants. Sorting members affects the outline.

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To sort members:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select the dimension for the members.
3. On **Dimensions**, select the members whose children or descendants you want to sort.
4. For **Sort**, select children or descendants.

   Sorting by children affects only members in the level immediately below the selected member. Sorting by descendants affects all descendants of the selected member.
5. Click (Sort Ascending) to sort by ascending order or (Sort Descending) to sort by descending order.
6. Click **OK**.

   The next time you create or refresh the database, the outline is generated with members in the order that is displayed.

Viewing a Member's Ancestors

To view a member's ancestors:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select a dimension.
3. Select the member in the dimension hierarchy.
4. Click (Show Ancestors).
5. Click **OK**.

Determining Where Members Are Used in an Application

To view where members are used in an application:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the dimension whose member's usage you want to view.
3. Click (Show Usage).

About Custom Dimensions, Entities, Accounts, Periods, and Cubes

See the following topics for more information:

- About Custom Dimensions
- About Entities
- About Accounts
- Accounts, Entities, Periods, and Cubes

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Adding or Editing User-Defined Custom Dimensions

User-defined custom dimensions must conform to guidelines listed in Naming Restrictions.

Table 21-3 Properties for User-Defined Custom Dimensions

Property	Value
Dimension	Enter a name that is unique across all dimensions.
Alias	**Optional**: Select an alias table. Enter an alternate name for the dimension. See About Aliases.
Description	**Optional**: Enter a description.
Valid for Cubes	Select cubes for which the dimension is valid. Clearing this option makes all members of the dimension invalid for the deselected cube.
Apply Security	Allow security to be set on the dimension members; must be selected before assigning access rights to dimension members. Otherwise, dimensions have no security and users can access members without restriction.
Data Storage	Select a data storage option. The default is **Never Share**.

To add or change user-defined dimensions:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Click (Add Dimension) or select an existing dimension and click (Edit Dimension).
3. Specify any of the properties listed above.
4. Click **Save**.
5. Click **OK**.

   Click **Refresh** to revert to the previous values and keep the page open.

- Setting Dimension Properties
- Setting Dimension Density and Order
- Setting the Evaluation Order

Setting Dimension Properties

Dimension properties must conform to guidelines listed in Naming Restrictions.

Table 21-4 Dimensions Properties

Property	Value
Dimension	Enter a name that is unique across all dimensions.
Description	**Optional**: Enter a description.
Alias Table and **Alias**	**Optional**: Select an alias table. Enter an alternate name for the dimension. See Administering Alias Tables.
Valid for Cubes	Select cubes for which the dimension is valid. Clearing this option makes all members of the dimension invalid for the deselected cube.

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Table 21-4 (Cont.) Dimensions Properties

Property	Value
Two Pass Calculation	Recalculate values of members based on values of parent members or other members. Available for Account and Entity members with Dynamic Calc or Dynamic Calc and Store properties.
Apply Security	Allow security to be set on the dimension members; must be selected before assigning access rights to dimension members. Otherwise, dimensions have no security and users can access members without restriction.
Data Storage	Select a data storage option. The default is **Never Share**.
Display Option	Set application default display options for the **Member Selection** dialog box. Select **Member Name** or **Alias** to display members or aliases. **Member Name:Alias** displays members on the left and aliases on the right. **Alias:Member Name** displays aliases on the left and members on the right.

Setting Dimension Density and Order

The **Performance Settings** tab enables you to set dimensions as sparse or dense and set their order of precedence.

To manage performance settings:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the **Performance Settings** tab.
3. For each dimension, set its **Density** as **Dense** or **Sparse**.

   See About Sparse and Dense Dimensions.

> **Note**
>
> The **Density** column is hidden for applications that contain only ASO cubes. If an application contains a BSO cube or a BSO and an ASO cube, the **Density** column displays for both cubes.

4. Set the order of precedence by selecting a dimension and clicking (Move Up) or (Move Down) next to the **Position** column heading.

Setting the Evaluation Order

The **Evaluation Order** tab enables you to specify which data type prevails when a data intersection has conflicting data types.

> **Note**
>
> Account members that require Text, Date, or Smart List data types, require the Account dimension to have an Evaluation Order set for the cube or cubes in question.

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To set evaluation order:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select **Evaluation Order**, and then select the cube.
3. Select an action to move selected dimensions into or out of **Selected Dimensions**:
   - (Move selected items to) moves selected dimensions
   - (Move all items to) moves all dimensions
   - (Remove selected items from) removes selected dimensions
   - (Remove all items from) removes all dimensions

   You need to select only dimensions whose members have specific data types (that is, their data type isn't "Unspecified"). The data type "Unspecified" doesn't conflict with another data type.
4. Set the order of precedence for dimensions in **Selected Dimensions** by clicking the move arrows to the right of the **Selected Dimensions** pane.
5. Click **Save**.

Working with Members

You can assign access rights to members, rearrange the dimension member hierarchy, share members of the Entity, Account, and user-defined custom dimensions, and enable creation of dynamic members "on-the-fly."

- Finding Dimension Members
- About Assigning Access to Members
- Adding or Editing Members
- Deleting Members
- Deleting Parent Members
- Working with Shared Members
- Creating Shared Members
- About Dynamic Members

Finding Dimension Members

To find dimension members in dimension hierarchies:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select the dimension for the member.
3. For **Search**, select **Name**, **Alias**, or **Both**.
4. Enter the search text (member name, alias, or partial string) for which to search.
5. Click (Search Down) or (Search Up).

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About Assigning Access to Members

Service Administrators can assign permissions to members.

You can assign permissions to members by selecting the dimension property **Apply Security**. If you omit or clear the **Apply Security** setting, all users can access the dimension's members. By default, the Account, Entity, Scenario, and Version dimensions are enabled for access permissions. Optionally, you can enable this option for Period, Years, and Custom dimensions.

For complete information, see Assigning Access to Dimension Members. To enable access to members, see Editing Dimension Properties.

Adding or Editing Members

Members must conform to guidelines listed in Naming Restrictions. Shared members must be consistent with Working with Shared Members.

Table 21-5 Member Properties

Property	Value
Name	Enter a name that is unique across all dimension members.
Description	**Optional**: Enter a description.
Alias Table	**Optional**: Select the alias table to store the alias name. Enter an alternate name for the member in **Alias**. See Administering Alias Tables
For Account members only: **Account Type**	Select **Expense**, **Revenue**, **Asset**, **Liability**, **Equity**, or **Saved Assumption**. For descriptions, see Account Types.
For Account members only: **Variance Reporting**	If the account type is **Saved Assumption**, select **Expense** or **Non-Expense**. Designate the saved assumption as a revenue, asset, liability, or equity account.
For Account members only: **Time Balance**	Select **Flow**, **First**, **Balance**, **Average**, **Fill**, **Weighted Average - Actual_Actual**, or **Weighted Average - Actual_365**. For descriptions, see Time Balance Property.
For Account members only: **Skip**	If the account type is **Asset**, **Equity**, or **Liability**, select **None**, **Missing**, **Zeros**, or **Missing and Zeros**. For descriptions, see Setting Account Calculations for Zeros and Missing Values.
For Account members only: **Exchange Rate Type**	Select **Average**, **Ending**, or **Historical**. For descriptions, see Data Type.
For Account members only: **Data Type**	Select **Percentage**, **Date**, or **Text**. For descriptions, see Data Type.
For Account members only: **Distribution**	Sets the weekly distribution. Available for leaf Account members if the option was selected when creating the application and the base time period is 12 months.

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Table 21-5 (Cont.) Member Properties

Property	Value
Hierarchy Type	Hierarchy Type is available for dimensions bound to an aggregate storage cube. Aggregate storage dimensions are automatically enabled to support multiple hierarchies. The first hierarchy in a multiple hierarchy dimension must be stored. **Note:** For members with a stored hierarchy type, the only valid cube aggregation options are Addition or Ignore. In a stored hierarchy, the first member must be set to Addition. For members with a dynamic hierarchy type, all cube aggregation options are valid. Stored hierarchy members that are not children of Label Only members must have Addition set as the consolidation operator. Children of Label Only members can be set to Ignore.
Data Storage	Select a data storage property. The default is Never Share for new custom dimension members (except root members).
Two Pass Calculation	Recalculate values of members based on values of parent members or other members. Available for Account and Entity members with Dynamic Calc or Dynamic Calc and Store properties.
Plan Type	Select the plan types (or cubes) for which the member is valid. **Note:** A member can belong to both aggregate storage and block storage cubes. Select an aggregation option for each selected cube. You can select a source cube only if multiple cubes are valid for the member. Only cubes and aggregation options for which the member's parent is valid are available. If the parent isn't valid for a cube or aggregation option, neither is the child member. Deselecting a cube for an account or entity parent member deselects it for all descendents of that parent. For members with a stored hierarchy type, the only valid aggregation options are Addition or Ignore. See Aggregation Options. **Caution:** Deselecting a cube for dimension members after data is entered into an application may result in loss of data when an application is refreshed. For account members, data is lost if the deselected cube is the source cube. Members of a custom dimension and a Period dimension can set usage by cube, similar to the Account and Entity dimensions.
For Account members only: **Source Cube**	Select the source cube for the member. A shared member is a pointer to the base member and isn't stored; this is disabled for shared members. The source cube of a shared Account member matches the source cube of the base member, even though Source Plan field is unavailable because it doesn't apply to shared members.
Smart Lists	**Optional**: Select a Smart List to associate with the member.
Enable for Dynamic Children	Enables users to create children for this member by entering a member name in the runtime prompt for a business rule that has been configured with a dynamic parent member (seeAbout Dynamic Members).
Number of Possible Dynamic Children	This option is available if **Enable for Dynamic Children** is selected. Enter the maximum number of dynamically-added members that users can create. The default is 10.

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Table 21-5 (Cont.) Member Properties

Property	Value
Access Granted to Member Creator	This option is available if **Enable for Dynamic Children** is selected. Determines the access that member creators have to dynamic members that they create with a runtime prompt: • **Inherit**: The member creator will inherit the closest parent's access to the newly-created member. • **None**: The member creator will not be assigned any access to the newly-created member. (A Service Administrator can later assign the member creator access to the members.) • **Read**: The member creator will be assigned Read access to the newly-created member. • **Write**: The member creator will be assigned Write access to the newly-created member. **Note:** If a Service Administrator changes these settings, they affect only future dynamic members; they don't retroactively affect dynamic members.

To add or edit members:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the dimension.
3. Perform one action:
   - To add a child member, select the parent level of the dimension hierarchy to which to add a member and click (Add Child).
   - To add a sibling, select the level of the dimension hierarchy to which to add a sibling and click (Add Sibling).
   - To edit a member, select that member from the dimension hierarchy and click (Edit).

   > **Note**
   >
   > To add an All Years parent member that includes all members of the Year dimension, select the Year dimension, click **Action**, and then click **Add All Years**. The All Years parent member enables users to view the accumulated data across multiple years, for example, a project's total cost up to its end date. The All Years member doesn't include the No Year member, if one is defined for the application.

4. On **Member Properties**, set or change member properties described in Table 1.

   If you don't see the new member on the page, click **Next**.
5. Click **Save** to save information to the relational database and see changes in the dimension hierarchy.
6. Refresh the database so edited members are visible to users entering data.
7. After creating a dimension member, you typically complete these tasks:

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- Assign access. See Assigning Access to Dimension Members.
- Specify attributes.

Deleting Members

Each data value is identified by a set of dimension member values and a cube. Deleting dimension members or deselecting the cube results in data loss when refreshing an application.

Before deleting members, understand where in the application they are used (in which forms, exchange rates, and so on) by using **Show Usage**. See Determining Where Members Are Used in an Application.

You must delete the entity member throughout the application before deleting it from Dimensions. For example, if the entity member is used in a form, you must delete it from the form before deleting it from Dimensions.

To delete members:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the dimension whose member you want to delete.
3. From the dimension hierarchy, select the entity member to delete.
4. Click (Delete).

   Deleting a base member also deletes its shared members.
5. Click **Yes**.
6. Update and validate business rules and reports.

Deleting Parent Members

Data values are identified by a set of dimension member values and a cube. Deleting dimension members or deselecting the cube results in data loss when refreshing the application.

To delete a parent member and all its descendants from the dimension hierarchy:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the dimension whose member and descendants you want to delete.
3. Select the member whose branch to delete.
4. Click (Delete).
5. Click **Yes**.

Working with Shared Members

Sharing members allow alternate rollup structures within an application. A base member must exist before you can create a shared member. You can create multiple shared members for the base member. A base member must display before its shared members in position from top to bottom.

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Shared members are available for Entity, Account, and user-defined custom dimensions. Shared member values can be ignored to avoid double-counting values when you roll up the outline.

Shared members share some property definitions with base members, such as member name, alias name, and cubes for which members are valid. Shared members must have unique parent members and different rollup aggregation settings. Custom attributes, custom attribute values, and member formulas are not allowed for shared members. Renaming base members renames all shared members.

Shared members can't be moved to another parent member. You must delete shared members and recreate them under different parent members. The base member need not be level zero. You can enter data in shared members, and values are stored with base members.

Shared members are displayed similarly to base members in the dimension hierarchy for member selection in Oracle Smart View for Office.

Creating Shared Members

You create shared members the same way as other members, with these differences:

- The base member can't be the parent of the shared member.
- You can't add a shared member as a sibling to the base member.
- You must give the shared member the same name as its base member. It can have a different description.
- You must select Shared as the Data Storage for the shared member.

About Dynamic Members

Dynamic members are members that users can create when working with business rules. They're sometimes referred to as "on the fly members".

A Service Administrator enables end users to create dynamic members under a parent member and must refresh the database to create the required placeholders in Essbase. In business rules with runtime prompts, users can then create members by entering desired member names in the runtime prompts. Subsequent database refreshes will rename used dynamic children to the names specified by the end users and recreate the required placeholders in Essbase. For information about working with business rules and dynamic members, see *Designing with Calculation Manager*.

If a parent member is enabled for adding dynamic children, users can create new members by entering their name in the runtime prompt.

> **Note**
>
> You can use the `HSP_NOMOF_<PlantypeName>` user defined attribute (UDA) to make sure that dynamic members are not created in the specified plan type, `<PlantypeName>`, even if the parent member is valid for this plan type. This prevents dynamic children from affecting aggregation on select cubes. See Working with UDAs.

Enabling Parent Members for Adding Dynamic Children

To enable a parent member for adding dynamic children:

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1. From the Home page, click **Navigator** , and then under **Create and Manage**, click **Dimensions**.
2. Edit the parent member and select the option **Enable for Dynamic Children** (see Adding or Editing Members).
   - **Optional**: Set the member property **Number of Possible Dynamic Children** (the default is 10). This setting determines the number of placeholders that are created for dynamically adding or loading members under the parent. If all placeholders are used, subsequently added children are added as normal members and can't be used until the database is refreshed.
   - **Optional**: Set the member property **Access Granted to Member Creator** (the default is Inherit).
3. Refresh the database to create the placeholders for dynamic members in the databases for the cubes in which the members are used.
4. In Calculation Manager:
   a. Create a business rule with a runtime prompt (of the Variable type Member). Under the **Dynamic Member Parent** column, use the **Member Selector** to select the parent member that you enabled for dynamic children in the application.
   b. Select the option **Create dynamic members**.
   c. Deploy the business rule.

      For information about working with business rules, see *Designing with Calculation Manager*.

> **Note**
>
> - In Calculation Manager, selecting both the **Create dynamic members** and the **Delete dynamic members** options enables temporary dynamic members to be created for calculations, and those temporary dynamic members will be deleted after the business rule completes.
> - Selecting the **Delete dynamic members** option alone presents you with the **Member Selector** in the runtime prompt so you can delete any member that you created dynamically under the parent (if you have write access to that member). This enables you to have full control over cleaning up and managing the members under the parent. The key is proper design to fit your requirements, setting the appropriate access rights to dynamic children with the **Access Granted to Member Creator** member property.

Results

If all the above conditions are met, when users run the business rule with the runtime prompt, they enter the name of the dynamic member, and then click **Launch**. If the business rule runs successfully, the member is created in the dimension hierarchy under the dynamic member's parent.

Child members that you import under parent members enabled for dynamic children are added as dynamic child members if there are dynamic member placeholders in Essbase. After the placeholders are full, any remaining children are added as normal members, and can't be used until the database is refreshed.

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> **Note**
>
> If you load a parent member that is enabled for dynamic children and its child members during the same import, the child members are loaded as normal members. This is because the database must be refreshed to create the placeholders in Essbase.

Considerations

When you create a dynamic member and data is captured against that member in multiple cubes (either through direct data entry, through calculations, data load, or data maps/Smart Push), then you must ensure that you clear the data from each of these areas before you remove the member using the **Delete dynamic members** operation. Performing the **Delete dynamic members** operation does not remove the data; the member is removed but the cube(s) will retain the data without the member showing up in the interface, and any rollup on the cube will reflect incorrect totals.

Working with Attributes

Use attributes to group members using the same criterion. You can assign attributes to sparse dimensions only. You can't assign attributes to label-only members. Attribute dimensions don't have aggregation properties because parents are dynamically calculated.

The Account dimension is usually defined as dense, so you can't assign attributes to it unless it's changed to sparse for all cubes. If you change a dimension from sparse to dense, all attributes and attribute values for that dimension are automatically deleted.

Attributes can have data types of text, date, Boolean, and numeric, as described in Understanding Attribute Data Types. Attribute names must conform to guidelines listed in Naming Restrictions. When attributes are defined, you can use the **Member Selection** dialog box to select attribute functions, such as **Equal** and **GreaterOrEqual**.

> **Note**
>
> This topic documents how to work with attributes using the Classic Dimension Editor. For information about using the Simplified Dimension Editor to work with attributes, see Working with Attributes.

To create and change attributes, attribute values, and aliases:

1. From the Home page, click **Navigator** , and then under **Create and Manage**, click **Dimensions**.
2. Select a sparse dimension for which to define an attribute, attribute value, or alias.

   > **Note**
   >
   > Only sparse dimensions can contain attributes.

3. Click **Actions**, then click **Custom Attributes** .

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> **Note**
>
> If the dimension isn't sparse, **Custom Attributes** isn't available.

4. Select options.

   - To create attributes, click (Create). Type an attribute name, and select a data type: **Text**, **Date**, **Boolean**, or **Numeric**.

     > **Note**
     >
     > – You can't modify the type after the attribute is created.
     >
     > – Before working with date attributes, you must select the **Attribute Dimension Date Format** in **Application Settings** and save it.

     See Understanding Attribute Data Types.

   - To modify attributes, click (Edit), and update the attribute name.
   - To set aliases for attributes, select an attribute and an attribute value, click (Alias). Select an alias table, type an alias name, and click **Save**.

5. Click **Close**.

   When you click **Close**, the hierarchy is validated and an error displays if issues are detected. For example, date attribute values must be entered in the correct format, and numeric and date attribute dimensions must have at least one attribute value defined.

6. Update and validate business rules and reports.

Related topics

- Understanding Attribute Data Types
- Deleting Attributes

> **Note**
>
> Instead of explicitly filtering by an attribute (like Red, for instance), you can create a user variable for the attribute dimension, and then use the user variable as the filter. Then you can enable the user variable as a dynamic user variable which would allow users to change the value of the filter at runtime. This is a useful technique that allows for dynamic filtering. See Managing User Variables.

Table 21-6 Tutorials

Your Goal	Learn How
Learn how to report on data with attribute hierarchies. You can view and report on data organized by attribute on the web on forms and dashboards, with MR reports, or in Oracle Smart View for Office with ad hoc analysis.	Reporting on Data with Attribute Hierarchies

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Understanding Attribute Data Types

Attribute dimensions can have a data type of text, numeric, Boolean, or date that enables different functions for grouping, selecting, or calculating data. The attribute type applies only to level 0 members of the attribute dimension.

- Text attributes enable basic attribute member selection and attribute comparisons in calculations. When you perform such comparisons, characters are compared. For example, a package type Bottle is less than a package type Can because B precedes C in the alphabet.
- Numeric attribute dimensions use numeric values for the names of level 0 members. You can include the names (values) of numeric attribute dimension members in calculations. For example, you can use the number of ounces specified in an Ounces attribute to calculate profit per ounce for each product. You can also associate numeric attributes with ranges of base dimension values, for example, to analyze product sales by market population groupings.
- Boolean attribute dimensions in a database contain only two members. When a Boolean attribute dimension is added in the business process, two attribute values, True and False, are created for this attribute dimension by default. A base dimension, such as Account or Entity, can be associated with only one attribute dimension that has the Boolean data type.
- Date attributes can specify the date format as month-day-year or day-month-year, and sequence information accordingly. You can use date attributes in calculations, for example, comparing dates in a calculation that selects product sales since 12-22-1998. Users can set the date format by selecting an option in **Attribute Dimension Date Format** in **Application Settings**.

> ⚠ **Caution**
>
> Before you can work with date attributes, you must toggle the **Attribute Dimension Date Format** selection in **Application Settings** and save it. From the Home page, click **Application**, and then click **Settings**. In **Attribute Dimension Date Format**, select a date format (**MM-dd-yyyy** or **dd-MM-yyyy**), and then click **Save**. After you select and save the date format, you must manually change any existing attribute date values to the supported format. If any date attributes in your application use an unsupported format, the system will provide you with a list of dimension attribute values that you will need to fix.
>
> Also, if the **Attribute Dimension Date Format** setting is changed in the application settings, you must change any date attributes in the original format to the new format before you can add or save a new attribute.

Deleting Attributes

When you delete an attribute, all attribute values associated with the attribute are also deleted. Attribute values are removed from members to which they had been assigned, and the attribute is removed from dimensions to which it was assigned.

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> **Note**
>
> This topic documents how to delete attributes using the Classic Dimension Editor which is accessed by using the **Dimensions** link in the Navigator menu. In a previous update, we released the Simplified Dimension Editor.
>
> For information about using the Simplified Dimension Editor to delete attributes, see Deleting Attributes.

To delete attributes:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the sparse dimension from which to delete an attribute, and click (Custom Attributes).
3. Select the attribute to delete.
4. Above the **Attribute** column, click (Delete).
5. Click **OK**.
6. Update and validate business rules and reports.

Working with Attribute Values

Attribute values provide users with another way of selecting dimensions members when using forms. Data values for attribute values are dynamically calculated but not stored.

> **Note**
>
> This topic documents how to work with attribute values using the Classic Dimension Editor. For information about using the Simplified Dimension Editor to work with attribute values, see Working with Attribute Values.

- Creating Attribute Values
- Assigning Attribute Values to Members
- Editing and Deleting Attribute Values

Creating Attribute Values

You can define attribute values for sparse dimensions, which are typically the Entity and user-defined custom dimensions. After you define an attribute value for a dimension, you can assign it to members of that dimension.

To create attribute values:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the sparse dimension for which to create an attribute value.

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3. Select the top level in the dimension hierarchy.
4. Click (Custom Attributes).
5. On the **Manage Attributes and Values** page, select the attribute for which to specify a value.
6. Above the **Attribute Values** column, click (Add Child) or (Add Sibling).
7. Enter a name and choose a data type.
8. Click **Save**.

Assigning Attribute Values to Members

You can assign attribute values members of a dimension that are defined as sparse for all cubes. Attribute values must be assigned to the same-level sparse dimension members. Otherwise, errors display during refresh.

To assign attribute values to members:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the sparse dimension for whose member you want to assign an attribute value.
3. In the **Dimension** hierarchy, select a member to which to assign an attribute value.
4. Click (Custom Attributes).
5. For members assigned attribute values, click (Edit) to change a member's attribute value.
6. Select attribute values to assign to the member.
7. Perform an action:
   - To assign the value to the selected member, click (Add).
   - To remove a value from the selected member, select the value to remove and click (Remove).
   - To remove all values from the selected member, click (Remove All).
8. Click **Save**.

Editing and Deleting Attribute Values

> **Note**
>
> When you delete an attribute value, it's removed from custom dimension members to which it's assigned.

To edit or delete attribute values:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.

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2. Select the sparse dimension containing the attribute value you want to modify or delete.
3. Select the top level in the dimension hierarchy.
4. Click (Custom Attributes).
5. For **Attributes**, select the attribute containing the value to modify or delete, and then select the attribute value.
6. Above **Attribute Values**, click (Edit) or (Delete).
7. If editing, enter a name. If deleting, confirm the deletion.
8. Click **Save**.
9. Update and validate business rules and reports.

Customizing Calendars

Use the Period dimension to work with the yearly calendar rollup structure. When creating the application, the Service Administrator specifies the base time periods that span the application database. Use the Years dimension to add years to the calendar.

Related Topics

- Defining How Calendars Roll Up
- Creating and Editing Summary Time Periods
- Deleting Summary Time Periods
- Working with the Years Dimension
- Adding Years to the Calendar
- Editing Year Information
- Renaming Time Periods
- Assigning Aliases to Summary Time Periods
- Editing the BegBalance Member

Defining How Calendars Roll Up

Table 21-7 Calendar Roll Up

Base Time Period	Roll Up
12 Months	Four quarters are created per year. Months roll up into parent quarters and quarters roll up into years.
Quarters	Quarters roll up into years.
Custom	No default rollup structures. A flat list of the custom base time periods displays.

After the application calendar is created, you can't change the base time period or reduce the number of years in the calendar. Service Administrators can change the names, descriptions, aliases, and ranges of the summary time periods in the hierarchy.

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Creating and Editing Summary Time Periods

You can change such aspects as name, description, alias, starting period, and ending period. However, you can't change the order of base time periods or skip base time periods. The range can't extend beyond the current fiscal year.

You must work from the top of the hierarchy to the bottom when creating summary time periods. (Otherwise, the application views the rollup structure as asymmetrical and you can't continue.) The summary time period displays in the hierarchy as a parent of the selected item. To enforce a balanced hierarchy, all base members must be the same number of levels from the root.

To create or edit summary time periods:

1. From the Home page, click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select **Period**.
3. Perform one action:
   - To add a time period, select the time period in the dimension hierarchy to which you want to add a child or sibling, and then click (Add Child) or (Add Sibling).
   - To edit a time period, select the time period and click (Edit).
4. For **Name**, enter or change the name for the summary time period.
5. **Optional**: For **Description**, enter a description.
6. **Optional**: For **Alias Table** and **Alias**, select an alias table to use. Enter an alias name.

   The default table is used if you don't select one.
7. For **Start Period**, select the starting period.

   The range can't extend beyond the current fiscal year. For summary time periods, **Start Period** displays the first child, or all children except the first child of the sibling above it.
8. For **End Period**, select the ending period.

   For summary time periods, **End Period** displays the last child, or all children from the Start Period through the next sibling's children, except the last child.
9. Click **Save**.

Deleting Summary Time Periods

When you remove a summary time period from the hierarchy, its children are moved into another summary time period:

- If you delete the first summary time period, children are moved into the next sibling of the summary time period.
- If you delete the last summary time period, children are moved into the previous sibling of the summary time period.
- If you delete a summary time period from the middle of a hierarchy, children are moved into the previous sibling of the summary time period.

To delete summary time periods:

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1. Click the **Navigator** icon, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select **Period**.
3. In the dimension hierarchy, select the summary time period to delete.

   You can't delete base time periods.
4. Click (Delete).
5. Click **OK**.

Working with the Years Dimension

Use the Years dimension to work with calendar years.

Table 21-8 Years Tasks

Task	See Topic
• Add years before the Start year or after the End year of the calendar. • Add an All Years parent member that includes all members of the Years dimension (except No Year, if that member exists).	Adding Members
Add or update the description and alias for a year.	Editing Members

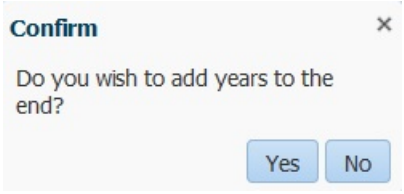
Adding Years to the Calendar

You can add years to the beginning or end of the calendar, but you can't reduce the number of calendar years without creating a database.

To add years to the calendar:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select **Year**.
3. Click (Add Years).
4. In **Add Years**, enter the number of years to add to the calendar.
5. Click **OK**.

   The confirmation dialog asks you to confirm the following:



   - To add years after the End year, click **Yes**.
   - To add years before the Start year, click **No**.

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> **Note**
>
> To add an All Years parent member that includes all members of the Year dimension, click (Add "All Years"). The All Years parent member enables users to view the accumulated data across multiple years, for example, a project's total cost up to its end date. (This parent member doesn't include the No Year member, if one exists.)

Editing Year Information

You can add or update the description and alias for a member of the Years dimension.

To edit a member of the Years dimension:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select **Year**.
3. Select the year you want to edit, and then click (Edit Year).
4. Update the information for that year, and then click **Save**.

Renaming Time Periods

You can rename root-level, base time periods, and user-defined summary time periods.

To rename time periods:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select **Period**.
3. Select the time period you want to rename, and then click (Edit).
4. Rename the time period, and then click **Save**.

Assigning Aliases to Summary Time Periods

You can assign and change aliases to base time periods and summary time periods.

To assign or change the alias:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select **Period**.
3. Select the summary time period.
4. Click (Edit).
5. For **Alias Table**, select the alias table to use.
6. For **Alias**, enter an alias name.
7. Click **Save**.

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Editing the BegBalance Member

You can edit the BegBalance member of the Period dimension. As the first time period in the application, the BegBalance member is useful for entering beginning data when you start a new application, fiscal year, or calendar year. You can rename and describe BegBalance and give it an alias.

To edit the BegBalance member:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Dimensions**.
2. For **Dimension**, select **Period**.
3. Select the first member, **BegBalance**.
4. Click (Edit).
5. For **Edit Member**:
   - Enter a name.
   - Enter a description.
   - Select an alias table to use for the BegBalance member, and enter an alias.
6. Click **Save**.

Setting up Dynamic Time Series Members

You can use Dynamic Time Series members to create reports that show period-to-date data, such as quarter-to-date expenses. Dynamic Time Series members are created automatically during application creation, and can be used with members of the Period dimension. To set up Dynamic Time Series, you enable a predefined Dynamic Time Series member and associate it with a generation number (and, optionally, an alias table and alias name). For example, to calculate quarter-to-date values, you can enable the Q-T-D (quarter-to-date) member and associate it with generation number 2. You can then use the Q-T-D Dynamic Time Series member to calculate monthly values up to the current month in the quarter.

> **Note**
>
> Dynamic Time Series isn't supported for the Period dimension in an aggregate storage application.

The Dynamic Time Series members provide up to eight levels of period-to-date reporting. Your data and database outline determine which members you can use. For example, if the database contains hourly, daily, weekly, monthly, quarterly, and yearly data, you can report D-T-D (day-to date), W-T-D (week-to-date), M-T-D (month-to-date), Q-T-D, and Y-T-D information. If the database contains monthly data for the past 5 years, you can report Y-T-D and H-T-D (history-to-date) information, up to a specific year. If the database tracks data for seasonal time periods, you can report P-T-D or S-T-D (season-to-date) information.

Oracle recommends that you avoid assigning time balance properties (such as First and Average) to members set for dynamic calculations if you plan to use the members in Dynamic Time Series calculations. Doing so may retrieve incorrect values for parent members in your accounts dimension.

To set up Dynamic Time Series members:

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1. Click **Navigator**, and then under **Create and Manage**, click **Dimensions**.
2. Select the Period dimension, and then click (Dynamic Time Series).
3. Select **Enabled** for the DTS series to use:
   - Y-T-D: Year-to-date
   - H-T-D: History-to-date
   - S-T-D: Season-to-date
   - Q-T-D: Quarter-to-date
   - P-T-D: Period-to-date
   - M-T-D: Month-to-date
   - W-T-D: Week-to-date
   - D-T-D: Day-to-date
4. Select a generation.

   The number of generations displayed depends on the number of generations in the time dimension. You can't associate Dynamic Time Series members with the highest generation (the dimension root).
5. **Optional**: Add an alias name. In the **Edit Alias Name** column, click (Alias), enter an alias name, and then click **OK**.
6. Click **Save**.

Additional Supported Application Features

Additional features are supported for application dimensions. You can add children and siblings to Scenario, Version, and Period dimensions, and you can use the cut, paste, expand, and collapse features to work with their dimension hierarchies (see Working with Dimension Hierarchies). You can also use shared members for these dimensions, and can set two pass calculations at the root level for all dimensions. For example, you can:

Table 21-9 Features

Feature	More Information
In the Period dimension, create alternate hierarchies and use shared descendants. Data Storage for all time periods can be set to any valid Data Storage value. The Consolidation operator for all time periods, including BegBalance, can be set to any valid consolidation operator. For example, it can be set to + instead of ~ (ignore).	See Working with the Years Dimension and Editing the BegBalance Member.
Turn on two pass calculation at the root level, for example, for Account. **Caution:** Two pass calculation is ignored on any non-Account member not set to Dynamic Calc.	See Adding or Editing Members.
For attributes, create hierarchies and assign aliases.	See Working with Attributes.

Considerations for Alternate Hierarchies in Period Dimensions

If you create an alternate hierarchy in the Period dimension, the alternate hierarchy must follow the YearTotal member in the outline.

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Working with UDAs

You can use user-defined attributes (UDAs), descriptive words or phrases, within calc scripts, member formulas, reports, and forms. UDAs return lists of members associated with the UDA. For example:

- You can use the `HSP_UDF` UDA to prevent a formula from being overwritten when the application is refreshed. You must log on to each database associated with the business process and create the member formula with a UDA. The syntax for this UDA is: `(UDAs: HSP_UDF)`.
- If you use the `@XREF` function to look up a data value in another database to calculate a value from the current database, you can add the `HSP_NOLINK` UDA to members to prevent the `@XREF` function from being created on all cubes that are not the source cube selected for that member.
- For a Product dimension with several product members, you can create a UDA called New Products and assign this UDA to the new products in the Product dimension hierarchy. Then you can base certain calculations on the designation New Products.
- When designing forms, you can use a UDA to select members for forms based on a common attribute. When you create forms with UDAs, any members that are assigned to the UDA are dynamically added to the form. For example, if you create a UDA called New Products and assign this UDA to the new products in the Product dimension hierarchy, the form will automatically display the new products at runtime. When selected in the form designer, a UDA is preceded by `UDA`; for example, `UDA(New Products)`.
- You can use the `HSP_NOMOF_<PlantypeName>` UDA to make sure that dynamic members (or "on the fly members") are not created in the specified plan type, `<PlantypeName>`, even if the parent member is valid for this plan type. This prevents dynamic children from affecting aggregation on select cubes.

UDAs are specific to dimensions. For example, creating a UDA for an Account member makes it available for non-shared Account members. Deleting it removes it for all Account members. To make UDAs available for multiple dimensions, create the same UDA for multiple dimensions. For example, create a UDA named New for Account and Entity dimensions to make the UDA named New available for Account and Entity members.

> **Note**
>
> This procedure describes how to work with UDAs in the Classic Dimension Editor. To work with UDAs in the Simplified dimension editor, see Working with UDAs in the Simplified Dimension Editor.

To work with UDAs:

1. Click the **Navigator** icon , and then under **Create and Manage**, click **Dimensions**.
2. Select the dimension for whose members to associate the UDA.
3. From the dimension hierarchy, select a member and click (Edit).
4. Select the **UDA** tab.
5. Select a task:

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- To create a UDA, click (Create), enter a name, and then click **Save**.

> **Note**
>
> Use no more than 60 characters when naming UDAs.

- To modify a UDA, click (Edit), change the name, and then click **Save**.
- To delete a UDA, select the UDA, click (Delete), and then confirm deletion.

> **Note**
>
> Deleting the UDA removes it for the whole dimension. If you delete UDAs, you must update all member formulas, calc scripts, and reports that reference them.

- To clear UDA selections, click (Clear).

6. To add or remove UDAs for the member, use the arrows to move UDAs to and from the **Selected UDA** panel.
7. Click **Save**.

Working with Member Formulas

You can define member formulas to combine operators, calculation functions, dimension and member names, and numeric constants to perform calculations on members. Member formulas can also include:

- Operator type, function, value, member name, UDA, and so on allowed in formulas.
- Predefined formula expressions, including Smart List values, that expand into a formula or value upon database refresh.

To define member formulas:

1. Click the **Navigator** icon , and then under **Create and Manage**, click **Dimensions**.
2. Select the dimension for whose member to add or change a formula.
3. Select the member and click (Edit).
4. Select the **Member Formula** tab.
5. Select options for the following fields:
   - **Cube**

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> **Note**
>
> With the exception of Account formulas, a formula entered for the default cube will be applied to all cubes unless it's overridden by a different formula entered for a specific cube.
>
> Also, a default formula on Account members will only be transferred to Oracle Essbase for the source cube and not for any of the other cubes. A default formula won't be transferred to any aggregate storage cubes.

- **Data Storage**: Select a data storage option. The default is **Store**.

> **Note**
>
> The cube specific data storage field will not display the **Shared** or **Label Only** options. This is because a member can't be set to Shared or Label Only in one cube and not another.

- **Solve Order**: Specifies the order in which formulas are evaluated. Enter a whole number between 0 and 100000 (or use arrows to increase or decrease the number). The formulas for members that have a specified solve order are calculated in order from the lowest solve order to the highest. The default is 0.

> **Note**
>
> Solve order is available for aggregate storage cubes and also for block storage cubes that are enabled for Hybrid. Solve order for block storage cubes is only editable using the Simplified dimension editor. See Accessing Edit Member Properties.
>
> You can set the Solve Order at the dimension root level in both Hybrid BSO and ASO cubes. This does not apply to non-hybrid BSO cubes.
>
> – A higher solve order means the member is calculated later. For example, a member with a solve order of 1 is solved before a member with a solve order of 2.
>
> – The minimum solver order you can set is 0, and the maximum is 127.
>
> – Dynamic members (with or without formulas) that do not have a specified solve order inherit the solve order of their dimension, unless they are tagged as two pass.
>
> – Member level solve order will overrule the setting at the root level it if is changed later at the member level.

6. In the text box, define formulas for the member.
7. **Optional**: To check the validity of the member formula, click **Validate**.
8. Click **Save**.

   Before you click **Save**, clicking **Reset** restores the previous member formula information.

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Viewing Details of Formula Validation

To view details of the member formula validation:

1. On **Member Formula**, click **Validate**.
2. Click **Save**.

Working with Formula Expressions

In the text box on the **Member Formula** tab, you can include predefined formula expressions in member formulas, and test them with the **Validate** button. You can also load them.

You can update the dimension outline without updating the business rules and calc scripts that depend on the outline. Calculations become more independent of specifics in the outline. You can use Smart Lists as objects in calculations. Performance isn't decreased when you use formula expressions because they are run only when you refresh the database.

To use a formula expression in a member formula:

1. Click the **Navigator** icon , and then under **Create and Manage**, click **Dimensions**.
2. Select the dimension with which you want to work.
3. Select the member and click (Edit).
4. Select the **Member Formula** tab.
5. Select options for the following fields:
   - **Cube**

     > **Note**
     >
     > With the exception of Account formulas, a formula entered for the default cube will be applied to all cubes unless it's overridden by a different formula entered for a specific cube.
     >
     > Also, a default formula on Account members will only be transferred to Oracle Essbase for the source cube and not for any of the other cubes. A default formula won't be transferred to any aggregate storage cubes.

   - **Data Storage**: Select a data storage option. The default is **Store**.

     > **Note**
     >
     > The cube specific data storage field will not display the **Shared** or **Label Only** options. This is because a member can't be set to Shared or Label Only in one cube and not another.

   - **Solve Order**: Specifies the order in which formulas are evaluated. Enter a whole number between 0 and 100000 (or use arrows to increase or decrease the number). The formulas for members that have a specified solve order are calculated in order from the lowest solve order to the highest. The default is 0.

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> **Note**
>
> Solve order is available for aggregate storage cubes and also for block storage cubes that are enabled for Hybrid. Solve order for block storage cubes is only editable using the Simplified dimension editor. See Accessing Edit Member Properties.

6. In the text box, define formulas for the member.

   You can include business process formula expressions and Essbase native formulas in the member formula.

   The business process provides predefined formula expressions that you can use in member formulas. You can't edit or create your own formula expressions.

7. **Optional**: To check the validity of the member formula, click **Validate**.
8. **Optional**: If there are errors in the member formula, click **Show Details** to view a description.
9. **Optional**: Click **Reset** to restore the previous member formula if you don't want to save the changes you made to the member formula.
10. Click **Save**.

Syntax

Member formula expressions support functions and variables. Follow these syntax rules for functions and variables when you create formula expressions:

- Enclose variables or properties with square brackets, [ ]. If you omit square brackets, the variable is treated as a native variable.
- Enclose member names with quotation marks.
- Characters in variables are case-insensitive, but can't have extra spaces or characters such as underscore (_).
- You can include subcalls to other functions within an expression.
- Don't enter text where a number is required.
- The order of the outline is important in a member formula. For example, don't reference a value that has not been calculated yet.

Including Smart List Values as Variables

You can include a Smart List as a variable in a formula expression, such as the formula expression, "Status"= [Status.Departed].

"Status" is the member name, Status is the Smart List name, and Departed is a Smart List entry. If the Smart List ID for Departed is 2, Status.Departed is replaced with a 2 in the member formula (the application treats Smart Lists as numbers). If the Smart List ID for Departed is 2, 2 is put in the calculation and 2 is stored in the database.

Write Smart Lists in this format: [SmartListName.SmartListEntry]

Formula Expressions

Formula expressions can include these predefined variables and functions.

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Table 21-10 Variables in Formula Expressions

Variable	Description
NumberOfPeriodsInYear	Returns the number of time periods in the year
NumberOfYears	Returns the number of years in the application

Table 21-11 Functions in Formula Expressions

Function	Description
Dimension(dimTag)	Returns the name of a predefined dimension. The dimtags are: • DIM_NAME_PERIOD • DIM_NAME_YEAR • DIM_NAME_ACCOUNT • DIM_NAME_ENTITY • DIM_NAME_SCENARIO • DIM_NAME_VERSION • DIM_NAME_CURRENCY
Period(periodName)	Returns the specified period. The periodName options are: • FIRST_QTR_PERIOD • SECOND_QTR_PERIOD • THIRD_QTR_PERIOD • FOURTH_QTR_PERIOD • FIRST_PERIOD • LAST_PERIOD
CrossRef(accountName)	Generates a cross-reference by adding the default prefix of "No" to each dimension name (except Currency, Period and Year), followed by the specified account. For example, in an application with the following dimensions: Account, Period, HSP_View, Year, Scenario, Version, Entity, and Product CrossRef("5800") returns: "BegBalance"->"No HSP_View"->"No Scenario"->"No Version"->"No Entity"->"No Product"->"5800";
CrossRef(accountName, prefix)	Generates a cross-reference by adding the specified prefix to each dimension name (except Currency, Period and Year), followed by the specified account. The prefix should be in double quotes. For example, in an application with the following dimensions: Account, Period, HSP_View, Year, Scenario, Version, Entity, and Product CrossRef("5800", "NoX") returns: "BegBalance"->"NoX HSP_View"->"NoX Scenario"->"NoX Version"->"NoX Entity"->"NoX Product"->"5800";
CrossRef(accountName, prefix, true)	Generates a cross-reference by adding the specified prefix to each dimension name, including Year (except Currency and Period), followed by the specified account. For example, CrossRef("5800", "NoX", true) returns: "BegBalance"->"NoX HSP_View"->"NoX Year"->"NoX Scenario"->"NoX Version"->"NoX Entity"->"NoX Product"->"5800";
getCalendarTPIndex()	Generates a member formula that returns an index for the time period; the index is based on the calendar year.
getFiscalTPIndex()	Generates a member formula that returns an index for the time period; the index is based on the fiscal year.

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Table 21-11 (Cont.) Functions in Formula Expressions

Function	Description
CYTD(memberName)	Generates a calendar year-to-date formula for the member
CYTD(memberName, calTpIndexName, fiscalTpIndexName)	Generates a calendar year-to-date formula for the member, and the time period index based on the calendar year and fiscal year. Use when members are renamed. The default member names are "Cal TP-Index" and "Fiscal TP-Index."

Understanding Common Errors

Follow the rules of syntax carefully. If formula expression syntax contains errors, error messages are returned after you validate the member formula. To get information about error messages, review the formula details on the **Member Formula** tab. The most common error message is "Failed to execute." This occurs when you use parameters in the expression incorrectly. These actions cause "Failed to execute" error messages:

- Entering the wrong number of parameters in the formula expression
- Misspelling member names, functions, or variable names
- Not surrounding member names with quotation marks
- Including numbers where text is required

Administering Forms

Create forms for entering data based on your planning needs.

Related Topics

- About Forms
- Creating Forms
- Working with Forms and Form Components
- Managing Forms and Folders

About Forms

Forms are grids for entering data. You can create forms to meet your needs and then you can use the forms and other artifacts to design dashboards to summarize the data.

See Designing Dashboards.

Troubleshooting

For help with resolving form functional and performance issues, see Resolving Form Functional and Performance Issues in *Operations Guide*.

Form Components

Point of View

Select point of view (POV) dimensions such as year, scenario, and version to define the context for pages, rows, and columns. For example, if Scenario is Budget, all data displayed

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and entered in pages, rows, and columns is written to the Budget scenario dimension member. Each POV dimension is set to one member, which users can't change.

To simplify a form's POV, or better tailor it to the needs and roles of users, you specify only relevant members or define user variables. See Defining Form Page and Point of View.

See also, Managing User Variables.

Page Axis

Use the page axis to specify member combinations that may span dimensions so that users can work with data in smaller, logical views. Each page axis item can have members selected from different dimensions. Users see only members they can access.

You can specify multiple page drop-down lists, and select members using relationship functions or attributes. Switch between member sets by selecting them from the page axis.

You display member names or aliases on the page axis. You can specify the number of members in a page dimension that enables a search drop-down list on the data entry page, useful if dimensions contain many members. See Defining Form Page and Point of View.

Rows and Columns

Rows and columns define the grid into which users enter data. For example, you can assign Unit Sales to the row axis and January to the column axis. When users access forms, they can enter data into the cell where the Unit Sales row intersects with the January column.

By default, forms have one set of rows and columns. You can add rows and columns to create asymmetrical combinations of members. See Creating Asymmetric Rows and Columns.

Attribute Dimensions

In addition to using attribute dimensions as filters, you can define attributes as unique dimensions within the application so that they can be placed on an axis other than their associated base dimension. This enables you to perform cross-dimensional rollups across attribute members.

Attribute dimensions are optional on form grids and are listed separately on the **Layout** tab of the Form Designer. To use an attribute dimension in a form, drag the attribute dimension to the point of view, page, row, or column.

> ⚠ **Caution**
>
> Don't drag the attribute on any axis when using it as a filter as you'll cause a cross-dimensional reference which will have a performance impact. When using an attribute dimension as a filter, there is a form option that shows the associated attribute (just like there is an option to show alias). See Defining the Layout.

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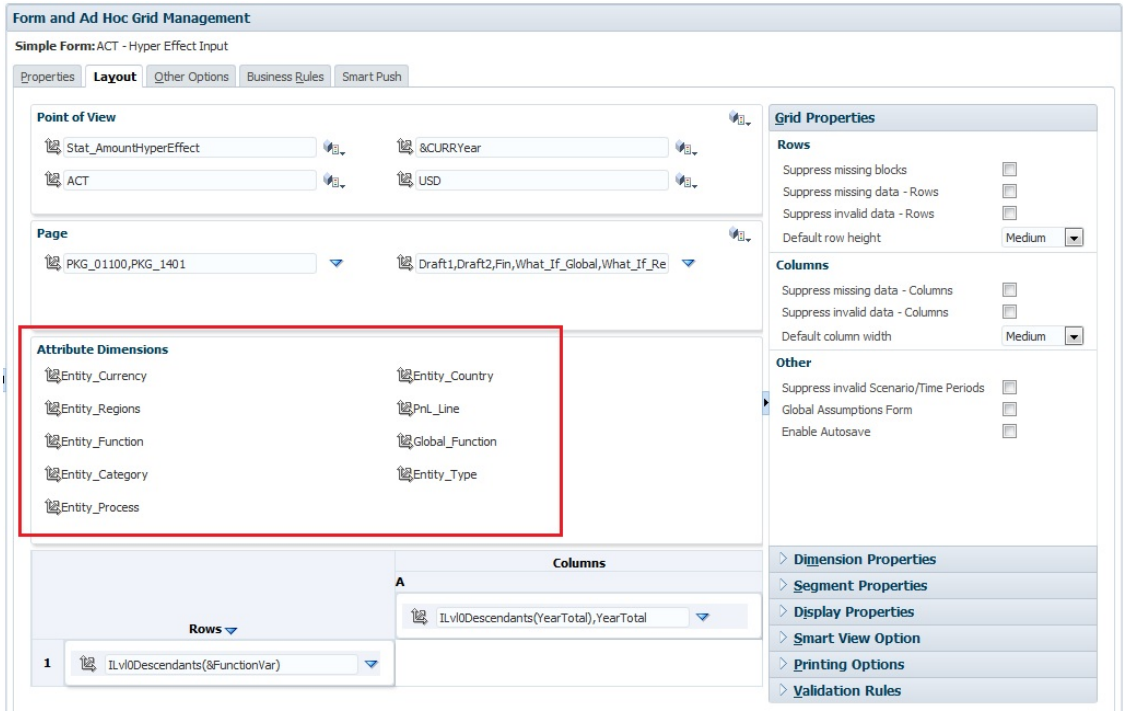

Form Design

Use these design guidelines to plan effective forms.

Form Design Overview

You'll build a number of forms to support data entry and summary-level reports. The form content is similar to the templates you use to collect and calculate data. The layout may differ from what you are accustomed to in spreadsheets.

To enhance usability, group forms within major categories such as Revenue, Compensation Expense, Other Expenses, and so on. You can create some forms to support data entry, and others for summary and review. You can also include charts to help users analyze results.

Form performance is based on several factors, including network and environmental factors, structure, and layout.

Design Considerations

Design forms to enable users to enter information such as revenue, expenses, and assumptions.

Best practices:

- Group accounts logically, but don't include too many accounts on a single form.
- Limit the number of entry forms to an amount comfortable for end users. A delicate balance needs to be achieved between the number of accounts on a single form and the number of forms required to support your process.
- Use detail forms to enable users to enter all related information. All accounts that require input should be found on a form. The accounts can be broken down into several different forms.

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- While building forms, ensure that you select all appropriate options to enhance the design of your form. For example, use settings to control precision, display, and menus, and to associate the proper rules with your form.
- Use Substitution Variables to reference dimensions such as Years.
- Suppress invalid Scenario/Time Period option, set Periods in row or column on the form to the Start and End Period set for the Scenario. Leveraging this functionality can be used instead of substitution variables for Years.
- Consider setting valid intersections to set relationships between different dimensions. Suppressing invalid combinations can be set in row or column to make only valid intersections available to end users. By default, only valid intersections will be available to the end users when the dimensions are set in the Page selection.
- Use relationships to incorporate members onto the forms instead of picking members individually.
- Consider using User Variables for dimensions such as Entity and Scenario to help reduce the dimension selection for end users.
- If your application supports multiple currencies, consider setting a User variable so users can define their base currency.
- Organize forms into folders.
- Use substitution variables to reduce the maintenance on forms.
- Put dense dimensions such as Account and Period on the row and column of a form. Put sparse dimensions such as Entity on the Page axis.
- Dimensions such as Scenario or Version and Year can reside on the POV, column, or row. It's important to properly gauge how columns or rows will be returned when a user opens the form.

Associate Rules with Forms

Associating rules with forms enables users with appropriate access to launch associated business rules from the form to calculate and derive values.

You can associate multiple business rules with a form by cube. Business rules associated with a form can be set to launch automatically when the form is opened or saved. You can select Use Members on Form to populate runtime prompts from the current form instead of prompting users for input when rules are launched.

Best practices:

- For rules that take longer to run, set them to launch from an Action menu or simply through association with the form.
- If a business rule has runtime prompts, limit the number of prompts to keep the user’s job simple.

In Forms, you can enable a Form Grid Properties option to Pivot Rows and Columns after a data load. If you have a business rule that was associated with a form before the Pivot option was enabled, you may need to adjust the rule to accommodate the change in layout. This is especially true if you use methods such as getColumns() or getRows() with assumptions of the dimensions they contain. If the rule is written generically, changes may not be required. Before using the form Pivot Rows and Columns After Load option, review any existing business rules associated with a form.

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Add Menus to Forms

You can associate menus with forms. Action menus allow users to click rows or columns in forms and select menu items. For example, they can launch a business rule, with or without runtime prompts, or move to another form.

Menus are context-sensitive. The menus that display depend on the form settings and where users right-click in the form.

Best practices:

- When designing forms, use Other Options to select menus available for Form menu item types.
- As you update applications, update the appropriate menus. For example, if you delete a business rule referenced by a menu, remove it from the menu.

Build Data Validation Forms

Data validation can serve as a visual clue to users that business policies have been met. You can add conditional color coding to forms, and validation messages can be generated if entered data violates validation rules or if a condition is met.

Defining data validation rules involves these main tasks:

- Identify the data cells or location that you want to display with validation messages or in different colors when conditions are met. Be mindful of accessibility concerns if color is the only differentiator.
- Identify the cell, column, or row that needs to participate during rule evaluation, and define the rule accordingly.
- Create the data validation rule at the location identified.

Organize Forms into Folders

Use folders as a way to organize the forms in your application. Forms can be grouped in folders by process or user type, or simply to help users readily find forms. You can move forms into folders, and you can create a folder hierarchy. Creating folders also simplifies assigning access because all forms in the folder will inherit the access permissions assigned.

Build Summary Level Forms

Summary level forms typically bring together all of the pieces of a user's plan or forecast. They enable users to review and analyze their results.

Using dashboards can also be an effective way to help users analyze their results.

Forms and Cubes

When you create a form, you associate it with a cube, which determines the form's valid members. For example, if you assign a form to the Revenue cube, you can add only accounts that are valid for the Revenue cube. Entered data is saved to the selected cube's database.

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> **Note**
>
> - You can't change a form's cube after assigning it.
> - You can edit form accounts only if their source cube matches the form's cube.
> - If you add an account to a form associated with a cube other than the account's source cube, the account is read-only on that form.

Forms and Permissions

Assign permissions to a form to determine which users can modify its design (for example, layout and instructions) and input data. Users can input data on forms only if they have permission to one secured dimension's member. For example, if users have read-only permission to the Europe entity, the rows and columns that include the Europe entity are read-only. Users can change data only for members to which they have write permission.

Forms and Versions

For bottom-up versions, rows and columns with level 0 members allow data entry. Rows or columns set to a parent member are read-only. The point of view must also be set to the level 0 member to allow data entry on a bottom-up version. Target versions allow data entry in parent and children members set to Store.

Filtering Form Members by Attributes

You can select members by using attributes. For example, on the Entity dimension you can select members by a specific Region such as South. The resulting grid will only contain members that have the South attribute (for example, TX, NM, and so on). Values can be entered and saved into rows and columns filtered by attributes.

Forms and Shared Members

Because you can't select shared members individually, select them using a relationship function. For example, select an alternate functional rollup to include all members under that rollup. Users can enter values in rows or columns that display shared members, and data is saved to the base members in the database.

Forms and Calculations

To optimize calculations, select row members using relationships (such as Descendants or Children) instead of selecting individual children. For example, calculating individual parent-level totals could take several passes, so use a relationship instead.

Understanding Implied Sharing in Forms

Some members are shared even if you don't explicitly set them as shared. These members are implied shared members.

When an implied share relationship is created, each implied member assumes the other member's value. The application assumes shared member relationships in these situations:

- A parent has only one child
- A parent has only one child that consolidates to the parent
- The data type of the parent is Label Only. The parent inherits the value of the first child, regardless of the child's aggregation settings.

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In a form that contains members with an implied sharing relationship, when a value is added for the parent, the child assumes the same value after the form is saved. Likewise, if a value is added for the child, the parent usually assumes the same value after a form is saved.

For example, when a calculation script or load rule populates an implied share member, the other implied share member assumes the value of the member populated by the calculation script or load rule. The last value calculated or imported takes precedence. The result is the same whether you refer to the parent or the child as a variable in a calculation script.

Creating Forms

Table 21-12 Form Creation Checklist

Task	Want to Know More?
Define the layout, including: • Adding rows and columns • Assigning dimensions to columns and rows • Assigning attribute dimensions to the point of view, page, columns, and rows • Selecting dimension members for users to work with • Setting grid properties • Setting dimension properties • Adding formula rows and columns • Setting display properties • Setting Smart View form display options • Setting Prediction options • Setting printing options • Adding and updating format/validation rules • Enabling drilling on shared members • Defining Parent First (PF) functions on forms	See Defining the Layout.
Define page axis and point of view	See Defining Form Page and Point of View.
Select members	See Using the Member Selector
Set form precision, context menus associations, and whether to enable dynamic user variables	See Setting Form Precision and Other Options.
Select business rules and set properties	See About Rules.
Using Smart Push	See Moving Data from One Cube to Another Cube Using Smart Push in *Administering Data Integration*.
Define access permissions	See Setting Up Access Permissions.
Design formula rows and columns	See Designing Forms with Formula Rows and Columns .
Design data validation rules and format rules	See Designing Forms with Data Validation Rules and Form Rules .
Set the evaluation order for dimensions	See Setting the Evaluation Order in the Simplified Interface

To create forms:

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1. Click the **Navigator** icon , then under **Create and Manage**, click **Forms**.
2. Under Form and Ad Hoc Grid Management, click (Create).
3. On the **Properties** tab, enter a name of up to 80 characters, and an optional description of up to 255 characters.
4. Select the **Cube** associated with the form. See Forms and Cubes.
5. Optionally, you can enter instructions describing how to work with the form.
6. Click **Next**, and see Defining the Layout.

Defining the Layout

When you create forms, they initially contain only one row and column, and all dimensions, except for the optional attribute dimensions, are in Point of View. Add rows and columns as needed. When setting row and column layout:

- Assign at least one dimension to the row and column axis.
- You can't select the same dimension for multiple axes. To use dimensions on multiple axes, set user variables in the point of view.
- You can select a dimension from any axis and drag it to another
- You can exclude members of dimensions from the form after a dimension is assigned to a row or column.

> **Note**
>
> Attribute dimensions are optional and are displayed in their own area on the **Layout** tab. Drag the attribute dimension to the point of view, page, row, or column to use it in the form.

> **Caution**
>
> When working with attributes as filters, don't drag the attribute on any axis as you'll cause a cross-dimensional reference which will have a performance impact. When using an attribute dimension as a filter, there is a form option that shows the associated attribute (just like there is an option to show alias).

To define the layout:

1. Open the form, and then click the **Layout** tab.

   See Selecting and Opening Forms and Folders.

2. Click the icon next to a dimension on the **Layout** tab, and then drag it to the desired axis (**Rows** or **Columns**). Optionally, you can also select the attribute dimensions to use on **Point of View**, **Page**, **Rows**, and **Columns**.
3. To edit dimensions added to rows and columns, click

   next to a dimension:

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- Select **Edit** to launch the member selector and define the dimension members (including member relationships) and variables you want to include on the form.

Table 21-13 Member Selection Options for Forms

Member Selection Option	Description
Members	Make selections by clicking a member in the member list. You can also select members by relationship in the **Function Selector**. See Selecting Members for Forms.
Variables	Make selections for the following options: – **User Variables** (see Selecting User Variables as Members) – **Substitution Variables** (see Selecting Substitution Variables as Members) – **Attributes** (see Selecting Attribute Values as Members) – **UDAs** (see Selecting UDAs as Members)

- Select **Exclude** or **Exclude All** to use the member selector to define the dimension members you want to exclude:
  - **Exclude**: Selecting this option excludes members by ID. Only the specified members (base or shared) will be excluded.
  - **Exclude All**: Selecting this option excludes members by name. If a base member is specified, then the base and all of its shared members will be excluded. If a shared member is specified, then this member, its base member, and all other shared members of this member will be excluded.

  Service Administrators can also choose to display or hide excluded members to users in the segment drop-down member selector list using the application setting, **Filter Out Excluded Members in Segment Drop-down**.

  See What Application and System Settings Can I Specify?.
- Select **Clear** to remove selections.

4. **Optional**: To change the order of dimensions in rows or columns, click , and then **Move Dimension Up** or **Move Dimension Down**.
5. Select a row header (such as 1 or 2) to specify row properties, or a column header (such as A or B) to set column properties described in Segment Properties.

   To enable users to transfer global assumptions from a test to a production environment for a form, select **Global Assumptions Form**, and then update the form to store global assumptions. See Designing Forms with Global Assumptions.
6. **Optional**: Add formula rows and columns. See Adding Formula Rows and Columns.
7. **Optional**: Define Parent First (PF) functions, which place the parent members above the children in forms. PF functions can only be defined for the following functions: `DescendantsPF`, `IDescendantsPF`, `IChildrenPF`.

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> **Note**
>
> PF functions are not yet available as selections in the member selector in the form designer or elsewhere in the application. They must be typed in manually on the **Layout** tab in a POV, page, column, or row.
>
> See Defining Parent First Functions on Forms.

Subtopics:

- Segment Properties
- Setting Form Grid Properties
- Setting Dimension Properties
- Setting Display Properties
- Setting Smart View Form Options
- Setting Prediction Options
- Setting Printing Options
- Including Data Validation and Format Rules in Forms
- Enabling Drilling on Shared Members
- Defining Parent First Functions on Forms

Segment Properties

To access **Segment Properties** for a form, click the row number in the form layout.

Table 21-14 Segment Properties

Option	Description
Apply to all rows	Clear to specify different properties for individual rows.
Apply to all columns	Clear to specify different properties for individual columns.
Hide	Conceal a column or row
Read-only	Create a read-only row or column, enabling users to compare old, read-only data with new, editable data.
Show separator	Create a bold border before the segment to visually distinguish it
Suppress hierarchy	Suppress indentation.
Suppress missing data	Hides empty rows or columns. Clear to display rows or columns with "#MISSING" in cells when data is missing.
Suppress invalid Scenario/ Time Periods	This option ties the column display to the start and end period for the scenario. If selected, it suppresses the display of time periods outside of the scenario time period range.
Column width	• **Default**: Use the column width defined at the grid level (under **Grid Properties**) • **Small**: 50px • **Medium**: 100px • **Large**: 300px • **Size-to-Fit**: Column will expand to fit data • **Custom**: Specify the number of pixels

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Table 21-14 (Cont.) Segment Properties

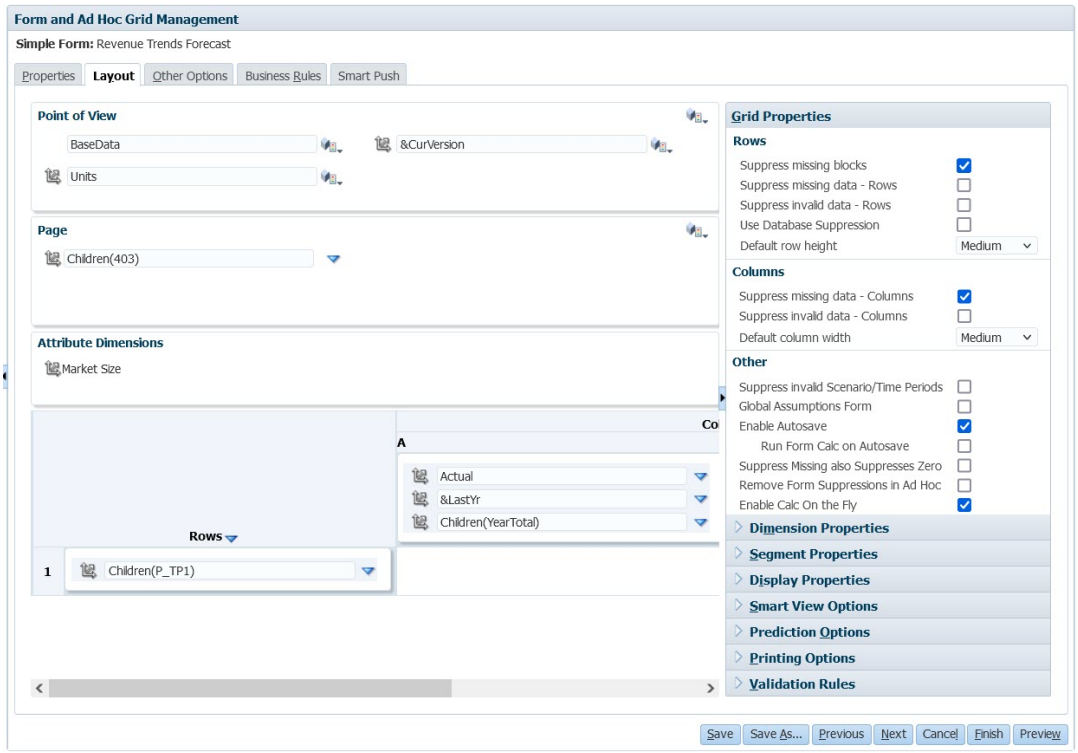
Option	Description
Row height	• **Default**: Use the row height defined at the grid level (under **Grid Properties**) • **Medium**: Use standard height • **Size-to-Fit**: Row will expand to fit data • **Custom**: Specify the number of pixels
Enable drop-down for dimensions	Allows you to set up drop-down member selectors on row dimensions in Oracle Smart View for Office grids and business process web forms. When these drop-down selectors are enabled, form users can directly select a member from the drop-down list on the row dimension in the form. For example, suppose there are two dimensions placed on the row axes, Entities and Line item; the member selector drop-down may be enabled for the Entity dimension members, for the Line item dimension members, or for both. Depending on the settings chosen by the Service Administrator, the drop-down member selector may also allow users to add data to member rows that may have otherwise been suppressed or excluded.

Setting Form Grid Properties

Grid properties define how rows and columns display.

To set grid properties:

1. Open the form, and then click **Layout**. See Selecting and Opening Forms and Folders.



2. In **Grid Properties**, set row and column properties as follows:

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Table 21-15 Form Grid Properties

Option	Description
Suppress missing blocks	On multiple rows: Greatly improves the efficiency of **Suppress missing data**. This option can degrade performance if no or few rows are suppressed. Test forms before and after using this setting to determine if performance is improved. Note that if you use this option: • Some suppressed blocks may ignore Dynamic Calc members • Row members may not indent
Suppress missing data	Hide rows or columns without data. Clear to display "#MISSING" in cells when data is missing.
Suppress invalid data	Hide rows or columns with invalid data. Cells with invalid data are read-only.
Use Database Suppression	Applies row suppression at the Oracle Essbase level instead of at the business process level, which reduces the amount of data on the business process side and eliminates the impact on query thresholds. **Note:** If you use this option and you have a formula column on the form that results in a #MISSING value, the row won't appear on the form because it will be suppressed in Essbase and won't be returned to the business process for formula evaluation.
Default row height	• **Medium**: Row will have medium height • **Size-to-Fit**: Row will expand to fit data • **Custom**: Specify the number of pixels
Default column width	Specify the width in pixels: • **Small**: 50px • **Medium**: 100px • **Large**: 300px • **Size-to-Fit**: Column will expand to fit data • **Custom**: Specify the number of pixels
Suppress invalid Scenario/ Time Periods	This option ties the form grid display to the start and end period for the scenario. If selected, it suppresses the display of time periods outside of the scenario time period range.
Global Assumptions Form	To transfer global assumptions from a test to a production environment for a form, select **Global Assumptions Form**, and update it to store assumptions. See Designing Forms with Global Assumptions.
Enable Autosave	This option lets users use successive undo actions with Ctrl+Z. It also automatically saves their changes when they navigate between cells. **Note:** For optimal performance, use only dense dimensions on rows and columns.
Run Form Calc on Autosave	Only available if **Enable Autosave** is selected. If **Run Form Calc on Autosave** is selected, dynamically calculated cells that depend on modified and saved values (for example, a row with a formula that calculates a percentage of an aggregated parent value) are updated and display on a green background.

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Table 21-15 (Cont.) Form Grid Properties

Option	Description
Suppress Missing also Suppresses Zero	When this option is selected along with the **Suppress missing data** option for forms, all rows or columns containing both #Missing and zeros are suppressed. When selected, this setting overrides the runtime suppression selections made for suppressing missing data or zeroes in Oracle Smart View for Office. **Note:** For Forms 2.0, filtering is not applied if the row or column has a combination of zeroes and #Missing.
Remove Form Suppressions in Ad Hoc	Select to allow Smart View users to perform ad hoc analysis on a form that has other suppression options specified. If there is missing data on the form, the base dimension members for the columns or rows are displayed on the ad hoc grid in Smart View. This allows users to continue ad hoc analysis on the form, even though suppression options were enabled in form design. **Note:** Selecting this option does not override any suppression options users may have set in the Smart View **Options** dialog, **Data Options** tab. If the **Remove Form Suppressions in Ad Hoc** option is enabled for the form, and users see an error when performing ad hoc analysis on the form, then they should clear the suppression options in Smart View.
Enable Calc On the Fly	When you select this option, in Forms 2.0, when you hover over a cell with a member formula attached, you see the member formula associated with the cell and you can perform calc on the fly. Calc on the fly allows you to perform calculations without having to submit your changes.
Pivot Rows and Columns after load	This option enables you to transpose the grid by pivoting the rows and columns after a data load. It is also available from the Actions menu Edit option at runtime.

3. Click **Save** to continue, or **Finish** to save and close the form.

Setting Dimension Properties

You can specify dimension display properties such as if to display the member names or aliases, and whether to permit users to view member formulas. These properties apply to row, column, page, and point of view dimensions.

To set dimension properties:

1. Open the form, and then click **Layout**.

   See Selecting and Opening Forms and Folders.

2. Click in a point of view, page, row, or column dimension, then under **Dimension Properties** apply properties such as:

Table 21-16 Dimension Properties

Property	Description
Apply to all row | column | page dimensions	Apply properties to all row, column, or page dimensions
Apply to all POV dimensions	Apply properties to all point of view dimensions

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Table 21-16 (Cont.) Dimension Properties

Property	Description
Start expanded	For row or column dimensions: Expand and display the dimension member list.
Enable custom attributes	For row or column dimensions in Forms 1.0 only: Use custom attributes.
Drill on shared members	For row or column dimensions: Enable drilling on shared members when the shared member is on a parent member for the main hierarchy.
Flex beyond form definition	For row and column dimensions in flex forms: Select to allow Oracle Smart View for Office users to add row or column members that are not included as part of the form definition. Enable this option on a per-row or per-column basis. You may also select **Apply to all row dimensions**, but note that all selected properties in the current row dimension will be applied to all row dimensions in the form. This option is displayed only when **Enable flex form for rows** or **Enable flex form for columns**, or both, is selected in the **Smart View Options** section of the form definition (see Setting Smart View Form Options).
Row Width	For row dimensions: Select from the following column sizing options: • **Small**: 50px • **Medium**: 100px • **Large**: 300px • **Size-to-Fit** (default): Column will expand to fit content • **Custom**: Specify the number of pixels Note that this setting is retained if the form is viewed on its own or within a dashboard.

3. Click **Save** to continue, or **Finish** to save and close the form.

Setting Display Properties

Define display options on the **Layout** tab.

> **Note**
>
> - Assign the Account dimension to a row axis.
> - Account, Entity, Versions, and Scenario dimensions can't be assigned to the column axis.
> - The Entity dimension can be assigned to any axis.
> - Assign Version and Scenario dimensions to the page or Point of View axis.

To set display properties:

1. Open the form, and then click **Layout**.

   See Selecting and Opening Forms and Folders.

2. Select **Display Properties**, and then options such as:

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Table 21-17 Display Properties

Option	Description
Make form read-only	Prevent users from editing the form.
Hide form	Conceal forms that are part of a dashboard, or accessed from menus or task lists.
Display missing values as blank	Display cells without data as empty cells. Clear to display "#MISSING". See Displaying #MISSING with Smart Lists.
Enable Mass Allocate	Users must have the Mass Allocate role to use this option. See Spreading Values Using Allocations in *Working with FreeForm*.
Enable Grid Spread	See Spreading Values Using Grid Spread in *Working with FreeForm*.
Enable cell-level document	Enable users to add, edit, and view documents in form cells, depending on access permissions. See Working with Attachments in *Working with FreeForm*.
Message for forms with no data	Enter custom text to display when invalid data exist. Leave blank to display `There are no valid rows of data for this form`.
Hide Save Confirmation Message	Prevents the form save confirmation message from being displayed to users.

3. Click **Save** to save your work and continue, or click **Finish** to save your work and close the form.

Setting Smart View Form Options

Specify options for form display in Oracle Smart View for Office.

To set Smart View form options:

1. Open the form, and then click **Layout**.

   See Selecting and Opening Forms and Folders.

2. Select **Smart View Options**, and then settings such as:

Table 21-18 Smart View Options

Option	Description
Disable spreading	Disables spreading options in Smart View, including default spreading, spreading data for time periods, spreading using grid spread, and spreading using mass allocations.
Disable formatting	When selected, the formatting definition for a form is not included when the form is opened in Smart View.
Repeat member labels	Allows repeated member labels in forms. If this setting is cleared, you may override it in Smart View to allow repeated members in forms by selecting **Repeat Member Labels** in the Smart View **Options** dialog, **Formatting** tab. For forms with **Enable drop-down for dimensions** selected (Segment property), the form layout will automatically behave as if the **Repeat member labels** option is selected, regardless of whether or not it is selected. For flex forms (see **Enable flex form for rows** and **Enable flex form for columns** below), the **Repeat member labels** option is automatically enabled and cannot be cleared.

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Table 21-18 (Cont.) Smart View Options

Option	Description
Disable sheet protection	Disables protection on data cells in forms in Smart View. Allows users to enter or paste data into cells, but does not allow users to submit the changed data. By default, this option is cleared.
Enable flex form for rows	Enables the form as a flex form on rows in Smart View. When this option is enabled, the **Repeat member labels** option is automatically enabled, and cannot be cleared. See Designing Flex Forms.
Enable flex form for columns	Enables the form as a flex form on columns in Smart View. When this option is enabled, the **Repeat member labels** option is automatically enabled, and cannot be cleared. See Designing Flex Forms.
Show invalid members for flex form	Retains invalid members on the flex form sheet after a refresh in Smart View. Invalid members can be the result of: • Mistyping or misspelling a member name • Errors when copying and pasting member names from one sheet to another, or within the same sheet • Entering a member name that is beyond the form definition When this check box is enabled, invalid members are retained on the sheet upon Refresh and highlighted, making them easy to find and correct. See Designing Flex Forms.

3. Click **Save** to continue, or **Finish** to save and close the form.

Setting Prediction Options

To set prediction options to use when using Predictive Planning on a form:

1. Open the form, and then click **Layout**.
   See Selecting and Opening Forms and Folders.
2. Select **Prediction Options**, and then settings such as:

Table 21-19 Prediction Options

Option	Description
Include Events	Include events associated with the selected calendar by default when performing predictions on the form.

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Table 21-19 (Cont.) Prediction Options

Option	Description
Select Calendar	Select the calendar to associate with the form by default. Any events associated with the calendar are considered for predictions if you select **Include Events**. In FreeForm applications, forms must have a calendar associated with them, and the calendar must map to the granularity defined in the form. If you don't associate the calendar with the form in the Form Designer, a user who opens the form and runs a prediction is prompted to select a calendar for use with the prediction on the form. This selection is saved with the form.
Prediction for Dynamic Calc	Specify to generate predictions at the parent level at run time and load the form with predicted values. Members on the form in Scenario, Version, Plan Element or relevant dimensions must be associated with the UDA **HSP_DYN_PREDICT**. For more information, see Using Dynamic Calc Members in Insights and Predictive Planning, Auto Predict, and Advanced Predictions.

For more information about calendars and events in predictions, see Configuring IPM Insights.

3. Click **Save** to continue, or **Finish** to save and close the form.
These selections are saved with the form, so it is the default option for all users of the form.

Setting Printing Options

To set printing options:

1. Open the form, and then click **Layout**.

   See Selecting and Opening Forms and Folders.

2. Select **Printing Options**, and then settings such as:

Table 21-20 Printing Options

Option	Description
Include supporting detail	Print supporting detail as extra rows in PDF files as follows: **Normal Order**: In the same order as on the **Supporting Detail** page **Reverse Order**: Before the member associated with it. Supporting detail for children displays above parents, and the order of siblings is preserved
Show comments	Display cell text notes
Format data	Apply number format settings
Apply precision	Print data using a specific number of decimal places

3. Click **Save** to continue, or **Finish** to save and close the form.

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Including Data Validation and Format Rules in Forms

You can add and update format and data validation rules for grids, columns, rows, or cells, on the **Layout** tab. When processed, rules may change the format of cells, and, display validation messages to users during data entry. Areas on the **Layout** tab with a green background, indicate locations where format and data validation rules have been defined.

> **Note**
>
> Before adding a format or data validation rule to a form, consider the function it will perform. For information and best practices about using format and data validation rules, see Managing Data Validation and Format Rules in Forms.

To include format and data validation rules in forms:

1. Open the form, and then click **Layout**.

   See Selecting and Opening Forms and Folders.

2. Select **Format/Validation Rules**, and then select rule options:

Table 21-21 Format/Validation Rules Options

Option	Description
Rules for location	Select a location to view rules that were defined at that location (grid, columns, rows, or cells). Select **All Rules** to view all of the rules that have been defined for a form.
(Add Validation Rule)	Launches the **Format/Validation Rule Builder** where you can specify new rule properties. For data validation rules, see Creating and Updating Data Validation Rules in Forms. For format rules, see Creating and Updating Format Rules in Forms.
(Edit Validation Rule)	Launches the **Format/Validation Rule Builder** where you can modify properties for the selected rule. For data validation rules, see Creating and Updating Data Validation Rules in Forms. For format rules, see Creating and Updating Format Rules in Forms.
(Move to Top, Move Up, Move Down, Move to Bottom)	If multiple rules are defined at the same location, you can change the order in which rules are processed when rules have the same precedence. To move a rule up, down, or to the top or bottom of the list, select the rule and click the arrows.
(Delete Selected Validation Rule(s))	Deletes the selected rule.

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> **Note**
>
> You can also right-click form elements in the **Layout** tab to view context menu options. The context menu options that display are context-sensitive, and will differ depending on the context location and if rules were already copied.
>
> Context menu options include:
>
> - **Add/Edit Format/Validation Rules**: Create or modify format or validation rules for individual data cells or all data cells in a grid, row, or column in the **Format/Validation Rule Builder**.
> - **Copy | Paste Format/Validation Rules**: Copy and paste the format/validation rules at the selected locations.
> - **Add/Edit Format/Validation Rules for Metadata**: Create or modify format or validation rules for the metadata cells only for a grid, row, or a column in the **Format/Validation Rule Builder**.
> - **Copy | Paste Format/Validation Rules for Metadata**: Copy and paste the format/validation rules for metadata at the selected locations.

3. For data validation rules, select from the following validation options that are displayed at the bottom of the **Format/Validation Rules** pane, then build and validate the rules:

Table 21-22 Validation Options

Option	Description
Validate only for pages with existing blocks	For data validation rules, run validations only on page combinations that have potential blocks. The only exceptions being page combination having any Dynamic Calc, Dynamic Calc and Store, Label only, or Store with one child member settings. In this case, the page is loaded.
Validate only for cells and pages the user has access to	For data validation rules, run validations as the current user, using their security, and, not a Service Administrator.

For more information, see Creating and Updating Data Validation Rules in Forms.

4. In the form, click **Next** to continue building the form, and then validate and save it.

Enabling Drilling on Shared Members

Form designers often use shared members to create alternate rollup structures within an application. To increase usability for these alternative hierarchies, the application provides the ability to drill on a shared member for a given dimension. The option to drill is a dimension property which provides the flexibility to focus the drill behavior to a particular dimension. The drilling option is only available for dimensions placed on the row or column axis during form design. Page or POV dimensions don't have drill capability.

For example, using the following hierarchy representing the Entity dimension:

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World			
	USA		
		West	
			CA
			AZ
			CO
		East	
			NY
			MA
			PA
		South	
			TX
			FL
			NM
		North	
			WA
			MI
			MN
	Southwest		
		South (Shared)	
		West (Shared)	
	Northeast		
		North (Shared)	
		East (Shared)	

A form with iDescendants(Southwest) defined on the row would return the following members:

- TX
- FL
- NM
- South
- CA
- AZ
- CO
- West
- Southwest

To enable drilling on shared members in a form:

1. Open the form, and then click **Layout**.

   See Selecting and Opening Forms and Folders.
2. Click in a row or column dimension to expand **Dimension Properties**.
3. Under **Dimension Properties**, select **Drill on shared members**.
4. Click **Save**.

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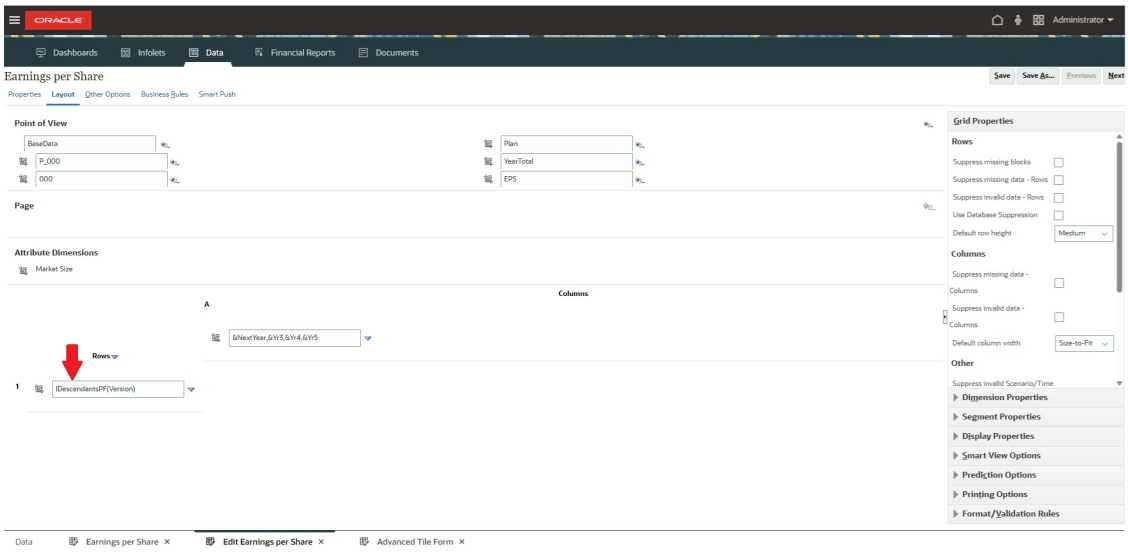
Defining Parent First Functions on Forms

You can define Parent First (PF) functions, which place the parent members above the children in forms, for the following functions:

- `DescendantsPF`
- `IDescendantsPF`
- `IChildrenPF`

PF functions are not available as selections in the member selector in the form designer or elsewhere in the application. You can only type them in on the **Layout** tab of the form designer.



PF Function Examples

The following examples illustrate how PF functions display on forms:

Table 21-23 Examples of PF Functions on Forms

Function	Without PF Function	With PF Function
`Descendants`	`Descendants(YearTotal)` Jan, Feb, Mar, **Q1**, Apr, May, Jun, **Q2**, Jul, Aug, Sep, **Q3**, Oct, Nov, Dec, **Q4**	`DescendantsPF(YearTotal)` **Q1**, Jan, Feb, Mar, **Q2**, Apr, May, Jun, **Q3**, Jul, Aug, Sep, **Q4**, Oct, Nov, Dec
`IDescendants`	`IDescendants(YearTotal)` Jan, Feb, Mar, **Q1**, Apr, May, Jun, **Q2**, Jul, Aug, Sep, **Q3**, Oct, Nov, Dec, **Q4**, **YearTotal**	`IDescendantsPF(YearTotal)` **YearTotal**, **Q1**, Jan, Feb, Mar, **Q2**, Apr, May, Jun, **Q3**, Jul, Aug, Sep, **Q4**, Oct, Nov, Dec
`IChildren`	`IChildren(YearTotal)` Q1, Q2, Q3, Q4, **YearTotal**	`IChildrenPF(YearTotal)` **YearTotal**, Q1, Q2, Q3, Q4

Manually Entering PF Functions

To create a PF function in the form designer:

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1. Open a form for editing or create a form, and then click the **Layout** tab.
2. In a POV, Page, Column, or Row, find an already defined `Descendants`, `IDescendants`, or `IChildren` function or define one.

   For a full list of member relationship functions and their descriptions, see Member Relationships.
3. Manually enter `PF` directly after `Descendants`, `IDescendants`, or `IChildren`; for example, `DescendantsPF(YearTotal)`, `IDescendantsPF(YearTotal)`, or `IChildrenPF(YearTotal)`.
4. Click **Save** and then view the form. The parent members will display first in the form's hierarchy.

About Precision Settings

In **Other Options**, you can control data precision by applying minimum and maximum values for different account types. For example, you can truncate and round the decimal portion of longer numbers. Precision settings affect just value display, not their stored values. For example, if **Minimum Precision** is 2, and the value 100 from Q1 is spread to Jan, Feb, and Mar, the month cells display 33.33 when they are not selected. When selected, they display their actual values (for example, 33.33333333333333). Because the number of decimal places for storing values is finite, when the values for Jan, Feb, and Mar aggregate to Q1, 33.33333333333333 is multiplied by 3, and Q1's value is 99.99999999999998.

Setting Form Precision and Other Options

To set precision and other options:

1. Open the form, and then click **Other Options**.

   See Selecting and Opening Forms and Folders.
2. In **Precision**, specify the number of decimal positions displayed in a cell for **Currency Values**, **Non-Currency Values**, and **Percentage Values**.

   See About Precision Settings.

   Specify **Minimum** values to add zeros to numbers with few decimal places. Specify **Maximum** values to truncate and round the decimal portion of large numbers. For example:

Table 21-24 Data Precision Examples

Value	Minimum Precision	Maximum Precision	Displayed Value
100	0	Any	100
100	3	Any number greater than or equal to 3 or None	100.000
100.12345	Any number less than or equal to 5	None	100.12345
100.12345	7	None	100.1234500
100.12345	Any number less than or equal to 3	3	100.123
100.12345	0	0	100
100.12345	2	4	100.1235
100	2	4	100.00

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> **Note**
>
> Your selections override the precision set for the currency member. To use the currency member's precision setting instead, select **Use Currency member precision setting**.

3. Under **Context Menus**, associate menus with the form by moving them from **Available Menus** to **Selected Menus**. For multiple menus, select a "move" option on the right to specify the display order.

   For information on how to create context menus (Action Menus), see Administering Action Menus.

4. Select **Enable dynamic user variables** to allow dynamic user variables in the form, and move **Available User Variables** to **Selected User Variables**.

   See Dynamically Setting User Variables in *Working with FreeForm*.

Moving Data Using Smart Push

For more meaningful and complete reporting, you can use Smart Push to instantly move comments, attachments, and supporting detail from source cubes to a reporting cube while working in forms. Users can then do more analysis on the data coming from the different cubes.

To learn how to move data using Smart Push, see Moving Data from One Cube to Another Cube Using Smart Push in *Administering Data Integration*.

Troubleshooting

For help with resolving issues with Smart Push, see Resolving Issues with Smart Push in *Operations Guide*.

Creating Asymmetric Rows and Columns

Asymmetric rows and columns contain different sets of members selected across the same dimensions. For example:

Row/Column A—Scenario = Actual, Time Period = Q1

Row/Column B—Scenario = Budget, Time Period = Q2, Q3, Q4

To create asymmetric rows or columns:

1. Open the form, and then click the **Layout** tab.

   See Selecting and Opening Forms and Folders.

2. Click to select the dimension to modify.

3. Click (Member Selector) to the right of the dimension name, and then modify the members selected for this dimension.

   See Using the Member Selector.

4. Click **Save** to continue, or click **Finish** to save and close.

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Adding Formula Rows and Columns

You can define formulas for rows and columns. For example, you can create a formula column to calculate the percentage variance between monthly sales columns. The formula for a row or column applies to all row or column dimensions. To define or assign existing formulas to forms, select the row or column on the **Layout** tab, and use the options under **Segment Properties**.

See Creating Formulas.

> **Tip**
>
> Consider adding a formula row between two other rows to create a blank row. Blank rows help you visually separate subtotals and totals in a form.

To add formula rows and columns:

1. Open the form, and then click the **Layout** tab.

   See Selecting and Opening Forms and Folders.
2. On the **Layout** tab, right-click **Rows** or **Columns**.
3. Select **Add Formula Row** or **Add Formula Column**.
4. Click the new **Formula Label**, and then enter the formula name.
5. Click the row or column number of the Formula Label, and specify **Segment Properties** such as **Display formula on form** so users can view the formula when they click (Formula).
6. Under **Segment Properties**, for each dimension in **Formula Data Type**, select how to display the formula result, such as a Percentage, Smart List, Date, or in a Text format.
7. Enter the formula, and then click .

   See Editing Formulas.
8. Click **Validate** to find and fix any errors.

Defining Form Page and Point of View

You can select dimensions and members for the page axis and point of view (POV). POV dimensions and members must be valid for the form cube, and not assigned to a page, column, or row axis. The POV sets the members that define data intersections and the form's context. If you use user variables, variable names displays in the POV.

See Managing User Variables.

To define page axis and point of view:

1. Open the form, and then click **Layout**.

   See Selecting and Opening Forms and Folders.
2. Click and then drag that dimension to **Page**.
3. Click (Member Selector) for each page axis dimension and select members.

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See Using the Member Selector.

> **Note**
>
> To shorten the number of members displayed in the member list so that the **Search** box can be easily seen, go to the Home page, click **Application**, and then click **Settings**. For the **Number of Items on the Page Drop-down** option, enter a lower value.

4. Repeat Steps 2 and 3 as needed. Assign multiple dimensions to the page axis to enable users to change dimensionality while entering data. They can use **Display** to use the most recent selection.
5. Specify **Dimension Properties**.

   See Editing Dimension Properties.
6. Click and drag a dimension to **Point of View** to add them, or members, to the form's point of view.
7. Specify **Dimension Properties**.

   See Editing Dimension Properties.
8. Click **Save** to continue, or **Finish** to save and close.

Designing Specific Types of Forms

- Designing Forms with Formula Rows and Columns
- Designing Forms with Data Validation Rules and Form Rules
- Designing Forms with Global Assumptions
- Designing Flex Forms

Designing Forms with Formula Rows and Columns

Formula rows and columns contain formulas that perform mathematical calculations on grid members. For example, you might want to rank values in a particular column or calculate variance between two rows. A formula consists of grid references, arithmetic operators, and mathematical functions. To define or assign existing formulas to forms, select the appropriate row or column on the **Layout** tab and then make selections under **Segment Properties**.

See Adding Formula Rows and Columns.

For information about creating formulas and using the mathematical functions, see Form Formula Functions.

Designing Forms with Data Validation Rules and Form Rules

You can design forms that include predefined data validation rules that help implement business policies and practices. You can specify cell colors and data validation messages that are generated on the form if entered data violates a validation rule. Data validation rules are saved as part of the form.

You can also define format rules for a form at design time, and then users will have the ability to view the format rules for a form at runtime when styles are applied.

See Including Data Validation and Format Rules in Forms.

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See also Managing Data Validation and Format Rules in Forms.

Designing Forms with Global Assumptions

Form designers typically create a specific-purpose form that stores global assumptions (also called "drivers" or "driver data"), such as a depreciation value, tax rate, or unit price. This is useful if you want to migrate the form's driver data using Migration, for example, from a test environment to a production environment. Such forms are typically small, and are used as "lookup tables" that drive calculations.

> **Note**
>
> Global assumptions can't be used with dashboards or forms with multiple segments.

To enable transferring global assumptions from a test to production environment during form design

1. In the **Layout** tab, click **Grid Properties**.
2. Under **Other**, select **Global Assumptions Form**.
3. Update the form to store driver data such as a tax rate. When migrating with Migration, this enables migrating the driver data contained in the tagged form.

Designing Flex Forms

Flex forms are a form type that provides flexible row and column management in Oracle Smart View for Office.

Flex forms retain all regular form properties and features, such as running business rules attached to the flex form, running rules on save or before save, and enabling some ad hoc-specific functionality. However, with flex forms, dimension and member row and column cells and all data cells are unprotected. This means that Smart View users can rearrange row and column members, and sort or move rows or columns. Modified row and column order is maintained on refresh and during submit. Smart View users can also filter data using Excel's filtering functionality.

> **Note**
>
> Flex forms are used only in Smart View and not in the web interface.

During form definition:

- Select **Enable flex form for rows** or **Enable flex form for columns**, or both, to enable flex form-specific features. These options are located in the **Layout** tab, under **Smart View Options**.

  See Setting Smart View Form Options.
- If you have selected **Enable flex form for rows** or **Enable flex form for columns**, or both, then, in **Dimension Properties**, you can select either **Member name** or **Alias**, but not both.

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- If you have selected **Enable flex form for rows** or **Enable flex form for columns**, or both, then you can enable the **Show invalid members on flex form** option. This option is located in the **Layout** tab, under **Smart View Options**.

  When entering member names in a flex form, errors may occur; for example, a user may mistype or misspell a member name; make an error when copying and pasting member names from one sheet to another, or within the same sheet; or enter a member name that is beyond the form definition, resulting in an invalid member. By enabling the **Show invalid members for flex form** option, the flex form will retain any invalid members on the form after refresh. Invalid members are highlighted making them easy to locate and edit. After users make necessary corrections to member names and refresh again, they can then enter and submit data on the flex form.

  See Setting Smart View Form Options.

- On a row-by-row basis, or column-by-column basis, or both, enable the **Flex beyond form definition** option. The **Flex beyond form definition** property is located under **Dimension Properties** in the **Layout tab** when a row or column is selected.

  When this option is enabled for a row or column dimension, Smart View users may add row or column members to the flex form that are not included as part of the form definition. Do this only for the rows or columns you require flex functionality.

  See Setting Dimension Properties.

- Define action menus to open another flex form in Smart View.

  To define action menus:

  See Administering Action Menus.

  To attach an action menu to a flex form:

  See Setting Form Precision and Other Options.

For information on using flex forms, see Flex Forms in *Working with Oracle Smart View for Office*.

Working with Forms and Form Components

Related Topics

- Selecting and Opening Forms and Folders
- Previewing Forms
- Printing Form Definitions
- Searching for Forms
- Editing Forms
- Moving, Deleting, and Renaming Forms
- How Cell Formatting in Smart View Persists in FreeForm

Selecting and Opening Forms and Folders

Use these procedures to select and open form folders and the forms they contain.

To select and open form folders or forms:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Forms**.
2. Perform one of the following steps:

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- To open a folder, expand Library beneath **Folders**, and then select a form folder.
- To open a form, expand Library beneath **Folders**, select a form folder, and then select a form from the list that displays in the **Form and Ad Hoc Grid Management** pane.

3. After you select a form folder, use the buttons next to **Folders** to create, delete, rename, move, and assign access to the folder.
4. To manage forms, select one of the following actions in the **Form and Ad Hoc Grid Management** pane:
   - To open the form, select a form, and then click **Edit**.
   - To create a form, click **Create simple form** .
   - To edit, delete, move, assign access, rename forms, or show usage (for composite forms only), select a form, and use the corresponding button.

   The icon next to the form name indicates the type of form:

   - Simple form
   - Composite form

     > **Note**
     >
     > Oracle has stopped supporting composite forms. However, your existing composite forms will continue to work. Oracle recommends that you shift your usage to dashboards instead of composite forms.

   - Ad hoc grid

For information about setting up ad hoc grids:

See Using Ad Hoc for Data Entry and Analysis in *Working with FreeForm*.

Previewing Forms

While you're designing forms, you can preview the dimensions that are assigned to the Point of View, columns, rows, and page axes. Previewing displays member attributes, alias, and data associated with forms, although new data can't be entered.

Previewing completes regular form design validation checks, and checks for proper evaluation of any data validation rules included in the form. Data validation rules must be properly completed before the form can be saved. In addition, data validation rules are saved as part of the form. If you don't save changes to a form, any data validation rule changes made after the form was last saved are lost.

To preview a form’s design:

1. Under **Form and Ad Hoc Grid Management**, select a form, and then click **Edit** to open the form.
2. With the form open, click **Preview**.

   The form opens in edit mode in a new tab.
3. Resolve any issues reported during the design validation checks, including any issues with data validation rules.

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4. Save the form to ensure that updates are saved, including any changes to data validation rules.

Printing Form Definitions

Service Administrators can print form definition reports that include information on dimension members, business rules, access permissions, and other form components.

To create and print form definition reports:

1. Click the **Navigator** icon, and then under **Monitor and Explore**, click **System Reports**.
2. Select the **Forms** tab.
3. Under **Select Report Options**, elect the form definitions to print by moving them from **Available Forms** to **Selected Forms**.
4. **Optional**: Select **Include Member Selection List** to include column and row members on the report.
5. **Optional**: Select **Include Business Rules** to include associated business rules.
6. Click **Create Report**.

   Adobe Acrobat generates a consolidated report, including:

   - Cube
   - Description
   - Column dimension and members and additional column definitions
   - Row dimension and members and additional row definitions
   - Page and Point of View dimensions
   - Form access permissions
   - Associated business rules
7. To print the report, select **File**, and then **Print** on the Adobe toolbar.

Searching for Forms

To search for forms:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Forms**.
2. Under Form and Ad Hoc Grid Management, enter part or all the form name in the **Search** box.
3. Click (Search Down) to search forward or (Search Up) to search previous.

Editing Forms

You can edit the layout, members, and properties of forms. For example, you can add formula rows or columns to a form.

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> **Note**
>
> - Previously, you could create composite forms (forms that comprise multiple simple forms) to summarize data. Oracle has stopped supporting composite forms. However your existing composite forms will continue to work. Oracle recommends that you shift your usage to dashboards instead of composite forms.
>
> See Designing Dashboards.
> - The standardized **Data** page does not support listing composite forms. To view composite forms on the **Data** page, you must revert to the legacy pages using the **Legacy List Pages** application setting. For more information about the standardized artifact pages and reverting to legacy pages, see About the Standardized Cloud EPM Platform Artifact Pages.

To edit forms:

1. Select the form, then click (Edit). See Selecting and Opening Forms and Folders.

   Alternatively, on the Home page, you can click **Data**. Expand a form folder, click the name of a form to open it, click **Actions**, and then select **Edit**.
2. Select:
   a. **Properties** to edit the description and instructions. See Creating Forms.
   b. **Layout** to edit form layout. See Defining the Layout.
   c. **Other Options** to edit form precision, change the context menus associated with the form, enable/disable dynamic user variables, and select user variables. See Setting Form Precision and Other Options.

      See also, Administering Action Menus and Dynamically Setting User Variables.
   d. **Business Rules** to change which business rules are associated with the form, or modify business rule properties.

      See Administering Rules.
3. Choose an option:
   - To save your work, click **Save**.
   - To save a variant of a form with a new name, click **Save As**, enter a **New Form Name**, and then click **OK**. The updated form with the new name is added to the form list and the newly named form opens in a new dynamic tab.

Moving, Deleting, and Renaming Forms

To move, delete, and rename forms:

1. Select the form.

   See Selecting and Opening Forms and Folders.
2. Perform any of the following tasks:
   - To move a form, click **Move**, and then select the destination folder.

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> **Note**
>
> You can move multiple forms simultaneously if they are in the same folder.

- To delete a form, click **Delete**.
- To rename a form, click **Rename**, and then enter the new name.

3. Click **OK**.

How Cell Formatting in Smart View Persists in FreeForm

> **Note**
>
> Before you perform the following steps:
>
> See Saving Native Excel Formatting to Cloud EPM in *Working with Oracle Smart View for Office*.

To select the formatting for a form:

1. In the form, right-click, and then select **Apply**.
2. Select:
   - **Cell Styles**: To use the business process formatting
   - **Custom Styles**: To use the formatting saved in Oracle Smart View for Office

For more information:

- On which Microsoft Excel formatting features are supported in Smart View and FreeForm:

  See About Smart View Formatting in Web Forms in *Working with FreeForm*.
- On saving Excel formatting, see Saving Excel Formatting on a Form in *Smart View for Office User's Guide*.

Managing Forms and Folders

Use the Form and Ad Hoc Grid Management and the Business Rule Folders pages to manage folders and forms. To access the Form and Ad Hoc Grid Management page, click the **Navigator** icon, and then under **Create and Manage**, click **Forms**. You can access the Business Rule Folders by clicking the **Navigator** icon, and then selecting **Rules Security** under **Create and Manage**.

Table 21-25 Tasks

Task	Topic
Create folders	Creating Folders
Move, delete, or rename folders	Working with Folders
Create forms	Creating Forms

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Table 21-25 (Cont.) Tasks

Task	Topic
Assign permissions to forms and folders	About Assigning Permissions to Artifacts, Rules, and Folders
Move and delete forms	Moving, Deleting, and Renaming Forms

To view all forms or business rules in a folder, click the folder's name in the left-hand folders area. To select all the forms, select the check box at the top of the forms list.

Creating Folders

Use folders to hierarchically organize forms and business rules. You can move folders within the hierarchy, and give folders the same name if they are on different hierarchical levels. You can't:

- Delete folders unless they are empty
- Select multiple folders
- Rename, move, or delete the top-level folder. For Forms, the top-level folder is called Library. For Business Rules, the top-level folder is called CalcMgrRules.
- Add files to the top-level folder unless you're an administrator

To create folders:

1. Perform an action:
   - For form folders: Click the **Navigator** icon, and then under **Create and Manage**, click **Forms**.
   - For Calculation Manager business rule folders: Click the **Navigator** icon, and then under **Create and Manage**, click **Rules Security**.
2. Select the folder under which to create the folder.
3. Above the folders list, click **Create**.
4. Enter the folder name.
5. Click **OK**.

Working with Folders

To move, delete, or rename folders:

1. For form folders: Click the **Navigator** icon, and then under **Create and Manage**, click **Forms**.

   For business rule folders: Click the **Navigator** icon, and then under **Create and Manage**, click **Rules Security**.
2. Select the folder to move, delete, or rename.
3. Perform a task:
   - To move, click **Move**. Select the destination folder to which to move the selected folder.

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> **Note**
>
> When you move folders, all nested folders, forms, and Calculation Manager business rules within them are also moved.

- To delete, click **Delete**.
- To rename, click **Rename**, and then enter the new name.

4. Click **OK**.

> **Tip**
>
> Did you know that folder names can be localized based on the user's browser locale? To localize a folder name (for example, from English to Japanese), you can customize the folder name on the **Artifact Labels** page in the **Tools** cluster.
>
> For more information, see Specifying Artifact Labels.

Administering Rules

- About Rules
- Adding and Removing Rules in Forms
- Setting Business Rule Properties
- Viewing Rules Usage
- About Runtime Prompts
- Using Groovy Rules

About Rules

With appropriate access, users can launch business rules from the application. Rules can also prompt users for input when rules are launched.

To launch business rules from within data forms or independently in the application, the rules must be deployed from Calculation Manager. For optimum performance, business rules that run in forms should be designed to execute within three minutes. For rules with a longer execution time, you can schedule batch processing or run the rules during non-peak hours.

You can diagnose performance issues with business rules and pinpoint and address any issues using activity reports. Activity reports identify which business rules are taking the longest to execute. You can then open the rules in Calculation Manager, review the log, and optimize the steps within the rule to improve performance.

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Viewing Business Rules - Standardized Interface

> **Note**
>
> This section provides details about working with the **Rules** page in the default standardized interface. To learn more about the standardized interface, see About the Standardized Cloud EPM Platform Artifact Pages.
>
> However, there are cases when the legacy artifact pages might still be in use. If you're viewing the legacy **Rules** page, see Viewing Business Rules - Legacy Interface.

To view the business rules in your application, on the Home page, click **Rules**.

Enter search criteria and the search results will display in a drop-down under the **Search** box. Click **Enter** to view the search results in the list region. Search results list all instances of rules that include the search criteria.

Filter chips display the search criteria that is entered in the **Search** box. Filter chips can be selected or removed to quickly filter information without changing the current view. For rules, we've provided an out-of-the-box filter chip called **Type** which filters rules by type. Click the **Type** filter chip and select **Rules**, **Rulesets**, **Templates**, or select all three options.

Clicking the name of a rule opens the rule in Calculation Manager within a new tab.

Click **System View** to display the default view when you launch Calculation Manager. Calculation Manager displays in a new tab and it lists all the applications and objects to which you have access. See Viewing Objects in Calculation Manager in *Designing with Calculation Manager*.

If you select an object in the list region of the **Rules** page, the (global) **Actions** menu displays, and you can select **System View** or you can assign permissions to the selected objects.

Each rules line item also has an item-specific **Actions** menu which includes the following options:

- **Edit**: Open the rule in Calculation Manager within a new tab.
- **Launch**: Run the rule.
- **Assign Permissions**: Assign access permissions to the rule.

Viewing Business Rules - Legacy Interface

> **Note**
>
> This section provides details about working with the **Rules** page in the legacy interface. Starting with the October 2025 (25.10) update, Oracle plans to de-support legacy artifact pages. For all new and existing environments, the **Legacy List Pages** application setting will be removed and the **Rules** page will only render as standardized.

To view the business rules in your application, on the Home page, click **Rules**.

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You can filter the rules by **Cube** and by **Rule Type**. Click (Filter) to select filter criteria.

To assign permissions to a rule, click (Permission) next to the rule.

To run a rule, click (Launch) next to the rule.

Working with Business Rules

For information on:

- Creating, updating, and deploying business rules, see Deploying Business Rules and Business Rulesets from the Deployment View in *Designing with Calculation Manager*
- Using runtime prompts, see About Runtime Prompts
- Generating a report that details where rules are used, see Viewing Rules Usage
- Viewing an activity report to diagnose rules performance issues, see Viewing Activity Reports
- Optimizing business rules, see Optimizing Business Rules in *Designing with Calculation Manager*

Troubleshooting

For help with designing rules to avoid common execution errors and optimizing slow rules, see Troubleshooting Business Rule Errors and Performance in *Operations Guide*.

Adding and Removing Rules in Forms

You can associate one or more rules with a form, by cube. Users can launch associated rules from the form to calculate and allocate values. You can set whether each rule associated with a form automatically launches when the form is opened or saved. You can also change the order in which the rules are launched or remove rules from a form.

Note the following:

- Rules must be deployed to the business process from Calculation Manager before you can add rules to business process forms.

  See About Rules.
- Likewise, if a rule is deleted in Calculation Manager, the rules must be redeployed to the business process in order for them to no longer be used in the business process. Optionally, you can manually remove a rule from a form using the procedure in this topic.
- Rules that are set to automatically run when a form is loaded or saved never run in the background.

To add, update, or remove business rules in forms:

1. Take an action:
   - To update the current form, click **Actions**, and then click **Business Rules**.
   - To open a form for editing so you can associate rules, go to the Home page and click the **Navigator** icon , and then under **Create and Manage**, click **Forms**. Select the form, click (Edit), and then click **Business Rules**.
2. From the **Cube** drop-down menu, select the cube.

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3. From the **Business Rules** list, select the rules to associate with the form, and add them to **Selected Business Rules**. To remove associated rules, select the rules under **Selected Business Rules** and remove them.

   By default, the Calculate Form rule is selected. Calculate Form is automatically created for forms to calculate subtotals. You can remove Calculate Form to prevent users from calculating data in forms.

4. To change the order of selected business rules (the order in which rules display and launch), select a business rule in **Selected Business Rules** and click the up or down arrow to move it up or down in the list. The rule listed first displays and launches first; the rule at the bottom of the list displays and launches last.

   > ⚠ **Caution**
   >
   > The order in which business rules launch is important and may affect data.

5. To set business rule properties, click **Properties**.

   See Setting Business Rule Properties.

6. Click **Save** to save your work and continue creating or editing the form, or click **Finish** to save your work and close the form.

Setting Business Rule Properties

You can specify whether business rules associated with forms launch automatically when users load (open) or save the form. If business rules have runtime prompts, you can set whether the default members in the runtime prompt match the members selected in the page and Point of View axes.

To set business rule properties:

1. Take an action:
   - To update the current form, click **Actions**, and then click **Business Rules**.
   - To open a form for editing, go to the Home page and click the **Navigator** icon , and then under **Create and Manage**, click **Forms**. Select the form, click (Edit), and then click **Business Rules**.
2. There are four phases in a form where a rule can be assigned and each may have information relevant to that phase. Select from the following:
   - **Run Before Load**
   - **Run After Load**
   - **Run Before Save**
   - **Run After Save**

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> **Note**
>
> - **Run After Load** and **Run Before Save** are enabled only for Groovy rules. You can't select these options for calc scripts or graphical rules. For more information about Groovy rules, see Using Groovy Rules.
> - Some actions such as saving the grid, changing the page, or launching an action menu item will also reload the page after the operation completes. This ensures the grid contains the newest data. When this happens, the before and after load actions will run as they do in normal page loads.
> - While hidden runtime prompts are not supported for calc script rules on the before and after load options, they are supported for Groovy rules.
> - For information about where Groovy rules are supported, see About Groovy Business Rules.

3. **Optional**: If a business rule has runtime prompts, select **Use Members on Form** to match the default member selection on the runtime prompt window to the current members in the page and Point of View axes of the open form.

   To learn how this option interacts with other settings and conditions, see Understanding Runtime Prompts.

4. **Optional**: To hide the runtime prompt value from the user, select **Hide Prompt**, which automatically selects **Use Members on Form**.

   After saving the form, the next time you return to this page, **Use Members on Form** displays as selected.

   You can hide runtime prompts if:

   - All runtime prompt member values are filled in (appropriate dimension members can be read from form's Page/Point of View)
   - No dimensions are repeated in the runtime prompt

5. Click **OK**.

Viewing Rules Usage

Business rules can be used in the following artifacts:

- Forms
- Rulesets
- Menus
- Task lists

Service Administrators can generate a report, called the Rules Usage Report, that details where rules are used. Filters enable you to refine the details by rule name, rule type, and cube.

> **Note**
>
> The only rules listed in the Rules Usage Report are those rules that are deployed in the business process.

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To view rules usage:

1. Click the **Navigator** icon , and then under **Monitor and Explore**, click **System Reports**.
2. Click the **Rules Details** tab.
3. If filtering, select filter criteria, and then click **Apply Filter**:
   - **Rule Type**: Choose one or more options: **All**, **Rules**, and **Rulesets**
   - **Cube**: Choose from a list of available cubes.
   - **Name Filter**: Enter a rule or ruleset name. Optionally, you can use wildcard characters.

   The **Rules** list displays the rules that are deployed in the business process and that match the filter criteria.
4. For **File Format**, select one of the following formats:
   - **XLSX** (default)
   - **PDF**
   - **HTML**
   - **XML**

   > **Note**
   >
   > Choosing XML format enables you to configure the Rules Usage Report with the help of Oracle BI Publisher Add-in for MS Office. You can have greater control over which columns to display in your report, the order in which the columns are displayed, whether to apply sorting, and so on.

5. Click **Create Report**.

The Rules Usage Reports provides the following information:

- Rule Name
- Rule Type
- Cube
- Primary Association (Type and Name)
- Secondary Association (Type and Name)

Primary association refers to the rule that is associated with the artifact directly; for example, a rule linked to a form or to a task. Secondary association refers to the rule that is associated with an artifact indirectly; for example, a rule that is linked to a menu item, and the menu item is linked to a form, or a rule that is linked to a task for a given task list.

Rules in the generated report are displayed in alphabetical order as follows:

- Rule name
- Primary association type
- Primary association name

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Example 21-1 Sample Rules Usage Report Showing Primary Associations (PDF Format)

ORACLE EPM Cloud

Rules Usage Report

Rule Name	Rule Type	Cube	Primary Association		Secondary Association	
			Type	Name	Type	Name
Act_Agg_Ek	Rules	PSPPlan1				
Adhoc - Clear 5+7	Rules	PSPPlan1				
Agg	Rules	PSPPlan1	Form	_XX_Agg		
Agg - Roll up E and O from drop down, Customer in row, all others at none	Rules	PSPPlan1	Form	1 - Corp Prof Fees		

Page 1 of 82

1/10/2018 18:00 PM

Example 21-2 Sample Rules Usage Report Showing Primary and Secondary Associations (PDF Format)

ORACLE EPM Cloud

Rule Name	Rule Type	Cube	Primary Association		Secondary Association	
			Type	Name	Type	Name
psp_Agg_budget	Rules	PSPPlan1				
psp_Agg_Fcst_All Dims	Rules	PSPPlan1	Task	Aggregation	Task List	Execute Royalty Calc
psp_Agg_FC@AER_All Dims	Rules	PSPPlan1				
psp_Agg_budget_All Dims	Rules	PSPPlan1				
psp_Rowfocused_Agg	Rules	PSPPlan1	Form	1.0 psp_Agg		
psp_Agg_Actuals_All Dims	Rules	PSPPlan1				

Page 81 of 82

1/10/2018 18:00 PM

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About Runtime Prompts

When launched, business rules can prompt users for such variables as members, text, dates, or numbers. Prompts should be specific and tell users what type of data is expected. For example:

- Select a month.
- Enter the expected number of customer visits per quarter.
- What percentage change in earnings do you expect next month?

If, in Calculation Manager, the option **Create dynamic members** is selected for the business rule and the parent member is enabled for adding dynamic children, then users can create new members by entering their name in the runtime prompt.

When launching business rules with runtime prompts, the application validates the value entered, but not the business rule. To set the default member selection in a runtime prompt, see Setting Business Rule Properties. To understand how other settings and conditions affect runtime prompts, see Understanding Runtime Prompts.

Understanding Runtime Prompts

The display and values of runtime prompts are affected by such aspects as:

- Whether the **Use as Override Value** property is set at the rule or the ruleset level at design-time
- Whether there are valid members on the form's Page/Point of View and whether the **Use Members on Form** and **Hide Prompt** options on the **Business Rule Properties** tab are selected (see Setting Business Rule Properties)
- Whether the **Runtime Prompt** option is set during form design or set when designing the runtime prompt (see *Designing with Calculation Manager*)
- Whether the **Use Last Value** property is set when designing the business rule
- Whether, in Calculation Manager, the option **Create dynamic members** is selected for the business rule and the parent member is enabled for adding dynamic children. If yes, then users can create new members by entering their name in the runtime prompt.

> **Note**
>
> In the business process, runtime prompt values aren't replaced inside double quoted text. Alternately, you can use Groovy rules to achieve the same. However, launching a rule from Calculation Manager will replace runtime prompt values in double quoted text.

Principles:

1. If the **Use as Override Value** property is set at the rule or the ruleset level at design-time, the value overridden at the rule level or the ruleset level will take precedence over the values of members in the Page/Point of View and the last saved value. This occurs regardless of where the rule is launched (from the form or from the **Rules** link on the **Navigator** menu), and regardless of whether the runtime prompt is hidden during design. The **Override Value** can be set as a user variable, in which case the rule will be launched with the current value of the variable.

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2. When launched from a form, the values of members in the Page/Point of View take precedence over the last saved value if the **Use Members on Form** option is selected, regardless of whether the runtime prompt is hidden during design. The business rule is run without displaying the hidden runtime prompt to users, and the runtime prompt values are taken from the Page/Point of View members.
3. If the **Use Last Value** option is selected for the runtime prompt at design time, and if any of these conditions exist:
   - **Use Members on Form** isn't selected
   - A runtime prompt is launched from the **Rules** link on the **Navigator** menu
   - Values can't be pre-filled from the context

   Then the precedence of runtime prompt values is determined by:

   a. The last saved value takes precedence.
   b. If a ruleset is launched, the value overridden at the ruleset level at design-time is used.
   c. If a business rule is launched, the value overridden at the rule-level at design-time is used. If it's not overridden at the rule-level, the runtime prompt value at design-time is used.

   Runtime prompts that are hidden at design time never use the last saved value. In these cases, the **Use Last Value** setting is ignored.
4. The **Use Members on Form** and **Hide Prompt** options apply only to Member and Cross Dimension runtime prompt types (Cross Dimension runtime prompt types are available only for business rules created with Calculation Manager).

   The value set at the rule or the ruleset level at design-time when the **Use as Override Value** property is set participates in the **Hide Prompt** behavior.
5. For Cross Dimension runtime prompts: the runtime prompt isn't hidden unless all the prompts in the runtime prompt can be pre-filled from the **Override Value** or Page/Point of View. The runtime prompt is displayed with some values pre-filled from the **Override Value** or Page/Point of View and others filled according to Principles 1, 2 and 3.

This table describes the result on runtime prompts of these settings and conditions:

Table 21-26 How Member Availability and Other Settings Affect Runtime Prompts

Availability of Override Value and member on the Page/ Point of View	Use Members on Form option is selected	Hide Runtime Prompt property is set during runtime prompt design	Hide Prompt option is selected for the form	Result on Runtime Prompt
Use as Override Value is set and **Override Value** is available or the member is available on the Page/Point of View to use as the runtime prompt value.	Yes	Yes	Yes or No Setting is ignored	The business rule runs without displaying the runtime prompt to users. Instead, the runtime prompt value is taken from the **Override Value** or Page/Point of View member.

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Table 21-26 (Cont.) How Member Availability and Other Settings Affect Runtime Prompts

Availability of Override Value and member on the Page/ Point of View	Use Members on Form option is selected	Hide Runtime Prompt property is set during runtime prompt design	Hide Prompt option is selected for the form	Result on Runtime Prompt
Use as Override Value is set and **Override Value** is available or the member is available on the Page/Point of View to use as the runtime prompt value.	Yes	No	Yes	If all runtime prompts can be pre-filled from the **Override Value** or Page/Point of View context and are valid and within limits, the runtime prompts are not displayed. However, if even one runtime prompt value can't be pre-filled from the **Override Value** or Page/Point of View context, then all runtime prompts display, with values pre-filled wherever possible. All others follow Principles 1 and 3.
Use as Override Value is set and **Override Value** is available or the member is available on the Page/Point of View to use as the runtime prompt value.	Yes	No	No	The runtime prompt is displayed to users, with values pre-filled from the **Override Value** or Page/Point of View.
Use as Override Value isn't set and the member isn't available on the Page/Point of View to use as the runtime prompt value.	Yes	Yes	Yes or No Setting is ignored	The business rule displays the runtime prompt to users, with values pre-filled according to Principle 3. For example, the form context can't be passed because the dimension of the runtime prompt is on rows or columns, so the **Hide Prompt** setting is ignored and the runtime prompt displayed.
Use as Override Value isn't set and the member isn't available on the Page/Point of View to use as the runtime prompt value.	Yes	No	Yes	The runtime prompt is displayed to users, with values pre-filled according to Principle 3.
Use as Override Value is set and **Override Value** is available, and the member isn't available on the Page/Point of View to use as the runtime prompt value.	Yes	No	No	If all runtime prompts can be pre-filled from the **Overide Value** and are valid and within limits, the runtime prompts are not displayed. However, if even one runtime prompt value can't be pre-filled from the **Overide Value**, then all runtime prompts display, with values pre-filled wherever possible. All others follow Principles 1 and 3.
Use as Override Value isn't set and the member isn't available on the Page/Point of View to use as the runtime prompt value.	Yes	No	No	The runtime prompt is displayed to users, with values pre-filled according to Principle 3.

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Table 21-26 (Cont.) How Member Availability and Other Settings Affect Runtime Prompts

Availability of Override Value and member on the Page/ Point of View	Use Members on Form option is selected	Hide Runtime Prompt property is set during runtime prompt design	Hide Prompt option is selected for the form	Result on Runtime Prompt
Use as Override Value is set and **Override Value** is available, and the member isn't available on the Page/Point of View to use as the runtime prompt value.	Yes	No	No	The runtime prompt is displayed to users, with values pre-filled according to Principles 1 and 3.
Use as Override Value is set and **Override Value** is available or the member is available on the Page/Point of View to use as the runtime prompt value.	No	Yes	Not available	The business rule runs without displaying the runtime prompt to users. Instead, the design-time values are used.
Use as Override Value is set and **Override Value** is available or the member is available on the Page/Point of View to use as the runtime prompt value.	No	No	Not available	The runtime prompt is displayed to users, with values pre-filled according to Principle 3.
Use as Override Value isn't set and the member isn't available on the Page/Point of View to use as the runtime prompt value.	No	Yes	Not available	The business rule runs without displaying the runtime prompt to users. Instead, the design-time values are used.
Use as Override Value isn't set and the member isn't available on the Page/Point of View to use as the runtime prompt value.	No	No	Not available	The runtime prompt is displayed to users, with values pre-filled according to Principle 3.

When hidden runtime prompt values are ambiguous, note:

- If the form context can't be passed in (because the dimension is on the row or column, for example), hidden runtime prompts are displayed.
- With hidden Cross Dimension runtime prompt types, if all prompts can't be passed in from the Override Value or context, the runtime prompt displays with values pre-filled from **Override Value** or context values and design time values. For example, if the Cross Dimension has runtime prompts for Period, Entity, and Scenario, and Entity is defined on the row and Scenario has an **Override Value** set, then the runtime prompt displays with the Override Scenario, then design time value for Entity, then Page Period.
- If the **Override Value** is present, or context can be passed in for the runtime prompt value but it's out of limits, then the runtime prompt is displayed with the context value pre-filled.
- If there is more than one runtime prompt of type Member or type Cross Dimension combined, then the runtime prompts are displayed with the **Override Value** or context values pre-filled. For example, if there is a member type runtime prompt for the Entity dimension and a Cross Dimension type runtime prompt with one prompt for the Entity dimension, then both runtime prompts are displayed. This rule doesn't apply to Calculation Manager rulesets.
- When launched from the **Rules** link on the **Navigator** menu, runtime prompts are hidden and the design-time value (overridden at the rule or ruleset level) is used to launch the business rule. If the provided design-time value is out of limits, then the runtime prompt is displayed with the design-time value pre-filled.

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- Runtime variables that are hidden during design never use the last saved value. The **Use Last Value** property is ignored, and the values are not saved to the database.

Using Groovy Rules

Create rules written in the Groovy scripting language that solve use cases that normal business rules can't solve.

Table 21-27 Where Can I Learn More About Groovy Rules?

Your Goal	Learn More
Learn the benefits of Groovy rules.	About Groovy Business Rules
View examples of business scenarios where Groovy business rules might be used.	Groovy Rule Business Scenarios
Watch videos and complete hands-on tutorials that teach best practices when implementing and using Groovy rules.	• Groovy Business Rule Tutorial Videos • Learning Groovy in Oracle Cloud EPM
Create Groovy business rules and a Groovy template using Calculation Manager.	• Creating a Groovy Business Rule • Creating a Groovy Template for a Planning BSO Cube
Connect to the Java APIs used for creating Groovy rules and view example Groovy scripts.	• Java API Reference for Groovy Rules • Groovy Business Rule Examples
Edit the script for a Groovy business rule or template using Calculation Manager.	Editing the Script For a Groovy Business Rule or Template
Secure Groovy templates for authorized users.	Assigning Access to Groovy Templates

About Groovy Business Rules

Groovy business rules allow you to design sophisticated rules that solve use cases that normal business rules can't solve; for example, rules to prevent users from saving data on forms if the data value is above a predefined threshold.

> **Note**
>
> Groovy is an advanced customizable rules framework that comes with Cloud EPM Platform and is available with Enterprise Performance Management Enterprise Cloud Service along with Enterprise PBCS and PBCS Plus One. You can create and edit Groovy rules in:
>
> - Planning (including these application types: Custom, Module, FreeForm, Sales Planning, Strategic Workforce Planning, and Cash Forecasting)
> - Enterprise Profitability and Cost Management
> - Financial Consolidation and Close
> - FreeForm
> - Tax Reporting

You create Groovy rules in Calculation Manager and execute them from any place that you can execute a calc script rule in an application; for example, on the Rules page, within the context of a form, in the job scheduler, in dashboards, in task lists, and so on.

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Groovy rules are also supported in rulesets. You can have a combination of calc script rules and Groovy rules within a ruleset.

Groovy rules are not supported in composite forms.

You can execute jobs of type rules, rulesets, and templates synchronously from a Groovy rule.

You can write Groovy scripts to run select EPM Automate commands directly in Oracle Fusion Cloud Enterprise Performance Management, without installing EPM Automate client on a client machine. Refer to Running Commands without Installing EPM Automate and Supported Commands in *Working with EPM Automate* for information on which EPM Automate commands can be run using Groovy and example scripts.

Oracle supports two types of Groovy rules:

- Rules that can dynamically generate calc scripts at runtime based on context other than the runtime prompts and return the calc script which is then executed against Oracle Essbase.

  For example, you could create a rule to calculate expenses for projects only for the duration (start and end dates) of the project.

  Another example is a trend-based calculation that restricts the calculation to the accounts available on the form. You could use this calculation for various forms in Revenue, Expense, Balance Sheet, and Cash Flow. This allows for optimization and reuse.
- Pure Groovy rules that can, for example, perform data validations and cancel the operation if the data entered violates company policies.

Video

Your Goal	Watch This Video
Learn about training options for creating Groovy rules in Oracle Fusion Cloud Enterprise Performance Management.	Learning Groovy in Oracle Cloud EPM

Java API Reference for Groovy Rules

For Enterprise applications, a Java API Reference is available to use as you create Groovy rules.

The Java API Reference includes examples that demonstrate the syntax and power of the EPM Groovy object model.

To view the Java API Reference, see the Java API Reference for Oracle Enterprise Performance Management Cloud Groovy Rules on the Cloud Help Center. You can also access this reference from the Oracle Enterprise Planning and Budgeting Cloud Academy. To access the Academy, sign in, and then click **Academy**.

Groovy Business Rule Examples

Example Groovy scripts are available.

To see example Groovy scripts:

1. See the Java API Reference for Oracle Enterprise Performance Management Cloud Groovy Rules, https://docs.oracle.com/cloud/latest/epm-common/GROOV/.
2. Do one of the following:

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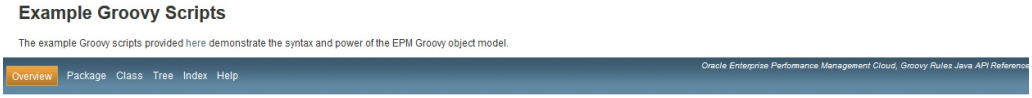
- Under **Example Groovy Scripts** on the main page, click the word "here" to view sample scripts:



- Under **All Classes** in the left pane, click a class to see the examples for that class.

  For example, to see Strategic Modeling examples, click the StrategicModel class in the left pane.

Groovy Business Rule Tutorial Videos

Watch these tutorial videos for details and best practices when implementing and using Groovy Business Rules.

Your Goal	Watch this Video
See the training options for creating Groovy rules in Oracle Fusion Cloud Enterprise Performance Management:	Learning Groovy in Oracle Cloud EPM
Use Groovy rules to calculate incrementally loaded data in Data Management.	Calculating Incrementally Loaded Data in Data Management Using Groovy Rules
Use Groovy templates to improve usability and calculation performance for user actions.	Customizing Actions to Improve Performance Using Groovy Templates

Groovy Business Rule Tutorials

Complete these tutorials for hands-on examples on implementing Groovy Business Rules.

Implement Groovy: Basic Tasks

Your Goal	Learn How
Learn about the Groovy scripting language and how to create a Groovy script for your business process in Oracle Fusion Cloud Enterprise Performance Management.	Introduction to Groovy Business Rules
Implement Groovy scripts to work with data grids and data grid iterators. • Use conditional logic to set the background colors in a data grid • Implement performance benchmarking • Set default data values in a grid	Working with Data Grids and Iterators in Groovy

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Your Goal	Learn How
Create a Groovy script to work with business process metadata, in this case, moving a member from one parent to another. The script includes RTPs to prompt users for input. You'll also learn how to create a right-click action menu with a menu item to call the script, and how to associate the action menu with a data form.	Moving Dimension Members with Groovy
Push data from a source location to a target location using Groovy rules and Smart Push.	Moving Modified Data Using Groovy and Smart Push

Implement Groovy: Advanced Tasks

Your Goal	Learn How
Implement a Groovy script that validates data entry against allowed ranges stored in a driver cube.	Validating Data Entry Rules with Groovy
Create a Groovy script to work with business process metadata, in this case, adding a member to a dimension. The script includes RTPs to prompt users for input. You'll also learn how to create a right-click action menu with a menu item to call the script, and how to associate the action menu with a data form.	Adding Dimension Members with Groovy
Generate focused calculation scripts in your business process to calculate only data that has been edited, instead of the entire data entry form.	Calculating Modified Data Using Groovy
Understand Strategic Modeling concepts and learn how to integrate data between strategic models and business process applications using Groovy scripts.	Integrating Data between Planning and Strategic Models Using Groovy
Create a Groovy template with interactive Design Time Prompts (DTPs).	Creating Groovy Templates

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Your Goal	Learn How
Learn how to use a Groovy business rule in the business process to export dimension metadata from Cloud EPM, and then import the metadata into the business process.	Integrating Dimension Metadata from Enterprise Data Management Cloud with Planning Using Groovy Rules

Implement Groovy: REST API

Your Goal	Learn How
Call a Data Management REST API to execute a data load rule that will load the latest product volumes for the user's entity into the business process.	Calling an Internal REST API Using Groovy
Call an external Rest API from a Groovy script in your business process. You'll also learn how to create a right-click action menu with a menu item to call the script, and how to associate the action menu with a data form.	Calling an External REST API Using Groovy

Groovy Rule Business Scenarios

This section provides examples of business scenarios where you can use Groovy business rules.

These topics contain scenarios that show you how to use Groovy rules to perform focused data movement and smart calculations on business process data:

- Moving Modified Data Using Groovy Rules and Smart Push
- Calculating Modified Data Using Groovy Rules

Moving Modified Data Using Groovy Rules and Smart Push

In this topic, we'll show you best practices on how you can use Groovy rules and Smart Push to move modified data into a reporting cube.

On a data form, users can modify employee salary and assign a Reporting Manager. After the form data is saved, Smart Push is run to copy data to the reporting cube for analysis. Then, dimensions are also consolidated, if necessary, as the data is being copied.

If the form contains hundreds of rows of data, all of that data is checked against the configured data map, processed for dimension consolidation, if any is set, and then pushed to the reporting cube. Depending on the amount of data on your form, this process may take a while to complete.

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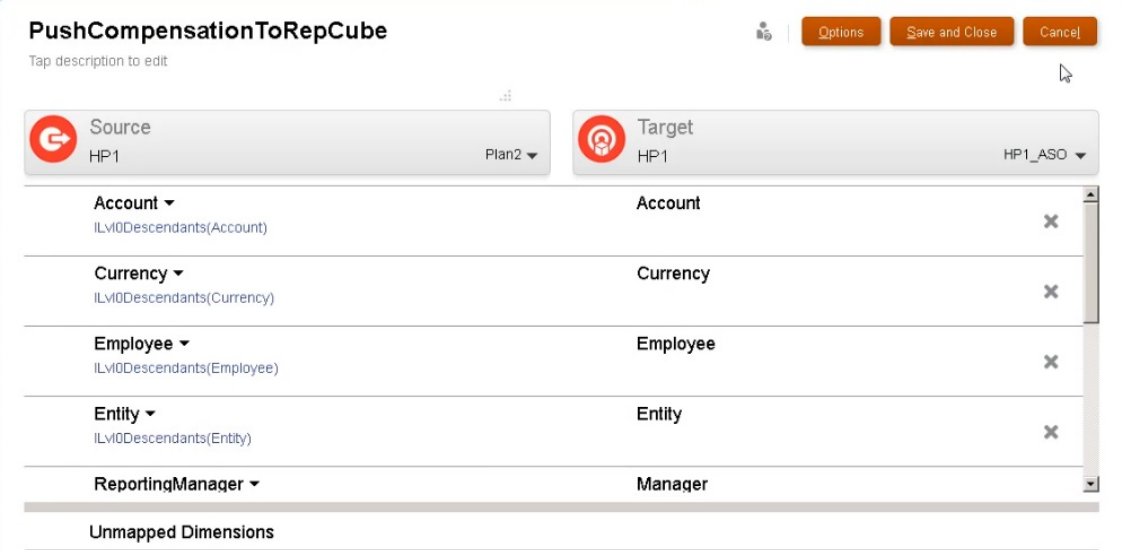
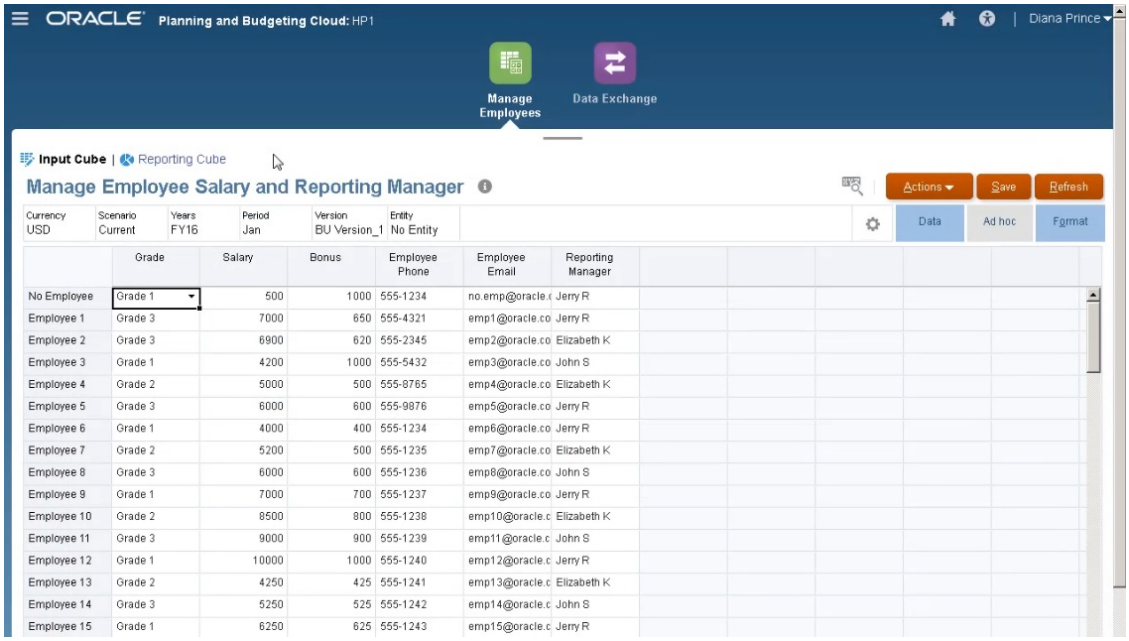
You can improve the performance of data synchronization between your input and reporting cubes by designing a Groovy rule that identifies and isolates the data you modified, and pushes only the isolated data into your reporting cube.

For example, we configured a simple data map for Smart Push that maps dimension members from our input cube to our reporting cube.



Then we prepared forms that display data from our input and reporting cubes.



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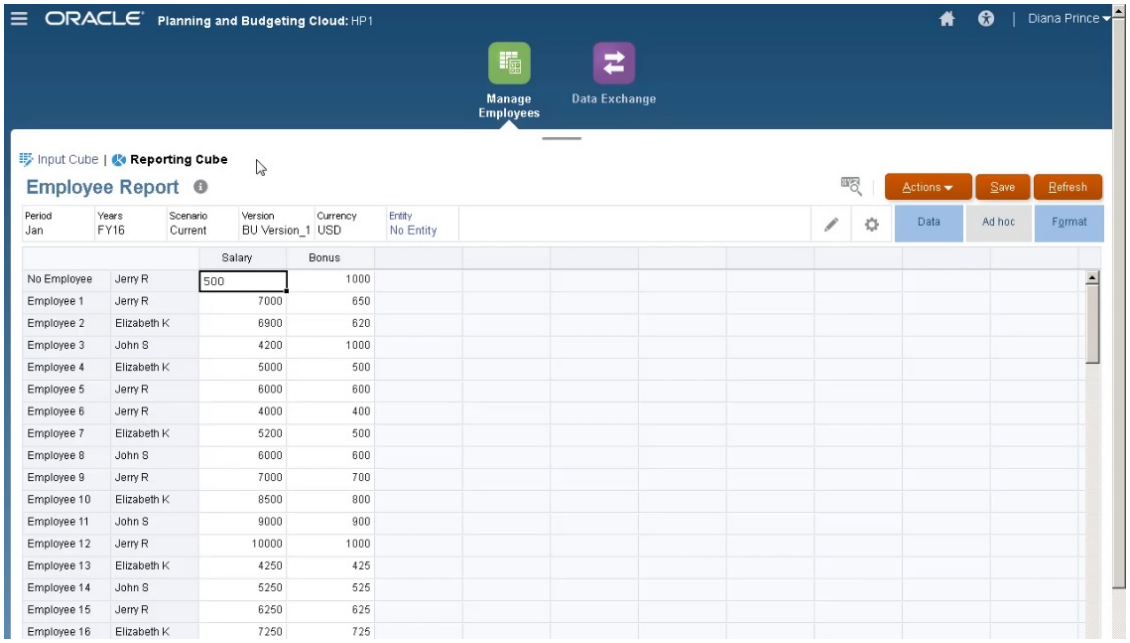
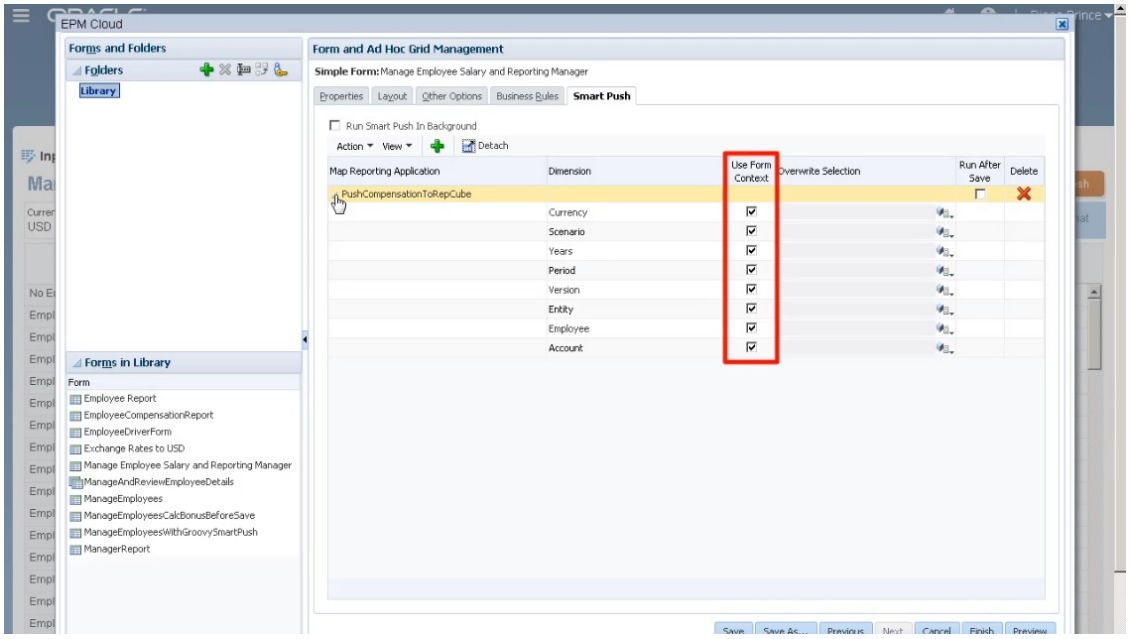


The form used for data entry has Smart Push enabled and uses the form's context to push data for all the members selected for the dimensions defined on the form.



For the Groovy rule, here's the entire script.

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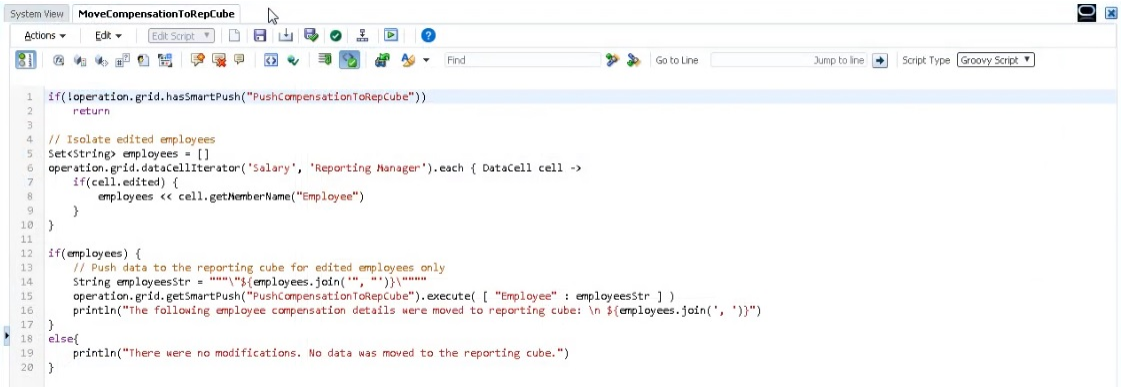

The script starts by identifying the associated data map for smart push.

```
if(!operation.grid.hasSmartPush("PushCompensationToRepCube"))
    return

// Isolate edited employees
Set<String> employees = []
operation.grid.dataCellIterator('Salary', 'Reporting Manager').each { DataCell cell ->
    if(cell.edited) {
        employees << cell.getMemberName("Employee")
    }
}

if(employees) {
    // Push data to the reporting cube for edited employees only
    String employeesStr = """\"${employees.join('", "')}\"""
    operation.grid.getSmartPush("PushCompensationToRepCube").execute( [ "Employee" : employeesStr ] )
    println("The following employee compensation details were moved to reporting cube: \n ${employees.join(', ')}")
}
else{
    println("There were no modifications. No data was moved to the reporting cube.")
}
```

This section of the script uses the `dataCellIterator` method to identify edited cells and isolates them.

```
if(!operation.grid.hasSmartPush("PushCompensationToRepCube"))
    return

// Isolate edited employees
Set<String> employees = []
operation.grid.dataCellIterator('Salary', 'Reporting Manager').each { DataCell cell ->
    if(cell.edited) {
        employees << cell.getMemberName("Employee")
    }
}

if(employees) {
    // Push data to the reporting cube for edited employees only
    String employeesStr = """\"${employees.join('", "')}\"""
    operation.grid.getSmartPush("PushCompensationToRepCube").execute( [ "Employee" : employeesStr ] )
    println("The following employee compensation details were moved to reporting cube: \n ${employees.join(', ')}")
}
else{
    println("There were no modifications. No data was moved to the reporting cube.")
}
```

Then, this section takes the isolated data and runs the focused data movement using Smart Push and the data map configured and associated with the form. A message displays when modified data is moved to the reporting cube.

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```
if(!operation.grid.hasSmartPush("PushCompensationToRepCube"))
    return

// Isolate edited employees
Set<String> employees = []
operation.grid.dataCellIterator('Salary', 'Reporting Manager').each { DataCell cell ->
    if(cell.edited) {
        employees << cell.getMemberName("Employee")
    }
}

if(employees) {
    // Push data to the reporting cube for edited employees only
    String employeesStr = """\"${employees.join('", "')}\""""
    operation.grid.getSmartPush("PushCompensationToRepCube").execute( [ "Employee" : employeesStr ] )
    println("The following employee compensation details were moved to reporting cube: \n ${employees.join(', ')}")
}
else{
    println("There were no modifications. No data was moved to the reporting cube.")
}
```

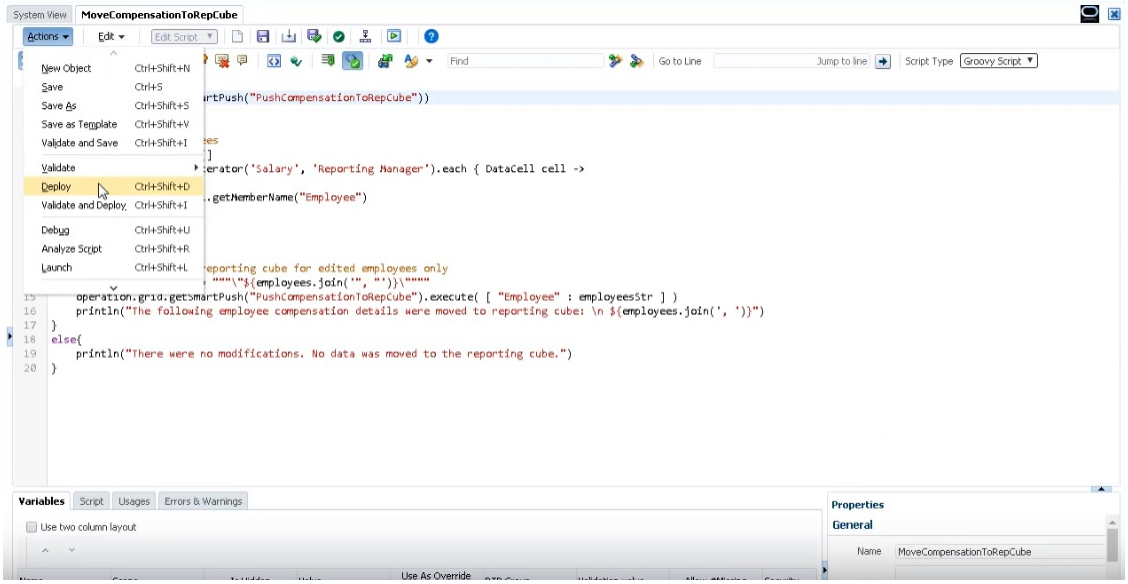
If there are no modifications, a message displays when there are no changes and data was not moved to the reporting cube.

```
if(!operation.grid.hasSmartPush("PushCompensationToRepCube"))
    return

// Isolate edited employees
Set<String> employees = []
operation.grid.dataCellIterator('Salary', 'Reporting Manager').each { DataCell cell ->
    if(cell.edited) {
        employees << cell.getMemberName("Employee")
    }
}

if(employees) {
    // Push data to the reporting cube for edited employees only
    String employeesStr = """\"${employees.join('", "')}\""""
    operation.grid.getSmartPush("PushCompensationToRepCube").execute( [ "Employee" : employeesStr ] )
    println("The following employee compensation details were moved to reporting cube: \n ${employees.join(', ')}")
}
else{
    println("There were no modifications. No data was moved to the reporting cube.")
}
```

After we created this Groovy rule, we deployed it the same way we deploy business rules.



Then, in form management, associated it with the form. The Groovy rule runs after the form data is saved.

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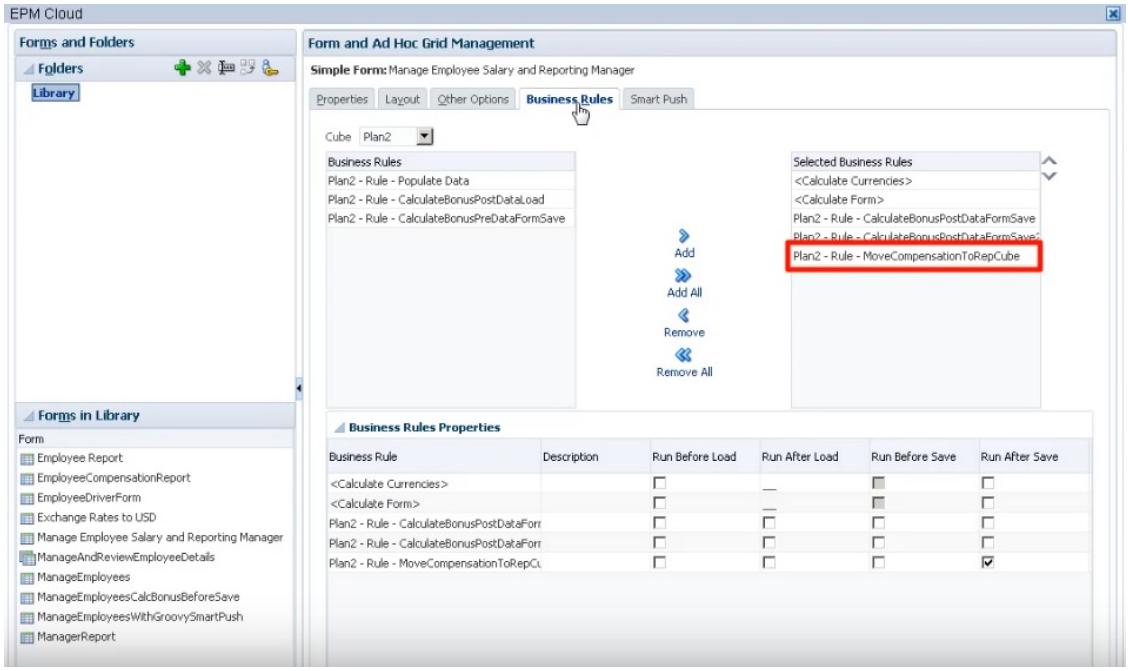

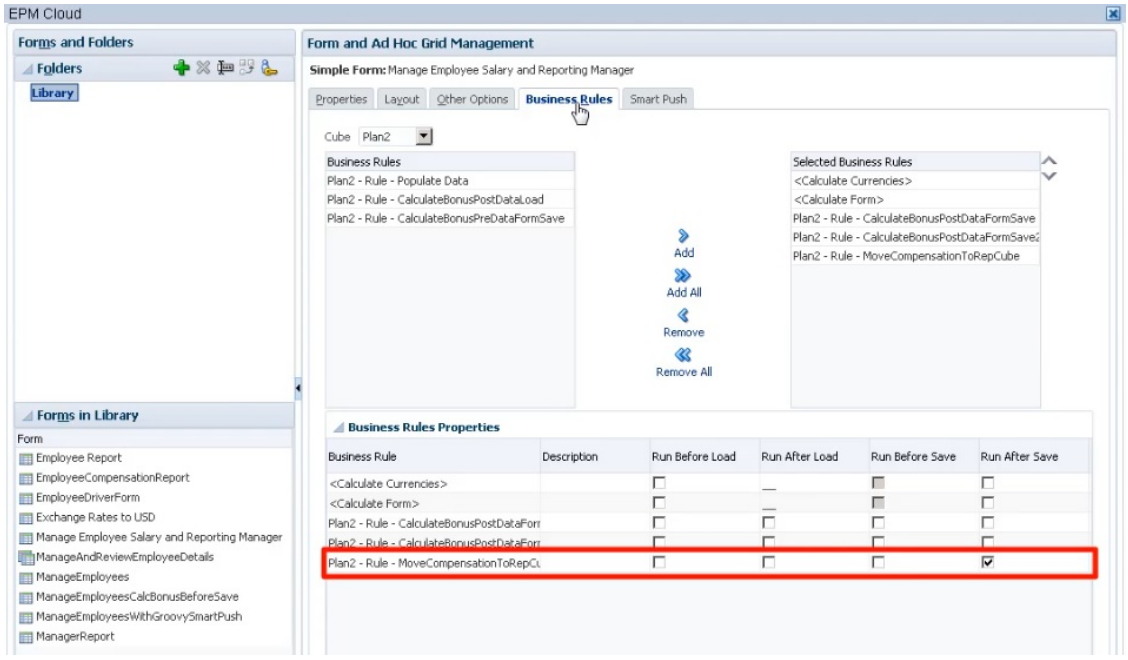

To run the Groovy rule, open the form in **Input Cube**, make some changes, and then click **Save**.

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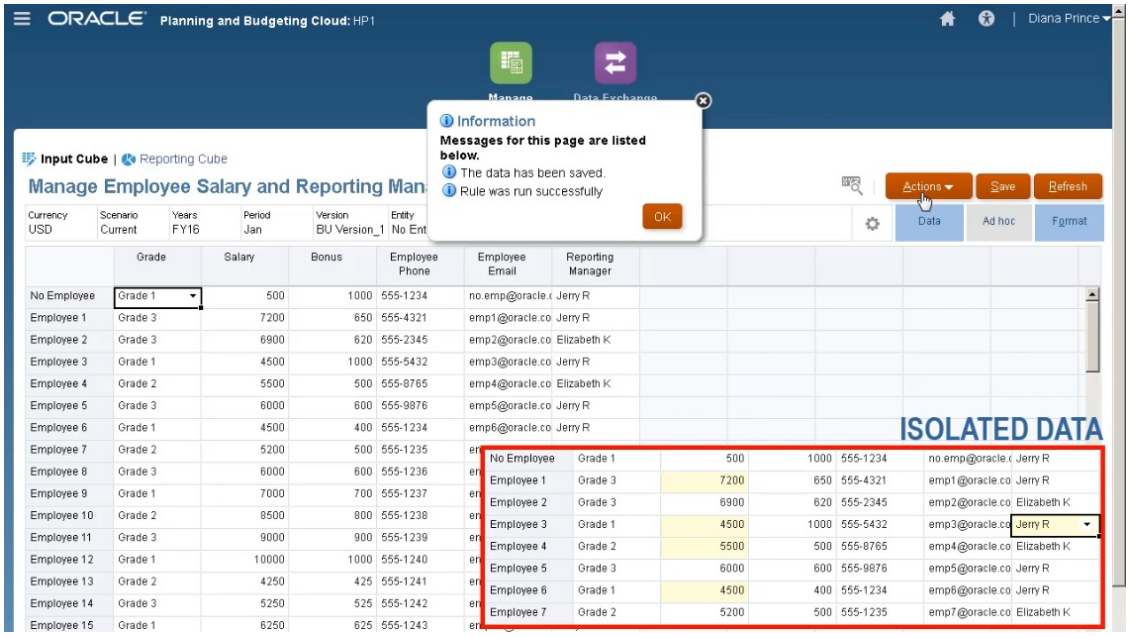

Once saved, the updated data will display in the reporting cube.

Calculating Modified Data Using Groovy Rules

In this topic, we'll show you how to improve business process calculation performance by using Groovy rules to calculate only the data that's been modified.

After you update data in a form, it needs to be calculated. When you use a business rule to calculate a standard calculation script, all the data for the form is calculated. With large forms or calculations that affect large areas of the database, you might see slower calculation performance.

You can manually calculate one row at a time by right-clicking on the row. This can provide faster calculation performance, but isn't a good option when making changes to multiple rows of data.

Groovy rules can dynamically focus on the cells that you update.

You can use Groovy rules on forms in two ways:

- When you calculate data on save, the Groovy rule generates a calculation script with a FIX statement based on the cells you updated. Then the calculation is run on the server against the modified data instead of all records on the form. As a best practice, use this option with larger forms where you are updating a lot of data.
- When you calculate data before saving, the Groovy rule calculates modified data in memory and displays the results on the form before saving them to the database. As a best practice, use this option when you have a small number of calculations to perform, or when you want to perform complex procedural calculations that are difficult to build into the outline.

In this example, bonus calculations are calculated as a percent of salary.

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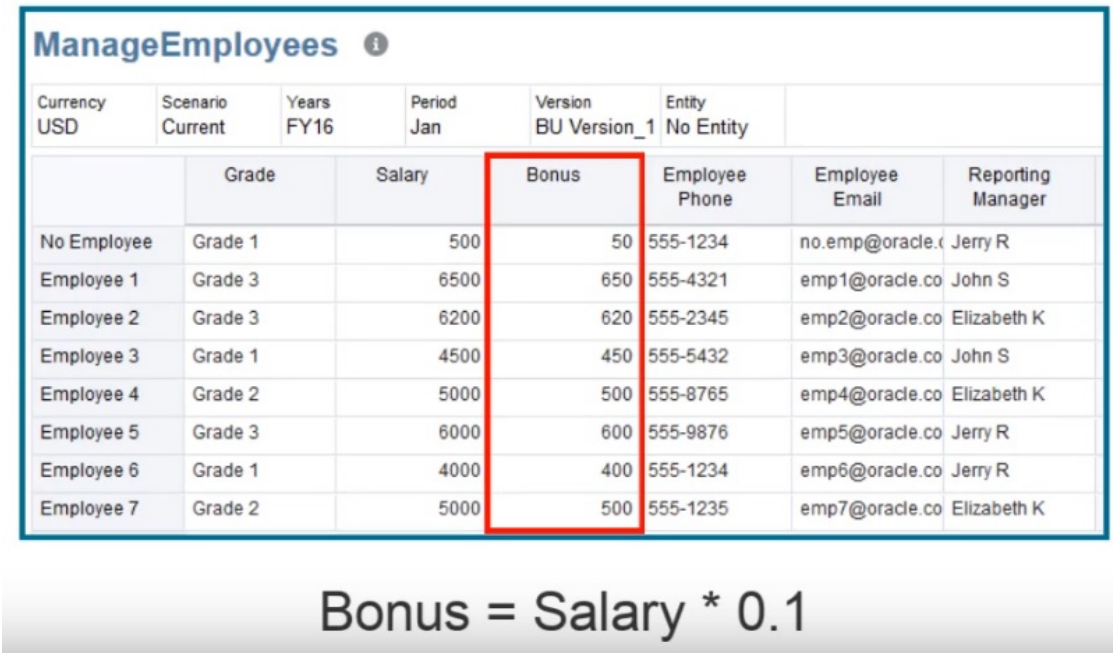
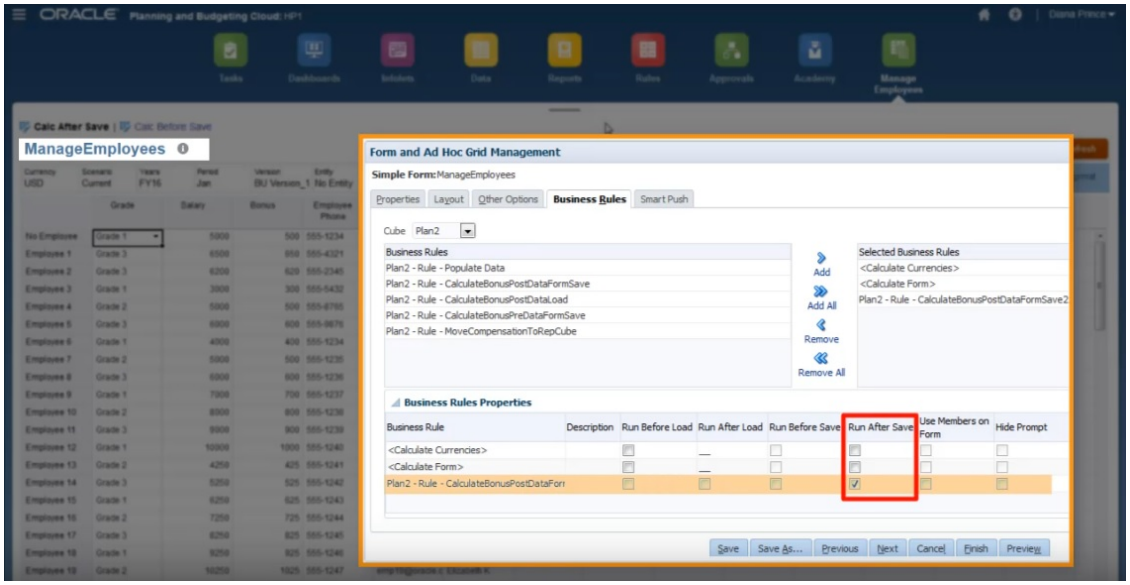


Let's look at calculating on save. Notice that this form is defined with a Groovy rule to calculate on save.



After updating some of the salary data and some of the phone numbers on the form, the phone numbers are saved but don't affect the calculation.

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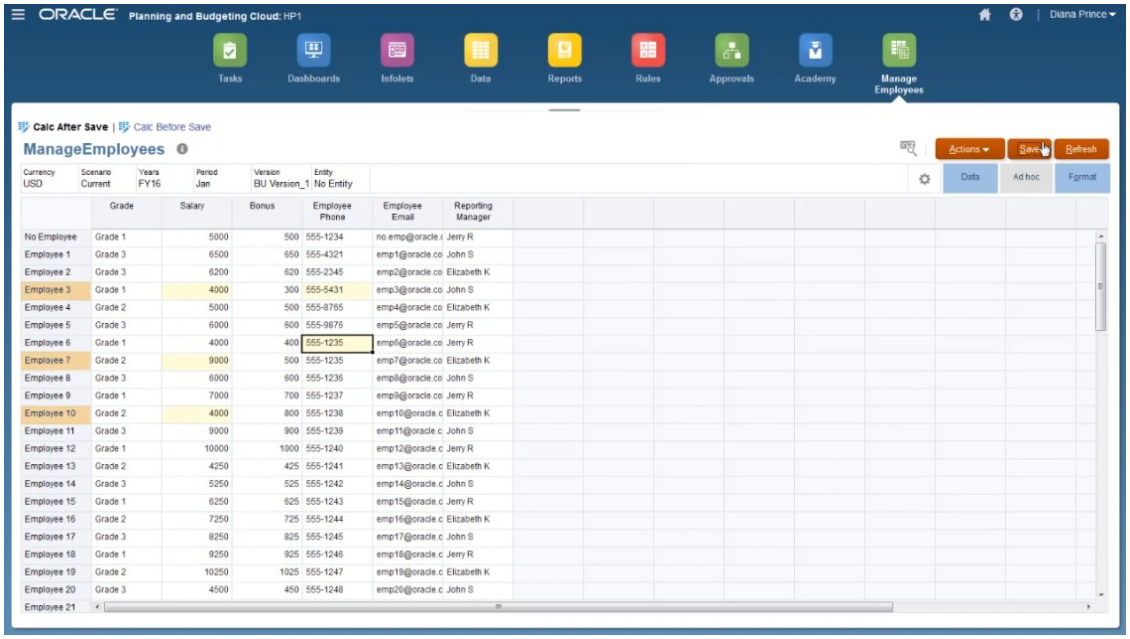
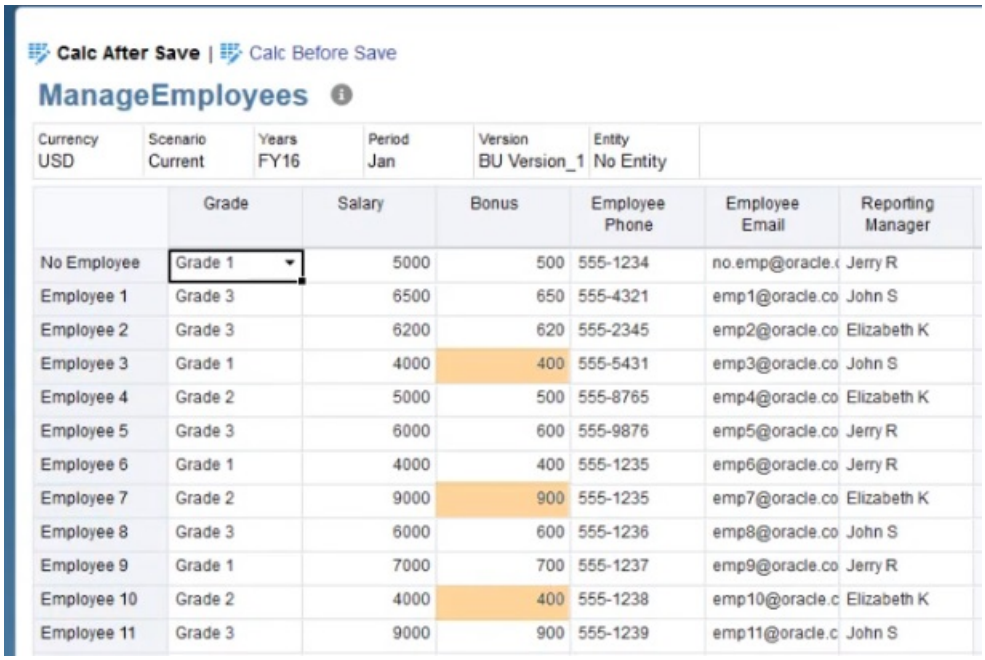


When you save the form to run the associated Groovy rule, the calculated bonus data is displayed on the form.



Viewing the job details, here's the calculation script that the Groovy rule created. Notice that the FIX statement includes only the modified employees.

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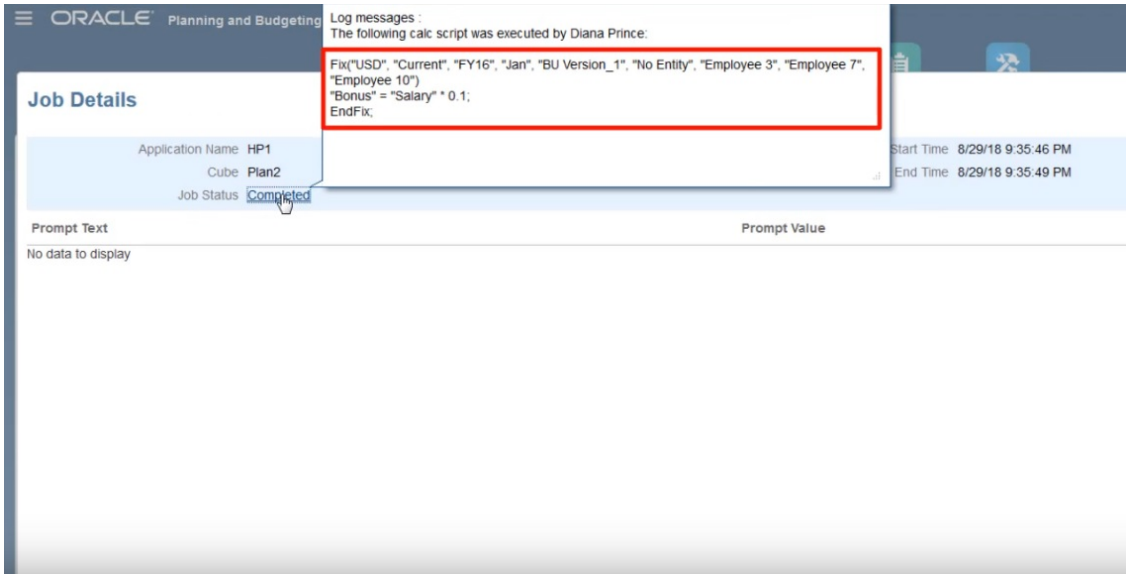
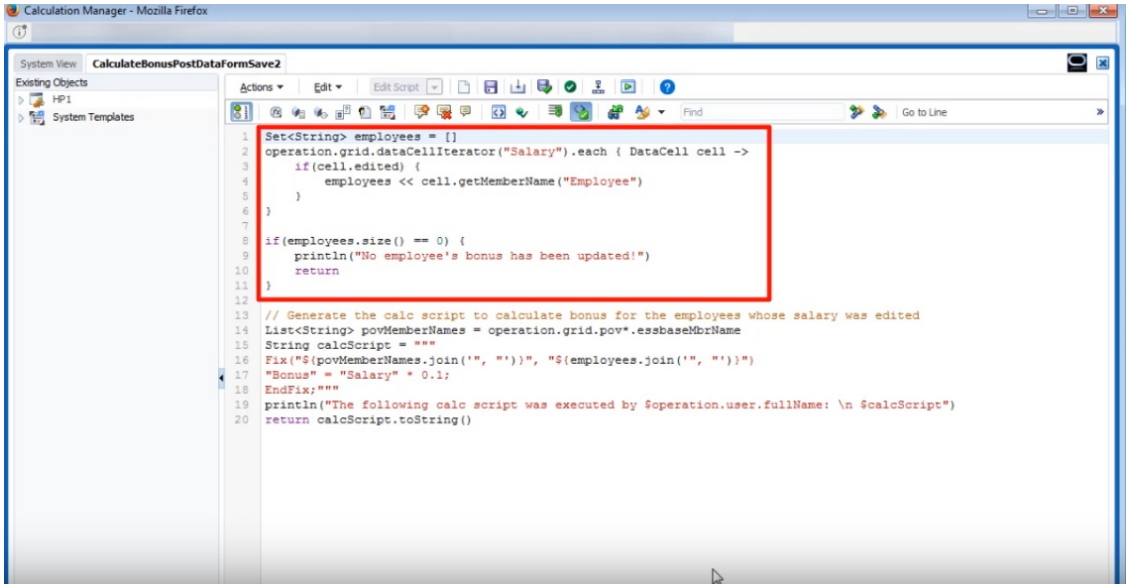


Let's open Calculation Manager to see the Groovy rule. This first section iterates over the grid in the form to find Salary cells that have been edited and collects the related employee names.



This section generates the calculation script; it finds the POV members from the form, then creates a FIX statement on the POV members and the edited employees.

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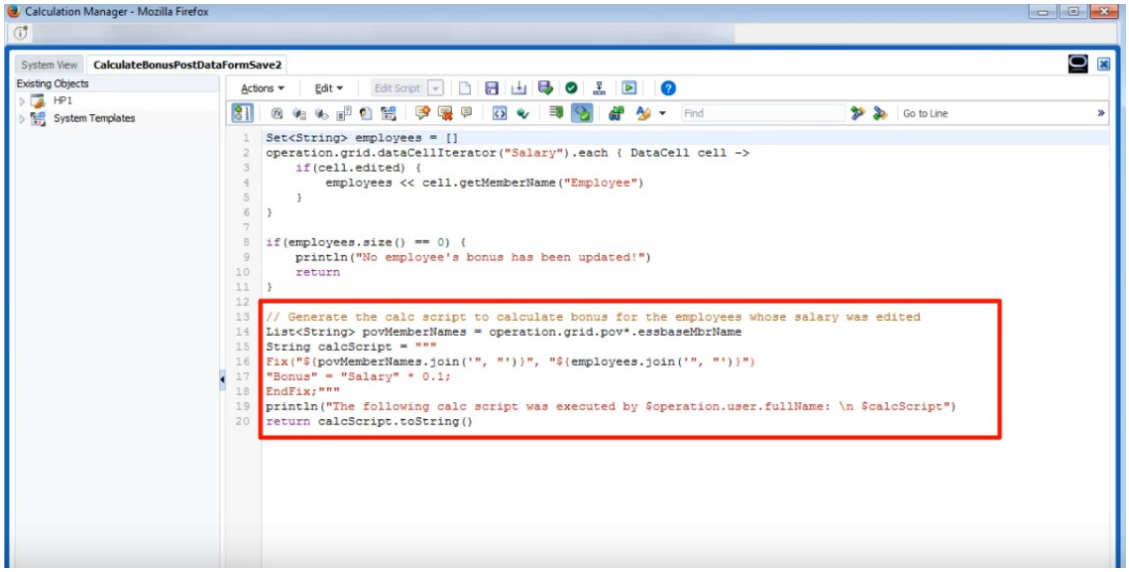
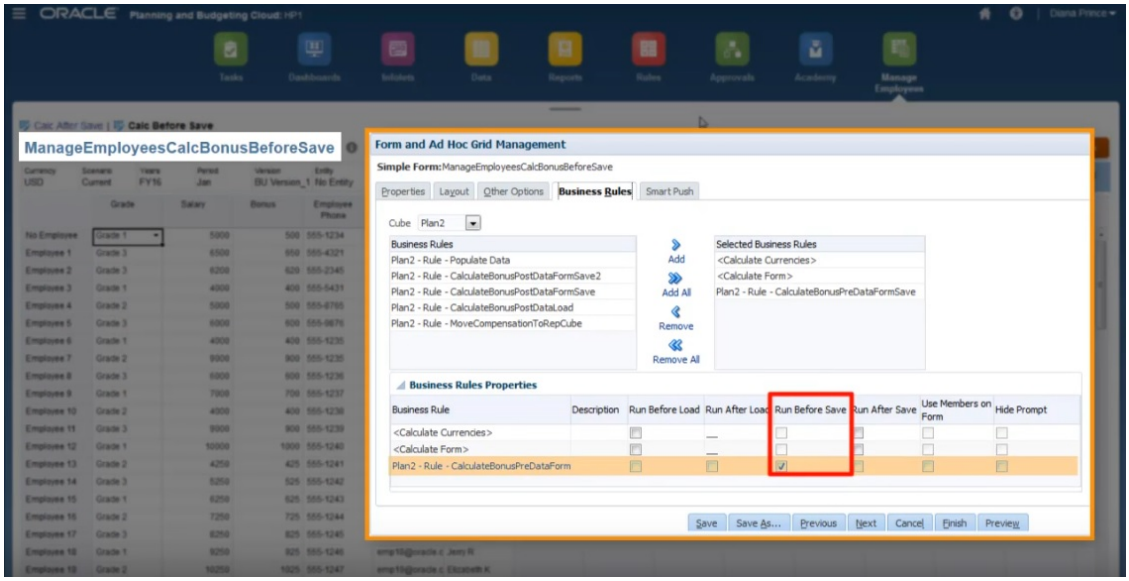


Now let's take a look at calculating before save.

This is the same form, but now it's defined with a different Groovy rule to calculate before save.



After updating the salary data and some of the phone numbers again, save the form to run the associated Groovy rule. The calculated bonus data is displayed on the form, but when you look at the job details, you can see that some of the bonus data was updated, but this time no calculation script was generated.

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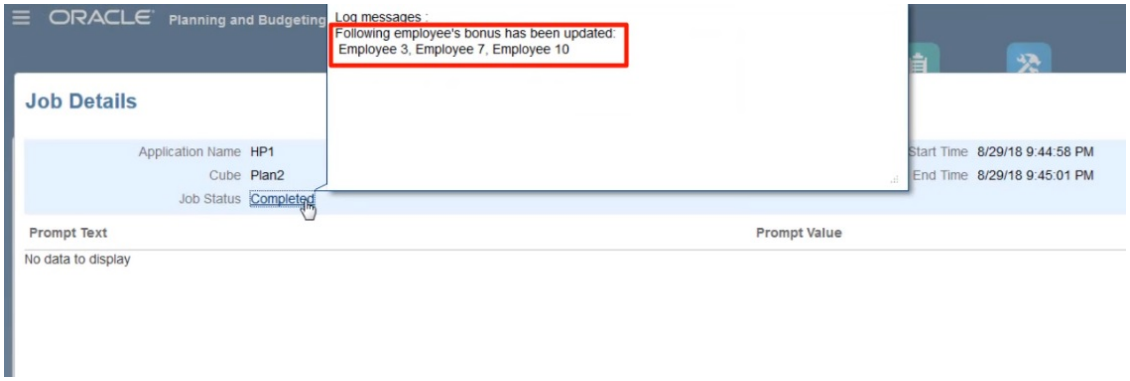
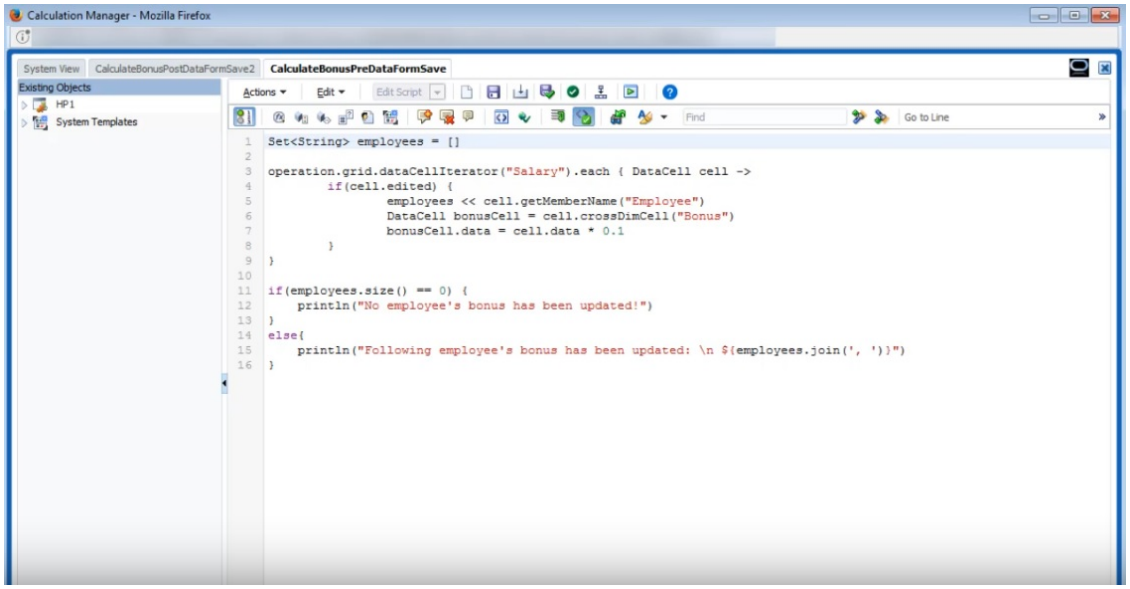


Open the Groovy rule in Calculation Manager.

Similar to the first rule, the script iterates over the grid in the form to find Salary cells that have been edited and collects the related employee names. Then it calculates bonus data for just the edited employees. Notice that no Oracle Essbase calculation script is generated.



Groovy rules also work with FreeForm forms in Oracle Smart View for Office. Smart View gives you all the power of Microsoft Excel to make mass adjustments your data, and any Groovy rules for the form set to run before or after saving are executed when you submit data.

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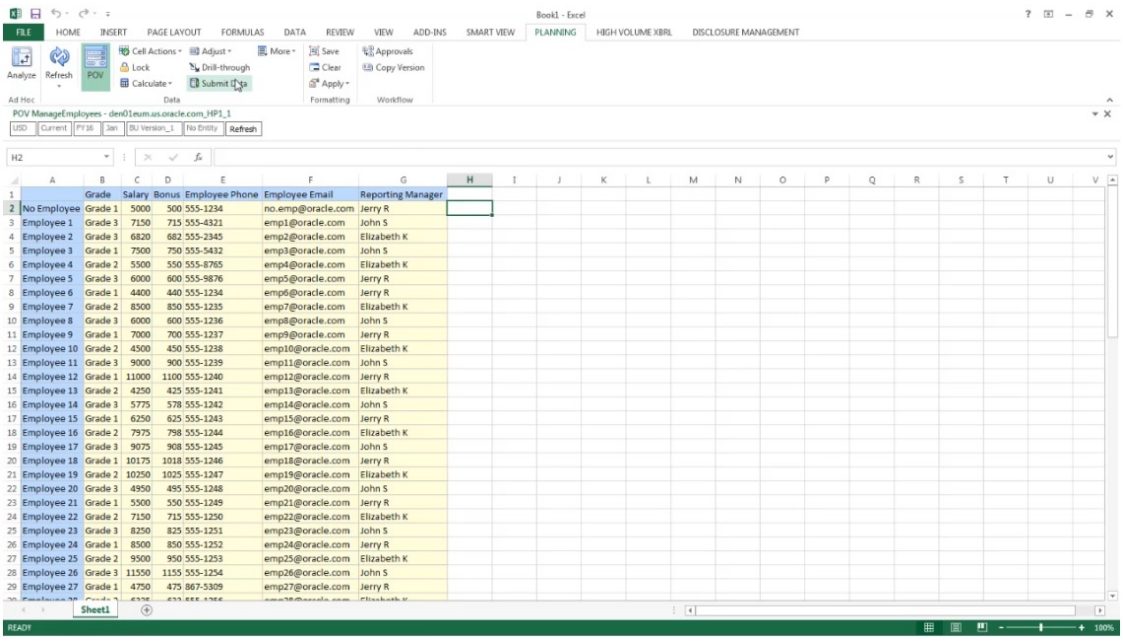

Administering Rules Security

Related Topics

- Assigning Access to Rules
- Adding, Editing, and Removing Access to Rules
- Assigning Access to Groovy Templates

Assigning Access to Rules

Service Administrators can assign access to business rules.

To assign access to Groovy business rule templates, see Assigning Access to Groovy Templates.

To assign access to rules:

1. Click the **Navigator** icon, and then under Create and Manage, click **Rules Security**.
2. Under **Business Rule Folders**, select the folder containing the rules, and then select the rules.
3. Click (Assign Access).
4. Add, edit, or remove access.

   See Adding, Editing, and Removing Access to Rules.

Adding, Editing, and Removing Access to Rules

You can specify which users and groups can access the selected rule.

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> **Note**
>
> Rulesets inherit launch permissions from the rules included in the ruleset.

To assign, edit, and remove access permissions to rules:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Rules Security**.
2. Under **Business Rule Folders**, select the folder containing the rules, and then select the rules.
3. Click (Assign Access).
4. Perform a task:
   - To add access, click (Add Access), and then select from the list of available users and groups.

     For **Type of Access**:
     - Select **Launch** to allow the selected users and groups to launch the selected rules.
     - Select **No Launch** to prevent the selected users and groups from launching the selected rules.
   - To edit access, click (Edit Access), and then select the applicable **Type of Access**.
   - To remove access, select the users or groups for which to remove access, and then click (Remove Access).

Assigning Access to Groovy Templates

To assign access to Groovy templates:

1. From the Home page, click **Rules**, then **Filter**, and then select a cube and artifact type.
2. Next to the rule or template, click **Permission**, and then click **Assign Permission**.
3. In **Assign Permission**, click **Permission**, and then assign permissions. See Setting Up Access Permissions for more information.

Administering Smart Lists

Smart Lists are custom drop-down lists that users access from form cells.

Related Topics

- Working with Smart Lists
- Synchronizing Smart Lists in Reporting Applications
- Setting Smart List Properties
- Defining Smart List Entries
- Previewing Smart Lists

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- Displaying #MISSING with Smart Lists

Working with Smart Lists

Service Administrators use Smart Lists to create custom drop-down lists that users access from form cells. When clicking in cells whose members are associated with a Smart List (as a member property), users select items from drop-down lists instead of entering data. Users can't type in cells that contain Smart Lists. Smart Lists display in cells as down arrows that expand when users click into the cells.

Perform these tasks to create and administer Smart Lists:

- Define Smart Lists.
- Associate Smart Lists with members.
- Select dimensions for which Smart Lists are displayed.
- Optionally:
  - Use Smart List values in member formulas.
  - Set how #MISSING cells associated with Smart Lists display in forms.
  - Synchronize Smart Lists in reporting applications

> **Note**
>
> For a Smart List to be viewable in forms, the dimension to which the Smart List type member belongs must be first in the evaluation order. See Managing Dimensions.

To create or work with Smart Lists:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Smart Lists**.
2. Perform one action:
   - To create a Smart List, click (Create), and then set Smart List properties. See Setting Smart List Properties.
   - To edit a Smart List, select it, click (Edit), and then set Smart List properties. See Setting Smart List Properties.
   - To delete Smart Lists, select them, click (Delete), and then click **OK**. Deleting Smart Lists also deletes any associated mappings with dimension members and reporting applications.

     Data cells can display only one Smart List. If multiple Smart Lists intersect at cells, set which one takes precedence.
   - **Optional**: Click (Synchronize) to synchronize Smart Lists in a reporting application. See Synchronizing Smart Lists in Reporting Applications.

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> **Note**
>
> When a member in a Smart List is deleted and then added back in again, the new member assumes a new unique ID. Any data that was associated with the deleted member will display as the deleted member's numeric ID.

Synchronizing Smart Lists in Reporting Applications

Synchronizing Smart Lists in reporting applications identifies dimensions in reporting applications to which Smart Lists are mapped, and adds level 0 members of the selected dimensions as new Smart List entries to the selected Smart Lists. To define data maps, see Defining Data Maps in *Administering Data Integration*.

To synchronize Smart Lists in reporting applications:

1. View the **Smart Lists** listing page.

   See Working with Smart Lists.

2. Click the Smart List you want to synchronize, and then select

   .

   During synchronization, values from reporting applications in all existing mappings are appended after the last Smart List item in the appropriate Smart list. If a Smart List is mapped to two dimensions, all members from the first mapping are inserted first, and then members from the second mapping are inserted. If a member already exists in a Smart List, it's not added again. Smart List members in the FreeForm Smart Lists are not deleted, even if the corresponding dimension members on the reporting application are deleted.

   > **Note**
   >
   > If Account is mapped as Smart List to Dimension, all of the level 0 members in the Account dimension are brought in as Smart List entries when the Smart List is synchronized. For example, Smart Lists may include entries such as HSP_Average and HSP_Ending. If this occurs, delete the extra entries from the Smart List.

3. Click **OK** in the dialog box that says "Smart List Synchronized Successfully."

> **Note**
>
> Smart List names can’t have spaces in them. If you're synchronizing Smart Lists in a reporting application, ensure that any new members don't have spaces in the name.

Setting Smart List Properties

Define Smart List properties on the Properties tab.

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> **Note**
>
> For a Smart List to be viewable in forms, the dimension to which the Smart List type member belongs must be first in the evaluation order. See Managing Dimensions.

Table 21-28 Smart List Properties

Property	Description
Smart List	Enter a unique name containing only alphanumeric and underscore characters (for example: Position) and no special characters or spaces. Smart List names can be referenced in formula expressions.
Label	Enter the text to display when the Smart List is selected. Spaces and special characters are allowed.
Display Order	How Smart Lists are sorted in the drop-down list: by ID, Name, or Label
#MISSING Drop-Down Label	Enter a label (for example, "No Justification") to be displayed as an entry in the Smart List whose value is #MISSING. Note the following: • It displays as the first selection in the Smart List drop-down, allowing #MISSING as a selection in the form. • When the cell isn't in focus, this label displays only if **Drop-Down Setting** is selected in the next option. Otherwise, #MISSING or a blank cell is displayed, depending on the **Display Missing Values As Blank** selection for the form. • #MISSING labels determine only the display of cells with #MISSING data; #MISSING remains the stored value.
#MISSING Form Label	Determines how #MISSING values are represented in cells associated with Smart Lists. The options are as follows: • **Drop-Down Setting**: Displays the label set in **#MISSING Drop-Down Label**. • **Form Setting**: Displays #MISSING or leaves cells blank, depending on the **Display Missing Values As Blank** selection for the form. This selection determines what is displayed in the cell when it's not the focus. When the cell is in focus, the Smart List item that is selected from the drop-down is displayed.
Automatically Generate ID	Generate a numeric ID for each Smart List entry. If you don't select this option, you can customize Smart List ID values.
Create from Members	Create a Smart List based on dimension hierarchies. Smart List values are dynamically updated when members are updated. Note that user security for the dimension is honored for Smart Lists created from the dimension's hierarchy.
Member Selection	Manually select the members to use in Smart Lists.

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Table 21-28 (Cont.) Smart List Properties

Property	Description
Minimum Access Required	Select the minimum access required for members added to Smart Lists. This access defines how the members are filtered and displayed in Smart Lists for users. The options are as follows: • **None**: All members added to the Smart List are visible in the Smart List drop-down entries. Access control is not applied. • **Display**: Members having Display, Read, or Write permissions are visible in the Smart List drop-down entries. However, the display of their data values in the grid depends on the member access permissions. For example, members with Minimum Access Required set as Display are visible in the Smart List drop-down entries but their data values remain hidden in the grid, if their member access permission is set as Display. • **Read**: Members having Read or Write permissions are visible in the Smart List drop-down entries. • **Write**: Only members with Write permission are visible in the Smart List drop-down entries.

Use the **Entries** tab to define selections on Smart Lists. See Defining Smart List Entries.

Defining Smart List Entries

To define Smart List entries:

1. On the **Entries** tab, define drop-down list items:
   - **For first items only**: enter information into the first row.
   - To add an item, click (Add) and enter the information.
   - To delete an item, select it and click (Delete).
   - To edit an entry, change the information in its row:

Table 21-29 Smart List Entries

Entry Property	Description
ID	Unique number that sets the order for the displayed entry. The ID is customizable only if **Automatically generate ID** isn't selected on the **Properties** tab.
Name	Unique alphanumeric name containing alphanumeric and underscore characters (for example: Customer_Feedback) and no special characters or spaces
Label	Displayed text for the Smart List entry on the drop-down list (for example: Customer Feedback).

> **Note**
>
> Items highlighted in red are duplicates.

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2. Click **Save**.

   Use the **Preview** tab to preview the defined Smart List. See Previewing Smart Lists.

Previewing Smart Lists

Preview the defined Smart List on the **Preview** tab. The tab shows the Smart List as displayed in a drop-down list or a table.

To preview a Smart List:

1. Click the **Navigator** icon , and then under **Create and Manage**, click **Smart Lists**.
2. Select a Smart List and click (Edit).
3. Click **Save**.

Displaying #MISSING with Smart Lists

Service Administrators set values displayed in Smart Lists and data cells, including the display when no data is in the cell. Cells can display no value, #MISSING, or (for cells associated with Smart Lists) a specified value.

Use these options to control the display of #MISSING when cells are not in focus:

Table 21-30 Display Options for #MISSING

Option	Guideline
Blank	When designing forms, select **Display Missing Values as Blank**. When setting Smart List properties, select **Form Setting**.
#MISSING	When designing forms, don't select **Display Missing Values as Blank**. When setting Smart List properties, select **Form Setting**.
A custom label, such as "No Change"	When setting Smart List properties, enter the custom label in the **#MISSING Drop-Down Label** field (for example, `No Change`). Select **Drop-Down Setting**.

Administering Task Lists

Guide users through the planning process with Task Lists.

Task Lists guide users through the planning process by listing tasks, instructions, and end dates. Service Administrators and Power Users create and manage tasks and task lists.

Related Topics

- Working with Task Lists
- Adding Instructions to Task Lists
- Adding Tasks to Task Lists
- Editing Task Lists
- Assigning Access to Task Lists

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Working with Task Lists

Task lists organize groups of tasks for users. You must create task lists before creating tasks.

To create and rename task lists:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Task Lists**.
2. Perform a task:
   - To create a task list, click (Create Task List).
   - To rename a task list, select the task list, then click (Rename Task List).
   - To move a task list, select the task list, then click (Move Task List).
   - To remove a task list, select the task list, then click (Remove Task List).
3. Click **OK**.

To define the task list, see:

- Adding Instructions to Task Lists
- Adding Tasks to Task Lists

Adding Instructions to Task Lists

To add instructions to task lists:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Task Lists**.
2. For **Task List**, select the task list to modify, and then click (Edit Task List).
3. Click the **Instructions** tab.
4. Enter instructions for the task list.
5. Click **Save**, and then click **Close**

Adding Tasks to Task Lists

You can set completion dates and alerts for tasks that users perform at runtime. You can also set up email messages, for example, to alert users that a task was not completed by its end date. Alert messages are sent after an "alert date" that you set, and are repeated until the end date is reached for a task.

To add tasks to task lists:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Task Lists**.
2. Perform a task:
   - Define a task list by clicking (Create Task List), and then entering a name.
   - Modify a task list to add a new task by selecting it, and then clicking (Edit Task List).

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3. Click (Add Child).
4. For **Task**, enter a task name.
5. From **Type**, select the kind of task you're creating and what it should enable users to do. For example, if you're creating a task that enables Service Administrators to modify all dimensions at runtime, select **Dimension Editor**.

   Specify the information required for the task type using this table:

Table 21-31 Task Information

Task Type	Action
URL	Opens a specified URL Enter a fully qualified URL to associate with this task, such as `https://www.company_name.com`, and then go to step 6.
Form	Opens a form Select the form for users to complete, and then go to step 6. **Optional**: Select **Set Page Member Defaults** to select the member from each dimension to display as the default when the task is first opened. After you select this option, you can select the members for the page dimensions. The page member defaults apply until a user updates the form and returns to the task in another session. Where page member defaults are set, they override the most recently used settings in each session.
Dashboard	Opens a dashboard Select the dashboard for users to work with, and then go to step 6.
Business Rule	Launches a business rule Perform these tasks: • From **Cube**, select the cube associated with the business rule to execute. • From **Business Rules**, select the business rule to execute. • Go to step 6.
Descriptive	Enables Service Administrators to add a task with no task properties.
Job Console	Opens the Job Console to enable users to view a list of their jobs by type (such as copy data and push data) and by completion status Perform these tasks: • In **Job Type** select the kind of job to be displayed in the console. • In **Status**, select the status of the task to be displayed in the console. • Go to step 6.
Dimension Editor	Enables Service Administrators to view and modify a dimension at runtime Perform these tasks, and then go to step 6: • To enable Service Administrators to view and modify all dimensions, select **All** from **Dimension**. • To enable Service Administrators to view and modify a particular dimension, select it.
Refresh Application	Enables Service Administrators to refresh the application at runtime to capture recent changes. After enabling, go to step 6.

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Table 21-31 (Cont.) Task Information

Task Type	Action
Import and Export	Enables Service Administrators to import and export data and metadata using flat files. From **Load Method,** specify what import or export task the user must perform, such as export data to a file or import metadata from a file, and then go to step 6.

> **Note**
>
> The Security Management, Process Management, and Form Management task types cannot be added to a task list or edited in the Simplified Interface. These task types are accessible only in desktop mode of the application.
>
> The Dashboard task type cannot be added to a task list or edited while in desktop mode. This task type is accessible only in the Simplified Interface.

6. For **Duration**, enter how long you expect it will take to complete the task. Enter a number in the first field and the unit of time in the second field; for example, if you think the task will take two weeks, type 2 in the first field and then in the second field, type a `w` or `W` to select **Week(s)**. Time unit options for the second field are as follows:
   - **Hour(s)**: Type `h` or `H` to select this option.
   - **Day(s)**: Type `d` or `D` to select this option.
   - **Week(s)**: Type `w` or `W` to select this option.
   - **Month(s)**: Type `m` or `M` to select this option.
   - **Year(s)**: Type `y` or `Y` to select this option.

> **Note**
>
> Unit of time options are not case-sensitive. If the language you are using is not English, these entries are localized. Type the first letter of the localized word and then select the whole word that is displayed.

7. To ensure that the task is started on a certain date and then completed by a certain date, perform these tasks:
   a. For **Start Date**, click (Select Date and Time), and then use the calendar to specify the year, month, day, and time by which the task must begin.
   b. For **End Date**, click (Select Date and Time), and then use the calendar to specify the year, month, day, and time by which the task should be completed.

      To send email messages that prompt users to finish uncompleted tasks, select **Repeat Every**, and then specify how often to send these emails, such as once a day.
8. To send email messages that prompt users that the date by which they must complete the task is approaching, select **Alert**, and then:
   a. Click (Select Date and Time) to use the calendar to specify the date and time on which to start sending alert reminders.

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b. Click **Repeat Every** and specify the frequency for email reminders.

For example, if a task's end date is June 9 2014 at 5 PM, and you want to remind users once a day starting the week before the task is due, select June 2 2014 at 5 PM, enter 1, and then select Weeks.

9. To make task completion dependent on completing another primary (or parent) task, click next to **Dependency**, select the primary task, and then click **OK**.

> **Note**
>
> You cannot make task completion dependent on completing a child task of another primary task. Dependencies across task list hierarchies are not supported when a task has children. For example, consider a case where Task 3 is dependent on Task 2 and Task 4 is, in turn, dependent on Task 3. When Task 3 is incomplete, Task 4 becomes incomplete automatically. However, when Task 2.1 is incomplete, its parent task (Task 2) changes to incomplete, but this is not recursive (Task 3 and Task 4 are not affected).
>
> - Task1
> - Task2
>   - Task2.1
>   - Task2.2
> - Task3
> - Task4

10. In the **Instructions** pane to the right, enter information that explains how to complete the task. Use the formatting buttons to customize the information. For example, you can modify the font, font size, and layout (as bullet points or numbered steps, for example).
11. Click **Save** and then click **OK**.

Editing Task Lists

Related Topics

- Editing Tasks
- Copying and Moving Tasks
- Moving and Reordering Task Lists
- Clearing Task Lists
- Deleting Tasks and Task Lists

Editing Tasks

Use the Edit Task dialog box to modify the type of task, its instructions, the end date and alert date, and email reminder messages.

To edit tasks:

1. Click the **Navigator** icon , and then under **Create and Manage**, click **Task Lists**.

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2. Select a task list, and then select (Edit Task List).
3. Select a task, and then select (Edit Task).
4. In **Edit Task**:
   - Modify the task name.
   - Select another task type.
   - **Optional**: Modify the values for **Duration**, **Start Date**, **End Date**, Alert, and Dependency options.

   For descriptions of the various options, see Adding Tasks to Task Lists.
5. For **Task - Instructions**, modify instructions for completing the task.
6. Click **Save**.

Copying and Moving Tasks

To copy or move tasks:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Task Lists**.
2. Perform an action:

   To copy tasks:

   a. Select a task list, and then select (Edit Task List).
   b. Select the task to copy, and then click **Save As**.
   c. Enter the name for the new task list, and then click **OK**.

   To move tasks:

   a. Select a task list, then select (Edit Task List).
   b. Select a task, and then click (Edit Task).
   c. Select the task that will be a child of another task, and then select (Cut).
   d. Select the task that will be the parent task, and then select (Paste).
   e. Click **Save**.

   To cut and paste tasks:

   a. Select a task list, and then click (Edit Task).
   b. Select a task, and then click (Cut).
   c. To move the task to a new position, select the task to appear above it.
   d. Click (Paste).
   e. Click **OK**.

Moving and Reordering Task Lists

To move or reorder task lists:

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1. Click the **Navigator** icon, and then under **Create and Manage**, click **Task Lists**.
2. Perform a task:

   To move a task list:

   a. Select a task list, and then click (Move Task List).
   b. Select the destination folder.
   c. Click **OK**.

   To reorder task lists:

   a. Select a task list.
   b. Click (Move Up) or (Move Down).

Clearing Task Lists

After a task list is completed, you can clear the completion status, end dates, and alerts for all tasks within a selected task list. This allows tasks to be reused for a future planning period.

Alerts are enabled only when task lists contains end dates. Clearing task lists disables alerts by clearing the check boxes for end dates. It doesn't delete the dates.

To clear task lists:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Task Lists**.
2. Select the task list you want to clear, and then click (Edit Task List).
3. In the **Clear** drop-down list, select an option:
   - **Completion Status**: Clears completion status only
   - **Due Dates and Alerts**: Clears any alerts that are enabled, based on the end date set for the task
   - **Both**: Clears completion status, end dates, and alerts
4. Click (Clear).
5. Click **OK**.

Deleting Tasks and Task Lists

To delete tasks and task lists:

1. Click the **Navigator** icon, and then under **Create and Manage**, click **Task Lists**.
2. Perform a task:
   - To delete a task, select the task list with the task to delete, and then click (Edit Task List). Select the tasks to delete, and then click (Delete).
   - To delete a task list, select the task list to delete, and then select (Delete Task List).
3. Click **OK**.

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Assigning Access to Task Lists

You can determine who can view and modify task lists. By default, Service Administrators can manage and assign access permissions for task lists.

> **Note**
>
> Being assigned to a task list means being able to access and complete tasks in the task list. It doesn't mean being able to assign tasks to someone else.

- Adding Access to Task Lists
- Changing and Removing Access to Task Lists

Adding Access to Task Lists

To assign access to task lists:

1. Click the **Navigator** icon , and then under **Create and Manage**, click **Task Lists**.
2. Select the task list to modify.
3. Click (Assign Access).
4. Click (Add Access).
5. Select the users or groups to access the task list.
   - Click **Users** to display all user names; click **Groups** to display all groups.
   - If there are multiple pages of users and groups, a pagination bar displays at the bottom of the page to help you navigate between the pages and to jump to a specific page.
6. For **Type of Access**, select how users or groups can use the task list:
   - **Assigned**: View and use
   - **Manage**: Modify
   - **Manage and Assign**: View, use, and modify
   - **None**: No access
7. Click **Add**.
8. Click **Close**.

Changing and Removing Access to Task Lists

To change or remove access to task lists:

1. Click the **Navigator** icon , and then under **Create and Manage**, click **Task Lists**.
2. Select the task list to modify, then click (Assign Access).
3. Perform a task:

   To change access to task lists:

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a. Select users or groups, and click (Edit Access).

b. For **Type of Access**, select:

- **Assigned**: View and use
- **Manage**: Modify
- **Manage and Assign**: View, use, and modify
- **None**: No access

c. Click **Set**.

To remove access to task lists:

a. Select the user or group, and click (Remove Access).

b. Click **OK**.

4. Click **Close**.

Clearing Cell Details

You can clear these cell details for a cube: account annotations, supporting detail, cell text, and cell-level documents. For information on creating and viewing account annotations, supporting detail, cell text, and cell-level documents, see *Working with FreeForm*.

> **Note**
>
> Because this function is for Service Administrators and Power Users, the application assumes that you have full access to details you delete.

To clear cell details:

1. Click the **Navigator** icon , and then under **Actions**, click **Clear Cell Details**.
2. For **Cube**, select a cube.
3. To select all dimensions in the cube, click **Add All Dimensions**.

   Select members for the displayed dimensions.
4. Specify the type of information to delete by selecting at least one option from **Clear Options**.
5. Select members for the data intersections:

   a. In **Dimension**, select at least one dimension with details to delete.

   b. For displayed dimensions, click (Member Selector). On the **Member Selection** page, make a selection that includes the details to delete.

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> **Note**
>
> For every dimension selected, you must select at least one member. If a dimension isn't selected, the application includes all its members when clearing cell details.

> **Caution**
>
> To clear exactly the data you want, select at least one Account member and members from all other dimensions. If you don't select at least one Account member, no data is cleared, even if you select the **Yes, set value to #Missing** option. If you don't select members from the other dimensions, all data for the selected Account member is cleared across all other dimensions.

6. **Optional**: To select another dimension so you can select its members, click (Add Dimension).
7. Click **Clear**.

   A Confirmation page displays your selections.
8. Click **Finish** to proceed, or **Back** to change your selections.

   If Clear Cell Details is successful, data is deleted from the cube. If data isn't deleted successfully, a message is displayed. You can also check the log file.
9. **Optional**: To view the execution status of Clear Cell Details and review the information that was deleted, view the Job . See Managing Jobs.
10. To clear cell details for another cube, select another cube in step 2, and repeat the procedure.

Copying Data

You can copy plans from one dimensional intersection to another, including relational data and supporting detail. For example, you can copy *Budget, FY10, Final* to *Forecast, FY11, First Draft*.

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> **Note**
>
> - Selected Copy Data settings are preserved for the current session only.
> - Copied dimension members must be present in the selected cubes.
> - Data must be copied into cells that can accept data. For example, you can't copy data into read-only or dynamic cells.
> - You can copy account annotations, supporting detail, and cell text.
> - You can't use this feature with attributes, so don't select attributes to be copied.
> - Because this is an administrative function, the application assumes you have full access to data you copy.
> - This feature doesn't calculate data. To perform calculations, such as increasing the forecast by 5%, apply the business rule after copying data.
> - For Copy Data to be successful, you must select at least one member for Scenario, Account, Entity, Period, and Version dimensions.
> - Copy Data is enabled for block storage and aggregate storage cubes.

To copy data:

1. Click the **Navigator** icon , and then under **Actions**, click **Copy Data**.
2. In **Cube**, select a cube.

   Dimensions are displayed for this cube. You can only copy from one cube at a time.
3. In **Copy Data Options**, select the type of information to copy.
4. Under **Static Dimensions**, select the members for the data intersections:
   a. For **Dimension**, select a dimension from which to copy.
   b. For **Members**, click (Member Selector) to make a selection from which to copy. You can select multiple members. You must select at least one member for Scenario, Account, Entity, Period, and Version dimensions.
5. **Optional**: To add another static dimension to the list, click (Add Dimension) and select dimension members.

   > **Note**
   >
   > To remove a dimension, select **Select Dimension**. The dimension moves to the **Dimensions with Source and Destination** area.

6. In **Dimensions with Source and Destination**, enter dimensions into which to copy data.

   In **Source** and **Destination** click (Member Selector).
7. Click **Copy Data**.

   Data is copied from one intersection to the other. If data isn't copied successfully, a message is displayed. You can also check the log file.
8. To copy data for another cube, select another cube in step 2, and repeat the procedure.

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Administering Application Monitoring

Related Topics

- About Application Monitoring
- Assumptions
- How Application Monitoring Works
- Using Application Monitoring Graphs
- Launching Application Monitoring
- Modifying Artifacts for Optimal Performance

About Application Monitoring

Application Monitoring enables Service Administrators, at design-time, to identify and resolve design flaws before an application is placed in production (and on an ongoing basis, as the application evolves with new members and data is added to the application) and used by end users. Service Administrators can use application diagnostics to evaluate the following:

- An entire application
- Types of artifacts such as forms
- Individual "cherry picked" artifacts

Application Monitoring identifies changes that Service Administrators can make at design-time to resolve artifacts with design flaws, and provides easy access to the editors required to modify them. At runtime, internal processes perform checks that may deny system request if artifacts need to be modified. Application Monitoring evaluates your application and artifacts based on the user a Service Administrator selects. This enables you to determine the potential problems that may occur at runtime given the individual user's variables and security access as part of the analysis.

Assumptions

Oracle can only provide general guidelines about how to design your application and artifacts to ensure reasonable performance. Oracle can't specify precise value ranges for artifacts and operations.

How Application Monitoring Works

Application Monitoring evaluates applications as follows:

- At design-time: Analyzes the application or specific artifacts in terms of all possible problems that could occur. To determine this worst possible situation, suppression options (such as those used with forms) are ignored. Application Monitoring provides information about how to fix the design flaws it encounters.
- At runtime: Internal governors perform checks that may prevent users from performing particular actions if the design flaws found are not resolved.

Use Application Monitoring to analyze the following:

- Block storage cubes
- Dimensions

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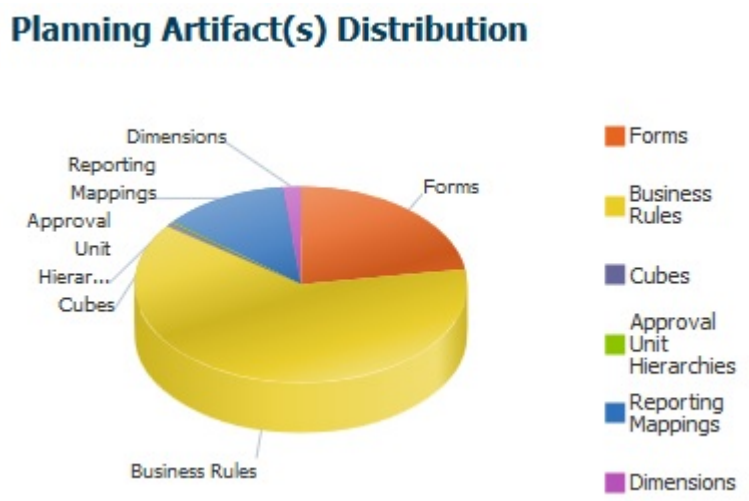
- Forms
- Standalone business rules
- Reporting mappings
- Export data functionality

Using Application Monitoring Graphs

Artifacts that may cause performance issues and that should be modified are yellow, and those that will cause performance issues and must be redesigned are red. Use the pie graphs as follows:

- **<Business Process> Artifacts Distribution** : Depicts how much of your application is comprised of various artifact types. Place your mouse over portions of the graph to view how many artifacts of each type were analyzed by application diagnostics. For example, if you mouse over forms and the pop-up value is 55, then 55 forms were analyzed. Click each portion of the graph to view the compliance status of each artifact type. This displays another pie graph that represents how many artifacts of each type are in acceptable, warning, or error status.

  Example Artifact Distribution Graph:



- **Filtered For <Artifact>**: Click the graph portions that represent the artifact status, and to view and drill down on artifacts with a particular status.

  Filtered Graph:

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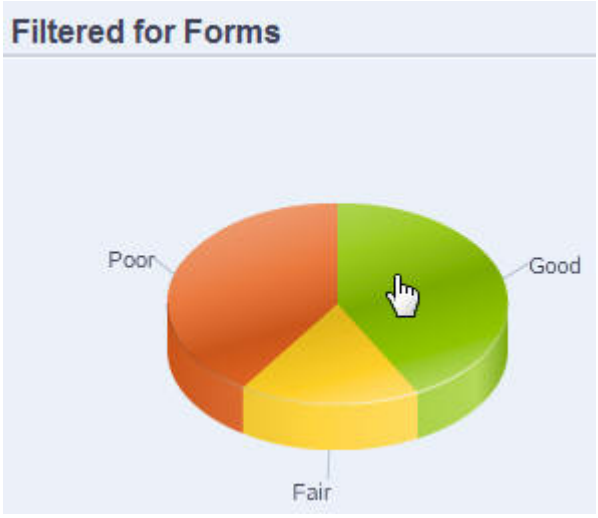

Launching Application Monitoring

To launch Application Monitoring:

1. Click the **Navigator** icon, and then under **Monitor and Explore**, click **Application Monitoring**.
2. In **Select User**, click (Select User) to select the user whose settings you want to use to perform the analysis, and then click (Go).
3. Perform a task:
   - A pie graph depicts the distribution of artifacts in the application. The lower grid represents the integrity of all application artifacts. A status of fair and poor indicate artifacts that you should or must modify for optimal performance.
   - To analyze specific types of artifacts, such as forms, select the slice for the type within the pie graph. For example, to ensure that business rules are designed well and will not cause performance issues, click the Business Rules slice of the pie graph. If a large portion of the graph is fair or poor, you should modify your rules. More information about all artifacts of that type display in the grid below.
   - To "cherry pick" and analyze one particular artifact, expand **Search** to find the artifact by specifying its name, start date, end date, and type, select the artifact, and then click (Go). Information about the artifact displays in the lower grid.

   Tip

   For information about what the pie graphs display and how to drill down to access more information, see Using Application Monitoring Graphs.
4. Click (Run Application Monitor) or select **Actions**, and then **Run Application Monitor**.
5. If the **View Details** column doesn't display in the grid, select **View**, then **Columns**, and then **View Details**. This also enables you to display or conceal different types of artifact data.

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6. To display information about how to fix artifacts, click (Show Details) in **View Details**. Then click artifacts to launch editors that enable you to modify them.

Modifying Artifacts for Optimal Performance

After reviewing the details provided by the application diagnostics that tell you how to modify your artifacts, edit them as follows:

- Forms: Clicking on the form artifact name launches the **Form Designer** in a new tab. Edit forms as described in Administering Forms.
- Business Rules: Clicking a business rule name launches the **System View** which displays all application artifacts that you can access. You can perform tasks such as:
  - Use the upper **Rule Designer** to view the individual components of the rule, such as conditions, commands, and scripts. Click each component to modify them in the lower portion of the page. You can also insert and remove components like formulas, system templates, and scripts by dragging and dropping them from the **Rule Pallet** and **Existing Object** panes to the left.
  - From **Designer**, select **Edit Script** to modify and format the script by performing tasks such as inserting functions, editing templates, inserting variables, inserting member ranges, use comments, and validate syntax.
  - Create, open, delete, and refresh artifacts, pages, and other documents.
  - Import and export business rules and other objects.
  - Determine how and where artifacts are used.

    See *Designing with Calculation Manager* and Understanding Runtime Prompts.
- Cubes: Clicking the cube artifact name launches the **Performance Settings** tab of the dimension editor in a new tab. Edit the dimensional outline or dimensional sparsity as described in Editing Dimensions in the Simplified Dimension Editor.
- Reporting Mappings: Clicking the reporting mapping artifact name launches the **Report Mapping Designer** in a new tab. Edit reporting mappings as described in Defining Data Maps in *Administering Data Integration*.

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A

Naming Restrictions

Understand important restrictions for names, including names for applications, databases, dimensions, members, aliases, users, and groups.

Related Topics

- Naming Restrictions for Applications and Databases
- Naming Restrictions for Dimensions, Members, and Aliases
- Dimension and Member Names in Calculation Scripts, Report Scripts, Formulas, Filters, and Substitution Variables
- Restrictions for User and Group Names

Naming Restrictions for Applications and Databases

When naming an application, follow these rules:

- Use no more than eight characters for an application's name.
- Don't use spaces or tabs.
- Don't use an underscore (_) as either the first or last character.
- Don't use these special characters:

Table A-1 Restricted Characters in Application Names

Character	Meaning
*	asterisk
\	backslash
[ ]	brackets
:	colon
,	comma
"	double quotation mark
=	equal sign
>	greater than sign
<	less than sign
.	period
+	plus sign
?	question mark
;	semicolon
'	single quotation mark
/	slash
|	vertical bar

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- For application names in relational database environments, don't use extended characters (except for underscores).
- For aggregate storage databases, don't use DEFAULT, LOG, METADATA, or TEMP as application names.

Enter names in the preferred case. Application names are created exactly as entered.

Naming Restrictions for Dimensions, Members, and Aliases

A list of the most common restricted characters is provided in this section. However, Oracle strongly suggests that you review the conventions described in Naming Conventions for Essbase in addition to the information in this section.

When naming dimensions, members, and aliases, follow these rules:

- Use no more than 80 characters for dimension, member, or alias names. The 80-character limit applies to all application types.
- Names are not case-sensitive for dimensions, non-shared members, and aliases. Do not use matching names with only case differences; for example, do not name two members "zzz" and "ZZZ," or a base member "zzz" and a shared member "ZZZ."

  In other words, shared member names must use the same name and case of the base member. When you rename a base member (even if all you do is change the case), the shared member names will be updated to match the base member names.
- Do not use dimension names that are the same name as a member property or a column header in the metadata load file; for example, Data Type.
- HTML tags cannot be used in member names, dimension names, aliases, and descriptions.
- The TAB key is restricted and cannot be used in dimension, member, and alias names.
- The following is a list of characters that are restricted and cannot be used in dimension, member, and alias names:

Table A-2 Restricted Characters in Dimension, Member, and Alias Names

Character	Meaning
"	quotation mark
[ ]	bracket
\	backslash
/	slash

> **Note**
>
> – Brackets [ ] are permitted but not recommended in block storage outlines. They cause errors when converting to aggregate storage outlines.
>
> – Do not use the ASCII substitute character (zero-width space character) `0x1A` in member names and aliases.

- The following is a list of characters that cannot be used at the beginning of dimension, member, and alias names:

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Table A-3 Restricted Characters at the Beginning of Dimension, Member, and Alias Names

Character	Meaning
@	at sign
&	ampersand
\	backslash
[ ]	brackets
,	comma
-	dash, hyphen, or minus sign
=	equal sign
<	less than sign
( )	parentheses
.	period
+	plus sign
"	quotation mark
/	slash
_	underscore
|	vertical bar

> **Note**
>
> Member names with the preceding characters are not allowed in business process data export jobs.

- The following characters are not recommended for names or aliases, and cannot be at the beginning or end of a member or alias name. All are in Unicode format.
  - 0x0009 - Horizontal Tab (common tab)
  - 0x000A - Line Feed
  - 0x000B - Vertical Tab
  - 0x000C - Form Feed
  - 0x000D - Carriage Return
  - 0x0020 - Space
  - 0x0085 - Next line
  - 0x00A0 - Non-breaking Space
  - 0x1680 - Ogham Space
  - 0x180E - Mongolian Vowel Separator
  - 0x2000 - EN Quad
  - 0x2001 - EM Quad
  - 0x2002 - EN Space
  - 0x2003 - EM Space
  - 0x2004 - Three Per EM Space

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  - 0x2005 - Four Per EM Space
  - 0x2006 - Six Per EM Space
  - 0x2007 - Figure Space
  - 0x2008 - Punctuation Space
  - 0x2009 - Thin Space
  - 0x200A - Hair Space
  - 0x200B - Zero Width Space
  - 0x2028 - Line Separator
  - 0x2029 - Paragraph Separator
  - 0x202F - Narrow Non-breaking Space
  - 0x205F - Medium math space
  - 0x3000 - Ideographic Space
- Spaces cannot be placed at the beginning or end of names.
- For time periods in custom calendars, don't use spaces in prefixes.
- The following cannot be used as dimension, member, or alias names:
  - Calculation script commands, operators, and keywords
  - Function names and function arguments
  - Names of other dimensions and members (unless the member is shared)
  - If Dynamic Time Series is enabled, don't use History, Year, Season, Period, Quarter, Month, Week, or Day.
  - In unique member outlines that contain an attribute dimension, you cannot use Sum, Count, Min, Max, and Avg as member or dimension names. Using these names causes duplicate name conflicts.

    Note

    The members that are created in the attribute calculations dimension (Sum, Count, Min, Max, and Avg) are not considered reserved words because you can change these names in the attribute calculations dimension and then use the standard name in an attribute or standard dimension.

    If the outline is tagged as a duplicate member outline, and an attribute dimension (and, therefore, the attribute calculations dimension) exists before you use Sum, Count, Min, Max, and Avg as a base member, the duplicate name is allowed. However, if, you use Sum, Count, Min, Max, and Avg as a base member before creating an attribute dimension, the duplicate name isn't allowed.
- The following is a list of words that are restricted and cannot be used in dimension, member, and alias names:

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Table A-4 Restricted Words in Dimension, Member, and Alias Names

Restricted Word	Restricted Word	Restricted Word
ALL	AND	ASSIGN
AVERAGE	CALC	CALCMBR
COPYFORWARD	COUNT	CROSSDIM
CURMBRNAME	DIM	DIMNAME
DIV	DYNAMIC	EMPTYPARM
EQ	EQOP	EXCEPT
EXP	EXPERROR	FLOAT
FUNCTION	GE	GEN
GENRANGE	GROUP	GT
ID	IDERROR	INTEGER
LE	LEVELRANGE	LOCAL
LOOPBLOCK	LOOPPARMS	LT
MBR	MBRNAME	MBRONLY
MINUS	MISSING	MUL
MULOP	NE	NON
NONINPUT	NOT	OR
PAREN	PARENPARM	PERCENT
PLUS	RELOP	REQUEST
SET	SKIPBOTH	SKIPMISSING
SKIPNONE	SKIPZERO	STATUS
SUM	TO	TOLOCALRATE
TRAILMISSING	TRAILSUM	TYPE
UMINUS	UPPER	VARORXMBR
XMBRONLY	$$UNIVERSE$$	#MISSING
#MI		

- The following characters should generally be avoided:
  - , (comma)
  - : (colon)
  - [ ] (square brackets)
  - { } (curly braces)
  - / (forward slash)
  - ( ) (parentheses)
  - * (asterisk)
  - ! (exclamation point)
  - > (greater than sign)
  - ; (semicolon)
  - % (percent sign)
  - ~ *(tilde)

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Dimension and Member Names in Calculation Scripts, Report Scripts, Formulas, Filters, and Substitution Variables

In substitution variable values, calculation scripts, report scripts, filter definitions, partition definitions, or formulas, you must enclose member names in brackets ([]) when used within MDX statements and in quotation marks (" ") for block storage databases, in these situations:

- The name starts with one or more numerals (for example, 100).
- The name contains spaces or these characters:
  - ampersand (&)
  - asterisk (*)
  - at sign (@)
  - backslash (\ )
  - braces ({ })
  - colon (:)
  - comma (,)
  - dash, hyphen, or minus sign (-)
  - exclamation point (!)
  - equal sign (=)
  - greater than sign (>)
  - less than sign (<)
  - parentheses ( )
  - percent sign (%)
  - period (.)
  - plus sign (+)
  - semicolon (;)
  - slash (/)
  - tilde (~)

In calculation scripts and formulas, enclose member names that are also Essbase keywords in quotation marks (" ") for block storage databases, and in brackets ([ ]) for aggregate storage databases including these member names:

```
BEGIN DOUBLE ELSE END FUNCTION GLOBAL IF MACRO MEMBER RANGE RETURN STRING THEN
```

Enclose in quotation marks names, in calculation scripts, report scripts, or formulas, that contain or start with spaces or these characters: + - * / ( ) : , @ ;) { } [ ] <

Also enclose in quotation marks names that start with numbers. For example:

```
@SUM(ac1, "12345");
```

Restrictions for User and Group Names

User and group names can have up to 80 characters.

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B

Form Formula Functions

Learn how to use formulas and functions in forms.

Related Topics

- About Form Formula Functions
- Working with Formulas
- Formula Functions

About Form Formula Functions

After you create a formula row or column, define the formula using the **Segment Properties** pane. Formulas include grid references, mathematical operators, and, optionally, mathematical functions. For a complete list, see Formula Functions.

> **Note**
>
> The application requires level 0 members that are Dynamic Calc to have a member formula. For Dynamic Calc members that don't have a formula, the application inserts a semicolon (;) when refreshed.

Working with Formulas

Related Topics

- Creating Formulas
- Editing Formulas
- Deleting Formulas

Creating Formulas

To create formulas:

1. Select the row or column for which to associate the formula.
2. If it doesn't automatically expand, select **Segment Properties**.
3. In Formula, enter the name for the formula and then click (Edit Formula).
4. In the **Formula** box that is displayed, select the operation or function that the formula will perform, such as COUNT(), MAX(), and IfThen(). See Formula Functions.
5. Click **Validate** to ensure that the formula doesn't contain any errors.

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Editing Formulas

To edit formulas:

1. Open a form with a formula row or column.
2. In **Layout**, select a numbered formula row or column.
3. Use **Segment Properties** to modify the formula properties.
4. Click (Edit Formula) to change the operation or function performed.
5. Save the form.

Deleting Formulas

To delete a formula:

1. Select the formula row or column.
2. Click **Delete**.
3. To verify formula deletion:
   a. Click the checkmark in the formula bar.
   b. Click another cell within the grid to reset the formula bar.
   c. Click the cell from which you deleted the formula to verify deletion.

Formula Functions

This section defines the mathematical functions available for creating formulas for form formula rows and columns. To insert formula rows and columns in forms, see Adding Formula Rows and Columns.

The syntax for mathematical functions:

```
FunctionName(arguments)
```

For more information about arguments, see Arguments.

Table B-1 Mathematical Function Syntax

Variable	Description
FunctionName	The name of a mathematical function
arguments	A numeric value, a row, column, or cell reference, or an embedded function

Mathematical Functions

Table B-2 Mathematical Functions

Function	Description
Abs	Returns the absolute value of numeric values or references. See Abs.

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Table B-2 (Cont.) Mathematical Functions

Function	Description
Average	Returns the average of a group of numeric values or references. See Average.
AverageA	Returns the average of a group of numeric values or references. The calculation includes #MISSING cells only for rows or columns that are not suppressed. See AverageA.
Count	Returns the number of values in a group of numeric values or references. See Count.
CountA	Returns the number of values in a group of numeric values or references. The calculation includes #MISSING cells only for rows or columns that are not suppressed. See CountA.
Difference	Returns the absolute value of a numeric value or reference subtracted from another numeric value or reference. See Difference.
Eval	Evaluates an expression. Eval is useful for embedding expressions as function arguments. See Eval.
IfThen, If	Returns one value if a condition equals true, and another value if a specified condition equals false. See IfThen, If.
Max	Returns the maximum value from a group of numeric values or references. See Max.
Min	Returns the minimum value from a group of numeric values or references. See Min.
Mod	Returns the remainder, modulus, from a division formula. See Mod.
PercentOfTotal	Returns the result of a numeric value or reference divided by another numeric value or reference, multiplied by 100. See PercentOfTotal.
Pi	Returns the number 3.14159265358979, to 15 digits. See Pi.
Product	Multiplies all numbers or references and returns the product. See Product.
Random	Returns a random number between 0.0 and 1.0. See Random.
Round	Rounds a number up or down by specified digits. See Round.
Sqrt	Returns the square root of a numeric value, row, column, or cell. See Sqrt.
Sum	Returns the sum of a group of numeric values or references. See Sum.
Truncate / Trunc	Removes the specified number of digits from numeric values. See Truncate/Trunc.
Variance / Var	Evaluates the difference between the specified values based on the account type for the current account. See Variance/Var.
VariancePercent / VarPer	Evaluates the percentage difference between the specified values based on account type for the current account. See VariancePercent/VarPer.

Arguments

Mathematical functions accept numeric values, row, column, or cell references, or embedded functions as arguments. There are four argument types:

- Numeric

  See Numeric Arguments.

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- Property

  See Property Arguments.
- Row, column, or cell reference

  See Row, Column, or Cell Reference Arguments.
- Embedded functions

  See Embedded Functions as Arguments.

Numeric Arguments

The syntax for a numeric argument is

```
(numeral1, numeral2,...numeraln)
```

where numerals *1* through *n* are any numbers including decimals and negative values. For example, the expression Average(10,20,30) returns the value 20.

Row, Column, or Cell Reference Arguments

The row, column, or cell argument identifies a row, column, or cell in a grid. The syntax:

```
FunctionName(GridName.GridElement[segment(range)].Property)
```

Table B-3 Argument Components

Argument	Description
GridName	The form name. For example: Difference (grid1.row[5], grid1.row[6]) returns the difference of two rows on form grid1. Optional. If GridName isn't specified, the default is the name of the current form.
GridElement	One of the following keywords: row, col, column, or cell. For example, Max(row[1], row[2], row[3]) returns the maximum value of three rows. GridElement is optional. However, a cell reference requires row and column segment identifiers. For example, cell[2, A] and [2, A] both refer to the cell that is the intersection between row 2 and column A. The keyword cell is optional. Cell references can use the [row, col] syntax or [col, row] syntax. Optional. If GridElement isn't specified, letters represent columns and numbers represent rows; for example: Max ([1, A], [2, A], [3, A]) refers to rows 1, 2 and 3 of column A.
segment	A row, column, or cell reference number. For an expanded row or column, you must specify the segment. For example, row[2] addresses row segment 2. Segments are enclosed in square brackets [ ]. Required.

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Table B-3 (Cont.) Argument Components

Argument	Description
range	The rows, columns, or cell that are expanded from the specified segment. If *range* is specified, the system calculates the formula using only the specified range. For example, row[2(3:5)] uses only the third through fifth rows of expanded segment 2. Optional. When *range* isn't provided, all expanded cells are used. **Note:** If a segment expands to only one row or column, don't use the range argument.

property	One of these keywords: average, averageA, count, countA, max, min, product, or sum. The property specifies how to aggregate the specified expanded rows, columns, or cells.
	Oracle recommends that property is *not* specified when a reference is an argument. By not specifying the property, the function calculates the reference in the most appropriate way. For example, the following expression returns the average of the cells within rows 1 and 2:
	`Average(row[1], row[2])`
	In contrast, the following example first calculates the average of row[1], then the average of row[2], adds these two results, then divides by 2:
	`Average(row[1].average, row[2].average)`
	The default property for a row, column, or cell reference is sum . For example, `row[2]` is equivalent to `Sum(row[2])`.

Because segment is the only required part of a reference, the following references are the same:

```
Grid1.row[1].sum
```

```
[1]
```

AverageA and CountA include #MISSING cells in the calculation. For example, if row 1 is a segment row that expands to Qtr1 = 100, Qtr2 = 200, Qtr3 = #MISSING, and Qtr4 = 400, the following function returns the value four (4):

```
row[1].CountA
```

All other functions exclude #MISSING data cells. For example, the previous example of row 1 that expands to Qtr1 = 100, Qtr2 = 200, Qtr3 = #MISSING, and Qtr4 = 400, returns three in this example:

```
row[1].Count
```

Property Arguments

Property arguments consolidate expanded references to a single value that is then used in the calculation. Use property arguments to perform calculations on an aggregate row, column, or cell. The two types of property arguments are:

- Aggregate

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See Aggregate Property Argument.

- Reference

  See Reference Property Argument.

Aggregate Property Argument

An aggregate row, column, or cell contains multiple rows, columns, or cells, respectively.

The aggregate property argument is the last argument in the following mathematical function syntax:

```
FunctionName(GridName.Gridelement[segment(range)].property)
```

You apply the following aggregate properties to a row, column, or cell reference.

Table B-4 Aggregate Properties

Properties	Description
Average	Returns the average of a row, column, or cell. The calculation excludes #MISSING and #ERROR values.
AverageA	Returns the average of a row, column, or cell. The calculation includes #MISSING and #ERROR values.
Count	Returns the number of values in a row, column, or cell. The calculation excludes #MISSING and #ERROR values.
CountA	Returns the number of values in a row, column, or cell. The calculation treats #MISSING and #ERROR values as zero (0).
Max	Returns the maximum value of a row, column, or cell
Min	Returns the minimum value of a row, column, or cell
Product	Returns the product of rows or columns
Sum	Returns the sum of a row, column, or cell

When used as a mathematical function argument, the default for property is the same as the function. In the following example the default property is Average:

```
Average(row[2])
```

When not used as a mathematical function argument, the default for property is sum. In the following example the default property is the sum of an aggregate row:

```
row[2]
```

Reference Property Argument

A reference property argument specifies how to treat formula reference results and is used in conjunction with the other properties.

There is one reference property argument:

```
IfNonNumber/IFFN
```


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IfNonNumber specifies a replacement of #MISSING and #ERROR values with a specific numeric value. The syntax:

```
AXIS[segment(range)].IfNonNumber(arg).AggregateProperty
```

Table B-5 Reference Property Arguments

Argument	Description
AXIS	One of these keywords: row, column, or cell Optional
Segment(range)	Indicates any valid axis reference, such as a row number, column letter
IfNonNumber	Indicates how to treat missing or error data within the Axis Ref
(arg)	Indicates what number to use if missing or error data is encountered within the AxisRef
AggregateProperty	The aggregate function is used for aggregate segments Optional

Example:

If cell[1,A] = 3 and

cell[1,B] = #MISSING,

The expression:

```
cell[1,A] / cell[1,B]
```

returns #ERROR.

The expression:

```
cell[1,A] / cell[1,B].ifNonnumber(1)
```

replaces cell[1,B] with 1 and returns a 3.

> **Note**
>
> If you use suppression for #MISSING in a grid, and the grid contains a formula row or column that uses the IfNonNumber property, #MISSING remains suppressed.

Embedded Functions as Arguments

You can embed functions as arguments within a function.

Example:

In this example, the function Average is embedded in the function Sum:

sum(row[3:5], avg(row[4:6], 40, 50), row[7; 9], 70, 80)

- Row segments 3, 4 and 5

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- The average of row segments 4, 5 and 6, with the numbers 40 and 50
- Row segments 7 and 9
- The numbers 70 and 80

Abs

Abs is a mathematical function that returns the absolute value of a numeric value, row, column, or cell. The absolute value of a number is that number without regard to sign. A negative number becomes positive, while a positive number doesn't change. The function syntax:

```
Abs (argument)
```

where *argument* is one of the following:

Table B-6 Arguments for the Abs Function

Argument	Description
numeric	A numeric value. For example, Abs (- 20) returns the value 20. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property.`
function	An embedded function

Examples:

- The following expression includes a numeric argument and returns the value 30:

  ```
  Abs ( -30 )
  ```

- The following example returns the absolute value of the value in row 1:

  ```
  Abs(row[1])
  ```

- The following example calculates the absolute value of the sum of column E:

  ```
  Abs( column[E].sum )
  ```

- The following example points to expanded rows 1 through 3 within design segment 3 of the form Grid1:

  ```
  Abs( Grid1.row[3(1:3)])
  ```

Average

Average is a mathematical function that returns the average of a group of numeric values, rows, columns, or cells. Average excludes #MISSING and #ERROR cells when obtaining the average.

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> **Note**
>
> The calculation doesn't include missing values regardless of whether they are suppressed or not.

The function syntax:

`Average(arguments) or Avg(arguments)`

where *arguments* is one or more of the following:

Table B-7 Arguments for the Average Function

Argument	Description
numeric	A numeric value. For example, Average (10, 20, 30) returns the value 20. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property` For example Avg(Grid1.row[4(3:5)]) returns the average of form grid1, row 4, range 3 through 5.
function	An embedded function

Examples:

- The following expression returns the value 20:

  ```
  Avg( 10, 30, 20)
  ```

- The following example returns the average of all numbers that are part of three aggregate rows:

  ```
  Average( row[1], row[6], row[8] )
  ```

- The following example calculates the average of three aggregate columns; E, G, and I. The calculation produces three numbers and calculates the average of the three numbers:

  ```
  Avg(column[E].avg, column[G].avg,column[I].avg)
  ```

- The following example calculates the average of aggregate row 3 and divides the average by 100:

  ```
  Avg(row[3])/100
  ```

AverageA

AverageA is a mathematical function that returns the average of a group of numeric values, rows, columns, or cells. AverageA includes #MISSING and #ERROR cells, which are treated as zero values when obtaining the average.

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> **Note**
>
> #MISSING and #ERROR are included only for rows or columns that are not suppressed.

The function syntax:

```
AverageA(arguments) or AvgA(arguments)
```

where *arguments* is one or more of these arguments:

Table B-8 Arguments for the AverageA Function

Argument	Description
numeric	A numeric value. For example, AverageA (10, 20, 30) returns the value 20. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)]. Property.` For example, AvgA(Grid1.row[4(3:5)] returns the average of form grid1, row segment 4, range 3 through 5.
function	An embedded function

Example

In the following example, a grid has 4 rows with the values 10, 20, 30, and, #ERROR. The following formula in the fifth row returns the value 15:

```
AverageA([1:4])
```

Count

Count is a mathematical function that returns the number of values in a group of numeric values, rows, columns, or cells. Count excludes #MISSING and #ERROR when obtaining the count. The function syntax:

```
Count (arguments)
```

where *arguments* is one or more of these arguments:

Table B-9 Arguments for the Count Function

Argument	Description
numeric	A numeric value. For example, Count (10, 20, 30) returns the value 3. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property`
function	An embedded function

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Examples:

- The following example returns the count of three rows, 1, 6, and 8:

```
Count(row[1], row[6], row[8])
```

- The following example returns the count of 3 columns:

```
Count( column[E], column[G], column[I] )
```

- The following example calculates the count of the cell located at row 4, column D:

```
Count(cell[D,4])
```

- The following example calculates the count of aggregate row 3 in grid 5:

```
Count(grid1.row[3])
```

CountA

CountA is a mathematical function that returns the number of values in a group of numeric values, rows, columns, or cells. CountA includes #MISSING and #ERROR cells when obtaining the count only for rows or columns that are not suppressed. The function syntax:

```
CountA(arguments)
```

where *arguments* is one or more of these arguments:

Table B-10 Arguments for the CountA Function

Argument	Description
numeric	A numeric value. For example, `CountA(10,20,30,50)` returns the value 4. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `CountA(GridName.GridElement[segment(range)]).property`
function	An embedded function

Examples:

- In the following example, if a grid has 4 rows with the values 10, 20, 30, and, #ERROR. The following formula in the fifth row returns the count of four rows:

```
CountA([1:4])
```

- The following example returns the count of four rows:

```
CountA(row[1], row[6], row[8] row[where data yields #ERROR])
```


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Difference

Difference is a mathematical function that returns the absolute value of the difference of a numeric value, row, or column subtracted from another numeric value, row, or column. This is also known as the variance. The function syntax:

```
Difference(arg1, arg2)
```

where *arg2* is subtracted from *arg1* and are one or more of the following arguments:

Table B-11 Arguments for the Difference Function

Argument	Description
numeric	A numeric value. For example, Difference (3, 5) returns the absolute value 2. Numeric values can include decimals and negative values.
row, column, or reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property` The following example returns the difference of two rows in form grid1: Difference(grid1.row[1], grid1.row[6])
function	An embedded function

> **Note**
>
> The Difference function returns the absolute value of arg2 subtracted from arg1, whereas the minus sign in subtraction negates a number.

Examples:

- The following example returns the absolute value of 8:

  ```
  Difference(3, -5)
  ```

- The following example calculates the difference of two aggregate columns:

  ```
  Difference( column[E], column[G] )
  ```

> **Note**
>
> You can type the text label "Difference" or "Variance."

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Eval

Eval is a mathematical function that evaluates an expression. You use Eval as an embedded function argument to consolidate multiple expressions into one expression. The function syntax:

```
Eval(expression)
```

where *expression* is one or more of these arguments:

Table B-12 Arguments for the Eval Function

Argument	Description
numeric	A numeric value. Numeric values can include decimals and negative values.
row, column, or reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property`
function	An embedded function
operators	Use any of the supported arithmetic operators (+, -, *, /, ^, %, and so on).

Example:

The following example divides row 1 by row 2 and then rounds the data to four places:

```
Round(Eval([1]/[2]),4)
```

IfThen, If

IfThen is a conditional function that returns a value when the condition equals True, and another value when the condition equals False.

The function syntax:

```
IfThen(Condition, TrueParameter, FalseParameter)
```

- *Condition* is a logical expression that evaluates to true or false. Full conditional logic can be used as well as complex Boolean operators (And, Not, and Or). A *condition* can also test for #MISSING and #ERROR values. See the following table for a list of valid conditional operators.
- *TrueParameter* and *FalseParameter* are any valid expression that are evaluated based on the outcome of the condition.

The following table describes the conditional operators that are fully supported. Alternate syntax is listed wherever it's supported by a conditional operator.

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Table B-13 Conditional Operators

Conditional Operator	Syntax	Logic
Equal To	*expression* = *expression*	Tests whether the left expression is equal to the right expression. The routine that evaluates the condition doesn't consider any rounding. If rounding is required, use the Round function. Example: `1= 4` Returns false
Greater Than	*expression* > *expression*	Tests whether the left expression is greater than the right expression. Example: `1 > 4` Returns false
Greater Than or Equal To	*expression* >= *expression*	Tests whether the left expression is greater than or equal to the right expression. The correct syntax is ">=". The syntax "=>" isn't supported. Example: `1 >= 4` Returns false
Less Than	*expression* < *expression*	Tests whether the left expression is less than the right expression. Example: `1 < 4` Returns true
Less Than or Equal To	*expression* <= *expression*	Tests whether the left expression is less than or equal to the right expression. The correct syntax is "<=". The syntax "=<" isn't supported. Example: `1 <= 4` Returns true

Not Equal To	*expression* <> *expression*
	expression != *expression*
		Tests whether the left expression isn't equal to the right expression.
		The routine that evaluates the condition doesn't consider any rounding. If rounding is required, use the Round function.
		Examples:
		`1 <> 4`
		Returns true
		`1 != 4`
		Returns true

IsMissing	`IsMissing (`*reference*`)` `IsMiss (`*reference*`)`	Tests whether the reference contains a #MISSING result. If the reference is an expanded row or column, then all resulting cells must be #MISSING in order for the condition to be true. Example: `IsMissing([1])` Returns true if row 1 has a #MISSING value.

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Table B-13 (Cont.) Conditional Operators

Conditional Operator	Syntax	Logic
IsError	`IsError (`*reference*`)` `IsErr (`*reference*`)`	Tests whether the reference contains an #ERROR result. If the reference is an expanded row or column, all resulting cells must be #ERROR in order for the condition to be true. Only formula rows and columns can result in #ERROR. Example: `IsError([2])` Returns true if row 2 has a #ERROR value.

IsNonNumeric	`IsNN (`*reference*`)`
	`IsNonNumerid (`*reference*`)`
	`IfNN (`*reference*`)`
	`IfNonNumber (`*reference*`)`
		Tests whether the reference contains a #MISSING or #ERROR results.
		If the reference is an expanded row or column, all resulting cells must be #MISSING and/or #ERROR in order for the condition to be true.
		Example:
		`IsNN([3])`
		Returns true if row 3 has a #MISSING or #ERROR value.

Parenthesis	`(`*condition*`)`	Used to group a condition. Used mostly for visual clarity. Example: `(1 > 4)` Returns false

Table B-14 Conditional Operators

Complex Conditions	Syntax	Logic
And	`(`*condition* `AND` *condition*`)` `(`*condition* `&` *condition)*	Complex condition used to compare two conditions. Returns true if all conditions result in true. Example: `(1 > 4 AND 5 > 2)` Returns false
Not	`NOT (`*condition*`)` `! (`*condition*`)`	Used to negate the result by reversing the result of the condition. Example: `Not (1 > 4)` Returns true
Or	`(`*condition* `OR` *condition*`)` `(`*condition* `|` *condition*`)`	Complex condition used to compare two conditions. Returns true if any of the conditions result in true. Example: `(1 > 4 OR 5 > 2)` Returns true

Notes on Conditions

- *Expression*: Any valid formula expression. The expression can be any combination of a constant (integer or real number), a reference, or another function.
- *Reference*: Any valid reference; thus the IFNN reference property can be used as part of the reference.

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- *Condition*: Any valid condition applied to the complex conditions And, Not, and Or. Those operators can have embedded conditions.

  > **Note**
  >
  > And, Not, and Or operators require surrounding parentheses.

- When any `expression` within the condition returns an #ERROR or #MISSING value, the If function returns #MISSING or #ERROR. This doesn't apply when you use the IsMissing, IsError, or IsNonNumeric conditions.

Complex Conditions

Complex conditions And, Or, and Not are fully supported. However, they must be surrounded by parentheses.

Valid example:

```
If ( ([A] > [B] and [A] > 1000), [A], [B])
```

Invalid example:

```
If ( [A] > [B] and [A] > 1000, [A], [B])
```

Max

Max is a mathematical function that returns the maximum value in a group of numeric values, rows, columns, or cells. The function syntax:

```
Max (arguments)
```

where *arguments* is one or more of these arguments:

Table B-15 Arguments for the Max Function

Argument	Description
numeric	A numeric value. For example, Max (10, 20, 30) returns the value 30. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property.`
function	An embedded function

Examples:

- The following example returns the maximum value in rows 1, 6, and 8:

  ```
  Max(row[1], row[6], row[8])
  ```


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- The following example calculates the maximum of the sums of aggregate rows:

```
Max(row[1].sum, row[2].sum, row[3].sum)
```

Min

Min is a mathematical function that returns the minimum value in a group of numeric values, rows, columns, or cells. The function syntax:

```
Min (arguments)
```

where *arguments* is one or more of these arguments:

Table B-16 Arguments for the Min Function

Argument	Description
numeric	A numeric value. For example, Min (10, 20, 30) returns the value 10. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property`
function	An embedded function

Examples:

- The following example returns the minimum value in rows 1, 6, and 8:

```
Min (row[1], row[6], row[8])
```

- The following example calculates the minimum of the sums of aggregate rows:

```
Min(row[1].sum, row[2].sum, row[3].sum)
```

Mod

Mod is a mathematical function that returns the remainder, or modulus, from a division. The function syntax:

```
Mod (arg1, arg2)
```

where *arg2* is the divisor and *arg1* and *arg2* are one of these arguments:

Table B-17 Arguments for the Mod Function

Argument	Description
numeric	A numeric value. For example, Mod (6, 4) returns the value 2. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property`

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Table B-17 (Cont.) Arguments for the Mod Function

Argument	Description
function	An embedded function

Example:

The following example divides 10 by 5 and returns the remainder of 0:

```
Mod (10,5) = 0
```

PercentOfTotal

PercentOfTotal is a mathematical function that returns the result of a numeric value, row, column, or cell divided by another numeric value, row, column, or cell which is multiplied by 100. The function syntax:

```
PercentOfTotal (arg1, arg2)
```

- where *arg1* is a component of the running total (*arg2*). Normally, this is a row or column reference.
- where *arg2* is the running total relative to *arg1.* Normally this is a cell reference containing the grand total.
- *arg1* is divided by *arg2*, with the result multiplied by 100. *arg1* and *arg2* are one or more of these arguments:

Table B-18 Arguments for the PercentOfTotal Function

Argument	Description
numeric	A numeric value. For example, PercentofTotal (100, 20) returns the value 500. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].Property`
function	An embedded function

> **Note**
>
> This function requires two arguments.

Examples:

- The following example returns the value of 5 percent.

  ```
  PercentofTotal(20, 400)
  ```


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- The following example divides the value of each cell in column A by the Total Mkt value in cell A5, multiplies the result by 100, and displays the resulting PercentOfTotal in column B. The formula:

  ```
  PercentOfTotal ([A], [A,5]),
  ```

Results for the PercentOfTotal Function

Using the above example, the following table shows the PercentOfTotal results in column B:

		A	B
1		Sales	% Total
2	Mkt1	60	**20%**
3	Mkt2	120	**40%**
4	Mkt3	120	**40%**
5	Total Mkt	300	**100%**

Tip

You enter the formula by clicking on the header for Column B and using the formula bar.

Pi

Pi is a mathematical function that returns the number 3.14159265358979, the mathematical constant, accurate to 15 digits. Pi is the ratio of the circumference of a circle to its diameter. The function syntax:

```
PI()
```

Example:

The following example returns row 3 divided by the product of Pi and 2:

```
row[3] / (PI() * 2)
```

Product

Product is a mathematical function that multiplies all numbers or references and returns the product. The function syntax:

```
Product(arguments)
```

where *arguments* is one or more of these arguments:

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Table B-19 Arguments for the Product Function

Argument	Description
numeric	A numeric value. For example, Product(2, 20) returns the value 40. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].property`
function	An embedded function

Example:

The following example returns 40:

```
Product(2,20)
```

Random

Random is a mathematical function that returns a random number between 0.0 and 1.0. The function syntax:

```
Random()
```

Example:

The following example returns a random number between 0.0 and 1.0 and multiplies it by 1000:

```
Random() * 1000
```

Round

Round is a mathematical function that rounds a number up or down by the specified digits. The function syntax:

Round (*arg1*, *integer*)

where *arg1* is one or more of these arguments:

Table B-20 Arguments for the Round Function

Argument	Description
numeric	A numeric value. For example, Round(81.3987,3) returns the value 81.399. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].property`
function	An embedded function

Integer specifies the number of digits to which you want to round the number:

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- If *integer* is greater than zero, the number rounds to the specified number of decimal places.
- If *integer* is zero, the number rounds to the nearest integer.
- If *integer* is less than zero, the number is rounded to the left of the decimal point.

Examples:

The following example rounds to three decimals:

```
Round(3594.5567,3) = 3594.557
```

The following example rounds to the nearest integer:

```
Round(3594.5567,0) = 3595
```

The following example rounds to the thousands. This is also known as scaling:

```
Round(3594.5567,-3) = 4000
```

Sqrt

Sqrt is a mathematical function that returns the square root of a numeric value, row, column, or cell. The syntax for the Sqrt function:

```
Sqrt(argument)
```

where *argument* is one of these arguments:

Table B-21 Arguments for the Sqrt Function

Argument	Description
numeric	A numeric value. For example, Sqrt(100) returns the value 10. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].property`
function	An embedded function

Example

The following example returns the value of 4:

```
Sqrt(16)
```

Sum

Sum is a mathematical function that returns the summation of a group of numeric values, rows, columns, or cells.

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The syntax for the Sum function:

```
Sum (arguments)
```

where *arguments* is one or more of these arguments:

Table B-22 Arguments for the Sum Function

Argument	Description
numeric	A numeric value. For example, Sum(10, 20, 30) returns the value 60. Numeric values can include decimals and negative values.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].property`
function	An embedded function

Examples:

- The following example returns the value 30:

  ```
  sum(10,20)
  ```

- The following example returns the sum of three rows:

  ```
  sum(row[1],row[6],row[8])
  ```

- The following example calculates the sum of three aggregate columns:

  ```
  sum(column[E], column[G], column[I])
  ```

When a formula row includes `IDESC`, the sum includes all of the parents and their descendants.

For example, a form is created with the following rows (and each member has children):

```
IDESC("Mem1"), IDESC("Mem2"), IDESC("Mem3"), IDESC("Mem4")
```

If a formula row is added with the following formula:

```
SUM(ROW[1],ROW[2],ROW[3],ROW[4])
```

When the form is opened for data entry, the formula row will return the sum of all the parents and their children.

Truncate/Trunc

Truncate is a mathematical function that removes the specified number of digits from numeric values.

Syntax:

```
Trunc (arg1, integer)
```


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- where *arg1* is one of these arguments:

Table B-23 Arguments for the Truncate/Trunc Function

Argument	Description
numeric	A numeric value; for example, 234.567.
row, column, or cell reference	A pointer to a row, column, or cell within a grid. References can be specified in several ways. The reference syntax: `GridName.GridElement[segment(range)].property`
function	An embedded function

- where *integer* specifies the number of digits you want to remove:
  - A positive *integer* determines the number of significant digits that remain to the right of the decimal point.
  - A zero (0) *integer* returns the integer located to the left of the decimal point.
  - A negative *integer* indicates the number of significant digits that are truncated to the left of the decimal point.

Examples:

The following statement uses a positive integer of 2. The first two digits to the right of the decimal point remain, and the following digit is removed:

```
Trunc(234.567, 2) = 234.56
```

The following statement uses a zero (0) integer. All digits to the right of the decimal point are removed:

```
Trunc(234.567, 0) = 234
```

The following statement uses a negative integer of -2. All digits to the right of the decimal point are removed and the last two digits of the integer are truncated.

```
Trunc(234.567, -2) = 200
```

> **Note**
>
> Formatting previously applied to a cell, column, or row is maintained when you use the Trunc function. The following example shows the results of a Trunc function where the cell value was previously formatted to display three decimal places: `Trunc(234.567, 0) = 234.000`

Variance/Var

Variance is a financial function that evaluates the difference between the specified values based on account type for the current account. For example, for Expense or Liability accounts, a positive result represents a decrease, so the result appears as a negative number. You can use this function with these UDA account types: Asset, Liability, Equity, Revenue, and Expense.

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Syntax:

```
Var (reference1, reference2)
```

where *reference1* and *reference2* are references to a row, column, or cell that correspond to members of the same Account dimension whose variance results are to be calculated.

Expected Results

The following table describes the expected results when using the Variance function with Accounts.

Table B-24 Expected Results when Using the Variance Function

Column A	Column B	Var ([A] , [B])=0	Var ([A] , [B])>0	Var ([A] , [B])<0
Asset	Asset	0	Returns a positive value	Returns a negative value
Liability	Liability	0	Returns a positive value	Returns a negative value
Equity	Equity	0	Returns a positive value	Returns a negative value
Revenue	Revenue	0	Returns a positive value	Returns a negative value
Expense	Expense	0	Returns a negative value	Returns a positive value

Variance Behavior

- The Variance function expects comparison of the same account type. When you compare two different account types, like Sales & Expense, the Variance function performs the straight math without applying the logic of the account type. For example:

  Table B-25 Results for the Variance Function When Comparing Two Different Account Types

Sales	Expense	Result
-400	100	-500

- When the Variance function is applied to a dimension that isn't tagged as an Accounts type, an #ERROR results at runtime.
- #MISSING is treated as zero (0), unless specified differently using the ifnonnumber property.

Examples

The Variance function accepts cell, column, or row references only.

Table B-26 Syntax Examples for the Variance Function

Syntax	Example
Sample syntax referencing a column:	`Var ([A], [B])`

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Table B-26 (Cont.) Syntax Examples for the Variance Function

Syntax	Example
Sample syntax referencing a row:	`Var ([3], [4])`
Sample syntax referencing a cell:	`Var (Cell [3,A], [3,B])`

In this example, the variance between column A **(Actual)** and column B (**Budget)** is calculated as:

```
Var([A],[B])
```

This example produces the following report:

```
                                Year   Product Market

                          Actual   Budget Variance

                        ======== ======== ========

Sales(Income)           400,855  373,080    27,775

COGS (Expense)          179,336  158,940   -20,396
```

VariancePercent/VarPer

VariancePercent is a financial function that evaluates the difference, in percent, between the specified values based on account type for the current account. For example, for an Income, Flow, Asset, or Balance account, a positive result represents an increase, so the result appears as a positive number. For Expense or Liability accounts, a positive result represents a decrease, so the result appears as a negative number.

Syntax:

`VarPer (`*`reference1, reference2`*`)`

where *reference1* and *reference2* are references to a row, column, or cell that correspond to members of the same Account dimension whose VariancePercent results are to be calculated.

Expected Results

The following table describes the expected results when using the VariancePercent function with Accounts tagged with the below UDAs.

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Table B-27 Expected Results when Using the VariancePercent Function

Col A	Col B	VarPer ([A] , [B])=0	VarPer ([A] , [B])>0	VaPer ([A] , [B])<0
Asset	Asset	0	Returns a positive value	Returns a negative value
Liability	Liability	0	Returns a negative value	Returns a positive value
Equity	Equity	0	Returns a positive value	Returns a negative value
Revenue	Revenue	0	Returns a positive value	Returns a negative value
Expense	Expense	0	Returns a negative value	Returns a positive value

VariancePercent Behavior

- The VariancePercent function expects comparison of the same account type. When you compare two account types, like Sales & Expense, the VariancePercent function performs the straight math without applying the logic of the account type. For example:

  Table B-28 Expected Results of the VariancePercent Function when Two Account Types are Compared

Sales	Expense	Result
-400	100	-5.

- When the VariancePercent function is applied to a dimension that isn't of type Accounts, an #ERROR results at runtime.
- #MISSING is treated as zero (0), unless specified differently using ifnonnumber property.

Examples

The VariancePercent function accepts, cell, column, or row references only.

Table B-29 Syntax Examples of the VariancePercent Function

Syntax	Example
Sample syntax referencing a column	`VarPer ([A], [B])`
Sample syntax referencing a row	`VarPer ([3], [4])`
Sample syntax referencing a cell	`VarPer (Cell [3,A], [3,B])`

In this example, the VariancePercent between column A **(Actual)** and column B (**Budget)** is calculated as follows:

```
VarPer([A],[B])
```

Example Result of the VariancePercent Function when Comparing Columns

This example produces the following report:

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	Year Actual	Product Budget	Market VariancePercent
Sales (Income)	400,855	373,080	7%
COGS (Expense)	179,336	158,940	-13%

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C

Using Smart View to Manage Applications

Understand how to manage applications in Smart View, including installation, controlling the display, downloading templates, and creating, updating, and deleting applications. Also learn about the Planning Admin Extension and Office AutoCorrect.

Related Topics

- About Managing Applications in Smart View
- Installing Smart View and the Admin Extension for Application Management
- Controlling the Display of Application Management Options in Smart View
- Downloading the Application Templates
- Creating an Application
- Working with Artifacts in the Application Template
- Updating an Application in Smart View
- Deleting an Application
- Planning Admin Extension and Office AutoCorrect

About Managing Applications in Smart View

Applies to: FreeForm, Planning

Service Administrators can use Excel to create and manage Planning, Planning Modules, and FreeForm applications in Oracle Smart View for Office and Oracle Smart View for Office (Mac and Browser).

> **Note**
>
> You must have Service Administrator privileges to create and manage applications in Smart View and Smart View (Mac and Browser).

Using a downloadable Excel template workbook, you can quickly create, model, and manage applications in Smart View. By modifying the worksheets within the application template workbook, you can define application properties, dimension and member properties, security (access permissions), and substitution variables to fit your needs. You can also load data from the template workbook.

Once the application is created, it's available to users from Oracle Smart View for Office, Oracle Smart View for Office (Mac and Browser), and the web application. Service Administrators can further update and refresh the application in Smart View as needed. You can also delete the application from Smart View.

Application management in Smart View is a quick and easy way to get an application up and running in your business process environment.

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To begin, see the topics in Installing Smart View and the Admin Extension for Application Management.

After installation, get started by downloading and then editing the application template workbook available in Smart View. Or, download the application template zip file from **Downloads** menu in your business process. The template zip file contains starter workbooks for Planning, Planning Modules, and FreeForm applications, and a sample Planning application workbook. See Downloading the Application Templates.

> **Note**
>
> Starting in 22.03, a single Excel template workbook, `PlanningApplicationTemplate_FreeForm.xlsx`, allows you to create a multi-cube application (any combination of up to 12 ASO and BSO cubes) for your FreeForm business process.
>
> The multi-cube FreeForm template replaces the two previously available templates that allowed you to create a single-cube FreeForm ASO or BSO application.

Then continue with Creating an Application, using the information in Working with Artifacts in the Application Template to modify the template according to your requirements.

> **Note**
>
> For information on creating applications in the web, see the documentation for your business process:
>
> - Creating a Planning Application in *Administering Planning*
> - Creating a FreeForm App in *Administering FreeForm*
> - Setting Up Your Application in *Administering Planning Modules*

Videos

Your Goal	Watch This Video
Learn about creating an application in Smart View. The video shows the process of creating a Planning application using Oracle Smart View for Office, but the concepts also apply to Oracle Smart View for Office (Mac and Browser), as well as the FreeForm and Planning Modules business processes.	Overview: Creating and Managing Planning Cloud Applications Using Smart View

Installing Smart View and the Admin Extension for Application Management

Most procedures described in this appendix are performed within Smart View, with the exception of Controlling the Display of Application Management Options in Smart View and Downloading the Application Template Zip File from the Web Interface.

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> **Note**
>
> Application management is supported for these Oracle Fusion Cloud Enterprise Performance Management business processes: Planning, Planning Modules, and FreeForm.

Before you begin, ensure the following depending on whether you plan to use Windows-based Oracle Smart View for Office, Oracle Smart View for Office (Mac and Browser), or both:

- **Windows-based Smart View:** Install Smart View.

  The Admin Extension is not required to use the application management features in Windows-based Smart View.

  See Installing Windows-based Oracle Smart View for Office

- **Smart View (Mac and Browser):** Deploy Smart View (Mac and Browser) with the Dimension Editor options selected.

  You must select the Dimension Editor options in order to work with the application templates in Smart View (Mac and Browser).

  See Deploying Oracle Smart View for Office (Mac and Browser) with the Admin Extension

Installing Windows-based Oracle Smart View for Office

To use the Windows client-based version of Smart View, see Downloading and Installing Clients in *Getting Started Guide for Administrators* for instructions on installing Smart View.

> **Note**
>
> The Admin Extension is not required with Windows-based Smart View when working with application management.

After installation, get started by downloading and then editing the application template workbook available in Smart View or from the template download zip file in the web.

- By default, you can download the Planning application workbook template directly from Smart View.
- Or, download the application template zip file from the Downloads page of the Planning, Planning Modules, or FreeForm business process. The template zip file contains starter application template workbooks for these business processes, along with a sample Planning application workbook.

Deploying Oracle Smart View for Office (Mac and Browser) with the Admin Extension

To use the Mac- or browser-based version of Smart View, you must deploy a Smart View (Mac and Browser) manifest file with the Dimension Editor options enabled.

See the following topics in *Deploying and Administering Oracle Smart View for Office (Mac and Browser)* for instructions on deploying Smart View (Mac and Browser) with the Dimension Editor options enabled:

- Creating and Saving the Manifest File
- Enabling the Admin Extension

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- Sideloading and Logging In or Deploying the Manifest File to Office 365 Users

Controlling the Display of Application Management Options in Smart View

When a new application is created in the web, in Oracle Smart View for Office, or in Oracle Smart View for Office (Mac and Browser), all application management options are displayed by default to Service Administrators in either version of Smart View.

You can choose to hide or display two of the options, **Update Application** and **Delete Application**, using the **Suppress Application Management Options in Smart View** option in Planning, Planning Modules, or FreeForm.

This setting, and the display of the application management options, applies to Service Administrators only.

To control the display of the **Update Application** and **Delete Application** options in Smart View:

1. From the Home page, click **Application**, and then click **Settings**.
2. Choose a setting for **Suppress Application Management Options in Smart View**:
   - **Yes**: Application options are not displayed in Smart View.
   - **No**: Default. Application options are displayed in Smart View.
3. Click **Save**.

> **Note**
>
> Any time a new application is created, whether in the web, in Oracle Smart View for Office, or in Oracle Smart View for Office (Mac and Browser), the **Suppress Application Management Options in Smart View** setting defaults to **No** and the **Update Application** and **Delete Application** options will automatically be displayed. To hide the options, perform the above procedure.

Downloading the Application Templates

Related Topics

- Downloading the Template in Smart View
- Downloading the Application Template Zip File from the Web Interface

Downloading the Template in Smart View

When you're logged into Smart View as a Service Administrator, the Download Template command is available in the Smart View Panel on Windows and in the Smart View Home panel on the Mac and in the browser.

The default template that you download can be used immediately to begin creating a Planning application. Or, you can modify the template to create your own custom application.

To download the default application template from Smart View:

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1. Log in to the applicable data source in Smart View.
2. Perform an action:
   - In the Smart View Panel (Windows), at the bottom of the panel, click the **Download Template** command.
   - In the Smart View Home panel (Mac and Browser), click , and select **Download Template** from the drop-down menu.

     If prompted, allow the file to be created in your cloud repository.

   The default application template, `PlanningApplicationTemplate.xlsx`, is opened. The default template contains the basic elements required to begin creating a standard Planning application.

   You can edit the default template to create a custom application, as described in Working with Artifacts in the Application Template.

   To download the Planning Modules, FreeForm, or sample Planning templates, download the application template zip file from the web, described in Downloading the Application Template Zip File from the Web Interface.

Downloading the Application Template Zip File from the Web Interface

When you're logged into Planning, Planning Modules, or FreeForm as Service Administrator, you can download a zip file containing these application template workbook files:

- `PlanningApplicationTemplate.xlsx`: Creates a Planning application.
- `PlanningApplicationTemplate_epbcs.xlsx`: Creates a Planning Modules application.
- `PlanningApplicationTemplate_FreeForm.xlsx`: Creates a FreeForm application.
- `PlanningApplicationTemplate_sample.xlsx`: Creates the Planning sample application, Sample.

> **Note**
>
> Starting in 22.03, a single Excel template workbook, `PlanningApplicationTemplate_FreeForm.xlsx`, allows you to create a multi-cube application (any combination of up to 12 ASO and BSO cubes) for your FreeForm business process.
>
> The multi-cube FreeForm template replaces the two previously-available templates that allowed you to create a single-cube FreeForm ASO or BSO application.

To download the application template zip file:

1. From the computer where you want to install or deploy the client software, access a business process instance.
2. On the Planning, Planning Modules, or FreeForm Home page, access **Settings and Actions** by clicking your user name at the top right corner of the screen.
3. Select **Downloads**.
4. In the **Downloads** page, Planning Extensions section, click **Download for Application Templates**.
5. Save the zip file locally or to a server location that you can access, and unzip the contents.

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You should now have these files saved:

- `PlanningApplicationTemplate.xlsx`
- `PlanningApplicationTemplate_epbcs.xlsx`
- `PlanningApplicationTemplate_FreeForm.xlsx`
- `PlanningApplicationTemplate_sample.xlsx`

6. Open the template file to work with in Excel and proceed with Creating an Application, using the information in Working with Artifacts in the Application Template to modify the template according to your requirements.

Creating an Application

From Oracle Smart View for Office and Oracle Smart View for Office (Mac and Browser), Service Administrators can create a new application based on either the downloaded template with no modifications or a modified template. In either case, the template must adhere to template rules and guidelines.

When you create an application, you can:

- Define cubes
- Define dimensions
- Add a limited amount of data
- Define security and access permissions
- Define substitution variables

Only Service Administrators can create an application and only when no application exists in the cloud business process. Once an application is created, the option is no longer available in the Smart View Panel or Smart View Home panel.

Before you begin, you should launch Excel and, from Smart View, log in to the applicable data source.

To create a new application in Smart View:

1. If you have not already done so, perform one of the following actions:
   - Download the application template for Planning as described in Downloading the Template in Smart View.

     Downloading the application template simply opens the template in Smart View.
   - Download and unzip the application template zip file as described in Downloading the Application Template Zip File from the Web Interface, then open one of the template files.
   - Open a saved template file.

   An Excel application template workbook must be open when creating an application.
2. **Optional:** Edit and then save the application template as needed.

   See Working with Artifacts in the Application Template for information on modifying the worksheets in the application template workbook.
3. With the application template workbook open in Smart View, select **Create New Application**:
   - Oracle Smart View for Office (Windows):

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In the tree in the Smart View Panel, select the server name to access the **Create New Application** command at the bottom of the panel. You can also right-click the server name to access the **Create New Application** command.

- Oracle Smart View for Office (Mac and Browser):
  In the Smart View Home panel, click the **Actions** button, , and then select **Create New Application** from the drop-down menu.

The application creation status appears in the lower left corner of Excel.

4. When the application creation process is complete, verify that the application appears in the Smart View Panel (Windows) or Smart View Home panel (Mac and Browser).
   - Oracle Smart View for Office (Windows):

     The application name is displayed in the tree in the Smart View Panel. Expand the application name to view folders and cubes.
   - Oracle Smart View for Office (Mac and Browser):

     In the Smart View Home panel, the application folders and cubes are displayed. The application name is not displayed.

> **Note**
>
> After application creation is complete, artifacts are displayed when the Home panel is reloaded. To reload the Home panel, right-click on the **Actions** button, , and select the **Reload frame** option in the active workbook.

Note that only one application per cloud business process is allowed. Once an application is created, the **Create New Application** command no longer appears in the Application Management popup window.

The application is now available to the applicable users.

Next steps include updating the application template workbook. You can also delete the application and create a new one. See:

- Working with Artifacts in the Application Template
- Updating an Application in Smart View
- Deleting an Application

Working with Artifacts in the Application Template

Related Topics

- About Working with Artifacts in the Application Template
- Application Definition
- Dimension Definition
- Attribute Dimension Definition
- Data Definition
- Substitution Variable Definition
- Security Definition
- Advanced Settings Definition

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About Working with Artifacts in the Application Template

Editing the application template involves editing Excel worksheets in the application workbook.

The application template workbook contains sheets for defining application properties, cubes, dimension and member properties, substitution variables, and security. You can also load data using the template.

Application Definition

In the application template workbook, the Definition worksheet defines your application and cubes. Using the application definition worksheet provided in the template zip file as your starting point, you can edit the worksheet provided to suit your requirements. You can also use the Sample application template file as your guideline for a standard application or adapt it to create a Planning Modules or FreeForm application.

Follow this workflow for setting up a data worksheet in your application template workbook:

1. Define the Worksheet Name
2. Define the Worksheet Type
3. Define Application Properties
4. Define Cubes

Define the Worksheet Name

The worksheet name is in the Excel worksheet tab. The first tab is in the workbook is for the application definition worksheet. Ensure that the application definition worksheet is named, `Definition`. For example:

Define the Worksheet Type

In cell A1 in the application worksheet, type `Application Definition` for the worksheet type. In Figure 1, cell A1 contains the worksheet type, `Application Definition`.

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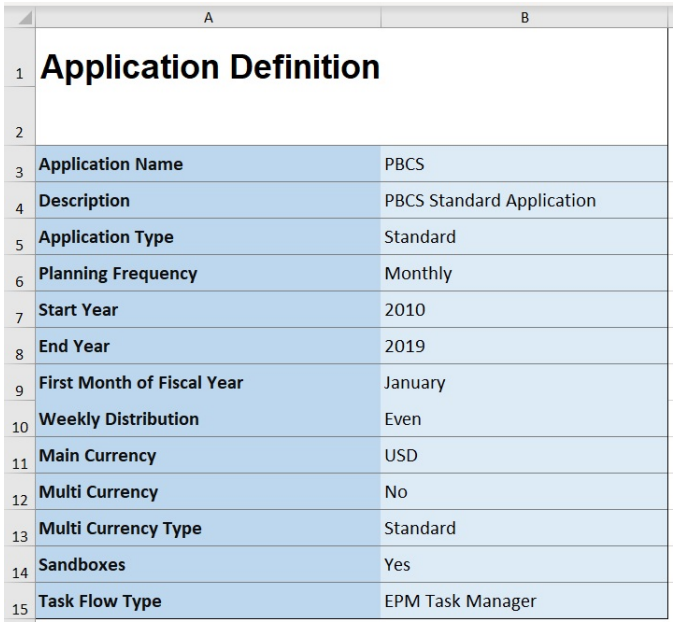
Figure C-1 Worksheet Type and Application Properties Shown in Application Definition Worksheet



	A	B
1	**Application Definition**	
2		
3	**Application Name**	PBCS
4	**Description**	PBCS Standard Application
5	**Application Type**	Standard
6	**Planning Frequency**	Monthly
7	**Start Year**	2010
8	**End Year**	2019
9	**First Month of Fiscal Year**	January
10	**Weekly Distribution**	Even
11	**Main Currency**	USD
12	**Multi Currency**	No
13	**Multi Currency Type**	Standard
14	**Sandboxes**	Yes
15	**Task Flow Type**	EPM Task Manager

Define Application Properties

Application properties are the basic information needed to create an application. In general, only the Application Name, Application Description, and Application Type are required properties. For all other properties, if not specified, the default values will be provided during application creation.

> **Note**
>
> You can create the FreeForm app with only the application properties and no dimensions, attribute dimensions, security (access permissions), or substitution variables in the workbook.

Table 1 shows the supported properties, their default values, and valid values.

Table C-1 Application Properties

Property	Required	Value Type	Default Value	Valid Values	Comments
Application Name	Yes	Text			Name of Planning application
Description	Yes	Text			Application description

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Table C-1 (Cont.) Application Properties

Property	Required	Value Type	Default Value	Valid Values	Comments
Application Type	Yes	Text	Standard	**Standard**: An advanced application with two cubes and the option to add more later **Enterprise**: Also referred to as EPBCS Advanced **FreeForm**: A FreeForm application	Similar to choices on web interface
Planning Frequency	No	Text	Monthly	Monthly Weekly Quarterly Custom	
Weekly Distribution	No	Text	445	Even 445 454 544	Applicable for Monthly Planning frequency
Periods Per Year	No	Numeric	13		Only for Custom Planning Frequency
Prefix	No	Text	TP		Only for Custom Planning Frequency
Start Year	No	Numeric	2010		Year in 4 digits
End Year	No	Numeric	2019		Year in 4 digits Calculated as (*Start Year + Number of Years + 1)*
First Month of Fiscal Year	No	Text	January	Jan through Dec	Only for Monthly Planning Frequency
First Fiscal Year Start Date	No	Date			Only for Quarterly or Custom Planning frequency

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Table C-1 (Cont.) Application Properties

Property	Required	Value Type	Default Value	Valid Values	Comments
Task Flow Type	No	Text	EPM Task Manager	EPM Task Manager Task List	Defaults to EPM Task Manager if entry is missing or typed incorrectly. **Note:** This option does no apply to FreeForm applications.

Define Cubes

In the application worksheet, the same sheet where you define the application properties, list all cubes.

To define cubes in the application worksheet, add a line with the label, `Cubes` after the application definition section. For example, in Table 2, a line with the word `Cubes` is added to line 18 of the worksheet.

In the next row, add the `Name` label to column A, and the `Type` label to column B.

Figure C-2 Cube Definition Section of Application Definition Worksheet

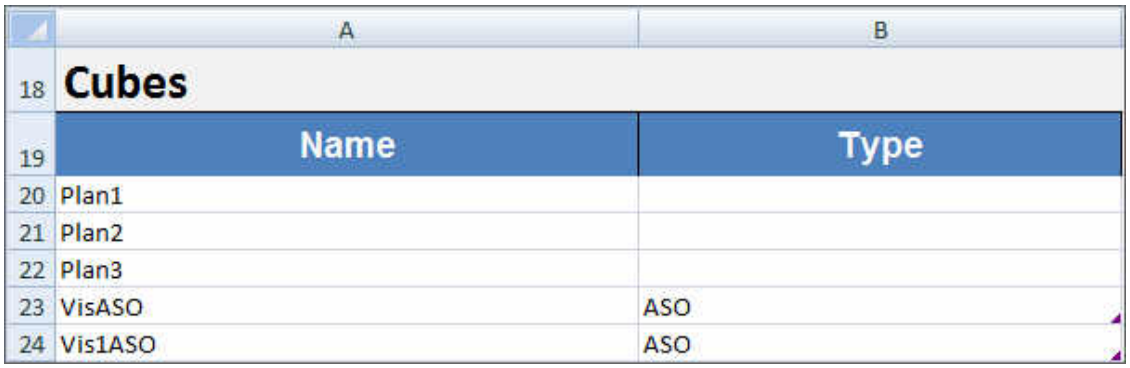

Table 2 shows the properties related to cube definition. Only the cube name is required. For the Type column, use BSO or ASO. Note that if the type is left blank, BSO is used by default.

All application types, Planning, Planning Modules, and FreeForm, support multiple cubes and any combination of BSO and ASO cubes.

Table C-2 Cube Properties

Header Name	Required	Comments
Name	Yes	Name of the cube

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Table C-2 (Cont.) Cube Properties

Header Name	Required	Comments
Type	BSO ASO	BSO is the default. If cell is left blank, BSO is used

Dimension Definition

In the template workbook, create a new worksheet for each dimension in your application. You can use the worksheets provided in the templates, including the Sample application template, as your guide.

Follow this workflow for setting up a dimension worksheet in your application template workbook:

1. Define the Worksheet Name
2. Define the Worksheet Type
3. Define the Dimension Name
4. Define Dimension Properties
5. Define Members

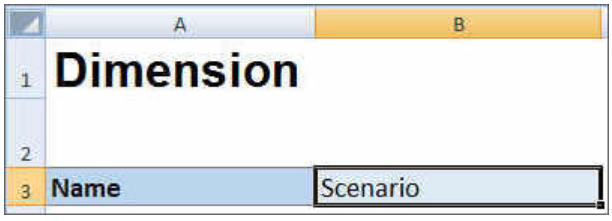
Define the Worksheet Name

Define the worksheet name in the Excel worksheet tab. For a dimension worksheet name, use the dimension name prefixed with "`Dim.`". For example, for the Scenario dimension, you would name the worksheet `Dim.Scenario`. For example:

Define the Worksheet Type

In cell A1 in the dimension worksheet, type `Dimension` for the worksheet type. In Figure 1, cell A1 contains the worksheet type, `Dimension`.

Figure C-3 Worksheet Type and Dimension Name Shown in a Dimension Worksheet of Application Template



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Define the Dimension Name

In the dimension worksheet, type the dimension name. In Figure 1, cell B3 contains the dimension name, `Scenario`.

If the dimension name that you provide doesn't exist, then a new custom type dimension is automatically created when you create the application.

Define Dimension Properties

You can define dimension properties directly on the dimension worksheet. Using the Type property, you can specify user-defined names for the Account, Year, and Entity dimensions and then map them to the Accounts, Year, or Entity dimension types. You can also define any other custom dimensions as you require. The templates in the template zip file includes example Accounts, Year, and custom dimensions and their properties to guide you.

> **Note**
>
> You can create the FreeForm app with no dimensions or any other application properties (such as attribute dimensions, security (access permissions), and substitution variables) in the workbook.

Table C-3 Dimension Properties

Property	Required	Value Type	Default Value	Valid Values	Comments
Name	Yes	Text			Name of the dimension
Type	No	Text		Account Period Entity	If this property is omitted, or no value is supplied, then this dimension is interpreted as a custom dimension.
Description	No	Text			An optional dimension description.
Alias	No	Text	None	The alias for the dimension	Aliases are alternate name for dimension members.
Alias Table	No	Text	No	The alias table	The default alias table for the application.

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Table C-3 (Cont.) Dimension Properties

Property	Required	Value Type	Default Value	Valid Values	Comments
Hierarchy Type	No	Text	Not Set	Not Set Stored Dynamic Multiple	Available for dimensions bound to an aggregate storage cube. Aggregate storage dimensions are automatically enabled to support multiple hierarchies. The first hierarchy in a multiple hierarchy dimension must be Stored.
Density	No	Boolean	Sparse	Dense | Sparse	Sparse dimensions lack data values for the majority of member combinations. Dense dimensions have data values for the majority of member combinations.
Two Pass Calculation	No	Boolean	No	Yes | No	Recalculate values of members based on values of parent members or other members. Available for Account and Entity members with Dynamic Calc or Dynamic Calc and Store properties.

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Table C-3 (Cont.) Dimension Properties

Property	Required	Value Type	Default Value	Valid Values	Comments
Apply Security	No	Boolean	No	Yes | No	Allow security to be set on the dimension members, per the security laid out in the Security worksheet in the application template.
Partial Share	No	Boolean	No	Yes | No	Alternate hierarchies are supported in the Entity dimension. One entity can have multiple parents, and contribute differently to each parent. These members are referred to as partially shared entities, where only a part of input data is shared across all instances of the entities.
Data Storage	No	Text	Store	Dynamic Calc and Store Dynamic Calc Store Never Share Shared Label Only	Data storage properties define where and when consolidations are stored. For example, by default, members are tagged as Store.
Performance Order	No	Numeric	None	A positive or negative number; for example: 1 -1	Evaluation order of dimensions in the application.

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Table C-3 (Cont.) Dimension Properties

Property	Required	Value Type	Default Value	Valid Values	Comments
Evaluation Order	No	Numeric	None	A positive or negative number; for example: 1 -1	The order of your dimensions determines how your data calculations will perform. Evaluation Order enables you to specify which data type prevails when a data intersection has conflicting data types.
Display Option	No	Text	Member Name or Alias	**Member Name or Alias** to display members or aliases. **Member Name:Alias** displays members on the left and aliases on the right. **Alias:Member Name** displays aliases on the left and members on the right.	Set application default display option.

Define Members

In the dimension worksheet, under the `Members` heading in (cell A5 in the Planning application template), add members and their properties. Your list of members should conform to the format described in Importing and Exporting Data and Metadata. You may add columns only for required properties. Any missing properties will be added based on application, cube, and dimension defaults.

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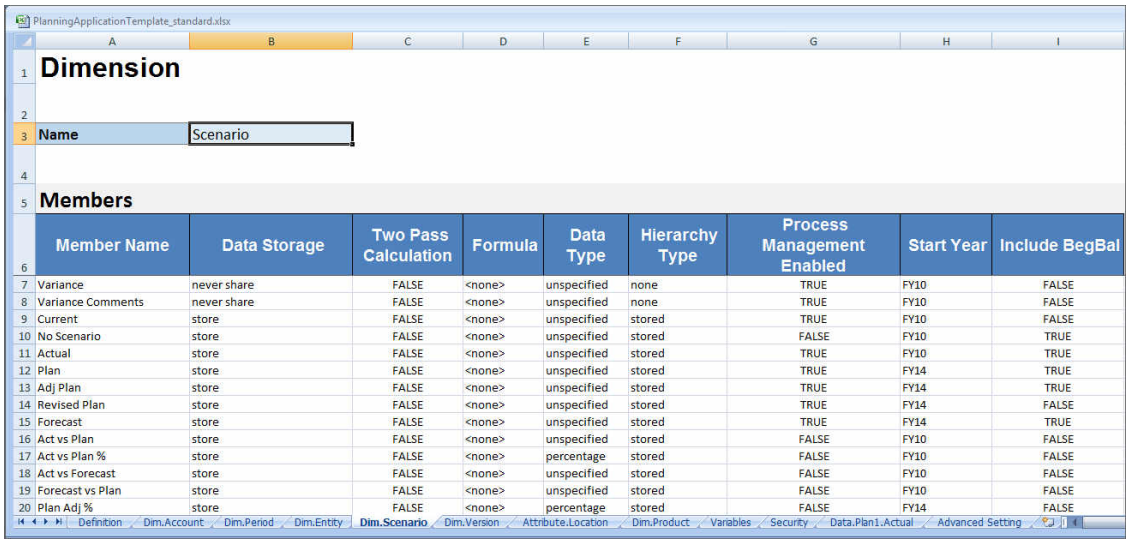
Figure C-4 Example Dimension Worksheet for the Scenario Dimension in a Standard Planning Application



PlanningApplicationTemplate_standard.xlsx

Dimension

Name	Scenario

Members

Member Name	Data Storage	Two Pass Calculation	Formula	Data Type	Hierarchy Type	Process Management Enabled	Start Year	Include BegBal
Variance	never share	FALSE	<none>	unspecified	none	TRUE	FY10	FALSE
Variance Comments	never share	FALSE	<none>	unspecified	none	TRUE	FY10	FALSE
Current	store	FALSE	<none>	unspecified	stored	TRUE	FY10	FALSE
No Scenario	store	FALSE	<none>	unspecified	stored	FALSE	FY10	TRUE
Actual	store	FALSE	<none>	unspecified	stored	TRUE	FY10	TRUE
Plan	store	FALSE	<none>	unspecified	stored	TRUE	FY14	TRUE
Adj Plan	store	FALSE	<none>	unspecified	stored	TRUE	FY14	TRUE
Revised Plan	store	FALSE	<none>	unspecified	stored	TRUE	FY14	FALSE
Forecast	store	FALSE	<none>	unspecified	stored	TRUE	FY14	TRUE
Act vs Plan	store	FALSE	<none>	unspecified	stored	FALSE	FY10	FALSE
Act vs Plan %	store	FALSE	<none>	percentage	stored	FALSE	FY10	FALSE
Act vs Forecast	store	FALSE	<none>	unspecified	stored	FALSE	FY10	FALSE
Forecast vs Plan	store	FALSE	<none>	unspecified	stored	FALSE	FY10	FALSE
Plan Adj %	store	FALSE	<none>	percentage	stored	FALSE	FY14	FALSE

Definition | Dim.Account | Dim.Period | Dim.Entity | Dim.Scenario | Dim.Version | Attribute.Location | Dim.Product | Variables | Security | Data.Plan1.Actual | Advanced Setting

Attribute Dimension Definition

In the template workbook, create a new worksheet for each attribute dimension in your application. You can use the attribute dimension worksheet provided in the template for the Vision application as your guideline.

Follow this workflow for setting up an attribute dimension worksheet in your application template workbook:

1. Define the Worksheet Name
2. Define the Worksheet Type
3. Define the Attribute Dimension Name
4. Define the Attribute Dimension Type
5. Define the Base Dimension
6. Define Attribute Members

Define the Worksheet Name

Define the worksheet name in the Excel worksheet tab. For the worksheet name, use the dimension name prefixed with "`Attribute.`". For example, for the Location attribute dimension, you would name the worksheet `Attribute.Location`. For example:

Define the Worksheet Type

In cell A1 in the attribute dimension worksheet, type `Attribute` for the worksheet type. In Figure 1, cell A1 contains the worksheet type, `Attribute`.

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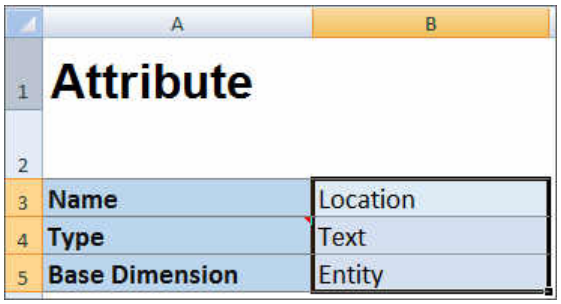
Figure C-5 Worksheet Type and Attribute Dimension Properties Shown in an Attribute Dimension Worksheet of Application Template



Define the Attribute Dimension Name

In the attribute dimension worksheet, type the dimension name. In Figure 1, cell B3 contains the attribute dimension name, `Location`.

Define the Attribute Dimension Type

In the attribute dimension worksheet, type the attribute dimension type. In Figure 1, cell B4 contains the attribute dimension type, `Text`.

Valid attribute dimension types are:

- Text
- Numeric
- Boolean
- Date

Define the Base Dimension

In the attribute dimension worksheet, type the base dimension for the attribute dimension. In Figure 1, cell B5 contains the base dimension, `Entity`.

Define Attribute Members

In the dimension worksheet, under the `Members` heading in cell A5, add attribute members and their properties. Your list of attribute members should conform to the format described in Importing and Exporting Data and Metadata. You may add columns only for required properties. Any missing properties will be added based on application, cube, and dimension defaults.

Data Definition

With the application creation template, you can load up to 1,000 rows of data using the Essbase data format.

In the template workbook, create a worksheet for your application data, using the data worksheet provided in the template for the Sample application as your guideline. Or, edit the worksheet provided in the template.

Follow this workflow for setting up a data worksheet in your application template workbook:

1. Define the Worksheet Name

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2. Define the Worksheet Type
3. Define the Cube Name
4. Define Data

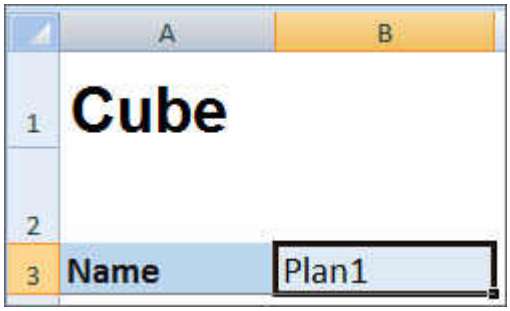
Define the Worksheet Name

Define the worksheet name in the Excel worksheet tab. For the worksheet name, use the cube name where data will be loaded, prefixed with `Data.`. For example, to load data to the Plan1 cube, you would name the worksheet `Data.Plan1` to indicate that you're loading data to Plan1. After the required `Data.` prefix, you may add any other indicators to the name to help you identify the data. For example, to show that you're loading actual data versus historical data to Plan 1, you would name the worksheet `Data.Plan1.Actual`.

Define the Worksheet Type

In cell A1 in the data worksheet, type `Cube` to signify that data in this sheet will be loaded to a cube. In Figure 1, cell A1 contains `Cube` to indicate that the worksheet will contain data for loading to a cube.

Figure C-6 Worksheet Type and Cube Name Shown in Data Worksheet of Application Template



Define the Cube Name

In the data worksheet, type the name of the cube to which to load data. In Figure 1, cell B3 contains the cube name, `Plan1`.

Define Data

Define data in the Essbase columnar format. Use "Dimension" as the header for each member intersection.

Substitution Variable Definition

In the application template workbook, include a worksheet for your substitution variables, using the substitution variable definition worksheet provided in the template for the Sample application as your guideline.

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Substitution variables can be defined for Standard, Enterprise, and FreeForm applications.

Follow this workflow for setting up a substitution variable worksheet in your application template workbook:

1. Define the Worksheet Name
2. Define the Worksheet Type
3. Define the Substitution Variables

Define the Worksheet Name

Define the worksheet name in the Excel worksheet tab. Name the substitution variable definition worksheet, `Variables`. For example:

Define the Worksheet Type

In cell A1 in the data worksheet, type Substitution Variables to signify that data in this sheet will be loaded to a cube. In Figure 1, cell A1 contains `Substitution Variables` to indicate that the worksheet will contain substitution variable information for loading to the specified cubes in the application.

Figure C-7 Worksheet Type and Substitution Variables Shown in Variables Worksheet of Application Template

	A	B	C
1	**Substitution Variables**		
2			
3	**Cube**	**Name**	**Value**
4	All Cubes	CurrYr	FY12
5	Plan3	CurrYr	FY11
6	Plan2	NextYr	FY13
7	All Cubes	CurVersion	Working

Define the Substitution Variables

In the Variables worksheet, using Figure 1 for reference, add information for each substitution variable in the following format:

- In the **Cube** column, enter the name of the cube to which the substitution variable will apply. For example:
  - If a variable applies to all cubes, enter `All Cubes`
  - If a variable applies to a specific cube, enter the cube name; for example, enter `Plan3`

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  - If a variable applies to more than one cube, but not all cubes, list each cube name on its own row individually; for example, if a variable applies to both Plan2 and Plan3 only, add an entry for `Plan2` on one row, and an entry for `Plan3` on another row.
- In the **Name** column, enter the substitution variable name. In Figure 1, the substitution variable names are `CurrYr`, `NextYr`, and `CurVersion`.
- In the **Value** column, enter the value to be used for the substitution variable.

Security Definition

In the application template workbook, include a worksheet for user permissions, using the security definition worksheet provided in the template for the Vision application as your guideline.

Follow this workflow for setting up a user permissions worksheet in your application template workbook:

1. Define the Worksheet Name
2. Define the Worksheet Type
3. Define Security Properties

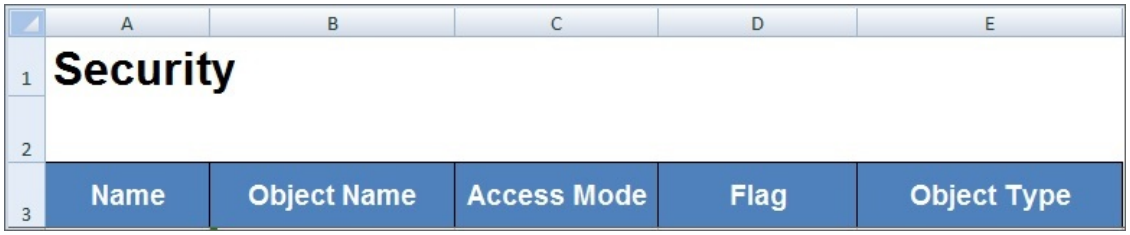
Define the Worksheet Name

Define the worksheet name in the Excel worksheet tab. Name the security worksheet where you'll define user permissions, `Security`. For example:

Define the Worksheet Type

In cell A1 in the data worksheet, type `Security` to signify that data in this sheet will be loaded to a cube. In Figure 1, cell A1 contains `Security` to indicate that the worksheet will list all user access permissions for the application.

Figure C-8 Worksheet Type Shown in Security Worksheet of Application Template



Define Security Properties

Security properties include the user name and other user access information.

Table 1 shows the supported security properties.

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Table C-4 Security Properties

Property Name	Description
Name	Name of the user or group
Object Name	Artifact name
Access Mode	Permission granted. Default is NONE. Valid values: • READ • WRITE • READWRITE • LAUNCH (valid when Object Type is Rule) • NOLAUNCH (valid when Object Type is Rule) • NONE: Default
Flag	Member function to be used while applying the access. Default is MEMBER. Valid values: • MEMBER: Default • CHILDREN • @ICHILDREN • @DESCENDANTS • @IDESCENDANTS
Object Type	Artifact object type. Default is SL_DIMENSION (Dimension/Member). Valid values: • SL_FORM - Form • SL_COMPOSITE - Composite Form, Infolet, or Dashboard • SL_TASKLIST - Tasklist • SL_CALCRULE - Rule • SL_FORMFOLDER - Form Folder • SL_CALCFOLDER - Rule Folder • SL_DIMENSION - Dimension/Member: Default • SL_CALCTEMPLATE - Template • SL_REPORT - Management Report • SL_REPORTSSHOT - Management Report Snapshot

Advanced Settings Definition

In the application template workbook, include a worksheet for advanced settings, using the advanced settings worksheet provided in the template for the Sample application as your guideline.

Advanced settings are:

- Evaluation order
- Performance settings
- Dimension properties such as hierarchy type, two-pass calculations, and data storage options
- Density (for aggregate storage applications only)
- Dimension assignments to cubes

Follow this workflow for setting up an advanced settings worksheet in your application template workbook:

1. Define the Worksheet Name

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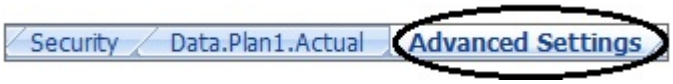
2. Define the Worksheet Name
3. Define Evaluation Order
4. Define Performance Settings
5. Define Dimension Settings
6. Define Density Settings
7. Assign Dimensions to Cubes

Define the Worksheet Name

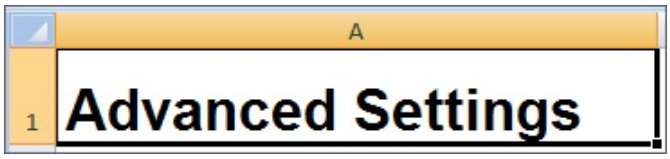
Name the advanced settings definition worksheet, `Advanced Settings`. For example:



Define the Worksheet Type

In cell A1 in the data worksheet, type `Advanced Settings` to signify that this sheet contains the advanced, or miscellaneous settings for the application. In Figure 1, cell A1 contains the worksheet type, **Advanced Settings**, to indicate that the worksheet will contain advanced settings for the application.

Figure C-9 Worksheet Type Shown in Advanced Settings Worksheet of Application Template



Define Evaluation Order

In the Advanced Settings worksheet, define the evaluation order of dimensions in the "Evaluation Order" section. Dimensions are listed in the first column under the heading "Dimensions". Cubes are listed in subsequent columns. The dimensions in each cube can have a different evaluation order.

3	Evaluation Order	Cubes	
4	Dimensions	Plan1	PBCS
5	Account	3	
6	Period		
7	Entity	1	
8	Year		
9	Scenario	2	
10	Version	1	
11	HSP_View		

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Define Performance Settings

For Planning applications only.

In the Advanced Settings worksheet, define dimension storage property of each dimension for each cube in the "Performance Settings" section. Valid values are Dense or Sparse. Each cube requires at least one dense dimension. Dimensions are listed in the first column under the heading "Dimensions". Cubes are listed in subsequent columns. The sparse or dense value of each dimension can differ between cubes.

14	Performance Setting	Cubes				
15	Dimensions	Plan1	Plan2	Plan3	VisASO	Vis1ASO
16	Account	Dense	Dense	Dense	Dense	Dense
17	Period	Dense	Dense	Dense	Dense	Dense
18	Entity	Sparse	Sparse	Sparse	Sparse	Sparse
19	Year	Sparse	Sparse	Sparse	Sparse	Sparse
20	Scenario	Sparse	Sparse	Sparse	Sparse	Sparse
21	Version	Sparse	Sparse	Sparse	Sparse	Sparse
22	HSP_View	Sparse	Sparse	Sparse	Sparse	Sparse

Define Dimension Settings

In the Advanced Settings worksheet, define dimensions properties in the "Dimension Settings" section. Dimensions are listed in the first column under the heading "Dimensions". Dimension properties are listed in subsequent columns. See Table 1 for the list of dimension properties

Table C-5 Dimension Properties Used in Dimension Settings Section of the Advanced Settings Worksheet

Dimension Property	Description or Valid Values
Description	Optional descriptive text
Alias Table	Optional alias table
Hierarchy Type	• Dynamic • Store
Two Pass Calculation	Yes | No
Apply Security	Yes | No
Data Storage	• Store • Dynamic Calc and Store • Dynamic Calc • Never Share • Label Only • Shared
Display Option	Set application default display options for the Member Selection dialog box. Select Member Name or Alias to display members or aliases. Member Name:Alias displays members on the left and aliases on the right. Alias:Member Name displays aliases on the left and members on the right. • Member Name • Alias • Member Name:Alias • Alias:Member Name

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You can also use Figure 2 for reference in adding information to the Dimension Settings section.

Figure C-10 Dimension Settings Section of the Advanced Settings Worksheet

25	Dimension Settings							
26	Properties	Description	Alias Table	Hierarchy Type	Two Pass Calculation	Apply Security	Data Storage	Display Option
27	Account							
28	Entity				Yes		Never Share	
29	Period			Dynamic				
30	Year							
31	Scenario							
32	Version							

Define Density Settings

For Planning Modules applications only.

In the Advanced Settings worksheet, define the dimension storage property of each dimension for each cube in the "Density" section. Each cube requires at least one dense dimension. Dimensions are listed in the first column under the heading "Dimensions". Cubes are listed in subsequent columns. The sparse or dense value of each dimension can differ between cubes.

25	Density	Cubes	
26	Dimensions	Plan1	EPBCS
27	Account	Dense	Dense
28	Currency	Sparse	Sparse
29	Entity	Sparse	Sparse
30	HSP_View	Sparse	Sparse
31	Period	Dense	Dense
32	Scenario	Sparse	Sparse
33	Version	Sparse	Sparse
34	Years	Sparse	Sparse

Assign Dimensions to Cubes

In the Advanced Settings worksheet, assign dimensions to cubes in the "Dimension Valid for" section. Dimensions are listed in the first column under the heading "Dimensions". Cubes are listed in subsequent columns.

For each dimension and cube intersection:

- Type `Yes` to include the dimension in the cube.
- Type `No` or leave the cell blank to exclude the dimension from the cube.

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Dimension Valid For	Cubes				
Dimensions	Plan1	Plan2	Plan3	VisASO	Vis1ASO
Account	Yes	Yes	Yes	Yes	Yes
Entity	Yes	Yes	Yes	Yes	Yes
Period	Yes	Yes	Yes	Yes	
Year	Yes	Yes	Yes	Yes	
Scenario	Yes	Yes	Yes		
Version	Yes	Yes	Yes		

Updating an Application in Smart View

From Oracle Smart View for Office and Oracle Smart View for Office (Mac and Browser), Service Administrators can update Planning, Planning Modules, or FreeForm applications based on either the downloaded template or a modified copy of one of the templates. The modified template must always adhere to template guidelines (described in Working with Artifacts in the Application Template).

When you update an application, you can:

- Define new or modify existing dimensions
- Add or modify a limited amount of data
- Define new or modify existing access permissions
- Define new or modify existing substitution variables
- Modify advanced application setting

Only Service Administrators can update an application.

Before you begin, you should launch Excel and, from Smart View, log in to the applicable data source.

To update an application in Smart View:

1. If you have not already done so, open the modified application template file.
2. In the Smart View Panel (Windows) or Smart View Home panel (Mac and Browser), perform a task:
   - Oracle Smart View for Office: Select the application name and, in the Action Panel, select **Application Management**.

     The application name is displayed in the tree in the Smart View Panel.
   - Oracle Smart View for Office (Mac and Browser): In the Smart View Home panel, click the Actions button, , and select the **Application Management** command in the drop-down list.

     You can click the Actions button, , from any folder in the library tree.
3. In the popup dialog, select **Update Application**.

   The application update status appears in the lower left corner of Excel.
4. When the application update process is complete, access the application, either in the web application or in Smart View, and verify your changes.

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Deleting an Application

Service Administrators may delete applications from Oracle Smart View for Office and Oracle Smart View for Office (Mac and Browser).

Before you begin, you should launch Excel and, from Smart View, log in to the applicable data source.

To delete an application from Smart View:

1. In the Smart View Panel (Windows) or Smart View Home panel (Mac and Browser), perform a task:
   - Oracle Smart View for Office: Select the application name and, in the Action Panel, select **Application Management**.

     The application name is displayed in the tree in the Smart View Panel.
   - Oracle Smart View for Office (Mac and Browser): In the Smart View Home panel, click the Actions button, , and select the **Application Management** command in the drop-down list.

     You can click the Actions button, , from any folder in the library tree.
2. In the popup dialog, select **Delete Application**, and confirm in the subsequent dialog.
3. In the Smart View Panel (Windows) or Smart View Home panel (Mac and Browser), verify that the application is removed.

Planning Admin Extension and Office AutoCorrect

> **Note**
>
> The information in this topic applies to the Planning Admin Extension on the Windows-based version of Oracle Smart View for Office; this information does not apply to the Planning Admin Extension in Oracle Smart View for Office (Mac and Browser).

In the Planning Admin Extension on Windows-based versions of Excel, Office AutoCorrect can affect member editing in Smart View.

When editing member properties, such as member names, member descriptions, aliases, or member formulas of the data type Text, you may have names that begin with two initial capital letters; for example

```
DGreen
```

Excel will autocorrect this name to one initial capital; for example:

```
Dgreen
```

In Excel, you can set an AutoCorrect option to retain two initial capital letters in a name.

To retain two initial capital letters in a name:

1. In Excel, access the **Excel Options** dialog.
2. Select **Proofing**, and then click the **AutoCorrect Options** button.

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3. In the **AutoCorrect** dialog, clear the **Correct TWo INitial CApitals** check box.
4. Click **OK** to close the **AutoCorrect** dialog, and then click **OK** to close the **Excel Options** dialog.

The change takes effect immediately. There is no need to restart Excel.

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D

Using Smart View to Import and Edit Application Metadata

Understand how to work with application metadata, install the Admin Extension, and use Smart View grids. Also learn about dimensions, members, application members, and shared members in Smart View as well as how to refresh databases.

Related Topics

- About Using Smart View to Work with Application Metadata
- Installing Smart View and the Admin Extension for Editing Dimensions
- Using Smart View Grids to Import and Edit Application Metadata
- Importing Dimensions in Smart View
- Editing Members in Smart View
- Adding Application Members in Smart View
- Moving Members in Smart View
- Working with Attribute Dimensions
- Designating Shared Members in Smart View
- Refreshing Databases
- Planning Admin Extension and Office AutoCorrect

About Using Smart View to Work with Application Metadata

Applies to: Enterprise Profitability and Cost Management, Financial Consolidation and Close, FreeForm, Planning, Tax Reporting

Service Administrators can use the Planning Admin Extension with Oracle Smart View for Office and Oracle Smart View for Office (Mac and Browser) to quickly import and edit dimension and member metadata.

> **Note**
>
> - You must have Service Administrator privileges to import and edit dimensions in Smart View and Smart View (Mac and Browser).
> - All procedures described in this appendix are performed within Smart View using the Planning Admin Extension.
>   - To use the Windows client-based version of Smart View, see *Getting Started Guide for Administrators* for instructions on installing the extension.
>   - To use the Mac- or browser-based version of Smart View, see the *Deploying and Administering Oracle Smart View for Office (Mac and Browser)* for instructions on deploying the extension.

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Smart View and Smart View are the Microsoft Office interfaces to Oracle Fusion Cloud Enterprise Performance Management business processes, where end users can perform their planning and forecasting activities. They can analyze business process data in Microsoft Excel, Word, PowerPoint, and Outlook. Ad hoc grids in Smart View enable users to personalize focused data slices that they frequently access and share them with others through Smart View or their Cloud EPM business process. See *Working with Oracle Smart View for Office*, *Working with Oracle Smart View for Office (Mac and Browser)*, and the chapter on using ad hoc grids in the *Working with* guide for your business process (for example, see Using Ad Hoc for Data Entry and Analysis in *Working with Planning*.

Installing Smart View and the Admin Extension for Editing Dimensions

Before you begin editing dimensions in Oracle Smart View for Office, ensure the following depending on whether you plan to use Windows-based Smart View, Oracle Smart View for Office (Mac and Browser), or both:

- **Windows-based Smart View**Install Smart View and the Planning Admin Extension.

  See Installing Oracle Smart View for Office and the Planning Admin Extension
- **Smart View (Mac and Browser):** Deploy Smart View (Mac and Browser) with the Dimension Editor options enabled.

  See Deploying Oracle Smart View for Office (Mac and Browser) with the Admin Extension

Installing Oracle Smart View for Office and the Planning Admin Extension

To use the Windows client-based version of Smart View, see *Getting Started Guide for Administrators*, Downloading and Installing Clients for instructions on installing Smart View.

Then install the Admin Extension.

> **Note**
>
> Prior to 22.07, the Admin Extension file was in MSI format. Starting in 22.07, the Admin Extension file is in SVEXT format.
>
> If you installed the Admin Extension prior to 22.07, then you must first uninstall the MSI version of the extension from the Windows Control Panel, Programs and Features. If you keep the MSI version on your machine and also install the SVEXT version, then the Smart View extension update process will not recognize updates to the SVEXT version of the Admin Extension. You may continue to use the MSI version of the Admin Extension that you previously installed. However, the MSI version of the extension will not be updated in the future so will not contain any future bug fixes or enhancements. For this reason, Oracle recommends moving to the Admin Extension in SVEXT format.

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Tip

- Before proceeding with installing the Admin Extension, Oracle recommends that you first clear your browser cache.
- Enterprise Profitability and Cost Management: The Admin Extension is supported starting with Smart View 22.100.

To install the Admin Extension:

- Download the extension from the Downloads page of your business process. See *Getting Started Guide for Administrators*, Downloading and Installing Clients for instructions on installing the Planning Admin Extension.
- Download and install using the extension installation and update process in Smart View. See Installing Extensions in *Working with Oracle Smart View for Office*.

Either method will download and install the Admin Extension in SVEXT format.

Deploying Oracle Smart View for Office (Mac and Browser) with the Admin Extension

To use the Mac- or browser-based version of Smart View, you must deploy a Smart View manifest file with the Admin Extension options enabled.

See the following topics in the *Deploying and Administering Oracle Smart View for Office (Mac and Browser)* for instructions on deploying Smart View with the Admin Extension features.

- Creating and Saving the Manifest File
- Enabling the Admin Extension
- Sideloading and Logging In or Deploying the Manifest File to Office 365 Users

Using Smart View Grids to Import and Edit Application Metadata

The Smart View grid allows you to rapidly add, edit, and move the members of a dimension.

- About the Smart View Grid
- The Smart View Grid and Ribbon Display
- Guidelines for Using the Smart View Grid
- Default Metadata Dimension Member Properties

About the Smart View Grid

In Oracle Smart View for Office and in Oracle Smart View for Office (Mac and Browser), the grid consists of two dimensions:

- A Oracle Fusion Cloud Enterprise Performance Management dimension on one axis (Planning, Financial Consolidation and Close, Tax Reporting)
- A metadata dimension on another axis

A metadata dimension represents a flat list of metadata members. For each dimension, a predefined set of metadata members is presented by default. Each metadata member corresponds to a specific member property valid for the dimension that was imported onto the grid. Instead of containing numeric values, the metadata grid data cells hold the specific

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property values. To add metadata dimension member properties that are not on the grid by default, use the **Member Selection** dialog in Smart View.

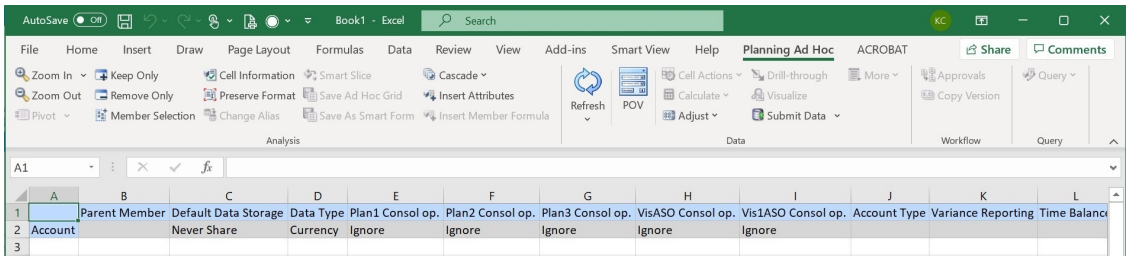
The Smart View Grid and Ribbon Display

The Smart View grid displays the dimension with member names in rows and member properties in columns. When you first import a dimension for editing, a predefined set of member properties is displayed by default in the grid.

Dimension Grid in Smart View

In Figure 1, you can see an example of the default member properties for the Vision Account dimension in Oracle Smart View for Office.

Figure D-1 Dimension Imported into Smart View for Office Grid with the Planning Ad Hoc Ribbon Displayed



Note that in Figure 1, the Planning Ad Hoc ribbon is displayed. Depending on the data source to which you are connected, the provider ad hoc ribbon for that data source is displayed. For example, if you are connected to an Enterprise Profitability and Cost Management data source, then the EPCM Ad Hoc ribbon is displayed.

The ad hoc ribbon that is displayed is the same as the provider ad hoc ribbon displayed for a conventional Smart View ad hoc grid, only with fewer options enabled.

For all dimensions, including attribute dimensions, these are the valid options in the provider ad hoc ribbon:

- Zoom In
- Zoom Out
- Keep Only
- Remove Only
- Member Selection
- Refresh
- Submit Data

The following options may appear enabled, but are not valid for metadata grids: Preserve Format, Cascade, Insert Attributes, POV, and Adjust.

You can add metadata dimension member properties using the **Member Selection** dialog box, accessed from the ribbon, or accessed by right-clicking a property in the grid, then selecting **Smart View**, and then **Member Selection**.

For information on using the **Member Selection** dialog box in Smart View, see Selecting Members from the Member Selector in *Working with Oracle Smart View for Office 22.100*.

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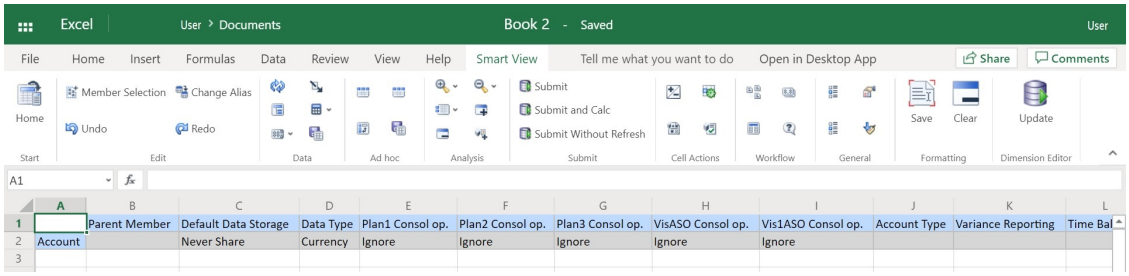
Dimension Grid in Smart View (Mac and Browser)

In Figure 2, you can see an example of the default member properties for the Vision Account dimension in Oracle Smart View for Office (Mac and Browser), running in the Chrome browser.

Figure D-2 Dimension Imported into Smart View (Mac and Browser) Grid with the Smart View Ribbon Displayed



Note that in Figure 2, the Smart View ribbon is displayed. This is the default ribbon name for an Oracle Smart View for Office (Mac and Browser) deployment. You may specify a different name for the ribbon. When importing a dimension on to a grid, all ribbon options appear enabled, but only a subset are applicable to a dimension grid.

For all dimensions, including attribute dimensions, these are the valid options in the ribbon:

- Zoom In
- Zoom Out
- Keep Only
- Remove Only
- Member Selection
- Refresh
- Update

You can add metadata dimension member properties using the **Member Selection** dialog box, accessed from the ribbon.

For information on using the **Member Selection** dialog box in Oracle Smart View for Office (Mac and Browser) to add metadata member properties to the grid, see Selecting Members from the Member Selector in *Working with Oracle Smart View for Office (Mac and Browser)*.

Guidelines for Using the Smart View Grid

The following guidelines will assist you in using the Smart View grid to edit dimension metadata:

- Use the Planning Admin Extension to add dimension members, and to edit and update dimension and member properties.
- The Planning Admin Extension cannot be used to:
  - Delete or remove dimensions (that is, the dimension root member) or members.
  - Edit dimension or member names.

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- Data cell values can be textual or enumeration or numerical.
- The Parent Member is used to specify or modify the parent/child relationship.
- The position of a member in a grid doesn't necessarily represent the actual position of siblings in the outline.
- Each metadata grid must be linked to a corresponding dimension.
- Columns for each dimension are based on the corresponding set of member properties available in the dimension editor.

  When you first import a dimension, a predefined set of metadata dimension member properties is displayed. See Default Metadata Dimension Member Properties for a list of those member properties.
- After a metadata grid is opened, it can't be relinked to a different dimension.
- The corresponding valid set of metadata members is specific to each dimension.
- Dimension members are valid for corresponding dimensions only.
- In Oracle Smart View for Office, the properties available in the dimension grid and in the **Member Selection** dialog for a selected dimension list only the Plan type- or cube-specific properties enabled for the cube. This includes properties such as consolidation operator, member formula, data storage, and others. When a dimension is not enabled for Plan types (or cubes), then dimension-specific properties are not listed on the grid or in the **Member Selection** dialog.

  For example, the Product dimension is not enabled for the Plan3 cube. Properties (such as Plan3 Data Storage, Plan3 Consol Op, Plan3 Formula, Plan3 Member Formula Description, Plan3 Solve Order) will not be listed in the grid or in the **Member Selection** dialog as they are not used for the cube.
- For the Time Period dimension, Data Storage is the only editable property.
- The following functionality isn't available in Smart View grids with dimension metadata:
  - Pivot
  - Pivot to POV
  - Cell Text
  - Cell Notes
  - Supporting Details

Default Metadata Dimension Member Properties

When you first import a dimension in the Planning Admin Extension, a predefined set of dimension metadata member properties is displayed by default. Use the Member Selection dialog box to add other metadata member properties to the grid.

The sections below list the default member properties for the Planning dimensions.

For information on using the Member Selection dialog box in Oracle Smart View for Office to add other member properties to the grid, see "Selecting Members from the Member Selector" in *Working with Oracle Smart View for Office*.

Account Dimension

- Parent Member
- Default Data Storage

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- Data Type
- Consol op. for all Cubes
- Account Type
- Variance Reporting
- Time Balance
- Default Alias Table
- Default Formula

Entity Dimension

- Parent Member
- Default Data Storage
- Data Type
- Consol op. for all Cubes
- Base Currency
- Default Alias Table

Custom Dimensions

- Parent Member
- Default Data Storage
- Data Type
- Consol op. for all Cubes
- Default Alias Table

Scenario Dimension

- Parent Member
- Default Data Storage
- Data Type
- Consol op. for all Cubes
- Start Yr.
- End Yr.
- Start Period
- End Period
- Default Alias Table
- Default Formula

Version Dimension

- Parent Member
- Default Data Storage
- Data Type
- Consol op. for all Cubes

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- Type
- Default Alias Table
- Default Formula

Period and Years Dimensions

- Leave as is or limit to Parent Member
- Default Data Storage – Note that for the Time Period dimension, Data Storage is the only editable property
- Data Type
- Default Alias Table

Currency Dimension

- Parent Member
- Default Data Storage
- Data Type
- Symbol
- Predefined Symbol
- Scale
- Precision
- Triangulation Currency
- Reporting Currency
- Thousands Separator
- Decimal Separator
- Negative Sign
- Negative Color
- Default Alias Table
- Default Formula

HSP_View Dimensions (Sandbox dimensions)

- Parent Member
- Default Data Storage
- Data Type
- Consol op. for all Cubes
- Default Alias Table

Importing Dimensions in Smart View

Importing dimensions into the Smart View grid refers to placing the dimension and its members, and their respective properties, on the grid in preparation for editing.

By importing a dimension into Oracle Smart View for Office or Oracle Smart View for Office (Mac and Browser), you can rapidly add, edit, and move the members of the dimension.

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> **Note**
>
> You must have Service Administrator privileges to import dimensions in Smart View.

See:

- Importing Dimensions in Smart View for Office
- Importing Dimensions in Smart View (Mac and Browser)

Importing Dimensions in Smart View for Office

> **Note**
>
> You must have Service Administrator privileges to import dimensions in Oracle Smart View for Office.

To import a dimension into a Smart View grid:

1. From the Smart View ribbon, click Panel.
2. In the Smart View Panel, click Shared Connections or Private Connections, and provide your user name, and password in the login windows.

   If you chose Private Connections, select the private connection in the drop-down list of connections, then provide your login credentials.
3. From the drop-down list in the Smart View Panel, select your business process (for example, **Planning**) or **EPM Cloud**.
4. In the tree in the Smart View Panel, expand the **Dimensions** folder to view the application dimensions in the folder.

   If attribute dimensions are defined, they are displayed in the **Attribute Dimensions** folder.

   If a Time Period dimension is defined, it's displayed in the Dimensions folder. Note that for the Time Period dimension, Data Storage is the only editable property.

   In the following example, the Dimensions folder is expanded and contains 10 dimensions.

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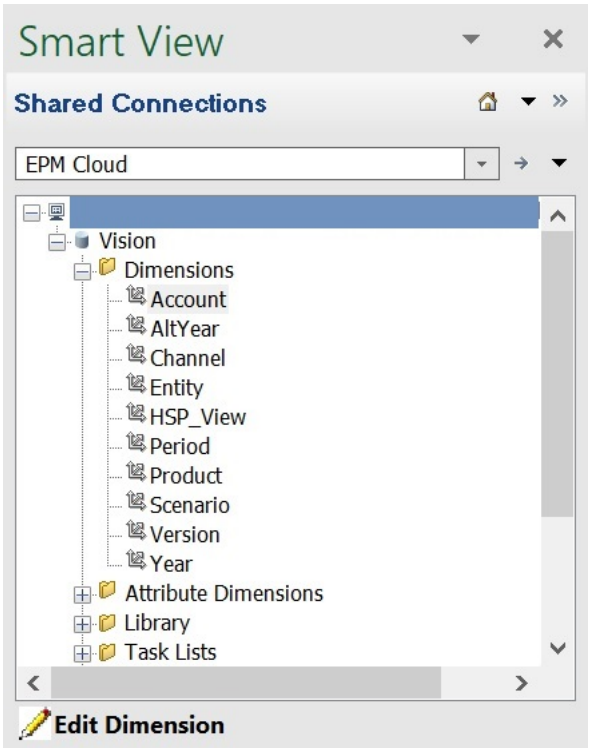

5. Right-click a dimension name and select **Edit Dimension**.

   Alternatively, select a dimension and then, in the Action Panel, select **Edit Dimension**.

   See Dimension Grid in Smart View for an illustration of the initial grid for the Account dimension.

Importing Dimensions in Smart View (Mac and Browser)

Before you begin to work with dimensions in Oracle Smart View for Office (Mac and Browser), you should either sideload or deploy Smart View in your environment. Instructions are in *Deploying and Administering Oracle Smart View for Office (Mac and Browser)*:

- Sideloading and Logging In
- Deploying the Manifest File to Office 365 Users

To import a dimension into an Oracle Smart View for Office (Mac and Browser) grid:

1. From the Smart View ribbon, click **Home**.

   In the **Home** panel, dimensions are displayed in the **Dimensions** folder. If attribute dimensions are defined, they are displayed in the **Attribute Dimensions** folder.

   If a Time Period dimension is defined, it's displayed in the Dimensions folder. Note that for the Time Period dimension, Data Storage is the only editable property.

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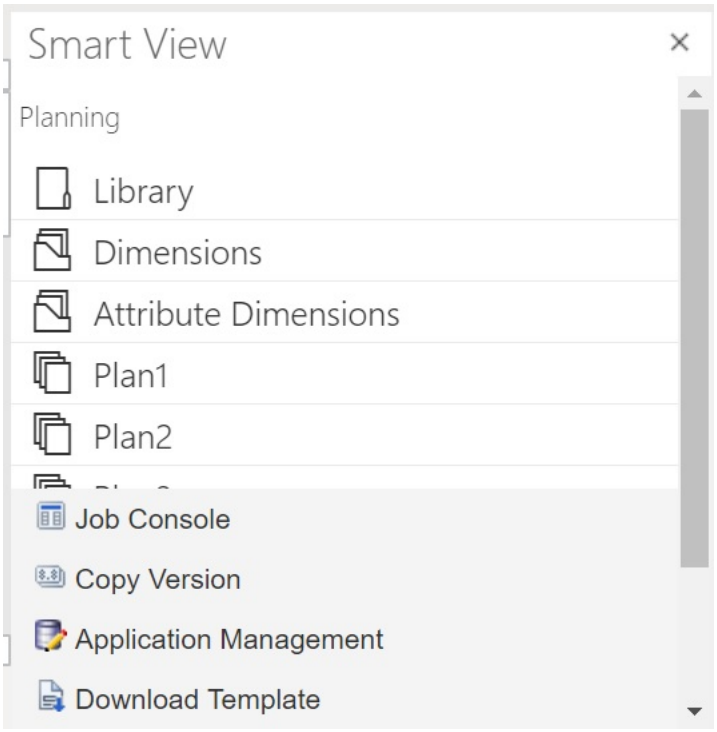
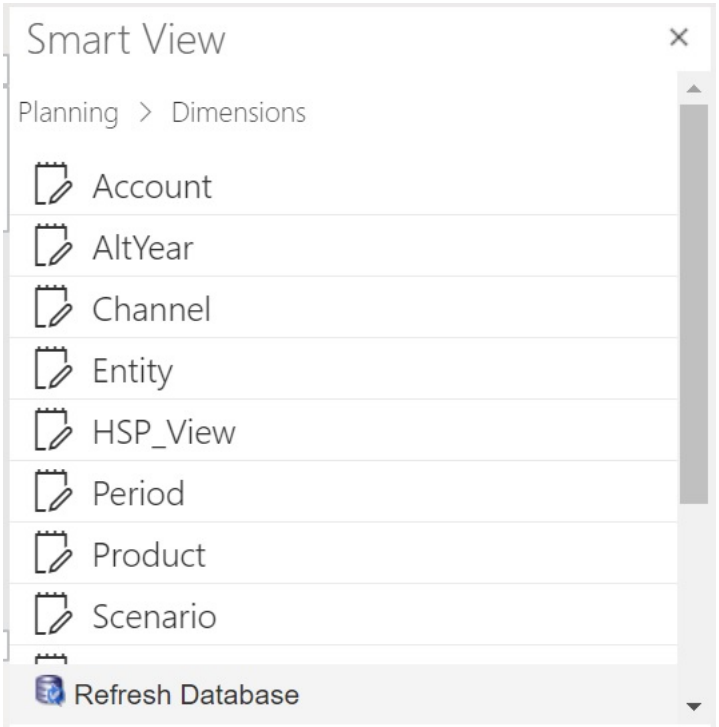


2. In the **Smart View Home** panel, click the **Dimensions** folder to expand it and view the application dimensions in the folder.

   In the following example, the Dimensions folder is expanded and shows eight of the 10 dimensions in the Vision application.



3. Click a dimension name to import the dimension on to the grid.

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See Dimension Grid in Smart View (Mac and Browser) for an illustration of the initial grid for the Account dimension.

4. **Optional:** Click the provider link in the path at the top of the panel to return to the main page of the Home panel

   For example, in the figure in step 2, you would click **Planning** to return to the main page of the Home panel.

Editing Members in Smart View

The Smart View grid allows you to rapidly edit the properties of members of a dimension.

To edit member properties in Smart View:

1. Within Oracle Smart View for Office or Oracle Smart View for Office (Mac and Browser), import a dimension into a Smart View grid (see Importing Dimensions in Smart View).
2. Highlight a member property in the grid.

   Note that for the Time Period dimension, Data Storage is the only editable property.
3. In the drop-down menu, select a value.

> **Note**
>
> Modified cells are displayed in a different color.

4. To save the grid, perform a task:
   - Oracle Smart View for Office: In the Planning Ad Hoc ribbon, click **Submit Data**.
   - In Oracle Smart View for Office (Mac and Browser), in the Smart View ribbon, click **Update**.

> **Note**
>
> You can save changes to multiple members and properties in the same **Submit Data** or **Update** operation. If the operation fails for one member, the application server will stop the operation and not save any changes.

Adding Application Members in Smart View

The Smart View grid allows you to rapidly add members to a dimension.

- Adding Members in Smart View
- Guidelines for Adding Members in Smart View

Videos

Your Goal	Watch This Video
Learn about adding Entity dimension members in Oracle Smart View for Office.	Managing Entity dimension members in Oracle® Smart View for Office

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Your Goal	Watch This Video
Learn about adding Account dimension members in Smart View.	Managing Account dimension members in Oracle® Smart View for Office
Learn about adding Scenario dimension members in Smart View.	Managing Scenario dimension members in Oracle® Smart View for Office

Adding Members in Smart View

The Smart View grid allows you to rapidly add members to a dimension.

To add members in Smart View:

1. Within Oracle Smart View for Office or Oracle Smart View for Office (Mac and Browser), import a dimension into a Smart View grid (see Importing Dimensions in Smart View).

   Note that Oracle Smart View for Office does not support adding members to the Time Period dimension.

2. Enter the name of a new member in the name column.

   > **Note**
   >
   > To add a member with a numeric name, place a single quote (') in front of the numeric member name to inform the system that the number is a member name and not data. For example, to add a member named 123, enter '123.

3. From the ribbon, click **Refresh**.

   New members are marked by an asterisk (*) in the grid after you perform the Refresh. A default set of properties is automatically applied to the new member. The default Parent Member is the root member of the dimension.

4. **Optional:** To change any property from the default value (the root member), highlight the appropriate cell in the grid, and then from the drop-down menu, select another value. (See Editing Members in Smart View.)

   > **Note**
   >
   > Perform a Refresh *before* you modify any properties in your new member. The Refresh operation will replace any changed values with default values from the server.

5. To save the grid, perform a task:

   - Oracle Smart View for Office: In the Planning Ad Hoc ribbon, click **Submit Data**.
   - In Oracle Smart View for Office (Mac and Browser), in the Smart View ribbon, click **Update**.

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> **Note**
>
> - To modify the properties of the new member, see Editing Members in Smart View.
> - In Oracle Smart View for Office, member properties, including member names, can be localized based on the particular Locale set by in the Smart View add-in.
> - You can save changes to multiple members and properties in the same **Submit Data** or **Update** operation. If the operation fails for one member, the application server will stop the operation and not save any changes.

Guidelines for Adding Members in Smart View

- A new member is added as the last sibling under the specified parent.
- Parent-child relationships are determined by the Parent Name column property.
- The relative position of a new member in the grid is insignificant.
- The relative position of a new member in the grid will not be changed after performing the **Submit Data** or **Update** operation. To see the actual position of a new member in the outline reflected in the ad hoc grid, perform **Zoom Out**, then **Zoom In**.
- A full validity check of a new member name, including a check for invalid characters and a check for duplicate names, is performed during the **Submit Data** or **Update** operation.
- Default properties are automatically applied by the application to a new member. The specific default properties are based upon those of the dimension.
- When adding new currency members to a Currency dimension in a simplified application, the parent member for the new currency defaults to the "`Currency`" dimension member. It should default to the "`Input Currencies`" member. You can correct this by doing the following:
  1. Replace "`Currency`" with "`Input Currencies`" in the `Parent Member` column for each new currency member that you added.
  2. Perform a **Submit Data** or **Update**.
  3. Click **Refresh**.
  4. Verify that for each new currency you added, the entry for the `Parent Member` column is "`Input Currencies`."

Moving Members in Smart View

The Smart View grid allows you to rapidly move members from one parent to another within a dimension.

To move a member in Oracle Smart View for Office:

1. Within Smart View or Oracle Smart View for Office (Mac and Browser), import a dimension into a Smart View grid (see Importing Dimensions in Smart View for Office.)
2. Highlight the member in the **Parent Member** column in the grid.

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> **Note**
>
> An empty Parent Member property causes the value of the Parent Member to default to the root member.

3. Enter a parent name at the intersection of the **member name** and **Parent** column.
4. Click **Submit Data** to save the grid.

> **Note**
>
> You can save moves of multiple members or subtrees in the same **Submit Data** operation. If the **Submit Data** operation fails for one member, the application server stops the operation and doesn't save changes.

Guidelines for Moving Members in Smart View

- An empty Parent Member in the grid denotes a root member in the dimension.
- The value of a Parent Member follows the rules applicable to the corresponding value in the metadata load file used by Smart View.

Working with Attribute Dimensions

Related Topics

- Adding Attribute Dimension Members in Smart View
- Associating Attribute Dimension Members with Dimension Members

Adding Attribute Dimension Members in Smart View

You create attribute dimensions in the web interface. You can then add members to attribute dimensions using the Admin Extension in Oracle Smart View for Office or Oracle Smart View for Office (Mac and Browser).

- Adding Attribute Dimension Members in Smart View for Office
- Adding Attribute Dimension Members in Smart View (Mac and Browser)

Adding Attribute Dimension Members in Smart View for Office

You create attribute dimensions in the web interface. You can then add members to attribute dimensions using the Admin Extension in Oracle Smart View for Office.

To add members to attribute dimension members:

1. From the Smart View ribbon, click Panel.
2. In the Smart View Panel, click Shared Connections or Private Connections, and provide your user name, and password in the login windows.

   If you chose Private Connections, select the private connection in the drop-down list of connections, then provide your login credentials.
3. From the Smart View Panel, select **EPM Cloud**.

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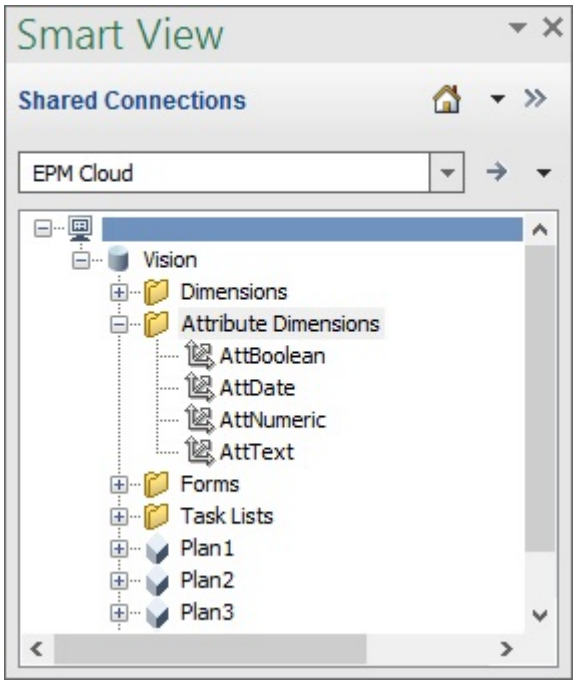
4. In the tree in the Smart View Panel, expand the Attribute Dimensions node.

   In the following example, the Attribute Dimensions node contains four attribute dimensions, one for each attribute type: text, date, Boolean, and numeric.



5. Select an attribute dimension, and then select **Edit Dimension**.

   To illustrate, we selected the AttText attribute dimension to edit. An example of the initial grid for the attribute dimension is shown. This dimension has no members yet, and just two properties, Parent member and Default Alias Table.

	A	B	C
1		Parent Member	Default Alias Table
2	AttText		

6. Type the attribute member names to add; for example:

	A	B	C
1		Parent Member	Default Alias Table
2	AttText		
3	Gold		
4	Silver		
5	Platinum		

7. From the Smart View ribbon, click **Submit**.

   The attribute members are submitted to the application.

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	A	B	C
1		Parent Member	Default Alias Table
2	AttText		
3	Gold	AttText	
4	Silver	AttText	
5	Platinum	AttText	

8. **Optional:** Continue editing the attribute members, entering data for other properties by typing directly in the grid, and then submit your changes.

Adding Attribute Dimension Members in Smart View (Mac and Browser)

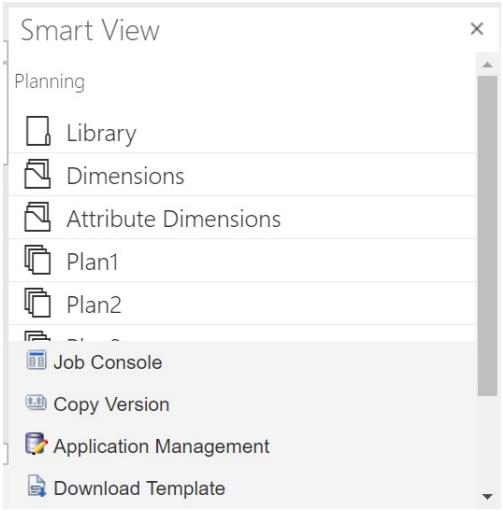
Before you begin adding attribute dimension members in Oracle Smart View for Office (Mac and Browser), you should either sideload or deploy Smart View in your environment. Instructions are in *Deploying and Administering Oracle Smart View for Office (Mac and Browser)*:

- Sideloading and Logging In
- Deploying the Manifest File to Office 365 Users

To add a member to an attribute dimension in a Oracle Smart View for Office (Mac and Browser) grid:

1. From the Smart View ribbon, click **Home**.

   In the **Home** panel, dimensions are displayed in the **Dimensions** folder. If attribute dimensions are defined, they are displayed in the **Attribute Dimensions** folder.



2. In the **Smart View Home** panel, click the **Attribute Dimensions** folder to expand it and view the attribute dimensions in the folder.

   In the following example, the Attribute Dimensions folder is expanded and shows one attribute dimension, Market Size, in the Vision application.

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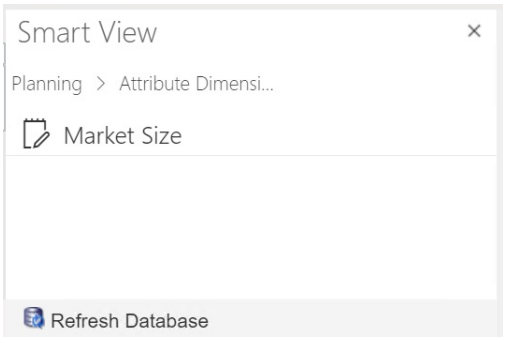

3. Select an attribute dimension to import it onto the grid.

   To illustrate, we selected the Market Size attribute dimension to edit. An example of the initial grid for the attribute dimension is shown. This dimension has no members yet, and just two properties, Parent member and Default Alias Table.

	A	B	C
1		Parent Member	Default Alias Table
2	Market Size		

4. In the attribute name column, under the attribute name, type the attribute member names to add.

	A	B	C
1		Parent Member	Default Alias Table
2	Market Size		
3	Large		
4	Medium		
5	Small		

5. From the Smart View ribbon, click **Submit**.

   The attribute members are submitted to the application.

	A	B	C
1		Parent Member	Default Alias Table
2	Market Size		
3	Large	Market Size	
4	Medium	Market Size	
5	Small	Market Size	

6. **Optional:** Continue editing the attribute members, entering data for other properties by typing directly in the grid, and then submit your changes.

Associating Attribute Dimension Members with Dimension Members

The illustrations in this topic are from Oracle Smart View for Office, but the concepts also apply to Oracle Smart View for Office (Mac and Browser).

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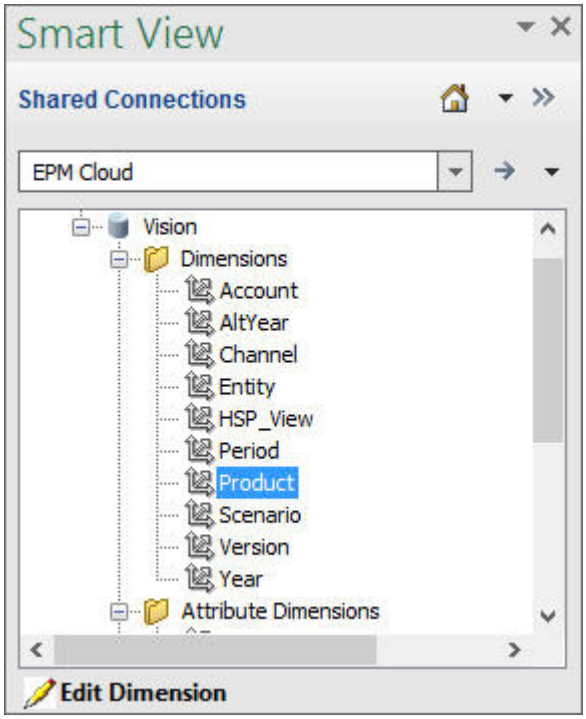
1. In the tree in the Smart View Panel (Windows) or Smart View Home (Mac and Browser), expand the **Dimensions** node, and select the dimension to work with.

   In the following example, the Dimensions node shows the standard dimensions in the Vision application along with some custom dimensions such as AltYear and Channel. The Product dimension is selected.



2. Select **Edit Dimension** and note the initial layout of the grid on the sheet.

   In Oracle Smart View for Office (Mac and Browser), just click the dimension to import it onto the grid.

   If the attribute dimensions are not displayed on the sheet, then use the Member Selector to add them. In the Planning Ad Hoc ribbon (Windows) or Smart View ribbon (Mac and Browser), select **Member Selection**, and then add the attribute dimensions.

3. In the grid, expand the dimension to view its members.

4. Associate the attribute member values to the dimension members using the cell-based member selector, as shown in Figure 1.

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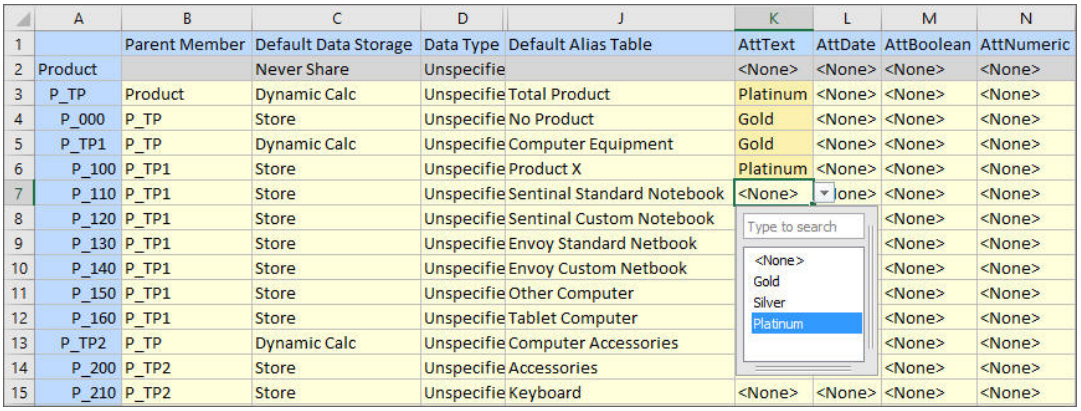
Figure D-3 Associating Attribute Values to Dimension Members



5. Repeat the previous step for any other dimension member and attribute value associations you want to define.
6. At any time, click **Submit Data** in the Planning Ad Hoc ribbon (Windows) or **Submit** in the Smart View ribbon (Mac and Browser) to submit your changes to the database.

   You can perform interim submits, or submit all changes at once.

Designating Shared Members in Smart View

The Smart View grid allows you to rapidly designate members for sharing within a dimension.

> **Note**
>
> In Smart View, the Shared data storage option isn't available for members of the Currency dimension or any attribute dimensions.

To designate a shared member in Oracle Smart View for Office:

1. Verify that the base member exists.
2. Highlight the base member in the **Parent Member** column in the grid.
3. Change the parent name value for the base member.
4. Highlight the base member in the **Data Storage** column.
5. In the drop-down menu, select **Shared**.
6. Click **Submit Data** (Windows) or **Submit** (Mac and Browser) to save the grid.

The submit operation will refresh the base member with its original Parent Member and Data Storage properties. The shared member will be added under the specified parent on the server. You must zoom in on the parent to see the newly added shared member.

> **Note**
>
> The new shared member will *not* be automatically added to the Smart View grid. The list of members displayed in the grid remains unchanged.

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Refreshing Databases

Related Topics

- Refreshing Databases in Smart View for Office
- Refreshing Databases in Smart View (Mac and Browser)

Refreshing Databases in Smart View for Office

The Oracle Smart View for Office grid allows you to quickly refresh a database.

To refresh a database in Smart View:

1. From the Smart View ribbon, click **Panel**.
2. From the Smart View Panel, select **Planning**.
3. Expand the **Dimensions** folder to view the application dimensions in the folder.

   If attribute dimensions are defined, they are displayed in the **Attributes** folder.
4. Right-click the *root* **Dimension** or **Attributes** folder and select **Refresh Database**.

   The Smart View **Refresh Database** dialog box is displayed.

   > **Tip**
   >
   > The database refresh cannot be completed while a rule type job is running or in progress. Check the Job Console, either in the web or in Smart View, for any jobs running against the database. When all jobs are complete, then perform the database refresh.
5. Click **Refresh**.

   A progress bar is displayed, indicating the percentage of steps completed for the refresh or create operation.

Refreshing Databases in Smart View (Mac and Browser)

The Smart View grid allows you to quickly refresh a database or create a new one.

To refresh a database in Oracle Smart View for Office (Mac and Browser):

1. From the Smart View ribbon, click **Home**.
2. From the Smart View Home panel, navigate to the root application **Dimensions** folder.

   If attributes are defined in the application, to refresh attribute dimensions, navigate to the **Attribute Dimensions** folder.
3. Select the *root* **Dimension** or **Attribute Dimensions** folder, click the **Actions** button, , and then select **Refresh Database** from the drop-down menu.

   The **Refresh Database** dialog box is displayed.

   In the following example, the **Dimensions** folder is selected, and in the **Actions** drop-down menu, **Refresh Database** is selected.

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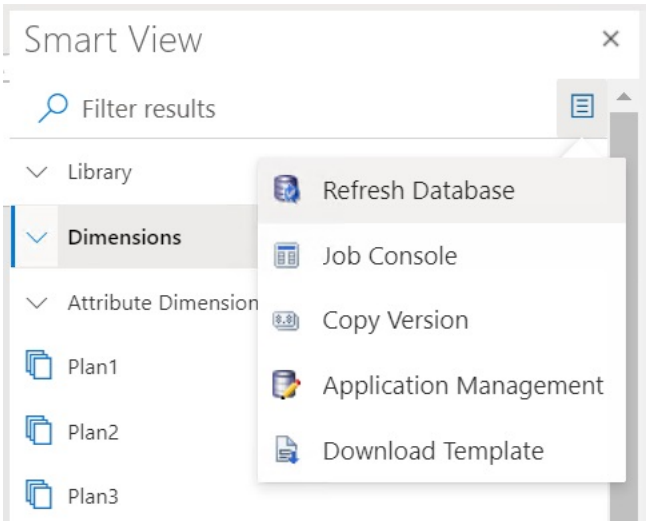

Tip

The database refresh cannot be completed while a rule type job is running or in progress. Check the Job Console, either in the web or in Oracle Smart View for Office (Mac and Browser), for any jobs running against the database. When all jobs are complete, then perform the database refresh.

4. Select the **Database** check box, and then click **Refresh**.

   In the following illustration, the **Database** and **Shared Members** check boxes are selected. Other options that can be selected for refresh are:

   - Update custom-defined functions
   - Security filters
   - Validate limits

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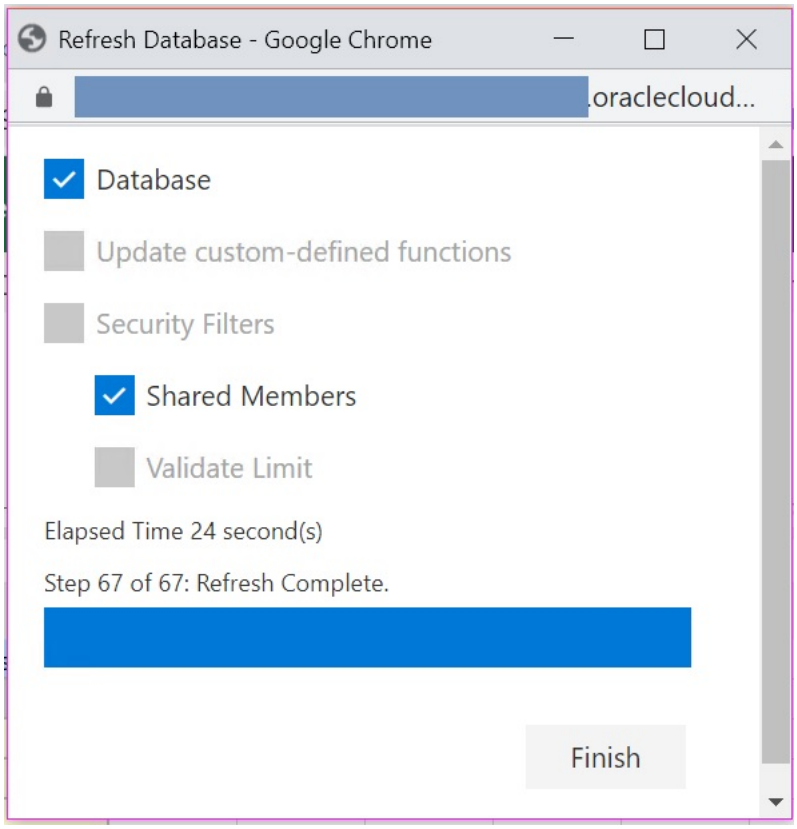

A progress bar is displayed, indicating the percentage of steps completed for the refresh or create operation.

5. Click **Finish** to close the dialog.

Planning Admin Extension and Office AutoCorrect

> **Note**
>
> The information in this topic applies to the Planning Admin Extension on the Windows-based version of Oracle Smart View for Office; this information does not apply to the Planning Admin Extension in Oracle Smart View for Office (Mac and Browser).

In the Planning Admin Extension on Windows-based versions of Excel, Office AutoCorrect can affect member editing in Smart View.

When editing member properties, such as member names, member descriptions, aliases, or member formulas of the data type Text, you may have names that begin with two initial capital letters; for example

```
DGreen
```

Excel will autocorrect this name to one initial capital; for example:

```
Dgreen
```

In Excel, you can set an AutoCorrect option to retain two initial capital letters in a name.

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To retain two initial capital letters in a name:

1. In Excel, access the **Excel Options** dialog.
2. Select **Proofing**, and then click the **AutoCorrect Options** button.
3. In the **AutoCorrect** dialog, clear the **Correct TWo INitial CApitals** check box.
4. Click **OK** to close the **AutoCorrect** dialog, and then click **OK** to close the **Excel Options** dialog.

The change takes effect immediately. There is no need to restart Excel.

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E

FreeForm Best Practices

Use these best practices for FreeForm.

This table provides links to the best practices mentioned in this guide.

Category	Best Practice For	See this Section
EPM Center of Excellence	Creating and running an EPM Center of Excellence	Creating and Running an EPM Center of Excellence
Application Launch	Resolving login issues	Resolving Login Issues in *Operations Guide*
Down Environments	Resolving issues with environments that are down	Dealing with Down Environments in *Operations Guide*
Dynamic Tabs	Using dynamic tabs	Using Dynamic Tabs in Redwood Experience in *Getting Started Guide for Administrators*
FreeForm App Creation	Creating FreeForm apps	Important Considerations for FreeForm Apps
Cube Optimization	Improving cube performance	• Improving Cube Performance • Optimize BSO Cubes in *Operations Guide* • Optimizing Aggregate Storage Option Cubes in *Operations Guide*
Data Export	Resolving issues if you get an Oracle Essbase query limit error when exporting a large number of data cells from ASO cubes	Handling Issues Related to Large Data Export from ASO Cubes in *Operations Guide*
Application Database Refresh	Resolving database refresh issues	Troubleshooting Database Refresh Issues in *Operations Guide*
Daily Maintenance	Deciding whether to perform a complete export during the daily maintenance process or to create an application backup	Exporting Smart List Textual Data During Daily Maintenance for Incremental Data Import in *Operations Guide*
Connecting Environments	Connecting Oracle Fusion Cloud Enterprise Performance Management environments and migrating Cloud EPM connections	• Considerations • Considerations for Migrating Cloud EPM Connections
Navigation Flows	Designing navigation flows and handling issues with navigation flows in connected environments	• Navigation Flow Design Best Practices and Naming Considerations • Handling Issues with Navigation Flows in *Oracle Fusion Cloud EPM Operations Guide*

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Category	Best Practice For	See this Section
Dashboards Containing Master Forms	Designing dashboards with master forms and target objects	Creating Dashboards Containing Master Forms and Details
Dashboards 2.0 POV Bar	Using POV bars in Dashboards 2.0	Dashboard 2.0 POV Bar Considerations
Dynamic Calc	Adding children under a dynamic calc parent	About Dynamic Calc
Aliases	Assigning aliases to dimension members	About Aliases
Dynamic Members	Removing dynamic members	Considerations
Period Dimensions	Creating an alternate hierarchy in the Period dimension	Considerations for Alternate Hierarchies in Period Dimensions
Form Design	Designing forms and resolving form functional and performance issues	• Form Design • Resolving Form Functional and Performance Issues in *Operations Guide*
Smart Push	Resolving issues with Smart Push	Resolving Issues with Smart Push in *Operations Guide*
Rules Design	Designing rules to avoid common execution errors and optimizing slow rules	Troubleshooting Business Rule Errors and Performance in *Operations Guide*
Groovy Rules	Implementing and using Groovy rules and using Groovy rules to calculate data on forms	• Groovy Business Rule Tutorial Videos • Learning Groovy in Oracle Cloud EPM • Calculating Modified Data Using Groovy Rules
Groovy Rules and Smart Push	Using Groovy rules and Smart Push to move modified data into a reporting cube	Moving Modified Data Using Groovy Rules and Smart Push

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